# Useful Tables for Googology

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G.26 2005 . . . 1139
G.27 2006 . . . 1140
G.28 2007 . . . 1141
G.29 2008 . . . 1142
G.30 2009 . . . 1143
G.31 2010 . . . 1144
G.32 2011 . . . 1145
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# Appendix A

# Recursive Ordinal Tables

The results in this appendix are primarily cited from [1-3]. In this appendix, we analyze the structure of recursive ordinals by comparing important notations. The scopes of each section may have some overlap. The results in this appendix are updated to 2025.

## A.1 Natural Numbers

Natural numbers are, of course, part of recursive ordinals. Listing natural numbers was one of the most important tasks in the early development of googology, but it is no longer significant; people now focus only on a few of the most important large numbers. Therefore, this table will only cover the simplest results. For a more complete list, refer to [4-5].

| Natural Number | Name |
|---|---|
| 1 | one |
| 2 | two |
| 3 | three |
| 4 | four |
| 5 | five |
| 6 | six |
| 7 | seven |
| 8 | eight |
| 9 | nine |
| 10 | ten |
| 20 | twenty |
| 50 | fifty |
| 100 | one hundred |
| 1000 | one thousand |
| $10^4$ | ten thousand |
| $10^5$ | one hundred thousand |
| $10^{10}$ | ten billion |
| $2^{64}-1$ | Archimedes' large number |
| $6.02 \times 10^{23}$ | Avogadro constant |
| $10^{100}$ | Googol |
| $10^{7\times 2^{122}}$ | Asaṃkhyeya |
| $10^{10^{100}}$ | Googolplex |
| $e^{e^{e^{79}}}$ | first Skewes number |
| $10^{10^{10^{10^2}}}$ | Poincaré recurrence time |
| $e^{e^{e^{e^{7.705}}}}$ | second Skewes number |
| $10 \uparrow^2 10$ | |
| $3 \uparrow^3 3$ | tritri |
| $G(1) = 3 \uparrow^4 3$ | |

| Natural Number | Name |
|---|---|
| $G(64)$ | Graham's number |
| TREE(3) | |
| SSCG(13) | |
| $D^5(99)$ | Loader's number |
| $\Sigma(1919)$ | |
| $\Xi(10^6)$ | |
| Rayo($10^{100}$) | Rayo's number |

## A.2 Cantor-style vs PrSS

The results in this section are primarily cited from[1].

| Cantor-style | PrSS |
|---|---|
| $\omega$ | $(0,1)$ |
| $\omega+1$ | $(0,1,0)$ |
| $\omega+2$ | $(0,1,0,0)$ |
| $\omega+5$ | $(0,1,0,0,0,0,0)$ |
| $\omega\cdot 2$ | $(0,1,0,1)$ |
| $\omega\cdot 2+1$ | $(0,1,0,1,0)$ |
| $\omega\cdot 3$ | $(0,1,0,1,0,1)$ |
| $\omega^2$ | $(0,1,1)$ |
| $\omega^2+1$ | $(0,1,1,0)$ |
| $\omega^2+2$ | $(0,1,1,0,0)$ |
| $\omega^2+\omega$ | $(0,1,1,0,1)$ |
| $\omega^2+\omega+1$ | $(0,1,1,0,1,0)$ |
| $\omega^2+\omega\cdot 2$ | $(0,1,1,0,1,0,1)$ |
| $\omega^2\cdot 2$ | $(0,1,1,0,1,1)$ |
| $\omega^2\cdot 2+\omega$ | $(0,1,1,0,1,1,0,1)$ |
| $\omega^2\cdot 3$ | $(0,1,1,0,1,1,0,1,1)$ |
| $\omega^3$ | $(0,1,1,1)$ |
| $\omega^3+1$ | $(0,1,1,1,0)$ |
| $\omega^3+\omega$ | $(0,1,1,1,0,1)$ |
| $\omega^3+\omega^2$ | $(0,1,1,1,0,1,1)$ |
| $\omega^3\cdot 2$ | $(0,1,1,1,0,1,1,1)$ |
| $\omega^3\cdot 3$ | $(0,1,1,1,0,1,1,1,0,1,1,1)$ |
| $\omega^4$ | $(0,1,1,1,1)$ |
| $\omega^4\cdot 2$ | $(0,1,1,1,1,0,1,1,1,1)$ |
| $\omega^5$ | $(0,1,1,1,1,1)$ |
| $\omega^\omega$ | $(0,1,2)$ |
| $\omega^\omega+1$ | $(0,1,2,0)$ |
| $\omega^\omega+2$ | $(0,1,2,0,0)$ |
| $\omega^\omega+\omega$ | $(0,1,2,0,1)$ |
| $\omega^\omega+\omega\cdot 2$ | $(0,1,2,0,1,0,1)$ |
| $\omega^\omega+\omega^2$ | $(0,1,2,0,1,1)$ |
| $\omega^\omega+\omega^3$ | $(0,1,2,0,1,1,1)$ |
| $\omega^\omega\cdot 2$ | $(0,1,2,0,1,2)$ |
| $\omega^\omega\cdot 3$ | $(0,1,2,0,1,2,0,1,2)$ |
| $\omega^{\omega+1}$ | $(0,1,2,1)$ |
| $\omega^{\omega+1}+1$ | $(0,1,2,1,0)$ |
| $\omega^{\omega+1}+\omega$ | $(0,1,2,1,0,1)$ |
| $\omega^{\omega+1}+\omega^2$ | $(0,1,2,1,0,1,1)$ |
| $\omega^{\omega+1}+\omega^3$ | $(0,1,2,1,0,1,1,1)$ |
| $\omega^{\omega+1}+\omega^\omega$ | $(0,1,2,1,0,1,2)$ |
| $\omega^{\omega+1}+\omega^\omega\cdot 2$ | $(0,1,2,1,0,1,2,0,1,2)$ |
| $\omega^{\omega+1}\cdot 2$ | $(0,1,2,1,0,1,2,1)$ |

| Cantor-style | PrSS |
|---|---|
| $\omega^{\omega+1} \cdot 3$ | $(0,1,2,1,0,1,2,1,0,1,2,1)$ |
| $\omega^{\omega+2}$ | $(0,1,2,1,1)$ |
| $\omega^{\omega+2} + \omega^{\omega}$ | $(0,1,2,1,1,0,1,2)$ |
| $\omega^{\omega+2} + \omega^{\omega+1}$ | $(0,1,2,1,1,0,1,2,1)$ |
| $\omega^{\omega+3}$ | $(0,1,2,1,1,1)$ |
| $\omega^{\omega+4}$ | $(0,1,2,1,1,1)$ |
| $\omega^{\omega\cdot 2}$ | $(0,1,2,1,2)$ |
| $\omega^{\omega\cdot 2} + \omega$ | $(0,1,2,1,2,0,1)$ |
| $\omega^{\omega\cdot 2} + \omega^{\omega}$ | $(0,1,2,1,2,0,1,2)$ |
| $\omega^{\omega\cdot 2} + \omega^{\omega+1}$ | $(0,1,2,1,2,0,1,2,1)$ |
| $\omega^{\omega\cdot 2} + \omega^{\omega+2}$ | $(0,1,2,1,2,0,1,2,1,1)$ |
| $\omega^{\omega\cdot 2} \cdot 2$ | $(0,1,2,1,2,0,1,2,1,2)$ |
| $\omega^{\omega\cdot 2+1}$ | $(0,1,2,1,2,1)$ |
| $\omega^{\omega\cdot 2+1} \cdot 2$ | $(0,1,2,1,2,1,0,1,2,1,2,1)$ |
| $\omega^{\omega\cdot 2+2}$ | $(0,1,2,1,2,1,1)$ |
| $\omega^{\omega\cdot 2+3}$ | $(0,1,2,1,2,1,1,1)$ |
| $\omega^{\omega\cdot 3}$ | $(0,1,2,1,2,1,2)$ |
| $\omega^{\omega\cdot 3+1}$ | $(0,1,2,1,2,1,2,1)$ |
| $\omega^{\omega\cdot 4}$ | $(0,1,2,1,2,1,2,1,2)$ |
| $\omega^{\omega^2}$ | $(0,1,2,2)$ |
| $\omega^{\omega^2} \cdot 2$ | $(0,1,2,2,0,1,2,2)$ |
| $\omega^{\omega^2+1}$ | $(0,1,2,2,1)$ |
| $\omega^{\omega^2+2}$ | $(0,1,2,2,1,1)$ |
| $\omega^{\omega^2+\omega}$ | $(0,1,2,2,1,2)$ |
| $\omega^{\omega^2+\omega+1}$ | $(0,1,2,2,1,2,1)$ |
| $\omega^{\omega^2+\omega\cdot 2}$ | $(0,1,2,2,1,2,1,2)$ |
| $\omega^{\omega^2+\omega\cdot 3}$ | $(0,1,2,2,1,2,1,2,1,2)$ |
| $\omega^{\omega^2\cdot 2}$ | $(0,1,2,2,1,2,2)$ |
| $\omega^{\omega^2\cdot 3}$ | $(0,1,2,2,1,2,2,1,2,2)$ |
| $\omega^{\omega^3}$ | $(0,1,2,2,2)$ |
| $\omega^{\omega^3+1}$ | $(0,1,2,2,2,1)$ |
| $\omega^{\omega^3+\omega}$ | $(0,1,2,2,2,1,2)$ |
| $\omega^{\omega^3+\omega^2}$ | $(0,1,2,2,2,1,2,2)$ |
| $\omega^{\omega^3\cdot 2}$ | $(0,1,2,2,2,1,2,2,2)$ |
| $\omega^{\omega^4}$ | $(0,1,2,2,2,2)$ |
| $\omega^{\omega^4\cdot 2}$ | $(0,1,2,2,2,2,1,2,2,2,2)$ |
| $\omega^{\omega^5}$ | $(0,1,2,2,2,2,2)$ |
| $\omega^{\omega^{\omega}}$ | $(0,1,2,3)$ |
| $\omega^{\omega^{\omega}} + \omega$ | $(0,1,2,3,0,1)$ |
| $\omega^{\omega^{\omega}} + \omega^{\omega}$ | $(0,1,2,3,0,1,2)$ |
| $\omega^{\omega^{\omega}} \cdot 2$ | $(0,1,2,3,0,1,2,3)$ |
| $\omega^{\omega^{\omega}+1}$ | $(0,1,2,3,1)$ |
| $\omega^{\omega^{\omega}+\omega}$ | $(0,1,2,3,1,2)$ |
| $\omega^{\omega^{\omega}\cdot 2}$ | $(0,1,2,3,1,2,3)$ |
| $\omega^{\omega^{\omega+1}}$ | $(0,1,2,3,2)$ |
| $\omega^{\omega^{\omega\cdot 2}}$ | $(0,1,2,3,2,3)$ |
| $\omega^{\omega^{\omega^2}}$ | $(0,1,2,3,3)$ |
| $\omega^{\omega^{\omega^{\omega}}}$ | $(0,1,2,3,4)$ |
| $\omega^{\omega^{\omega^{\omega}}} + 1$ | $(0,1,2,3,4,0)$ |
| $\omega^{\omega^{\omega^{\omega}}+1}$ | $(0,1,2,3,4,1)$ |
| $\omega^{\omega^{\omega^{\omega}+1}}$ | $(0,1,2,3,4,2)$ |

| Cantor-style | PrSS |
|---|---|
| $\omega^{\omega^{\omega^{\omega+1}}}$ | $(0,1,2,3,4,3)$ |
| $\omega^{\omega^{\omega^{\omega^2}}}$ | $(0,1,2,3,4,4)$ |
| $\omega^{\omega^{\omega^{\omega^{\omega}}}}$ | $(0,1,2,3,4,5)$ |
| $\varepsilon_0 = \omega^{\omega^{\omega^{\cdots}}}$ | $(0,1,2,3,\ldots)$ |

## A.3 Veblen Function vs Cantor-style

| Veblen function | Cantor-style |
|---|---|
| $\varphi(1)$ | $\omega$ |
| $\varphi(1)+1$ | $\omega+1$ |
| $\varphi(1)+2$ | $\omega+2$ |
| $\varphi(1)\cdot 2$ | $\omega\cdot 2$ |
| $\varphi(1)\cdot 2+1$ | $\omega\cdot 2+1$ |
| $\varphi(1)\cdot 3$ | $\omega\cdot 3$ |
| $\varphi(2)$ | $\omega^2$ |
| $\varphi(2)+1$ | $\omega^2+1$ |
| $\varphi(2)+2$ | $\omega^2+2$ |
| $\varphi(2)+\varphi(1)$ | $\omega^2+\omega$ |
| $\varphi(2)+\varphi(1)+1$ | $\omega^2+\omega+1$ |
| $\varphi(2)+\varphi(1)\cdot 2$ | $\omega^2+\omega\cdot 2$ |
| $\varphi(2)\cdot 2$ | $\omega^2\cdot 2$ |
| $\varphi(2)\cdot 2+\varphi(1)$ | $\omega^2\cdot 2+\omega$ |
| $\varphi(2)\cdot 3$ | $\omega^2\cdot 3$ |
| $\varphi(3)$ | $\omega^3$ |
| $\varphi(3)+1$ | $\omega^3+1$ |
| $\varphi(3)+\varphi(1)$ | $\omega^3+\omega$ |
| $\varphi(3)+\varphi(2)$ | $\omega^3+\omega^2$ |
| $\varphi(3)\cdot 2$ | $\omega^3\cdot 2$ |
| $\varphi(3)\cdot 3$ | $\omega^3\cdot 3$ |
| $\varphi(4)$ | $\omega^4$ |
| $\varphi(4)\cdot 2$ | $\omega^4\cdot 2$ |
| $\varphi(5)$ | $\omega^5$ |
| $\varphi(\varphi(1))$ | $\omega^\omega$ |
| $\varphi(\varphi(1))+1$ | $\omega^\omega+1$ |
| $\varphi(\varphi(1))+2$ | $\omega^\omega+2$ |
| $\varphi(\varphi(1))+\varphi(1)$ | $\omega^\omega+\omega$ |
| $\varphi(\varphi(1))+\varphi(1)\cdot 2$ | $\omega^\omega+\omega\cdot 2$ |
| $\varphi(\varphi(1))+\varphi(2)$ | $\omega^\omega+\omega^2$ |
| $\varphi(\varphi(1))+\varphi(3)$ | $\omega^\omega+\omega^3$ |
| $\varphi(\varphi(1))\cdot 2$ | $\omega^\omega\cdot 2$ |
| $\varphi(\varphi(1))\cdot 3$ | $\omega^\omega\cdot 3$ |
| $\varphi(\varphi(1)+1)$ | $\omega^{\omega+1}$ |
| $\varphi(\varphi(1)+1)+1$ | $\omega^{\omega+1}+1$ |
| $\varphi(\varphi(1)+1)+\varphi(1)$ | $\omega^{\omega+1}+\omega$ |
| $\varphi(\varphi(1)+1)+\varphi(2)$ | $\omega^{\omega+1}+\omega^2$ |
| $\varphi(\varphi(1)+1)+\varphi(3)$ | $\omega^{\omega+1}+\omega^3$ |
| $\varphi(\varphi(1)+1)+\varphi(\varphi(1))$ | $\omega^{\omega+1}+\omega^\omega$ |
| $\varphi(\varphi(1)+1)+\varphi(\varphi(1))\cdot 2$ | $\omega^{\omega+1}+\omega^\omega\cdot 2$ |
| $\varphi(\varphi(1)+1)\cdot 2$ | $\omega^{\omega+1}\cdot 2$ |
| $\varphi(\varphi(1)+1)\cdot 3$ | $\omega^{\omega+1}\cdot 3$ |
| $\varphi(\varphi(1)+2)$ | $\omega^{\omega+2}$ |
| $\varphi(\varphi(1)+2)+\varphi(\varphi(1))$ | $\omega^{\omega+2}+\omega^\omega$ |

| Veblen function | Cantor-style |
|---|---|
| $\varphi(\varphi(1)+2)+\varphi(\varphi(1)+1)$ | $\omega^{\omega+2}+\omega^{\omega+1}$ |
| $\varphi(\varphi(1)+3)$ | $\omega^{\omega+3}$ |
| $\varphi(\varphi(1)+4)$ | $\omega^{\omega+4}$ |
| $\varphi(\varphi(1)\cdot 2)$ | $\omega^{\omega\cdot 2}$ |
| $\varphi(\varphi(1)\cdot 2)+\varphi(1)$ | $\omega^{\omega\cdot 2}+\omega$ |
| $\varphi(\varphi(1)\cdot 2)+\varphi(\varphi(1))$ | $\omega^{\omega\cdot 2}+\omega^{\omega}$ |
| $\varphi(\varphi(1)\cdot 2)+\varphi(\varphi(1)+1)$ | $\omega^{\omega\cdot 2}+\omega^{\omega+1}$ |
| $\varphi(\varphi(1)\cdot 2)+\varphi(\varphi(1)+2)$ | $\omega^{\omega\cdot 2}+\omega^{\omega+2}$ |
| $\varphi(\varphi(1)\cdot 2)\cdot 2$ | $\omega^{\omega\cdot 2}\cdot 2$ |
| $\varphi(\varphi(1)\cdot 2+1)$ | $\omega^{\omega\cdot 2+1}$ |
| $\varphi(\varphi(1)\cdot 2+1)\cdot 2$ | $\omega^{\omega\cdot 2+1}\cdot 2$ |
| $\varphi(\varphi(1)\cdot 2+2)$ | $\omega^{\omega\cdot 2+2}$ |
| $\varphi(\varphi(1)\cdot 2+3)$ | $\omega^{\omega\cdot 2+3}$ |
| $\varphi(\varphi(1)\cdot 3)$ | $\omega^{\omega\cdot 3}$ |
| $\varphi(\varphi(1)\cdot 3+1)$ | $\omega^{\omega\cdot 3+1}$ |
| $\varphi(\varphi(1)\cdot 4)$ | $\omega^{\omega\cdot 4}$ |
| $\varphi(\varphi(2))$ | $\omega^{\omega^2}$ |
| $\varphi(\varphi(2))\cdot 2$ | $\omega^{\omega^2}\cdot 2$ |
| $\varphi(\varphi(2)+1)$ | $\omega^{\omega^2+1}$ |
| $\varphi(\varphi(2)+2)$ | $\omega^{\omega^2+2}$ |
| $\varphi(\varphi(2)+\varphi(1))$ | $\omega^{\omega^2+\omega}$ |
| $\varphi(\varphi(2)+\varphi(1)+1)$ | $\omega^{\omega^2+\omega+1}$ |
| $\varphi(\varphi(2)+\varphi(1)\cdot 2)$ | $\omega^{\omega^2+\omega\cdot 2}$ |
| $\varphi(\varphi(2)+\varphi(1)\cdot 3)$ | $\omega^{\omega^2+\omega\cdot 3}$ |
| $\varphi(\varphi(2)\cdot 2)$ | $\omega^{\omega^2\cdot 2}$ |
| $\varphi(\varphi(2)\cdot 3)$ | $\omega^{\omega^2\cdot 3}$ |
| $\varphi(\varphi(3))$ | $\omega^{\omega^3}$ |
| $\varphi(\varphi(3)+1)$ | $\omega^{\omega^3+1}$ |
| $\varphi(\varphi(3)+\varphi(1))$ | $\omega^{\omega^3+\omega}$ |
| $\varphi(\varphi(3)+\varphi(2))$ | $\omega^{\omega^3+\omega^2}$ |
| $\varphi(\varphi(3)\cdot 2)$ | $\omega^{\omega^3\cdot 2}$ |
| $\varphi(\varphi(4))$ | $\omega^{\omega^4}$ |
| $\varphi(\varphi(4)\cdot 2)$ | $\omega^{\omega^4\cdot 2}$ |
| $\varphi(\varphi(5))$ | $\omega^{\omega^5}$ |
| $\varphi(\varphi(\varphi(1)))$ | $\omega^{\omega^{\omega}}$ |
| $\varphi(\varphi(\varphi(1)))+\varphi(1)$ | $\omega^{\omega^{\omega}}+\omega$ |
| $\varphi(\varphi(\varphi(1)))+\varphi(\varphi(1))$ | $\omega^{\omega^{\omega}}+\omega^{\omega}$ |
| $\varphi(\varphi(\varphi(1)))\cdot 2$ | $\omega^{\omega^{\omega}}\cdot 2$ |
| $\varphi(\varphi(\varphi(1))+1)$ | $\omega^{\omega^{\omega}+1}$ |
| $\varphi(\varphi(\varphi(1))+\varphi(1))$ | $\omega^{\omega^{\omega}+\omega}$ |
| $\varphi(\varphi(\varphi(1))\cdot 2)$ | $\omega^{\omega^{\omega}\cdot 2}$ |
| $\varphi(\varphi(\varphi(1)+1))$ | $\omega^{\omega^{\omega+1}}$ |
| $\varphi(\varphi(\varphi(1)\cdot 2))$ | $\omega^{\omega^{\omega\cdot 2}}$ |
| $\varphi(\varphi(\varphi(2)))$ | $\omega^{\omega^{\omega^2}}$ |
| $\varphi(\varphi(\varphi(\varphi(1))))$ | $\omega^{\omega^{\omega^{\omega}}}$ |
| $\varphi(\varphi(\varphi(\varphi(1))))+1$ | $\omega^{\omega^{\omega^{\omega}}}+1$ |
| $\varphi(\varphi(\varphi(\varphi(1)))+1)$ | $\omega^{\omega^{\omega^{\omega}}+1}$ |
| $\varphi(\varphi(\varphi(\varphi(1))+1))$ | $\omega^{\omega^{\omega^{\omega}+1}}$ |
| $\varphi(\varphi(\varphi(\varphi(1)+1)))$ | $\omega^{\omega^{\omega^{\omega+1}}}$ |

| Veblen function | Cantor-style |
|---|---|
| $\varphi(\varphi(\varphi(\varphi(2))))$ | $\omega^{\omega^{\omega^{\omega^2}}}$ |
| $\varphi(\varphi(\varphi(\varphi(\varphi(1)))))$ | $\omega^{\omega^{\omega^{\omega^{\omega}}}}$ |
| $\varphi(1,0)$ | $\varepsilon_0$ |
| $\varphi(1,0)+1$ | $\varepsilon_0+1$ |
| $\varphi(1,0)+\varphi(1)$ | $\varepsilon_0+\omega$ |
| $\varphi(1,0)+\varphi(\varphi(1))$ | $\varepsilon_0+\omega^{\omega}$ |
| $\varphi(1,0)+\varphi(\varphi(\varphi(1)))$ | $\varepsilon_0+\omega^{\omega^{\omega}}$ |
| $\varphi(1,0)\cdot 2$ | $\varepsilon_0\cdot 2$ |
| $\varphi(1,0)\cdot 3$ | $\varepsilon_0\cdot 3$ |
| $\varphi(\varphi(1,0)+1)$ | $\varepsilon_0\cdot\omega$<br>$\omega^{\varepsilon_0+1}$ |
| $\varphi(\varphi(1,0)+1)+\varphi(1)$ | $\varepsilon_0\cdot\omega+\omega$<br>$\omega^{\varepsilon_0+1}+\omega$ |
| $\varphi(\varphi(1,0)+1)+\varphi(\varphi(1))$ | $\varepsilon_0\cdot\omega+\omega^{\omega}$<br>$\omega^{\varepsilon_0+1}+\omega^{\omega}$ |
| $\varphi(\varphi(1,0)+1)\cdot 2$ | $\varepsilon_0\cdot\omega\cdot 2$<br>$\omega^{\varepsilon_0+1}\cdot 2$ |
| $\varphi(\varphi(1,0)+2)$ | $\varepsilon_0\cdot\omega^2$<br>$\omega^{\varepsilon_0+2}$ |
| $\varphi(\varphi(1,0)+2)\cdot 2$ | $\varepsilon_0\cdot\omega^2\cdot 2$<br>$\omega^{\varepsilon_0+2}\cdot 2$ |
| $\varphi(\varphi(1,0)+3)$ | $\varepsilon_0\cdot\omega^3$<br>$\omega^{\varepsilon_0+3}$ |
| $\varphi(\varphi(1,0)+\varphi(1))$ | $\varepsilon_0\cdot\omega^{\omega}$<br>$\omega^{\varepsilon_0+\omega}$ |
| $\varphi(\varphi(1,0)+\varphi(2))$ | $\varepsilon_0\cdot\omega^{\omega^2}$<br>$\omega^{\varepsilon_0+\omega^2}$ |
| $\varphi(\varphi(1,0)+\varphi(\varphi(1)))$ | $\varepsilon_0\cdot\omega^{\omega^{\omega}}$<br>$\omega^{\varepsilon_0+\omega^{\omega}}$ |
| $\varphi(\varphi(1,0)\cdot 2)$ | ${\varepsilon_0}^2$<br>$\omega^{\varepsilon_0\cdot 2}$ |
| $\varphi(\varphi(1,0)\cdot 2)+\varphi(1)$ | ${\varepsilon_0}^2+\omega$<br>$\omega^{\varepsilon_0\cdot 2}+\omega$ |
| $\varphi(\varphi(1,0)\cdot 2)+\varphi(\varphi(1))$ | ${\varepsilon_0}^2+\omega^{\omega}$<br>$\omega^{\varepsilon_0\cdot 2}+\omega^{\omega}$ |
| $\varphi(\varphi(1,0)\cdot 2)+\varphi(1,0)$ | ${\varepsilon_0}^2+\varepsilon_0$<br>$\omega^{\varepsilon_0\cdot 2}+\varepsilon_0$ |
| $\varphi(\varphi(1,0)\cdot 2)+\varphi(\varphi(1,0)+1)$ | ${\varepsilon_0}^2+\varepsilon_0\cdot\omega$<br>$\omega^{\varepsilon_0\cdot 2}+\omega^{\varepsilon_0+1}$ |
| $\varphi(\varphi(1,0)\cdot 2)\cdot 2$ | ${\varepsilon_0}^2\cdot 2$<br>$\omega^{\varepsilon_0\cdot 2}\cdot 2$ |
| $\varphi(\varphi(1,0)\cdot 2+1)$ | ${\varepsilon_0}^2\cdot\omega$<br>$\omega^{\varepsilon_0\cdot 2+1}$ |
| $\varphi(\varphi(1,0)\cdot 2+\varepsilon(1))$ | ${\varepsilon_0}^2\cdot\omega^{\omega}$<br>$\omega^{\varepsilon_0\cdot 2+\omega}$ |
| $\varphi(\varphi(1,0)\cdot 3)$ | ${\varepsilon_0}^3$<br>$\omega^{\varepsilon_0\cdot 3}$ |
| $\varphi(\varphi(1,0)\cdot 4)$ | ${\varepsilon_0}^4$<br>$\omega^{\varepsilon_0\cdot 4}$ |

| Veblen function | Cantor-style |
|---|---|
| $\varphi(\varphi(\varphi(1,0)+1))$ | ${\varepsilon_0}^{\omega}$<br>$\omega^{\omega^{\varepsilon_0+1}}$ |
| $\varphi(\varphi(\varphi(1,0)+1))\cdot 2$ | ${\varepsilon_0}^{\omega}\cdot 2$<br>$\omega^{\omega^{\varepsilon_0+1}}\cdot 2$ |
| $\varphi(\varphi(\varphi(1,0)+1)+1)$ | ${\varepsilon_0}^{\omega}\cdot\omega$<br>$\omega^{\omega^{\varepsilon_0+1}+1}$ |
| $\varphi(\varphi(\varphi(1,0)+1)+\varphi(1))$ | ${\varepsilon_0}^{\omega}\cdot\omega^{\omega}$<br>$\omega^{\omega^{\varepsilon_0+1}+\omega}$ |
| $\varphi(\varphi(\varphi(1,0)+1)+\varphi(1,0))$ | ${\varepsilon_0}^{\omega+1}$<br>$\omega^{\omega^{\varepsilon_0+1}+\varepsilon_0}$ |
| $\varphi(\varphi(\varphi(1,0)+1)+\varphi(1,0)\cdot 2)$ | ${\varepsilon_0}^{\omega+2}$<br>$\omega^{\omega^{\varepsilon_0+1}+\varepsilon_0\cdot 2}$ |
| $\varphi(\varphi(\varphi(1,0)+1)\cdot 2)$ | ${\varepsilon_0}^{\omega\cdot 2}$<br>$\omega^{\omega^{\varepsilon_0+1}\cdot 2}$ |
| $\varphi(\varphi(\varphi(1,0)+2))$ | ${\varepsilon_0}^{\omega^2}$<br>$\omega^{\omega^{\varepsilon_0+2}}$ |
| $\varphi(\varphi(\varphi(1,0)+\varphi(1)))$ | $\omega^{\omega^{\varepsilon_0+\omega}}$ |
| $\varphi(\varphi(\varphi(1,0)+\varphi(\varphi(1))))$ | ${\varepsilon_0}^{\omega^{\omega}}$<br>$\omega^{\omega^{\varepsilon_0+\omega^{\omega}}}$ |
| $\varphi(\varphi(\varphi(1,0)\cdot 2))$ | ${\varepsilon_0}^{\varepsilon_0}$<br>$\omega^{\omega^{\varepsilon_0\cdot 2}}$ |
| $\varphi(\varphi(\varphi(1,0)\cdot 2)+1)$ | ${\varepsilon_0}^{\varepsilon_0}\cdot\omega$<br>$\omega^{\omega^{\varepsilon_0\cdot 2}+1}$ |
| $\varphi(\varphi(\varphi(1,0)\cdot 2)+\varphi(1,0))$ | ${\varepsilon_0}^{\varepsilon_0+1}$<br>$\omega^{\omega^{\varepsilon_0\cdot 2}+\varepsilon_0}$ |
| $\varphi(\varphi(\varphi(1,0)\cdot 2)+\varphi(\varphi(1,0)+1))$ | ${\varepsilon_0}^{\varepsilon_0+\omega}$<br>$\omega^{\omega^{\varepsilon_0\cdot 2}+\omega^{\varepsilon_0+1}}$ |
| $\varphi(\varphi(\varphi(1,0)\cdot 2)\cdot 2)$ | ${\varepsilon_0}^{\varepsilon_0\cdot 2}$<br>$\omega^{\omega^{\varepsilon_0\cdot 2}\cdot 2}$ |
| $\varphi(\varphi(\varphi(1,0)\cdot 2+1))$ | ${\varepsilon_0}^{\varepsilon_0\cdot\omega}$<br>$\omega^{\omega^{\varepsilon_0\cdot 2+1}}$ |
| $\varphi(\varphi(\varphi(1,0)\cdot 3))$ | ${\varepsilon_0}^{{\varepsilon_0}^2}$<br>$\omega^{\omega^{\varepsilon_0\cdot 3}}$ |
| $\varphi(\varphi(\varphi(\varphi(1,0)+1)))$ | ${\varepsilon_0}^{{\varepsilon_0}^{\omega}}$<br>$\omega^{\omega^{\omega^{\varepsilon_0+1}}}$ |
| $\varphi(\varphi(\varphi(\varphi(1,0)\cdot 2)))$ | ${\varepsilon_0}^{{\varepsilon_0}^{\varepsilon_0}}$<br>$\omega^{\omega^{\omega^{\varepsilon_0\cdot 2}}}$ |
| $\varphi(1,1)$ | $\varepsilon_1$ |
| $\varphi(1,1)+\varphi(1,0)$ | $\varepsilon_1+\varepsilon_0$ |
| $\varphi(1,1)+\varphi(\varphi(1,0))$ | $\varepsilon_1+\varepsilon_0\cdot\omega$<br>$\varepsilon_1+\omega^{\varepsilon_0+1}$ |
| $\varphi(1,1)+\varphi(\varphi(\varphi(1,0)))$ | $\varepsilon_1+{\varepsilon_0}^{\omega}$<br>$\varepsilon_1+\omega^{\omega^{\varepsilon_0+1}}$ |
| $\varphi(1,1)\cdot 2$ | $\varepsilon_1\cdot 2$ |
| $\varphi(\varphi(1,1)+1)$ | $\varepsilon_1\cdot\omega$<br>$\omega^{\varepsilon_1+1}$ |

| Veblen function | Cantor-style |
|---|---|
| $\varphi(\varphi(1,1)+\varphi(1,0))$ | $\varepsilon_1 \cdot \varepsilon_0$<br>$\omega^{\varepsilon_1+\varepsilon_0}$ |
| $\varphi(\varphi(1,1)+\varphi(\varphi(1,0)\cdot 2))$ | ${\varepsilon_1 \cdot \varepsilon_0}^2$<br>$\omega^{\varepsilon_1+\omega^{\varepsilon_0\cdot 2}}$ |
| $\varphi(\varphi(1,1)\cdot 2)$ | ${\varepsilon_1}^2$<br>$\omega^{\varepsilon_1\cdot 2}$ |
| $\varphi(\varphi(1,1)\cdot 3)$ | ${\varepsilon_1}^3$<br>$\omega^{\varepsilon_1\cdot 3}$ |
| $\varphi(\varphi(\varphi(1,1)+1))$ | ${\varepsilon_1}^\omega$<br>$\omega^{\omega^{\varepsilon_1+1}}$ |
| $\varphi(\varphi(\varphi(1,1)+\varphi(1,0)))$ | ${\varepsilon_1}^{\varepsilon_0}$<br>$\omega^{\omega^{\varepsilon_1+\varepsilon_0}}$ |
| $\varphi(\varphi(\varphi(1,1)\cdot 2))$ | ${\varepsilon_1}^{\varepsilon_1}$<br>$\omega^{\omega^{\varepsilon_1\cdot 2}}$ |
| $\varphi(1,2)$ | $\varepsilon_2$ |
| $\varphi(\varphi(1,2)+1)$ | $\varepsilon_2 \cdot \omega$<br>$\omega^{\varepsilon_2+1}$ |
| $\varphi(\varphi(1,2)+\varphi(1,0))$ | $\varepsilon_2 \cdot \varepsilon_0$<br>$\omega^{\varepsilon_2+\varepsilon_0}$ |
| $\varphi(\varphi(1,2)+\varphi(1,1))$ | $\varepsilon_2 \cdot \varepsilon_1$<br>$\omega^{\varepsilon_2+\varepsilon_1}$ |
| $\varphi(\varphi(1,2)\cdot 2)$ | ${\varepsilon_2}^2$<br>$\omega^{\varepsilon_2\cdot 2}$ |
| $\varphi(\varphi(\varphi(1,2)+1))$ | ${\varepsilon_2}^\omega$<br>$\omega^{\omega^{\varepsilon_2+1}}$ |
| $\varphi(\varphi(\varphi(1,2)\cdot 2))$ | ${\varepsilon_2}^{\varepsilon_2}$<br>$\omega^{\omega^{\varepsilon_2\cdot 2}}$ |
| $\varphi(1,3)$ | $\varepsilon_3$ |
| $\varphi(\varphi(1,3)\cdot 2)$ | ${\varepsilon_3}^2$<br>$\omega^{\varepsilon_3\cdot 2}$ |
| $\varphi(1,4)$ | $\varepsilon_4$ |
| $\varphi(1,5)$ | $\varepsilon_5$ |
| $\varphi(1,\varphi(1))$ | $\varepsilon_\omega$ |
| $\varphi(\varphi(1,\varphi(1))\cdot 2)$ | ${\varepsilon_\omega}^2$<br>$\omega^{\varepsilon_\omega\cdot 2}$ |
| $\varphi(1,\varphi(1)+1)$ | $\varepsilon_{\omega+1}$ |
| $\varphi(1,\varphi(1)+2)$ | $\varepsilon_{\omega+2}$ |
| $\varphi(1,\varphi(1)\cdot 2)$ | $\varepsilon_{\omega\cdot 2}$ |
| $\varphi(1,\varphi(2))$ | $\varepsilon_{\omega^2}$ |
| $\varphi(1,\varphi(\varphi(1)))$ | $\varepsilon_{\omega^\omega}$ |
| $\varphi(1,\varphi(\varphi(\varphi(1))))$ | $\varepsilon_{\omega^{\omega^\omega}}$ |
| $\varphi(1,\varphi(1,0))$ | $\varepsilon_{\varepsilon_0}$ |
| $\varphi(1,\varphi(1,0)+1)$ | $\varepsilon_{\varepsilon_0+1}$ |
| $\varphi(1,\varphi(1,0)\cdot 2)$ | $\varepsilon_{\varepsilon_0\cdot 2}$ |
| $\varphi(1,\varphi(\varphi(1,0)+1))$ | $\varepsilon_{\varepsilon_0\cdot\omega}$<br>$\varepsilon_{\omega^{\varepsilon_0}+1}$ |
| $\varphi(1,\varphi(\varphi(1,0)\cdot 2))$ | ${\varepsilon_0}^2$<br>$\varepsilon_{\omega^{\varepsilon_0\cdot 2}}$ |
| $\varphi(1,\varphi(1,1))$ | $\varepsilon_{\varepsilon_1}$ |
| $\varphi(1,\varphi(1,2))$ | $\varepsilon_{\varepsilon_2}$ |

| Veblen function | Cantor-style |
|---|---|
| $\varphi(1,\varphi(1,\varphi(0)))$ | $\varepsilon_{\varepsilon_\omega}$ |
| $\varphi(1,\varphi(1,\varphi(1,0)))$ | $\varepsilon_{\varepsilon_{\varepsilon_0}}$ |
| $\varphi(2,0)$ | $\zeta_0$ |
| $\varphi(2,0)+\varphi(1)$ | $\zeta_0+\omega$ |
| $\varphi(2,0)+\varphi(1,0)$ | $\zeta_0+\varepsilon_0$ |
| $\varphi(2,0)+\varphi(1,\varphi(1,0))$ | $\zeta_0+\varepsilon_{\varepsilon_0}$ |
| $\varphi(2,0)\cdot 2$ | $\zeta_0\cdot 2$ |
| $\varphi(\varphi(2,0)+1)$ | $\zeta_0\cdot\omega$<br>$\omega^{\zeta_0+1}$ |
| $\varphi(\varphi(2,0)+\varphi(1,0))$ | $\zeta_0\cdot\varepsilon_0$<br>$\omega^{\zeta_0+\varepsilon_0}$ |
| $\varphi(\varphi(2,0)+\varphi(1,\varphi(1,0)))$ | $\zeta_0\cdot\varepsilon_{\varepsilon_0}$<br>$\omega^{\zeta_0+\varepsilon_{\varepsilon_0}}$ |
| $\varphi(\varphi(2,0)\cdot 2)$ | ${\zeta_0}^2$<br>$\omega^{\zeta_0\cdot 2}$ |
| $\varphi(\varphi(\varphi(2,0)+1))$ | ${\zeta_0}^\omega$<br>$\omega^{\omega^{\zeta_0+1}}$ |
| $\varphi(\varphi(\varphi(2,0)+\varphi(1,0)))$ | ${\zeta_0}^{\varepsilon_0}$<br>$\omega^{\omega^{\zeta_0+\varepsilon_0}}$ |
| $\varphi(\varphi(\varphi(2,0)\cdot 2))$ | ${\zeta_0}^{\zeta_0}$<br>$\omega^{\omega^{\zeta_0\cdot 2}}$ |
| $\varphi(\varphi(\varphi(\varphi(2,0)\cdot 2)))$ | ${\zeta_0}^{{\zeta_0}^{\zeta_0}}$<br>$\omega^{\omega^{\omega^{\zeta_0\cdot 2}}}$ |
| $\varphi(1,\varphi(2,0)+1)$ | $\varepsilon_{\zeta_0+1}$ |
| $\varphi(\varphi(1,\varphi(2,0)+1)\cdot 2)$ | ${\varepsilon_{\zeta_0+1}}^2$<br>$\omega^{\varepsilon_{\zeta_0+1}\cdot 2}$ |
| $\varphi(\varphi(\varphi(1,\varphi(2,0)+1)\cdot 2))$ | ${\varepsilon_{\zeta_0+1}}^{\varepsilon_{\zeta_0+1}}$<br>$\omega^{\omega^{\varepsilon_{\zeta_0+1}\cdot 2}}$ |
| $\varphi(1,\varphi(2,0)+2)$ | $\varepsilon_{\zeta_0+2}$ |
| $\varphi(1,\varphi(2,0)+3)$ | $\varepsilon_{\zeta_0+3}$ |
| $\varphi(1,\varphi(2,0)+\varphi(0))$ | $\varepsilon_{\zeta_0+\omega}$ |
| $\varphi(1,\varphi(2,0)+\varphi(1,0))$ | $\varepsilon_{\zeta_0+\varepsilon_0}$ |
| $\varphi(1,\varphi(2,0)+\varphi(1,\varphi(1,0)))$ | $\varepsilon_{\zeta_0+\varepsilon_{\varepsilon_0}}$ |
| $\psi(\Omega+\varphi(2,0))$ | $\varepsilon_{\zeta_0\cdot 2}$ |
| $\varphi(1,\varphi(2,0)\cdot 2)$ | $\varepsilon_{\zeta_0\cdot 3}$ |
| $\varphi(1,\varphi(\varphi(2,0)+1))$ | $\varepsilon_{\zeta_0\cdot\omega}$<br>$\varepsilon_{\omega^{\zeta_0+1}}$ |
| $\varphi(1,\varphi(\varphi(2,0)\cdot 2))$ | $\varepsilon_{{\zeta_0}^2}$<br>$\varepsilon_{\omega^{\zeta_0\cdot 2}}$ |
| $\varphi(1,\varphi(\varphi(\varphi(2,0)\cdot 2)))$ | $\varepsilon_{{\zeta_0}^{\zeta_0}}$<br>$\varepsilon_{\omega^{\omega^{\zeta_0\cdot 2}}}$ |
| $\varphi(1,\varphi(1,\varphi(2,0)+1))$ | $\varepsilon_{\varepsilon_{\zeta_0+1}}$ |
| $\varphi(1,\varphi(1,\varphi(2,0)+\varphi(1,0)))$ | $\varepsilon_{\varepsilon_{\zeta_0+\varepsilon_0}}$ |
| $\varphi(1,\varphi(1,\varphi(2,0)\cdot 2)))$ | $\varepsilon_{\varepsilon_{\zeta_0\cdot 2}}$ |
| $\varphi(1,\varphi(1,\varphi(1,\varphi(2,0)+1))))$ | $\varepsilon_{\varepsilon_{\varepsilon_{\zeta_0+1}}}$ |
| $\varphi(2,1)$ | $\zeta_1$ |
| $\varphi(\varphi(2,1)+1)$ | $\zeta_1\cdot\omega$<br>$\omega^{\zeta_1+1}$ |
| $\varphi(\varphi(2,1)+\varphi(2,0))$ | $\zeta_1\cdot\zeta_0$<br>$\omega^{\zeta_1+\zeta_0}$ |

| Veblen function | Cantor-style |
|---|---|
| $\varphi(\varphi(2,1)\cdot 2)$ | ${\zeta_1}^2$<br>$\omega^{\zeta_1\cdot 2}$ |
| $\varphi(\varphi(\varphi(2,1)\cdot 2))$ | ${\zeta_1}^{\zeta_1}$<br>$\omega^{\omega^{\zeta_1\cdot 2}}$ |
| $\varphi(1,\varphi(2,1)+1)$ | $\varepsilon_{\zeta_1+1}$ |
| $\varphi(1,\varphi(2,1)+\varphi(2,0))$ | $\varepsilon_{\zeta_1+\zeta_0}$ |
| $\varphi(1,\varphi(2,1)\cdot 2)$ | $\varepsilon_{\zeta_1\cdot 2}$ |
| $\varphi(1,\varphi(\varphi(2,1)\cdot 2))$ | $\varepsilon_{{\zeta_1}^2}$<br>$\varepsilon_{\omega^{\zeta_1\cdot 2}}$ |
| $\varphi(1,\varphi(1,\varphi(2,1)+1))$ | $\varepsilon_{\varepsilon_{\zeta_1+1}}$ |
| $\varphi(2,2)$ | $\zeta_2$ |
| $\varphi(\varphi(2,2)+1)\cdot\omega$ | $\zeta_2\cdot\omega$<br>$\omega^{\zeta_2+1}$ |
| $\varphi(1,\varphi(2,2)+1)$ | $\varepsilon_{\zeta_2+1}$ |
| $\varphi(1,\varphi(2,2)\cdot 2)$ | $\varepsilon_{\zeta_2\cdot 2}$ |
| $\varphi(1,\varphi(1,\varphi(2,2)+1))$ | $\varepsilon_{\varepsilon_{\zeta_2+1}}$ |
| $\varphi(2,3)$ | $\zeta_3$ |
| $\varphi(2,4)$ | $\zeta_4$ |
| $\varphi(2,\varphi(1))$ | $\zeta_\omega$ |
| $\varphi(2,\varphi(1,0))$ | $\zeta_{\varepsilon_0}$ |
| $\varphi(2,\varphi(1,\varphi(1,0)))$ | $\zeta_{\varepsilon_{\varepsilon_0}}$ |
| $\varphi(2,\varphi(2,0))$ | $\zeta_{\zeta_0}$ |
| $\varphi(2,\varphi(2,1))$ | $\zeta_{\zeta_1}$ |
| $\varphi(2,\varphi(2,\varphi(1,0)))$ | $\zeta_{\zeta_{\varepsilon_0}}$ |
| $\varphi(2,\varphi(2,\varphi(2,0)))$ | $\zeta_{\zeta_{\zeta_0}}$ |
| $\varphi(3,0)$ | $\eta_0$ |
| $\varphi(\varphi(3,0)+1)$ | $\eta_0\cdot\omega$<br>$\omega^{\eta_0+1}$ |
| $\varphi(\varphi(\varphi(\varphi(3,0)\cdot 2)))$ | ${\eta_0}^{\eta_0}$<br>$\omega^{\omega^{\eta_0\cdot 2}}$ |
| $\varphi(1,\varphi(3,0)+1)$ | $\varepsilon_{\eta_0+1}$ |
| $\varphi(1,\varphi(3,0)\cdot 2)$ | $\varepsilon_{\eta_0\cdot 2}$ |
| $\varphi(1,\varphi(1,\varphi(3,0)\cdot 2))$ | $\varepsilon_{\varepsilon_{\eta_0\cdot 2}}$ |
| $\varphi(2,\varphi(3,0)+1)$ | $\zeta_{\eta_0+1}$ |
| $\varphi(2,\varphi(3,0)\cdot 2)$ | $\zeta_{\eta_0\cdot 2}$ |
| $\varphi(2,\varphi(2,\varphi(3,0)\cdot 2))$ | $\zeta_{\zeta_{\eta_0\cdot 2}}$ |
| $\varphi(3,1)$ | $\eta_1$ |
| $\varphi(\varphi(3,1)+1)$ | $\eta_1\cdot\omega$<br>$\omega^{\eta_1+1}$ |
| $\varphi(1,\varphi(3,1)+1)$ | $\varepsilon_{\eta_1+1}$ |
| $\varphi(2,\varphi(3,1)+1)$ | $\zeta_{\eta_1+1}$ |
| $\varphi(2,\varphi(2,\varphi(3,1)+1))$ | $\zeta_{\zeta_{\eta_1+1}}$ |
| $\varphi(3,2)$ | $\eta_2$ |
| $\varphi(3,3)$ | $\eta_3$ |
| $\varphi(3,\varphi(1))$ | $\eta_\omega$ |
| $\varphi(3,\varphi(1,0))$ | $\eta_{\varepsilon_0}$ |
| $\varphi(3,\varphi(2,0))$ | $\eta_{\zeta_0}$ |
| $\varphi(3,\varphi(3,0))$ | $\eta_{\eta_0}$ |
| $\varphi(3,\varphi(3,\varphi(3,0)))$ | $\eta_{\eta_{\eta_0}}$ |
| $\varphi(4,0)$ | |
| $\varphi(\varphi(4,0)+1)$ | $\omega^{\varphi(4,0)+1}$ |
| $\varphi(1,\varphi(4,0)+1)$ | $\varepsilon_{\varphi(4,0)+1}$ |
| $\varphi(2,\varphi(4,0)+1)$ | $\zeta_{\varphi(4,0)+1}$ |

| Veblen function | Cantor-style |
|---|---|
| $\varphi(3, \varphi(4, 0) + 1)$ | $\eta_{\varphi(4,0)+1}$ |
| $\varphi(4, 1)$ | |
| $\varphi(4, 2)$ | |
| $\varphi(4, \omega)$ | |
| $\varphi(4, \varphi(4, 0))$ | |
| $\varphi(5, 0)$ | |
| $\varphi(4, \varphi(5, 0) + 1)$ | |
| $\varphi(5, 1)$ | |
| $\varphi(5, \varphi(5, 0))$ | |
| $\varphi(6, 0)$ | |
| $\varphi(7, 0)$ | |
| $\varphi(\varphi(1), 0)$ | $\varphi(\omega, 0)$ |
| $\varphi(\varphi(\varphi(1), 0) + 1)$ | $\omega^{\varphi(\omega,0)+1}$ |
| $\varphi(1, \varphi(\varphi(1), 0) + 1)$ | $\varepsilon_{\varphi(\omega,0)+1}$ |
| $\varphi(2, \varphi(\varphi(1), 0) + 1)$ | $\zeta_{\varphi(\omega,0)+1}$ |
| $\varphi(3, \varphi(\varphi(1), 0) + 1)$ | $\eta_{\varphi(\omega,0)+1}$ |
| $\varphi(4, \varphi(\varphi(1), 0) + 1)$ | $\varphi(4, \varphi(\omega, 0) + 1)$ |
| $\varphi(5, \varphi(\varphi(1), 0) + 1)$ | $\varphi(5, \varphi(\omega, 0) + 1)$ |
| $\varphi(6, \varphi(\varphi(1), 0) + 1)$ | $\varphi(6, \varphi(\omega, 0) + 1)$ |
| $\varphi(\varphi(1), 1)$ | $\varphi(\omega, 1)$ |
| $\varphi(\varphi(1), 2)$ | $\varphi(\omega, 2)$ |
| $\varphi(\varphi(1), \varphi(1))$ | $\varphi(\omega, \omega)$ |
| $\varphi(\varphi(1), \varphi(\varphi(1), 0))$ | $\varphi(\omega, \varphi(\omega, 0))$ |
| $\varphi(\varphi(1) + 1, 0)$ | $\varphi(\omega + 1, 0)$ |
| $\varphi(\varphi(1), \varphi(\varphi(1) + 1, 0) + 1)$ | $\varphi(\omega, \varphi(\omega + 1, 0) + 1)$ |
| $\varphi(\varphi(1) + 1, 1)$ | $\varphi(\omega + 1, 1)$ |
| $\varphi(\varphi(1) + 2, 0)$ | $\varphi(\omega + 2, 0)$ |
| $\varphi(\varphi(1) + 3, 0)$ | $\varphi(\omega + 3, 0)$ |
| $\varphi(\varphi(1) \cdot 2, 0)$ | $\varphi(\omega \cdot 2, 0)$ |
| $\varphi(\varphi(1) \cdot 3, 0)$ | $\varphi(\omega \cdot 3, 0)$ |
| $\varphi(\varphi(2), 0)$ | $\varphi(\omega^2, 0)$ |
| $\varphi(\varphi(\varphi(1)), 0)$ | $\varphi(\omega^\omega, 0)$ |
| $\varphi(\varphi(1, 0), 0)$ | $\varphi(\varepsilon_0, 0)$ |
| $\varphi(\varphi(1, 1), 0)$ | $\varphi(\varepsilon_1, 0)$ |
| $\varphi(\varphi(2, 0), 0)$ | $\varphi(\zeta_0, 0)$ |
| $\varphi(\varphi(3, 0), 0)$ | $\varphi(\eta_0, 0)$ |
| $\varphi(\varphi(\varphi(1), 0), 0)$ | $\varphi(\varphi(\omega, 0), 0)$ |
| $\varphi(\varphi(\varphi(1, 0), 0), 0)$ | $\varphi(\varphi(\varepsilon_0, 0), 0)$ |
| $\varphi(1, 0, 0)$ | $\Gamma_0$ |
| $\varphi(1, 0, 0) \cdot 2$ | $\Gamma_0 \cdot 2$ |
| $\varphi(\varphi(1, 0, 0) + 1)$ | $\Gamma_0 \cdot \omega$<br>$\omega^{\Gamma_0+1}$ |
| $\varphi(\varphi(1, 0, 0) \cdot 2)$ | ${\Gamma_0}^{\Gamma_0}$<br>$\omega^{\Gamma_0 \cdot 2}$ |
| $\varphi(1, \varphi(1, 0, 0) + 1)$ | $\varepsilon_{\Gamma_0+1}$ |
| $\varphi(1, \varphi(1, 0, 0) \cdot 2)$ | $\varepsilon_{\Gamma_0 \cdot 2}$ |
| $\varphi(2, \varphi(1, 0, 0) + 1)$ | $\zeta_{\Gamma_0+1}$ |
| $\varphi(3, \varphi(1, 0, 0) + 1)$ | $\eta_{\Gamma_0+1}$ |
| $\varphi(4, \varphi(1, 0, 0) + 1)$ | $\varphi(4, \Gamma_0 + 1)$ |
| $\varphi(5, \varphi(1, 0, 0) + 1)$ | $\varphi(5, \Gamma_0 + 1)$ |
| $\varphi(\varphi(1), \varphi(1, 0, 0) + 1)$ | $\varphi(\omega, \Gamma_0 + 1)$ |
| $\varphi(\varphi(1, 0), \varphi(1, 0, 0) + 1)$ | $\varphi(\varepsilon_0, \Gamma_0 + 1)$ |
| $\varphi(\varphi(\varphi(1), 0), \varphi(1, 0, 0) + 1)$ | $\varphi(\varphi(\omega, 0), \Gamma_0 + 1)$ |
| $\varphi(\varphi(\varphi(1, 0), 0), \varphi(1, 0, 0) + 1)$ | $\varphi(\varphi(\varepsilon_0, 0), \Gamma_0 + 1)$ |

| Veblen function | Cantor-style |
|---|---|
| $\varphi(\varphi(1,0,0),1)$ | $\varphi(\Gamma_0,1)$ |
| $\varphi(\varphi(1,0,0),2)$ | $\varphi(\Gamma_0,2)$ |
| $\varphi(\varphi(1,0,0),\varphi(1,0,0))$ | $\varphi(\Gamma_0,\Gamma_0)$ |
| $\varphi(\varphi(1,0,0)+1,0)$ | $\varphi(\Gamma_0+1,0)$ |
| $\varphi(\varphi(1,0,0)+\omega,0)$ | $\varphi(\Gamma_0+\omega,0)$ |
| $\varphi(\varphi(1,0,0)\cdot 2,0)$ | $\varphi(\Gamma_0\cdot 2,0)$ |
| $\varphi(\varphi(1,\varphi(1,0,0)+1),0)$ | $\varphi(\varepsilon_{\Gamma_0+1},0)$ |
| $\varphi(\varphi(\varphi(1,0,0),1),0)$ | $\varphi(\varphi(\Gamma_0,1),0)$ |
| $\varphi(\varphi(\varphi(1,0,0)+1,0),0)$ | $\varphi(\varphi(\Gamma_0+1,0),0)$ |
| $\varphi(1,0,1)$ | $\Gamma_1$ |
| $\varphi(\varphi(1,0,0),\varphi(1,0,1)+1)$ | $\varphi(\Gamma_0,\Gamma_1+1)$ |
| $\varphi(\varphi(1,0,1),1)$ | $\varphi(\Gamma_1,1)$ |
| $\varphi(1,0,2)$ | $\Gamma_2$ |
| $\varphi(1,0,3)$ | $\Gamma_3$ |
| $\varphi(1,0,\varphi(1))$ | $\Gamma_\omega$ |
| $\varphi(1,0,\varphi(1,0))$ | $\Gamma_{\varepsilon_0}$ |
| $\varphi(1,0,\varphi(1,0,0))$ | $\Gamma_{\Gamma_0}$ |
| $\varphi(1,1,0)$ | $\alpha\mapsto\Gamma_\alpha$ fp. |

## A.4 MOCF vs Veblen Function

The results in this section are primarily cited from[1].

| Madore's OCF | Veblen function |
|---|---|
| $\psi(0)$ | $\varepsilon_0$ |
| $\psi(0)+1$ | $\varepsilon_0+1$ |
| $\psi(0)+\omega$ | $\varepsilon_0+\omega$ |
| $\psi(0)+\omega^\omega$ | $\varepsilon_0+\omega^\omega$ |
| $\psi(0)+\omega^{\omega^\omega}$ | $\varepsilon_0+\omega^{\omega^\omega}$ |
| $\psi(0)\cdot 2$ | $\varepsilon_0\cdot 2$ |
| $\psi(0)\cdot 3$ | $\varepsilon_0\cdot 3$ |
| $\psi(0)\cdot\omega$ | $\varepsilon_0\cdot\omega$<br>$\omega^{\varepsilon_0+1}$ |
| $\psi(0)\cdot\omega+\omega$ | $\varepsilon_0\cdot\omega+\omega$<br>$\omega^{\varepsilon_0+1}+\omega$ |
| $\psi(0)\cdot\omega+\omega^\omega$ | $\varepsilon_0\cdot\omega+\omega^\omega$<br>$\omega^{\varepsilon_0+1}+\omega^\omega$ |
| $\psi(0)\cdot\omega\cdot 2$ | $\varepsilon_0\cdot\omega\cdot 2$<br>$\omega^{\varepsilon_0+1}\cdot 2$ |
| $\psi(0)\cdot\omega^2$ | $\varepsilon_0\cdot\omega^2$<br>$\omega^{\varepsilon_0+2}$ |
| $\psi(0)\cdot\omega^2\cdot 2$ | $\varepsilon_0\cdot\omega^2\cdot 2$<br>$\omega^{\varepsilon_0+2}\cdot 2$ |
| $\psi(0)\cdot\omega^3$ | $\varepsilon_0\cdot\omega^3$<br>$\omega^{\varepsilon_0+3}$ |
| $\psi(0)\cdot\omega^\omega$ | $\varepsilon_0\cdot\omega^\omega$<br>$\omega^{\varepsilon_0+\omega}$ |
| $\psi(0)\cdot\omega^{\omega^2}$ | $\varepsilon_0\cdot\omega^{\omega^2}$<br>$\omega^{\varepsilon_0+\omega^2}$ |
| $\psi(0)\cdot\omega^{\omega^\omega}$ | $\varepsilon_0\cdot\omega^{\omega^\omega}$<br>$\omega^{\varepsilon_0+\omega^\omega}$ |

| Madore's OCF | Veblen function |
|---|---|
| $\psi(0)^2$ | $\omega^{\varepsilon_0\cdot 2}$<br>${\varepsilon_0}^2$ |
| $\psi(0)^2+\omega$ | ${\varepsilon_0}^2+\omega$<br>$\omega^{\varepsilon_0\cdot 2}+\omega$ |
| $\psi(0)^2+\omega^\omega$ | ${\varepsilon_0}^2+\omega^\omega$<br>$\omega^{\varepsilon_0\cdot 2}+\omega^\omega$ |
| $\psi(0)^2+\psi(0)$ | ${\varepsilon_0}^2+\varepsilon_0$<br>$\omega^{\varepsilon_0\cdot 2}+\varepsilon_0$ |
| $\psi(0)^2+\psi(0)\cdot\omega$ | ${\varepsilon_0}^2+\varepsilon_0\cdot\omega$<br>$\omega^{\varepsilon_0\cdot 2}+\omega^{\varepsilon_0+1}$ |
| $\psi(0)^2\cdot 2$ | ${\varepsilon_0}^2\cdot 2$<br>$\omega^{\varepsilon_0\cdot 2}\cdot 2$ |
| $\psi(0)^2\cdot\omega$ | ${\varepsilon_0}^2\cdot\omega$<br>$\omega^{\varepsilon_0\cdot 2+1}$ |
| $\psi(0)^2\cdot\omega^\omega$ | ${\varepsilon_0}^2\cdot\omega^\omega$<br>$\omega^{\varepsilon_0\cdot 2+\omega}$ |
| $\psi(0)^3$ | ${\varepsilon_0}^3$<br>$\omega^{\varepsilon_0\cdot 3}$ |
| $\psi(0)^4$ | ${\varepsilon_0}^4$<br>$\omega^{\varepsilon_0\cdot 4}$ |
| $\psi(0)^\omega$ | ${\varepsilon_0}^\omega$<br>$\omega^{\omega^{\varepsilon_0+1}}$ |
| $\psi(0)^\omega\cdot 2$ | ${\varepsilon_0}^\omega\cdot 2$<br>$\omega^{\omega^{\varepsilon_0+1}}\cdot 2$ |
| $\psi(0)^\omega\cdot\omega$ | $\omega^{\omega^{\varepsilon_0+1}+1}$ |
| $\psi(0)^\omega\cdot\omega^\omega$ | ${\varepsilon_0}^\omega\cdot\omega^\omega$<br>$\omega^{\omega^{\varepsilon_0+1}+\omega}$ |
| $\psi(0)^{\omega+1}$ | ${\varepsilon_0}^{\omega+1}$<br>$\omega^{\omega^{\varepsilon_0+1}+\varepsilon_0}$ |
| $\psi(0)^{\omega+2}$ | ${\varepsilon_0}^{\omega+2}$<br>$\omega^{\omega^{\varepsilon_0+1}+\varepsilon_0\cdot 2}$ |
| $\psi(0)^{\omega\cdot 2}$ | ${\varepsilon_0}^{\omega\cdot 2}$<br>$\omega^{\omega^{\varepsilon_0+1}\cdot 2}$ |
| $\psi(0)^{\omega^2}$ | ${\varepsilon_0}^{\omega^2}$<br>$\omega^{\omega^{\varepsilon_0+2}}$ |
| $\psi(0)^{\omega^\omega}$ | $\omega^{\omega^{\varepsilon_0+\omega}}$ |
| $\psi(0)^{\omega^{\omega^\omega}}$ | ${\varepsilon_0}^{\omega^\omega}$<br>$\omega^{\omega^{\varepsilon_0+\omega^\omega}}$ |
| $\psi(0)^{\psi(0)}$ | $\omega^{\omega^{\varepsilon_0\cdot 2}}$<br>${\varepsilon_0}^{\varepsilon_0}$ |
| $\psi(0)^{\psi(0)}\cdot\omega$ | ${\varepsilon_0}^{\varepsilon_0}\cdot\omega$<br>$\omega^{\omega^{\varepsilon_0\cdot 2}+1}$ |
| $\psi(0)^{\psi(0)+1}$ | ${\varepsilon_0}^{\varepsilon_0+1}$<br>$\omega^{\omega^{\varepsilon_0\cdot 2}+\varepsilon_0}$ |
| $\psi(0)^{\psi(0)+\omega}$ | ${\varepsilon_0}^{\varepsilon_0+\omega}$<br>$\omega^{\omega^{\varepsilon_0\cdot 2}+\omega^{\varepsilon_0+1}}$ |

| Madore's OCF | Veblen function |
|---|---|
| $\psi(0)^{\psi(0)\cdot 2}$ | ${\varepsilon_0}^{\varepsilon_0\cdot 2}$<br>$\omega^{\omega^{\varepsilon_0\cdot 2}\cdot 2}$ |
| $\psi(0)^{\psi(0)\cdot\omega}$ | ${\varepsilon_0}^{\varepsilon_0\cdot\omega}$<br>$\omega^{\omega^{\varepsilon_0\cdot 2+1}}$ |
| $\psi(0)^{\psi(0)^2}$ | ${\varepsilon_0}^{{\varepsilon_0}^2}$<br>$\omega^{\omega^{\varepsilon_0\cdot 3}}$ |
| $\psi(0)^{\psi(0)^\omega}$ | ${\varepsilon_0}^{{\varepsilon_0}^\omega}$<br>$\omega^{\omega^{\omega^{\varepsilon_0+1}}}$ |
| $\psi(0)^{\psi(0)^{\psi(0)}}$ | ${\varepsilon_0}^{{\varepsilon_0}^{\varepsilon_0}}$<br>$\omega^{\omega^{\omega^{\varepsilon_0\cdot 2}}}$ |
| $\psi(1)$ | $\varepsilon_1$ |
| $\psi(1)+\psi(0)$ | $\varepsilon_1+\varepsilon_0$ |
| $\psi(1)+\psi(0)\cdot\omega$ | $\varepsilon_1+\varepsilon_0\cdot\omega$<br>$\varepsilon_1+\omega^{\varepsilon_0+1}$ |
| $\psi(1)+\psi(0)^\omega$ | $\varepsilon_1+{\varepsilon_0}^\omega$<br>$\varepsilon_1+\omega^{\omega^{\varepsilon_0}}$ |
| $\psi(1)\cdot 2$ | $\varepsilon_1\cdot 2$ |
| $\psi(1)\cdot\omega$ | $\varepsilon_1\cdot\omega$<br>$\omega^{\varepsilon_1+1}$ |
| $\psi(1)\cdot\psi(0)$ | $\varepsilon_1\cdot\varepsilon_0$<br>$\omega^{\varepsilon_1+\varepsilon_0}$ |
| $\psi(1)\cdot\psi(0)^2$ | $\varepsilon_1\cdot{\varepsilon_0}^2$<br>$\omega^{\varepsilon_1+\omega^{\varepsilon_0\cdot 2}}$ |
| $\psi(1)^2$ | ${\varepsilon_1}^2$<br>$\omega^{\varepsilon_1\cdot 2}$ |
| $\psi(1)^3$ | ${\varepsilon_1}^3$<br>$\omega^{\varepsilon_1\cdot 3}$ |
| $\psi(1)^\omega$ | ${\varepsilon_1}^\omega$<br>$\omega^{\omega^{\varepsilon_1+1}}$ |
| $\psi(1)^{\psi(0)}$ | ${\varepsilon_1}^{\varepsilon_0}$<br>$\omega^{\omega^{\varepsilon_1+\varepsilon_0}}$ |
| $\psi(1)^{\psi(1)}$ | ${\varepsilon_1}^{\varepsilon_1}$<br>$\omega^{\omega^{\varepsilon_1\cdot 2}}$ |
| $\psi(2)$ | $\varepsilon_2$ |
| $\psi(2)\cdot\omega$ | $\varepsilon_2\cdot\omega$<br>$\omega^{\varepsilon_2+1}$ |
| $\psi(2)\cdot\psi(0)$ | $\varepsilon_2\cdot\varepsilon_0$<br>$\omega^{\varepsilon_2+\varepsilon_0}$ |
| $\psi(2)\cdot\psi(1)$ | $\varepsilon_2\cdot\varepsilon_1$<br>$\omega^{\varepsilon_2+\varepsilon_1}$ |
| $\psi(2)^2$ | ${\varepsilon_2}^2$<br>$\omega^{\varepsilon_2\cdot 2}$ |
| $\psi(2)^\omega$ | ${\varepsilon_2}^\omega$<br>$\omega^{\omega^{\varepsilon_2+1}}$ |
| $\psi(2)^{\psi(2)}$ | ${\varepsilon_2}^{\varepsilon_2}$<br>$\omega^{\omega^{\varepsilon_2\cdot 2}}$ |
| $\psi(3)$ | $\varepsilon_3$ |

| Madore's OCF | Veblen function |
|---|---|
| $\psi(3)^2$ | ${\varepsilon_3}^2$<br>$\omega^{\varepsilon_3 \cdot 2}$ |
| $\psi(4)$ | $\varepsilon_4$ |
| $\psi(5)$ | $\varepsilon_5$ |
| $\psi(\omega)$ | $\varepsilon_\omega$ |
| $\psi(\omega)^2$ | ${\varepsilon_\omega}^2$<br>$\omega^{\varepsilon_\omega \cdot 2}$ |
| $\psi(\omega+1)$ | $\varepsilon_{\omega+1}$ |
| $\psi(\omega+2)$ | $\varepsilon_{\omega+2}$ |
| $\psi(\omega \cdot 2)$ | $\varepsilon_{\omega \cdot 2}$ |
| $\psi(\omega^2)$ | $\varepsilon_{\omega^2}$ |
| $\psi(\omega^\omega)$ | $\varepsilon_{\omega^\omega}$ |
| $\psi(\omega^{\omega^\omega})$ | $\varepsilon_{\omega^{\omega^\omega}}$ |
| $\psi(\psi(0))$ | $\varepsilon_{\varepsilon_0}$ |
| $\psi(\psi(0)+1)$ | $\varepsilon_{\varepsilon_0+1}$ |
| $\psi(\psi(0) \cdot 2)$ | $\varepsilon_{\varepsilon_0 \cdot 2}$ |
| $\psi(\psi(0) \cdot \omega)$ | $\varepsilon_{\varepsilon_0 \cdot \omega}$<br>$\varepsilon_{\omega^{\varepsilon_0+1}}$ |
| $\psi(\psi(0)^2)$ | ${\varepsilon_{\varepsilon_0}}^2$<br>$\varepsilon_{\omega^{\varepsilon_0 \cdot 2}}$ |
| $\psi(\psi(1))$ | $\varepsilon_{\varepsilon_1}$ |
| $\psi(\psi(2))$ | $\varepsilon_{\varepsilon_2}$ |
| $\psi(\psi(\omega))$ | $\varepsilon_{\varepsilon_\omega}$ |
| $\psi(\psi(\psi(0)))$ | $\varepsilon_{\varepsilon_{\varepsilon_0}}$ |
| $\psi(\Omega)$ | $\zeta_0$ |
| $\psi(\Omega)+\omega$ | $\zeta_0+\omega$ |
| $\psi(\Omega)+\psi(0)$ | $\zeta_0+\varepsilon_0$ |
| $\psi(\Omega)+\psi(\psi(0))$ | $\zeta_0+\varepsilon_{\varepsilon_0}$ |
| $\psi(\Omega) \cdot 2$ | $\zeta_0 \cdot 2$ |
| $\psi(\Omega) \cdot \omega$ | $\zeta_0 \cdot \omega$<br>$\omega^{\zeta_0+1}$ |
| $\psi(\Omega) \cdot \psi(0)$ | $\zeta_0 \cdot \varepsilon_0$<br>$\omega^{\zeta_0+\varepsilon_0}$ |
| $\psi(\Omega) \cdot \psi(\psi(0))$ | $\zeta_0 \cdot \varepsilon_{\varepsilon_0}$<br>$\omega^{\zeta_0+\varepsilon_{\varepsilon_0}}$ |
| $\psi(\Omega)^2$ | ${\zeta_0}^2$<br>$\omega^{\zeta_0 \cdot 2}$ |
| $\psi(\Omega)^\omega$ | ${\zeta_0}^\omega$<br>$\omega^{\omega^{\zeta_0+1}}$ |
| $\psi(\Omega)^{\psi(0)}$ | ${\zeta_0}^{\varepsilon_0}$<br>$\omega^{\omega^{\zeta_0+\varepsilon_0}}$ |
| $\psi(\Omega)^{\psi(\Omega)}$ | $\omega^{\omega^{\zeta_0 \cdot 2}}$ |
| $\psi(\Omega)^{\psi(\Omega)^{\psi(\Omega)}}$ | ${\zeta_0}^{{\zeta_0}^{\zeta_0}}$<br>$\omega^{\omega^{\omega^{\zeta_0 \cdot 2}}}$ |
| $\psi(\Omega+1)$ | $\varepsilon_{\zeta_0+1}$ |
| $\psi(\Omega+1)^2$ | ${\varepsilon_{\zeta_0+1}}^2$<br>$\omega^{\varepsilon_{\zeta_0+1} \cdot 2}$ |
| $\psi(\Omega+1)^{\psi(\Omega+1)}$ | ${\varepsilon_{\zeta_0+1}}^{\varepsilon_{\zeta_0+1}}$<br>$\omega^{\omega^{\varepsilon_{\zeta_0+1} \cdot 2}}$ |
| $\psi(\Omega+2)$ | $\varepsilon_{\zeta_0+2}$ |
| $\psi(\Omega+3)$ | $\varepsilon_{\zeta_0+3}$ |

| Madore's OCF | Veblen function |
|---|---|
| $\psi(\Omega+\omega)$ | $\varepsilon_{\zeta_0+\omega}$ |
| $\psi(\Omega+\psi(0))$ | $\varepsilon_{\zeta_0+\varepsilon_0}$ |
| $\psi(\Omega+\psi(\psi(0)))$ | $\varepsilon_{\zeta_0+\varepsilon_{\varepsilon_0}}$ |
| $\psi(\Omega+\psi(\Omega))$ | $\varepsilon_{\zeta_0\cdot 2}$ |
| $\psi(\Omega+\psi(\Omega)\cdot 2)$ | $\varepsilon_{\zeta_0\cdot 3}$ |
| $\psi(\Omega+\psi(\Omega)\cdot\omega)$ | $\varepsilon_{\zeta_0\cdot\omega}$<br>$\varepsilon_{\omega^{\zeta_0+1}}$ |
| $\psi(\Omega+\psi(\Omega)^2)$ | $\varepsilon_{\zeta_0{}^2}$<br>$\varepsilon_{\omega^{\zeta_0\cdot 2}}$ |
| $\psi(\Omega+\psi(\Omega)^{\psi(\Omega)})$ | $\varepsilon_{\zeta_0{}^{\zeta_0}}$<br>$\varepsilon_{\omega^{\omega^{\zeta_0\cdot 2}}}$ |
| $\psi(\Omega+\psi(\Omega+1))$ | $\varepsilon_{\varepsilon_{\zeta_0+1}}$ |
| $\psi(\Omega+\psi(\Omega+\psi(0)))$ | $\varepsilon_{\varepsilon_{\zeta_0+\varepsilon_0}}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega)))$ | $\varepsilon_{\varepsilon_{\varepsilon_{\zeta_0\cdot 2}}}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega+1)))$ | $\varepsilon_{\varepsilon_{\varepsilon_{\varepsilon_{\zeta_0+1}}}}$ |
| $\psi(\Omega\cdot 2)$ | $\zeta_1$ |
| $\psi(\Omega\cdot 2)\cdot\omega$ | $\zeta_1\cdot\omega$<br>$\omega^{\zeta_1+1}$ |
| $\psi(\Omega\cdot 2)\cdot\psi(\Omega)$ | $\zeta_1\cdot\zeta_0$<br>$\omega^{\zeta_1+\zeta_0}$ |
| $\psi(\Omega\cdot 2)^2$ | $\zeta_1{}^2$<br>$\omega^{\zeta_1\cdot 2}$ |
| $\psi(\Omega\cdot 2)^{\psi(\Omega\cdot 2)}$ | $\zeta_1{}^{\zeta_1}$<br>$\omega^{\omega^{\zeta_1\cdot 2}}$ |
| $\psi(\Omega\cdot 2+1)$ | $\varepsilon_{\zeta_1+1}$ |
| $\psi(\Omega\cdot 2+\psi(\Omega+\psi(\Omega)))$ | $\varepsilon_{\zeta_1+\zeta_0}$ |
| $\psi(\Omega\cdot 2+\psi(\Omega\cdot 2))$ | $\varepsilon_{\zeta_1\cdot 2}$ |
| $\psi(\Omega\cdot 2+\psi(\Omega\cdot 2)^2)$ | $\varepsilon_{\omega^{\zeta_1\cdot 2}}$<br>$\varepsilon_{\zeta_1^2}$ |
| $\psi(\Omega\cdot 2+\psi(\Omega\cdot 2+1))$ | $\varepsilon_{\varepsilon_{\zeta_1+1}}$ |
| $\psi(\Omega\cdot 2+\psi(\Omega\cdot 2+1))$ | $\varepsilon_{\varepsilon_{\zeta_1+1}}$ |
| $\psi(\Omega\cdot 3)$ | $\zeta_2$ |
| $\psi(\Omega\cdot 3)\cdot\omega$ | $\zeta_2\cdot\omega$<br>$\omega^{\zeta_2+1}$ |
| $\psi(\Omega\cdot 3+1)$ | $\varepsilon_{\zeta_2+1}$ |
| $\psi(\Omega\cdot 3+\psi(\Omega\cdot 3))$ | $\varepsilon_{\zeta_2\cdot 2}$ |
| $\psi(\Omega\cdot 3+\psi(\Omega\cdot 3+1))$ | $\varepsilon_{\varepsilon_{\zeta_2+1}}$ |
| $\psi(\Omega\cdot 4)$ | $\zeta_3$ |
| $\psi(\Omega\cdot 5)$ | $\zeta_4$ |
| $\psi(\Omega\cdot\omega)$ | $\zeta_\omega$ |
| $\psi(\Omega\cdot\psi(0))$ | $\zeta_{\varepsilon_0}$ |
| $\psi(\Omega\cdot\psi(\psi(0)))$ | $\zeta_{\varepsilon_{\varepsilon_0}}$ |
| $\psi(\Omega\cdot\psi(\Omega))$ | $\zeta_{\zeta_0}$ |
| $\psi(\Omega\cdot\psi(\Omega\cdot 2))$ | $\zeta_{\zeta_1}$ |
| $\psi(\Omega\cdot\psi(\Omega\cdot\psi(0)))$ | $\zeta_{\zeta_{\varepsilon_0}}$ |
| $\psi(\Omega\cdot\psi(\Omega\cdot\psi(\Omega)))$ | $\zeta_{\zeta_{\zeta_0}}$ |
| $\psi(\Omega^2)$ | $\eta_0$ |
| $\psi(\Omega^2)\cdot\omega$ | $\eta_0\cdot\omega$<br>$\omega^{\eta_0+1}$ |
| $\psi(\Omega^2)^{\psi(\Omega^2)}$ | $\eta_0{}^{\eta_0}$<br>$\omega^{\omega^{\eta_0\cdot 2}}$ |
| $\psi(\Omega^2+1)$ | $\varepsilon_{\eta_0+1}$ |

| Madore's OCF | Veblen function |
|---|---|
| $\psi(\Omega^2+\psi(\Omega^2))$ | $\varepsilon_{\eta_0\cdot 2}$ |
| $\psi(\Omega^2+\psi(\Omega^2+\psi(\Omega^2)))$ | $\varepsilon_{\varepsilon_{\eta_0\cdot 2}}$ |
| $\psi(\Omega^2+\Omega)$ | $\zeta_{\eta_0+1}$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2))$ | $\zeta_{\eta_0\cdot 2}$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+\Omega\cdot\psi(\Omega^2)))$ | $\zeta_{\zeta_{\eta_0\cdot 2}}$ |
| $\psi(\Omega^2\cdot 2)$ | $\eta_1$ |
| $\psi(\Omega^2\cdot 2)\cdot\omega$ | $\omega^{\eta_1+1}$ |
| $\psi(\Omega^2\cdot 2+1)$ | $\varepsilon_{\eta_1+1}$ |
| $\psi(\Omega^2\cdot 2+\Omega)$ | $\zeta_{\eta_1+1}$ |
| $\psi(\Omega^2\cdot 2+\Omega\cdot\psi(\Omega^2\cdot 2+\Omega))$ | $\zeta_{\zeta_{\eta_1+1}}$ |
| $\psi(\Omega^2\cdot 3)$ | $\eta_2$ |
| $\psi(\Omega^2\cdot 4)$ | $\eta_3$ |
| $\psi(\Omega^2\cdot\omega)$ | $\eta_\omega$ |
| $\psi(\Omega^2\cdot\psi(0))$ | $\eta_{\varepsilon_0}$ |
| $\psi(\Omega^2\cdot\psi(\Omega))$ | $\eta_{\zeta_0}$ |
| $\psi(\Omega^2\cdot\psi(\Omega^2))$ | $\eta_{\eta_0}$ |
| $\psi(\Omega^2\cdot\psi(\Omega^2\cdot\psi(\Omega^2)))$ | $\eta_{\eta_{\eta_0}}$ |
| $\psi(\Omega^3)$ | $\varphi(4,0)$ |
| $\psi(\Omega^3)\cdot\omega$ | $\omega^{\varphi(4,0)+1}$ |
| $\psi(\Omega^3+1)$ | $\varepsilon_{\varphi(4,0)+1}$ |
| $\psi(\Omega^3+\Omega^2)$ | $\eta_{\varphi(4,0)+1}$ |
| $\psi(\Omega^3\cdot 2)$ | $\varphi(4,1)$ |
| $\psi(\Omega^3\cdot 3)$ | $\varphi(4,2)$ |
| $\psi(\Omega^3\cdot\omega)$ | $\varphi(4,\omega)$ |
| $\psi(\Omega^3\cdot\psi(\Omega^3))$ | $\varphi(4,\varphi(4,0))$ |
| $\psi(\Omega^4)$ | $\varphi(5,0)$ |
| $\psi(\Omega^4+\Omega^3)$ | $\varphi(4,\varphi(5,0)+1)$ |
| $\psi(\Omega^4\cdot 2)$ | $\varphi(5,1)$ |
| $\psi(\Omega^4\cdot\psi(\Omega^4))$ | $\varphi(5,\varphi(5,0))$ |
| $\psi(\Omega^5)$ | $\varphi(6,0)$ |
| $\psi(\Omega^6)$ | $\varphi(7,0)$ |
| $\psi(\Omega^\omega)$ | $\varphi(\omega,0)$ |
| $\psi(\Omega^\omega)\cdot\omega$ | $\omega^{\varphi(\omega,0)+1}$ |
| $\psi(\Omega^\omega+1)$ | $\varepsilon_{\varphi(\omega,0)+1}$ |
| $\psi(\Omega^\omega+\Omega)$ | $\zeta_{\varphi(\omega,0)+1}$ |
| $\psi(\Omega^\omega+\Omega^2)$ | $\eta_{\varphi(\omega,0)+1}$ |
| $\psi(\Omega^\omega+\Omega^3)$ | $\varphi(4,\varphi(\omega,0)+1)$ |
| $\psi(\Omega^\omega+\Omega^4)$ | $\varphi(5,\varphi(\omega,0)+1)$ |
| $\psi(\Omega^\omega+\Omega^5)$ | $\varphi(6,\varphi(\omega,0)+1)$ |
| $\psi(\Omega^\omega\cdot 2)$ | $\varphi(\omega,1)$ |
| $\psi(\Omega^\omega\cdot 3)$ | $\varphi(\omega,2)$ |
| $\psi(\Omega^\omega\cdot\omega)$ | $\varphi(\omega,\omega)$ |
| $\psi(\Omega^\omega\cdot\psi(\Omega^\omega))$ | $\varphi(\omega,\varphi(\omega,0))$ |
| $\psi(\Omega^{\omega+1})$ | $\varphi(\omega+1,0)$ |
| $\psi(\Omega^{\omega+1}+\Omega^\omega)$ | $\varphi(\omega,\varphi(\omega+1,0)+1)$ |
| $\psi(\Omega^{\omega+1}\cdot 2)$ | $\varphi(\omega+1,1)$ |
| $\psi(\Omega^{\omega+2})$ | $\varphi(\omega+2,0)$ |
| $\psi(\Omega^{\omega+3})$ | $\varphi(\omega+3,0)$ |
| $\psi(\Omega^{\omega\cdot 2})$ | $\varphi(\omega\cdot 2,0)$ |
| $\psi(\Omega^{\omega\cdot 3})$ | $\varphi(\omega\cdot 3,0)$ |
| $\psi(\Omega^{\omega^2})$ | $\varphi(\omega^2,0)$ |
| $\psi(\Omega^{\omega^\omega})$ | $\varphi(\omega^\omega,0)$ |
| $\psi(\Omega^{\psi(0)})$ | $\varphi(\varepsilon_0,0)$ |
| $\psi(\Omega^{\psi(1)})$ | $\varphi(\varepsilon_1,0)$ |
| $\psi(\Omega^{\psi(\Omega)})$ | $\varphi(\zeta_0,0)$ |

| Madore's OCF | Veblen function |
|---|---|
| $\psi(\Omega^{\psi(\Omega^2)})$ | $\varphi(\eta_0, 0)$ |
| $\psi(\Omega^{\psi(\Omega^\omega)})$ | $\varphi(\varphi(\omega, 0), 0)$ |
| $\psi(\Omega^{\psi(\Omega^{\psi(0)})})$ | $\varphi(\varphi(\varepsilon_0, 0), 0)$ |
| $\psi(\Omega^\Omega)$ | $\Gamma_0$<br>$\varphi(1, 0, 0)$ |
| $\psi(\Omega^\Omega) \cdot 2$ | $\Gamma_0 \cdot 2$<br>$\varphi(1, 0, 0) \cdot 2$ |
| $\psi(\Omega^\Omega) \cdot \omega$ | $\omega^{\Gamma_0+1}$<br>$\varphi(\varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega)^2$ | $\omega^{\Gamma_0 \cdot 2}$<br>$\varphi(\varphi(1, 0, 0) \cdot 2)$ |
| $\psi(\Omega^\Omega + 1)$ | $\varepsilon_{\Gamma_0+1}$<br>$\varphi(1, \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \psi(\Omega^\Omega))$ | $\varepsilon_{\Gamma_0 \cdot 2}$<br>$\varphi(1, \varphi(1, 0, 0) \cdot 2)$ |
| $\psi(\Omega^\Omega + \Omega)$ | $\zeta_{\Gamma_0+1}$<br>$\varphi(2, \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^2)$ | $\eta_{\Gamma_0+1}$<br>$\varphi(3, \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^3)$ | $\varphi(4, \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^4)$ | $\varphi(5, \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^\omega)$ | $\varphi(\omega, \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(0)})$ | $\varphi(\varepsilon_0, \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\omega)})$ | $\varphi(\varphi(\omega, 0), \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^{\psi(0)})})$ | $\varphi(\varphi(\varepsilon_0, 0), \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)})$ | $\varphi(\Gamma_0, 1)$<br>$\varphi(\varphi(1, 0, 0), 1)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)} \cdot 2)$ | $\varphi(\Gamma_0, 2)$<br>$\varphi(\varphi(1, 0, 0), 2)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)} \cdot \psi(\Omega^\Omega))$ | $\varphi(\Gamma_0, \Gamma_0)$<br>$\varphi(\varphi(1, 0, 0), \varphi(1, 0, 0))$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)+1})$ | $\varphi(\Gamma_0 + 1, 0)$<br>$\varphi(\varphi(1, 0, 0) + 1, 0)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)+\omega})$ | $\varphi(\Gamma_0 + \omega, 0)$<br>$\varphi(\varphi(1, 0, 0) + \omega, 0)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)\cdot 2})$ | $\varphi(\Gamma_0 \cdot 2, 0)$<br>$\varphi(\varphi(1, 0, 0) \cdot 2, 0)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+1)})$ | $\varphi(\varepsilon_{\Gamma_0+1}, 0)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+\Omega^{\psi(\Omega^\Omega)})})$ | $\varphi(\varphi(\Gamma_0, 1), 0)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+\Omega^{\psi(\Omega^\Omega)}+1)})$ | $\varphi(\varphi(\Gamma_0 + 1, 0), 0)$ |
| $\psi(\Omega^\Omega \cdot 2)$ | $\Gamma_1$ |
| $\psi(\Omega^\Omega \cdot 2 + \Omega^{\psi(\Omega^\Omega)})$ | $\varphi(\Gamma_0, \Gamma_1 + 1)$ |
| $\psi(\Omega^\Omega \cdot 2 + \Omega^{\psi(\Omega^\Omega \cdot 2)})$ | $\varphi(\Gamma_1, 1)$ |
| $\psi(\Omega^\Omega \cdot 3)$ | $\Gamma_2$ |
| $\psi(\Omega^\Omega \cdot 4)$ | $\Gamma_3$ |
| $\psi(\Omega^\Omega \cdot \omega)$ | $\Gamma_\omega$ |
| $\psi(\Omega^\Omega \cdot \psi(0))$ | $\Gamma_{\varepsilon_0}$ |
| $\psi(\Omega^\Omega \cdot \psi(\Omega^\Omega))$ | $\Gamma_{\Gamma_0}$ |
| $\psi(\Omega^{\Omega+1})$ | $\varphi(1, 1, 0)$ |
| $\psi(\Omega^{\Omega+1} + \Omega^\omega)$ | $\varphi(\omega, \varphi(1, 1, 0) + 1)$ |

| Madore's OCF | Veblen function |
|---|---|
| $\psi(\Omega^{\Omega+1}+\Omega^{\psi(\Omega^{\Omega})})$ | $\varphi(\Gamma_0,\varphi(1,1,0)+1)$ |
| $\psi(\Omega^{\Omega+1}+\Omega^{\psi(\Omega^{\Omega+1})})$ | $\varphi(\varphi(1,1,0),1)$ |
| $\psi(\Omega^{\Omega+1}+\Omega^{\psi(\Omega^{\Omega+1})+1})$ | $\varphi(\varphi(1,1,0)+1,1)$ |
| $\psi(\Omega^{\Omega+1}+\Omega^{\Omega})$ | $\Gamma_{\varphi(1,1,0)+1}$ |
| $\psi(\Omega^{\Omega+1}+\Omega^{\Omega}\cdot\psi(\Omega^{\Omega+1}+\Omega^{\Omega}))$ | $\Gamma_{\Gamma_{\varphi(1,1,0)+1}}$ |
| $\psi(\Omega^{\Omega+1}\cdot 2)$ | $\varphi(1,1,1)$ |
| $\psi(\Omega^{\Omega+1}\cdot 3)$ | $\varphi(1,1,2)$ |
| $\psi(\Omega^{\Omega+1}\cdot\omega)$ | $\varphi(1,1,\omega)$ |
| $\psi(\Omega^{\Omega+2})$ | $\varphi(1,2,0)$ |
| $\psi\left(\Omega^{\Omega+2}+\Omega^{\psi\left(\Omega^{\Omega+2}\right)+1}\right)$ | $\varphi(\varphi(1,2,0)+1,0)$ |
| $\psi\left(\Omega^{\Omega+2}+\Omega^{\Omega}\right)$ | $\Gamma_{\varphi(1,2,0)+1}$ |
| $\psi\left(\Omega^{\Omega+2}+\Omega^{\Omega}\psi\left(\Omega^{\Omega+2}+\Omega^{\Omega}\right)\right)$ | $\Gamma_{\Gamma_{\varphi(1,2,0)+1}}$ |
| $\psi\left(\Omega^{\Omega+2}+\Omega^{\Omega+1}\right)$ | $\varphi(1,1,\varphi(1,2,0)+1)$ |
| $\psi\left(\Omega^{\Omega+2}\cdot 2\right)$ | $\varphi(1,2,1)$ |
| $\psi\left(\Omega^{\Omega+2}\cdot\psi\left(\Omega^{\Omega+2}\right)\right)$ | $\varphi(1,2,\varphi(1,2,0))$ |
| $\psi\left(\Omega^{\Omega+3}\right)$ | $\varphi(1,3,0)$ |
| $\psi\left(\Omega^{\Omega+3}\cdot 2\right)$ | $\varphi(1,3,1)$ |
| $\psi\left(\Omega^{\Omega+4}\right)$ | $\varphi(1,4,0)$ |
| $\psi\left(\Omega^{\Omega+\omega}\right)$ | $\varphi(1,\omega,0)$ |
| $\psi\left(\Omega^{\Omega+\psi(\Omega^{\Omega})}\right)$ | $\varphi(1,\varphi(1,0,0),0)$ |
| $\psi\left(\Omega^{\Omega+\psi(\Omega^{\Omega+\psi(\Omega^{\Omega})})}\right)$ | $\varphi(1,\varphi(1,\varphi(1,0,0),0),0)$ |
| $\psi\left(\Omega^{\Omega\cdot 2}\right)$ | $\varphi(2,0,0)$ |
| $\psi\left(\Omega^{\Omega\cdot}+\Omega^{\psi\left(\Omega^{\Omega\cdot 2}\right)+1}\right)$ | $\varphi(\varphi(2,0,0)+1,0)$ |
| $\psi\left(\Omega^{\Omega\cdot 2}+\Omega^{\Omega}\right)$ | $\varphi(1,0,\varphi(2,0,0)+1)$ |
| $\psi\left(\Omega^{\Omega\cdot 2}+\Omega^{\Omega+\omega}\right)$ | $\varphi(1,\omega,\varphi(2,0,0)+1)$ |
| $\psi\left(\Omega^{\Omega\cdot 2}+\Omega^{\Omega+\psi(\Omega^{\Omega}\cdot 2)}\right)$ | $\varphi(1,\varphi(2,0,0),1)$ |
| $\psi\left(\Omega^{\Omega\cdot 2}+\Omega^{\Omega+\psi(\Omega^{\Omega}\cdot 2)+1}\right)$ | $\varphi(1,\varphi(2,0,0)+1,1)$ |
| $\psi\left(\Omega^{\Omega\cdot 2}\cdot 2\right)$ | $\varphi(2,0,1)$ |
| $\psi\left(\Omega^{\Omega\cdot 2}\cdot\psi\left(\Omega^{\Omega\cdot 2}\right)\right)$ | $\varphi(2,0,\varphi(2,0,0))$ |
| $\psi\left(\Omega^{\Omega\cdot 2+1}\right)$ | $\varphi(2,1,0)$ |
| $\psi\left(\Omega^{\Omega\cdot 2+1}\cdot 2\right)$ | $\varphi(2,1,1)$ |
| $\psi\left(\Omega^{\Omega\cdot 2+\psi(\Omega^{\Omega}\cdot 2)}\right)$ | $\varphi(2,\varphi(2,2,0),0)$ |
| $\psi\left(\Omega^{\Omega\cdot 3}\right)$ | $\varphi(2,\varphi(3,0,0),0)$ |
| $\psi\left(\Omega^{\Omega\cdot 3}+\Omega^{\psi(\Omega^{\Omega}\cdot 3)}\right)$ | $\varphi(\varphi(3,0,0),1)$ |
| $\psi\left(\Omega^{\Omega\cdot 3}+\Omega^{\Omega+\psi(\Omega^{\Omega}\cdot 3)}\right)$ | $\varphi(1,\varphi(3,0,0),1)$ |
| $\psi\left(\Omega^{\Omega\cdot 3}+\Omega^{\Omega\cdot 2+\psi(\Omega^{\Omega}\cdot 3)}\right)$ | $\varphi(2,\varphi(3,0,0),1)$ |
| $\psi\left(\Omega^{\Omega\cdot 3}\cdot 2\right)$ | $\varphi(3,0,1)$ |
| $\psi\left(\Omega^{\Omega\cdot 3+1}\right)$ | $\varphi(3,1,0)$ |
| $\psi\left(\Omega^{\Omega\cdot 4}\right)$ | $\varphi(4,0,0)$ |
| $\psi\left(\Omega^{\Omega\cdot 5}\right)$ | $\varphi(5,0,0)$ |
| $\psi\left(\Omega^{\Omega\cdot\omega}\right)$ | $\varphi(\omega,0,0)$ |
| $\psi\left(\Omega^{\Omega\cdot\psi(0)}\right)$ | $\varphi(\varepsilon_0,0,0)$ |
| $\psi\left(\Omega^{\Omega\cdot\psi(\Omega^{\Omega})}\right)$ | $\varphi(\varphi(1,0,0),0,0)$ |
| $\psi\left(\Omega^{\Omega^2}\right)$ | $\varphi(1,0,0,0)$<br>$\varphi(1@3)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\varphi(\varphi(1,0,0,0),1)$<br>$\varphi(\varphi(1@3)@1,1)$ |

| Madore's OCF | Veblen function |
|---|---|
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega}\right)$ | $\varphi(1,0,\varphi(1,0,0,0)+1)$<br>$\varphi(1@2,\varphi(1@3)+1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega+1}\right)$ | $\varphi(1,1,\varphi(1,0,0,0)+1)$<br>$\varphi(1@2,1@1,\varphi(1@3)+1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega+\omega}\right)$ | $\varphi(1,\omega,\varphi(1,0,0,0)+1)$<br>$\varphi(1@2,\omega@1,\varphi(1@3)+1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega+\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\varphi(1,\varphi(1,0,0,0),1)$<br>$\varphi(1@2,\varphi(1@3)@1,1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot 2}\right)$ | $\varphi(2,0,\varphi(1,0,0,0)+1)$<br>$\varphi(2@2,\varphi(1@3)+1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot 2+\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\varphi(2,\varphi(1,0,0,0),1)$<br>$\varphi(2@2,\varphi(1@3)@1,1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot 3}\right)$ | $\varphi(3,0,\varphi(1,0,0,0)+1)$<br>$\varphi(3@2,\varphi(1@3)+1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega}\right)}\right)$ | $\varphi\left(\Gamma_0,0,\varphi(1,0,0,0)+1\right)$<br>$\varphi\left(\Gamma_0@2,\varphi(1@3)+1\right)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\varphi(\varphi(1,0,0,0),0,1)$<br>$\varphi(\varphi(1@3)@2,1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)+1}\right)$ | $\varphi(\varphi(1,0,0,0),1,0)$<br>$\varphi(\varphi(1@3)@2,1@1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)+\Omega}\right)$ | $\varphi(\varphi(1,0,0,0)+1,0,0)$<br>$\varphi(\varphi(1@3)+1@2)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)\cdot 2}\right)$ | $\varphi(\varphi(1,0,0,0)2,0,0)$<br>$\varphi(\varphi(1@3)2@2)$ |
| $\psi\left(\Omega^{\Omega^2}\cdot 2\right)$ | $\varphi(1,0,0,1)$<br>$\varphi(1@3,1)$ |
| $\psi\left(\Omega^{\Omega^2+1}\right)$ | $\varphi(1,0,1,0)$<br>$\varphi(1@3,1@1)$ |
| $\psi\left(\Omega^{\Omega^2+\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\varphi(1,0,\varphi(1,0,0,0),0)$<br>$\varphi(1@3,\varphi(1@3)@1)$ |
| $\psi\left(\Omega^{\Omega^2+\Omega}\right)$ | $\varphi(1,1,0,0)$<br>$\varphi(1@3,1@2)$ |
| $\psi\left(\Omega^{\Omega^2+\Omega}+\Omega^{\Omega^2+\psi\left(\Omega^{\Omega^2+\Omega}\right)\cdot 2}\right)$ | $\varphi(1,0,\varphi(1,1,0,0)2,0)$<br>$\varphi(1@3,\varphi(1@3,1@2)2@1)$ |
| $\psi\left(\Omega^{\Omega^2+\Omega}\cdot 2\right)$ | $\varphi(1,1,0,1)$<br>$\varphi(1@3,1@2,1)$ |
| $\psi\left(\Omega^{\Omega^2+\Omega+1}\right)$ | $\varphi(1,1,1,0)$<br>$\varphi(1@3,1@2,1@1)$ |
| $\psi\left(\Omega^{\Omega^2+\Omega\cdot 2}\right)$ | $\varphi(1,2,0,0)$<br>$\varphi(1@3,2@2)$ |
| $\psi\left(\Omega^{\Omega^2+\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\varphi(1,\varphi(1,0,0,0),0,0)$<br>$\varphi(1@3,\varphi(1@3)@2)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 2}\right)$ | $\varphi(2,0,0,0)$<br>$\varphi(2@3)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega^2+\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\varphi(1,\varphi(2,0,0,0),0,1)$<br>$\varphi(1@3,\varphi(2@3)@2,1)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 2}\cdot 2\right)$ | $\varphi(2,0,0,1)$<br>$\varphi(2@3,1)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 2+1}\right)$ | $\varphi(2,0,1,0)$<br>$\varphi(2@3,1@1)$ |

| Madore's OCF | Veblen function |
|---|---|
| $\psi\left(\Omega^{\Omega^2\cdot 2+1}\cdot 2\right)$ | $\varphi(2,0,2,1)$<br>$\varphi(2@3,2@1,1)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 2+\Omega}\right)$ | $\varphi(2,1,0,0)$<br>$\varphi(2@3,1@2)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 3}\right)$ | $\varphi(2,1,0,0)$<br>$\varphi(2@3,1@2)$ |
| $\psi\left(\Omega^{\Omega^2\cdot\psi(\Omega^\Omega)}\right)$ | $\varphi(\varphi(1,0,0,0),0,0,0)$<br>$\varphi(\varphi(1@3)@3)$ |
| $\psi\left(\Omega^{\Omega^3}\right)$ | $\varphi(1,0,0,0,0)$<br>$\varphi(1@4)$ |
| $\psi\left(\Omega^{\Omega^3}+\Omega^{\psi(\Omega^{\Omega^3})}\right)$ | $\varphi(\varphi(1,0,0,0,0),1)$<br>$\varphi(\varphi(1@4)@1,1)$ |
| $\psi\left(\Omega^{\Omega^3}+\Omega^{\Omega\cdot\psi(\Omega^{\Omega^3})}\right)$ | $\varphi(\varphi(1,0,0,0,0),0,1)$<br>$\varphi(\varphi(1@4)@2,1)$ |
| $\psi\left(\Omega^{\Omega^3}+\Omega^{\Omega^2\cdot\psi(\Omega^{\Omega^3})}\right)$ | $\varphi(\varphi(1,0,0,0,0),0,0,1)$<br>$\varphi(\varphi(1@4)@3,1)$ |
| $\psi\left(\Omega^{\Omega^3}\cdot 2\right)$ | $\varphi(1,0,0,0,1)$<br>$\varphi(1@4,1)$ |
| $\psi\left(\Omega^{\Omega^3+1}\right)$ | $\varphi(1,0,0,1,0)$<br>$\varphi(1@4,1@1)$ |
| $\psi\left(\Omega^{\Omega^3+\Omega}\right)$ | $\varphi(1,0,1,0,0)$<br>$\varphi(1@4,1@2)$ |
| $\psi\left(\Omega^{\Omega^3+\Omega^2}\right)$ | $\varphi(1,1,0,0,0)$<br>$\varphi(1@4,1@3)$ |
| $\psi\left(\Omega^{\Omega^3\cdot 2}\right)$ | $\varphi(2,0,0,0,0)$<br>$\varphi(2@4)$ |
| $\psi\left(\Omega^{\Omega^4}\right)$ | $\varphi(1,0,0,0,0,0)$<br>$\varphi(1@5)$ |
| $\psi\left(\Omega^{\Omega^5}\right)$ | $\varphi(1,0,0,0,0,0,0)$<br>$\varphi(1@6)$ |
| $\psi\left(\Omega^{\Omega^6}\right)$ | $\varphi(1,0,0,0,0,0,0,0)$<br>$\varphi(1@7)$ |
| $\psi\left(\Omega^{\Omega^\omega}\right)$ | $\varphi(1@\omega)$ |
| $\psi\left(\Omega^{\Omega^\omega}+1\right)$ | $\varphi(1@1,\varphi(1@\omega)+1)$ |
| $\psi\left(\Omega^{\Omega^\omega}+\Omega^\Omega\right)$ | $\varphi(1@2,\varphi(1@\omega)+1)$ |
| $\psi\left(\Omega^{\Omega^\omega}+\Omega^{\Omega^2}\right)$ | $\varphi(1@3,\varphi(1@\omega)+1)$ |
| $\psi\left(\Omega^{\Omega^\omega}+\Omega^{\Omega^3}\right)$ | $\varphi(1@4,\varphi(1@\omega)+1)$ |
| $\psi\left(\Omega^{\Omega^\omega}+\Omega^{\Omega^4}\right)$ | $\varphi(1@5,\varphi(1@\omega)+1)$ |
| $\psi\left(\Omega^{\Omega^\omega}+\Omega^{\Omega^5}\right)$ | $\varphi(1@6,\varphi(1@\omega)+1)$ |
| $\psi\left(\Omega^{\Omega^\omega}\cdot 2\right)$ | $\varphi(1@\omega,1)$ |
| $\psi\left(\Omega^{\Omega^\omega+1}\right)$ | $\varphi(1@\omega,1@1)$ |
| $\psi\left(\Omega^{\Omega^\omega+\psi(\Omega^{\Omega^\omega})}\right)$ | $\varphi(1@\omega,\varphi(1@\omega)@1)$ |
| $\psi\left(\Omega^{\Omega^\omega+\Omega}\right)$ | $\varphi(1@\omega,1@2)$ |
| $\psi\left(\Omega^{\Omega^\omega+\Omega\cdot\psi(\Omega^{\Omega^\omega})}\right)$ | $\varphi(1@\omega,\varphi(1@\omega)@2)$ |
| $\psi\left(\Omega^{\Omega^\omega+\Omega^2}\right)$ | $\varphi(1@\omega,1@3)$ |
| $\psi\left(\Omega^{\Omega^\omega+\Omega^3}\right)$ | $\varphi(1@\omega,1@4)$ |
| $\psi\left(\Omega^{\Omega^\omega+\Omega^4}\right)$ | $\varphi(1@\omega,1@5)$ |

| Madore's OCF | Veblen function |
|---|---|
| $\psi\left(\Omega^{\Omega^{\omega}\cdot 2}\right)$ | $\varphi(2@\omega)$ |
| $\psi\left(\Omega^{\Omega^{\omega}\cdot\psi(\Omega^{\Omega^{\omega}})}\right)$ | $\varphi(\varphi(1@\omega)@\omega)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}}\right)$ | $\varphi(1@\omega+1)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}}+\Omega^{\Omega^{\omega}}\right)$ | $\varphi(\varphi(1@\omega+1)@\omega,1)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}}\cdot 2\right)$ | $\varphi(1@\omega+1,1)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}+1}\right)$ | $\varphi(1@\omega+1,1@1)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}+\Omega}\right)$ | $\varphi(1@\omega+1,1@2)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}+\Omega^{\omega}}\right)$ | $\varphi(1@\omega+1,1@\omega)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}\cdot 2}\right)$ | $\varphi(2@\omega+1)$ |
| $\psi\left(\Omega^{\Omega^{\omega+2}}\right)$ | $\varphi(1@\omega+2)$ |
| $\psi\left(\Omega^{\Omega^{\omega+3}}\right)$ | $\varphi(1@\omega+3)$ |
| $\psi\left(\Omega^{\Omega^{\omega\cdot 2}}\right)$ | $\varphi(1@\omega\cdot 2)$ |
| $\psi\left(\Omega^{\Omega^{\omega^2}}\right)$ | $\varphi(1@\omega^2)$ |
| $\psi\left(\Omega^{\Omega^{\psi(0)}}\right)$ | $\varphi(1@\varepsilon_0)$ |
| $\psi\left(\Omega^{\Omega^{\psi(\Omega)}}\right)$ | $\varphi(1@\zeta_0)$ |
| $\psi\left(\Omega^{\Omega^{\psi(\Omega^{\Omega})}}\right)$ | $\varphi(1@\Gamma_0)$ |
| $\psi\left(\Omega^{\Omega^{\psi(\Omega^{\Omega^2})}}\right)$ | $\varphi(1@\varphi(1@3))$ |
| $\psi\left(\Omega^{\Omega^{\psi(\Omega^{\Omega^{\omega}})}}\right)$ | $\varphi(1@\varphi(1@\omega))$ |
| $\psi\left(\Omega^{\Omega^{\Omega}}\right)$ | $\varphi(1@(1,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega}}+\Omega^{\psi\left(\Omega^{\Omega^{\Omega}}\right)}\right)$ | $\varphi(\varphi(1@(1,0))@1,1)$ |
| $\psi\left(\Omega^{\Omega^{\Omega}}+\Omega^{\Omega^{\omega}\cdot\psi\left(\Omega^{\Omega^{\Omega}}\right)}\right)$ | $\varphi(\varphi(1@(1,0))@\omega,1)$ |
| $\psi\left(\Omega^{\Omega^{\Omega}}\cdot 2\right)$ | $\varphi(1@(1,0),1)$ |
| $\psi\left(\Omega^{\Omega^{\Omega}+1}\right)$ | $\varphi(1@(1,0),1@1)$ |
| $\psi\left(\Omega^{\Omega^{\Omega}+\Omega^{\omega}}\right)$ | $\varphi(1@(1,0),1@\omega)$ |
| $\psi\left(\Omega^{\Omega^{\Omega}\cdot 2}\right)$ | $\varphi(2@(1,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega}\cdot\psi\left(\Omega^{\Omega^{\Omega}}\right)}\right)$ | $\varphi(\varphi(1@(1,0))@(1,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega+1}}\right)$ | $\varphi(1@(1,1))$ |
| $\psi\left(\Omega^{\Omega^{\Omega+1}}+\Omega^{\Omega^{\Omega}\cdot\psi(\Omega^{\Omega^{\Omega+1}})}\right)$ | $\varphi(\varphi(1@(1,1))@(1,0),1)$ |
| $\psi\left(\Omega^{\Omega^{\Omega+1}\cdot 2}\right)$ | $\varphi(2@(1,1))$ |
| $\psi\left(\Omega^{\Omega^{\Omega+2}}\right)$ | $\varphi(1@(1,2))$ |
| $\psi\left(\Omega^{\Omega^{\Omega+\omega}}\right)$ | $\varphi(1@(1,\omega))$ |
| $\psi\left(\Omega^{\Omega^{\Omega+\psi\left(\Omega^{\Omega^{\Omega}}\right)}}\right)$ | $\varphi(1@(1,\varphi(1@(1,0))))$ |
| $\psi\left(\Omega^{\Omega^{\Omega\cdot 2}}\right)$ | $\varphi(1@(2,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega\cdot 2+\psi\left(\Omega^{\Omega^{\Omega\cdot 2}}\right)}}\right)$ | $\varphi(1@(2,\varphi(1@(2,0))))$ |

| Madore's OCF | Veblen function |
|---|---|
| $\psi\left(\Omega^{\Omega^{\Omega\cdot 3}}\right)$ | $\varphi(1@(3,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega\cdot\omega}}\right)$ | $\varphi(1@(\omega,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^{\Omega}}\right)}}\right)$ | $\varphi(1@(\varphi(1@(1,0)),0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^2}}\right)$ | $\varphi(1@(1,0,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^2\cdot 2}}\right)$ | $\varphi(2@(1,0,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^2+1}}\right)$ | $\varphi(1@(1,0,1))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^2+\Omega}}\right)$ | $\varphi(1@(1,1,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^2\cdot 2}}\right)$ | $\varphi(1@(2,0,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^3}}\right)$ | $\varphi(1@(1,0,0,0))$<br>$\varphi(1@(1@3))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^3\cdot 2}}\right)$ | $\varphi(1@(2@3))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^4}}\right)$ | $\varphi(1@(1@4))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^\omega}}\right)$ | $\varphi(1@(1@\omega))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega}}}\right)$ | $\varphi(1@(1@(1,0)))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega}}}+1\right)$ | $\varphi(1@1,\varphi(1@(1@(1,0)))+1)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega}}}\cdot 2\right)$ | $\varphi(1@(1@(1,0)),1)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega}}+1}\right)$ | $\varphi(1@(1@(1,0)),1@1)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega}}\cdot 2}\right)$ | $\varphi(2@(1@(1,0)))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega+1}}}\right)$ | $\varphi(1@(1@(1,0),1))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega\cdot 2}}}\right)$ | $\varphi(1@(2@(1,0)))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega+1}}}\right)$ | $\varphi(1@(1@(1,1)))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega\cdot 2}}}\right)$ | $\varphi(1@(1@(2,0)))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega^2}}}\right)$ | $\varphi(1@(1@(1,0,0)))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega^{\Omega}}}}\right)$ | $\varphi(1@(1@(1@(1,0))))$ |
| $\psi\left(\psi_1(0)\right)$ | $\varphi(1@(1,,0))$ |

## A.5 BOCF vs Cantor-style/Veblen Function

The results in this section are primarily cited from[1].

| Buchholz's OCF | Cantor-style/Veblen function |
|---|---|
| $\psi(0)$ | $1$ |
| $\psi(0)\cdot 2$ | $2$ |
| $\psi(0)\cdot 3$ | $3$ |
| $\psi(1)$ | $\omega$ |
| $\psi(1)+\psi(0)$ | $\omega+1$ |
| $\psi(1)+\psi(0)\cdot 2$ | $\omega+2$ |
| $\psi(1)+\psi(0)\cdot 5$ | $\omega+5$ |
| $\psi(1)\cdot 2$ | $\omega\cdot 2$ |
| $\psi(1)\cdot 2+\psi(0)$ | $\omega\cdot 2+1$ |
| $\psi(1)\cdot 3$ | $\omega\cdot 3$ |

| Buchholz's OCF | Cantor-style/Veblen function |
|---|---|
| $\psi(2)$ | $\omega^2$ |
| $\psi(2)+1$ | $\omega^2+1$ |
| $\psi(2)+2$ | $\omega^2+2$ |
| $\psi(2)+\psi(1)$ | $\omega^2+\omega$ |
| $\psi(2)+\psi(1)+\psi(0)$ | $\omega^2+\omega+1$ |
| $\psi(2)+\psi(1)\cdot 2$ | $\omega^2+\omega\cdot 2$ |
| $\psi(2)\cdot 2+\psi(1)$ | $\omega^2\cdot 2+\omega$ |
| $\psi(2)\cdot 3$ | $\omega^2\cdot 3$ |
| $\psi(3)$ | $\omega^3$ |
| $\psi(3)+1$ | $\omega^3+1$ |
| $\psi(3)+\psi(1)$ | $\omega^3+\omega$ |
| $\psi(3)+\psi(2)$ | $\omega^3+\omega^2$ |
| $\psi(3)\cdot 2$ | $\omega^3\cdot 2$ |
| $\psi(3)\cdot 3$ | $\omega^3\cdot 3$ |
| $\psi(4)$ | $\omega^4$ |
| $\psi(4)\cdot 2$ | $\omega^4\cdot 2$ |
| $\psi(5)$ | $\omega^5$ |
| $\psi(\psi(1))$ | $\omega^\omega$ |
| $\psi(\psi(1))+\psi(0)$ | $\omega^\omega+1$ |
| $\psi(\psi(1))+\psi(0)\cdot 2$ | $\omega^\omega+2$ |
| $\psi(\psi(1))+\psi(1)$ | $\omega^\omega+\omega$ |
| $\psi(\psi(1))+\psi(1)\cdot 2$ | $\omega^\omega+\omega\cdot 2$ |
| $\psi(\psi(1))+\psi(2)$ | $\omega^\omega+\omega^2$ |
| $\psi(\psi(1))+\psi(3)$ | $\omega^\omega+\omega^3$ |
| $\psi(\psi(1))\cdot 2$ | $\omega^\omega\cdot 2$ |
| $\psi(\psi(1))\cdot 3$ | $\omega^\omega\cdot 3$ |
| $\psi(\psi(1)+1)$ | $\omega^{\omega+1}$ |
| $\psi(\psi(1)+1)+\psi(0)$ | $\omega^{\omega+1}+1$ |
| $\psi(\psi(1)+1)+\psi(1)$ | $\omega^{\omega+1}+\omega$ |
| $\psi(\psi(1)+1)+\psi(2)$ | $\omega^{\omega+1}+\omega^2$ |
| $\psi(\psi(1)+1)+\psi(3)$ | $\omega^{\omega+1}+\omega^3$ |
| $\psi(\psi(1)+1)+\psi(\psi(1))$ | $\omega^{\omega+1}+\omega^\omega$ |
| $\psi(\psi(1)+1)+\psi(\psi(1))\cdot 2$ | $\omega^{\omega+1}+\omega^\omega\cdot 2$ |
| $\psi(\psi(1)+1)\cdot 2$ | $\omega^{\omega+1}\cdot 2$ |
| $\psi(\psi(1)+1)\cdot 3$ | $\omega^{\omega+1}\cdot 3$ |
| $\psi(\psi(1)+2)$ | $\omega^{\omega+2}$ |
| $\psi(\psi(1)+2)+\psi(\psi(1))$ | $\omega^{\omega+2}+\omega^\omega$ |
| $\psi(\psi(1)+2)+\psi(\psi(1)+1)$ | $\omega^{\omega+2}+\omega^{\omega+1}$ |
| $\psi(\psi(1)+3)$ | $\omega^{\omega+3}$ |
| $\psi(\psi(1)+4)$ | $\omega^{\omega+4}$ |
| $\psi(\psi(1)\cdot 2)$ | $\omega^{\omega\cdot 2}$ |
| $\psi(\psi(1)\cdot 2)+\psi(1)$ | $\omega^{\omega\cdot 2}+\omega$ |
| $\psi(\psi(1)\cdot 2)+\psi(\psi(1))$ | $\omega^{\omega\cdot 2}+\omega^\omega$ |
| $\psi(\psi(1)\cdot 2)+\psi(\psi(1)+1)$ | $\omega^{\omega\cdot 2}+\omega^{\omega+1}$ |
| $\psi(\psi(1)\cdot 2)+\psi(\psi(1)+2)$ | $\omega^{\omega\cdot 2}+\omega^{\omega+2}$ |
| $\psi(\psi(1)\cdot 2)\cdot 2$ | $\omega^{\omega\cdot 2}\cdot 2$ |
| $\psi(\psi(1)\cdot 2+1)$ | $\omega^{\omega\cdot 2+1}$ |
| $\psi(\psi(1)\cdot 2+1)\cdot 2$ | $\omega^{\omega\cdot 2+1}\cdot 2$ |
| $\psi(\psi(1)\cdot 2+2)$ | $\omega^{\omega\cdot 2+2}$ |
| $\psi(\psi(1)\cdot 2+3)$ | $\omega^{\omega\cdot 2+3}$ |
| $\psi(\psi(1)\cdot 3)$ | $\omega^{\omega\cdot 3}$ |
| $\psi(\psi(1)\cdot 3+1)$ | $\omega^{\omega\cdot 3+1}$ |
| $\psi(\psi(1)\cdot 4)$ | $\omega^{\omega\cdot 4}$ |
| $\psi(\psi(2))$ | $\omega^{\omega^2}$ |
| $\psi(\psi(2))\cdot 2$ | $\omega^{\omega^2}\cdot 2$ |

| Buchholz's OCF | Cantor-style/Veblen function |
|---|---|
| $\psi(\psi(2)+1)$ | $\omega^{\omega^2+1}$ |
| $\psi(\psi(2)+2)$ | $\omega^{\omega^2+2}$ |
| $\psi(\psi(2)+\psi(1))$ | $\omega^{\omega^2+\omega}$ |
| $\psi(\psi(2)+\psi(1)+1)$ | $\omega^{\omega^2+\omega+1}$ |
| $\psi(\psi(2)+\psi(1)\cdot 2)$ | $\omega^{\omega^2+\omega\cdot 2}$ |
| $\psi(\psi(2)+\psi(1)\cdot 3)$ | $\omega^{\omega^2+\omega\cdot 3}$ |
| $\psi(\psi(2)\cdot 2)$ | $\omega^{\omega^2\cdot 2}$ |
| $\psi(\psi(2)\cdot 3)$ | $\omega^{\omega^2\cdot 3}$ |
| $\psi(\psi(3))$ | $\omega^{\omega^3}$ |
| $\psi(\psi(3)+1)$ | $\omega^{\omega^3+1}$ |
| $\psi(\psi(3)+\psi(1))$ | $\omega^{\omega^3+\omega}$ |
| $\psi(\psi(3)+\psi(2))$ | $\omega^{\omega^3+\omega^2}$ |
| $\psi(\psi(3)\cdot 2)$ | $\omega^{\omega^3\cdot 2}$ |
| $\psi(\psi(4))$ | $\omega^{\omega^4}$ |
| $\psi(\psi(4)\cdot 2)$ | $\omega^{\omega^4\cdot 2}$ |
| $\psi(\psi(5))$ | $\omega^{\omega^5}$ |
| $\psi(\psi(\psi(1)))$ | $\omega^{\omega^\omega}$ |
| $\psi(\psi(\psi(1)))+\psi(1)$ | $\omega^{\omega^\omega}+\omega$ |
| $\psi(\psi(\psi(1)))+\psi(\psi(1))$ | $\omega^{\omega^\omega}+\omega^\omega$ |
| $\psi(\psi(\psi(1)))\cdot 2$ | $\omega^{\omega^\omega}\cdot 2$ |
| $\psi(\psi(\psi(1))+1)$ | $\omega^{\omega^\omega+1}$ |
| $\psi(\psi(\psi(1))+\psi(1))$ | $\omega^{\omega^\omega+\omega}$ |
| $\psi(\psi(\psi(1))\cdot 2)$ | $\omega^{\omega^\omega\cdot 2}$ |
| $\psi(\psi(\psi(1)+1))$ | $\omega^{\omega^{\omega+1}}$ |
| $\psi(\psi(\psi(1)\cdot 2))$ | $\omega^{\omega^\omega\cdot 2}$ |
| $\psi(\psi(\psi(2)))$ | $\omega^{\omega^{\omega^2}}$ |
| $\psi(\psi(\psi(\psi(1))))$ | $\omega^{\omega^{\omega^\omega}}$ |
| $\psi(\psi(\psi(\psi(1))))+1$ | $\omega^{\omega^{\omega^\omega}}+1$ |
| $\psi(\psi(\psi(\psi(1)))+1)$ | $\omega^{\omega^{\omega^\omega}+1}$ |
| $\psi(\psi(\psi(\psi(1))+1))$ | $\omega^{\omega^{\omega^\omega+1}}$ |
| $\psi(\psi(\psi(\psi(1)+1)))$ | $\omega^{\omega^{\omega^{\omega+1}}}$ |
| $\psi(\psi(\psi(\psi(2))))$ | $\omega^{\omega^{\omega^{\omega^2}}}$ |
| $\psi(\psi(\psi(\psi(\psi(1)))))$ | $\omega^{\omega^{\omega^{\omega^\omega}}}$ |
| $\psi(\Omega)$ | $\varepsilon_0$ |
| $\psi(\Omega)+\psi(0)$ | $\varepsilon_0+1$ |
| $\psi(\Omega)+\psi(1)$ | $\varepsilon_0+\omega$ |
| $\psi(\Omega)+\psi(\psi(1))$ | $\varepsilon_0+\omega^\omega$ |
| $\psi(\Omega)+\psi(\psi(\psi(1)))$ | $\varepsilon_0+\omega^{\omega^\omega}$ |
| $\psi(\Omega)\cdot 2$ | $\varepsilon_0\cdot 2$ |
| $\psi(\Omega)\cdot 3$ | $\varepsilon_0\cdot 3$ |
| $\psi(\Omega+1)$ | $\varepsilon_0\cdot\omega$<br>$\omega^{\varepsilon_0+1}$ |
| $\psi(\Omega+1)+\psi(1)$ | $\varepsilon_0\cdot\omega+\omega$<br>$\omega^{\varepsilon_0+1}+\omega$ |
| $\psi(\Omega+1)+\psi(\psi(1))$ | $\varepsilon_0\cdot\omega+\omega^\omega$<br>$\omega^{\varepsilon_0+1}+\omega^\omega$ |
| $\psi(\Omega+1)\cdot 2$ | $\varepsilon_0\cdot\omega\cdot 2$<br>$\omega^{\varepsilon_0+1}\cdot 2$ |
| $\psi(\Omega+2)$ | $\varepsilon_0\cdot\omega^2$<br>$\omega^{\varepsilon_0+2}$ |

| Buchholz's OCF | Cantor-style/Veblen function |
|---|---|
| $\psi(\Omega+2)\cdot 2$ | $\varepsilon_0\cdot\omega^2\cdot 2$<br>$\omega^{\varepsilon_0+2}\cdot 2$ |
| $\psi(\Omega+3)$ | $\varepsilon_0\cdot\omega^3$<br>$\omega^{\varepsilon_0+3}$ |
| $\psi(\Omega+\psi(1))$ | $\varepsilon_0\cdot\omega^\omega$<br>$\omega^{\varepsilon_0+\omega}$ |
| $\psi(\Omega+\psi(2))$ | $\varepsilon_0\cdot\omega^{\omega^2}$<br>$\omega^{\varepsilon_0+\omega^2}$ |
| $\psi(\Omega+\psi(\psi(1)))$ | $\varepsilon_0\cdot\omega^{\omega^\omega}$<br>$\omega^{\varepsilon_0+\omega^\omega}$ |
| $\psi(\Omega+\psi(\Omega))$ | ${\varepsilon_0}^2$<br>$\omega^{\varepsilon_0\cdot 2}$ |
| $\psi(\Omega+\psi(\Omega))+\psi(1)$ | ${\varepsilon_0}^2+\omega$<br>$\omega^{\varepsilon_0\cdot 2}+\omega$ |
| $\psi(\Omega+\psi(\Omega))+\psi(\psi(1))$ | ${\varepsilon_0}^2+\omega^\omega$<br>$\omega^{\varepsilon_0\cdot 2}+\omega^\omega$ |
| $\psi(\Omega+\psi(\Omega))+\psi(\Omega)$ | ${\varepsilon_0}^2+\varepsilon_0$<br>$\omega^{\varepsilon_0\cdot 2}+\varepsilon_0$ |
| $\psi(\Omega+\psi(\Omega))+\psi(\Omega+1)$ | ${\varepsilon_0}^2+\varepsilon_0\cdot\omega$<br>$\omega^{\varepsilon_0\cdot 2}+\omega^{\varepsilon_0+1}$ |
| $\psi(\Omega+\psi(\Omega))\cdot 2$ | ${\varepsilon_0}^2\cdot 2$<br>$\omega^{\varepsilon_0\cdot 2}\cdot 2$ |
| $\psi(\Omega+\psi(\Omega)+1)$ | ${\varepsilon_0}^2\cdot\omega$<br>$\omega^{\varepsilon_0\cdot 2+1}$ |
| $\psi(\Omega+\psi(\Omega)+\psi(1))$ | ${\varepsilon_0}^2\cdot\omega^\omega$<br>$\omega^{\varepsilon_0\cdot 2+\omega}$ |
| $\psi(\Omega+\psi(\Omega)\cdot 2)$ | ${\varepsilon_0}^3$<br>$\omega^{\varepsilon_0\cdot 3}$ |
| $\psi(\Omega+\psi(\Omega)\cdot 3)$ | ${\varepsilon_0}^4$<br>$\omega^{\varepsilon_0\cdot 4}$ |
| $\psi(\Omega+\psi(\Omega+1))$ | ${\varepsilon_0}^\omega$<br>$\omega^{\omega^{\varepsilon_0+1}}$ |
| $\psi(\Omega+\psi(\Omega+1))\cdot 2$ | ${\varepsilon_0}^\omega\cdot 2$<br>$\omega^{\omega^{\varepsilon_0+1}}\cdot 2$ |
| $\psi(\Omega+\psi(\Omega+1)+1)$ | ${\varepsilon_0}^\omega\cdot\omega$<br>$\omega^{\omega^{\varepsilon_0+1}+1}$ |
| $\psi(\Omega+\psi(\Omega+1)+\psi(1))$ | ${\varepsilon_0}^\omega\cdot\omega^\omega$<br>$\omega^{\omega^{\varepsilon_0+1}+\omega}$ |
| $\psi(\Omega+\psi(\Omega+1)+\psi(\Omega))$ | ${\varepsilon_0}^{\omega+1}$<br>$\omega^{\omega^{\varepsilon_0+1}+\varepsilon_0}$ |
| $\psi(\Omega+\psi(\Omega+1)+\psi(\Omega)\cdot 2)$ | ${\varepsilon_0}^{\omega+2}$<br>$\omega^{\omega^{\varepsilon_0+1}+\varepsilon_0\cdot 2}$ |
| $\psi(\Omega+\psi(\Omega+1)\cdot 2)$ | ${\varepsilon_0}^{\omega\cdot 2}$<br>$\omega^{\omega^{\varepsilon_0+1}\cdot 2}$ |
| $\psi(\Omega+\psi(\Omega+2))$ | ${\varepsilon_0}^{\omega^2}$<br>$\omega^{\omega^{\varepsilon_0+2}}$ |
| $\psi(\Omega+\psi(\Omega+\psi(1)))$ | $\omega^{\omega^{\varepsilon_0+\omega}}$ |

| Buchholz's OCF | Cantor-style/Veblen function |
|---|---|
| $\psi(\Omega+\psi(\Omega+\psi(\psi(1))))$ | ${\varepsilon_0}^{\omega^\omega}$<br>$\omega^{\omega^{\varepsilon_0+\omega^\omega}}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega)))$ | $\omega^{\omega^{\varepsilon_0\cdot 2}}$<br>${\varepsilon_0}^{\varepsilon_0}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega))+1)$ | ${\varepsilon_0}^{\varepsilon_0}\cdot\omega$<br>$\omega^{\omega^{\varepsilon_0\cdot 2}+1}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega))+\psi(\Omega))$ | ${\varepsilon_0}^{\varepsilon_0+1}$<br>$\omega^{\omega^{\varepsilon_0\cdot 2}+\varepsilon_0}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega))+\psi(\Omega+1))$ | ${\varepsilon_0}^{\varepsilon_0+\omega}$<br>$\omega^{\omega^{\varepsilon_0\cdot 2}+\omega^{\varepsilon_0+1}}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega))\cdot 2)$ | ${\varepsilon_0}^{\varepsilon_0\cdot 2}$<br>$\omega^{\omega^{\varepsilon_0\cdot 2}\cdot 2}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega)+1))$ | ${\varepsilon_0}^{\varepsilon_0\cdot\omega}$<br>$\omega^{\omega^{\varepsilon_0\cdot 2+1}}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega)\cdot 2))$ | ${\varepsilon_0}^{{\varepsilon_0}^2}$<br>$\omega^{\omega^{\varepsilon_0\cdot 3}}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega+1)))$ | ${\varepsilon_0}^{{\varepsilon_0}^\omega}$<br>$\omega^{\omega^{\omega^{\varepsilon_0+1}}}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega+\psi(\Omega))))$ | ${\varepsilon_0}^{{\varepsilon_0}^{\varepsilon_0}}$<br>$\omega^{\omega^{\omega^{\varepsilon_0\cdot 2}}}$ |
| $\psi(\Omega\cdot 2)$ | $\varepsilon_1$ |
| $\psi(\Omega\cdot 2)+\psi(\Omega)$ | $\varepsilon_1+\varepsilon_0$ |
| $\psi(\Omega\cdot 2)+\psi(\Omega+1)$ | $\varepsilon_1+\varepsilon_0\cdot\omega$<br>$\varepsilon_1+\omega^{\varepsilon_0+1}$ |
| $\psi(\Omega\cdot 2)+\psi(\Omega+\psi(\Omega+1))$ | $\varepsilon_1+{\varepsilon_0}^\omega$<br>$\varepsilon_1+\omega^{\omega^{\varepsilon_0}}$ |
| $\psi(\Omega\cdot 2)\cdot 2$ | $\varepsilon_1\cdot 2$ |
| $\psi(\Omega\cdot 2+1)$ | $\varepsilon_1\cdot\omega$<br>$\omega^{\varepsilon_1+1}$ |
| $\psi(\Omega\cdot 2+\psi(\Omega))$ | $\varepsilon_1\cdot\varepsilon_0$<br>$\omega^{\varepsilon_1+\varepsilon_0}$ |
| $\psi(\Omega\cdot 2+\psi(\Omega+\psi(\Omega)))$ | $\varepsilon_1\cdot{\varepsilon_0}^2$<br>$\omega^{\varepsilon_1+\omega^{\varepsilon_0\cdot 2}}$ |
| $\psi(\Omega\cdot 2+\psi(\Omega\cdot 2)\cdot 2)$ | ${\varepsilon_1}^2$<br>$\omega^{\varepsilon_1\cdot 2}$ |
| $\psi(\Omega\cdot 2+\psi(\Omega\cdot 2)\cdot 3)$ | ${\varepsilon_1}^3$<br>$\omega^{\varepsilon_1\cdot 3}$ |
| $\psi(\Omega\cdot 2+\psi(\Omega\cdot 2+1))$ | ${\varepsilon_1}^\omega$<br>$\omega^{\omega^{\varepsilon_1+1}}$ |
| $\psi(\Omega\cdot 2+\psi(\Omega\cdot 2+\psi(\Omega)))$ | ${\varepsilon_1}^{\varepsilon_0}$<br>$\omega^{\omega^{\varepsilon_1+\varepsilon_0}}$ |
| $\psi(\Omega\cdot 2+\psi(\Omega\cdot 2+\psi(\Omega\cdot 2)))$ | ${\varepsilon_1}^{\varepsilon_1}$<br>$\omega^{\omega^{\varepsilon_1\cdot 2}}$ |
| $\psi(\Omega\cdot 3)$ | $\varepsilon_2$ |
| $\psi(\Omega\cdot 3+1)$ | $\varepsilon_2\cdot\omega$<br>$\omega^{\varepsilon_2+1}$ |

| Buchholz's OCF | Cantor-style/Veblen function |
|---|---|
| $\psi(\Omega \cdot 3 + \psi(\Omega))$ | $\varepsilon_2 \cdot \varepsilon_0$<br>$\omega^{\varepsilon_2+\varepsilon_0}$ |
| $\psi(\Omega \cdot 3 + \psi(\Omega \cdot 2))$ | $\varepsilon_2 \cdot \varepsilon_1$<br>$\omega^{\varepsilon_2+\varepsilon_1}$ |
| $\psi(\Omega \cdot 3 + \psi(\Omega \cdot 3))$ | ${\varepsilon_2}^2$<br>$\omega^{\varepsilon_2 \cdot 2}$ |
| $\psi(\Omega \cdot 3 + \psi(\Omega \cdot 3 + 1))$ | ${\varepsilon_2}^\omega$<br>$\omega^{\omega^{\varepsilon_2+1}}$ |
| $\psi(\Omega \cdot 3 + \psi(\Omega \cdot 3 + \psi(\Omega \cdot 3)))$ | ${\varepsilon_2}^{\varepsilon_2}$<br>$\omega^{\omega^{\varepsilon_2 \cdot 2}}$ |
| $\psi(\Omega \cdot 4)$ | $\varepsilon_3$ |
| $\psi(\Omega \cdot 4 + \psi(\Omega \cdot 4))$ | ${\varepsilon_3}^2$<br>$\omega^{\varepsilon_3 \cdot 2}$ |
| $\psi(\Omega \cdot 5)$ | $\varepsilon_4$ |
| $\psi(\Omega \cdot 6)$ | $\varepsilon_5$ |
| $\psi(\Omega \cdot \psi(1))$ | $\varepsilon_\omega$ |
| $\psi(\Omega \cdot \psi(1) + \psi(\Omega \cdot \psi(1)))$ | ${\varepsilon_\omega}^2$<br>$\omega^{\varepsilon_\omega \cdot 2}$ |
| $\psi(\Omega \cdot \psi(1) + \Omega)$ | $\varepsilon_{\omega+1}$ |
| $\psi(\Omega \cdot \psi(1) + \Omega \cdot 2)$ | $\varepsilon_{\omega+2}$ |
| $\psi(\Omega \cdot \psi(1) \cdot 2)$ | $\varepsilon_{\omega \cdot 2}$ |
| $\psi(\Omega \cdot \psi(2))$ | $\varepsilon_{\omega^2}$ |
| $\psi(\Omega \cdot \psi(\psi(1)))$ | $\varepsilon_{\omega^\omega}$ |
| $\psi(\Omega \cdot \psi(\psi(\psi(1))))$ | $\varepsilon_{\omega^{\omega^\omega}}$ |
| $\psi(\Omega \cdot \psi(\Omega))$ | $\varepsilon_{\varepsilon_0}$ |
| $\psi(\Omega \cdot \psi(\Omega) + \Omega)$ | $\varepsilon_{\varepsilon_0+1}$ |
| $\psi(\Omega \cdot \psi(\Omega) \cdot 2)$ | $\varepsilon_{\varepsilon_0 \cdot 2}$ |
| $\psi(\Omega \cdot \psi(\Omega + 1))$ | $\varepsilon_{\varepsilon_0 \cdot \omega}$<br>$\varepsilon_{\omega^{\varepsilon_0+1}}$ |
| $\psi(\Omega \cdot \psi(\Omega + \psi(\Omega)))$ | ${\varepsilon_{\varepsilon_0}}^2$<br>$\varepsilon_{\omega^{\varepsilon_0 \cdot 2}}$ |
| $\psi(\Omega \cdot \psi(\Omega \cdot 2))$ | $\varepsilon_{\varepsilon_1}$ |
| $\psi(\Omega \cdot \psi(\Omega \cdot 3))$ | $\varepsilon_{\varepsilon_2}$ |
| $\psi(\Omega \cdot \psi(\Omega \cdot \psi(1)))$ | $\varepsilon_{\varepsilon_\omega}$ |
| $\psi(\Omega \cdot \psi(\Omega \cdot \psi(\Omega)))$ | $\varepsilon_{\varepsilon_{\varepsilon_0}}$ |
| $\psi(\Omega^2)$ | $\zeta_0$ |
| $\psi(\Omega^2) + \psi(1)$ | $\zeta_0 + \omega$ |
| $\psi(\Omega^2) + \psi(\Omega)$ | $\zeta_0 + \varepsilon_0$ |
| $\psi(\Omega^2) + \psi(\Omega \cdot \psi(\Omega))$ | $\zeta_0 + \varepsilon_{\varepsilon_0}$ |
| $\psi(\Omega^2) \cdot 2$ | $\zeta_0 \cdot 2$ |
| $\psi(\Omega^2 + 1)$ | $\zeta_0 \cdot \omega$<br>$\omega^{\zeta_0+1}$ |
| $\psi(\Omega^2 + \psi(\Omega))$ | $\zeta_0 \cdot \varepsilon_0$<br>$\omega^{\zeta_0+\varepsilon_0}$ |
| $\psi(\Omega^2 + \psi(\Omega \cdot \psi(\Omega)))$ | $\zeta_0 \cdot \varepsilon_{\varepsilon_0}$<br>$\omega^{\zeta_0+\varepsilon_{\varepsilon_0}}$ |
| $\psi(\Omega^2 + \psi(\Omega^2))$ | ${\zeta_0}^2$<br>$\omega^{\zeta_0 \cdot 2}$ |
| $\psi(\Omega^2 + \psi(\Omega^2 + 1))$ | ${\zeta_0}^\omega$<br>$\omega^{\omega^{\zeta_0+1}}$ |

| Buchholz's OCF | Cantor-style/Veblen function |
|---|---|
| $\psi(\Omega^2+\psi(\Omega^2+\psi(\Omega)))$ | ${\zeta_0}^{\varepsilon_0}$<br>$\omega^{\omega^{\zeta_0+\varepsilon_0}}$ |
| $\psi(\Omega^2+\psi(\Omega^2+\psi(\Omega^2)))$ | ${\zeta_0}^{\zeta_0}$<br>$\omega^{\omega^{\zeta_0\cdot 2}}$ |
| $\psi(\Omega^2+\psi(\Omega^2+\psi(\Omega^2+\psi(\Omega^2))))$ | ${\zeta_0}^{{\zeta_0}^{\zeta_0}}$<br>$\omega^{\omega^{\omega^{\zeta_0\cdot 2}}}$ |
| $\psi(\Omega^2+\Omega)$ | $\varepsilon_{\zeta_0+1}$ |
| $\psi(\Omega^2+\Omega+\psi(\Omega^2+\Omega))$ | ${\varepsilon_{\zeta_0+1}}^2$<br>$\omega^{\varepsilon_{\zeta_0+1}\cdot 2}$ |
| $\psi(\Omega^2+\Omega+\psi(\Omega^2+\Omega+\psi(\Omega^2+\Omega)))$ | ${\varepsilon_{\zeta_0+1}}^{\varepsilon_{\zeta_0+1}}$<br>$\omega^{\omega^{\varepsilon_{\zeta_0+1}\cdot 2}}$ |
| $\psi(\Omega^2+\Omega\cdot 2)$ | $\varepsilon_{\zeta_0+2}$ |
| $\psi(\Omega^2+\Omega\cdot 3)$ | $\varepsilon_{\zeta_0+3}$ |
| $\psi(\Omega^2+\Omega\cdot\omega)$ | $\varepsilon_{\zeta_0+\omega}$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega))$ | $\varepsilon_{\zeta_0+\varepsilon_0}$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega\cdot\psi(\Omega)))$ | $\varepsilon_{\zeta_0+\varepsilon_{\varepsilon_0}}$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2))$ | $\varepsilon_{\zeta_0\cdot 2}$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2)\cdot 2)$ | $\varepsilon_{\zeta_0\cdot 3}$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+1))$ | $\varepsilon_{\zeta_0\cdot\omega}$<br>$\varepsilon_{\omega^{\zeta_0+1}}$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+\psi(\Omega^2)))$ | $\varepsilon_{\omega^{\zeta_0\cdot 2}}$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+\psi(\Omega^2+\psi(\Omega^2))))$ | $\varepsilon_{{\zeta_0}^{\zeta_0}}$<br>$\varepsilon_{\omega^{\omega^{\zeta_0\cdot 2}}}$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+\Omega))$ | $\varepsilon_{\varepsilon_{\zeta_0+1}}$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+\Omega\cdot\psi(\Omega)))$ | $\varepsilon_{\varepsilon_{\zeta_0+\varepsilon_0}}$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+\Omega\cdot\psi(\Omega^2)))$ | $\varepsilon_{\varepsilon_{\zeta_0\cdot 2}}$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+\Omega)))$ | $\varepsilon_{\varepsilon_{\varepsilon_{\zeta_0+1}}}$ |
| $\psi(\Omega^2\cdot 2)$ | $\zeta_1$ |
| $\psi(\Omega^2\cdot 2+1)$ | $\zeta_1\cdot\omega$<br>$\omega^{\zeta_1+1}$ |
| $\psi(\Omega^2\cdot 2+\psi(\Omega^2))$ | $\zeta_1\cdot\zeta_0$<br>$\omega^{\zeta_1+\zeta_0}$ |
| $\psi(\Omega^2\cdot 2+\psi(\Omega^2\cdot 2))$ | ${\zeta_1}^2$<br>$\omega^{\zeta_1\cdot 2}$ |
| $\psi(\Omega^2\cdot 2+\psi(\Omega^2\cdot 2+\psi(\Omega^2\cdot 2)))$ | ${\zeta_1}^{\zeta_1}$<br>$\omega^{\omega^{\zeta_1\cdot 2}}$ |
| $\psi(\Omega^2\cdot 2+\Omega)$ | $\varepsilon_{\zeta_1+1}$ |
| $\psi(\Omega^2\cdot 2+\Omega\cdot\psi(\Omega^2))$ | $\varepsilon_{\zeta_1+\zeta_0}$ |
| $\psi(\Omega^2\cdot 2+\Omega\cdot\psi(\Omega^2\cdot 2))$ | $\varepsilon_{\zeta_1\cdot 2}$ |
| $\psi(\Omega^2\cdot 2+\Omega\cdot\psi(\Omega^2\cdot 2+\psi(\Omega^2\cdot 2)))$ | $\varepsilon_{\omega^{\zeta_1\cdot 2}}$<br>$\varepsilon_{{\zeta_1}^2}$ |
| $\psi(\Omega^2\cdot 2+\Omega\cdot\psi(\Omega^2\cdot 2+\Omega))$ | $\varepsilon_{\varepsilon_{\zeta_1+1}}$ |
| $\psi(\Omega^2\cdot 3)$ | $\zeta_2$ |
| $\psi(\Omega^2\cdot 3+1)$ | $\zeta_2\cdot\omega$<br>$\omega^{\zeta_2+1}$ |
| $\psi(\Omega^2\cdot 3+\Omega)$ | $\varepsilon_{\zeta_2+1}$ |
| $\psi(\Omega^2\cdot 3+\Omega\cdot\psi(\Omega^2\cdot 3))$ | $\varepsilon_{\zeta_2\cdot 2}$ |
| $\psi(\Omega^2\cdot 3+\Omega\cdot\psi(\Omega^2\cdot 3+\Omega))$ | $\varepsilon_{\varepsilon_{\zeta_2+1}}$ |
| $\psi(\Omega^2\cdot 4)$ | $\zeta_3$ |
| $\psi(\Omega^2\cdot 5)$ | $\zeta_4$ |

| Buchholz's OCF | Cantor-style/Veblen function |
|---|---|
| $\psi(\Omega^2 \cdot \psi(1))$ | $\zeta_\omega$ |
| $\psi(\Omega^2 \cdot \psi(\Omega))$ | $\zeta_{\varepsilon_0}$ |
| $\psi(\Omega^2 \cdot \psi(\Omega \cdot \psi(\Omega)))$ | $\zeta_{\varepsilon_{\varepsilon_0}}$ |
| $\psi(\Omega^2 \cdot \psi(\Omega^2))$ | $\zeta_{\zeta_0}$ |
| $\psi(\Omega^2 \cdot \psi(\Omega^2 \cdot 2))$ | $\zeta_{\zeta_1}$ |
| $\psi(\Omega^2 \cdot \psi(\Omega^2 \cdot \psi(\Omega)))$ | $\zeta_{\zeta_{\varepsilon_0}}$ |
| $\psi(\Omega^2 \cdot \psi(\Omega^2 \cdot \psi(\Omega^2)))$ | $\zeta_{\zeta_{\zeta_0}}$ |
| $\psi(\Omega^3)$ | $\eta_0$ |
| $\psi(\Omega^3 + 1)$ | $\eta_0 \cdot \omega$<br>$\omega^{\eta_0+1}$ |
| $\psi(\Omega^3 + \psi(\Omega^3 + \psi(\Omega^3)))$ | ${\eta_0}^{\eta_0}$<br>$\omega^{\omega^{\eta_0 \cdot 2}}$ |
| $\psi(\Omega^3 + \Omega)$ | $\varepsilon_{\eta_0+1}$ |
| $\psi(\Omega^3 + \Omega \cdot \psi(\Omega^3))$ | $\varepsilon_{\eta_0 \cdot 2}$ |
| $\psi(\Omega^3 + \Omega \cdot \psi(\Omega^3 + \Omega\psi(\Omega^3)))$ | $\varepsilon_{\varepsilon_{\eta_0 \cdot 2}}$ |
| $\psi(\Omega^3 + \Omega^2)$ | $\zeta_{\eta_0+1}$ |
| $\psi(\Omega^3 + \Omega^2 \cdot \psi(\Omega^3))$ | $\zeta_{\eta_0 \cdot 2}$ |
| $\psi(\Omega^3 + \Omega^2\psi(\Omega^3 + \Omega^2 \cdot \psi(\Omega^3)))$ | $\zeta_{\zeta_{\eta_0 \cdot 2}}$ |
| $\psi(\Omega^3 \cdot 2)$ | $\eta_1$ |
| $\psi(\Omega^3 \cdot 2 + 1)$ | $\eta_1 \cdot \omega$<br>$\omega^{\eta_1+1}$ |
| $\psi(\Omega^3 \cdot 2 + \Omega)$ | $\varepsilon_{\eta_1+1}$ |
| $\psi(\Omega^3 \cdot 2 + \Omega^2)$ | $\zeta_{\eta_1+1}$ |
| $\psi(\Omega^3 \cdot 2 + \Omega^2 \cdot \psi(\Omega^3 \cdot 2 + \Omega^2))$ | $\zeta_{\zeta_{\eta_1+1}}$ |
| $\psi(\Omega^3 \cdot 3)$ | $\eta_2$ |
| $\psi(\Omega^3 \cdot 4)$ | $\eta_3$ |
| $\psi(\Omega^3 \cdot \psi(1))$ | $\eta_\omega$ |
| $\psi(\Omega^3 \cdot \psi(\Omega))$ | $\eta_{\varepsilon_0}$ |
| $\psi(\Omega^3 \cdot \psi(\Omega^2))$ | $\eta_{\zeta_0}$ |
| $\psi(\Omega^3 \cdot \psi(\Omega^3))$ | $\eta_{\eta_0}$ |
| $\psi(\Omega^3 \cdot \psi(\Omega^3 \cdot \psi(\Omega^3)))$ | $\eta_{\eta_{\eta_0}}$ |
| $\psi(\Omega^4)$ | $\varphi(4, 0)$ |
| $\psi(\Omega^4 + 1)$ | $\omega^{\varphi(4,0)+1}$ |
| $\psi(\Omega^4 + \Omega)$ | $\varepsilon_{\varphi(4,0)+1}$ |
| $\psi(\Omega^4 + \Omega^3)$ | $\eta_{\varphi(4,0)+1}$ |
| $\psi(\Omega^4 \cdot 2)$ | $\varphi(4, 1)$ |
| $\psi(\Omega^4 \cdot 3)$ | $\varphi(4, 2)$ |
| $\psi(\Omega^4 \cdot \psi(1))$ | $\varphi(4, \omega)$ |
| $\psi(\Omega^4 \cdot \psi(\Omega^4))$ | $\varphi(4, \varphi(4, 0))$ |
| $\psi(\Omega^5)$ | $\varphi(5, 0)$ |
| $\psi(\Omega^5 + \Omega^4)$ | $\varphi(4, \varphi(5, 0) + 1)$ |
| $\psi(\Omega^5 \cdot 2)$ | $\varphi(5, 1)$ |
| $\psi(\Omega^5 \cdot \psi(\Omega^5))$ | $\varphi(5, \varphi(5, 0))$ |
| $\psi(\Omega^6)$ | $\varphi(6, 0)$ |
| $\psi(\Omega^7)$ | $\varphi(7, 0)$ |
| $\psi(\Omega^\omega)$ | $\varphi(\omega, 0)$ |
| $\psi(\Omega^\omega + 1)$ | $\omega^{\varphi(\omega,0)+1}$ |
| $\psi(\Omega^\omega + \Omega)$ | $\varepsilon_{\varphi(\omega,0)+1}$ |
| $\psi(\Omega^\omega + \Omega^2)$ | $\zeta_{\varphi(\omega,0)+1}$ |
| $\psi(\Omega^\omega + \Omega^3)$ | $\eta_{\varphi(\omega,0)+1}$ |
| $\psi(\Omega^\omega + \Omega^4)$ | $\varphi(4, \varphi(\omega, 0) + 1)$ |
| $\psi(\Omega^\omega + \Omega^5)$ | $\varphi(5, \varphi(\omega, 0) + 1)$ |
| $\psi(\Omega^\omega + \Omega^6)$ | $\varphi(6, \varphi(\omega, 0) + 1)$ |
| $\psi(\Omega^\omega \cdot 2)$ | $\varphi(\omega, 1)$ |

| Buchholz's OCF | Cantor-style/Veblen function |
|---|---|
| $\psi(\Omega^\omega \cdot 3)$ | $\varphi(\omega, 2)$ |
| $\psi(\Omega^\omega \cdot \psi(1))$ | $\varphi(\omega, \omega)$ |
| $\psi(\Omega^\omega \cdot \psi(\Omega^\omega))$ | $\varphi(\omega, \varphi(\omega, 0))$ |
| $\psi(\Omega^{\omega+1})$ | $\varphi(\omega + 1, 0)$ |
| $\psi(\Omega^{\omega+1} + \Omega^\omega)$ | $\varphi(\omega, \varphi(\omega + 1, 0) + 1)$ |
| $\psi(\Omega^{\omega+1} \cdot 2)$ | $\varphi(\omega + 1, 1)$ |
| $\psi(\Omega^{\omega+2})$ | $\varphi(\omega + 2, 0)$ |
| $\psi(\Omega^{\omega+3})$ | $\varphi(\omega + 3, 0)$ |
| $\psi(\Omega^{\omega \cdot 2})$ | $\varphi(\omega \cdot 2, 0)$ |
| $\psi(\Omega^{\omega \cdot 3})$ | $\varphi(\omega \cdot 3, 0)$ |
| $\psi(\Omega^{\psi(2)})$ | $\varphi(\omega^2, 0)$ |
| $\psi(\Omega^{\psi(\psi(1))})$ | $\varphi(\omega^\omega, 0)$ |
| $\psi(\Omega^{\psi(\Omega)})$ | $\varphi(\varepsilon_0, 0)$ |
| $\psi(\Omega^{\psi(\Omega \cdot 2)})$ | $\varphi(\varepsilon_1, 0)$ |
| $\psi(\Omega^{\psi(\Omega^2)})$ | $\varphi(\zeta_0, 0)$ |
| $\psi(\Omega^{\psi(\Omega^3)})$ | $\varphi(\eta_0, 0)$ |
| $\psi(\Omega^{\psi(\Omega^\omega)})$ | $\varphi(\varphi(\omega, 0), 0)$ |
| $\psi(\Omega^{\psi(\Omega^{\psi(\Omega)})})$ | $\varphi(\varphi(\varepsilon_0, 0), 0)$ |
| $\psi(\Omega^\Omega)$ | $\Gamma_0$<br>$\varphi(1, 0, 0)$ |
| $\psi(\Omega^\Omega) \cdot 2$ | $\Gamma_0 \cdot 2$<br>$\varphi(1, 0, 0) \cdot 2$ |
| $\psi(\Omega^\Omega + 1)$ | $\omega^{\Gamma_0 + 1}$<br>$\varphi(\varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \psi(\Omega^\Omega))$ | $\omega^{\Gamma_0 \cdot 2}$<br>$\varphi(\varphi(1, 0, 0) \cdot 2)$ |
| $\psi(\Omega^\Omega + \Omega)$ | $\varepsilon_{\Gamma_0 + 1}$<br>$\varphi(1, \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)})$ | $\varepsilon_{\Gamma_0 \cdot 2}$<br>$\varphi(1, \varphi(1, 0, 0) \cdot 2)$ |
| $\psi(\Omega^\Omega + \Omega^2)$ | $\zeta_{\Gamma_0 + 1}$<br>$\varphi(2, \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^3)$ | $\eta_{\Gamma_0 + 1}$<br>$\varphi(3, \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^4)$ | $\varphi(4, \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^5)$ | $\varphi(5, \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^\omega)$ | $\varphi(\omega, \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega)})$ | $\varphi(\varepsilon_0, \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\omega)})$ | $\varphi(\varphi(\omega, 0), \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^{\psi(0)})})$ | $\varphi(\varphi(\varepsilon_0, 0), \varphi(1, 0, 0) + 1)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)})$ | $\varphi(\Gamma_0, 1)$<br>$\varphi(\varphi(1, 0, 0), 1)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)} \cdot 2)$ | $\varphi(\Gamma_0, 2)$<br>$\varphi(\varphi(1, 0, 0), 2)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)} \cdot \psi(\Omega^\Omega))$ | $\varphi(\Gamma_0, \Gamma_0)$<br>$\varphi(\varphi(1, 0, 0), \varphi(1, 0, 0))$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)+1})$ | $\varphi(\Gamma_0 + 1, 0)$<br>$\varphi(\varphi(1, 0, 0) + 1, 0)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)+\omega})$ | $\varphi(\Gamma_0 + \omega, 0)$<br>$\varphi(\varphi(1, 0, 0) + \omega, 0)$ |

| Buchholz's OCF | Cantor-style/Veblen function |
|---|---|
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)\cdot 2})$ | $\varphi(\Gamma_0 \cdot 2, 0)$<br>$\varphi(\varphi(1,0,0)\cdot 2, 0)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+\Omega)})$ | $\varphi(\varepsilon_{\Gamma_0+1}, 0)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+\Omega^{\psi(\Omega^\Omega)})})$ | $\varphi(\varphi(\Gamma_0, 1), 0)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+\Omega^{\psi(\Omega^\Omega)}+\Omega)})$ | $\varphi(\varphi(\Gamma_0+1, 0), 0)$ |
| $\psi(\Omega^\Omega \cdot 2)$ | $\Gamma_1$ |
| $\psi(\Omega^\Omega \cdot 2 + \Omega^{\psi(\Omega^\Omega)})$ | $\varphi(\Gamma_0, \Gamma_1 + 1)$ |
| $\psi(\Omega^\Omega \cdot 2 + \Omega^{\psi(\Omega^\Omega\cdot 2)})$ | $\varphi(\Gamma_1, 1)$ |
| $\psi(\Omega^\Omega \cdot 3)$ | $\Gamma_2$ |
| $\psi(\Omega^\Omega \cdot 4)$ | $\Gamma_3$ |
| $\psi(\Omega^\Omega \cdot \omega)$ | $\Gamma_\omega$ |
| $\psi(\Omega^\Omega \cdot \psi(0))$ | $\Gamma_{\varepsilon_0}$ |
| $\psi(\Omega^\Omega \cdot \psi(\Omega^\Omega))$ | $\Gamma_{\Gamma_0}$ |
| $\psi(\Omega^{\Omega+1})$ | $\varphi(1,1,0)$ |
| $\psi(\Omega^{\Omega+1} + \Omega^\omega)$ | $\varphi(\omega, \varphi(1,1,0)+1)$ |
| $\psi(\Omega^{\Omega+1} + \Omega^{\psi(\Omega^\Omega)})$ | $\varphi(\Gamma_0, \varphi(1,1,0)+1)$ |
| $\psi(\Omega^{\Omega+1} + \Omega^{\psi(\Omega^{\Omega+1})})$ | $\varphi(\varphi(1,1,0), 1)$ |
| $\psi(\Omega^{\Omega+1} + \Omega^{\psi(\Omega^{\Omega+1})+1})$ | $\varphi(\varphi(1,1,0)+1, 1)$ |
| $\psi(\Omega^{\Omega+1} + \Omega^\Omega)$ | $\Gamma_{\varphi(1,1,0)+1}$ |
| $\psi(\Omega^{\Omega+1} + \Omega^\Omega \cdot \psi(\Omega^{\Omega+1} + \Omega^\Omega))$ | $\Gamma_{\Gamma_{\varphi(1,1,0)+1}}$ |
| $\psi(\Omega^{\Omega+1} \cdot 2)$ | $\varphi(1,1,1)$ |
| $\psi(\Omega^{\Omega+1} \cdot 3)$ | $\varphi(1,1,2)$ |
| $\psi(\Omega^{\Omega+1} \cdot \omega)$ | $\varphi(1,1,\omega)$ |
| $\psi(\Omega^{\Omega+2})$ | $\varphi(1,2,0)$ |
| $\psi\left(\Omega^{\Omega+2} + \Omega^{\psi\left(\Omega^{\Omega+2}\right)+1}\right)$ | $\varphi(\varphi(1,2,0)+1, 0)$ |
| $\psi\left(\Omega^{\Omega+2} + \Omega^\Omega\right)$ | $\Gamma_{\varphi(1,2,0)+1}$ |
| $\psi\left(\Omega^{\Omega+2} + \Omega^\Omega \psi\left(\Omega^{\Omega+2} + \Omega^\Omega\right)\right)$ | $\Gamma_{\Gamma_{\varphi(1,2,0)+1}}$ |
| $\psi\left(\Omega^{\Omega+2} + \Omega^{\Omega+1}\right)$ | $\varphi(1,1,\varphi(1,2,0)+1)$ |
| $\psi\left(\Omega^{\Omega+2} \cdot 2\right)$ | $\varphi(1,2,1)$ |
| $\psi\left(\Omega^{\Omega+2} \cdot \psi\left(\Omega^{\Omega+2}\right)\right)$ | $\varphi(1,2,\varphi(1,2,0))$ |
| $\psi\left(\Omega^{\Omega+3}\right)$ | $\varphi(1,3,0)$ |
| $\psi\left(\Omega^{\Omega+3} \cdot 2\right)$ | $\varphi(1,3,1)$ |
| $\psi\left(\Omega^{\Omega+4}\right)$ | $\varphi(1,4,0)$ |
| $\psi\left(\Omega^{\Omega+\omega}\right)$ | $\varphi(1,\omega,0)$ |
| $\psi\left(\Omega^{\Omega+\psi(\Omega^\Omega)}\right)$ | $\varphi(1,\varphi(1,0,0),0)$ |
| $\psi\left(\Omega^{\Omega+\psi(\Omega^{\Omega+\psi(\Omega^\Omega)})}\right)$ | $\varphi(1,\varphi(1,\varphi(1,0,0),0),0)$ |
| $\psi\left(\Omega^{\Omega\cdot 2}\right)$ | $\varphi(2,0,0)$ |
| $\psi\left(\Omega^{\Omega\cdot} + \Omega^{\psi\left(\Omega^{\Omega\cdot 2}\right)+1}\right)$ | $\varphi(\varphi(2,0,0)+1, 0)$ |
| $\psi\left(\Omega^{\Omega\cdot 2} + \Omega^\Omega\right)$ | $\varphi(1,0,\varphi(2,0,0)+1)$ |
| $\psi\left(\Omega^{\Omega\cdot 2} + \Omega^{\Omega+\omega}\right)$ | $\varphi(1,\omega,\varphi(2,0,0)+1)$ |
| $\psi\left(\Omega^{\Omega\cdot 2} + \Omega^{\Omega+\psi(\Omega^\Omega\cdot 2)}\right)$ | $\varphi(1,\varphi(2,0,0),1)$ |
| $\psi\left(\Omega^{\Omega\cdot 2} + \Omega^{\Omega+\psi(\Omega^\Omega\cdot 2)+1}\right)$ | $\varphi(1,\varphi(2,0,0)+1,1)$ |
| $\psi\left(\Omega^{\Omega\cdot 2} \cdot 2\right)$ | $\varphi(2,0,1)$ |
| $\psi\left(\Omega^{\Omega\cdot 2} \cdot \psi\left(\Omega^{\Omega\cdot 2}\right)\right)$ | $\varphi(2,0,\varphi(2,0,0))$ |
| $\psi\left(\Omega^{\Omega\cdot 2+1}\right)$ | $\varphi(2,1,0)$ |
| $\psi\left(\Omega^{\Omega\cdot 2+1} \cdot 2\right)$ | $\varphi(2,1,1)$ |
| $\psi\left(\Omega^{\Omega\cdot 2+\psi(\Omega^\Omega\cdot 2)}\right)$ | $\varphi(2,\varphi(2,2,0),0)$ |
| $\psi\left(\Omega^{\Omega\cdot 3}\right)$ | $\varphi(3,0,0)$ |
| $\psi\left(\Omega^{\Omega\cdot 3} + \Omega^{\psi(\Omega^\Omega\cdot 3)}\right)$ | $\varphi(\varphi(3,0,0),1)$ |

| Buchholz's OCF | Cantor-style/Veblen function |
|---|---|
| $\psi\left(\Omega^{\Omega\cdot 3}+\Omega^{\Omega+\psi(\Omega^{\Omega}\cdot 3)}\right)$ | $\varphi(1,\varphi(3,0,0),1)$ |
| $\psi\left(\Omega^{\Omega\cdot 3}+\Omega^{\Omega\cdot 2+\psi(\Omega^{\Omega}\cdot 3)}\right)$ | $\varphi(2,\varphi(3,0,0),1)$ |
| $\psi\left(\Omega^{\Omega\cdot 3}\cdot 2\right)$ | $\varphi(3,0,1)$ |
| $\psi\left(\Omega^{\Omega\cdot 3+1}\right)$ | $\varphi(3,1,0)$ |
| $\psi\left(\Omega^{\Omega\cdot 4}\right)$ | $\varphi(4,0,0)$ |
| $\psi\left(\Omega^{\Omega\cdot 5}\right)$ | $\varphi(5,0,0)$ |
| $\psi\left(\Omega^{\Omega\cdot\omega}\right)$ | $\varphi(\omega,0,0)$ |
| $\psi\left(\Omega^{\Omega\cdot\psi(0)}\right)$ | $\varphi(\varepsilon_0,0,0)$ |
| $\psi\left(\Omega^{\Omega\cdot\psi(\Omega^{\Omega})}\right)$ | $\varphi(\varphi(1,0,0),0,0)$ |
| $\psi\left(\Omega^{\Omega^2}\right)$ | $\varphi(1,0,0,0)$<br>$\varphi(1@3)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\varphi(\varphi(1,0,0,0),1)$<br>$\varphi(\varphi(1@3)@1,1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega}\right)$ | $\varphi(1,0,\varphi(1,0,0,0)+1)$<br>$\varphi(1@2,\varphi(1@3)+1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega+1}\right)$ | $\varphi(1,1,\varphi(1,0,0,0)+1)$<br>$\varphi(1@2,1@1,\varphi(1@3)+1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega+\omega}\right)$ | $\varphi(1,\omega,\varphi(1,0,0,0)+1)$<br>$\varphi(1@2,\omega@1,\varphi(1@3)+1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega+\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\varphi(1,\varphi(1,0,0,0),1)$<br>$\varphi(1@2,\varphi(1@3)@1,1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot 2}\right)$ | $\varphi(2,0,\varphi(1,0,0,0)+1)$<br>$\varphi(2@2,\varphi(1@3)+1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot 2+\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\varphi(2,\varphi(1,0,0,0),1)$<br>$\varphi(2@2,\varphi(1@3)@1,1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot 3}\right)$ | $\varphi(3,0,\varphi(1,0,0,0)+1)$<br>$\varphi(3@2,\varphi(1@3)+1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi(\Omega^{\Omega})}\right)$ | $\varphi\left(\Gamma_0,0,\varphi(1,0,0,0)+1\right)$<br>$\varphi\left(\Gamma_0@2,\varphi(1@3)+1\right)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\varphi(\varphi(1,0,0,0),0,1)$<br>$\varphi(\varphi(1@3)@2,1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)+1}\right)$ | $\varphi(\varphi(1,0,0,0),1,0)$<br>$\varphi(\varphi(1@3)@2,1@1)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)+\Omega}\right)$ | $\varphi(\varphi(1,0,0,0)+1,0,0)$<br>$\varphi(\varphi(1@3)+1@2)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)\cdot 2}\right)$ | $\varphi(\varphi(1,0,0,0)2,0,0)$<br>$\varphi(\varphi(1@3)2@2)$ |
| $\psi\left(\Omega^{\Omega^2}\cdot 2\right)$ | $\varphi(1,0,0,1)$<br>$\varphi(1@3,1)$ |
| $\psi\left(\Omega^{\Omega^2+1}\right)$ | $\varphi(1,0,1,0)$<br>$\varphi(1@3,1@1)$ |
| $\psi\left(\Omega^{\Omega^2+\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\varphi(1,0,\varphi(1,0,0,0),0)$<br>$\varphi(1@3,\varphi(1@3)@1)$ |
| $\psi\left(\Omega^{\Omega^2+\Omega}\right)$ | $\varphi(1,1,0,0)$<br>$\varphi(1@3,1@2)$ |
| $\psi\left(\Omega^{\Omega^2+\Omega}+\Omega^{\Omega^2+\psi\left(\Omega^{\Omega^2+\Omega}\right)\cdot 2}\right)$ | $\varphi(1,0,\varphi(1,1,0,0)2,0)$<br>$\varphi(1@3,\varphi(1@3,1@2)2@1)$ |
| $\psi\left(\Omega^{\Omega^2+\Omega}\cdot 2\right)$ | $\varphi(1,1,0,1)$<br>$\varphi(1@3,1@2,1)$ |

| Buchholz's OCF | Cantor-style/Veblen function |
|---|---|
| $\psi\left(\Omega^{\Omega^2+\Omega+1}\right)$ | $\varphi(1,1,1,0)$<br>$\varphi(1@3,1@2,1@1)$ |
| $\psi\left(\Omega^{\Omega^2+\Omega\cdot 2}\right)$ | $\varphi(1,2,0,0)$<br>$\varphi(1@3,2@2)$ |
| $\psi\left(\Omega^{\Omega^2+\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\varphi(1,\varphi(1,0,0,0),0,0)$<br>$\varphi(1@3,\varphi(1@3)@2)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 2}\right)$ | $\varphi(2,0,0,0)$<br>$\varphi(2@3)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega^2+\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\varphi(1,\varphi(2,0,0,0),0,1)$<br>$\varphi(1@3,\varphi(2@3)@2,1)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 2}\cdot 2\right)$ | $\varphi(2,0,0,1)$<br>$\varphi(2@3,1)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 2+1}\right)$ | $\varphi(2,0,1,0)$<br>$\varphi(2@3,1@1)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 2+1}\cdot 2\right)$ | $\varphi(2,0,2,1)$<br>$\varphi(2@3,2@1,1)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 2+\Omega}\right)$ | $\varphi(2,1,0,0)$<br>$\varphi(2@3,1@2)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 3}\right)$ | $\varphi(2,1,0,0)$<br>$\varphi(2@3,1@2)$ |
| $\psi\left(\Omega^{\Omega^2\cdot\psi(\Omega^\Omega)}\right)$ | $\varphi(\varphi(1,0,0,0),0,0,0)$<br>$\varphi(\varphi(1@3)@3)$ |
| $\psi\left(\Omega^{\Omega^3}\right)$ | $\varphi(1,0,0,0,0)$<br>$\varphi(1@4)$ |
| $\psi\left(\Omega^{\Omega^3}+\Omega^{\psi(\Omega^{\Omega^3})}\right)$ | $\varphi(\varphi(1,0,0,0,0),1)$<br>$\varphi(\varphi(1@4)@1,1)$ |
| $\psi\left(\Omega^{\Omega^3}+\Omega^{\Omega\cdot\psi(\Omega^{\Omega^3})}\right)$ | $\varphi(\varphi(1,0,0,0,0),0,1)$<br>$\varphi(\varphi(1@4)@2,1)$ |
| $\psi\left(\Omega^{\Omega^3}+\Omega^{\Omega^2\cdot\psi(\Omega^{\Omega^3})}\right)$ | $\varphi(\varphi(1,0,0,0,0),0,0,1)$<br>$\varphi(\varphi(1@4)@3,1)$ |
| $\psi\left(\Omega^{\Omega^3}\cdot 2\right)$ | $\varphi(1,0,0,0,1)$<br>$\varphi(1@4,1)$ |
| $\psi\left(\Omega^{\Omega^3+1}\right)$ | $\varphi(1,0,0,1,0)$<br>$\varphi(1@4,1@1)$ |
| $\psi\left(\Omega^{\Omega^3+\Omega}\right)$ | $\varphi(1,0,1,0,0)$<br>$\varphi(1@4,1@2)$ |
| $\psi\left(\Omega^{\Omega^3+\Omega^2}\right)$ | $\varphi(1,1,0,0,0)$<br>$\varphi(1@4,1@3)$ |
| $\psi\left(\Omega^{\Omega^3\cdot 2}\right)$ | $\varphi(2,0,0,0,0)$<br>$\varphi(2@4)$ |
| $\psi\left(\Omega^{\Omega^4}\right)$ | $\varphi(1,0,0,0,0,0)$<br>$\varphi(1@5)$ |
| $\psi\left(\Omega^{\Omega^5}\right)$ | $\varphi(1,0,0,0,0,0,0)$<br>$\varphi(1@6)$ |
| $\psi\left(\Omega^{\Omega^6}\right)$ | $\varphi(1,0,0,0,0,0,0,0)$<br>$\varphi(1@7)$ |
| $\psi\left(\Omega^{\Omega^\omega}\right)$ | $\varphi(1@\omega)$ |
| $\psi\left(\Omega^{\Omega^\omega}+\Omega\right)$ | $\varphi(1@1,\varphi(1@\omega)+1)$ |
| $\psi\left(\Omega^{\Omega^\omega}+\Omega^\Omega\right)$ | $\varphi(1@2,\varphi(1@\omega)+1)$ |
| $\psi\left(\Omega^{\Omega^\omega}+\Omega^{\Omega^2}\right)$ | $\varphi(1@3,\varphi(1@\omega)+1)$ |

| Buchholz's OCF | Cantor-style/Veblen function |
|---|---|
| $\psi\left(\Omega^{\Omega^\omega}+\Omega^{\Omega^3}\right)$ | $\varphi(1@4,\varphi(1@\omega)+1)$ |
| $\psi\left(\Omega^{\Omega^\omega}+\Omega^{\Omega^4}\right)$ | $\varphi(1@5,\varphi(1@\omega)+1)$ |
| $\psi\left(\Omega^{\Omega^\omega}+\Omega^{\Omega^5}\right)$ | $\varphi(1@6,\varphi(1@\omega)+1)$ |
| $\psi\left(\Omega^{\Omega^\omega}\cdot 2\right)$ | $\varphi(1@\omega,1)$ |
| $\psi\left(\Omega^{\Omega^\omega+1}\right)$ | $\varphi(1@\omega,1@1)$ |
| $\psi\left(\Omega^{\Omega^\omega+\psi(\Omega^{\Omega^\omega})}\right)$ | $\varphi(1@\omega,\varphi(1@\omega)@1)$ |
| $\psi\left(\Omega^{\Omega^\omega+\Omega}\right)$ | $\varphi(1@\omega,1@2)$ |
| $\psi\left(\Omega^{\Omega^\omega+\Omega\cdot\psi(\Omega^{\Omega^\omega})}\right)$ | $\varphi(1@\omega,\varphi(1@\omega)@2)$ |
| $\psi\left(\Omega^{\Omega^\omega+\Omega^2}\right)$ | $\varphi(1@\omega,1@3)$ |
| $\psi\left(\Omega^{\Omega^\omega+\Omega^3}\right)$ | $\varphi(1@\omega,1@4)$ |
| $\psi\left(\Omega^{\Omega^\omega+\Omega^4}\right)$ | $\varphi(1@\omega,1@5)$ |
| $\psi\left(\Omega^{\Omega^\omega\cdot 2}\right)$ | $\varphi(2@\omega)$ |
| $\psi\left(\Omega^{\Omega^\omega\cdot\psi(\Omega^{\Omega^\omega})}\right)$ | $\varphi(\varphi(1@\omega)@\omega)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}}\right)$ | $\varphi(1@\omega+1)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}}+\Omega^{\Omega^\omega}\right)$ | $\varphi(\varphi(1@\omega+1)@\omega,1)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}}\cdot 2\right)$ | $\varphi(1@\omega+1,1)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}+1}\right)$ | $\varphi(1@\omega+1,1@1)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}+\Omega}\right)$ | $\varphi(1@\omega+1,1@2)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}+\Omega^\omega}\right)$ | $\varphi(1@\omega+1,1@\omega)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}\cdot 2}\right)$ | $\varphi(2@\omega+1)$ |
| $\psi\left(\Omega^{\Omega^{\omega+2}}\right)$ | $\varphi(1@\omega+2)$ |
| $\psi\left(\Omega^{\Omega^{\omega+3}}\right)$ | $\varphi(1@\omega+3)$ |
| $\psi\left(\Omega^{\Omega^{\omega\cdot 2}}\right)$ | $\varphi(1@\omega\cdot 2)$ |
| $\psi\left(\Omega^{\Omega^{\omega^2}}\right)$ | $\varphi(1@\omega^2)$ |
| $\psi\left(\Omega^{\Omega^{\psi(0)}}\right)$ | $\varphi(1@\varepsilon_0)$ |
| $\psi\left(\Omega^{\Omega^{\psi(\Omega)}}\right)$ | $\varphi(1@\zeta_0)$ |
| $\psi\left(\Omega^{\Omega^{\psi(\Omega^\Omega)}}\right)$ | $\varphi(1@\Gamma_0)$ |
| $\psi\left(\Omega^{\Omega^{\psi(\Omega^{\Omega^2})}}\right)$ | $\varphi(1@\varphi(1@3))$ |
| $\psi\left(\Omega^{\Omega^{\psi(\Omega^{\Omega^\omega})}}\right)$ | $\varphi(1@\varphi(1@\omega))$ |
| $\psi\left(\Omega^{\Omega^\Omega}\right)$ | $\varphi(1@(1,0))$ |
| $\psi\left(\Omega^{\Omega^\Omega}+\Omega^{\psi\left(\Omega^{\Omega^\Omega}\right)}\right)$ | $\varphi(\varphi(1@(1,0))@1,1)$ |
| $\psi\left(\Omega^{\Omega^\Omega}+\Omega^{\Omega^\omega\cdot\psi\left(\Omega^{\Omega^\Omega}\right)}\right)$ | $\varphi(\varphi(1@(1,0))@\omega,1)$ |
| $\psi\left(\Omega^{\Omega^\Omega}\cdot 2\right)$ | $\varphi(1@(1,0),1)$ |
| $\psi\left(\Omega^{\Omega^\Omega+1}\right)$ | $\varphi(1@(1,0),1@1)$ |
| $\psi\left(\Omega^{\Omega^\Omega+\Omega^\omega}\right)$ | $\varphi(1@(1,0),1@\omega)$ |
| $\psi\left(\Omega^{\Omega^\Omega\cdot 2}\right)$ | $\varphi(2@(1,0))$ |
| $\psi\left(\Omega^{\Omega^\Omega\cdot\psi\left(\Omega^{\Omega^\Omega}\right)}\right)$ | $\varphi(\varphi(1@(1,0))@(1,0))$ |

| Buchholz's OCF | Cantor-style/Veblen function |
|---|---|
| $\psi\left(\Omega^{\Omega^{\Omega+1}}\right)$ | $\varphi(1@(1,1))$ |
| $\psi\left(\Omega^{\Omega^{\Omega+1}}+\Omega^{\Omega^{\Omega}\cdot\psi(\Omega^{\Omega^{\Omega+1}})}\right)$ | $\varphi(\varphi(1@(1,1))@(1,0),1)$ |
| $\psi\left(\Omega^{\Omega^{\Omega+1}\cdot 2}\right)$ | $\varphi(2@(1,1))$ |
| $\psi\left(\Omega^{\Omega^{\Omega+2}}\right)$ | $\varphi(1@(1,2))$ |
| $\psi\left(\Omega^{\Omega^{\Omega+\omega}}\right)$ | $\varphi(1@(1,\omega))$ |
| $\psi\left(\Omega^{\Omega^{\Omega+\psi\left(\Omega^{\Omega^{\Omega}}\right)}}\right)$ | $\varphi(1@(1,\varphi(1@(1,0))))$ |
| $\psi\left(\Omega^{\Omega^{\Omega\cdot 2}}\right)$ | $\varphi(1@(2,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega\cdot 2+\psi\left(\Omega^{\Omega^{\Omega\cdot 2}}\right)}}\right)$ | $\varphi(1@(2,\varphi(1@(2,0))))$ |
| $\psi\left(\Omega^{\Omega^{\Omega\cdot 3}}\right)$ | $\varphi(1@(3,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega\cdot\omega}}\right)$ | $\varphi(1@(\omega,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^{\Omega}}\right)}}\right)$ | $\varphi(1@(\varphi(1@(1,0)),0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{2}}}\right)$ | $\varphi(1@(1,0,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{2}}\cdot 2}\right)$ | $\varphi(2@(1,0,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{2}+1}}\right)$ | $\varphi(1@(1,0,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{2}+\Omega}}\right)$ | $\varphi(1@(1,1,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{2}\cdot 2}}\right)$ | $\varphi(1@(2,0,0))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{3}}}\right)$ | $\varphi(1@(1,0,0,0))$<br>$\varphi(1@(1@3))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{3}\cdot 2}}\right)$ | $\varphi(1@(2@3))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{4}}}\right)$ | $\varphi(1@(2@3))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\omega}}}\right)$ | $\varphi(1@(1@\omega))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega}}}\right)$ | $\varphi(1@(1@(1,0)))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega}}}+\Omega\right)$ | $\varphi(1@1,\varphi(1@(1@(1,0)))+1)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega}}}\cdot 2\right)$ | $\varphi(1@(1@(1,0)),1)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega}}+1}\right)$ | $\varphi(1@(1@(1,0)),1@1)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega}}\cdot 2}\right)$ | $\varphi(2@(1@(1,0)))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega}+1}}\right)$ | $\varphi(1@(1@(1,0),1))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega}\cdot 2}}\right)$ | $\varphi(1@(2@(1,0)))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega+1}}}\right)$ | $\varphi(1@(1@(1,1)))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega\cdot 2}}}\right)$ | $\varphi(1@(1@(2,0)))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega^{2}}}}\right)$ | $\varphi(1@(1@(1,0,0)))$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega^{\Omega}}}}\right)$ | $\varphi(1@(1@(1@(1,0))))$ |
| $\psi\left(\Omega_2\right)$<br>$\psi\left(\varepsilon_{\Omega+1}\right)$ | $\varphi(1@(1,,0))$ |

## A.6 BOCF vs MOCF

The results in this section are primarily cited from [1-2] 。

| Buchholz's OCF | Madore's OCF |
|---|---|
| $\psi(\Omega)$ | $\psi(0)$ |
| $\psi(\Omega)+\psi(0)$ | $\psi(0)+1$ |
| $\psi(\Omega)+\psi(1)$ | $\psi(0)+\omega$ |
| $\psi(\Omega)+\psi(\psi(1))$ | $\psi(0)+\omega^{\omega}$ |
| $\psi(\Omega)+\psi(\psi(\psi(1)))$ | $\psi(0)+\omega^{\omega^{\omega}}$ |
| $\psi(\Omega)\cdot 2$ | $\psi(0)\cdot 2$ |
| $\psi(\Omega)\cdot 3$ | $\psi(0)\cdot 3$ |
| $\psi(\Omega+1)$ | $\psi(0)\cdot\omega$ |
| $\psi(\Omega+1)+\psi(1)$ | $\psi(0)\cdot\omega+\omega$ |
| $\psi(\Omega+1)+\psi(\psi(1))$ | $\psi(0)\cdot\omega+\omega^{\omega}$ |
| $\psi(\Omega+1)\cdot 2$ | $\psi(0)\cdot\omega\cdot 2$ |
| $\psi(\Omega+2)$ | $\psi(0)\cdot\omega^2$ |
| $\psi(\Omega+2)\cdot 2$ | $\psi(0)\cdot\omega^2\cdot 2$ |
| $\psi(\Omega+3)$ | $\psi(0)\cdot\omega^3$ |
| $\psi(\Omega+\psi(1))$ | $\psi(0)\cdot\omega^{\omega}$ |
| $\psi(\Omega+\psi(2))$ | $\psi(0)\cdot\omega^{\omega^2}$ |
| $\psi(\Omega+\psi(\psi(1)))$ | $\psi(0)\cdot\omega^{\omega^{\omega}}$ |
| $\psi(\Omega+\psi(\Omega))$ | $\psi(0)^2$ |
| $\psi(\Omega+\psi(\Omega))+\psi(1)$ | $\psi(0)^2+\omega$ |
| $\psi(\Omega+\psi(\Omega))+\psi(\psi(1))$ | $\psi(0)^2+\omega^{\omega}$ |
| $\psi(\Omega+\psi(\Omega))+\psi(\Omega)$ | $\psi(0)^2+\psi(0)$ |
| $\psi(\Omega+\psi(\Omega))+\psi(\Omega+1)$ | $\psi(0)^2+\psi(0)\cdot\omega$ |
| $\psi(\Omega+\psi(\Omega))\cdot 2$ | $\psi(0)^2\cdot 2$ |
| $\psi(\Omega+\psi(\Omega)+1)$ | $\psi(0)^2\cdot\omega$ |
| $\psi(\Omega+\psi(\Omega)+\psi(1))$ | $\psi(0)^2\cdot\omega^{\omega}$ |
| $\psi(\Omega+\psi(\Omega)\cdot 2)$ | $\psi(0)^3$ |
| $\psi(\Omega+\psi(\Omega)\cdot 3)$ | $\psi(0)^4$ |
| $\psi(\Omega+\psi(\Omega+1))$ | $\psi(0)^{\omega}$ |
| $\psi(\Omega+\psi(\Omega+1))\cdot 2$ | $\psi(0)^{\omega}\cdot 2$ |
| $\psi(\Omega+\psi(\Omega+1)+1)$ | $\psi(0)^{\omega}\cdot\omega$ |
| $\psi(\Omega+\psi(\Omega+1)+\psi(1))$ | $\psi(0)^{\omega}\cdot\omega^{\omega}$ |
| $\psi(\Omega+\psi(\Omega+1)+\psi(\Omega))$ | $\psi(0)^{\omega+1}$ |
| $\psi(\Omega+\psi(\Omega+1)+\psi(\Omega)\cdot 2)$ | $\psi(0)^{\omega+2}$ |
| $\psi(\Omega+\psi(\Omega+1)\cdot 2)$ | $\psi(0)^{\omega\cdot 2}$ |
| $\psi(\Omega+\psi(\Omega+2))$ | $\psi(0)^{\omega^2}$ |
| $\psi(\Omega+\psi(\Omega+\psi(1)))$ | $\psi(0)^{\omega^{\omega}}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\psi(1))))$ | $\psi(0)^{\omega^{\omega^{\omega}}}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega)))$ | $\psi(0)^{\psi(0)}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega))+1)$ | $\psi(0)^{\psi(0)}\cdot\omega$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega))+\psi(\Omega))$ | $\psi(0)^{\psi(0)+1}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega))+\psi(\Omega+1))$ | $\psi(0)^{\psi(0)+\omega}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega))\cdot 2)$ | $\psi(0)^{\psi(0)\cdot 2}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega)+1))$ | $\psi(0)^{\psi(0)\cdot\omega}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega)\cdot 2))$ | $\psi(0)^{\psi(0)^2}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega+1)))$ | $\psi(0)^{\psi(0)^{\omega}}$ |
| $\psi(\Omega+\psi(\Omega+\psi(\Omega+\psi(\Omega))))$ | $\psi(0)^{\psi(0)^{\psi(0)}}$ |
| $\psi(\Omega\cdot 2)$ | $\psi(1)$ |
| $\psi(\Omega\cdot 2)+\psi(\Omega)$ | $\psi(1)+\psi(0)$ |
| $\psi(\Omega\cdot 2)+\psi(\Omega+1)$ | $\psi(1)+\psi(0)\cdot\omega$ |
| $\psi(\Omega\cdot 2)+\psi(\Omega+\psi(\Omega+1))$ | $\psi(1)+\psi(0)^{\omega}$ |

| Buchholz's OCF | Madore's OCF |
|---|---|
| $\psi(\Omega \cdot 2) \cdot 2$ | $\psi(1) \cdot 2$ |
| $\psi(\Omega \cdot 2 + 1)$ | $\psi(1) \cdot \omega$ |
| $\psi(\Omega \cdot 2 + \psi(\Omega))$ | $\psi(1) \cdot \psi(0)$ |
| $\psi(\Omega \cdot 2 + \psi(\Omega + \psi(\Omega)))$ | $\psi(1) \cdot \psi(0)^2$ |
| $\psi(\Omega \cdot 2 + \psi(\Omega \cdot 2) \cdot 2)$ | $\psi(1)^3$ |
| $\psi(\Omega \cdot 2 + \psi(\Omega \cdot 2 + 1))$ | $\psi(1)^\omega$ |
| $\psi(\Omega \cdot 2 + \psi(\Omega \cdot 2 + \psi(\Omega)))$ | $\psi(1)^{\psi(0)}$ |
| $\psi(\Omega \cdot 2 + \psi(\Omega \cdot 2 + \psi(\Omega \cdot 2)))$ | $\psi(1)^{\psi(1)}$ |
| $\psi(\Omega \cdot 3)$ | $\psi(2)$ |
| $\psi(\Omega \cdot 3 + 1)$ | $\psi(2) \cdot \omega$ |
| $\psi(\Omega \cdot 3 + \psi(\Omega))$ | $\psi(2) \cdot \psi(0)$ |
| $\psi(\Omega \cdot 3 + \psi(\Omega \cdot 2))$ | $\psi(2) \cdot \psi(1)$ |
| $\psi(\Omega \cdot 3 + \psi(\Omega \cdot 3))$ | $\psi(2)^2$ |
| $\psi(\Omega \cdot 3 + \psi(\Omega \cdot 3 + 1))$ | $\psi(2)^\omega$ |
| $\psi(\Omega \cdot 3 + \psi(\Omega \cdot 3 + \psi(\Omega \cdot 3)))$ | $\psi(2)^{\psi(2)}$ |
| $\psi(\Omega \cdot 4)$ | $\psi(3)$ |
| $\psi(\Omega \cdot 4 + \psi(\Omega \cdot 4))$ | $\psi(3)^2$ |
| $\psi(\Omega \cdot 5)$ | $\psi(4)$ |
| $\psi(\Omega \cdot 6)$ | $\psi(5)$ |
| $\psi(\Omega \cdot \psi(1))$ | $\psi(\omega)$ |
| $\psi(\Omega \cdot \psi(1) + \psi(\Omega \cdot \psi(1)))$ | $\psi(\omega)^2$ |
| $\psi(\Omega \cdot \psi(1) + \Omega)$ | $\psi(\omega + 1)$ |
| $\psi(\Omega \cdot \psi(1) + \Omega \cdot 2)$ | $\psi(\omega + 2)$ |
| $\psi(\Omega \cdot \psi(1) \cdot 2)$ | $\psi(\omega \cdot 2)$ |
| $\psi(\Omega \cdot \psi(2))$ | $\psi(\omega^2)$ |
| $\psi(\Omega \cdot \psi(\psi(1)))$ | $\psi(\omega^\omega)$ |
| $\psi(\Omega \cdot \psi(\psi(\psi(1))))$ | $\psi(\omega^{\omega^\omega})$ |
| $\psi(\Omega \cdot \psi(\Omega))$ | $\psi(\psi(0))$ |
| $\psi(\Omega \cdot \psi(\Omega) + \Omega)$ | $\psi(\psi(0) + 1)$ |
| $\psi(\Omega \cdot \psi(\Omega) \cdot 2)$ | $\psi(\psi(0) \cdot 2)$ |
| $\psi(\Omega \cdot \psi(\Omega + 1))$ | $\psi(\psi(0) \cdot \omega)$ |
| $\psi(\Omega \cdot \psi(\Omega + \psi(\Omega)))$ | $\psi(\psi(0)^2)$ |
| $\psi(\Omega \cdot \psi(\Omega \cdot 2))$ | $\psi(\psi(1))$ |
| $\psi(\Omega \cdot \psi(\Omega \cdot 3))$ | $\psi(\psi(2))$ |
| $\psi(\Omega \cdot \psi(\Omega \cdot \psi(1)))$ | $\psi(\psi(\omega))$ |
| $\psi(\Omega \cdot \psi(\Omega \cdot \psi(\Omega)))$ | $\psi(\psi(\psi(0)))$ |
| $\psi(\Omega^2)$ | $\psi(\Omega)$ |
| $\psi(\Omega^2) + \psi(1)$ | $\psi(\Omega) + \omega$ |
| $\psi(\Omega^2) + \psi(\Omega)$ | $\psi(\Omega) + \psi(0)$ |
| $\psi(\Omega^2) + \psi(\Omega \cdot \psi(\Omega))$ | $\psi(\Omega) + \psi(\psi(0))$ |
| $\psi(\Omega^2) \cdot 2$ | $\psi(\Omega) \cdot 2$ |
| $\psi(\Omega^2 + 1)$ | $\psi(\Omega) \cdot \omega$ |
| $\psi(\Omega^2 + \psi(\Omega))$ | $\psi(\Omega) \cdot \psi(0)$ |
| $\psi(\Omega^2 + \psi(\Omega \cdot \psi(\Omega)))$ | $\psi(\Omega) \cdot \psi(\psi(0))$ |
| $\psi(\Omega^2 + \psi(\Omega^2))$ | $\psi(\Omega)^2$ |
| $\psi(\Omega^2 + \psi(\Omega^2 + 1))$ | $\psi(\Omega)^\omega$ |
| $\psi(\Omega^2 + \psi(\Omega^2 + \psi(\Omega)))$ | $\psi(\Omega)^{\psi(0)}$ |
| $\psi(\Omega^2 + \psi(\Omega^2 + \psi(\Omega^2)))$ | $\psi(\Omega)^{\psi(\Omega)}$ |
| $\psi(\Omega^2 + \psi(\Omega^2 + \psi(\Omega^2 + \psi(\Omega^2))))$ | $\psi(\Omega)^{\psi(\Omega)^{\psi(\Omega)}}$ |
| $\psi(\Omega^2 + \Omega)$ | $\psi(\Omega + 1)$ |
| $\psi(\Omega^2 + \Omega + \psi(\Omega^2 + \Omega))$ | $\psi(\Omega + 1)^2$ |
| $\psi(\Omega^2 + \Omega + \psi(\Omega^2 + \Omega + \psi(\Omega^2 + \Omega)))$ | $\psi(\Omega + 1)^{\psi(\Omega+1)}$ |
| $\psi(\Omega^2 + \Omega \cdot 2)$ | $\psi(\Omega + 2)$ |
| $\psi(\Omega^2 + \Omega \cdot 3)$ | $\psi(\Omega + 3)$ |
| $\psi(\Omega^2 + \Omega \cdot \omega)$ | $\psi(\Omega + \omega)$ |

| Buchholz's OCF | Madore's OCF |
|---|---|
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega))$ | $\psi(\Omega+\psi(0))$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega\cdot\psi(\Omega)))$ | $\psi(\Omega+\psi(\psi(0)))$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2))$ | $\psi(\Omega+\psi(\Omega))$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2)\cdot 2)$ | $\psi(\Omega+\psi(\Omega)\cdot 2)$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+1))$ | $\psi(\Omega+\psi(\Omega)\cdot\omega)$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+\psi(\Omega^2)))$ | $\psi(\Omega+\psi(\Omega)^2)$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+\psi(\Omega^2+\psi(\Omega^2))))$ | $\psi(\Omega+\psi(\Omega)^{\psi(\Omega)})$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+\Omega))$ | $\psi(\Omega+\psi(\Omega+1))$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+\Omega\cdot\psi(\Omega)))$ | $\psi(\Omega+\psi(\Omega+\psi(0)))$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+\Omega\cdot\psi(\Omega^2)))$ | $\psi(\Omega+\psi(\Omega+\psi(\Omega)))$ |
| $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+\Omega)))$ | $\psi(\Omega+\psi(\Omega+\psi(\Omega+1)))$ |
| $\psi(\Omega^2\cdot 2)$ | $\psi(\Omega\cdot 2)$ |
| $\psi(\Omega^2\cdot 2+1)$ | $\psi(\Omega\cdot 2)\cdot\omega$ |
| $\psi(\Omega^2\cdot 2+\psi(\Omega^2))$ | $\psi(\Omega\cdot 2)\cdot\psi(\Omega)$ |
| $\psi(\Omega^2\cdot 2+\psi(\Omega^2\cdot 2))$ | $\psi(\Omega\cdot 2)^2$ |
| $\psi(\Omega^2\cdot 2+\psi(\Omega^2\cdot 2+\psi(\Omega^2\cdot 2)))$ | $\psi(\Omega\cdot 2)^{\psi(\Omega\cdot 2)}$ |
| $\psi(\Omega^2\cdot 2+\Omega)$ | $\psi(\Omega\cdot 2+1)$ |
| $\psi(\Omega^2\cdot 2+\Omega\cdot\psi(\Omega^2))$ | $\psi(\Omega\cdot 2+\psi(\Omega+\psi(\Omega)))$ |
| $\psi(\Omega^2\cdot 2+\Omega\cdot\psi(\Omega^2\cdot 2))$ | $\psi(\Omega\cdot 2+\psi(\Omega\cdot 2))$ |
| $\psi(\Omega^2\cdot 2+\Omega\cdot\psi(\Omega^2\cdot 2+\psi(\Omega^2\cdot 2)))$ | $\psi(\Omega\cdot 2+\psi(\Omega\cdot 2)^2)$ |
| $\psi(\Omega^2\cdot 2+\Omega\cdot\psi(\Omega^2\cdot 2+\Omega))$ | $\psi(\Omega\cdot 2+\psi(\Omega\cdot 2+1))$ |
| $\psi(\Omega^2\cdot 2+\Omega\cdot\psi(\Omega^2\cdot 2+\Omega))$ | $\psi(\Omega\cdot 2+\psi(\Omega\cdot 2+1))$ |
| $\psi(\Omega^2\cdot 3)$ | $\psi(\Omega\cdot 3)$ |
| $\psi(\Omega^2\cdot 3+1)$ | $\psi(\Omega\cdot 3)\cdot\omega$ |
| $\psi(\Omega^2\cdot 3+\Omega)$ | $\psi(\Omega\cdot 3+1)$ |
| $\psi(\Omega^2\cdot 3+\Omega\cdot\psi(\Omega^2\cdot 3))$ | $\psi(\Omega\cdot 3+\psi(\Omega\cdot 3))$ |
| $\psi(\Omega^2\cdot 3+\Omega\cdot\psi(\Omega^2\cdot 3+\Omega))$ | $\psi(\Omega\cdot 3+\psi(\Omega\cdot 3+1))$ |
| $\psi(\Omega^2\cdot 4)$ | $\psi(\Omega\cdot 4)$ |
| $\psi(\Omega^2\cdot 5)$ | $\psi(\Omega\cdot 5)$ |
| $\psi(\Omega^2\cdot\psi(1))$ | $\psi(\Omega\cdot\omega)$ |
| $\psi(\Omega^2\cdot\psi(\Omega))$ | $\psi(\Omega\cdot\psi(0))$ |
| $\psi(\Omega^2\cdot\psi(\Omega\cdot\psi(\Omega)))$ | $\psi(\Omega\cdot\psi(\psi(0)))$ |
| $\psi(\Omega^2\cdot\psi(\Omega^2))$ | $\psi(\Omega\cdot\psi(\Omega))$ |
| $\psi(\Omega^2\cdot\psi(\Omega^2\cdot 2))$ | $\psi(\Omega\cdot\psi(\Omega\cdot 2))$ |
| $\psi(\Omega^2\cdot\psi(\Omega^2\cdot\psi(\Omega)))$ | $\psi(\Omega\cdot\psi(\Omega\cdot\psi(0)))$ |
| $\psi(\Omega^2\cdot\psi(\Omega^2\cdot\psi(\Omega^2)))$ | $\psi(\Omega\cdot\psi(\Omega\cdot\psi(\Omega)))$ |
| $\psi(\Omega^3)$ | $\psi(\Omega^2)$ |
| $\psi(\Omega^3+1)$ | $\psi(\Omega^2)\cdot\omega$ |
| $\psi(\Omega^3+\psi(\Omega^3+\psi(\Omega^3)))$ | $\psi(\Omega^2)^{\psi(\Omega^2)}$ |
| $\psi(\Omega^3+\Omega)$ | $\psi(\Omega^2+1)$ |
| $\psi(\Omega^3+\Omega\cdot\psi(\Omega^3))$ | $\psi(\Omega^2+\psi(\Omega^2))$ |
| $\psi(\Omega^3+\Omega\cdot\psi(\Omega^3+\Omega\psi(\Omega^3)))$ | $\psi(\Omega^2+\psi(\Omega^2+\psi(\Omega^2)))$ |
| $\psi(\Omega^3+\Omega^2)$ | $\psi(\Omega^2+\Omega)$ |
| $\psi(\Omega^3+\Omega^2\cdot\psi(\Omega^3))$ | $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2))$ |
| $\psi(\Omega^3+\Omega^2\psi(\Omega^3+\Omega^2\cdot\psi(\Omega^3)))$ | $\psi(\Omega^2+\Omega\cdot\psi(\Omega^2+\Omega\cdot\psi(\Omega^2)))$ |
| $\psi(\Omega^3\cdot 2)$ | $\psi(\Omega^2\cdot 2)$ |
| $\psi(\Omega^3\cdot 2+1)$ | $\psi(\Omega^2\cdot 2)\cdot\omega$ |
| $\psi(\Omega^3\cdot 2+\Omega)$ | $\psi(\Omega^2\cdot 2+1)$ |
| $\psi(\Omega^3\cdot 2+\Omega^2)$ | $\psi(\Omega^2\cdot 2+\Omega)$ |
| $\psi(\Omega^3\cdot 2+\Omega^2\cdot\psi(\Omega^3\cdot 2+\Omega^2))$ | $\psi(\Omega^2\cdot 2+\Omega\cdot\psi(\Omega^2\cdot 2+\Omega))$ |
| $\psi(\Omega^3\cdot 3)$ | $\psi(\Omega^2\cdot 3)$ |
| $\psi(\Omega^3\cdot 4)$ | $\psi(\Omega^2\cdot 4)$ |
| $\psi(\Omega^3\cdot\psi(1))$ | $\psi(\Omega^2\cdot\omega)$ |
| $\psi(\Omega^3\cdot\psi(\Omega))$ | $\psi(\Omega^2\cdot\psi(0))$ |
| $\psi(\Omega^3\cdot\psi(\Omega^2))$ | $\psi(\Omega^2\cdot\psi(\Omega))$ |

| Buchholz's OCF | Madore's OCF |
|---|---|
| $\psi(\Omega^3 \cdot \psi(\Omega^3))$ | $\psi(\Omega^2 \cdot \psi(\Omega^2))$ |
| $\psi(\Omega^3 \cdot \psi(\Omega^3 \cdot \psi(\Omega^3)))$ | $\psi(\Omega^2 \cdot \psi(\Omega^2 \cdot \psi(\Omega^2)))$ |
| $\psi(\Omega^4)$ | $\psi(\Omega^3)$ |
| $\psi(\Omega^4 + 1)$ | $\psi(\Omega^3) \cdot \omega$ |
| $\psi(\Omega^4 + \Omega)$ | $\psi(\Omega^3 + 1)$ |
| $\psi(\Omega^4 + \Omega^3)$ | $\psi(\Omega^3 + \Omega^2)$ |
| $\psi(\Omega^4 \cdot 2)$ | $\psi(\Omega^3 \cdot 2)$ |
| $\psi(\Omega^4 \cdot 3)$ | $\psi(\Omega^3 \cdot 3)$ |
| $\psi(\Omega^4 \cdot \psi(1))$ | $\psi(\Omega^3 \cdot \omega)$ |
| $\psi(\Omega^4 \cdot \psi(\Omega^4))$ | $\psi(\Omega^3 \cdot \psi(\Omega^3))$ |
| $\psi(\Omega^5)$ | $\psi(\Omega^4)$ |
| $\psi(\Omega^5 + \Omega^4)$ | $\psi(\Omega^4 + \Omega^3)$ |
| $\psi(\Omega^5 \cdot 2)$ | $\psi(\Omega^4 \cdot 2)$ |
| $\psi(\Omega^5 \cdot \psi(\Omega^5))$ | $\psi(\Omega^4 \cdot \psi(\Omega^4))$ |
| $\psi(\Omega^6)$ | $\psi(\Omega^5)$ |
| $\psi(\Omega^7)$ | $\psi(\Omega^6)$ |
| $\psi(\Omega^\omega)$ | $\psi(\Omega^\omega)$ |
| $\psi(\Omega^\omega + 1)$ | $\psi(\Omega^\omega) \cdot \omega$ |
| $\psi(\Omega^\omega + \Omega)$ | $\psi(\Omega^\omega + 1)$ |
| $\psi(\Omega^\omega + \Omega^2)$ | $\psi(\Omega^\omega + \Omega)$ |
| $\psi(\Omega^\omega + \Omega^3)$ | $\psi(\Omega^\omega + \Omega^2)$ |
| $\psi(\Omega^\omega + \Omega^4)$ | $\psi(\Omega^\omega + \Omega^3)$ |
| $\psi(\Omega^\omega + \Omega^5)$ | $\psi(\Omega^\omega + \Omega^4)$ |
| $\psi(\Omega^\omega + \Omega^6)$ | $\psi(\Omega^\omega + \Omega^5)$ |
| $\psi(\Omega^\omega \cdot 2)$ | $\psi(\Omega^\omega \cdot 2)$ |
| $\psi(\Omega^\omega \cdot 3)$ | $\psi(\Omega^\omega \cdot 3)$ |
| $\psi(\Omega^\omega \cdot \psi(1))$ | $\psi(\Omega^\omega \cdot \omega)$ |
| $\psi(\Omega^\omega \cdot \psi(\Omega^\omega))$ | $\psi(\Omega^\omega \cdot \psi(\Omega^\omega))$ |
| $\psi(\Omega^{\omega+1})$ | $\psi(\Omega^{\omega+1})$ |
| $\psi(\Omega^{\omega+1} + \Omega^\omega)$ | $\psi(\Omega^{\omega+1} + \Omega^\omega)$ |
| $\psi(\Omega^{\omega+1} \cdot 2)$ | $\psi(\Omega^{\omega+1} \cdot 2)$ |
| $\psi(\Omega^{\omega+2})$ | $\psi(\Omega^{\omega+2})$ |
| $\psi(\Omega^{\omega+3})$ | $\psi(\Omega^{\omega+3})$ |
| $\psi(\Omega^{\omega \cdot 2})$ | $\psi(\Omega^{\omega \cdot 2})$ |
| $\psi(\Omega^{\omega \cdot 3})$ | $\psi(\Omega^{\omega \cdot 3})$ |
| $\psi(\Omega^{\psi(2)})$ | $\psi(\Omega^{\omega^2})$ |
| $\psi(\Omega^{\psi(\psi(1))})$ | $\psi(\Omega^{\omega^\omega})$ |
| $\psi(\Omega^{\psi(\Omega)})$ | $\psi(\Omega^{\psi(0)})$ |
| $\psi(\Omega^{\psi(\Omega \cdot 2)})$ | $\psi(\Omega^{\psi(1)})$ |
| $\psi(\Omega^{\psi(\Omega^2)})$ | $\psi(\Omega^{\psi(\Omega)})$ |
| $\psi(\Omega^{\psi(\Omega^3)})$ | $\psi(\Omega^{\psi(\Omega^2)})$ |
| $\psi(\Omega^{\psi(\Omega^\omega)})$ | $\psi(\Omega^{\psi(\Omega^\omega)})$ |
| $\psi(\Omega^{\psi(\Omega^{\psi(\Omega)})})$ | $\psi(\Omega^{\psi(\Omega^{\psi(0)})})$ |
| $\psi(\Omega^\Omega)$ | $\psi(\Omega^\Omega)$ |
| $\psi(\Omega^\Omega) \cdot 2$ | $\psi(\Omega^\Omega) \cdot 2$ |
| $\psi(\Omega^\Omega + 1)$ | $\psi(\Omega^\Omega) \cdot \omega$ |
| $\psi(\Omega^\Omega + \psi(\Omega^\Omega))$ | $\psi(\Omega^\Omega)^2$ |
| $\psi(\Omega^\Omega + \Omega)$ | $\psi(\Omega^\Omega + 1)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)})$ | $\psi(\Omega^\Omega + \psi(\Omega^\Omega))$ |
| $\psi(\Omega^\Omega + \Omega^2)$ | $\psi(\Omega^\Omega + \Omega)$ |
| $\psi(\Omega^\Omega + \Omega^3)$ | $\psi(\Omega^\Omega + \Omega^2)$ |
| $\psi(\Omega^\Omega + \Omega^4)$ | $\psi(\Omega^\Omega + \Omega^3)$ |
| $\psi(\Omega^\Omega + \Omega^5)$ | $\psi(\Omega^\Omega + \Omega^4)$ |
| $\psi(\Omega^\Omega + \Omega^\omega)$ | $\psi(\Omega^\Omega + \Omega^\omega)$ |

| Buchholz's OCF | Madore's OCF |
|---|---|
| $\psi(\Omega^\Omega + \Omega^{\psi(0)})$ | $\psi(\Omega^\Omega + \Omega^{\psi(0)})$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\omega)})$ | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\omega)})$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^{\psi(0)})})$ | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^{\psi(0)})})$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)})$ | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)})$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)} \cdot 2)$ | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)} \cdot 2)$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)} \cdot \psi(\Omega^\Omega))$ | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)} \cdot \psi(\Omega^\Omega))$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)+1})$ | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)+1})$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)+\omega})$ | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)+\omega})$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)\cdot 2})$ | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)\cdot 2})$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+\Omega)})$ | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+1)})$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+\Omega^{\psi(\Omega^\Omega)})})$ | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+\Omega^{\psi(\Omega^\Omega)})})$ |
| $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+\Omega^{\psi(\Omega^\Omega)}+\Omega)})$ | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+\Omega^{\psi(\Omega^\Omega)}+1)})$ |
| $\psi(\Omega^\Omega \cdot 2)$ | $\psi(\Omega^\Omega \cdot 2)$ |
| $\psi(\Omega^\Omega \cdot 2 + \Omega^{\psi(\Omega^\Omega)})$ | $\psi(\Omega^\Omega \cdot 2 + \Omega^{\psi(\Omega^\Omega)})$ |
| $\psi(\Omega^\Omega \cdot 2 + \Omega^{\psi(\Omega^\Omega\cdot 2)})$ | $\psi(\Omega^\Omega \cdot 2 + \Omega^{\psi(\Omega^\Omega\cdot 2)})$ |
| $\psi(\Omega^\Omega \cdot 3)$ | $\psi(\Omega^\Omega \cdot 3)$ |
| $\psi(\Omega^\Omega \cdot 4)$ | $\psi(\Omega^\Omega \cdot 4)$ |
| $\psi(\Omega^\Omega \cdot \omega)$ | $\psi(\Omega^\Omega \cdot \omega)$ |
| $\psi(\Omega^\Omega \cdot \psi(0))$ | $\psi(\Omega^\Omega \cdot \psi(0))$ |
| $\psi(\Omega^\Omega \cdot \psi(\Omega^\Omega))$ | $\psi(\Omega^\Omega \cdot \psi(\Omega^\Omega))$ |
| $\psi(\Omega^{\Omega+1})$ | $\psi(\Omega^{\Omega+1})$ |
| $\psi(\Omega^{\Omega+1} + \Omega^\omega)$ | $\psi(\Omega^{\Omega+1} + \Omega^\omega)$ |
| $\psi(\Omega^{\Omega+1} + \Omega^{\psi(\Omega^\Omega)})$ | $\psi(\Omega^{\Omega+1} + \Omega^{\psi(\Omega^\Omega)})$ |
| $\psi(\Omega^{\Omega+1} + \Omega^{\psi(\Omega^{\Omega+1})})$ | $\psi(\Omega^{\Omega+1} + \Omega^{\psi(\Omega^{\Omega+1})})$ |
| $\psi(\Omega^{\Omega+1} + \Omega^{\psi(\Omega^{\Omega+1})+1})$ | $\psi(\Omega^{\Omega+1} + \Omega^{\psi(\Omega^{\Omega+1})+1})$ |
| $\psi(\Omega^{\Omega+1} + \Omega^\Omega)$ | $\psi(\Omega^{\Omega+1} + \Omega^\Omega)$ |
| $\psi(\Omega^{\Omega+1} + \Omega^\Omega \cdot \psi(\Omega^{\Omega+1} + \Omega^\Omega))$ | $\psi(\Omega^{\Omega+1} + \Omega^\Omega \cdot \psi(\Omega^{\Omega+1} + \Omega^\Omega))$ |
| $\psi(\Omega^{\Omega+1} \cdot 2)$ | $\psi(\Omega^{\Omega+1} \cdot 2)$ |
| $\psi(\Omega^{\Omega+1} \cdot 3)$ | $\psi(\Omega^{\Omega+1} \cdot 3)$ |
| $\psi(\Omega^{\Omega+1} \cdot \omega)$ | $\psi(\Omega^{\Omega+1} \cdot \omega)$ |
| $\psi(\Omega^{\Omega+2})$ | $\psi(\Omega^{\Omega+2})$ |
| $\psi\left(\Omega^{\Omega+2} + \Omega^{\psi(\Omega^{\Omega+2})+1}\right)$ | $\psi\left(\Omega^{\Omega+2} + \Omega^{\psi(\Omega^{\Omega+2})+1}\right)$ |
| $\psi\left(\Omega^{\Omega+2} + \Omega^\Omega\right)$ | $\psi\left(\Omega^{\Omega+2} + \Omega^\Omega\right)$ |
| $\psi\left(\Omega^{\Omega+2} + \Omega^\Omega \psi\left(\Omega^{\Omega+2} + \Omega^\Omega\right)\right)$ | $\psi\left(\Omega^{\Omega+2} + \Omega^\Omega \psi\left(\Omega^{\Omega+2} + \Omega^\Omega\right)\right)$ |
| $\psi\left(\Omega^{\Omega+2} + \Omega^{\Omega+1}\right)$ | $\psi\left(\Omega^{\Omega+2} + \Omega^{\Omega+1}\right)$ |
| $\psi\left(\Omega^{\Omega+2} \cdot 2\right)$ | $\psi\left(\Omega^{\Omega+2} \cdot 2\right)$ |
| $\psi\left(\Omega^{\Omega+2} \cdot \psi\left(\Omega^{\Omega+2}\right)\right)$ | $\psi\left(\Omega^{\Omega+2} \cdot \psi\left(\Omega^{\Omega+2}\right)\right)$ |
| $\psi\left(\Omega^{\Omega+3}\right)$ | $\psi\left(\Omega^{\Omega+3}\right)$ |
| $\psi\left(\Omega^{\Omega+3} \cdot 2\right)$ | $\psi\left(\Omega^{\Omega+3} \cdot 2\right)$ |
| $\psi\left(\Omega^{\Omega+4}\right)$ | $\psi\left(\Omega^{\Omega+4}\right)$ |
| $\psi\left(\Omega^{\Omega+\omega}\right)$ | $\psi\left(\Omega^{\Omega+\omega}\right)$ |
| $\psi\left(\Omega^{\Omega+\psi(\Omega^\Omega)}\right)$ | $\psi\left(\Omega^{\Omega+\psi(\Omega^\Omega)}\right)$ |
| $\psi\left(\Omega^{\Omega+\psi(\Omega^{\Omega+\psi(\Omega^\Omega)})}\right)$ | $\psi\left(\Omega^{\Omega+\psi(\Omega^{\Omega+\psi(\Omega^\Omega)})}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 2}\right)$ | $\psi\left(\Omega^{\Omega\cdot 2}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot} + \Omega^{\psi\left(\Omega^{\Omega\cdot 2}\right)+1}\right)$ | $\psi\left(\Omega^{\Omega\cdot} + \Omega^{\psi\left(\Omega^{\Omega\cdot 2}\right)+1}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 2} + \Omega^\Omega\right)$ | $\psi\left(\Omega^{\Omega\cdot 2} + \Omega^\Omega\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 2} + \Omega^{\Omega+\omega}\right)$ | $\psi\left(\Omega^{\Omega\cdot 2} + \Omega^{\Omega+\omega}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 2} + \Omega^{\Omega+\psi(\Omega^\Omega\cdot 2)}\right)$ | $\psi\left(\Omega^{\Omega\cdot 2} + \Omega^{\Omega+\psi(\Omega^\Omega\cdot 2)}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 2} + \Omega^{\Omega+\psi(\Omega^\Omega\cdot 2)+1}\right)$ | $\psi\left(\Omega^{\Omega\cdot 2} + \Omega^{\Omega+\psi(\Omega^\Omega\cdot 2)+1}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 2} \cdot 2\right)$ | $\psi\left(\Omega^{\Omega\cdot 2} \cdot 2\right)$ |

| Buchholz's OCF | Madore's OCF |
|---|---|
| $\psi\left(\Omega^{\Omega\cdot 2}\cdot\psi\left(\Omega^{\Omega\cdot 2}\right)\right)$ | $\psi\left(\Omega^{\Omega\cdot 2}\cdot\psi\left(\Omega^{\Omega\cdot 2}\right)\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 2+1}\right)$ | $\psi\left(\Omega^{\Omega\cdot 2+1}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 2+1}\cdot 2\right)$ | $\psi\left(\Omega^{\Omega\cdot 2+1}\cdot 2\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 2+\psi(\Omega^{\Omega}\cdot 2)}\right)$ | $\psi\left(\Omega^{\Omega\cdot 2+\psi(\Omega^{\Omega}\cdot 2)}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 3}\right)$ | $\psi\left(\Omega^{\Omega\cdot 3}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 3}+\Omega^{\psi(\Omega^{\Omega}\cdot 3)}\right)$ | $\psi\left(\Omega^{\Omega\cdot 3}+\Omega^{\psi(\Omega^{\Omega}\cdot 3)}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 3}+\Omega^{\Omega+\psi(\Omega^{\Omega}\cdot 3)}\right)$ | $\psi\left(\Omega^{\Omega\cdot 3}+\Omega^{\Omega+\psi(\Omega^{\Omega}\cdot 3)}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 3}+\Omega^{\Omega\cdot 2+\psi(\Omega^{\Omega}\cdot 3)}\right)$ | $\psi\left(\Omega^{\Omega\cdot 3}+\Omega^{\Omega\cdot 2+\psi(\Omega^{\Omega}\cdot 3)}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 3}\cdot 2\right)$ | $\psi\left(\Omega^{\Omega\cdot 3}\cdot 2\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 3+1}\right)$ | $\psi\left(\Omega^{\Omega\cdot 3+1}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 4}\right)$ | $\psi\left(\Omega^{\Omega\cdot 4}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot 5}\right)$ | $\psi\left(\Omega^{\Omega\cdot 5}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot\omega}\right)$ | $\psi\left(\Omega^{\Omega\cdot\omega}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot\psi(0)}\right)$ | $\psi\left(\Omega^{\Omega\cdot\psi(0)}\right)$ |
| $\psi\left(\Omega^{\Omega\cdot\psi(\Omega^{\Omega})}\right)$ | $\psi\left(\Omega^{\Omega\cdot\psi(\Omega^{\Omega})}\right)$ |
| $\psi\left(\Omega^{\Omega^2}\right)$ | $\psi\left(\Omega^{\Omega^2}\right)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\psi\left(\Omega^{\Omega^2}+\Omega^{\psi\left(\Omega^{\Omega^2}\right)}\right)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega}\right)$ | $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega}\right)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega+1}\right)$ | $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega+1}\right)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega+\omega}\right)$ | $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega+\omega}\right)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega+\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega+\psi\left(\Omega^{\Omega^2}\right)}\right)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot 2}\right)$ | $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot 2}\right)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot 2+\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot 2+\psi\left(\Omega^{\Omega^2}\right)}\right)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot 3}\right)$ | $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot 3}\right)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega}\right)}\right)$ | $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega}\right)}\right)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)}\right)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)+1}\right)$ | $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)+1}\right)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)+\Omega}\right)$ | $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)+\Omega}\right)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)\cdot 2}\right)$ | $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)\cdot 2}\right)$ |
| $\psi\left(\Omega^{\Omega^2}\cdot 2\right)$ | $\psi\left(\Omega^{\Omega^2}\cdot 2\right)$ |
| $\psi\left(\Omega^{\Omega^2+1}\right)$ | $\psi\left(\Omega^{\Omega^2+1}\right)$ |
| $\psi\left(\Omega^{\Omega^2+\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\psi\left(\Omega^{\Omega^2+\psi\left(\Omega^{\Omega^2}\right)}\right)$ |
| $\psi\left(\Omega^{\Omega^2+\Omega}\right)$ | $\psi\left(\Omega^{\Omega^2+\Omega}\right)$ |
| $\psi\left(\Omega^{\Omega^2+\Omega}+\Omega^{\Omega^2+\psi\left(\Omega^{\Omega^2+\Omega}\right)\cdot 2}\right)$ | $\psi\left(\Omega^{\Omega^2+\Omega}+\Omega^{\Omega^2+\psi\left(\Omega^{\Omega^2+\Omega}\right)\cdot 2}\right)$ |
| $\psi\left(\Omega^{\Omega^2+\Omega}\cdot 2\right)$ | $\psi\left(\Omega^{\Omega^2+\Omega}\cdot 2\right)$ |
| $\psi\left(\Omega^{\Omega^2+\Omega+1}\right)$ | $\psi\left(\Omega^{\Omega^2+\Omega+1}\right)$ |
| $\psi\left(\Omega^{\Omega^2+\Omega\cdot 2}\right)$ | $\psi\left(\Omega^{\Omega^2+\Omega\cdot 2}\right)$ |

| Buchholz's OCF | Madore's OCF |
|---|---|
| $\psi\left(\Omega^{\Omega^2+\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\psi\left(\Omega^{\Omega^2+\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)}\right)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 2}\right)$ | $\psi\left(\Omega^{\Omega^2\cdot 2}\right)$ |
| $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega^2+\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)}\right)$ | $\psi\left(\Omega^{\Omega^2}+\Omega^{\Omega^2+\Omega\cdot\psi\left(\Omega^{\Omega^2}\right)}\right)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 2}\cdot 2\right)$ | $\psi\left(\Omega^{\Omega^2\cdot 2}\cdot 2\right)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 2+1}\right)$ | $\psi\left(\Omega^{\Omega^2\cdot 2+1}\right)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 2+1}\cdot 2\right)$ | $\psi\left(\Omega^{\Omega^2\cdot 2+1}\cdot 2\right)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 2+\Omega}\right)$ | $\psi\left(\Omega^{\Omega^2\cdot 2+\Omega}\right)$ |
| $\psi\left(\Omega^{\Omega^2\cdot 3}\right)$ | $\psi\left(\Omega^{\Omega^2\cdot 3}\right)$ |
| $\psi\left(\Omega^{\Omega^2\cdot\psi(\Omega^\Omega)}\right)$ | $\psi\left(\Omega^{\Omega^2\cdot\psi(\Omega^\Omega)}\right)$ |
| $\psi\left(\Omega^{\Omega^3}\right)$ | $\psi\left(\Omega^{\Omega^3}\right)$ |
| $\psi\left(\Omega^{\Omega^3}+\Omega^{\psi(\Omega^{\Omega^3})}\right)$ | $\psi\left(\Omega^{\Omega^3}+\Omega^{\psi(\Omega^{\Omega^3})}\right)$ |
| $\psi\left(\Omega^{\Omega^3}+\Omega^{\Omega\cdot\psi(\Omega^{\Omega^3})}\right)$ | $\psi\left(\Omega^{\Omega^3}+\Omega^{\Omega\cdot\psi(\Omega^{\Omega^3})}\right)$ |
| $\psi\left(\Omega^{\Omega^3}+\Omega^{\Omega^2\cdot\psi(\Omega^{\Omega^3})}\right)$ | $\psi\left(\Omega^{\Omega^3}+\Omega^{\Omega^2\cdot\psi(\Omega^{\Omega^3})}\right)$ |
| $\psi\left(\Omega^{\Omega^3}\cdot 2\right)$ | $\psi\left(\Omega^{\Omega^3}\cdot 2\right)$ |
| $\psi\left(\Omega^{\Omega^3+1}\right)$ | $\psi\left(\Omega^{\Omega^3+1}\right)$ |
| $\psi\left(\Omega^{\Omega^3+\Omega}\right)$ | $\psi\left(\Omega^{\Omega^3+\Omega}\right)$ |
| $\psi\left(\Omega^{\Omega^3+\Omega^2}\right)$ | $\psi\left(\Omega^{\Omega^3+\Omega^2}\right)$ |
| $\psi\left(\Omega^{\Omega^3\cdot 2}\right)$ | $\psi\left(\Omega^{\Omega^3\cdot 2}\right)$ |
| $\psi\left(\Omega^{\Omega^4}\right)$ | $\psi\left(\Omega^{\Omega^4}\right)$ |
| $\psi\left(\Omega^{\Omega^5}\right)$ | $\psi\left(\Omega^{\Omega^5}\right)$ |
| $\psi\left(\Omega^{\Omega^6}\right)$ | $\psi\left(\Omega^{\Omega^6}\right)$ |
| $\psi\left(\Omega^{\Omega^\omega}\right)$ | $\psi\left(\Omega^{\Omega^\omega}\right)$ |
| $\psi\left(\Omega^{\Omega^\omega}+\Omega\right)$ | $\psi\left(\Omega^{\Omega^\omega}+1\right)$ |
| $\psi\left(\Omega^{\Omega^\omega}+\Omega^\Omega\right)$ | $\psi\left(\Omega^{\Omega^\omega}+\Omega^\Omega\right)$ |
| $\psi\left(\Omega^{\Omega^\omega}+\Omega^{\Omega^2}\right)$ | $\psi\left(\Omega^{\Omega^\omega}+\Omega^{\Omega^2}\right)$ |
| $\psi\left(\Omega^{\Omega^\omega}+\Omega^{\Omega^3}\right)$ | $\psi\left(\Omega^{\Omega^\omega}+\Omega^{\Omega^3}\right)$ |
| $\psi\left(\Omega^{\Omega^\omega}+\Omega^{\Omega^4}\right)$ | $\psi\left(\Omega^{\Omega^\omega}+\Omega^{\Omega^4}\right)$ |
| $\psi\left(\Omega^{\Omega^\omega}+\Omega^{\Omega^5}\right)$ | $\psi\left(\Omega^{\Omega^\omega}+\Omega^{\Omega^5}\right)$ |
| $\psi\left(\Omega^{\Omega^\omega}\cdot 2\right)$ | $\psi\left(\Omega^{\Omega^\omega}\cdot 2\right)$ |
| $\psi\left(\Omega^{\Omega^\omega+1}\right)$ | $\psi\left(\Omega^{\Omega^\omega+1}\right)$ |
| $\psi\left(\Omega^{\Omega^\omega+\psi(\Omega^{\Omega^\omega})}\right)$ | $\psi\left(\Omega^{\Omega^\omega+\psi(\Omega^{\Omega^\omega})}\right)$ |
| $\psi\left(\Omega^{\Omega^\omega+\Omega}\right)$ | $\psi\left(\Omega^{\Omega^\omega+\Omega}\right)$ |
| $\psi\left(\Omega^{\Omega^\omega+\Omega\cdot\psi(\Omega^{\Omega^\omega})}\right)$ | $\psi\left(\Omega^{\Omega^\omega+\Omega\cdot\psi(\Omega^{\Omega^\omega})}\right)$ |
| $\psi\left(\Omega^{\Omega^\omega+\Omega^2}\right)$ | $\psi\left(\Omega^{\Omega^\omega+\Omega^2}\right)$ |
| $\psi\left(\Omega^{\Omega^\omega+\Omega^3}\right)$ | $\psi\left(\Omega^{\Omega^\omega+\Omega^3}\right)$ |
| $\psi\left(\Omega^{\Omega^\omega+\Omega^4}\right)$ | $\psi\left(\Omega^{\Omega^\omega+\Omega^4}\right)$ |
| $\psi\left(\Omega^{\Omega^\omega\cdot 2}\right)$ | $\psi\left(\Omega^{\Omega^\omega\cdot 2}\right)$ |
| $\psi\left(\Omega^{\Omega^\omega\cdot\psi(\Omega^{\Omega^\omega})}\right)$ | $\psi\left(\Omega^{\Omega^\omega\cdot\psi(\Omega^{\Omega^\omega})}\right)$ |

| Buchholz's OCF | Madore's OCF |
|---|---|
| $\psi\left(\Omega^{\Omega^{\omega+1}}\right)$ | $\psi\left(\Omega^{\Omega^{\omega+1}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}}+\Omega^{\Omega^{\omega}}\right)$ | $\psi\left(\Omega^{\Omega^{\omega+1}}+\Omega^{\Omega^{\omega}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}}\cdot 2\right)$ | $\psi\left(\Omega^{\Omega^{\omega+1}}\cdot 2\right)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}+1}\right)$ | $\psi\left(\Omega^{\Omega^{\omega+1}+1}\right)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}+\Omega}\right)$ | $\psi\left(\Omega^{\Omega^{\omega+1}+\Omega}\right)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}+\Omega^{\omega}}\right)$ | $\psi\left(\Omega^{\Omega^{\omega+1}+\Omega^{\omega}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\omega+1}\cdot 2}\right)$ | $\psi\left(\Omega^{\Omega^{\omega+1}\cdot 2}\right)$ |
| $\psi\left(\Omega^{\Omega^{\omega+2}}\right)$ | $\psi\left(\Omega^{\Omega^{\omega+2}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\omega+3}}\right)$ | $\psi\left(\Omega^{\Omega^{\omega+3}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\omega\cdot 2}}\right)$ | $\psi\left(\Omega^{\Omega^{\omega\cdot 2}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\omega^2}}\right)$ | $\psi\left(\Omega^{\Omega^{\omega^2}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\psi(0)}}\right)$ | $\psi\left(\Omega^{\Omega^{\psi(0)}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\psi(\Omega)}}\right)$ | $\psi\left(\Omega^{\Omega^{\psi(\Omega)}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\psi(\Omega^{\Omega})}}\right)$ | $\psi\left(\Omega^{\Omega^{\psi(\Omega^{\Omega})}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\psi(\Omega^{\Omega^2})}}\right)$ | $\psi\left(\Omega^{\Omega^{\psi(\Omega^{\Omega^2})}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\psi(\Omega^{\Omega^{\omega}})}}\right)$ | $\psi\left(\Omega^{\Omega^{\psi(\Omega^{\Omega^{\omega}})}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega}}+\Omega^{\psi\left(\Omega^{\Omega^{\Omega}}\right)}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega}}+\Omega^{\psi\left(\Omega^{\Omega^{\Omega}}\right)}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega}}+\Omega^{\Omega^{\omega}\cdot\psi\left(\Omega^{\Omega^{\Omega}}\right)}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega}}+\Omega^{\Omega^{\omega}\cdot\psi\left(\Omega^{\Omega^{\Omega}}\right)}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega}}\cdot 2\right)$ | $\psi\left(\Omega^{\Omega^{\Omega}}\cdot 2\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega}+1}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega}+1}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega}+\Omega^{\omega}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega}+\Omega^{\omega}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega}\cdot 2}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega}\cdot 2}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega}\cdot\psi\left(\Omega^{\Omega^{\Omega}}\right)}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega}\cdot\psi\left(\Omega^{\Omega^{\Omega}}\right)}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega+1}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega+1}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega+1}\cdot 2}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega+1}\cdot 2}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega+2}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega+2}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega+\omega}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega+\omega}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega+\psi\left(\Omega^{\Omega^{\Omega}}\right)}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega+\psi\left(\Omega^{\Omega^{\Omega}}\right)}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega\cdot 2}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega\cdot 2}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega\cdot 2+\psi\left(\Omega^{\Omega^{\Omega\cdot 2}}\right)}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega\cdot 2+\psi\left(\Omega^{\Omega^{\Omega\cdot 2}}\right)}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega\cdot 3}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega\cdot 3}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega\cdot\omega}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega\cdot\omega}}\right)$ |

| Buchholz's OCF | Madore's OCF |
|---|---|
| $\psi\left(\Omega^{\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^{\Omega}}\right)}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega\cdot\psi\left(\Omega^{\Omega^{\Omega}}\right)}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^2}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^2}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^2\cdot 2}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^2\cdot 2}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^2+1}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^2+1}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^2+\Omega}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^2+\Omega}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^2\cdot 2}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^2\cdot 2}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^3}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^3}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^3\cdot 2}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^3\cdot 2}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^4}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^4}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^\omega}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^\omega}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^\Omega}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^\Omega}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^\Omega}}+\Omega\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^\Omega}}+1\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^\Omega}}\cdot 2\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^\Omega}}\cdot 2\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^\Omega}+1}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^\Omega}+1}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^\Omega}\cdot 2}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^\Omega}\cdot 2}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^\Omega+1}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^\Omega+1}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^\Omega\cdot 2}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^\Omega\cdot 2}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega+1}}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^{\Omega+1}}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega\cdot 2}}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^{\Omega\cdot 2}}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega^2}}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^{\Omega^2}}}\right)$ |
| $\psi\left(\Omega^{\Omega^{\Omega^{\Omega^\Omega}}}\right)$ | $\psi\left(\Omega^{\Omega^{\Omega^{\Omega^\Omega}}}\right)$ |
| $\psi\left(\Omega_2\right)$ | $\psi(\psi_1(0))$ |
| $\psi\left(\Omega_2+\Omega\right)$ | $\psi(\psi_1(0)+1)$ |
| $\psi\left(\Omega_2+\Omega^\Omega\right)$ | $\psi(\psi_1(0)+\Omega^\Omega)$ |
| $\psi\left(\Omega_2+\Omega^{\Omega^\Omega}\right)$ | $\psi(\psi_1(0)+\Omega^{\Omega^\Omega})$ |
| $\psi\left(\Omega_2+\psi_1(\Omega_2)\right)$ | $\psi(\psi_1(0)\cdot 2)$ |
| $\psi\left(\Omega_2+\psi_1(\Omega_2)\cdot 2\right)$ | $\psi(\psi_1(0)\cdot 3)$ |
| $\psi\left(\Omega_2+\psi_1(\Omega_2+1)\right)$ | $\psi(\psi_1(0)\cdot\omega)$ |
| $\psi\left(\Omega_2+\psi_1(\Omega_2+\psi(\Omega_2))\right)$ | $\psi(\psi_1(0)\cdot\psi(\psi_1(0)))$ |
| $\psi\left(\Omega_2+\psi_1(\Omega_2+\Omega)\right)$ | $\psi(\psi_1(0)\cdot\Omega)$ |
| $\psi\left(\Omega_2+\psi_1(\Omega_2+\Omega^{\Omega^\Omega})\right)$ | $\psi(\psi_1(0)\cdot\Omega^{\Omega^\Omega})$ |
| $\psi\left(\Omega_2+\psi_1(\Omega_2+\psi_1(\Omega_2))\right)$ | $\psi(\psi_1(0)^2)$ |
| $\psi\left(\Omega_2+\psi_1(\Omega_2+\psi_1(\Omega_2))\cdot 2\right)$ | $\psi(\psi_1(0)^2\cdot 2)$ |
| $\psi\left(\Omega_2+\psi_1(\Omega_2+\psi_1(\Omega_2))\cdot\Omega\right)$ | $\psi(\psi_1(0)^2\cdot\Omega)$ |
| $\psi\left(\Omega_2+\psi_1(\Omega_2+\psi_1(\Omega_2)\cdot 2)\right)$ | $\psi(\psi_1(0)^3)$ |
| $\psi\left(\Omega_2+\psi_1(\Omega_2+\psi_1(\Omega_2+1))\right)$ | $\psi(\psi_1(0)^\omega)$ |
| $\psi\left(\Omega_2+\psi_1(\Omega_2+\psi_1(\Omega_2+\Omega))\right)$ | $\psi(\psi_1(0)^\Omega)$ |
| $\psi\left(\Omega_2+\psi_1(\Omega_2+\psi_1(\Omega_2+\Omega^\Omega))\right)$ | $\psi(\psi_1(0)^{\Omega^\Omega})$ |
| $\psi\left(\Omega_2+\psi_1(\Omega_2+\psi_1(\Omega_2+\psi_1(\Omega_2)))\right)$ | $\psi(\psi_1(0)^{\psi_1(0)})$ |
| $\psi\left(\Omega_2\cdot 2\right)$ | $\psi(\psi_1(1))$ |
| $\psi\left(\Omega_2\cdot 2+\psi_1(\Omega_2)\right)$ | $\psi(\psi_1(1)+\psi_1(0))$ |

| Buchholz's OCF | Madore's OCF |
|---|---|
| $\psi\left(\Omega_2 \cdot 2 + \psi_1(\Omega_2 \cdot 2)\right)$ | $\psi(\psi_1(1) \cdot 2)$ |
| $\psi\left(\Omega_2 \cdot 2 + \psi_1(\Omega_2 \cdot 2 + \psi_1(\Omega_2 \cdot 2))\right)$ | $\psi(\psi_1(1)^2)$ |
| $\psi\left(\Omega_2 \cdot 3\right)$ | $\psi(\psi_1(2))$ |
| $\psi\left(\Omega_2 \cdot \omega\right)$ | $\psi(\psi_1(\omega))$ |
| $\psi\left(\Omega_2 \cdot \psi(\Omega_2)\right)$ | $\psi(\psi_1(\psi(\psi_1(0))))$ |
| $\psi\left(\Omega_2 \cdot \Omega\right)$ | $\psi(\psi_1(\Omega))$ |
| $\psi\left(\Omega_2 \cdot \Omega^2\right)$ | $\psi(\psi_1(\Omega^2))$ |
| $\psi\left(\Omega_2 \cdot \psi_1(\Omega_2)\right)$ | $\psi(\psi_1(\psi_1(0)))$ |
| $\psi\left({\Omega_2}^2\right)$ | $\psi(\Omega_2)$ |
| $\psi\left({\Omega_2}^2 + \psi_1({\Omega_2}^2)\right)$ | $\psi(\Omega_2 + \psi_1(\Omega_2))$ |
| $\psi\left({\Omega_2}^2 + \Omega_2\right)$ | $\psi(\Omega_2 + \psi_1(\Omega_2 + 1))$ |
| $\psi\left({\Omega_2}^2 + \Omega_2 \cdot \psi_1({\Omega_2}^2)\right)$ | $\psi(\Omega_2 + \psi_1(\Omega_2 + \psi_1(\Omega_2)))$ |
| $\psi\left({\Omega_2}^2 \cdot 2\right)$ | $\psi(\Omega_2 \cdot 2)$ |
| $\psi\left({\Omega_2}^2 \cdot \Omega\right)$ | $\psi(\Omega_2 \cdot \Omega)$ |
| $\psi\left({\Omega_2}^2 \cdot \psi_1({\Omega_2}^2)\right)$ | $\psi(\Omega_2 \cdot \psi_1(\Omega_2))$ |
| $\psi\left({\Omega_2}^3\right)$ | $\psi({\Omega_2}^2)$ |
| $\psi\left({\Omega_2}^4\right)$ | $\psi({\Omega_2}^3)$ |
| $\psi\left({\Omega_2}^\omega\right)$ | $\psi({\Omega_2}^\omega)$ |
| $\psi\left({\Omega_2}^{\psi({\Omega_2}^\omega)}\right)$ | $\psi({\Omega_2}^{\psi({\Omega_2}^\omega)})$ |
| $\psi\left({\Omega_2}^\Omega\right)$ | $\psi({\Omega_2}^\Omega)$ |
| $\psi\left({\Omega_2}^{\psi_1({\Omega_2}^\Omega)}\right)$ | $\psi({\Omega_2}^{\psi_1({\Omega_2}^\omega)})$ |
| $\psi\left({\Omega_2}^{\Omega_2}\right)$ | $\psi({\Omega_2}^{\Omega_2})$ |
| $\psi\left({\Omega_2}^{\Omega_2}\right)$ | $\psi({\Omega_2}^{\Omega_2})$ |
| $\psi\left({\Omega_2}^{\Omega_2} + \Omega\right)$ | $\psi({\Omega_2}^{\Omega_2} + 1)$ |
| $\psi\left({\Omega_2}^{\Omega_2} + \psi_1(\Omega_2)\right)$ | $\psi({\Omega_2}^{\Omega_2} + \psi_1(0))$ |
| $\psi\left({\Omega_2}^{\Omega_2} + \psi_1({\Omega_2}^{\Omega_2})\right)$ | $\psi({\Omega_2}^{\Omega_2} + \psi_1({\Omega_2}^{\Omega_2}))$ |
| $\psi\left({\Omega_2}^{\Omega_2} + \Omega_2\right)$ | $\psi({\Omega_2}^{\Omega_2} + \Omega_2)$ |
| $\psi\left({\Omega_2}^{\Omega_2} + {\Omega_2}^\omega\right)$ | $\psi({\Omega_2}^{\Omega_2} + {\Omega_2}^\omega)$ |
| $\psi\left({\Omega_2}^{\Omega_2} + {\Omega_2}^\Omega\right)$ | $\psi({\Omega_2}^{\Omega_2} + {\Omega_2}^\Omega)$ |
| $\psi\left({\Omega_2}^{\Omega_2} + {\Omega_2}^{\psi_1(\Omega_2)}\right)$ | $\psi({\Omega_2}^{\Omega_2} + {\Omega_2}^{\psi_1(\Omega_2)})$ |
| $\psi\left({\Omega_2}^{\Omega_2} + {\Omega_2}^{\psi_1({\Omega_2}^{\Omega_2})}\right)$ | $\psi({\Omega_2}^{\Omega_2} + {\Omega_2}^{\psi_1({\Omega_2}^{\Omega_2})})$ |
| $\psi\left({\Omega_2}^{\Omega_2} \cdot 2\right)$ | $\psi({\Omega_2}^{\Omega_2} \cdot 2)$ |
| $\psi\left(\psi_2\left(\psi_2\left(\psi_2(0)\right) + \Omega\right)\right)$ | $\psi({\Omega_2}^{\Omega_2} \cdot \Omega^2)$ |
| $\psi\left(\psi_2\left(\psi_2\left(\psi_2(0)\right) + \psi_1\left(\psi_2(0)\right)\right)\right)$ | $\psi({\Omega_2}^{\Omega_2} \cdot \psi_1(0))$ |
| $\psi\left({\Omega_2}^{\Omega_2+1}\right)$ | $\psi({\Omega_2}^{\Omega_2+1})$ |
| $\psi\left({\Omega_2}^{\Omega_2 \cdot 2}\right)$ | $\psi({\Omega_2}^{\Omega_2 \cdot 2})$ |
| $\psi\left({\Omega_2}^{\Omega_2 \cdot \Omega}\right)$ | $\psi({\Omega_2}^{\Omega_2 \cdot \Omega})$ |
| $\psi\left({\Omega_2}^{{\Omega_2}^2}\right)$ | $\psi({\Omega_2}^{{\Omega_2}^2})$ |
| $\psi\left({\Omega_2}^{{\Omega_2}^2 \cdot 2}\right)$ | $\psi({\Omega_2}^{{\Omega_2}^2 \cdot 2})$ |
| $\psi\left({\Omega_2}^{{\Omega_2}^3}\right)$ | $\psi({\Omega_2}^{{\Omega_2}^3})$ |
| $\psi\left({\Omega_2}^{{\Omega_2}^\omega}\right)$ | $\psi({\Omega_2}^{{\Omega_2}^\omega})$ |
| $\psi\left({\Omega_2}^{{\Omega_2}^\Omega}\right)$ | $\psi({\Omega_2}^{{\Omega_2}^\Omega})$ |
| $\psi\left({\Omega_2}^{{\Omega_2}^{\Omega_2}}\right)$ | $\psi({\Omega_2}^{{\Omega_2}^{\Omega_2}})$ |
| $\psi\left({\Omega_2}^{{\Omega_2}^{\Omega_2} \cdot 2}\right)$ | $\psi({\Omega_2}^{{\Omega_2}^{\Omega_2} \cdot 2})$ |
| $\psi\left({\Omega_2}^{{\Omega_2}^{\Omega_2 \cdot 2}}\right)$ | $\psi({\Omega_2}^{{\Omega_2}^{\Omega_2 \cdot 2}})$ |
| $\psi\left({\Omega_2}^{{\Omega_2}^{\Omega_2 \cdot 2}}\right)$ | $\psi({\Omega_2}^{{\Omega_2}^{\Omega_2 \cdot 2}})$ |
| $\psi\left({\Omega_2}^{{\Omega_2}^{{\Omega_2}^2}}\right)$ | $\psi({\Omega_2}^{{\Omega_2}^{{\Omega_2}^2}})$ |
| $\psi\left({\Omega_2}^{{\Omega_2}^{{\Omega_2}^\omega}}\right)$ | $\psi({\Omega_2}^{{\Omega_2}^{{\Omega_2}^\omega}})$ |

| Buchholz's OCF | Madore's OCF |
|---|---|
| $\psi(\Omega_3)$ | $\psi(\psi_2(0))$ |
| $\psi\left(\Omega_3+\Omega\right)$ | $\psi(\psi_2(0)+1)$ |
| $\psi\left(\Omega_3+\Omega^{\Omega}\right)$ | $\psi(\psi_2(0)+\Omega^{\Omega})$ |
| $\psi\left(\Omega_3+\psi_1\left(\Omega_2\right)\right)$ | $\psi(\psi_2(0)+\psi_1(0))$ |
| $\psi\left(\Omega_3+\psi_1\left(\Omega_3\right)\right)$ | $\psi(\psi_2(0)+\psi_1(\psi_2(0)))$ |
| $\psi\left(\Omega_3+\Omega_2\right)$ | $\psi(\psi_2(0)+\Omega_2)$ |
| $\psi\left(\Omega_3+{\Omega_2}^{\Omega_2}\right)$ | $\psi(\psi_2(0)+{\Omega_2}^{\Omega_2})$ |
| $\psi\left(\Omega_3+\psi_2\left(\Omega_3\right)\right)$ | $\psi(\psi_2(0)\cdot 2)$ |
| $\psi\left(\Omega_3\cdot 2\right)$ | $\psi(\psi_2(1))$ |
| $\psi\left({\Omega_3}^2\right)$ | $\psi(\Omega_3)$ |
| $\psi\left({\Omega_3}^{\omega}\right)$ | $\psi({\Omega_3}^{\omega})$ |
| $\psi\left({\Omega_3}^{\Omega}\right)$ | $\psi({\Omega_3}^{\Omega})$ |
| $\psi\left({\Omega_3}^{\Omega_2}\right)$ | $\psi({\Omega_3}^{\Omega_2})$ |
| $\psi\left({\Omega_3}^{\Omega_3}\right)$ | $\psi({\Omega_3}^{\Omega_3})$ |
| $\psi\left({\Omega_3}^{\Omega_3}\cdot 2\right)$ | $\psi({\Omega_3}^{\Omega_3}\cdot 2)$ |
| $\psi\left({\Omega_3}^{\Omega_3+1}\right)$ | $\psi({\Omega_3}^{\Omega_3+1})$ |
| $\psi\left({\Omega_3}^{\Omega_3\cdot 2}\right)$ | $\psi({\Omega_3}^{\Omega_3\cdot 2})$ |
| $\psi\left({\Omega_3}^{\Omega_3\cdot\omega}\right)$ | $\psi({\Omega_3}^{\Omega_3\cdot\omega})$ |
| $\psi\left({\Omega_3}^{{\Omega_3}^2}\right)$ | $\psi({\Omega_3}^{{\Omega_3}^2})$ |
| $\psi\left({\Omega_3}^{{\Omega_3}^{\omega}}\right)$ | $\psi({\Omega_3}^{{\Omega_3}^{\omega}})$ |
| $\psi\left({\Omega_3}^{{\Omega_3}^{\Omega}}\right)$ | $\psi({\Omega_3}^{{\Omega_3}^{\Omega}})$ |
| $\psi\left({\Omega_3}^{{\Omega_3}^{\Omega_2}}\right)$ | $\psi({\Omega_3}^{{\Omega_3}^{\Omega_2}})$ |
| $\psi\left({\Omega_3}^{{\Omega_3}^{\Omega_3}}\right)$ | $\psi({\Omega_3}^{{\Omega_3}^{\Omega_3}})$ |
| $\psi\left(\Omega_4\right)$ | $\psi(\psi_3(0))$ |
| $\psi\left(\Omega_4+\Omega\cdot\psi\left(\Omega_4\right)\right)$ | $\psi(\psi_3(0)+\psi(\psi_3(0)))$ |
| $\psi\left(\Omega_4+\Omega^2\right)$ | $\psi(\psi_3(0)+\psi(\psi_3(0)))$ |
| $\psi\left(\Omega_4+\Omega^2\right)$ | $\psi(\psi_3(0)+\Omega)$ |
| $\psi\left(\Omega_4+\Omega_2\cdot\psi_1\left(\Omega_4\right)\right)$ | $\psi(\psi_3(0)+\psi_1(\psi_3(0)))$ |
| $\psi\left(\Omega_4+{\Omega_2}^2\right)$ | $\psi(\psi_3(0)+\Omega_2)$ |
| $\psi\left(\Omega_4+\Omega_3\cdot\psi_2\left(\Omega_4\right)\right)$ | $\psi(\psi_3(0)+\psi_2(\psi_3(0)))$ |
| $\psi\left(\Omega_4\cdot 2\right)$ | $\psi(\psi_3(0)+\Omega_3)$ |
| $\psi\left(\Omega_4+{\Omega_3}^2\right)$ | $\psi(\psi_3(1))$ |
| $\psi\left(\Omega_4\cdot\psi_3\left(\Omega_4\right)\right)$ | $\psi(\psi_3(\psi_3(0)))$ |
| $\psi\left({\Omega_4}^2\right)$ | $\psi(\Omega_4)$ |
| $\psi\left({\Omega_4}^{\omega}\right)$ | $\psi({\Omega_4}^{\omega})$ |
| $\psi\left({\Omega_4}^{\Omega_4}\right)$ | $\psi({\Omega_4}^{\Omega_4})$ |
| $\psi\left(\Omega_5\right)$ | $\psi(\psi_4(0))$ |
| $\psi\left({\Omega_5}^2\right)$ | $\psi(\Omega_5)$ |
| $\psi\left({\Omega_5}^{\Omega_5}\right)$ | $\psi({\Omega_5}^{\Omega_5})$ |
| $\psi\left(\Omega_6\right)$ | $\psi(\psi_5(0))$ |
| $\psi\left(\Omega_7\right)$ | $\psi(\psi_6(0))$ |
| $\psi\left(\Omega_\omega\right)$ | $\psi\left(\Omega_\omega\right)$ |
| $\psi\left(\Omega_\omega+1\right)$ | $\psi\left(\Omega_\omega\right)\cdot\omega$ |
| $\psi\left(\Omega_\omega+\psi(\Omega)\right)$ | $\psi\left(\Omega_\omega\right)\cdot\psi(0)$ |
| $\psi\left(\Omega_\omega+\psi(\Omega_\omega)\right)$ | $\psi\left(\Omega_\omega\right)^2$ |
| $\psi\left(\Omega_\omega+\Omega\right)$ | $\psi\left(\Omega_\omega+1\right)$ |
| $\psi\left(\Omega_\omega+\Omega\cdot\psi(\Omega_\omega)\right)$ | $\psi\left(\Omega_\omega+\psi(\Omega_\omega)\right)$ |
| $\psi\left(\Omega_\omega+\Omega^2\right)$ | $\psi\left(\Omega_\omega+\Omega\right))$ |
| $\psi\left(\Omega_\omega+\psi_1(\Omega_2)\right)$ | $\psi\left(\Omega_\omega+\psi_1(0)\right)$ |
| $\psi\left(\Omega_\omega+\psi_1(\Omega_3)\right)$ | $\psi\left(\Omega_\omega+\psi_1(\psi_2(0))\right)$ |
| $\psi\left(\Omega_\omega+\psi_1(\Omega_\omega)\right)$ | $\psi\left(\Omega_\omega+\psi_1(\Omega_\omega)\right)$ |
| $\psi\left(\Omega_\omega+\Omega_2\right)$ | $\psi\left(\Omega_\omega+\psi_1(\Omega_\omega+1)\right)$ |
| $\psi\left(\Omega_\omega+\psi_2(\Omega_\omega)\right)$ | $\psi\left(\Omega_\omega+\psi_2(\Omega_\omega)\right)$ |

| Buchholz's OCF | Madore's OCF |
|---|---|
| $\psi\left(\Omega_\omega \cdot 2\right)$ | $\psi\left(\Omega_\omega \cdot 2\right)$ |
| $\psi\left(\Omega_\omega \cdot \omega\right)$ | $\psi\left(\Omega_\omega \cdot \omega\right)$ |
| $\psi\left(\Omega_\omega \cdot \psi(\Omega)\right)$ | $\psi\left(\Omega_\omega \cdot \psi(0)\right)$ |
| $\psi\left(\Omega_\omega \cdot \psi(\Omega_\omega)\right)$ | $\psi\left(\Omega_\omega \cdot \psi(\Omega_\omega)\right)$ |
| $\psi\left(\Omega_\omega \cdot \Omega\right)$ | $\psi\left(\Omega_\omega \cdot \Omega\right)$ |
| $\psi\left(\Omega_\omega \cdot \Omega_2\right)$ | $\psi\left(\Omega_\omega \cdot \Omega_2\right)$ |
| $\psi\left({\Omega_\omega}^2\right)$ | $\psi\left({\Omega_\omega}^2\right)$ |
| $\psi\left({\Omega_\omega}^3\right)$ | $\psi\left({\Omega_\omega}^3\right)$ |
| $\psi\left({\Omega_\omega}^\omega\right)$ | $\psi\left({\Omega_\omega}^\omega\right)$ |
| $\psi\left({\Omega_\omega}^\Omega\right)$ | $\psi\left({\Omega_\omega}^\Omega\right)$ |
| $\psi\left({\Omega_\omega}^{\Omega_2}\right)$ | $\psi\left({\Omega_\omega}^{\Omega_2}\right)$ |
| $\psi\left({\Omega_\omega}^{\Omega_\omega}\right)$ | $\psi\left({\Omega_\omega}^{\Omega_\omega}\right)$ |
| $\psi\left({\Omega_\omega}^{{\Omega_\omega}^{\Omega_\omega}}\right)$ | $\psi\left({\Omega_\omega}^{{\Omega_\omega}^{\Omega_\omega}}\right)$ |
| $\psi\left(\Omega_{\omega+1}\right)$ | $\psi\left(\psi_\omega(0)\right)$ |
| $\psi\left(\Omega_{\omega+1} + \Omega_\omega\right)$ | $\psi\left(\psi_\omega(0) + \Omega_\omega\right)$ |
| $\psi\left(\Omega_{\omega+1} + \psi_\omega(\Omega_{\omega+1})\right)$ | $\psi\left(\psi_\omega(0) \cdot 2\right)$ |
| $\psi\left(\Omega_{\omega+1} + \psi_\omega(\Omega_{\omega+1} + 1)\right)$ | $\psi\left(\psi_\omega(0) \cdot \omega\right)$ |
| $\psi\left(\Omega_{\omega+1} + \psi_\omega(\Omega_{\omega+1} \cdot 2)\right)$ | $\psi\left(\psi_\omega(0)^2\right)$ |
| $\psi\left(\Omega_{\omega+1} \cdot 2\right)$ | $\psi\left(\psi_\omega(1)\right)$ |
| $\psi\left(\Omega_{\omega+1} \cdot \omega\right)$ | $\psi\left(\psi_\omega(\omega)\right)$ |
| $\psi\left(\Omega_{\omega+1} \cdot \Omega_\omega\right)$ | $\psi\left(\psi_\omega(\Omega_\omega)\right)$ |
| $\psi\left({\Omega_{\omega+1}}^2\right)$ | $\psi\left(\Omega_{\omega+1}\right)$ |
| $\psi\left({\Omega_{\omega+1}}^\omega\right)$ | $\psi\left({\Omega_{\omega+1}}^\omega\right)$ |
| $\psi\left({\Omega_{\omega+1}}^{\Omega_\omega}\right)$ | $\psi\left({\Omega_{\omega+1}}^{\Omega_\omega}\right)$ |
| $\psi\left({\Omega_{\omega+1}}^{\Omega_{\omega+1}}\right)$ | $\psi\left({\Omega_{\omega+1}}^{\Omega_{\omega+1}}\right)$ |
| $\psi\left(\Omega_{\omega+2}\right)$ | $\psi\left(\psi_{\omega+1}(0)\right)$ |
| $\psi\left({\Omega_{\omega+2}}^2\right)$ | $\psi\left(\Omega_{\omega+2}\right)$ |
| $\psi\left({\Omega_{\omega+2}}^\omega\right)$ | $\psi\left({\Omega_{\omega+2}}^\omega\right)$ |
| $\psi\left(\Omega_{\omega+3}\right)$ | $\psi\left(\psi_{\omega+2}(0)\right)$ |
| $\psi\left(\Omega_{\omega\cdot 2}\right)$ | $\psi\left(\Omega_{\omega\cdot 2}\right)$ |
| $\psi\left(\Omega_{\omega\cdot 3}\right)$ | $\psi\left(\Omega_{\omega\cdot 3}\right)$ |
| $\psi\left(\Omega_{\omega^2}\right)$ | $\psi\left(\Omega_{\omega^2}\right)$ |
| $\psi\left(\Omega_{\omega^\omega}\right)$ | $\psi\left(\Omega_{\omega^\omega}\right)$ |
| $\psi\left(\Omega_{\psi(\Omega)}\right)$ | $\psi\left(\Omega_{\psi(0)}\right)$ |
| $\psi\left(\Omega_{\psi(\Omega^\omega)}\right)$ | $\psi\left(\Omega_{\psi(\Omega^\omega)}\right)$ |
| $\psi\left(\Omega_{\psi(\Omega_2)}\right)$ | $\psi\left(\Omega_{\psi(\psi_1(0))}\right)$ |
| $\psi\left(\Omega_{\psi(\Omega_\omega)}\right)$ | $\psi\left(\Omega_{\psi(\Omega_\omega)}\right)$ |
| $\psi\left(\Omega_{\psi(\Omega_{\psi(\Omega_\omega)})}\right)$ | $\psi\left(\Omega_{\psi(\Omega_{\psi(\Omega_\omega)})}\right)$ |
| $\psi\left(\Omega_\Omega\right)$ | $\psi\left(\Omega_\Omega\right)$ |
| $\psi\left(\Omega_{\Omega+1}\right)$ | $\psi\left(\psi_\Omega(0)\right)$ |
| $\psi\left(\Omega_{\Omega+\omega}\right)$ | $\psi\left(\Omega_{\Omega+\omega}\right)$ |
| $\psi\left(\Omega_{\Omega\cdot 2}\right)$ | $\psi\left(\Omega_{\Omega\cdot 2}\right)$ |
| $\psi\left(\Omega_{\Omega^2}\right)$ | $\psi\left(\Omega_{\Omega^2}\right)$ |
| $\psi\left(\Omega_{\Omega^\Omega}\right)$ | $\psi\left(\Omega_{\Omega^\Omega}\right)$ |
| $\psi\left(\Omega_{\psi_1(\Omega_2)}\right)$ | $\psi\left(\Omega_{\psi_1(0)}\right)$ |
| $\psi\left(\Omega_{\psi_1(\Omega_\omega)}\right)$ | $\psi\left(\Omega_{\psi_1(\Omega_\omega)}\right)$ |
| $\psi\left(\Omega_{\psi_1(\Omega_\Omega)}\right)$ | $\psi\left(\Omega_{\psi_1(\Omega_\Omega)}\right)$ |
| $\psi\left(\Omega_{\psi_1(\Omega_{\psi_1(\Omega_\omega)})}\right)$ | $\psi\left(\Omega_{\psi_1(\Omega_{\psi_1(\Omega_\omega)})}\right)$ |
| $\psi\left(\Omega_{\Omega_2}\right)$ | $\psi\left(\Omega_{\Omega_2}\right)$ |
| $\psi\left(\Omega_{\Omega_\omega}\right)$ | $\psi\left(\Omega_{\Omega_\omega}\right)$ |
| $\psi\left(\Omega_{\Omega_\Omega}\right)$ | $\psi\left(\Omega_{\Omega_\Omega}\right)$ |
| $\psi\left(I\right)$ | $\psi\left(\psi_I(0)\right)$ |

## A.7 BMS vs Cantor-style/Veblen Function

The results in this section are primarily cited from [6-7].

| BMS | Cantor-style/Veblen function |
|---|---|
| (0) | $1$ |
| (0)(0) | $2$ |
| (0)(0)(0) | $3$ |
| (0)(1) | $\omega$ |
| (0)(1)(0) | $\omega+1$ |
| (0)(1)(0)(1) | $\omega\cdot 2$ |
| (0)(1)(0)(1)(0)(1) | $\omega\cdot 3$ |
| (0)(1)(1) | $\omega^2$ |
| (0)(1)(1)(0)(1) | $\omega^2+\omega$ |
| (0)(1)(1)(0)(1)(1) | $\omega^2\cdot 2$ |
| (0)(1)(1)(1) | $\omega^3$ |
| (0)(1)(1)(1)(1) | $\omega^4$ |
| (0)(1)(2) | $\omega^\omega$ |
| (0)(1)(2)(1) | $\omega^{\omega+1}$ |
| (0)(1)(2)(1)(2) | $\omega^{\omega\cdot 2}$ |
| (0)(1)(2)(2) | $\omega^{\omega^2}$ |
| (0)(1)(2)(2)(2) | $\omega^{\omega^3}$ |
| (0)(1)(2)(3) | $\omega^{\omega^\omega}$ |
| (0)(1)(2)(3)(1) | $\omega^{\omega^\omega+1}$ |
| (0)(1)(2)(3)(2) | $\omega^{\omega^{\omega+1}}$ |
| (0)(1)(2)(3)(3) | $\omega^{\omega^{\omega^2}}$ |
| (0)(1)(2)(3)(4) | $\omega^{\omega^{\omega^\omega}}$ |
| (0,0)(1,1) | $\varepsilon_0$ |
| (0,0)(1,1)(0,0) | $\varepsilon_0+1$ |
| (0,0)(1,1)(0,0)(1,1) | $\varepsilon_0\cdot 2$ |
| (0,0)(1,1)(0,0)(1,1)(0,0)(1,1) | $\varepsilon_0\cdot 3$ |
| (0,0)(1,1)(1,0) | $\varepsilon_0\cdot\omega$ |
| (0,0)(1,1)(1,0)(1,0) | $\varepsilon_0\cdot\omega^2$ |
| (0,0)(1,1)(1,0)(2,0) | $\varepsilon_0\cdot\omega^\omega$ |
| (0,0)(1,1)(1,0)(2,0)(1,0) | $\varepsilon_0\cdot\omega^{\omega+1}$ |
| (0,0)(1,1)(1,0)(2,0)(2,0) | $\varepsilon_0\cdot\omega^{\omega^2}$ |
| (0,0)(1,1)(1,0)(2,0)(3,0) | $\varepsilon_0\cdot\omega^{\omega^\omega}$ |
| (0,0)(1,1)(1,0)(2,1) | ${\varepsilon_0}^2$ |
| (0,0)(1,1)(1,0)(2,1)(1,0) | ${\varepsilon_0}^2\cdot\omega$ |
| (0,0)(1,1)(1,0)(2,1)(1,0)(2,1) | ${\varepsilon_0}^3$ |
| (0,0)(1,1)(1,0)(2,1)(2,0) | ${\varepsilon_0}^\omega$ |
| (0,0)(1,1)(1,0)(2,1)<br>(2,0)(1,0)(2,1) | ${\varepsilon_0}^{\omega+1}$ |
| (0,0)(1,1)(1,0)(2,1)(2,0)(2,0) | ${\varepsilon_0}^{\omega^2}$ |
| (0,0)(1,1)(1,0)(2,1)(2,0)(3,0) | ${\varepsilon_0}^{\omega^\omega}$ |
| (0,0)(1,1)(1,0)(2,1)(2,0)(3,1) | ${\varepsilon_0}^{\varepsilon_0}$ |
| (0,0)(1,1)(1,0)(2,1)<br>(2,0)(3,1)(1,0)(2,1) | ${\varepsilon_0}^{\varepsilon_0+1}$ |
| (0,0)(1,1)(1,0)(2,1)<br>(2,0)(3,1)(2,0) | ${\varepsilon_0}^{\varepsilon_0\cdot\omega}$ |
| (0,0)(1,1)(1,0)(2,1)<br>(2,0)(3,1)(2,0)(3,1) | ${\varepsilon_0}^{{\varepsilon_0}^2}$ |
| (0,0)(1,1)(1,0)(2,1)<br>(2,0)(3,1)(3,0) | ${\varepsilon_0}^{{\varepsilon_0}^\omega}$ |

| BMS | Cantor-style/Veblen function |
|---|---|
| (0,0)(1,1)(1,0)(2,1)<br>(2,0)(3,1)(3,0)(4,1) | ${\varepsilon_0}^{{\varepsilon_0}^{\varepsilon_0}}$ |
| (0,0)(1,1)(1,1) | $\varepsilon_1$ |
| (0,0)(1,1)(1,1)(1,1) | $\varepsilon_2$ |
| (0,0)(1,1)(2,0) | $\varepsilon_\omega$ |
| (0,0)(1,1)(2,0)(1,1) | $\varepsilon_{\omega+1}$ |
| (0,0)(1,1)(2,0)(1,1)(2,0) | $\varepsilon_{\omega\cdot 2}$ |
| (0,0)(1,1)(2,0)(2,0) | $\varepsilon_{\omega^2}$ |
| (0,0)(1,1)(2,0)(2,0)(2,0) | $\varepsilon_{\omega^3}$ |
| (0,0)(1,1)(2,0)(3,0) | $\varepsilon_{\omega^\omega}$ |
| (0,0)(1,1)(2,0)(3,1) | $\varepsilon_{\varepsilon_0}$ |
| (0,0)(1,1)(2,0)(3,1)(1,1) | $\varepsilon_{\varepsilon_0+1}$ |
| (0,0)(1,1)(2,0)(3,1)(2,0) | $\varepsilon_{\varepsilon_0\cdot\omega}$ |
| (0,0)(1,1)(2,0)(3,1)(3,1) | $\varepsilon_{\varepsilon_1}$ |
| (0,0)(1,1)(2,0)(3,1)(4,0) | $\varepsilon_{\varepsilon_\omega}$ |
| (0,0)(1,1)(2,0)(3,1)(4,0)(5,1) | $\varepsilon_{\varepsilon_{\varepsilon_0}}$ |
| (0,0)(1,1)(2,1) | $\zeta_0$ |
| (0,0)(1,1)(2,1)(1,1) | $\varepsilon_{\zeta_0+1}$ |
| (0,0)(1,1)(2,1)(1,1)(2,0) | $\varepsilon_{\zeta_0+\omega}$ |
| (0,0)(1,1)(2,1)(1,1)(2,0)(3,1) | $\varepsilon_{\zeta_0+\varepsilon_0}$ |
| (0,0)(1,1)(2,1)(1,1)<br>(2,0)(3,1)(2,0) | $\varepsilon_{\zeta_0+\varepsilon_0\cdot\omega}$ |
| (0,0)(1,1)(2,1)(1,1)<br>(2,0)(3,1)(3,1) | $\varepsilon_{\zeta_0+\varepsilon_1}$ |
| (0,0)(1,1)(2,1)(1,1)<br>(2,0)(3,1)(4,1) | $\varepsilon_{\zeta_0\cdot 2}$ |
| (0,0)(1,1)(2,1)(1,1)<br>(2,0)(3,1)(4,1)(3,0) | $\varepsilon_{\zeta_0\cdot\omega}$ |
| (0,0)(1,1)(2,1)(1,1)<br>(2,0)(3,1)(4,1)(3,1) | $\varepsilon_{\varepsilon_{\zeta_0+1}}$ |
| (0,0)(1,1)(2,1)(1,1)(2,1) | $\zeta_1$ |
| (0,0)(1,1)(2,1)(1,1)<br>(2,1)(1,1)(2,1) | $\zeta_2$ |
| (0,0)(1,1)(2,1)(2,0) | $\zeta_\omega$ |
| (0,0)(1,1)(2,1)(2,0)(3,0) | $\zeta_{\omega^\omega}$ |
| (0,0)(1,1)(2,1)(2,0)(3,1) | $\zeta_{\varepsilon_0}$ |
| (0,0)(1,1)(2,1)(2,0)(3,1)(4,1) | $\zeta_{\zeta_0}$ |
| (0,0)(1,1)(2,1)(2,1) | $\eta_0$ |
| (0,0)(1,1)(2,1)(2,1)(1,1) | $\varepsilon_{\eta_0+1}$ |
| (0,0)(1,1)(2,1)(2,1)(1,1)(2,1) | $\zeta_{\eta_0+1}$ |
| (0,0)(1,1)(2,1)(2,1)<br>(1,1)(2,1)(2,1) | $\eta_1$ |
| (0,0)(1,1)(2,1)(2,1)(2,0) | $\eta_\omega$ |
| (0,0)(1,1)(2,1)(2,1)<br>(2,0)(3,1)(4,1)(4,1) | $\eta_{\eta_0}$ |
| (0,0)(1,1)(2,1)(2,1)(2,1) | $\varphi(4,0)$ |
| (0,0)(1,1)(2,1)(3,0) | $\varphi(\omega,0)$ |
| (0,0)(1,1)(2,1)(3,0)(1,1) | $\varphi(1,\varphi(\omega,0)+1)$ |
| (0,0)(1,1)(2,1)(3,0)(1,1)(2,1) | $\varphi(2,\varphi(\omega,0)+1)$ |
| (0,0)(1,1)(2,1)(3,0)<br>(1,1)(2,1)(3,0) | $\varphi(\omega,1)$ |
| (0,0)(1,1)(2,1)(3,0)(2,0) | $\varphi(\omega,\omega)$ |
| (0,0)(1,1)(2,1)(3,0)(2,0)(3,1) | $\varphi(\omega,\varphi(1,0))$ |
| (0,0)(1,1)(2,1)(3,0)<br>(2,0)(3,1)(4,1)(5,0) | $\varphi(\omega,\varphi(\omega,0))$ |

| BMS | Cantor-style/Veblen function |
|---|---|
| (0,0)(1,1)(2,1)(3,0)(2,1) | $\varphi(\omega+1,0)$ |
| (0,0)(1,1)(2,1)(3,0)(2,1)(3,0) | $\varphi(\omega\cdot 2,0)$ |
| (0,0)(1,1)(2,1)(3,0)(3,0) | $\varphi(\omega^2,0)$ |
| (0,0)(1,1)(2,1)(3,0)(4,1) | $\varphi(\varphi(1,0),0)$ |
| (0,0)(1,1)(2,1)(3,0)<br>(4,1)(5,1)(6,0) | $\varphi(\varphi(\omega,0),0)$ |
| (0,0)(1,1)(2,1)(3,1) | $\varphi(1,0,0)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1) | $\varphi(1,\varphi(1,0,0)+1)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1) | $\varphi(2,\varphi(1,0,0)+1)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(3,0) | $\varphi(\omega,\varphi(1,0,0)+1)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(3,0)(4,1) | $\varphi(\varphi(1,0),\varphi(1,0,0)+1)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1) | $\varphi(\varphi(2,0),\varphi(1,0,0)+1)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1) | $\varphi(\varphi(1,0,0),1)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(2,0) | $\varphi(\varphi(1,0,0),\omega)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(2,1) | $\varphi(\varphi(1,0,0)+1,0)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(2,1)(3,0) | $\varphi(\varphi(1,0,0)+\omega,0)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1) | $\varphi(\varphi(1,0,0)\cdot 2,0)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(3,0) | $\varphi(\varphi(1,0,0)\cdot\omega,0)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(4,0) | $\varphi(\varphi(1,0,0)^\omega,0)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(4,1) | $\varphi(\varphi(1,\varphi(1,0,0)+1),0)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)<br>(6,1)(4,1)(5,1)(6,0) | $\varphi(\varphi(\omega,\varphi(1,0,0)+1),0)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(3,1) | $\varphi(1,0,1)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,1)(1,1)(2,1)(3,1) | $\varphi(1,0,2)$ |
| (0,0)(1,1)(2,1)(3,1)(2,0) | $\varphi(1,0,\omega)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(2,0)(3,1)(4,1)(5,1) | $\varphi(1,0,\varphi(1,0,0))$ |
| (0,0)(1,1)(2,1)(3,1)(2,1) | $\varphi(1,1,0)$ |
| (0,0)(1,1)(2,1)(3,1)(2,1)(3,0) | $\varphi(1,\omega,0)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1) | $\varphi(1,\varphi(1,0,0),0)$ |
| (0,0)(1,1)(2,1)(3,1)(2,1)(3,1) | $\varphi(2,0,0)$ |
| (0,0)(1,1)(2,1)(3,1)(2,1)<br>(3,1)(1,1)(2,1)(3,1)(2,1)(3,1) | $\varphi(2,0,1)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(2,1)(3,1)(2,1) | $\varphi(2,1,0)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(2,1)(3,1)(2,1)(3,1) | $\varphi(3,0,0)$ |
| (0,0)(1,1)(2,1)(3,1)(3,0) | $\varphi(\omega,0,0)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(3,0)(4,1)(5,1)(6,1) | $\varphi(\varphi(1,0,0),0,0)$ |

| BMS | Cantor-style/Veblen function |
|---|---|
| (0,0)(1,1)(2,1)(3,1)(3,1) | $\varphi(1,0,0,0)$ |
| (0,0)(1,1)(2,1)(3,1)(3,1)(2,1) | $\varphi(1,0,1,0)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(3,1)(2,1)(3,1) | $\varphi(1,1,0,0)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(3,1)(2,1)(3,1)(3,1) | $\varphi(2,0,0,0)$ |
| (0,0)(1,1)(2,1)(3,1)(3,1)(3,1) | $\varphi(1,0,0,0,0)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(3,1)(3,1)(3,1) | $\varphi(1,0,0,0,0,0)$ |
| (0,0)(1,1)(2,1)(3,1)(4,0) | $\varphi(1@\omega)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,0)(1,1)(2,1)(3,1)(4,0) | $\varphi(1@\omega,1@0)$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(2,1) | $\varphi(1@\omega,1@1)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,0)(2,1)(3,1) | $\varphi(1@\omega,1@2)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,0)(2,1)(3,1)(4,0) | $\varphi(2@\omega)$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(3,0) | $\varphi(\omega@\omega)$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(3,1) | $\varphi(1@\omega+1)$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(4,0) | $\varphi(1@\omega^2)$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(5,1) | $\varphi(1@\varphi(1,0))$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,0)(5,1)(6,1)(7,1) | $\varphi(1@\varphi(1,0,0))$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,0)(5,1)(6,1)(7,1)(8,0) | $\varphi(1@\varphi(1@\omega))$ |
| (0,0)(1,1)(2,1)(3,1)(4,1) | $\varphi(1@(1,0))$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)<br>(1,1)(2,1)(3,1)(4,1) | $\varphi(1@(1,0),1)$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(2,1) | $\varphi(1@(1,0),1@1)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,1)(2,1)(3,1)(4,1) | $\varphi(2@(1,0))$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(3,1) | $\varphi(1@(1,1))$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,1)(3,1)(4,1) | $\varphi(1@(2,0))$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(4,0) | $\varphi(1@(\omega,0))$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(4,1) | $\varphi(1@(1,0,0))$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(5,0) | $\varphi(1@(1@\omega))$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(5,1) | $\varphi(1@(1@(1,0)))$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,1)(5,1)(6,1) | $\varphi(1@(1@(1@(1,0))))$ |
| (0,0)(1,1)(2,2) | $\varphi(1@(1,,0))$ |

## A.8 BMS vs MOCF

The results in this section are primarily cited from [6,8-9] 。

| BMS | Madore's OCF |
|---|---|
| (0,0)(1,1) | $\psi(0)$ |
| (0,0)(1,1)(0,0) | $\psi(0)+1$ |
| (0,0)(1,1)(0,0)(1,1) | $\psi(0)\cdot 2$ |
| (0,0)(1,1)(0,0)(1,1)(0,0)(1,1) | $\psi(0)\cdot 3$ |
| (0,0)(1,1)(1,0) | $\psi(0)\cdot\omega$ |
| (0,0)(1,1)(1,0)(1,0) | $\psi(0)\cdot\omega^2$ |
| (0,0)(1,1)(1,0)(2,0) | $\psi(0)\cdot\omega^\omega$ |
| (0,0)(1,1)(1,0)(2,0)(1,0) | $\psi(0)\cdot\omega^{\omega+1}$ |

| BMS | Madore's OCF |
|---|---|
| (0,0)(1,1)(1,0)(2,0)(2,0) | $\psi(0)\cdot\omega^{\omega^2}$ |
| (0,0)(1,1)(1,0)(2,0)(3,0) | $\psi(0)\cdot\omega^{\omega^\omega}$ |
| (0,0)(1,1)(1,0)(2,1) | $\psi(0)^2$ |
| (0,0)(1,1)(1,0)(2,1)(1,0) | $\psi(0)^2\cdot\omega$ |
| (0,0)(1,1)(1,0)(2,1)(1,0)(2,1) | $\psi(0)^3$ |
| (0,0)(1,1)(1,0)(2,1)(2,0) | $\psi(0)^\omega$ |
| (0,0)(1,1)(1,0)(2,1)<br>(2,0)(1,0)(2,1) | $\psi(0)^{\omega+1}$ |
| (0,0)(1,1)(1,0)(2,1)(2,0)(2,0) | $\psi(0)^{\omega^2}$ |
| (0,0)(1,1)(1,0)(2,1)(2,0)(3,0) | $\psi(0)^{\omega^\omega}$ |
| (0,0)(1,1)(1,0)(2,1)(2,0)(3,1) | $\psi(0)^{\psi(0)}$ |
| (0,0)(1,1)(1,0)(2,1)<br>(2,0)(3,1)(1,0)(2,1) | $\psi(0)^{\psi(0)+1}$ |
| (0,0)(1,1)(1,0)(2,1)<br>(2,0)(3,1)(2,0) | $\psi(0)^{\psi(0)+\omega}$ |
| (0,0)(1,1)(1,0)(2,1)<br>(2,0)(3,1)(2,0)(3,1) | $\psi(0)^{\psi(0)^2}$ |
| (0,0)(1,1)(1,0)(2,1)<br>(2,0)(3,1)(3,0) | $\psi(0)^{\psi(0)^\omega}$ |
| (0,0)(1,1)(1,0)(2,1)<br>(2,0)(3,1)(3,0)(4,1) | $\psi(0)^{\psi(0)^{\psi(0)}}$ |
| (0,0)(1,1)(1,1) | $\psi(1)$ |
| (0,0)(1,1)(1,1)(1,1) | $\psi(2)$ |
| (0,0)(1,1)(2,0) | $\psi(\omega)$ |
| (0,0)(1,1)(2,0)(1,1) | $\psi(\omega+1)$ |
| (0,0)(1,1)(2,0)(1,1)(2,0) | $\psi(\omega\cdot 2)$ |
| (0,0)(1,1)(2,0)(2,0) | $\psi(\omega^2)$ |
| (0,0)(1,1)(2,0)(2,0)(2,0) | $\psi(\omega^3)$ |
| (0,0)(1,1)(2,0)(3,0) | $\psi(\omega^\omega)$ |
| (0,0)(1,1)(2,0)(3,1) | $\psi(\psi(0))$ |
| (0,0)(1,1)(2,0)(3,1)(1,1) | $\psi(\psi(0)+1)$ |
| (0,0)(1,1)(2,0)(3,1)(2,0) | $\psi(\psi(0)\cdot\omega)$ |
| (0,0)(1,1)(2,0)(3,1)(3,1) | $\psi(\psi(1))$ |
| (0,0)(1,1)(2,0)(3,1)(4,0) | $\psi(\psi(\omega))$ |
| (0,0)(1,1)(2,0)(3,1)(4,0)(5,1) | $\psi(\psi(\psi(0)))$ |
| (0,0)(1,1)(2,1) | $\psi(\Omega)$ |
| (0,0)(1,1)(2,1)(1,1) | $\psi(\Omega+1)$ |
| (0,0)(1,1)(2,1)(1,1)(2,0) | $\psi(\Omega+\omega)$ |
| (0,0)(1,1)(2,1)(1,1)(2,0)(3,1) | $\psi(\Omega+\psi(0))$ |
| (0,0)(1,1)(2,1)(1,1)<br>(2,0)(3,1)(2,0) | $\psi(\psi(\Omega+\psi(1)))$ |
| (0,0)(1,1)(2,1)(1,1)<br>(2,0)(3,1)(3,1) | $\psi(\Omega+\psi(2))$ |
| (0,0)(1,1)(2,1)(1,1)<br>(2,0)(3,1)(4,1) | $\psi(\Omega+\psi(\Omega))$ |
| (0,0)(1,1)(2,1)(1,1)<br>(2,0)(3,1)(4,1)(3,0) | $\psi(\Omega+\psi(\Omega)\cdot\omega)$ |
| (0,0)(1,1)(2,1)(1,1)<br>(2,0)(3,1)(4,1)(3,1) | $\psi(\Omega+\psi(\Omega)^2)$ |
| (0,0)(1,1)(2,1)(1,1)(2,1) | $\psi(\Omega\cdot 2)$ |
| (0,0)(1,1)(2,1)(1,1)<br>(2,1)(1,1)(2,1) | $\psi(\Omega\cdot 3)$ |
| (0,0)(1,1)(2,1)(2,0) | $\psi(\Omega\cdot\omega)$ |
| (0,0)(1,1)(2,1)(2,0)(3,0) | $\psi(\Omega^2\cdot\psi(2))$ |
| (0,0)(1,1)(2,1)(2,0)(3,1) | $\psi(\Omega\cdot\psi(0))$ |

| BMS | Madore's OCF |
|---|---|
| (0,0)(1,1)(2,1)(2,0)(3,1)(4,1) | $\psi(\Omega \cdot \psi(\Omega))$ |
| (0,0)(1,1)(2,1)(2,1) | $\psi(\Omega^2)$ |
| (0,0)(1,1)(2,1)(2,1)(1,1) | $\psi(\Omega^2 + 1)$ |
| (0,0)(1,1)(2,1)(2,1)(1,1)(2,1) | $\psi(\Omega^2 + \Omega)$ |
| (0,0)(1,1)(2,1)(2,1)<br>(1,1)(2,1)(2,1) | $\psi(\Omega^2 \cdot 2)$ |
| (0,0)(1,1)(2,1)(2,1)(2,0) | $\psi(\Omega^2 \cdot \omega)$ |
| (0,0)(1,1)(2,1)(2,1)<br>(2,0)(3,1)(4,1)(4,1) | $\psi(\Omega^2 \cdot \psi(\Omega^2))$ |
| (0,0)(1,1)(2,1)(2,1)(2,1) | $\psi(\Omega^3)$ |
| (0,0)(1,1)(2,1)(3,0) | $\psi(\Omega^\omega)$ |
| (0,0)(1,1)(2,1)(3,0)(1,1) | $\psi(\Omega^\omega + 1)$ |
| (0,0)(1,1)(2,1)(3,0)(1,1)(2,1) | $\psi(\Omega^\omega + \Omega)$ |
| (0,0)(1,1)(2,1)(3,0)<br>(1,1)(2,1)(3,0) | $\psi(\Omega^\omega \cdot 2)$ |
| (0,0)(1,1)(2,1)(3,0)(2,0) | $\psi(\Omega^\omega \cdot \omega)$ |
| (0,0)(1,1)(2,1)(3,0)(2,0)(3,1) | $\psi(\Omega^\omega \cdot \psi(1))$ |
| (0,0)(1,1)(2,1)(3,0)<br>(2,0)(3,1)(4,1)(5,0) | $\psi(\Omega^\omega \cdot \psi(\Omega^\omega))$ |
| (0,0)(1,1)(2,1)(3,0)(2,1) | $\psi(\Omega^{\omega+1})$ |
| (0,0)(1,1)(2,1)(3,0)(2,1)(3,0) | $\psi(\Omega^{\omega \cdot 2})$ |
| (0,0)(1,1)(2,1)(3,0)(3,0) | $\psi(\Omega^{\omega^2})$ |
| (0,0)(1,1)(2,1)(3,0)(4,1) | $\psi(\Omega^{\psi(0)})$ |
| (0,0)(1,1)(2,1)(3,0)<br>(4,1)(5,1)(6,0) | $\psi(\Omega^{\psi(\Omega^\omega)})$ |
| (0,0)(1,1)(2,1)(3,1) | $\psi(\Omega^\Omega)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1) | $\psi(\Omega^\Omega + 1)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1) | $\psi(\Omega^\Omega + \Omega)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(3,0) | $\psi(\Omega^\Omega + \Omega^\omega)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(3,0)(4,1) | $\psi(\Omega^\Omega + \Omega^{\psi(1)})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega)})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(2,0) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)} \cdot \omega)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(2,1) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)+1})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(2,1)(3,0) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)+\omega})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega) \cdot 2})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(3,0) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega) \cdot \omega})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(4,0) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)^\omega})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(4,1) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+1)})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)<br>(6,1)(4,1)(5,1)(6,0) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+\Omega \cdot \omega)})$ |

| BMS | Madore's OCF |
|---|---|
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(3,1) | $\psi(\Omega^{\Omega} \cdot 2)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,1)(1,1)(2,1)(3,1) | $\psi(\Omega^{\Omega} \cdot 3)$ |
| (0,0)(1,1)(2,1)(3,1)(2,0) | $\psi(\Omega^{\Omega} \cdot \omega)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(2,0)(3,1)(4,1)(5,1) | $\psi(\Omega^{\Omega} \cdot \psi(\Omega^{\Omega}))$ |
| (0,0)(1,1)(2,1)(3,1)(2,1) | $\psi(\Omega^{\Omega+1})$ |
| (0,0)(1,1)(2,1)(3,1)(2,1)(3,0) | $\psi(\Omega^{\Omega+\omega})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1) | $\psi(\Omega^{\Omega+\psi(\Omega^{\Omega})})$ |
| (0,0)(1,1)(2,1)(3,1)(2,1)(3,1) | $\psi(\Omega^{\Omega\cdot 2})$ |
| (0,0)(1,1)(2,1)(3,1)(2,1)<br>(3,1)(1,1)(2,1)(3,1)(2,1)(3,1) | $\psi(\Omega^{\Omega\cdot 2} \cdot 2)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(2,1)(3,1)(2,1) | $\psi(\Omega^{\Omega\cdot 2+1})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(2,1)(3,1)(2,1)(3,1) | $\psi(\Omega^{\Omega\cdot 3})$ |
| (0,0)(1,1)(2,1)(3,1)(3,0) | $\psi(\Omega^{\Omega\cdot\omega})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(3,0)(4,1)(5,1)(6,1) | $\psi(\Omega^{\Omega\cdot\psi(\Omega^{\Omega})})$ |
| (0,0)(1,1)(2,1)(3,1)(3,1) | $\psi(\Omega^{\Omega^2})$ |
| (0,0)(1,1)(2,1)(3,1)(3,1)(2,1) | $\psi(\Omega^{\Omega^2+1})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(3,1)(2,1)(3,1) | $\psi(\Omega^{\Omega^2+\Omega})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(3,1)(2,1)(3,1)(3,1) | $\psi(\Omega^{\Omega^2\cdot 2})$ |
| (0,0)(1,1)(2,1)(3,1)(3,1)(3,1) | $\psi(\Omega^{\Omega^3})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(3,1)(3,1)(3,1) | $\psi(\Omega^{\Omega^4})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0) | $\psi(\Omega^{\Omega^{\omega}})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,0)(1,1)(2,1)(3,1)(4,0) | $\psi(\Omega^{\Omega^{\omega}} \cdot 2)$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(2,1) | $\psi(\Omega^{\Omega^{\omega}+1})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,0)(2,1)(3,1) | $\psi(\Omega^{\Omega^{\omega}+\Omega})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,0)(2,1)(3,1)(4,0) | $\psi(\Omega^{\Omega^{\omega}\cdot 2})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(3,0) | $\psi(\Omega^{\Omega^{\omega}\cdot\omega})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(3,1) | $\psi(\Omega^{\Omega^{\omega+1}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(4,0) | $\psi(\Omega^{\Omega^{\omega^2}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(5,1) | $\psi(\Omega^{\Omega^{\psi(1)}})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,0)(5,1)(6,1)(7,1) | $\psi(\Omega^{\Omega^{\psi(\Omega^{\Omega})}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1) | $\psi(\Omega^{\Omega^{\Omega}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(1,1) | $\psi(\Omega^{\Omega^{\Omega}} + 1)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,1)(1,1)(1,1) | $\psi(\Omega^{\Omega^{\Omega}} + 2)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,1)(1,1)(2,0) | $\psi(\Omega^{\Omega^{\Omega}} + \omega)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,1)(1,1)(2,0)(3,1) | $\psi(\Omega^{\Omega^{\Omega}} + \psi(0))$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)<br>(1,1)(2,0)(3,1)(4,1)(5,1)(6,1) | $\psi(\Omega^{\Omega^{\Omega}} + \psi(\Omega^{\Omega^{\Omega}}))$ |

| BMS | Madore's OCF |
|---|---|
| (0,0)(1,1)(2,1)(3,1)(4,1)(1,1)(2,1) | $\psi(\Omega^{\Omega^{\Omega}}+\Omega)$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)<br>(1,1)(2,1)(3,1)(4,1) | $\psi(\Omega^{\Omega^{\Omega}}\cdot 2)$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(2,0) | $\psi(\Omega^{\Omega^{\Omega}}\cdot\omega)$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(2,1) | $\psi(\Omega^{\Omega^{\Omega}+1})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,1)(2,1)(3,1) | $\psi(\Omega^{\Omega^{\Omega}+\Omega})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,1)(2,1)(3,1)(4,1) | $\psi(\Omega^{\Omega^{\Omega}\cdot 2})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(3,0) | $\psi(\Omega^{\Omega^{\Omega}\cdot\omega})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(3,1) | $\psi(\Omega^{\Omega^{\Omega+1}})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,1)(3,1)(4,1) | $\psi(\Omega^{\Omega^{\Omega\cdot 2}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(4,0) | $\psi(\Omega^{\Omega^{\Omega\cdot\omega}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(4,1) | $\psi(\Omega^{\Omega^{\Omega^2}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(5,0) | $\psi(\Omega^{\Omega^{\Omega^{\omega}}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(5,1) | $\psi(\Omega^{\Omega^{\Omega^{\Omega}}})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,1)(5,1)(6,1) | $\psi(\Omega^{\Omega^{\Omega^{\Omega^{\Omega}}}})$ |
| (0,0)(1,1)(2,2) | $\psi(\psi_1(0))$ |
| (0,0)(1,1)(2,2)(1,1) | $\psi(\psi_1(0)+1)$ |
| (0,0)(1,1)(2,2)(1,1)(2,2) | $\psi(\psi_1(0)\cdot 2)$ |
| (0,0)(1,1)(2,2)(2,0) | $\psi(\psi_1(0)\cdot\omega)$ |
| (0,0)(1,1)(2,2)(2,1) | $\psi(\psi_1(0)\cdot\Omega)$ |
| (0,0)(1,1)(2,2)(2,1)(3,1) | $\psi(\psi_1(0)\cdot\Omega^{\Omega})$ |
| (0,0)(1,1)(2,2)(2,1)(3,2) | $\psi(\psi_1(0)^2)$ |
| (0,0)(1,1)(2,2)(2,1)(3,2)(3,1) | $\psi(\psi_1(0)^{\Omega})$ |
| (0,0)(1,1)(2,2)(2,1)<br>(3,2)(3,1)(4,2) | $\psi(\psi_1(0)^{\psi_1(0)})$ |
| (0,0)(1,1)(2,2)(2,2) | $\psi(\psi_1(1))$ |
| (0,0)(1,1)(2,2)(3,0) | $\psi(\psi_1(\omega))$ |
| (0,0)(1,1)(2,2)(3,0)(4,1) | $\psi(\psi_1(\psi(0)))$ |
| (0,0)(1,1)(2,2)(3,0)(4,1)(5,2) | $\psi(\psi_1(\psi(\psi_1(0))))$ |
| (0,0)(1,1)(2,2)(3,1) | $\psi(\psi_1(\Omega))$ |
| (0,0)(1,1)(2,2)(3,1)(4,2) | $\psi(\psi_1(\psi_1(0)))$ |
| (0,0)(1,1)(2,2)(3,2) | $\psi(\Omega_2)$ |
| (0,0)(1,1)(2,2)(3,2)(1,1)(2,2) | $\psi(\Omega_2+\psi_1(0))$ |
| (0,0)(1,1)(2,2)(3,2)<br>(1,1)(2,2)(3,2) | $\psi(\Omega_2+\psi_1(\Omega_2))$ |
| (0,0)(1,1)(2,2)(3,2)<br>(1,1)(2,2)(3,2)(1,1)(2,2)(3,2) | $\psi(\Omega_2+\psi_1(\Omega_2)\cdot 2)$ |
| (0,0)(1,1)(2,2)(3,2)(2,0) | $\psi(\Omega_2+\psi_1(\Omega_2)\cdot\omega)$ |
| (0,0)(1,1)(2,2)(3,2)(2,1) | $\psi(\Omega_2+\psi_1(\Omega_2)\cdot\Omega)$ |
| (0,0)(1,1)(2,2)(3,2)(2,1)(3,1) | $\psi(\Omega_2+\psi_1(\Omega_2)\cdot\Omega^{\Omega})$ |
| (0,0)(1,1)(2,2)(3,2)(2,1)(3,2) | $\psi(\Omega_2+\psi_1(\Omega_2)\cdot\psi_1(0))$ |
| (0,0)(1,1)(2,2)(3,2)<br>(2,1)(3,2)(3,2) | $\psi(\Omega_2+\psi_1(\Omega_2)\cdot\psi_1(1))$ |
| (0,0)(1,1)(2,2)(3,2)<br>(2,1)(3,2)(4,1) | $\psi(\Omega_2+\psi_1(\Omega_2)\cdot\psi_1(\Omega))$ |
| (0,0)(1,1)(2,2)(3,2)<br>(2,1)(3,2)(4,2) | $\psi(\Omega_2+\psi_1(\Omega_2)^2)$ |
| (0,0)(1,1)(2,2)(3,2)(2,1)<br>(3,2)(4,2)(2,1)(3,2)(4,2) | $\psi(\Omega_2+\psi_1(\Omega_2)^3)$ |

| BMS | Madore's OCF |
|---|---|
| (0,0)(1,1)(2,2)(3,2)<br>(2,1)(3,2)(4,2)(3,0) | $\psi(\Omega_2 + \psi_1(\Omega_2)^\omega)$ |
| (0,0)(1,1)(2,2)(3,2)<br>(2,1)(3,2)(4,2)(3,1) | $\psi(\Omega_2 + \psi_1(\Omega_2)^\Omega)$ |
| (0,0)(1,1)(2,2)(3,2)<br>(2,1)(3,2)(4,2)(3,1)(4,2) | $\psi(\Omega_2 + \psi_1(\Omega_2)^{\psi_1(0)})$ |
| (0,0)(1,1)(2,2)(3,2)(2,1)<br>(3,2)(4,2)(3,1)(4,2)(5,2) | $\psi(\Omega_2 + \psi_1(\Omega_2)^{\psi_1(\Omega_2)})$ |
| (0,0)(1,1)(2,2)(3,2)(2,2) | $\psi(\Omega_2 + \psi_1(\Omega_2 + 1))$ |
| (0,0)(1,1)(2,2)(3,2)(2,2)(3,0) | $\psi(\Omega_2 + \psi_1(\Omega_2 + \omega))$ |
| (0,0)(1,1)(2,2)(3,2)(2,2)(3,1) | $\psi(\Omega_2 + \psi_1(\Omega_2 + \Omega))$ |
| (0,0)(1,1)(2,2)(3,2)<br>(2,2)(3,1)(4,2) | $\psi(\Omega_2 + \psi_1(\Omega_2 + \psi_1(0)))$ |
| (0,0)(1,1)(2,2)(3,2)<br>(2,2)(3,1)(4,2)(5,2) | $\psi(\Omega_2 + \psi_1(\Omega_2 + \psi_1(\Omega_2)))$ |
| (0,0)(1,1)(2,2)(3,2)(2,2)(3,2) | $\psi(\Omega_2 \cdot 2)$ |
| (0,0)(1,1)(2,2)(3,2)(3,0) | $\psi(\Omega_2 \cdot \omega)$ |
| (0,0)(1,1)(2,2)(3,2)(3,1) | $\psi(\Omega_2 \cdot \Omega)$ |
| (0,0)(1,1)(2,2)(3,2)(3,2) | $\psi({\Omega_2}^2)$ |
| (0,0)(1,1)(2,2)(3,2)(3,2)(3,2) | $\psi({\Omega_2}^3)$ |
| (0,0)(1,1)(2,2)(3,2)(4,0) | $\psi({\Omega_2}^\omega)$ |
| (0,0)(1,1)(2,2)(3,2)(4,1) | $\psi({\Omega_2}^\Omega)$ |
| (0,0)(1,1)(2,2)(3,2)(4,2) | $\psi({\Omega_2}^{\Omega_2})$ |
| (0,0)(1,1)(2,2)(3,2)(4,2)(5,2) | $\psi({\Omega_2}^{{\Omega_2}^{\Omega_2}})$ |
| (0,0)(1,1)(2,2)(3,3) | $\psi(\psi_2(0))$ |
| (0,0)(1,1)(2,2)(3,3)(4,3) | $\psi(\Omega_3)$ |
| (0,0)(1,1)(2,2)(3,3)<br>(4,3)(3,3)(4,3) | $\psi(\Omega_3 \cdot 2)$ |
| (0,0)(1,1)(2,2)(3,3)(4,3)(4,3) | $\psi({\Omega_3}^2)$ |
| (0,0)(1,1)(2,2)(3,3)(4,3)(5,3) | $\psi({\Omega_3}^{\Omega_3})$ |
| (0,0)(1,1)(2,2)(3,3)(4,4) | $\psi(\psi_3(0))$ |
| (0,0)(1,1)(2,2)(3,3)(4,4)(5,4) | $\psi(\Omega_4)$ |
| (0,0)(1,1)(2,2)(3,3)(4,4)(5,5) | $\psi(\psi_4(0))$ |
| (0,0,0)(1,1,1) | $\psi(\Omega_\omega)$ |
| (0,0,0)(1,1,1)(1,0,0) | $\psi(\Omega_\omega) \cdot \omega$ |
| (0,0,0)(1,1,1)(1,0,0)(2,0,0) | $\psi(\Omega_\omega) \cdot \omega^\omega$ |
| (0,0,0)(1,1,1)(1,0,0)(2,1,0) | $\psi(\Omega_\omega) \cdot \psi(0)$ |
| (0,0,0)(1,1,1)(1,0,0)(2,1,0)(3,1,0) | $\psi(\Omega_\omega) \cdot \psi(\Omega)$ |
| (0,0,0)(1,1,1)(1,0,0)(2,1,0)(3,2,0) | $\psi(\Omega_\omega) \cdot \psi(\psi_1(0))$ |
| (0,0,0)(1,1,1)(1,0,0)<br>(2,1,0)(3,2,0)(4,3,0) | $\psi(\Omega_\omega) \cdot \psi(\psi_2(0))$ |
| (0,0,0)(1,1,1)(1,0,0)(2,1,1) | $\psi(\Omega_\omega)^2$ |
| (0,0,0)(1,1,1)(1,0,0)<br>(2,1,1)(1,0,0)(2,1,1) | $\psi(\Omega_\omega)^3$ |
| (0,0,0)(1,1,1)(1,0,0)(2,1,1)(2,0,0) | $\psi(\Omega_\omega)^\omega$ |
| (0,0,0)(1,1,1)(1,0,0)<br>(2,1,1)(2,0,0)(3,1,1) | $\psi(\Omega_\omega)^{\psi(\Omega_\omega)}$ |
| (0,0,0)(1,1,1)(1,0,0)(2,1,1)<br>(2,0,0)(3,1,1)(3,0,0)(4,1,1) | $\psi(\Omega_\omega)^{\psi(\Omega_\omega)^{\psi(\Omega_\omega)}}$ |
| (0,0,0)(1,1,1)(1,1,0) | $\psi(\Omega_\omega + 1)$ |
| (0,0,0)(1,1,1)(1,1,0)(1,1,0) | $\psi(\Omega_\omega + 2)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,0,0) | $\psi(\Omega_\omega + \omega)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,0,0)(3,0,0) | $\psi(\Omega_\omega + \omega^\omega)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,0,0)(3,1,0) | $\psi(\Omega_\omega + \psi(0))$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(1,1,0)<br>(2,0,0)(3,1,0)(3,1,0) | $\psi(\Omega_\omega + \psi(1))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,0,0)(3,1,0)(4,1,0) | $\psi(\Omega_\omega + \psi(\Omega))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,0,0)(3,1,0)(4,2,0) | $\psi(\Omega_\omega + \psi(\psi_1(0)))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,0,0)(3,1,1) | $\psi(\Omega_\omega + \psi(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,0,0)(3,1,1)(3,1,0) | $\psi(\Omega_\omega + \psi(\Omega_\omega + 1))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,0,0)<br>(3,1,1)(3,1,0)(4,0,0)(5,1,1) | $\psi(\Omega_\omega + \psi(\Omega_\omega + \psi(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,1,0) | $\psi(\Omega_\omega + \Omega)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,1,0)(2,0,0) | $\psi(\Omega_\omega + \Omega \cdot \omega)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,1,0)(2,1,0) | $\psi(\Omega_\omega + \Omega^2)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,1,0)(3,0,0) | $\psi(\Omega_\omega + \Omega^\omega)$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,1,0)(3,0,0)(4,1,1) | $\psi(\Omega_\omega + \Omega^{\psi(\Omega_\omega)})$ |
| (0,0,0)(1,1,1)(1,1,0)(2,1,0)(3,1,0) | $\psi(\Omega_\omega + \Omega^\Omega)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0) | $\psi(\Omega_\omega + \psi_1(0))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)(2,0,0) | $\psi(\Omega_\omega + \psi_1(0) \cdot \omega)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)(2,1,0) | $\psi(\Omega_\omega + \psi_1(0) \cdot \Omega)$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(2,1,0)(3,2,0) | $\psi(\Omega_\omega + \psi_1(0)^2)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(2,1,0)(3,2,0)(3,0,0) | $\psi(\Omega_\omega + \psi_1(0)^\omega)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(2,1,0)(3,2,0)(3,1,0) | $\psi(\Omega_\omega + \psi_1(0)^\Omega)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(2,1,0)(3,2,0)(3,1,0)(3,1,0) | $\psi(\Omega_\omega + \psi_1(0)^{\Omega^2})$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(2,1,0)(3,2,0)(3,1,0)(4,1,0) | $\psi(\Omega_\omega + \psi_1(0)^{\Omega^\Omega})$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(2,1,0)(3,2,0)(3,1,0)(4,2,0) | $\psi(\Omega_\omega + \psi_1(0)^{\psi_1(0)})$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)(2,2,0) | $\psi(\Omega_\omega + \psi_1(1))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)(3,0,0) | $\psi(\Omega_\omega + \psi_1(\omega))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)(3,1,0) | $\psi(\Omega_\omega + \psi_1(\Omega))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,1,0)(3,0,0) | $\psi(\Omega_\omega + \psi_1(\Omega \cdot \omega))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,1,0)(3,1,0) | $\psi(\Omega_\omega + \psi_1(\Omega^2))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,1,0)(4,0,0) | $\psi(\Omega_\omega + \psi_1(\Omega^\omega))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,1,0)(4,1,0) | $\psi(\Omega_\omega + \psi_1(\Omega^\Omega))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,1,0)(4,2,0) | $\psi(\Omega_\omega + \psi_1(\psi_1(0)))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(3,1,0)(4,2,0)(2,2,0) | $\psi(\Omega_\omega + \psi_1(\psi_1(0) + 1))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(3,1,0)(4,2,0)(4,2,0) | $\psi(\Omega_\omega + \psi_1(\psi_1(1)))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(3,1,0)(4,2,0)(5,0,0) | $\psi(\Omega_\omega + \psi_1(\psi_1(\omega)))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(3,1,0)(4,2,0)(5,1,0) | $\psi(\Omega_\omega + \psi_1(\psi_1(\Omega)))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)(3,2,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2))$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,2,0)(2,0,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2) \cdot \omega)$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,2,0)(2,1,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2) \cdot \Omega)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(3,2,0)(2,1,0)(3,2,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2) \cdot \psi_1(0))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(3,2,0)(2,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2)^2)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(3,2,0)(2,1,0)(3,2,0)<br>(4,2,0)(3,1,0)(4,2,0)(5,2,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2)^{\psi_1(\Omega_2)})$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,2,0)(2,2,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2 + 1))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(3,2,0)(2,2,0)(3,1,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2 + \Omega))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(3,2,0)(2,2,0)(3,1,0)(4,0,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2 + \Omega^\omega))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(3,2,0)(2,2,0)(3,1,0)(4,1,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2 + \Omega^\Omega))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(3,2,0)(2,2,0)(3,1,0)(4,2,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2 + \psi_1(0)))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)(3,2,0)<br>(2,2,0)(3,1,0)(4,2,0)(5,1,0)(6,2,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2 + \psi_1(\psi_1(0))))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)(3,2,0)<br>(2,2,0)(3,1,0)(4,2,0)(5,2,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2 + \psi_1(\Omega_2)))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(3,2,0)(2,2,0)(3,2,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2 \cdot 2))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,2,0)(3,0,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,2,0)(3,1,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2 \cdot \Omega))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(3,2,0)(3,1,0)(4,2,0) | $\psi(\Omega_\omega + \psi_1(\Omega_2 \cdot \psi_1(0)))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,2,0)(3,2,0) | $\psi(\Omega_\omega + \psi_1({\Omega_2}^2))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,2,0)(4,1,0) | $\psi(\Omega_\omega + \psi_1({\Omega_2}^\Omega))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,2,0)(4,2,0) | $\psi(\Omega_\omega + \psi_1({\Omega_2}^{\Omega_2}))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)(3,3,0) | $\psi(\Omega_\omega + \psi_1(\psi_2(0)))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,3,0)(3,3,0) | $\psi(\Omega_\omega + \psi_1(\psi_2(1)))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,3,0)(4,0,0) | $\psi(\Omega_\omega + \psi_1(\psi_2(\omega)))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,3,0)(4,1,0) | $\psi(\Omega_\omega + \psi_1(\psi_2(\Omega)))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(3,3,0)(4,1,0)(5,2,0) | $\psi(\Omega_\omega + \psi_1(\psi_2(\psi_1(0))))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,3,0)(4,2,0) | $\psi(\Omega_\omega + \psi_1(\psi_2(\Omega_2)))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)<br>(3,3,0)(4,2,0)(5,3,0) | $\psi(\Omega_\omega + \psi_1(\psi_2(\psi_2(0))))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,3,0)(4,3,0) | $\psi(\Omega_\omega + \psi_1(\Omega_3))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,0)(3,3,0)(4,4,0) | $\psi(\Omega_\omega + \psi_1(\psi_3(0)))$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(1,1,0)(2,2,1) | $\psi(\Omega_\omega + \psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)(2,2,0) | $\psi(\Omega_\omega + \psi_1(\Omega_\omega + 1))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,1)(2,2,0)(3,1,0) | $\psi(\Omega_\omega + \psi_1(\Omega_\omega + \Omega))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,2,0)(3,1,0)(4,2,0) | $\psi(\Omega_\omega + \psi_1(\Omega_\omega + \psi_1(0)))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,2,0)(3,1,0)(4,2,1) | $\psi(\Omega_\omega + \psi_1(\Omega_\omega + \psi_1(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,1)(2,2,0)(3,2,0) | $\psi(\Omega_\omega + \Omega_2)$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,1)(2,2,0)(3,3,0) | $\psi(\Omega_\omega + \psi_2(0))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,2,0)(3,3,0)(3,3,0) | $\psi(\Omega_\omega + \psi_2(1))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,1)(2,2,0)(3,3,1) | $\psi(\Omega_\omega + \psi_2(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,2,0)(3,3,1)(3,3,0) | $\psi(\Omega_\omega + \psi_2(\Omega_\omega + 1))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)(2,2,0)<br>(3,3,1)(3,3,0)(4,2,0)(5,3,0) | $\psi(\Omega_\omega + \psi_2(\Omega_\omega + \psi_2(0)))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)(2,2,0)<br>(3,3,1)(3,3,0)(4,2,0)(5,3,1) | $\psi(\Omega_\omega + \psi_2(\Omega_\omega + \psi_2(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,2,0)(3,3,1)(3,3,0)(4,3,0) | $\psi(\Omega_\omega + \Omega_3)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,2,0)(3,3,1)(3,3,0)(4,4,0) | $\psi(\Omega_\omega + \psi_3(0))$ |
| (0,0,0)(1,1,1)(1,1,1) | $\psi(\Omega_\omega \cdot 2)$ |
| (0,0,0)(1,1,1)(1,1,1)(1,1,1) | $\psi(\Omega_\omega \cdot 3)$ |
| (0,0,0)(1,1,1)(2,0,0) | $\psi(\Omega_\omega \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,0,0)(1,1,1) | $\psi(\Omega_\omega \cdot \omega + \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,0,0)(2,0,0) | $\psi(\Omega_\omega \cdot \omega^2)$ |
| (0,0,0)(1,1,1)(2,0,0)(3,1,0) | $\psi(\Omega_\omega \cdot \psi(0))$ |
| (0,0,0)(1,1,1)(2,0,0)(3,1,1) | $\psi(\Omega_\omega \cdot \psi(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,0,0)(3,1,1)(3,1,1) | $\psi(\Omega_\omega \cdot \psi(\Omega_\omega \cdot 2))$ |
| (0,0,0)(1,1,1)(2,0,0)<br>(3,1,1)(4,0,0)(5,1,1) | $\psi(\Omega_\omega \cdot \psi(\Omega_\omega \cdot \psi(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0) | $\psi(\Omega_\omega \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0) | $\psi(\Omega_\omega \cdot \Omega + 1)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,0,0)(3,1,1) | $\psi(\Omega_\omega \cdot \Omega + \psi(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,1,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,2,1)(3,1,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,2,1)(3,1,0)(2,2,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \Omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,0)(3,1,0)(4,2,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \Omega + \psi_1(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0)(3,2,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,0)(3,2,0)(1,1,0)(2,2,1) | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 + \psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,1,0)(2,2,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega + 1))$ |

| BMS | Madore's OCF |
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| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,1,0)(2,2,0)(3,2,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega + \Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0)(3,2,0)(2,2,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega + \Omega_2 + 1))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0)(3,2,0)<br>(2,2,0)(3,1,0)(4,2,1) | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega + \Omega_2 + \psi_1(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,0)(3,2,0)(2,2,0)(3,2,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,0)(3,2,0)(3,0,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,0)(3,2,0)(3,2,0) | $\psi(\Omega_\omega \cdot \Omega + {\Omega_2}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0)(3,3,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_2(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0)(3,3,1) | $\psi(\Omega_\omega \cdot \Omega + \psi_2(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0)(3,3,1)(4,1,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_2(\Omega_\omega \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0)(3,3,1)<br>(4,1,0)(3,3,0)(4,3,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_3)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,1)(1,1,0)(2,2,1) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,1)(1,1,0)(2,2,1)(3,1,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_1(\Omega_\omega \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,1)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,0)(3,1,0)(4,2,1) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_1(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0)(3,2,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_1(\Omega_\omega \cdot \Omega + \Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0)(3,3,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_1(\Omega_\omega \cdot \Omega + \psi_2(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0)(3,3,1) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_1(\Omega_\omega \cdot \Omega + \psi_2(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_1(\Omega_\omega \cdot \Omega + \Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,0,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_1(\Omega_\omega \cdot \Omega + \Omega_\omega) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_1(\Omega_\omega \cdot \Omega + \Omega_\omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,1)(2,2,0)(3,1,0)(4,2,1) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_1(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_1(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,0)(3,2,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,0)<br>(3,2,0)(2,2,0)(3,2,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \Omega_2 \cdot 2)$ |

| BMS | Madore's OCF |
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| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)<br>(2,2,0)(3,2,0)(3,2,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + {\Omega_2}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,0)(3,3,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_2(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,1)(2,2,0)(3,3,1)(4,1,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_2(\Omega_\omega \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,0)<br>(3,3,1)(4,1,0)(3,3,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_2(\Omega_\omega \cdot \Omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,1)(2,2,0)(3,3,1)<br>(4,1,0)(3,3,0)(4,1,0)(5,2,1) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_2(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,0)<br>(3,3,1)(4,1,0)(3,3,0)(4,2,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_2(\Omega_\omega \cdot \Omega + \Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,0)(3,3,1)<br>(4,1,0)(3,3,0)(4,2,0)(5,3,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_2(\Omega_\omega \cdot \Omega + \psi_2(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,0)(3,3,1)<br>(4,1,0)(3,3,0)(4,2,0)(5,3,1) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_2(\Omega_\omega \cdot \Omega + \psi_2(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,0)<br>(3,3,1)(4,1,0)(3,3,0)(4,3,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_2(\Omega_\omega \cdot \Omega + \Omega_3))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,0)<br>(3,3,1)(4,1,0)(3,3,0)(4,4,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_2(\Omega_\omega \cdot \Omega + \psi_3(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,0)<br>(3,3,1)(4,1,0)(3,3,0)(4,4,1) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_2(\Omega_\omega \cdot \Omega + \psi_3(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,0)<br>(3,3,1)(4,1,0)(3,3,1) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_2(\Omega_\omega \cdot \Omega + \Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,0)<br>(3,3,1)(4,1,0)(3,3,1)(3,3,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_2(\Omega_\omega \cdot \Omega + \Omega_\omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,0)(3,3,1)<br>(4,1,0)(3,3,1)(3,3,0)(4,3,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \Omega_3)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,1) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,1)(2,2,1) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega \cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(3,0,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(3,1,0) | $\psi(\Omega_\omega \cdot \Omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,1)(3,1,0)(1,1,0)(2,2,1) | $\psi(\Omega_\omega \cdot \Omega \cdot 2 + \psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(3,1,0) | $\psi(\Omega_\omega \cdot \Omega \cdot 2 + \psi_1(\Omega_\omega \cdot \Omega \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,1)(3,1,0)(2,2,0) | $\psi(\Omega_\omega \cdot \Omega \cdot 2 + \psi_1(\Omega_\omega \cdot \Omega \cdot 2 + 1))$ |

| BMS | Madore's OCF |
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| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,1)(3,1,0)(2,2,0)(3,2,0) | $\psi(\Omega_\omega \cdot \Omega \cdot 2 + \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(3,1,0)(2,2,1) | $\psi(\Omega_\omega \cdot \Omega \cdot 2 + \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(3,0,0) | $\psi(\Omega_\omega \cdot \Omega \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(3,1,0) | $\psi(\Omega_\omega \cdot \Omega^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,0,0) | $\psi(\Omega_\omega \cdot \Omega^\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,1,0) | $\psi(\Omega_\omega \cdot \Omega^\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,0) | $\psi(\Omega_\omega \cdot \psi_1(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,0)(4,2,0) | $\psi(\Omega_\omega \cdot \psi_1(1))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,0)(5,1,0) | $\psi(\Omega_\omega \cdot \psi_1(\Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,0)(5,1,0)(6,2,0) | $\psi(\Omega_\omega \cdot \psi_1(\psi_1(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,0)(5,2,0) | $\psi(\Omega_\omega \cdot \psi_1(\Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,0)(5,3,0) | $\psi(\Omega_\omega \cdot \psi_1(\psi_2(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,1) | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,1)(4,2,1) | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,1)(5,1,0) | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,1)(5,1,0)<br>(4,2,0)(5,1,0)(6,2,1) | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(4,2,1)(5,1,0)(4,2,0)(5,2,0) | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \Omega + \Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(4,2,1)(5,1,0)(4,2,0)(5,3,1) | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \Omega + \psi_2(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,1)(5,1,0)(4,2,1) | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \Omega + \Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,1)(5,1,0)(4,2,1)(5,1,0) | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \Omega \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(4,2,1)(5,1,0)(6,2,0) | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \psi_1(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(4,2,1)(5,1,0)(6,2,1) | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \psi_1(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,2,1)(3,2,0) | $\psi(\Omega_\omega \cdot \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(1,1,0)(2,2,1) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(1,1,0)(2,2,1)(3,1,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(1,1,0)(2,2,1)(3,1,0)(2,2,1) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega + \Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,1)(3,1,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega \cdot 2))$ |

| BMS | Madore's OCF |
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| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(1,1,0)(2,2,1)(3,1,0)(4,2,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \psi_1(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(1,1,0)(2,2,1)(3,1,0)(4,2,1) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \psi_1(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(1,1,0)(2,2,1)(3,2,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,0,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega_2) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(2,1,0)(3,2,1)(4,2,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega_2)^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega_2 + 1))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(2,2,0)(3,1,0)(4,2,1) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(2,2,0)(3,1,0)(4,2,1)(5,2,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega_2)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)<br>(3,1,0)(4,2,1)(5,2,0)(4,2,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega_2 + 1)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,2,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_2(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,1) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_2(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(2,2,0)(3,3,1)(4,2,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_2(\Omega_\omega \cdot \Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(2,2,0)(3,3,1)(4,2,0)(3,3,0)<br>(4,2,0)(5,3,1)(6,2,0)(5,3,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_2(\Omega_\omega \cdot \Omega_2 + \psi_2(\Omega_\omega \cdot \Omega_2 + 1)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,1)<br>(4,2,0)(3,3,0)(4,3,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \Omega_3)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,1)<br>(4,2,0)(3,3,0)(4,4,1) | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_3(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(2,2,0)(3,3,1)(4,2,0)(3,3,0)<br>(4,4,1)(5,2,0)(4,4,0)(5,4,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \Omega_4)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(2,2,0)(3,3,1)(4,2,0)(3,3,1) | $\psi(\Omega_\omega \cdot \Omega_2 + \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,1)<br>(4,2,0)(3,3,1)(3,3,1) | $\psi(\Omega_\omega \cdot \Omega_2 + \Omega_\omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,1)<br>(4,2,0)(3,3,1)(4,0,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \Omega_\omega \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,1)<br>(4,2,0)(3,3,1)(4,1,0) | $\psi(\Omega_\omega \cdot \Omega_2 + \Omega_\omega \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,1)<br>(4,2,0)(3,3,1)(4,2,0) | $\psi(\Omega_\omega \cdot \Omega_2 \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(2,2,0)(3,3,1)(4,2,0)(4,0,0) | $\psi(\Omega_\omega \cdot \Omega_2 \cdot \omega)$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(2,2,0)(3,3,1)(4,2,0)(4,2,0) | $\psi(\Omega_\omega \cdot {\Omega_2}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(2,2,0)(3,3,1)(4,2,0)(5,2,0) | $\psi(\Omega_\omega \cdot {\Omega_2}^{\Omega_2})$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(2,2,0)(3,3,1)(4,2,0)(5,3,0) | $\psi(\Omega_\omega \cdot \psi_2(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,1)<br>(4,2,0)(5,3,0)(5,3,0) | $\psi(\Omega_\omega \cdot \psi_2(1))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(2,2,0)(3,3,1)(4,2,0)(5,3,1) | $\psi(\Omega_\omega \cdot \psi_2(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,1)<br>(4,2,0)(5,3,1)(6,1,0) | $\psi(\Omega_\omega \cdot \psi_2(\Omega_\omega \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,1)<br>(4,2,0)(5,3,1)(6,2,0) | $\psi(\Omega_\omega \cdot \psi_2(\Omega_\omega \cdot \Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(2,2,0)(3,3,1)(4,3,0) | $\psi(\Omega_\omega \cdot \Omega_3)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,1)<br>(4,3,0)(3,3,0)(4,4,1)(5,4,0) | $\psi(\Omega_\omega \cdot \Omega_4)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1) | $\psi({\Omega_\omega}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0) | $\psi({\Omega_\omega}^2 + 1)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,1)(1,1,0)(2,1,0) | $\psi({\Omega_\omega}^2 + \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,1)(1,1,0)(2,2,0) | $\psi({\Omega_\omega}^2 + \psi_1(0))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,1)(1,1,0)(2,2,1) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,1,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,1,0)(2,2,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,1,0)(2,2,0)(3,2,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega + \Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,1,0)(2,2,0)(3,3,1) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega + \psi_2(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0)(3,3,1)(4,1,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega + \psi_2(\Omega_\omega \cdot \Omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,1,0)(2,2,0)<br>(3,3,1)(4,1,0)(3,3,0)(4,3,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega + \Omega_3))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,1,0)(2,2,1) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega + \Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,1,0)(2,2,1)(3,1,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,1,0)(3,1,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega^2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,1,0)(4,2,1) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \psi_1(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,1)(5,1,0)(6,2,1) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \psi_1(\Omega_\omega))))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega_2))$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,2,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega_2 + \Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,1)(4,2,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega_2 + \psi_2(\Omega_\omega \cdot \Omega_2)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,0)<br>(3,3,1)(4,2,0)(3,3,0)(4,3,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega_2 + \Omega_3))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,0)<br>(3,3,1)(4,2,0)(3,3,1) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega_2 + \Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,0)<br>(3,3,1)(4,2,0)(3,3,1)(3,3,1) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega_2 + \Omega_\omega \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,0)<br>(3,3,1)(4,2,0)(3,3,1)(4,2,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega_2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,0)<br>(3,3,1)(4,2,0)(4,2,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot {\Omega_2}^2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,0)<br>(3,3,1)(4,2,0)(5,3,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \psi_2(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,0)<br>(3,3,1)(4,2,0)(5,3,1) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \psi_2(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,1)(4,3,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega_3))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1) | $\psi({\Omega_\omega}^2 + \psi_1({\Omega_\omega}^2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,1)(2,2,0)(3,1,0)<br>(4,2,1)(5,2,0)(4,2,1) | $\psi({\Omega_\omega}^2 + \psi_1({\Omega_\omega}^2 + \psi_1({\Omega_\omega}^2)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,1)(2,2,0)(3,2,0) | $\psi({\Omega_\omega}^2 + \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,1)(2,2,0)(3,3,1) | $\psi({\Omega_\omega}^2 + \psi_2(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,1) | $\psi({\Omega_\omega}^2 + \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,1)(1,1,1)(1,1,1) | $\psi({\Omega_\omega}^2 + \Omega_\omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(2,0,0) | $\psi({\Omega_\omega}^2 + \Omega_\omega \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(2,1,0) | $\psi({\Omega_\omega}^2 + \Omega_\omega \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1)(3,1,0) | $\psi({\Omega_\omega}^2 + \Omega_\omega \cdot \Omega + \psi_1({\Omega_\omega}^2 + \Omega_\omega \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,0)(2,2,1) | $\psi({\Omega_\omega}^2 + \Omega_\omega \cdot \Omega + \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1)(3,2,0) | $\psi({\Omega_\omega}^2 + \Omega_\omega \cdot \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1)(3,2,0)<br>(2,2,0)(3,3,1)(4,3,0)(3,3,1)(4,3,0) | $\psi({\Omega_\omega}^2 + \Omega_\omega \cdot \Omega_3)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,1)(2,1,0)(1,1,1) | $\psi({\Omega_\omega}^2 \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(2,0,0) | $\psi({\Omega_\omega}^2 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(2,1,0) | $\psi({\Omega_\omega}^2 \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(2,1,0)(1,1,1) | $\psi({\Omega_\omega}^3)$ |

| BMS | Madore's OCF |
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| (0,0,0)(1,1,1)(2,1,0)(2,1,0)<br>(1,1,1)(2,1,0)(2,1,0)(1,1,1) | $\psi({\Omega_\omega}^3 \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(2,1,0)(2,1,0)(1,1,1) | $\psi({\Omega_\omega}^4)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,0,0) | $\psi({\Omega_\omega}^\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0) | $\psi({\Omega_\omega}^\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,1,0) | $\psi(\Omega_\omega^\Omega + \psi_1(\Omega_\omega \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1) | $\psi(\Omega_\omega^\Omega + \psi_1(\Omega_\omega^2))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(3,2,0)(2,2,1) | $\psi({\Omega_\omega}^\Omega + \psi_1({\Omega_\omega}^3))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,0,0) | $\psi({\Omega_\omega}^\Omega + \psi_1({\Omega_\omega}^\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,1,0) | $\psi({\Omega_\omega}^\Omega + \psi_1({\Omega_\omega}^\Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(4,1,0)(2,2,0)(3,2,0) | $\psi({\Omega_\omega}^\Omega + \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(4,1,0)(2,2,1) | $\psi(\Omega_\omega^\Omega + \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(4,1,0)(2,2,1)(2,2,1) | $\psi({\Omega_\omega}^\Omega + \Omega_\omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(4,1,0)(2,2,1)(3,1,0) | $\psi({\Omega_\omega}^\Omega + \Omega_\omega \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,1,0)<br>(2,2,1)(3,2,0)(2,2,1) | $\psi({\Omega_\omega}^\Omega + {\Omega_\omega}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,1,0)<br>(2,2,1)(3,2,0)(4,0,0) | $\psi({\Omega_\omega}^\Omega + {\Omega_\omega}^\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,1,0)<br>(2,2,1)(3,2,0)(4,1,0) | $\psi({\Omega_\omega}^\Omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(4,1,0)(3,0,0) | $\psi({\Omega_\omega}^\Omega \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(4,1,0)(3,1,0) | $\psi({\Omega_\omega}^\Omega \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(4,1,0)(3,2,0)(2,2,1) | $\psi({\Omega_\omega}^{\Omega+1})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(4,1,0)(3,2,0)(4,1,0) | $\psi({\Omega_\omega}^{\Omega \cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(4,1,0)(4,1,0) | $\psi({\Omega_\omega}^{\Omega^2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(4,1,0)(5,1,0) | $\psi({\Omega_\omega}^{\Omega^\Omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(4,1,0)(5,2,0) | $\psi({\Omega_\omega}^{\psi_1(0)})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(4,1,0)(5,2,1) | $\psi({\Omega_\omega}^{\psi_1(\Omega_\omega)})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,1,0)<br>(5,2,1)(6,2,0)(5,2,1) | $\psi({\Omega_\omega}^{\psi_1({\Omega_\omega}^2)})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,1,0)<br>(5,2,1)(6,2,0)(7,1,0)(8,2,1) | $\psi({\Omega_\omega}^{\psi_1({\Omega_\omega}^{\psi_1(\Omega_\omega)})})$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,2,0) | $\psi({\Omega_\omega}^{\Omega_2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,2,0)<br>(2,2,0)(3,3,1)(4,3,0)(5,3,0) | $\psi({\Omega_\omega}^{\Omega_3})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(1,1,1) | $\psi({\Omega_\omega}^{\Omega_\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,1)(2,1,0)(1,1,1) | $\psi({\Omega_\omega}^{\Omega_\omega} + {\Omega_\omega}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,1)(2,1,0)(3,1,0)(1,1,1) | $\psi(\Omega_\omega^{\Omega_\omega} \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(2,1,0) | $\psi({\Omega_\omega}^{\Omega_\omega} \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(2,1,0)(1,1,1) | $\psi({\Omega_\omega}^{\Omega_\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(2,1,0)(3,1,0) | $\psi({\Omega_\omega}^{\Omega_\omega+\Omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(2,1,0)(3,1,0)(1,1,1) | $\psi({\Omega_\omega}^{\Omega_\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(3,1,0) | $\psi({\Omega_\omega}^{\Omega_\omega\cdot\Omega})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(3,1,0)(1,1,1) | $\psi({\Omega_\omega}^{\Omega_\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(4,1,0) | $\psi({\Omega_\omega}^{{\Omega_\omega}^{\Omega}})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(4,1,0)(1,1,1) | $\psi({\Omega_\omega}^{{\Omega_\omega}^{\Omega_\omega}})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0) | $\psi(\psi_\omega(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(1,1,1) | $\psi(\psi_\omega(0) + \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(1,1,1)(2,1,0) | $\psi(\psi_\omega(0) + \Omega_\omega \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,1)(2,1,0)(1,1,1) | $\psi(\psi_\omega(0) + {\Omega_\omega}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,1)(2,1,0)(3,1,0) | $\psi(\psi_\omega(0) + {\Omega_\omega}^{\Omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,1)(2,1,0)(3,1,0)(1,1,1) | $\psi(\psi_\omega(0) + {\Omega_\omega}^{\Omega_\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,1)(2,1,0)(3,2,0) | $\psi(\psi_\omega(0) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(2,0,0) | $\psi(\psi_\omega(0) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(2,1,0) | $\psi(\psi_\omega(0) \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(2,1,0)(1,1,1) | $\psi(\psi_\omega(0) \cdot \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(2,1,0)(3,2,0) | $\psi(\psi_\omega(0)^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(3,0,0) | $\psi(\psi_\omega(0)^\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(3,1,0) | $\psi(\psi_\omega(0)^\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(3,1,0)(1,1,1) | $\psi(\psi_\omega(0)^{\Omega_\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(3,1,0)(4,2,0) | $\psi(\psi_\omega(0)^{\psi_\omega(0)})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(3,1,0)(4,2,0)(4,1,0)(5,2,0) | $\psi(\psi_\omega(0)^{\psi_\omega(0)^{\psi_\omega(0)}})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(3,2,0) | $\psi(\psi_\omega(1))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,0,0) | $\psi(\psi_\omega(\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,0,0)(5,1,1)(6,1,0)(7,2,0) | $\psi(\psi_\omega(\psi(\psi_\omega(0))))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0) | $\psi(\psi_\omega(\Omega))$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,1,0)(1,1,0) | $\psi(\psi_\omega(\Omega)+1)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)(4,3,0) | $\psi(\psi_\omega(\Omega)+\psi_1(\psi_\omega(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)(5,1,0) | $\psi(\psi_\omega(\Omega)+\psi_1(\psi_\omega(\Omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(2,2,0)(3,2,0) | $\psi(\psi_\omega(\Omega)+\Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(2,2,1) | $\psi(\psi_\omega(\Omega)+\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(2,2,1)(3,1,0) | $\psi(\psi_\omega(\Omega)+\Omega_\omega\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(2,2,1)(3,2,0)(2,2,1) | $\psi(\psi_\omega(\Omega)+{\Omega_\omega}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(2,2,1)(3,2,0)(4,3,0) | $\psi(\psi_\omega(\Omega)+\psi_\omega(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)(5,1,0)<br>(2,2,1)(3,2,0)(4,3,0)(4,3,0) | $\psi(\psi_\omega(\Omega)+\psi_\omega(1))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)(5,1,0)<br>(2,2,1)(3,2,0)(4,3,0)(5,1,0) | $\psi(\psi_\omega(\Omega)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)(5,1,0)<br>(2,2,1)(3,2,0)(4,3,0)(5,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(4,3,0)(5,1,0)<br>(2,2,1)(3,2,0)(4,3,0)(5,1,0) | $\psi(\psi_\omega(\Omega)\cdot 2+\psi_1(\psi_\omega(\Omega)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)(5,1,0)<br>(2,2,1)(3,2,0)(4,3,0)(5,1,0)(2,2,1) | $\psi(\psi_\omega(\Omega)\cdot 2+\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(3,0,0) | $\psi(\psi_\omega(\Omega)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(3,1,0) | $\psi(\psi_\omega(\Omega)\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(3,1,0)(4,2,1) | $\psi(\psi_\omega(\Omega)\cdot\psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)(5,1,0)<br>(3,1,0)(4,2,1)(5,2,0)(6,3,0)(7,1,0) | $\psi(\psi_\omega(\Omega)\cdot\psi_1(\psi_\omega(\Omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(4,3,0)(5,1,0)(3,2,0) | $\psi(\psi_\omega(\Omega)\cdot\Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(3,2,0)(1,1,0) | $\psi(\psi_\omega(\Omega)\cdot\Omega_2+1)$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(3,2,0)(1,1,0)(2,2,1)<br>(3,2,0)(4,3,0)(5,1,0)(3,2,0) | $\psi(\psi_\omega(\Omega)\cdot\Omega_2+\psi_1(\psi_\omega(\Omega)\cdot\Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(3,2,0)(2,2,0) | $\psi(\psi_\omega(\Omega)\cdot\Omega_2+\psi_1(\psi_\omega(\Omega)\cdot\Omega_2+1))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(3,2,0)(2,2,0)(3,2,0) | $\psi(\psi_\omega(\Omega)\cdot\Omega_2+\Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(3,2,0)(2,2,0)(3,3,1) | $\psi(\psi_\omega(\Omega)\cdot\Omega_2+\psi_2(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(3,2,0)(2,2,0)(3,3,1)<br>(4,3,0)(5,4,0)(6,1,0)(4,2,0) | $\psi(\psi_\omega(\Omega)\cdot\Omega_2+\psi_2(\psi_\omega(\Omega)\cdot\Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)(5,1,0)<br>(3,2,0)(2,2,0)(3,3,1)(4,3,0)(5,4,0)<br>(6,1,0)(4,2,0)(3,3,0)(4,3,0) | $\psi(\psi_\omega(\Omega)\cdot\Omega_2+\Omega_3)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)(5,1,0)<br>(3,2,0)(2,2,0)(3,3,1)(4,3,0)<br>(5,4,0)(6,1,0)(4,2,0)(3,3,1) | $\psi(\psi_\omega(\Omega)\cdot\Omega_2+\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(3,2,0)(2,2,0)(3,3,1)(4,3,0)<br>(5,4,0)(6,1,0)(4,2,0)(3,3,1)<br>(4,3,0)(5,4,0)(6,1,0) | $\psi(\psi_\omega(\Omega)\cdot\Omega_2+\psi_\omega(\Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)(5,1,0)<br>(3,2,0)(2,2,0)(3,3,1)(4,3,0)(5,4,0)<br>(6,1,0)(4,2,0)(3,3,1)(4,3,0)<br>(5,4,0)(6,1,0)(4,2,0) | $\psi(\psi_\omega(\Omega)\cdot\Omega_2\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)(5,1,0)<br>(3,2,0)(2,2,0)(3,3,1)(4,3,0)<br>(5,4,0)(6,1,0)(4,2,0)(4,2,0) | $\psi(\psi_\omega(\Omega)\cdot{\Omega_2}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)(5,1,0)<br>(3,2,0)(2,2,0)(3,3,1)(4,3,0)<br>(5,4,0)(6,1,0)(4,2,0)(5,3,0) | $\psi(\psi_\omega(\Omega)\cdot\psi_2(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(3,2,0)(2,2,0)(3,3,1)<br>(4,3,0)(5,4,0)(6,1,0)(4,3,0) | $\psi(\psi_\omega(\Omega)\cdot\Omega_3)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(3,2,0)(2,2,1) | $\psi(\psi_\omega(\Omega)\cdot\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(3,2,0)(2,2,1)(2,2,1) | $\psi(\psi_\omega(\Omega)\cdot\Omega_\omega+\Omega_\omega)$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(3,2,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(3,2,0) | $\psi(\psi_\omega(\Omega)\cdot\Omega_\omega+\psi_\omega(\Omega_\omega)\cdot\Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)(5,1,0)<br>(3,2,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(3,2,0)(2,2,1) | $\psi(\psi_\omega(\Omega)\cdot\Omega_\omega\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(3,2,0)(3,2,0)(2,2,1) | $\psi(\psi_\omega(\Omega)\cdot\Omega_\omega^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(3,2,0)(4,2,0)(2,2,1) | $\psi(\psi_\omega(\Omega)\cdot{\Omega_\omega}^{\Omega_\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(3,2,0)(4,3,0) | $\psi(\psi_\omega(\Omega)\cdot\psi_\omega(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(3,2,0)(4,3,0)(4,3,0) | $\psi(\psi_\omega(\Omega)\cdot\psi_\omega(1))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(3,2,0)(4,3,0)(5,1,0) | $\psi(\psi_\omega(\Omega)^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(4,0,0) | $\psi(\psi_\omega(\Omega)^\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(4,2,0) | $\psi(\psi_\omega(\Omega)^{\Omega_2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(4,2,0)(2,2,1) | $\psi(\psi_\omega(\Omega)^{\Omega_\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(4,2,0)(5,2,0)(2,2,1) | $\psi(\psi_\omega(\Omega)^{\Omega_\omega^{\Omega_\omega}})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(4,2,0)(5,3,0) | $\psi(\psi_\omega(\Omega)^{\psi_\omega(0)})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(4,2,0)(5,3,0)(6,1,0) | $\psi(\psi_\omega(\Omega)^{\psi_\omega(\Omega)})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(4,3,0) | $\psi(\psi_\omega(\Omega+1))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(4,3,0)(4,3,0) | $\psi(\psi_\omega(\Omega+2))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(4,3,0)(5,1,0) | $\psi(\psi_\omega(\Omega\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(5,1,0) | $\psi(\psi_\omega(\Omega^2))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)<br>(4,3,0)(5,1,0)(6,2,0) | $\psi(\psi_\omega(\psi_1(0)))$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(5,1,0)(6,2,1)(7,2,0)(8,3,0) | $\psi(\psi_\omega(\psi_1(\psi_\omega(0))))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)(5,2,0) | $\psi(\psi_\omega(\Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,1,0)(1,1,1) | $\psi(\psi_\omega(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(1,1,1)(1,1,1) | $\psi(\psi_\omega(\Omega_\omega)+\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,1,0)(2,0,0) | $\psi(\psi_\omega(\Omega_\omega)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,1,0)(2,1,0)(1,1,1) | $\psi(\psi_\omega(\Omega_\omega)\cdot\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(2,1,0)(3,2,0)(4,1,0) | $\psi(\psi_\omega(\Omega_\omega)\cdot\psi_\omega(\Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0)<br>(2,1,0)(3,2,0)(4,1,0)(1,1,1) | $\psi(\psi_\omega(\Omega_\omega)^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(2,1,0)(3,2,0)(4,1,0) | $\psi(\psi_\omega(\Omega_\omega)^2)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,1,0)(3,2,0) | $\psi(\psi_\omega(\Omega_\omega+1))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0) | $\psi(\psi_\omega(\Omega_\omega+\Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0)(1,1,1) | $\psi(\psi_\omega(\Omega_\omega\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,1,0)(4,1,0) | $\psi(\psi_\omega(\Omega_\omega\cdot\Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(4,1,0)(1,1,1) | $\psi(\psi_\omega({\Omega_\omega}^2))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,1,0)(5,1,0)(1,1,1) | $\psi(\psi_\omega({\Omega_\omega}^{\Omega_\omega}))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,1,0)(5,2,0) | $\psi(\psi_\omega(\psi_\omega(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(1,1,1) | $\psi(\Omega_{\omega+1}+\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(1,1,1)(2,1,0)(3,2,0) | $\psi(\Omega_{\omega+1}+\psi_\omega(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,2,0)<br>(1,1,1)(2,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(2,0,0) | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(2,1,0) | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(2,1,0)(1,1,1) | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})\cdot\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(2,1,0)(3,2,0) | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})\cdot\psi_\omega(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(2,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})^2)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(3,1,0) | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})^\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(3,1,0)(1,1,1) | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})^{\Omega_\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(3,1,0)(4,2,0)(5,2,0) | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})^{\psi_\omega(\Omega_{\omega+1})})$ |

| BMS | Madore's OCF |
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| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(3,2,0) | $\psi(\Omega_{\omega+1} + \psi_\omega(\Omega_{\omega+1} + 1))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(3,2,0)(4,1,0) | $\psi(\Omega_{\omega+1} + \psi_\omega(\Omega_{\omega+1} + \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(3,2,0)(4,1,0)(1,1,1) | $\psi(\Omega_{\omega+1} + \psi_\omega(\Omega_{\omega+1} + \Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(3,2,0)(4,1,0)(5,2,0) | $\psi(\Omega_{\omega+1} + \psi_\omega(\Omega_{\omega+1} + \psi_\omega(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(3,2,0)(4,1,0)(5,2,0)(6,2,0) | $\psi(\Omega_{\omega+1} + \psi_\omega(\Omega_{\omega+1} + \psi_\omega(\Omega_{\omega+1})))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\omega+1} \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,2,0)<br>(3,2,0)(4,2,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\omega+1} \cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(4,0,0) | $\psi(\Omega_{\omega+1} \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(4,1,0) | $\psi(\Omega_{\omega+1} \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(4,1,0)(1,1,1) | $\psi(\Omega_{\omega+1} \cdot \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(4,1,0)(5,2,0) | $\psi(\Omega_{\omega+1} \cdot \psi_\omega(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(4,1,0)(5,2,0)(6,2,0) | $\psi(\Omega_{\omega+1} \cdot \psi_\omega(\Omega_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(4,2,0) | $\psi({\Omega_{\omega+1}}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(5,1,0) | $\psi({\Omega_{\omega+1}}^\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(5,1,0)(1,1,1) | $\psi({\Omega_{\omega+1}}^{\Omega_\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(5,1,0)(6,2,0)(7,2,0) | $\psi({\Omega_{\omega+1}}^{\psi_\omega(\Omega_{\omega+1})})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(5,2,0) | $\psi({\Omega_{\omega+1}}^{\Omega_{\omega+1}})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,3,0) | $\psi(\psi_{\omega+1}(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,3,0)(1,1,1)(2,1,0)(3,2,0)(4,3,0) | $\psi(\psi_{\omega+1}(0) + \psi_\omega(\psi_{\omega+1}(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,3,0)(3,2,0)(4,3,0) | $\psi(\psi_{\omega+1}(0) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(4,3,0) | $\psi(\psi_{\omega+1}(1))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,1,0) | $\psi(\psi_{\omega+1}(\Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,3,0)(5,1,0)(1,1,1) | $\psi(\psi_{\omega+1}(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,3,0)(5,1,0)(6,2,0) | $\psi(\psi_{\omega+1}(\psi_\omega(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,3,0)(5,1,0)(6,2,0)(7,3,0) | $\psi(\psi_{\omega+1}(\psi_\omega(\psi_{\omega+1}(0))))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,2,0) | $\psi(\psi_{\omega+1}(\Omega_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,2,0)(6,3,0) | $\psi(\psi_{\omega+1}(\psi_{\omega+1}(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,3,0) | $\psi(\Omega_{\omega+2})$ |

| BMS | Madore's OCF |
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| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,3,0)(5,3,0)(6,3,0) | $\psi({\Omega_{\omega+2}}^{\Omega_{\omega+2}})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,4,0) | $\psi(\psi_{\omega+2}(0))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,4,0)(6,4,0) | $\psi(\Omega_{\omega+3})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1) | $\psi(\Omega_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(1,1,1) | $\psi(\Omega_{\omega\cdot 2}+\Omega_{\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(3,2,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega}(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega}(\Omega_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(3,2,0)(4,3,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega}(\psi_{\omega+1}(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(3,2,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega}(\Omega_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(2,1,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega}(\Omega_{\omega\cdot 2})\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(2,1,0)(1,1,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega}(\Omega_{\omega\cdot 2})\cdot\Omega_{\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(2,1,0)(3,2,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega}(\Omega_{\omega\cdot 2})\cdot\psi_{\omega}(0))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(2,1,0)(3,2,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega}(\Omega_{\omega\cdot 2})^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(3,1,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega}(\Omega_{\omega\cdot 2})^{\Omega})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(3,1,0)(4,2,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega}(\Omega_{\omega\cdot 2})^{\psi_{\omega}(\Omega_{\omega\cdot 2})})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(3,2,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega}(\Omega_{\omega\cdot 2}+1))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(3,2,0)(4,1,0)(5,2,1)(5,2,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega}(\Omega_{\omega\cdot 2}+\psi_{\omega}(\Omega_{\omega\cdot 2}+1)))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(3,2,0)(4,2,0) | $\psi(\Omega_{\omega\cdot 2}+\Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(3,2,0)(4,2,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\omega\cdot 2}+\Omega_{\omega+1}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(3,2,0)(4,2,0)(4,1,0)(1,1,1) | $\psi(\Omega_{\omega\cdot 2}+\Omega_{\omega+1}\cdot\Omega_{\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(3,2,0)(4,2,0)(4,2,0) | $\psi(\Omega_{\omega\cdot 2}+{\Omega_{\omega+1}}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(3,2,0)(4,2,0)(5,2,0) | $\psi(\Omega_{\omega\cdot 2}+{\Omega_{\omega+1}}^{\Omega_{\omega+1}})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(3,2,0)(4,3,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega+1}(0))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(3,2,0)(4,3,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega+1}(\Omega_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(3,2,0)(4,3,1)(4,3,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega+1}(\Omega_{\omega\cdot 2}+1))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(3,2,0)(4,3,1)(4,3,0)(5,3,0) | $\psi(\Omega_{\omega\cdot 2}+\Omega_{\omega+2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(3,2,0)(4,3,1)(4,3,0)(5,4,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega+2}(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(3,2,0)(4,3,1)(4,3,0)(5,4,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega+2}(\Omega_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(3,2,1) | $\psi(\Omega_{\omega\cdot 2}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(1,1,1) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega})$ |

| BMS | Madore's OCF |
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| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,1,0)(1,1,1) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\psi_\omega(\Omega_{\omega\cdot 2}\cdot\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(2,1,0)(3,2,0) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\psi_\omega(\Omega_{\omega\cdot 2}\cdot\Omega_\omega)\cdot\psi_\omega(0))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(3,2,0) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\psi_\omega(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+1))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,2,0)(4,3,0) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\psi_{\omega+1}(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,2,0)(4,3,1)(5,1,0)(1,1,1) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\psi_{\omega+1}(\Omega_{\omega\cdot 2}\cdot\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0)<br>(3,2,0)(4,3,1)(5,1,0)(4,3,0)(5,3,0) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\Omega_{\omega+2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0)<br>(3,2,0)(4,3,1)(5,1,0)(4,3,0)<br>(5,4,1)(6,1,0)(5,4,0)(6,4,0) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\Omega_{\omega+3})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(3,2,1) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\Omega_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,2,1)(3,2,1) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\Omega_{\omega\cdot 2}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,2,1)(4,1,0)(1,1,1) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(4,1,0)(1,1,1) | $\psi(\Omega_{\omega\cdot 2}\cdot{\Omega_\omega}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(5,2,0) | $\psi(\Omega_{\omega\cdot 2}\cdot\psi_\omega(0))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(5,2,1) | $\psi(\Omega_{\omega\cdot 2}\cdot\psi_\omega(\Omega_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(5,2,1)(6,1,0)(7,2,1) | $\psi(\Omega_{\omega\cdot 2}\cdot\psi_\omega(\Omega_{\omega\cdot 2}\cdot\psi_\omega(\Omega_{\omega\cdot 2})))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,2,0) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}+\Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(3,2,0)(4,3,1)(5,2,0) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}+\psi_{\omega+1}(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,2,0)<br>(3,2,0)(4,3,1)(5,2,0)(4,3,0) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}+\psi_{\omega+1}(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}+1))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,2,0)<br>(3,2,0)(4,3,1)(5,2,0)(4,3,0)(5,3,0) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}+\Omega_{\omega+2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,2,0)<br>(3,2,0)(4,3,1)(5,2,0)(4,3,1) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}+\Omega_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,2,0)<br>(3,2,0)(4,3,1)(5,2,0)(4,3,1)(5,2,0) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,2,0)<br>(3,2,0)(4,3,1)(5,2,0)(5,2,0) | $\psi(\Omega_{\omega\cdot 2}\cdot{\Omega_{\omega+1}}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,2,0)<br>(3,2,0)(4,3,1)(5,2,0)(6,3,1) | $\psi(\Omega_{\omega\cdot 2}\cdot\psi_{\omega+1}(\Omega_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(3,2,0)(4,3,1)(5,3,0) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(3,2,0)(4,3,1)(5,3,0)<br>(4,3,0)(5,4,1)(6,4,0) | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+3})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(3,2,1) | $\psi({\Omega_{\omega\cdot 2}}^2)$ |

| BMS | Madore's OCF |
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| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(4,2,0)(3,2,1) | $\psi({\Omega_{\omega\cdot 2}}^3)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,1,0)(1,1,1) | $\psi({\Omega_{\omega\cdot 2}}^{\Omega_\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,2,0)(3,2,1) | $\psi({\Omega_{\omega\cdot 2}}^{\Omega_{\omega\cdot 2}})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,0) | $\psi(\psi_{\omega\cdot 2}(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,0)(5,3,0) | $\psi(\psi_{\omega\cdot 2}(1))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,0)(6,1,0) | $\psi(\psi_{\omega\cdot 2}(\Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,0)(6,1,0)(1,1,1) | $\psi(\psi_{\omega\cdot 2}(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,0)(6,2,0)(3,2,1) | $\psi(\psi_{\omega\cdot 2}(\Omega_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,0)(6,2,0)(7,3,0) | $\psi(\psi_{\omega\cdot 2}(\psi_{\omega\cdot 2}(0)))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,0)(6,3,0) | $\psi(\Omega_{\omega\cdot 2+1})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,0)(6,4,0) | $\psi(\psi_{\omega\cdot 2+1}(0))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,1) | $\psi(\Omega_{\omega\cdot 3})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,1)(5,3,1) | $\psi(\Omega_{\omega\cdot 3}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,1)(6,1,0)(1,1,1) | $\psi(\Omega_{\omega\cdot 3}\cdot\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,1)(6,2,0)(3,2,1) | $\psi(\Omega_{\omega\cdot 3}\cdot\Omega_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,1)(6,3,0)(5,3,1) | $\psi({\Omega_{\omega\cdot 3}}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,1)(6,3,0)(7,4,0) | $\psi(\psi_{\omega\cdot 3}(0))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,1)(6,3,0)(7,4,1) | $\psi(\Omega_{\omega\cdot 4})$ |
| (0,0,0)(1,1,1)(2,1,1) | $\psi(\Omega_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1) | $\psi(\Omega_{\omega^2}+\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(1,1,1)(2,1,0)(3,2,0) | $\psi(\Omega_{\omega^2}+\psi_\omega(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\omega^2}+\psi_\omega(\Omega_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(1,1,1)(2,1,0)(3,2,1) | $\psi(\Omega_{\omega^2}+\psi_\omega(\Omega_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,0)(5,3,1) | $\psi(\Omega_{\omega^2}+\psi_\omega(\Omega_{\omega\cdot 3}))$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1) | $\psi(\Omega_{\omega^2}+\psi_\omega(\Omega_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(3,2,0)<br>(4,1,0)(5,2,1)(6,2,1) | $\psi(\Omega_{\omega^2}+\psi_\omega(\Omega_{\omega^2}+\psi_\omega(\Omega_{\omega^2})))$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(3,2,0)(4,2,0) | $\psi(\Omega_{\omega^2}+\Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(3,2,1) | $\psi(\Omega_{\omega^2}+\Omega_{\omega\cdot 2})$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(3,2,1)(4,2,0)(5,3,0) | $\psi(\Omega_{\omega^2} + \psi_{\omega\cdot 2}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(3,2,1)<br>(3,2,0)(4,3,1)(5,3,1)(4,3,1) | $\psi(\Omega_{\omega^2} + \Omega_{\omega\cdot 3})$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1)(2,1,1) | $\psi(\Omega_{\omega^2} \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,0,0) | $\psi(\Omega_{\omega^2} \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0) | $\psi(\Omega_{\omega^2} \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(1,1,1)(1,1,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,0) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \Omega_\omega \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(1,1,1)(2,1,0)(1,1,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \Omega_\omega^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(1,1,1)(2,1,0)(3,2,0) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \psi_\omega(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \psi_\omega(\Omega_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,1,0)(1,1,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \psi_\omega(\Omega_{\omega^2} \cdot \Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,1,0)(3,2,0)<br>(4,1,0)(5,2,1)(6,2,1)(6,1,0)(1,1,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \psi_\omega(\Omega_{\omega^2} \cdot \Omega_\omega + \psi_\omega(\Omega_{\omega^2} \cdot \Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(4,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(5,1,0)(1,1,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \psi_{\omega+1}(\Omega_{\omega^2} \cdot \Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(5,1,0)(4,3,0)(5,3,0) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \Omega_{\omega+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,1,0)(3,2,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \Omega_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(4,1,0)(3,2,1)(4,2,0)(5,3,0) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \psi_{\omega\cdot 2}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,1,0)(3,2,1)<br>(4,2,0)(5,3,1)(6,3,1)(6,1,0)(1,1,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \psi_{\omega\cdot 2}(\Omega_{\omega^2} \cdot \Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,1,0)<br>(3,2,1)(4,2,0)(5,3,1)(6,3,1)<br>(6,1,0)(5,3,0)(6,3,0) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \Omega_{\omega\cdot 2+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,1,0)(3,2,1)<br>(4,2,0)(5,3,1)(6,3,1)(6,1,0)(5,3,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \Omega_{\omega\cdot 3})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(4,1,0)(3,2,1)(4,2,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \Omega_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,1,0)<br>(3,2,1)(4,2,1)(3,2,1)(4,2,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega + \Omega_{\omega^2} \cdot 2)$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,1,0)<br>(3,2,1)(4,2,1)(4,1,0)(1,1,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(4,1,0)(4,1,0)(1,1,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,1,0)(5,2,0) | $\psi(\Omega_{\omega^2} \cdot \psi_\omega(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(4,1,0)(5,2,1)(6,2,1) | $\psi(\Omega_{\omega^2} \cdot \psi_\omega(\Omega_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,2,0) | $\psi(\Omega_{\omega^2} \cdot \Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,2,0)(3,2,0)<br>(4,3,1)(5,3,1)(5,2,0)(4,3,1)(5,3,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_{\omega+1} + \Omega_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,2,0)<br>(3,2,0)(4,3,1)(5,3,1)(5,3,0) | $\psi(\Omega_{\omega^2} \cdot \Omega_{\omega+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,2,0)(3,2,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(4,2,0)(3,2,1)(4,2,0)<br>(5,3,1)(6,3,1)(6,2,0)(5,3,1)(6,3,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_{\omega\cdot 2} + \Omega_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,2,0)(3,2,1)<br>(4,2,0)(5,3,1)(6,3,1)(6,3,0) | $\psi(\Omega_{\omega^2} \cdot \Omega_{\omega\cdot 2+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,2,0)(3,2,1)<br>(4,2,0)(5,3,1)(6,3,1)(6,3,0)(5,3,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_{\omega\cdot 3})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1) | $\psi({\Omega_{\omega^2}}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(1,1,1)(2,1,1)(1,1,1)(2,1,1) | $\psi({\Omega_{\omega^2}}^2 + \Omega_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,1)(2,1,0)(1,1,1)(2,1,1) | $\psi({\Omega_{\omega^2}}^2 \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(2,1,0)(1,1,1)(2,1,1) | $\psi({\Omega_{\omega^2}}^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(3,1,0) | $\psi({\Omega_{\omega^2}}^\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(3,1,0)(1,1,1) | $\psi({\Omega_{\omega^2}}^{\Omega_\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(3,1,0)(1,1,1)(2,1,1) | $\psi({\Omega_{\omega^2}}^{\Omega_{\omega^2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(3,2,0) | $\psi(\psi_{\omega^2}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(3,2,0)(3,2,0) | $\psi(\psi_{\omega^2}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(3,2,0)(4,1,0) | $\psi(\psi_{\omega^2}(\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(3,2,0)(4,1,0)(1,1,1)(2,1,1) | $\psi(\psi_{\omega^2}(\Omega_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(3,2,0)(4,1,0)(5,2,0) | $\psi(\psi_{\omega^2}(\psi_{\omega^2}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\omega^2+1})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(3,2,0)(4,3,0) | $\psi(\psi_{\omega^2+1}(0))$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(3,2,1) | $\psi(\Omega_{\omega^2+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,1) | $\psi(\Omega_{\omega^2+\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1) | $\psi(\Omega_{\omega^2\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(4,2,0)(5,3,1)(6,3,1) | $\psi(\Omega_{\omega^2\cdot 3})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,1) | $\psi(\Omega_{\omega^3})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(1,1,1)(2,1,1)(2,1,1) | $\psi(\Omega_{\omega^3}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,1)(2,1,0) | $\psi(\Omega_{\omega^3}\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(2,1,1)(2,1,0)(1,1,1) | $\psi(\Omega_{\omega^3}\cdot\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1) | $\psi(\Omega_{\omega^3}\cdot\Omega_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(2,1,1) | $\psi({\Omega_{\omega^3}}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(2,1,1)(2,1,0)(3,2,0) | $\psi(\psi_{\omega^3}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,1)(2,1,1) | $\psi(\Omega_{\omega^4})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,0,0) | $\psi(\Omega_{\omega^\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,0,0) | $\psi(\Omega_{\omega^\omega}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,0,0)<br>(2,1,0)(1,1,1)(2,1,1)(3,0,0) | $\psi({\Omega_{\omega^\omega}}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,0,0)(2,1,0)(3,2,0) | $\psi(\psi_{\omega^\omega}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,0,0)<br>(2,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\omega^\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,0,0)(2,1,0)(3,2,1) | $\psi(\Omega_{\omega^\omega+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1) | $\psi(\Omega_{\omega^\omega+\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,0,0) | $\psi(\Omega_{\omega^\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,0,0)(2,1,1) | $\psi(\Omega_{\omega^{\omega+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,0,0)(2,1,1)(3,0,0) | $\psi(\Omega_{\omega^{\omega\cdot 2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,0,0)(3,0,0) | $\psi(\Omega_{\omega^{\omega^2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,0,0)(4,0,0) | $\psi(\Omega_{\omega^{\omega^\omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,0,0)(4,1,0) | $\psi(\Omega_{\psi(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,0,0)<br>(4,1,0)(2,1,0)(3,2,0) | $\psi(\psi_{\psi(0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,0,0)(4,1,0)(5,1,0) | $\psi(\Omega_{\psi(\Omega)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,0,0)(4,1,1) | $\psi(\Omega_{\psi(\Omega_\omega)})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,0,0)(4,1,1)(5,1,1) | $\psi(\Omega_{\psi(\Omega_{\omega^2})})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,0,0)<br>(4,1,1)(5,1,1)(6,0,0)(7,1,0) | $\psi(\Omega_{\psi(\Omega_{\psi(0)})})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0) | $\psi(\Omega_\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0) | $\psi(\Omega_\Omega+1)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1) | $\psi(\Omega_\Omega+\psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1) | $\psi(\Omega_\Omega+\psi_1(\Omega_{\omega^2}))$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0) | $\psi(\Omega_\Omega + \psi_1(\Omega_\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,0) | $\psi(\Omega_\Omega + \psi_1(\Omega_\Omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,0)<br>(3,1,0)(4,2,1)(5,2,1)(6,1,0) | $\psi(\Omega_\Omega + \psi_1(\Omega_\Omega + \psi_1(\Omega_\Omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,0)(3,2,0) | $\psi(\Omega_\Omega + \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,0)(3,3,1)(4,3,1)(5,1,0) | $\psi(\Omega_\Omega + \psi_2(\Omega_\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)(2,2,0)<br>(3,3,1)(4,3,1)(5,1,0)(3,3,0)(4,3,0) | $\psi(\Omega_\Omega + \Omega_3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1) | $\psi(\Omega_\Omega + \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)(2,2,1) | $\psi(\Omega_\Omega + \Omega_\omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,1)(3,2,0)(2,2,1) | $\psi(\Omega_\Omega + {\Omega_\omega}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,1)(3,2,0)(4,3,0) | $\psi(\Omega_\Omega + \psi_\omega(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,1)(3,2,0)(4,3,1)(5,3,1)(6,1,0) | $\psi(\Omega_\Omega + \psi_\omega(\Omega_\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0)(4,3,0)(5,3,0) | $\psi(\Omega_\Omega + \Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0)(4,3,1) | $\psi(\Omega_\Omega + \Omega_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)(3,2,1) | $\psi(\Omega_\Omega + \Omega_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,1)(3,2,1)(2,2,1)(3,2,1) | $\psi(\Omega_\Omega + \Omega_{\omega^2} \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,1)(3,2,1)(3,2,0)(2,2,1) | $\psi(\Omega_\Omega + {\Omega_{\omega^2}}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,1)(3,2,1)(3,2,0)(4,3,0) | $\psi(\Omega_\Omega + \psi_{\omega^2}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)<br>(3,2,1)(3,2,0)(4,3,1)(5,3,1)<br>(6,1,0)(4,3,0)(5,3,0) | $\psi(\Omega_\Omega + \Omega_{\omega^2+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)(3,2,1)<br>(3,2,0)(4,3,1)(5,3,1)(6,1,0)(4,3,1) | $\psi(\Omega_\Omega + \Omega_{\omega^2+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)<br>(3,2,1)(3,2,0)(4,3,1)(5,3,1)<br>(6,1,0)(4,3,1)(5,3,1) | $\psi(\Omega_\Omega + \Omega_{\omega^2\cdot 2})$ |

| BMS | Madore's OCF |
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| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,1)(3,2,1)(3,2,1) | $\psi(\Omega_\Omega + \Omega_{\omega^3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,1)(3,2,1)(4,0,0) | $\psi(\Omega_\Omega + \Omega_{\omega^\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,1)(3,2,1)(4,0,0)(5,1,1) | $\psi(\Omega_\Omega + \Omega_{\psi(\Omega_\omega)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)(3,2,1)<br>(4,0,0)(5,1,1)(6,1,1)(7,1,0)(5,1,0)<br>(6,2,1)(7,2,1)(8,1,0)(6,2,0)(7,2,0) | $\psi(\Omega_\Omega + \Omega_{\psi(\Omega_\Omega+\Omega_2)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,1)(3,2,1)(4,1,0) | $\psi(\Omega_\Omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)(3,2,1)<br>(4,1,0)(2,2,1)(3,2,1)(4,1,0) | $\psi(\Omega_\Omega \cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,1,0) | $\psi(\Omega_\Omega \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(3,1,0)(4,2,1)(5,2,1)(6,1,0) | $\psi(\Omega_\Omega \cdot \psi_1(\Omega_\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0) | $\psi(\Omega_\Omega \cdot \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(2,2,1) | $\psi(\Omega_\Omega \cdot \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,0)(2,2,1)(3,2,1) | $\psi(\Omega_\Omega \cdot \Omega_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,0)(2,2,1)(3,2,1)(4,1,0) | $\psi({\Omega_\Omega}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(3,2,0)(2,2,1)(3,2,1)(4,1,0) | $\psi({\Omega_\Omega}^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,1,0) | $\psi({\Omega_\Omega}^\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(4,2,0)(2,2,1)(3,2,1)(4,1,0) | $\psi({\Omega_\Omega}^{\Omega_\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,0) | $\psi(\psi_\Omega(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,0)(4,3,0)(4,3,0) | $\psi(\psi_\Omega(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,0)(4,3,0)(5,2,0) | $\psi(\psi_\Omega(\Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,0)<br>(5,2,0)(2,2,1)(3,2,1)(4,1,0) | $\psi(\psi_\Omega(\Omega_\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,0)(4,3,0)(5,2,0)(6,3,0) | $\psi(\psi_\Omega(\psi_\Omega(0)))$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,0)(4,3,0)(5,3,0) | $\psi(\Omega_{\Omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,1) | $\psi(\Omega_{\Omega+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,0)(4,3,1)(5,3,0)(6,4,1) | $\psi(\Omega_{\Omega+\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)<br>(4,1,0)(3,2,0)(4,3,1)(5,3,1) | $\psi(\Omega_{\Omega+\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,0,0)(7,1,0) | $\psi(\Omega_{\Omega+\psi(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,0)(4,3,1)(5,3,1)(6,1,0) | $\psi(\Omega_{\Omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,1)<br>(5,3,1)(6,1,0)(2,2,0)(3,2,0) | $\psi(\Omega_{\Omega\cdot 2}+\Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,1)<br>(5,3,1)(6,1,0)(2,2,1)(3,2,1)(4,1,0) | $\psi(\Omega_{\Omega\cdot 2}+\Omega_\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,1)<br>(5,3,1)(6,1,0)(2,2,1)(3,2,1)<br>(4,1,0)(3,2,0)(4,3,0) | $\psi(\Omega_{\Omega\cdot 2}+\psi_\Omega(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,1)<br>(5,3,1)(6,1,0)(2,2,1)(3,2,1)<br>(4,1,0)(3,2,0)(4,3,0)(5,2,0)<br>(2,2,1)(3,2,1)(4,1,0) | $\psi(\Omega_{\Omega\cdot 2}+\psi_\Omega(\Omega_\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,1)<br>(5,3,1)(6,1,0)(2,2,1)(3,2,1)<br>(4,1,0)(3,2,0)(4,3,0)(5,3,0) | $\psi(\Omega_{\Omega\cdot 2}+\psi_\Omega(\Omega_{\Omega+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,1)<br>(5,3,1)(6,1,0)(2,2,1)(3,2,1)<br>(4,1,0)(3,2,0)(4,3,1)(5,3,1)(6,1,0) | $\psi(\Omega_{\Omega\cdot 2}+\psi_\Omega(\Omega_{\Omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0)(3,1,0) | $\psi(\Omega_{\Omega\cdot 2}+\psi_\Omega(\Omega_{\Omega\cdot 2})\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)(2,2,1)<br>(3,2,1)(4,1,0)(3,2,0)(4,3,1)(5,3,1)<br>(6,1,0)(3,2,0)(2,2,1)(3,2,1)(4,1,0) | $\psi(\Omega_{\Omega\cdot 2}+\psi_\Omega(\Omega_{\Omega\cdot 2})\cdot\Omega_\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0)(3,2,0)<br>(3,2,0)(2,2,1)(3,2,1)(4,1,0) | $\psi(\Omega_{\Omega\cdot 2}+\psi_\Omega(\Omega_{\Omega\cdot 2})\cdot\Omega_\Omega^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0)(3,2,0)(4,3,0) | $\psi(\Omega_{\Omega\cdot 2}+\psi_\Omega(\Omega_{\Omega\cdot 2})\cdot\psi_\Omega(0))$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0) | $\psi(\Omega_{\Omega\cdot 2}+\psi_\Omega(\Omega_{\Omega\cdot 2})^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0)(4,3,0) | $\psi(\Omega_{\Omega\cdot 2}+\psi_\Omega(\Omega_{\Omega\cdot 2}+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,1)<br>(5,3,1)(6,1,0)(4,3,0)(5,3,0) | $\psi(\Omega_{\Omega\cdot 2}+\Omega_{\Omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0)(4,3,1) | $\psi(\Omega_{\Omega\cdot 2}+\Omega_{\Omega+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,1)<br>(5,3,1)(6,1,0)(4,3,1)(5,3,1)(6,1,0) | $\psi(\Omega_{\Omega\cdot 2}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0)(5,2,0) | $\psi(\Omega_{\Omega\cdot 2}\cdot \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0)(5,2,0)<br>(2,2,1)(3,2,1)(4,1,0) | $\psi(\Omega_{\Omega\cdot 2}\cdot \Omega_\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0)(5,3,0)<br>(4,3,1)(5,3,1)(6,1,0) | $\psi({\Omega_{\Omega\cdot 2}}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0)(5,3,0)<br>(6,3,0)(4,3,1)(5,3,1)(6,1,0) | $\psi({\Omega_{\Omega\cdot 2}}^{\Omega_{\Omega\cdot 2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,1)<br>(5,3,1)(6,1,0)(5,3,0)(6,4,0) | $\psi(\psi_{\Omega\cdot 2}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,1)<br>(5,3,1)(6,1,0)(5,3,0)(6,4,0)(7,4,0) | $\psi(\Omega_{\Omega\cdot 2+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,1)<br>(5,3,1)(6,1,0)(5,3,0)(6,4,1) | $\psi(\Omega_{\Omega\cdot 2+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0)<br>(5,3,0)(6,4,1)(7,4,1) | $\psi(\Omega_{\Omega\cdot 2+\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0)(5,3,0)<br>(6,4,1)(7,4,1)(8,1,0) | $\psi(\Omega_{\Omega\cdot 3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)(3,2,1) | $\psi(\Omega_{\Omega\cdot\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,1)(3,2,1) | $\psi(\Omega_{\Omega\cdot\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,1)(4,1,0) | $\psi(\Omega_{\Omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(4,0,0) | $\psi(\Omega_{\Omega^\omega})$ |

| BMS | Madore's OCF |
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| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(4,1,0) | $\psi(\Omega_{\Omega^\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(4,1,0)(3,2,1)(4,1,0) | $\psi(\Omega_{\Omega^{\Omega+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(4,1,0)(3,2,1)(4,1,0)(4,1,0) | $\psi(\Omega_{\Omega^{\Omega\cdot 2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(4,1,0)(4,1,0) | $\psi(\Omega_{\Omega^{\Omega^2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(5,1,0) | $\psi(\Omega_{\Omega^{\Omega^\Omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(5,2,0) | $\psi(\Omega_{\psi_1(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(5,2,0)(6,2,0) | $\psi(\Omega_{\psi_1(\Omega_2)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(5,2,1) | $\psi(\Omega_{\psi_1(\Omega_\omega)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,1,0) | $\psi(\Omega_{\psi_1(\Omega_\Omega)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,1,0)(8,2,0) | $\psi(\Omega_{\psi_1(\Omega_{\psi_1(0)})})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,0) | $\psi(\Omega_{\Omega_2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(2,2,0)(3,2,0) | $\psi(\Omega_{\Omega_2}+\Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(2,2,0)(3,3,1) | $\psi(\Omega_{\Omega_2}+\psi_2(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,0)<br>(2,2,0)(3,3,1)(4,3,1)(5,1,0) | $\psi(\Omega_{\Omega_2}+\psi_2(\Omega_\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,0)<br>(2,2,0)(3,3,1)(4,3,1)(5,2,0) | $\psi(\Omega_{\Omega_2}+\psi_2(\Omega_{\Omega_2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,0)(2,2,0)<br>(3,3,1)(4,3,1)(5,2,0)(3,3,0)(4,3,0) | $\psi(\Omega_{\Omega_2}+\Omega_3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(2,2,0)<br>(3,3,1)(4,3,1)(5,2,0)(3,3,1) | $\psi(\Omega_{\Omega_2}+\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(2,2,0)(3,3,1)<br>(4,3,1)(5,2,0)(3,3,1)(4,3,1)(5,2,0) | $\psi(\Omega_{\Omega_2}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(2,2,0)<br>(3,3,1)(4,3,1)(5,2,0)(4,3,0)<br>(3,3,1)(4,3,1)(5,2,0) | $\psi({\Omega_{\Omega_2}}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(2,2,0)<br>(3,3,1)(4,3,1)(5,2,0)(4,3,0)(5,4,0) | $\psi(\psi_{\Omega_2}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(2,2,0)(3,3,1)<br>(4,3,1)(5,2,0)(4,3,0)(5,4,0)(6,4,0) | $\psi(\Omega_{\Omega_2+1})$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(2,2,0)(3,3,1)<br>(4,3,1)(5,2,0)(4,3,0)(5,4,1) | $\psi(\Omega_{\Omega_2+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(2,2,0)<br>(3,3,1)(4,3,1)(5,2,0)(4,3,0)<br>(5,4,1)(6,4,1)(7,2,0) | $\psi(\Omega_{\Omega_2\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(2,2,0)<br>(3,3,1)(4,3,1)(5,2,0)(4,3,1) | $\psi(\Omega_{\Omega_2\cdot\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(2,2,0)(3,3,1)<br>(4,3,1)(5,2,0)(4,3,1)(5,2,0) | $\psi(\Omega_{\Omega_2{}^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(2,2,0)<br>(3,3,1)(4,3,1)(5,2,0)(5,2,0) | $\psi(\Omega_{\Omega_2{}^{\Omega_2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(2,2,0)<br>(3,3,1)(4,3,1)(5,2,0)(6,3,0) | $\psi(\Omega_{\psi_2(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(2,2,0)<br>(3,3,1)(4,3,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,2,0)(9,3,0) | $\psi(\Omega_{\psi_2(\Omega_{\psi_2(0)})})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,0)<br>(2,2,0)(3,3,1)(4,3,1)(5,3,0) | $\psi(\Omega_{\Omega_3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1) | $\psi(\Omega_{\Omega_\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(1,1,1) | $\psi(\Omega_{\Omega_\omega}+\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,0) | $\psi(\Omega_{\Omega_\omega}+\psi_\omega(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0) | $\psi(\Omega_{\Omega_\omega}+\psi_\omega(\Omega_\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(1,1,1) | $\psi(\Omega_{\Omega_\omega}+\psi_\omega(\Omega_{\Omega_\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\Omega_\omega}+\Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0)(1,1,1) | $\psi(\Omega_{\Omega_\omega}+\psi_{\omega+1}(\Omega_{\Omega_\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(3,2,1) | $\psi(\Omega_{\Omega_\omega}+\Omega_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(1,1,1) | $\psi(\Omega_{\Omega_\omega}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(4,1,0) | $\psi(\Omega_{\Omega_\omega}\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(4,1,0)(1,1,1) | $\psi(\Omega_{\Omega_\omega}\cdot\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(4,1,0)(3,2,1)(4,2,1)(5,1,0)(1,1,1) | $\psi(\Omega_{\Omega_\omega}\cdot\Omega_\omega+\Omega_{\Omega_\omega})$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(4,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,1,0)(1,1,1) | $\psi(\Omega_{\Omega_\omega} \cdot \Omega_\omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(4,1,0)(4,1,0)(1,1,1) | $\psi(\Omega_{\Omega_\omega} \cdot {\Omega_\omega}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(4,1,0)(5,2,0) | $\psi(\Omega_{\Omega_\omega} \cdot \psi_\omega(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(4,2,0) | $\psi(\Omega_{\Omega_\omega} \cdot \Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(4,2,0)(3,2,1) | $\psi(\Omega_{\Omega_\omega} \cdot \Omega_{\omega \cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(4,2,0)(3,2,1)(4,2,1) | $\psi(\Omega_{\Omega_\omega} \cdot \Omega_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(4,2,0)<br>(3,2,1)(4,2,1)(5,1,0)(1,1,1) | $\psi({\Omega_{\Omega_\omega}}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(4,2,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,0)<br>(3,2,1)(4,2,1)(5,1,0)(1,1,1) | $\psi({\Omega_{\Omega_\omega}}^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(4,2,0)(4,1,0)(1,1,1) | $\psi({\Omega_{\Omega_\omega}}^{\Omega_\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(4,2,0)(5,3,0) | $\psi(\psi_{\Omega_\omega}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,1,0)(1,1,1) | $\psi(\psi_{\Omega_\omega}(\Omega_{\Omega_\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,2,0) | $\psi(\Omega_{\Omega_\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(4,2,0)(5,3,1)(6,3,1)<br>(7,2,0)(6,3,0)(7,4,0) | $\psi(\psi_{\Omega_\omega+1}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,2,0)(6,3,0)<br>(7,4,1)(8,4,1)(9,1,0)(1,1,1) | $\psi(\psi_{\Omega_\omega \cdot 2}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,1,0)(4,2,1) | $\psi(\Omega_{\Omega_\omega \cdot \omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(4,2,1)(5,1,0) | $\psi(\Omega_{\Omega_\omega \cdot \Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(4,2,1)(5,1,0)(1,1,1) | $\psi(\Omega_{{\Omega_\omega}^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(5,1,0)(1,1,1) | $\psi(\Omega_{{\Omega_\omega}^{\Omega_\omega}})$ |

| BMS | Madore's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(6,2,0) | $\psi(\Omega_{\psi_\omega(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(6,2,1)(7,2,1)(8,1,0)(1,1,1) | $\psi(\Omega_{\psi_\omega(\Omega_{\Omega_\omega})})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0) | $\psi(\Omega_{\Omega_{\omega+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(3,2,0)(4,3,1)(5,3,1)(6,3,0) | $\psi(\Omega_{\Omega_{\omega+2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(3,2,1) | $\psi(\Omega_{\Omega_{\omega\cdot 2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(3,2,1)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,0)(5,3,1) | $\psi(\Omega_{\Omega_{\omega\cdot 3}})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1) | $\psi(\Omega_{\Omega_{\omega 2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(2,1,1) | $\psi(\Omega_{\Omega_{\omega 3}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,0,0) | $\psi(\Omega_{\Omega_{\omega\omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0) | $\psi(\Omega_{\Omega_\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(1,1,1) | $\psi(\Omega_{\Omega_{\Omega_\omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,0) | $\psi(\Omega_{\Omega_{\Omega_\Omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_I(0))$ |

## A.9 BMS vs BOCF

The results in this section are primarily cited from[2] 。

| BMS | Buchholz's OCF |
|---|---|
| (0) | $\psi(0)$ |
| (0)(0) | $\psi(0)\cdot 2$ |
| (0)(0)(0) | $\psi(0)\cdot 3$ |
| (0)(1) | $\psi(1)$ |
| (0)(1)(0) | $\psi(1)+\psi(0)$ |
| (0)(1)(0)(1) | $\psi(1)\cdot 2$ |
| (0)(1)(0)(1)(0)(1) | $\psi(1)\cdot 3$ |
| (0)(1)(1) | $\psi(2)$ |
| (0)(1)(1)(0)(1) | $\psi(2)+\psi(1)$ |
| (0)(1)(1)(0)(1)(1) | $\psi(2)\cdot 2$ |
| (0)(1)(1)(1) | $\psi(3)$ |
| (0)(1)(1)(1)(1) | $\psi(4)$ |
| (0)(1)(2) | $\psi(\psi(1))$ |
| (0)(1)(2)(1) | $\psi(\psi(1)+\psi(0))$ |
| (0)(1)(2)(1)(2) | $\psi(\psi(1)\cdot 2)$ |
| (0)(1)(2)(2) | $\psi(\psi(2))$ |
| (0)(1)(2)(2)(2) | $\psi(\psi(3))$ |
| (0)(1)(2)(3) | $\psi(\psi(\psi(1)))$ |
| (0)(1)(2)(3)(1) | $\psi(\psi(\psi(1))+\psi(0))$ |
| (0)(1)(2)(3)(2) | $\psi(\psi(\psi(1)+\psi(0)))$ |
| (0)(1)(2)(3)(3) | $\psi(\psi(\psi(2)))$ |

| BMS | Buchholz's OCF |
|---|---|
| (0)(1)(2)(3)(4) | $\psi(\psi(\psi(\psi(1))))$ |
| (0,0)(1,1) | $\psi(\Omega)$ |
| (0,0)(1,1)(0,0) | $\psi(\Omega)+1$ |
| (0,0)(1,1)(0,0)(1,1) | $\psi(\Omega)\cdot 2$ |
| (0,0)(1,1)(1,0) | $\psi(\Omega+1)$ |
| (0,0)(1,1)(1,0)(2,0) | $\psi(\Omega+\omega)$ |
| (0,0)(1,1)(1,0)(2,1) | $\psi(\Omega+\psi(\Omega))$ |
| (0,0)(1,1)(1,0)(2,1)(1,0) | $\psi(\Omega+\psi(\Omega)+1)$ |
| (0,0)(1,1)(1,0)(2,1)(2,0) | $\psi(\Omega+\psi(\Omega+1))$ |
| (0,0)(1,1)(1,0)(2,1)(2,0)(3,1) | $\psi(\Omega+\psi(\Omega+\psi(\Omega)))$ |
| (0,0)(1,1)(1,1) | $\psi(\Omega\cdot 2)$ |
| (0,0)(1,1)(1,1)(1,0) | $\psi(\Omega\cdot 2+1)$ |
| (0,0)(1,1)(1,1)(1,0)(2,0) | $\psi(\Omega\cdot 2+\omega)$ |
| (0,0)(1,1)(1,1)(1,0)(2,1) | $\psi(\Omega\cdot 2+\psi(\Omega))$ |
| (0,0)(1,1)(1,1)(1,0)(2,1)(2,1) | $\psi(\Omega\cdot 2+\psi(\Omega\cdot 2))$ |
| (0,0)(1,1)(1,1)(1,0)(2,1)(2,1)(2,0) | $\psi(\Omega\cdot 2+\psi(\Omega\cdot 2+1))$ |
| (0,0)(1,1)(1,1)(1,0)<br>(2,1)(2,1)(2,0)(3,1)(3,1) | $\psi(\Omega\cdot 2+\psi(\Omega\cdot 2+\psi(\Omega\cdot 2)))$ |
| (0,0)(1,1)(1,1)(1,1) | $\psi(\Omega\cdot 3)$ |
| (0,0)(1,1)(1,1)(1,1)<br>(1,0)(2,1)(2,1)(2,1) | $\psi(\Omega\cdot 3+\psi(\Omega\cdot 3))$ |
| (0,0)(1,1)(1,1)(1,1)(1,1) | $\psi(\Omega\cdot 4)$ |
| (0,0)(1,1)(1,1)(1,1)(1,1)(1,1) | $\psi(\Omega\cdot 5)$ |
| (0,0)(1,1)(2,0) | $\psi(\Omega\cdot\omega)$ |
| (0,0)(1,1)(2,0)(1,0) | $\psi(\Omega\cdot\omega+1)$ |
| (0,0)(1,1)(2,0)(1,0)(2,0) | $\psi(\Omega\cdot\omega+\omega)$ |
| (0,0)(1,1)(2,0)(1,0)(2,1) | $\psi(\Omega\cdot\omega+\psi(\Omega))$ |
| (0,0)(1,1)(2,0)(1,0)(2,1)(2,1) | $\psi(\Omega\cdot\omega+\psi(\Omega\cdot 2))$ |
| (0,0)(1,1)(2,0)(1,0)(2,1)(3,0) | $\psi(\Omega\cdot\omega+\psi(\Omega\cdot\omega))$ |
| (0,0)(1,1)(2,0)(1,0)(2,1)(3,0)(2,0) | $\psi(\Omega\cdot\omega+\psi(\Omega\cdot\omega+1))$ |
| (0,0)(1,1)(2,0)(1,0)<br>(2,1)(3,0)(2,0)(3,1) | $\psi(\Omega\cdot\omega+\psi(\Omega\cdot\omega+\psi(\Omega)))$ |
| (0,0)(1,1)(2,0)(1,1) | $\psi(\Omega\cdot(\omega+1))$ |
| (0,0)(1,1)(2,0)(1,1)(1,0) | $\psi(\Omega\cdot(\omega+1)+1)$ |
| (0,0)(1,1)(2,0)(1,1)(1,0)(2,1) | $\psi(\Omega\cdot(\omega+1)+\psi(\Omega))$ |
| (0,0)(1,1)(2,0)(1,1)<br>(1,0)(2,1)(3,0)(2,1) | $\psi(\Omega\cdot(\omega+1)+\psi(\Omega\cdot(\omega+1)))$ |
| (0,0)(1,1)(2,0)(1,1)(1,1) | $\psi(\Omega\cdot(\omega+2))$ |
| (0,0)(1,1)(2,0)(1,1)(1,1)(1,1) | $\psi(\Omega\cdot(\omega+3))$ |
| (0,0)(1,1)(2,0)(1,1)(2,0) | $\psi(\Omega\cdot(\omega\cdot 2))$ |
| (0,0)(1,1)(2,0)(1,1)(2,0)(1,1) | $\psi(\Omega\cdot(\omega\cdot 2+1))$ |
| (0,0)(1,1)(2,0)(1,1)(2,0)(1,1)(2,0) | $\psi(\Omega\cdot(\omega\cdot 3))$ |
| (0,0)(1,1)(2,0)(2,0) | $\psi(\Omega\cdot\omega^2)$ |
| (0,0)(1,1)(2,0)(2,0)(1,1) | $\psi(\Omega\cdot(\omega^2+1))$ |
| (0,0)(1,1)(2,0)(2,0)(1,1)(2,0)(2,0) | $\psi(\Omega\cdot\omega^2\cdot 2)$ |
| (0,0)(1,1)(2,0)(2,0)(2,0) | $\psi(\Omega\cdot\omega^3)$ |
| (0,0)(1,1)(2,0)(3,0) | $\psi(\Omega\cdot\omega^\omega)$ |
| (0,0)(1,1)(2,0)(3,0)(4,0) | $\psi(\Omega\cdot\omega^{\omega^\omega})$ |
| (0,0)(1,1)(2,0)(3,1) | $\psi(\Omega\cdot\psi(\Omega))$ |
| (0,0)(1,1)(2,0)(3,1)(3,1) | $\psi(\Omega\cdot\psi(\Omega\cdot 2))$ |
| (0,0)(1,1)(2,0)(3,1)(4,0) | $\psi(\Omega\cdot\psi(\Omega\cdot\omega))$ |
| (0,0)(1,1)(2,0)(3,1)(4,0)(5,1) | $\psi(\Omega\cdot\psi(\Omega\cdot\psi(\Omega)))$ |
| (0,0)(1,1)(2,1) | $\psi(\Omega^2)$ |
| (0,0)(1,1)(2,1)(1,0) | $\psi(\Omega^2+1)$ |
| (0,0)(1,1)(2,1)(1,0)(2,1) | $\psi(\Omega^2+\psi(\Omega))$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0)(1,1)(2,1)(1,0)(2,1)(3,1) | $\psi(\Omega^2 + \psi(\Omega^2))$ |
| (0,0)(1,1)(2,1)(1,1) | $\psi(\Omega^2 + \Omega)$ |
| (0,0)(1,1)(2,1)(1,1)(1,1) | $\psi(\Omega^2 + \Omega \cdot 2)$ |
| (0,0)(1,1)(2,1)(1,1)(2,0) | $\psi(\Omega^2 + \Omega \cdot \omega)$ |
| (0,0)(1,1)(2,1)(1,1)(2,0)(3,1) | $\psi(\Omega^2 + \Omega \cdot \psi(\Omega))$ |
| (0,0)(1,1)(2,1)(1,1)(2,0)(3,1)(4,1) | $\psi(\Omega^2 + \Omega \cdot \psi(\Omega^2))$ |
| (0,0)(1,1)(2,1)(1,1)(2,1) | $\psi(\Omega^2 \cdot 2)$ |
| (0,0)(1,1)(2,1)(1,1)(2,1)(1,1) | $\psi(\Omega^2 \cdot 2 + \Omega)$ |
| (0,0)(1,1)(2,1)(1,1)(2,1)(1,1)(2,0) | $\psi(\Omega^2 \cdot 2 + \Omega \cdot \omega)$ |
| (0,0)(1,1)(2,1)(1,1)(2,1)(1,1)(2,1) | $\psi(\Omega^2 \cdot 3)$ |
| (0,0)(1,1)(2,1)(2,0) | $\psi(\Omega^2 \cdot \omega)$ |
| (0,0)(1,1)(2,1)(2,0)(1,1) | $\psi(\Omega^2 \cdot \omega + \Omega)$ |
| (0,0)(1,1)(2,1)(2,0)(1,1)(2,1) | $\psi(\Omega^2 \cdot (\omega + 1))$ |
| (0,0)(1,1)(2,1)(2,0)(2,0) | $\psi(\Omega^2 \cdot \omega \cdot 2)$ |
| (0,0)(1,1)(2,1)(2,0)(3,0) | $\psi(\Omega^2 \cdot \omega^2)$ |
| (0,0)(1,1)(2,1)(2,0)(3,0)(4,0) | $\psi(\Omega^2 \cdot \omega^\omega)$ |
| (0,0)(1,1)(2,1)(2,0)(3,1) | $\psi(\Omega^2 \cdot \psi(\Omega))$ |
| (0,0)(1,1)(2,1)(2,0)(3,1)(3,1) | $\psi(\Omega^2 \cdot \psi(\Omega \cdot 2))$ |
| (0,0)(1,1)(2,1)(2,0)(3,1)(4,1) | $\psi(\Omega^2 \cdot \psi(\Omega^2))$ |
| (0,0)(1,1)(2,1)(2,0)(3,1)(4,1)(4,0) | $\psi(\Omega^2 \cdot \psi(\Omega^2 \cdot \omega))$ |
| (0,0)(1,1)(2,1)(2,1) | $\psi(\Omega^3)$ |
| (0,0)(1,1)(2,1)(2,1)(1,0) | $\psi(\Omega^3 + 1)$ |
| (0,0)(1,1)(2,1)(2,1)(1,0)(2,1) | $\psi(\Omega^3 + \psi(\Omega))$ |
| (0,0)(1,1)(2,1)(2,1)<br>(1,0)(2,1)(3,1)(3,1) | $\psi(\Omega^3 + \psi(\Omega^3))$ |
| (0,0)(1,1)(2,1)(2,1)(1,1) | $\psi(\Omega^3 + \Omega)$ |
| (0,0)(1,1)(2,1)(2,1)(1,1)(1,1) | $\psi(\Omega^3 + \Omega \cdot 2)$ |
| (0,0)(1,1)(2,1)(2,1)(1,1)(2,0) | $\psi(\Omega^3 + \Omega \cdot \omega)$ |
| (0,0)(1,1)(2,1)(2,1)(1,1)(2,0)(3,1) | $\psi(\Omega^3 + \Omega \cdot \psi(\Omega))$ |
| (0,0)(1,1)(2,1)(2,1)(1,1)<br>(2,0)(3,1)(4,1)(4,1) | $\psi(\Omega^3 + \Omega \cdot \psi(\Omega^3))$ |
| (0,0)(1,1)(2,1)(2,1)(1,1)(2,1) | $\psi(\Omega^3 + \Omega^2)$ |
| (0,0)(1,1)(2,1)(2,1)(1,1)(2,1)(2,1) | $\psi(\Omega^3 \cdot 2)$ |
| (0,0)(1,1)(2,1)(2,1)(2,0) | $\psi(\Omega^3 \cdot \omega)$ |
| (0,0)(1,1)(2,1)(2,1)<br>(2,0)(1,1)(2,1)(2,1) | $\psi(\Omega^3 \cdot (\omega + 1))$ |
| (0,0)(1,1)(2,1)(2,1)(2,0)(2,0) | $\psi(\Omega^3 \cdot \omega \cdot 2)$ |
| (0,0)(1,1)(2,1)(2,1)(2,0)(3,0) | $\psi(\Omega^3 \cdot \omega^2)$ |
| (0,0)(1,1)(2,1)(2,1)(2,0)(3,1) | $\psi(\Omega^3 \cdot \psi(\Omega))$ |
| (0,0)(1,1)(2,1)(2,1)<br>(2,0)(3,1)(4,1)(4,1) | $\psi(\Omega^3 \cdot \psi(\Omega^3))$ |
| (0,0)(1,1)(2,1)(2,1)(2,1) | $\psi(\Omega^4)$ |
| (0,0)(1,1)(2,1)(2,1)(2,1)(1,1) | $\psi(\Omega^4 + \Omega)$ |
| (0,0)(1,1)(2,1)(2,1)(2,1)(2,0) | $\psi(\Omega^4 \cdot \omega)$ |
| (0,0)(1,1)(2,1)(2,1)(2,1)(2,1) | $\psi(\Omega^5)$ |
| (0,0)(1,1)(2,1)(3,0) | $\psi(\Omega^\omega)$ |
| (0,0)(1,1)(2,1)(3,0)(1,0) | $\psi(\Omega^\omega + 1)$ |
| (0,0)(1,1)(2,1)(3,0)(1,0)(2,0) | $\psi(\Omega^\omega + \omega)$ |
| (0,0)(1,1)(2,1)(3,0)(1,0)(2,1) | $\psi(\Omega^\omega + \psi(\Omega))$ |
| (0,0)(1,1)(2,1)(3,0)(1,0)(2,1)(3,1) | $\psi(\Omega^\omega + \psi(\Omega^2))$ |
| (0,0)(1,1)(2,1)(3,0)<br>(1,0)(2,1)(3,1)(4,0) | $\psi(\Omega^\omega + \psi(\Omega^2))$ |
| (0,0)(1,1)(2,1)(3,0)(1,1) | $\psi(\Omega^\omega + \Omega)$ |
| (0,0)(1,1)(2,1)(3,0)(1,1)(1,1) | $\psi(\Omega^\omega + \Omega \cdot 2)$ |
| (0,0)(1,1)(2,1)(3,0)(1,1)(2,0) | $\psi(\Omega^\omega + \Omega \cdot \omega)$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0)(1,1)(2,1)(3,0)(1,1)(2,0)(3,1) | $\psi(\Omega^\omega + \Omega \cdot \psi(\Omega))$ |
| (0,0)(1,1)(2,1)(3,0)(1,1)(2,1) | $\psi(\Omega^\omega + \Omega^2)$ |
| (0,0)(1,1)(2,1)(3,0)(1,1)(2,1)(3,0) | $\psi(\Omega^\omega \cdot 2)$ |
| (0,0)(1,1)(2,1)(3,0)(2,0) | $\psi(\Omega^\omega \cdot \omega)$ |
| (0,0)(1,1)(2,1)(3,0)(2,0)(3,1) | $\psi(\Omega^\omega \cdot \psi(\Omega))$ |
| (0,0)(1,1)(2,1)(3,0)(2,1) | $\psi(\Omega^{\omega+1})$ |
| (0,0)(1,1)(2,1)(3,0)(2,1)(2,0) | $\psi(\Omega^{\omega+1} \cdot \omega)$ |
| (0,0)(1,1)(2,1)(3,0)(2,1)(2,1) | $\psi(\Omega^{\omega+2})$ |
| (0,0)(1,1)(2,1)(3,0)(2,1)(3,0) | $\psi(\Omega^{\omega \cdot 2})$ |
| (0,0)(1,1)(2,1)(3,0)(2,1)(3,0)(2,1) | $\psi(\Omega^{\omega \cdot 2+1})$ |
| (0,0)(1,1)(2,1)(3,0)(3,0) | $\psi(\Omega^{\omega \cdot 3})$ |
| (0,0)(1,1)(2,1)(3,0)(4,0) | $\psi(\Omega^{\omega^2})$ |
| (0,0)(1,1)(2,1)(3,0)(4,1) | $\psi(\Omega^{\psi(\Omega)})$ |
| (0,0)(1,1)(2,1)(3,0)(4,1)(5,1) | $\psi(\Omega^{\psi(\Omega^2)})$ |
| (0,0)(1,1)(2,1)(3,0)(4,1)(5,1)(6,0) | $\psi(\Omega^{\psi(\Omega^\omega)})$ |
| (0,0)(1,1)(2,1)(3,0)<br>(4,1)(5,1)(6,0)(7,1)(8,1) | $\psi(\Omega^{\psi(\Omega^{\psi(\Omega^\omega)})})$ |
| (0,0)(1,1)(2,1)(3,1) | $\psi(\Omega^\Omega)$ |
| (0,0)(1,1)(2,1)(3,1)(1,0) | $\psi(\Omega^\Omega + 1)$ |
| (0,0)(1,1)(2,1)(3,1)(1,0)(2,0) | $\psi(\Omega^\Omega + \omega)$ |
| (0,0)(1,1)(2,1)(3,1)(1,0)(2,1) | $\psi(\Omega^\Omega + \psi(\Omega))$ |
| (0,0)(1,1)(2,1)(3,1)(1,0)(2,1)(3,1) | $\psi(\Omega^\Omega + \psi(\Omega^2))$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,0)(2,1)(3,1)(4,0) | $\psi(\Omega^\Omega + \psi(\Omega^2))$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,0)(2,1)(3,1)(4,1) | $\psi(\Omega^\Omega + \psi(\Omega^\Omega))$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,0)(2,1)(3,1)(4,1)(2,0) | $\psi(\Omega^\Omega + \psi(\Omega^\Omega + 1))$ |
| (0,0)(1,1)(2,1)(3,1)(1,0)<br>(2,1)(3,1)(4,1)(2,0)(3,1) | $\psi(\Omega^\Omega + \psi(\Omega^\Omega + \psi(\Omega)))$ |
| (0,0)(1,1)(2,1)(3,1)(1,0)<br>(2,1)(3,1)(4,1)(2,0)(3,1)(4,1) | $\psi(\Omega^\Omega + \psi(\Omega^\Omega + \psi(\Omega^2)))$ |
| (0,0)(1,1)(2,1)(3,1)(1,0)(2,1)<br>(3,1)(4,1)(2,0)(3,1)(4,1)(5,0) | $\psi(\Omega^\Omega + \psi(\Omega^\Omega + \psi(\Omega^\omega)))$ |
| (0,0)(1,1)(2,1)(3,1)(1,0)(2,1)<br>(3,1)(4,1)(2,0)(3,1)(4,1)(5,1) | $\psi(\Omega^\Omega + \psi(\Omega^\Omega + \psi(\Omega^\Omega)))$ |
| (0,0)(1,1)(2,1)(3,1)(1,1) | $\psi(\Omega^\Omega + \Omega)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(1,0) | $\psi(\Omega^\Omega + \Omega + 1)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(1,0)(2,1) | $\psi(\Omega^\Omega + \Omega + \psi(\Omega))$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(1,0)(2,1)(3,1)(4,1) | $\psi(\Omega^\Omega + \Omega + \psi(\Omega^\Omega))$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(1,1) | $\psi(\Omega^\Omega + \Omega \cdot 2)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,0) | $\psi(\Omega^\Omega + \Omega \cdot \omega)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,0)(3,1) | $\psi(\Omega^\Omega + \Omega \cdot \psi(\Omega))$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,0)(3,1)(4,1)(5,1) | $\psi(\Omega^\Omega + \Omega \cdot \psi(\Omega^\Omega))$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1) | $\psi(\Omega^\Omega + \Omega^2)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(1,0)(2,1) | $\psi(\Omega^\Omega + \Omega^2 + \psi(\Omega))$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(1,0)(2,1)(3,1)(4,1) | $\psi(\Omega^\Omega + \Omega^2 + \psi(\Omega^\Omega))$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)(1,1) | $\psi(\Omega^\Omega + \Omega^2 + \Omega)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(1,1)(1,1) | $\psi(\Omega^\Omega + \Omega^2 + \Omega \cdot 2)$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(1,1)(2,1) | $\psi(\Omega^\Omega + \Omega^2 \cdot 2)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(1,1)(2,1)(1,1)(2,1) | $\psi(\Omega^\Omega + \Omega^2 \cdot 3)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)(2,0) | $\psi(\Omega^\Omega + \Omega^2 \cdot \omega)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(2,0)(3,1) | $\psi(\Omega^\Omega + \Omega^2 \cdot \psi(\Omega))$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(2,0)(3,1)(4,1)(5,1) | $\psi(\Omega^\Omega + \Omega^2 \cdot \psi(\Omega^\Omega))$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)(2,1) | $\psi(\Omega^\Omega + \Omega^3)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(2,1)(1,1) | $\psi(\Omega^\Omega + \Omega^3 + \Omega)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(2,1)(1,1)(2,1)(2,1) | $\psi(\Omega^\Omega + \Omega^3 \cdot 2)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(2,1)(2,0) | $\psi(\Omega^\Omega + \Omega^3 \cdot \omega)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(2,1)(2,1) | $\psi(\Omega^\Omega + \Omega^4)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)(3,0) | $\psi(\Omega^\Omega + \Omega^\omega)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(1,1)(2,1)(3,0) | $\psi(\Omega^\Omega + \Omega^\omega \cdot 2)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(3,0)(2,0) | $\psi(\Omega^\Omega + \Omega^\omega \cdot \omega)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(3,0)(2,1) | $\psi(\Omega^\Omega + \Omega^{\omega+1})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(2,1)(3,0) | $\psi(\Omega^\Omega + \Omega^{\omega\cdot 2})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(3,0)(3,0) | $\psi(\Omega^\Omega + \Omega^{\omega^2})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(3,0)(4,1) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega)})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1)(2,0) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)} \cdot \omega)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1)(2,1) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)+1})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1)(3,0) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)+\omega})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1)(4,0) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega)\cdot\omega})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1)(4,1) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+\Omega)})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(4,1)<br>(5,1)(6,0)(7,1)(8,1)(9,1) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+\Omega^{\psi(\Omega^\Omega)})})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(4,1)(5,1)<br>(6,0)(7,1)(8,1)(9,1)(5,1) | $\psi(\Omega^\Omega + \Omega^{\psi(\Omega^\Omega+\Omega^{\psi(\Omega^\Omega)+1})})$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)(3,1) | $\psi(\Omega^\Omega \cdot 2)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(3,1)(1,1) | $\psi(\Omega^\Omega \cdot 2 + \Omega)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,1)(1,1)(2,1) | $\psi(\Omega^\Omega \cdot 2 + \Omega^2)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,1)(1,1)(2,1)(2,1) | $\psi(\Omega^\Omega \cdot 2 + \Omega^3)$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,1)(1,1)(2,1)(3,1) | $\psi(\Omega^\Omega \cdot 3)$ |
| (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,1)(1,1)(2,1)(3,1)(1,1)(2,1)(3,1) | $\psi(\Omega^\Omega \cdot 4)$ |
| (0,0)(1,1)(2,1)(3,1)(2,0) | $\psi(\Omega^\Omega \cdot \omega)$ |
| (0,0)(1,1)(2,1)(3,1)(2,0)(3,1) | $\psi(\Omega^\Omega \cdot \psi(\Omega))$ |
| (0,0)(1,1)(2,1)(3,1)(2,0)(3,1)(4,1) | $\psi(\Omega^\Omega \cdot \psi(\Omega^2))$ |
| (0,0)(1,1)(2,1)(3,1)<br>(2,0)(3,1)(4,1)(5,1) | $\psi(\Omega^\Omega \cdot \psi(\Omega^\Omega))$ |
| (0,0)(1,1)(2,1)(3,1)(2,0)(3,1)<br>(4,1)(5,1)(4,0)(5,1)(6,1)(7,1) | $\psi(\Omega^\Omega \cdot \psi(\Omega^\Omega \cdot \psi(\Omega^\Omega)))$ |
| (0,0)(1,1)(2,1)(3,1)(2,1) | $\psi(\Omega^{\omega+1})$ |
| (0,0)(1,1)(2,1)(3,1)(2,1)(1,1) | $\psi(\Omega^{\omega+1} + \Omega)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(2,1)(1,1)(2,1)(3,1) | $\psi(\Omega^{\omega+1} + \Omega^\Omega)$ |
| (0,0)(1,1)(2,1)(3,1)(2,1)<br>(1,1)(2,1)(3,1)(2,1) | $\psi(\Omega^{\omega+1} \cdot 2)$ |
| (0,0)(1,1)(2,1)(3,1)(2,1)(2,0) | $\psi(\Omega^{\omega+1} \cdot \omega)$ |
| (0,0)(1,1)(2,1)(3,1)(2,1)(2,0)(3,1) | $\psi(\Omega^{\omega+1} \cdot \psi(\Omega))$ |
| (0,0)(1,1)(2,1)(3,1)<br>(2,1)(2,0)(3,1)(4,1)(5,1) | $\psi(\Omega^{\omega+1} \cdot \psi(\Omega^\Omega))$ |
| (0,0)(1,1)(2,1)(3,1)(2,1)(2,1) | $\psi(\Omega^{\omega+2})$ |
| (0,0)(1,1)(2,1)(3,1)(2,1)(3,0) | $\psi(\Omega^{\Omega+\omega})$ |
| (0,0)(1,1)(2,1)(3,1)(2,1)(3,0)(4,1) | $\psi(\Omega^{\Omega+\psi(\Omega)})$ |
| (0,0)(1,1)(2,1)(3,1)(2,1)(3,1) | $\psi(\Omega^{\omega\cdot 2})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(2,1)(3,1)(2,1)(3,1) | $\psi(\Omega^{\omega\cdot 3})$ |
| (0,0)(1,1)(2,1)(3,1)(3,0) | $\psi(\Omega^{\Omega\cdot\omega})$ |
| (0,0)(1,1)(2,1)(3,1)(3,0)(4,1) | $\psi(\Omega^{\Omega\cdot\psi(\Omega)})$ |
| (0,0)(1,1)(2,1)(3,1)(3,1) | $\psi(\Omega^{\Omega^2})$ |
| (0,0)(1,1)(2,1)(3,1)(3,1)(1,1) | $\psi(\Omega^{\Omega^2} + \Omega)$ |
| (0,0)(1,1)(2,1)(3,1)(3,1)<br>(1,1)(2,1)(3,0)(4,1)(5,1)(6,1)(6,1) | $\psi(\Omega^{\Omega^2} + \Omega^{\psi(\Omega^{\Omega^2})})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(3,1)(1,1)(2,1)(3,1) | $\psi(\Omega^{\Omega^2} + \Omega^\Omega)$ |
| (0,0)(1,1)(2,1)(3,1)(3,1)(1,1)<br>(2,1)(3,1)(3,0)(4,1)(5,1)(6,1)(6,1) | $\psi(\Omega^{\Omega^2} + \Omega^{\Omega\cdot\psi(\Omega^{\Omega^2})})$ |
| (0,0)(1,1)(2,1)(3,1)(3,1)<br>(1,1)(2,1)(3,1)(3,1) | $\psi(\Omega^{\Omega^2} \cdot 2)$ |
| (0,0)(1,1)(2,1)(3,1)(3,1)<br>(2,0)(3,1)(4,1)(5,1)(5,1) | $\psi(\Omega^{\Omega^2} \cdot \psi(\Omega^{\Omega^2}))$ |
| (0,0)(1,1)(2,1)(3,1)(3,1)(2,1) | $\psi(\Omega^{\Omega^2+1})$ |
| (0,0)(1,1)(2,1)(3,1)(3,1)(2,1)(3,1) | $\psi(\Omega^{\Omega^2+\Omega})$ |
| (0,0)(1,1)(2,1)(3,1)(3,1)<br>(2,1)(3,1)(4,0)(5,1)(6,1)(7,1)(7,1) | $\psi(\Omega^{\Omega^2+\Omega\cdot\psi(\Omega^{\Omega^2})})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(3,1)(2,1)(3,1)(3,1) | $\psi(\Omega^{\Omega^2\cdot 2})$ |
| (0,0)(1,1)(2,1)(3,1)(3,1)(3,0) | $\psi(\Omega^{\Omega^2\cdot\omega})$ |
| (0,0)(1,1)(2,1)(3,1)(3,1)<br>(3,0)(4,1)(5,1)(6,1)(6,1) | $\psi(\Omega^{\Omega^2\cdot\psi(\Omega^{\Omega^2})})$ |
| (0,0)(1,1)(2,1)(3,1)(3,1)(3,1) | $\psi(\Omega^{\Omega^3})$ |
| (0,0)(1,1)(2,1)(3,1)(3,1)(3,1)(3,1) | $\psi(\Omega^{\Omega^4})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0) | $\psi(\Omega^{\Omega^\omega})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(1,1) | $\psi(\Omega^{\Omega^\omega} + \Omega)$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0)(1,1)(2,1)(3,1)(4,0)(1,1)(2,1) | $\psi(\Omega^{\Omega^\omega}+\Omega^2)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,0)(1,1)(2,1)(2,1) | $\psi(\Omega^{\Omega^\omega}+\Omega^3)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,0)(1,1)(2,1)(3,0) | $\psi(\Omega^{\Omega^\omega}+\Omega^\omega)$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)<br>(1,1)(2,1)(3,0)(4,1)(5,1)(6,1)(7,0) | $\psi(\Omega^{\Omega^\omega}+\Omega^{\psi(\Omega^{\Omega^\omega})})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,0)(1,1)(2,1)(3,1) | $\psi(\Omega^{\Omega^\omega}+\Omega^\Omega)$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,0)(1,1)(2,1)(3,1)(4,0) | $\psi(\Omega^{\Omega^\omega}\cdot 2)$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(2,0) | $\psi(\Omega^{\Omega^\omega}\cdot\omega)$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(2,1) | $\psi(\Omega^{\Omega^\omega+1})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(2,1)(3,0) | $\psi(\Omega^{\Omega^\omega+\omega})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(2,1)(3,1) | $\psi(\Omega^{\Omega^\omega+\Omega})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)<br>(2,1)(3,1)(3,0)(4,1)(5,1)(6,1)(7,0) | $\psi(\Omega^{\Omega^\omega+\Omega\cdot\psi(\Omega^{\Omega^\omega})})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,0)(2,1)(3,1)(3,1) | $\psi(\Omega^{\Omega^\omega+\Omega^2})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,0)(2,1)(3,1)(4,0) | $\psi(\Omega^{\Omega^\omega\cdot 2})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(3,0) | $\psi(\Omega^{\Omega^\omega\cdot\omega})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)<br>(3,0)(4,1)(5,1)(6,1)(7,0) | $\psi(\Omega^{\Omega^\omega\cdot\psi(\Omega^{\Omega^\omega})})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(3,1) | $\psi(\Omega^{\Omega^{\omega+1}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(3,1)(3,1) | $\psi(\Omega^{\Omega^{\omega+2}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(3,1)(4,0) | $\psi(\Omega^{\Omega^{\omega\cdot 2}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(4,0) | $\psi(\Omega^{\Omega^{\omega^2}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(5,0) | $\psi(\Omega^{\Omega^{\omega^\omega}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(5,1) | $\psi(\Omega^{\Omega^{\psi(\Omega)}})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,0)(5,1)(6,1)(7,1) | $\psi(\Omega^{\Omega^{\psi(\Omega^\Omega)}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)<br>(5,1)(6,1)(7,1)(8,0) | $\psi(\Omega^{\Omega^\psi(\Omega^{\Omega^\omega})})$ |
| (0,0)(1,1)(2,1)(3,1)(4,0)(5,1)(6,1)<br>(7,1)(8,0)(9,1)(11,1)(12,1)(13) | $\psi(\Omega^{\Omega^{\psi(\Omega^{\Omega^{\psi(\Omega^{\Omega^\omega})}})}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1) | $\psi(\Omega^{\Omega^\Omega})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(1,1) | $\psi(\Omega^{\Omega^\Omega}+\Omega)$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1)(7,1) | $\psi(\Omega^{\Omega^\Omega}+\Omega^{\psi(\Omega^{\Omega^\Omega})})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(1,1)(2,1)(3,1) | $\psi(\Omega^{\Omega^\Omega}+\Omega^\Omega)$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(1,1)(2,1)(3,1)(4,0) | $\psi(\Omega^{\Omega^\Omega}+\Omega^{\Omega^\omega})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(1,1)<br>(2,1)(3,1)(4,0)(5,1)(6,1)(7,1)(8,1) | $\psi(\Omega^{\Omega^\Omega}+\Omega^{\psi(\Omega^{\Omega^\Omega})})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)<br>(1,1)(2,1)(3,1)(4,1) | $\psi(\Omega^{\Omega^\Omega}\cdot 2)$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(2,0) | $\psi(\Omega^{\Omega^\Omega}\cdot\omega)$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(2,1) | $\psi(\Omega^{\Omega^\Omega+1})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,1)(2,1)(3,1)(4,1) | $\psi(\Omega^{\Omega^\Omega\cdot 2})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(3,0) | $\psi(\Omega^{\Omega^\Omega\cdot\omega})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)<br>(3,0)(4,1)(5,1)(6,1)(7,1) | $\psi(\Omega^{\Omega^{\Omega\cdot\psi(\Omega^{\Omega^\Omega})}})$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0)(1,1)(2,1)(3,1)(4,1)(3,1) | $\psi(\Omega^{\Omega^{\omega+1}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(3,1)(4,0) | $\psi(\Omega^{\Omega^{\Omega+\omega}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(3,1)(4,1) | $\psi(\Omega^{\Omega^{\omega\cdot 2}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(4,0) | $\psi(\Omega^{\Omega^{\Omega\cdot\omega}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(4,1) | $\psi(\Omega^{\Omega^{\Omega^2}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(4,1)(4,1) | $\psi(\Omega^{\Omega^{\Omega^3}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(5,0) | $\psi(\Omega^{\Omega^{\Omega^\omega}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(5,0)(6,1) | $\psi(\Omega^{\Omega^{\Omega^{\psi(\Omega)}}})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,1)(5,0)(6,1)(7,1) | $\psi(\Omega^{\Omega^{\Omega^{\psi(\Omega^\Omega)}}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)<br>(5,0)(6,1)(7,1)(8,1) | $\psi(\Omega^{\Omega^{\Omega^{\psi(\Omega^{\Omega^\Omega})}}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(5,1) | $\psi(\Omega^{\Omega^{\Omega^\Omega}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(5,1)(6,0) | $\psi(\Omega^{\Omega^{\Omega^{\Omega^\omega}}})$ |
| (0,0)(1,1)(2,1)(3,1)(4,1)(5,1)(6,1) | $\psi(\Omega^{\Omega^{\Omega^{\Omega^\Omega}}})$ |
| (0,0)(1,1)(2,1)(3,1)<br>(4,1)(5,1)(6,1)(7,1) | $\psi(\Omega^{\Omega^{\Omega^{\Omega^{\Omega^\Omega}}}})$ |
| (0,0)(1,1)(2,2) | $\psi(\Omega_2)$ |
| (0,0)(1,1)(2,2)(1,0) | $\psi(\Omega_2+1)$ |
| (0,0)(1,1)(2,2)(1,0)(2,1) | $\psi(\Omega_2+\psi(\Omega))$ |
| (0,0)(1,1)(2,2)(1,0)(2,1)(3,1) | $\psi(\Omega_2+\psi(\Omega^2))$ |
| (0,0)(1,1)(2,2)(1,0)(2,1)(3,2) | $\psi(\Omega_2+\psi(\Omega_2))$ |
| (0,0)(1,1)(2,2)(1,0)(2,1)<br>(3,2)(2,0)(3,1)(4,2) | $\psi(\Omega_2+\psi(\Omega_2+\psi(\Omega_2)))$ |
| (0,0)(1,1)(2,2)(1,1) | $\psi(\Omega_2+\Omega)$ |
| (0,0)(1,1)(2,2)(1,1)(2,1) | $\psi(\Omega_2+\Omega^2)$ |
| (0,0)(1,1)(2,2)(1,1)<br>(2,1)(3,0)(4,1)(5,2) | $\psi(\Omega_2+\Omega^\psi(\Omega_2))$ |
| (0,0)(1,1)(2,2)(1,1)(2,1)(3,1) | $\psi(\Omega_2+\Omega^\Omega)$ |
| (0,0)(1,1)(2,2)(1,1)(2,2) | $\psi(\Omega_2+\psi_1(\Omega_2))$ |
| (0,0)(1,1)(2,2)(1,1)(2,2)(1,1) | $\psi(\Omega_2+\psi_1(\Omega_2)+\Omega)$ |
| (0,0)(1,1)(2,2)(1,1)(2,2)(1,1)(2,2) | $\psi(\Omega_2+\psi_1(\Omega_2)\cdot 2)$ |
| (0,0)(1,1)(2,2)(2,0) | $\psi(\Omega_2+\psi_1(\Omega_2+1))$ |
| (0,0)(1,1)(2,2)(2,0)(3,1) | $\psi(\Omega_2+\psi_1(\Omega_2+\psi(\Omega)))$ |
| (0,0)(1,1)(2,2)(2,0)(3,1)(4,2) | $\psi(\Omega_2+\psi_1(\Omega_2+\psi(\Omega_2)))$ |
| (0,0)(1,1)(2,2)(2,1) | $\psi(\Omega_2+\psi_1(\Omega_2+\Omega))$ |
| (0,0)(1,1)(2,2)(2,1)(3,0)(4,1)(5,2) | $\psi(\Omega_2+\psi_1(\Omega_2+\Omega^2))$ |
| (0,0)(1,1)(2,2)(2,1)<br>(3,0)(4,1)(5,2)(4,1)(5,2) | $\psi(\Omega_2+\psi_1(\Omega_2+\Omega^\psi(\Omega_2+\psi_1(\Omega_2))))$ |
| (0,0)(1,1)(2,2)(2,1)(3,1) | $\psi(\Omega_2+\psi_1(\Omega_2+\Omega^\Omega))$ |
| (0,0)(1,1)(2,2)(2,1)(3,2) | $\psi(\Omega_2+\psi_1(\Omega_2+\psi_1(\Omega_2)))$ |
| (0,0)(1,1)(2,2)(2,1)(3,2)(1,0) | $\psi(\Omega_2+\psi_1(\Omega_2+\psi_1(\Omega_2))+1)$ |
| (0,0)(1,1)(2,2)(2,1)(3,2)(2,0) | $\psi(\Omega_2+\psi_1(\Omega_2+\psi_1(\Omega_2)+1))$ |
| (0,0)(1,1)(2,2)(2,1)(3,2)(3,0) | $\psi(\Omega_2+\psi_1(\Omega_2+\psi_1(\Omega_2+1)))$ |
| (0,0)(1,1)(2,2)(2,1)(3,2)(3,1) | $\psi(\Omega_2+\psi_1(\Omega_2+\psi_1(\Omega_2+\Omega)))$ |
| (0,0)(1,1)(2,2)(2,1)(3,2)(3,1)(4,2) | $\psi(\Omega_2+\psi_1(\Omega_2+\psi_1(\Omega_2+\psi_1(\Omega_2))))$ |
| (0,0)(1,1)(2,2)(2,2) | $\psi(\Omega_2\cdot 2)$ |
| (0,0)(1,1)(2,2)(2,2)(1,1)(2,2) | $\psi(\Omega_2\cdot 2+\psi_1(\Omega_2))$ |
| (0,0)(1,1)(2,2)(2,2)(1,1)(2,2)(2,2) | $\psi(\Omega_2\cdot 2+\psi_1(\Omega_2\cdot 2))$ |
| (0,0)(1,1)(2,2)(2,2)(2,0) | $\psi(\Omega_2\cdot 2+\psi_1(\Omega_2\cdot 2+1))$ |
| (0,0)(1,1)(2,2)(2,2)(2,1) | $\psi(\Omega_2\cdot 2+\psi_1(\Omega_2\cdot 2+\Omega))$ |
| (0,0)(1,1)(2,2)(2,2)(2,2) | $\psi(\Omega_2\cdot 3)$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0)(1,1)(2,2)(3,0) | $\psi(\Omega_2 \cdot \omega)$ |
| (0,0)(1,1)(2,2)(3,0)(4,1) | $\psi(\Omega_2 \cdot \psi(\Omega))$ |
| (0,0)(1,1)(2,2)(3,0)(4,1)(5,1) | $\psi(\Omega_2 \cdot \psi(\Omega^2))$ |
| (0,0)(1,1)(2,2)(3,0)(4,1)(5,2) | $\psi(\Omega_2 \cdot \psi(\Omega_2))$ |
| (0,0)(1,1)(2,2)(3,1) | $\psi(\Omega_2 \cdot \Omega)$ |
| (0,0)(1,1)(2,2)(3,1)(4,2) | $\psi(\Omega_2 \cdot \psi_1(\Omega_2))$ |
| (0,0)(1,1)(2,2)(3,2) | $\psi({\Omega_2}^2)$ |
| (0,0)(1,1)(2,2)(3,2)(2,2) | $\psi({\Omega_2}^2 + \Omega_2)$ |
| (0,0)(1,1)(2,2)(3,2)(2,2)(3,1) | $\psi({\Omega_2}^2 + \Omega_2 \cdot \Omega)$ |
| (0,0)(1,1)(2,2)(3,2)<br>(2,2)(3,1)(4,2)(5,2) | $\psi({\Omega_2}^2 + \Omega_2 \cdot \psi_1({\Omega_2}^2))$ |
| (0,0)(1,1)(2,2)(3,2)(2,2)(3,2) | $\psi({\Omega_2}^2 \cdot 2)$ |
| (0,0)(1,1)(2,2)(3,2)(3,0) | $\psi({\Omega_2}^2 \cdot \omega)$ |
| (0,0)(1,1)(2,2)(3,2)(3,1) | $\psi({\Omega_2}^2 \cdot \Omega)$ |
| (0,0)(1,1)(2,2)(3,2)(3,2) | $\psi({\Omega_2}^3)$ |
| (0,0)(1,1)(2,2)(3,2)(4,0) | $\psi({\Omega_2}^\omega)$ |
| (0,0)(1,1)(2,2)(3,2)(4,1) | $\psi({\Omega_2}^\Omega)$ |
| (0,0)(1,1)(2,2)(3,2)(4,2) | $\psi({\Omega_2}^{\Omega_2})$ |
| (0,0)(1,1)(2,2)(3,2)(4,2)(2,2) | $\psi({\Omega_2}^{\Omega_2} + \Omega_2)$ |
| (0,0)(1,1)(2,2)(3,2)(4,2)(2,2)(3,2) | $\psi({\Omega_2}^{\Omega_2} + {\Omega_2}^2)$ |
| (0,0)(1,1)(2,2)(3,2)(4,2)<br>(2,2)(3,2)(4,1)(5,2)(6,2)(7,2) | $\psi({\Omega_2}^{\Omega_2} + {\Omega_2}^{\psi_1({\Omega_2}^{\Omega_2})})$ |
| (0,0)(1,1)(2,2)(3,2)<br>(4,2)(2,2)(3,2)(4,2) | $\psi({\Omega_2}^{\Omega_2} \cdot 2)$ |
| (0,0)(1,1)(2,2)(3,2)(4,2)(3,0) | $\psi({\Omega_2}^{\Omega_2} \cdot \omega)$ |
| (0,0)(1,1)(2,2)(3,2)(4,2)(3,1) | $\psi({\Omega_2}^{\Omega_2} \cdot \Omega)$ |
| (0,0)(1,1)(2,2)(3,2)(4,2)(3,2) | $\psi({\Omega_2}^{\Omega_2+1})$ |
| (0,0)(1,1)(2,2)(3,2)(4,2)(3,2)(4,2) | $\psi({\Omega_2}^{\Omega_2 \cdot 2})$ |
| (0,0)(1,1)(2,2)(3,2)(4,2)(4,0) | $\psi({\Omega_2}^{\Omega_2 \cdot \omega})$ |
| (0,0)(1,1)(2,2)(3,2)(4,2)(4,1) | $\psi({\Omega_2}^{\Omega_2 \cdot \Omega})$ |
| (0,0)(1,1)(2,2)(3,2)(4,2)(4,2) | $\psi({\Omega_2}^{{\Omega_2}^2})$ |
| (0,0)(1,1)(2,2)(3,2)(4,2)(4,2)(4,2) | $\psi({\Omega_2}^{{\Omega_2}^3})$ |
| (0,0)(1,1)(2,2)(3,2)(4,2)(5,0) | $\psi({\Omega_2}^{{\Omega_2}^\omega})$ |
| (0,0)(1,1)(2,2)(3,2)(4,2)(5,1) | $\psi({\Omega_2}^{{\Omega_2}^\Omega})$ |
| (0,0)(1,1)(2,2)(3,2)(4,2)(5,2) | $\psi({\Omega_2}^{{\Omega_2}^{\Omega_2}})$ |
| (0,0)(1,1)(2,2)(3,2)(4,2)(5,2)(6,2) | $\psi({\Omega_2}^{{\Omega_2}^{{\Omega_2}^{\Omega_2}}})$ |
| (0,0)(1,1)(2,2)(3,3) | $\psi(\Omega_3)$ |
| (0,0)(1,1)(2,2)(3,3)(1,1) | $\psi(\Omega_3 + \Omega)$ |
| (0,0)(1,1)(2,2)(3,3)(1,1)(2,1)(3,1) | $\psi(\Omega_3 + \Omega^\Omega)$ |
| (0,0)(1,1)(2,2)(3,3)(1,1)(2,2) | $\psi(\Omega_3 + \psi_1(\Omega_2))$ |
| (0,0)(1,1)(2,2)(3,3)(2,1)(3,2)(4,3) | $\psi(\Omega_3 + \psi_1(\Omega_3))$ |
| (0,0)(1,1)(2,2)(3,3)(2,2) | $\psi(\Omega_3 + \Omega_2)$ |
| (0,0)(1,1)(2,2)(3,3)(2,2)(3,2) | $\psi(\Omega_3 + {\Omega_2}^2)$ |
| (0,0)(1,1)(2,2)(3,3)(2,2)(3,3) | $\psi(\Omega_3 + \psi_2(\Omega_3))$ |
| (0,0)(1,1)(2,2)(3,3)(3,0) | $\psi(\Omega_3 + \psi_2(\Omega_3 + 1))$ |
| (0,0)(1,1)(2,2)(3,3)(3,1) | $\psi(\Omega_3 + \psi_2(\Omega_3 + \Omega))$ |
| (0,0)(1,1)(2,2)(3,3)(3,1)(4,2)(5,3) | $\psi(\Omega_3 + \psi_2(\Omega_3 + \psi_1(\Omega_3)))$ |
| (0,0)(1,1)(2,2)(3,3)(3,2) | $\psi(\Omega_3 + \psi_2(\Omega_3 + \Omega_2))$ |
| (0,0)(1,1)(2,2)(3,3)(3,2)(4,3) | $\psi(\Omega_3 + \psi_2(\Omega_3 + \psi_2(\Omega_3)))$ |
| (0,0)(1,1)(2,2)(3,3)(3,3) | $\psi(\Omega_3 \cdot 2)$ |
| (0,0)(1,1)(2,2)(3,3)(4,0) | $\psi(\Omega_3 \cdot \omega)$ |
| (0,0)(1,1)(2,2)(3,3)(4,3) | $\psi({\Omega_3}^2)$ |
| (0,0)(1,1)(2,2)(3,3)(4,3)(4,3) | $\psi({\Omega_3}^3)$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0)(1,1)(2,2)(3,3)(4,3)(5,0) | $\psi({\Omega_3}^{\omega})$ |
| (0,0)(1,1)(2,2)(3,3)(4,3)(5,3) | $\psi({\Omega_3}^{\Omega_3})$ |
| (0,0)(1,1)(2,2)(3,3)(4,4) | $\psi(\Omega_4)$ |
| (0,0)(1,1)(2,2)(3,3)(4,4)(5,5) | $\psi(\Omega_5)$ |
| (0,0)(1,1)(2,2)(3,3)(4,4)(5,5)(6,6) | $\psi(\Omega_6)$ |
| (0,0,0)(1,1,1) | $\psi(\Omega_\omega)$ |
| (0,0,0)(1,1,1)(0,0,0) | $\psi(\Omega_\omega)+1$ |
| (0,0,0)(1,1,1)(1,0,0) | $\psi(\Omega_\omega+1)$ |
| (0,0,0)(1,1,1)(1,0,0)(2,1,0) | $\psi(\Omega_\omega+\psi(\Omega))$ |
| (0,0,0)(1,1,1)(1,0,0)(2,1,0)(3,2,0) | $\psi(\Omega_\omega+\psi(\Omega_2))$ |
| (0,0,0)(1,1,1)(1,0,0)(2,1,1) | $\psi(\Omega_\omega+\psi(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(1,0,0)(2,1,1)(2,0,0) | $\psi(\Omega_\omega+\psi(\Omega_\omega+1))$ |
| (0,0,0)(1,1,1)(1,0,0)<br>(2,1,1)(2,0,0)(3,1,1) | $\psi(\Omega_\omega+\psi(\Omega_\omega+\psi(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(1,1,0) | $\psi(\Omega_\omega+\Omega)$ |
| (0,0,0)(1,1,1)(1,1,0)(1,1,0) | $\psi(\Omega_\omega+\Omega\cdot 2)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,0,0) | $\psi(\Omega_\omega+\Omega\cdot\omega)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,0,0)(3,1,1) | $\psi(\Omega_\omega+\Omega\cdot\psi(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,1,0) | $\psi(\Omega_\omega+\Omega^2)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0) | $\psi(\Omega_\omega+\psi_1(\Omega_2))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,0)(3,3,0) | $\psi(\Omega_\omega+\psi_1(\Omega_3))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1) | $\psi(\Omega_\omega+\psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)(1,0,0) | $\psi(\Omega_\omega+\psi_1(\Omega_\omega)+1)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)(1,1,0) | $\psi(\Omega_\omega+\psi_1(\Omega_\omega)+\Omega)$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,1)(1,1,0)(2,2,0) | $\psi(\Omega_\omega+\psi_1(\Omega_\omega)+\psi_1(\Omega_2))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,1)(1,1,0)(2,2,1) | $\psi(\Omega_\omega+\psi_1(\Omega_\omega)\cdot 2)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)(2,0,0) | $\psi(\Omega_\omega+\psi_1(\Omega_\omega+1))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,1)(2,0,0)(3,1,1) | $\psi(\Omega_\omega+\psi_1(\Omega_\omega+\psi(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)(2,1,0) | $\psi(\Omega_\omega+\psi_1(\Omega_\omega+\Omega))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)(2,2,0) | $\psi(\Omega_\omega+\Omega_2)$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)(2,2,0)(3,3,1) | $\psi(\Omega_\omega+\psi_2(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(1,1,0)<br>(2,2,1)(2,2,0)(3,3,1)(3,2,0) | $\psi(\Omega_\omega+\psi_2(\Omega_\omega+\Omega_2))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,2,0)(3,3,1)(3,2,0)(4,3,1) | $\psi(\Omega_\omega+\psi_2(\Omega_\omega+\psi_2(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,2,0)(3,3,1)(3,3,0) | $\psi(\Omega_\omega+\Omega_3)$ |
| (0,0,0)(1,1,1)(1,1,1) | $\psi(\Omega_\omega\cdot 2)$ |
| (0,0,0)(1,1,1)(1,1,1)(1,1,0) | $\psi(\Omega_\omega\cdot 2+\Omega)$ |
| (0,0,0)(1,1,1)(1,1,1)<br>(1,1,0)(2,2,1)(2,2,1) | $\psi(\Omega_\omega\cdot 2+\psi_1(\Omega_\omega\cdot 2))$ |
| (0,0,0)(1,1,1)(1,1,1)(1,1,0)<br>(2,2,1)(2,2,1)(2,2,0)(3,3,1)(3,3,1) | $\psi(\Omega_\omega\cdot 2+\psi_1(\Omega_\omega\cdot 2+\psi_2(\Omega_\omega\cdot 2)))$ |
| (0,0,0)(1,1,1)(1,1,1)(1,1,1) | $\psi(\Omega_\omega\cdot 3)$ |
| (0,0,0)(1,1,1)(1,1,1)(1,1,1)(1,1,1) | $\psi(\Omega_\omega\cdot 4)$ |
| (0,0,0)(1,1,1)(2,0,0) | $\psi(\Omega_\omega\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,0,0)(1,1,0) | $\psi(\Omega_\omega\cdot\omega+\Omega)$ |
| (0,0,0)(1,1,1)(2,0,0)(1,1,0)(2,2,1) | $\psi(\Omega_\omega\cdot\omega+\psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,0,0)<br>(1,1,0)(2,2,1)(3,0,0) | $\psi(\Omega_\omega\cdot\omega+\psi_1(\Omega_\omega\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,0,0)(1,1,0)<br>(2,2,1)(3,0,0)(2,2,0) | $\psi(\Omega_\omega\cdot\omega+\Omega_2)$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,0,0)(1,1,0)<br>(2,2,1)(3,0,0)(2,2,0)(3,3,1) | $\psi(\Omega_\omega \cdot \omega + \psi_2(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,0,0)(1,1,0)(2,2,1)<br>(3,0,0)(2,2,0)(3,3,1)(4,0,0) | $\psi(\Omega_\omega \cdot \omega + \psi_2(\Omega_\omega \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,0,0)(1,1,1) | $\psi(\Omega_\omega \cdot (\omega + 1))$ |
| (0,0,0)(1,1,1)(2,0,0)(1,1,1)(1,1,1) | $\psi(\Omega_\omega \cdot (\omega + 2))$ |
| (0,0,0)(1,1,1)(2,0,0)(1,1,1)(2,0,0) | $\psi(\Omega_\omega \cdot (\omega \cdot 2))$ |
| (0,0,0)(1,1,1)(2,0,0)(2,0,0) | $\psi(\Omega_\omega \cdot \omega^2)$ |
| (0,0,0)(1,1,1)(2,0,0)(3,1,0) | $\psi(\Omega_\omega \cdot \psi(\Omega))$ |
| (0,0,0)(1,1,1)(2,0,0)(3,1,1) | $\psi(\Omega_\omega \cdot \psi(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0) | $\psi(\Omega_\omega \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,0,0) | $\psi(\Omega_\omega \cdot \Omega + 1)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(1,1,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,1,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)(2,0,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)(2,2,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega + \Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(2,2,0)(3,3,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega + \psi_2(\Omega_3)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(2,2,0)(3,3,1) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega + \psi_2(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,2,1)(2,2,1) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,2,1)(3,0,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,2,1)(3,1,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,0,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \Omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,1,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \Omega + \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,1,0)(3,2,1) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \Omega + \psi_2(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,1,0)(3,2,1)(4,1,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \Omega + \psi_2(\Omega_\omega \cdot \Omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0)(3,3,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_2(\Omega_3))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0)(3,3,1) | $\psi(\Omega_\omega \cdot \Omega + \psi_2(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,0)(3,3,1)(3,3,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_2(\Omega_\omega + \Omega_3))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,0)(3,3,1)(3,3,1) | $\psi(\Omega_\omega \cdot \Omega + \psi_2(\Omega_\omega \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,0)(3,3,1)(4,0,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_2(\Omega_\omega \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,0)(3,3,1)(4,1,0) | $\psi(\Omega_\omega \cdot \Omega + \psi_2(\Omega_\omega \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,0)(3,3,1)(4,1,0)(3,3,0) | $\psi(\Omega_\omega \cdot \Omega + \Omega_3)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1) | $\psi(\Omega_\omega \cdot (\Omega + 1))$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(1,1,0) | $\psi(\Omega_\omega \cdot (\Omega+1)+\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1) | $\psi(\Omega_\omega \cdot (\Omega+1)+\psi_1(\Omega_\omega \cdot (\Omega+1)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,0,0) | $\psi(\Omega_\omega \cdot (\Omega+1)+\psi_1(\Omega_\omega \cdot (\Omega+1)+1))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,1,0) | $\psi(\Omega_\omega \cdot (\Omega+1)+\psi_1(\Omega_\omega \cdot (\Omega+1)+\Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,1,0)<br>(3,2,1)(4,1,0)(3,2,1) | $\psi(\Omega_\omega \cdot (\Omega+1)+\psi_1(\Omega_\omega \cdot (\Omega+1)$<br>$+\psi_1(\Omega_\omega \cdot (\Omega+1))))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,0) | $\psi(\Omega_\omega \cdot (\Omega+1)+\Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,0)<br>(3,3,1)(4,1,0)(3,3,1) | $\psi(\Omega_\omega \cdot (\Omega+1)+\Omega_3)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(2,2,1) | $\psi(\Omega_\omega \cdot (\Omega+2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,1)(2,2,1)(2,2,1) | $\psi(\Omega_\omega \cdot (\Omega+3))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(3,0,0) | $\psi(\Omega_\omega \cdot (\Omega+\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1)(3,1,0) | $\psi(\Omega_\omega \cdot \Omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(3,0,0) | $\psi(\Omega_\omega \cdot \Omega \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(3,1,0) | $\psi(\Omega_\omega \cdot \Omega^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,0,0) | $\psi(\Omega_\omega \cdot \Omega^\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,0) | $\psi(\Omega_\omega \cdot \psi_1(\Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,1) | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,1)(5,1,0) | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(4,2,1)(5,1,0)(6,2,1) | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \psi_1(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,2,1)(3,2,0) | $\psi(\Omega_\omega \cdot \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(1,0,0) | $\psi(\Omega_\omega \cdot \Omega_2+1)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(1,1,0) | $\psi(\Omega_\omega \cdot \Omega_2+\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(1,1,0)(2,2,1) | $\psi(\Omega_\omega \cdot (\Omega_2+1))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(1,1,0)(2,2,1)(3,2,0) | $\psi(\Omega_\omega \cdot \Omega_2 \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,0,0) | $\psi(\Omega_\omega \cdot \Omega_2 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,1,0) | $\psi(\Omega_\omega \cdot \Omega_2 \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0) | $\psi(\Omega_\omega \cdot {\Omega_2}^2)$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,1) | $\psi(\Omega_\omega \cdot \psi_2(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(2,2,0)(3,3,1)(4,2,0) | $\psi(\Omega_\omega \cdot \psi_2(\Omega_\omega \cdot \Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(2,2,0)(3,3,1)(4,3,0) | $\psi(\Omega_\omega \cdot \Omega_3)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1) | $\psi({\Omega_\omega}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0) | $\psi({\Omega_\omega}^2 + \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,1)(1,1,0)(2,2,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,1)(1,1,0)(2,2,1) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0) | $\psi({\Omega_\omega}^2 + \psi_1(\Omega_\omega \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1) | $\psi({\Omega_\omega}^2 + \psi_1({\Omega_\omega}^2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,1)(2,1,0)(3,2,1) | $\psi({\Omega_\omega}^2 + \psi_1({\Omega_\omega}^2 + \psi_1(\Omega_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1)<br>(2,1,0)(3,2,1)(4,2,0)(3,2,1) | $\psi({\Omega_\omega}^2 + \psi_1({\Omega_\omega}^2 + \psi_1({\Omega_\omega}^2)))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,1)(2,2,0) | $\psi({\Omega_\omega}^2 + \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,1)(2,2,0)(3,3,1) | $\psi({\Omega_\omega}^2 + \psi_2(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1)<br>(2,2,0)(3,3,1)(4,3,0)(3,3,1) | $\psi({\Omega_\omega}^2 + \psi_2({\Omega_\omega}^2))$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,1) | $\psi({\Omega_\omega}^2 + \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,1)(1,1,1)(1,1,1) | $\psi({\Omega_\omega}^2 + \Omega_\omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(2,0,0) | $\psi({\Omega_\omega}^2 + \Omega_\omega \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)(2,1,0) | $\psi({\Omega_\omega}^2 + \Omega_\omega \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(1,1,1)(2,1,0)(1,1,1) | $\psi({\Omega_\omega}^2 \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(1,1,1)(2,0,0) | $\psi({\Omega_\omega}^2 \cdot 2 + \Omega_\omega \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(1,1,1)(2,1,0) | $\psi({\Omega_\omega}^2 \cdot 2 + \Omega_\omega \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(1,1,1)(2,1,0)(1,1,1) | $\psi({\Omega_\omega}^2 \cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,0)(2,0,0) | $\psi({\Omega_\omega}^2 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(2,1,0) | $\psi({\Omega_\omega}^2 \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(2,1,0)<br>(1,1,0)(2,2,1)(3,1,0)(2,2,1) | $\psi({\Omega_\omega}^2 \cdot \Omega + \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(3,1,0)(2,2,1) | $\psi({\Omega_\omega}^2 \cdot (\Omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,0)(2,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(3,2,0) | $\psi({\Omega_\omega}^2 \cdot \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(2,1,0)(1,1,1) | $\psi({\Omega_\omega}^3)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(2,1,0)(2,1,0)(1,1,1) | $\psi({\Omega_\omega}^4)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,0,0) | $\psi({\Omega_\omega}^\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,0,0)(4,1,1) | $\psi({\Omega_\omega}^{\psi(\Omega_\omega)})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0) | $\psi({\Omega_\omega}^\Omega)$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,2,0) | $\psi({\Omega_\omega}^{\Omega_2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(1,1,1) | $\psi({\Omega_\omega}^{\Omega_\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(2,0,0) | $\psi({\Omega_\omega}^{\Omega_\omega \cdot \omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(2,1,0) | $\psi({\Omega_\omega}^{\Omega_\omega \cdot \Omega})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(2,1,0)(1,1,1) | $\psi({\Omega_\omega}^{\Omega_\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(2,1,0)(2,1,0)(1,1,1) | $\psi({\Omega_\omega}^{\Omega_\omega+2})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(2,1,0)(3,0,0) | $\psi({\Omega_\omega}^{\Omega_\omega+\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(2,1,0)(3,1,0)(1,1,1) | $\psi({\Omega_\omega}^{\Omega_\omega \cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(3,0,0) | $\psi({\Omega_\omega}^{\Omega_\omega \cdot \omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(3,1,0) | $\psi({\Omega_\omega}^{\Omega_\omega \cdot \Omega})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(3,1,0)(1,1,1) | $\psi({\Omega_\omega}^{{\Omega_\omega}^2})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(3,1,0)(3,1,0)(1,1,1) | $\psi({\Omega_\omega}^{{\Omega_\omega}^3})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,1,0)(4,0,0) | $\psi({\Omega_\omega}^{{\Omega_\omega}^\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(4,1,0)(1,1,1) | $\psi({\Omega_\omega}^{{\Omega_\omega}^{\Omega_\omega}})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(4,1,0)(5,1,0)(1,1,1) | $\psi({\Omega_\omega}^{{\Omega_\omega}^{{\Omega_\omega}^{\Omega_\omega}}})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0) | $\psi(\Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(1,0,0) | $\psi(\Omega_{\omega+1}+1)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(1,1,0) | $\psi(\Omega_{\omega+1}+\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(1,1,0)(2,2,0) | $\psi(\Omega_{\omega+1}+\psi_1(\Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(1,1,0)(2,2,1) | $\psi(\Omega_{\omega+1}+\psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,0,0) | $\psi(\Omega_{\omega+1}+\psi_1(\Omega_\omega \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,1,0) | $\psi(\Omega_{\omega+1}+\psi_1(\Omega_\omega \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,2,0) | $\psi(\Omega_{\omega+1}+\psi_1(\Omega_\omega \cdot \Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1) | $\psi(\Omega_{\omega+1}+\psi_1({\Omega_\omega}^2))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0) | $\psi(\Omega_{\omega+1}+\psi_1(\Omega_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(1,1,0)<br>(2,2,1)(3,2,0)(4,3,0)(2,2,0) | $\psi(\Omega_{\omega+1}+\Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(1,1,0)<br>(2,2,1)(3,2,0)(4,3,0)(2,2,0)(3,3,0) | $\psi(\Omega_{\omega+1}+\psi_2(\Omega_3))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(1,1,0)<br>(2,2,1)(3,2,0)(4,3,0)(2,2,0)(3,3,1) | $\psi(\Omega_{\omega+1}+\psi_2(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(2,2,0)(3,3,1)(4,3,0) | $\psi(\Omega_{\omega+1}+\psi_2(\Omega_\omega+\Omega_3))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(2,2,0)(3,3,1)(4,3,0)(3,3,1) | $\psi(\Omega_{\omega+1}+\psi_2({\Omega_\omega}^2))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(2,2,0)(3,3,1)(4,3,0)(5,4,0) | $\psi(\Omega_{\omega+1}+\psi_2(\Omega_{\omega+1}))$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)(2,2,0)<br>(3,3,1)(4,3,0)(5,4,0)(3,3,0) | $\psi(\Omega_{\omega+1}+\Omega_3)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(1,1,1) | $\psi(\Omega_{\omega+1}+\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,1)(2,1,0)(3,2,0) | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(2,0,0) | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1}+1))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(2,1,0) | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1}+\Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(2,1,0)(3,2,0) | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(3,0,0) | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1}+1)))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(3,1,0) | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1}+\Omega)))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(3,2,0) | $\psi(\Omega_{\omega+1}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,0,0) | $\psi(\Omega_{\omega+1}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,1,0) | $\psi(\Omega_{\omega+1}\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,1,0)(1,1,1) | $\psi(\Omega_{\omega+1}\cdot\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,1,0)(5,2,0) | $\psi(\Omega_{\omega+1}\cdot\psi_\omega(\Omega_{\omega+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,2,0) | $\psi({\Omega_{\omega+1}}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(4,2,0) | $\psi({\Omega_{\omega+1}}^3)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(5,0,0) | $\psi({\Omega_{\omega+1}}^\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(5,1,0) | $\psi({\Omega_{\omega+1}}^\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(5,1,0)(1,1,1) | $\psi({\Omega_{\omega+1}}^{\Omega_\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(5,2,0) | $\psi({\Omega_{\omega+1}}^{\Omega_{\omega+1}})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(5,2,0)(6,2,0) | $\psi({\Omega_{\omega+1}}^{{\Omega_{\omega+1}}^{\Omega_{\omega+1}}})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,3,0) | $\psi(\Omega_{\omega+2})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(4,3,0) | $\psi(\Omega_{\omega+2}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,0,0) | $\psi(\Omega_{\omega+2}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,1,0) | $\psi(\Omega_{\omega+2}\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,3,0)(5,1,0)(1,1,1) | $\psi(\Omega_{\omega+2}\cdot\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,2,0) | $\psi(\Omega_{\omega+2}\cdot\Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,3,0) | $\psi({\Omega_{\omega+2}}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,3,0)(5,3,0)(6,3,0) | $\psi({\Omega_{\omega+2}}^{\Omega_{\omega+2}})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,4,0) | $\psi(\Omega_{\omega+3})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1) | $\psi(\Omega_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(1,1,0) | $\psi(\Omega_{\omega\cdot 2}+\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(1,1,0)(2,2,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_1(\Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(1,1,0)(2,2,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_1(\Omega_\omega))$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_1(\Omega_\omega\cdot\Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_1({\Omega_\omega}^2))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(1,1,0)<br>(2,2,1)(3,2,0)(3,2,0)(2,2,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_1({\Omega_\omega}^3))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_1(\Omega_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_1(\Omega_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(1,1,0)<br>(2,2,1)(3,2,0)(4,3,1)(2,2,0) | $\psi(\Omega_{\omega\cdot 2}+\Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,1)<br>(2,2,0)(3,3,1)(4,3,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_2(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,1)<br>(2,2,0)(3,3,1)(4,3,0)(5,4,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_2(\Omega_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,1)(2,2,0)<br>(3,3,1)(4,3,0)(5,4,1)(3,3,0) | $\psi(\Omega_{\omega\cdot 2}+\Omega_3)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(1,1,1) | $\psi(\Omega_{\omega\cdot 2}+\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(1,1,1)(1,1,1) | $\psi(\Omega_{\omega\cdot 2}+\Omega_\omega\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(1,1,1)(2,0,0) | $\psi(\Omega_{\omega\cdot 2}+\Omega_\omega\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(1,1,1)(2,1,0) | $\psi(\Omega_{\omega\cdot 2}+\Omega_\omega\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(1,1,1) | $\psi(\Omega_{\omega\cdot 2}+{\Omega_\omega}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(3,2,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(3,2,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(2,0,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2}+1))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(2,1,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2}+\Omega))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(2,1,0)(1,1,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2}+\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(2,1,0)(1,1,1)(2,1,0)(3,2,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2}+\Omega_\omega)+\psi_\omega(\Omega_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(2,1,0)(3,2,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega+1})))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(2,1,0)(3,2,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2})))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(3,1,0)(4,2,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2}$<br>$+\psi_\omega(\Omega_{\omega\cdot 2}))))$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(3,2,0) | $\psi(\Omega_{\omega\cdot 2}+\Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(3,2,0)(4,2,0) | $\psi(\Omega_{\omega\cdot 2}+{\Omega_{\omega+1}}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(3,2,0)(4,2,0)(5,2,0) | $\psi(\Omega_{\omega\cdot 2}+\Omega_{\omega+1}{}^{\Omega}_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(3,2,0)(4,3,0) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega+1}(\Omega_{\omega+2}))$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(3,2,0)(4,3,1) | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega+1}(\Omega_{\omega\cdot 2}))$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(3,2,0)(4,3,1)(4,3,0) | $\psi(\Omega_{\omega\cdot 2}+\Omega_{\omega+2})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(3,2,1) | $\psi(\Omega_{\omega\cdot 2}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,0,0) | $\psi(\Omega_{\omega\cdot 2}\cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0) | $\psi(\Omega_{\omega\cdot 2}\cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(1,1,1) | $\psi(\Omega_{\omega\cdot 2}\cdot \Omega_{\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,2,0) | $\psi(\Omega_{\omega\cdot 2}\cdot \Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(3,2,1) | $\psi({\Omega_{\omega\cdot 2}}^2)$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(4,2,0)(3,2,1) | $\psi({\Omega_{\omega\cdot 2}}^3)$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(5,0,0) | $\psi({\Omega_{\omega\cdot 2}}^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,2,0)(3,2,1) | $\psi(\Omega_{\omega\cdot 2^{\Omega}_{\omega\cdot 2}})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,0) | $\psi(\Omega_{\omega\cdot 2+1})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,0)(6,4,0) | $\psi(\Omega_{\omega\cdot 2+2})$ |
| (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,1) | $\psi(\Omega_{\omega\cdot 3})$ |
| (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,1)(6,4,0)(7,5,1) | $\psi(\Omega_{\omega\cdot 4})$ |
| (0,0,0)(1,1,1)(2,1,1) | $\psi(\Omega_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,0) | $\psi(\Omega_{\omega^2}+\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,0)(2,2,0) | $\psi(\Omega_{\omega^2}+\psi_1(\Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,0)(2,2,1) | $\psi(\Omega_{\omega^2}+\psi_1(\Omega_{\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(1,1,0)(2,2,1)(3,2,1) | $\psi(\Omega_{\omega^2}+\psi_1(\Omega_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,0)<br>(2,2,1)(3,2,1)(2,2,0) | $\psi(\Omega_{\omega^2}+\Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,0)(2,2,1)<br>(3,2,1)(2,2,0)(3,3,1)(4,3,1) | $\psi(\Omega_{\omega^2}+\psi_2(\Omega_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1) | $\psi(\Omega_{\omega^2}+\Omega_{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1)(2,0,0) | $\psi(\Omega_{\omega^2}+\Omega_{\omega}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1)(2,1,0) | $\psi(\Omega_{\omega^2}+\Omega_{\omega}\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(1,1,1)(2,1,0)(3,1,0) | $\psi(\Omega_{\omega^2}+{\Omega_{\omega}}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(1,1,1)(2,1,0)(3,2,0) | $\psi(\Omega_{\omega^2}+\psi_{\omega}(\Omega_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,0)(3,2,0)(4,3,0) | $\psi(\Omega_{\omega^2}+\psi_{\omega}(\Omega_{\omega+2}))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(1,1,1)(2,1,0)(3,2,1) | $\psi(\Omega_{\omega^2}+\psi_{\omega}(\Omega_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1) | $\psi(\Omega_{\omega^2}+\psi_{\omega}(\Omega_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(3,2,0) | $\psi(\Omega_{\omega^2}+\Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(3,2,1) | $\psi(\Omega_{\omega^2}+\Omega_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(1,1,1)(2,1,1) | $\psi(\Omega_{\omega^2}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,1)(1,1,1)(2,1,1) | $\psi(\Omega_{\omega^2}\cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,0,0) | $\psi(\Omega_{\omega^2}\cdot \omega)$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)<br>(2,0,0)(1,1,1)(2,1,1) | $\psi(\Omega_{\omega^2} \cdot (\omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(2,0,0)<br>(1,1,1)(2,1,1)(2,0,0) | $\psi(\Omega_{\omega^2} \cdot (\omega \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(2,0,0)(2,0,0) | $\psi(\Omega_{\omega^2} \cdot \omega^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,0,0)(3,0,0) | $\psi(\Omega_{\omega^2} \cdot \omega^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,0,0)(3,1,0) | $\psi(\Omega_{\omega^2} \cdot \psi(\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(2,0,0)(3,1,1) | $\psi(\Omega_{\omega^2} \cdot \psi(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(2,0,0)(3,1,1)(4,1,1) | $\psi(\Omega_{\omega^2} \cdot \psi(\Omega_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0) | $\psi(\Omega_{\omega^2} \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1) | $\psi(\Omega_{\omega^2} \cdot \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1) | $\psi({\Omega_{\omega^2}}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(2,0,0) | $\psi({\Omega_{\omega^2}}^2 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(2,1,0) | $\psi({\Omega_{\omega^2}}^2 \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(2,1,0)(1,1,1) | $\psi({\Omega_{\omega^2}}^2 \cdot \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(2,1,0)(1,1,1)(2,1,1) | $\psi({\Omega_{\omega^2}}^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(2,1,0)(2,1,0)(1,1,1)(2,1,1) | $\psi({\Omega_{\omega^2}}^4)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(3,0,0) | $\psi({\Omega_{\omega^2}}^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(3,1,0) | $\psi({\Omega_{\omega^2}}^\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(3,1,0)(1,1,1) | $\psi({\Omega_{\omega^2}}^{\Omega_\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(3,1,0)(1,1,1)(2,1,1) | $\psi({\Omega_{\omega^2}}^{\Omega_{\omega^2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(3,2,0) | $\psi(\Omega_{\omega^2+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,0)(3,2,1) | $\psi(\Omega_{\omega^2+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1) | $\psi(\Omega_{\omega^2 \cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,1) | $\psi(\Omega_{\omega^3})$ |
| (0,0,0)(1,1,1)(2,1,1)(2,1,1)(2,1,1) | $\psi(\Omega_{\omega^4})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,0,0) | $\psi(\Omega_{\omega^\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,0,0)(4,1,1) | $\psi(\Omega_{\psi(\Omega_\omega)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0) | $\psi(\Omega_\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,0,0) | $\psi(\Omega_\Omega + 1)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0) | $\psi(\Omega_\Omega + \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,0) | $\psi(\Omega_\Omega + \psi_1(\Omega_2))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1) | $\psi(\Omega_\Omega + \psi_1(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1) | $\psi(\Omega_\Omega + \psi_1(\Omega_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0) | $\psi(\Omega_\Omega + \psi_1(\Omega_\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)(2,2,0) | $\psi(\Omega_\Omega + \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1,?) | $\psi(\Omega_\Omega + \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)(3,2,1) | $\psi(\Omega_\Omega + \Omega_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,1)(3,2,1)(4,1,0) | $\psi(\Omega_\Omega \cdot 2)$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,0,0) | $\psi(\Omega_\Omega \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,1,0) | $\psi(\Omega_\Omega \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0) | $\psi(\Omega_\Omega \cdot \Omega_2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(2,2,1) | $\psi(\Omega_\Omega \cdot \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,0)(2,2,1)(3,2,1)(4,1,0) | $\psi({\Omega_\Omega}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,0) | $\psi(\Omega_{\Omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,1) | $\psi(\Omega_{\Omega+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,0)(4,3,1)(5,3,1) | $\psi(\Omega_{\Omega+\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,0)(4,3,1)(5,3,1)(6,1,0) | $\psi(\Omega_{\Omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,1) | $\psi(\Omega_{\Omega\cdot\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,1)(4,1,0) | $\psi(\Omega_{\Omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(4,0,0) | $\psi(\Omega_{\Omega^\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(4,1,0) | $\psi(\Omega_{\Omega^\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(5,1,0) | $\psi(\Omega_{\Omega^{\Omega^\Omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(5,2,0) | $\psi(\Omega_{\psi_1(\Omega_2)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(5,2,1) | $\psi(\Omega_{\psi_1(\Omega_\omega)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,1,0) | $\psi(\Omega_{\psi_1(\Omega_\Omega)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,0) | $\psi(\Omega_{\Omega_2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,0)<br>(2,2,0)(3,3,1)(4,3,1)(5,3,0) | $\psi(\Omega_{\Omega_3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1) | $\psi(\Omega_{\Omega_\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1) | $\psi(\Omega_{\Omega_\omega} + \psi_\omega(\Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1) | $\psi(\Omega_{\Omega_\omega} + \psi_\omega(\Omega_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)(5,1,0) | $\psi(\Omega_{\Omega_\omega} + \psi_\omega(\Omega_\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(1,1,1) | $\psi(\Omega_{\Omega_\omega} + \psi_\omega(\Omega_{\Omega_\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(3,2,0) | $\psi(\Omega_{\Omega_\omega} + \Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(3,2,1) | $\psi(\Omega_{\Omega_\omega} + \Omega_{\omega\cdot 2})$ |

| BMS | Buchholz's OCF |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(3,2,1)(4,2,1) | $\psi(\Omega_{\Omega_\omega} + \Omega_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(3,2,1)(4,2,1)(5,1,0) | $\psi(\Omega_{\Omega_\omega} + \Omega_\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(1,1,1) | $\psi(\Omega_{\Omega_\omega} \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(4,0,0) | $\psi(\Omega_{\Omega_\omega} \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(4,1,0)(5,2,0) | $\psi(\Omega_{\Omega_\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0) | $\psi(\Omega_{\Omega_{\omega+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1) | $\psi(\Omega_{\Omega_{\omega^2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,0,0) | $\psi(\Omega_{\Omega_{(\omega^\omega)}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,0,0)(4,1,1) | $\psi(\Omega_{\Omega_{\psi(\Omega_\omega)}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0) | $\psi(\Omega_{\Omega_\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(4,2,0)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,0)(4,3,1)(5,3,1)(6,3,0) | $\psi(\Omega_{\Omega_{\Omega+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(1,1,1) | $\psi(\Omega_{\Omega_{\Omega_\omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,0) | $\psi(\Omega_{\Omega_{\Omega_\Omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I)$ |

## A.10 BMS vs Reflection-Collapsing Notation (Madore-like)

The results in this section are primarily cited from[6,10-17]. The reflection-collapsing notation used is the Madore-like version defined by Meitian Li.

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_I(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_I(0) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(2,0,0) | $\psi(\psi_I(0) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0)<br>(3,1,1)(4,1,1)(5,1,0)(4,0,0) | $\psi(\psi_I(0) \cdot \psi(\psi_I(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0) | $\psi(\psi_I(0) \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(1,1,1) | $\psi(\psi_I(0) \cdot \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(1,1,1)(2,1,1) | $\psi(\psi_I(0) \cdot \Omega_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,0) | $\psi(\psi_I(0) \cdot \Omega_\Omega)$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_I(0)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0) | $\psi(\psi_I(0)^2 + \psi_I(0) \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_I(0)^2 \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(2,1,0) | $\psi(\psi_I(0)^2 \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_I(0)^3)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,1,0) | $\psi(\psi_I(0)^\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_I(0)^{\psi_I(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,0) | $\psi(\psi_{\Omega_{\psi_I(0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,0)(3,2,0) | $\psi(\psi_{\Omega_{\psi_I(0)+1}}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,0)(4,1,0) | $\psi(\psi_{\Omega_{\psi_I(0)+1}}(\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,0)(4,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{\Omega_{\psi_I(0)+1}}(\psi_I(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,0)(4,1,0)(5,2,0) | $\psi(\psi_{\Omega_{\psi_I(0)+1}}(\psi_{\Omega_{\psi_I(0)+1}}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\psi_I(0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1) | $\psi(\Omega_{\psi_I(0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1) | $\psi(\Omega_{\psi_I(0)+\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0) | $\psi(\Omega_{\psi_I(0)+\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_I(0)\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} + \psi_I(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} + \psi_I(0) \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} + \psi_{\Omega_{\psi_I(0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} + \psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(2,1,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} + \psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2}) \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} + \psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2}) \cdot \psi_I(0))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(2,1,0)(3,2,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} + \psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2})$<br>$\cdot\psi_{\Omega_{\psi_I(0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} + \psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2})^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(3,1,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} + \psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2})^{\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(3,2,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} + \psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2} + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} + \Omega_{\psi_I(0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(3,2,0)(4,3,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} + \psi_{\Omega_{\psi_I(0)+2}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} + \psi_{\Omega_{\psi_I(0)+2}}(\Omega_{\psi_I(0)\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(3,2,0)(4,3,1)<br>(5,3,1)(6,1,0)(4,3,0)(5,3,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} + \Omega_{\psi_I(0)+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(3,2,1) | $\psi(\Omega_{\psi_I(0)\cdot 2} + \Omega_{\psi_I(0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(3,2,1)(4,2,1) | $\psi(\Omega_{\psi_I(0)\cdot 2} + \Omega_{\psi_I(0)+\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(4,0,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} \cdot \psi_I(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,1,0)(5,2,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} \cdot \psi_{\Omega_{\psi_I(0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,1,0)<br>(5,2,1)(6,2,1)(7,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} \cdot \psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,0) | $\psi(\Omega_{\psi_I(0)\cdot 2} \cdot \Omega_{\psi_I(0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,0)(3,2,1) | $\psi(\Omega_{\psi_I(0)\cdot 2} \cdot \Omega_{\psi_I(0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,0)<br>(3,2,1)(4,2,1)(5,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi({\Omega_{\psi_I(0)\cdot 2}}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,0)(3,2,1)<br>(4,2,1)(5,1,0)(4,2,0)(3,2,1)(4,2,1)<br>(5,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi({\Omega_{\psi_I(0)\cdot 2}}^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,0)<br>(4,2,0)(3,2,1)(4,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi({\Omega_{\psi_I(0)\cdot 2}}^{\Omega_{\psi_I(0)\cdot 2}})$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,0)<br>(5,2,0)(3,2,1)(4,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_I(0)\cdot 2^{\Omega_{\psi_I(0)\cdot 2^{\Omega_{\psi_I(0)\cdot 2}}}}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,0)(5,3,0) | $\psi(\psi_{\Omega_{\psi_I(0)\cdot 2+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(4,2,0)(5,3,0)(6,3,0) | $\psi(\Omega_{\psi_I(0)\cdot 2+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,0)(5,3,1) | $\psi(\Omega_{\psi_I(0)\cdot 2+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_I(0)\cdot 3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1) | $\psi(\Omega_{\psi_I(0)\cdot\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1)(3,2,0) | $\psi(\Omega_{\psi_I(0)\cdot\omega}+\psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot\omega}+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(4,2,1)(3,2,0)(4,2,0) | $\psi(\Omega_{\psi_I(0)\cdot\omega}+\Omega_{\psi_I(0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1)(3,2,1) | $\psi(\Omega_{\psi_I(0)\cdot\omega}+\Omega_{\psi_I(0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1) | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1)(4,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\psi_I(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)<br>(4,2,1)(4,1,0)(3,2,0) | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\psi_I(0)$<br>$+\psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot\omega}\cdot\psi_I(0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(4,2,1)(4,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\psi_I(0)+\Omega_{\psi_I(0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1)(4,1,0)<br>(3,2,0)(4,3,1)(5,3,1)(6,1,0)<br>(5,1,0)(5,3,1)(4,3,0)(5,3,0) | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\psi_I(0)+\Omega_{\psi_I(0)+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(4,2,1)(4,1,0)(3,2,1) | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\psi_I(0)+\Omega_{\psi_I(0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1)(4,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1)(4,1,0) | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\psi_I(0)+\Omega_{\psi_I(0)\cdot\omega}\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1)(4,1,0)<br>(4,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\psi_I(0)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(4,2,1)(4,1,0)(5,2,0) | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\psi_{\Omega_{\psi_I(0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1)(4,2,0) | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\Omega_{\psi_I(0)+1})$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1)(4,2,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1) | $\psi(\Omega_{\psi_I(0)\cdot\omega}{}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(4,2,1)(4,2,0)(5,3,0) | $\psi(\psi_{\Omega_{\psi_I(0)\cdot\omega+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(4,2,1)(4,2,0)(5,3,0)(6,3,0) | $\psi(\Omega_{\psi_I(0)\cdot\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(4,2,1)<br>(4,2,0)(5,3,1)(6,3,1)(7,1,0)(6,3,1) | $\psi(\Omega_{\psi_I(0)\cdot\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1)(4,2,1) | $\psi(\Omega_{\psi_I(0)\cdot\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1)(5,1,0) | $\psi(\Omega_{\psi_I(0)\cdot\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_I(0)^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(5,1,0) | $\psi(\Omega_{\psi_I(0)^\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_I(0)^{\psi_I(0)}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(6,2,0) | $\psi(\Omega_{\psi_{\Omega_{\psi_I(0)+1}}(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(6,2,0)(6,2,0) | $\psi(\Omega_{\psi_{\Omega_{\psi_I(0)+1}}(1)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(6,2,1)(7,2,1)(8,1,0)(9,2,0) | $\psi(\Omega_{\psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_{\Omega_{\psi_I(0)+1}}(0)})})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0) | $\psi(\Omega_{\Omega_{\psi_I(0)+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(3,2,1) | $\psi(\Omega_{\Omega_{\psi_I(0)+\omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(3,2,1)(4,2,1) | $\psi(\Omega_{\Omega_{\psi_I(0)+\omega^2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(3,2,1)(4,2,1)<br>(5,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\Omega_{\psi_I(0)\cdot 2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1) | $\psi(\Omega_{\Omega_{\psi_I(0)\cdot\omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(3,2,1)(4,2,1)(5,2,0) | $\psi(\Omega_{\Omega_{\Omega_{\psi_I(0)+1}}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_I(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_I(1)+\psi_I(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,0) | $\psi(\psi_I(1)+\psi_{\Omega_{\psi_I(0)+1}}(0))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_I(1) + \psi_{\Omega_{\psi_I(0)+1}}(\psi_I(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0)(2,1,0) | $\psi(\psi_I(1) + \psi_{\Omega_{\psi_I(0)+1}}(\psi_I(1)) \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0)(3,2,0) | $\psi(\psi_I(1) + \psi_{\Omega_{\psi_I(0)+1}}(\psi_I(1) + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,0,0)(3,2,0)(4,2,0) | $\psi(\psi_I(1) + \Omega_{\psi_I(0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0)(3,2,1) | $\psi(\psi_I(1) + \Omega_{\psi_I(0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_I(1) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,1,0) | $\psi(\psi_I(1) \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_I(1) \cdot \psi_I(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_I(1) \cdot \psi_{\Omega_{\psi_I(0)+1}}(\psi_I(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,1,0)(3,2,0)(4,2,0) | $\psi(\psi_I(1) \cdot \Omega_{\psi_I(0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_I(1)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,1,0)(4,1,0) | $\psi(\psi_I(1)^\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0) | $\psi(\psi_I(1)^{\Omega_{\psi_I(0)+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_I(1)^{\psi_I(1)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)(5,3,0) | $\psi(\psi_{\Omega_{\psi_I(1)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,0)(5,3,0)(6,3,0) | $\psi(\Omega_{\psi_I(1)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_I(1)+\psi_I(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,2,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{\psi_I(1)\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,0) | $\psi(\Omega_{\Omega_{\psi_I(1)+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(\psi_I(2))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(\psi_I(\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(2,1,0) | $\psi(\psi_I(\omega)\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(\psi_I(\omega)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,0)(3,2,0) | $\psi(\psi_{\Omega_{\psi_I(\omega)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(\Omega_{\psi_I(\omega)\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(5,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(\Omega_{\psi_I(\omega)^{\psi_I(\omega)}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)(6,2,0) | $\psi(\Omega_{\psi_{\Omega_{\psi_I(\omega)+1}}(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0) | $\psi(\Omega_{\Omega_{\psi_I(\omega)+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_I(\omega+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(\psi_I(\omega\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(2,1,1) | $\psi(\psi_I(\omega^2))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0) | $\psi(\psi_I(\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_I(\psi_I(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0) | $\psi(\psi_I(\psi_I(0))+\psi_I(0)\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,0)(4,2,0) | $\psi(\psi_I(\psi_I(0))+\Omega_{\psi_I(0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_I(\psi_I(0))+\psi_I(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(\psi_I(\psi_I(0))+\psi_I(\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0) | $\psi(\psi_I(\psi_I(0))+\psi_I(\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_I(\psi_I(0))\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,1,0)(2,1,0) | $\psi(\psi_I(\psi_I(0))\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,1,0)(4,2,0)(5,3,0) | $\psi(\psi_{\Omega_{\psi_I(\psi_I(0))+1}}(0))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,1,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,0)(6,0,0) | $\psi(\psi_I(\psi_I(0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,0)(6,3,1) | $\psi(\psi_I(\psi_I(0)+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)<br>(5,1,0)(4,2,0)(5,3,1)(6,3,1)<br>(7,3,0)(6,3,1)(7,1,0) | $\psi(\psi_I(\psi_I(0)+\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,1,0)(4,2,1) | $\psi(\psi_I(\psi_I(0)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,1,0)(6,2,0) | $\psi(\psi_I(\psi_{\Omega_{\psi_I(0)+1}}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,2,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_I(\psi_I(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(\psi_I(\psi_I(\omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0) | $\psi(\psi_I(\psi_I(\Omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0) | $\psi(\psi_I(\psi_I(\psi_I(\Omega))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I)$<br>$\psi(\psi_I(I))$<br>$\psi(\Phi(2,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,0) | $\psi(I+1)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,0)(2,1,0) | $\psi(I+\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\psi_I(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\psi_I(I))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0) | $\psi(I+\psi_I(I)\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\psi_I(I)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(0))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I+\psi_I(I+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1) | $\psi(I+\psi_I(I+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\psi_I(I+\psi_I(I)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,1,0)(6,2,0) | $\psi(I+\psi_I(I+\psi_{\Omega_{\psi_I(I)+1}}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(5,2,0) | $\psi(I+\psi_I(I+\Omega_{\psi_I(I)+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I+\psi_I(I+\psi_I(I+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,2,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\psi_I(I+\psi_I(I+\psi_I(I))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(I\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,2,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(I\cdot 2+\psi_I(I\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(I\cdot 2+\psi_I(I\cdot 2+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,2,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,0)(6,3,1)(7,3,0)(6,0,0) | $\psi(I\cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1) | $\psi(I\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1)(2,1,1) | $\psi(I\cdot\omega^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0) | $\psi(I\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1)(3,1,0) | $\psi(I\cdot\psi_I(I\cdot\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1) | $\psi(I^2\cdot\omega)$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,0,0) | $\psi(I^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I^{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,0)(3,0,0) | $\psi(I^{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,1,0) | $\psi(I^\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(3,1,0) | $\psi(I^{\psi_I(I^\Omega)})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(I^I)$<br>$\psi(\Phi(1,0,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I^{I+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,1,0)<br>(2,1,1)(3,1,0)(3,1,0)(2,0,0) | $\psi(I^{I\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(3,0,0) | $\psi(I^{I\cdot\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(I^{I^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,0,0) | $\psi(I^{I^\omega})$<br>$\psi(\Phi(1@\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,1,0) | $\psi(I^{I^\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(2,0,0) | $\psi(I^{I^I})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(5,1,0)(2,0,0) | $\psi(I^{I^{I^I}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{I+1}}(0))$<br>$\psi(\varepsilon_{I+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{I+1}}(0)+\psi_I(\psi_{\Omega_{I+1}}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,0) | $\psi(\psi_{\Omega_{I+1}}(0)+\psi_{\Omega_{\psi_I(\psi_{\Omega_{I+1}}(0))+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,0)(4,2,0) | $\psi(\psi_{\Omega_{I+1}}(0)+\Omega_{\psi_I(\psi_{\Omega_{I+1}}(0))+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{\Omega_{I+1}}(0)+\psi_I(\psi_{\Omega_{I+1}}(0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,1,0) | $\psi(\psi_{\Omega_{I+1}}(0)+\psi_I(\psi_{\Omega_{I+1}}(0)+\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1)<br>(5,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{I+1}}(0)+\psi_I(\psi_{\Omega_{I+1}}(0)+\psi_I(\psi_{\Omega_{I+1}}(0))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{\Omega_{I+1}}(0)+I)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1)<br>(5,2,0)(4,2,0)(5,3,1)(6,3,1)<br>(7,3,0)(6,3,1)(7,3,0)(6,0,0) | $\psi(\psi_{\Omega_{I+1}}(0)+I\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(4,2,1) | $\psi(\psi_{\Omega_{I+1}}(0)+I\cdot\omega)$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{\Omega_{I+1}}(0) + I^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(5,0,0) | $\psi(\psi_{\Omega_{I+1}}(0) + I^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(5,2,0)(4,0,0) | $\psi(\psi_{\Omega_{I+1}}(0) + I^I)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(6,3,0) | $\psi(\psi_{\Omega_{I+1}}(0) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(6,3,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,0)(8,4,0) | $\psi(\psi_{\Omega_{I+1}}(0) \cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(2,1,1) | $\psi(\psi_{\Omega_{I+1}}(0) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0) | $\psi(\psi_{\Omega_{I+1}}(0) \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0) | $\psi(\psi_{\Omega_{I+1}}(0) \cdot \psi_I(\psi_{\Omega_{I+1}}(0) \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{\Omega_{I+1}}(0) \cdot I)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{I+1}}(0)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(3,1,0) | $\psi(\psi_{\Omega_{I+1}}(0)^\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(3,1,0)(2,0,0) | $\psi(\psi_{\Omega_{I+1}}(0)^I)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{I+1}}(0)^{\psi_{\Omega_{I+1}}(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(4,1,0) | $\psi(\psi_{\Omega_{I+1}}(0)^{\psi_{\Omega_{I+1}}(0)^\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(4,1,0)(5,2,0) | $\psi(\psi_{\Omega_{I+1}}(0)^{\psi_{\Omega_{I+1}}(0)^{\psi_{\Omega_{I+1}}(0)}})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(4,2,0) | $\psi(\psi_{\Omega_{I+1}}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(5,1,0) | $\psi(\psi_{\Omega_{I+1}}(\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(5,1,0) | $\psi(\psi_{\Omega_{I+1}}(\psi_I(\psi_{\Omega_{I+1}}(\Omega))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,1,0)(2,0,0) | $\psi(\psi_{\Omega_{I+1}}(I))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(5,1,0)(4,2,0)(5,1,0)(2,0,0) | $\psi(\psi_{\Omega_{I+1}}(I \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(5,1,0)(5,1,0)(2,0,0) | $\psi(\psi_{\Omega_{I+1}}(I^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,1,0)(6,2,0) | $\psi(\psi_{\Omega_{I+1}}(\psi_{\Omega_{I+1}}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,2,0) | $\psi(\Omega_{I+1})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(5,3,0) | $\psi(\psi_{\Omega_{I+2}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,3,0)(6,3,0) | $\psi(\Omega_{I+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1) | $\psi(\Omega_{I+\omega})$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1) | $\psi(\Omega_{I+\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0) | $\psi(\Omega_{I+\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,1,0) | $\psi(\Omega_{I+\psi_I(\Omega_{I+\Omega})})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(2,0,0) | $\psi(\Omega_{I\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(2,1,0) | $\psi(\Omega_{I\cdot 2}+\psi_I(\Omega_{I\cdot 2})\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{I\cdot 2}+\psi_I(\Omega_{I\cdot 2}+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{I\cdot 2}+I)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(6,3,1)(7,3,1)(8,2,0)(4,0,0) | $\psi(\Omega_{I\cdot 2}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(2,1,1) | $\psi(\Omega_{I\cdot 2}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{I\cdot 2}\cdot I)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,1,0)(3,1,0)(2,0,0) | $\psi({\Omega_{I\cdot 2}}^{I})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(2,0,0) | $\psi({\Omega_{I\cdot 2}}^{\psi_{\Omega_{I+1}}(\Omega_{I\cdot 2})})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(4,1,0) | $\psi({\Omega_{I\cdot 2}}^{\psi_{\Omega_{I+1}}(\Omega_{I\cdot 2})\cdot\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(4,2,0) | $\psi({\Omega_{I\cdot 2}}^{\psi_{\Omega_{I+1}}(\Omega_{I\cdot 2}+1)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,1,0)(5,2,0) | $\psi({\Omega_{I\cdot 2}}^{\Omega_{I+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(4,2,1) | $\psi({\Omega_{I\cdot 2}}^{\Omega_{I+\omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(4,2,1)<br>(5,2,1)(6,1,0)(2,0,0) | $\psi({\Omega_{I\cdot 2}}^{\Omega_{I\cdot 2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,1,0)(5,1,0)(2,0,0) | $\psi({\Omega_{I\cdot 2}}^{{\Omega_{I\cdot 2}}^{I}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,1,0)(5,2,0)(6,3,0) | $\psi(\psi_{\Omega_{I\cdot 2+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,1,0)(5,2,0)(6,3,0)(7,3,0) | $\psi(\Omega_{I\cdot 2+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(5,2,0)<br>(6,3,1)(7,3,1)(8,1,0)(2,0,0) | $\psi(\Omega_{I\cdot 3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(5,2,1) | $\psi(\Omega_{I\cdot\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,1,0)(5,2,1)(6,1,0)(2,0,0) | $\psi(\Omega_{I^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,1,0)(6,1,0)(2,0,0) | $\psi(\Omega_{I^I})$ |

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|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(7,2,0) | $\psi(\Omega_{\psi_{\Omega_{I+1}}(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0) | $\psi(\Omega_{\Omega_{I+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(4,2,1) | $\psi(\Omega_{\Omega_{I+\omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(4,2,1)(5,2,1)(6,2,0) | $\psi(\Omega_{\Omega_{\Omega_{I+1}}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(\psi_{I_2}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(\psi_{I_2}(0)+\psi_I(\psi_{I_2}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{I_2}(0)+I)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(\psi_{I_2}(0)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(2,1,1) | $\psi(\psi_{I_2}(0)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(3,1,0) | $\psi(\psi_{I_2}(0)^{\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(\psi_{I_2}(0)^{\psi_I(\psi_{I_2}(0))})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(3,1,0)(2,0,0) | $\psi(\psi_{I_2}(0)^{I})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(3,1,0)(4,2,0) | $\psi(\psi_{I_2}(0)^{\psi_{\Omega_{I+1}}(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(\psi_{I_2}(0)^{\psi_{\Omega_{I+1}}(\psi_{I_2}(0))})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0)(4,2,0) | $\psi(\psi_{I_2}(0)^{\psi_{\Omega_{I+1}}(\psi_{I_2}(0)+1)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(\psi_{I_2}(0)^{\psi_{I_2}(0)})?$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,0) | $\psi(\psi_{I_2}(0)^{\psi_{I_2}(0)^{\Omega_{\psi_I(0)+1}}})?$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,0)(6,3,0) | $\psi(\psi_{\Omega_{\psi_{I_2}(0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,0)(6,3,0)(6,3,0) | $\psi(\psi_{\Omega_{\psi_{I_2}(0)+1}}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,0)(6,3,0)(7,3,0) | $\psi(\Omega_{\psi_{I_2}(0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,0) | $\psi(\Omega_{\Omega_{\psi_{I_2}(0)+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,0)(7,0,0) | $\psi(\psi_{I_2}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,1) | $\psi(\psi_{I_2}(\omega))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,1,0) | $\psi(\psi_{I_2}(\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,1,0)(2,0,0) | $\psi(\psi_{I_2}(I))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,1)<br>(6,1,0)(5,2,1)(6,1,0)(2,0,0) | $\psi(\psi_{I_2}(I^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,1,0)(7,2,0) | $\psi(\psi_{I_2}(\psi_{\Omega_{I+1}}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0) | $\psi(\psi_{I_2}(\Omega_{I+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0)(4,2,1) | $\psi(\psi_{I_2}(\Omega_{I+\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,1)<br>(6,2,0)(4,2,1)(5,2,1)(6,2,0) | $\psi(\psi_{I_2}(\Omega_{\Omega_{I+1}}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,1)(6,2,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(\psi_{I_2}(\psi_{I_2}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0) | $\psi(\psi_{I_2}(\psi_{I_2}(\Omega_{I+1})))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(I_2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0)<br>(5,2,0)(6,3,1)(7,3,1)(8,3,0)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(I_2 \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0)(5,2,1) | $\psi(I_2 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi({I_2}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(6,1,0)(2,0,0) | $\psi({I_2}^I)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(6,2,0)(5,0,0) | $\psi({I_2}^{I_2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,0) | $\psi(\psi_{\Omega_{I_2+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,0)(7,3,0) | $\psi(\psi_{\Omega_{I_2+1}}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,0)(8,3,0) | $\psi(\Omega_{I_2+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,1)(8,3,1)(9,3,0) | $\psi(\Omega_{\Omega_{I_2+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,1)<br>(8,3,1)(9,3,0)(8,0,0) | $\psi(\psi_{I_3}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,0)(8,3,1)(9,3,0)(8,0,0) | $\psi(I_3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1) | $\psi(I_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I_\omega + \psi_I(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I_\omega + \psi_I(I))$ |

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|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(I_\omega + \psi_I(\psi_{\Omega_{I+1}}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,0)(5,2,0) | $\psi(I_\omega + \psi_I(\Omega_{I+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I_\omega + \psi_I(\psi_{I_2}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1) | $\psi(I_\omega + \psi_I(I_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(2,1,0)(3,2,0) | $\psi(I_\omega + \psi_{\Omega_{\psi_I(I_\omega)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,0)(4,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1) | $\psi(I_\omega + \psi_{\Omega_{\psi_I(I_\omega)+1}}(I_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(2,1,0)(3,2,0)(4,2,0) | $\psi(I_\omega + \Omega_{\psi_I(I_\omega)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I_\omega + \psi_I(I_\omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(I_\omega + \psi_I(I_\omega + \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(I_\omega + I)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(6,3,0) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(2,1,0)(3,2,1)(4,2,1)(5,2,1) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(2,1,0)(3,2,1)(4,2,1)(5,2,1) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega)$<br>$+\psi_{\Omega_{\psi_I(I_\omega)+1}}(I_\omega + \psi_{\Omega_{I+1}}(I_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) + \psi_I(I_\omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega)$<br>$+\psi_I(I_\omega + \psi_{\Omega_{I+1}}(I_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega)$<br>$+\psi_I(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) + 1))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,0)(6,3,1) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) + \psi_I(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) + \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,0)(5,3,1)(6,3,1)<br>(7,3,0)(6,3,1)(7,3,0)(6,0,0) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) + I)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,1) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(2,1,1) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) \cdot I)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(4,2,0)(5,2,0) | $\psi(I_\omega + \Omega_{I+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(I_\omega + \psi_{I_2}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(7,3,1)(8,3,1)(9,3,0)(8,0,0) | $\psi(I_\omega + \psi_{I_3}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,1) | $\psi(I_\omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0) | $\psi(I_\omega \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,1) | $\psi({I_\omega}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,0)(3,2,0) | $\psi(\psi_{\Omega_{I_\omega+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0) | $\psi(\Omega_{\Omega_{I_\omega+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{I_{\omega+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,0)(7,3,1)(8,3,0)(7,0,0) | $\psi(I_{\omega+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) | $\psi(I_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,1) | $\psi(I_{\omega\cdot 3})$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1) | $\psi(I_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(I_{\omega^2+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) | $\psi(I_{\omega^2+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(4,2,1) | $\psi(I_{\omega^2\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(2,1,1) | $\psi(I_{\omega^3})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0) | $\psi(I_{\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I_{\psi_I(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(I_{\psi_I(\psi_{\Omega_{I+1}}(0))})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) | $\psi(I_{\psi_I(I_\omega)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0) | $\psi(I_{\psi_I(I_\Omega)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,1,0) | $\psi(I_{\psi_I(I_{\psi_I(I_\Omega)})})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(2,0,0) | $\psi(I_I)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,2,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(8,3,1)(9,2,0)(5,0,0) | $\psi(I_{I_2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,1) | $\psi(I_{I_\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0) | $\psi(I_{I_\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0) | $\psi(I_{I_{I_\Omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I_{I_{I_{\cdots}}})$<br>$\psi(\psi_{I(1,0)}(0))$<br>$\psi(\mathrm{IFP})$<br>$\psi((1-)^{1,0}\ 2\ 1-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0) | $\psi(\psi_{I(1,0)}(0)\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(1,1,1)(2,1,1)(3,1,1) | $\psi(\psi_{I(1,0)}(0)\cdot I_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{I(1,0)}(0)^2)$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(2,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{I(1,0)}(0)^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,1,0) | $\psi(\psi_{I(1,0)}(0)^\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{I(1,0)}(0)^{\psi_{I(1,0)}(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,0) | $\psi(\psi_{\Omega_{\psi_{I(1,0)}(0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{\psi_{I(1,0)}(0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{\psi_{I(1,0)}(0)\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0) | $\psi(\Omega_{\Omega_{\psi_{I(1,0)}(0)+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(3,2,1)(4,2,1)(5,2,0) | $\psi(\Omega_{\Omega_{\Omega_{\psi_{I(1,0)}(0)+1}}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{I_{\psi_{I(1,0)}(0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(\psi_{I_{\psi_{I(1,0)}(0)+1}}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1) | $\psi(\psi_{I_{\psi_{I(1,0)}(0)+1}}(\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(I_{\psi_{I(1,0)}(0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,0)(7,3,0) | $\psi(\Omega_{I_{\psi_{I(1,0)}(0)+1}+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(6,3,1)(7,3,1)(8,3,0)(7,0,0) | $\psi(\psi_{I_{\psi_{I(1,0)}(0)+1}+2}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(6,3,1)(7,3,1)(8,3,0)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(I_{\psi_{I(1,0)}(0)+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1) | $\psi(I_{\psi_{I(1,0)}(0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1) | $\psi(I_{\psi_{I(1,0)}(0)+\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,0)(3,1,0)(2,0,0) | $\psi(I_{\psi_{I(1,0)}(0)\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0) | $\psi(I_{\Omega_{\psi_{I(1,0)}(0)+1}})$ |

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|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0) | $\psi(I_{I_{\Omega_{\psi_{I(1,0)}(0)+1}}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{I(1,0)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(\psi_{I(1,0)}(2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1) | $\psi(\psi_{I(1,0)}(\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(2,1,1) | $\psi(\psi_{I(1,0)}(\omega^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0) | $\psi(\psi_{I(1,0)}(\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,1)(3,1,0) | $\psi(\psi_{I(1,0)}(\psi_{I(1,0)}(\Omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0)+\psi_{I(1,0)}(I(1,0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,1,0) | $\psi(I(1,0)+\psi_{I(1,0)}(I(1,0))\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,0) | $\psi(I(1,0)+\psi_{\Omega_{\psi_{I(1,0)}(I(1,0))+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,0) | $\psi(I(1,0)+\psi_{\Omega_{\psi_{I(1,0)}(I(1,0))+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,0)(4,2,0) | $\psi(I(1,0)+\Omega_{\psi_{I(1,0)}(I(1,0))+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0)+\Omega_{\psi_{I(1,0)}(I(1,0))\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0) | $\psi(I(1,0)+\Omega_{\Omega_{\psi_{I(1,0)}(I(1,0))+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0)+\psi_{I_{\psi_{I(1,0)}(I(1,0))+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0)+I_{\psi_{I(1,0)}(I(1,0))+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) | $\psi(I(1,0)+I_{\psi_{I(1,0)}(I(1,0))+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(2,0,0) | $\psi(I(1,0)+I_{\psi_{I(1,0)}(I(1,0))\cdot 2})$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0)+\psi_{I(1,0)}(I(1,0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(I(1,0)+\psi_{I(1,0)}(I(1,0)+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)(4,2,1)<br>(5,1,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0)+\psi_{I(1,0)}(I(1,0)$<br>$+\psi_{I(1,0)}(I(1,0))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1) | $\psi(I(1,0)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1)(3,1,0) | $\psi(I(1,0)\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0) | $\psi(I(1,0)\cdot\psi_{I(1,0)}(I(1,0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1) | $\psi(I(1,0)^2\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0)^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(3,0,0) | $\psi(I(1,0)^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(3,1,0)(2,0,0) | $\psi(I(1,0)^{I(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{I(1,0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,0)(5,2,0) | $\psi(\Omega_{I(1,0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,1,0) | $\psi(\Omega_{I(1,0)+\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{I(1,0)\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,1,0)(5,2,1) | $\psi(\Omega_{I(1,0)\cdot\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,1,0)(7,2,0) | $\psi(\Omega_{\psi_{\Omega_{I(1,0)+1}}(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,0) | $\psi(\Omega_{\Omega_{I(1,0)+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(\psi_{I_{I(1,0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(I_{I(1,0)+1})$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) | $\psi(I_{I(1,0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I_{I(1,0)+\psi_{I(1,0)}(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,1,0)(2,0,0) | $\psi(I_{I(1,0)\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,1,0)(5,2,1) | $\psi(I_{I(1,0)\cdot\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,1,0)(6,0,0) | $\psi(I_{I(1,0)^\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,1,0)(7,2,0) | $\psi(I_{\psi_{\Omega_{I(1,0)+1}}(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0) | $\psi(I_{\Omega_{I(1,0)+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I_{\psi_{I_{I(1,0)+1}}(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(4,2,1)(5,2,1)<br>(6,2,0)(5,2,1)(6,1,0)(2,0,0) | $\psi(I_{I_{I(1,0)+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(4,2,1)(5,2,1)(6,2,1) | $\psi(I_{I_{I(1,0)+\omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0) | $\psi(I_{I_{\Omega_{I(1,0)+1}}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(\psi_{I(1,1)}(0))$<br>$\psi((1-)^{(1,0)}\ 2\ 1-2 \text{ aft } (2\ 1-)^2\ 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(5,2,0)(6,3,0) | $\psi(\psi_{\Omega_{\psi_{I(1,1)}(0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(\psi_{I_{\psi_{I(1,1)}(0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,2,0)(6,3,1)(7,3,1)(8,3,1) | $\psi(I_{\psi_{I(1,1)}(0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,2,0)(6,3,1)(7,3,1)<br>(8,3,1)(7,3,1)(8,3,0)(7,0,0) | $\psi(\psi_{I(1,1)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(5,2,1) | $\psi(\psi_{I(1,1)}(\omega))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,2,1)(6,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{I(1,1)}(\psi_{I(1,0)}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1,1))$<br>$\psi(\text{2nd } (2\ 1-)^2\ 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(7,3,1)(8,3,1)(9,3,1)<br>(8,3,1)(9,3,0)(8,3,1)(9,3,0)(8,0,0) | $\psi(I(1,2))$<br>$\psi(\text{3rd } (2\ 1-)^2\ 2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1) | $\psi(I(1,\omega))$<br>$\psi(1-(2\ 1-)^2\ 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1) | $\psi(I(1,\omega^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0) | $\psi(I(1,\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,\psi_I(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0) | $\psi(I(1,\psi_I(I(1,\Omega))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,1,0)(2,0,0) | $\psi(I(1,I))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,1) | $\psi(I(1,I_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,\psi_{I(1,0)}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,1,0)(2,0,0) | $\psi(I(1,I(1,0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,0) | $\psi(I(1,I(1,\Omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0) | $\psi(I(1,I(1,I(1,\Omega))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{I(2,0)}(0))$<br>$\psi((1-)^{1,0}\ (2\ 1-)^2\ 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,0) | $\psi(\psi_{\Omega_{\psi_{I(2,0)}(0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{I_{\psi_{I(2,0)}(0)+1}}(0))$ |

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|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{I(1,\psi_{I(2,0)}(0)+1)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{I(2,0)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(\psi_{I(2,0)}(\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I(2,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{I(2,0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,0)(5,2,0) | $\psi(\Omega_{I(2,0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(I_{I(2,0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,2,0)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1,I(2,0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(I(2,1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(I(2,\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,0) | $\psi(I(2,\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{I(3,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0) | $\psi(I(\omega,0))$<br>$\psi((2\ 1-)^{\omega}\ 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0) | $\psi(I(\omega,0)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(2,1,0) | $\psi(I(\omega,0)\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0) | $\psi(I(\omega,0)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,0)(3,2,0) | $\psi(\psi_{\Omega_{I(\omega,0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{I(\omega,0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(I_{I(\omega,0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,I(\omega,0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I(2,I(\omega,0)+1))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,0,0) | $\psi(I(\omega,1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,0,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,1)(7,0,0) | $\psi(I(\omega,2))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(2,1,1) | $\psi(I(\omega,\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0) | $\psi(I(\omega,\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(2,1,1)(3,1,0) | $\psi(I(\omega,I(\omega,\Omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{I(\omega+1,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{I(\omega+2,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,1)(3,0,0) | $\psi(I(\omega\cdot 2,0))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(3,0,0) | $\psi(I(\omega^2,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0) | $\psi(I(\Omega,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_I(0),0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,1,0)(2,0,0) | $\psi(I(I,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,1,0)(2,0,0) | $\psi(I(I(1,0),0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0) | $\psi(I(I(\Omega,0),0))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{I(1,0,0)}(0))$<br>$\psi(\psi_{\psi_M(M^M)}(0))$<br>$\psi((2\ 1-)^{1,0}\ 2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,0) | $\psi(\psi_{I(1,0,0)}(0)\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,0) | $\psi(\psi_{\Omega_{\psi_{I(1,0,0)}(0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{I_{\psi_{I(1,0,0)}(0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,0,0) | $\psi(I(\omega,\psi_{I(1,0,0)}(0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0) | $\psi(I(\Omega,\psi_{I(1,0,0)}(0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0),1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)+\psi_{I(1,0,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(2,1,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)+\psi_{I(1,0,0)}(0)\cdot\Omega)$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)+\psi_{\Omega_{\psi_{I(1,0,0)}(0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)$<br>$+\psi_{\Omega_{\psi_{I(1,0,0)}(0)+1}}(I(\psi_{I(1,0,0)}(0),1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(2,1,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)$<br>$+\psi_{\Omega_{\psi_{I(1,0,0)}(0)+1}}(I(\psi_{I(1,0,0)}(0),1))\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(3,2,0)(4,2,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)+\Omega_{\psi_{I(1,0,0)}(0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(3,2,1) | $\psi(I(\psi_{I(1,0,0)}(0),1)+\Omega_{\psi_{I(1,0,0)}(0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(3,2,1)(4,2,1)(5,2,1) | $\psi(I(\psi_{I(1,0,0)}(0),1)+I_{\psi_{I(1,0,0)}(0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,0,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)+I(\omega,\psi_{I(1,0,0)}(0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,1,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)+I(\Omega,\psi_{I(1,0,0)}(0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,1,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)\cdot\psi_{I(1,0,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(4,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)\cdot\psi_{I(1,0,0)}(0)$<br>$+I(\psi_{I(1,0,0)}(0),1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)(<br>2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,1,0)(4,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)\cdot\psi_{I(1,0,0)}(0)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,1,0)(5,2,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)\cdot\psi_{\Omega_{\psi_{I(1,0,0)}(0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(4,1,0)(5,2,1)(6,2,1)(7,2,1) | $\psi(I(\psi_{I(1,0,0)}(0),1)\cdot I_{\psi_{I(1,0,0)}(0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(4,1,0)(5,2,1)(6,2,1)(7,2,1)(7,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)^2)$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(4,1,0)(5,2,1)(6,2,1)(7,2,1)(7,1,0)<br>(4,2,1)(5,2,1)(5,1,0)(4,1,0)(5,2,1)<br>(6,2,1)(7,2,1)(7,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,1,0)(5,2,1)(6,2,1)(7,2,1)<br>(7,1,0)(5,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0),1)^{\psi_{I(1,0,0)}(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,0) | $\psi(\psi_{\Omega_{I(\psi_{I(1,0,0)}(0),1)+1}}(0))?$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,0)(5,3,1) | $\psi(\Omega_{I(\psi_{I(1,0,0)}(0),1)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,1) | $\psi(I_{I(\psi_{I(1,0,0)}(0),1)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,1)(7,1,0) | $\psi(I(\Omega, I(\psi_{I(1,0,0)}(0),1)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,1)(7,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0),2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,1) | $\psi(I(\psi_{I(1,0,0)}(0),\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(4,2,1)(5,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0),\psi_{I(1,0,0)}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,1,0)(4,2,1) | $\psi(I(\psi_{I(1,0,0)}(0),\psi_{I(1,0,0)}(0)+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(4,2,1)(5,1,0)(4,2,1)(5,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0),\psi_{I(1,0,0)}(0)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,1,0)(6,2,0) | $\psi(I(\psi_{I(1,0,0)}(0),\psi_{\Omega_{\psi_{I(1,0,0)}(0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,1)(5,2,0) | $\psi(I(\psi_{I(1,0,0)}(0),\Omega_{\psi_{I(1,0,0)}(0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(4,2,1)(5,2,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0),I(\psi_{I(1,0,0)}(0),1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{I(\psi_{I(1,0,0)}(0)+1,0)}(0))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(4,2,1)(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(\psi_{I(1,0,0)}(0)+1,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I(\psi_{I(1,0,0)}(0)+2,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,2,1)(5,0,0) | $\psi(I(\psi_{I(1,0,0)}(0)+\omega,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(4,2,1)(5,2,1)(5,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0)\cdot 2,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(5,0,0) | $\psi(I(\psi_{I(1,0,0)}(0)\cdot\omega,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(5,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0)^2,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(6,2,0) | $\psi(I(\psi_{\Omega_{\psi_{I(1,0,0)}(0)+1}}(0),0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,2,0) | $\psi(I(\Omega_{\psi_{I(1,0,0)}(0)+1},0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(3,2,1)(4,2,1)(5,2,1) | $\psi(I(I_{\psi_{I(1,0,0)}(0)+\omega},0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(3,2,1)(4,2,1)(5,2,1)(5,2,0) | $\psi(I(I(\Omega_{\psi_{I(1,0,0)}(0)+1},0),0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{I(1,0,0)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(4,2,0)(5,3,1)(6,3,1)<br>(7,3,1)(7,3,0)(6,0,0) | $\psi(\psi_{I(1,0,0)}(2))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1) | $\psi(\psi_{I(1,0,0)}(\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0) | $\psi(\psi_{I(1,0,0)}(\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,0) | $\psi(\psi_{I(1,0,0)}(\psi_{I(1,0,0)}(\Omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,0))$<br>$\psi(\psi_M(M^M))$<br>$\psi(M^M)$<br>$\psi((2\ 1-)^{1,1}\ 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,0)+\psi_{I(1,0,0)}(I(1,0,0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0) | $\psi(I(1,0,0)+\psi_{I(1,0,0)}(I(1,0,0))\cdot\Omega)$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,0) | $\psi(I(1,0,0)+\psi_{\Omega_{\psi_{I(1,0,0)}(I(1,0,0))+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,1,0) | $\psi(I(1,0,0)+I(\Omega,\psi_{I(1,0,0)}(I(1,0,0))+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)}(I(1,0,0)),1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0,0)+\psi_{I(1,0,0)}(I(1,0,0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(4,2,1) | $\psi(I(1,0,0)+\psi_{I(1,0,0)}(I(1,0,0)+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,0)$<br>$+\psi_{I(1,0,0)}(I(1,0,0)+\psi_{I(1,0,0)}(I(1,0,0))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,2,1)(5,2,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0,0)+\psi_{I(1,0,0)}(I(1,0,0)$<br>$+\psi_{I(1,0,0)}(I(1,0,0)+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0,0)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1) | $\psi(I(1,0,0)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,0)\cdot\psi_{I(1,0,0)}(I(1,0,0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,0)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(3,0,0) | $\psi(I(1,0,0)^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(3,1,0)(2,0,0) | $\psi(I(1,0,0)^{I(1,0,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{I(1,0,0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,0)(5,2,0) | $\psi(\Omega_{I(1,0,0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(\psi_{I_{I(1,0,0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(I_{I(1,0,0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1,I(1,0,0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0) | $\psi(I(\omega,I(1,0,0)+1))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,1,0) | $\psi(I(\Omega, I(1,0,0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(0), I(1,0,0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(I(1,0,0)), I(1,0,0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,1,0) | $\psi(I(\psi_{I(1,0,0)}(I(\Omega, I(1,0,0)+1)), I(1,0,0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,1,0) | $\psi(I(\psi_{I(1,0,0)}(I(\psi_{I(1,0,0)}(I(\Omega, I(1,0,0)+1))$<br>$, I(1,0,0)+1)), I(1,0,0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,1,0)(2,0,0) | $\psi(I(I(1,0,0),1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(2,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,1,0)(2,0,0) | $\psi(I(I(1,0,0),1) + \psi_{I(1,0,0)}(I(I(1,0,0),1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(2,1,0) | $\psi(I(I(1,0,0),1) + \psi_{I(1,0,0)}(I(I(1,0,0),1)) \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,0,0) | $\psi(I(I(1,0,0),1) + \psi_{I(1,0,0)}(I(I(1,0,0),1)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(4,2,1)(5,1,0)(2,0,0) | $\psi(I(I(1,0,0),1) + I(1,0,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,1)(8,2,0)(4,0,0) | $\psi(I(I(1,0,0),1) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(2,1,1) | $\psi(I(I(1,0,0),1) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,1,0)(2,0,0) | $\psi(I(I(1,0,0),1) \cdot \psi_{I(1,0,0)}(I(I(1,0,0),1)))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(I(1,0,0),1)\cdot I(1,0,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,1,0) | $\psi(I(I(1,0,0),1)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(3,1,0) | $\psi(I(I(1,0,0),1)^\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(4,2,0) | $\psi(\psi_{\Omega_{I(I(1,0,0),1)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(4,2,1)(5,2,1)(6,2,1)(6,0,0) | $\psi(I(\omega,I(I(1,0,0),1)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(2,0,0) | $\psi(I(I(1,0,0),2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(5,0,0) | $\psi(I(I(1,0,0),\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(5,1,0)(2,0,0) | $\psi(\psi_{I(I(1,0,0)+1,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,1,0)(5,2,1) | $\psi(\psi_{I(I(1,0,0)+1,0)}(\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(I(I(1,0,0)+1,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I(I(1,0,0)+2,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(5,2,1)(6,2,1)(6,1,0)(2,0,0) | $\psi(I(I(1,0,0)\cdot 2,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(6,1,0)(2,0,0) | $\psi(I(I(1,0,0)^2,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,1,0)(7,2,0) | $\psi(I(\psi_{\Omega_{I(1,0,0)+1}}(0),0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0) | $\psi(I(\Omega_{I(1,0,0)+1},0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(4,2,1)(5,2,1)(6,2,1)(6,2,0) | $\psi(I(I(\Omega_{I(1,0,0)+1},0),0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(5,0,0) | $\psi(\psi_{I(1,0,1)}(0))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(9,3,0)(8,3,1)(9,3,0)(8,0,0) | $\psi(\psi_{I(1,0,2)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1) | $\psi(I(1,0,\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1) | $\psi(I(1,0,\omega^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0) | $\psi(I(1,0,\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{I(1,1,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(\psi_{I(1,1,0)}(\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,1,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(I(1,1,\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{I(1,2,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,0,0) | $\psi(I(1,\omega,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0) | $\psi(I(1,\Omega,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{I(2,0,0)}(0))$<br>$\psi((2\ 1-)^{2,0}\ 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1) | $\psi(\psi_{I(2,0,0)}(\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I(2,0,0))$<br>$\psi(\psi_M(M^{M\cdot 2}))$<br>$\psi(M^{M\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,1) | $\psi(I(2,0,\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{I(2,1,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0) | $\psi(I(2,\omega,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{I(3,0,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,0,0) | $\psi(I(\omega,0,0))$<br>$\psi(\psi_M(M^{M\cdot\omega}))$<br>$\psi((2\ 1-)^{\omega,0}\ 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(\psi_{I(1,0,0,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,0,0))$<br>$\psi(\psi_M(M^{M^2}))$<br>$\psi(M^{M^2})$<br>$\psi((2\ 1-)^{1,0,1}\ 2)$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1) | $\psi(I(1,0,0,\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,1,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,1,0,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(2,0,0,0))$<br>$\psi(M^{M^2\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(3,0,0) | $\psi(I(\omega,0,0,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(3,1,0)(2,0,0) | $\psi(\psi_{I(1,0,0,0,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,0,0,0))$<br>$\psi(M^{M^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(3,1,0)(3,1,0)(2,0,0) | $\psi(\psi_{I(1,0,0,0,0,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,0,0) | $\psi(I(1@\omega))$<br>$\psi(M^{M^\omega})$<br>$\psi((2\ 1-)^{1@\omega}\ 2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,1,0) | $\psi(I(1@\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(7,1,0)(2,0,0) | $\psi(I(1@I))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,0,0) | $\psi(I(1@(1,0)))$<br>$\psi(M^{M^M})$<br>$\psi((2\ 1-)^{1@(1,0)}\ 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,1,1) | $\psi(I(1@(1,0),\omega@0))$<br>$\psi(M^{M^M}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1@(1,0),1@1))$<br>$\psi(M^{M^M+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1@(1,0),2@1))$<br>$\psi(M^{M^M+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1@(1,0),1@2))$<br>$\psi(M^{M^M+M})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1@(1,0),1@3))$<br>$\psi(M^{M^M+M^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,1)(3,1,0)(4,0,0) | $\psi(I(1@(1,0),1@\omega))$<br>$\psi(M^{M^M+M^\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,0,0) | $\psi(I(2@(1,0)))$<br>$\psi(M^{M^M\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(3,0,0) | $\psi(I(\omega@(1,0)))$<br>$\psi(M^{M^M\cdot\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1@(1,1)))$<br>$\psi(M^{M^{M+1}})$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(3,1,0)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(3,1,0)(2,0,0) | $\psi(I(2@(1,1)))$<br>$\psi(M^{M^{M+1}\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(3,1,0)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1@(1,2)))$<br>$\psi(M^{M^{M+2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(3,1,0)(4,1,0)(2,0,0) | $\psi(I(1@(2,0)))$<br>$\psi(M^{M^{M\cdot 2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(3,1,0)(4,1,0)<br>(3,1,0)(4,1,0)(2,0,0) | $\psi(I(1@(3,0)))$<br>$\psi(M^{M^{M\cdot 3}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(4,0,0) | $\psi(I(1@(\omega,0)))$<br>$\psi(M^{M^{M\cdot\omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(4,1,0)(2,0,0) | $\psi(I(1@(1,0,0)))$<br>$\psi(M^{M^{M^2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(4,1,0)(3,1,0)(4,1,0)(2,0,0) | $\psi(I(1@(1,1,0)))$<br>$\psi(M^{M^{M^2+M}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(4,1,0)(3,1,0)<br>(4,1,0)(4,1,0)(2,0,0) | $\psi(I(1@(2,0,0)))$<br>$\psi(M^{M^{M^2\cdot 2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(4,1,0)(4,0,0) | $\psi(I(1@(\omega,0,0)))$<br>$\psi(M^{M^{M^2\cdot\omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(4,1,0)(4,1,0)(2,0,0) | $\psi(I(1@(1,0,0,0)))$<br>$\psi(M^{M^{M^3}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(5,0,0) | $\psi(I(1@(1@\omega)))$<br>$\psi(M^{M^{M^\omega}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(5,1,0)(2,0,0) | $\psi(I(1@(1@(1,0))))$<br>$\psi(M^{M^{M^M}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(5,1,0)(6,1,0)(2,0,0) | $\psi(I(1@(1@(1@(1,0)))))$<br>$\psi(M^{M^{M^{M^M}}})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\varepsilon_{M+1})$<br>$\psi(\psi_{\Omega_{M+1}}(0))$<br>$\psi((1-)^{1,0}\text{ aft }2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,1) | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(M\cdot\omega))$<br>$\psi(\psi_{\Omega_{M+1}}(0)+I_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{\Omega_{M+1}}(0)+I(1,0,0))$<br>$\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(M^M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0) | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0))\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0))^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,0) | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_{\Omega_{\psi_M(\psi_{\Omega_{M+1}}(0))+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,0)(4,2,0) | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0)+1))$<br>$\psi(\psi_{\Omega_{M+1}}(0)+\Omega_{\psi_M(\psi_{\Omega_{M+1}}(0))+1})$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1) | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0)+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,1,0) | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0)+\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0))))$<br>$\psi(\psi_{\Omega_{M+1}}(0)+\Omega_{\psi_M(\psi_{\Omega_{M+1}}(0))\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,1,0)(4,2,1) | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0))\cdot\omega))$<br>$\psi(\psi_{\Omega_{M+1}}(0)+\Omega_{\psi_M(\psi_{\Omega_{M+1}}(0))\cdot\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(5,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{M+1}}(0)$<br>$+\psi_M(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0))^{\psi_M(\psi_{\Omega_{M+1}}(0))}))$<br>$\psi(\psi_{\Omega_{M+1}}(0)+\Omega_{\psi_M(\psi_{\Omega_{M+1}}(0))^{\psi_M(\psi_{\Omega_{M+1}}(0))}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0) | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0)+1)))$<br>$\psi(\psi_{\Omega_{M+1}}(0)+\Omega_{\Omega_{\psi_M(\psi_{\Omega_{M+1}}(0))+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(\alpha\to\psi_{\Omega_{M+1}}(0)+\psi_M(\alpha)\text{ FP})$<br>$\psi(\psi_{\Omega_{M+1}}(0)+\psi_{I_{\psi_M(\psi_{\Omega_{M+1}}(0))+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{\Omega_{M+1}}(0)+M)$<br>$\psi(\psi_{\Omega_{M+1}}(0)+I_{\psi_M(\psi_{\Omega_{M+1}}(0))+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1) | $\psi(\psi_{\Omega_{M+1}}(0)+M\cdot\omega)$<br>$\psi(\psi_{\Omega_{M+1}}(0)+I_{\psi_M(\psi_{\Omega_{M+1}}(0))+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\alpha\to(\psi_{\Omega_{M+1}}(0)+M\cdot\alpha)\text{ FP})$<br>$\psi(\psi_{\Omega_{M+1}}(0)+\psi_{I(1,\psi_M(\psi_{\Omega_{M+1}}(0))+1)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{\Omega_{M+1}}(0)+M^2)$<br>$\psi(\psi_{\Omega_{M+1}}(0)+I(1,\psi_M(\psi_{\Omega_{M+1}}(0))+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\alpha\to(\psi_{\Omega_{M+1}}(0)+M^\alpha)\text{ FP})$<br>$\psi(\psi_{\Omega_{M+1}}(0)+\psi_{I(1,0,\psi_M(\psi_{\Omega_{M+1}}(0))+1)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{\Omega_{M+1}}(0)+M^M)$<br>$\psi(\psi_{\Omega_{M+1}}(0)+I(1,0,\psi_M(\psi_{\Omega_{M+1}}(0))+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(6,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_{\Omega_{M+1}}(0)+M^{M^M})$<br>$\psi(\psi_{\Omega_{M+1}}(0)+I(1@(1,0),\psi_M(\psi_{\Omega_{M+1}}(0))+1@0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,0) | $\psi(\psi_{\Omega_{M+1}}(0)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(6,3,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,1)(7,3,0)(8,4,0) | $\psi(\psi_{\Omega_{M+1}}(0)\cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1) | $\psi(\psi_{\Omega_{M+1}}(0)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0) | $\psi(\psi_{\Omega_{M+1}}(0)\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0)(2,1,1)(3,1,0) | $\psi(\psi_{\Omega_{M+1}}(0)\cdot\psi_M(\psi_{\Omega_{M+1}}(0)\cdot\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\alpha\to\psi_{\Omega_{M+1}}(0)\cdot\psi_M(\alpha)\text{ FP})$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{\Omega_{M+1}}(0) \cdot M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1) | $\psi(\psi_{\Omega_{M+1}}(0) \cdot M \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{\Omega_{M+1}}(0) \cdot M^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)(3,0,0) | $\psi(\psi_{\Omega_{M+1}}(0) \cdot M^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(\alpha \to \psi_{\Omega_{M+1}}(0) \cdot M^{\psi_M(\alpha)} \text{ FP})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{\Omega_{M+1}}(0) \cdot M^M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{M+1}}(0)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(3,0,0) | $\psi(\psi_{\Omega_{M+1}}(0)^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{\Omega_{M+1}}(0)^M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{M+1}}(0)^{\psi_{\Omega_{M+1}}(0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(4,0,0) | $\psi(\psi_{\Omega_{M+1}}(0)^{\psi_{\Omega_{M+1}}(0)^\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(4,1,0)(5,2,0) | $\psi(\psi_{\Omega_{M+1}}(0)^{\psi_{\Omega_{M+1}}(0)^{\psi_{\Omega_{M+1}}(0)}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(4,2,0) | $\psi(\psi_{\Omega_{M+1}}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,1,0) | $\psi(\psi_{\Omega_{M+1}}(\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,1,0)(2,0,0) | $\psi(\alpha \to \psi_{\Omega_{M+1}}(\psi_M(\alpha)) \text{ FP})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(5,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{\Omega_{M+1}}(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,1,0)(6,2,0) | $\psi(\psi_{\Omega_{M+1}}(\psi_{\Omega_{M+1}}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,2,0) | $\psi(\psi_{M_2}(0))$<br>$\psi(\Omega_{M+1})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,1) | $\psi(\Omega_{M+\omega})$<br>$\psi(\psi_{M_2}(\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,1)(5,2,1)(6,1,0) | $\psi(\Omega_{M+\psi_M(\Omega_{M+\Omega})})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,1,0)(2,0,0) | $\psi(\alpha \to \Omega_{M+\psi_M(\alpha)} \text{ FP})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)<br>(6,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{M\cdot 2})$<br>$\psi(\psi_{M_2}(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,0) | $\psi(\Omega_{\Omega_{M+1}})$<br>$\psi(\psi_{M_2}(\psi_{M_2}(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(\psi_{I_{M+1}}(0))$<br>$\psi(\alpha \to \psi_{M_2}(\alpha) \text{ FP})$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,0)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I_{M+1})$<br>$\psi(\psi_{M_2}(M_2))$<br>$\psi(M_2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) | $\psi(I_{M+\omega})$<br>$\psi(\psi_{M_2}(M_2 \cdot \omega))$<br>$\psi(M_2 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1, M+1))$<br>$\psi(\psi_{M_2}({M_2}^2))$<br>$\psi({M_2}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,1,0)(2,0,0) | $\psi(I(M, 1))$<br>$\psi(\psi_{M_2}({M_2}^M))$<br>$\psi({M_2}^M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1, 0, M+1))$<br>$\psi(\psi_{M_2}({M_2}^{M_2}))$<br>$\psi({M_2}^{M_2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)<br>(7,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1@(1,0), M+1@0))$<br>$\psi(M_2^{M_2^{M_2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)(7,3,0) | $\psi(\psi_{\Omega_{M_2+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(7,3,0)(7,3,0) | $\psi(\psi_{\Omega_{M_2+1}}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(7,3,0)(8,3,0) | $\psi(\Omega_{M_2+1})$<br>$\psi(\psi_{M_3}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)(7,3,1)<br>(8,3,1)(9,3,0)(8,3,1)(9,3,0)(8,0,0) | $\psi(I_{M_2+1})$<br>$\psi(M_3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(M_\omega)$<br>$\psi(1-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(2,1,0) | $\psi(M_\omega \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi({M_\omega}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,0)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi({M_\omega}^{M_\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,0)(3,2,0) | $\psi(\psi_{\Omega_{M_\omega+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{M_\omega+1})$<br>$\psi(\psi_{M_{\omega+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I_{M_\omega+1})$<br>$\psi(M_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,0) | $\psi(\psi_{\Omega_{M_{\omega+1}+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,1)(8,3,0)(9,4,0) | $\psi(\psi_{\Omega_{M_{\omega+2}+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,2,1) | $\psi(M_{\omega \cdot 2})$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,2,1)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,1)(7,3,1) | $\psi(M_{\omega\cdot 3})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(2,1,1) | $\psi(M_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(2,1,1) | $\psi(M_{\omega^3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0) | $\psi(M_{\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,1) | $\psi(M_{I_\omega})$<br>$\psi(M_{\psi_M(M\cdot\omega)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1) | $\psi(M_{\psi_M(M_\omega)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,1,0) | $\psi(M_{\psi_M(M_\Omega)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,1,0) | $\psi(M_{\psi_M(M_{\psi_M(M_\Omega)})})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(5,2,1)(6,1,0)(2,0,0) | $\psi(\alpha \to M_{\psi_M(\alpha)}\ \mathrm{FP})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(5,2,1)(6,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(M_M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(M_{M_\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(2,1,1)(3,1,0) | $\psi(M_{M_\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{M(1,0)}(0))$<br>$\psi((1-)^{1,0}\ 2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1,0))$<br>$\psi(\psi_{M(1;0)}(0))$<br>$\psi(2\ 1-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(3,0,0) | $\psi(M(1,0)^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{M(1,0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,0)(5,2,0) | $\psi(\Omega_{M(1,0)+1})$<br>$\psi(\psi_{M_{M(1,0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(I_{M(1,0)+1})$<br>$\psi(M_{M(1,0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1) | $\psi(I_{M(1,0)+\omega})$<br>$\psi(M_{M(1,0)+1}\cdot\omega)$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1, M(1,0)+1))$<br>$\psi({M_{M(1,0)+1}}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(7,3,0) | $\psi(\psi_{\Omega_{M_{M(1,0)+1}+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1) | $\psi(M_{M(1,0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(5,2,1) | $\psi(M_{M(1,0)+\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,0) | $\psi(M_{\Omega_{M(1,0)+1}})$? |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,0) | $\psi(M_{M_{\Omega_{M(1,0)+1}}})$? |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(\psi_{M(1,1)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,0)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(M(1,1))$<br>$\psi(\psi_{M(1;0)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1) | $\psi(M(1,\omega))$<br>$\psi(\psi_{M(1;0)}(\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(2,1,1) | $\psi(M(1,\omega^2))$<br>$\psi(\psi_{M(1;0)}(\omega^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,0) | $\psi(M(1,\Omega))$<br>$\psi(\psi_{M(1;0)}(\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{M(2,0)}(0))$<br>$\psi(\alpha \to \psi_{M(1;0)}(\alpha)\ \mathrm{FP})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(2,0))$<br>$\psi(\psi_{M(1;0)}(M(1;0)))$<br>$\psi(M(1;0))$<br>$\psi(2-2\ 1-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(2,0)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(3,0,0) | $\psi(M(2,0)^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(2,0)^{M(2,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{M(2,0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(5,2,0) | $\psi(\Omega_{M(2,0)+1})$<br>$\psi(\psi_{M_{M(2,0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(I_{M(2,0)+1})$<br>$\psi(M_{M(2,0)+1})$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1) | $\psi(M_{M(2,0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(M(1, M(2,0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(M(2,1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(M(2,\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1)(2,1,1) | $\psi(M(2,\omega^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0) | $\psi(M(2,\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{M(3,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(3,0))$<br>$\psi(\psi_{M(1;0)}(M(1;0)^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1) | $\psi(M(3,\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,0,0) | $\psi(M(\omega,0))$<br>$\psi(\psi_{M(1;0)}(M(1;0)^\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,0) | $\psi(M(\Omega,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{M(1,0,0)}(0))$<br>$\psi((2\ 1-)^{1,0}\ 2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1,0,0))$<br>$\psi((2\ 1-)^{1,1}\ 2-2)$<br>$\psi(\psi_{M(1;0)}(M(1;0)^{M(1;0)}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,1) | $\psi(M(1,0,\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1,1,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0) | $\psi(M(1,\omega,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(2,0,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(3,0,0) | $\psi(M(\omega,0,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1,0,0,0))$<br>$\psi(\psi_{M(1;0)}(M(1;0)^{M(1;0)^2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,0,0) | $\psi(M(1@\omega))$<br>$\psi(\psi_{M(1;0)}(M(1;0)^{M(1;0)^\omega}))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,1,0)(2,0,0) | $\psi(M(1@(1,0)))$<br>$\psi(\psi_{M(1;0)}(M(1;0)^{M(1;0)^{M(1;0)}}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{M(1;0)+1}}(0))$<br>$\psi((1-)^{1,0}\ 2-2\ 1-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,0)(5,2,0) | $\psi(\Omega_{M(1;0)+1})$<br>$\psi(2 \text{ aft } 2-2\ 1-2-2)$<br>$\psi(\psi_{M_{M(1;0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(I_{M(1;0)+1})$<br>$\psi(2\ 1-2 \text{ aft } 2-2\ 1-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1) | $\psi(M_{M(1;0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(\psi_{M(1,M(1;0)+1)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(M(1,M(1;0)+1))$<br>$\psi(\psi_{M(1;1)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(M(2,M(1;0)+1))$<br>$\psi(\psi_{M(1;1)}(M(1;1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,1)(6,1,0) | $\psi(M(\Omega,M(1;0)+1))$<br>$\psi(\psi_{M(1;1)}(M(1;1)^{\Omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(M(1,0,M(1;0)+1))$<br>$\psi(\psi_{M(1;1)}(M(1;1)^{M(1;1)}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(7,3,0) | $\psi(\psi_{\Omega_{M(1;1)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(7,3,0)(8,3,0) | $\psi(\Omega_{M(1;1)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,1)(6,2,0)<br>(7,3,1)(8,3,1)(9,3,1)(9,3,1)(8,3,1)<br>(9,3,1)(9,3,0)(10,4,0)(11,4,0) | $\psi(\Omega_{M(1;2)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(1;\omega))$<br>$\psi(1-2-2\ 1-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1) | $\psi(M(1;\omega^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{M(1;1,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1;1,0))$<br>$\psi(2\ 1-2-2\ 1-2-2)$<br>$\psi(\psi_{M(2;0)}(0))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(M(1;1,\omega))$<br>$\psi(1-2\ 1-2-2\ 1-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,1)(2,0,0) | $\psi(M(1;2,0))$<br>$\psi(2\ 1-2\ 1-2-2\ 1-2-2)$<br>$\psi(\psi_{M(2;0)}(M(2;0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,0,0) | $\psi(M(1;1,0,0))$<br>$\psi((2\ 1-)^{1,1}\ 2-2\ 1-2-2)$<br>$\psi(\psi_{M(2;0)}(M(2;0)^{M(2;0)}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,0,0) | $\psi(M(1;1@\omega))$<br>$\psi(\psi_{M(2;0)}(M(2;0)^{M(2;0)^{\omega}}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,1,0)(2,0,0) | $\psi(M(1;1@(1,0)))$<br>$\psi(\psi_{M(2;0)}(M(2;0)^{M(2;0)^{M(2;0)}}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{M(2;0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0)(5,2,0) | $\psi(\Omega_{M(2;0)+1})$<br>$\psi(2\ \text{aft}\ 2-2\ 1-2-2\ 1-2-2)$<br>$\psi(\psi_{M_{M(2;0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(6,2,0)(7,3,0)(8,3,0) | $\psi(\Omega_{M(2;1)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1) | $\psi(M(2;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,0,0) | $\psi(M(\omega;0))$<br>$\psi((2-2\ 1-)^{\omega}\ 2-2)$<br>$\psi(\psi_N(\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,1,0) | $\psi(M(\Omega;0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{M(1,0;0)}(0))$<br>$\psi((2-2\ 1-)^{1,0}\ 2-2)$<br>$\psi(\alpha \to \psi_N(\alpha)\ \text{FP})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1,0;0))$<br>$\psi(2\ 1-(2-2\ 1-)^{1,0}\ 2-2)$<br>$\psi(\alpha \to \psi_N(\alpha)\ \text{AP})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1) | $\psi(M(1,0;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1,0;1,0))$<br>$\psi(\psi_{M(1,1;0)}(0))$<br>$\psi(\psi_{\psi_N(N)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1,0;1,0,0))$<br>$\psi(\psi_{M(1,1;0)}(M(1,1;0)^{M(1,1;0)}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(1,1;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1) | $\psi(M(1,2;\omega))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1)(3,0,0) | $\psi(M(1,\omega;0))$<br>$\psi(\psi_N(N+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_{M(2,0;0)}(0))$<br>$\psi(\alpha \to \psi_N(N+\alpha)\ \mathrm{FP})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(2,0;0))$<br>$\psi(\alpha \to \psi_N(N+\alpha)\ \mathrm{AP})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(M(2,0;1,0))$<br>$\psi(\psi_{M(2;1,0)}(0))$<br>$\psi(\psi_{\psi_N(N\cdot 2)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(2,1;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(3,0,0) | $\psi(M(\omega,0;0))$<br>$\psi(\psi_N(N\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(\psi_{M(1,0,0;0)}(0))$<br>$\psi(\alpha \to \psi_N(N\cdot\alpha)\ \mathrm{FP})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1,0,0;0))$<br>$\psi(\alpha \to \psi_N(N\cdot\alpha)\ \mathrm{AP})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1,0,0;1,0))$<br>$\psi(\psi_{\psi_N(N^2)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(1,0,1;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1,1,0;0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(2,0,0;0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(3,0,0) | $\psi(M(\omega,0,0;0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1,0,0,0;0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,0,0) | $\psi(M(1@\omega;0))$<br>$\psi(\psi_N(N^\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,1,0)(2,0,0) | $\psi(M(1@(1,0);0))$<br>$\psi((2-2\ 1-)^{1@(1,0)}\ 2-2)$<br>$\psi(\alpha \to \psi_N(N^\alpha)\ \mathrm{FP})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1@(1,0);1,0))$<br>$\psi(2\ 1-(2-2\ 1-)^{1@(1,0)}\ 2-2)$<br>$\psi(\alpha \to \psi_N(N^\alpha)\ \mathrm{AP})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,1,0)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1@(1,1);0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,1,0)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1@(2,0);0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,1,0)(4,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1@(1,0,0);0))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,1,0)(5,0,0) | $\psi(M(1@(1@\omega);0))$<br>$\psi(\psi_N(N^{N^\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(5,1,0)(2,0,0) | $\psi(M(1@(1@(1,0));0))$<br>$\psi(\psi_N(N^{N^N}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\psi_{\Omega_{N+1}}(0))$<br>$\psi((1-)^{1,0}\text{ aft }2-2-2)$<br>$\psi(\psi_N(\varepsilon_{N+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0)(4,2,0) | $\psi(\text{2nd }(1-)^{1,0}\text{ aft }2-2-2)$<br>$\psi(\psi_{\Omega_{N+1}}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0)(5,2,0) | $\psi(2\text{ aft }2-2-2)$<br>$\psi(\Omega_{N+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,1) | $\psi(1-2\text{ aft }2-2-2)$<br>$\psi(\Omega_{N+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi((1-)^{1,0}\ 2\text{ aft }2-2-2)$<br>$\psi(\psi_{I_{N+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,0)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(2\ 1-2\text{ aft }2-2-2)$<br>$\psi(I_{N+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) | $\psi(1-2\ 1-2\text{ aft }2-2-2)$<br>$\psi(I_{N+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1) | $\psi(1-2-2\text{ aft }2-2-2)$<br>$\psi(M_{N+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(5,2,1)(6,2,1) | $\psi(1-2\ 1-2-2\text{ aft }2-2-2)$<br>$\psi(M(1,N+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(5,2,1)(6,2,1)(6,2,1) | $\psi(1-2-2\ 1-2-2\text{ aft }2-2-2)$<br>$\psi(M(1;N+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(6,0,0) | $\psi((2-2\ 1-)^\omega\ 2-2\text{ aft }2-2-2)$<br>$\psi(M(\omega;N+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(6,1,0)(2,0,0) | $\psi((2-2\ 1-)^{(2-2-2)}\ 2-2\text{ aft }2-2-2)$<br>$\psi(M(N;1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(6,2,0)(5,0,0) | $\psi((2-2\ 1-)^{1,0}\ 2-2\text{ aft }2-2-2)$<br>$\psi(M(1,0;N+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(6,2,0)(7,2,0) | $\psi((2-2\ 1-)^{1@(1,0)}\ 2-2\text{ aft }2-2-2)$<br>$\psi(M(1@(1,0);N+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(6,2,0)(7,3,0) | $\psi((1-)^{1,0}\text{ aft 2nd }2-2-2)$<br>$\psi(\psi_{\Omega_{N_2+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(6,2,0)(7,3,1)(8,3,1)(9,3,1)<br>(9,3,1)(9,3,0)(10,4,0) | $\psi((1-)^{1,0}\text{ aft 3rd }2-2-2)$<br>$\psi(\psi_{\Omega_{N_3+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2-2)$<br>$\psi(N_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(2,1,1) | $\psi(1-1-2-2-2)$<br>$\psi(N_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(2,1,1)(3,1,0) | $\psi((1-)^{(2)}\ 2-2-2)$<br>$\psi(N_\Omega)$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(3,1,1) | $\psi((1-)^{(1-2-2-2)}\ 2-2-2)$<br>$\psi(N_{N_\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi((1-)^{1,0}\ 2-2-2)$<br>$\psi(\psi_{N(1,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(1-2\ 1-2-2-2)$<br>$\psi(N(1,\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi((2\ 1-)^{1,0}\ 2-2-2)$<br>$\psi(\psi_{N(1,0,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi((1-)^{1,0}\ \text{aft}\ 2-2\ 1-2-2-2)$<br>$\psi(\psi_{\Omega_{N(1;0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2\ 1-2-2-2)$<br>$\psi(N(1;\omega))$<br>$\psi(M(1;1;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2\ 1-2-2\ 1-2-2-2)$<br>$\psi(N(2;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)(3,1,0) | $\psi((2-2\ 1-)^{(2)}\ 2-2-2)$<br>$\psi(N(\Omega;0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi((2-2\ 1-)^{1,0}\ 2-2-2)$<br>$\psi(\psi_{N(1,0;0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi((1-)^{1,0}\ \text{aft}\ 2-2-2\ 1-2-2-2)$<br>$\psi(\psi_{\Omega_{N(1;0;0)+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2-2\ 1-2-2-2)$<br>$\psi(N(1;0;\omega))$<br>$\psi(M(2;0;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2-2\ 1-2-2-2\ 1-2-2-2)$<br>$\psi(N(2;0;\omega))$<br>$\psi(M(3;0;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,0,0) | $\psi((2-2-2\ 1-)^{\omega}1-2-2-2)$<br>$\psi(M(\omega;0;0))$<br>$\psi(N(\omega;0;0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,0) | $\psi((2-2-2\ 1-)^{(2)}1-2-2-2)$<br>$\psi(M(\Omega;0;0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi((2-2-2\ 1-)^{1,0}\ 1-2-2-2)$<br>$\psi(\psi_{M(1,0;0;0)}(0))$<br>$\psi(\psi_Q(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,1,0)(2,0,0) | $\psi((2-2-2\ 1-)^{1@(1,0)}1-2-2-2)$<br>$\psi(\psi_Q(Q^{Q^Q}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi((1-)^{1,0}\ \text{aft}\ 2-2-2-2)$<br>$\psi(\psi_{\Omega_{Q+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2-2-2)$<br>$\psi(Q_\omega)$<br>$\psi(M(1;0;0;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(1-2\ 1-2-2-2-2)$<br>$\psi(Q(1,\omega))$<br>$\psi(M(1;0;0;1,\omega))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2\ 1-2-2-2\ 1-2-2-2-2)$<br>$\psi(Q(1;1;\omega))$<br>$\psi(M(1;1;1;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2-2-2-2)$<br>$\psi(Q\{2\}_\omega)$<br>$\psi(M(1;0;0;0;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,0,0) | $\psi((2-)^\omega)$<br>$\psi(Q\{\omega\})$<br>$\psi(M(1;@\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0) | $\psi((2-)^{(2)})$<br>$\psi(Q\{\Omega\})$<br>$\psi(M(1;@\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0) | $\psi((2-)^{(2-)^{(2)}})$<br>$\psi(M(1;@M(1;@\Omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(2,0,0) | $\psi((2-)^{1,0})$<br>$\psi(\psi_K(0))$<br>$\psi(M(1;@(1,0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(\text{2nd } (2-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(2,1,1) | $\psi(1-(2-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,1)(3,1,1) | $\psi(1-2\ 1-(2-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2\ 1-(2-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,1)(3,1,1)(4,1,0) | $\psi((2-)^{(2)}\ 1-(2-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(2,1,1)(3,1,1)(4,1,0)(2,0,0) | $\psi((2-)^{1,0}\ 1-(2-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(3,1,0)(2,0,0) | $\psi(((2-)^{1,0}\ 1-)^{1,0}\ (2-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(3,1,1) | $\psi(1-(2-)^{1,1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(3,1,1)(3,1,1) | $\psi(1-(2-)^{1,2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(3,1,1)(4,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(4,1,0)(2,0,0) | $\psi((2-)^{1,(2-)^{1,0}})$<br>$\psi(\psi_K(\psi_K(0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(3,1,1)(4,1,0)(2,0,0) | $\psi((2-)^{2,0})$<br>$\psi(\psi_K(K))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(3,1,1)(4,1,0)(3,1,1)(4,0,0)(2,0,0) | $\psi((2-)^{3,0})$<br>$\psi(\psi_K(K^2))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(4,0,0) | $\psi((2-)^{\omega,0})$<br>$\psi(\psi_K(K^\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(4,1,0)(2,0,0) | $\psi((2-)^{1,0,0})$<br>$\psi(\psi_K(K^K))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(4,1,0)(3,1,1)(4,1,0)(4,1,0)(2,0,0) | $\psi((2-)^{2,0,0})$<br>$\psi(\psi_K(K^{K\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(4,1,0)(4,1,0)(2,0,0) | $\psi((2-)^{1,0,0,0})$<br>$\psi(\psi_K(K^{K^2}))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,1,0)(2,0,0) | $\psi((2-)^{1@(1,0)})$<br>$\psi(\psi_K(K^{K^K}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,1,0)(6,1,0)(2,0,0) | $\psi((2-)^{1@(1@(1,0))})$<br>$\psi(\psi_K(K^{K^{K^K}}))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(5,2,0) | $\psi((1-)^{1,0}$ aft 3)<br>$\psi(\psi_{\Omega_{K+1}}(0))$<br>$\psi(\psi_K(\varepsilon_{K+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(6,2,0) | $\psi(2$ aft 3)<br>$\psi(\Omega_{K+1})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(5,2,1) | $\psi(1-2$ aft 3)<br>$\psi(\Omega_{K+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,0)(6,0,0) | $\psi((1-)^{1,0}$ 2 aft 3)<br>$\psi(\psi_{I_{K+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,1)(6,2,1)(7,2,1) | $\psi(1-2\ 1-2$ aft 3)<br>$\psi(I_{K+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,1)(6,2,1)(7,2,1)(7,2,1) | $\psi(1-2-2$ aft 3)<br>$\psi(M_{K+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(7,2,1)(7,2,1) | $\psi(1-2-2-2$ aft 3)<br>$\psi(N_{K+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,0,0) | $\psi((2-)^{\omega}$ aft 3) |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,1,0) | $\psi((2-)^{(2)}$ aft 3) |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi((2-)^{(1-2-2)}$ aft 3) |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(4,0,0) | $\psi((2-)^{(2-)^{\omega}}$ aft 3) |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(4,1,0)(2,0,0) | $\psi((2-)^{(2-)^{1,0}}$ aft 3)<br>$\psi((2-)^{\psi_K(0)}$ aft 3) |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(4,1,0)(3,1,1) | $\psi((2-)^{(2-)^{1,1}}$ aft 3) |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(4,1,0)(5,2,0) | $\psi((2-)^{\psi_K(\psi_{\Omega_{K+1}}(0))}$ aft 3) |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,1,0)(2,0,0) | $\psi((2-)^{(3)}$ aft 3) |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,2,0)(6,0,0) | $\psi((2-)^{1,0}$ aft 3)<br>$\psi(\psi_{K_2}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,2,0)(9,3,0) | $\psi((1-)^{1,0}$ aft 2nd 3)<br>$\psi(\psi_{\Omega_{K_2+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,2,0)(9,3,1)<br>(10,3,1)(11,3,1)(12,3,0)(13,4,0) | $\psi((2-)^{1,0}$ aft 2nd 3)<br>$\psi(\psi_{K_3}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1) | $\psi(1-3)$<br>$\psi(K_\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(6,2,1) | $\psi(2$nd $1-3)$<br>$\psi(K_{\omega\cdot 2})$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(2,1,1) | $\psi(1-1-3)$<br>$\psi(K_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(2,1,1)(3,1,0) | $\psi((1-)^{(2)}\ 3)$<br>$\psi(K_\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,1) | $\psi((1-)^{(1-2\ 1-2)}\ 3)$<br>$\psi(K_{I_\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(5,2,1)(6,2,1)<br>(7,2,1)(8,1,0)(2,0,0) | $\psi((1-)^{(3)}\ 3)$<br>$\psi(K_K)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi((1-)^{1,0}\ 3)$<br>$\psi(\psi_{K(1,0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(2,1,1)(3,1,1) | $\psi(1-2\ 1-3)$<br>$\psi(K(1,\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2\ 1-3)$<br>$\psi(K(1;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2-2\ 1-3)$<br>$\psi(K(1;0;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(2,1,1)(3,1,1)(4,0,0) | $\psi((2-)^{\omega}\ 1-3)$<br>$\psi(K(1;@\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(2,1,1)(3,1,1)(4,1,0)(2,0,0) | $\psi((2-)^{1,0}\ 1-3)$<br>$\psi(K(1;@(1,0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(2,1,1)(3,1,1)(4,1,1) | $\psi(1-3\ 1-3)$<br>$\psi(K(1;;\omega))$<br>$\psi(M(2;;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(3,0,0) | $\psi((3\ 1-)^{\omega}\ 3)$<br>$\psi(K(\omega;;0))$<br>$\psi(M(\omega;;0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,0)(2,0,0) | $\psi((3\ 1-)^{1,0}\ 3)$<br>$\psi(\psi_{K(1,0;;0)}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(3,1,1) | $\psi(1-2-3)$<br>$\psi(K(1;0;;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,1)(2,1,1)(3,1,1)(4,1,1) | $\psi(1-3\ 1-2-3)$<br>$\psi(K(1;1;;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(4,1,1)(3,1,1) | $\psi(1-2-3\ 1-2-3)$<br>$\psi(K(2;0;;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2-3)$<br>$\psi(K(1;0;0;;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,1)(4,0,0) | $\psi((2-)^{\omega}\ 3)$<br>$\psi(K(\{1;@\omega\};;0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,1)(4,1,0)(2,0,0) | $\psi((2-)^{1,0}\ 3)$<br>$\psi(K(\{1;@(1,0)\};;0))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(3,1,1)(4,1,1) | $\psi(1-3\ 2-3)$<br>$\psi(K(1;;0;;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,1)(4,1,1)(3,1,1)(4,1,1) | $\psi(1-3\ 2-3\ 2-3)$<br>$\psi(K(1;;0;;0;;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(4,0,0) | $\psi((3\ 2-)^{\omega}\ 3)$<br>$\psi(K(1;;@\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(4,1,0)(2,0,0) | $\psi((3\ 2-)^{1,0}\ 3)$<br>$\psi(K(1;;@(1,0)))$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(4,1,1) | $\psi(1-3-3)$<br>$\psi(K(1;;;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,1)(2,1,1)(3,1,1)(4,1,1) | $\psi(1-3\ 1-3-3)$<br>$\psi(K(1;;;1;;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(4,1,1)(2,1,1)(3,1,1)(4,1,1)(4,1,1) | $\psi(1-3-3\ 1-3-3)$<br>$\psi(K(2;;;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,1)(3,1,1) | $\psi(1-2-3-3)$<br>$\psi(K(1;0;;;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,1)(3,1,1)(4,1,1) | $\psi(1-3\ 2-3-3)$<br>$\psi(K(1;;0;;;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(4,1,1)(3,1,1)(4,1,1)(4,1,1) | $\psi(1-3-3\ 2-3-3)$<br>$\psi(K(1;;;0;;;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,1)(4,1,1) | $\psi(1-3-3-3)$<br>$\psi(K(1;;;;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,1)(4,1,1)(4,1,1) | $\psi(1-3-3-3-3)$<br>$\psi(K(1;;;;;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(5,0,0) | $\psi((3-)^{\omega})$<br>$\psi(K(1[\omega]0))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(5,1,0) | $\psi((3-)^{(2)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(5,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,1)(6,2,1)(7,2,1)<br>(8,2,1)(9,1,0)(2,0,0) | $\psi((3-)^{(3)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,0)(2,0,0) | $\psi((3-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,0)(6,2,0) | $\psi(2\text{ aft }4)$<br>$\psi(\psi_{\Omega_{\kappa+1}}(0))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(5,1,1) | $\psi(1-4)$<br>$\psi(\kappa_{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(2,1,1) | $\psi(1-1-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(2,1,1)(3,1,1) | $\psi(1-2\ 1-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(5,1,1)(2,1,1)(3,1,1)(4,1,1) | $\psi(1-3\ 1-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(5,1,1)(2,1,1)(3,1,1)(4,1,1)(5,1,1) | $\psi(1-4\ 1-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(3,1,1) | $\psi(1-2-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(3,1,1)(4,1,1) | $\psi(1-3\ 2-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(3,1,1)(4,1,1)(5,1,1) | $\psi(1-4\ 2-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(4,1,1) | $\psi(1-3-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(4,1,1)(5,1,1) | $\psi(1-4\ 3-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(5,0,0) | $\psi((4\ 3-)^{\omega}\ 4)$ |

| BMS | Reflection-Collapsing Notation (Madore-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(5,1,1) | $\psi(1-4-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(5,1,1)(5,1,1) | $\psi(1-4-4-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,0,0) | $\psi((4-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,1) | $\psi(1-5)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(5,1,1)(6,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,1) | $\psi(1-5\ 1-5)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,1)(3,1,0) | $\psi((5\ 1-)^{(2)}\ 5)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,1)(3,1,1) | $\psi(1-2-5)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,1)(4,1,1) | $\psi(1-3-5)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(5,1,1)(6,1,1)(5,1,1) | $\psi(1-4-5)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(5,1,1)(6,1,1)(6,1,1) | $\psi(1-5-5)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(5,1,1)(6,1,1)(7,1,1) | $\psi(1-6)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(5,1,1)(6,1,1)(7,1,1)(8,1,1) | $\psi(1-7)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(5,1,1)(6,1,1)(7,1,1)(8,1,1)(9,1,1) | $\psi(1-8)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,1)(7,1,1)<br>(8,1,1)(9,1,1)(10,1,1) | $\psi(1-9)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,1)(7,1,1)<br>(8,1,1)(9,1,1)(10,1,1)(11,1,1) | $\psi(1-10)$ |
| (0,0,0)(1,1,1)(2,2,0) | $\psi(\text{psd. }\Pi_{\omega})$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0)$ |

## A.11 BMS vs Reflection-Collapsing Notation (Buchholz-like)

The results in this section are primarily cited from [18-23]. The reflection-collapsing notation used is the Buchholz-like version defined by Patchouli.

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0)(1,1) | $\psi(\Omega)$<br>$\psi(M)$ |
| (0,0,0)(1,1,1) | $\psi(\Omega_{\omega})$<br>$\psi(M\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1) | $\psi(\Omega_{\omega^2})$<br>$\psi(M\cdot\omega^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0) | $\psi(\Omega_{\Omega})$<br>$\psi(M\cdot\psi_M(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1) | $\psi(\Omega_{\Omega_{\omega}})$<br>$\psi(M\cdot\psi_M(M\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0) | $\psi(\Omega_{\Omega_{\Omega}})$<br>$\psi(M\cdot\psi_M(M\cdot\psi_M(M)))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,0) | $\psi(\Omega_{\Omega_{\Omega_\Omega}})$<br>$\psi(M \cdot \psi_M(M \cdot \psi_M(M \cdot \psi_M(M))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I)$<br>$\psi(M^2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,0,0) | $\psi(I+1)$<br>$\psi(M^2+1)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,0) | $\psi(I+\Omega)$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,0,0)(1,1,0)(2,2,1)(3,2,1)<br>(4,2,0)(3,0,0)(2,2,0) | $\psi(I+\Omega_2)$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,0,0)(1,1,1) | $\psi(I+\Omega_\omega)$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,0,0)(1,1,1)(2,1,1)(3,1,0) | $\psi(I+\Omega_\Omega)$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(1,1,1) | $\psi(I+\Omega_\Omega)$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0) | $\psi(I+\Omega_{\Omega_\omega})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M\psi_M(M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(1,1,1) | $\psi(I+\Omega_{\Omega_\Omega})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M\psi_M(M\psi_M(M))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\psi_I(I))$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,0,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(1))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,0,0)(2,0,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(2))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(\Omega))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(1,1,1) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(\Omega_\omega))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I)+\psi_I(I))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2)+\psi_{\psi_M(M^2)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I)+\psi_{\Omega_{\psi_I(I)+1}}(\Omega))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I)\cdot 2)$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+1))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(2,1,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_M(M^2)}(1)))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(2,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_I(I)))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_M(M^2)}(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(2,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_I(I))+\psi_{\Omega_{\psi_I(I)+1}}(\Omega))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2))+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(2,1,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(2,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_I(I))\cdot 2)$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2))\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(2,1,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_I(I)+1))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(2,1,0)(2,1,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_I(I)\cdot 2+1))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2)\cdot 2+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(1)))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,0,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(1)+1))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(1)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,0,0)(2,1,0)(3,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(1)\cdot 2))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(1)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,0,0)(3,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(2)))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,1,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(M)))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(I)))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,1,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(I)+1))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,1,0)(2,1,0)(3,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(I)+\psi_{\Omega_{\psi_I(I)+1}}(1)))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2)$<br>$+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,1,0)(2,1,0)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(I)\cdot 2))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2$<br>$+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2)\cdot 2))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,1,0)(2,1,0)(3,1,0)(2,1,0)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(I)\cdot 3))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2$<br>$+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2)\cdot 3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,1,0)(3,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(I+1)))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2$<br>$+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,1,0)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_I(I))))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_M(M^2)}(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,1,0)(4,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(1))))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2$<br>$+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,0) | $\psi(I+\Omega_{\psi_I(I)+1})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,0)(1,1,1)<br>(2,1,1)(3,1,1)(2,0,0) | $\psi(I+\Omega_{\psi_I(I)+1}+\psi_I(I))$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+1)+\psi_{\psi_M(M^2)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,0)(3,2,0) | $\psi(I+\Omega_{\psi_I(I)+1}+\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+1}))$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+1)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_M(M^2)}(M^2+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,0)(2,0,0) | $\psi(I+\Omega_{\psi_I(I)+1}+\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+1}+1))$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+1)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_M(M^2)}(M^2+1)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,0)(2,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\Omega_{\psi_I(I)+1}+\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+1}+\psi_I(I)))$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+1)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+1)+\psi_{\psi_M(M^2)}(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,0)(2,1,0)(3,2,0) | $\psi(I+\Omega_{\psi_I(I)+1}+\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+1}$<br>$+\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+1})))$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+1)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+1)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2$<br>$+\psi_{\psi_M(M^2)}(M^2+1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,0)(3,0,0) | $\psi(I+\Omega_{\psi_I(I)+1}+\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+1}+$<br>$\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+1}+1)))$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+1)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+1)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2$<br>$+\psi_{\psi_M(M^2)}(M^2+1)+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(3,1,0)(4,2,0) | $\psi(I+\Omega_{\psi_I(I)+1}+\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+1}+\psi_{\Omega_{\psi_I(I)+1}}(I+$<br>$\Omega_{\psi_I(I)+1}+\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+1}))))$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+1)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+1)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2$<br>$+\psi_{\psi_M(M^2)}(M^2+1)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+1)))))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(3,2,0) | $\psi(I+\Omega_{\psi_I(I)+1}\cdot 2)$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+1)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,1,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(\Omega))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(I))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,1,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(I)+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+2}}(I)+1))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,1,0)(2,1,0)(3,2,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(I)+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+2}}(I)+$<br>$\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+1})))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2)$<br>$+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_M(M^2)}(M^2+1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,1,0)(2,1,0)(3,2,0)(4,1,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(I)+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+2}}(I)+$<br>$\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+2}}(\Omega))))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,1,0)(3,1,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(I)+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+2}}(I)+$<br>$\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+2}}(I)+\Omega)))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2)+\psi_{\psi_M(M^2)}(1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,1,0)(3,1,0)(4,2,0)(5,1,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(I)+\psi_{\Omega_{\psi_I(I)+1}}(I+$<br>$\psi_{\Omega_{\psi_I(I)+2}}(I)+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+2}}(I)+$<br>$\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+2}}(\Omega)))))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2$<br>$+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,1,0)(3,2,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(I)+\Omega_{\psi_I(I)+1})$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2)+\psi_{\psi_M(M^2)}(M^2+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,1,0)(3,2,0)(4,1,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(I)+\psi_{\Omega_{\psi_I(I)+2}}(\Omega))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2)+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,1,0)(4,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(I+1))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,1,0)(4,1,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(I+\Omega))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2+\psi_{\psi_M(M^2)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,1,0)(4,1,0)(4,1,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(I+\psi_I(I)+\Omega))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2)+\psi_{\psi_M(M^2)}(1)))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,1,0)(5,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(I+\psi_{\Omega_{\psi_I(I)+1}}(1)))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,1,0)(5,2,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+1})))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_M(M^2)}(M^2+1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,2,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(I+\Omega_{\psi_I(I)+1}))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2+\psi_{\psi_M(M^2)}(M^2+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,2,0)(4,2,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(I+\Omega_{\psi_I(I)+1}\cdot 2))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2+\psi_{\psi_M(M^2)}(M^2+1)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,2,0)(5,2,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+2}}(I+\psi_{\Omega_{\psi_I(I)+2}}(I+\Omega_{\psi_I(I)+1})))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2+\psi_{\psi_M(M^2)}(M^2+1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,3,0) | $\psi(I+\Omega_{\psi_I(I)+2})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,3,0)(2,0,0) | $\psi(I+\Omega_{\psi_I(I)+2}+\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+2}+1))$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+2)+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+2)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,3,0)(3,2,0) | $\psi(I+\Omega_{\psi_I(I)+2}+\Omega_{\psi_I(I)+1})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+2)+\psi_{\psi_M(M^2)}(M^2+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,3,0)(3,2,0)(4,3,0) | $\psi(I+\Omega_{\psi_I(I)+2}+\psi_{\Omega_{\psi_I(I)+2}}(I+\Omega_{\psi_I(I)+2}))$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+2)+\psi_{\psi_{\psi_M(M^2)}(M^2+2)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,3,0)(4,3,0) | $\psi(I+\Omega_{\psi_I(I)+2}\cdot 2)$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+2)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,3,0)(5,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+3}}(1))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+3)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,0)(4,3,0)(5,4,0) | $\psi(I+\Omega_{\psi_I(I)+3})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1) | $\psi(I+\Omega_{\psi_I(I)+\omega})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(3,2,0) | $\psi(I+\Omega_{\psi_I(I)+\omega}+\Omega_{\psi_I(I)+1})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\omega)+\psi_{\psi_M(M^2)}(M^2+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(3,2,0)(4,3,1)(4,3,0) | $\psi(I+\Omega_{\psi_I(I)+\omega}+\Omega_{\psi_I(I)+2})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\omega)+\psi_{\psi_M(M^2)}(M^2+2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(3,2,1) | $\psi(I+\Omega_{\psi_I(I)+\omega}\cdot 2)$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\omega)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,1,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+\omega+1}}(\Omega))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+\omega+1)}(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+\omega+1}}(I))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+\omega+1)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,1,0)(3,2,1) | $\psi(I+\psi_{\Omega_{\psi_I(I)+\omega+1}}(I)+\Omega_{\psi_I(I)+1})$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+\omega+1)}(M^2)$<br>$+\psi_{\psi_M(M^2)}(M^2+\omega))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,1,0)(3,2,1)(4,1,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+\omega+1}}(I)+\psi_{\Omega_{\psi_I(I)+\omega+1}}(\Omega))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+\omega+1)}(M^2)$<br>$+\psi_{\psi_{\psi_M(M^2)}(M^2+\omega+1)}(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,1,0)(4,1,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+\omega+1}}(I+\Omega))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+\omega+1)}(M^2+\psi_{\psi_M(M^2)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,1,0)(4,1,0)<br>(3,2,1)(4,1,0)(4,1,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+\omega+1}}(I+\psi_I(I))+$<br>$\psi_{\Omega_{\psi_I(I)+\omega+1}}(I+\Omega))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+\omega+1)}(M^2+\psi_{\psi_M(M^2)}(M^2))+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+\omega+1)}$<br>$(M^2+\psi_{\psi_M(M^2)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,1,0)(4,1,0)(4,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+\omega+1}}(I+\psi_I(I)+1))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+\omega+1)}$<br>$(M^2+\psi_{\psi_M(M^2)}(M^2)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,1,0)(5,0,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+\omega+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(1)))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+\omega+1)}$<br>$(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,1,0)(5,2,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+\omega+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+1})))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+\omega+1)}$<br>$(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_M(M^2)}(M^2+1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,1,0)(5,2,1) | $\psi(I+\psi_{\Omega_{\psi_I(I)+\omega+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+\omega)}))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+\omega+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_M(M^2)}(M^2+\omega))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,0) | $\psi(I+\psi_{\Omega_{\psi_I(I)+\omega+1}}(I+\Omega_{\psi_I(I)+1}))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+\omega+1)}$<br>$(M^2+\psi_{\psi_M(M^2)}(M^2+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,0)(3,2,1) | $\psi(I+\psi_{\Omega_{\psi_I(I)+\omega+1}}(I+\Omega_{\psi_I(I)+\omega}))$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+\omega+1)}$<br>$(M^2+\psi_{\psi_M(M^2)}(M^2+\omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,0)(3,2,1)(4,2,0)(3,2,1) | $\psi(I+\psi_{\Omega_{\psi_I(I)+\omega+1}}(I+\Omega_{\psi_I(I)+\omega})\cdot 2)$<br>$\psi(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+\omega+1)}(M^2+\psi_{\psi_M(M^2)}$<br>$(M^2+\omega))\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,0) | $\psi(I+\Omega_{\psi_I(I)+\omega+1})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\omega+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,0)(6,4,0) | $\psi(I+\Omega_{\psi_I(I)+\omega+2})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\omega+2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,1) | $\psi(I+\Omega_{\psi_I(I)+\omega\cdot 2})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\omega\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,1)(6,3,0)(7,4,1) | $\psi(I+\Omega_{\psi_I(I)+\omega\cdot 3})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\omega\cdot 3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1) | $\psi(I+\Omega_{\psi_I(I)+\omega^2})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi(2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(4,2,0)(5,3,0) | $\psi(I+\Omega_{\psi_I(I)+\omega^2+1})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi(2)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(4,2,0)(5,3,1) | $\psi(I+\Omega_{\psi_I(I)+\omega^2+\omega})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi(2)+\omega))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(4,2,0)<br>(5,3,1)(6,3,0)(7,4,1) | $\psi(I+\Omega_{\psi_I(I)+\omega^2+\omega\cdot 2})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi(2)+\omega\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(4,2,0)(5,3,1)(6,3,1) | $\psi(I+\Omega_{\psi_I(I)+\omega^2\cdot 2})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi(2)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(4,2,0)(5,3,1)<br>(6,3,1)(6,3,0)(7,4,1)(8,4,1) | $\psi(I+\Omega_{\psi_I(I)+\omega^2\cdot 3})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi(2)\cdot 3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(4,2,0)(5,3,1)<br>(6,3,1)(6,3,0)(7,4,1)(8,4,1) | $\psi(I+\Omega_{\psi_I(I)+\omega^2\cdot 3})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi(2)\cdot 3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(4,2,1) | $\psi(I+\Omega_{\psi_I(I)+\omega^3})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi(3)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0) | $\psi(I+\Omega_{\psi_I(I)+\Omega})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\Omega_{\psi_I(I)\cdot 2})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2)}(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,0)(5,3,0) | $\psi(I+\Omega_{\psi_I(I)\cdot 2+1})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2)}(M^2)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I+\Omega_{\psi_I(I)\cdot 3})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2)}(M^2)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1) | $\psi(I+\Omega_{\psi_I(I)\cdot\omega})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(4,2,1)(4,2,1) | $\psi(I+\Omega_{\psi_I(I)\cdot\omega^2})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)<br>(4,2,1)(5,1,0)(4,2,1) | $\psi(I+\Omega_{\psi_{\Omega_{\psi_I(I)+1}}(I+1)})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(5,0,0) | $\psi(I+\Omega_{\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_I(I)\cdot\omega)})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(5,1,0)(4,2,1) | $\psi(I+\Omega_{\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(I)+1)})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2)+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(5,1,0)(5,0,0) | $\psi(I+\Omega_{\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(I+1))})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(6,0,0) | $\psi(I+\Omega_{\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_{\Omega_{\psi_I(I)+1}}(I+\psi_I(I)\cdot\omega))})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(1)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(6,2,0) | $\psi(I+\Omega_{\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+1})})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+1))))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(6,2,1) | $\psi(I+\Omega_{\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+\omega})})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+\omega))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)<br>(6,2,1)(7,2,1)(8,1,0) | $\psi(I+\Omega_{\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_{\Omega_{\psi_I(I)+1}}(\Omega)})})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)<br>(6,2,1)(7,2,1)(8,1,0) | $\psi(I+\Omega_{\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_{\Omega_{\psi_I(I)+1}}(\Omega)})})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(6,2,1)<br>(7,2,1)(8,1,0)(9,2,0) | $\psi(I+\Omega_{\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+1}))})})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+1))))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0)(6,2,1)(7,2,1)<br>(8,1,0)(9,2,1)(10,2,1)(11,1,0) | $\psi(I+\Omega_{\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_{\Omega_{\psi_I(I)+1}}(\Omega)})})})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M)))))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0) | $\psi(I+\Omega_{\Omega_{\psi_I(I)+1}})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2)}(M^2+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(3,2,1) | $\psi(I+\Omega_{\Omega_{\psi_I(I)+\omega}})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2)}(M^2+\omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)<br>(3,2,1)(4,2,1)(5,2,0) | $\psi(I+\Omega_{\Omega_{\Omega_{\psi_I(I)+1}}})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2)}(M^2+1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(3,2,1)(4,2,1)<br>(5,2,0)(3,2,1)(4,2,1)(5,2,0) | $\psi(I+\Omega_{\Omega_{\Omega_{\Omega_{\psi_I(I)+1}}}})$<br>$\psi(M^2+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2)}(M^2+1)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I\cdot 2)$<br>$\psi(M^2+\psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I\cdot 2+\psi_I(I))$<br>$\psi(M^2+\psi_M(M^2)+\psi_{\psi_M(M^2)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0)(3,2,0) | $\psi(I\cdot 2+\Omega_{\psi_I(I)+1})$<br>$\psi(M^2+\psi_M(M^2)+\psi_{\psi_M(M^2)}(M^2+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0)(3,2,1) | $\psi(I\cdot 2+\Omega_{\psi_I(I)+\omega})$<br>$\psi(M^2+\psi_M(M^2)+\psi_{\psi_M(M^2)}(M^2+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0)<br>(3,2,1)(4,2,1)(5,2,0) | $\psi(I\cdot 2+\Omega_{\Omega_{\psi_I(I)+1}})$<br>$\psi(M^2+\psi_M(M^2)+\psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I\cdot 2+\psi_I(I\cdot 2))$<br>$\psi(M^2+\psi_M(M^2)+\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0)(4,0,0) | $\psi(I\cdot 2+\psi_{\Omega_{\psi_I(I\cdot 2)+1}}(1))$<br>$\psi(M^2+\psi_M(M^2)+\psi_{\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+1)}(1))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I\cdot 2+\psi_{\Omega_{\psi_I(I\cdot 2)+1}}(I))$<br>$\psi(M^2+\psi_M(M^2)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+1)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0) | $\psi(I\cdot 2+\psi_{\Omega_{\psi_I(I\cdot 2)+1}}(I+\Omega_{\psi_I(I)+1}))$<br>$\psi(M^2+\psi_M(M^2)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+1)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I\cdot 2+\psi_{\Omega_{\psi_I(I\cdot 2)+1}}(I\cdot 2))$<br>$\psi(M^2+\psi_M(M^2)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+1)}(M^2+\psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)(4,0,0) | $\psi(I\cdot 2+\psi_{\Omega_{\psi_I(I\cdot 2)+1}}(I\cdot 2+1))$<br>$\psi(M^2+\psi_M(M^2)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+1)}(M^2+\psi_M(M^2)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)(5,3,0) | $\psi(I\cdot 2+\Omega_{\psi_I(I\cdot 2)+1})$<br>$\psi(M^2+\psi_M(M^2)+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)<br>(4,2,0)(5,3,0)(6,4,0) | $\psi(I\cdot 2+\Omega_{\psi_I(I\cdot 2)+2})$<br>$\psi(M^2+\psi_M(M^2)+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)(5,3,1) | $\psi(I\cdot 2+\Omega_{\psi_I(I\cdot 2)+\omega})$<br>$\psi(M^2+\psi_M(M^2)+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)<br>(4,2,0)(5,3,1)(6,3,1) | $\psi(I\cdot 2+\Omega_{\psi_I(I\cdot 2)+\omega^2})$<br>$\psi(M^2+\psi_M(M^2)+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+\psi(2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,0) | $\psi(I\cdot 2+\Omega_{\Omega_{\psi_I(I\cdot 2)+1}})$<br>$\psi(M^2+\psi_M(M^2)+\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,0)(5,3,1)(6,3,1)(7,3,0) | $\psi(I\cdot 2+\Omega_{\Omega_{\Omega_{\psi_I(I\cdot 2)+1}}})$<br>$\psi(M^2+\psi_M(M^2)+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+\psi_{\psi_M(M^2)}(M^2+$<br>$\psi_M(M^2)+\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(I\cdot 3)$<br>$\psi(M^2+\psi_M(M^2)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,0)(6,3,0)(7,4,1)<br>(8,4,1)(9,4,0)(8,0,0) | $\psi(I\cdot 4)$<br>$\psi(M^2+\psi_M(M^2)\cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(I\cdot\omega)$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I\cdot\omega+\psi_I(I))$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+\psi_{\psi_M(M^2)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,0) | $\psi(I\cdot\omega+\psi_{\Omega_{\psi_I(I)+1}}(I+\Omega_{\psi_I(I)+1}))$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_M(M^2)}(M^2+1)))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I\cdot\omega+\psi_{\Omega_{\psi_I(I)+1}}(I\cdot 2))$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,0)(6,0,0) | $\psi(I\cdot\omega+\psi_{\Omega_{\psi_I(I)+1}}(I\cdot 3))$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_M(M^2)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(I\cdot\omega+\psi_{\Omega_{\psi_I(I)+1}}(I\cdot\omega))$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2+\psi_{\psi_M(M^2+M)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(3,2,0) | $\psi(I\cdot\omega+\Omega_{\psi_I(I)+1})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+\psi_{\psi_M(M^2)}(M^2+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I\cdot\omega+\psi_I(I\cdot 2))$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,0)(5,3,0) | $\psi(I\cdot\omega+\psi_{\Omega_{\psi_I(I\cdot 2)+1}}(I\cdot 2+\Omega_{\psi_I(I\cdot 2)+1}))$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+1)}(M^2+\psi_M(M^2)$<br>$+\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,0)(6,0,0) | $\psi(I\cdot\omega+\psi_{\Omega_{\psi_I(I\cdot 2)+1}}(I\cdot 3))$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+1)}(M^2+\psi_M(M^2)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,0)(6,3,1) | $\psi(I\cdot\omega+\psi_{\Omega_{\psi_I(I\cdot 2)+1}}(I\cdot\omega))$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+1)}(M^2+\psi_{\psi_M(M^2+M)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,0)(5,3,1)(6,3,1)<br>(7,3,0)(6,3,1)(5,3,0) | $\psi(I\cdot\omega+\Omega_{\psi_I(I\cdot 2)+1})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,0)(5,3,1)(6,3,1)(7,3,0)<br>(6,3,1)(5,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(I\cdot\omega+\psi_I(I\cdot 3))$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,0)(5,3,1)(6,3,1)(7,3,0)<br>(6,3,1)(5,3,1)(6,3,1)(7,3,0)(6,3,0)<br>(7,4,1)(8,4,1)(9,4,0)(8,4,1)(7,4,1)<br>(8,4,1)(9,4,0)(8,0,0) | $\psi(I\cdot\omega+\psi_I(I\cdot 4))$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)\cdot 3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(I\cdot\omega+\psi_I(I\cdot\omega))$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(1)))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(I\cdot\omega+\psi_I(I\cdot\omega)\cdot 2)$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(1))\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,0,0) | $\psi(I\cdot\omega+\psi_I(I\cdot\omega)\cdot\omega)$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(1)+1)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(I\cdot\omega+\psi_{\Omega_{\psi_I(I\cdot\omega)+1}}(I\cdot\omega))$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(1)+1)}$<br>$(M^2+\psi_{\psi_M(M^2+M)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,0)(3,2,0) | $\psi(I\cdot\omega+\Omega_{\psi_I(I\cdot\omega)+1})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(1)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,0)(3,2,1) | $\psi(I\cdot\omega+\Omega_{\psi_I(I\cdot\omega)+\omega})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(1)+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0) | $\psi(I\cdot\omega+\Omega_{\Omega_{\psi_I(I\cdot\omega)+1}})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(1)+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(1)+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I\cdot\omega+I)$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+\psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I\cdot\omega+I)$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+\psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,0)(5,3,0) | $\psi(I\cdot\omega+I+\Omega_{\psi_I(I\cdot\omega+I)+1})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+\psi_M(M^2)+$<br>$\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(1)+\psi_M(M^2)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(I\cdot\omega+I\cdot 2)$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)+\psi_M(M^2)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1) | $\psi(I\cdot\omega\cdot 2)$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,0)(6,3,1) | $\psi(I\cdot\omega\cdot 3)$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(1)\cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(2,1,1) | $\psi(I\cdot\omega^2)$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,1)(2,1,1) | $\psi(I\cdot\omega^3)$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(3))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0) | $\psi(I\cdot\Omega)$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I\cdot\psi_I(I))$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M\cdot\psi_{\psi_M(M^2)}(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,1,0)(3,2,0) | $\psi(I\cdot\psi_I(I)+\Omega_{\psi_I(I)+1})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M\cdot\psi_{\psi_M(M^2)}(M^2))+$<br>$\psi_{\psi_M(M^2)}(M^2+1))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I \cdot \psi_I(I) + \psi_I(I \cdot 2))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2))+$<br>$\psi_{\psi_M(M^2)}(M^2 + \psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(I \cdot \psi_I(I) + \psi_I(I \cdot \omega))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2))+$<br>$\psi_{\psi_M(M^2)}(M^2 + \psi_{\psi_M(M^2+M)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I \cdot \psi_I(I) + \psi_I(I \cdot \psi_I(I)))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2))+$<br>$\psi_{\psi_M(M^2)}(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)<br>(5,1,0)(4,2,0)(5,3,0) | $\psi(I \cdot \psi_I(I) + \Omega_{\psi_I(I \cdot \psi_I(I))+1})$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2))+$<br>$\psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2)) + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(I \cdot \psi_I(I) + I)$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2))$<br>$+\psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,0)(6,3,1) | $\psi(I \cdot \psi_I(I) + I \cdot \omega)$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2))$<br>$+\psi_{\psi_M(M^2+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,0)(6,3,1)<br>(7,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I \cdot \psi_I(I) \cdot 2)$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2)) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,0)(6,3,1)<br>(7,1,0)(6,3,0)(7,4,1)<br>(8,4,1)(9,4,0)(8,0,0) | $\psi(I \cdot \psi_I(I) \cdot 2 + I)$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot$<br>$\psi_{\psi_M(M^2)}(M^2)) \cdot 2 + \psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,0)(6,3,1)<br>(7,1,0)(6,3,0)(7,4,1)(8,4,1)<br>(9,4,0)(8,4,1)(9,0,0) | $\psi(I \cdot \psi_I(I) \cdot 3)$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2)) \cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,1,0)(4,2,1) | $\psi(I \cdot \psi_I(I) \cdot \omega)$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2) + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)<br>(5,1,0)(4,2,1)(5,1,0) | $\psi(I \cdot \psi_I(I) \cdot \Omega)$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2)$<br>$+\psi_{\psi_M(M^2)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0)<br>(4,2,1)(5,1,0)(4,2,1) | $\psi(I \cdot \psi_{\Omega_{\psi_I(I)+1}}(I+1))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2)$<br>$+\psi_{\psi_M(M^2)}(M^2) + 1))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,2,0) | $\psi(I \cdot \Omega_{\psi_I(I)+1})$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,2,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I \cdot \psi_I(I \cdot 2))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot$<br>$\psi_{\psi_M(M^2)}(M^2 + \psi_M(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,2,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,0)(6,3,1)(7,2,0)(6,3,0) | $\psi(I \cdot \psi_I(I \cdot 2) + \Omega_{\psi_I(I \cdot 2)+1})$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot$<br>$\psi_{\psi_M(M^2)}(M^2 + \psi_M(M^2)))+$<br>$\psi_{\psi_M(M^2)}(M^2 + \psi_M(M^2) + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,2,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,0)(6,3,1)(7,3,0) | $\psi(I \cdot \Omega_{\psi_I(I \cdot 2)+1})$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot$<br>$\psi_{\psi_M(M^2)}(M^2 + \psi_M(M^2) + 1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,2,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,0)(6,3,1)(7,3,0)(5,3,1)<br>(6,3,1)(7,3,0)(6,3,0)(7,4,1)(8,4,1)<br>(9,4,0)(8,4,1)(9,3,0)(8,4,0) | $\psi(I \cdot \Omega_{\psi_I(I \cdot 3)+1})$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot$<br>$\psi_{\psi_M(M^2)}(M^2 + \psi_M(M^2) \cdot 2 + 1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(I \cdot \psi_I(I \cdot \omega))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot$<br>$\psi_{\psi_M(M^2)}(M^2 + \psi_{\psi_M(M^2+M)}(1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1<br>)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(2,1,1) | $\psi(I \cdot \psi_I(I \cdot \omega^2))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot$<br>$\psi_{\psi_M(M^2)}(M^2 + \psi_{\psi_M(M^2+M)}(2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0) | $\psi(I \cdot \psi_I(I \cdot \Omega))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I \cdot \psi_I(I \cdot \psi_I(I)))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot$<br>$\psi_{\psi_M(M^2)}(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(I \cdot \psi_I(I \cdot \psi_I(I \cdot \omega)))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2 + \psi_{\psi_M(M^2+M)}(1))))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0) | $\psi(I \cdot \psi_I(I \cdot \psi_I(I \cdot \Omega)))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M \cdot \psi_{\psi_M(M^2)}(M^2 + \psi_{\psi_M(M^2+M)}(M))))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I^2)$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I^2 + \psi_I(I))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2) + \psi_{\psi_M(M^2)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(I^2 + \psi_I(I \cdot \omega))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2) + \psi_{\psi_M(M^2)}$<br>$(M^2 + \psi_{\psi_M(M^2+M)}(1)))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I^2+\psi_I(I^2))$ $\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2)+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I^2+\psi_I(I^2)\cdot\omega)$ $\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2)+\psi_{\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(M^2)+1)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,0) | $\psi(I^2+\Omega_{\psi_I(I^2)+1})$ $\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2)+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(M^2)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I^2+I)$ $\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2)+\psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,0)(5,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(I^2+I\cdot 2)$ $\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2)+\psi_M(M^2)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(I^2+I\cdot\omega)$ $\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2)+\psi_{\psi_M(M^2+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I^2+I\cdot\psi_I(I))$ $\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2)+\psi_{\psi_M(M^2+M)}(M\psi_{\psi_M(M^2)}(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(I^2+I\cdot\psi_I(I\cdot\omega))$ $\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2)+\psi_{\psi_M(M^2+M)}(M\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I^2+I\cdot\psi_I(I^2))$ $\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2)+\psi_{\psi_M(M^2+M)}(M\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I^2+I\cdot\psi_I(I^2))$ $\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2)+\psi_{\psi_M(M^2+M)}(M\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,2,0)(4,2,1)(5,2,0) | $\psi(I^2+I\cdot\Omega_{\psi_I(I^2)+1})$ $\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2)+\psi_{\psi_M(M^2+M)}(M\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(M^2)+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,2,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I^2+I\cdot\psi_I(I^2+I))$ $\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2)+\psi_{\psi_M(M^2+M)}(M\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(M^2)+\psi_M(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1)(5,2,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(I^2+I\cdot\psi_I(I^2+I\cdot\omega))$ $\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2)+\psi_{\psi_M(M^2+M)}(M\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(M^2)+\psi_{\psi_M(M^2+M)}(1))))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,2,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0) | $\psi(I^2 + I \cdot \psi_I(I^2 + I \cdot \omega))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2)+$<br>$\psi_{\psi_M(M^2+M)}(M\psi_{\psi_M(M^2)}(M^2 + \psi_{\psi_M(M^2+M)}(M^2)+$<br>$\psi_{\psi_M(M^2+M)}(M\psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2) + 1)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(I^2 \cdot 2)$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,2,0)(4,2,0)(5,3,1)(6,3,1)<br>(7,3,0)(6,3,1)(7,3,0)(6,0,0) | $\psi(I^2 \cdot 3)$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2) \cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1) | $\psi(I^2 \cdot \omega)$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2 + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0) | $\psi(I^2 \cdot \Omega)$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2 + \psi_{\psi_M(M^2)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I^2 \cdot \psi_I(I))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2 + \psi_{\psi_M(M^2)}(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,1) | $\psi(I^2 \cdot \psi_I(I \cdot \omega))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2 + \psi_{\psi_M(M^2+M)}(1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I^2 \cdot \psi_I(I^2))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2 + \psi_{\psi_M(M^2+M)}(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1) | $\psi(I^2 \cdot \psi_I(I^2 \cdot \omega))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2 + \psi_{\psi_M(M^2+M)}(M^2 + 1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1)(3,1,0) | $\psi(I^2 \cdot \psi_I(I^2 \cdot \Omega))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2)}(M^2 + \psi_{\psi_M(M^2+M)}(M^2 + \psi_{\psi_M(M^2)}(1)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0) | $\psi(I^2 \cdot \psi_I(I^2 \cdot \psi_I(I^2 \cdot \Omega)))$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2 + \psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2 + \psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2 + \psi_{\psi_M(M^2)}(1)))))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I^3)$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2 + \psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1) | $\psi(I^3 \cdot \omega)$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2 + \psi_M(M^2) + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I^4)$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2 + \psi_M(M^2) \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,0,0) | $\psi(I^\omega)$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2 + \psi_{\psi_M(M^2+M)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I^{\omega+1})$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(1) + \psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,0)(3,0,0) | $\psi(I^{\omega \cdot 2})$<br>$\psi(M^2 + \psi_{\psi_M(M^2+M)}(M^2 + \psi_{\psi_M(M^2+M)}(1) \cdot 2))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,0,0)(3,0,0) | $\psi(I^{\omega^2})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,1,0) | $\psi(I^{\Omega})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I^{\psi_I(I)})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M\psi_{\psi_M(M)}(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I^{\psi_I(I^2)})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M\psi_{\psi_M(M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(3,0,0) | $\psi(I^{\psi_I(I^{\omega})})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M\psi_{\psi_M(M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(1))))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(3,1,0) | $\psi(I^{\psi_I(I^{\Omega})})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M\psi_{\psi_M(M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M))))))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(I^{I})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(2,1,1) | $\psi(I^{I}\cdot\omega)$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I^{I+1})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2)+\psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(3,1,0)(2,0,0) | $\psi(I^{I\cdot 2})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(3,1,0)<br>(2,1,1)(3,1,0)(3,1,0)(2,0,0) | $\psi(I^{I\cdot 3})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2)\cdot 3))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(3,0,0) | $\psi(I^{I\cdot\omega})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(3,1,0) | $\psi(I^{I\cdot\Omega})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2)}(1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(I^{I^2})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_M(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(3,1,0)(2,1,1) | $\psi(I^{I^2}\cdot\omega)$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_M(M^2))+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(3,1,0)(3,0,0) | $\psi(I^{I^2\cdot\omega})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_M(M^2)+1)))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(3,1,0)(3,1,0)(2,0,0) | $\psi(I^{I^3})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_M(M^2)\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,1,0)<br>(3,1,0)(3,1,0)(3,1,0)(2,0,0) | $\psi(I^{I^4})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_M(M^2)\cdot 3)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,0,0) | $\psi(I^{I^\omega})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,1,0) | $\psi(I^{I^\Omega})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M))))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(2,0,0) | $\psi(I^{I^I})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(2,1,1) | $\psi(I^{I^I}\cdot\omega)$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2))+1))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(3,0,0) | $\psi(I^{I^{I\cdot\omega}})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2)+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(3,1,0)(2,0,0) | $\psi(I^{I^{I+1}})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2)+\psi_M(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(3,1,0)(4,0,0) | $\psi(I^{I^{I+\omega}})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2)+\psi_{\psi_M(M^2+M)}(1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(3,1,0)(4,1,0)(2,0,0) | $\psi(I^{I^{I\cdot 2}})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2)\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(4,0,0) | $\psi(I^{I^{I\cdot\omega}})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2+1))))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(4,1,0)(2,0,0) | $\psi(I^{I^{I^2}})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2+\psi_M(M^2)))))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(5,0,0) | $\psi(I^{I^{I^\omega}})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(1)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(5,1,0)(2,0,0) | $\psi(I^{I^{I^I}})$<br>$\psi(M^2+\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{I+1})$<br>$\psi(M^2+M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{I+1})$<br>$\psi(M^2+M)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{I+1}+\psi_I(I))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{I+1}+\psi_I(I^2))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{I+1}+\psi_I(I^3))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_M(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(1,1,1)(2,1,1)(3,1,0)(3,0,0) | $\psi(\Omega_{I+1}+\psi_I(I^\omega))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2)}(M^2+\psi_{\psi_M(M^2+M)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(1,1,1)(2,1,1)(3,1,0)(3,1,0)(2,0,0) | $\psi(\Omega_{I+1}+\psi_I(I^I))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{I+1}+\psi_I(\Omega_{I+1}))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2)}(M^2+M))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(2,0,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{\psi_I(\Omega_{I+1})+1}}(1))$<br>$\psi(M^2+M+\psi_{\psi_{\psi_M(M^2)}(M^2+M+1)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{\psi_I(\Omega_{I+1})+1}}(\Omega_{I+1}))$<br>$\psi(M^2+M+\psi_{\psi_{\psi_M(M^2)}(M^2+M+1)}(M^2+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,0) | $\psi(\Omega_{I+1}+\Omega_{\psi_I(\Omega_{I+1})+1})$<br>$\psi(M^2+M+\psi_{\psi_M(M^2)}(M^2+M+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,0)(4,3,0) | $\psi(\Omega_{I+1}+\Omega_{\psi_I(\Omega_{I+1})+2})$<br>$\psi(M^2+M+\psi_{\psi_M(M^2)}(M^2+M+2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1) | $\psi(\Omega_{I+1}+\Omega_{\psi_I(\Omega_{I+1})+\omega})$<br>$\psi(M^2+M+\psi_{\psi_M(M^2)}(M^2+M+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{I+1}+I)$<br>$\psi(M^2+M+\psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(\Omega_{I+1}+I\cdot 2)$<br>$\psi(M^2+M+\psi_M(M^2)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(\Omega_{I+1}+I\cdot\omega)$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{I+1}+I^2)$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{I+1}+I^3)$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+\psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(5,0,0) | $\psi(\Omega_{I+1}+I^\omega)$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(5,2,0)(4,0,0) | $\psi(\Omega_{I+1}+I^I)$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(6,3,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1})+I)$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M)+\psi_M(M^2))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(6,3,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,0)(8,4,0) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1}) \cdot 2)$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}(M^2 + M) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(2,1,1) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1} + 1))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}(M^2 + M + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1} + 1) + \psi_{\Omega_{I+1}}(\Omega_{I+1}))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}(M^2 + M + 1) +$<br>$\psi_{\psi_M(M^2+M)}(M^2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,0)(4,2,1) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1} + 1) \cdot 2)$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}(M^2 + M + 1) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(2,1,1) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1} + 2))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}(M^2 + M + 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(2,1,1)(2,1,1) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1} + 3))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}(M^2 + M + 3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,0,0) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1} + \omega))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}(M^2 + M + \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1} + \Omega))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}$<br>$(M^2 + M + \psi_{\psi_M(M^2)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1} + I))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}$<br>$(M^2 + M + \psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(2,1,1) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1} + I + 1))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}$<br>$(M^2 + M + \psi_M(M^2) + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(2,1,1) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1} + I + 1))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}$<br>$(M^2 + M + \psi_M(M^2) + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1} + I \cdot 2))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}(M^2 +$<br>$M + \psi_M(M^2) \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(3,0,0) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1} + \psi_{\Omega_{I+1}}(1)))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}(M^2 +$<br>$M + \psi_{\psi_M(M^2+M)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,1)(3,1,0)(3,0,0)(3,0,0) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1} + \psi_{\Omega_{I+1}}(2)))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}(M^2 +$<br>$M + \psi_{\psi_M(M^2+M)}(2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,1)(3,1,0)(3,1,0)(2,0,0) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1} + \psi_{\Omega_{I+1}}(I)))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}(M^2 +$<br>$M + \psi_{\psi_M(M^2+M)}(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,1)(3,1,0)(3,1,0)(2,1,1)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1} + \psi_{\Omega_{I+1}}(I) \cdot 2))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}(M^2 +$<br>$M + \psi_{\psi_M(M^2+M)}(M^2) \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(4,0,0) | $\psi(\Omega_{I+1} + \psi_{\Omega_{I+1}}(\Omega_{I+1} + \psi_{\Omega_{I+1}}(I \cdot \omega)))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M)}(M^2 + M +$<br>$\psi_{\psi_M(M^2+M)}(M^2 + \psi_{\psi_M(M^2+M)}(1))))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(4,0,0)<br>(2,1,1)(3,1,0)(4,0,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(I\cdot\omega)\cdot 2))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(1))\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,1)(3,1,0)(4,0,0)(3,0,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(I\cdot\omega+1)))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(1)+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,1)(3,1,0)(4,0,0)(3,1,0)(2,0,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(I\cdot\omega+I)))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(1)+\psi_M(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,1)(3,1,0)(4,0,0)(3,1,0)(4,0,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(I\cdot\omega\cdot 2)))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(1)\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(4,0,0)(4,0,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\psi_{\Omega_{I+1}}(2))))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(4,1,0)(2,0,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\psi_{\Omega_{I+1}}(I))))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+$<br>$\psi_{\psi_M(M^2+M)}(M^2+\psi_{\psi_M(M^2+M)}(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1})))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+$<br>$M+\psi_{\psi_M(M^2+M)}(M^2+M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1})\cdot 2))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+$<br>$M+\psi_{\psi_M(M^2+M)}(M^2+M)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(3,0,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+1)))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+$<br>$M+\psi_{\psi_M(M^2+M)}(M^2+M+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(3,1,0)(2,0,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+I)))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M+\psi_M(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(3,1,0)(3,0,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+I+1)))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M+\psi_M(M^2)+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(3,1,0)(3,1,0)(2,0,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+I\cdot 2)))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M+\psi_M(M^2)\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(3,1,0)(4,0,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+I\cdot 2)))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M+\psi_M(M^2)\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(3,1,0)(4,1,0)(2,0,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+I^2)))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M+\psi_{\psi_M(M^2+M)}(M^2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(3,1,0)(4,2,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+$<br>$\psi_{\Omega_{I+1}}(\Omega_{I+1}))))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M))))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(4,0,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+$<br>$\psi_{\Omega_{I+1}}(\Omega_{I+1}+1))))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+1))))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(4,1,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+$<br>$\psi_{\Omega_{I+1}}(\Omega_{I+1}+I))))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+$<br>$\psi_{\psi_M(M^2+M)}(M^2+$<br>$M+\psi_{\psi_M(M^2+M)}(M^2+M+\psi_M(M^2)))))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(4,1,0)(5,2,0) | $\psi(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1}+$<br>$\psi_{\Omega_{I+1}}(\Omega_{I+1}+\psi_{\Omega_{I+1}}(\Omega_{I+1})))))$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M+\psi_{\psi_M(M^2+M)}(M^2+M+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M)))))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(4,2,0) | $\psi(\Omega_{I+1}\cdot 2)$<br>$\psi(M^2+M+\psi_M(M^2+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{I+1}\cdot 2+I)$<br>$\psi(M^2+M+\psi_M(M^2+M)+\psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{I+1}\cdot 2+I)$<br>$\psi(M^2+M+\psi_M(M^2+M)+\psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{I+1}\cdot 2+\psi_{\Omega_{I+1}}(\Omega_{I+1}))$<br>$\psi(M^2+M+\psi_M(M^2+M)+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,0)(6,3,0) | $\psi(\Omega_{I+1}\cdot 2+\psi_{\Omega_{I+1}}(\Omega_{I+1}\cdot 2))$<br>$\psi(M^2+M+\psi_M(M^2+M)+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M+\psi_M(M^2+M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(4,2,0)(2,1,1) | $\psi(\Omega_{I+1}\cdot 2+\psi_{\Omega_{I+1}}(\Omega_{I+1}\cdot 2+1))$<br>$\psi(M^2+M+\psi_M(M^2+M)+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M+\psi_M(M^2+M)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(4,2,0)(3,0,0) | $\psi(\Omega_{I+1}\cdot 2+\psi_{\Omega_{I+1}}(\Omega_{I+1}\cdot 2+\psi_{\Omega_{I+1}}(\Omega_{I+1}\cdot 2+1)))$<br>$\psi(M^2+M+\psi_M(M^2+M)+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M+\psi_M(M^2+M)+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M+\psi_M(M^2+M)+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(4,2,0)(4,0,0) | $\psi(\Omega_{I+1}\cdot 2+\psi_{\Omega_{I+1}}(\Omega_{I+1}\cdot 2+$<br>$\psi_{\Omega_{I+1}}(\Omega_{I+1}\cdot 2+\psi_{\Omega_{I+1}}(\Omega_{I+1}\cdot 2+1))))$<br>$\psi(M^2+M+\psi_M(M^2+M)+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M+\psi_M(M^2+M)+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M+\psi_M(M^2+M)+$<br>$\psi_{\psi_M(M^2+M)}(M^2+M+\psi_M(M^2+M)+1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(4,2,0)(4,2,0) | $\psi(\Omega_{I+1}\cdot 3)$<br>$\psi(M^2+M+\psi_M(M^2+M)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(5,0,0) | $\psi(\Omega_{I+1}\cdot\omega)$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M\cdot 2)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,1,0)(2,0,0) | $\psi(\Omega_{I+1}\cdot I)$<br>$\psi(M^2+M+\psi_{\psi_M(M^2+M\cdot 2)}(M^2))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,1,0)(6,2,0) | $\psi(\Omega_{I+1} \cdot \psi_{\Omega_{I+1}}(\Omega_{I+1}))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M\cdot 2)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2 + M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,1,0)(6,2,0)(7,0,0) | $\psi(\Omega_{I+1} \cdot \psi_{\Omega_{I+1}}(\Omega_{I+1} \cdot \omega))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M\cdot 2)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2 + M + \psi_{\psi_M(M^2+M)}(1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(5,1,0)(6,2,0)(7,1,0)(8,2,0) | $\psi(\Omega_{I+1} \cdot \psi_{\Omega_{I+1}}(\Omega_{I+1} \cdot \psi_{\Omega_{I+1}}(\Omega_{I+1})))$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M\cdot 2)}(M^2+$<br>$\psi_{\psi_M(M^2+M)}(M^2 + M+$<br>$\psi_{\psi_M(M^2+M)}(M^2 + \psi_{\psi_M(M^2+M)}(M^2 + M)))))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(5,2,0) | $\psi(\Omega_{I+1}^2)$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M\cdot 2)}(M^2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,2,0)(5,0,0) | $\psi(\Omega_{I+1}^2 \cdot \omega)$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M\cdot 2)}(M^2 + M + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,2,0)(5,2,0) | $\psi(\Omega_{I+1}^3)$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M\cdot 2)}$<br>$(M^2 + M + \psi_M(M^2 + M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,2,0)(6,0,0) | $\psi(\Omega_{I+1}^\omega)$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M\cdot 2)}$<br>$(M^2 + M + \psi_{\psi_M(M^2+M\cdot 2)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,2,0)(6,2,0) | $\psi(\Omega_{I+1}^{\Omega_{I+1}})$<br>$\psi(M^2 + M + \psi_{\psi_M(M^2+M\cdot 2)}$<br>$(M^2 + M + \psi_{\psi_M(M^2+M\cdot 2)}(M^2 + M)))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(5,3,0) | $\psi(\Omega_{I+2})$<br>$\psi(M^2 + M \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,3,0)(4,2,0) | $\psi(\Omega_{I+2} + \Omega_{I+1})$<br>$\psi(M^2 + M \cdot 2 + \psi_M(M^2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,3,0)(4,2,0)(5,3,0) | $\psi(\Omega_{I+2} + \psi_{\Omega_{I+2}}(\Omega_{I+2}))$<br>$\psi(M^2 + M \cdot 2 + \psi_{\psi_M(M^2+M\cdot 2)}(M^2 + M \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,3,0)(5,3,0) | $\psi(\Omega_{I+2} \cdot 2)$<br>$\psi(M^2 + M \cdot 2 + \psi_M(M^2 + M \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,3,0)(6,4,0) | $\psi(\Omega_{I+3})$<br>$\psi(M^2 + M \cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,3,0)(6,4,0)(7,5,0) | $\psi(\Omega_{I+4})$<br>$\psi(M^2 + M \cdot 4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1) | $\psi(\Omega_{I+\omega})$<br>$\psi(M^2 + M + \psi_M(M^2 + M \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(4,2,1) | $\psi(\Omega_{I+\omega} \cdot 2)$<br>$\psi(M^2 + M \cdot \omega + \psi_M(M^2 + M \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,0) | $\psi(\Omega_{I+\omega} \cdot \Omega_{I+1})$<br>$\psi(M^2 + M \cdot \omega + \psi_{\psi_M(M^2+M\cdot\omega+M)}(M^2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,0)(4,2,1) | $\psi(\Omega_{I+\omega}^2)$<br>$\psi(M^2 + M \cdot \omega + \psi_{\psi_M(M^2+M\cdot\omega+M)}(M^2 + M \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{I+\omega+1})$<br>$\psi(M^2 + M \cdot \omega + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,0)(6,3,0)(7,4,0) | $\psi(\Omega_{I+\omega+2})$<br>$\psi(M^2 + M \cdot \omega + M \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,0)(6,3,1) | $\psi(\Omega_{I+\omega\cdot 2})$<br>$\psi(M^2 + M \cdot \omega \cdot 2)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,0)(6,3,1)(7,3,0)(8,4,1) | $\psi(\Omega_{I+\omega\cdot 3})$<br>$\psi(M^2+M\cdot\omega\cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1) | $\psi(\Omega_{I+\omega^2})$<br>$\psi(M^2+M\cdot\psi(2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(5,2,1) | $\psi(\Omega_{I+\omega^3})$<br>$\psi(M^2+M\cdot\psi(3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0) | $\psi(\Omega_{I+\Omega})$<br>$\psi(M^2+M\cdot\psi_{\psi_M(M^2)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(2,0,0) | $\psi(\Omega_{I\cdot 2})$<br>$\psi(M^2+M\cdot\psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(7,2,0) | $\psi(\Omega_{\psi_{\Omega_{I+1}}(\Omega_{I+1})})$<br>$\psi(M^2+M\cdot\psi_{\psi_M(M^2+M)}(M^2+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0) | $\psi(\Omega_{\Omega_{I+1}})$<br>$\psi(M^2+M\cdot\psi_M(M^2+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(4,2,1)(5,2,1)(6,2,0) | $\psi(\Omega_{\Omega_{\Omega_{I+1}}})$<br>$\psi(M^2+M\cdot\psi_M(M^2+M\cdot\psi_M(M^2+M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(I_2)$<br>$\psi(M^2\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I_2+I)$<br>$\psi(M^2\cdot 2+\psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1<br>)(5,2,1)(6,2,0)(5,0,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,0) | $\psi(I_2+\psi_{\Omega_{I+1}}(\Omega_{I+1}))$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2+M)}(M^2+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(I_2+\psi_{\Omega_{I+1}}(I_2))$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2+M)}(M^2\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,1)(7,3,1)(8,3,0)<br>(7,0,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,0)(6,0,0) | $\psi(I_2+\psi_{\Omega_{I+1}}(I_2)+I)$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2+M)}(M^2\cdot 2)+\psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,1)(7,3,1)(8,3,0)<br>(7,0,0)(4,2,0)(5,3,1)(6,3,1)(7,3,0)<br>(8,4,1)(9,4,1)(10,4,0)(9,0,0) | $\psi(I_2+\psi_{\Omega_{I+1}}(I_2)\cdot 2)$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2+M)}(M^2\cdot 2)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(2,1,1) | $\psi(I_2+\psi_{\Omega_{I+1}}(I_2+1))$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2+M)}(M^2\cdot 2+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I_2+\psi_{\Omega_{I+1}}(I_2+I))$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2+M)}(M^2\cdot 2+\psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I_2+\psi_{\Omega_{I+1}}(I_2+I))$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2+M)}(M^2\cdot 2+\psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(I_2+\psi_{\Omega_{I+1}}(I_2+\psi_{\Omega_{I+1}}(I_2)))$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2+M)}(M^2\cdot 2+$<br>$\psi_{\psi_M(M^2+M)}(M^2\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(3,0,0) | $\psi(I_2+\psi_{\Omega_{I+1}}(I_2+\psi_{\Omega_{I+1}}(I_2+1)))$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2+M)}(M^2\cdot 2+$<br>$\psi_{\psi_M(M^2+M)}(M^2\cdot 2+1)))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I_2+\psi_{\Omega_{I+1}}(I_2+\psi_{\Omega_{I+1}}(I_2+\psi_{\Omega_{I+1}}(I_2))))$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2+M)}(M^2\cdot 2+$<br>$\psi_{\psi_M(M^2+M)}(M^2\cdot 2+\psi_{\psi_M(M^2+M)}(M^2\cdot 2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(4,0,0) | $\psi(I_2+\psi_{\Omega_{I+1}}(I_2+\psi_{\Omega_{I+1}}(I_2+\psi_{\Omega_{I+1}}(I_2+1))))$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2+M)}(M^2\cdot 2+$<br>$\psi_{\psi_M(M^2+M)}(M^2\cdot 2+\psi_{\psi_M(M^2+M)}(M^2\cdot 2+1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,0)(6,0,0) | $\psi(I_2+\psi_{\Omega_{I+1}}(I_2+\psi_{\Omega_{I+1}}(I_2+$<br>$\psi_{\Omega_{I+1}}(I_2+\psi_{\Omega_{I+1}}(I_2)))))$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2+M)}(M^2\cdot 2+$<br>$\psi_{\psi_M(M^2+M)}(M^2\cdot 2+$<br>$\psi_{\psi_M(M^2+M)}(M^2\cdot 2+\psi_{\psi_M(M^2+M)}(M^2\cdot 2)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(4,2,0) | $\psi(I_2+\Omega_{I+1})$<br>$\psi(M^2\cdot 2+\psi_M(M^2+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(4,2,1) | $\psi(I_2+\Omega_{I+\omega})$<br>$\psi(M^2\cdot 2+\psi_M(M^2+M\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I_2+\psi_{I_2}(I_2))$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2\cdot 2)}(M^2\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I_2+\psi_{\Omega_{\psi_{I_2}(I_2)+1}}(I_2))$<br>$\psi(M^2\cdot 2+\psi_{\psi_{\psi_M(M^2\cdot 2)}(M^2\cdot 2+1)}(M^2\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,0)(6,3,0) | $\psi(I_2+\Omega_{\psi_{I_2}(I_2)+1})$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2\cdot 2)}(M^2\cdot 2+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,0)(6,3,1) | $\psi(I_2+\Omega_{\psi_{I_2}(I_2)+\omega})$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2\cdot 2)}(M^2\cdot 2+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(I_2\cdot 2)$<br>$\psi(M^2\cdot 2+\psi_M(M^2\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1) | $\psi(I_2\cdot\omega)$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2\cdot 2+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,1,0)(2,0,0) | $\psi(I_2\cdot I)$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2\cdot 2+M)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0) | $\psi(I_2\cdot\Omega_{I+1})$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2\cdot 2+M)}(M^2+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0)(4,2,1) | $\psi(I_2\cdot\Omega_{I+\omega})$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2\cdot 2+M)}(M^2+M\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(I_2\cdot\psi_{I_2}(I_2))$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2\cdot 2+M)}(M^2+$<br>$M\cdot\psi_{\psi_M(M^2\cdot 2)}(M^2\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,1) | $\psi(I_2\cdot\psi_{I_2}(I_2\cdot\omega))$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2\cdot 2+M)}(M^2+$<br>$M\cdot\psi_{\psi_M(M^2\cdot 2)}(M^2\cdot 2+\psi_{\psi_M(M^2\cdot 2+M)}(1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0) | $\psi(I_2\cdot\psi_{I_2}(I_2\cdot\Omega_{I+1}))$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2\cdot 2+M)}(M^2+M\cdot$<br>$\psi_{\psi_M(M^2\cdot 2)}(M^2\cdot 2+\psi_{\psi_M(M^2\cdot 2+M)}(M^2+M))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(I_2^2)$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2\cdot 2+M)}(M^2\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I_2^3)$<br>$\psi(M^2\cdot 2+\psi_{\psi_M(M^2\cdot 2+M)}(M^2\cdot 2+\psi_M(M^2\cdot 2)))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(6,2,0)(5,0,0) | $\psi(I_2^{I_2})$<br>$\psi(M^2 \cdot 2 + \psi_{\psi_M(M^2 \cdot 2+M)}(M^2 \cdot 2+$<br>$\psi_{\psi_M(M^2 \cdot 2+M)}(M^2 \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,2,0)(5,0,0) | $\psi(I_2^{I_2^{I_2}})$<br>$\psi(M^2 \cdot 2 + \psi_{\psi_M(M^2 \cdot 2+M)}(M^2 \cdot 2+$<br>$\psi_{\psi_M(M^2 \cdot 2+M)}(M^2 \cdot 2 + \psi_{\psi_M(M^2 \cdot 2+M)}(M^2 \cdot 2))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{I_2+1})$<br>$\psi(M^2 \cdot 2 + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,1) | $\psi(\Omega_{I_2+\omega})$<br>$\psi(M^2 \cdot 2 + M \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,1)(8,3,1)(9,3,0) | $\psi(\Omega_{\Omega_{I_2+1}})$<br>$\psi(M^2 \cdot 2 + M \cdot \psi_M(M^2 \cdot 2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,1)<br>(8,3,1)(9,3,0)(8,0,0) | $\psi(I_3)$<br>$\psi(M^2 \cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,1)(8,3,1)(9,3,0)<br>(8,3,0)(9,4,1)(10,4,1)<br>(11,4,0)(10,0,0) | $\psi(I_3 \cdot 2)$<br>$\psi(M^2 \cdot 3 + \psi_M(M^2 \cdot 3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,1)<br>(8,3,1)(9,3,0)(8,3,1) | $\psi(I_3 \cdot \omega)$<br>$\psi(M^2 \cdot 3 + \psi_{\psi_M(M^2 \cdot 3+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,1)<br>(8,3,1)(9,3,0)(10,4,0) | $\psi(\Omega_{I_3+1})$<br>$\psi(M^2 \cdot 3 + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,1)(8,3,1)(9,3,0)<br>(10,4,1)(11,4,1)(12,4,0)(11,0,0) | $\psi(I_4)$<br>$\psi(M^2 \cdot 4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1) | $\psi(I_\omega)$<br>$\psi(M^2 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1) | $\psi(I_\omega + \Omega_\omega)$<br>$\psi(M^2 \cdot \omega + \psi_M(M \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I_\omega + \psi_I(I))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,0) | $\psi(I_\omega + \psi_{\Omega_{\psi_I(I)+1}}(I + \Omega_{\psi_I(I)+1}))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_{\psi_M(M^2)}(M^2+1)}$<br>$(M^2 + \psi_{\psi_M(M^2)}(M^2+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I_\omega + \psi_{\Omega_{\psi_I(I)+1}}(I \cdot 2))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2 + \psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,0)(6,0,0) | $\psi(I_\omega + \psi_{\Omega_{\psi_I(I)+1}}(I \cdot 3))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2 + \psi_M(M^2) \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1) | $\psi(I_\omega + \psi_{\Omega_{\psi_I(I)+1}}(I \cdot \omega))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2 + \psi_{\psi_M(M^2+M)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,0) | $\psi(I_\omega + \psi_{\Omega_{\psi_I(I)+1}}(\Omega_{I+1}))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2 + M))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(6,3,1)(7,3,1)(8,3,0)(7,0,0) | $\psi(I_\omega + \psi_{\Omega_{\psi_I(I)+1}}(I_2))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(6,3,1)(7,3,1)(8,3,0)(9,4,1)<br>(10,4,1)(11,4,0)(10,0,0) | $\psi(I_\omega + \psi_{\Omega_{\psi_I(I)+1}}(I_3))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2 \cdot 3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1) | $\psi(I_\omega + \psi_{\Omega_{\psi_I(I)+1}}(I_\omega))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_{\psi_M(M^2)}(M^2+1)}(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(3,2,0) | $\psi(I_\omega + \Omega_{\psi_I(I)+1})$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2)}(M^2 + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(3,2,0)(4,3,1)<br>(5,3,1)(6,3,1)(4,3,0) | $\psi(I_\omega + \Omega_{\psi_I(I)+2})$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2)}(M^2 + 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(3,2,1) | $\psi(I_\omega + \Omega_{\psi_I(I)+\omega})$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2)}(M^2 + \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I_\omega + \psi_I(I \cdot 2))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2)}(M^2 + \psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(3,2,1)(4,2,1)(5,2,0)(6,3,0) | $\psi(I_\omega + \psi_{\Omega_{\psi_I(I\cdot 2)+1}}(I \cdot 2 + \Omega_{\psi_I(I\cdot 2)+1}))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+1)}(M^2+$<br>$\psi_M(M^2) + \psi_{\psi_M(M^2)}(M^2 + \psi_M(M^2) + 1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(3,2,1)(4,2,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(I_\omega + \psi_{\Omega_{\psi_I(I\cdot 2)+1}}(I \cdot 3))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+1)}$<br>$(M^2 + \psi_M(M^2) \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(3,2,1)(4,2,1)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,1) | $\psi(I_\omega + \psi_{\Omega_{\psi_I(I\cdot 2)+1}}(I_\omega))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_{\psi_M(M^2)}(M^2+\psi_M(M^2)+1)}(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(3,2,1)(4,2,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,1)(6,3,0) | $\psi(I_\omega + \Omega_{\psi_I(I\cdot 2)+1})$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2)}(M^2 + \psi_M(M^2) + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(3,2,1)(4,2,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,1)(6,3,1)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(I_\omega + \psi_I(I \cdot 3))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2)}(M^2 + \psi_M(M^2) \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(I_\omega + \psi_I(I \cdot \omega))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2)}(M^2 + \psi_{\psi_M(M^2+M)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(I_\omega + \psi_I(\Omega_{I+1}))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2)}(M^2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I_\omega + \psi_I(I_2))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2)}(M^2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,0) | $\psi(I_\omega + \psi_I(\Omega_{I_2+1}))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2)}(M^2 \cdot 2 + M))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,0)<br>(7,3,1)(8,3,1)(9,3,0)(8,0,0) | $\psi(I_\omega + \psi_I(I_3))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2)}(M^2 \cdot 3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1) | $\psi(I_\omega + \psi_I(I_\omega))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2)}(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1) | $\psi(I_\omega + \psi_I(I_\omega) \cdot 2)$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2)}(M^2 \cdot \omega) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(2,1,0) | $\psi(I_\omega + \psi_{\Omega_{\psi_I(I_\omega)+1}}(\Omega))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_{\psi_M(M^2)}(M^2 \cdot \omega+1)}(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1) | $\psi(I_\omega + \psi_{\Omega_{\psi_I(I_\omega)+1}}(I_\omega))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_{\psi_M(M^2)}(M^2 \cdot \omega+1)}(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(2,1,0)(3,2,0) | $\psi(I_\omega + \Omega_{\psi_I(I_\omega)+1})$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2)}(M^2 \cdot \omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I_\omega + I)$<br>$\psi(M^2 \cdot \omega + \psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(I_\omega + I \cdot 2)$<br>$\psi(M^2 \cdot \omega + \psi_M(M^2) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(I_\omega + I \cdot \omega)$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(6,3,0) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(\Omega_{I+1}))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(3,2,0) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) + \Omega_{\psi_I(I_\omega)+1})$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega)$<br>$+\psi_{\psi_M(M^2)}(M^2 \cdot \omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) + \psi_I(I_\omega + I))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega)+$<br>$\psi_{\psi_M(M^2)}(M^2 \cdot \omega + \psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,0) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) + \psi_I(I_\omega + \psi_{\Omega_{I+1}}(\Omega_{I+1})))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega)+$<br>$\psi_{\psi_M(M^2)}(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 + M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,1)(7,3,1)(8,3,1) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) + \psi_I(I_\omega + \psi_{\Omega_{I+1}}(I_\omega)))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega)+$<br>$\psi_{\psi_M(M^2)}(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,1)(7,3,1)<br>(8,3,1)(4,2,0)(5,3,0) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) + \Omega_{\psi_I(I_\omega+\psi_{\Omega_{I+1}}(I_\omega))+1})$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega)$<br>$+\psi_{\psi_M(M^2)}(M^2 \cdot \omega+$<br>$\psi_{\psi_M(M^2+M)}(M^2 \cdot \omega) + 1))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,1)(7,3,1)(8,3,1)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) + I)$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega) + \psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(2,1,1) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega + 1))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega + I))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega + \psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega + \psi_{\Omega_{I+1}}(\Omega_{I+1})))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega +$<br>$\psi_{\psi_M(M^2+M)}(M^2 + M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega + \psi_{\Omega_{I+1}}(I_\omega)))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega$<br>$+\psi_{\psi_M(M^2+M)}(M^2 \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) \cdot 2))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega +$<br>$\psi_{\psi_M(M^2+M)}(M^2 \cdot \omega) \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(3,0,0) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega + \psi_{\Omega_{I+1}}(I_\omega + 1)))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega +$<br>$\psi_{\psi_M(M^2+M)}(M^2 \cdot \omega + 1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(3,1,0)(4,2,0) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega + \psi_{\Omega_{I+1}}(I_\omega + \psi_{\Omega_{I+1}}(\Omega_{I+1}))))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}$<br>$(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 + M))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega + \psi_{\Omega_{I+1}}(I_\omega + \psi_{\Omega_{I+1}}(I_\omega))))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}$<br>$(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(4,0,0) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega + \psi_{\Omega_{I+1}}(I_\omega + \psi_{\Omega_{I+1}}(I_\omega + 1))))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}$<br>$(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega + 1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(4,1,0)(5,2,1)(6,2,1)(7,2,1) | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega +$<br>$\psi_{\Omega_{I+1}}(I_\omega + \psi_{\Omega_{I+1}}(I_\omega + \psi_{\Omega_{I+1}}(I_\omega)))))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}$<br>$(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega +$<br>$\psi_{\psi_M(M^2+M)}(M^2 \cdot \omega + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(4,2,0) | $\psi(I_\omega + \Omega_{I+1})$<br>$\psi(M^2 \cdot \omega + \psi_M(M^2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,1)(5,3,0) | $\psi(I_\omega + \Omega_{I+2})$<br>$\psi(M^2 \cdot \omega + \psi_M(M^2 + M \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(4,2,1) | $\psi(I_\omega + \Omega_{I+\omega})$<br>$\psi(M^2 \cdot \omega + \psi_M(M^2 + M \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(I_\omega + \psi_{I_2}(I_2))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot 2)}(M^2 \cdot 2))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,0) | $\psi(I_\omega + \psi_{I_2}(\Omega_{I_2+1}))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot 2)}(M^2 \cdot 2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(4,2,1)(5,2,1)(6,2,0)<br>(7,3,1)(8,3,1)(9,3,1) | $\psi(I_\omega + \psi_{I_2}(I_\omega))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot 2)}(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(5,2,0)(6,3,0) | $\psi(I_\omega + \Omega_{\psi_{I_2}(I_\omega)+1})$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot 2)}(M^2 \cdot \omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(I_\omega + I_2)$<br>$\psi(M^2 \cdot \omega + \psi_M(M^2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(5,2,0)(6,3,1)(7,3,1)(8,3,1) | $\psi(I_\omega + \psi_{\Omega_{I_2+1}}(I_\omega))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot 2+M)}(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(5,2,0)(6,3,1)(7,3,1)(8,3,1)<br>(6,3,1)(7,3,1)(8,3,0)(7,0,0) | $\psi(I_\omega + \psi_{\Omega_{I_2+1}}(I_\omega) + \psi_{I_2}(I_\omega + I_2))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot 2+M)}(M^2 \cdot \omega) +$<br>$\psi_{\psi_M(M^2 \cdot 2)}(M^2 \cdot \omega + \psi_M(M^2 \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(5,2,0)(6,3,1)(7,3,1)(8,3,1)<br>(6,3,1)(7,3,1)(8,3,0)(9,4,1)(10,4,1)<br>(11,4,1)(7,3,0)(8,4,1)<br>(9,4,1)(10,4,0)(9,0,0) | $\psi(I_\omega + \psi_{\Omega_{I_2+1}}(I_\omega) + I_2)$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot 2+M)}(M^2 \cdot \omega) + \psi_M(M^2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(5,2,0)(6,3,1)(7,3,1)(8,3,1)<br>(6,3,1)(7,3,1)(8,3,0)(9,4,1)(10,4,1)<br>(11,4,1)(7,3,0)(8,4,1)(9,4,1)(10,4,0)<br>(11,5,1)(12,5,1)(13,5,1) | $\psi(I_\omega + \psi_{\Omega_{I_2+1}}(I_\omega) \cdot 2)$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot 2+M)}(M^2 \cdot \omega) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,1)<br>(8,3,1)(9,3,1)(5,2,1) | $\psi(I_\omega + \psi_{\Omega_{I_2+1}}(I_\omega + 1))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot 2+M)}(M^2 \cdot \omega + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,1)<br>(8,3,1)(9,3,1)(7,3,0) | $\psi(I_\omega + \Omega_{I_2+1})$<br>$\psi(M^2 \cdot \omega + \psi_M(M^2 \cdot 2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(7,3,1)(8,3,1)(9,3,0)(10,4,1)<br>(11,4,1)(12,4,1) | $\psi(I_\omega + \psi_{I_3}(I_\omega))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot 3)}(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(7,3,1)(8,3,1)(9,3,0)(10,4,1)<br>(11,4,1)(12,4,1)(10,4,0) | $\psi(I_\omega + \Omega_{I_3+1})$<br>$\psi(M^2 \cdot \omega + \psi_M(M^2 \cdot 3 + M))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1) | $\psi(I_\omega \cdot 2)$<br>$\psi(M^2 \cdot \omega + \psi_M(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,1)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)<br>(4,2,1)(5,2,1)(6,2,1) | $\psi(I_\omega \cdot 2 + \psi_I(I_\omega \cdot 2))$<br>$\psi(M^2 \cdot \omega + \psi_M(M^2 \cdot \omega)+$<br>$\psi_{\psi_M(M^2)}(M^2 \cdot \omega + \psi_M(M^2 \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,1)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(4,2,1)<br>(5,2,1)(6,2,1)(4,2,0) | $\psi(I_\omega \cdot 2 + \Omega_{I+1})$<br>$\psi(M^2 \cdot \omega + \psi_M(M^2 \cdot \omega) + \psi_M(M^2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,1)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(4,2,1)(5,2,1)<br>(6,2,1)(4,2,1)(5,2,1)(6,2,0)(7,3,1)<br>(8,3,1)(9,3,1)(7,3,1)(8,3,1)(9,3,1) | $\psi(I_\omega \cdot 2 + \psi_{I_2}(I_\omega \cdot 2))$<br>$\psi(M^2 \cdot \omega + \psi_M(M^2 \cdot \omega)+$<br>$\psi_{\psi_M(M^2 \cdot 2)}(M^2 \cdot \omega + \psi_M(M^2 \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,1)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(4,2,1)(5,2,1)<br>(6,2,1)(4,2,1)(5,2,1)(6,2,0)(7,3,1)<br>(8,3,1)(9,3,1)(7,3,1)<br>(8,3,1)(9,3,1)(7,3,0) | $\psi(I_\omega \cdot 2 + \Omega_{I_2+1})$<br>$\psi(M^2 \cdot \omega + \psi_M(M^2 \cdot \omega) + \psi_M(M^2 \cdot 2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,1)(1,1,1)(2,1,1)(3,1,1) | $\psi(I_\omega \cdot 3)$<br>$\psi(M^2 \cdot \omega + \psi_M(M^2 \cdot \omega) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,0,0) | $\psi(I_\omega \cdot \omega)$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot \omega + M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0) | $\psi(I_\omega \cdot \Omega)$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot \omega + M)}(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I_\omega \cdot \psi_I(I))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot \omega + M)}(M \cdot \psi_{\psi_M(M^2)}(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1) | $\psi(I_\omega \cdot \psi_I(I_\omega))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot \omega + M)}(M \cdot \psi_{\psi_M(M^2)}(M^2 \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,1,0)(2,0,0) | $\psi(I_\omega \cdot I)$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot \omega + M)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,1,0)(6,2,1)(7,2,1)(8,2,1) | $\psi(I_\omega \cdot \psi_{\Omega_{I+1}}(I_\omega))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot \omega + M)}$<br>$(M^2 + \psi_{\psi_M(M^2 + M)}(M^2 \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,1,0)(6,2,1)(7,2,1)(8,2,1)<br>(7,1,0)(8,2,1)(9,2,1)(10,2,1) | $\psi(I_\omega \cdot \psi_{\Omega_{I+1}}(I_\omega \cdot \psi_{\Omega_{I+1}}(I_\omega)))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot \omega + M)}(M^2 +$<br>$\psi_{\psi_M(M^2 + M)}(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot \omega + M)}$<br>$(M^2 + \psi_{\psi_M(M^2 + M)}(M^2 \cdot \omega)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,0) | $\psi(I_\omega \cdot \Omega_{I+1})$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot \omega + M)}(M^2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,0)(4,2,1)(5,2,1)(6,2,0)<br>(7,3,1)(8,3,1)(9,3,1) | $\psi(I_\omega \cdot \psi_{I_2}(I_\omega))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot \omega + M)}(M^2 +$<br>$M \cdot \psi_{\psi_M(M^2 \cdot 2)}(M^2 \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,0)(4,2,1)(5,2,1)(6,2,0)<br>(7,3,1)(8,3,1)(9,3,1)(8,2,0)(5,0,0) | $\psi(I_\omega \cdot I_2)$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2 \cdot \omega + M)}(M^2 \cdot 2))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,0)(4,2,1)(5,2,1)(6,2,0)<br>(7,3,1)(8,3,1)(9,3,1)(8,3,0) | $\psi(I_\omega \cdot \Omega_{I_2+1})$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2\cdot\omega+M)}(M^2 \cdot 2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,0)(4,2,1)(5,2,1)(6,2,0)<br>(7,3,1)(8,3,1)(9,3,1)(8,3,0)(7,3,1)<br>(8,3,1)(9,3,0)(10,4,1)<br>(11,4,1)(12,4,1) | $\psi(I_\omega \cdot \psi_{I_3}(I_\omega))$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2\cdot\omega+M)}(M^2 \cdot 2+$<br>$M \cdot \psi_{\psi_M(M^2\cdot 3)}(M^2 \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,1) | $\psi(I_\omega^2)$<br>$\psi(M^2 \cdot \omega + \psi_{\psi_M(M^2\cdot\omega+M)}(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,0)(3,2,0) | $\psi(\Omega_{I_\omega+1})$<br>$\psi(M^2 \cdot \omega + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I_{\omega+1})$<br>$\psi(M^2 \cdot \omega + M^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(I_{\omega+1} \cdot 2)$<br>$\psi(M^2 \cdot \omega + M^2 + \psi_M(M^2 \cdot \omega + M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(I_{\omega+1} \cdot \omega)$<br>$\psi(M^2 \cdot \omega + M^2 + \psi_{\psi_M(M^2\cdot\omega+M^2+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{I_{\omega+1}})$<br>$\psi(M^2 \cdot \omega + M^2 + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(I_{\omega+2})$<br>$\psi(M^2 \cdot \omega + M^2 \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1) | $\psi(I_{\omega\cdot 2})$<br>$\psi(M^2 \cdot \omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(4,2,0)(5,3,0) | $\psi(\Omega_{I_{\omega\cdot 2+1}})$<br>$\psi(M^2 \cdot \omega \cdot 2 + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(I_{\omega\cdot 2+1})$<br>$\psi(M^2 \cdot \omega \cdot 2 + M \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,1) | $\psi(I_{\omega\cdot 3})$<br>$\psi(M^2 \cdot \omega \cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1) | $\psi(I_{\psi(2)})$<br>$\psi(M^2 \cdot \psi(2))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(2,1,1) | $\psi(I_{\psi(3)})$<br>$\psi(M^2 \cdot \psi(3))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0) | $\psi(I_\Omega)$<br>$\psi(M^2 \cdot \psi_M(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,0) | $\psi(I_\Omega + \Omega)$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_M(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,1,0)(2,2,0) | $\psi(I_\Omega + \Omega_2)$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_M(M \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,1,0)(2,2,1) | $\psi(I_\Omega + \Omega_\omega)$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_M(M \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)(3,2,1)(4,1,0) | $\psi(I_\Omega + \Omega_\Omega)$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_M(M \cdot \psi_M(M)))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)(3,2,1)(4,2,0) | $\psi(I_\Omega + \Omega_{\Omega_2})$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_M(M \cdot \psi_M(M \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)<br>(3,2,1)(4,2,0)(3,0,0) | $\psi(I_\Omega + \psi_I(I))$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_{\psi_M(M^2)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)(3,2,1)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,1) | $\psi(I_\Omega + \psi_I(I_\omega))$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_{\psi_M(M^2)}(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)(3,2,1)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,1)(6,3,1)(7,1,0) | $\psi(I_\Omega + \psi_I(I_\Omega))$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_{\psi_M(M^2)}(M^2 \cdot \psi_M(M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)(3,2,1)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,1)(6,3,1)(7,1,0)<br>(3,2,0)(4,3,1)(5,3,1)(6,3,0)(5,0,0) | $\psi(I_\Omega + I)$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)(3,2,1)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,1)(6,3,1)(7,1,0)<br>(3,2,0)(4,3,1)(5,3,1)(6,3,0)(7,4,1)<br>(8,4,1)(9,4,1)(8,4,1)(9,1,0) | $\psi(I_\Omega + \psi_{\Omega_{I+1}}(I_\Omega))$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_{\psi_M(M^2+M)}(M^2 \cdot \psi_M(M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)(3,2,1)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,1)<br>(6,3,1)(7,1,0)(3,2,1) | $\psi(I_\Omega + \psi_{\Omega_{I+1}}(I_\Omega + 1))$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_{\psi_M(M^2+M)}(M^2 \cdot \psi_M(M) + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)(3,2,1)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,1)<br>(6,3,1)(7,1,0)(5,3,0) | $\psi(I_\Omega + \Omega_{I+1})$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_M(M^2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)(3,2,1)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,1)(6,3,1)<br>(7,1,0)(5,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(I_\Omega + \psi_{I_2}(I_2))$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_{\psi_M(M^2 \cdot 2)}(M^2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)(3,2,1)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,1)(6,3,1)(7,1,0)<br>(5,3,1)(6,3,1)(7,3,0)(8,4,1)(9,4,1)<br>(10,4,1)(9,4,1)(10,1,0)(8,4,0) | $\psi(I_\Omega + \Omega_{I_2+1})$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_M(M^2 \cdot 2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)(3,2,1)(4,2,1) | $\psi(I_\Omega + I_\omega)$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_M(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(4,3,1)(5,3,1)(6,3,1)(5,3,1)<br>(6,1,0)(4,3,0) | $\psi(I_\Omega + \Omega_{I_\omega+1})$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_M(M^2 \cdot \omega + M))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(4,3,1)(5,3,1)(6,3,1)(5,3,1)<br>(6,1,0)(4,3,1)(5,3,1)(6,3,0)(5,0,0) | $\psi(I_\Omega + I_{\omega+1})$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_M(M^2 \cdot \omega + M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(4,3,1)(5,3,1)(6,3,1)(5,3,1)<br>(6,1,0)(4,3,1)(5,3,1)(6,3,1) | $\psi(I_\Omega + I_{\omega\cdot 2})$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_M(M^2 \cdot \omega \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)<br>(3,2,1)(4,2,1)(3,2,1) | $\psi(I_\Omega + I_{\psi(2)})$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_M(M^2 \cdot \psi(2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,1,0) | $\psi(I_\Omega \cdot 2)$<br>$\psi(M^2 \cdot \psi_M(M) + \psi_M(M^2 \cdot \psi_M(M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(3,2,0)(4,3,0) | $\psi(\Omega_{I_\Omega+1})$<br>$\psi(M^2 \cdot \psi_M(M) + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(3,2,0)(4,3,1)<br>(5,3,1)(6,3,0)(5,0,0) | $\psi(I_{\Omega+1})$<br>$\psi(M^2 \cdot \psi_M(M) + M^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(3,2,0)<br>(4,3,1)(5,3,1)(6,3,1) | $\psi(I_{\Omega+\omega})$<br>$\psi(M^2 \cdot \psi_M(M) + M^2 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(3,2,0)(4,3,1)(5,3,1)<br>(6,3,1)(5,3,1)(6,1,0) | $\psi(I_{\Omega\cdot 2})$<br>$\psi(M^2 \cdot \psi_M(M) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(3,2,0)(4,3,1)(5,3,1)<br>(6,3,1)(5,3,1)(6,1,0)(5,3,0)(6,4,1)<br>(7,4,1)(8,4,1)(7,4,1)(8,1,0) | $\psi(I_{\Omega\cdot 3})$<br>$\psi(M^2 \cdot \psi_M(M) \cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,1,0)(3,2,1) | $\psi(I_{\Omega\cdot\omega})$<br>$\psi(M^2 \cdot \psi_{\psi_M(M\cdot 2)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(3,2,1)(4,1,0) | $\psi(I_{\Omega^2})$<br>$\psi(M^2 \cdot \psi_{\psi_M(M\cdot 2)}(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(5,2,0) | $\psi(I_{\psi_{\Omega_2}(\Omega_2)})$<br>$\psi(M^2 \cdot \psi_{\psi_M(M\cdot 2)}(M \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,2,0) | $\psi(I_{\Omega_2})$<br>$\psi(M^2 \cdot \psi_M(M \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,0)(2,2,0)(3,3,1)(4,3,1)<br>(5,3,1)(4,3,1)(5,2,0)(3,3,1)(4,3,1)<br>(5,3,1)(4,3,1)(5,2,0) | $\psi(I_{\Omega_2} \cdot 2)$<br>$\psi(M^2 \cdot \psi_M(M \cdot 2) + \psi_M(M^2 \cdot \psi_M(M \cdot 2)))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,0)(2,2,0)(3,3,1)(4,3,1)<br>(5,3,1)(4,3,1)(5,2,0)(4,3,0)(5,4,1)<br>(6,4,1)(7,4,1) | $\psi(I_{\Omega_2+\omega})$<br>$\psi(M^2 \cdot \psi_M(M \cdot 2) + M^2 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,0)(2,2,0)(3,3,1)(4,3,1)<br>(5,3,1)(4,3,1)(5,2,0)(4,3,0)(5,4,1)<br>(6,4,1)(7,4,1)(6,4,1)(7,2,0) | $\psi(I_{\Omega_2 \cdot 2})$<br>$\psi(M^2 \cdot \psi_M(M \cdot 2) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,0)(2,2,0)(3,3,1)(4,3,1)<br>(5,3,1)(4,3,1)(5,2,0)(4,3,1) | $\psi(I_{\Omega_2 \cdot \omega})$<br>$\psi(M^2 \cdot \psi_{\psi_M(M \cdot 3)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,0)(2,2,0)(3,3,1)(4,3,1)<br>(5,3,1)(4,3,1)(5,3,0) | $\psi(I_{\Omega_3})$<br>$\psi(M^2 \cdot \psi_M(M \cdot 3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1) | $\psi(I_{\Omega_\omega})$<br>$\psi(M^2 \cdot \psi_M(M \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,1,0)(3,2,0) | $\psi(I_{\Omega_\omega} + \Omega_{\omega+1})$<br>$\psi(M^2 \cdot \psi_M(M \cdot \omega) + \psi_M(M \cdot \omega + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,1,0)(1,1,1) | $\psi(I_{\Omega_\omega} \cdot 2)$<br>$\psi(M^2 \cdot \psi_M(M \cdot \omega) + \psi_M(M^2 \cdot \psi_M(M \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,1,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,1)(6,3,1)(7,1,0)(1,1,1) | $\psi(I_{\Omega_\omega \cdot 2})$<br>$\psi(M^2 \cdot \psi_M(M \cdot \omega) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,1,0)(4,2,1) | $\psi(I_{\Omega_\omega \cdot \omega})$<br>$\psi(M^2 \cdot \psi_{\psi_M(M \cdot \omega + M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0) | $\psi(I_{\Omega_{\omega+1}})$<br>$\psi(M^2 \cdot \psi_M(M \cdot \omega + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(3,2,0)(4,3,1)<br>(5,3,1)(6,3,1)(5,3,1)(6,3,0) | $\psi(I_{\Omega_{\omega+2}})$<br>$\psi(M^2 \cdot \psi_M(M \cdot \omega + M \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(3,2,1) | $\psi(I_{\Omega_{\omega \cdot 2}})$<br>$\psi(M^2 \cdot \psi_M(M \cdot \omega \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(3,2,1)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,1)(6,3,1)(7,3,0) | $\psi(I_{\Omega_{\omega \cdot 2+1}})$<br>$\psi(M^2 \cdot \psi_M(M \cdot \omega \cdot 2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(3,2,1)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,1)<br>(6,3,1)(7,3,0)(5,3,1) | $\psi(I_{\Omega_{\omega \cdot 3}})$<br>$\psi(M^2 \cdot \psi_M(M \cdot \omega \cdot 3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1) | $\psi(I_{\Omega_{\psi(2)}})$<br>$\psi(M^2 \cdot \psi_M(M \cdot \psi(2)))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I_{\psi_I(I)})$<br>$\psi(M^2 \cdot \psi_{\psi_M(M^2)}(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1) | $\psi(I_{\psi_I(I_\omega)})$<br>$\psi(M^2 \cdot \psi_{\psi_M(M^2)}(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0) | $\psi(I_{\psi_I(I_\Omega)})$<br>$\psi(M^2 \cdot \psi_{\psi_M(M^2)}(M^2 \cdot \psi_M(M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(2,0,0) | $\psi(I_I)$<br>$\psi(M^2 \cdot \psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I_I + I)$<br>$\psi(M^2 \cdot \psi_M(M^2) + \psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(6,3,1)(7,3,1)<br>(8,3,0)(7,0,0) | $\psi(I_I + \psi_{\Omega_{I+1}}(I_2))$<br>$\psi(M^2 \cdot \psi_M(M^2) + \psi_{\psi_M(M^2+M)}(M^2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1) | $\psi(I_I + \psi_{\Omega_{I+1}}(I_\omega))$<br>$\psi(M^2 \cdot \psi_M(M^2) + \psi_{\psi_M(M^2+M)}(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(4,2,1)<br>(5,1,0)(2,0,0) | $\psi(I_I + \psi_{\Omega_{I+1}}(I_I))$<br>$\psi(M^2 \cdot \psi_M(M^2) + \psi_{\psi_M(M^2+M)}(M^2 \cdot \psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(2,1,1) | $\psi(I_I + \psi_{\Omega_{I+1}}(I_I + 1))$<br>$\psi(M^2 \cdot \psi_M(M^2) + \psi_{\psi_M(M^2+M)}(M^2 \cdot \psi_M(M^2) + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(4,2,0) | $\psi(I_I + \Omega_{I+1})$<br>$\psi(M^2 \cdot \psi_M(M^2) + \psi_M(M^2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(2,0,0) | $\psi(I_I \cdot 2)$<br>$\psi(M^2 \cdot \psi_M(M^2) + \psi_M(M^2 \cdot \psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,0)(7,0,0) | $\psi(I_{I+1})$<br>$\psi(M^2 \cdot \psi_M(M^2) + M^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,1)<br>(7,3,1)(8,1,0)(2,0,0) | $\psi(I_{I \cdot 2})$<br>$\psi(M^2 \cdot \psi_M(M^2) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(5,2,1) | $\psi(I_{I \cdot \omega})$<br>$\psi(M^2 \cdot \psi_M(M^2) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(7,2,0) | $\psi(I_{\psi_{\Omega_{I+1}}(\Omega_{I+1})})$<br>$\psi(M^2 \cdot \psi_{\psi_M(M^2+M)}(M^2 + M))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,2,0) | $\psi(I_{\Omega_{I+1}})$<br>$\psi(M^2 \cdot \psi_M(M^2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,2,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,1)(8,3,1)(9,3,1)<br>(8,3,1)(9,2,0)(5,0,0) | $\psi(I_{I_2})$<br>$\psi(M^2 \cdot \psi_M(M^2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,2,0)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,1)(8,3,1)(9,3,1)<br>(8,3,1)(9,3,0) | $\psi(I_{\Omega_{I_2+1}})$<br>$\psi(M^2 \cdot \psi_M(M^2 \cdot 2 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,1) | $\psi(I_{I_\omega})$<br>$\psi(M^2 \cdot \psi_M(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0) | $\psi(I_{I_\Omega})$<br>$\psi(M^2 \cdot \psi_M(M^2 \cdot \psi_M(M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,2,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(2,0,0) | $\psi(I_{I_I})$<br>$\psi(M^2 \cdot \psi_M(M^2 \cdot \psi_M(M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,2,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,2,0) | $\psi(I_{I_{\Omega_{I+1}}})$<br>$\psi(M^2 \cdot \psi_M(M^2 \cdot \psi_M(M^2 + M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,1) | $\psi(I_{I_{I_\omega}})$<br>$\psi(M^2 \cdot \psi_M(M^2 \cdot \psi_M(M^2 \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0) | $\psi(I_{I_{I_\Omega}})$<br>$\psi(M^2 \cdot \psi_M(M^2 \cdot \psi_M(M^2 \cdot \psi_M(M))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0))$<br>$\psi(M^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,0)(4,3,0) | $\psi(I(1,0) + \Omega_{\psi_{I(1,0)}(I(1,0))+2})$<br>$\psi(M^3 + \psi_M(M^2 \cdot \psi_{\psi_M(M^3)}(M^3) + M \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1) | $\psi(I(1,0) + \Omega_{\psi_{I(1,0)}(I(1,0))+\omega})$<br>$\psi(M^3 + \psi_M(M^2 \cdot \psi_{\psi_M(M^3)}(M^3) + M \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0) + I_{\psi_{I(1,0)}(I(1,0))+1})$<br>$\psi(M^3 + \psi_M(M^2 \cdot \psi_{\psi_M(M^3)}(M^3) + M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1) | $\psi(I(1,0) + I_{\psi_{I(1,0)}(I(1,0))+\omega})$<br>$\psi(M^3 + \psi_M(M^2 \cdot \psi_{\psi_M(M^3)}(M^3) + M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,1,0)(2,0,0) | $\psi(I(1,0) + I_{\psi_{I(1,0)}(I(1,0))\cdot 2})$<br>$\psi(M^3 + \psi_{\psi_M(M^3)}(M^3 + \psi_{\psi_M(M^3)}(M^3)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,1,0)(4,2,1)(5,1,0)(2,0,0) | $\psi(I(1,0) + I_{\psi_{I(1,0)}(I(1,0))\cdot 3})$<br>$\psi(M^3 + \psi_{\psi_M(M^3)}(M^3 + \psi_{\psi_M(M^3)}(M^3) \cdot 2))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,1,0)(5,0,0) | $\psi(I(1,0)+I_{\psi_{I(1,0)}(I(1,0))\cdot\omega})$<br>$\psi(M^3+\psi_{\psi_M(M^3)}(M^3+$<br>$\psi_{\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+M)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0) | $\psi(I(1,0)+I_{\Omega_{\psi_{I(1,0)}(I(1,0))}+1})$<br>$\psi(M^3+\psi_{\psi_M(M^3)}(M^3+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0)+I_{\psi_{I_{\psi_{I(1,0)}(I(1,0))+1}}(I(1,0)+I_{\psi_{I(1,0)}(I(1,0))+1})})$<br>$\psi(M^3+\psi_{\psi_M(M^3)}(M^3+\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+$<br>$M\cdot\psi_{\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+M^2)}(M^3+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+M^2)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,0) | $\psi(I(1,0)+I_{\psi_{I_{\psi_{I(1,0)}(I(1,0))+1}}(I(1,0)+\Omega_{I_{\psi_{I(1,0)}(I(1,0))+1}+1})})$<br>$\psi(M^3+\psi_{\psi_M(M^3)}(M^3+\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+$<br>$M\cdot\psi_{\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+M^2)}(M^3+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+M^2+M)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(3,2,1)(4,2,1)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,1) | $\psi(I(1,0)+I_{\psi_{I_{\psi_{I(1,0)}(I(1,0))+1}}(I(1,0)+I_{\psi_{I(1,0)}(I(1,0))+\omega})})$<br>$\psi(M^3+\psi_{\psi_M(M^3)}(M^3+\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+$<br>$M\cdot\psi_{\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+M^2)}(M^3+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+M^2\cdot\omega)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(3,2,1)(4,2,1)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,1)(7,3,1)(8,1,0)<br>(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0)+I_{\psi_{I_{\psi_{I(1,0)}(I(1,0))+1}}(I(1,0)+I_{\psi_{I(1,0)}(I(1,0))\cdot 2})})$<br>$\psi(M^3+\psi_{\psi_M(M^3)}(M^3+\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+$<br>$M\cdot\psi_{\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+M^2)}(M^3+$<br>$\psi_{\psi_M(M^3)}(M^3+\psi_{\psi_M(M^3)}(M^3))))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(3,2,1)(4,2,1)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,1)(7,3,1)(8,2,0) | $\psi(I(1,0)+I_{\psi_{I_{\psi_{I(1,0)}(I(1,0))+1}}(I(1,0)+I_{\Omega_{\psi_{I(1,0)}(I(1,0))+1}})})$<br>$\psi(M^3+\psi_{\psi_M(M^3)}(M^3+\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+$<br>$M\cdot\psi_{\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+M^2)}(M^3+$<br>$\psi_{\psi_M(M^3)}(M^3+\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+M))))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(3,2,1)(4,2,1)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,1)<br>(7,3,1)(8,2,0)(4,0,0) | $\psi(I(1,0)+I_{I_{\psi_{I(1,0)}(I(1,0))+1}})$<br>$\psi(M^3+\psi_{\psi_M(M^3)}(M^3+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+M^2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(3,2,1)(4,2,1)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,1)(7,3,1)(8,3,0) | $\psi(I(1,0)+I_{\Omega_{I_{\psi_{I(1,0)}(I(1,0))+1}}+1})$<br>$\psi(M^3+\psi_{\psi_M(M^3)}(M^3+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+M^2+M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(3,2,1)(4,2,1)(5,2,1) | $\psi(I(1,0)+I_{I_{\psi_{I(1,0)}(I(1,0))+\omega}})$<br>$\psi(M^3+\psi_{\psi_M(M^3)}(M^3+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+M^2\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0) | $\psi(I(1,0)+I_{I_{\Omega_{\psi_{I(1,0)}(I(1,0))}+1}})$<br>$\psi(M^3+\psi_{\psi_M(M^3)}(M^3+$<br>$\psi_{\psi_M(M^3)}(M^3+\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3)+M))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0)\cdot 2)$<br>$\psi(M^3+\psi_M(M^3))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,2,0)(5,3,0) | $\psi(I(1,0)\cdot 2+\Omega_{\psi_{I(1,0)}(I(1,0)\cdot 2)+1})$<br>$\psi(M^3+\psi_M(M^3)+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3+\psi_M(M^3))+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,0)(6,0,0) | $\psi(I(1,0)\cdot 2+I_{\psi_{I(1,0)}(I(1,0)\cdot 2)+1})$<br>$\psi(M^3+\psi_M(M^3)+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3+\psi_M(M^3))+M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,1) | $\psi(I(1,0)\cdot 2+I_{\psi_{I(1,0)}(I(1,0)\cdot 2)+\omega})$<br>$\psi(M^3+\psi_M(M^3)+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3+\psi_M(M^3))+M^2\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,2,0)(5,3,1)(6,3,1)<br>(7,3,1)(6,3,1)(7,3,0) | $\psi(I(1,0)\cdot 2+I_{\Omega_{\psi_{I(1,0)}(I(1,0)\cdot 2)+1}})$<br>$\psi(M^3+\psi_M(M^3)+\psi_{\psi_M(M^3)}(M^3+$<br>$\psi_M(M^3)+\psi_M(M^2\cdot\psi_{\psi_M(M^3)}$<br>$(M^3+\psi_M(M^3))+M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,2,0)(5,3,1)(6,3,1)<br>(7,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(I(1,0)\cdot 3)$<br>$\psi(M^3+\psi_M(M^3)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,2,0)(5,3,1)(6,3,1)<br>(7,3,1)(6,3,1)(7,3,0)(6,3,0)(7,4,1)<br>(8,4,1)(9,4,1)(8,4,1)(9,4,0)(8,0,0) | $\psi(I(1,0)\cdot 4)$<br>$\psi(M^3+\psi_M(M^3)\cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1) | $\psi(I(1,0)\cdot\omega)$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0)\cdot\omega+I(1,0))$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(1)+\psi_M(M^3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,1)(6,3,1)(7,3,0)(6,0,0) | $\psi(I(1,0)\cdot\omega+I(1,0)\cdot 2)$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(1)+\psi_M(M^3)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(I(1,0)\cdot\omega\cdot 2)$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(1)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(2,1,1) | $\psi(I(1,0)\cdot\psi(2))$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0) | $\psi(I(1,0)\cdot\Omega)$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,1) | $\psi(I(1,0)\cdot I_\omega)$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^2\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0)\cdot\psi_{I(1,0)}(I(1,0)))$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^2\cdot\psi_{\psi_M(M^3)}(M^3)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(I(1,0)\cdot\psi_{I(1,0)}(I(1,0)\cdot\omega))$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^2\cdot\psi_{\psi_M(M^3)}$<br>$(M^3+\psi_{\psi_M(M^3+M)}(1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,1)(3,1,0) | $\psi(I(1,0)\cdot\psi_{I(1,0)}(I(1,0)\cdot\Omega))$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^2\cdot\psi_{\psi_M(M^3)}$<br>$(M^3+\psi_{\psi_M(M^3+M)}(M))))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0)^2)$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,0) | $\psi(I(1,0)^2+\Omega_{\psi_{I(1,0)}(I(1,0)^2)+1})$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^3)+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3))+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0)^2+I_{\psi_{I(1,0)}(I(1,0)^2)+1})$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^3)+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3))+M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1) | $\psi(I(1,0)^2+I_{\psi_{I(1,0)}(I(1,0)^2)+\omega})$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^3)+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3))+M^2\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0) | $\psi(I(1,0)^2+I_{\Omega_{\psi_{I(1,0)}(I(1,0)^2)+1}})$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^3)+$<br>$\psi_{\psi_M(M^3)}(M^3+\psi_{\psi_M(M^3+M)}(M^3)+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3))+M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0)^2+I(1,0))$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^3)+\psi_M(M^3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(I(1,0)^2+I(1,0)\cdot\omega)$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^3)+\psi_{\psi_M(M^3+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0) | $\psi(I(1,0)^2+I(1,0)\cdot\Omega_{\psi_{I(1,0)}(I(1,0)^2)+1})$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^3)+$<br>$\psi_{\psi_M(M^3+M)}(M^2\cdot\psi_{\psi_M(M^3)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3))+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)(4,2,1)<br>(5,2,0)(3,2,1)(4,2,1)(5,2,1) | $\psi(I(1,0)^2+I(1,0)\cdot I_{\psi_{I(1,0)}(I(1,0)^2)+\omega})$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^3)+$<br>$\psi_{\psi_M(M^3+M)}(M^2\cdot\psi_{\psi_M(M^3)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3))+M^2\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)(4,2,1)<br>(5,2,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0) | $\psi(I(1,0)^2+I(1,0)\cdot I_{\Omega_{\psi_{I(1,0)}(I(1,0)^2)+1}})$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^3)+$<br>$\psi_{\psi_M(M^3+M)}(M^2\cdot\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3))+M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)(4,2,1)<br>(5,2,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0)^2+I(1,0)\cdot$<br>$\psi_{I(1,0)}(I(1,0)^2+I(1,0)))$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^3)+$<br>$\psi_{\psi_M(M^3+M)}(M^2\cdot\psi_{\psi_M(M^3)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3)+\psi_M(M^3))))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)(4,2,1)<br>(5,2,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,2,1) | $\psi(I(1,0)^2 + I(1,0)\cdot$<br>$\psi_{I(1,0)}(I(1,0)^2 + I(1,0)\cdot\omega))$<br>$\psi(M^3 + \psi_{\psi_M(M^3+M)}(M^3)+$<br>$\psi_{\psi_M(M^3+M)}(M^2 \cdot \psi_{\psi_M(M^3)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3) + \psi_{\psi_M(M^3+M)}(1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)(4,2,1)<br>(5,2,0)(3,2,1)(4,2,1)(5,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0) | $\psi(I(1,0)^2 + I(1,0)\cdot\psi_{I(1,0)}(I(1,0)^2+$<br>$I(1,0)\cdot\Omega_{\psi_{I(1,0)}(I(1,0)^2)+1}))$<br>$\psi(M^3 + \psi_{\psi_M(M^3+M)}(M^3)+$<br>$\psi_{\psi_M(M^3+M)}(M^2\cdot$<br>$\psi_{\psi_M(M^3)}(M^3 + \psi_{\psi_M(M^3+M)}(M^3)+$<br>$\psi_{\psi_M(M^3+M)}(M^2 \cdot \psi_{\psi_M(M^3)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3)) + M))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0)^2 \cdot 2)$<br>$\psi(M^3 + \psi_{\psi_M(M^3+M)}(M^3)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1) | $\psi(I(1,0)^2 \cdot \omega)$<br>$\psi(M^3 + \psi_{\psi_M(M^3+M)}(M^3+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0)^3)$<br>$\psi(M^3 + \psi_{\psi_M(M^3+M)}(M^3 + \psi_M(M^3)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0) | $\psi(I(1,0)^4)$<br>$\psi(M^3 + \psi_{\psi_M(M^3+M)}(M^3 + \psi_M(M^3)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(3,0,0) | $\psi(I(1,0)^\omega)$<br>$\psi(M^3 + \psi_{\psi_M(M^3+M)}(M^3 + \psi_{\psi_M(M^3+M)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(3,1,0) | $\psi(I(1,0)^\Omega)$<br>$\psi(M^3 + \psi_{\psi_M(M^3+M)}(M^3 + \psi_{\psi_M(M^3+M)}(M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0)^{\psi_{I(1,0)}(I(1,0))})$<br>$\psi(M^3 + \psi_{\psi_M(M^3+M)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^2 \cdot \psi_{\psi_M(M^3)}(M^3))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0)^{\psi_{I(1,0)}(I(1,0)^2)})$<br>$\psi(M^3 + \psi_{\psi_M(M^3+M)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^2 \cdot \psi_{\psi_M(M^3)}(M^3 + \psi_M(M^3)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(3,0,0) | $\psi(I(1,0)^{\psi_{I(1,0)}(I(1,0)^\omega)})$<br>$\psi(M^3 + \psi_{\psi_M(M^3+M)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^2 \cdot \psi_{\psi_M(M^3)}$<br>$(M^3 + \psi_{\psi_M(M^3+M)}(1)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(3,1,0) | $\psi(I(1,0)^{\psi_{I(1,0)}(I(1,0)^\Omega)})$<br>$\psi(M^3 + \psi_{\psi_M(M^3+M)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^2 \cdot \psi_{\psi_M(M^3)}$<br>$(M^3 + \psi_{\psi_M(M^3+M)}(M)))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(I(1,0)^{I(1,0)})$<br>$\psi(M^3 + \psi_{\psi_M(M^3+M)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0)^{I(1,0)+1})$<br>$\psi(M^3 + \psi_{\psi_M(M^3+M)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3) + \psi_M(M^3)))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(2,1,1)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(I(1,0)^{I(1,0)\cdot 2})$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(3,0,0) | $\psi(I(1,0)^{I(1,0)\cdot\omega})$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(3,1,0)(2,0,0) | $\psi(I(1,0)^{I(1,0)^2})$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3+\psi_M(M^3))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,0,0) | $\psi(I(1,0)^{I(1,0)^\omega})$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3+\psi_{\psi_M(M^3+M)}(1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(2,0,0) | $\psi(I(1,0)^{I(1,0)^{I(1,0)}})$<br>$\psi(M^3+\psi_{\psi_M(M^3+M)}(M^3+$<br>$\psi_{\psi_M(M^3+M)}(M^3+\psi_{\psi_M(M^3+M)}(M^3))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{I(1,0)+1})$<br>$\psi(M^3+M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0) | $\psi(\Omega_{I(1,0)+1}+I_\Omega)$<br>$\psi(M^3+M+\psi_M(M^2\cdot\psi_M(M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{I(1,0)+1}+\psi_{I(1,0)}(I(1,0)))$<br>$\psi(M^3+M+\psi_{\psi_M(M^3)}(M^3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(\Omega_{I(1,0)+1}+\psi_{I(1,0)}(I(1,0)\cdot\omega))$<br>$\psi(M^3+M+\psi_{\psi_M(M^3)}(M^3+\psi_{\psi_M(M^3+M)}(1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{I(1,0)+1}+\psi_{I(1,0)}(\Omega_{I(1,0)+1}))$<br>$\psi(M^3+M+\psi_{\psi_M(M^3)}(M^3+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{I(1,0)+1}+\psi_{I(1,0)}(\Omega_{I(1,0)+1}))$<br>$\psi(M^3+M+\psi_{\psi_M(M^3)}(M^3+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(2,0,0) | $\psi(\Omega_{I(1,0)+1}+\psi_{\Omega_{\psi_{I(1,0)}(\Omega_{I(1,0)+1})+1}}(1))$<br>$\psi(M^3+M+$<br>$\psi_{\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3+M)+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{I(1,0)+1}+\psi_{\Omega_{\psi_{I(1,0)}(\Omega_{I(1,0)+1})+1}}(\Omega_{I(1,0)+1}))$<br>$\psi(M^3+M+$<br>$\psi_{\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3+M)+M)}(M^3+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,0) | $\psi(\Omega_{I(1,0)+1}+\Omega_{\psi_{I(1,0)}(\Omega_{I(1,0)+1})+1})$<br>$\psi(M^3+M+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3+M)+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{I(1,0)+1}+I_{\psi_{I(1,0)}(\Omega_{I(1,0)+1})+1})$<br>$\psi(M^3+M+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3+M)+M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1) | $\psi(\Omega_{I(1,0)+1}+I_{\psi_{I(1,0)}(\Omega_{I(1,0)+1})+\omega})$<br>$\psi(M^3+M+$<br>$\psi_M(M^2\cdot\psi_{\psi_M(M^3)}(M^3+M)+M^2\cdot\omega))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0) | $\psi(\Omega_{I(1,0)+1} + I_{\Omega_{\psi_{I(1,0)}(\Omega_{I(1,0)+1})+1}})$<br>$\psi(M^3 + M + \psi_{\psi_M(M^3)}(M^3 + M +$<br>$\psi_M(M^2 \cdot \psi_{\psi_M(M^3)}(M^3 + M) + M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{I(1,0)+1} + I(1,0))$<br>$\psi(M^3 + M + \psi_M(M^3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(\Omega_{I(1,0)+1} + I(1,0) \cdot \omega)$<br>$\psi(M^3 + M + \psi_{\psi_M(M^3+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{I(1,0)+1} + \psi_{\Omega_{I(1,0)+1}}(\Omega_{I(1,0)+1}))$<br>$\psi(M^3 + M + \psi_{\psi_M(M^3+M)}(M^3 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(2,1,1) | $\psi(\Omega_{I(1,0)+1} + \psi_{\Omega_{I(1,0)+1}}(\Omega_{I(1,0)+1} + 1))$<br>$\psi(M^3 + M + \psi_{\psi_M(M^3+M)}(M^3 + M + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(4,2,0) | $\psi(\Omega_{I(1,0)+1} \cdot 2)$<br>$\psi(M^3 + M + \psi_M(M^3 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(5,3,0) | $\psi(\Omega_{I(1,0)+2})$<br>$\psi(M^3 + M \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1) | $\psi(\Omega_{I(1,0)+\omega})$<br>$\psi(M^3 + M \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(I_{I(1,0)+1})$<br>$\psi(M^3 + M^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{I_{I(1,0)+1}+1})$<br>$\psi(M^3 + M^2 + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) | $\psi(I_{I(1,0)+\omega})$<br>$\psi(M^3 + M^2 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) | $\psi(I_{I(1,0)+\omega})$<br>$\psi(M^3 + M^2 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,1,0)(2,0,0) | $\psi(I_{I(1,0)\cdot 2})$<br>$\psi(M^3 + M^2 \cdot \psi_M(M^3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0) | $\psi(I_{\Omega_{I(1,0)+1}})$<br>$\psi(M^3 + M^2 \cdot \psi_M(M^3 + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1,1))$<br>$\psi(M^3 \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(5,2,1)(6,2,0)<br>(7,3,1)(8,3,1)(9,3,1)(8,3,1)(9,3,0)<br>(8,0,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,1) + I)$<br>$\psi(M^3 \cdot 2 + \psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,0,0)(1,1,1)(2,1,1)(3,1,1) | $\psi(I(1,1) + I_\omega)$<br>$\psi(M^3 \cdot 2 + \psi_M(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1,1) + \psi_{I(1,0)}(I(1,1)))$<br>$\psi(M^3 \cdot 2 + \psi_{\psi_M(M^3)}(M^3 \cdot 2))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,0,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,1)+I(1,0))$<br>$\psi(M^3\cdot 2+\psi_M(M^3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,0,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(6,3,1)(7,3,1)<br>(8,3,1)(7,3,1)(8,3,0)(7,0,0) | $\psi(I(1,1)+\psi_{\Omega_{I(1,0)+1}}(I(1,1)))$<br>$\psi(M^3\cdot 2+\psi_{\psi_M(M^2\cdot\psi_M(M^3)+M)}(M^3\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(2,1,1) | $\psi(I(1,1)+\psi_{\Omega_{I(1,0)+1}}(I(1,1)+1))$<br>$\psi(M^3\cdot 2+\psi_{\psi_M(M^2\cdot\psi_M(M^3)+M)}(M^3\cdot 2+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(4,2,0) | $\psi(I(1,1)+\Omega_{I(1,0)+1})$<br>$\psi(M^3\cdot 2+\psi_M(M^2\cdot\psi_M(M^3)+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,0,0)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1,1)+\psi_{I(1,1)}(I(1,1)))$<br>$\psi(M^3\cdot 2+\psi_{\psi_M(M^3\cdot 2)}(M^3\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,2,0)(6,3,0) | $\psi(I(1,1)+\Omega_{\psi_{I(1,1)}(I(1,1))+1})$<br>$\psi(M^3\cdot 2+\psi_M(M^2\cdot\psi_{\psi_M(M^3\cdot 2)}(M^3\cdot 2)+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,2,0)(6,3,1)(7,3,1)(8,3,1)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(I(1,1)\cdot 2)$<br>$\psi(M^3\cdot 2+\psi_M(M^3\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,2,1) | $\psi(I(1,1)\cdot\omega)$<br>$\psi(M^3\cdot 2+\psi_{\psi_M(M^3\cdot 2+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{I(1,1)+1})$<br>$\psi(M^3\cdot 2+M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(7,3,1)(8,3,1)(9,3,1)<br>(8,3,1)(9,3,0)(8,0,0) | $\psi(I(1,2))$<br>$\psi(M^3\cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(7,3,1)(8,3,1)(9,3,1)(8,3,1)<br>(9,3,0)(8,0,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,2)+I(1,0))$<br>$\psi(M^3\cdot 3+\psi_M(M^3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(7,3,1)(8,3,1)(9,3,1)(8,3,1)<br>(9,3,0)(8,0,0)(5,2,0)(6,3,1)(7,3,1)<br>(8,3,1)(7,3,1)(8,3,0)(7,0,0) | $\psi(I(1,2)+I(1,1))$<br>$\psi(M^3\cdot 3+\psi_M(M^3\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(7,3,1)(8,3,1)(9,3,1)(8,3,1)<br>(9,3,0)(8,0,0)(8,3,0)(9,4,1)(10,4,1)<br>(11,4,1)(10,4,1)(11,4,0)(10,0,0) | $\psi(I(1,2)\cdot 2)$<br>$\psi(M^3\cdot 3+\psi_M(M^3\cdot 3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(7,3,1)(8,3,1)(9,3,1)(8,3,1)<br>(9,3,0)(10,4,1)(11,4,1)(12,4,1)<br>(11,4,1)(12,4,0)(11,0,0) | $\psi(I(1,3))$<br>$\psi(M^3\cdot 4)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1) | $\psi(I(1,\omega))$<br>$\psi(M^3\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,1) | $\psi(I(1,\omega)+\psi_{I(1,0)}(I(1,\omega)))$<br>$\psi(M^3\cdot\omega+\psi_{\psi_M(M^3)}(M^3\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,\omega)+I(1,0))$<br>$\psi(M^3\cdot\omega+\psi_M(M^3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,1)(4,2,0) | $\psi(I(1,\omega)+\Omega_{I(1,0)+1})$<br>$\psi(M^3\cdot\omega+\psi_M(M^3+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(8,3,1)(9,3,1) | $\psi(I(1,\omega)+\psi_{I(1,1)}(I(1,\omega)))$<br>$\psi(M^3\cdot\omega+\psi_{\psi_M(M^3\cdot 2)}(M^3\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(7,3,1)(8,3,1)(9,3,1)(8,3,1)<br>(9,3,1)(5,2,0)(6,3,1)(7,3,1)<br>(8,3,1)(7,3,1)(8,3,0)(7,0,0) | $\psi(I(1,\omega)+I(1,1))$<br>$\psi(M^3\cdot\omega+\psi_M(M^3\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1) | $\psi(I(1,\omega)\cdot 2)$<br>$\psi(M^3\cdot\omega+\psi_M(M^3\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,0)(3,2,0) | $\psi(\Omega_{I(1,\omega)+1})$<br>$\psi(M^3\cdot\omega+M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,\omega+1))$<br>$\psi(M^3\cdot\omega+M^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(6,3,1)(7,3,1)(8,3,1)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(I(1,\omega+2))$<br>$\psi(M^3\cdot\omega+M^3\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,1) | $\psi(I(1,\omega\cdot 2))$<br>$\psi(M^3\cdot\omega\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1) | $\psi(I(1,\psi(2)))$<br>$\psi(M^3\cdot\psi(2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0) | $\psi(I(1,\Omega))$<br>$\psi(M^3\cdot\psi_M(M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,1,0)(2,0,0) | $\psi(I(1,I))$<br>$\psi(M^3\cdot\psi_M(M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0) | $\psi(I(1,\Omega_{I+1}))$<br>$\psi(M^3\cdot\psi_M(M^2+M))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,1) | $\psi(I(1, I_\omega))$<br>$\psi(M^3 \cdot \psi_M(M^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,1,0)(2,0,0) | $\psi(I(1, I(1, 0)))$<br>$\psi(M^3 \cdot \psi_M(M^3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(I(1, I(1, \omega)))$<br>$\psi(M^3 \cdot \psi_M(M^3 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(2, 0))$<br>$\psi(M^4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(I(2, 0) \cdot \omega)$<br>$\psi(M^4 + \psi_{\psi_M(M^4+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{I(2,0)+1})$<br>$\psi(M^4 + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I_{I(2,0)+1})$<br>$\psi(M^4 + M^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1, I(2, 0) + 1))$<br>$\psi(M^4 + M^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1<br>)(6,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I(2, 1))$<br>$\psi(M^4 \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(I(2, \omega))$<br>$\psi(M^4 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I(3, 0))$<br>$\psi(M^5)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(4, 0))$<br>$\psi(M^6)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0) | $\psi(I(\omega, 0))$<br>$\psi(M^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0) | $\psi(I(\omega, 0) \cdot 2)$<br>$\psi(M^\omega + \psi_M(M^\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,0)(3,2,0) | $\psi(\Omega_{I(\omega,0)+1})$<br>$\psi(M^\omega + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I_{I(\omega,0)+1})$<br>$\psi(M^\omega + M^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1, I(\omega, 0) + 1))$<br>$\psi(M^\omega + M^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(2, I(\omega, 0) + 1))$<br>$\psi(M^\omega + M^4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,0,0) | $\psi(I(\omega, 1))$<br>$\psi(M^\omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,0,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,1)(7,0,0) | $\psi(I(\omega, 2))$<br>$\psi(M^\omega \cdot 3)$ |

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|---|---|
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(2,1,1) | $\psi(I(\omega,\omega))$<br>$\psi(M^{\omega}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(\omega+1,0))$<br>$\psi(M^{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,1,1) | $\psi(I(\omega+1,0)\cdot\omega)$<br>$\psi(M^{\omega+1}+\psi_M(M^{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{I(\omega+1,0)+1})$<br>$\psi(M^{\omega+1}+M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,0,0) | $\psi(I(\omega,I(\omega+1,0)+1))$<br>$\psi(M^{\omega+1}+M^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,0,0)(5,2,1) | $\psi(I(\omega,I(\omega+1,0)+\omega))$<br>$\psi(M^{\omega+1}+M^{\omega}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,0,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(I(\omega+1,1))$<br>$\psi(M^{\omega+1}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,0,0)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(9,0,0)(8,3,1)(9,3,0)(8,0,0) | $\psi(I(\omega+1,2))$<br>$\psi(M^{\omega+1}\cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,1) | $\psi(I(\omega+1,\omega))$<br>$\psi(M^{\omega+1}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(\omega+2,0))$<br>$\psi(M^{\omega+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I(\omega+3,0))$<br>$\psi(M^{\omega+3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,1)(3,0,0) | $\psi(I(\omega\cdot 2,0))$<br>$\psi(M^{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,1)(3,0,0) | $\psi(I(\omega\cdot 3,0))$<br>$\psi(M^{\omega\cdot 3})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(3,0,0) | $\psi(I(\psi(2),0))$<br>$\psi(M^{\psi(2)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0) | $\psi(I(\Omega,0))$<br>$\psi(M^{\psi_M(M)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0) | $\psi(I(I(\Omega,0),0))$<br>$\psi(M^{\psi_M(M^{\psi_M(M)})})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,0))$<br>$\psi(M^M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,0)+\psi_{I(1,0,0)}(I(1,0,0)))$<br>$\psi(M^M+\psi_{\psi_M(M^M)}(M^M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,0) | $\psi(I(1,0,0)+\Omega_{\psi_{I(1,0,0)}(I(1,0,0))+1})$<br>$\psi(M^M+\psi_M(M^{\psi_{\psi_M(M^M)}(M^M)}+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0,0)+I_{\psi_{I(1,0,0)}(I(1,0,0))+1})$<br>$\psi(M^M+\psi_M(M^{\psi_{\psi_M(M^M)}(M^M)}+M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0,0)+I(1,\psi_{I(1,0,0)}(I(1,0,0))+1))$<br>$\psi(M^M+\psi_M(M^{\psi_{\psi_M(M^M)}(M^M)}+M^3))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0,0)+I(2,\psi_{I(1,0,0)}(I(1,0,0))+1))$<br>$\psi(M^M+\psi_M(M^{\psi_{\psi_M(M^M)}(M^M)}+M^4))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,0,0) | $\psi(I(1,0,0)+I(\omega,\psi_{I(1,0,0)}(I(1,0,0))+1))$<br>$\psi(M^M+\psi_M(M^{\psi_{\psi_M(M^M)}(M^M)}+M^\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)}(I(1,0,0)),1))$<br>$\psi(M^M+\psi_M(M^{\psi_{\psi_M(M^M)}(M^M)}\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)}(I(1,0,0)),1)\cdot 2)$<br>$\psi(M^M+\psi_M(M^{\psi_{\psi_M(M^M)}(M^M)}\cdot 2)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,0)(5,3,0) | $\psi(I(1,0,0)+\Omega_{I(\psi_{I(1,0,0)}(I(1,0,0)),1)+1})$<br>$\psi(M^M+\psi_M(M^{\psi_{\psi_M(M^M)}(M^M)}\cdot 2+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,1)(7,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)}(I(1,0,0)),2))$<br>$\psi(M^M+\psi_M(M^{\psi_{\psi_M(M^M)}(M^M)}\cdot 3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,1) | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)}(I(1,0,0)),\omega))$<br>$\psi(M^M+\psi_M(M^{\psi_{\psi_M(M^M)}(M^M)}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,2,0) | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)}(I(1,0,0)),$<br>$\Omega_{\psi_{I(1,0,0)}(I(1,0,0))+1}))$<br>$\psi(M^M+\psi_M(M^{\psi_{\psi_M(M^M)}(M^M)}\cdot$<br>$\psi_M(M^{\psi_{\psi_M(M^M)}(M^M)}+M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)}(I(1,0,0))+1,0))$<br>$\psi(M^M+\psi_{\psi_M(M^M)}(M^M+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,2,1) | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)}(I(1,0,0))+1,\omega))$<br>$\psi(M^M+\psi_M(M^{\psi_{\psi_M(M^M)}(M^M)+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)}(I(1,0,0))+2,0))$<br>$\psi(M^M+\psi_{\psi_M(M^M)}(M^M+2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,2,1)(5,0,0) | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)}(I(1,0,0))+\omega,0))$<br>$\psi(M^M+\psi_{\psi_M(M^M)}(M^M+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(4,2,1)(5,2,1)(5,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)}(I(1,0,0))\cdot 2,0))$<br>$\psi(M^M+\psi_{\psi_M(M^M)}(M^M+\psi_{\psi_M(M^M)}(M^M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,2,0) | $\psi(I(1,0,0)+I(\Omega_{\psi_{I(1,0,0)}(I(1,0,0))+1},0))$<br>$\psi(M^M+\psi_{\psi_M(M^M)}(M^M+$<br>$\psi_M(M^{\psi_{\psi_M(M^M)}(M^M)}+M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,2,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,0) | $\psi(I(1,0,0)+I(I(\Omega_{\psi_{I(1,0,0)}(I(1,0,0))+1},0),0))$<br>$\psi(M^M+\psi_{\psi_M(M^M)}(M^M+\psi_{\psi_M(M^M)}(M^M+$<br>$\psi_M(M^{\psi_{\psi_M(M^M)}(M^M)}+M))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,0,0) | $\psi(I(1,0,0)\cdot 2)$<br>$\psi(M^M+\psi_M(M^M))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(4,2,0)(5,3,0) | $\psi(I(1,0,0)\cdot 2+\Omega_{\psi_{I(1,0,0)}(I(1,0,0)\cdot 2)+1})$<br>$\psi(M^M+\psi_M(M^M)+$<br>$\psi_M(M^{\psi_{\psi_M(M^M)}(M^M+\psi_M(M^M))}+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,2,0)<br>(4,2,0)(5,3,1)(6,3,1)<br>(7,3,1)(7,3,0)(6,0,0) | $\psi(I(1,0,0)\cdot 3)$<br>$\psi(M^M+\psi_M(M^M)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1) | $\psi(I(1,0,0)\cdot\omega)$<br>$\psi(M^M+\psi_{\psi_M(M^M+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{I(1,0,0)+1})$<br>$\psi(M^M+M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I_{I(1,0,0)+1})$<br>$\psi(M^M+M^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1,I(1,0,0)+1))$<br>$\psi(M^M+M^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0) | $\psi(I(\omega,I(1,0,0)+1))$<br>$\psi(M^M+M^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,1,0) | $\psi(I(\Omega,I(1,0,0)+1))$<br>$\psi(M^M+M^{\psi_M(M)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(I(\psi_{I(1,0,0)}(I(1,0,0)),I(1,0,0)+1))$<br>$\psi(M^M+M^{\psi_{\psi_M(M^M)}(M^M)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,0) | $\psi(I(\psi_{I(1,0,0)}(\Omega_{I(1,0,0)+1}),I(1,0,0)+1))$<br>$\psi(M^M+M^{\psi_{\psi_M(M^M)}(M^M+M)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,1,0) | $\psi(I(\psi_{I(1,0,0)}(I(\Omega,I(1,0,0)+1)),I(1,0,0)+1))$<br>$\psi(M^M+M^{\psi_{\psi_M(M^M)}(M^M+M^{\psi_M(M)})})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(2,0,0) | $\psi(I(I(1,0,0),1))$<br>$\psi(M^M+M^{\psi_M(M^M)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(4,2,0) | $\psi(I(I(1,0,0),1)+\Omega_{I(1,0,0)+1})$<br>$\psi(M^M+M^{\psi_M(M^M)}+\psi_M(M^M+M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(4,2,1)(5,2,1)(6,2,1)(6,1,0) | $\psi(I(I(1,0,0),1)+I(\Omega,I(1,0,0)+1))$<br>$\psi(M^M+M^{\psi_M(M^M)}+\psi_M(M^M+M^{\psi_M(M)}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(2,0,0) | $\psi(I(I(1,0,0),1)\cdot 2)$<br>$\psi(M^M+M^{\psi_M(M^M)}+\psi_M(M^M+M^{\psi_M(M^M)}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(5,2,0)(6,3,0) | $\psi(\Omega_{I(I(1,0,0),1)+1})$<br>$\psi(M^M+M^{\psi_M(M^M)}+M)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,1)(8,1,0) | $\psi(I(\Omega, I(I(1,0,0),1)+1))$<br>$\psi(M^M + M^{\psi_M(M^M)} + M^{\psi_M(M)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(5,2,0)(6,3,1)(7,3,1)<br>(8,3,1)(8,1,0)(2,0,0) | $\psi(I(I(1,0,0),2))$<br>$\psi(M^M + M^{\psi_M(M^M)} \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(5,2,0)(6,3,1)(7,3,1)(8,3,1)<br>(8,1,0)(7,3,0)(8,4,1)(9,4,1)<br>(10,4,1)(10,1,0)(2,0,0) | $\psi(I(I(1,0,0),3))$<br>$\psi(M^M + M^{\psi_M(M^M)} \cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(5,2,1) | $\psi(I(I(1,0,0),\omega))$<br>$\psi(M^M + M^{\psi_M(M^M)} \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(5,2,1)(6,1,0)(2,0,0) | $\psi(I(I(1,0,0),I(1,0,0)))$<br>$\psi(M^M + M^{\psi_M(M^M)} \cdot \psi_M(M^M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(5,2,1)(6,2,0) | $\psi(I(I(1,0,0),\Omega_{I(1,0,0)+1}))$<br>$\psi(M^M + M^{\psi_M(M^M)} \cdot \psi_M(M^M + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(I(I(1,0,0)+1,0))$<br>$\psi(M^M + M^{\psi_M(M^M)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(5,2,1)(6,2,0)(5,2,1) | $\psi(I(I(1,0,0)+1,0)\cdot\omega)$<br>$\psi(M^M + M^{\psi_M(M^M)+1} \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(5,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{I(I(1,0,0)+1,0)+1})$<br>$\psi(M^M + M^{\psi_M(M^M)+1} + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(5,2,1)(6,2,1) | $\psi(I(I(1,0,0)+1,\omega))$<br>$\psi(M^M + M^{\psi_M(M^M)+1} \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I(I(1,0,0)+2,0))$<br>$\psi(M^M + M^{\psi_M(M^M)+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(5,2,1)(6,2,1)(6,1,0) | $\psi(I(I(1,0,0)+\Omega,0))$<br>$\psi(M^M + M^{\psi_M(M^M)+\psi_M(M)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,1,0)(5,2,1)(6,2,1)(6,1,0)(2,0,0) | $\psi(I(I(1,0,0)\cdot 2,0))$<br>$\psi(M^M + M^{\psi_M(M^M)\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(6,0,0) | $\psi(I(I(1,0,0)\cdot\omega,0))$<br>$\psi(M^M + M^{\psi_{\psi_M(M^M+M)}(1)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,1,0)(7,2,0) | $\psi(I(\psi_{\Omega_{I(1,0,0)+1}}(\Omega_{I(1,0,0)+1}),0))$<br>$\psi(M^M + M^{\psi_{\psi_M(M^M+M)}(M^M+M)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0) | $\psi(I(\Omega_{I(1,0,0)+1},0))$<br>$\psi(M^M + M^{\psi_M(M^M+M)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(4,2,1)(5,2,1)(6,2,1)(6,2,0) | $\psi(I(I(\Omega_{I(1,0,0)+1},0),0))$<br>$\psi(M^M + M^{\psi_M(M^M+M^{\psi_M(M^M+M)})})$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(5,0,0) | $\psi(I(1,0,1))$<br>$\psi(M^M \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(5,2,1) | $\psi(I(1,0,1) \cdot \omega)$<br>$\psi(M^M \cdot 2 + \psi_{\psi_M(M^M \cdot 2+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{I(1,0,1)+1})$<br>$\psi(M^M \cdot 2 + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(9,1,0)(2,0,0) | $\psi(I(I(1,0,0), I(1,0,1)+1))$<br>$\psi(M^M \cdot 2 + M^{\psi_M(M^M)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(9,2,0)(5,0,0) | $\psi(I(I(1,0,1),1))$<br>$\psi(M^M \cdot 2 + M^{\psi_M(M^M \cdot 2)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(9,3,0) | $\psi(I(\Omega_{I(1,0,1)+1}, 0))$<br>$\psi(M^M \cdot 2 + M^{\psi_M(M^M \cdot 2+M)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(9,3,0)(8,0,0) | $\psi(I(1,0,2))$<br>$\psi(M^M \cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(9,3,0)(8,3,1)(9,3,0)(10,4,1)<br>(11,4,1)(12,4,1)(12,4,0)(11,0,0) | $\psi(I(1,0,3))$<br>$\psi(M^M \cdot 4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1) | $\psi(I(1,0,\omega))$<br>$\psi(M^M \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,1,0))$<br>$\psi(M^{M+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{I(1,1,0)+1})$<br>$\psi(M^{M+1} + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,1,0)(2,0,0) | $\psi(I(I(1,1,0),1))$<br>$\psi(M^{M+1} + M^{\psi_M(M^{M+1})})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0) | $\psi(I(\Omega_{I(1,1,0)+1}, 0))$<br>$\psi(M^{M+1} + M^{\psi_M(M^{M+1}+M)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(5,0,0) | $\psi(I(1,0,I(1,1,0)+1))$<br>$\psi(M^{M+1} + M^M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0) | $\psi(I(1,0,\Omega_{I(1,1,0)+1}))$<br>$\psi(M^{M+1} + M^M \cdot \psi_M(M^{M+1} + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1,1,1))$<br>$\psi(M^{M+1} \cdot 2)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(7,3,1)(8,3,1)(9,3,1)<br>(9,1,0)(2,0,0) | $\psi(I(I(1,1,0),I(1,1,1)+1))$<br>$\psi(M^{M+1}\cdot 2+M^{\psi_M(M^{M+1})})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(9,2,0)(5,0,0) | $\psi(I(I(1,1,1),1))$<br>$\psi(M^{M+1}\cdot 2+M^{\psi_M(M^{M+1}\cdot 2)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(7,3,1)(8,3,1)(9,3,1)<br>(9,3,0)(8,0,0) | $\psi(I(1,0,I(1,1,1)+1))$<br>$\psi(M^{M+1}\cdot 2+M^{M})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(7,3,1)(8,3,1)(9,3,1)<br>(9,3,0)(8,3,1)(9,3,1)<br>(8,3,1)(9,3,0)(8,0,0) | $\psi(I(1,1,2))$<br>$\psi(M^{M+1}\cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(I(1,1,\omega))$<br>$\psi(M^{M+1}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,2,0))$<br>$\psi(M^{M+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(5,0,0) | $\psi(I(1,0,I(1,2,0)+1))$<br>$\psi(M^{M+2}+M^{M})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,0)<br>(5,2,1)(6,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1,1,I(1,2,0)+1))$<br>$\psi(M^{M+2}+M^{M+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,0)<br>(5,2,1)(6,2,1)(5,2,1)(6,2,1)(5,2,1)<br>(6,2,0)(5,0,0) | $\psi(I(1,2,1))$<br>$\psi(M^{M+2}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1) | $\psi(I(1,2,\omega))$<br>$\psi(M^{M+2}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,3,0))$<br>$\psi(M^{M+3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,4,0))$<br>$\psi(M^{M+4})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0) | $\psi(I(1,\omega,0))$<br>$\psi(M^{M+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(2,0,0))$<br>$\psi(M^{M\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1) | $\psi(I(2,0,0)\cdot\omega)$<br>$\psi(M^{M\cdot 2}+\psi_{\psi_M(M^{M\cdot 2}+M)}(1))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{I(2,0,0)+1})$<br>$\psi(M^{M\cdot 2}+M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1,0,I(2,0,0)+1))$<br>$\psi(M^{M\cdot 2}+M^{M})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1,1,I(2,0,0)+1))$<br>$\psi(M^{M\cdot 2}+M^{M+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1,2,I(2,0,0)+1))$<br>$\psi(M^{M\cdot 2}+M^{M+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,1)<br>(6,2,1)(6,2,0) | $\psi(I(1,\Omega_{I(2,0,0)+1},0))$<br>$\psi(M^{M\cdot 2}+M^{M+\psi_M(M^{M\cdot 2}+M)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,1)<br>(6,2,1)(6,2,0)(5,0,0) | $\psi(I(2,0,1))$<br>$\psi(M^{M\cdot 2}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,1)(6,2,1)<br>(6,2,0)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(9,3,0)(8,3,1)<br>(9,3,1)(9,3,0)(8,0,0) | $\psi(I(2,0,2))$<br>$\psi(M^{M\cdot 2}\cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,1) | $\psi(I(2,0,\omega))$<br>$\psi(M^{M\cdot 2}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(2,1,0))$<br>$\psi(M^{M\cdot 2+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(2,2,0))$<br>$\psi(M^{M\cdot 2+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0) | $\psi(I(2,\omega,0))$<br>$\psi(M^{M\cdot 2+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(I(3,0,0))$<br>$\psi(M^{M\cdot 3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(4,0,0))$<br>$\psi(M^{M\cdot 4})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,0,0) | $\psi(I(1,0,0,0))$<br>$\psi(M^{M^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{I(1,0,0,0)+1})$<br>$\psi(M^{M^2}+M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(5,0,0) | $\psi(I(1,0,I(1,0,0,0)+1))$<br>$\psi(M^{M^2}+M^{M})$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(I(1,1,I(1,0,0,0)+1))$<br>$\psi(M^{M^2}+M^{M+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(5,2,1)<br>(6,2,1)(6,2,0)(5,0,0) | $\psi(I(2,0,I(1,0,0,0)+1))$<br>$\psi(M^{M^2}+M^{M\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(5,2,1)(6,2,1)(6,2,0)<br>(5,2,1)(6,2,1)(6,2,0)(5,0,0) | $\psi(I(3,0,I(1,0,0,0)+1))$<br>$\psi(M^{M^2}+M^{M\cdot 3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(6,2,0)(5,0,0) | $\psi(I(1,0,0,1))$<br>$\psi(M^{M^2}\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1) | $\psi(I(1,0,0,\omega))$<br>$\psi(M^{M^2}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,1,0))$<br>$\psi(M^{M^2+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(I(1,0,1,\omega))$<br>$\psi(M^{M^2+1}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,0,2,0))$<br>$\psi(M^{M^2+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(1,1,0,0))$<br>$\psi(M^{M^2+M})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(I(1,1,1,0))$<br>$\psi(M^{M^2+M+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(I(1,2,0,0))$<br>$\psi(M^{M^2+M\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(3,0,0) | $\psi(I(1,\omega,0,0))$<br>$\psi(M^{M^2+M\cdot\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(I(2,0,0,0))$<br>$\psi(M^{M^2\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(3,1,0)(2,0,0) | $\psi(I(3,0,0,0))$<br>$\psi(M^{M^2\cdot 3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(3,0,0) | $\psi(I(\omega,0,0,0))$<br>$\psi(M^{M^2\cdot\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(I(1,0,0,0,0))$<br>$\psi(M^{M^3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0) | $\psi(I(1@\omega))$<br>$\psi(M^{M^\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,1) | $\psi(M^{M^\omega}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M^{M^\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,1)(3,0,0) | $\psi(M^{M^\omega+\omega})$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(M^{M^{\omega}+M})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1) | $\psi(M^{M^{\omega}+M}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(M^{M^{\omega}+M+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(M^{M^{\omega}+M\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,1)(3,1,0)(3,0,0) | $\psi(M^{M^{\omega}+M\cdot\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(M^{M^{\omega}+M^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(3,1,0)(2,0,0) | $\psi(M^{M^{\omega}+M^3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,1)(3,1,0)(4,0,0) | $\psi(M^{M^{\omega}\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(3,0,0) | $\psi(M^{M^{\omega}\cdot\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(3,1,0)(2,0,0) | $\psi(M^{M^{\omega+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(3,1,0)(3,1,0)(2,0,0) | $\psi(M^{M^{\omega+2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(3,1,0)(4,0,0) | $\psi(M^{M^{\omega\cdot 2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(4,0,0) | $\psi(M^{M^{\psi(2)}})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,1,0) | $\psi(M^{M^{\psi_M(M)}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,1,1)(3,1,1) | $\psi(M^{M^M}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(M^{M^M+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(M^{M^M+M})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(M^{M^M+M^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,0,0) | $\psi(M^{M^M\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(3,1,0)(2,0,0) | $\psi(M^{M^{M+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(3,1,0)(4,1,0)(2,0,0) | $\psi(M^{M^{M\cdot 2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(4,1,0)(2,0,0) | $\psi(M^{M^{M^2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(4,1,0)(4,1,0)(2,0,0) | $\psi(M^{M^{M^3}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(5,0,0) | $\psi(M^{M^{M^{\omega}}})$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(5,1,0)(2,0,0) | $\psi(M^{M^{M^M}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(5,1,0)(6,1,0)(2,0,0) | $\psi(M^{M^{M^{M^M}}})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\varepsilon_{M+1})$<br>$\psi(\Omega_{M+1})$<br>$\psi(M_2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M+1}+\psi_M(\Omega_{M+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,0) | $\psi(\Omega_{M+1}+M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1) | $\psi(\Omega_{M+1}+M\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{M+1}+M^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1) | $\psi(\Omega_{M+1}+M^2\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{M+1}+M^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0) | $\psi(\Omega_{M+1}+M^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M+1}+M^{\psi_M(\Omega_{M+1})})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,1,0) | $\psi(\Omega_{M+1}+M^{\psi_M(\Omega_{M+1})}+$<br>$\psi_{\psi_M(\Omega_{M+1}+M)}(\Omega_{M+1}+M^\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(2,0,0) | $\psi(\Omega_{M+1}+M^{\psi_M(\Omega_{M+1})}+$<br>$\psi_{\psi_M(\Omega_{M+1}+M)}(\Omega_{M+1}+M^{\psi_M(\Omega_{M+1})}+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(2,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M+1}+M^{\psi_M(\Omega_{M+1})}+$<br>$\psi_{\psi_M(\Omega_{M+1}+M)}(\Omega_{M+1}+M^{\psi_M(\Omega_{M+1})}+\psi_M(\Omega_{M+1})))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(2,1,0)(3,2,0) | $\psi(\Omega_{M+1}+M^{\psi_M(\Omega_{M+1})}+\psi_{\psi_M(\Omega_{M+1}+M)}(\Omega_{M+1}+$<br>$M^{\psi_M(\Omega_{M+1})}+\psi_{\psi_M(\Omega_{M+1}+M)}(\Omega_{M+1}+M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,1,0) | $\psi(\Omega_{M+1}+M^{\psi_M(\Omega_{M+1})}+$<br>$\psi_{\psi_M(\Omega_{M+1}+M)}(\Omega_{M+1}+M^{\psi_M(\Omega_{M+1})}$<br>$+\psi_{\psi_M(\Omega_{M+1}+M)}(\Omega_{M+1}+M^\Omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(3,0,0) | $\psi(\Omega_{M+1}+M^{\psi_M(\Omega_{M+1})}+\psi_{\psi_M(\Omega_{M+1}+M)}(\Omega_{M+1}+$<br>$M^{\psi_M(\Omega_{M+1})}+\psi_{\psi_M(\Omega_{M+1}+M)}(\Omega_{M+1}$<br>$+M^{\psi_M(\Omega_{M+1})}+1)))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(3,2,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})} + \psi_M(\Omega_{M+1} + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(3,2,1) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})} + \psi_M(\Omega_{M+1} + M \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})} + \psi_M(\Omega_{M+1} + M^{\Omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,1,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})} +$<br>$\psi_M(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})}) + \psi_M(\Omega_{M+1} + M^{\Omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,0,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})} +$<br>$\psi_{\psi_M(\Omega_{M+1}+M^{\psi_M(\Omega_{M+1})}+M)}(1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})} +$<br>$\psi_{\psi_M(\Omega_{M+1}+M^{\psi_M(\Omega_{M+1})}+M)}(\Omega_{M+1} + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})} +$<br>$\psi_{\psi_M(\Omega_{M+1}+M^{\psi_M(\Omega_{M+1})}+M)}(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,0)(5,3,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})} + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,0)(5,3,1) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})} + M \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,0)(5,3,1)(6,3,1)(7,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})} \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,1) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})} \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})} \cdot \psi_M(\Omega_{M+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,1)(5,2,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})} \cdot \psi_M(\Omega_{M+1} + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,2,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})} \cdot \psi_M(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,2,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,1)(5,2,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})} \cdot \psi_M(\Omega_{M+1} +$<br>$M^{\psi_M(\Omega_{M+1})} \cdot \psi_M(\Omega_{M+1} + M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})+1})$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,2,0)(4,2,0)(5,3,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})+1} + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,1)(5,2,1) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})+1} \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,2,1)(5,0,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,2,1)(5,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1})\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0) | $\psi(\Omega_{M+1} + M^{\psi_M(\Omega_{M+1}+M)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{M+1} + M^M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{M+1} + M^M + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(6,3,1)(7,3,1)<br>(8,3,1)(8,3,0)(7,0,0) | $\psi(\Omega_{M+1} + M^M \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,2,1)(5,2,1) | $\psi(\Omega_{M+1} + M^M \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{M+1} + M^{M+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(4,2,1)(5,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{M+1} + M^{M\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(5,2,0)(4,0,0) | $\psi(\Omega_{M+1} + M^{M^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,2,0)(4,0,0) | $\psi(\Omega_{M+1} + M^{M^M})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(6,3,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1}) + \psi_M(\Omega_{M+1}))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(6,3,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1}) +$<br>$\psi_{\psi_M(\Omega_{M+1}+M)}(\Omega_{M+1} + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(6,3,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1}) +$<br>$\psi_{\psi_M(\Omega_{M+1}+M)}(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1})))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,0)(2,0,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1}) +$<br>$\psi_{\psi_M(\Omega_{M+1}+M)}(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1}) + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,0)(3,2,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1}) + \psi_M(\Omega_{M+1} + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(6,3,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1}) +$<br>$\psi_M(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1})))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(6,3,0)(4,2,0)(5,3,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1}) + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(6,3,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,0)(7,0,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1}) + M^M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(6,3,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,0)(8,4,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1}) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + M) +$<br>$\psi_M(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + \psi_M(M))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + M) +$<br>$\psi_{\psi_M(\Omega_{M+1}+\psi_{\Omega_{M+1}}(\Omega_{M+1}+M))}(\Omega_{M+1} +$<br>$\psi_{\Omega_{M+1}}(\Omega_{M+1} + M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(6,3,0)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + M) +$<br>$\psi_M(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + M)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + M) + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + M + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + M \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + M \cdot 2 + 1))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)(3,0,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + M \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + M^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + M^2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + M^3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,0,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + M^M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1})))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(3,0,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(3,1,0)(4,2,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} +$<br>$\psi_{\Omega_{M+1}}(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1}))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(4,0,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} +$<br>$\psi_{\Omega_{M+1}}(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + 1))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(4,1,0)(5,2,0) | $\psi(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1} +$<br>$\psi_{\Omega_{M+1}}(\Omega_{M+1} + \psi_{\Omega_{M+1}}(\Omega_{M+1})))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(4,2,0) | $\psi(\Omega_{M+1} \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(4,2,0)(2,1,1) | $\psi(\Omega_{M+1} \cdot 2 + \psi_{\Omega_{M+1}}(\Omega_{M+1} \cdot 2 + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(4,2,0)(4,2,0) | $\psi(\Omega_{M+1} \cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,3,0) | $\psi(\Omega_{M+2})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,1) | $\psi(\Omega_{M+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{M+\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1) | $\psi(\Omega_{M+\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(2,0,0) | $\psi(\Omega_{M\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0) | $\psi(\Omega_{\Omega_{M+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(I_{M+1})$<br>$\psi({M_2}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0)(2,1,1) | $\psi({M_2}^2 + \psi_{\Omega_{M+1}}({M_2}^2 + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0)(4,2,0) | $\psi({M_2}^2 + \Omega_{M+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi({M_2}^2 + \psi_{\psi_{M_2}({M_2}^2)}({M_2}^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,0)(7,0,0) | $\psi({M_2}^2 + \psi_{M_2}({M_2}^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,0) | $\psi({M_2}^2 + M_2)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,1)<br>(8,3,1)(9,3,0)(8,0,0) | $\psi({M_2}^2 \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) | $\psi({M_2}^2 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi({M_2}^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,0,0) | $\psi({M_2}^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,1,0)(2,0,0) | $\psi({M_2}^M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)(5,0,0) | $\psi({M_2}^{M_2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)<br>(5,2,1)(6,2,0)(7,3,0) | $\psi({M_2}^{M_2} + M_2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(5,2,1)(6,2,1) | $\psi({M_2}^{M_2} \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi({M_2}^{M_2+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)(5,2,1)<br>(6,2,1)(6,2,0)(5,0,0) | $\psi({M_2}^{M_2 \cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(6,2,0)(5,0,0) | $\psi({M_2}^{{M_2}^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{M_2+1})$<br>$\psi(M_3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)<br>(7,3,1)(8,3,1)(9,3,0)(8,0,0) | $\psi({M_3}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)(7,3,1)<br>(8,3,1)(9,3,1)(9,3,0)(8,0,0) | $\psi({M_3}^{M_3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)(7,3,1)<br>(8,3,1)(9,3,1)(9,3,0)(10,4,0) | $\psi(M_4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(M_\omega)$<br>$\psi(N \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(M_\omega + \psi_M(M_2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(7,3,0) | $\psi(M_\omega + \psi_M(M_3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1) | $\psi(M_\omega + \psi_M(M_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,0)(3,2,0) | $\psi(M_\omega + M)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(6,3,0) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(6,3,1)(7,3,1)<br>(8,3,1)(8,3,0)(9,4,0) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega) + \psi_M(M_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(2,1,0)(3,2,0) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega) + \psi_{\psi_M(M_\omega+M)}(M_\omega + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega)+$<br>$\psi_{\psi_M(M_\omega+M)}(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1)(3,2,0) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega) + \psi_M(M_\omega + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,0) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega)+$<br>$\psi_M(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,1)(8,3,1) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega)+$<br>$\psi_M(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,1)(7,3,1)<br>(8,3,1)(8,3,1)(4,2,0)(5,3,0) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega) + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,1) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega + 1))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega + M))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,0) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega + M) + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)<br>(2,1,1)(3,1,0)(4,2,0) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega + M) + M)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,1)(3,1,1) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega + M + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega + M \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1) | $\psi(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega + \psi_{\psi_{M_2}(M_2)}(M_\omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(4,2,0) | $\psi(M_\omega + \psi_{M_2}(M_2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(7,3,1)<br>(8,3,1)(9,3,1)(9,3,1) | $\psi(M_\omega + \psi_{M_2}(M_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(7,3,1)(8,3,1)(9,3,1)<br>(9,3,1)(5,2,0)(6,3,0) | $\psi(M_\omega + M_2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(9,3,1)(7,3,0) | $\psi(M_\omega + \psi_{M_3}(M_3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(M_\omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,0)(3,2,0) | $\psi(\Omega_{M_\omega+1})$<br>$\psi(M_{\omega+1})$<br>$\psi(N \cdot \omega + N)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi({M_{\omega+1}}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,0) | $\psi(M_{\omega+2})$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,1)(8,3,0)(9,4,0) | $\psi(M_{\omega+3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,2,1) | $\psi(M_{\omega\cdot 2})$<br>$\psi(N\cdot\omega\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,2,1)<br>(4,2,0)(5,3,1)(6,3,1)(7,3,1)(7,3,1) | $\psi(M_{\omega\cdot 3})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(2,1,1) | $\psi(M_{\psi(2)})$<br>$\psi(N\cdot\psi(2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0) | $\psi(M_{\psi_M(M)})$<br>$\psi(N\cdot\Omega)$<br>$\psi(N\cdot\psi_{\psi_N(N)}(N))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1) | $\psi(M_{\psi_M(M_\omega)})$<br>$\psi(N\cdot\psi_M(M_\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(5,2,1)(6,1,0) | $\psi(M_{\psi_M(M_{\psi_M(M)})})$<br>$\psi(N\cdot\psi_M(M_\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,1,0)(2,0,0) | $\psi(M_M)$<br>$\psi(N\cdot\psi_N(N))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,0) | $\psi(M_{\Omega_{M+1}})$<br>$\psi(N\cdot\Omega_{M+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,1) | $\psi(M_{M_\omega})$<br>$\psi(N\cdot\psi_N(N\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(2,1,1)(3,1,0) | $\psi(M_{M_{\psi_M(M)}})$<br>$\psi(N\cdot\psi_N(N\cdot\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1,0))$<br>$\psi(M(1;0))$<br>$\psi(N^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1,0)+\psi_{M(1,0)}(M(1,0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,0) | $\psi(M(1,0)+\Omega_{\psi_{M(1,0)}(M(1,0))+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1) | $\psi(M(1,0)+M_{\psi_{M(1,0)}(M(1,0))+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(M(1,0)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M(1,0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1) | $\psi(M_{M(1,0)+1})$<br>$\psi(\psi_{M(1,1)}(1))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,0) | $\psi(M_{\Omega_{M(1,0)+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(M(1,1))$<br>$\psi(M(1;0)\cdot 2)$<br>$\psi(N^2+\psi_N(N^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{M(1,1)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(7,3,1)(8,3,1)<br>(9,3,1)(9,3,1)(8,3,1)(9,3,0)(8,0,0) | $\psi(M(1,2))$<br>$\psi(M(1;0)\cdot 3)$<br>$\psi(N^2+\psi_N(N^2)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1) | $\psi(M(1,\omega))$<br>$\psi(M(1;0)\cdot\omega)$<br>$\psi(N^2+\psi_N(N^2)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0) | $\psi(M(1,\Omega))$<br>$\psi(M(1;0)\cdot\Omega)$<br>$\psi(N^2+\psi_N(N^2)\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0) | $\psi(M(1,M(1,\Omega)))$<br>$\psi(M(1;0)\cdot\psi_{M(1;0)}(M(1;0)\cdot\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(2,0))$<br>$\psi(M(1;0)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(2,0)+\psi_{M(2,0)}(M(2,0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,0) | $\psi(M(2,0)+\Omega_{\psi_{M(2,0)}(M(2,0))+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,1) | $\psi(M(2,0)+M_{\psi_{M(2,0)}(M(2,0))+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(M(2,0)+M(1,\psi_{M(2,0)}(M(2,0))+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)<br>(5,2,1)(4,2,1)(5,2,1) | $\psi(M(2,0)+M(1,\psi_{M(2,0)}(M(2,0))+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0) | $\psi(M(2,0)+M(1,\Omega_{\psi_{M(2,0)}(M(2,0))+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(M(2,0)\cdot 2)$<br>$\psi(M(1;0)^2+\psi_{M(1;0)}(M(1;0)^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(M(2,0)\cdot 2)$<br>$\psi(M(1;0)^2+\psi_{M(1;0)}(M(1;0)^2))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(M(2,0)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M(2,0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1) | $\psi(M_{M(2,0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(M(1,M(2,0)+1))$<br>$\psi(M(1;0)^2+M(1;0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(5,2,1)(6,2,1) | $\psi(M(1,M(2,0)+\omega))$<br>$\psi(M(1;0)^2+M(1;0)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(M(2,1))$<br>$\psi(M(1;0)^2\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{M(2,1)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(7,3,1)(8,3,1)(9,3,1)<br>(9,3,1)(8,3,1)(9,3,1)<br>(8,3,1)(9,3,0)(8,0,0) | $\psi(M(2,2))$<br>$\psi(M(1;0)^2\cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(M(2,\omega))$<br>$\psi(M(1;0)^2\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(M(3,0))$<br>$\psi(M(1;0)^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(4,0))$<br>$\psi(M(1;0)^4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,0,0) | $\psi(M(\omega,0))$<br>$\psi(M(1;0)^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(M(1,0,0))$<br>$\psi(M(1;0)^{M(1;0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1) | $\psi(M(1,0,\omega))$<br>$\psi(M(1;0)^{M(1;0)}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1,1,0))$<br>$\psi(M(1;0)^{M(1;0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(M(2,0,0))$<br>$\psi(M(1;0)^{M(1;0)\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(3,1,0)(2,0,0) | $\psi(M(1,0,0,0))$<br>$\psi(M(1;0)^{M(1;0)^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M(1;0)+1})$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1) | $\psi(M_{M(1;0)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(M(1, M(1;0)+1))$<br>$\psi(M(1;1))$<br>$\psi(N^2 \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(M(2, M(1;0)+1))$<br>$\psi(M(1;1)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(5,0,0) | $\psi(M(1, 0, M(1;0)+1))$<br>$\psi(M(1;1)^{M(1;1)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{M(1;1)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,1)(6,2,0)<br>(7,3,1)(8,3,1)(9,3,1)(9,3,1)<br>(8,3,1)(9,3,1)(9,3,0)(8,0,0) | $\psi(M(1;2))$<br>$\psi(N^2 \cdot 3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(1;\omega))$<br>$\psi(N^2 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1;1,0))$<br>$\psi(M(2;0))$<br>$\psi(N^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(M(1;1,0)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M(1;1,0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(6,0,0) | $\psi(M(1, M(1;1,0)+1))$<br>$\psi(M(1; M(1;1,0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(6,0,0) | $\psi(M(1, 0, M(1;1,0)+1))$<br>$\psi(M(1; M(1;1,0)+1)^{M(1;M(1;1,0)+1)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{M(1;M(1;1,0)+1)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(7,3,1)(8,3,1)(9,3,1)<br>(9,3,1)(8,3,1)(9,3,0)(9,0,0) | $\psi(M(1, M(1; M(1;1,0)+1)+1))$<br>$\psi(M(1; M(1;1,0)+2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,1)(6,2,1) | $\psi(M(1; M(1;1,0)+\omega))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,0) | $\psi(M(1;\Omega_{M(1;1,0)+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(M(1;1,1))$<br>$\psi(M(2;0)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(M(1;1,\omega))$<br>$\psi(M(2;0)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1;2,0))$<br>$\psi(M(2;0)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(M(1;1,0,0))$<br>$\psi(M(2;0)^{M(2;0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M(2;0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(M(1,M(2;0)+1))$<br>$\psi(M(1;M(2;0)+1))$<br>$\psi(N^3+N^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{M(1;M(2;0)+1)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,1)(6,2,1) | $\psi(M(1;M(2;0)+\omega))$<br>$\psi(N^3+N^2\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(M(1;1,M(2;0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(7,3,0) | $\psi(M(2;1))$<br>$\psi(N^3\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1) | $\psi(M(2;\omega))$<br>$\psi(N^3\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(2;1,0))$<br>$\psi(M(3;0))$<br>$\psi(N^4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(M(1,M(2;1,0)+1))$<br>$\psi(M(1;M(2;1,0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{M(1;M(2;1,0)+1)+1})$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,1)(6,2,1) | $\psi(M(1;M(2;1,0)+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(M(1;1,M(2;1,0)+1))$<br>$\psi(M(2;M(2;1,0)+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{M(2;M(2;1,0)+1)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,1)(6,2,1) | $\psi(M(2;M(2;1,0)+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(M(2;1,1))$<br>$\psi(M(3;0)\cdot 2)$<br>$\psi(N^3+\psi_N(N^4))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(M(2;1,\omega))$<br>$\psi(M(3;0)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M(3;0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(3;\omega))$<br>$\psi(N^4\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,0,0) | $\psi(M(\omega;0))$<br>$\psi(N^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,0,0)(2,1,1) | $\psi(M(\omega;\omega))$<br>$\psi(N^\omega\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(\omega;1,0))$<br>$\psi(M(\omega+1;0))$<br>$\psi(N^{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,0,0)(2,1,1)(3,1,1) | $\psi(M(\omega;1,\omega))$<br>$\psi(M(\omega+1;0)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M(\omega+1;0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(\omega+1;\omega))$<br>$\psi(N^{\omega+1}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,1)(3,1,1)(3,0,0) | $\psi(M(\omega\cdot 2;0))$<br>$\psi(N^{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,1,0) | $\psi(M(\Omega;0))$<br>$\psi(N^\Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(M(1,0;0))$<br>$\psi(M(1,1;0))$<br>$\psi(N^N)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,1,1) | $\psi(M(1,0;0)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M(1,0;0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1) | $\psi(M(1,0;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1,0;1,0))$<br>$\psi(M(1,1;0)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M(1,1;0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1) | $\psi(M_{M(1,1;0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(6,0,0) | $\psi(M(\omega;M(1,1;0)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(6,2,0)(5,0,0) | $\psi(M(1,0;M(1,1;0)+1))$<br>$\psi(M(1,1;1))$<br>$\psi(N^N\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)<br>(6,2,0)(5,2,1)(6,2,1) | $\psi(M(1,0;M(1,1;0)+\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(6,2,0)(5,2,1)<br>(6,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{M(1,1;1)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(1,1;\omega))$<br>$\psi(N^N\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1,1;1,0))$<br>$\psi(M(1,2;0))$<br>$\psi(N^{N+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M(1,2;0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1) | $\psi(M(1,2;\omega))$<br>$\psi(N^{N+1}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(1,3;\omega))$<br>$\psi(N^{N+2}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1)(3,1,0) | $\psi(M(1,\Omega;0))$<br>$\psi(N^{N+\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(M(2,0;0))$<br>$\psi(M(2,1;0))$<br>$\psi(N^{N\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1) | $\psi(M(2,0;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(2,1;\omega))$<br>$\psi(N^{N\cdot 2}\cdot\omega)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1) | $\psi(M(2,2;\omega))$<br>$\psi(N^{N\cdot 2+1}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(M(3,0;0))$<br>$\psi(M(3,1;0))$<br>$\psi(N^{N\cdot 3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(3,0,0) | $\psi(M(\omega,0;0))$<br>$\psi(N^{N\cdot\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(3,0,0)(2,1,1) | $\psi(M(\omega,0;\omega))$<br>$\psi(N^{N\cdot\omega}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,1)(3,1,1) | $\psi(M(\omega,0;1,\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(\omega,1;\omega))$<br>$\psi(N^{N\cdot\omega+1}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(M(\omega+1,1;0))$<br>$\psi(N^{N\cdot\omega+N+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(3,0,0) | $\psi(M(\omega\cdot 2,0;0))$<br>$\psi(N^{N\cdot\omega+N+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(3,0,0) | $\psi(M(\psi(2),0;0))$<br>$\psi(N^{N\cdot\psi(2)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(M(1,0,0;0))$<br>$\psi(M(1,0,1;0))$<br>$\psi(N^{N^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1) | $\psi(M(1,0,0;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(1,0,1;\omega))$<br>$\psi(N^{N^2}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(1,0,2;\omega))$<br>$\psi(N^{N^2+1}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(M(1,1,1;0))$<br>$\psi(N^{N^2+N})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(1,1,1;\omega))$<br>$\psi(N^{N^2+N}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(M(1,2,1;0))$<br>$\psi(N^{N^2+N\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(2,0,0) | $\psi(M(2,0,1;0))$<br>$\psi(N^{N^2\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(3,1,0)(2,0,0) | $\psi(M(1,0,0,1;0))$<br>$\psi(N^{N^3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,0,0) | $\psi(M(1@\omega;0))$<br>$\psi(N^{N^\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,1,0)(2,0,0) | $\psi(N^{N^N})$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(5,1,0)(2,0,0) | $\psi(N^{N^{N^{N}}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{N+1})$<br>$\psi(\Pi_2 \text{ aft } \Pi_2 - \Pi_2 - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{N+1})$<br>$\psi(2 \text{ aft } 2 - 2 - 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{N+1} + \psi_N(\Omega_{N+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0)(2,0,0) | $\psi(\Omega_{N+1} + \psi_N(\Omega_{N+1}) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0)(2,1,0) | $\psi(\Omega_{N+1} + \psi_N(\Omega_{N+1}) \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{N+1} + \psi_N(\Omega_{N+1})^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,1,0)(4,2,0)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{N+1} + \psi_N(\Omega_{N+1})^2 \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0)(2,1,0)(2,1,0) | $\psi(\Omega_{N+1} + \psi_N(\Omega_{N+1})^2 \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(2,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{N+1} + \psi_N(\Omega_{N+1})^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{N+1} + \psi_N(\Omega_{N+1})^{\psi_N(\Omega_{N+1})})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,0) | $\psi(\Omega_{N+1} + \Omega_{\psi_N(\Omega_{N+1})+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)<br>(3,2,0)(2,1,0)(3,2,0) | $\psi(\Omega_{N+1} + \Omega_{\psi_N(\Omega_{N+1})+1} +$<br>$\psi_{\Omega_{\psi_N(\Omega_{N+1})+1}}(N + \Omega_{\psi_N(\Omega_{N+1})+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,0)(3,2,0) | $\psi(\Omega_{N+1} + \Omega_{\psi_N(\Omega_{N+1})+1} \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,0)(4,2,0) | $\psi(\Omega_{N+1} + \Omega_{\psi_N(\Omega_{N+1})+1}{}^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,0)(4,3,0) | $\psi(\Omega_{N+1} + \Omega_{\psi_N(\Omega_{N+1})+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1) | $\psi(\Omega_{N+1} + \Omega_{\psi_N(\Omega_{N+1})+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,0) | $\psi(\Omega_{N+1} + \Omega_{\psi_N(\Omega_{N+1})+\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,1) | $\psi(\Omega_{N+1} + \Omega_{\psi_N(\Omega_{N+1})+\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1) | $\psi(\Omega_{N+1} + \Omega_{\psi_N(\Omega_{N+1})+\omega^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,1,0) | $\psi(\Omega_{N+1} + \Omega_{\psi_N(\Omega_{N+1})+\Omega})$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,1,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{N+1} + \Omega_{\psi_N(\Omega_{N+1})\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0) | $\psi(\Omega_{N+1} + \Omega_{\Omega_{\psi_N(\Omega_{N+1})+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{N+1} + I_{\psi_N(\Omega_{N+1})+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,0)(5,3,0) | $\psi(\Omega_{N+1} + I_{\psi_N(\Omega_{N+1})+1} +$<br>$\Omega_{\psi_{I_{\psi_N(\Omega_{N+1})+1}}(I_{\psi_N(\Omega_{N+1})+1})+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1) | $\psi(\Omega_{N+1} + I_{\psi_N(\Omega_{N+1})+1} \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{N+1} + \Omega_{I_{\psi_N(\Omega_{N+1})+1}+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(6,3,1)(7,3,1)(8,3,0)(7,0,0) | $\psi(\Omega_{N+1} + I_{\psi_N(\Omega_{N+1})+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1) | $\psi(\Omega_{N+1} + I_{\psi_N(\Omega_{N+1})+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+1}{}^{3})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,1) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+1}{}^{3} \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+1}{}^{4})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+1}{}^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0)(4,2,1) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+1}{}^{\omega} \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+1}{}^{M_{\psi_N(\Omega_{N+1})+1}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,2,1)(5,2,1) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+1}{}^{M_{\psi_N(\Omega_{N+1})+1}} \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,2,1)(5,2,1) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+1}{}^{M_{\psi_N(\Omega_{N+1})+1}} \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,2,1)<br>(5,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+1}{}^{M_{\psi_N(\Omega_{N+1})+1}\cdot 2})$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(5,0,0) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+1}{}^{M_{\psi_N(\Omega_{N+1})+1}\cdot\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(5,2,0)(4,0,0) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+1}{}^{M_{\psi_N(\Omega_{N+1})+1}{}^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,2,0)(4,0,0) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+1}{}^{M_{\psi_N(\Omega_{N+1})+1}{}^{M_{\psi_N(\Omega_{N+1})+1}}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{N+1} + \Omega_{M_{\psi_N(\Omega_{N+1})+1}+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(\Omega_{N+1} + I_{M_{\psi_N(\Omega_{N+1})+1}+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,1)(7,3,1)(8,3,1)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(\Omega_{N+1} + M^3_{\psi_N(\Omega_{N+1})+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,1)(8,0,0) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+2}{}^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,1)(7,3,1)<br>(8,3,1)(8,3,0)(7,0,0) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+2}{}^{M_{\psi_N(\Omega_{N+1})+2}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,1)(7,3,1)(8,3,1)<br>(8,3,0)(9,3,0)(7,0,0) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+2}{}^{M_{\psi_N(\Omega_{N+1})+2}{}^{M_{\psi_N(\Omega_{N+1})+2}}})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,1)(7,3,1)<br>(8,3,1)(8,3,0)(9,4,0) | $\psi(\Omega_{N+1} + \Omega_{M_{\psi_N(\Omega_{N+1})+2}+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+\omega})$<br>$\psi(\Omega_{N+1} + N\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,0)(5,3,0) | $\psi(\Omega_{N+1} + \Omega_{M_{\psi_N(\Omega_{N+1})+\omega}+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,0)(5,3,1)(6,3,1)<br>(7,3,1)(7,3,0)(8,4,0) | $\psi(\Omega_{N+1} + \Omega_{M_{\psi_N(\Omega_{N+1})+\omega+1}+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,1)(7,3,1) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+\omega\cdot 2})$<br>$\psi(\Omega_{N+1} + N\cdot\omega\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1)(4,2,1) | $\psi(\Omega_{N+1} + M_{\psi_N(\Omega_{N+1})+\omega^2})$<br>$\psi(\Omega_{N+1} + N\cdot\omega^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,0) | $\psi(\Omega_{N+1} + M_{\Omega_{\psi_N(\Omega_{N+1})+1}})$<br>$\psi(\Omega_{N+1} + N\cdot\Omega_{\psi_N(\Omega_{N+1})+1})$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1) | $\psi(\Omega_{N+1} + M_{M_{\psi_N(\Omega_{N+1})+\omega}})$<br>$\psi(\Omega_{N+1} + N \cdot \psi_N(\Omega_{N+1} + N \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{N+1} + M(1, \psi_N(\Omega_{N+1}) + 1))$<br>$\psi(\Omega_{N+1} + M(1; \psi_N(\Omega_{N+1}) + 1))$<br>$\psi(\Omega_{N+1} + N^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(\Omega_{N+1} + M(1, \psi_N(\Omega_{N+1}) + 1) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{N+1} + \Omega_{M(1,\psi_N(\Omega_{N+1})+1)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(\Omega_{N+1} + I_{M(1,\psi_N(\Omega_{N+1})+1)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,1)(8,3,0)(9,4,0) | $\psi(\Omega_{N+1} + \Omega_{M_{M(1,\psi_N(\Omega_{N+1})+1)+1}+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,1)(8,3,1) | $\psi(\Omega_{N+1} + M_{M(1,\psi_N(\Omega_{N+1})+1)+\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,0)(6,3,1)<br>(7,3,1)(8,3,1)(8,3,1)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(\Omega_{N+1} + M(1, \psi_N(\Omega_{N+1}) + 2))$<br>$\psi(\Omega_{N+1} + M(1; \psi_N(\Omega_{N+1}) + 1) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,1) | $\psi(\Omega_{N+1} + M(1, \psi_N(\Omega_{N+1}) + \omega))$<br>$\psi(\Omega_{N+1} + M(1; \psi_N(\Omega_{N+1}) + 1) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,1)(4,2,1)<br>(5,2,0)(4,0,0) | $\psi(\Omega_{N+1} + M(2, \psi_N(\Omega_{N+1}) + 1))$<br>$\psi(\Omega_{N+1} + M(1; \psi_N(\Omega_{N+1}) + 1)^2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{N+1} + M(1, 0, \psi_N(\Omega_{N+1}) + 1))$<br>$\psi(\Omega_{N+1} + M(1; \psi_N(\Omega_{N+1}) + 1)^{M(1;\psi_N(\Omega_{N+1})+1)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{N+1} + \Omega_{M(1;\psi_N(\Omega_{N+1})+1)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,1)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,1)(8,3,1) | $\psi(\Omega_{N+1} + M_{M(1;\psi_N(\Omega_{N+1})+1)+\omega})$<br>$\psi(\Omega_{N+1} + N^2 + N \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,1)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,1)(8,3,1)<br>(7,3,1)(8,3,0)(7,0,0) | $\psi(\Omega_{N+1} + M(1, M(1; \psi_N(\Omega_{N+1}) + 1) + 1))$<br>$\psi(\Omega_{N+1} + M(1; \psi_N(\Omega_{N+1}) + 2))$<br>$\psi(\Omega_{N+1} + N^2 \cdot 2)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,1)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,1)<br>(8,3,1)(7,3,1)(8,3,1) | $\psi(\Omega_{N+1} + M(1, M(1; \psi_N(\Omega_{N+1}) + 1) + \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,1)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,1)(8,3,1)(7,3,1)<br>(8,3,1)(8,3,0)(7,0,0) | $\psi(\Omega_{N+1} + M(1, 0, M(1; \psi_N(\Omega_{N+1}) + 1) + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,1)(5,2,0)<br>(6,3,1)(7,3,1)(8,3,1)(8,3,1)(7,3,1)<br>(8,3,1)(8,3,0)(9,4,0) | $\psi(\Omega_{N+1} + \Omega_{M(1;\psi_N(\Omega_{N+1})+2)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,1)(5,2,1) | $\psi(\Omega_{N+1} + M(1; \psi_N(\Omega_{N+1}) + \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{N+1} + M(1; 1, \psi_N(\Omega_{N+1}) + 1))$<br>$\psi(\Omega_{N+1} + M(2; \psi_N(\Omega_{N+1}) + 1))$<br>$\psi(\Omega_{N+1} + N^3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,2,1) | $\psi(\Omega_{N+1} + M(1; 1, \psi_N(\Omega_{N+1}) + 1) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,1)(5,2,1)<br>(4,2,1)(5,2,1) | $\psi(\Omega_{N+1} + M(1; 1, \psi_N(\Omega_{N+1}) + \omega))$<br>$\psi(\Omega_{N+1} + M(2; \psi_N(\Omega_{N+1}) + 1) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,1)(5,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{N+1} + M(1; 1, 0, \psi_N(\Omega_{N+1}) + 1))$<br>$\psi(\Omega_{N+1} + M(2; \psi_N(\Omega_{N+1}) + 1)^{M(2;\psi_N(\Omega_{N+1})+1)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,1)(5,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{N+1} + \Omega_{M(2;\psi_N(\Omega_{N+1})+1)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,1)(5,2,1)<br>(4,2,1)(5,2,1)(5,2,1) | $\psi(\Omega_{N+1} + M(2; \psi_N(\Omega_{N+1}) + \omega))$<br>$\psi(\Omega_{N+1} + N^3 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,1)(5,2,1)<br>(4,2,1)(5,2,1)(5,2,1)<br>(4,2,1)(5,2,1)(5,2,1) | $\psi(\Omega_{N+1} + M(3; \psi_N(\Omega_{N+1}) + \omega))$<br>$\psi(\Omega_{N+1} + N^4 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1)(5,0,0) | $\psi(\Omega_{N+1} + M(\omega; \psi_N(\Omega_{N+1}) + 1))$<br>$\psi(\Omega_{N+1} + N^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{N+1} + M(1, 0; \psi_N(\Omega_{N+1}) + 1))$<br>$\psi(\Omega_{N+1} + M(1, 1; \psi_N(\Omega_{N+1}) + 1))$<br>$\psi(\Omega_{N+1} + N^N)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(5,2,0)(4,2,1)(5,2,1) | $\psi(\Omega_{N+1} + M(1, 0; \psi_N(\Omega_{N+1}) + \omega))$<br>$\psi(\Omega_{N+1} + M(1, 1; \psi_N(\Omega_{N+1}) + 1) \cdot \omega)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(5,2,0)(4,2,1)(5,2,1)<br>(5,2,0)(6,3,0) | $\psi(\Omega_{N+1}+\Omega_{M(1,1;\psi_N(\Omega_{N+1})+1)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(5,2,0)<br>(4,2,1)(5,2,1)(5,2,1) | $\psi(\Omega_{N+1}+M(1,1;\psi_N(\Omega_{N+1})+\omega))$<br>$\psi(\Omega_{N+1}+N^N\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(5,2,0)(4,2,1)(5,2,1)<br>(5,2,1)(4,2,1)(5,2,1)(5,2,1) | $\psi(\Omega_{N+1}+M(1,2;\psi_N(\Omega_{N+1})+\omega))$<br>$\psi(\Omega_{N+1}+N^{N+1}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(5,2,0)(4,2,1)(5,2,1)<br>(5,2,1)(5,2,0)(4,2,1)(5,2,1)(5,2,1) | $\psi(\Omega_{N+1}+M(2,1;\psi_N(\Omega_{N+1})+\omega))$<br>$\psi(\Omega_{N+1}+N^{N\cdot 2}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(5,2,0)(5,2,0)(5,2,0)<br>(4,2,1)(5,2,1)(5,2,1) | $\psi(\Omega_{N+1}+M(1,0,0,1;\psi_N(\Omega_{N+1})+\omega))$<br>$\psi(\Omega_{N+1}+N^{N^3}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(5,2,0)(6,0,0) | $\psi(\Omega_{N+1}+M(1@\omega;\psi_N(\Omega_{N+1})+\omega))$<br>$\psi(\Omega_{N+1}+N^{N^\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(5,2,0)(6,2,0)(4,2,1)<br>(5,2,1)(5,2,1) | $\psi(\Omega_{N+1}+N^{N^N}\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(5,2,0)(6,3,0)(4,2,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1})+$<br>$\Omega_{\psi_N(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}))+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(5,2,0)(6,3,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,1)(7,3,1) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1})+$<br>$M_{\psi_N(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}))+\omega})$<br>$\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1})+N\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(5,2,0)(6,3,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,1)(7,3,1)(7,0,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1})+N^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(5,2,0)(6,3,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,1)<br>(7,3,1)(7,3,0)(8,4,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1})\cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0)(2,1,1) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+\Omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)+\psi_N(\Omega_{N+1}))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)+\Omega_{\psi_N(\Omega_{N+1})+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)+\psi_N(\Omega_{N+1}+N\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,1)(5,2,0)(6,3,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)+$<br>$\psi_N(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1})))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)+\psi_N(\Omega_{N+1}+$<br>$\psi_{\Omega_{N+1}}(\Omega_{N+1}+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)+\psi_N(\Omega_{N+1}+$<br>$\psi_{\Omega_{N+1}}(\Omega_{N+1}+\Omega)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,1)(5,2,0)<br>(6,3,0)(4,2,1)(5,2,0)(4,0,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)+$<br>$\psi_{\psi_N(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N))}(\Omega_{N+1}+$<br>$\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)+$<br>$\Omega_{\psi_{\psi_N(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N))}(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N))+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)+$<br>$\psi_N(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)+$<br>$\psi_{\psi_N(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N))}(\Omega_{N+1}+$<br>$\psi_{\Omega_{N+1}}(\Omega_{N+1}+N))+\psi_{\Omega_{N+1}}(\Omega_{N+1})))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(6,2,0)(7,3,0)<br>(5,2,1)(6,2,0)(5,0,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)+$<br>$\psi_N(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)+\psi_N(\Omega_{N+1}+$<br>$\psi_{\Omega_{N+1}}(\Omega_{N+1}+N))\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)+$<br>$\Omega_{\psi_N(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N))+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(6,2,1)(6,2,1)(6,2,1) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)+N\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)+\psi_{\Omega_{N+1}}(\Omega_{N+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)<br>(7,3,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(\Omega_{N+1}+\psi_{\Omega_{N+1}}(\Omega_{N+1}+N)\cdot 2)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1)(6,2,0)<br>(7,3,0)(5,2,1)(6,2,1)(6,2,0)<br>(7,3,1)(8,3,1)(9,3,1)(9,3,1) | $\psi(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + N) \cdot 2 + N \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + N + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + N \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + N \cdot 2 + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + N^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + N^2 + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(2,0,0) | $\psi(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + N^3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1})))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1}) + N + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1}) \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(3,1,0)(2,0,0) | $\psi(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + N)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + N) + N))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(3,1,0)(3,1,0)(2,0,0) | $\psi(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + N \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(3,1,0)(4,2,0) | $\psi(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} +$<br>$\psi_{\Omega_{N+1}}(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1}))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(4,1,0)(5,2,0) | $\psi(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1} +$<br>$\psi_{\Omega_{N+1}}(\Omega_{N+1} + \psi_{\Omega_{N+1}}(\Omega_{N+1})))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(4,2,0) | $\psi(\Omega_{N+1} \cdot 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,3,0) | $\psi(\Omega_{N+2})$<br>$\psi(\text{2nd 2 aft } 2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,1) | $\psi(\Omega_{N+\omega})$<br>$\psi(1-2 \text{ aft } 2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(I_{N+1})$<br>$\psi(2\ 1-2 \text{ aft } 2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) | $\psi(I_{N+\omega})$<br>$\psi(1-2\ 1-2 \text{ aft } 2-2-2)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{M_{N+1}+1})$<br>$\psi(2 \text{ aft } 2-2 \text{ aft } 2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1) | $\psi(M_{N+\omega})$<br>$\psi(N_2 \cdot \omega)$<br>$\psi(1-2-2 \text{ aft } 2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(5,2,1)(6,2,1)(6,2,1) | $\psi(M(1;N+\omega))$<br>$\psi(N_2^2 \cdot \omega)$<br>$\psi(1-2-2\ 1-2-2 \text{ aft } 2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(6,0,0) | $\psi(M(\omega;N+\omega))$<br>$\psi(N_2^\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(6,2,0)(5,2,1)(6,2,1)(6,2,1) | $\psi(M(1,1;N+\omega))$<br>$\psi(N_2^{N_2} \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{N_2+1})$<br>$\psi(2 \text{ aft 2nd } 2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(6,2,0)(7,3,1)(8,3,1)(9,3,1)(9,3,1) | $\psi(M_{N_2+\omega})$<br>$\psi(1-2-2 \text{ aft 2nd } 2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(6,2,0)(7,3,1)(8,3,1)(9,3,1)<br>(9,3,1)(9,3,0)(10,4,0) | $\psi(\Omega_{N_3+1})$<br>$\psi(2 \text{ aft 3rd } 2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,1,1) | $\psi(M(1;0;\omega))$<br>$\psi(N_\omega)$<br>$\psi(1-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(M(1;0;1,0))$<br>$\psi(2\ 1-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1) | $\psi(M(1;0;1,\omega))$<br>$\psi(1-2\ 1-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(M(1;0;2,\omega))$<br>$\psi(1-2\ 1-2\ 1-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(M(1;0;1,0,0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M(1;1;0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(1;1;\omega))$<br>$\psi(1-2-2\ 1-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1) | $\psi(M(1;2;\omega))$<br>$\psi(1-2-2\ 1-2-2\ 1-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M(2;0;0)+1})$<br>$\psi(2 \text{ aft } 2-2-2\ 1-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1)(6,2,1) | $\psi(M(1;0;M(2;0;0)+\omega))$<br>$\psi(1-2-2-2 \text{ aft } 2-2-2\ 1-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,1)(6,2,1) | $\psi(M(1;1;M(2;0;0)+\omega))$<br>$\psi(1-2-2\ 1-2-2-2 \text{ aft}$<br>$2-2-2\ 1-2-2-2)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1)(6,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{M(2;0;1)+1})$<br>$\psi(2 \text{ aft } 2\text{nd } 2-2-2\ 1-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,1,1) | $\psi(M(2;0;\omega))$<br>$\psi(1-2-2-2\ 1-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(2;1;\omega))$<br>$\psi(1-2-2\ 1-2-2-2\ 1-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)(3,1,1) | $\psi(M(3;0;\omega))$<br>$\psi(1-2-2-2\ 1-2-2-2\ 1-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,0,0) | $\psi(M(\omega;0;0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(M(1,0;0;0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1) | $\psi(M(1,0;0;\omega))$<br>$\psi(M(1,0;1;0)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(M(1,0;1;\omega))$<br>$\psi(M(1,1;0;0)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{M(1,1;0;0)+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,1,1)<br>(3,1,1)(3,1,1)(3,1,1) | $\psi(M(1,1;0;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)(3,1,1) | $\psi(M(1,2;0;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(M(2,0;0;0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,1,1)<br>(3,1,1)(3,1,1)(3,1,1) | $\psi(M(2,1;0;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(2,1,1)<br>(3,1,1)(3,1,1)(3,1,1) | $\psi(M(1,0,1;0;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,1)(3,1,1) | $\psi(M(1,0,0,1;0;\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,0,0) | $\psi(M(1@\omega;0;0))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(2 \text{ aft } 2-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(6,2,1)(6,2,0)(7,3,0) | $\psi(2 \text{ aft } 2\text{nd } 2-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,1) | $\psi(1-2-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(1-2\ 1-2-2-2-2)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2-2-2\ 1-2-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,0,0) | $\psi((2-2-2-2\ 1-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(2\ 1-(2-2-2-2\ 1-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi((2-2-2-2\ 1-)^{(2,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,0)(3,1,0)(2,0,0) | $\psi((2-2-2-2\ 1-)^{(1,0,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,0)(4,0,0) | $\psi((2-2-2-2\ 1-)^{(1@\omega)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(2\text{ aft }2-2-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2-2-2-2-2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,0,0) | $\psi((2-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,0,0)(2,1,0)(3,2,0) | $\psi(2\text{ aft }(2-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(6,0,0) | $\psi(2\text{nd }(2-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,0,0)(2,1,1) | $\psi(1-(2-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,0,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(2\ 1-(2-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,0,0)(2,1,1)(3,1,1) | $\psi(1-2\ 1-(2-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,0,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2\ 1-(2-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,0,0)(2,1,1)(3,1,1)(4,0,0) | $\psi((2-)^{\omega}\ 1-(2-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,0,0)(3,0,0) | $\psi(((2-)^{\omega}\ 1-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,0,0)(3,1,0)(2,0,0) | $\psi(((2-)^{\omega}\ 1-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,0,0)<br>(3,1,0)(2,1,1)(3,1,1)(4,0,0) | $\psi(((2-)^{\omega}\ 1-)^{(1,1)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,0,0)(3,1,0)(3,1,0)(2,0,0) | $\psi(((2-)^{\omega}\ 1-)^{(1,0,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,0,0)(3,1,0)(4,2,0) | $\psi(2\text{ aft }(2-)^{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,0,0)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(7,0,0)(6,2,0)(7,3,0) | $\psi(2\text{ aft 2nd }(2-)^{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,0,0)(3,1,1) | $\psi(1-(2-)^{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,0,0)<br>(3,1,1)(2,1,1)(3,1,1)(4,0,0)(3,1,1) | $\psi(1-(2-)^{\omega+1}\ 1-(2-)^{\omega+1})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,0,0)(3,1,1)(3,1,0)(4,2,0) | $\psi(2\text{ aft }(2-)^{\omega+2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,0,0)(3,1,1)(3,1,1) | $\psi(1-(2-)^{\omega+2})$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,0,0)(3,1,1)(4,0,0) | $\psi((2-)^{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0) | $\psi((2-)^{\Omega})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(2,0,0) | $\psi((2-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(2,1,1) | $\psi(1-(2-)^{(1,0)})$<br>$\psi((2\ 1-(2-)^{(1,0)})\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,1)(3,1,1) | $\psi(1-2\ 1-(2-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(2,1,1)(3,1,1)(4,1,0)(2,0,0) | $\psi((2-)^{(1,0)}\ 1-(2-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(3,1,0)(2,0,0) | $\psi(((2-)^{(1,0)}\ 1-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(3,1,0)(4,2,0) | $\psi(2\text{ aft }(2-)^{(1,1)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(3,1,1) | $\psi(1-(2-)^{(1,1)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(3,1,1)(3,1,1) | $\psi(1-(2-)^{(1,2)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(3,1,1)(4,1,0)(2,0,0) | $\psi((2-)^{(2,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(4,1,0)(2,0,0) | $\psi((2-)^{(1,0,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,0,0) | $\psi((2-)^{(1@\omega)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,1,0)(2,0,0) | $\psi((2-)^{(1@(1,0))})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0) | $\psi(\Pi_2\text{ aft }\Pi_3)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(5,2,0) | $\psi(\Omega_{K+1})$<br>$\psi(2\text{ aft }3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(5,2,0) | $\psi(\Omega_{K+1}+\psi_K(\Omega_{K+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(2,1,0)(3,2,0) | $\psi(\Omega_{K+1}+K)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1) | $\psi(\Omega_{K+1}+K^2\times\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(6,0,0) | $\psi(\Omega_{K+1}+K^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(6,0,0)(4,2,1) | $\psi(\Omega_{K+1}+K^{\omega}\times\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(6,0,0)(5,2,1) | $\psi(\Omega_{K+1}+K^{\omega+1}\times\omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(6,0,0)(5,2,1)(6,0,0) | $\psi(\Omega_{K+1}+K^{\omega\times 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(6,2,0) | $\psi(\Omega_{K+1}+K^{\psi_K(\Omega_{K+1}+K)})$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(4,0,0) | $\psi(\Omega_{K+1} + K^K)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(4,2,1) | $\psi(\Omega_{K+1} + K^K \times \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(6,2,0)(4,2,1)(5,2,1) | $\psi(1 - 2\ 1 - \{\psi_K(\Omega_{K+1} + K^K \times x)\})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,0)(6,3,0) | $\psi(\Omega_{K+1} + K^{K+1} + K)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(5,2,1) | $\psi(\Omega_{K+1} + K^{K+1} \times \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1)(6,2,0)(4,0,0) | $\psi(\Omega_{K+1} + K^{K\times 2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(6,2,0)(6,2,0)(4,0,0) | $\psi(\Omega_{K+1} + K^{K^2})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(6,2,0)(7,2,0)(4,0,0) | $\psi(\Omega_{K+1} + K^{K^K})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,0) | $\psi(\Omega_{K+1} + \psi_{\Omega_{K+1}}(\Omega_{K+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,0)(4,2,0)(5,3,0) | $\psi(\Omega_{K+1} + \psi_{\Omega_{K+1}}(\Omega_{K+1}) + K)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(6,2,0)(7,3,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,1)(8,3,0)(6,0,0) | $\psi(\Omega_{K+1} + \psi_{\Omega_{K+1}}(\Omega_{K+1}) + K^K)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(6,2,0)(7,3,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,1)(8,3,0)(9,3,0)(6,0,0) | $\psi(\Omega_{K+1} + \psi_{\Omega_{K+1}}(\Omega_{K+1}) + K^{K^K})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(6,2,0)(7,3,0)(4,2,0)(5,3,1)<br>(6,3,1)(7,3,1)(8,3,0)(9,4,0) | $\psi(\Omega_{K+1} + \psi_{\Omega_{K+1}}(\Omega_{K+1}) \times 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(2,1,1) | $\psi(\Omega_{K+1} + \psi_{\Omega_{K+1}}(\Omega_{K+1} + 1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(2,1,1)(3,1,1) | $\psi(1 - 2\ 1 - \{\Omega_{K+1} + \psi_{\Omega_{K+1}}(\Omega_{K+1} + x)\})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(3,1,0)(4,2,0) | $\psi(\Omega_{K+1} + \psi_{\Omega_{K+1}}(\Omega_{K+1} + K) + K)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(7,2,0)(8,3,0) | $\psi(\Omega_{K+1} + \psi_{\Omega_{K+1}}(\Omega_{K+1} + K) +$<br>$\psi_{\Omega_{K+1}}(\Omega_{K+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(7,2,0)(8,3,0)(6,2,0)(7,3,0) | $\psi(\Omega_{K+1} + \psi_{\Omega_{K+1}}(\Omega_{K+1} + K) \times 2 + K)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(3,1,1) | $\psi(\Omega_{K+1} + \psi_{\Omega_{K+1}}(\Omega_{K+1} + K + 1))$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(3,1,1)(2,1,1) | $\psi(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1}+K+2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(3,1,1)(3,1,0)(4,2,0) | $\psi(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1}+K\times 2)+K)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(3,1,1)(3,1,1) | $\psi(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1}+K\times 2+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(3,1,1)(4,1,0)(2,0,0) | $\psi(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1}+K^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,0)(3,1,1)(4,1,0)(3,1,0)(4,2,0) | $\psi(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1}+K^2)+K)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(3,1,1)(4,1,0)(3,1,1) | $\psi(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1}+K^2+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(3,1,1)(4,1,0)<br>(3,1,1)(4,1,0)(2,0,0) | $\psi(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1}+K^2\times 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,0)(3,1,1)(4,1,0)(4,1,0)(2,0,0) | $\psi(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1}+K^3))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,0)(3,1,1)(4,1,0)(5,1,0)(2,0,0) | $\psi(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1}+K^K))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,0)(3,1,1)(4,1,0)(5,2,0) | $\psi(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1})))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(4,0,0) | $\psi(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1}+$<br>$\psi_{\Omega_{K+1}}(\Omega_{K+1}+1)))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(4,1,0)(5,2,0) | $\psi(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1}+$<br>$\psi_{\Omega_{K+1}}(\Omega_{K+1}))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(5,1,0)(6,2,0) | $\psi(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1}+$<br>$\psi_{\Omega_{K+1}}(\Omega_{K+1}+\psi_{\Omega_{K+1}}(\Omega_{K+1})))))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(5,2,0) | $\psi(\Omega_{K+1}\times 2)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0)(6,3,0) | $\psi(\text{2nd 2 aft 3})$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(5,2,1) | $\psi(1-2\text{ aft }3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(7,2,1) | $\psi(1-2-2\text{ aft }3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,0,0) | $\psi((2-)^{\omega}\text{ aft }3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,0,0)(7,2,1) | $\psi(1-(2-)^{\omega+1}\text{ aft }3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,2,0)(7,2,1) | $\psi(1-(2-)^{(1,1)}\text{ aft }3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,1)(6,2,1)(7,2,1)<br>(8,2,0)(7,2,1)(8,2,0)(7,2,1) | $\psi(1-(2-)^{(2,1)}\text{ aft }3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,2,0)(9,3,0) | $\psi(2\text{ aft 2nd }3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,2,0)(9,3,1)<br>(10,3,1)(11,3,1)(12,3,0)(13,4,0) | $\psi(2\text{ aft 3rd }3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1) | $\psi(1-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(6,2,0)(7,3,0) | $\psi(2\text{ aft }3\text{ aft }1-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(6,2,1) | $\psi(\text{2nd }1-3)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(2,1,1) | $\psi(1-1-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(2,1,1)(3,1,1) | $\psi(1-2\ 1-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2\ 1-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(2,1,1)(3,1,1)(4,1,0)(2,0,0) | $\psi((2-)^{(1,0)}\ 1-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(2,1,1)(3,1,1)(4,1,0)(2,1,1) | $\psi(1-(2-)^{(1,0)}\ 1-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(2,1,1)(3,1,1)(4,1,0)(3,1,1) | $\psi(1-(2-)^{(1,1)}\ 1-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(3,1,1)(4,1,0)(2,0,0) | $\psi((2-)^{(2,0)}\ 1-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(2,1,1)(3,1,1)(4,1,0)(5,2,0) | $\psi(2\text{ aft }3\ 1-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(2,1,1)(3,1,1)(4,1,1) | $\psi(1-3\ 1-3)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(3,0,0) | $\psi((3\ 1-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(4,1,1) | $\psi(1-(3\ 1-)^{(1,1)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,0)(4,2,0) | $\psi(2\text{ aft }2-3)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(3,1,1) | $\psi(1-2-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,1)(3,1,1) | $\psi(1-2-2-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,1)(4,1,0)(2,0,0) | $\psi((2-)^{(1,0)}3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(3,1,1)(4,1,0)(3,1,0)(2,0,0) | $\psi(((2-)^{(1,0)}3\ 1-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,1)(4,1,0)(3,1,1) | $\psi(1-(2-)^{(1,1)}3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,1)(4,1,0)(5,2,0) | $\psi(2\text{ aft }3\ 2-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,1)(4,1,1) | $\psi(1-3\ 2-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,1)(4,1,1)(3,1,1) | $\psi(1-2-3\ 2-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(3,1,1)(4,1,1)(3,1,1)(4,1,1) | $\psi(1-3\ 2-3\ 2-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,0)(2,0,0) | $\psi((3\ 2-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(4,1,0)(2,1,1)(3,1,1)(4,1,1) | $\psi(1-3\ 1-(3\ 2-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,0)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,0)(2,0,0) | $\psi((3\ 2-)^{(1,0)}1-(3\ 2-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,0)(3,1,0)(2,0,0) | $\psi(((3\ 2-)^{(1,0)}\ 1-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,0)(3,1,1) | $\psi(1-2-(3\ 2-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,0)(3,1,1)(4,1,1) | $\psi(1-(3\ 2-)^{(1,1)})$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,0)(5,2,0) | $\psi(2 \text{ aft } 3-3)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(4,1,1) | $\psi(1-3-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,1)(2,1,1) | $\psi(1-1-3-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(4,1,1)(2,1,1)(3,1,1)(4,1,1)(4,1,1) | $\psi(3-3\ 1-3-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,1)(3,1,1) | $\psi(1-2-3-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,1)(3,1,1)(4,1,1) | $\psi(1-3\ 2-3-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(4,1,1)(3,1,1)(4,1,1)(4,1,1) | $\psi(1-3-3\ 2-3-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,1)(3,1,1)(4,1,1)<br>(4,1,1)(3,1,1)(4,1,1)(4,1,1) | $\psi(1-3-3\ 2-3-3\ 2-3-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,1)(4,1,0)(2,0,0) | $\psi((3-3\ 2-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(4,1,1)(4,1,0)(3,1,1)(4,1,1)(4,1,1) | $\psi(1-(3-3\ 2-)^{(1,1)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,1)(4,1,0)(5,2,0) | $\psi(2 \text{ aft } 3-3-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,1)(4,1,1) | $\psi(1-3-3-3)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,1)(4,1,1)(4,1,1) | $\psi(1-3-3-3-3)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(5,0,0) | $\psi((3-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,0)(2,0,0) | $\psi((3-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,0)(2,1,1) | $\psi(1-(3-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,0)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,0)(2,0,0) | $\psi((3-)^{(1,0)}\ 1-(3-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,0)(3,1,0)(2,0,0) | $\psi(((3-)^{(1,0)}\ 1-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,0)(3,1,1) | $\psi(1-2-(3-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(5,1,0)(3,1,1)(4,1,1)(5,1,0)(2,0,0) | $\psi((3-)^{(1,0)}\ 2-(3-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,0)(4,1,0)(2,0,0) | $\psi(((3-)^{(1,0)}\ 2-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,0)(4,1,1) | $\psi(1-(3-)^{(1,1)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,0)(4,1,1)(4,1,1) | $\psi(1-(3-)^{(1,2)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(5,1,0)(4,1,1)(5,1,0)(2,0,0) | $\psi((3-)^{(2,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,0)(6,2,0) | $\psi(2 \text{ aft } 4)$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(5,1,1) | $\psi(1-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(2,1,1) | $\psi(1-1-4)$ |

| BMS | Reflection-Collapsing Notation (Buchholz-like) |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(5,1,1)(2,1,1)(3,1,1)(4,1,1)(5,1,1) | $\psi(1-4\ 1-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(3,1,0)(4,2,0) | $\psi(2 \text{ aft } 2-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(3,1,1) | $\psi(1-2-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(5,1,1)(3,1,1)(4,1,1)(5,1,1) | $\psi(1-4\ 2-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(4,1,0)(5,2,0) | $\psi(2 \text{ aft } 3-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(4,1,1) | $\psi(1-3-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(4,1,1)(5,1,1) | $\psi(1-4\ 3-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(5,1,0)(6,2,0) | $\psi(2 \text{ aft } 4-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(5,1,1) | $\psi(1-4-4)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,0,0) | $\psi((4-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,0)(2,0,0) | $\psi((4-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,0)(2,1,1) | $\psi(1-(4-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,0)(3,1,1) | $\psi(1-2-(4-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,0)(4,1,1) | $\psi(1-3-(4-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,0)(5,1,1) | $\psi(1-(4-)^{(1,1)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(5,1,1)(6,1,0)(5,1,1)(6,1,0)(2,0,0) | $\psi((4-)^{(2,0)})$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,0)(7,2,0) | $\psi(2 \text{ aft } 5)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,1) | $\psi(1-5)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(5,1,1)(6,1,1)(7,1,0)(8,2,0) | $\psi(2 \text{ aft } 6)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,1)(7,1,1) | $\psi(1-6)$ |
| (0,0,0)(1,1,1)(2,2,0) | $\psi(\text{psd}.\Pi_\omega)$<br>$\psi(\lambda\alpha.\alpha+1-\Pi_0)$ |

## A.12 BMS vs Stable-Collapsing Notation (Meitianli.ver)

The results in this section are primarily cited from[6,24-28].

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0) | $\psi(\Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(1,1,1) | $\psi((\Pi_1-)^{(\min\ \Pi_1-\Pi_2)} \text{ aft } \Pi_\omega)$<br>$\psi((\lambda\alpha.(\alpha+1)-\Pi_0)+\Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(1,1,1)(2,2,0) | $\psi(2\text{nd } (\Pi_1-)^{(\min\ \Pi_\omega)})$<br>$\psi((\lambda\alpha.(\alpha+1)-\Pi_0)\cdot 2)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(2,1,0) | $\psi((\Pi_1-)^{(\Pi_1-)^{(\Pi_2 \text{ aft } \Pi_\omega)}})$<br>$\psi((\lambda\alpha.(\alpha+1)-\ _0)\cdot\Omega)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,0)(1,1,1)(2,2,0) | $\psi((\Pi_1-)^{(\Pi_1-)^{(2\text{nd } \Pi_\omega)}})$<br>$\psi((\lambda\alpha.(\alpha+1)-\Pi_0)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,0) | $\psi((\Pi_1-)^{1,0} \text{ aft } \Pi_\omega)$<br>$\psi(\psi_{\Omega_{(\lambda\alpha.(\alpha+1)-\Pi_0)}+1}(0))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,0)(3,2,0)(4,2,0) | $\psi(\Pi_2 \text{ aft } \Pi_\omega)$<br>$\psi(\Omega_{(\lambda\alpha.(\alpha+1)-\Pi_0)+1})$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,1) | $\psi(\Pi_1-\Pi_2 \text{ aft } \Pi_\omega)$<br>$\psi(\Omega_{(\lambda\alpha.(\alpha+1)-\Pi_0)+\omega})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,0)(3,2,1)(4,2,1) | $\psi(\Pi_1-\Pi_1-\Pi_2 \text{ aft } \Pi_\omega)$<br>$\psi(\Omega_{(\lambda\alpha.(\alpha+1)-\Pi_0)+\omega^2})$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1) | $\psi(\Pi_1-\Pi_2\ \Pi_1-\Pi_2 \text{ aft } \Pi_\omega)$<br>$\psi(I_{(\lambda\alpha.(\alpha+1)-\Pi_0)+\omega})$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(6,2,1) | $\psi(\Pi_1-\Pi_3 \text{ aft } \Pi_\omega)$<br>$\psi(K_{(\lambda\alpha.(\alpha+1)-\Pi_0)+\omega})$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(6,2,1)(7,2,1) | $\psi(\Pi_1-\Pi_4 \text{ aft } \Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,0)(3,2,1)(4,3,0) | $\psi(2\text{nd } \Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,1)<br>(4,3,0)(4,2,0)(5,3,1)(6,4,0) | $\psi(3\text{rd } \Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1) | $\psi(\Pi_1-\Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(2,1,1) | $\psi(\Pi_1-\Pi_1-\Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,0) | $\psi((\Pi_1-)^{(\min\ \Pi_2)\ \Pi_\omega})$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,2,0) | $\psi((\Pi_1-)^{(\min\ \Pi_\omega)}\ \Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi((\Pi_1-)^{1,0}\ \Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\Pi_2\ \Pi_1-\Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,1) | $\psi(\Pi_1-\Pi_2\ \Pi_1-\Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,1)(3,1,1)(4,1,1) | $\psi(\Pi_1-\Pi_3\ \Pi_1-\Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0) | $\psi(\Pi_\omega\ \Pi_1-\Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(3,1,0)(2,0,0) | $\psi((\Pi_\omega\ \Pi_1-)^\omega\ \Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,1)(3,2,0)(3,1,1) | $\psi(\Pi_1-\Pi_2-\Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(3,1,1)(4,1,1) | $\psi(\Pi_1-\Pi_3\ \Pi_2-\Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(3,1,1)(4,2,0) | $\psi(\Pi_\omega\ \Pi_2-\Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(3,1,1)(4,2,0)(4,1,1) | $\psi(\Pi_1-\Pi_3-\Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,2,0)(4,1,1)(5,2,0)(5,1,1) | $\psi(\Pi_1-\Pi_4-\Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,2,0) | $\psi(\Pi_\omega-\Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,2,0)(2,2,0) | $\psi(\Pi_\omega-\Pi_\omega-\Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,0,0) | $\psi((\Pi_\omega-)^\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,0,0) | $\psi((\Pi_\omega-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,1,1) | $\psi(1-(\Pi_\omega-)^{1,0})$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)<br>(2,1,1)(3,2,0)(4,1,0)(2,0,0) | $\psi((\Pi_\omega-)^{1,0}\ 1-(\ _\omega-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,1,1)<br>(3,2,0)(4,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(((\Pi_\omega-)^{1,0}\ 1-)^{1,0}\ (\Pi_\omega-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)<br>(2,1,1)(3,2,0)(4,1,0)(3,1,1) | $\psi(2-(\Pi_\omega-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,2,0) | $\psi((\Pi_\omega-)^{1,1})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(4,2,0) | $\psi((\Pi_1-)^{1,0}\ \mathrm{aft}\ \Pi_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1) | $\psi(\Pi_1-\Pi_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(2,1,1) | $\psi(\Pi_1-\Pi_1-\Pi_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(2,1,1)(3,1,1) | $\psi(\Pi_1-\Pi_2\ \Pi_1-\Pi_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(2,1,1)(3,2,0) | $\psi(\Pi_\omega\ \Pi_1-\Pi_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(2,1,1)(3,2,0)(4,1,1) | $\psi(\Pi_1-\Pi_{\omega+1}\ \Pi_1-\Pi_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(2,1,1)(3,2,0)(4,1,1)(3,1,1) | $\psi(\Pi_1-\Pi_2-\Pi_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(2,1,1)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,1,1)(4,1,1) | $\psi(\Pi_1-\Pi_3-\Pi_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(2,2,0) | $\psi(\Pi_\omega-\Pi_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(2,2,0)(2,2,0) | $\psi(\Pi_\omega-\Pi_\omega-\Pi_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(2,2,0)(3,1,0)(2,0,0) | $\psi((\Pi_\omega-)^{1,0}\ \Pi_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(2,2,0)(3,1,1) | $\psi(\Pi_1-\Pi_{\omega+1}\ \Pi_\omega-\Pi_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(3,1,1) | $\psi(\Pi_1-\Pi_{\omega+1}-\Pi_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,0,0) | $\psi((\Pi_{\omega+1}-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,1,1) | $\psi(\Pi_1-\Pi_{\omega+2})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(4,1,1)(5,1,1) | $\psi(\Pi_1-\Pi_{\omega+3})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,2,0) | $\psi(\Pi_{\omega\cdot 2})$<br>$\psi(\lambda\alpha.(\alpha+2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(4,2,0)(2,2,0) | $\psi(\Pi_\omega-\Pi_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,2,0)(2,2,0)(3,1,1) | $\psi(\Pi_1-\Pi_{\omega+1}\ \Pi_\omega-\Pi_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,2,0)(2,2,0)(3,1,1)(4,2,0) | $\psi(\Pi_{\omega\cdot 2}\ \Pi_\omega-\Pi_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(4,2,0)(3,1,1) | $\psi(\Pi_1-\Pi_{\omega+1}-\Pi_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(4,2,0)(4,1,1) | $\psi(\Pi_1-\Pi_{\omega+2}-\Pi_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,2,0)(4,1,1)(5,2,0)(5,1,1) | $\psi(\Pi_1-\Pi_{\omega+3}-\Pi_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(4,2,0)(4,2,0) | $\psi(\Pi_{\omega\cdot 2}-\Pi_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(4,2,0)(5,1,1) | $\psi(\Pi_1-\Pi_{\omega\cdot 2+1})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,2,0)(5,1,1)(6,1,1) | $\psi(\Pi_1-\Pi_{\omega\cdot 2+2})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(4,2,0)(5,1,1)(6,2,0) | $\psi(\Pi_{\omega\cdot 3})$<br>$\psi(\lambda\alpha.(\alpha+3)-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0) | $\psi(\Pi_{\omega^2})$<br>$\psi(\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(2,1,1)(3,2,0)(4,2,0)(3,1,1) | $\psi(\Pi_1-\Pi_2-\Pi_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0) | $\psi(\Pi_\omega-\Pi_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(2,2,0)(3,1,1)(4,2,0) | $\psi(\Pi_{\omega\cdot 2}-\Pi_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(2,2,0)(3,1,1)(4,2,0)(5,2,0)<br>(4,2,0)(5,1,1)(6,2,0) | $\psi(\Pi_{\omega\cdot 3}-\Pi_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(2,2,0)(3,2,0) | $\psi(\Pi_{\omega^2}-\Pi_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,0,0) | $\psi((\Pi_{\omega^2}-)^\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(3,1,0)(2,0,0) | $\psi((\Pi_{\omega^2}-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1) | $\psi(\Pi_1-\Pi_{\omega^2+1})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(3,1,1)(4,1,1) | $\psi(\Pi_1-\Pi_{\omega^2+2})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(3,1,1)(4,2,0) | $\psi(\Pi_{\omega^2+\omega})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(3,1,1)(4,2,0)(5,1,1) | $\psi(\Pi_1-\Pi_{\omega^2+\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(3,1,1)(4,2,0)(5,1,1)(6,2,0) | $\psi(\Pi_{\omega^2+\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(3,1,1)(4,2,0)(5,2,0) | $\psi(\Pi_{\omega^2\cdot 2})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,2,0) | $\psi(\Pi_{\omega^3})$<br>$\psi(\lambda\alpha.(\alpha+\omega^2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(3,2,0)(3,2,0) | $\psi(\Pi_{\omega^4})$<br>$\psi(\lambda\alpha.(\alpha+\omega^3)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,0,0) | $\psi(\Pi_{\omega^\omega})$<br>$\psi(\lambda\alpha.(\alpha+\omega^\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,0,0)(2,1,1) | $\psi(\Pi_1-\Pi_{\omega^\omega})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,0,0)(2,2,0) | $\psi(\Pi_\omega-\Pi_{\omega^\omega})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,0,0)(2,2,0)(3,2,0)(4,0,0) | $\psi(\Pi_{\omega^\omega}-\Pi_{\omega^\omega})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,0,0)(3,1,1) | $\psi(\Pi_1-\Pi_{\omega^\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,0,0)(3,1,1)(4,2,0)(5,2,0) | $\psi(\Pi_{\omega^\omega+\omega^2})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,0,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,0,0) | $\psi(\Pi_{\omega^\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,0,0)(3,2,0) | $\psi(\Pi_{\omega^{\omega+1}})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,0,0)(3,2,0)(3,2,0) | $\psi(\Pi_{\omega^{\omega+2}})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,0,0)(3,2,0)(4,0,0) | $\psi(\Pi_{\omega^{\omega\cdot 2}})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,0,0)(4,0,0) | $\psi(\Pi_{\omega^{\omega^2}})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,0,0)(5,1,0) | $\psi(\Pi_{\psi(0)})$<br>$\psi(\lambda\alpha.(\alpha+\varepsilon_0)-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,0,0)(5,1,1)(6,2,0) | $\psi(\Pi_{\psi(\Pi_\omega)})$<br>$\psi(\lambda\alpha.(\alpha+\psi(\lambda\alpha.(\alpha+1)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0) | $\psi(\Pi_{\Pi_2})$<br>$\psi(\lambda\alpha.(\alpha+\Omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(1,1,1) | $\psi(\Pi_{\Pi_1-\Pi_2})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(1,1,1)(2,2,0) | $\psi(\Pi_{\Pi_\omega})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0) | $\psi(\Pi_{\Pi_\Omega})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(2,0,0) | $\psi(\alpha\to\Pi_\alpha)$<br>$\psi(\Pi_{1,0})$<br>$\psi(\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(2,1,0)(3,2,1)(4,3,0) | $\psi(\lambda\alpha.(\alpha+1)-\Pi_0 \text{ aft } \lambda\alpha.(\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(2,1,0)(3,2,1)(4,3,0)(5,3,0)(6,1,0) | $\psi(\lambda\alpha.(\alpha+\Omega)-\Pi_0 \text{ aft } \lambda\alpha.(\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(2,1,0)(3,2,1)(4,3,0)(5,3,0)(6,1,0)<br>(1,1,1)(2,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha+\lambda\alpha.(\alpha\cdot 2)$<br>$-\Pi_0)-\Pi_0 \text{ aft } \lambda\alpha.(\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(2,1,0)(3,2,1)(4,3,0)<br>(5,3,0)(6,2,0)(4,0,0) | $\psi(\text{2nd } \lambda\alpha.(\alpha\cdot 2)-\Pi_0)$<br>$\psi(\lambda\alpha.(\alpha\cdot 2)-\Pi_0 \text{ aft } \lambda\alpha.(\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(2,1,1) | $\psi(1-(\lambda\alpha.(\alpha\cdot 2)-\ _0))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(2,1,1)(3,1,1) | $\psi(1-2\ 1-(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(2,1,1)(3,2,0) | $\psi(\lambda\alpha.(\alpha+1)-\Pi_0\ 1-(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))$<br>$\psi(\Pi_\omega\ \Pi_1-\Pi_{1,0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(2,1,1)(3,2,0)(4,2,0)(5,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha\cdot 2)-\Pi_0\ 1-(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(2,1,1)(3,2,0)(4,2,0)(5,1,0)(3,1,1) | $\psi(1-2-(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(2,2,0) | $\psi((\lambda\alpha.(\alpha+1)-\Pi_0)-(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))$<br>$\psi(\Pi_\omega-\Pi_{1,0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(2,2,0)(3,1,1) | $\psi(1-(\lambda\alpha.(\alpha+1)-\Pi_1)\ (\lambda\alpha.(\alpha+1)-\Pi_0)$<br>$(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(2,2,0)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,0)(2,0,0) | $\psi((\lambda\alpha.(\alpha\cdot 2)-\Pi_0)(\lambda\alpha.(\alpha+1)-\Pi_0)$<br>$-(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(2,2,0)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,0)(3,1,1) | $\psi(1-(\lambda\alpha.(\alpha+1)-\Pi_1)$<br>$-(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(2,2,0)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,0)(4,2,0) | $\psi((\lambda\alpha.(\alpha+2)-\Pi_0)$<br>$-(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(2,2,0)(3,2,0) | $\psi((\lambda\alpha.(\alpha+\omega)-\Pi_0)-(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(2,2,0)(3,2,0)(3,2,0) | $\psi((\lambda\alpha.(\alpha+\omega^2)-\Pi_0)-(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(2,2,0)(3,2,0)(4,1,0) | $\psi((\lambda\alpha.(\alpha+\Omega)-\Pi_0)-(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(2,2,0)(3,2,0)(4,1,0)<br>(1,1,1)(2,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi((\lambda\alpha.(\alpha+\lambda\alpha.(\alpha\cdot 2)-\Pi_0)-\Pi_0)-(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(2,2,0)(3,2,0)(4,1,0)(1,1,1)<br>(2,2,0)(3,2,0)(4,1,0)(2,2,0)<br>(3,2,0)(4,1,0)(2,0,0) | $\psi((\lambda\alpha.(\alpha+(\lambda\alpha.(\alpha+\Omega)-\Pi_0)$<br>$-(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))-\Pi_0)-(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(2,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi((\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$<br>$-(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(3,0,0) | $\psi((\lambda\alpha.(\alpha\cdot 2)-\Pi_0-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,1,0)(2,0,0) | $\psi((\lambda\alpha.(\alpha\cdot 2)-\Pi_0-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(3,1,1) | $\psi(1-(\lambda\alpha.(\alpha\cdot 2)-\Pi_1))$<br>$\psi(\Pi_1-\Pi_{1,1})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,1,1)(4,1,1) | $\psi(1-(\lambda\alpha.(\alpha\cdot 2)-\Pi_2))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\alpha\cdot 2+1)-\Pi_0)$<br>$\psi(\Pi_{1,\omega})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,1,1)(4,2,0)(5,1,1)(6,2,0) | $\psi(\lambda\alpha.(\alpha\cdot 2+2)-\Pi_0)$<br>$\psi(\Pi_{1,\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,1,1)(4,2,0)(5,2,0) | $\psi(\lambda\alpha.(\alpha\cdot 2+\omega)-\Pi_0)$<br>$\psi(\Pi_{1,\omega^2})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(5,2,0) | $\psi(\lambda\alpha.(\alpha\cdot 2+\omega^2)-\Pi_0)$<br>$\psi(\Pi_{1,\omega^3})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0) | $\psi(\lambda\alpha.(\alpha\cdot 2+\Omega)-\Pi_0)$<br>$\psi(\Pi_{1,\Omega})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(1,1,1)<br>(2,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha\cdot 2+\lambda\alpha.(\alpha\cdot 2)-\Pi_0)-\Pi_0)$<br>$\psi(\Pi_{1,\Pi_{1,0}})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(1,1,1)<br>(2,2,0)(3,2,0)(4,1,0)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,0)(1,1,1)<br>(2,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha\cdot 2+\lambda\alpha.(\alpha\cdot 2$<br>$+\lambda\alpha.(\alpha\cdot 2)-\Pi_0)-\Pi_0)-\Pi_0)$<br>$\psi(\Pi_{1,\Pi_{1,\Pi_{1,0}}})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha\cdot 3)-\Pi_0)$<br>$\psi(\Pi_{2,0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)<br>(4,2,0)(5,2,0)(6,1,0)(2,0,0) | $\psi((\lambda\alpha.(\alpha\cdot 3)-\Pi_0)-(\lambda\alpha.(\alpha\cdot 3)-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(5,1,1) | $\psi(1-(\lambda\alpha.(\alpha\cdot 3)-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,1,1)(4,2,0)(5,2,0)<br>(6,1,0)(5,1,1)(6,2,0) | $\psi(\lambda\alpha.(\alpha\cdot 3+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,1,1)(4,2,0)(5,2,0)(6,1,0)<br>(5,1,1)(6,2,0)(7,2,0)(8,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha\cdot 4)-\Pi_0)$<br>$\psi(\Pi_{3,0})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(3,2,0) | $\psi(\lambda\alpha.(\alpha\cdot\omega)-\Pi_0)$<br>$\psi(\Pi_{\omega,0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(3,1,1) | $\psi(1-(\lambda\alpha.(\alpha\cdot\omega)-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\alpha\cdot\omega+1)-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha\cdot\omega+\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,0)(5,2,0) | $\psi(\lambda\alpha.(\alpha\cdot\omega\cdot 2)-{}_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(3,2,0) | $\psi(\lambda\alpha.(\alpha\cdot\omega^2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,0,0) | $\psi(\lambda\alpha.(\alpha\cdot\omega^\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0) | $\psi(\lambda\alpha.(\alpha\cdot\Omega)-\Pi_0)$<br>$\psi(\Pi_{\Pi_2,0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(4,1,0)(1,1,1)(2,2,0) | $\psi(\lambda\alpha.(\alpha\cdot\lambda\alpha.(\alpha+1)-\Pi_0)-\Pi_0)$<br>$\psi(\Pi_{\Pi_\omega,0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0)(1,1,1)<br>(2,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha\cdot\lambda\alpha.(\alpha\cdot 2)-\Pi_0)-\Pi_0)$<br>$\psi(\Pi_{\Pi_{\Pi_{1,0},0},0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(4,1,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0)(1,1,1)<br>(2,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha\cdot\lambda\alpha.(\alpha\cdot\lambda\alpha.(\alpha\cdot 2)-\Pi_0)-\Pi_0)-\Pi_0)$<br>$\psi(\Pi_{\Pi_{\Pi_{\Pi_{1,0},0},0},0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha^2)-\Pi_0)$<br>$\psi(\Pi_{1,0,0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0)(3,1,1) | $\psi(1-(\lambda\alpha.(\alpha^2)-\Pi_1))$<br>$\psi(\Pi_1-\Pi_{1,0,1})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(4,1,0)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\alpha^2+1)-\Pi_0)$<br>$\psi(\Pi_{1,0,\omega})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha^2+\alpha)-\Pi_0)$<br>$\psi(\Pi_{1,1,0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,0)(5,2,0)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha^2\cdot 2)-\Pi_0)$<br>$\psi(\Pi_{2,0,0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0)(3,2,0) | $\psi(\lambda\alpha.(\alpha^2\cdot\omega)-\Pi_0)$<br>$\psi(\Pi_{\omega,0,0})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(4,1,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha^3)-\Pi_0)$<br>$\psi(\Pi_{1,0,0,0})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(4,0,0) | $\psi(\lambda\alpha.(\alpha^\omega)-\Pi_0)$<br>$\psi(\Pi_{1@\omega})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha^\alpha)-\Pi_0)$<br>$\psi(\Pi_{1@(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(4,1,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha^{\alpha+1})-\Pi_0)$<br>$\psi(\Pi_{1@(1,1)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,1,0)(3,2,0)(4,1,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha^{\alpha\cdot 2})-\Pi_0)$<br>$\psi(\Pi_{1@(2,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(4,1,0)(4,0,0) | $\psi(\lambda\alpha.(\alpha^{\alpha\cdot\omega})-\Pi_0)$<br>$\psi(\Pi_{1@(\omega,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(4,1,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha^{\alpha^2})-\Pi_0)$<br>$\psi(\Pi_{1@(1,0,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha^{\alpha^\alpha})-\Pi_0)$<br>$\psi(\Pi_{1@(1@(1,0))})$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,1,0)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha^{\alpha^{\alpha^{\alpha}}})-\Pi_0)$<br>$\psi(\Pi_{1@(1@(1@(1,0)))})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(5,2,0) | $\psi(\lambda\alpha.(\varepsilon_{\alpha+1})-\Pi_0)$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0))-\Pi_0)$<br>$\psi(\Pi_{1@(1@(1@(\ldots)))})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0)(3,1,1) | $\psi(1-(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0))-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0)+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0)+\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,0)(7,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0)\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0)(3,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0)\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0)\cdot\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,0)(3,2,0)(4,1,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0)^2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0)(4,1,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0)^{\psi_{\Omega_{\alpha+1}}(0)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0)(6,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\omega))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0)(6,1,0)(7,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(0)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}))-\Pi_0)$<br>$\psi(\lambda\alpha.(\zeta_{\alpha+1})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0)(6,2,0)(7,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}}))-\Pi_0)$<br>$\psi(\lambda\alpha.(\Gamma_{\alpha+1})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0)(6,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+2}}(0)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0)(6,3,0)(7,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+2}))-\Pi_0)$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(2\ aft\ \Omega_{\alpha+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(5,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+\omega}))-\Pi_0)$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(1-2\ aft\ \Omega_{\alpha+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,1)(6,2,1)(7,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(I_{\alpha+\omega}))-\Pi_0)$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(1-2\ 1-2\ \text{aft}\ \Omega_{\alpha+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(K_{\alpha+\omega}))-\Pi_0)$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(1-3\ \text{aft}\ \Omega_{\alpha+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,1)(6,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\beta+1)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,3,0)(7,3,0)(8,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\beta+\alpha)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,3,0)(7,3,0)(8,1,0)(9,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\beta+\psi_{\Omega_{\alpha+1}}(0))-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,3,0)(7,3,0)(8,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\beta+\Omega_{\alpha+1})-\Pi_0))-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,3,0)(7,3,0)(8,2,0)(5,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\beta+\Omega_{\alpha+\omega})-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,3,0)(7,3,0)(8,2,0)(6,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\beta\cdot 2)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,1)(6,3,0)(7,3,0)(8,2,0)<br>(6,3,0)(7,3,0)(8,2,0)(6,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\beta\cdot 3)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,3,0)(7,3,0)(8,2,0)<br>(7,3,0)(8,2,0)(6,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\beta^2)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,1)(6,3,0)(7,3,0)<br>(8,2,0)(8,2,0)(6,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\beta^\beta)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,3,0)(7,3,0)(8,2,0)(9,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\psi_{\Omega_{\beta+1}}(0))-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,1)(6,3,0)(7,3,0)<br>(8,2,0)(9,3,0)(10,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\Omega_{\beta+1}))-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,3,0)(7,3,0)(8,2,0)(9,3,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\Omega_{\beta+\omega}))-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,1)(6,3,0)(7,3,0)<br>(8,2,0)(9,3,1)(10,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\gamma.(\gamma+1)-\Pi_0))-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,3,0)(7,3,0)(8,2,0)(9,3,1)<br>(10,4,0)(11,4,0)(12,3,0)(10,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\gamma.(\gamma\cdot 2)-\Pi_0))-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1)$<br>$\psi(\lambda\alpha.(2 \text{ aft } \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,1)(2,1,1) | $\psi(1-(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,1)(2,2,0) | $\psi((\lambda\alpha.(\alpha+1)-\Pi_0)-(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(2,2,0)(3,2,0) | $\psi((\lambda\alpha.(\alpha+\omega)-\Pi_0)-(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(2,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi((\lambda\alpha.(\alpha\cdot 2)-\Pi_0)-(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(2,2,0)(3,2,0)(4,1,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1)-(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,1)(3,0,0) | $\psi((\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1-)^\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,0)(2,0,0) | $\psi((\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,1)(3,1,1) | $\psi(1-(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_2))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(4,1,1) | $\psi(1-(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_3))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(4,2,0)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(5,1,1) | $\psi(1-(\lambda\alpha.(\Omega_{\alpha+1}+\alpha\cdot 2)-\Pi_1))$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(5,1,1)<br>(6,2,0)(7,2,0)(8,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\alpha\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(4,2,0)(5,2,0)<br>(6,1,0)(5,2,0)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\alpha^2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(4,2,0)(5,2,0)<br>(6,1,0)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\alpha^\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(7,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(4,2,0)(5,2,0)<br>(6,1,0)(7,2,0)(7,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(4,2,0)(5,2,0)<br>(6,1,0)(7,2,0)(8,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(7,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+\omega}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(4,2,0)(5,2,0)<br>(6,1,0)(7,2,1)(8,3,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\beta+1)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(4,2,0)(5,2,0)(6,1,0)<br>(7,2,1)(8,3,0)(9,3,0)(10,2,0)(8,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\beta\cdot 2)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)<br>(7,2,1)(8,3,0)(9,3,0)(10,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\Omega_{\beta+1})-\Pi_1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)<br>(7,2,1)(8,3,0)(9,3,0)(10,2,1)(9,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(1-(\lambda\beta.(\Omega_{\beta+1})-\Pi_2)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(7,2,1)<br>(8,3,0)(9,3,0)(10,2,1)(9,2,1)<br>(10,3,0)(11,3,0)(12,2,0)(8,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\Omega_{\beta+1}+\beta)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(7,2,1)<br>(8,3,0)(9,3,0)(10,2,1)(9,2,1)<br>(10,3,0)(11,3,0)(12,2,0)(13,3,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\Omega_{\beta+1}$<br>$+\psi_{\Omega_{\beta+1}}(0))-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,1)<br>(5,1,1)(6,2,0)(7,2,0)(8,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot 3)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,1)(3,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,2,0)(4,1,0)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\psi_{\Omega_{\alpha+1}}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,2,0)(4,1,0)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,2,0)(4,1,0)(5,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+\omega}))-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,2,0)(4,1,0)(5,2,1)(6,3,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\beta+1)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,2,0)(4,1,0)<br>(5,2,1)(6,3,0)(7,3,0)(8,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\Omega_{\beta+1})-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,2,0)(4,1,0)(5,2,1)<br>(6,3,0)(7,3,0)(8,2,1)(7,3,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\Omega_{\beta+1}\cdot\omega)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,2,0)(4,1,0)(5,2,1)(6,3,0)(7,3,0)<br>(8,2,1)(7,3,0)(8,2,0)(9,3,1)(10,4,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\Omega_{\beta+1}$<br>$\cdot\psi_{\Omega_{\beta+1}}(\lambda\gamma.(\gamma+1)-\Pi_0))-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.({\Omega_{\alpha+1}}^2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,2,0)(4,1,1)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.({\Omega_{\alpha+1}}^3)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,1)(4,0,0) | $\psi(\lambda\alpha.({\Omega_{\alpha+1}}^\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.({\Omega_{\alpha+1}}^\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,1)(4,1,0)(5,2,0) | $\psi(\lambda\alpha.({\Omega_{\alpha+1}}^{\psi_{\Omega_{\alpha+1}}(0)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,1)(4,1,1) | $\psi(\lambda\alpha.({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(4,1,1)(4,1,1) | $\psi(\lambda\alpha.({\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,0,0) | $\psi(\lambda\alpha.({\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^\omega})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,1,1) | $\psi(\lambda\alpha.({\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{\Omega_{\alpha+1}}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(5,2,0)(6,2,0)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}\cdot 2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,2,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}({\Omega_{\alpha+2}}^2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(5,2,0)(6,2,0)(7,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}({\Omega_{\alpha+2}}^\alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,2,0)(7,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}({\Omega_{\alpha+2}}^{\Omega_{\alpha+1}}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(5,2,0)(6,2,0)(7,1,1)(8,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}({\Omega_{\alpha+2}}^{\psi_{\Omega_{\alpha+2}}(0)}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,2,0)(7,1,1)<br>(8,2,0)(9,2,0)(10,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}({\Omega_{\alpha+2}}^{\psi_{\Omega_{\alpha+2}}({\Omega_{\alpha+2}}^{\Omega_{\alpha+1}})}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}({\Omega_{\alpha+2}}^{\Omega_{\alpha+2}}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,2,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}({\Omega_{\alpha+2}}^{{\Omega_{\alpha+2}}^{\Omega_{\alpha+2}}}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(0)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,0)(4,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+3}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+\omega}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,3,1)(4,3,1)(5,3,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(I_{\alpha+\omega}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)(4,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\beta.(\beta+1)-\Pi_0))-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,4,0)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\beta.(\beta+\alpha)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,4,0)(6,3,0)(4,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\beta.(\beta\cdot 2)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,4,0)(6,3,0)(7,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\beta.(\psi_{\Omega_{\beta+1}}(0))-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,4,0)(6,3,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\beta.(\Omega_{\beta+1})-\Pi_1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,4,0)(6,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\beta.({\Omega_{\beta+1}}^{\Omega_{\beta+1}})-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,3,1)(4,4,0)(5,5,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\beta.(\psi_{\Omega_{\beta+2}}(0))-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,5,1)(6,6,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\lambda\gamma.(\gamma+1)-\Pi_0))-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1) | $\psi(1-(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0) | $\psi((\lambda\alpha.(\alpha+1)-\Pi_0)-(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,0)(2,0,0) | $\psi((\lambda\alpha.(\alpha\cdot 2)-\Pi_0)-(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(2,2,0)(3,2,0)(4,1,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1)-(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,0) | $\psi((\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(0))-\Pi_1)-(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,2,0)(7,2,0) | $\psi((\lambda\alpha.(\psi_{\Omega_{\alpha+2}}({\Omega_{\alpha+2}}^{\Omega_{\alpha+2}}))-\Pi_1)-(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,0)(6,3,1) | $\psi((\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+\omega}))-\Pi_1)-(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)-(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)-(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$<br>$-(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,0)(2,0,0) | $\psi((\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,1,1) | $\psi(1-(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_2))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,0)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,0)(7,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+1}}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\Omega_{\alpha+1})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,1)(7,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,0)(8,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}))-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,1,1)<br>(4,2,0)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}({\Omega_{\alpha+2}}^{\Omega_{\alpha+2}}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,0)(5,3,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(0)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,0)<br>(5,3,1)(6,4,0)(7,4,0)(8,3,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\lambda\beta.(\Omega_{\beta+1})-\Pi_1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,1,1)<br>(4,2,0)(5,3,1)(6,4,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\lambda\beta.(\Omega_{\beta+2})-\Pi_1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,1,1)<br>(4,2,0)(5,3,1)(6,4,1)(6,4,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}((\lambda\beta.(\beta+1)-\Pi_0)$<br>$-(\lambda\beta.(\Omega_{\beta+2})-\Pi_1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,1,1)(4,2,0)<br>(5,3,1)(6,4,1)(6,4,0)(7,4,0)(8,3,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}((\lambda\beta.(\Omega_{\beta+1})-\Pi_1)$<br>$-(\lambda\beta.(\Omega_{\beta+2})-\Pi_1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,0)(5,3,1)<br>(6,4,1)(6,4,0)(7,4,0)(8,3,1)(9,4,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}((\lambda\beta.(\Omega_{\beta+2})-\Pi_1)$<br>$-(\lambda\beta.(\Omega_{\beta+2})-\Pi_1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,0)<br>(5,3,1)(6,4,1)(6,4,0)(7,4,0)<br>(8,3,1)(9,4,1)(7,3,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(1-(\lambda\beta.(\Omega_{\beta+2})-\Pi_2)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,1,1)(6,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot 3)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(3,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega^2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(4,1,0)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\psi_{\Omega_{\alpha+1}}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\Omega_{\alpha+1})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(4,1,1)(5,2,1) | $\psi(\lambda\alpha.({\Omega_{\alpha+2}}^2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.({\Omega_{\alpha+2}}^{\alpha})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(4,1,1)(5,2,1) | $\psi(\lambda\alpha.({\Omega_{\alpha+2}}^{\Omega_{\alpha+2}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,1,1)(6,2,1) | $\psi(\lambda\alpha.({\Omega_{\alpha+2}}^{{\Omega_{\alpha+2}}^{\Omega_{\alpha+2}}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,0)<br>(6,2,0)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3}\cdot 2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,2,0)(6,2,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}({\Omega_{\alpha+3}}^2))-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,0)<br>(6,2,0)(7,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}({\Omega_{\alpha+3}}^{\alpha})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,0)<br>(6,2,0)(7,1,1)(8,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}({\Omega_{\alpha+3}}^{\Omega_{\alpha+2}})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,2,0)(6,2,0)(7,1,1)<br>(8,2,1)(8,2,0)(9,2,0)(10,1,1)(11,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3}^{\psi_{\Omega_{\alpha+3}}({\Omega_{\alpha+3}}^{\Omega_{\alpha+2}})})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(2,2,0)(3,2,0)(4,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}({\Omega_{\alpha+3}}^{\Omega_{\alpha+3}})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+4}}(0))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(2,2,0)(3,3,1)(4,4,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\lambda\beta.(\Omega_{\beta+2}) - \Pi_0)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+3}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+4}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,1,1) | $\psi(1 - (\lambda\alpha.(\Omega_{\alpha+\omega}) - \Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0) | $\psi((\lambda\alpha.(\alpha+1) - \Pi_0) - (\lambda\alpha.(\Omega_{\alpha+\omega}) - \Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi((\lambda\alpha.(\alpha\cdot 2) - \Pi_0) - (\lambda\alpha.(\Omega_{\alpha+\omega}) - \Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+1}) - \Pi_1) - (\lambda\alpha.(\Omega_{\alpha+\omega}) - \Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1) - (\lambda\alpha.(\Omega_{\alpha+\omega}) - \Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)(6,0,0) | $\psi((\lambda\alpha.(\Omega_{\alpha+\omega}) - \Pi_0) - (\lambda\alpha.(\Omega_{\alpha+\omega}) - \Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)<br>(6,0,0)(3,1,0)(2,0,0) | $\psi((\lambda\alpha.(\Omega_{\alpha+\omega}) - \Pi_0-)^{1,0})$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,0,0)(3,1,1) | $\psi(1 - (\lambda\alpha.(\Omega_{\alpha+\omega}) - \Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)<br>(6,0,0)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega} + 1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)(6,0,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega} + \alpha) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)(6,0,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega} + \Omega_{\alpha+1}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)<br>(6,0,0)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega} + \Omega_{\alpha+2}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,0,0)<br>(3,1,1)(4,2,1)(5,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}\cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,0,0)(3,1,1)<br>(4,2,1)(5,0,0)(4,2,0)(5,2,0)(6,1,1)<br>(7,2,1)(8,0,0)(5,1,1)(6,2,1)(7,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}\cdot 3) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,0,0)(3,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}\cdot\omega) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)<br>(6,0,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}\cdot\alpha) - \Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,1)<br>(5,2,1)(6,0,0)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}\cdot\Omega_{\alpha+1})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)<br>(6,0,0)(3,2,0)(4,1,1)(5,2,1)(6,0,0) | $\psi(\lambda\alpha.({\Omega_{\alpha+\omega}}^2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)<br>(6,0,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.({\Omega_{\alpha+\omega}}^\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,0,0)(4,1,1) | $\psi(\lambda\alpha.({\Omega_{\alpha+\omega}}^{\Omega_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)<br>(6,0,0)(4,1,1)(5,2,1)(6,0,0) | $\psi(\lambda\alpha.({\Omega_{\alpha+\omega}}^{\Omega_{\alpha+\omega}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,0,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+\omega+1}}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)<br>(6,0,0)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+\omega+1}}(\Omega_{\alpha+\omega+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,0,0)(5,2,0)<br>(6,2,0)(7,1,1)(8,2,1)(9,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+\omega+1}}({\Omega_{\alpha+\omega+1}}^{\Omega_{\alpha+\omega}}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+\omega+1}}({\Omega_{\alpha+\omega+1}}^{\Omega_{\alpha+\omega+1}}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,0,0)(2,2,0)(3,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+\omega+1}}(\psi_{\Omega_{\alpha+\omega+2}}(0)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,3,1)(4,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+\omega+1}}(\lambda\beta.(\beta+1)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega+1})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,0,0)(2,2,1)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,0,0)(2,2,1)(3,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega\cdot 2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\Omega})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot 2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,1,1) | $\psi(1-(\lambda\alpha.(\Omega_{\alpha\cdot 2})-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,0) | $\psi((\lambda\alpha.(\alpha+1)-\Pi_0)-(\lambda\alpha.(\Omega_{\alpha\cdot 2})-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,1,0)(2,0,0) | $\psi((\lambda\alpha.(\Omega_{\alpha\cdot 2})-\Pi_0)-(\lambda\alpha.(\Omega_{\alpha\cdot 2})-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,1,0)(3,1,1) | $\psi(1-(\lambda\alpha.(\Omega_{\alpha\cdot 2})-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,1,0)(3,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot 2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)<br>(6,1,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot 2}\cdot\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)<br>(6,1,0)(4,1,1)(5,2,1)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.({\Omega_{\alpha\cdot 2}}^2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,1,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(2,2,0)(3,2,0)(4,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}({\Omega_{\alpha\cdot 2+1}}^{\Omega_{\alpha\cdot 2+1}}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(2,2,0)(3,3,1)(4,4,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}(\lambda\alpha'.(\Omega_{\alpha'+2})-\Pi_1))-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(2,2,0)(3,3,1)(4,4,1)(5,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}(\lambda\alpha'.(\Omega_{\alpha'+\alpha})-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(2,2,0)(3,3,1)(4,4,1)(5,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}(\lambda\alpha'.(\Omega_{\alpha'+\Omega_{\alpha+1}})-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(2,2,0)(3,3,1)(4,4,1)(5,1,1)<br>(6,2,1)(7,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}(\lambda\alpha'.(\Omega_{\alpha'+\Omega_{\alpha\cdot 2}})-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(2,2,0)(3,3,1)(4,4,1)(5,1,1)<br>(6,2,1)(7,1,0)(6,2,0)(7,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}(\lambda\alpha'.(\Omega_{\alpha'+\psi_{\Omega_{\alpha\cdot 2+1}}(0)})-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(2,2,0)(3,3,1)(4,4,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}(\lambda\alpha'.(\Omega_{\alpha'+\Omega_{\alpha\cdot 2+1}})-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,0)<br>(3,3,1)(4,4,1)(5,2,0)(6,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}(\lambda\alpha'.(\Omega_{\alpha'+\psi_{\Omega_{\alpha\cdot 2+2}}(0)})-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(2,2,0)(3,3,1)(4,4,1)(5,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}(\lambda\alpha'.(\Omega_{\alpha'+\Omega_{\alpha\cdot 2+2}})-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,0)<br>(3,3,1)(4,4,1)(5,3,0)(3,3,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}(\lambda\alpha'.(\Omega_{\alpha'+\Omega_{\alpha\cdot 2+\omega}})-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,0)<br>(3,3,1)(4,4,1)(5,3,0)(4,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}(\lambda\alpha'.(\Omega_{\alpha'\cdot 2})-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot 2+1})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,1,0)(2,2,1)(3,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot 2+\omega})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(2,2,1)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot 3})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(3,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot\omega})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha^2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,1,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha^\alpha})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(0)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,1,0)(4,2,1)(5,3,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\beta+1)-\Pi_0)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,1,0)(4,2,1)(5,3,1) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\Omega_{\beta+1})-\Pi_1)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(4,2,1)(5,3,1)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\Omega_{\beta+\alpha})-\Pi_0)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(4,2,1)(5,3,1)(6,2,0)(5,0,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\Omega_{\beta\cdot 2})-\Pi_0)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(4,2,1)(5,3,1)(6,2,0)(7,3,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\Omega_{\psi_{\Omega_{\beta+1}}(0)})-\Pi_0)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(4,2,1)<br>(5,3,1)(6,2,0)(7,3,1)(8,4,1) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\beta.(\Omega_{\psi_{\Omega_{\beta+1}}(\lambda\gamma.(\Omega_{\gamma+1})-\Pi_1)})-\Pi_0)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}+1})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(2,2,1)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}+\alpha})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(2,2,1)(3,1,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}\cdot 2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}\cdot\alpha})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(3,1,1) | $\psi(\lambda\alpha.(\Omega_{\Omega^2_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+2}}(0)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,0)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2})})-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,0)(5,3,1)(6,4,1) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+2}}(\lambda\beta.(\Omega_{\beta+1})-\Pi_1)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,0)(5,3,1)(6,4,1)(7,3,1) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+2}}(\lambda\beta.(\Omega_{\Omega_{\beta+1}})-\Pi_0)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,0)<br>(5,3,1)(6,4,1)(7,3,1)(8,4,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+2}}(\lambda\beta.(\Omega_{\psi_{\Omega_{\beta+1}}(0)})-\Pi_0)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+2}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+3}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,0,0) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+\omega}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha\cdot 2}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,1,0)(6,2,0) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\psi_{\Omega_{\alpha+1}}(0)}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,1,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\Omega_{\alpha+1}}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,1,1)(6,2,1)(7,1,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\Omega_{\Omega_{\alpha+1}}}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)+1})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(2,2,1)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)+\omega})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(2,2,1)(3,1,1) | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)+\Omega_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(2,2,1)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)+\Omega_{\alpha+2}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)+\Omega_{\alpha\cdot 2}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,1,1) | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)+\Omega_{\Omega_{\alpha+1}}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)\cdot 2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)\cdot 3})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)\cdot\alpha})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(3,1,1) | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)\cdot\Omega_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1)(4,2,1)<br>(5,2,0)(3,1,1)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)^2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\psi_{I_{\alpha+1}}(0)+1}}(0)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\psi_{I_{\alpha+1}}(0)+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,1,1)(6,2,1)(7,2,0)(6,2,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\Omega_{\psi_{I_{\alpha+1}}(0)+1}}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(2,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,2,0)(2,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,0,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\omega))-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\alpha)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\Omega_{\alpha+1})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,1,1)(4,2,1)(5,0,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\Omega_{\alpha+\omega})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,1,1)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\psi_{I_{\alpha+1}}(0))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,2,0)(2,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+1)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,2,0)(2,2,1)(3,2,0)(3,1,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\Omega_{\alpha+1})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,2,0)(2,2,1)(3,2,0)(3,1,1)<br>(4,2,1)(5,2,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\psi_{I_{\alpha+1}}(I_{\alpha+1}))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,2,0)(2,2,1)(3,2,0)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot 2)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,2,0)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot \alpha)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(3,2,0)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}({I_{\alpha+1}}^2)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}({I_{\alpha+1}}^{\Omega_{\alpha+1}})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}({I_{\alpha+1}}^{I_{\alpha+1}})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,3,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\psi_{\Omega_{I_{\alpha+1}}+1}(0))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(4,3,0)(5,3,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\Omega_{I_{\alpha+1}+1})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,3,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\Omega_{I_{\alpha+1}+\omega})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,3,1)(6,3,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+\omega})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(4,3,1)(5,4,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\lambda\beta.(\beta+1) - \Pi_0)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,4,1)(6,4,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\lambda\beta.(\psi_{I_{\beta+1}}(0)) - \Pi_0)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1) | $\psi(\lambda\alpha.(I_{\alpha+1}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}+1}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}+\psi_{I_{\alpha+1}}(0)}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}\cdot 2}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,1)(4,1,1) | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}\cdot\Omega_{\alpha+1}}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,1)<br>(4,1,1)(5,2,1)(6,2,1) | $\psi(\lambda\alpha.(\Omega_{I^2_{\alpha+1}}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,1)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{I_{\alpha+1}}}(0)}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{I_{\alpha+1}}}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,1,1)(6,2,1)(7,2,1)(6,2,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\Omega_{I_{\alpha+1}}}}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+2}}(0)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,0)(2,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+2}}(1)) - \Pi_0)$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,0)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+2}}(I_{\alpha+2}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,0)(4,3,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+2}}(\psi_{\Omega_{I_{\alpha+2}+1}}(0)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(4,3,1)(5,3,1)(6,3,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+2}}(I_{\alpha+\omega}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,2,1) | $\psi(\lambda\alpha.(I_{\alpha+2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,1)(2,2,1)(3,2,1) | $\psi(\lambda\alpha.(I_{\alpha+3})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(I_{\alpha\cdot 2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1) | $\psi(\lambda\alpha.(I_{\Omega_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(I_{\Omega_{\alpha+2}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,1,1)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.(I_{\psi_{I_{\alpha+1}}(0)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,1,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(I_{I_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(I_{I_{\alpha+2}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1)<br>(4,2,1)(5,2,1)(5,1,1)(6,2,1)(7,2,1) | $\psi(\lambda\alpha.(I_{I_{I_{\alpha+1}}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I(1,\alpha+1)}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,2,0)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\psi_{I(1,\alpha+1)}(0)+1})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,0)(2,2,1)(3,2,1) | $\psi(\lambda\alpha.(I_{\psi_{I(1,\alpha+1)}(0)+1})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,0)(2,2,1)(3,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I(1,\alpha+1)}(1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,2,0)(3,0,0) | $\psi(\lambda\alpha.(\psi_{I(1,\alpha+1)}(\omega))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,2,0)(3,1,1) | $\psi(\lambda\alpha.(\psi_{I(1,\alpha+1)}(\Omega_{\alpha+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(3,1,1)(4,2,1)(5,2,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{I(1,\alpha+1)}(\psi_{I(1,\alpha+1)}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,2,0)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I(1,\alpha+1)}(I(1,\alpha+1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,2,0)(4,3,0) | $\psi(\lambda\alpha.(\psi_{I(1,\alpha+1)}(\psi_{\Omega_{I(1,\alpha+1)+1}}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1) | $\psi(\lambda\alpha.(I(1,\alpha+1))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,1)(2,2,1)(3,2,1)(3,2,1) | $\psi(\lambda\alpha.(I(1,\alpha+2))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,1)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(I(1,\alpha\cdot 2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,2,1)(3,1,1) | $\psi(\lambda\alpha.(I(1,\Omega_{\alpha+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,1)(3,1,1)(4,2,1)(5,2,1)(5,2,1) | $\psi(\lambda\alpha.(I(1,I(1,\alpha+1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I(2,\alpha+1)}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,2,1)(3,2,1) | $\psi(\lambda\alpha.(I(2,\alpha+1))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,0,0) | $\psi(\lambda\alpha.(I(\omega,\alpha+1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(I(\alpha,1))-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,1,0)(3,2,1) | $\psi(\lambda\alpha.(I(\alpha+1,0))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,1,0)(3,2,1)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(I(\alpha\cdot 2,0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,1,0)(5,2,0) | $\psi(\lambda\alpha.(I(\psi_{\Omega_{\alpha+1}}(0),0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,1) | $\psi(\lambda\alpha.(I(\Omega_{\alpha+1},0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,1,1)(5,2,1)(6,2,1)(7,1,1) | $\psi(\lambda\alpha.(I(I(\Omega_{\alpha+1},0),0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0) | $\psi(\lambda\alpha.(\psi_{I(1,0,\alpha+1)}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(2,2,1)(3,2,1) | $\psi(\lambda\alpha.(I_{\psi_{I(1,0,\alpha+1)}(0)+1})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(2,2,1)(3,2,1)(3,2,1) | $\psi(\lambda\alpha.(I(1,\psi_{I(1,0,\alpha+1)}(0)+1))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(I(\alpha,\psi_{I(1,0,\alpha+1)}(0)+1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(2,2,1)(3,2,1)(4,1,1) | $\psi(\lambda\alpha.(I(\Omega_{\alpha+1},\psi_{I(1,0,\alpha+1)}(0)+1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(2,2,1)(3,2,1)(4,1,1)<br>(5,2,1)(6,2,1)(7,2,0) | $\psi(\lambda\alpha.(I(\psi_{I(1,0,\alpha+1)}(0),1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(2,2,1)(3,2,1)(4,2,0) | $\psi(\lambda\alpha.(\psi_{I(1,0,\alpha+1)}(1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{I(1,0,\alpha+1)}(\alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(3,1,1)(4,2,1)(5,2,1)(6,2,0) | $\psi(\lambda\alpha.(\psi_{I(1,0,\alpha+1)}(\psi_{I(1,0,\alpha+1)}(0)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,0)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I(1,0,\alpha+1)}(I(1,0,\alpha+1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,0)(3,2,1) | $\psi(\lambda\alpha.(I(1,0,\alpha+1))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,0)(3,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I(1,1,\alpha+1)}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(3,2,1)(3,2,1) | $\psi(\lambda\alpha.(I(1,1,\alpha+1))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(3,2,1)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(I(1,\alpha,1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(3,2,1)(4,2,0) | $\psi(\lambda\alpha.(\psi_{I(2,0,\alpha+1)}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(3,2,1)(4,2,0)(3,2,1) | $\psi(\lambda\alpha.(I(2,0,\alpha+1))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,0)(4,0,0) | $\psi(\lambda\alpha.(I(\omega,0,\alpha+1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(I(\alpha,0,1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,0)(4,2,0) | $\psi(\lambda\alpha.(\psi_{I(1,0,0,\alpha+1)}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(4,2,0)(3,2,1) | $\psi(\lambda\alpha.(I(1,0,0,\alpha+1))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(4,2,0)(3,2,1)(4,2,0)(4,2,0)(3,2,1) | $\psi(\lambda\alpha.(I(2,0,0,\alpha+1))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(4,2,0)(4,0,0) | $\psi(\lambda\alpha.(I(\omega,0,0,\alpha+1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(4,2,0)(4,2,0)(3,2,1) | $\psi(\lambda\alpha.(I(1,0,0,0,\alpha+1))-\Pi_1)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,0)(5,0,0) | $\psi(\lambda\alpha.(I(1@\omega,\alpha+1@0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(5,1,0)(2,0,0) | $\psi(\lambda\alpha.(I(1@\alpha,1@0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,0)(5,2,0) | $\psi(\lambda\alpha.(I(1@(1,0),\alpha+1@0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,0)(5,3,0) | $\psi(\lambda\alpha.(\psi_{M_{\alpha+1}}(\varepsilon_{M_{\alpha+1}}))-\Pi_0)$<br>$\psi(\lambda\alpha.(\psi_{M_{\alpha+1}}(\psi_{\Omega_{M_{\alpha+1}+1}}(0)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(5,3,0)(6,3,0) | $\psi(\lambda\alpha.(\psi_{M_{\alpha+1}}(\Omega_{M_{\alpha+1}+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(5,3,1)(6,3,1)(6,3,1) | $\psi(\lambda\alpha.(\psi_{M_{\alpha+1}}(M_{\alpha+\omega}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,1)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{M_{\alpha+1}+1})-\Pi_1)$<br>$\psi(\lambda\alpha.(2\text{ aft }2-2\text{ aft }\alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(2,2,1)(3,2,1) | $\psi(\lambda\alpha.(I_{M_{\alpha+1}+1})-\Pi_1)$<br>$\psi(\lambda\alpha.(2\ 1-2\text{ aft }2-2\text{ aft }\alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(2,2,1)(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(M_{\alpha+2})-\Pi_1)$<br>$\psi(\lambda\alpha.(2\text{nd }2-2\text{ aft }\alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,1)(3,0,0) | $\psi(\lambda\alpha.(M_{\alpha+\omega})-\Pi_0)$<br>$\psi(\lambda\alpha.(1-2-2\text{ aft }\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,1)(3,1,1) | $\psi(\lambda\alpha.(M_{\Omega_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,1,1)(4,2,1)(5,2,1)(6,2,1) | $\psi(\lambda\alpha.(M_{M_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{M(1,\alpha+1)}(0))-\Pi_0)$<br>$\psi(\lambda\alpha.((1-)^{1,0}\ 2-2\text{ aft }\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,1)(3,2,1) | $\psi(\lambda\alpha.(M(1,\alpha+1))-\Pi_1)$<br>$\psi(\lambda\alpha.(2\ 1-2-2\text{ aft }\alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,2,0) | $\psi(\lambda\alpha.(\psi_{M(1,0,\alpha+1)}(0))-\Pi_0)$<br>$\psi(\lambda\alpha.((2\ 1-)^{1,0}\ 2-2\text{ aft }\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,2,0)(3,2,1) | $\psi(\lambda\alpha.(M(1,0,\alpha+1))-\Pi_1)$<br>$\psi(\lambda\alpha.((2\ 1-)^{1,1}\ 2-2\text{ aft }\alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(M(1;\alpha+1))-\Pi_1)$<br>$\psi(\lambda\alpha.(2-2\ 1-2-2\text{ aft }\alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,2,1)(3,2,1) | $\psi(\lambda\alpha.(M(1;1,\alpha+1))-\Pi_1)$<br>$\psi(\lambda\alpha.(2\ 1-2-2\ 1-2-2\text{ aft }\alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,2,1)(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(M(2;\alpha+1))-\Pi_1)$<br>$\psi(\lambda\alpha.(2-2\ 1-2-2\ 1-2-2\text{ aft }\alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(M(\alpha;1))-\Pi_0)$<br>$\psi(\lambda\alpha.((2-2\ 1-)^{\alpha}2-2\text{ aft }\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,0)(3,2,1) | $\psi(\lambda\alpha.(M(1,0;\alpha+1))-\Pi_1)$<br>$\psi(\lambda\alpha.((2-2\ 1-)^{1,0}\ 2-2\text{ aft }\alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,0)(5,3,0) | $\psi(\lambda\alpha.(\psi_{N_{\alpha+1}}(\psi_{\Omega_{N_{\alpha+1}+1}}(0)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(N_{\alpha+1})-\Pi_1)$<br>$\psi(\lambda\alpha.(2-2-2\text{ aft }\alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,1)(2,2,1)(3,2,1) | $\psi(\lambda\alpha.(2\ 1-2\text{ aft }2-2-2\text{ aft }\alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,1)(2,2,1)(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2\text{ aft }2-2-2\text{ aft }\alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(4,2,1)(2,2,1)(3,2,1)(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(2\text{nd }2-2-2\text{ aft }\alpha)-\Pi_1)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,1)(3,0,0) | $\psi(\lambda\alpha.(1-2-2-2 \text{ aft } \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,1)(3,2,1) | $\psi(\lambda\alpha.(2\ 1-2-2-2 \text{ aft } \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,1)(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2\ 1-2-2-2 \text{ aft } \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(4,2,1)(3,2,1)(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2-2\ 1-2-2-2 \text{ aft } \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,1)(4,2,0) | $\psi(\lambda\alpha.((2-2-2\ 1-)^{1,0}\ 2-2-2 \text{ aft } \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2-2-2 \text{ aft } \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,1)(5,0,0) | $\psi(\lambda\alpha.((2-)^{\omega} \text{ aft } \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.((2-)^{1,0} \text{ aft } \alpha)-\Pi_0)$<br>$\psi(\lambda\alpha.(\psi_{K_{\alpha+1}}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(3 \text{ aft } \alpha)-\Pi_2)$<br>$\psi(\lambda\alpha.(K_{\alpha+1})-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(5,2,1)(2,2,1)(3,2,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(2\text{nd } 3 \text{ aft } \alpha)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(5,2,1)(3,2,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(3\ 1-3 \text{ aft } \alpha)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(3\ 2-3 \text{ aft } \alpha)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1) | $\psi(\lambda\alpha.(3-3 \text{ aft } \alpha)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(6,0,0) | $\psi(\lambda\alpha.((3-)^{\omega} \text{ aft } \alpha)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(6,2,1) | $\psi(\lambda\alpha.(4 \text{ aft } \alpha)-\Pi_3)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(6,2,1)(7,2,1) | $\psi(\lambda\alpha.(5 \text{ aft } \alpha)-\Pi_4)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0[\alpha+1])-\Pi_0)$<br>$\psi(\lambda\alpha.(\Pi_{\omega} \text{ aft } \alpha)-\Pi_0)$<br>$\psi(2-\pi-(+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0) | $\psi((\lambda\alpha.(\alpha+1)-\Pi_0)-(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(2,2,0)(3,2,0)(4,1,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1)-(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)(6,3,0) | $\psi((\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$<br>$-(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,3,0)(3,1,1) | $\psi(1-(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)<br>(6,3,0)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,3,0)(3,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,3,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,3,0) | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1)-\Pi_0)^2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,3,0)(5,2,0) | $\psi(\lambda\alpha.((1-)^{1,0} \text{ aft } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)<br>(6,3,0)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.((1-)^{2,0} \text{ aft } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,0)(3,2,0)(4,2,0) | $\psi(\lambda\alpha.((1-)^{1,0,0} \text{ aft } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,1) | $\psi(\lambda\alpha.(2 \text{ aft } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,1)(3,2,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(3 \text{ aft } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(2,2,1)(3,3,0) | $\psi(\lambda\alpha.(2\text{nd } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,0,0) | $\psi(\lambda\alpha.(1-(\lambda\beta.(\beta+1)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.((1-)^{\alpha}\ (\lambda\beta.(\beta+1)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,1,1) | $\psi(\lambda\alpha.((1-)^{2 \text{ aft } \alpha}\ (\lambda\beta.(\beta+1)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,2,0) | $\psi(\lambda\alpha.((1-)^{1,0}\ (\lambda\beta.(\beta+1)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,2,1) | $\psi(\lambda\alpha.(2\ 1-(\lambda\beta.(\beta+1)-\Pi_0))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(3,2,1)(4,3,0) | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1)-\Pi_0)\ 1-(\lambda\beta.(\beta+1)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(3,2,1)(4,3,0)(4,2,1) | $\psi(\lambda\alpha.(2-(\lambda\beta.(\beta+1)-\Pi_0))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,3,0) | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1)-\Pi_0)-(\lambda\beta.(\beta+1)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1)-\Pi_0-)^{\alpha})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,2,0) | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1)-\Pi_0-)^{1,0})-\Pi_0)$<br>$\psi(\lambda\alpha.(\psi_{\lambda\beta.(\beta+1)-\Pi_1}(0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,2,0)(4,2,0) | $\psi(\lambda\alpha.(\psi_{\lambda\beta.(\beta+1)-\Pi_1}(1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,2,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\lambda\beta.(\beta+1)-\Pi_1}(\lambda\beta.(\beta+1)-\Pi_1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,0)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\lambda\beta.(\beta+1)-\Pi_1}(2 \text{ aft } \lambda\beta.(\beta+1)-\Pi_1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,2,0)(5,3,1) | $\psi(\lambda\alpha.(\psi_{\lambda\beta.(\beta+1)-\Pi_1}(1-2 \text{ aft } \lambda\beta.(\beta+1)-\Pi_1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,0)(5,3,1)(6,4,0) | $\psi(\lambda\alpha.(\psi_{\lambda\beta.(\beta+1)-\Pi_1}(\lambda\alpha'.(\alpha'+1)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,0)(5,3,1)(6,4,1)(7,5,0) | $\psi(\lambda\alpha.(\psi_{\lambda\beta.(\beta+1)-\Pi_1}(\lambda\alpha'.(\lambda\beta'.$<br>$(\beta'+1)-\Pi_0)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,0)(5,3,1)(6,4,1)(7,5,0)(8,4,0) | $\psi(\lambda\alpha.(\psi_{\lambda\beta.(\beta+1)-\Pi_1}$<br>$(\lambda\alpha'.(\psi_{\lambda\beta'.(\beta'+1)-\Pi_1}(0))-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,2,1)(3,3,0) | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1)-\Pi_0)-(\lambda\beta.(\beta+1)-\Pi_1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,1)(3,3,0)(4,2,1) | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1)-\Pi_1)\ (\lambda\beta.(\beta+1)-\Pi_0)$<br>$-(\lambda\beta.(\beta+1)-\Pi_1))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,2,1)(4,2,1) | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1)-\Pi_1)-(\lambda\beta.(\beta+1)-\Pi_1))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_2)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,2,1)(5,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+2)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,2,1)(5,3,0)(5,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+2)-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,1)(5,3,0)(5,2,1)(6,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+3)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\omega)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,3,0)(4,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\omega^2)-\Pi_0)-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,1,0)(2,0,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\alpha)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,3,0)(5,1,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\Omega_{\alpha+1})-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,3,0)(5,1,1)(6,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\psi_{\Omega_{\alpha+2}}(0))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,3,0)(5,1,1)(6,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\Omega_{\alpha+2})-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,1,1)(6,2,1)(7,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+I_{\alpha+1})-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,1,1)(6,2,1)(7,2,1)(8,2,1)(9,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+K_{\alpha+1})-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,1,1)(6,2,1)(7,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,1,1)(6,2,1)(7,3,0)(8,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\lambda\beta.(\beta+\omega)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,1,1)(6,2,1)(7,3,0)(8,3,0)<br>(9,1,1)(10,2,1)(11,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\lambda\beta.(\beta+\lambda\beta.(\beta+1)$<br>$-\Pi_0)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,3,0)(5,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta\cdot 2)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(4,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta\cdot 2)-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,2,0)(4,2,1)(5,3,0)(6,3,0)(7,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta\cdot 3)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(4,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta\cdot\omega)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(4,3,0)(5,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta^2)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(5,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta^\beta)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(6,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(0))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(6,3,0)(7,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\Omega_{\beta+1}))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(6,3,1)(7,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\alpha'.(\alpha'+1)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(6,3,1)(7,4,1)(8,5,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\alpha'.(\lambda\beta'.(\beta'+1)$<br>$-\Pi_0)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,2,0)(6,3,1)(7,4,1)(8,5,0)<br>(9,5,0)(10,4,0)(11,5,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\alpha'.(\lambda\beta'.(\psi_{\Omega_{\beta'+1}}(0))$<br>$-\Pi_0)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,2,0)(6,3,1)(7,4,1)(8,5,0)(9,5,0)<br>(10,4,0)(11,5,1)(12,6,1)(13,7,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\alpha'.(\lambda\beta'.(\psi_{\Omega_{\beta'+1}}$<br>$(\lambda\alpha''.(\lambda\beta''.(\beta''+1)-\Pi_0)-\Pi_0))$<br>$-\Pi_0)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,3,0)(5,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+1})-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,3,0)(5,2,1)(4,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+1}\cdot\omega)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,1)(4,3,0)(5,2,1) | $\psi(\lambda\alpha.(\lambda\beta.({\Omega_{\beta+1}}^2)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,3,0)(5,2,1)(6,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(0))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,1)(6,3,0)(7,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\Omega_{\beta+2}))-\Pi_0)-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,1)(6,3,0)(7,3,0)(7,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\Omega^2_{\beta+2})) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,1)(6,3,0)(7,3,0)(8,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\Omega_{\beta+2}{}^{\beta})) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,2,1)(6,3,0)(7,3,0)(8,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\Omega_{\beta+2}{}^{\Omega_{\beta+1}})) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,3,0)(5,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\Omega_{\beta+2}{}^{\Omega_{\beta+2}})) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\psi_{\Omega_{\beta+3}}(0))) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+2})) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+3})) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,0,0) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+\omega})) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+\alpha})) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta\cdot 2})) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,2,0)(3,3,1)(4,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta\cdot 3})) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,2,0)(4,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta^2})) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,2,0)(5,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\psi_{\Omega_{\beta+1}}(0)})) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\Omega_{\beta+1}})) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,2,1)(5,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\Omega_{\beta+2}})) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,2,1)(5,3,1)(6,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\Omega_{\Omega_{\beta+1}}})) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{I_{\beta+1}}(0))) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(I_{\beta+1})) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,3,1)(4,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(I(1,\beta+1))) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(5,1,0)(2,0,0) | $\psi(\lambda\alpha.(\lambda\beta.(I(\alpha,\beta+1))) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,3,1)(5,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(I(\beta,1))) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,3,1)(5,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{I(1,0,\beta+1)}(0))) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,3,1)(5,3,0)(4,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(I(1,0,\beta+1))) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,3,1)(5,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(M_{\beta+1}) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,3,1)(5,3,1)(5,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(N_{\beta+1}) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,3,1)(5,3,1)(6,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(K_{\beta+1}) - \Pi_2) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(5,3,1)(6,3,1)(7,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(4 \text{ aft } \beta) - \Pi_3) - \Pi_3)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_0)$<br>$\psi(3 - \pi - (+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(2,2,0)(3,2,0)(4,1,1)<br>(5,2,1)(6,3,1)(7,4,0)(3,1,1) | $\psi(1 - (\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_1))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(2,2,1)(3,3,1)(4,4,0) | $\psi(\lambda\alpha.(2\text{nd } \lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,4,0)(3,3,0) | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1) - \Pi_0)$<br>$-(\lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0)) - \Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(3,3,0)(4,3,0)(5,2,0)<br>(6,3,1)(7,4,1)(8,5,0) | $\psi(\lambda\alpha.((\lambda\beta.(\lambda\gamma.(\gamma+1)-\Pi_0)-\Pi_0)$<br>$-(\lambda\beta.(\lambda\gamma.(\gamma+1)-\Pi_0)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(3,3,0)(4,3,0)(5,2,0)<br>(6,3,1)(7,4,1)(8,5,0)(4,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+1)-\Pi_0)-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,4,0)(3,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(2\text{ aft }\lambda\gamma.(\gamma+1)-\Pi_0)-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(3,3,1)(4,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(2\text{nd }\lambda\gamma.(\gamma+1)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,4,0)(4,4,0) | $\psi(\lambda\alpha.(\lambda\beta.((\lambda\gamma.(\gamma+1)-\Pi_0)$<br>$-(\lambda\gamma.(\gamma+1)-\Pi_0))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,4,0)(5,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+1)-\Pi_1)-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(5,3,1)(6,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+2)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,4,0)(5,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+\omega)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(5,4,0)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+\alpha)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(5,4,0)(6,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+\beta)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(5,4,0)(6,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma\cdot 2)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(5,4,0)(6,3,0)(7,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\psi_{\Omega_{\gamma+1}}(0))-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(5,4,0)(6,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Omega_{\gamma+1})-\Pi_1)-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,4,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Omega_{\gamma+2})-\Pi_1)-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,4,1)(5,0,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Omega_{\gamma+\omega})-\Pi_1)-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,4,1)(5,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Omega_{\gamma+\beta})-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,4,1)(5,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Omega_{\gamma\cdot 2})-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,4,1)(5,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\psi_{I_{\gamma+1}}(0))-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,4,1)(5,4,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(I_{\gamma+1})-\Pi_1)-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,4,1)(5,4,1)(6,4,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(M_{\gamma+1})-\Pi_1)-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,1)(5,4,1)(6,4,1)(7,4,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(K_{\gamma+1})-\Pi_2)-\Pi_2)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,4,1)(5,5,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\lambda\delta.(\delta+1)-\Pi_0)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,4,1)(5,5,0)(6,4,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\lambda\delta.(\delta+1)-\Pi_1)-\Pi_1)-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,1)(5,5,1)(6,6,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\lambda\delta.(\lambda\epsilon.(\epsilon+1)$<br>$-\Pi_0)-\Pi_0)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,2) | $\psi(\omega-\pi-\Pi_0)$ |

## A.13 BMS vs Stable-Collapsing Notation (Patchouli.ver)

The results in this section are primarily cited from[29-31].

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0) | $\psi(\Pi_\omega)$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(1,1,1)(2,2,0) | $\psi(\Pi_\omega\cdot 2)$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0\cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,0) | $\psi(\Pi_2 \text{ aft } \Pi_\omega)$<br>$\psi(\Pi_2 \text{ aft } \lambda\alpha.(\alpha+1)-\Pi_0\cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1) | $\psi(\Pi_1-\Pi_2\ \Pi_1-\Pi_2 \text{ aft } \Pi_\omega)$<br>$\psi(\Pi_1-\Pi_2\ \Pi_1-\Pi_2 \text{ aft } \lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1) | $\psi(\Pi_1-\Pi_2-\Pi_2 \text{ aft } \Pi_\omega)$<br>$\psi(\Pi_1-\Pi_2-\Pi_2 \text{ aft } \lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(6,2,1) | $\psi(\Pi_1-\Pi_3 \text{ aft } \Pi_\omega)$<br>$\psi(\Pi_1-\Pi_3 \text{ aft } \lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)<br>(3,2,1)(4,3,0) | $\psi(\text{2nd } \Pi_\omega)$<br>$\psi(\text{2nd } \lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,1)<br>(4,3,0)(3,2,0) | $\psi(\text{2nd } \Pi_\omega+\Pi_2 \text{ aft } \Pi_\omega)$<br>$\psi(\text{2nd } \lambda\alpha.(\alpha+1)-\Pi_0+\Pi_2 \text{ aft } \lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,1)<br>(4,3,0)(3,2,1)(4,3,0) | $\psi(\text{2nd } \Pi_\omega\cdot 2)$<br>$\psi(\text{2nd } \lambda\alpha.(\alpha+1)-\Pi_0\cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,1)<br>(4,3,0)(4,2,0)(5,3,0) | $\psi(\Pi_2 \text{ aft 2nd } \Pi_\omega)$<br>$\psi(\Pi_2 \text{ aft 2nd } \lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,1)<br>(4,3,0)(4,2,0)(5,3,1)(6,4,0) | $\psi(\text{3rd } \Pi_\omega)$<br>$\psi(\text{3rd } \lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,1)<br>(4,3,0)(4,2,0)(5,3,1)(6,4,0)<br>(6,3,0)(7,4,1)(8,5,0) | $\psi(\text{4th } \Pi_\omega)$<br>$\psi(\text{4th } \lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1) | $\psi(\Pi_1-\Pi_\omega)$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,1,0)(2,0,0) | $\psi((\Pi_1-)^{(1,0)}\Pi_\omega)$<br>$\psi((\Pi_1-)^{(1,0)}\lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,1,0)(2,1,1) | $\psi((\Pi_1-)^{(1,1)}\Pi_\omega)$<br>$\psi((\Pi_1-)^{(1,1)}\lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi((\Pi_1-)^{(2,0)}\Pi_\omega)$<br>$\psi((\Pi_1-)^{(2,0)}\lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,1,0)(4,2,0) | $\psi(\Pi_2 \text{ aft } \Pi_2\ \Pi_1-\Pi_\omega)$<br>$\psi(\Pi_2 \text{ aft } \Pi_2\ \Pi_1-\lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,1) | $\psi(\Pi_1-\Pi_2\ \Pi_1-\Pi_\omega)$<br>$\psi(\Pi_1-\Pi_2\ \Pi_1-\lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1) | $\psi(\Pi_1-\Pi_2\ \Pi_1-\Pi_2\ \Pi_1-\Pi_\omega)$<br>$\psi(\Pi_1-\Pi_2\ \Pi_1-\Pi_2\ \Pi_1-\lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,1)<br>(3,1,0)(2,0,0) | $\psi((\Pi_2\ \Pi_1-)^{(1,0)}\Pi_\omega)$<br>$\psi((\Pi_2\ \Pi_1-)^{(1,0)}\lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0) | $\psi(\Pi_2 \text{ aft } \Pi_2-\Pi_2\ \Pi_1-\Pi_\omega)$<br>$\psi(\Pi_2 \text{ aft } \Pi_2-\Pi_2\ \Pi_1-\lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,1,1)(3,1,1) | $\psi(\Pi_1-\Pi_2-\Pi_2\ \Pi_1-\Pi_\omega)$<br>$\psi(\Pi_1-\Pi_2-\Pi_2\ \Pi_1-\lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1) | $\psi(\Pi_1-\Pi_2-\Pi_2-\Pi_2\ \Pi_1-\Pi_\omega)$<br>$\psi(\Pi_1-\Pi_2-\Pi_2-\Pi_2\ \Pi_1-\lambda\alpha.(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,1)<br>(4,1,0)(2,0,0) | $\psi((\Pi_2-)^{(1,0)}\ \Pi_1-\Pi_\omega)$<br>$\psi((\Pi_2-)^{(1,0)}\ \Pi_1-\lambda\alpha.(\alpha+1)-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,0) | $\psi(\Pi_2 \text{ aft } \Pi_3\ \Pi_1 - \Pi_\omega)$<br>$\psi(\Pi_2 \text{ aft } \Pi_3\ \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,1,1)(4,1,1) | $\psi(\Pi_1 - \Pi_3\ \Pi_1 - \Pi_\omega)$<br>$\psi(\Pi_1 - \Pi_3\ \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,1) | $\psi(\Pi_1 - \Pi_2 - \Pi_3\ \Pi_1 - \Pi_\omega)$<br>$\psi(\Pi_1 - \Pi_2 - \Pi_3\ \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,1)(4,1,1) | $\psi(\Pi_1 - \Pi_3\ \Pi_2 - \Pi_3\ \Pi_1 - \Pi_\omega)$<br>$\psi(\Pi_1 - \Pi_3\ \Pi_2 - \Pi_3\ \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,1) | $\psi(\Pi_1 - \Pi_3 - \Pi_3\ \Pi_1 - \Pi_\omega)$<br>$\psi(\Pi_1 - \Pi_3 - \Pi_3\ \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1) | $\psi(\Pi_1 - \Pi_4\ \Pi_1 - \Pi_\omega)$<br>$\psi(\Pi_1 - \Pi_4\ \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0) | $\psi(\Pi_\omega\ \Pi_1 - \Pi_\omega)$<br>$\psi(\lambda\alpha.(\alpha+1) - \Pi_0\ \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(2,1,1) | $\psi(\Pi_1 - \Pi_\omega\ \Pi_1 - \Pi_\omega)$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_0\ \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(2,1,1)(3,2,0) | $\psi(\Pi_\omega\ \Pi_1 - \Pi_\omega\ \Pi_1 - \Pi_\omega)$<br>$\psi(\lambda\alpha.(\alpha+1) - \Pi_0\ \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_0$<br>$\Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,0)(2,0,0) | $\psi((\Pi_\omega\ \Pi_1 -)^{(1,0)})$<br>$\psi((\lambda\alpha.(\alpha+1) - \Pi_0\ \Pi_1 -)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,0)(4,2,0) | $\psi(\Pi_2 \text{ aft } \Pi_2 - \Pi_\omega)$<br>$\psi(\Pi_2 \text{ aft } \Pi_2 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(3,1,1) | $\psi(\Pi_1 - \Pi_2 - \Pi_\omega)$<br>$\psi(\Pi_1 - \Pi_2 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(2,1,1)(3,2,0)(3,1,1) | $\psi(\Pi_1 - \Pi_2 - \Pi_\omega\ \Pi_1 - \Pi_2 - \Pi_\omega)$<br>$\psi(\Pi_1 - \Pi_2 - \lambda\alpha.(\alpha+1) - \Pi_0$<br>$\Pi_1 - \Pi_2 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi((\Pi_2 - \Pi_\omega\ \Pi_1 -)^{(1,0)})$<br>$\psi((\Pi_1 - \Pi_2 - \lambda\alpha.(\alpha+1) - \Pi_0\ \Pi_1 -)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\Pi_2 \text{ aft } \Pi_2 - \Pi_2 - \Pi_\omega)$<br>$\psi(\Pi_2 \text{ aft } \Pi_2 - \Pi_2 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(3,1,1) | $\psi(\Pi_1 - \Pi_2 - \Pi_2 - \Pi_\omega)$<br>$\psi(\Pi_1 - \Pi_2 - \Pi_2 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,1,1) | $\psi(\Pi_1 - \Pi_3\ \Pi_2 - \Pi_\omega)$<br>$\psi(\Pi_1 - \Pi_3\ \Pi_2 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,1,1)(5,1,1) | $\psi(\Pi_1 - \Pi_4\ \Pi_2 - \Pi_\omega)$<br>$\psi(\Pi_1 - \Pi_4\ \Pi_2 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,2,0) | $\psi(\Pi_\omega\ \Pi_2 - \Pi_\omega)$<br>$\psi(\lambda\alpha.(\alpha+1) - \Pi_0\ \Pi_2 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,2,0)(3,1,1)(4,2,0) | $\psi(\Pi_\omega\ \Pi_2 - \Pi_\omega\ \Pi_2 - \Pi_\omega)$<br>$\psi(\lambda\alpha.(\alpha+1) - \Pi_0\ \Pi_2 - \lambda\alpha.(\alpha+1) - \Pi_0$<br>$\Pi_2 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,2,0)(4,1,1) | $\psi(\Pi_1 - \Pi_3 - \Pi_\omega)$<br>$\psi(\Pi_1 - \Pi_3 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,2,0)(4,1,1)(5,2,0) | $\psi(\Pi_\omega\ \Pi_3 - \Pi_\omega)$<br>$\psi(\lambda\alpha.(\alpha+1) - \Pi_0\ \Pi_3 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,2,0)(4,1,1)(5,2,0)(5,1,1) | $\psi(\Pi_1 - \Pi_4 - \Pi_\omega)$<br>$\psi(\Pi_1 - \Pi_4 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,0)(2,2,0) | $\psi(\Pi_\omega - \Pi_\omega)$<br>$\psi(\lambda\alpha.(\alpha+1) - \Pi_0 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,2,0)(2,1,1) | $\psi(\Pi_1 - \Pi_\omega - \Pi_\omega)$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_0 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,2,0)(2,1,1)<br>(3,2,0)(3,2,0)(3,1,1) | $\psi(\Pi_1 - \Pi_2 - \Pi_\omega - \Pi_\omega)$<br>$\psi(\Pi_1 - \Pi_2 - \lambda\alpha.(\alpha+1) - \Pi_0 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,2,0)(2,2,0) | $\psi(\Pi_\omega - \Pi_\omega - \Pi_\omega)$<br>$\psi(\lambda\alpha.(\alpha+1) - \Pi_0 - \lambda\alpha.(\alpha+1)$<br>$-\Pi_0 - \lambda\alpha.(\alpha+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,0,0) | $\psi((\Pi_\omega-)^\omega)$<br>$\psi((\lambda\alpha.(\alpha+1) - \Pi_0-)^\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,0,0) | $\psi((\Pi_\omega-)^{(1,0)})$<br>$\psi((\lambda\alpha.(\alpha+1) - \Pi_0-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,1,1) | $\psi(\Pi_1 - (\Pi_\omega-)^{(1,0)})$<br>$\psi(\Pi_1 - (\lambda\alpha.(\alpha+1) - \Pi_0-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)<br>(2,1,1)(3,2,0) | $\psi(\Pi_\omega\ \Pi_1 - (\Pi_\omega-)^{(1,0)})$<br>$\psi(\lambda\alpha.(\alpha+1) - \Pi_0\ \Pi_1$<br>$-(\lambda\alpha.(\alpha+1) - \Pi_0-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,1,1)<br>(3,2,0)(4,1,0)(2,0,0) | $\psi((\Pi_\omega-)^{(1,0)}\ \Pi_1 - (\Pi_\omega-)^{(1,0)})$<br>$\psi((\lambda\alpha.(\alpha+1) - \Pi_0-)^{(1,0)}\ \Pi_1$<br>$-(\lambda\alpha.(\alpha+1) - \Pi_0-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,1,1)<br>(3,2,0)(4,1,0)(3,1,0)(2,0,0) | $\psi(((\Pi_\omega-)^{(1,0)}\ \Pi_1-)^{(1,0)})$<br>$\psi(((\lambda\alpha.(\alpha+1) - \Pi_0-)^{(1,0)}\ \Pi_1-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,1,1)<br>(3,2,0)(4,1,0)(3,1,1) | $\psi(\Pi_1 - \Pi_2 - (\Pi_\omega-)^{(1,0)})$<br>$\psi(\Pi_1 - \Pi_2 - (\lambda\alpha.(\alpha+1) - \Pi_0-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,1,1)<br>(3,2,0)(4,1,0)(3,1,1)<br>(4,2,0)(5,1,0)(4,1,1) | $\psi(\Pi_1 - \Pi_3 - (\Pi_\omega-)^{(1,0)})$<br>$\psi(\Pi_1 - \Pi_3 - (\lambda\alpha.(\alpha+1) - \Pi_0-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,2,0) | $\psi((\Pi_\omega-)^{(1,1)})$<br>$\psi((\lambda\alpha.(\alpha+1) - \Pi_0-)^{(1,1)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,2,0)<br>(3,1,0)(2,0,0) | $\psi((\Pi_\omega-)^{(2,0)})$<br>$\psi((\lambda\alpha.(\alpha+1) - \Pi_0-)^{(2,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)<br>(3,1,0)(2,0,0) | $\psi((\Pi_\omega-)^{(1,0,0)})$<br>$\psi((\lambda\alpha.(\alpha+1) - \Pi_0-)^{(1,0,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(4,2,0) | $\psi(\Pi_2 \text{ aft } \Pi_{\omega+1})$<br>$\psi(\Pi_2 \text{ aft } \lambda\alpha.(\alpha+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1) | $\psi(\Pi_1 - \Pi_{\omega+1})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(2,1,0)(3,2,0) | $\psi(\Pi_1 \text{ aft } \Pi_1 - \Pi_{\omega+1})$<br>$\psi(\Pi_1 \text{ aft } \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(2,1,1) | $\psi(\Pi_1 - \Pi_1 - \Pi_{\omega+1})$<br>$\psi(\Pi_1 - \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(2,1,1)(3,2,0) | $\psi(\Pi_\omega\ \Pi_1 - \Pi_{\omega+1})$<br>$\psi(\lambda\alpha.(\alpha+1) - \Pi_0\ \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(2,1,1)<br>(3,2,0)(3,2,0) | $\psi(\Pi_\omega - \Pi_\omega\ \Pi_1 - \Pi_{\omega+1})$<br>$\psi((\lambda\alpha.(\alpha+1) - \Pi_0-)^2\ \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(2,1,1)<br>(3,2,0)(4,1,0)(5,2,0) | $\psi(\Pi_2 \text{ aft } \Pi_{\omega+1}\ \Pi_1 - \Pi_{\omega+1})$<br>$\psi(\Pi_2 \text{ aft } \lambda\alpha.(\alpha+1) - \Pi_1\ \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_1)$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(2,1,1)<br>(3,2,0)(4,1,1) | $\psi(\Pi_1 - \Pi_{\omega+1}\ \Pi_1 - \Pi_{\omega+1})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_1\ \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(2,1,1)<br>(3,2,0)(4,1,1)(3,1,1) | $\psi(\Pi_1 - \Pi_2 - \Pi_{\omega+1})$<br>$\psi(\Pi_1 - \Pi_2 - \lambda\alpha.(\alpha+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(2,1,1)<br>(3,2,0)(4,1,1)(3,1,1)<br>(4,2,0)(5,1,1)(4,1,1) | $\psi(\Pi_1 - \Pi_3 - \Pi_{\omega+1})$<br>$\psi(\Pi_1 - \Pi_3 - \lambda\alpha.(\alpha+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(2,2,0) | $\psi(\Pi_\omega - \Pi_{\omega+1})$<br>$\psi(\lambda\alpha.(\alpha+1) - \Pi_0 - \lambda\alpha.(\alpha+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(2,2,0)<br>(3,1,0)(2,0,0) | $\psi((\Pi_\omega-)^{(1,0)}\Pi_{\omega+1})$<br>$\psi((\lambda\alpha.(\alpha+1) - \Pi_0-)^{(1,0)}\lambda\alpha.(\alpha+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(2,2,0)<br>(3,1,0)(4,2,0) | $\psi(\Pi_2 \text{ aft } \Pi_{\omega+1}\ \Pi_\omega - \Pi_{\omega+1})$<br>$\psi(\Pi_2 \text{ aft } \lambda\alpha.(\alpha+1) - \Pi_1$<br>$\lambda\alpha.(\alpha+1) - \Pi_0 - \lambda\alpha.(\alpha+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(2,2,0)(3,1,1) | $\psi(\Pi_1 - \Pi_{\omega+1}\ \Pi_\omega - \Pi_{\omega+1})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_1$<br>$\lambda\alpha.(\alpha+1) - \Pi_0 - \lambda\alpha.(\alpha+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(3,0,0) | $\psi((\Pi_{\omega+1}\ \Pi_\omega-)^\omega)$<br>$\psi((\lambda\alpha.(\alpha+1) - \Pi_1\ \lambda\alpha.(\alpha+1) - \Pi_0-)^\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(3,1,0)(2,0,0) | $\psi((\Pi_{\omega+1}\ \Pi_\omega-)^{(1,0)})$<br>$\psi((\lambda\alpha.(\alpha+1) - \Pi_1\ \lambda\alpha.(\alpha+1) - \Pi_0-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(3,1,0)(4,2,0) | $\psi(\Pi_2 \text{ aft } \Pi_{\omega+1} - \Pi_{\omega+1})$<br>$\psi(\Pi_2 \text{ aft } \lambda\alpha.(\alpha+1) - \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(3,1,1) | $\psi(\Pi_1 - \Pi_{\omega+1} - \Pi_{\omega+1})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(3,1,1)(3,1,1) | $\psi(\Pi_1 - \Pi_{\omega+1} - \Pi_{\omega+1} - \Pi_{\omega+1})$<br>$\psi(\Pi_1 - (\lambda\alpha.(\alpha+1) - \Pi_1-)^3)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,0,0) | $\psi((\Pi_{\omega+1}-)^\omega)$<br>$\psi((\lambda\alpha.(\alpha+1) - \Pi_1-)^\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,1,0)(2,0,0) | $\psi((\Pi_{\omega+1}-)^{(1,0)})$<br>$\psi((\lambda\alpha.(\alpha+1) - \Pi_1-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,1,0)(2,1,1) | $\psi(\Pi_1 - (\Pi_{\omega+1}-)^{(1,0)})$<br>$\psi(\Pi_1 - (\lambda\alpha.(\alpha+1) - \Pi_1-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,1,0)(2,2,0) | $\psi(\Pi_\omega - (\Pi_{\omega+1}-)^{(1,0)})$<br>$\psi(\lambda\alpha.(\alpha+1) - \Pi_0 - (\lambda\alpha.(\alpha+1) - \Pi_1-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,1,0)(2,2,0)(3,1,1) | $\psi(\Pi_{\omega+1}\ \Pi_\omega - (\Pi_{\omega+1}-)^{(1,0)})$<br>$\psi(\lambda\alpha.(\alpha+1) - \Pi_1\ \lambda\alpha.(\alpha+1)$<br>$-\Pi_0 - (\lambda\alpha.(\alpha+1) - \Pi_1-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,1,0)(3,1,1) | $\psi((\Pi_{\omega+1}-)^{(1,1)})$<br>$\psi((\lambda\alpha.(\alpha+1) - \Pi_1-)^{(1,1)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,1,0)(3,1,1)(4,1,0)(2,0,0) | $\psi((\Pi_{\omega+1}-)^{(2,0)})$<br>$\psi((\lambda\alpha.(\alpha+1) - \Pi_1-)^{(2,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,1,0)(5,2,0) | $\psi(\Pi_2 \text{ aft } \Pi_{\omega+2})$<br>$\psi(\Pi_2 \text{ aft } \lambda\alpha.(\alpha+1) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,1,1) | $\psi(\Pi_1 - \Pi_{\omega+2})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,1,1)(4,1,1) | $\psi(\Pi_1 - \Pi_{\omega+2} - \Pi_{\omega+2})$<br>$\psi(\Pi_1 - (\lambda\alpha.(\alpha+1) - \Pi_2-)^2)$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,1,1)<br>(5,1,0)(6,2,0) | $\psi(\Pi_2 \text{ aft } \Pi_{\omega+3})$<br>$\psi(\Pi_2 \text{ aft } \lambda\alpha.(\alpha+1)-\Pi_3)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,1,1)(5,1,1) | $\psi(\Pi_1-\Pi_{\omega+3})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha+1)-\Pi_3)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,2,0) | $\psi(\Pi_{\omega\cdot 2})$<br>$\psi(\lambda\alpha.(\alpha+2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,2,0)(2,1,1) | $\psi(\Pi_1-\Pi_{\omega\cdot 2})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha+2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,2,0)(2,2,0) | $\psi(\Pi_\omega-\Pi_{\omega\cdot 2})$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\alpha+2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,2,0)(3,1,1) | $\psi(\Pi_{\omega+1}-\Pi_{\omega\cdot 2})$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_1-\lambda\alpha.(\alpha+2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,2,0)<br>(3,1,1)(4,2,0) | $\psi(\Pi_{\omega\cdot 2}\ \Pi_{\omega+1}-\Pi_{\omega\cdot 2})$<br>$\psi(\lambda\alpha.(\alpha+2)-\Pi_0$<br>$\lambda\alpha.(\alpha+1)-\Pi_1-\lambda\alpha.(\alpha+2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,2,0)(4,1,1) | $\psi(\Pi_{\omega+2}-\Pi_{\omega\cdot 2})$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_2-\lambda\alpha.(\alpha+2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,2,0)(4,2,0) | $\psi(\Pi_{\omega\cdot 2}-\Pi_{\omega\cdot 2})$<br>$\psi((\lambda\alpha.(\alpha+2)-\Pi_0-)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,2,0)<br>(5,1,0)(6,2,0) | $\psi(\Pi_2 \text{ aft } \Pi_{\omega\cdot 2+1})$<br>$\psi(\Pi_2 \text{ aft } \lambda\alpha.(\alpha+2)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)<br>(4,2,0)(5,1,1) | $\psi(\Pi_1-\Pi_{\omega\cdot 2+1})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha+2)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,2,0)<br>(5,1,1)(5,1,1) | $\psi(\Pi_1-\Pi_{\omega\cdot 2+1}-\Pi_{\omega\cdot 2+1})$<br>$\psi(\Pi_1-(\lambda\alpha.(\alpha+2)-\Pi_1-)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,2,0)<br>(5,1,1)(6,1,1) | $\psi(\Pi_1-\Pi_{\omega\cdot 2+2})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha+2)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,2,0)<br>(5,1,1)(6,1,1)(7,1,1) | $\psi(\Pi_1-\Pi_{\omega\cdot 2+3})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha+2)-\Pi_3)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,2,0)<br>(5,1,1)(6,2,0) | $\psi(\Pi_{\omega\cdot 3})$<br>$\psi(\lambda\alpha.(\alpha+3)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,2,0)<br>(5,1,1)(6,2,0)(7,1,1)(8,2,0) | $\psi(\Pi_{\omega\cdot 4})$<br>$\psi(\lambda\alpha.(\alpha+4)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0) | $\psi(\Pi_{\omega^2})$<br>$\psi(\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,1,1) | $\psi(\Pi_1-\Pi_{\omega^2})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0) | $\psi(\Pi_\omega-\Pi_{\omega^2})$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(2,2,0)(3,1,1) | $\psi(\Pi_1-\Pi_{\omega+1}\ \Pi_\omega-\Pi_{\omega^2})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha+1)-\Pi_1$<br>$\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,1,1) | $\psi(\Pi_1-\Pi_{\omega+2}\ \Pi_\omega-\Pi_{\omega^2})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha+1)-\Pi_2$<br>$\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0) | $\psi(\Pi_{\omega\cdot 2}\ \Pi_\omega-\Pi_{\omega^2})$<br>$\psi(\lambda\alpha.(\alpha+2)-\Pi_0$<br>$\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,1,1)(6,2,0) | $\psi(\Pi_{\omega\cdot 3}\ \Pi_{\omega}-\Pi_{\omega^2})$<br>$\psi(\lambda\alpha.(\alpha+3)-\Pi_0$<br>$\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,2,0) | $\psi((\Pi_{\omega^2}\ \Pi_{\omega}-)^2\Pi_{\omega^2})$<br>$\psi((\lambda\alpha.(\alpha+\omega)-\Pi_0$<br>$\lambda\alpha.(\alpha+1)-\Pi_0-)^2\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,2,0)(3,1,1) | $\psi(\Pi_1-\Pi_{\omega+1}-\Pi_{\omega^2})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha+1)-\Pi_1-\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,2,0)(3,1,1)(4,2,0) | $\psi(\Pi_{\omega\cdot 2}\ \Pi_{\omega+1}-\Pi_{\omega^2})$<br>$\psi(\lambda\alpha.(\alpha+2)-\Pi_0$<br>$\lambda\alpha.(\alpha+1)-\Pi_1-\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,2,0)<br>(3,1,1)(4,2,0)(5,2,0) | $\psi(\Pi_{\omega^2}\ \Pi_{\omega+1}-\Pi_{\omega^2})$<br>$\psi(\lambda\alpha.(\alpha+\omega)-\Pi_0$<br>$\lambda\alpha.(\alpha+1)-\Pi_1-\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,2,0)(4,1,1) | $\psi(\Pi_1-\Pi_{\omega+2}-\Pi_{\omega^2})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha+1)-\Pi_2-\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,2,0)(4,2,0) | $\psi(\Pi_{\omega\cdot 2}-\Pi_{\omega^2})$<br>$\psi(\lambda\alpha.(\alpha+2)-\Pi_0-\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,2,0)(4,2,0)(5,1,1) | $\psi(\Pi_{\omega\cdot 2+1}\ \Pi_{\omega\cdot 2}-\Pi_{\omega^2})$<br>$\psi(\lambda\alpha.(\alpha+2)-\Pi_1$<br>$\lambda\alpha.(\alpha+2)-\Pi_0-\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,2,0)(4,2,0)(5,1,1)<br>(6,2,0)(7,2,0)(5,1,1) | $\psi(\Pi_1-\Pi_{\omega\cdot 2+1}-\Pi_{\omega^2})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha+2)-\Pi_1-\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,2,0)(4,2,0)(5,1,1)<br>(6,2,0)(7,2,0)(6,1,1) | $\psi(\Pi_1-\Pi_{\omega\cdot 2+2}-\Pi_{\omega^2})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha+2)-\Pi_2-\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,2,0)(4,2,0)<br>(5,1,1)(6,2,0)(7,2,0)(6,2,0) | $\psi(\Pi_{\omega\cdot 3}-\Pi_{\omega^2})$<br>$\psi(\lambda\alpha.(\alpha+3)-\Pi_0-\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(2,2,0)(3,2,0) | $\psi(\Pi_{\omega^2}-\Pi_{\omega^2})$<br>$\psi((\lambda\alpha.(\alpha+\omega)-\Pi_0-)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,0,0) | $\psi((\Pi_{\omega^2}-)^{\omega})$<br>$\psi((\lambda\alpha.(\alpha+\omega)-\Pi_0-)^{\omega})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(3,1,0)(2,0,0) | $\psi((\Pi_{\omega^2}-)^{(1,0)})$<br>$\psi((\lambda\alpha.(\alpha+\omega)-\Pi_0-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(3,1,0)(4,2,0) | $\psi(\Pi_2\ \text{aft}\ \Pi_{\omega^2+1})$<br>$\psi(\Pi_2\ \text{aft}\ \lambda\alpha.(\alpha+\omega)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1) | $\psi(\Pi_1-\Pi_{\omega^2+1})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha+\omega)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(3,1,1)(2,2,0) | $\psi(\Pi_{\omega}-\Pi_{\omega^2+1})$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\alpha+\omega)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(2,2,0)(3,1,1) | $\psi(\Pi_1-\Pi_{\omega+1}\ \Pi_{\omega}-\Pi_{\omega^2+1})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha+1)-\Pi_1$<br>$\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\alpha+\omega)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(2,2,0)(3,1,1)(4,2,0)(5,2,0) | $\psi(\Pi_{\omega^2}\ \Pi_{\omega}-\Pi_{\omega^2+1})$<br>$\psi(\lambda\alpha.(\alpha+\omega)-\Pi_0$<br>$\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\alpha+\omega)-\Pi_1)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(2,2,0)(3,1,1)(4,2,0)(5,2,0)(5,1,1) | $\psi(\Pi_1 - \Pi_{\omega^2+1}\ \Pi_\omega - \Pi_{\omega^2+1})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+\omega) - \Pi_1$<br>$\lambda\alpha.(\alpha+1) - \Pi_0 - \lambda\alpha.(\alpha+\omega) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(2,2,0)(3,1,1)(4,2,0)<br>(5,2,0)(5,1,1)(3,1,1) | $\psi(\Pi_1 - \Pi_{\omega+1} - \Pi_{\omega^2+1})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+1) - \Pi_1 - \lambda\alpha.(\alpha+\omega) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(2,2,0)(3,1,1)(4,2,0)<br>(5,2,0)(5,1,1)(4,2,0) | $\psi(\Pi_{\omega\cdot 2} - \Pi_{\omega^2+1})$<br>$\psi(\lambda\alpha.(\alpha+2) - \Pi_0 - \lambda\alpha.(\alpha+\omega) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(2,2,0)(3,1,1)(4,2,0)(5,2,0)(5,1,1)<br>(4,2,0)(5,1,1)(6,2,0)(7,2,0)<br>(7,1,1)(5,1,1) | $\psi(\Pi_1 - \Pi_{\omega\cdot 2+1} - \Pi_{\omega^2+1})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+2) - \Pi_1 - \lambda\alpha.(\alpha+\omega) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(2,2,0)(3,1,1)(4,2,0)(5,2,0)(5,1,1)<br>(4,2,0)(5,1,1)(6,2,0)(7,2,0)<br>(7,1,1)(6,2,0) | $\psi(\Pi_{\omega\cdot 3} - \Pi_{\omega^2+1})$<br>$\psi(\lambda\alpha.(\alpha+3) - \Pi_0 - \lambda\alpha.(\alpha+\omega) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(2,2,0)(3,2,0) | $\psi(\Pi_{\omega^2} - \Pi_{\omega^2+1})$<br>$\psi(\lambda\alpha.(\alpha+\omega) - \Pi_0 - \lambda\alpha.(\alpha+\omega) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(2,2,0)(3,2,0)(2,2,0)(3,2,0) | $\psi(\Pi_{\omega^2} - \Pi_{\omega^2} - \Pi_{\omega^2+1})$<br>$\psi((\lambda\alpha.(\alpha+\omega) - \Pi_0-)^2\lambda\alpha.(\alpha+\omega) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(2,2,0)(3,2,0)(3,1,1) | $\psi(\Pi_1 - \Pi_{\omega^2+1}\ \Pi_{\omega^2} - \Pi_{\omega^2+1})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+\omega) - \Pi_1$<br>$\lambda\alpha.(\alpha+\omega) - \Pi_0 - \lambda\alpha.(\alpha+\omega) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(2,2,0)(3,2,0)(3,1,1)(2,2,0)(3,2,0) | $\psi(\Pi_{\omega^2} - \Pi_{\omega^2+1}\ \Pi_{\omega^2} - \Pi_{\omega^2+1})$<br>$\psi(\lambda\alpha.(\alpha+\omega) - \Pi_0 - \lambda\alpha.(\alpha+\omega) - \Pi_1$<br>$\lambda\alpha.(\alpha+\omega) - \Pi_0 - \lambda\alpha.(\alpha+\omega) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(3,1,1)(3,1,1) | $\psi(\Pi_1 - \Pi_{\omega^2+1} - \Pi_{\omega^2+1})$<br>$\psi(\Pi_1 - (\lambda\alpha.(\alpha+\omega) - \Pi_1-)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(3,1,1)(4,1,1) | $\psi(\Pi_1 - \Pi_{\omega^2+2})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+\omega) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(3,1,1)(4,2,0) | $\psi(\Pi_{\omega^2+\omega})$<br>$\psi(\lambda\alpha.(\alpha+\omega+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(4,2,0)(5,1,1) | $\psi(\Pi_1 - \Pi_{\omega^2+\omega+1})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+\omega+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(4,2,0)(5,1,1)(6,2,0) | $\psi(\Pi_{\omega^2+\omega\cdot 2})$<br>$\psi(\lambda\alpha.(\alpha+\omega+2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(4,2,0)(5,1,1)(6,2,0)(7,1,1)(8,2,0) | $\psi(\Pi_{\omega^2+\omega\cdot 3})$<br>$\psi(\lambda\alpha.(\alpha+\omega+3) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(3,1,1)(4,2,0)(5,2,0) | $\psi(\Pi_{\omega^2\cdot 2})$<br>$\psi(\lambda\alpha.(\alpha+\omega\cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(4,2,0)(5,2,0)(4,2,0)(5,2,0) | $\psi(\Pi_{\omega^2\cdot 2} - \Pi_{\omega^2\cdot 2})$<br>$\psi((\lambda\alpha.(\alpha+\omega\cdot 2) - \Pi_0-)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(4,2,0)(5,2,0)(5,1,1) | $\psi(\Pi_1 - \Pi_{\omega^2\cdot 2+1})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+\omega\cdot 2) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)<br>(4,2,0)(5,2,0)(5,1,1)(6,2,0)(7,2,0) | $\psi(\Pi_{\omega^2\cdot 3})$<br>$\psi(\lambda\alpha.(\alpha+\omega\cdot 3) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,2,0) | $\psi(\Pi_{\omega^3})$<br>$\psi(\lambda\alpha.(\alpha+\omega^2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(3,2,0)(2,2,0) | $\psi(\Pi_\omega - \Pi_{\omega^3})$<br>$\psi(\lambda\alpha.(\alpha+1) - \Pi_0 - \lambda\alpha.(\alpha+\omega^2) - \Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,2,0)<br>(2,2,0)(3,2,0) | $\psi(\Pi_{\omega^2} - \Pi_{\omega^3})$<br>$\psi(\lambda\alpha.(\alpha+\omega) - \Pi_0 - \lambda\alpha.(\alpha+\omega^2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,2,0)<br>(2,2,0)(3,2,0)(3,2,0) | $\psi(\Pi_{\omega^3} - \Pi_{\omega^3})$<br>$\psi((\lambda\alpha.(\alpha+\omega^2) - \Pi_0-)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(3,2,0)(3,1,1) | $\psi(\Pi_1 - \Pi_{\omega^3+1})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+\omega^2) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(3,2,0)(3,1,1) | $\psi(\Pi_1 - \Pi_{\omega^3+1})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+\omega^2) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,2,0)<br>(3,1,1)(4,2,0) | $\psi(\Pi_{\omega^3+\omega})$<br>$\psi(\lambda\alpha.(\alpha+\omega^2+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,2,0)<br>(3,1,1)(4,2,0)(5,2,0)(5,2,0) | $\psi(\Pi_{\omega^3\cdot 2})$<br>$\psi(\lambda\alpha.(\alpha+\omega^2\cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,2,0)<br>(3,1,1)(4,2,0)(5,2,0)(5,2,0)<br>(5,1,1)(6,2,0)(7,2,0)(7,2,0) | $\psi(\Pi_{\omega^3\cdot 3})$<br>$\psi(\lambda\alpha.(\alpha+\omega^2\cdot 3) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(3,2,0)(3,2,0) | $\psi(\Pi_{\omega^4})$<br>$\psi(\lambda\alpha.(\alpha+\omega^3) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,2,0)<br>(3,2,0)(3,2,0) | $\psi(\Pi_{\omega^5})$<br>$\psi(\lambda\alpha.(\alpha+\omega^4) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,0,0) | $\psi(\Pi_{\omega^\omega})$<br>$\psi(\lambda\alpha.(\alpha+\omega^\omega) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,0,0)<br>(2,2,0)(3,2,0)(4,0,0) | $\psi(\Pi_{\omega^\omega} - \Pi_{\omega^\omega})$<br>$\psi((\lambda\alpha.(\alpha+\omega^\omega) - \Pi_0-)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,0,0)(3,1,1) | $\psi(\Pi_1 - \Pi_{\omega^\omega+1})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+\omega^\omega) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,0,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,0,0) | $\psi(\Pi_{\omega^\omega\cdot 2})$<br>$\psi(\lambda\alpha.(\alpha+\omega^\omega\cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,0,0)(3,2,0) | $\psi(\Pi_{\omega^{\omega+1}})$<br>$\psi(\lambda\alpha.(\alpha+\omega^{\omega+1}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,0,0)(4,0,0) | $\psi(\Pi_{\omega^{\omega^2}})$<br>$\psi(\lambda\alpha.(\alpha+\omega^{\omega^2}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,0,0)(5,1,0) | $\psi(\Pi_{\psi(\Omega)})$<br>$\psi(\lambda\alpha.(\alpha+\psi(\Omega)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0) | $\psi(\Pi_{\Pi_2})$<br>$\psi(\lambda\alpha.(\alpha+\Pi_2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,3,0)(4,3,0)<br>(5,1,0)(3,3,0)(4,3,0)(5,1,0) | $\psi(\Pi_{\Pi_2} - \Pi_{\Pi_2})$<br>$\psi((\lambda\alpha.(\alpha+\Pi_2) - \Pi_0-)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,3,0)<br>(4,3,0)(5,1,0)(4,2,1) | $\psi(\Pi_1 - \Pi_{\Pi_2+1})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha+\Pi_2) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,3,0)(4,3,0)(5,1,0)<br>(4,2,1)(5,3,0)(6,3,0)(7,1,0) | $\psi(\Pi_{\Pi_2\cdot 2})$<br>$\psi(\lambda\alpha.(\alpha+\Pi_2\cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,3,0)<br>(4,3,0)(5,1,0)(4,3,0) | $\psi(\Pi_{\Pi_2\cdot\omega})$<br>$\psi(\lambda\alpha.(\alpha+\Pi_2\cdot\omega) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(1,1,0)(2,2,1)(3,3,0)(4,3,0)(5,2,0) | $\psi(\Pi_{2\text{nd }\Pi_2})$<br>$\psi(\lambda\alpha.(\alpha+2\text{nd }\Pi_2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(1,1,1)(2,2,0) | $\psi(\Pi_{\Pi_\omega})$<br>$\psi(\lambda\alpha.(\alpha+\lambda\alpha.(\alpha+1) - \Pi_0) - \Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(1,1,1)(2,2,0)(3,2,0)(4,1,0) | $\psi(\Pi_{\Pi_{\Pi_2}})$<br>$\psi(\lambda\alpha.(\alpha+\lambda\alpha.(\alpha+\Pi_2)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(1,1,1)(2,2,0)(3,2,0)(4,1,0) | $\psi(\Pi_{\Pi_{\Pi_{\Pi_2}}})$<br>$\psi(\lambda\alpha.(\alpha+\lambda\alpha.(\alpha+\lambda\alpha.(\alpha+\Pi_2)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(2,0,0) | $\psi(\Pi_{(1,0)})$<br>$\psi(\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(2,1,0)(3,2,0) | $\psi(\Pi_2 \text{ aft } \Pi_{(1,0)})$<br>$\psi(\Pi_2 \text{ aft } \lambda\alpha.(\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(2,1,1) | $\psi(\Pi_1-\Pi_{(1,0)})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(2,1,1)(3,2,0)(4,2,0)(5,1,0)(2,0,0) | $\psi(\Pi_{(1,0)}\ \Pi_1-\Pi_{(1,0)})$<br>$\psi(\lambda\alpha.(\alpha\cdot 2)-\Pi_0\ \Pi_1-\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(2,1,1)(3,2,0)(4,2,0)(5,1,0)(3,1,1) | $\psi(\Pi_1-\Pi_2-\Pi_{(1,0)})$<br>$\psi(\Pi_1-\Pi_2-\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(2,1,1)(3,2,0)(4,2,0)(5,1,0)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,0)(4,1,1) | $\psi(\Pi_1-\Pi_3-\Pi_{(1,0)})$<br>$\psi(\Pi_1-\Pi_3-\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(2,2,0) | $\psi(\Pi_\omega-\Pi_{(1,0)})$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(2,2,0)(3,2,0) | $\psi(\Pi_{\omega^2}-\Pi_{(1,0)})$<br>$\psi(\lambda\alpha.(\alpha+\omega)-\Pi_0-\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(2,2,0)(3,2,0)(4,1,0) | $\psi(\Pi_{\Pi_2}-\Pi_{(1,0)})$<br>$\psi(\lambda\alpha.(\alpha+\Pi_2)-\Pi_0-\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(2,2,0)(3,2,0)(4,1,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(2,2,0)(3,2,0)(4,1,0) | $\psi(\Pi_{\Pi_{\Pi_2-\Pi_{(1,0)}}}-\Pi_{(1,0)})$<br>$\psi(\lambda\alpha.(\alpha+\lambda\alpha.(\alpha+\Pi_2)-\Pi_0-\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$<br>$-\Pi_0-\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(2,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1,0)}-\Pi_{(1,0)})$<br>$\psi((\lambda\alpha.(\alpha\cdot 2)-\Pi_0-)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,1,0)(4,2,0) | $\psi(\Pi_2 \text{ aft } \Pi_{(1,1)})$<br>$\psi(\Pi_2 \text{ aft } \lambda\alpha.(\alpha\cdot 2)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(3,1,1) | $\psi(\Pi_1-\Pi_{(1,1)})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha\cdot 2)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(2,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1,0)}-\Pi_{(1,1)})$<br>$\psi(\lambda\alpha.(\alpha\cdot 2)-\Pi_0-\lambda\alpha.(\alpha\cdot 2)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(2,2,0)(3,2,0)(4,1,0)(3,1,1) | $\psi(\Pi_1-\Pi_{(1,1)}\ \Pi_{(1,0)}-\Pi_{(1,1)})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha\cdot 2)-\Pi_1$<br>$\lambda\alpha.(\alpha\cdot 2)-\Pi_0-\lambda\alpha.(\alpha\cdot 2)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,1,1)(3,1,0)(2,0,0) | $\psi((\Pi_{(1,1)}\ \Pi_{(1,0)}-)^{(1,0)})$<br>$\psi((\lambda\alpha.(\alpha\cdot 2)-\Pi_1\ \lambda\alpha.(\alpha\cdot 2)-\Pi_0-)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,1,1)(3,1,1) | $\psi(\Pi_1-\Pi_{(1,1)}-\Pi_{(1,1)})$<br>$\psi(\Pi_1-(\lambda\alpha.(\alpha\cdot 2)-\Pi_1-)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,1,1)(4,1,1) | $\psi(\Pi_1-\Pi_{(1,2)})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha\cdot 2)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,1,1)(4,2,0) | $\psi(\Pi_{(1,\omega)})$<br>$\psi(\lambda\alpha.(\alpha\cdot 2+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,1,1)(4,2,0)(3,1,1) | $\psi(\Pi_1-\Pi_{(1,1)}-\Pi_{(1,\omega)})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha\cdot 2)-\Pi_1-\lambda\alpha.(\alpha\cdot 2+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,1,1)(4,2,0)(4,1,1) | $\psi(\Pi_1-\Pi_{(1,2)}-\Pi_{(1,\omega)})$<br>$\psi(\Pi_1-\lambda\alpha.(\alpha\cdot 2)-\Pi_2-\lambda\alpha.(\alpha\cdot 2+1)-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,1,1)(4,2,0)(4,2,0) | $\psi(\Pi_{(1,\omega)} - \Pi_{(1,\omega)})$<br>$\psi((\lambda\alpha.(\alpha \cdot 2 + 1) - \Pi_0 -)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,1,1)(4,2,0)(5,1,1) | $\psi(\Pi_{(1,\omega+1)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 2 + 1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,1,1)(6,2,0) | $\psi(\Pi_{(1,\omega\cdot 2)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 2 + 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0) | $\psi(\Pi_{(1,\omega^2)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 2 + \omega) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0) | $\psi(\Pi_{(1,\Pi_2)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 2 + \Pi_2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(2,0,0) | $\psi(\Pi_{(2,0)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 3) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(2,1,1) | $\psi(\Pi_1 - \Pi_{(2,0)})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha \cdot 3) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(2,2,0)<br>(3,2,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1,0)} - \Pi_{(2,0)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 2) - \Pi_0 - \lambda\alpha.(\alpha \cdot 3) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(2,2,0)<br>(3,2,0)(4,1,0)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,0)(2,0,0) | $\psi(\Pi_{(2,0)}\ \Pi_{(1,0)} - \Pi_{(2,0)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 3) - \Pi_0$<br>$\lambda\alpha.(\alpha \cdot 2) - \Pi_0 - \lambda\alpha.(\alpha \cdot 3) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(3,1,1) | $\psi(\Pi_1 - \Pi_{(1,1)} - \Pi_{(2,0)})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha \cdot 2) - \Pi_1 - \lambda\alpha.(\alpha \cdot 3) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,0)(2,0,0) | $\psi(\Pi_{(2,0)}\ \Pi_{(1,1)} - \Pi_{(2,0)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 3) - \Pi_0$<br>$\lambda\alpha.(\alpha \cdot 2) - \Pi_1 - \lambda\alpha.(\alpha \cdot 3) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(4,1,1) | $\psi(\Pi_1 - \Pi_{(1,2)} - \Pi_{(2,0)})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha \cdot 2) - \Pi_2 - \lambda\alpha.(\alpha \cdot 3) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(4,2,0) | $\psi(\Pi_{(1,\omega)} - \Pi_{(2,0)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 2 + 1) - \Pi_0 - \lambda\alpha.(\alpha \cdot 3) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)<br>(4,2,0)(5,2,0)(6,1,0)(2,0,0) | $\psi(\Pi_{(2,0)} - \Pi_{(2,0)})$<br>$\psi((\lambda\alpha.(\alpha \cdot 3) - \Pi_0 -)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(5,1,1) | $\psi(\Pi_1 - \Pi_{(2,1)})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha \cdot 3) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(5,1,1)<br>(6,2,0)(7,2,0)(8,1,0)(2,0,0) | $\psi(\Pi_{(3,0)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 4) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0) | $\psi(\Pi_{(\omega,0)})$<br>$\psi(\lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(2,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1,0)} - \Pi_{(\omega,0)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 2) - \Pi_0 - \lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(2,2,0)(3,2,0)(4,1,0)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,0)(2,0,0) | $\psi(\Pi_{(2,0)}\ \Pi_{(1,0)} - \Pi_{(\omega,0)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 3) - \Pi_0$<br>$\lambda\alpha.(\alpha \cdot 2) - \Pi_0 - \lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(2,2,0)(3,2,0)(4,1,0)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,0)(5,1,1)(6,2,0)<br>(7,2,0)(8,1,0)(2,0,0) | $\psi(\Pi_{(3,0)}\ \Pi_{(1,0)} - \Pi_{(\omega,0)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 4) - \Pi_0$<br>$\lambda\alpha.(\alpha \cdot 2) - \Pi_0 - \lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(2,2,0)(3,2,0)(4,1,0)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,0)(5,2,0) | $\psi(\Pi_{(\omega,0)}\ \Pi_{(1,0)} - \Pi_{(\omega,0)})$<br>$\psi(\lambda\alpha.\psi_{\Omega_{\alpha+1}}(1)$<br>$-\Pi_0\ \lambda\alpha.(\alpha \cdot 2) - \Pi_0 - \lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(2,2,0)(3,2,0)(4,1,0)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,0)(5,2,0)(3,1,1) | $\psi(\Pi_1 - \Pi_{(1,1)} - \Pi_{(\omega,0)})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha \cdot 2) - \Pi_1 - \lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(2,2,0)(3,2,0)(4,1,0)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,0)(5,2,0)(4,2,0) | $\psi(\Pi_{(1,\omega)} - \Pi_{(\omega,0)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 2 + 1) - \Pi_0 - \lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(2,2,0)(3,2,0)(4,1,0)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,0)<br>(5,2,0)(4,2,0)(5,1,1) | $\psi(\Pi_1 - \Pi_{(1,\omega+1)} - \Pi_{(\omega,0)})$<br>$\psi(\Pi_1 - \lambda\alpha.(\alpha \cdot 2 + 1) - \Pi_1 - \lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(2,2,0)(3,2,0)(4,1,0)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,0)<br>(5,2,0)(4,2,0)(5,2,0) | $\psi(\Pi_{(1,\omega^2)} - \Pi_{(\omega,0)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 2 + \omega) - \Pi_1 - \lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(2,2,0)(3,2,0)(4,1,0)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,0)(5,2,0)(4,2,0)<br>(5,2,0)(6,1,0)(5,1,1)(6,2,0)<br>(7,2,0)(8,1,0)(2,0,0) | $\psi(\Pi_{(3,0)}\ \Pi_{(2,0)} - \Pi_{(\omega,0)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 4) - \Pi_0$<br>$\lambda\alpha.(\alpha \cdot 3) - \Pi_0 - \lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(2,2,0)(3,2,0)(4,1,0)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,0)(5,2,0)(4,2,0)<br>(5,2,0)(6,1,0)(5,1,1)(6,2,0)<br>(7,2,0)(8,1,0)(7,2,0) | $\psi(\Pi_{(\omega,0)}\ \Pi_{(2,0)} - \Pi_{(\omega,0)})$<br>$\psi(\lambda\alpha.\psi_{\Omega_{\alpha+1}}(1)$<br>$-\Pi_0\ \lambda\alpha.(\alpha \cdot 3) - \Pi_0 - \lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(2,2,0)(3,2,0)(4,1,0)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,0)(5,2,0)(4,2,0)<br>(5,2,0)(6,1,0)(5,1,1)(6,2,0)<br>(7,2,0)(8,1,0)(7,2,0)(6,2,0) | $\psi(\Pi_{(2,\omega)} - \Pi_{(\omega,0)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 3 + 1) - \Pi_0 - \lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(2,2,0)(3,2,0)(4,1,0)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,0)(5,2,0)(4,2,0)<br>(5,2,0)(6,1,0)(5,1,1)(6,2,0)(7,2,0)<br>(8,1,0)(7,2,0)(6,2,0)(7,2,0)<br>(8,1,0)(2,0,0) | $\psi(\Pi_{(3,0)} - \Pi_{(\omega,0)})$<br>$\psi(\lambda\alpha.(\alpha \cdot 4) - \Pi_0 - \lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(2,2,0)(3,2,0)(4,1,0)(3,2,0) | $\psi(\Pi_{(\omega,0)} - \Pi_{(\omega,0)})$<br>$\psi(\lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_0 - \lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(2,2,0)(3,2,0)(4,1,0)(3,2,0)<br>(2,2,0)(3,2,0)(4,1,0)(3,2,0) | $\psi(\Pi_{(\omega,0)} - \Pi_{(\omega,0)} - \Pi_{(\omega,0)})$<br>$\psi((\lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_0 -)^3)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(3,1,1) | $\psi(\Pi_1 - \Pi_{(\omega,1)})$<br>$\psi(\Pi_1 - \lambda\alpha.\psi_{\Omega_{\alpha+1}}(1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(3,1,1)(4,2,0) | $\psi(\Pi_{(\omega,\omega)})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(1) + 1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(3,1,1)(4,2,0)(5,2,0)(6,1,0)<br>(5,1,1)(6,2,0)(7,2,0)(8,1,0)(2,0,0) | $\psi(\Pi_{(\omega+2,0)})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(1) + \alpha \cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,0)(5,2,0) | $\psi(\Pi_{(\omega\cdot 2,0)})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(1) \cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(3,1,1)(4,2,0)(5,2,0)(6,1,0)<br>(5,2,0)(5,1,1)(6,2,0)(7,2,0)<br>(8,1,0)(7,2,0) | $\psi(\Pi_{(\omega\cdot 3,0)})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(1) \cdot 3) - \Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(3,2,0) | $\psi(\Pi_{(\omega^2,0)})$<br>$\psi(\lambda\alpha.\psi_{\Omega_{\alpha+1}}(2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(3,2,0)(3,2,0) | $\psi(\Pi_{(\omega^3,0)})$<br>$\psi(\lambda\alpha.\psi_{\Omega_{\alpha+1}}(3)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0) | $\psi(\Pi_{(\Pi_2,0)})$<br>$\psi(\lambda\alpha.\psi_{\Omega_{\alpha+1}}(\Omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(4,1,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(3,2,0)(4,1,0) | $\psi(\Pi_{(\Pi_{(\Pi_2,0)},0)})$<br>$\psi(\lambda\alpha.\psi_{\Omega_{\alpha+1}}(\lambda\alpha.\psi_{\Omega_{\alpha+1}}(\Omega)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1,0,0)})$<br>$\psi(\lambda\alpha.\psi_{\Omega_{\alpha+1}}(\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0)(2,1,1) | $\psi(\Pi_1-\Pi_{(1,0,0)})$<br>$\psi(\Pi_1-\lambda\alpha.\psi_{\Omega_{\alpha+1}}(\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0)(2,2,0) | $\psi(\Pi_\omega-\Pi_{(1,0,0)})$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.\psi_{\Omega_{\alpha+1}}(\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(4,1,0)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1,0,0)}-\Pi_{(1,0,0)})$<br>$\psi((\lambda\alpha.\psi_{\Omega_{\alpha+1}}(\alpha)-\Pi_0-)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0)(3,1,1) | $\psi(\Pi_1-\Pi_{(1,0,1)})$<br>$\psi(\Pi_1-\lambda\alpha.\psi_{\Omega_{\alpha+1}}(\alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(4,1,0)(3,1,1)(4,2,0) | $\psi(\Pi_{(1,0,\omega)})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\alpha)+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(4,1,0)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,0)(2,0,0) | $\psi(\Pi_{(1,1,0)})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\alpha)+\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(4,1,0)(3,1,1)(4,2,0)(5,2,0)<br>(6,1,0)(5,1,1)(6,2,0)(7,2,0)<br>(8,1,0)(2,0,0) | $\psi(\Pi_{(1,2,0)})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\alpha)+\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(4,1,0)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,0)(5,2,0) | $\psi(\Pi_{(1,\omega,0)})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\alpha)+\psi_{\Omega_{\alpha+1}}(1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(4,1,0)(3,1,1)(4,2,0)(5,2,0)<br>(6,1,0)(5,2,0)(6,1,0)(2,0,0) | $\psi(\Pi_{(2,0,0)})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\alpha)\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0)(3,2,0) | $\psi(\Pi_{(\omega,0,0)})$<br>$\psi(\lambda\alpha.\psi_{\Omega_{\alpha+1}}(\alpha+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(3,2,0)(4,1,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1,0,0,0)})$<br>$\psi(\lambda\alpha.\psi_{\Omega_{\alpha+1}}(\alpha\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(4,0,0) | $\psi(\Pi_{(1@\omega)})$<br>$\psi(\lambda\alpha.\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(4,0,0)(3,1,1)(4,2,0) | $\psi(\Pi_{(1@\omega,\omega@0)})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(1))+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,0,0)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,0)(2,0,0) | $\psi(\Pi_{(1@\omega,1@1)})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(1))+\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,0,0)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,0)(6,0,0) | $\psi(\Pi_{(2@\omega)})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(1))\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(4,0,0)(3,2,0) | $\psi(\Pi_{(\omega@\omega)})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(1)+1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,0,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1@(\omega+1))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(1)+\alpha))-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,0,0)(3,2,0)(4,1,0)<br>(3,2,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1@(\omega+2))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(1)+\alpha\cdot 2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,0,0)(3,2,0)(4,1,0)(4,0,0) | $\psi(\Pi_{(1@(\omega\cdot 2))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(1)\cdot 2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,0,0)(3,2,0)(4,1,0)(4,0,0)<br>(3,2,0)(4,1,0)(4,0,0) | $\psi(\Pi_{(1@(\omega\cdot 3))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(1)\cdot 3))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(4,0,0)(4,0,0) | $\psi(\Pi_{(1@(\omega^2))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(2)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(4,0,0)(5,0,0) | $\psi(\Pi_{(1@(\omega^\omega))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\omega)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(4,1,0) | $\psi(\Pi_{(1@\Omega)})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\Omega)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1@(1,0))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(4,1,0)(3,2,0) | $\psi(\Pi_{(\omega@(1,0))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha)+1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,1,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1@(1,1))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha)+\alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,1,0)(3,2,0)(4,1,0)(4,0,0) | $\psi(\Pi_{(1@(1,\omega))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha)+\psi_{\Omega_{\alpha+1}}(1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,1,0)(3,2,0)(4,1,0)(4,0,0)(4,0,0) | $\psi(\Pi_{(1@(1,\omega^2))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha)+\psi_{\Omega_{\alpha+1}}(2)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,1,0)(3,2,0)(4,1,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1@(2,0))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha)\cdot 2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,1,0)(4,0,0) | $\psi(\Pi_{(1@(\omega,0))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha+1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,1,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1@(1,0,0))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha\cdot 2)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,1,0)(4,1,0)(3,2,0) | $\psi(\Pi_{(\omega@(1,0,0))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha\cdot 2)+1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,1,0)(4,1,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1@(1,0,1))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha\cdot 2)+\alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,1,0)(4,1,0)(3,2,0)<br>(4,1,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1@(1,1,0))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha\cdot 2)+\psi_{\Omega_{\alpha+1}}(\alpha)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,1,0)(4,1,0)(3,2,0)(4,1,0)<br>(4,1,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1@(2,0,0))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha\cdot 2)\cdot 2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,1,0)(4,1,0)(4,0,0) | $\psi(\Pi_{(1@(\omega,0,0))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha\cdot 2+1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,1,0)(4,1,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1@(1,0,0,0))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha\cdot 3)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,1,0)(4,1,0)(4,1,0)(4,0,0) | $\psi(\Pi_{(1@(\omega,0,0,0))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha\cdot 3+1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(4,1,0)(4,1,0)(4,1,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1@(1,0,0,0,0))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha\cdot 4)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,0,0) | $\psi(\Pi_{(1@(1@\omega))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(1))))-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,0,0)(4,1,0)(2,0,0) | $\psi(\Pi_{(1@(1@(\omega+1)))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(1)+\alpha)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,0,0)(4,1,0)(5,0,0) | $\psi(\Pi_{(1@(1@(\omega\cdot 2)))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(1)\cdot 2)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,0,0)(5,0,0) | $\psi(\Pi_{(1@(1@(\omega^2)))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(2))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,1,0)(2,0,0) | $\psi(\Pi_{(1@(1@(1,0)))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,1,0)(4,1,0)(5,1,0)(2,0,0) | $\psi(\Pi_{(1@(1@(2,0)))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha)\cdot 2)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,1,0)(5,1,0)(2,0,0) | $\psi(\Pi_{(1@(1@(1,0,0)))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha\cdot 2))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,1,0)(5,1,0)(5,1,0)(2,0,0) | $\psi(\Pi_{(1@(1@(1,0,0,0)))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha\cdot 3))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,1,0)(6,1,0)(2,0,0) | $\psi(\Pi_{(1@(1@(1@(1,0))))})$<br>$\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha)))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0)(3,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,0)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1})+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,0)(3,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}+1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}+\alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,0)(3,2,0)(4,1,0)(4,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,0)(3,2,0)(4,1,0)(5,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(1))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,0)(3,2,0)(4,1,0)(5,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(\alpha))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,0)(3,2,0)(4,1,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1})))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,0)(4,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}+1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}+\alpha)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,0)(4,1,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}$<br>$+\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,0)(5,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}$<br>$+\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}+1))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,0)(5,1,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}$<br>$+\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1})))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}\cdot 2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,0)(6,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+2}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,0)(5,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+\omega}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,2,1)(7,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha\cdot 2}))-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,0)(6,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(I_{\alpha+1})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'+1) - \Pi_0 \text{ aft } \alpha)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,3,0)(7,3,0)(8,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'+\alpha) - \Pi_0 \text{ aft } \alpha)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,3,0)(7,3,0)(8,2,0)(6,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'\cdot 2) - \Pi_0 \text{ aft } \alpha)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,3,0)(7,3,0)(8,2,0)(9,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+1}}(\Omega_{\alpha'+1}))$<br>$-\Pi_0 \text{ aft } \alpha)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,3,0)(7,3,0)<br>(8,2,0)(9,3,1)(10,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+1}}(\lambda\alpha''.(\alpha''+1)$<br>$-\Pi_0 \text{ aft } \alpha')) - \Pi_0 \text{ aft } \alpha)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)<br>(5,2,1)(6,3,0)(7,3,0)(8,2,0)(9,3,1)<br>(10,4,0)(11,4,0)(12,3,0)(13,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+1}}(\lambda\alpha''.(\psi_{\Omega_{\alpha''+1}}(\Omega_{\alpha''+1}))$<br>$-\Pi_0 \text{ aft } \alpha')) - \Pi_0 \text{ aft } \alpha)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(2,2,0)(3,2,0)(4,1,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1})) - \Pi_1 - \lambda\alpha.(\Omega_{\alpha+1}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(2,2,0)(3,2,0)(4,1,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+1}) - \Pi_1 -)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,0)(2,0,0) | $\psi((\lambda\alpha.(\Omega_{\alpha+1}) - \Pi_1 -)^{(1,0)})$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,0)(4,2,0) | $\psi(\Pi_2 \text{ aft } \lambda\alpha.(\Omega_{\alpha+1}) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,0)(4,2,1)(5,3,0)(6,3,0)(7,1,0) | $\psi(\lambda\alpha.(\alpha+\Omega) - \Pi_0 \text{ aft } \lambda\alpha.(\Omega_{\alpha+1}) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,0)(4,2,1)(5,3,0)(6,3,0)(7,2,0) | $\psi(\lambda\alpha.(\alpha + \Pi_2 \text{ aft } \lambda\alpha.(\Omega_{\alpha+1}) - \Pi_2) - \Pi_0$<br>$\text{aft } \lambda\alpha.(\Omega_{\alpha+1}) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,0)(4,2,1)(5,3,0)<br>(6,3,0)(7,2,0)(5,0,0) | $\psi(\lambda\alpha.(\alpha\cdot 2) - \Pi_0 \text{ aft } \lambda\alpha.(\Omega_{\alpha+1}) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,0)(4,2,1)(5,3,0)<br>(6,3,0)(7,2,0)(8,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1})) - \Pi_0 \text{ aft } \lambda\alpha.(\Omega_{\alpha+1}) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,0)(4,2,1)(5,3,0)(6,3,0)(7,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}) - \Pi_1 \text{ aft } \lambda\alpha.(\Omega_{\alpha+1}) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,0)(4,2,1)(5,3,0)(6,3,0)<br>(7,2,1)(6,2,0)(7,3,0) | $\psi(\Pi_2 \text{ aft 2nd } \lambda\alpha.(\Omega_{\alpha+1}) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1) | $\psi(\Pi_1 - \lambda\alpha.(\Omega_{\alpha+1}) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\Pi_2 \text{ aft } (\lambda\alpha.(\Omega_{\alpha+1}) - \Pi_2 -)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(3,1,1) | $\psi(\Pi_1 - (\lambda\alpha.(\Omega_{\alpha+1}) - \Pi_2 -)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(4,1,1) | $\psi(\Pi_1 - \lambda\alpha.(\Omega_{\alpha+1}) - \Pi_3)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(4,2,0)(5,1,1) | $\psi(\Pi_1 - \lambda\alpha.(\Omega_{\alpha+1}+1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(4,2,0)(5,1,1)(6,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+2) - \Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(4,2,0)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\alpha\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)<br>(5,2,0)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\alpha^2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)<br>(6,1,0)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\alpha^\alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(7,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)<br>(6,1,0)(7,2,1)(8,3,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'+1)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)<br>(7,2,1)(8,3,0)(9,3,0)(10,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\alpha'+1})-\Pi_1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,0)(7,2,1)<br>(8,3,0)(9,3,0)(10,2,1)(9,2,1)(10,3,0)<br>(11,3,0)(12,2,0)(13,3,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\alpha'+1}$<br>$+\psi_{\Omega_{\alpha'+1}}(\Omega_{\alpha'+1}))-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,1,1)(4,2,0)(5,2,0)(6,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,1)<br>(4,2,0)(5,2,0)(6,1,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+1}\cdot 2)-\Pi_0-)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)<br>(6,1,1)(5,1,0)(6,2,0) | $\psi(\Pi_2 \text{ aft } \lambda\alpha.(\Omega_{\alpha+1}\cdot 2)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,1)(5,1,1) | $\psi(\Pi_1-\lambda\alpha.(\Omega_{\alpha+1}\cdot 2)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)<br>(6,1,1)(5,1,1)(6,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot 2+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,1)<br>(5,1,1)(6,2,0)(7,2,0)(8,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot 3)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,2,0)(3,1,1)(4,2,0)(5,2,0)(6,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\omega+\Omega_{\alpha+1})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,2,0)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,1)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\omega\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,2,0)(3,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,2,0)(4,1,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1})))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+1}))-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(3,2,0)(4,1,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+1}+1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,2,0)(4,1,1)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+1}\cdot 2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(3,2,0)(4,1,1)(3,2,0)<br>(4,1,1)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+1}\cdot 3))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(4,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(4,0,0)(3,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(1)+1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(4,0,0)(3,2,0)(4,1,1)(4,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(1)\cdot 2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(4,0,0)(4,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(2)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(4,1,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\Omega)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\alpha)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(4,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+1})))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(4,1,1)(3,2,0)(4,1,1)(4,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+1})\cdot 2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(4,1,1)(4,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+1}+1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(4,1,1)(4,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+1}\cdot 2)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(1))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,1,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1})))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+1}))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,1,1)(6,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+1})))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(3,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(3,2,0)(4,1,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2})))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(4,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(4,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+\Omega_{\alpha+1})))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(4,1,1)(5,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}$<br>$+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(1))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(4,1,1)(5,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}$<br>$+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+1}))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(4,1,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}$<br>$+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(5,2,0)(4,1,1)(5,2,0)(4,1,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}$<br>$+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2})\cdot 2)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(5,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}$<br>$+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+1))))-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(5,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}$<br>$+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+\Omega_{\alpha+1}))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(5,1,1)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}$<br>$+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2})))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}\cdot 2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+1})))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+2})))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,2,0)(7,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+1))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,2,0)(7,1,1)(8,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2})))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(5,2,0)(6,2,0)(7,1,1)(8,2,0)(9,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}$<br>$(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+2}))))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)<br>(5,2,0)(6,2,0)(7,1,1)<br>(8,2,0)(9,2,0)(10,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+2}}$<br>$(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+1}))))))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+2))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,2,0)(4,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+2}\cdot 2))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,2,0)(5,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+1})))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,2,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+2})))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)<br>(4,2,0)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}$<br>$(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+2}))))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+3}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,0)(4,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+4}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+\omega}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)(4,3,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+\omega^2}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,3,1)(5,3,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Pi_1-\Pi_2\ \Pi_1-\Pi_2 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,3,1)(5,3,1)(6,3,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Pi_1-\Pi_3 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)(4,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'+1)-\Pi_0 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}((\lambda\alpha'.(\alpha'+1)-\Pi_0-)^{\omega} \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}((\lambda\alpha'.(\alpha'+1)-\Pi_0-)^{\alpha} \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}((\lambda\alpha'.(\alpha'+1)-\Pi_0-)^{\Omega_{\alpha+1}} \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}((\lambda\alpha'.(\alpha'+1)-\Pi_0-)^{\Omega_{\alpha+2}} \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}((\lambda\alpha'.(\alpha'+1)-\Pi_0-)^{\Omega_{\alpha+3}} \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,3,0)(6,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Pi_2 \text{ aft } \lambda\alpha'.(\alpha'+1)-\Pi_1 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,3,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Pi_1-\lambda\alpha'.(\alpha'+1)-\Pi_1 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,3,1)(6,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'+2)-\Pi_0 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'+\omega)-\Pi_0 \text{ aft } \alpha))-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,4,0)(6,3,0)(4,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'\cdot 2)-\Pi_0 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,4,0)(6,3,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\Omega_{\alpha'+1})-\Pi_1 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,4,0)(6,3,1)(7,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+2}}(\Omega_{\alpha'+2}))-\Pi_0 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,5,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+2}}(\Omega_{\alpha'+3}))-\Pi_0 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,5,1)(6,6,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+2}}(\lambda\alpha''.(\alpha''+1)$<br>$-\Pi_0 \text{ aft } \alpha'))-\Pi_0 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1) | $\psi(\Pi_1-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0) | $\psi(\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,1,1) | $\psi(\Pi_1-\lambda\alpha.(\alpha+1)-\Pi_1-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha\cdot 2)-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}))-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}))-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2})))-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,0)(6,2,0)(7,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}))))$<br>$-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,0)(6,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+3}))-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,0)(6,3,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+\omega}))-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,0)(6,3,1)(7,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'+1)$<br>$-\Pi_0 \text{ aft } \alpha))-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,0)(6,3,1)<br>(7,4,0)(8,4,0)(9,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'+\Omega_{\alpha+1})$<br>$-\Pi_0 \text{ aft } \alpha))-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,3,1)<br>(7,4,0)(8,4,0)(9,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'+\Omega_{\alpha+2})$<br>$-\Pi_0 \text{ aft } \alpha))-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,3,1)<br>(7,4,0)(8,4,0)(9,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'+\Omega_{\alpha+3})$<br>$-\Pi_0 \text{ aft } \alpha))-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,3,1)(7,4,0)<br>(8,4,0)(9,3,0)(8,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'\cdot 2)$<br>$-\Pi_0 \text{ aft } \alpha))-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,3,1)<br>(7,4,0)(8,4,0)(9,3,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\Omega_{\alpha'+1})$<br>$-\Pi_0 \text{ aft } \alpha))-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,3,1)(7,4,0)<br>(8,4,0)(9,3,1)(10,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+2}}(\Omega_{\alpha'+2}))$<br>$-\Pi_0 \text{ aft } \alpha))-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,3,1)(7,4,0)(8,4,0)<br>(9,3,1)(10,4,0)(11,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+2}}(\psi_{\Omega_{\alpha'+3}}(\Omega_{\alpha'+2})))$<br>$-\Pi_0 \text{ aft } \alpha))-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,3,1)(7,4,0)(8,4,0)<br>(9,3,1)(10,4,0)(11,4,0)(12,3,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+2}}(\psi_{\Omega_{\alpha'+3}}(\psi_{\Omega_{\alpha'+3}}(\Omega_{\alpha'+1}))))$<br>$-\Pi_0 \text{ aft } \alpha)) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,3,1)<br>(7,4,0)(8,4,0)(9,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+2}}(\psi_{\Omega_{\alpha'+3}}(\psi_{\Omega_{\alpha'+3}}(\Omega_{\alpha'+2}))))$<br>$-\Pi_0 \text{ aft } \alpha)) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,3,1)(7,4,0)(8,5,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+2}}(\Omega_{\alpha'+3}))$<br>$-\Pi_0 \text{ aft } \alpha)) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,3,1)(7,4,0)(8,5,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+2}}(\Omega_{\alpha'+\omega}))$<br>$-\Pi_0 \text{ aft } \alpha)) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,3,1)<br>(7,4,0)(8,5,1)(9,6,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+2}}(\lambda\alpha''.(\alpha''+1) - \Pi_0 \text{ aft } \alpha'))$<br>$-\Pi_0 \text{ aft } \alpha)) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,3,1)(7,4,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\Omega_{\alpha'+2})$<br>$-\Pi_1 \text{ aft } \alpha)) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,3,1)(7,4,1)(7,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'+1) - \Pi_0 - \lambda\alpha'.(\Omega_{\alpha'+2})$<br>$-\Pi_0 \text{ aft } \alpha)) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,3,1)(7,4,1)<br>(7,4,0)(8,4,0)(9,3,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\Omega_{\alpha'+1}) - \Pi_1 - \lambda\alpha'.(\Omega_{\alpha'+2})$<br>$-\Pi_0 \text{ aft } \alpha)) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,0)(6,3,1)(7,4,1)(7,4,0)<br>(8,4,0)(9,3,1)(10,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+2}}(\Omega_{\alpha'+2})) - \Pi_0 - \lambda\alpha'.(\Omega_{\alpha'+2})$<br>$-\Pi_0 \text{ aft } \alpha)) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1 -)^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1 -)^3)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,1,1) | $\psi(\Pi_1 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)<br>(4,2,0)(5,2,0)(6,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\Omega_{\alpha+1}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,0)(8,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+2}))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,0)(5,2,0)<br>(6,1,1)(7,2,0)(8,2,0)(9,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+1})))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)<br>(4,2,0)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+2})))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,0)(5,3,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+3})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)<br>(4,2,0)(5,3,1)(6,4,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'+1) - \Pi_0 \text{ aft } \alpha)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)<br>(4,2,0)(5,3,1)(6,4,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\Omega_{\alpha'+1}) - \Pi_1 \text{ aft } \alpha)) - \Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,0)(5,3,1)<br>(6,4,1)(6,4,0)(7,4,0)<br>(8,3,1)(9,4,1)(7,3,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}$<br>$+\psi_{\Omega_{\alpha+2}}(\Pi_1 - \lambda\alpha'.(\Omega_{\alpha'+2}) - \Pi_2 \text{ aft } \alpha)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,0)(5,3,1)<br>(6,4,1)(6,4,0)(7,4,0)(8,3,1)(9,4,1)<br>(7,3,1)(8,4,0)(9,5,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2} + \psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\Omega_{\alpha'+2} + \psi_{\Omega_{\alpha'+2}}(\Omega_{\alpha'+3}))$<br>$-\Pi_0 \text{ aft } \alpha)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2} \cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)<br>(2,2,0)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}) - \Pi_1 - \lambda\alpha.(\Omega_{\alpha+2} \cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1 - \lambda\alpha.(\Omega_{\alpha+2} \cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,1,1) | $\psi(\Pi_1 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_2\ \lambda\alpha.(\Omega_{\alpha+2})$<br>$-\Pi_1 - \lambda\alpha.(\Omega_{\alpha+2} \cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2} \cdot 2) - \Pi_0\ \lambda\alpha.(\Omega_{\alpha+2})$<br>$-\Pi_1 - \lambda\alpha.(\Omega_{\alpha+2} \cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(3,1,1) | $\psi(\Pi_1 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_2 - \lambda\alpha.(\Omega_{\alpha+2} \cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)<br>(4,2,1)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2} + 1) - \Pi_0\ \lambda\alpha.(\Omega_{\alpha+2})$<br>$-\Pi_2 - \lambda\alpha.(\Omega_{\alpha+2} \cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)<br>(4,2,1)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2} \cdot 2) - \Pi_0\ \lambda\alpha.(\Omega_{\alpha+2})$<br>$-\Pi_2 - \lambda\alpha.(\Omega_{\alpha+2} \cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,1,1) | $\psi(\Pi_1 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_3 - \lambda\alpha.(\Omega_{\alpha+2} \cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2} + 1) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2} \cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)<br>(4,2,1)(4,2,0)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2} + \omega) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2} \cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)<br>(4,2,0)(5,2,0)(6,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2} + \Omega_{\alpha+1}) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2} \cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)<br>(4,2,0)(5,2,0)(6,1,1)(7,2,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+2} \cdot 2) - \Pi_0 -)^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)<br>(4,2,0)(5,2,0)(6,1,1)(7,2,1)(5,1,1) | $\psi(\Pi_1 - \lambda\alpha.(\Omega_{\alpha+2} \cdot 2) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,1,1)(6,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2} \cdot 2 + 1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,1,1)(6,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2} \cdot 3) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)<br>(5,1,1)(6,2,1)(6,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2} \cdot 2 + 1) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2} \cdot 3) - \Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,1,1)(6,2,1)<br>(6,2,0)(7,2,0)(8,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot 2+\Omega_{\alpha+1})-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2}\cdot 3)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,1,1)(6,2,1)<br>(6,2,0)(7,2,0)(8,1,1)(9,2,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+2}\cdot 3)-\Pi_0-)^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,1,1)(6,2,1)<br>(6,2,0)(7,2,0)(8,1,1)(9,2,1)(7,1,1) | $\psi(\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot 3)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,1,1)(6,2,1)<br>(6,2,0)(7,2,0)(8,1,1)<br>(9,2,1)(7,1,1)(8,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot 4)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)<br>(2,2,0)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,1,1) | $\psi(\Pi_1-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_2\ \lambda\alpha.(\Omega_{\alpha+2})$<br>$-\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot 2)-\Pi_0\ \lambda\alpha.(\Omega_{\alpha+2})$<br>$-\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot 2)-\Pi_0\ \lambda\alpha.(\Omega_{\alpha+2})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot 2)$<br>$-\Pi_0\ \lambda\alpha.(\Omega_{\alpha+2})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(3,1,1) | $\psi(\Pi_1-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_2-\lambda\alpha.(\Omega_{\alpha+2}\cdot 2)$<br>$-\Pi_0\ \lambda\alpha.(\Omega_{\alpha+2})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)<br>(4,2,1)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot 2)-\Pi_0\ \lambda\alpha.(\Omega_{\alpha+2})$<br>$-\Pi_2-\lambda\alpha.(\Omega_{\alpha+2}\cdot 2)-\Pi_0\ \lambda\alpha.(\Omega_{\alpha+2})$<br>$-\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,1,1) | $\psi(\Pi_1-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_3-\lambda\alpha.(\Omega_{\alpha+2}\cdot 2)-\Pi_0$<br>$\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+1)-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2}\cdot 2)-\Pi_0$<br>$\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+2}\cdot 2)-\Pi_0-)^2\ \lambda\alpha.(\Omega_{\alpha+2})$<br>$-\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,1,1) | $\psi(\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot 2)-\Pi_1\ \lambda\alpha.(\Omega_{\alpha+2})$<br>$-\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_1\ \lambda\alpha.(\Omega_{\alpha+2})$<br>$-\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,2,0)(3,1,1) | $\psi(\Pi_1-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_2-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,2,0)(4,1,1) | $\psi(\Pi_1-\lambda\alpha.(\Omega_{\alpha+2})-\Pi_3-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,2,0)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+1)-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,2,0)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot 2)-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,2,0)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,1,1)(6,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot 3)-\Pi_0\ \lambda\alpha.(\Omega_{\alpha+2}\cdot 2)$<br>$-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,2,0)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,1,1)(6,2,1)<br>(6,2,0)(7,2,0)(8,1,1)(9,2,1)(7,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0\ \lambda\alpha.(\Omega_{\alpha+2}\cdot 2)$<br>$-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,2,0)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,1,1)(6,2,1)<br>(6,2,0)(7,2,0)(8,1,1)(9,2,1)<br>(7,2,0)(6,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot 2+1)-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,1,1)(4,2,1)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,2,0)(4,2,0)<br>(5,2,0)(6,1,1)(7,2,1)(5,1,1)(6,2,1)<br>(6,2,0)(7,2,0)(8,1,1)(9,2,1)(7,2,0)<br>(6,2,0)(7,2,0)(8,1,1)(9,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot 3)-\Pi_0-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(3,2,0) | $\psi((\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0-)^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0) | $\psi((\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0-)^3)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(3,1,1) | $\psi(\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega+1)-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega+\Omega_{\alpha+2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(3,1,1)(4,2,1)<br>(4,2,0)(5,2,0)(6,1,1)(7,2,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega+\Omega_{\alpha+2})-\Pi_0-)^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(3,1,1)(4,2,1)<br>(4,2,0)(5,2,0)(6,1,1)(7,2,1)<br>(5,1,1)(6,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega+\Omega_{\alpha+2}\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(3,1,1)(4,2,1)<br>(4,2,0)(5,2,0)(6,1,1)(7,2,1)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(3,1,1)(4,2,1)<br>(4,2,0)(5,2,0)(6,1,1)(7,2,1)<br>(5,2,0)(5,1,1) | $\psi(\Pi_1-\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega\cdot 2)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(3,1,1)(4,2,1)<br>(4,2,0)(5,2,0)(6,1,1)(7,2,1)<br>(5,2,0)(5,1,1)(6,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega\cdot 2+\Omega_{\alpha+2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(3,1,1)(4,2,1)<br>(4,2,0)(5,2,0)(6,1,1)(7,2,1)(5,2,0)<br>(5,1,1)(6,2,1)(6,2,0)(7,2,0)<br>(8,1,1)(9,2,1)(7,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega\cdot 3)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(3,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(4,1,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2})))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(4,1,1)<br>(5,2,0)(3,2,0)(4,1,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2})\cdot 2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)<br>(4,1,1)(5,2,0)(4,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+\Omega_{\alpha+1})))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(4,1,1)<br>(5,2,0)(4,1,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)<br>(4,1,1)(5,2,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}\cdot 2)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)<br>(4,1,1)(5,2,0)(6,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+1))))})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)<br>(4,1,1)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2})))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)<br>(4,1,1)(5,2,0)(6,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+3})))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(4,1,1)<br>(5,2,0)(6,3,1)(7,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'+1)-\Pi_0 \text{ aft } \alpha)))-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(4,1,1)(5,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+2})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(4,1,1)<br>(5,2,1)(3,2,0)(4,1,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+2} + \psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(3,2,0)(4,1,1)<br>(5,2,1)(3,2,0)(4,1,1)(5,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+2} \cdot 2)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(4,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+1}))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(4,1,1)(5,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+2}))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(4,1,1)<br>(5,2,1)(4,1,1)(5,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+2} \cdot 2))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+1})))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,1,1)(6,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+2})))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,2,0)(3,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3} + 1)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3} + \psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3}))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,2,0)<br>(4,1,1)(5,2,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3}$<br>$+\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3} + \psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3})))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,2,0)<br>(5,1,1)(6,2,1)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3} + \psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3}$<br>$+\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3} + \psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3}))))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,2,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3} \cdot 2)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+4}}(\Omega_{\alpha+3}))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,2,0)<br>(6,2,0)(7,1,1)(8,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+4}}(\psi_{\Omega_{\alpha+4}}(\Omega_{\alpha+2})))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,2,0)(6,2,0)<br>(7,1,1)(8,2,1)(8,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+4}}(\psi_{\Omega_{\alpha+4}}(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3}))))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)<br>(4,1,1)(5,2,1)(5,2,0)(6,2,0)(7,1,1)<br>(8,2,1)(8,2,0)(9,2,0)(10,1,1)(11,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+4}}(\psi_{\Omega_{\alpha+4}}$<br>$(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+4}}(\psi_{\Omega_{\alpha+4}}(\Omega_{\alpha+2}))))))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+4}}(\psi_{\Omega_{\alpha+4}}(\Omega_{\alpha+3})))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,2,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+4}}(\psi_{\Omega_{\alpha+4}}(\psi_{\Omega_{\alpha+4}}(\Omega_{\alpha+3}))))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+4})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,3,1)(4,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\lambda\alpha'.(\alpha' + 1) - \Pi_0 \text{ aft } \alpha)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,3,1)(4,4,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\lambda\alpha'.(\Omega_{\alpha'+2}) - \Pi_1 \text{ aft } \alpha)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,3,1)(4,4,1)(4,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+3}}(\Omega_{\alpha'+4})) - \Pi_0 \text{ aft } \alpha)) - \Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,3,1)<br>(4,4,1)(4,4,0)(5,5,1)(6,6,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+3}}(\lambda\alpha''.(\Omega_{\alpha''+2})$<br>$-\Pi_1 \text{ aft } \alpha')) - \Pi_1 \text{ aft } \alpha)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+3}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)<br>(2,2,0)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}) - \Pi_1 - \lambda\alpha.(\Omega_{\alpha+3}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1 - \lambda\alpha.(\Omega_{\alpha+3}) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+3}) - \Pi_1-)^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)<br>(5,2,1)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+3} + 1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,1)<br>(3,1,1)(4,2,0)(5,2,0)(6,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+3} + \Omega_{\alpha+1}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)<br>(5,2,1)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+3} + \Omega_{\alpha+2}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,1)<br>(3,1,1)(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+3} \cdot 2) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,1)(3,1,1)<br>(4,2,1)(4,2,1)(4,2,0)(5,2,0)<br>(6,1,1)(7,2,1)(7,2,1) | $\psi((\lambda\alpha.(\Omega_{\alpha+3} \cdot 2) - \Pi_0-)^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,1)(3,1,1)<br>(4,2,1)(4,2,1)(4,2,0)(5,2,0)(6,1,1)<br>(7,2,1)(7,2,1)(5,1,1)(6,2,1)(6,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+3} \cdot 3) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)<br>(2,2,0)(3,2,0)(4,1,1)<br>(5,2,1)(5,2,1)(3,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+3} \cdot \omega) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1)(5,2,1)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+4}}(\Omega_{\alpha_3})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,1)<br>(4,1,1)(5,2,1)(5,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+4}}(\psi_{\Omega_{\alpha+4}}(\Omega_{\alpha_3}))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,1)<br>(5,1,1)(6,2,1)(6,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+4}}(\psi_{\Omega_{\alpha+4}}(\psi_{\Omega_{\alpha+4}}(\Omega_{\alpha_3})))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+4}}(\Omega_{\alpha+4})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)<br>(5,2,1)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+4}}(\psi_{\Omega_{\alpha+5}}(\Omega_{\alpha+4}))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,1)(5,2,0)<br>(6,2,0)(7,1,1)(8,2,1)(8,2,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+4}}(\psi_{\Omega_{\alpha+5}}(\psi_{\Omega_{\alpha+5}}(\Omega_{\alpha+3})))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,1)(5,2,0)<br>(6,2,0)(7,1,1)(8,2,1)(8,2,1)(8,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+4}}(\psi_{\Omega_{\alpha+5}}(\psi_{\Omega_{\alpha+5}}(\psi_{\Omega_{\alpha+4}}(\Omega_{\alpha+4}))))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)<br>(2,2,0)(3,2,0)(4,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+4}}(\psi_{\Omega_{\alpha+5}}(\psi_{\Omega_{\alpha+5}}(\psi_{\Omega_{\alpha+5}}(\Omega_{\alpha+4}))))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)<br>(2,2,0)(3,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+4}}(\Omega_{\alpha+5})) - \Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)<br>(2,2,0)(3,3,1)(4,4,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+4}}(\lambda\alpha'.(\alpha'+1)-\Pi_0 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)<br>(2,2,0)(3,3,1)(4,4,1)(4,4,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+4}}(\lambda\alpha'.(\Omega_{\alpha'+3})-\Pi_1 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,0)<br>(3,3,1)(4,4,1)(4,4,1)(4,4,0)(5,5,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+4}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+4}}(\Omega_{\alpha'+4}))-\Pi_0 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+4})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)<br>(2,2,1)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+5})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0) | $\psi(\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+3})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,1)(5,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+4})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,0,0) | $\psi((\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_0-)^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,0,0)(3,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,0,0)<br>(3,1,1)(4,2,1)(5,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,0,0)(3,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,0,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+\omega+1}}(\Omega_{\alpha+\omega}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,0,0)<br>(4,1,1)(5,2,1)(6,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+\omega+1}}(\psi_{\Omega_{\alpha+\omega+1}}(\Omega_{\alpha+\omega})))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,0,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+\omega+1}}(\Omega_{\alpha+\omega+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+\omega+1}}(\Omega_{\alpha+\omega+2}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,0)(3,3,1)(4,3,1)(5,0,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+\omega+1}}(\lambda\alpha'.(\Omega_{\alpha'+\omega})-\Pi_0 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega+1})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,1)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,1)(3,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega\cdot 2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)<br>(2,2,1)(3,0,0)(2,2,1)(3,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega\cdot 3})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(4,1,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\psi(\Omega)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\Omega})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot 2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,0) | $\psi(\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\Omega_{\alpha\cdot 2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,1,0)(2,0,0) | $\psi((\lambda\alpha.(\Omega_{\alpha\cdot 2})-\Pi_0-)^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,1,0)(3,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot 2})-\Pi_1)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,1,0)(3,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot 2}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(2,2,0)(3,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}(\Omega_{\alpha\cdot 2+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot 2+1})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(2,2,1)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot 3})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha^2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1})})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(4,2,1)(5,3,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'+1)-\Pi_0 \text{ aft } \alpha)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(4,2,1)(5,3,1) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\alpha'+1})-\Pi_1 \text{ aft } \alpha)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(4,2,1)(5,3,1)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\alpha'+\alpha})-\Pi_0 \text{ aft } \alpha)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(4,2,1)(5,3,1)(6,2,0)(5,0,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\alpha'\cdot 2})-\Pi_0 \text{ aft } \alpha)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(4,2,1)(5,3,1)(6,2,0)(7,3,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\psi_{\Omega_{\alpha'+1}}(\Omega_{\alpha'+1})})-\Pi_0 \text{ aft } \alpha)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(4,2,1)(5,3,1)(6,2,0)(7,3,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\psi_{\Omega_{\alpha'+1}}(\Omega_{\alpha'+1})})-\Pi_0 \text{ aft } \alpha)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}+1})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(2,2,1)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}+\alpha})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(2,2,1)(3,1,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}\cdot 2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(2,2,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}\cdot 3})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(3,0,0) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}\cdot\omega})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}\cdot\alpha})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(3,1,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}{}^2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,0,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(1))})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,1,1) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+1}))})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,1,1)(5,1,1) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+1})))})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2})})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,0)(5,3,1)(6,4,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'+1)-\Pi_0 \text{ aft } \alpha)})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+2}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(2,2,1)(3,1,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+2}+\Omega_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(2,2,1)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+2}\cdot 2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(3,1,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+2}\cdot\Omega_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+2}{}^2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3})})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+3}})-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+4}}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,0,0) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+\omega}}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha\cdot 2}}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,1,0)(6,2,0) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1})}}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,1,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\Omega_{\alpha+1}}}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,1,1)(6,2,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\Omega_{\alpha+2}}}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,1,1)(6,2,1)(7,0,0) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\Omega_{\alpha+\omega}}}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,1,1)(6,2,1)(7,1,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\Omega_{\Omega_{\alpha+1}}}}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+1)) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+1)) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\Omega_{\alpha+1})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\Omega_{\alpha+2})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,0,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\Omega_{\alpha+\omega})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,1,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\Omega_{\Omega_{\alpha+1}})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,1,1)(6,2,1)(7,1,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\Omega_{\Omega_{\Omega_{\alpha+1}}})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\psi_{I_{\alpha+1}}(I_{\alpha+1}))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\psi_{I_{\alpha+1}}(I_{\alpha+1})\cdot 2)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(3,0,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\psi_{I_{\alpha+1}}(I_{\alpha+1})\cdot\omega)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(3,1,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\psi_{I_{\alpha+1}}(I_{\alpha+1})\cdot\Omega_{\alpha+1})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)<br>(3,1,1)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\psi_{I_{\alpha+1}}(I_{\alpha+1})^2)) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)<br>(4,1,1)(5,2,1)(6,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\psi_{I_{\alpha+1}}(I_{\alpha+1})^{\psi_{I_{\alpha+1}}(I_{\alpha+1})})) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(4,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\psi_{\psi_{I_{\alpha+1}}(I_{\alpha+1}+1)}(I_{\alpha+1}$<br>$+\psi_{I_{\alpha+1}}(I_{\alpha+1}+1)))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\psi_{I_{\alpha+1}}(I_{\alpha+1}+1))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\psi_{I_{\alpha+1}}(I_{\alpha+1}+2))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(4,2,1)(5,1,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\psi_{I_{\alpha+1}}(I_{\alpha+1}+\Omega_{\alpha+1}))) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(4,2,1)<br>(5,1,1)(6,2,1)(7,2,0)(6,2,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}+\psi_{I_{\alpha+1}}(I_{\alpha+1}$<br>$+\psi_{I_{\alpha+1}}(I_{\alpha+1}+1)))) - \Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot 2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,2,0)(2,2,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot 2+1))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,2,0)(2,2,1)(3,1,1)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot 2+\psi_{I_{\alpha+1}}(I_{\alpha+1})))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,2,0)(2,2,1)(3,1,1)<br>(4,2,1)(5,2,0)(4,2,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot 2+\psi_{I_{\alpha+1}}(I_{\alpha+1}+1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,2,0)(2,2,1)(3,1,1)<br>(4,2,1)(5,2,0)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot 2+\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot 2)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,2,0)(2,2,1)(3,1,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(4,2,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot 2+\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot 2+1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,2,0)(2,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot 3))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,2,0)(2,2,1)(3,2,0)(2,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot 4))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,0,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot\omega))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot\alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot\alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot\Omega_{\alpha+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,1,1)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot\psi_{I_{\alpha+1}}(I_{\alpha+1})))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,1,1)(4,2,1)(5,2,0)(5,1,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot\Omega_{\alpha+1})))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,1,1)(4,2,1)(5,2,0)(5,1,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot\Omega_{\alpha+1})))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,1)<br>(4,2,1)(5,2,0)(5,1,1)<br>(6,2,1)(7,2,0)(7,1,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot\psi_{I_{\alpha+1}}(I_{\alpha+1}$<br>$\cdot\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot\Omega_{\alpha+1}))))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}({I_{\alpha+1}}^2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,0,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}({I_{\alpha+1}}^\omega))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}({I_{\alpha+1}}^{\Omega_{\alpha+1}}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}({I_{\alpha+1}}^{I_{\alpha+1}}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,2,0)(4,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}({I_{\alpha+1}}^{{I_{\alpha+1}}^2}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,2,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}({I_{\alpha+1}}^{{I_{\alpha+1}}^{I_{\alpha+1}}}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,3,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\Omega_{I_{\alpha+1}+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,4,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\lambda\alpha'.(\alpha'+1)-\Pi_0 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,4,1)(6,4,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\lambda\alpha'.(\psi_{I_{\alpha'+1}}(I_{\alpha'+1}))-\Pi_0 \text{ aft } \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1) | $\psi(\lambda\alpha.(I_{\alpha+1})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}))-\Pi_0-\lambda\alpha.(I_{\alpha+1})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,2,1) | $\psi((\lambda\alpha.(I_{\alpha+1})-\Pi_1-)^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,2,1)(3,1,1) | $\psi(\Pi_1-\lambda\alpha.(I_{\alpha+1})-\Pi_2)$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)<br>(6,2,1)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(I_{\alpha+1}+\Omega_{\alpha+2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,2,1)<br>(3,1,1)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.(I_{\alpha+1}+\psi_{I_{\alpha+1}}(I_{\alpha+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,2,1)<br>(3,1,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(I_{\alpha+1}\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,2,1)(3,2,0) | $\psi(\lambda\alpha.(I_{\alpha+1}\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,0)(3,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{I_{\alpha+1}+1}}(\Omega_{I_{\alpha+1}+2}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}+1})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1) | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}+\Omega_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}+\psi_{I_{\alpha+1}}(I_{\alpha+1})})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(6,3,0) | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}+\psi_{I_{\alpha+1}}(\Omega_{I_{\alpha+1}+1})})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}\cdot 2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}\cdot 3})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(3,1,1) | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}\cdot\Omega_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)<br>(3,1,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(\Omega_{{I_{\alpha+1}}^2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)<br>(4,1,1)(5,2,1)(6,2,1) | $\psi(\lambda\alpha.(\Omega_{{I_{\alpha+1}}^{I_{\alpha+1}}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{I_{\alpha+1}+1}}(\Omega_{I_{\alpha+1}+1})})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(4,2,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{I_{\alpha+1}+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(4,2,1)(5,1,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{I_{\alpha+1}+\Omega_{\alpha+1}}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(4,2,1)<br>(5,1,1)(6,2,1)(7,2,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{I_{\alpha+1}\cdot 2}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+2}}(I_{\alpha+2}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,0)(2,2,1) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+2}}(I_{\alpha+2}+1))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,0)(2,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+2}}(I_{\alpha+2}\cdot 2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,0)(3,0,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+2}}(I_{\alpha+2}\cdot\omega))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,0)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+2}}({I_{\alpha+2}}^2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,0)(4,3,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+2}}(\Omega_{I_{\alpha+2}+1}))-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,1) | $\psi(\lambda\alpha.(I_{\alpha+2})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,1)(2,2,1)(3,2,1) | $\psi(\lambda\alpha.(I_{\alpha+3})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,2,1)(2,2,1)(3,2,1) | $\psi(\lambda\alpha.(I_{\alpha+4})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,0,0) | $\psi(\lambda\alpha.(I_{\alpha+\omega})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,0,0)(2,2,1) | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+\omega}+1})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,0,0)(2,2,1)(3,2,1) | $\psi(\lambda\alpha.(I_{\alpha+\omega+1})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,0,0)(2,2,1)(3,2,1)(3,0,0) | $\psi(\lambda\alpha.(I_{\alpha+\omega\cdot 2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(I_{\alpha\cdot 2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,1,0)(2,2,1)(3,2,1)(3,0,0) | $\psi(\lambda\alpha.(I_{\alpha\cdot 2+1})-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,1,0)(2,2,1)(3,2,1)(3,1,0)(2,0,0) | $\psi(\lambda\alpha.(I_{\alpha\cdot 3})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1) | $\psi(\lambda\alpha.(I_{\Omega_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,1,1)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.(I_{\psi_{I_{\alpha+1}}(I_{\alpha+1})})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,1,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(I_{I_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1)<br>(4,2,1)(5,2,1)(2,2,1)(3,2,1)<br>(3,1,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(I_{I_{\alpha+1}\cdot 2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1)<br>(4,2,1)(5,2,1)(3,1,1) | $\psi(\lambda\alpha.(I_{I_{\alpha+1}\cdot\Omega_{\alpha+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1)<br>(4,2,1)(5,2,1)(4,2,1) | $\psi(\lambda\alpha.(I_{\Omega_{I_{\alpha+1}+1}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(I_{I_{\alpha+2}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1)<br>(4,2,1)(5,2,1)(5,1,1) | $\psi(\lambda\alpha.(I_{I_{\Omega_{\alpha+1}}})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I(1,\alpha+1)}(I(1,\alpha+1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(2,2,1)(3,2,1) | $\psi(\lambda\alpha.(\psi_{I(1,\alpha+1)}(I(1,\alpha+1)+1))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(2,2,1)(3,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I(1,\alpha+1)}(I(1,\alpha+1)\cdot 2))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,0)(3,0,0) | $\psi(\lambda\alpha.(\psi_{I(1,\alpha+1)}(I(1,\alpha+1)\cdot\omega))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,0)(4,3,0) | $\psi(\lambda\alpha.(\psi_{I(1,\alpha+1)}(\Omega_{I(1,\alpha+1)+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1) | $\psi(\lambda\alpha.(I(1,\alpha+1))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,1)(2,2,1)(3,2,1)(3,2,1) | $\psi(\lambda\alpha.(I(1,\alpha+2))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,1)(3,0,0) | $\psi(\lambda\alpha.(I(1,\alpha+\omega))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,1)(3,1,1) | $\psi(\lambda\alpha.(I(1,\Omega_{\alpha+1}))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,1)(3,2,0) | $\psi(\lambda\alpha.(\psi_{I(2,\alpha+1)}(I(2,\alpha+1)))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,1)(3,2,1) | $\psi(\lambda\alpha.(I(2,\alpha+1))-\Pi_1)$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,1)(3,2,1)(3,2,1) | $\psi(\lambda\alpha.(I(3,\alpha+1))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,0,0) | $\psi(\lambda\alpha.(I(\omega,\alpha+1))-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,1,0)(2,0,0) | $\psi(\lambda\alpha.((2\ 1-)^{\alpha}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,1,0)(3,2,1) | $\psi(\lambda\alpha.((2\ 1-)^{\alpha+1}\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,1,0)(3,2,1)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.((2\ 1-)^{\alpha\cdot 2}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,1,0)(5,2,0) | $\psi(\lambda\alpha.((2\ 1-)^{\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1})}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,1) | $\psi(\lambda\alpha.((2\ 1-)^{\Omega_{\alpha+1}}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,1,1)(5,2,1)(6,2,1) | $\psi(\lambda\alpha.((2\ 1-)^{2\ 1-2\ \text{aft}\ \alpha}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,1,1)(5,2,1)(6,2,1)(7,1,1) | $\psi(\lambda\alpha.((2\ 1-)^{(2\ 1-)^{\Omega_{\alpha+1}}}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0) | $\psi(\lambda\alpha.((2\ 1-)^{(1,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(2,2,1) | $\psi(\lambda\alpha.(2\ \text{aft}\ (2\ 1-)^{(1,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(2,2,1)(3,2,1) | $\psi(\lambda\alpha.(2\ 1-2\ \text{aft}\ (2\ 1-)^{(1,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(2,2,1)(3,2,1)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.((2\ 1-)^{\alpha}\ \text{aft}\ (2\ 1-)^{(1,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(2,2,1)(3,2,1)(4,1,1) | $\psi(\lambda\alpha.((2\ 1-)^{\Omega_{\alpha+1}}\ \text{aft}\ (2\ 1-)^{(1,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(2,2,1)(3,2,1)(4,2,0) | $\psi(\lambda\alpha.(2\text{nd}\ (2\ 1-)^{(1,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(3,0,0) | $\psi(\lambda\alpha.(1-(2\ 1-)^{(1,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(3,2,0) | $\psi(\lambda\alpha.((1-)^{(1,0)}(2\ 1-)^{(1,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(3,2,1) | $\psi(\lambda\alpha.((2\ 1-)^{(1,1)}\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(3,2,1)(3,2,1) | $\psi(\lambda\alpha.((2\ 1-)^{(1,2)}\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(3,2,1)(4,2,0) | $\psi(\lambda\alpha.((2\ 1-)^{(2,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(3,2,1)(4,2,0)(3,2,1)(4,2,0) | $\psi(\lambda\alpha.((2\ 1-)^{(3,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(4,0,0) | $\psi(\lambda\alpha.((2\ 1-)^{(\omega,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(4,1,1) | $\psi(\lambda\alpha.((2\ 1-)^{(\Omega_{\alpha+1},0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(4,2,0) | $\psi(\lambda\alpha.((2\ 1-)^{(1,0,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(4,2,0)(3,2,1) | $\psi(\lambda\alpha.((2\ 1-)^{(1,0,1)}\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(4,2,0)(3,2,1)(4,2,0) | $\psi(\lambda\alpha.((2\ 1-)^{(1,1,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(4,2,0)(3,2,1)(4,2,0)(4,2,0) | $\psi(\lambda\alpha.((2\ 1-)^{(2,0,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(4,2,0)(4,0,0) | $\psi(\lambda\alpha.((2\ 1-)^{(\omega,0,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(4,2,0)(4,2,0) | $\psi(\lambda\alpha.((2\ 1-)^{(1,0,0,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(5,0,0) | $\psi(\lambda\alpha.((2\ 1-)^{(1@\omega)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(5,3,0) | $\psi(\lambda\alpha.(\psi_{2-2\ \text{aft}\ \alpha}(2\ \text{aft}\ 2-2\ \text{aft}\ \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(5,3,1)(6,4,0) | $\psi(\lambda\alpha.(\psi_{2-2\ \text{aft}\ \alpha}(\lambda\alpha'.(\alpha'+1)-\Pi_0\ \text{aft}\ \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(2,2,1) | $\psi(\lambda\alpha.(2\ \text{aft}\ 2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(2,2,1)(3,2,1) | $\psi(\lambda\alpha.(2\ 1-2\ \text{aft}\ 2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(2,2,1)(3,2,1)(4,2,0) | $\psi(\lambda\alpha.((2\ 1-)^{(1,0)}\ \text{aft}\ 2-2\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(2,2,1)(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(2\text{nd}\ 2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(3\text{rd}\ 2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,0,0) | $\psi(\lambda\alpha.(1-2-2\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,1,0) | $\psi(\lambda\alpha.((1-)^{\alpha}2-2\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,1,1) | $\psi(\lambda\alpha.((1-)^{\Omega_{\alpha+1}}2-2\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,0) | $\psi(\lambda\alpha.((1-)^{(1,0)}2-2\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,0)(4,3,0) | $\psi(\lambda\alpha.(\psi_{2\ 1-2-2\ \text{aft}\ \alpha}(2\ \text{aft}\ 2\ 1-2-2\ \text{aft}\ \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1) | $\psi(\lambda\alpha.(2\ 1-2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(3,2,1) | $\psi(\lambda\alpha.(2\ 1-2\ 1-2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,2,0) | $\psi(\lambda\alpha.((2\ 1-)^{(1,0)}2-2\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2\ 1-2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,2,1)(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2\ 1-2-2\ 1-2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,0,0) | $\psi(\lambda\alpha.((2-2\ 1-)^{\omega}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,0) | $\psi(\lambda\alpha.((2-2\ 1-)^{(1,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,1)(2,2,1) | $\psi(\lambda\alpha.(2\ \text{aft}\ 2-2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(4,2,1)(2,2,1)(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2\ \text{aft}\ 2-2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(4,2,1)(2,2,1)(3,2,1)(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(2\text{nd}\ 2-2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,1)(3,0,0) | $\psi(\lambda\alpha.(1-2-2-2\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,1)(3,2,1) | $\psi(\lambda\alpha.(2\ 1-2-2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,1)(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2\ 1-2-2-2\ \text{aft}\ \alpha)-\Pi_1)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,1)(3,2,1)(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2-2\ 1-2-2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(4,2,1)(3,2,1)(4,2,1)(4,2,1)<br>(3,2,1)(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2-2\ 1-2-2-2$<br>$1-2-2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,1)(4,2,0) | $\psi(\lambda\alpha.((2-2-2\ 1-)^{(1,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2-2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(4,2,1)(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2-2-2-2\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,0,0) | $\psi(\lambda\alpha.((2-)^{\omega}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,0) | $\psi(\lambda\alpha.((2-)^{(1,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,0)(3,2,1) | $\psi(\lambda\alpha.(2\ 1-(2-)^{(1,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,0)(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2\ 1-(2-)^{(1,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,0)(3,2,1)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.((2-)^{(1,0)}\ 1-(2-)^{(1,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,0) | $\psi(\lambda\alpha.(((2-)^{(1,0)}\ 1-)^{(1,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1) | $\psi(\lambda\alpha.((2-)^{(1,1)}\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(5,2,0) | $\psi(\lambda\alpha.((2-)^{(2,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,0)(5,2,0) | $\psi(\lambda\alpha.((2-)^{(1,0,0)}\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,0)(6,3,0) | $\psi(\lambda\alpha.(\psi_{3\ \text{aft}\ \alpha}(2\ \text{aft}\ 3\ \text{aft}\ \alpha))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(3\ \text{aft}\ \alpha)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(2,2,1) | $\psi(\lambda\alpha.(2\ \text{aft}\ 3\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(5,2,1)(2,2,1)(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2\ \text{aft}\ 3\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(5,2,1)(2,2,1)(3,2,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(2\text{nd}\ 3\ \text{aft}\ \alpha)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(3,0,0) | $\psi(\lambda\alpha.(1-3\ \text{aft}\ \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(3,2,1) | $\psi(\lambda\alpha.(2\ 1-3\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2\ 1-3\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(3,2,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(3\ 1-3\ \text{aft}\ \alpha)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(5,2,1)(3,2,1)(4,2,1)(5,2,1)<br>(3,2,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(3\ 1-3\ 1-3\ \text{aft}\ \alpha)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-3\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(2-2-3\ \text{aft}\ \alpha)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(3\ 2-3\ \text{aft}\ \alpha)-\Pi_2)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(3\ 2-3\ 2-3 \text{ aft } \alpha)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1) | $\psi(\lambda\alpha.(3-3 \text{ aft } \alpha)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1)(5,2,1) | $\psi(\lambda\alpha.(3-3-3 \text{ aft } \alpha)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(6,2,1) | $\psi(\lambda\alpha.(4 \text{ aft } \alpha)-\Pi_3)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(5,2,1)(6,2,1)(7,2,1) | $\psi(\lambda\alpha.(5 \text{ aft } \alpha)-\Pi_4)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0 \text{ aft } \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0 \text{ aft } \alpha)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,1,0)(3,2,0) | $\psi(\Pi_2 \text{ aft } \lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,1,1) | $\psi(\Pi_1-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,1,1)(3,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1\ \Pi_1-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,1,1)(3,2,1)(4,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)$<br>$-\Pi_0\ \Pi_1-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,1,1)(3,2,1)(4,3,0)(3,1,1) | $\psi(\Pi_1-\Pi_2-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0) | $\psi(\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,0)(3,1,1)(4,2,1)(5,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0\ \lambda\alpha.(\alpha+1)$<br>$-\Pi_1-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,0)(3,1,1)(4,2,1)(5,3,0)(4,2,0) | $\psi(\lambda\alpha.(\alpha+2)-\Pi_0-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,1,1)(4,2,1)(5,3,0)(4,2,0)<br>(5,1,1)(6,2,1)(7,3,0)(6,2,0) | $\psi(\lambda\alpha.(\alpha+3)-\Pi_0-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,0)(3,2,0) | $\psi(\lambda\alpha.(\alpha+\omega)-\Pi_0-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,0)(3,2,0)(3,2,0) | $\psi(\lambda\alpha.(\alpha+\omega^2)-\Pi_0-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\alpha\cdot 2)-\Pi_0-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,0)(3,2,0)(4,1,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}))$<br>$-\Pi_0-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,0)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(3,1,1)(4,2,1)(5,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0\ \lambda\alpha.(\Omega_{\alpha+1})-\Pi_2$<br>$-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(3,1,1)<br>(4,2,1)(5,3,0)(4,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+1)-\Pi_0-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(3,1,1)(4,2,1)<br>(5,3,0)(4,2,0)(5,2,0)(6,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot 2)-\Pi_0-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(3,1,1)(4,2,1)(5,3,0)<br>(4,2,0)(5,2,0)(6,1,1)(5,1,1)(6,2,1)<br>(7,3,0)(6,2,0)(7,2,0)(8,1,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot 3)-\Pi_0-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,0)(3,2,0)(4,1,1)(3,2,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\omega)-\Pi_0-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,0)(3,2,0)(4,1,1)(3,2,0)(4,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+1}))$<br>$-\Pi_0-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
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| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,0)(3,2,0)(4,1,1)(4,1,1) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+1})))$<br>$-\Pi_0 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,0)(3,2,0)(4,1,1)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}))$<br>$-\Pi_0 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2})))$<br>$-\Pi_0 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,0)(6,2,0)(7,2,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}))))$<br>$-\Pi_0 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,0)(6,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+3}))$<br>$-\Pi_0 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,0)(6,3,0) | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3}))$<br>$-\Pi_0 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(5,2,1) | $\psi(\lambda\alpha.(\Omega_{\alpha+3}) - \Pi_1 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}) - \Pi_0 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,1,0)(2,0,0) | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot 2}) - \Pi_0 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,1,1) | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}}) - \Pi_0 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,2,0) | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}))$<br>$-\Pi_0 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,2,1) | $\psi(\lambda\alpha.(I_{\alpha+1}) - \Pi_1 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,2,1)(5,2,1) | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}+1}) - \Pi_1 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,1) | $\psi(\lambda\alpha.(I_{\alpha+2}) - \Pi_1 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,2,1)(6,2,1) | $\psi(\lambda\alpha.(2\ 1 - 2\ 1 - 2 \text{ aft } \alpha)$<br>$-\Pi_1 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)<br>(6,2,1)(6,2,1)(6,2,1) | $\psi(\lambda\alpha.(2\ 1 - 2\ 1 - 2\ 1 - 2 \text{ aft } \alpha)$<br>$-\Pi_1 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,2,1)(7,1,0) | $\psi(\lambda\alpha.((2\ 1-)^{\alpha} \text{ aft } \alpha)$<br>$-\Pi_0 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,2,1)(7,2,0) | $\psi(\lambda\alpha.((2\ 1-)^{(1,0)} \text{ aft } \alpha)$<br>$-\Pi_0 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,2,1)(7,2,1) | $\psi(\lambda\alpha.(2 - 2 \text{ aft } \alpha) - \Pi_1 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)<br>(6,2,1)(7,2,1)(6,2,1) | $\psi(\lambda\alpha.(2\ 1 - 2 - 2 \text{ aft } \alpha)$<br>$-\Pi_1 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,2,1)<br>(7,2,1)(6,2,1)(7,2,1) | $\psi(\lambda\alpha.(2 - 2\ 1 - 2 - 2 \text{ aft } \alpha)$<br>$-\Pi_1 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)<br>(6,2,1)(7,2,1)(7,2,1) | $\psi(\lambda\alpha.(2 - 2 - 2 \text{ aft } \alpha)$<br>$-\Pi_1 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)<br>(6,2,1)(7,2,1)(7,2,1)(7,2,1) | $\psi(\lambda\alpha.(2 - 2 - 2 - 2 \text{ aft } \alpha)$<br>$-\Pi_1 - \lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)<br>(6,2,1)(7,2,1)(8,2,0) | $\psi(\lambda\alpha.((2-)^{(1,0)} \text{ aft } \alpha)$<br>$-\Pi_0 - \lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)<br>(6,2,1)(7,2,1)(8,2,1) | $\psi(\lambda\alpha.(3 \text{ aft } \alpha)-\Pi_2-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,2,1)<br>(7,2,1)(8,2,1)(9,2,1) | $\psi(\lambda\alpha.(4 \text{ aft } \alpha)-\Pi_3-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0$<br>$-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)<br>(6,3,0)(3,1,1)(4,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0+1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)<br>(6,3,0)(3,1,1)(4,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0+\Omega_{\alpha+2})-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,3,0)<br>(3,1,1)(4,2,1)(5,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0\cdot 2)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,3,0)(3,1,1)<br>(4,2,1)(5,3,0)(4,2,0)(5,2,0)<br>(6,1,1)(7,2,1)(8,3,0) | $\psi((\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0\cdot 2)-\Pi_0-)^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,3,0)(3,1,1)<br>(4,2,1)(5,3,0)(4,2,0)(5,2,0)(6,1,1)<br>(7,2,1)(8,3,0)(5,1,1)(6,2,1)(7,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0\cdot 3)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,3,0)(3,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0\cdot\omega)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0)<br>(3,2,0)(4,1,1)(5,2,1)(6,3,0)(5,2,0) | $\psi(\lambda\alpha.(\psi_{\Pi_2 \text{ aft } \lambda\beta.(\beta+1)-\Pi_0}$<br>$(\Pi_2 \text{ aft } \lambda\beta.(\beta+1)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(2,2,0)(3,3,0) | $\psi(\lambda\alpha.(\psi_{\Pi_2 \text{ aft } \lambda\beta.(\beta+1)-\Pi_0}$<br>$(2\text{nd } \Pi_2 \text{ aft } \lambda\beta.(\beta+1)-\Pi_0))-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,1) | $\psi(\lambda\alpha.(\Pi_2 \text{ aft } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,1)(2,2,1) | $\psi(\lambda\alpha.(2\text{nd } \Pi_2 \text{ aft } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,1)(3,0,0) | $\psi(\lambda\alpha.(\Pi_1-\Pi_2 \text{ aft } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,1)(3,1,1) | $\psi(\lambda\alpha.((\Pi_1-)^{\Omega_{\alpha+1}}\Pi_2 \text{ aft } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,1)(3,2,0) | $\psi(\lambda\alpha.((\Pi_1-)^{(1,0)}\Pi_2 \text{ aft } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,1)(3,2,1) | $\psi(\lambda\alpha.(\Pi_2\ \Pi_1-\Pi_2 \text{ aft } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,1)(3,2,1)(3,2,1) | $\psi(\lambda\alpha.(\Pi_2\ \Pi_1-\Pi_2$<br>$\Pi_1-\Pi_2 \text{ aft } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,1)(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(\Pi_2-\Pi_2 \text{ aft } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,1)(3,2,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(\Pi_3 \text{ aft } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,1)(3,3,0) | $\psi(\lambda\alpha.(2\text{nd } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(2,2,1)(3,3,0)(2,2,1)(3,3,0) | $\psi(\lambda\alpha.(3\text{rd } \lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,0,0) | $\psi(\lambda\alpha.(\Pi_1 - \lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(3,1,0)(2,0,0) | $\psi(\lambda\alpha.((\Pi_1-)^{\alpha}\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,1,1) | $\psi(\lambda\alpha.((\Pi_1-)^{\Omega_{\alpha+1}}\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,2,0) | $\psi(\lambda\alpha.((\Pi_1-)^{(1,0)}\lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,2,1) | $\psi(\lambda\alpha.(\Pi_2\ \Pi_1 - \lambda\beta.(\beta+1) - \Pi_0) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(3,2,1)(3,0,0) | $\psi(\lambda\alpha.(\Pi_1 - \Pi_2\ \Pi_1 - \lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(3,2,1)(3,2,1) | $\psi(\lambda\alpha.(\Pi_2\ \Pi_1 - \Pi_2\ \Pi_1 - \lambda\beta.(\beta+1) - \Pi_0) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(3,2,1)(4,2,1) | $\psi(\lambda\alpha.(\Pi_2 - \Pi_2\ \Pi_1 - \lambda\beta.(\beta+1) - \Pi_0) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(3,2,1)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(\Pi_3\ \Pi_1 - \lambda\beta.(\beta+1) - \Pi_0) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(3,2,1)(4,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0\ \Pi_1 - \lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(3,2,1)(4,3,0)(3,2,1) | $\psi(\lambda\alpha.(\Pi_2\ \Pi_1 - \lambda\beta.(\beta+1)$<br>$-\Pi_0\ \Pi_1 - \lambda\beta.(\beta+1) - \Pi_0) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(3,2,1)(4,3,0)(3,2,1)(4,3,0) | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1) - \Pi_0\ \Pi_1-)^2\lambda\beta.(\beta+1) - \Pi_0) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(3,2,1)(4,3,0)(4,2,0) | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1) - \Pi_0\ \Pi_1-)^{(1,0)} - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(3,2,1)(4,3,0)(4,2,1) | $\psi(\lambda\alpha.(\Pi_2 - \lambda\beta.(\beta+1) - \Pi_0) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(3,2,1)(4,3,0)(4,2,1)(3,2,1) | $\psi(\lambda\alpha.(\Pi_2\ \Pi_1 - \Pi_2 - \lambda\beta.(\beta+1) - \Pi_0) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(3,2,1)(4,3,0)(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(\Pi_2 - \Pi_2 - \lambda\beta.(\beta+1) - \Pi_0) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(3,2,1)(4,3,0)(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(\Pi_3 - \Pi_2 - \lambda\beta.(\beta+1) - \Pi_0) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(3,2,1)(4,3,0)(4,2,1)(5,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0\ \Pi_2 - \lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,2,1)<br>(4,3,0)(4,2,1)(5,3,0)(5,2,1) | $\psi(\lambda\alpha.(\Pi_3 - \lambda\beta.(\beta+1) - \Pi_0) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0 - \lambda\beta.(\beta+1) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(3,3,0)(3,3,0) | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1) - \Pi_0-)^3) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,0,0) | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1) - \Pi_0-)^{\omega}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,1,0)(2,0,0) | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1) - \Pi_0-)^{\alpha}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,1,1) | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1) - \Pi_0-)^{\Omega_{\alpha+1}}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,2,0) | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1) - \Pi_0-)^{(1,0)}) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,1)(3,2,1) | $\psi(\lambda\alpha.(\Pi_2\ \Pi_1 - \lambda\beta.(\beta+1) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,1)(3,2,1)(4,3,0)(5,2,1)(4,2,1) | $\psi(\lambda\alpha.(\Pi_2 - \lambda\beta.(\beta+1) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,1)(3,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0 - \lambda\beta.(\beta+1) - \Pi_1) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,1)(3,3,0)(4,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_1\ \lambda\beta.(\beta+1)$<br>$-\Pi_0 - \lambda\beta.(\beta+1) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,1)(4,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_1 - \lambda\beta.(\beta+1) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,1)(5,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_2) - \Pi_2)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,1)(5,2,1)(6,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_3)-\Pi_3)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,1)(5,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+2)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,1)(5,3,0)(5,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+2)-\Pi_0-\lambda\beta.(\beta+2)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,1)(5,3,0)(6,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+2)-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,2,1)(5,3,0)(6,2,1)(7,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+3)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\omega)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(3,3,0)(4,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\omega)-\Pi_0-\lambda\beta.(\beta+\omega)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(4,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\omega)-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(4,2,1)(5,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\omega+1)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(4,2,1)(5,3,0)(6,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\omega\cdot 2)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(4,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\omega^2)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,1,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\Omega)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,1,0)(2,0,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\alpha)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,1,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\Omega_{\alpha+1})-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,1,1)(6,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\Omega_{\alpha+2})-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,1,1)(6,2,1)(7,0,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\Omega_{\alpha+\omega})-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,1,1)(6,2,1)(7,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+I_{\alpha+1})-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,1,1)(6,2,1)(7,2,1)(8,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\Pi_2-\Pi_2 \text{ aft } \alpha)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,1,1)(6,2,1)(7,2,1)(8,2,1)(9,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\Pi_3 \text{ aft } \alpha)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,1,1)(6,2,1)(7,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\lambda\beta.(\beta+1)-\Pi_0 \text{ aft } \alpha)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,1,1)(6,2,1)(7,3,0)(8,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\lambda\beta.(\beta+\omega)-\Pi_0 \text{ aft } \alpha)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,1,1)(6,2,1)(7,3,0)(8,3,0)(9,1,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\lambda\beta.(\beta+\alpha)-\Pi_0 \text{ aft } \alpha)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,1,1)(6,2,1)(7,3,0)(8,3,0)<br>(9,1,1)(10,2,1)(11,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\lambda\beta.(\beta+\lambda\beta.(\beta+1)$<br>$-\Pi_0 \text{ aft } \alpha)-\Pi_0 \text{ aft } \alpha)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta\cdot 2)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(4,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\beta\cdot 2)-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,2,0)(4,2,1)(5,3,0)(6,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta\cdot 3)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(4,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta\cdot\omega)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(4,3,0)(5,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta^2)-\Pi_0)-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,2,0)(4,3,0)(5,2,0)(4,3,0)(5,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta^3)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(5,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta^\beta)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(6,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta^{\beta^\beta})-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(6,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\Omega_{\beta+1}))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(6,3,1)(7,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\alpha'.(\alpha'+1)-\Pi_0 \text{ aft } \beta))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(6,3,1)(7,4,1) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\alpha'.(\Omega_{\alpha'+2})-\Pi_1 \text{ aft } \beta))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,0)(6,3,1)(7,4,1)(8,5,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\alpha'.(\lambda\beta'.(\beta'+1)-\Pi_0)$<br>$-\Pi_0 \text{ aft } \beta))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,2,0)(6,3,1)(7,4,1)(8,5,0)(9,5,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\alpha'.(\lambda\beta'.(\beta'+\omega)-\Pi_0)$<br>$-\Pi_0 \text{ aft } \beta))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,2,0)(6,3,1)(7,4,1)<br>(8,5,0)(9,5,0)(10,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\alpha'.(\lambda\beta'.(\beta'\cdot 2)-\Pi_0)$<br>$-\Pi_0 \text{ aft } \beta))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,2,0)(6,3,1)(7,4,1)(8,5,0)<br>(9,5,0)(10,4,0)(11,5,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\alpha'.(\lambda\beta'.(\psi_{\Omega_{\beta'+1}}(\Omega_{\beta'+1}))-\Pi_0)$<br>$-\Pi_0 \text{ aft } \beta))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,2,0)(6,3,1)(7,4,1)(8,5,0)(9,5,0)<br>(10,4,0)(11,5,1)(12,6,1)(13,7,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\alpha'.(\lambda\beta'.(\psi_{\Omega_{\beta'+1}}$<br>$(\lambda\alpha''.(\lambda\beta''.(\beta''+1)-\Pi_0)$<br>$-\Pi_0 \text{ aft } \beta'))-\Pi_0)-\Pi_0 \text{ aft } \beta))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+1})-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,1)(4,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+1})-\Pi_2)-\Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,2,1)(4,2,1)(5,3,0)(6,3,0)(7,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+1}\cdot 2)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,2,1)(4,2,1)(5,3,0)(6,3,0)(7,2,1)<br>(6,2,1)(7,3,0)(8,3,0)(9,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+1}\cdot 3)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,1)(4,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+1}\cdot\omega)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,1)(4,3,0)(5,2,1) | $\psi(\lambda\alpha.(\lambda\beta.({\Omega_{\beta+1}}^2)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,1)(6,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\Omega_{\beta+2}))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,1)(6,3,0)(7,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\psi_{\Omega_{\beta+3}}(\Omega_{\beta+2})))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,2,1)(6,3,0)(7,3,0)(8,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\psi_{\Omega_{\beta+3}}(\psi_{\Omega_{\beta+3}}(\Omega_{\beta+1}))))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0)<br>(5,2,1)(6,3,0)(7,3,0)(8,2,1)(9,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\psi_{\Omega_{\beta+3}}(\psi_{\Omega_{\beta+3}}$<br>$(\psi_{\Omega_{\beta+2}}(\Omega_{\beta+2})))))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,3,0)(5,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\psi_{\Omega_{\beta+3}}(\psi_{\Omega_{\beta+3}}(\Omega_{\beta+2}))))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\Omega_{\beta+3}))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+2})-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(2,2,1) | $\psi(\lambda\alpha.(\Pi_2 \text{ aft } \lambda\beta.(\Omega_{\beta+2})-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0-\lambda\beta.(\Omega_{\beta+2})-\Pi_1)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(3,3,0)(4,3,0)(5,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+1})-\Pi_1-\lambda\beta.(\Omega_{\beta+2})-\Pi_1)-\Pi_1)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(3,3,0)(4,3,0)(5,2,1)(6,3,1) | $\psi(\lambda\alpha.((\lambda\beta.(\Omega_{\beta+2})-\Pi_1-)^2)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,3,0)<br>(4,3,0)(5,2,1)(6,3,1)(4,2,1)(5,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+2}\cdot 2)-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,3,0)<br>(4,3,0)(5,2,1)(6,3,1)(4,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+2}\cdot\omega)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,3,0)<br>(4,3,0)(5,2,1)(6,3,1)(6,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+3}}(\Omega_{\beta+3}))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(3,3,0)(4,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+3}}(\Omega_{\beta+4}))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+3})-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(3,3,1)(3,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+4})-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,0,0) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+\omega})-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+\alpha})-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,1,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+\Omega_{\alpha+1}})-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,2,0) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta\cdot 2})-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\Omega_{\beta+1}})-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,2,1)(5,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\Omega_{\beta+2}})-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,2,1)(5,3,1)(6,2,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\Omega_{\Omega_{\beta+1}}})-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{I_{\beta+1}}(I_{\beta+1}))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,0)(3,3,1)(4,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{I_{\beta+1}}(I_{\beta+1}\cdot 2))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,0)(5,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{I_{\beta+1}}(\Omega_{I_{\beta+1}+1}))-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(I_{\beta+1})-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(3,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{I_{\beta+1}+1})-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(3,3,1)(4,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(I_{\beta+2})-\Pi_1)-\Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(4,0,0) | $\psi(\lambda\alpha.(\lambda\beta.(\Pi_1-\Pi_2\ \Pi_1-\Pi_2\ \text{aft}\ \beta)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(4,1,0)(2,0,0) | $\psi(\lambda\alpha.(\lambda\beta.((\Pi_1-)^{\alpha}\Pi_2\ \Pi_1-\Pi_2\ \text{aft}\ \beta)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(4,2,0) | $\psi(\lambda\alpha.(\lambda\beta.((\Pi_1-)^{\beta}\Pi_2\ \Pi_1-\Pi_2\ \text{aft}\ \beta)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(4,2,1) | $\psi(\lambda\alpha.(\lambda\beta.((\Pi_1-)^{\Omega_{\beta+1}}\Pi_2\ \Pi_1-\Pi_2\ \text{aft}\ \beta)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(4,2,1)(5,3,1)(6,3,1) | $\psi(\lambda\alpha.(\lambda\beta.((\Pi_1-)^{\Pi_2\ \Pi_1-\Pi_2\ \text{aft}\ \beta}$<br>$\Pi_2\ \Pi_1-\Pi_2\ \text{aft}\ \beta)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(4,3,0) | $\psi(\lambda\alpha.(\lambda\beta.((\Pi_1-)^{(1,0)}\Pi_2\ \Pi_1-\Pi_2\ \text{aft}\ \beta)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(4,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Pi_2\ \Pi_1-\Pi_2\ \Pi_1-\Pi_2\ \text{aft}\ \beta)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(4,3,1)(4,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Pi_2\ \Pi_1-\Pi_2$<br>$\Pi_1-\Pi_2\ \Pi_1-\Pi_2\ \text{aft}\ \beta)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(5,1,0)(2,0,0) | $\psi(\lambda\alpha.(\lambda\beta.((\Pi_2\ \Pi_1-)^{\alpha}\ \text{aft}\ \beta)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(5,2,0) | $\psi(\lambda\alpha.(\lambda\beta.((\Pi_2\ \Pi_1-)^{\beta}\ \text{aft}\ \beta)-\Pi_0)-\Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(5,3,0) | $\psi(\lambda\alpha.(\lambda\beta.((\Pi_2\ \Pi_1-)^{(1,0)}\ \text{aft}\ \beta)-\Pi_0)-\Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(5,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Pi_2 - \Pi_2 \text{ aft } \beta) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(5,3,1)(5,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Pi_2 - \Pi_2 - \Pi_2 \text{ aft } \beta) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(5,3,1)(6,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Pi_3 \text{ aft } \beta) - \Pi_2) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,3,1)(5,3,1)(6,3,1)(7,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Pi_4 \text{ aft } \beta) - \Pi_3) - \Pi_3)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(2,2,0) | $\psi(\lambda\alpha.(\alpha+1) - \Pi_0$<br>$-\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(2,2,1) | $\psi(\lambda\alpha.(\Pi_2 \text{ aft } \lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(3,0,0) | $\psi(\lambda\alpha.(\Pi_1 - \lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(3,3,0) | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0$<br>$-\lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(3,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Pi_2 \text{ aft } \lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(3,3,1)(4,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(2\text{nd } \lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(4,0,0) | $\psi(\lambda\alpha.(\lambda\beta.(\Pi_1 - \lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(4,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Pi_2\ \Pi_1 - \lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(4,3,1)(5,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+1)$<br>$-\Pi_0\ \Pi_1 - \lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(4,3,1)(5,4,0)(5,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\Pi_2 - \lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(4,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_0$<br>$-\lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(5,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_1) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(5,4,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+2) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,0)(5,4,0)(6,3,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Omega_{\gamma+1}) - \Pi_1) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,4,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Omega_{\gamma+2}) - \Pi_1) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,1)(4,4,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Omega_{\gamma+3}) - \Pi_1) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,1)(5,4,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(I_{\gamma+1}) - \Pi_1) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,1)(5,4,1)(5,4,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Pi_2\ \Pi_1 - \Pi_2\ \Pi_1 - \Pi_2$<br>$\text{aft } \gamma) - \Pi_1) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,1)(5,4,1)(6,4,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Pi_2 - \Pi_2 \text{ aft } \gamma) - \Pi_1) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,1)(5,4,1)(6,4,1)(7,4,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Pi_3 \text{ aft } \gamma) - \Pi_2) - \Pi_2) - \Pi_2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,1)(5,5,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\lambda\delta.(\delta+1) - \Pi_0) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,1)(5,5,1) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\lambda\delta.(\Omega_{\delta+2}) - \Pi_1) - \Pi_1) - \Pi_1) - \Pi_1)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)<br>(4,4,1)(5,5,1)(6,6,0) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\lambda\delta.(\lambda\epsilon.(\epsilon+1)$<br>$-\Pi_0) - \Pi_0) - \Pi_0) - \Pi_0) - \Pi_0)$ |

| BMS | Stable-Collapsing Notation |
|---|---|
| (0,0,0)(1,1,1)(2,2,2) | $\psi(\omega - \pi - \Pi_0)$<br>p.f.e.c.LRO |

## A.14 BMS vs Projection

The results in this section are primarily cited from the analysis by Zuicai Mengxin (最菜萌新).

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha}+1))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha}+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha}+\Omega_{\alpha+1}{}^{\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha+\Omega_{\alpha+1}{}^{\alpha}}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha}+\Omega_{\alpha+1}{}^{\varepsilon_{\alpha+1}}))$<br>$=\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot(\alpha+1)}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot(\alpha+\omega)}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot(\alpha+\omega^2)}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot(\alpha+\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha}+\Omega_{\alpha+1}))}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,1)(4,2,1)(5,2,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,1)(7,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot(\alpha\cdot 2+\omega)}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha\cdot\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(2,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha\cdot\omega^2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha\cdot\Omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha^2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\varepsilon_{\alpha+1}}))$<br>$=\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}+1}+\Omega_{\alpha+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+1} + {\Omega_{\alpha+1}}^{\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+1} + {\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(5,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+1} \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+1} \cdot \omega^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+2}))$<br>$= \psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+1} \cdot \alpha))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\alpha}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0)(2,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\alpha}) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,1)(4,2,1)(5,2,1)<br>(5,2,0)(4,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\alpha} \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\alpha} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\alpha} \cdot \varepsilon_{\alpha+1}))$<br>$= \psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\alpha+1} + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,1) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\alpha\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(3,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\alpha\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\varepsilon_{\alpha+1}}))$<br>$= \psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot 2} + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot 2} + \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot 2} + {\Omega_{\alpha+1}}^{\Omega_{\alpha+1}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(5,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot 2} + {\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\alpha}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(7,3,0) | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot 2} \cdot 2 + \Omega_{\alpha+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2}\cdot\omega^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2}\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2+\alpha}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2+\varepsilon_{\alpha+1}}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 3}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot(\omega+1)}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,0,0)(3,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\omega^2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,2,1)(5,2,0)(4,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha}\cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot(\alpha+1)}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(3,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha^2}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\varepsilon_{\alpha+1}}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2}+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1)<br>(6,2,0)(7,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2}\cdot 2+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2}\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2}\cdot\alpha\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2+1}+\Omega_{\alpha+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(6,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2+1} + \Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(2,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2+\Omega_{\alpha+1}} + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2+\Omega_{\alpha+1}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2\cdot 2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2\cdot\alpha}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2\cdot\varepsilon_{\alpha+1}}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(6,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^3} + \Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^3} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^\Omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(4,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^\omega})}}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^\alpha}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^\alpha} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,1)(3,1,1)(4,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^\alpha+\Omega_{\alpha+1}{}^\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^\alpha\cdot\alpha}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\alpha+1}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(3,1,1)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\alpha\cdot 2}}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\alpha^2}}))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(5,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\varepsilon_{\alpha+1}}}))$<br>$= \psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(5,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} + \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,0)(6,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} + \Omega_{\alpha+1} \cdot \alpha))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} + \Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^\omega}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,2,1)(8,2,0)(9,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} \cdot 2 + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(6,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} \cdot \omega \cdot 2))$ |
| (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(2,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} \cdot \omega^2))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} \cdot \alpha))$ |

| BMS | Projection |
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| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(2,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}+1}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(2,1,1)(3,1,1)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}+\Omega_{\alpha+1}{}^{\alpha}}}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)<br>(2,1,1)(3,1,1)(4,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}+1}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,1)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}+\alpha}}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(3,1,1)(4,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\omega}}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha}}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(4,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{2}}}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\alpha}}}))$ |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}))$<br>$= \psi(\Pi_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(1,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) + \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(1,1,1)<br>(2,1,1)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) + I_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(1,1,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(1,1,0)<br>(2,2,1)(3,3,0)(3,1,0)(3,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(1,1,0)<br>(2,2,1)(3,3,0)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_2)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(1,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,0)(1,1,1)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_\omega + \Omega_\omega \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_\omega + \psi_\omega(\Omega_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,3,0)(4,1,0)(1,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_\omega + \psi_\omega(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,3,0)(4,1,0)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_\omega + \Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,3,0)<br>(4,1,0)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot (\Omega_\omega + 1))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,3,0)(4,1,0)<br>(3,2,1)(4,3,0)(4,1,0)(1,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_\omega \cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,3,0)(4,1,0)(4,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_\omega \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,3,0)<br>(4,1,0)(4,1,0)(1,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot {\Omega_\omega}^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,3,0)(4,1,0)(5,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \psi_\omega(\Omega_{\omega+1}))$ |

| BMS | Projection |
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| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,3,0)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_{\omega+1})$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,3,0)(4,2,0)(3,2,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_{\omega\cdot 2})$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,0)(1,1,1)(2,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_{\omega^2})$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \psi_I(I))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)<br>(5,3,0)(5,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot I)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,3,0)(5,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot \Omega_{I+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot I_\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(4,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,0)(1,1,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1})^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)<br>(1,1,1)(2,2,0)(1,1,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1})^2 + \psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(1,1,1)<br>(2,2,0)(2,1,0)(1,1,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1})^2 \cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)<br>(2,1,0)(1,1,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1})^3)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1})^\omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)<br>(3,1,0)(1,1,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1})^{\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1})})$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,0)(3,2,0)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} + \Omega_{\alpha+1}) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} + \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} + \Omega_{\alpha+1}{}^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,0)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)<br>(3,2,1)(4,3,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot 2) +$<br>$\psi_{\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}+\Omega_{\alpha+1})}(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot 2) + 1))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)<br>(3,2,1)(4,3,0)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot 2) + \psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)<br>(3,2,1)(4,3,0)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot 2) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)<br>(3,2,1)(4,3,0)(4,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot 2) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)<br>(3,2,1)(4,3,0)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot 2) \cdot \psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,1)<br>(4,3,0)(4,2,0)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot 2)^2)$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)<br>(3,2,1)(4,3,0)(4,2,0)(5,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot 2 + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,1)<br>(4,3,0)(4,2,0)(5,3,1)(6,4,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot 3))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(1,1,1)(2,2,0)(2,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot \omega) \cdot 2)$ |

| BMS | Projection |
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| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,1)(2,1,0)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot \omega + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(2,1,0)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot (\omega + 1)))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(2,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot \omega^2))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot \Omega))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot \alpha))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,3,0)<br>(4,2,1)(5,2,0)(4,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot \alpha \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,1)(3,1,0)(2,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot \alpha \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot \varepsilon_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,1)(3,1,0)(4,2,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot \Omega_{\alpha+1} + \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,1,0)(4,2,1)(5,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot \Omega_{\alpha+1} + \varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,0)<br>(4,2,1)(5,3,0)(5,2,1)(6,2,0)(5,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot (\Omega_{\alpha+1} + \alpha)))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,0)<br>(4,2,1)(5,3,0)(5,2,1)(6,2,0)(7,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot \Omega_{\alpha+1} \cdot 2 + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,1)(3,1,1)(2,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot \Omega_{\alpha+1} \cdot \omega^2))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot \Omega_{\alpha+1} \cdot \alpha))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot {\Omega_{\alpha+1}}^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,1)(3,1,1)(3,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot {\Omega_{\alpha+1}}^\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot {\Omega_{\alpha+1}}^\alpha))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot {\Omega_{\alpha+1}}^{\varepsilon_{\alpha+1}}))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,1)(3,1,1)(4,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot {\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^\omega}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^2))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(2,1,0)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^2 + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(2,1,0)(3,2,1)(4,3,0)(4,2,1)(5,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,1,1)(3,2,0)(2,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^2 \cdot \alpha))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(2,1,1)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^2 \cdot \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(2,1,1)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^3))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^\alpha))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(3,1,0)(2,0,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^\alpha) \cdot \omega)$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(3,1,0)(2,1,0)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\alpha} + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(3,1,0)(2,1,0)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\alpha} + \varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,0)(2,1,0)(3,2,1)(4,3,0)(4,2,1)<br>(5,3,0)(5,1,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\alpha} + \varepsilon_{\Omega_{\alpha+1}+1}{}^{\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\alpha})}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,0)(2,1,0)(3,2,1)(4,3,0)<br>(4,2,1)(5,3,0)(5,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\alpha} + \varepsilon_{\Omega_{\alpha+1}+1}{}^{\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\alpha}+\Omega_{\alpha+1})}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,0)(2,1,0)(3,2,1)(4,3,0)(4,2,1)<br>(5,3,0)(5,2,0)(4,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\alpha} \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(3,1,0)(2,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\alpha} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,0)(2,1,1)(3,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\alpha} \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\alpha^2}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\varepsilon_{\alpha+1}}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,0)(4,2,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}} + \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(2,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}} \cdot \omega^2))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(2,1,1)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(2,1,1)(3,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(2,1,1)(3,2,0)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}\cdot 2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(3,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}\cdot\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}\cdot\varepsilon_{\alpha+1}}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}{}^2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(3,1,1)(2,1,1)<br>(3,2,0)(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}{}^2\cdot 2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}{}^2\cdot\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}{}^3} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}{}^\omega}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}{}^\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,1,0)(3,1,1)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}{}^{\alpha\cdot 2}}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,1,0)(5,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1^{\Omega_{\alpha+1}^{\varepsilon_{\alpha+1}}}}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1^{\Omega_{\alpha+1}^{\Omega_{\alpha+1}}}\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1^{\varepsilon_{\Omega_{\alpha+1}+1}}}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,2,0)(2,1,1)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1^{\varepsilon_{\Omega_{\alpha+1}+1}+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,2,0)(2,1,1)<br>(3,2,0)(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1^{\varepsilon_{\Omega_{\alpha+1}+1}\cdot 2}}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,2,0)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1^{\varepsilon_{\Omega_{\alpha+1}+1}\cdot\Omega_{\alpha+1}}\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,2,0)(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1^{\varepsilon_{\Omega_{\alpha+1}+1}^2}}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,2,0)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1^{\varepsilon_{\Omega_{\alpha+1}+1}^\alpha}}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,2,0)(4,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1^{\varepsilon_{\Omega_{\alpha+1}+1}^{\Omega_{\alpha+1}}}\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0)<br>(3,1,1)(4,2,0)(4,1,1)(5,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1^{\varepsilon_{\Omega_{\alpha+1}+1}^{\varepsilon_{\Omega_{\alpha+1}+1}}}}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+2}))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,2,0)(2,1,0)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+2}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,2,0)(2,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(2,2,0)(2,1,1)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+2}\cdot\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,2,0)<br>(2,1,1)(3,2,0)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+2^2}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,2,0)(2,1,1)<br>(3,2,0)(3,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+2^\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,2,0)(2,1,1)<br>(3,2,0)(3,2,0)(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+2^{\varepsilon_{\Omega_{\alpha+1}+1}}}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,2,0)(2,1,1)<br>(3,2,0)(3,2,0)(3,1,1)(4,2,0)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+2^{\varepsilon_{\Omega_{\alpha+1}+2}}}))$ |
| (0,0,0)(1,1,1)(2,2,0)(2,2,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+3}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+\alpha}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,1,1)<br>(3,2,0)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+\alpha^2}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(2,2,0)<br>(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+\alpha\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+\alpha^2}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,0)(4,2,1)(5,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}+\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(4,2,1)<br>(5,3,0)(6,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}+\varepsilon_{\Omega_{\alpha+1}+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,0)(4,2,1)<br>(5,3,0)(6,2,0)(5,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}+\varepsilon_{\Omega_{\alpha+1}+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(2,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}\cdot\omega^2))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(2,1,1)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}\cdot\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(2,1,1)<br>(3,2,0)(4,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2^2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(2,2,0)<br>(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(2,2,0)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 3}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}^2}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}^2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(3,1,1)<br>(2,2,0)(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}^2\cdot 2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}^3}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}^\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}^{\Omega_{\alpha+1}}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\Omega_{\alpha+1}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,2,0)<br>(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\Omega_{\alpha+1}+1}\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(4,2,0)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\Omega_{\alpha+1}+2}}))$ |
| (0,0,0)(1,1,1)(2,2,0)<br>(3,1,1)(4,2,0)(5,1,1) | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\Omega_{\alpha+1}\cdot 2}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,1,1)(4,2,0)<br>(5,1,1)(6,2,0) | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\varepsilon_{\Omega_{\alpha+1}+1}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,1,0)<br>(3,2,0) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+1}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,1,0)<br>(3,2,1)(4,3,0)(5,3,0) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+1}\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,1,1) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,1,1)<br>(3,2,0) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+1}\cdot\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,1,1)<br>(3,2,0)(4,2,0) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+1}{}^2))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+1}+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,2,0) | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+1}\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+1}\cdot\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,2,0)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+1}\cdot\Omega_{\alpha+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,2,0)<br>(3,1,1)(4,2,0)(5,2,0) | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+1}{}^2}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)<br>(3,1,1)(4,2,0)(5,2,0)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\zeta_{\Omega_{\alpha+1}+1}+1}+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)(3,2,0) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+2}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)3,2,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+2}+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)(3,2,0)(2,2,0)(3,1,1)(4,2,0)(5,2,0)(4,2,0)(5,2,0) | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+2}\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(2,2,0)(3,2,0)(2,2,0)(3,2,0) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+3}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,0)(2,1,1) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+\alpha}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+\alpha}+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,0)(2,2,0)(3,2,0) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,0)(2,2,0)(3,2,0)(3,1,0) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+\alpha\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}\cdot 2}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}\cdot 2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)(3,0,0) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}{}^2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\zeta_{\varepsilon_{\Omega_{\alpha+1}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,1,1)(4,2,0)(5,2,0) | $\psi(\psi_\alpha(\zeta_{\zeta_{\Omega_{\alpha+1}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(3,2,0) | $\psi(\psi_\alpha(\eta_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,0,0) | $\psi(\psi_\alpha(\varphi(\omega,\Omega_{\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\varphi(\alpha,\Omega_{\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)(2,2,0)(3,2,0)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\varphi(\alpha,\Omega_{\alpha+1}+2)))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)(3,2,0) | $\psi(\psi_\alpha(\varphi(\alpha+1,\Omega_{\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)(5,2,0) | $\psi(\psi_\alpha(\varphi(\Omega_{\alpha+1},1)+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1) | $\psi(\psi_\alpha(\varphi(\Omega_{\alpha+1},1)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)(5,2,0) | $\psi(\psi_\alpha(\varphi(\varepsilon_{\Omega_{\alpha+1}+1},0)))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)(5,2,0)(6,2,0)(7,0,0) | $\psi(\psi_\alpha(\varphi(\varphi(\omega,\Omega_{\alpha+1}+1),0)))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,2,0) | $\psi(\psi_\alpha(\Gamma\Omega_{\alpha+1}+1))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha+2}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha+2}+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)(2,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+2}+\Omega_{\alpha+1}\cdot\omega)+\psi_\alpha(\Omega_{\alpha+2}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)(3,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha+2}+\Omega_{\alpha+1}\cdot\omega)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,0)(3,3,1)(4,4,0) | $\psi(\psi_\alpha(\Omega_{\alpha+2}+\varepsilon_{\Omega_{\alpha+1}+1}))$ $=\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot 2+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(1,1,1)(2,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega)+\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(1,1,1)(2,2,0)<br>(3,3,1)(4,4,0) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega)+$<br>$\psi_{\psi_\alpha(\Omega_{\alpha+2})}(\psi_\alpha(\Omega_{\alpha+2}\cdot 2+\Omega_{\alpha+1})))$ |
| (0,0,0)(1,1,1)(2,2,1)(1,1,1)(2,2,0)<br>(3,3,1)(4,4,1) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega)+\psi_{\psi_\alpha(\Omega_{\alpha+2})}(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,1)(1,1,1)(2,2,0)<br>(3,3,1)(4,4,1)(2,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega)+\psi_\alpha(\Omega_{\alpha+2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(1,1,1)(2,2,0)<br>(3,3,1)(4,4,1)(3,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega)+\psi_\alpha(\Omega_{\alpha+2}+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(1,1,1)(2,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega)\cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,0)(1,1,1)<br>(2,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega)^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,0)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,0)(3,2,1)<br>(4,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot(\omega+1)))$<br>$=\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega+\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,0)(3,2,1)<br>(4,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,1,0)<br>(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,1,0)<br>(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\Omega_{\alpha+1}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot(\Omega_{\alpha+1}+\omega)))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,0)<br>4,3,1)(5,4,1) | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^2+\Omega_{\alpha+2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,0)<br>(4,3,1)(5,4,1)(5,3,1) | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^2+\Omega_{\alpha+2}\cdot\omega^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,0)<br>(4,3,1)(5,4,1)(5,3,1)(6,4,0) | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^2\cdot 2+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^2\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,1)<br>(2,1,0)(3,2,0) | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^2\cdot\omega+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,1)<br>(2,1,1) | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^2\cdot\omega^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,1)<br>(2,1,1)(3,2,0) | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^2\cdot\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,1)<br>(2,1,1)(3,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^3\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,1)<br>(3,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,1)<br>(3,1,0)(2,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^\alpha))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,1)<br>(3,1,0)(4,2,0) | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^{\Omega_{\alpha+1}}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,1)<br>(3,1,1) | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^{\Omega_{\alpha+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,1)<br>(3,1,1)(4,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^{\Omega_{\alpha+2}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,1)<br>(3,1,1)(4,2,1)(4,1,1)(5,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^{{\Omega_{\alpha+2}}^{\Omega_{\alpha+2}}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(2,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+1}\cdot\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(2,1,1)(3,2,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+1} \cdot \Omega_{\alpha+2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(2,1,1)(3,2,1)(3,2,0) | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+2}+1}}^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(2,1,1)(3,2,1)(3,2,0)(3,0,0) | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+2}+1}}^\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(2,1,1)(3,2,1)(3,2,0)(3,1,1)(4,2,1)(4,2,0) | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+2}+1}}^{\varepsilon_{\Omega_{\alpha+2}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+\Omega_{\alpha+1}} + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+\Omega_{\alpha+1}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,1,1)(4,2,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}\cdot 2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,1,1)(4,2,1)(2,2,0)(3,1,1)(4,2,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}\cdot 3} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,1,1)(4,2,1)(3,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}\cdot\Omega_{\alpha+1}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,1,1)(4,2,1)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\Omega_{\alpha+2}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+2}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\zeta_(\Omega_{\alpha+2}+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)(2,2,0)(3,1,1)(4,2,1)(4,2,0)(5,2,0)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\zeta_{\Omega_{\alpha+2}+1}}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)(2,2,0)(3,2,0) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+2}+2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+2}+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)(3,1,1) | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+2}+\Omega_{\alpha+1}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)(3,1,1)(4,2,1)(4,2,0)(5,2,0) | $\psi(\psi_\alpha(\zeta_{\zeta_{\Omega_{\alpha+2}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0)(3,2,0) | $\psi(\psi_\alpha(\eta_{\Omega_{\alpha+2}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha+3} + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,3,1)(4,4,1) | $\psi(\psi_\alpha(\Omega_{\alpha+3} + \Omega_{\alpha+2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,3,1)(4,4,1)(4,4,0) | $\psi(\psi_\alpha(\Omega_{\alpha+3} + \varepsilon_{\Omega_{\alpha+2}+1})) = \psi(\psi_\alpha(\Omega_{\alpha+3} \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+3} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+3} \cdot \omega^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,1,1)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+3} \cdot \Omega_{\alpha+2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,1,1)(3,2,1)(3,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha+3}}^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+3}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,0)(3,3,1)(4,4,1)(4,4,1)(4,4,0) | $\psi(\psi_\alpha(\Omega_{\alpha+4} \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+4} \cdot \omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega})\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,1,0)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,1,0)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega}+\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,1,0)(3,2,1)(4,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega}+\Omega_{\alpha+2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,1,0)(3,2,1)(4,3,1)(5,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega}\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega}\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,1,1)(3,1,0)(2,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega}\cdot\alpha\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,1,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega}\cdot\varepsilon_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega}\cdot\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,1,1)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega}\cdot\Omega_{\alpha+2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,1,1)(3,2,1)(4,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha+\omega}}^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_(\Omega_{\alpha+\omega}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)(3,2,0) | $\psi(\psi_\alpha(\zeta_(\Omega_{\alpha+\omega}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)(3,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)(3,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0)(3,3,1)(4,4,1)(5,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega+1}+\Omega_{\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+\omega+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,1)(2,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega+2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,1)(3,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,0,0)(3,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha+\omega^2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0) | $\psi(\psi_\alpha(\Omega_{\alpha+\Omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(1,1,1)(2,2,1)(3,0,0) | $\psi(\psi_\alpha(\Omega_(\alpha+\psi_\alpha(\Omega_{\alpha+\omega}))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha\cdot 2+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha\cdot 3}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(3,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha^2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\varepsilon_{\alpha+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}}+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,0)(4,2,1)(5,3,1)(6,0,0) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}}+\Omega_{\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\Omega_{\alpha+1}}+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(2,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(2,2,1) (3,1,0)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}+\varepsilon_{\alpha+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(2,2,1) (3,1,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}\cdot 2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(3,1,0) (2,0,0) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}\cdot\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(3,1,0) (4,2,0) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}{}^2}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}{}^2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,1,0) (2,0,0) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}{}^\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\varepsilon_{\Omega_{\alpha+1}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,0) (5,3,1)(6,4,1)(7,0,0) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+2}}+\Omega_{\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+2}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,1) (2,2,1)(3,1,1)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+2}\cdot 2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,1) (3,0,0) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+2}\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,1) (3,1,1)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+2}{}^2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,1) (4,2,0) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+3}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,1) (5,0,0) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+\omega}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1)))$ $=\psi(\psi_\alpha(I_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,0,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1))\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,1,0) (3,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1)+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,1,0) (3,2,1)(4,3,1)(5,0,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1)+\Omega_{\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,1,1) | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,1,1) (2,1,1) | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1)\cdot\omega^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,1,1) (3,1,0)(2,0,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1)\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,1,1) (3,2,1)(4,0,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1)\cdot\Omega_{\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,1,1) (3,2,1)(4,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1)^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\mathrm{OFP}(\alpha+1)+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,0) (3,3,1) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,0) (3,3,1)(4,4,1)(5,0,0) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)+1}+\Omega_{\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1) (3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1}+\alpha)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1) (3,1,0)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)+\Omega_{\alpha+1}}+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1) (3,1,1) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)+\Omega_{\alpha+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1) (3,1,1)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)+\varepsilon_{\Omega_{\alpha+1}+1}}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,0,0) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)+\Omega_{\alpha+\omega}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,1,1) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)+\Omega_{\Omega_{\alpha+1}}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)\cdot 3}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(3,0,0) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)\cdot\Omega_{\alpha+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)<br>(3,1,1)(4,2,1)(5,0,0) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)\cdot\Omega_{\alpha+\omega}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)<br>(3,1,1)(4,2,1)(5,2,0) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)^2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)<br>(4,1,1)(5,2,1)(6,2,0) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)^{\mathrm{OFP}(\alpha+1)}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\varepsilon_{\mathrm{OFP}(\alpha+1)+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{\mathrm{OFP}(\alpha+1)+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(4,2,1)(5,0,0) | $\psi(\psi_\alpha(\Omega_{\Omega_{\mathrm{OFP}(\alpha+1)+\omega}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,2,0)(2,2,1) | $\psi(\psi_\alpha(\Omega_(\mathrm{OFP}(\alpha+2)+1)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,2,0)(2,2,1)(3,1,1)(4,2,1)(5,2,0) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+2)+\mathrm{OFP}(\alpha+1)}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,2,0)(2,2,1)(3,1,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,2,0) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+2)\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,2,0)(2,2,1)(3,1,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,2,0)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{\mathrm{OFP}(\alpha+2)+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,2,0)(2,2,1)(3,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+3)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,0,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+\omega)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,0)<br>(2,0,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,0)<br>(2,2,1)(3,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha\cdot 3)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,0)<br>(4,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\varepsilon_{\alpha+1})))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,0)<br>(4,2,1) | $\psi(\psi_\alpha(\mathrm{OFP}(\Omega_{\alpha+1})+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,1) | $\psi(\psi_\alpha(\mathrm{OFP}(\Omega_{\alpha+1})\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,1)<br>(2,2,1)(3,2,0)(3,0,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\Omega_{\alpha+1}+\omega)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,1)<br>(2,2,1)(3,2,0)(3,1,1) | $\psi(\psi_\alpha(\mathrm{OFP}(\Omega_{\alpha+1}\cdot 2)\cdot\omega))$ |

| BMS | Projection |
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| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,1)(3,0,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\Omega_{\alpha+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\mathrm{OFP}({\Omega_{\alpha+1}}^2)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\varepsilon_{\Omega_{\alpha+1}+1})))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,1)(4,2,1)(5,0,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\Omega_{\alpha+\omega})))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,1)(4,2,1)(5,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\mathrm{OFP}(\alpha+1))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0) | $\psi(\psi_\alpha(I_{\alpha+1})^2)$ $= \psi(\psi_\alpha(\psi_{I_{\alpha+1}}({I_{\alpha+1}}^2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_(\Phi(2,\alpha+1)+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0)(2,2,1) | $\psi(\psi_\alpha(\Omega_(\Phi(2,\alpha+1)+1)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0)(2,2,1)(3,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\Phi(2,\alpha+1)+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0)(2,2,1)(3,2,0)(3,1,1) | $\psi(\psi_\alpha(\mathrm{OFP}(\Phi(2,\alpha+1)+\Omega_{\alpha+1})\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0)(2,2,1)(3,2,0)(3,1,1)(4,2,1)(5,2,0)(5,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\Phi(2,\alpha+1)\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0)(2,2,1)(3,2,0)(3,2,0) | $\psi(\psi_\alpha(\Phi(2,\alpha+2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0)(3,0,0) | $\psi(\psi_\alpha(\Phi(2,\alpha+\omega)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Phi(2,\alpha\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\Phi(2,\Omega_{\alpha+1})+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0)(3,1,1) | $\psi(\psi_\alpha(\Phi(2,\Omega_{\alpha+1})\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0)(3,1,1)(4,2,1)(5,2,0)(5,2,0) | $\psi(\psi_\alpha(\Phi(2,\Phi(2,\alpha+1))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0)(3,2,0) | $\psi(\psi_\alpha(\Phi(3,\alpha+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,0,0) | $\psi(\psi_\alpha(\Phi(\omega,\alpha+1)))$ $= \psi(\psi_\alpha(I_{\alpha+1})^\omega)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,1,1) | $\psi(\psi_\alpha(\psi_{I_{\alpha+1}}({I_{\alpha+1}}^{\Omega_{\alpha+1}})\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,1,1)(5,2,1)(6,2,0)(7,0,0) | $\psi(\psi_\alpha(\psi_{I_{\alpha+1}}({I_{\alpha+1}}^{\psi_{I_{\alpha+1}}({I_{\alpha+1}}^\omega)})))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,2,0) | $\psi(\psi_\alpha(I_{\alpha+1})^{\psi_\alpha(I_{\alpha+1})})$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(I_{\alpha+1}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,3,1) | $\psi(\psi_\alpha(I_{\alpha+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,3,1)(5,4,1)(6,0,0) | $\psi(\psi_\alpha(I_{\alpha+1}+\Omega_{\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,3,1)(5,4,1)(6,4,0)(7,5,0) | $\psi(\psi_\alpha(I_{\alpha+1}\cdot 2+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1) | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,0,0) | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\omega)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,0)(3,2,0) | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\omega+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1) | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\omega^2))$ |

| BMS | Projection |
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| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1)(3,1,1) | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1)(3,2,0) | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\Omega_{\alpha+2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1)(3,2,1)(4,0,0) | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\Omega_{\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1)(3,2,1)(4,2,0) | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\mathrm{OFP}(\alpha+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1)(3,2,1)(4,2,0)(5,0,0) | $\psi(\psi_\alpha({I_{\alpha+1}}^2)^\omega)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1)(3,2,1)(4,2,0)(5,3,0) | $\psi(\psi_\alpha({I_{\alpha+1}}^2+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1)(3,2,1)(4,2,0)(5,3,1)(6,4,1)(7,4,1) | $\psi(\psi_\alpha({I_{\alpha+1}}^2+I_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1)(3,2,1)(4,2,1) | $\psi(\psi_\alpha({I_{\alpha+1}}^2\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1)(3,2,1)(4,2,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha({I_{\alpha+1}}^2\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1)(3,2,1)(4,2,1)(2,1,1)(3,2,1)(4,2,0)(5,3,0) | $\psi(\psi_\alpha({I_{\alpha+1}}^3+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1)(3,2,1)(4,2,1)(2,1,1)(3,2,1)(4,2,0)(5,3,1)(6,4,1)(7,4,1)(6,3,1)(7,4,1)(8,4,1) | $\psi(\psi_\alpha({I_{\alpha+1}}^3+I_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1)(3,2,1)(4,2,1)(2,1,1)(3,2,1)(4,2,1) | $\psi(\psi_\alpha({I_{\alpha+1}}^3\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1)(3,2,1)(4,2,1)(3,0,0) | $\psi(\psi_\alpha({I_{\alpha+1}}^\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1)(3,2,1)(4,2,1)(3,1,1) | $\psi(\psi_\alpha({I_{\alpha+1}}^{\Omega_{\alpha+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,1,1)(3,2,1)(4,2,1)(3,1,1)(4,2,1)(5,2,1) | $\psi(\psi_\alpha({I_{\alpha+1}}^{I_{\alpha+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{I_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{I_{\alpha+1}+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,0)(3,1,1)(4,2,1)(5,2,1)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{I_{\alpha+1}+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,0)(3,2,0) | $\psi(\psi_\alpha(\zeta_{I_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,0)(3,3,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+1}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,0)(3,3,1)(4,4,1)(5,4,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+1}+I_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)(3,0,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+\Omega_{\alpha+1}}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+\Omega_{\alpha+1}}\cdot\omega))$ |

| BMS | Projection |
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| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,0,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+\Omega_{\alpha+\omega}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+\mathrm{OFP}(\alpha+1)}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(6,3,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 2}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(2,0,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 2}\cdot\omega)\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(2,1,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 2}\cdot\omega^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(2,1,1)<br>(3,2,1)(4,2,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 2}\cdot I_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(2,1,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,1,1)(5,2,1)(6,2,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 2^2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{I_{\alpha+1}\cdot 2}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(2,2,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 2+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)<br>(2,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 2+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,2,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 3}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot\Omega_{\alpha+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(3,1,1)<br>(4,2,1)(5,0,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot\Omega_{\alpha+\omega}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)<br>(3,1,1)(4,2,1)(5,2,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}{}^2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(4,0,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}{}^\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)<br>(4,1,1)(5,2,1)(6,2,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}{}^{I_{\alpha+1}}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\varepsilon_{I_{\alpha+1}+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(4,2,0)(5,3,0) | $\psi(\psi_\alpha(\Omega_{\Omega_{I_{\alpha+1}+1}+1}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{I_{\alpha+1}+1}+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(4,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{I_{\alpha+1}+1}+1+I_{\alpha+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(4,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{I_{\alpha+1}+1}+1}\cdot 2\cdot\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(4,2,1)(4,2,0) | $\psi(\psi_\alpha(\Omega_{(\varepsilon_{\Omega_{I_{\alpha+1}+1}+1})}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(4,2,1)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{I_{\alpha+1}+2}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(4,2,1)(5,1,1)<br>(6,2,1)(7,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{I_{\alpha+1}}\cdot 2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(4,2,1)(5,1,1)<br>(6,2,1)(7,2,1)(6,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{\Omega_{I_{\alpha+1}+1}}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0) | $\psi(\psi_\alpha(I_{\alpha+2}))$<br>$= \psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(2,1,1) | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha+1}+1)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{(}\mathrm{OFP}(I_{\alpha+1}+1)+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(2,2,1) | $\psi(\psi_\alpha(\Omega_{(}\mathrm{OFP}(I_{\alpha+1}+1)+1)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(2,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(I_{\alpha+1}+1)+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(2,2,1)(3,1,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(I_{\alpha+1}+1)\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(2,2,1)(3,1,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,0)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{\mathrm{OFP}(I_{\alpha+1}+1)+1}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(2,2,1)(3,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha+1}+2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha+1}+\alpha)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(3,1,1) | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha+1}+\Omega_{\alpha+1})\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(3,1,1)(4,2,1)(5,2,1) | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha+1}\cdot 2)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(3,1,1)(4,2,1)(5,2,1)(3,1,1)<br>(4,2,1)(5,2,1) | $\psi(\psi_\alpha(\mathrm{OFP}({I_{\alpha+1}}^2)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(3,1,1)(4,2,1)(5,2,1)(4,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\varepsilon_{I_{\alpha+1}+1})))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(3,1,1)(4,2,1)(5,2,1)(4,2,1) | $\psi(\psi_\alpha(\mathrm{OFP}(\Omega_{I_{\alpha+1}+1})\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(3,1,1)(4,2,1)(5,2,1)(4,2,1)<br>(5,1,1)(6,2,1)(7,2,1) | $\psi(\psi_\alpha(\mathrm{OFP}(\Omega_{\Omega_{I_{\alpha+1}+1}})\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(3,1,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\mathrm{OFP}(I_{\alpha+1}+1))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(3,2,0) | $\psi(\psi_\alpha(\Phi(2, I_{\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(4,0,0) | $\psi(\psi_\alpha(I_{\alpha+2})^\omega)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(4,3,0) | $\psi(\psi_\alpha(I_{\alpha+2}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(4,3,1) | $\psi(\psi_\alpha(I_{\alpha+2}+\Omega_{\alpha+1}\cdot\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(4,3,1)(5,4,1)(6,0,0) | $\psi(\psi_\alpha(I_{\alpha+2} + \Omega_{\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(4,3,1)(5,4,1)(6,4,1) | $\psi(\psi_\alpha(I_{\alpha+2} + I_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,0)(4,3,1)(5,4,1)<br>(6,4,1)(5,4,1)(6,4,0) | $\psi(\psi_\alpha(I_{\alpha+2} \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,1) | $\psi(\psi_\alpha(I_{\alpha+2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,1,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,2,1) | $\psi(\psi_\alpha({I_{\alpha+2}}^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_(I_{\alpha+2} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+2}+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,0,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+2}+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,1,1)(4,2,1)(5,2,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+2}+I_{\alpha+1}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,1,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha+2}\cdot 2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,1,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,1)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\varepsilon_{I_{\alpha+2}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,1,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,2,1)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{I_{\alpha+2}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha+2} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,2,0)(2,2,1)(3,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha+2} + 2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,2,0)(3,1,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,1)(4,2,1)(5,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\mathrm{OFP}(I_{\alpha+2} + 1))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,2,0)(3,2,0) | $\psi(\psi_\alpha(I_{\alpha+3})^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,2,1) | $\psi(\psi_\alpha(I_{\alpha+3} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,0,0) | $\psi(\psi_\alpha(I_{\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,0,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,0,0)(2,0,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2}) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2})^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,0)(3,2,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2} + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,3,1)(5,3,1)(5,1,0)<br>(1,1,1)(2,2,1)(3,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2} + I(\alpha + \psi_\alpha(I_{\alpha\cdot 2}))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,3,1)<br>(5,3,1)(5,1,0)(2,0,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2} + I(\alpha + \psi_\alpha(I_{\alpha\cdot 2})) \cdot \omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,3,1)(5,3,1)(5,2,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2} + I(\alpha + \psi_\alpha(I_{\alpha\cdot 2} + \Omega_{\alpha+1}))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,3,1)<br>(5,3,1)(5,2,0)(3,2,1) | $\psi(\psi_\alpha(I_{\alpha\cdot 2} + I(\alpha + \psi_\alpha(I_{\alpha+1} + \Omega_{\alpha+1}\cdot\omega))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,3,1)<br>(5,3,1)(5,2,0)(4,0,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2}\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,1) | $\psi(\psi_\alpha(I_{\alpha\cdot 2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2}\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2}\cdot\varepsilon_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,1)(3,1,1) | $\psi(\psi_\alpha(I_{\alpha\cdot 2}\cdot\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,1)(3,2,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2}\cdot\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,1)(3,2,1)(4,0,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2}\cdot\Omega_{\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,1)(3,2,1)(4,2,1)(5,0,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2}\cdot I(\alpha+\omega)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,1)(3,2,1)(4,2,1)(5,1,0)(2,0,0) | $\psi(\psi_\alpha({I_{\alpha\cdot 2}}^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,1)(3,2,1)(4,2,1)(5,1,0)(3,0,0) | $\psi(\psi_\alpha({I_{\alpha\cdot 2}}^\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,1,1)(3,2,1)(4,2,1)(5,1,0)(3,1,1)<br>(4,2,1)(5,2,1)(6,1,0)(2,0,0) | $\psi(\psi_\alpha({I_{\alpha\cdot 2}}^{I_{\alpha\cdot 2}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{I_{\alpha\cdot 2}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{I_{\alpha\cdot 2}+2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,0)(3,3,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}+1} + \Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,0)(3,3,1)(4,4,1)(5,4,1)(5,0,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}+1} + I(\alpha+\omega)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(2,1,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}+1}\cdot\omega^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(2,1,1)(3,2,1)<br>(4,2,1)(5,1,0)(3,2,1) | $\psi(\psi_\alpha({\Omega_{I_{\alpha\cdot 2}+1}}^2\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{I_{\alpha\cdot 2}+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(2,2,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}+2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}+\Omega_{\alpha+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,0,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}+\Omega_{\alpha+\omega}}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,1,1)(4,2,1)<br>(5,2,1)(5,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,1)(5,1,0)<br>(2,2,1)(3,0,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}\cdot 2+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,1)<br>(5,1,0)(3,0,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,1,1)(4,2,1)<br>(5,2,1)(5,1,0)(3,1,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}\cdot\Omega_{\alpha+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,1)(5,1,0)<br>(3,1,1)(4,2,1)(5,2,1)(5,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}{}^2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,1,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\varepsilon_{I_{\alpha\cdot 2}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,1,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{I_{\alpha\cdot 2}+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha\cdot 2}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\mathrm{OFP}(I_{\alpha\cdot 2}+1)+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,0)(2,2,1) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(I_{\alpha\cdot 2}+1)+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)(<br>2,2,1)(3,2,0)(2,2,1)(3,1,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,1)(5,2,0) | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(I_{\alpha\cdot 2}+1)\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,0)(2,2,1)(3,1,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,1)(5,2,0)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{\mathrm{OFP}(I_{\alpha\cdot 2}+1)}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,0)(2,2,1)(3,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha\cdot 2}+2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,0)(3,0,0) | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha\cdot 2}+\omega)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,0)(3,1,1) | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha\cdot 2}+\Omega_{\alpha+1})\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,0)(3,1,1)(4,2,1)(5,2,1)<br>(5,1,0)(2,0,0) | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha\cdot 2}\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,0)(3,1,1)(4,2,1)(5,2,1)<br>(5,1,0)(2,2,1)(3,2,0)(3,1,1)(4,2,1)<br>(5,2,1)(5,1,0)(2,0,0) | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha\cdot 2}\cdot 3)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,0)(3,1,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\varepsilon_{I_{\alpha\cdot 2}+1})))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,0)(3,1,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\mathrm{OFP}(I_{\alpha\cdot 2}+1))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,0)(3,2,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2+1})^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2+1}+\Omega_{\alpha+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,0)(4,3,1)(5,4,1)(6,4,1)<br>(6,3,0)(5,0,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2+1} + I_{\alpha\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,1) | $\psi(\psi_\alpha(I_{\alpha\cdot 2+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_(I_{\alpha\cdot 2+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,1)(2,2,1) | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2+1}+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,1)(2,2,1)(3,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha\cdot 2+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,1)(2,2,1)(3,2,1) | $\psi(\psi_\alpha(I_{\alpha\cdot 2+2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,1)(3,0,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 2+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(2,2,1)(3,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(I_{\alpha\cdot 3}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(3,0,0) | $\psi(\psi_\alpha(I_{\alpha\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(3,1,0)(2,0,0) | $\psi(\psi_\alpha(I_{\alpha^2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(4,2,0) | $\psi(\psi_\alpha(I_{\varepsilon_{\alpha+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(4,2,1) | $\psi(\psi_\alpha(I_{\Omega_{\alpha+1}} + \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1) | $\psi(\psi_\alpha(I_{\Omega_{\alpha+1}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1)<br>(2,2,1)(3,2,1) | $\psi(\psi_\alpha(I_{\Omega_{\alpha+1}+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1)<br>(2,2,1)(3,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(I_{\Omega_{\alpha+1}+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1)<br>(2,2,1)(3,2,1)(3,1,1) | $\psi(\psi_\alpha(I_{\Omega_{\alpha+1}\cdot 2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1)<br>(3,0,0) | $\psi(\psi_\alpha(I_{\Omega_{\alpha+1}\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1)<br>(3,1,1) | $\psi(\psi_\alpha(I_{\Omega_{\alpha+1}{}^2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1)<br>(4,2,0) | $\psi(\psi_\alpha(I_{\varepsilon_{\Omega_{\alpha+1}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1)<br>(4,2,1) | $\psi(\psi_\alpha(I_{\Omega_{\alpha+2}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1)<br>(4,2,1)(5,0,0) | $\psi(\psi_\alpha(I_{\Omega_{\alpha+\omega}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1)<br>(4,2,1)(5,2,1)(5,0,0) | $\psi(\psi_\alpha(I_{I_{\alpha+\omega}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0) | $\psi(\psi_\alpha(\psi_{I(1,\alpha+1)}(I(1,\alpha+1))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(2,2,1) | $\psi(\psi_\alpha(\Omega_{\mathrm{IFP}(\alpha+1)+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(2,2,1)(3,2,1) | $\psi(\psi_\alpha(I_{\mathrm{IFP}(\alpha+1)+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(2,2,1)(3,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(I_{\mathrm{IFP}(\alpha+1)+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(2,2,1)(3,2,1)(3,1,1)<br>(4,2,1)(5,2,1)(5,0,0) | $\psi(\psi_\alpha(I_{\mathrm{IFP}(\alpha+1)+I_{\alpha+1}}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(2,2,1)(3,2,1)(3,1,1)<br>(4,2,1)(5,2,1)(5,2,0) | $\psi(\psi_\alpha(I_{\mathrm{IFP}(\alpha+1)\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(2,2,1)(3,2,1)(3,1,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,2,1) | $\psi(\psi_\alpha(I_{\Omega_{\mathrm{IFP}(\alpha+1)+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(2,2,1)(3,2,1)(3,1,1)(4,2,1)(5,2,1)<br>(5,2,0)(4,2,1)(5,2,1) | $\psi(\psi_\alpha(I_{I_{\mathrm{IFP}(\alpha+1)+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(2,2,1)(3,2,1)(3,2,0) | $\psi(\psi_\alpha(\mathrm{IFP}(\alpha+2)))$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,2,0)(3,0,0) | $\psi(\psi_\alpha(\mathrm{IFP}(\alpha+\omega)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(3,1,1)(4,2,1)(5,2,1)(5,2,0) | $\psi(\psi_\alpha(\mathrm{IFP}(\mathrm{IFP}(\alpha+1))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(3,2,0) | $\psi(\psi_\alpha(\psi_{I(1,\alpha+1)}(I(1,\alpha+1)^2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(3,2,0)(2,2,1)(3,2,1)(3,2,0)(3,2,0) | $\psi(\psi_\alpha(I(1,\alpha+1))^2\cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(3,2,0)(3,0,0) | $\psi(\psi_\alpha(I(1,\alpha+1))^2\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(3,2,0)(3,1,1)(4,2,1)<br>(5,2,1)(5,2,0)(5,2,0) | $\psi(\psi_\alpha(I(1,\alpha+1))^2\cdot$<br>$\psi_{\psi_\alpha(I(1,\alpha+1))}(\psi_\alpha(I(1,\alpha+1))^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,0)(3,2,0)(3,2,0) | $\psi(\psi_\alpha(I(1,\alpha+1))^3)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(I(1,\alpha+1)+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0)<br>(4,3,1)(5,4,1)(6,4,1)(6,4,0) | $\psi(\psi_\alpha(I(1,\alpha+1)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1) | $\psi(\psi_\alpha(I(1,\alpha+1)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)<br>(2,1,0)(3,2,0) | $\psi(\psi_\alpha(I(1,\alpha+1)\cdot\omega+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)<br>(2,1,0)(3,2,1)(4,3,1)(5,3,1)(5,3,1) | $\psi(\psi_\alpha(I(1,\alpha+1)\cdot\omega\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)<br>(2,1,1) | $\psi(\psi_\alpha(I(1,\alpha+1)\cdot\omega^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(I(1,\alpha+1)\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)<br>(2,1,1)(3,1,1) | $\psi(\psi_\alpha(I(1,\alpha+1)\cdot\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)<br>(2,1,1)(3,2,0) | $\psi(\psi_\alpha(I(1,\alpha+1)\cdot\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)<br>(2,1,1)(3,2,1)(4,2,1)(4,2,0) | $\psi(\psi_\alpha(I(1,\alpha+1)^2))$<br>$=\psi(\psi_\alpha(I(1,\alpha+1)\cdot\mathrm{IFP}(\alpha+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)<br>(2,1,1)(3,2,1)(4,2,1)(4,2,1) | $\psi(\psi_\alpha(I(1,\alpha+1)^2\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)<br>(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{I(1,\alpha+1)+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)<br>(2,2,0)(3,3,1) | $\psi(\psi_\alpha(\Omega_(I(1,\alpha+1)+1)+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)<br>(2,2,1) | $\psi(\psi_\alpha(\Omega_{I(1,\alpha+1)+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,1)(5,2,1) | $\psi(\psi_\alpha(\Omega_{I(1,\alpha+1)\cdot 2}\cdot\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(2,2,1)(3,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(I(1,\alpha+1)+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(2,2,1)(3,2,1) | $\psi(\psi_\alpha(I_{I(1,\alpha+1)+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(2,2,1)(3,2,1)(3,2,0) | $\psi(\psi_\alpha(\mathrm{IFP}(I(1,\alpha+1)+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(2,2,1)(3,2,1)(3,2,1) | $\psi(\psi_\alpha(I(1,\alpha+2)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,0,0) | $\psi(\psi_\alpha(I(1,\alpha+\omega)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(I(1,\alpha\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(I(1,\varepsilon_{\alpha+1})))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,1,1) | $\psi(\psi_\alpha(I(1,\Omega_{\alpha+1})\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,1,1)(4,2,1)(5,2,1)(5,2,1) | $\psi(\psi_\alpha(I(1,I(1,\alpha+1))\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,2,0) | $\psi(\psi_\alpha(I(2,\alpha+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,2,0)(2,2,1)(3,2,1)(3,2,1) | $\psi(\psi_\alpha(I(1,\psi_{I(2,\alpha+1)}(I(2,\alpha+1))+1)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,2,0)(2,2,1)(3,2,1)(3,2,1)(3,1,1)(4,2,1)(5,2,1)(5,2,1)(5,2,0)(3,0,0) | $\psi(\psi_\alpha(I(1,\psi_{I(2,\alpha+1)}(I(2,\alpha+1))\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,2,0)(2,2,1)(3,2,1)(3,2,1)(3,1,1)(4,2,1)(5,2,1)(5,2,1)(5,2,0)(3,1,1)(4,2,1)(5,2,1)(5,2,1)(5,2,0) | $\psi(\psi_\alpha(I(1,\psi_{I(2,\alpha+1)}(I(2,\alpha+1))^2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,2,0)(2,2,1)(3,2,1)(3,2,1)(3,1,1)(4,2,1)(5,2,1)(5,2,1)(5,2,0)(4,2,1)(5,2,1)(5,2,1) | $\psi(\psi_\alpha(I(1,I(1,\psi_{I(2,\alpha+1)}(I(2,\alpha+1))+1))\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,2,0)(2,2,1)(3,2,1)(3,2,1)(3,2,0) | $\psi(\psi_\alpha(I(2,\alpha+1))\cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\psi_{I(2,\alpha+1)}(I(2,\alpha+1)\cdot\alpha)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,2,0)(3,2,0) | $\psi(\psi_\alpha(I(2,\alpha+1))^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(I(2,\alpha+1)+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,2,1) | $\psi(\psi_\alpha(I(2,\alpha+1)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,2,1)(2,2,1)(3,2,1)(3,2,1)(3,2,1) | $\psi(\psi_\alpha(I(2,\alpha+2)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,2,1)(3,2,0) | $\psi(\psi_\alpha(I(3,\alpha+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1)(3,2,1)(3,2,1) | $\psi(\psi_\alpha(I(3,\alpha+1)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,0,0) | $\psi(\psi_\alpha(I(\omega,\alpha+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(I(\alpha,1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,0)(2,1,0)(3,2,0) | $\psi(\psi_\alpha(I(\alpha,1)+\Omega_{\alpha+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,0)<br>(2,1,0)(3,2,1)(4,3,1)(5,3,1)<br>(6,2,0)(4,0,0) | $\psi(\psi_\alpha(I(\alpha,1)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,0)<br>(2,1,1) | $\psi(\psi_\alpha(I(\alpha,1)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,0) | $\psi(\psi_\alpha(\varepsilon_(I(\alpha,1)+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,1) | $\psi(\psi_\alpha(\Omega_(I(\alpha,1)+1)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(I(\alpha,2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,0)<br>(3,0,0) | $\psi(\psi_\alpha(I(\alpha,\omega)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,0)<br>(3,1,0)(2,0,0) | $\psi(\psi_\alpha(I(\alpha,\alpha)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,0)<br>(3,1,1)(4,2,1)(5,2,1)(6,1,0)(2,0,0) | $\psi(\psi_\alpha(I(\alpha,I(\alpha,1))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,0) | $\psi(\psi_\alpha(I(\alpha+1,0)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,1) | $\psi(\psi_\alpha(I(\alpha+1,0)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,1)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(I(\alpha\cdot 2,0)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,0)<br>(4,1,0)(2,0,0) | $\psi(\psi_\alpha(I(\alpha^2,0)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,0)<br>(5,2,0) | $\psi(\psi_\alpha(I(\varepsilon_{\alpha+1},0)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,1) | $\psi(\psi_\alpha(I(\Omega_{\alpha+1},0)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,1,1)<br>(5,2,1)(6,2,1)(7,1,0)(2,0,0) | $\psi(\psi_\alpha(I(I(\alpha,1),0)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0) | $\psi(\psi_\alpha(I(1,0,\alpha+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(2,2,1)(3,2,1)(4,0,0) | $\psi(\psi_\alpha(I(\omega,\psi_{I(1,0,\alpha+1)}(I(1,0,\alpha+1))+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(2,2,1)(3,2,1)(4,1,1)<br>(5,2,1)(6,2,1)(7,2,0) | $\psi(\psi_\alpha(I(\psi_{I(1,0,\alpha+1)}(I(1,0,\alpha+1)),1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(2,2,1)(3,2,1)(4,2,0) | $\psi(\psi_\alpha(I(1,0,\alpha+1))\cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,0)(3,1,1) | $\psi(\psi_\alpha(\psi_{I(1,0,\alpha+1)}(I(1,0,\alpha+1)\cdot\Omega_{\alpha+1})\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(3,1,1)(4,2,1)(5,2,1)(6,2,0) | $\psi(\psi_\alpha(\psi_{I(1,0,\alpha+1)}(I(1,0,\alpha+1)\cdot$<br>$\psi_{I(1,0,\alpha+1)}(I(1,0,\alpha+1)))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(3,2,0) | $\psi(\psi_\alpha(I(1,0,\alpha+1))^2)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(3,2,0)(4,3,0) | $\psi(\psi_\alpha(I(1,0,\alpha+1)+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(3,2,1) | $\psi(\psi_\alpha(I(1,0,\alpha+1)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(3,2,1)(2,1,1) | $\psi(\psi_\alpha(I(1,0,\alpha+1)\cdot\omega^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(3,2,1)(2,1,1)(3,2,1)(4,2,1)(5,2,0) | $\psi(\psi_\alpha(I(1,0,\alpha+1)\cdot$<br>$\psi_{I(1,0,\alpha+1)}(I(1,0,\alpha+1))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(3,2,1)(2,1,1)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1) | $\psi(\psi_\alpha(I(1,0,\alpha+1)^2\cdot\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{I(1,0,\alpha+1)+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(2,2,1) | $\psi(\psi_\alpha(\Omega_{I(1,0,\alpha+1)+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(2,2,1)(3,2,1)(4,0,0) | $\psi(\psi_\alpha(I(\omega,I(1,0,\alpha+1)+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(2,2,1)(3,2,1)(4,1,1)(5,2,1)(6,2,1)(7,2,0) | $\psi(\psi_\alpha(I(I(1,0,\alpha+1),1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(2,2,1)(3,2,1)(4,1,1)(5,2,1)(6,2,1)(7,2,0)(6,2,1)(5,2,0) | $\psi(\psi_\alpha(I(\varepsilon_{I(1,0,\alpha+1)+1},0)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(2,2,1)(3,2,1)(4,2,0) | $\psi(\psi_\alpha(I(1,0,\alpha+2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1) | $\psi(\psi_\alpha(I(1,0,\alpha+2)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(3,0,0) | $\psi(\psi_\alpha(I(1,0,\alpha+\omega)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(I(1,0,\alpha\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(3,1,1) | $\psi(\psi_\alpha(I(1,0,\Omega_{\alpha+1})\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(3,1,1)(4,2,1)(5,2,1)(6,2,0) | $\psi(\psi_\alpha(I(1,0,I(1,0,\alpha+1))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(3,2,0) | $\psi(\psi_\alpha(I(1,1,\alpha+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(I(1,1,\alpha+1)+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(3,2,1) | $\psi(\psi_\alpha(I(1,1,\alpha+1)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(4,0,0) | $\psi(\psi_\alpha(I(1,\omega,\alpha+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(I(1,\alpha,1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(4,1,0)(5,2,0) | $\psi(\psi_\alpha(I(1,\varepsilon_{\alpha+1},0)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(4,1,1) | $\psi(\psi_\alpha(I(1,\Omega_{\alpha+1},0)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(4,2,0) | $\psi(\psi_\alpha(I(2,0,\alpha+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(4,2,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\psi_{I(2,0,\alpha+1)}(I(2,0,\alpha+1))+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(4,2,0)(3,2,1) | $\psi(\psi_\alpha(I(2,0,\alpha+1)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(4,2,0)(3,2,1)(3,2,1) | $\psi(\psi_\alpha(I(2,1,\alpha+1)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(3,2,1)(4,2,0)(3,2,1)(4,2,0) | $\psi(\psi_\alpha(I(3,0,\alpha+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(4,0,0) | $\psi(\psi_\alpha(I(\omega,0,\alpha+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(I(\alpha,0,1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(4,1,0)(5,2,0) | $\psi(\psi_\alpha(I(\varepsilon_{\alpha+1},0,0)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)(4,1,1) | $\psi(\psi_\alpha(I(\Omega_{\alpha+1},0,0)\cdot\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(4,1,1)(5,2,1)(6,2,1)(7,2,0)(7,0,0) | $\psi(\psi_\alpha(I(I(\omega,0,\alpha+1),0,0)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(4,2,0) | $\psi(\psi_\alpha(I(1,0,0,\alpha+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(4,2,0)(3,2,1)(4,2,0)(4,2,0) | $\psi(\psi_\alpha(I(2,0,0,\alpha+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(4,2,0)(4,2,0) | $\psi(\psi_\alpha(I(1,0,0,0,\alpha+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,0,0) | $\psi(\psi_\alpha(I(1@\omega,\alpha+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,0)(5,1,0)(2,0,0) | $\psi(\psi_\alpha(I(1@\alpha,1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,1,0)(2,2,1)(3,2,1)<br>(4,2,0)(5,1,0)(2,0,0) | $\psi(\psi_\alpha(I(1@\alpha,2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,1,0)(3,2,1) | $\psi(\psi_\alpha(I(1@\alpha,1,0)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,1,0)(3,2,1)(4,2,0)(5,1,0)(2,0,0) | $\psi(\psi_\alpha(I(2@\alpha)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,1,0)(4,0,0) | $\psi(\psi_\alpha(I(\omega@\alpha)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,1,0)(4,2,0) | $\psi(\psi_\alpha(I(1@(\alpha+1))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,1,0)(4,2,0)(5,1,0)(2,0,0) | $\psi(\psi_\alpha(I(1@(\alpha\cdot 2))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,1,0)(5,1,0)(2,0,0) | $\psi(\psi_\alpha(I(1@(\alpha^2))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,1,0)(6,2,0) | $\psi(\psi_\alpha(I(1@\varepsilon_{\alpha+1})))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,1,1) | $\psi(\psi_\alpha(I(1@\Omega_{\alpha+1})\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,2,0) | $\psi(\psi_\alpha(M_{\alpha+1})^{\psi_\alpha(M_{\alpha+1})^{\psi_\alpha(M_{\alpha+1})}})$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,3,0) | $\psi(\psi_\alpha(M_{\alpha+1}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,3,1) | $\psi(\psi_\alpha(M_{\alpha+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,3,1)(6,3,1)(6,3,1) | $\psi(\psi_\alpha(\psi_{M_{\alpha+1}}(M_{\alpha+\omega})))$<br>$=\psi(\psi_\alpha(M_{\alpha+1}+{\Omega_{\alpha+1}}^{\Omega_{\alpha+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,3,1)(6,4,0) | $\psi(\psi_\alpha(M_{\alpha+1}+\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,3,1)(6,4,1)(7,0,0) | $\psi(\psi_\alpha(M_{\alpha+1}+\Omega_{\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0)<br>(5,3,1)(6,4,1)(7,4,1)(8,4,0)(9,5,1) | $\psi(\psi_\alpha(M_{\alpha+1}\cdot 2+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1) | $\psi(\psi_\alpha(M_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,1,0)(3,2,0) | $\psi(\psi_\alpha(M_{\alpha+1}\cdot\omega+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,1,1) | $\psi(\psi_\alpha(M_{\alpha+1}\cdot\omega^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(M_{\alpha+1}\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,1,1)(3,2,0) | $\psi(\psi_\alpha(M_{\alpha+1}\cdot\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,1,1)(3,2,1)(4,0,0) | $\psi(\psi_\alpha(M_{\alpha+1}\cdot\Omega_{\alpha+\omega}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,1,1)(3,2,1)(4,2,1)(5,2,1) | $\psi(\psi_\alpha({M_{\alpha+1}}^2 \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{M_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,0)(3,3,1) | $\psi(\psi_\alpha(\Omega_{M_{\alpha+1}+1} + \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1) | $\psi(\psi_\alpha(\Omega_{M_{\alpha+1}+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,0,0) | $\psi(\psi_\alpha(\Omega_{M_{\alpha+1}+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{M_{\alpha+1}+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\Omega_{M_{\alpha+1}+\varepsilon_{\Omega_{\alpha+1}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,1)(6,2,1) | $\psi(\psi_\alpha(\Omega_{M_{\alpha+1}\cdot 2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,1,1)(4,2,1)<br>(5,2,1)(6,2,1)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\varepsilon_{M_{\alpha+1}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,1,1)(4,2,1)<br>(5,2,1)(6,2,1)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{M_{\alpha+1}+1}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(2,2,1)(3,2,0) | $\psi(\psi_\alpha(\mathrm{OFP}(M_{\alpha+1})))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,0)(4,3,1) | $\psi(\psi_\alpha(I(M_{\alpha+1}+1) + \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1) | $\psi(\psi_\alpha(I(M_{\alpha+1}+1) \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(3,2,0) | $\psi(\psi_\alpha(\mathrm{IFP}(M_{\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(3,2,1) | $\psi(\psi_\alpha(I(1, M_{\alpha+1}+1) \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,0,0) | $\psi(\psi_\alpha(I(\omega, M_{\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(I(\alpha, M_{\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,1,1)<br>(5,2,1)(6,2,1)(7,2,1) | $\psi(\psi_\alpha(I(M_{\alpha+1}, 1) \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,1,1)(5,2,1)(6,2,1)<br>(7,2,1)(3,2,1) | $\psi(\psi_\alpha(I(M_{\alpha+1}+1, 0) \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,1,1)(5,2,1)(6,2,1)<br>(7,2,1)(5,2,1)(6,0,0) | $\psi(\psi_\alpha(I(\Omega_{M_{\alpha+1}+\omega}, 0)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,0) | $\psi(\psi_\alpha(I(1, 0, M_{\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(I(1, 0, M_{\alpha+1}+1)) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,0)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(I(1, 0, M_{\alpha+1}+1) + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,0)(3,2,1) | $\psi(\psi_\alpha(I(1, 0, M_{\alpha+1}+1) \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,0)(3,2,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{I(1,0,M_{\alpha+1}+1)+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,0)(3,2,1)(2,2,1) | $\psi(\psi_\alpha(\Omega_{I(1,0,M_{\alpha+1}+1)\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,0)(3,2,1)(2,2,1)<br>(3,2,1)(4,1,1)(5,2,1)(6,2,1)(7,2,1)<br>(5,2,1)(6,2,1)(7,2,0) | $\psi(\psi_\alpha(I(I(1,0,M_{\alpha+1}+1),1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,0)(3,2,1)(2,2,1)<br>(3,2,1)(4,2,0) | $\psi(\psi_\alpha(I(1,0,M_{\alpha+1}+2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,0)(3,2,1)(2,2,1)<br>(3,2,1)(4,2,0)(3,2,1) | $\psi(\psi_\alpha(I(1,0,M_{\alpha+1}+2)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,0)<br>(3,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(I(1,0,M_{\alpha+1}+\alpha)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,0)(3,2,1)(3,1,1)<br>(4,2,1)(5,2,1)(6,2,1)<br>(4,2,1)(5,2,1)(6,2,0) | $\psi(\psi_\alpha(I(1,0,I(1,0,M_{\alpha+1}+1))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,0)(3,2,1)(3,2,0) | $\psi(\psi_\alpha(I(1,1,M_{\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,0)(3,2,1)(4,2,0) | $\psi(\psi_\alpha(I(2,0,M_{\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,0)(4,2,0) | $\psi(\psi_\alpha(I(1,0,0,M_{\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,0)(5,3,0) | $\psi(\psi_\alpha(M_{\alpha+2}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,1) | $\psi(\psi_\alpha(M_{\alpha+2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,1)(2,2,1) | $\psi(\psi_\alpha(\Omega_{M_{\alpha+2}+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,0,0) | $\psi(\psi_\alpha(I(\omega,M_{\alpha+2}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,0) | $\psi(\psi_\alpha(I(1,0,M_{\alpha+2}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,1)(2,2,1)(3,2,1)<br>(4,2,0)(5,3,0) | $\psi(\psi_\alpha(M_{\alpha+3}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,1)<br>(2,2,1)(3,2,1)(4,2,1) | $\psi(\psi_\alpha(M_{\alpha+3}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,0,0) | $\psi(\psi_\alpha(M_{\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,0)(2,0,0) | $\psi(\psi_\alpha(M_{\alpha\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,0)(2,0,0)(2,0,0) | $\psi(\psi_\alpha(M_{\alpha\cdot 2})\cdot\omega)$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,0)(2,1,0)(3,2,0) | $\psi(\psi_\alpha(M_{\alpha\cdot 2}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,3,1)(5,3,1)<br>(6,3,1)(5,2,0)(4,0,0) | $\psi(\psi_\alpha(M_{\alpha\cdot 2}\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,0)(2,1,1) | $\psi(\psi_\alpha(M_{\alpha\cdot 2}\cdot\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,0)(2,1,1)(3,2,0) | $\psi(\psi_\alpha(M_{\alpha\cdot 2}\cdot\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,0)(2,1,1)(3,2,1)(4,2,1)(5,2,1)<br>(4,1,0)(2,0,0) | $\psi(\psi_\alpha({M_{\alpha\cdot 2}}^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{M_{\alpha\cdot 2}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,0)(2,2,1) | $\psi(\psi_\alpha(\Omega_{M_{\alpha\cdot 2}+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,0)(2,2,1)(3,2,1)(4,2,0) | $\psi(\psi_\alpha(I(1,0,M_{\alpha\cdot 2}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,0)(2,2,1)(3,2,1)(4,2,1) | $\psi(\psi_\alpha(M_{\alpha\cdot 2+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,0)(2,2,1)(3,2,1)<br>(4,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(M_{\alpha\cdot 3}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,1,0)(3,1,0) | $\psi(\psi_\alpha(M_{\alpha^2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(M_{\varepsilon_{\alpha+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,1,1) | $\psi(\psi_\alpha(M_{\Omega_{\alpha+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,1)(2,2,1)(3,2,1)(4,2,1) | $\psi(\psi_\alpha(M_{\Omega_{\alpha+1}+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,1)(3,0,0) | $\psi(\psi_\alpha(M_{\Omega_{\alpha+1}\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,1)(4,2,0) | $\psi(\psi_\alpha(M_{\varepsilon_{\Omega_{\alpha+1}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,1)(4,2,1)(5,0,0) | $\psi(\psi_\alpha(M_{\Omega_{\alpha+\omega}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(6,2,1) | $\psi(\psi_\alpha(M_{M_{\alpha+1}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,1,1)(4,2,1)(5,2,1)(6,2,1)(5,1,1) | $\psi(\psi_\alpha(M_{M_{\Omega_{\alpha+1}}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0) | $\psi(\psi_\alpha(\mathrm{MFP}(\alpha+1)))$<br>$=\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(2,1,1) | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+$<br>$\psi_{\alpha_2}({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(2,2,0) | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(2,2,0)(3,3,0) | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+\varepsilon_{\alpha_2+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(2,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+\Omega_{\alpha_2+1}+$<br>$\psi_{\alpha_2}({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+\Omega_{\alpha_2+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(2,2,1)(3,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+\Omega_{\alpha_2+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(2,2,1)(3,2,1)(4,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+{\Omega_{\alpha_2+1}}^{\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(2,2,1)(3,2,1)(4,2,0) | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+{\Omega_{\alpha_2+1}}^{\alpha_2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(2,2,1)(3,2,1)(4,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot(\alpha_2+1)+$<br>$\psi_{\alpha_2}({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot(\alpha_2+1)+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(2,2,1)(3,2,1)(4,2,1)(3,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot(\alpha_2+\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(2,2,1)(3,2,1)<br>(4,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}} \cdot (\alpha_2+\alpha)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(2,2,1)(3,2,1)(4,2,1)(3,1,1)<br>(4,2,1)(5,2,1)(6,2,1)(5,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}} \cdot (\alpha_2+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}} \cdot \alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(2,2,1)(3,2,1)(4,2,1)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}} \cdot \alpha_2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}} \cdot \alpha_2 \cdot \alpha))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(3,1,1)(4,2,1)(5,2,1)(6,2,1)<br>(5,2,0)(5,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}} \cdot \alpha_2 \cdot$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}} \cdot \alpha_2 \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}} \cdot (\alpha_2{}^2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}} \cdot \varepsilon_{\alpha_2+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,0)(4,3,1)(5,4,1)<br>(6,4,1)(7,4,1)(6,4,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}} \cdot \alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(2,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(2,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}+\Omega_{\alpha_2+1}$<br>$+\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}+\Omega_{\alpha_2+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(2,2,1)(3,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}+\Omega_{\alpha_2+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(2,2,1)(3,2,1)(4,2,1)(3,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}$<br>$+\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(2,2,1)(3,2,1)(4,2,1)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} \cdot 2+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} \cdot 2+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} \cdot \alpha))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} \cdot \alpha \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} \cdot \varepsilon_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} \cdot \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,1,1)(4,2,1)(5,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} \cdot \Omega_{\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} \cdot \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,2,0)(2,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} \cdot \alpha_2+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} \cdot \alpha_2+\Omega_{\alpha_2+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,2,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} \cdot (\alpha_2+\alpha)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,2,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,1,1)(4,2,1)(5,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} \cdot (\alpha_2+\Omega_{\alpha+\omega})))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,2,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} \cdot \alpha_2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} \cdot \varepsilon_{\alpha_2+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(3,2,0)(4,3,1)(5,4,1)(6,4,1)<br>(7,4,1)(6,4,1)(6,4,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+2}+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} \cdot \alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+2}+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+2}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha+1}+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,0)(5,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\varepsilon_{\alpha+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,1,1)(5,2,1)(6,2,1)(7,2,1)<br>(6,2,1)(7,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\omega})}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,0)(2,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2}+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2}+\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,0)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2} \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,0)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2} \cdot \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,0)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2} \cdot \varepsilon_{\alpha_2+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,0)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2+1}+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,0)(3,2,1)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2 \cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,0)(5,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\varepsilon_{\alpha_2+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1} \cdot 2}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1} \cdot 2}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,1)(3,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1} \cdot 2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(3,2,1)(4,2,1)(3,2,1)(4,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1} \cdot 2+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1} \cdot \alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1} \cdot \alpha_2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(4,2,0)(2,2,1)(3,2,1)(4,2,1)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1} \cdot \alpha_2} \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)<br>(4,2,0)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1} \cdot \alpha_2} \cdot \alpha_2))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)(4,2,0)(3,2,1)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}\cdot(\alpha_2+1)}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}\cdot(\alpha_2+1)}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)(4,2,0)(3,2,1)(4,2,1)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}\cdot\alpha_2\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)(4,2,0)(5,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}\cdot\varepsilon_{\alpha_2+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^2}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^2}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)(4,2,1)(3,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^2}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)(4,2,1)(3,2,1)(4,2,1)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^2\cdot 2}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^2\cdot 2}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)(4,2,1)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^3}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^3}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)(5,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)(5,1,1)(6,2,1)(7,2,1)(8,2,1)(9,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^{\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^\omega})}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)(5,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^{\alpha_2}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,1)(5,2,1)(3,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}}}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,1,0)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}+\Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}+\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,1,1)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}+\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}+1}+\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}+1}))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}+\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}+1}+\Omega_{\alpha_2+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(2,2,1)(3,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,1,1)(4,2,1)(5,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}\cdot\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}+1})))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}\cdot\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,2,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}\cdot\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}+1}\cdot\Omega_{\alpha_2+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}{}^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,2,1)(4,3,0)(2,2,1)(3,3,0)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}{}^2\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,2,1)(4,3,0)(3,2,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}{}^2\cdot\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}+1}{}^2\cdot\Omega_{\alpha_2+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,2,1)(4,3,0)(4,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}{}^\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,2,1)(4,3,0)(4,2,1)(5,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}{}^{\varepsilon_{\Omega_{\alpha_2+1}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(3,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+2}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+\alpha}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+\alpha_2}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,2,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2}+\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,2,1)(2,1,1) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2}+\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2}+\omega)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,2,1)(2,1,1)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2}+\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2}+\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha+1}+1}))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,2,1)(2,2,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2}+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,2,1)(2,2,1)(3,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2}+\varepsilon_{\Omega_{\alpha_2+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,2,1)(3,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,2,1)(4,0,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,2,1)(5,3,0) | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\Omega_{\alpha_2+1}+1}}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,3,0) | $\psi(\psi_\alpha({\alpha_3}^2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,4,0) | $\psi(\psi_\alpha(\varepsilon_{\alpha_3+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,4,1)(5,5,1)(6,6,0) | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\psi_{\alpha_2}(\alpha_3)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\psi_{\alpha_2}(\Omega_{\alpha_3+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(2,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(2,2,1)(3,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\psi_{\alpha_3}(\Omega_{\alpha_3+1})+\psi_{\alpha_2}(\Omega_{\alpha_3+1}+\psi_{\alpha_3}(\Omega_{\alpha_3+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\psi_{\alpha_3}(\Omega_{\alpha_3+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\psi_{\alpha_3}(\Omega_{\alpha_3+1}+\alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\psi_{\alpha_3}(\Omega_{\alpha_3+1}+\psi_{\alpha_3}(\alpha_3))))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,3,0)(4,4,0) | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\varepsilon_{\alpha_3+1}))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}\cdot 2+\psi_{\alpha_2}(\Omega_{\alpha_3+1}\cdot 2+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}\cdot\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,2,1)(5,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}\cdot\psi_{\alpha_3}(\alpha_3)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}\cdot\alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,3,1) | $\psi(\psi_\alpha({\Omega_{\alpha_3+1}}^2+\psi_{\alpha_2}({\Omega_{\alpha_3+1}}^2+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,3,1)(2,2,0) | $\psi(\psi_\alpha({\Omega_{\alpha_3+1}}^2+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,3,1)(3,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha_3+1}}^2+\psi_{\alpha_3}({\Omega_{\alpha_3+1}}^2+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,3,1)(3,3,0) | $\psi(\psi_\alpha({\Omega_{\alpha_3+1}}^2+\alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,3,1)(3,3,1)(4,3,1) | $\psi(\psi_\alpha({\Omega_{\alpha_3+1}}^2\cdot 2+\psi_{\alpha_2}({\Omega_{\alpha_3+1}}^2\cdot 2+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,3,1)(4,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha_3+1}}^2\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,3,1)(4,3,1) | $\psi(\psi_\alpha({\Omega_{\alpha_3+1}}^3+\psi_{\alpha_2}({\Omega_{\alpha_3+1}}^3+1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,3,1)(5,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha_3+1}}^\omega))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,4,0) | $\psi(\psi_\alpha(\alpha_4))$ |

| BMS | Projection |
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| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,4,0)<br>(3,3,1)(4,4,0) | $\psi(\psi_\alpha(\alpha_4 + \psi_{\alpha_4}(\alpha_4)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,4,0)<br>(4,4,0) | $\psi(\psi_\alpha(\alpha_4 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,4,1) | $\psi(\psi_\alpha(\Omega_{\alpha_4+1} + \psi_{\alpha_2}(\Omega_{\alpha_4+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,4,1)<br>(5,5,0) | $\psi(\psi_\alpha(\alpha_5))$ |
| (0,0,0)(1,1,1)(2,2,2) | $\psi(\psi_\alpha(\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(1,1,0) | $\psi(\psi_\alpha(\alpha_\omega) + \Omega)$ |
| (0,0,0)(1,1,1)(2,2,2)(1,1,1) | $\psi(\psi_\alpha(\alpha_\omega) + \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,2,2)(1,1,1)(2,2,0) | $\psi(\psi_\alpha(\alpha_\omega) + \psi_\alpha(\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(1,1,1)(2,2,1) | $\psi(\psi_\alpha(\alpha_\omega) + \psi_\alpha(\Omega_{\alpha+2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(1,1,1)(2,2,2) | $\psi(\psi_\alpha(\alpha_\omega) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,0) | $\psi(\psi_\alpha(\alpha_\omega) \cdot \Omega)$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,0)<br>(1,1,0)(2,2,1)(3,3,2)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega) \cdot \Omega_2)$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,0)(1,1,1) | $\psi(\psi_\alpha(\alpha_\omega) \cdot \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(2,1,0)(1,1,1)(2,2,0) | $\psi(\psi_\alpha(\alpha_\omega) \cdot \psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,0)(1,1,1)<br>(2,2,0)(3,3,1)(4,4,2)(4,3,0) | $\psi(\psi_\alpha(\alpha_\omega) \cdot \psi_\alpha(\Omega_{\alpha+2} + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(2,1,0)(1,1,1)(2,2,1) | $\psi(\psi_\alpha(\alpha_\omega) \cdot \psi_\alpha(\Omega_{\alpha+2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,0)(1,1,1)<br>(2,2,1)(3,3,0)(4,4,1)(5,5,2)(5,4,0) | $\psi(\psi_\alpha(\alpha_\omega) \cdot \psi_\alpha(\Omega_{\alpha_2+2} + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,0)<br>(1,1,1)(2,2,1)(3,3,1) | $\psi(\psi_\alpha(\alpha_\omega) \cdot \psi_\alpha(\Omega_{\alpha_2+2} + \psi_{\alpha_2}(\Omega_{\alpha_2+2} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(2,1,0)(1,1,1)(2,2,2) | $\psi(\psi_\alpha(\alpha_\omega)^2)$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,0)<br>(1,1,1)(2,2,2)(1,1,1)(2,2,2) | $\psi(\psi_\alpha(\alpha_\omega)^2 + \psi_\alpha(\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega)^2 \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,0)(3,0,0) | $\psi(\psi_\alpha(\alpha_\omega)^\omega)$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,0)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(2,1,0)(3,2,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha+1}) +$<br>$\psi_{\psi_\alpha(\alpha_\omega+\Omega_{\alpha+1})}(\psi_\alpha(\alpha_\omega + \Omega_{\alpha+1}) + 1))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(2,1,0)(3,2,0)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha+1}) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,0)(3,2,1) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(2,1,0)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\alpha_\omega + \varepsilon_{\Omega_{\alpha+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(2,1,0)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(2,1,1)(1,1,1)(2,2,2) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + 1)) + \psi_\alpha(\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)(1,1,1)<br>(2,2,2)(2,1,0)(3,2,1)(4,3,2)(4,2,1) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + 1)) +$<br>$\psi_{\psi_\alpha(\alpha_\omega+\Omega_{\alpha+1})}(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + 1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(1,1,1)(2,2,2)(2,1,0)(3,2,1)<br>(4,3,2)(4,2,1)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + 1)) + \psi_\alpha(\alpha_\omega + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)(1,1,1)<br>(2,2,2)(2,1,0)(3,2,1)(4,3,2)<br>(4,2,1)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + 1)) + \psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(1,1,1)(2,2,2)(2,1,1) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + 1)) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(2,1,0)(1,1,1) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + 1)) \cdot \Omega_\omega)$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(2,1,0)(1,1,1)(2,2,2) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + 1)) \cdot \psi_\alpha(\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(2,1,0)(1,1,1)(2,2,2)(2,1,0)<br>(3,2,1)(4,3,2)(4,2,1)(4,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + 1)) \cdot \psi_\alpha(\alpha_\omega + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(2,1,0)(1,1,1)(2,2,2)(2,1,1) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + 1))^2)$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(2,1,0)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + 1) + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(2,1,0)(3,2,1)(4,3,2)(4,2,1) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + 1) \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)(2,1,1) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)(3,0,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \varepsilon_{\alpha+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)(3,1,1) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \Omega_{\alpha+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,1,1)(2,1,0)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \Omega_{\alpha+1} + 1) + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \Omega_{\alpha+1} + \alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \Omega_{\alpha+1} \cdot 2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \Omega_{\alpha+1} \cdot \alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \Omega_{\alpha+1} \cdot \alpha \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + {\Omega_{\alpha+1}}^2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,1,1)(4,0,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + {\Omega_{\alpha+1}}^\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \psi_{\alpha_2}(\alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,2,0)(2,1,1)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \psi_{\alpha_2}(\alpha_2) \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,2,0)(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \psi_{\alpha_2}(\alpha_2 + \psi_{\alpha_2}(\alpha_2)))))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,2,0)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \psi_{\alpha_2}(\alpha_2 \cdot 2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,2,0)(4,3,1) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \psi_{\alpha_2}(\Omega_{\alpha_2+1})) + \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)(3,2,1) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \psi_{\alpha_2}(\Omega_{\alpha_2+1}) + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,2,1)(2,1,1)(3,2,1) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \psi_{\alpha_2}(\Omega_{\alpha_2+1}) \cdot 2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,2,1)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \psi_{\alpha_2}(\Omega_{\alpha_2+1} + \alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \psi_{\alpha_2}(\alpha_3))))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)(3,2,2) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,2,2)(2,1,1)(3,2,2) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega) \cdot 2)))$ |

| BMS | Projection |
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| (0,0,0)(1,1,1)(2,2,2)(2,1,1)<br>(3,2,2)(3,1,1)(4,2,2) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega)))))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)<br>(2,1,0)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \alpha_2 + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)<br>(2,1,0)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\alpha_\omega + \alpha_2 + \psi_{\alpha_2}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)<br>(2,1,0)(3,2,1)(4,3,2)(4,3,0) | $\psi(\psi_\alpha(\alpha_\omega + \alpha_2 + \psi_{\alpha_2}(\alpha_\omega + \alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)(2,1,1) | $\psi(\psi_\alpha(\alpha_\omega + \alpha_2 + \psi_{\alpha_2}(\alpha_\omega + \alpha_2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)<br>(2,1,1)(3,1,1) | $\psi(\psi_\alpha(\alpha_\omega + \alpha_2 + \psi_{\alpha_2}(\alpha_\omega + \alpha_2 + \Omega_{\alpha+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)<br>(2,1,1)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \alpha_2 + \psi_{\alpha_2}(\alpha_\omega + \alpha_2 + \psi_{\alpha_2}(\alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)<br>(2,1,1)(3,2,2)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \alpha_2 + \psi_{\alpha_2}(\alpha_\omega + \alpha_2 + \psi_{\alpha_2}(\alpha_\omega + \alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)(2,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \alpha_2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)<br>(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega + \alpha_2 \cdot \alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)<br>(3,1,0)(4,2,1)(5,3,2) | $\psi(\psi_\alpha(\alpha_\omega + \alpha_2 \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)(3,1,1) | $\psi(\psi_\alpha(\alpha_\omega + \alpha_2 \cdot \Omega_{\alpha+1} +$<br>$\psi_{\alpha_2}(\alpha_\omega + \alpha_2 \cdot \Omega_{\alpha+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)<br>(3,1,1)(2,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \alpha_2 \cdot (\Omega_{\alpha+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)<br>(3,1,1)(2,2,0)(3,1,1) | $\psi(\psi_\alpha(\alpha_\omega + \alpha_2 \cdot \Omega_{\alpha+1} \cdot 2 +$<br>$\psi_{\alpha_2}(\alpha_\omega + \alpha_2 \cdot \Omega_{\alpha+1} \cdot 2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)<br>(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \alpha_2 \cdot \psi_{\alpha_2}(\alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)<br>(3,1,1)(4,2,2) | $\psi(\psi_\alpha(\alpha_\omega + \alpha_2 \cdot \psi_{\alpha_2}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega + {\alpha_2}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)<br>(3,2,0)(2,2,0)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega + {\alpha_2}^2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)(3,3,0) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_2+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)(3,3,1) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_2+1} + \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,0)<br>(3,3,1)(4,4,2) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_2+1} + \psi_{\alpha_2}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_2+1} + \psi_{\alpha_2}(\alpha_\omega + \Omega_{\alpha_2+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)(2,1,1) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_2+1} + \psi_{\alpha_2}(\alpha_\omega + \Omega_{\alpha_2+1} + 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)<br>(2,1,1)(3,2,2) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_2+1} +$<br>$\psi_{\alpha_2}(\alpha_\omega + \Omega_{\alpha_2+1} + \psi_{\alpha_2}(\alpha_\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)(2,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_2+1} + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)<br>(2,2,0)(3,3,0) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_2+1} \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)<br>(2,2,0)(3,3,1)(4,4,2) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_2+1} \cdot 2 + \psi_{\alpha_2}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)(2,2,1) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_2+1} \cdot 2 +$<br>$\psi_{\alpha_2}(\alpha_\omega + \Omega_{\alpha_2+1} \cdot 2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)(3,0,0) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_2+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)<br>(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_2+1} \cdot \alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)(3,1,1) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_2+1} \cdot \Omega_{\alpha+1} +$<br>$\psi_{\alpha_2}(\alpha_\omega + \Omega_{\alpha_2+1} \cdot \Omega_{\alpha+1} + 1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)<br>(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_2+1} \cdot \psi_{\alpha_2}(\alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)<br>(3,1,1)(4,2,2) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_2+1} \cdot \psi_{\alpha_2}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_2+1} \cdot \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)<br>(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\alpha_\omega + {\Omega_{\alpha_2+1}}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)(3,2,1) | $\psi(\psi_\alpha(\alpha_\omega + {\Omega_{\alpha_2+1}}^2 +$<br>$\psi_{\alpha_2}(\alpha_\omega + {\Omega_{\alpha_2+1}}^2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)(3,3,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_3}(\alpha_3)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)<br>(3,3,2)(3,3,1) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_3+1} + \psi_{\alpha_2}(\alpha_\omega + \Omega_{\alpha_3+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)<br>(3,3,2)(3,3,1)(4,3,0) | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_3+1} \cdot \alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,1)<br>(3,3,2)(3,3,1)(4,4,0) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_4}(\alpha_4)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,2)(2,1,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \psi_{\alpha_2}(\alpha_\omega \cdot 2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,2)<br>(2,1,1)(3,2,2)(3,2,2) | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega \cdot 2 + \psi_{\alpha_2}(\alpha_\omega \cdot 2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,2)(2,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,2)<br>(2,2,0)(3,3,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \Omega_{\alpha_2+1} + \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,2)(2,2,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \Omega_{\alpha_2+1} +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot 2 + \Omega_{\alpha_2+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,2)<br>(2,2,1)(3,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \Omega_{\alpha_2+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,2)<br>(2,2,1)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \Omega_{\alpha_2+1} \cdot \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,2)<br>(2,2,1)(3,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \psi_{\alpha_3}(\alpha_3)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,2)<br>(2,2,1)(3,3,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \psi_{\alpha_3}(\Omega_{\alpha_3+1}) +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot 2 + \psi_{\alpha_3}(\Omega_{\alpha_3+1}) + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,2)<br>(2,2,1)(3,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \psi_{\alpha_3}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,2)<br>(2,2,1)(3,3,2)(3,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \psi_{\alpha_3}(\alpha_\omega \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,2)(2,2,1)<br>(3,3,2)(3,3,2)(3,2,1)(4,3,2)(4,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \psi_{\alpha_3}(\alpha_\omega \cdot 2 + \psi_{\alpha_3}(\alpha_\omega \cdot 2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,2)<br>(2,2,1)(3,3,2)(3,3,2)(3,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(2,2,2)(2,2,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot 3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)<br>(2,1,0)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)<br>(2,1,0)(3,2,1)(4,3,2)(5,1,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,1,0)(1,1,1)<br>(2,2,2)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,1,0)(1,1,1)(2,2,2)<br>(3,1,0)(2,1,0)(3,2,1)(4,3,2)(5,1,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha))) +$<br>$\psi_{\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha+1})}(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega))))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)<br>(2,1,0)(3,2,1)(4,3,2)(5,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha)))+$<br>$\psi_{\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha+1})}(\psi_\alpha(\alpha_\omega \cdot \alpha+$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha))) + 1))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,1,0)(3,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha)))+$<br>$\psi_{\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha+1})}(\psi_\alpha(\alpha_\omega \cdot \alpha+$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha)))+$<br>$\psi_{\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha+1})}(\psi_\alpha(\alpha_\omega \cdot \alpha+$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha))) + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,1,0)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha)))+$<br>$\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)<br>(2,1,0)(3,2,1)(4,3,2)(5,1,0)<br>(3,2,1)(4,3,2)(5,1,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha)))+$<br>$\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,1,0)(4,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha))) \cdot \omega)$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,1,0)(4,2,0)(3,2,1)<br>(4,3,2)(5,1,0)(1,1,1)<br>(2,2,2)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha)))^2)$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,1,0)(4,2,0)(5,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha)) + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,1,0)(4,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha) + \alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,1,0)(4,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot (\psi_\alpha(\alpha_\omega \cdot \alpha) + 1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,2,0)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha+$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega)))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,2,0)(4,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,1,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)<br>(2,1,1)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,1,1)<br>(3,2,2)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)<br>(2,2,0)(3,3,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha_2+1} + \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,2,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha_2+1}+$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \alpha + \Omega_{\alpha_2+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)<br>(2,2,1)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha_2+1} \cdot \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)<br>(2,2,1)(3,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_3}(\alpha_3)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)<br>(2,2,1)(3,3,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)<br>(2,2,1)(3,3,2)(4,2,0)(3,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(2,2,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)<br>(2,2,2)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(3,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha \cdot \omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)<br>(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)(4,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \varepsilon_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)<br>(4,2,1)(5,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,0)<br>(4,2,1)(5,3,2)(6,2,0)(5,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)<br>(2,1,0)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} + 1) + \Omega_{\alpha+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)(2,1,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} + 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)<br>(2,1,1)(3,1,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} + \Omega_{\alpha+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)<br>(2,1,1)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)<br>(2,1,1)(3,2,2)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)<br>(2,1,1)(3,2,2)(4,1,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1}) + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)(2,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)<br>(2,2,0)(3,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \varepsilon_{\alpha_2+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)(2,2,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \Omega_{\alpha_2+1} +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} + \Omega_{\alpha_2+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)<br>(2,2,1)(3,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_3}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)(2,2,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\Omega_{\alpha+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)<br>(2,2,2)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\Omega_{\alpha+1} + \alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,1,1)(2,2,2)(3,1,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} \cdot 2 + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} \cdot 2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)(3,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\Omega_{\alpha+1}{}^2) + \psi_{\alpha_2}(\alpha_\omega \cdot (\Omega_{\alpha+1}{}^2) + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \psi_{\alpha_2}(\alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)(4,2,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot \psi_{\alpha_2}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,1,1)<br>(4,2,2)(5,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \psi_{\alpha_2}(\alpha_\omega \cdot \alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,1,0)(3,2,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,1,0)(3,2,1)(4,3,2)(5,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,1,0)(3,2,1)(4,3,2)(5,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha+1}))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,2,0)(3,2,0)<br>(4,3,1)(5,4,2)(6,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha+1} \cdot 2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,2,0)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega)))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,2,0)(4,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,2,0)(4,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot (\alpha + 1))))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,2,0)(6,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \varepsilon_{\alpha+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,2,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,1,0)<br>(3,2,1)(4,3,2)(5,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,1,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,1,1)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,1,1)(3,2,2)(4,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,0)(3,3,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_2+1} + \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_2+1} +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_2+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(2,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_2+1} + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_2+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_3)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,1,0)(5,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \varepsilon_{\alpha+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,1,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \Omega_{\alpha+1}) +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \Omega_{\alpha+1}) + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,1,1)(5,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \psi_{\alpha_2}(\alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha_2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,0)(3,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha_2 + \alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,0)(3,2,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_2+1}) +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_2+1}) + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,0)(3,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,0)(3,3,0)(4,4,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \varepsilon_{\alpha_3+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,0)(3,3,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 +$<br>$\Omega_{\alpha_3+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_3+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,0)(3,3,1)(4,4,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_4}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,1)(4,4,2)(5,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_4}(\alpha_\omega \cdot \alpha_2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,0)(3,3,1)<br>(4,4,2)(5,2,0)(4,4,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \alpha_4))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,0)(3,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,2)(2,1,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 1) +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot (\alpha_2 + 1) + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,2)(2,1,1)(3,2,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 1) +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot (\alpha_2 + 1) + \psi_{\alpha_2}(\alpha_\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,2)(2,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 1) + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,2)(2,2,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 1) + \Omega_{\alpha_2+1} +$<br>$\psi_{\alpha_2}(\alpha_\omega \cdot (\alpha_2 + 1) + \Omega_{\alpha_2+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,2)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 1) + \psi_{\alpha_3}(\alpha_\omega \cdot (\alpha_2 + 1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,0)<br>(3,3,2)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 1) +$<br>$\psi_{\alpha_3}(\alpha_\omega \cdot (\alpha_2 + 1) + \psi_{\alpha_3}(\alpha_\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,2)(3,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 1) + \alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,2)(3,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,2)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + \alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,2)(4,1,0)(2,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + \alpha) + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,2)(4,1,0)(3,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + \alpha) + \alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,2)(4,1,0)(3,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + \alpha + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,0)(3,3,2)<br>(4,1,0)(3,3,2)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + \alpha \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,2)(4,1,0)(5,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + \varepsilon_{\alpha+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,2)(4,1,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + \Omega_{\alpha+1}) + \psi_{\alpha_2}(\alpha_\omega \cdot (\alpha_2 + \Omega_{\alpha+1}) + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,2)(4,1,1)(5,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + \psi_{\alpha_2}(\alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(3,3,2)(4,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 \cdot \alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,0)(4,1,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 \cdot \Omega_{\alpha+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,0)(4,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot {\alpha_2}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,0)(5,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \varepsilon_{\alpha_2+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha_2+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha_2+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,1)(2,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha_2+1} + \alpha_2))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,1)(3,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\Omega_{\alpha_2+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,1)(3,3,2)(4,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\Omega_{\alpha_2+1} + \alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,2,1)(3,3,2)(4,2,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha_2+1} \cdot 2 + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha_2+1} \cdot 2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,1)(5,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \psi_{\alpha_3}(\alpha_3)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,2,1)(5,3,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot \psi_{\alpha_3}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,0)(2,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_3 + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,0)(3,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_3 + \alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,0)(3,3,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_3 + \Omega_{\alpha_3+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_3 + \Omega_{\alpha_3+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,3,0)(3,3,1)(4,4,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_3 + \psi_{\alpha_4}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,3,0)(3,3,1)(4,4,2)(5,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_3 + \psi_{\alpha_4}(\alpha_\omega \cdot \alpha_3)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,0)(3,3,1)<br>(4,4,2)(5,3,0)(4,4,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_3 + \alpha_4))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,0)(3,3,1)<br>(4,4,2)(5,3,0)(4,4,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_3 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,0)(3,3,1)<br>(4,4,2)(5,3,0)(4,4,2)(5,2,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_3 + \alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,0)(3,3,1)<br>(4,4,2)(5,3,0)(4,4,2)(5,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_3 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,0)(3,3,1)<br>(4,4,2)(5,3,0)(5,3,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot {\alpha_3}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,0)(3,3,1)<br>(4,4,2)(5,3,0)(6,4,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \varepsilon_{\alpha_3+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,3,0)(3,3,1)(4,4,2)(5,3,1) | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha_3+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha_3+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,0)(3,3,1)<br>(4,4,2)(5,3,1)(6,4,2) | $\psi(\psi_\alpha(\alpha_\omega \cdot \psi_{\alpha_4}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,1)<br>(3,3,2)(4,3,0)(3,3,1)(4,4,2)(5,4,0) | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_4))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,2) | $\psi(\psi_\alpha({\alpha_\omega}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(2,1,0)(3,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \Omega_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(2,1,0)(3,2,1)(4,3,2)(5,3,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,2)<br>(2,1,0)(3,2,1)(4,3,2)(5,3,0)(4,2,1) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + 1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(2,1,0)(3,2,1)(4,3,2)<br>(5,3,0)(4,2,1)(5,3,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,2)<br>(2,1,0)(3,2,1)(4,3,2)(5,3,0)(4,3,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(2,1,0)(3,2,1)(4,3,2)<br>(5,3,0)(4,3,1)(5,4,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(2,1,0)(3,2,1)(4,3,2)<br>(5,3,0)(4,3,1)(5,4,2)(6,3,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(2,1,0)(3,2,1)(4,3,2)(5,3,0)<br>(4,3,1)(5,4,2)(6,3,0)(5,4,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}(\alpha_\omega \cdot (\alpha_2 + 1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(2,1,0)(3,2,1)(4,3,2)(5,3,0)<br>(4,3,1)(5,4,2)(6,3,0)(5,4,2)(6,3,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,2)<br>(2,1,0)(3,2,1)(4,3,2)(5,3,0)<br>(4,3,1)(5,4,2)(6,4,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_3)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,2)<br>(2,1,0)(3,2,1)(4,3,2)(5,3,0)(4,3,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}({\alpha_\omega}^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(2,1,0)(3,2,1)(4,3,2)<br>(5,3,0)(4,3,2)(4,0,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}({\alpha_\omega}^2) + 1))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(2,1,0)(3,2,1)(4,3,2)<br>(5,3,0)(4,3,2)(4,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}({\alpha_\omega}^2) + \psi_\alpha({\alpha_\omega}^2 + \Omega_{\alpha+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,2)<br>(2,1,0)(3,2,1)(4,3,2)(5,3,0)(4,3,2)<br>(4,2,0)(3,2,1)(4,3,2)(5,3,0)(4,3,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}({\alpha_\omega}^2) + \psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}({\alpha_\omega}^2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,2)<br>(2,1,0)(3,2,1)(4,3,2)(5,3,0)<br>(4,3,2)(4,2,0)(5,3,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}({\alpha_\omega}^2) + \alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(2,1,1) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}({\alpha_\omega}^2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(2,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}({\alpha_\omega}^2 + \alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(2,1,1)(3,1,1) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}({\alpha_\omega}^2 + \Omega_{\alpha+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(2,1,1)(3,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}({\alpha_\omega}^2 + \psi_{\alpha_2}(\alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,2)<br>(2,1,1)(3,2,2)(4,2,0)(3,2,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}({\alpha_\omega}^2 + \psi_{\alpha_2}({\alpha_\omega}^2))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,0)(2,2,2)(2,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(2,2,0)(3,3,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \varepsilon_{\alpha_2+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,0)(2,2,2)(2,2,1) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \Omega_{\alpha_2+1} + \psi_{\alpha_2}({\alpha_\omega}^2 + \Omega_{\alpha_2+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(2,2,1)(3,3,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_3}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(2,2,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,0,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,1,0)(2,0,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,1,0)(2,2,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot (\alpha + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,1,0)(4,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \varepsilon_{\alpha+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,1,0)(4,2,1)(5,3,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,1,1) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \Omega_{\alpha+1} +$<br>$\psi_{\alpha_2}({\alpha_\omega}^2 + \alpha_\omega \cdot \Omega_{\alpha+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,1,1)(2,1,1) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \Omega_{\alpha+1} +$<br>$\psi_{\alpha_2}({\alpha_\omega}^2 + \alpha_\omega \cdot \Omega_{\alpha+1} + 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,1,1)(2,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \Omega_{\alpha+1} + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,1,1)(2,2,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot (\Omega_{\alpha+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,1,1)(2,2,2)(3,1,1) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \Omega_{\alpha+1} \cdot 2 +$<br>$\psi_{\alpha_2}({\alpha_\omega}^2 + \alpha_\omega \cdot \Omega_{\alpha+1} \cdot 2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,1,1)(4,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \psi_{\alpha_2}(\alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,0)(2,2,2)(3,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,2,0)(2,1,1) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \alpha_2 +$<br>$\psi_{\alpha_2}({\alpha_\omega}^2 + \alpha_\omega \cdot \alpha_2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,2,0)(2,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \alpha_2 + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,2)<br>(3,2,0)(2,2,1)(3,3,2)(4,3,0)(3,3,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}({\alpha_\omega}^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,2,0)(2,2,1)(3,3,2)<br>(4,3,0)(3,3,2)(4,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}({\alpha_\omega}^2 + \alpha_\omega \cdot \alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,2)<br>(3,2,0)(2,2,1)(3,3,2)(4,3,0)<br>(3,3,2)(4,2,0)(3,3,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \alpha_2 + \alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,2)<br>(3,2,0)(2,2,1)(3,3,2)(4,3,0)<br>(3,3,2)(4,2,0)(3,3,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot (\alpha_2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(2,2,2)<br>(3,2,0)(2,2,1)(3,3,2)(4,3,0)<br>(3,3,2)(4,2,0)(3,3,2)(4,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \alpha_2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,2,0)(2,2,1)(3,3,2)<br>(4,3,0)(3,3,2)(4,2,0)(5,3,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \varepsilon_{\alpha_2+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,2,0)(2,2,1)(3,3,2)<br>(4,3,0)(3,3,2)(4,3,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(2,2,2)(3,2,0)(2,2,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(3,1,0)(2,0,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(3,1,0)(2,2,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha + \alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(3,1,0)(2,2,2)(3,2,0)(2,2,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot (\alpha + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(3,1,1) | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}({\alpha_\omega}^2 \cdot \Omega_{\alpha+1} + 1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(3,1,1)(4,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \psi_{\alpha_2}(\alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(3,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(3,2,0)(2,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha_2 + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,0)(4,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha_2 + \psi_{\alpha_3}({\alpha_\omega}^2 \cdot \alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,0)(4,2,0)(3,3,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha_2 + \alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(3,2,0)(2,2,1)(3,3,2)(4,3,0)<br>(4,2,0)(3,3,2)(4,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha_2 + \alpha_\omega \cdot \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(3,2,0)(2,2,1)(3,3,2)(4,3,0)<br>(4,2,0)(3,3,2)(4,3,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha_2 + \alpha_\omega \cdot \alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(3,2,0)(2,2,1)(3,3,2)(4,3,0)<br>(4,2,0)(3,3,2)(4,3,0)(3,3,2) | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot (\alpha_2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(3,2,0)(2,2,1)(3,3,2)(4,3,0)<br>(4,2,0)(3,3,2)(4,3,0)(4,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha_2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,0)(4,2,0)(5,3,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \varepsilon_{\alpha_2+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,0)(4,3,0) | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,0)(3,2,0)(2,2,2) | $\psi(\psi_\alpha({\alpha_\omega}^3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(3,2,0)(2,2,2)(2,2,2) | $\psi(\psi_\alpha({\alpha_\omega}^3 + \alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(3,2,0)(2,2,2)(3,2,0)(2,2,2) | $\psi(\psi_\alpha({\alpha_\omega}^3 + {\alpha_\omega}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(3,2,0)(3,2,0)(2,2,2) | $\psi(\psi_\alpha({\alpha_\omega}^4))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(4,1,0)(2,0,0) | $\psi(\psi_\alpha({\alpha_\omega}^\alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(4,1,0)(2,2,2) | $\psi(\psi_\alpha({\alpha_\omega}^\alpha + \alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(4,1,0)(2,2,2)(3,2,0)(4,1,0)(2,0,0) | $\psi(\psi_\alpha({\alpha_\omega}^\alpha \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(4,1,1) | $\psi(\psi_\alpha({\alpha_\omega}^{\Omega_{\alpha+1}} + \psi_{\alpha_2}({\alpha_\omega}^{\Omega_{\alpha+1}} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(4,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^{\alpha_2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(4,2,0)(2,2,1)(3,3,2)(4,3,0)<br>(5,2,0)(4,3,0)(3,3,2) | $\psi(\psi_\alpha({\alpha_\omega}^(\alpha_2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(4,2,0)(2,2,1)(3,3,2)(4,3,0)<br>(5,2,0)(4,3,0)(5,2,0) | $\psi(\psi_\alpha({\alpha_\omega}^(\alpha_2 \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(4,2,0)<br>(2,2,1)(3,3,2)(4,3,0)(5,3,0) | $\psi(\psi_\alpha({\alpha_\omega}^{\alpha_3}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(4,2,0)(2,2,2) | $\psi(\psi_\alpha({\alpha_\omega}^{\alpha_\omega}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\varepsilon_{\alpha_\omega+1})) = \psi(\psi_\alpha(\Omega_{\alpha_\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(4,3,0)(4,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}) \cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,0)<br>(4,3,1)(5,4,2) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1} + \psi_{\alpha_2}(\alpha_\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(2,1,0)(3,2,1)(4,3,2)(5,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}+1)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(2,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}+2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(2,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}+\Omega_{\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(2,1,1)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(2,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(2,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\Omega_{\alpha_2+1}+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}+\Omega_{\alpha_2+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(2,2,1)(3,3,2)(4,3,0)(3,3,2) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\psi_{\alpha_3}({\alpha_\omega}^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(2,2,1)(3,3,2)(4,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\psi_{\alpha_3}(\Omega_{\alpha_\omega+1})+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}+\psi_{\alpha_3}(\Omega_{\alpha_\omega+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(2,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(2,2,2)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\alpha_\omega\cdot\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(2,2,2)(3,2,0)(2,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+{\alpha_\omega}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(2,2,2)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\varepsilon_{\alpha_\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(2,2,2)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot 2+\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}\cdot 2+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\Omega_{\alpha+1}+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}\cdot\Omega_{\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,1,1)(2,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\Omega_{\alpha+1}+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,1,1)(2,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\Omega_{\alpha+1}+\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,1,1)(2,2,2)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot(\Omega_{\alpha+1}+1)+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}\cdot(\Omega_{\alpha+1}+1)+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,1,1)(3,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,1,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\Omega_{\alpha+1}\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,1,1)(3,1,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot({\Omega_{\alpha+1}}^2)+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}\cdot({\Omega_{\alpha+1}}^2)+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,1,1)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot({\Omega_{\alpha+1}}^\alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\psi_{\alpha_2}(\alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,1)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_2+\psi_{\alpha_3}(\Omega_{\alpha_\omega+1}\cdot\alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,1)(4,2,0)(3,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_2+\alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,1)(4,2,0)(3,3,2) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_2+\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,1)(4,2,0)<br>(3,3,2)(4,3,0)(3,3,2) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_2+{\alpha_\omega}^2))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,1)(4,2,0)<br>(3,3,2)(4,3,0)(5,4,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_2+\varepsilon_{\alpha_\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,2,0)(2,2,1)(3,3,2)(4,3,1)<br>(4,2,0)(3,3,2)(4,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot(\alpha_2+1)+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}\cdot(\alpha_2+1)+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,2,0)(2,2,1)(3,3,2)(4,3,1)<br>(4,2,0)(3,3,2)(4,3,1)(4,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_2\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,1)(4,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,2,0)(2,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,2,0)(2,2,2)(2,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_\omega+\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,2,0)(2,2,2)(3,2,0)(2,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_\omega+{\alpha_\omega}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,2,0)(2,2,2)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot(\alpha_\omega+1)+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}\cdot(\alpha_\omega+1)+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,2,0)(2,2,2)(3,2,1)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot(\alpha_\omega+\alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(3,2,0)<br>(2,2,2)(3,2,1)(3,2,0)(2,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_\omega\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(3,2,0)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_\omega\cdot\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,2,0)(3,2,0)(2,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot{\alpha_\omega}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\varepsilon_{\alpha_\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(3,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^2+\psi_{\alpha_2}({\Omega_{\alpha_\omega+1}}^2+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,2,1)(2,2,2)(3,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^2+\Omega_{\alpha_\omega+1}+$<br>$\psi_{\alpha_2}({\Omega_{\alpha_\omega+1}}^2+\Omega_{\alpha_\omega+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(3,2,1)(3,2,0) | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^2\cdot\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,2,1)(2,2,1)(3,3,2)<br>(4,3,1)(4,3,1)(4,3,0) | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^2\cdot\alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(3,2,1)(3,2,0)(2,2,2) | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^2\cdot\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(3,2,1)(3,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^3+\psi_{\alpha_2}({\Omega_{\alpha_\omega+1}}^3+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,1,0)(2,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^\alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(4,1,1) | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^{\Omega_{\alpha+1}}+\psi_{\alpha_2}({\Omega_{\alpha_\omega+1}}^{\Omega_{\alpha+1}}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,1,1)(5,2,0) | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^{\psi_{\alpha_2}(\alpha_2)}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(4,2,0) | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^{\alpha_2}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,2,0)(2,2,2) | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^{\alpha_\omega}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,2,0)(5,3,0) | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^{\varepsilon_{\alpha_\omega+1}}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(4,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^{\Omega_{\alpha_\omega+1}}+\psi_{\alpha_2}({\Omega_{\alpha_\omega+1}}^{\Omega_{\alpha_\omega+1}}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_\omega+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,0)(4,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_\omega+1}+2}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,0)(5,4,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,0)(5,4,1)(6,5,2) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}+\psi_{\alpha_2}(\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(4,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}+\psi_{\alpha_2}(\Omega_{\alpha_\omega+2}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,1)(2,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,1)(2,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}+\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(2,2,2)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}+\Omega_{\alpha_\omega+1}+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_\omega+2}+\Omega_{\alpha_\omega+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(2,2,2)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}+\varepsilon_{\Omega_{\alpha_\omega+1}+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(2,2,2)(3,2,1)(4,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}\cdot 2+\psi_{\alpha_2}(\Omega_{\alpha_\omega+2}\cdot 2+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,1)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}\cdot\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(3,2,0)(2,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}\cdot\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,1)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}\cdot\Omega_{\alpha_\omega+1}+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_\omega+2}\cdot\Omega_{\alpha_\omega+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(3,2,1)(4,3,1) | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+2}}^2+\psi_{\alpha_2}({\Omega_{\alpha_\omega+2}}^2+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(4,2,0)(2,2,2) | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+2}}^{\alpha_\omega}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(4,2,1)(5,3,1) | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+2}}^{\Omega_{\alpha_\omega+2}}+\psi_{\alpha_2}({\Omega_{\alpha_\omega+2}}^{\Omega_{\alpha_\omega+2}}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,1)(4,3,0) | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_\omega+2}+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,1)(4,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+3}+\psi_{\alpha_2}(\Omega_{\alpha_\omega+3}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,1)(5,0,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,1)(5,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+\alpha_2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(5,2,0)(2,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,1)(5,2,1) | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha_\omega+1}}+\psi_{\alpha_2}(\Omega_{\Omega_{\alpha_\omega+1}}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,1)(5,3,0) | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha_\omega+1)))$<br>$=\psi(\psi_\alpha(\Omega_{\alpha_{\omega+1}+1}\cdot\alpha_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,1)(5,3,1) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2+\psi_{\alpha_2}({\Omega_{\alpha_{\omega+1}+1}}^2+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(5,3,1)(2,2,2) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2+\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(5,3,1)(2,2,2)(3,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2+\Omega_{\alpha_\omega+1}+$<br>$\psi_{\alpha_2}({\Omega_{\alpha_{\omega+1}+1}}^2+\Omega_{\alpha_\omega+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(4,3,1)<br>(5,3,1)(2,2,2)(3,2,1)(4,3,1)(5,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2+\psi_{\alpha_{\omega+1}}(\Omega_{\alpha_{\omega+1}+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(5,3,1)(3,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2+\psi_{\alpha_{\omega+1}}({\Omega_{\alpha_{\omega+1}+1}}^2+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(5,3,1)(3,2,0)(2,2,2) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2+\psi_{\alpha_{\omega+1}}({\Omega_{\alpha_{\omega+1}+1}}^2+\alpha_\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(4,3,1)<br>(5,3,1)(3,2,1)(4,3,1)(5,3,1) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2 + \psi_{\alpha_{\omega+1}}({\Omega_{\alpha_{\omega+1}+1}}^2 +$<br>$\psi_{\alpha_{\omega+1}}({\Omega_{\alpha_{\omega+1}+1}}^2)) + \psi_{\alpha_2}({\Omega_{\alpha_{\omega+1}+1}}^2 +$<br>$\psi_{\alpha_{\omega+1}}({\Omega_{\alpha_{\omega+1}+1}}^2 + \psi_{\alpha_{\omega+1}}({\Omega_{\alpha_{\omega+1}+1}}^2))) + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(5,3,1)(4,3,0) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2 + \alpha_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(5,3,1)(4,3,1) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2 + \Omega_{\alpha_{\omega+1}+1} +$<br>$\psi_{\alpha_2}({\Omega_{\alpha_{\omega+1}+1}}^2 + \Omega_{\alpha_{\omega+1}+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(5,3,1)(4,3,1)(5,3,1) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2 \cdot 2 + \psi_{\alpha_2}({\Omega_{\alpha_{\omega+1}+1}}^2 \cdot 2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(5,3,1)(5,3,0) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2 \cdot \alpha_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(5,3,1)(6,0,0) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(5,3,1)(6,2,0)(2,2,2) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^{\alpha_\omega}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,1)(5,3,1)(6,3,1) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^{\Omega_{\alpha_{\omega+1}+1}+1} +$<br>$\psi_{\alpha_2}({\Omega_{\alpha_{\omega+1}+1}}^{\Omega_{\alpha_{\omega+1}+1}} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,1)(5,4,0) | $\psi(\psi_\alpha(\alpha_{\omega+2}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,1)(5,4,1) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega+2}+1} + \psi_{\alpha_2}(\Omega_{\alpha_{\omega+2}+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,2)(2,2,2)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2} + \psi_{\alpha_{\omega+1}}(\alpha_{\omega\cdot 2})))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,2)(4,3,0) | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2} + \alpha_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,2)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2} \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,2)(5,2,0) | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2} \cdot \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,2)(5,2,0)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2} \cdot \alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,2)(5,2,0)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2} \cdot (\alpha_\omega + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,2)(5,2,1) | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2} \cdot \Omega_{\alpha_\omega+1} + \psi_{\alpha_2}(\alpha_{\omega\cdot 2} \cdot \Omega_{\alpha_\omega+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,2)(5,3,0) | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2} \cdot \alpha_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,2)(5,3,0)(4,3,2) | $\psi(\psi_\alpha({\alpha_{\omega\cdot 2}}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,2)(5,3,0)(6,0,0) | $\psi(\psi_\alpha({\alpha_{\omega\cdot 2}}^\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,2)(5,3,0)(6,4,0) | $\psi(\psi_\alpha(\varepsilon_{\alpha_{\omega\cdot 2}+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,1)(4,3,2)(5,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega\cdot 2}+1} + \psi_{\alpha_2}(\Omega_{\alpha_{\omega\cdot 2}+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,2)(5,3,1)(4,3,2) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega\cdot 2}+1} + \alpha_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,2)(5,3,1)(4,3,2)(5,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega\cdot 2}+1} \cdot 2 + \psi_{\alpha_2}(\Omega_{\alpha_{\omega\cdot 2}+1} \cdot 2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,2)(5,3,1)(6,4,0) | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,2)(5,3,1)(6,4,1) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega\cdot 2+1}+1} + \psi_{\alpha_2}(\Omega_{\alpha_{\omega\cdot 2+1}+1} + 1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,2)(5,3,1)(6,4,1)(7,5,0) | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2+2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,1)<br>(4,3,2)(5,3,1)(6,4,2) | $\psi(\psi_\alpha(\alpha_{\omega\cdot 3}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,1,1) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_2}(\alpha_{\omega^2}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(2,1,1)(3,2,2)(4,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_2}(\alpha_{\omega^2}+\psi_{\alpha_2}(\alpha_{\omega^2}))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,1)(3,3,2)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_3}(\alpha_{\omega^2})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,1)(3,3,2)(4,3,2)(3,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_\omega\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,0)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+{\alpha_\omega}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,0)(4,3,1) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,0)(4,3,1)(5,4,2)(6,4,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\alpha_{\omega^2})))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(2,2,2)(3,2,1) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(2,1,1) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(2,1,1)(3,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+$<br>$\psi_{\alpha_2}(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\alpha_\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(2,2,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(2,2,1) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+\Omega_{\alpha_2+1}+$<br>$\psi_{\alpha_2}(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+\Omega_{\alpha_2+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(2,2,1)(3,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+\psi_{\alpha_3}(\alpha_3)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(2,2,1)(3,3,2)<br>(4,3,2)(3,3,2)(4,3,1) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+\psi_{\alpha_3}(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1})+$<br>$\psi_{\alpha_2}(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+\psi_{\alpha_3}(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(2,2,1)(3,3,2)<br>(4,3,2)(3,3,2)(4,3,1)(3,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+\alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(2,2,2)(3,2,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}+\alpha_\omega\cdot\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(2,2,2)(3,2,1) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot 2+$<br>$\psi_{\alpha_2}(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot 2+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(2,2,2)(3,2,1)(2,2,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot 2+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(2,2,2)(3,2,1)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot 2+\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(3,0,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot\alpha))$ |

| BMS | Projection |
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| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(3,1,1) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot\Omega_{\alpha+1}+$<br>$\psi_{\alpha_2}(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot\Omega_{\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot\psi_{\alpha_2}(\alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(3,2,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(3,2,0)(2,2,1)(3,3,2)(4,3,2)<br>(3,3,2)(4,3,1)(4,2,0)(3,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot\alpha_2+\alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(3,2,0)(2,2,1)(3,3,2)(4,3,2)<br>(3,3,2)(4,3,1)(4,2,0)(3,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot\alpha_2+\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(3,2,0)(2,2,1)(3,3,2)(4,3,2)<br>(3,3,2)(4,3,1)(4,2,0)(3,3,2)(4,3,1) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot(\alpha_2+1)+$<br>$\psi_{\alpha_2}(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot(\alpha_2+1)+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(3,2,0)(2,2,1)(3,3,2)(4,3,2)<br>(3,3,2)(4,3,1)(4,2,0)<br>(3,3,2)(4,3,1)(4,2,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot\alpha_2\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(3,2,0)(2,2,1)(3,3,2)(4,3,2)<br>(3,3,2)(4,3,1)(4,2,0)(4,2,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot{\alpha_2}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(3,2,0)(2,2,1)(3,3,2)<br>(4,3,2)(3,3,2)(4,3,1)(4,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot\alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(3,2,0)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(3,2,0)(2,2,2)(3,2,1) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot(\alpha_\omega+1)+$<br>$\psi_{\alpha_2}(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot(\alpha_\omega+1)+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(3,2,0)(2,2,2)<br>(3,2,1)(3,2,0)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot\alpha_\omega\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(3,2,0)(3,2,0)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot{\alpha_\omega}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot\varepsilon_{\alpha_\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(3,2,1) | $\psi(\psi_\alpha(\alpha_{\omega^2}+{\Omega_{\alpha_\omega+1}}^2+$<br>$\psi_{\alpha_2}(\alpha_{\omega^2}+{\Omega_{\alpha_\omega+1}}^2+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,2,0)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+{\Omega_{\alpha_\omega+1}}^{\alpha_\omega}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,2,1) | $\psi(\psi_\alpha(\alpha_{\omega^2}+{\Omega_{\alpha_\omega+1}}^{\Omega_{\alpha_\omega+1}}+$<br>$\psi_{\alpha_2}(\alpha_{\omega^2}+{\Omega_{\alpha_\omega+1}}^{\Omega_{\alpha_\omega+1}}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\alpha_{\omega+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,0)(4,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\alpha_{\omega+1}\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,1) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\Omega_{\alpha_{\omega+1}+1})+$<br>$\psi_{\alpha_2}(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\Omega_{\alpha_{\omega+1}+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\alpha_{\omega\cdot 2})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\alpha_{\omega^2})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)(3,0,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\alpha_{\omega^2}+1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)<br>(3,2,1)(4,3,2)(5,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\alpha_{\omega^2}))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(4,3,2)(5,3,2)(4,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(4,3,2)(5,3,2)(4,3,0)(5,4,1) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_{\omega+1}+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(4,3,2)(5,3,2)(4,3,1) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_{\omega+1}+1}+$<br>$\psi_{\alpha_2}(\alpha_{\omega^2}+\Omega_{\alpha_{\omega+1}+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(4,3,2)(5,3,2)(4,3,1)(4,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_{\omega+1}+1}+\alpha_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(4,3,2)(5,3,2)(4,3,1)(5,0,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_{\omega+1}+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(4,3,2)(5,3,2)(4,3,1)(5,4,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+2}}(\alpha_{\omega+2})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)<br>(4,3,1)(5,4,2)(6,4,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+2}}(\alpha_{\omega^2})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)(4,3,1)<br>(5,4,2)(6,4,2)(5,4,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega+2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(4,3,2)(5,3,2)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(4,3,2)(5,3,2)(4,3,2)(4,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega\cdot 2}+\alpha_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(4,3,2)(5,3,2)(4,3,2)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega\cdot 2}\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(4,3,2)(5,3,2)(4,3,2)(5,2,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega\cdot 2}\cdot\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)<br>(4,3,2)(5,2,0)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega\cdot 2}\cdot\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(4,3,2)(5,3,2)(4,3,2)(5,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega\cdot 2}\cdot\alpha_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(4,3,2)(5,3,2)(4,3,2)(5,3,0)<br>(4,3,1)(5,4,2)(6,4,2)(5,4,2)(6,4,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega\cdot 2}\cdot\alpha_{\omega+2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)<br>(4,3,2)(5,3,0)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+{\alpha_{\omega\cdot 2}}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)<br>(4,3,2)(5,3,0)(6,4,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\varepsilon_{\alpha_{\omega\cdot 2}+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,1)(4,3,2)(5,3,2)(4,3,2)(5,3,1) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_{\omega\cdot 2}+1}+$<br>$\psi_{\alpha_2}(\alpha_{\omega^2}+\Omega_{\alpha_{\omega\cdot 2}+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)<br>(4,3,2)(5,3,1)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_{\omega\cdot 2}+1}+\alpha_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)<br>(4,3,2)(5,3,1)(5,0,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_{\omega\cdot 2}+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)<br>(4,3,2)(5,3,1)(6,4,0) | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega\cdot 2+1}}(\alpha_{\omega\cdot 2+1})))$ |

| BMS | Projection |
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| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)<br>(4,3,2)(5,3,1)(6,4,2)(7,4,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} + \psi_{\alpha_{\omega\cdot 2+1}}(\alpha_{\omega^2})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)<br>(4,3,2)(5,3,1)(6,4,2)(7,4,2)(6,4,0) | $\psi(\psi_\alpha(\alpha_{\omega^2} + \alpha_{\omega\cdot 2+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)<br>(4,3,2)(5,3,1)(6,4,2)(7,4,2)(6,4,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} + \alpha_{\omega\cdot 3}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,2)(2,1,1) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 2 + \psi_{\alpha_2}(\alpha_{\omega^2} \cdot 2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,2)(2,2,0) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 2 + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,2)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 2 + \alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(2,2,2)<br>(3,2,2)(2,2,2)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 2 + \psi_{\alpha_{\omega+1}}(\alpha_{\omega+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,2)(2,2,2)(3,2,1)<br>(4,3,2)(5,3,2)(4,3,2)(5,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 2 + \psi_{\alpha_{\omega+1}}(\alpha_{\omega^2} \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,2)(2,2,2)(3,2,1)<br>(4,3,2)(5,3,2)(4,3,2)(5,3,2)(4,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 2 + \alpha_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,2)(2,2,2)(3,2,1)(4,3,2)<br>(5,3,2)(4,3,2)(5,3,2)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 2 + \alpha_{\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(2,2,2)(3,2,2)(2,2,2)(3,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,0,0) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,1,1) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_{\omega^2} \cdot \Omega_{\alpha+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,0) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(2,2,0) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_2 + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(2,2,1)(3,3,2)(4,3,2)(4,2,0) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_{\omega^2} \cdot \alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,2)(4,2,0)(3,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_2 + \alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,2)(4,2,0)(3,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_2 + \alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(2,2,1)(3,3,2)(4,3,2)<br>(4,2,0)(3,3,2)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot (\alpha_2 + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(2,2,1)(3,3,2)(4,3,2)<br>(4,2,0)(3,3,2)(4,3,2)(4,2,0) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,2)(4,2,1) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \Omega_{\alpha_2+1} + \psi_{\alpha_2}(\alpha_{\omega^2} \cdot \Omega_{\alpha_2+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,0)<br>(2,2,1)(3,3,2)(4,3,2)(4,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(3,2,0)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_\omega))$ |

| BMS | Projection |
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| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(2,2,2)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_\omega + \alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(2,2,2)(3,2,1) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_\omega + \Omega_{\alpha_\omega+1} +$<br>$\psi_{\alpha_2}(\alpha_{\omega^2} \cdot \alpha_\omega + \Omega_{\alpha_\omega+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(2,2,2)(3,2,1)(4,3,2)(5,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_\omega + \psi_{\alpha_{\omega+1}}(\alpha_{\omega^2})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(2,2,2)(3,2,1)(4,3,2)<br>(5,3,2)(5,2,0)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_\omega + \psi_{\alpha_{\omega+1}}(\alpha_{\omega^2} \cdot \alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(2,2,2)(3,2,1)(4,3,2)<br>(5,3,2)(5,2,0)(4,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_\omega + \alpha_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(2,2,2)(3,2,1)(4,3,2)<br>(5,3,2)(5,2,0)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_\omega + \alpha_{\omega \cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(2,2,2)(3,2,1)(4,3,2)<br>(5,3,2)(5,2,0)(4,3,2)(5,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot (\alpha_\omega + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(2,2,2)(3,2,1)(4,3,2)(5,3,2)<br>(5,2,0)(4,3,2)(5,3,2)(5,2,0)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_\omega \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(2,2,2)(3,2,1)(4,3,2)<br>(5,3,2)(5,2,0)(6,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \varepsilon_{\alpha_\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,0)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)(5,2,1) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \Omega_{\alpha_\omega+1} + \psi_{\alpha_2}(\alpha_{\omega^2} \cdot \Omega_{\alpha_\omega+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(2,2,2)(3,2,1)(4,3,2)<br>(5,3,2)(5,2,1)(6,3,2)(7,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \psi_{\alpha_{\omega+1}}(\alpha_{\omega^2})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,0)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)(5,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(2,2,2)(3,2,1)(4,3,2)<br>(5,3,2)(5,3,0)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_{\omega \cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(2,2,2)(3,2,2) | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,0)<br>(2,2,2)(3,2,2)(2,2,2)(3,2,2) | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2 + \alpha_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,0)<br>(2,2,2)(3,2,2)(3,2,0) | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2 + \alpha_{\omega^2} \cdot \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,0)<br>(2,2,2)(3,2,2)(3,2,0)(2,2,2) | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2 + \alpha_{\omega^2} \cdot \alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,0)<br>(2,2,2)(3,2,2)(3,2,0)(2,2,2)(3,2,1)<br>(4,3,2)(5,3,2)(5,3,0)<br>(4,3,2)(5,3,2)(5,3,0) | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2 + \alpha_{\omega^2} \cdot \alpha_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,0)<br>(2,2,2)(3,2,2)(3,2,0)(2,2,2)(3,2,1)<br>(4,3,2)(5,3,2)(5,3,0)(4,3,2)<br>(5,3,2)(5,3,0)(4,3,2) | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2 + \alpha_{\omega^2} \cdot \alpha_{\omega \cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,0)<br>(2,2,2)(3,2,2)(3,2,0)(2,2,2)(3,2,2) | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2 \cdot \alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(3,2,0)(3,2,0) | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2 \cdot \alpha_2))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(3,2,0)(2,2,2)(3,2,2) | $\psi(\psi_\alpha({\alpha_{\omega^2}}^3))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(3,2,0)(4,0,0) | $\psi(\psi_\alpha({\alpha_{\omega^2}}^\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(4,2,0) | $\psi(\psi_\alpha({\alpha_{\omega^2}}^{\alpha_2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(4,2,0)(2,2,2) | $\psi(\psi_\alpha({\alpha_{\omega^2}}^{\alpha_\omega}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,0)<br>(4,2,0)(2,2,2)(3,2,1)(4,3,2)<br>(5,3,2)(5,3,0)(6,3,0)(4,3,2) | $\psi(\psi_\alpha({\alpha_{\omega^2}}^{\alpha_{\omega\cdot 2}}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(4,2,0)(2,2,2)(3,2,2) | $\psi(\psi_\alpha({\alpha_{\omega^2}}^{\alpha_{\omega^2}}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}+\psi_{\alpha_2}(\Omega_{\alpha_{\omega^2}+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(2,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(2,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}+\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(2,2,2)(3,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}+\alpha_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,1)<br>(2,2,2)(3,2,2)(3,2,0)(2,2,2)(3,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}+{\alpha_{\omega^2}}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,1)<br>(2,2,2)(3,2,2)(3,2,0)(4,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}\cdot 2+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(2,2,2)(3,2,2)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}\cdot 2+\psi_{\alpha_2}(\Omega_{\alpha_{\omega^2}+1}\cdot 2+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}\cdot\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(3,2,0)(2,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}\cdot\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(3,2,0)(2,2,2)(3,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}\cdot\alpha_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,1)<br>(3,2,0)(2,2,2)(3,2,2)(3,2,1)(3,2,0) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}\cdot(\alpha_{\omega^2}+\alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(3,2,0)(4,3,0) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega^2}+1}}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(3,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega^2}+1}}^2+\psi_{\alpha_2}({\Omega_{\alpha_{\omega^2}+1}}^2+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(3,2,1)(2,2,2)(3,2,2) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega^2}+1}}^2+\alpha_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(3,2,1)(3,2,0) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega^2}+1}}^2\cdot\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(3,2,1)(3,2,1) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega^2}+1}}^3+\psi_{\alpha_2}({\Omega_{\alpha_{\omega^2}+1}}^3+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(4,2,0) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega^2}+1}}^{\alpha_2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(4,2,0)(2,2,2) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega^2}+1}}^{\alpha_\omega}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(4,2,0)(2,2,2)(3,2,1)(4,3,2)<br>(5,3,2)(5,3,1)(6,3,0)(4,3,2) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega^2}+1}}^{\alpha_{\omega\cdot 2}}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(4,2,0)(2,2,2)(3,2,2) | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega^2}+1}}^{\alpha_{\omega^2}}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,1)<br>(4,2,0)(3,2,1)(4,2,0)(2,2,2)(3,2,2) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}}{}^{(}\alpha_{\omega^2}\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(4,2,0)(5,3,0) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}}{}^{\varepsilon_{\alpha_{\omega^2+1}}}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(4,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}}{}^{\Omega_{\alpha_{\omega^2+1}}}+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_{\omega^2+1}}{}^{\Omega_{\alpha_{\omega^2+1}}}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(4,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}+1}+\psi_{\alpha_2}(\Omega_{\alpha_{\omega^2+1}+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(4,3,2)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2+\omega}\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(4,3,2)(5,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^2+\omega}\cdot\alpha_{\omega^2+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(4,3,2)(5,3,0)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2+\omega}{}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(4,3,2)(5,3,0)(6,4,0) | $\psi(\psi_\alpha(\varepsilon_{\alpha_{\omega^2+\omega}+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(4,3,2)(5,3,1) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+\omega}+1}+\psi_{\alpha_2}(\Omega_{\alpha_{\omega^2+\omega}+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(4,3,2)(5,3,1)(6,4,2) | $\psi(\psi_\alpha(\alpha_{\omega^2+\omega\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,1)(4,3,2)(5,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^2\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^3}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,2)(2,2,2)(3,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^3}+\alpha_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,2)<br>(2,2,2)(3,2,2)(3,2,1)(4,3,2)<br>(5,3,2)(5,3,2)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^3}+\alpha_{\omega^2+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,2)<br>(2,2,2)(3,2,2)(3,2,1)(4,3,2)<br>(5,3,2)(5,3,2)(4,3,2)(5,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^3}+\alpha_{\omega^2\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,2)<br>(2,2,2)(3,2,2)(3,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^3}\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,2)<br>(3,2,0)(2,2,2)(3,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^3}\cdot\alpha_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,2)<br>(3,2,0)(2,2,2)(3,2,2)(3,2,1)<br>(4,3,2)(5,3,2)(5,3,2)(5,3,0) | $\psi(\psi_\alpha(\alpha_{\omega^3}\cdot\alpha_{\omega^2+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,2)<br>(3,2,0)(2,2,2)(3,2,2)(3,2,1)(4,3,2)<br>(5,3,2)(5,3,2)(5,3,0)(4,3,2)(5,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^3}\cdot\alpha_{\omega^2\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(3,2,2)<br>(3,2,0)(2,2,2)(3,2,2)(3,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^3}{}^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,2)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\varepsilon_{\alpha_{\omega^3}+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,2)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^3}+1}+\psi_{\alpha_2}(\Omega_{\alpha_{\omega^3}+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,2)(3,2,1)(4,3,2)(5,3,2)(5,3,2) | $\psi(\psi_\alpha(\alpha_{\omega^3\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(3,2,2)(3,2,2) | $\psi(\psi_\alpha(\alpha_{\omega^4}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,1,0)<br>(2,1,1)(3,2,2)(4,2,2)(5,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\alpha + \psi_{\alpha_2}(\alpha_\alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,1,0)(2,2,0) | $\psi(\psi_\alpha(\alpha_\alpha + \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,1,0)(2,2,2) | $\psi(\psi_\alpha(\alpha_\alpha + \alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,1,0)(2,2,2)(3,2,2)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_\alpha \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,1,0)(3,2,1) | $\psi(\psi_\alpha(\Omega_{\alpha_\alpha+1} + \psi_{\alpha_2}(\Omega_{\alpha_\alpha+1} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,1,0)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\alpha+\omega}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,1,0)<br>(3,2,1)(4,3,2)(5,3,2)(6,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_{\alpha\cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,1,0)(3,2,2) | $\psi(\psi_\alpha(\alpha_{\alpha\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,1,0)(3,2,2)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\alpha_{\alpha^2}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,1,0)(5,2,0) | $\psi(\psi_\alpha(\alpha_{\varepsilon_{\alpha+1}}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,1,1) | $\psi(\psi_\alpha(\alpha_{\Omega_{\alpha+1}} + \psi_{\alpha_2}(\alpha_{\Omega_{\alpha+1}} + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,1,1)(5,2,2) | $\psi(\psi_\alpha(\alpha_{\psi_{\alpha_2}(\alpha_\omega)}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0) | $\psi(\psi_\alpha(\alpha_{\alpha_2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(2,2,1)(3,3,2)(4,3,2)(5,2,0)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\alpha_2\cdot\omega}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(2,2,1)(3,3,2)(4,3,2)(5,3,0) | $\psi(\psi_\alpha(\alpha_{\alpha_3}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,0)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\alpha_\omega}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(2,2,2)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\alpha_\omega} + \alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(2,2,2)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\alpha_\omega} + \psi_{\alpha_{\omega+1}}(\alpha_{\omega\cdot 2})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)(6,2,0) | $\psi(\psi_\alpha(\alpha_{\alpha_\omega} + \psi_{\alpha_{\omega+1}}(\alpha_{\alpha_2})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(2,2,2)(3,2,1)(4,3,2)<br>(5,3,2)(6,2,0)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\alpha_\omega} + \psi_{\alpha_{\omega+1}}(\alpha_{\alpha_\omega})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(2,2,2)(3,2,1)(4,3,2)<br>(5,3,2)(6,2,0)(4,3,0) | $\psi(\psi_\alpha(\alpha_{\alpha_\omega} + \alpha_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(2,2,2)(3,2,1)(4,3,2)(5,3,2)<br>(6,2,0)(4,3,2)(5,3,2)(6,2,0) | $\psi(\psi_\alpha(\alpha_{\alpha_\omega} + \alpha_{\alpha_2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(2,2,2)(3,2,1)(4,3,2)(5,3,2)<br>(6,2,0)(4,3,2)(5,3,2)(6,2,0)(2,2,2) | $\psi(\psi_\alpha(\alpha_{\alpha_\omega} \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(2,2,2)(3,2,1)(4,3,2)(5,3,2)(6,3,0) | $\psi(\psi_\alpha(\alpha_{\alpha_{\omega+1}}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(2,2,2)(3,2,1)<br>(4,3,2)(5,3,2)(6,3,0)(4,3,2) | $\psi(\psi_\alpha(\alpha_{\alpha_{\omega\cdot 2}}))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(2,2,2)(3,2,2) | $\psi(\psi_\alpha(\alpha_{\alpha_{\omega^2}}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(2,2,2)(3,2,2)(4,2,0) | $\psi(\psi_\alpha(\alpha_{\alpha_{\alpha_2}}))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\alpha\mathrm{fp})) = \psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,0,0)(2,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)+\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,0,0)(2,2,2)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)+\alpha_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(3,0,0)(2,2,2)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,0,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,1,0)(2,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)\cdot\alpha))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,1,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)\cdot\Omega_{\alpha+1}+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\beta)\cdot\Omega_{\alpha+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,1,1)(4,2,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)\cdot\psi_{\alpha_2}(\alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)\cdot\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,0)(2,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)\cdot\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(3,2,0)(2,2,2)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,0)(3,2,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)^2\cdot\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,0)(3,2,0)(2,2,2)<br>(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)^3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,0)(4,2,0)(2,2,2)<br>(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)^{\psi_\beta(\alpha_{\beta+1}\cdot\beta)}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta+\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta+\Omega_{\beta+1})+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\beta+\Omega_{\beta+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(3,2,1)(2,2,2)(3,2,2)(4,2,0)(3,2,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta+\Omega_{\beta+1})\cdot 2+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\beta+\Omega_{\beta+1})\cdot 2+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,1)(3,2,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta+\Omega_{\beta+1})^2+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\beta+\Omega_{\beta+1})^2+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta+\Omega_{\beta+1}+\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,1)(4,3,2)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega))\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,1)(4,3,2)(5,3,0)(6,4,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega)+\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,1)(4,3,2)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega^2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(3,2,1)(4,3,2)(5,3,2)(6,1,0)(2,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\alpha))))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(3,2,1)(4,3,2)(5,3,2)(6,2,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(3,2,1)(4,3,2)(5,3,2)(6,2,0)(2,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\alpha_\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(3,2,1)(4,3,2)(5,3,2)(6,2,0)<br>(2,2,2)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\psi_\beta(\alpha_{\beta+1}\cdot\beta)))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(3,2,1)(4,3,2)(5,3,2)(6,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\psi_\beta(\alpha_{\beta+1}\cdot(\beta+1))))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(3,2,1)(4,3,2)(5,3,2)(6,3,0)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega))))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(3,2,1)(4,3,2)(5,3,2)(6,3,0)(5,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(3,2,2)(2,2,2)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(3,2,2)(2,2,2)(3,2,2)(4,2,0)(3,2,1)<br>(4,3,2)(5,3,2)(6,3,0)(5,3,2)(4,3,2)<br>(5,3,2)(6,3,0)(5,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(3,2,2)(2,2,2)(3,2,2)(4,2,0)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,2)(3,2,0)(2,2,2)<br>(3,2,2)(4,2,0)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega)^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,2)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega+\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,2)(3,2,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega+\Omega_{\beta+1})+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega+\Omega_{\beta+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,2)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta\cdot\omega+\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,2)(3,2,1)(4,3,2)<br>(5,3,2)(6,3,0)(5,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot(\omega+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,2)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,2)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,2)(4,1,0)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\alpha\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,2)(4,2,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,2)(4,2,0)(2,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\alpha_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,2)(4,2,0)(2,2,2)<br>(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\psi_\beta(\alpha_{\beta+1}\cdot\beta))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,2)(4,2,0)(2,2,2)<br>(3,2,2)(4,2,0)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(3,2,2)(4,2,0)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta^2+\beta)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(3,2,2)(4,2,0)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta^2+\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(3,2,2)(4,2,0)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta^2\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(3,2,2)(4,2,0)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta^3)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta^\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,0)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\varepsilon_{\beta+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(2,2,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(2,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,1)<br>(2,2,2)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+\psi_\beta(\alpha_{\beta+1}\cdot\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(2,2,2)(3,2,2)(4,2,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})\cdot 2+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})\cdot 2+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1}+\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(3,2,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1}+\Omega_{\beta+1})+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1}+\Omega_{\beta+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\Omega_{\beta+1}+\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(3,2,1)(4,3,2)(5,3,2)(6,3,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1}\cdot 2)+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1}\cdot 2)+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1}\cdot\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(3,2,2)(4,2,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot{\Omega_{\beta+1}}^2)+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot{\Omega_{\beta+1}}^2)+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot{\Omega_{\beta+1}}^\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\varepsilon_{\Omega_{\beta+1}+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(5,3,1)(6,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+\omega})))$<br>$=\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2+\Omega_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(5,3,2)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(5,3,2)(6,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(5,3,2)(6,3,2)(7,2,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(5,3,2)(6,3,2)(7,2,0)(2,2,2)<br>(3,2,2)(4,2,1)(5,3,2)(6,3,2)(7,2,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2+$<br>$\alpha_{\beta+1}\cdot\psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\alpha_2)))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,1)<br>(5,3,2)(6,3,2)(7,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\psi_\beta({\alpha_{\beta+1}}^2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,1)(5,3,2)(6,3,2)(7,3,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}))))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,1)<br>(5,3,2)(6,3,2)(7,3,0)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2+$<br>$\alpha_{\beta+1}\cdot\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,1)<br>(5,3,2)(6,3,2)(7,3,0)(6,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,1)<br>(5,3,2)(6,3,2)(7,3,0)(6,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\beta\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,1)<br>(5,3,2)(6,3,2)(7,3,0)(8,4,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\varepsilon_{\beta+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,1)<br>(5,3,2)(6,3,2)(7,3,1)(8,4,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot 2+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(2,2,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(2,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)+\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(2,2,2)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)+\alpha_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(2,2,2)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(2,2,2)(3,2,2)(4,2,0)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\varepsilon_{\beta+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(2,2,2)(3,2,2)(4,2,1) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+$<br>$\psi_{\alpha_2}(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(2,2,2)(3,2,2)(4,2,1)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)+$<br>$\psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(2,2,2)(3,2,2)(4,2,1)<br>(5,3,2)(6,3,2)(7,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)+$<br>$\psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(2,2,2)(3,2,2)(4,2,1)<br>(5,3,2)(6,3,2)(7,3,2)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)+$<br>$\psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega))\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(2,2,2)(3,2,2)(4,2,1)(5,3,2)(6,3,2)<br>(7,3,2)(3,2,1)(4,3,2)(5,3,2)(6,2,0)<br>(2,2,2)(3,2,2)(4,2,1)<br>(5,3,2)(6,3,2)(7,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)+$<br>$\psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)$<br>$+\psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(2,2,2)(3,2,2)(4,2,1)(5,3,2)(6,3,2)<br>(7,3,2)(3,2,1)(4,3,2)<br>(5,3,2)(6,3,0)(5,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)+\psi_\beta({\alpha_{\beta+1}}^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(2,2,2)(3,2,2)(4,2,1)(5,3,2)<br>(6,3,2)(7,3,2)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)+\psi_\beta({\alpha_{\beta+1}}^2)\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(2,2,2)(3,2,2)(4,2,1)(5,3,2)<br>(6,3,2)(7,3,2)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)+\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(2,2,2)(3,2,2)(4,2,1)(5,3,2)<br>(6,3,2)(7,3,2)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)+$<br>$\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(2,2,2)(3,2,2)(4,2,1)(5,3,2)(6,3,2)<br>(7,3,2)(5,3,2)(6,3,2)(7,3,1)<br>(8,4,2)(9,4,2)(10,4,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)+$<br>$\psi_{\psi_\beta({\alpha_{\beta+1}}^2\cdot 2)}(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(2,2,2)(3,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)\cdot 2))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,2)(3,2,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,0)(2,2,2)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot \alpha_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(3,2,0)(2,2,2)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot \psi_\beta(\alpha_{\beta+1} \cdot \beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(3,2,0)(2,2,2)(3,2,2)(4,2,0)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot \psi_\beta(\alpha_{\beta+1} \cdot \beta \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(3,2,0)(2,2,2)(3,2,2)(4,2,1)<br>(5,3,2)(6,3,2)(7,3,2)(6,2,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot$<br>$\psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(3,2,0)(2,2,2)(3,2,2)(4,2,1)(5,3,2)<br>(6,3,2)(7,3,2)(6,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot \psi_\beta({\alpha_{\beta+1}}^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(3,2,0)(2,2,2)(3,2,2)(4,2,1)<br>(5,3,2)(6,3,2)(7,3,2)(6,3,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot \psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(3,2,0)(2,2,2)(3,2,2)(4,2,1)(5,3,2)<br>(6,3,2)(7,3,2)(6,3,0)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot \psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,0)(2,2,2)(3,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega)^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,0)(3,2,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega)^2 \cdot \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(3,2,0)(3,2,0)(2,2,2)(3,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega)^3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega + \beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,1) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega + \Omega_{\beta+1})+$<br>$\psi_{\alpha_2}(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega + \Omega_{\beta+1}) + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega + \varepsilon_{\Omega_{\beta+1}+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega + \alpha_{\beta+1} \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(3,2,1)(4,3,2)(5,3,2)(6,3,0)(5,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega + \alpha_{\beta+1} \cdot \beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(3,2,1)(4,3,2)(5,3,2)(6,3,1)(7,4,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot (\omega + 1) + \alpha_{\beta+1} \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(3,2,1)(4,3,2)(5,3,2)(6,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,2)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,2)(4,2,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,2)(4,2,0)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \beta + \beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(3,2,2)(4,2,0)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \beta + \alpha_{\beta+1} \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,2)(4,2,0)(3,2,1)<br>(4,3,2)(5,3,2)(6,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot (\beta + \omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,2)(4,2,0)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \beta \cdot \omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(3,2,2)(4,2,0)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \beta^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,2)(4,2,0)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \varepsilon_{\beta+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,2)(4,2,1) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \Omega_{\beta+1})+$<br>$\psi_{\alpha_2}(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \Omega_{\beta+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,2)(4,2,1)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \Omega_{\beta+1}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(3,2,2)(4,2,1)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \Omega_{\beta+1}+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,2)(4,2,1)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \Omega_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(3,2,2)(4,2,1)(3,2,2)(4,2,1) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot {\Omega_{\beta+1}}^2)+$<br>$\psi_{\alpha_2}(\psi_\beta({\alpha_{\beta+1}}^2 \cdot {\Omega_{\beta+1}}^2)+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,2)(4,2,1)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \varepsilon_{\Omega_{\beta+1}+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,2)(4,2,1)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^3+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(3,2,2)(4,2,1)(5,3,2)(6,3,2)(7,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^3+{\alpha_{\beta+1}}^2\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(3,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^3 \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,2)(4,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,2)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\alpha)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,1,0)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\alpha+\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,1,0)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\alpha\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,1,0)(3,2,2)(4,2,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\alpha\cdot\alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,1,0)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\alpha\cdot\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,1,0)(3,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^(\alpha+1)\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,1,0)(3,2,2)(4,2,2)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^(\alpha\cdot 2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,1,0)(5,2,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\varepsilon_{\alpha+1}})))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,2)(4,2,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\alpha_2})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,0)(2,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\alpha_\omega})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,0)(2,2,2)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\psi_\beta(\alpha_{\beta+1}\cdot\beta)})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,0)(2,2,2)(3,2,2)(4,2,2)(4,2,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\psi_\beta({\alpha_{\beta+1}}^{\alpha_2})})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,0)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\beta+\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,0)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\beta\cdot\omega)))$ |

| BMS | Projection |
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| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,0)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\beta}\cdot\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,0)(3,2,2)(4,2,0)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\beta}\cdot\varepsilon_{\beta+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,0)(3,2,2)(4,2,1)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\beta+1)+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,0)(3,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\beta+1)\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,0)(3,2,2)(4,2,2)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\beta+\alpha))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,0)(3,2,2)(4,2,2)(4,2,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\beta+\alpha_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,0)(3,2,2)(4,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\beta\cdot 2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,0)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\beta^2})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,0)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\varepsilon_{\beta+1}})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\Omega_{\beta+1}})+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}{}^{\Omega_{\beta+1})}+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,1)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\Omega_{\beta+1}}+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,1)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\Omega_{\beta+1}}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,1)(3,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\Omega_{\beta+1}+1)\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,1)(3,2,2)(4,2,2)(4,2,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\Omega_{\beta+1}\cdot 2))+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}{}^{(}\Omega_{\beta+1}\cdot 2))+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,1)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\Omega_{\beta+1}\cdot\beta))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,1)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\varepsilon_{\Omega_{\beta+1}+1}})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,1)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,1)(5,3,2)(6,3,2)(7,3,2)(7,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}+\alpha_{\beta+1}{}^{\omega})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,1)(5,3,2)(6,3,2)(7,3,2)(7,2,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}+\alpha_{\beta+1}{}^{\alpha_2})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,1)(5,3,2)(6,3,2)<br>(7,3,2)(7,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}+\alpha_{\beta+1}{}^{\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}})})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,1)(5,3,2)(6,3,2)(7,3,2)(7,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}+$<br>$\alpha_{\beta+1}{}^{\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}+\alpha_{\beta+1})})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,1)(5,3,2)(6,3,2)<br>(7,3,2)(7,3,0)(6,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}+\alpha_{\beta+1}{}^{\beta})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,1)(5,3,2)(6,3,2)<br>(7,3,2)(7,3,1)(8,4,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}\cdot 2+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,2)(2,2,2)(3,2,2)(4,2,2)(4,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}\cdot\omega)+\psi_\beta(\alpha_{\beta+1}{}^{\omega})))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,2)(2,2,2)(3,2,2)(4,2,2)(4,2,1)<br>(5,3,2)(6,3,2)(7,3,2)(7,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\alpha_{\beta+1}}\cdot\omega)+$<br>$\psi_{\psi_\beta({\alpha_{\beta+1}}^{\alpha_{\beta+1}})}(\psi_\beta({\alpha_{\beta+1}}^{\alpha_{\beta+1}}\cdot\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,2)(2,2,2)(3,2,2)(4,2,2)(4,2,1)<br>(5,3,2)(6,3,2)(7,3,2)(7,3,2)<br>(3,2,2)(4,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\alpha_{\beta+1}}\cdot\omega)+\psi_\beta({\alpha_{\beta+1}}^{\alpha_{\beta+1}})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,2)(2,2,2)(3,2,2)(4,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\alpha_{\beta+1}}\cdot\omega)\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,2)(3,2,0)(2,2,2)<br>(3,2,2)(4,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\alpha_{\beta+1}}\cdot\omega)^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,2)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\alpha_{\beta+1}}\cdot\omega+\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,2)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\alpha_{\beta+1}}\cdot\omega+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,2)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\alpha_{\beta+1}}\cdot\omega^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,2)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\alpha_{\beta+1}}\cdot\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,2)(3,2,2)(4,2,1)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}+1)+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,2)(3,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}+1)\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,2)(3,2,2)(4,2,2)(4,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}+\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,2)(3,2,2)(4,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}+\beta))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,2)(3,2,2)(4,2,2)(4,2,0)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}+\varepsilon_{\beta+1}))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,2)(3,2,2)(4,2,2)(4,2,1) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}+\Omega_{\beta+1})+$<br>$\psi_{\alpha_2}(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}+\Omega_{\beta+1}))+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,2)(3,2,2)(4,2,2)(4,2,1)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}\cdot 2)+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,2)(3,2,2)(4,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}\cdot 2)\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,2)(4,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}\cdot\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,2)(4,1,0)(2,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}\cdot\alpha))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}\cdot\beta))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,2)(4,2,0)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}\cdot\beta)\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,2)(4,2,0)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}\cdot\beta)\cdot\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(4,2,2)(4,2,0)(3,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}\cdot\beta+1)\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,2)(4,2,0)(3,2,2)<br>(4,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}\cdot\beta+\beta))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,2)(4,2,0)(3,2,2)<br>(4,2,2)(4,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{(}\alpha_{\beta+1}\cdot\beta\cdot 2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(4,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{{\alpha_{\beta+1}}^2}\cdot\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,2)(5,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{{\alpha_{\beta+1}}^{\omega}})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(5,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{{\alpha_{\beta+1}}^{\beta}})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)<br>(5,2,0)(3,2,2)(4,2,2)(5,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{({\alpha_{\beta+1}}^{\beta}\cdot 2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(5,2,0)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{({\alpha_{\beta+1}}^{\beta}\cdot \beta))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(5,2,0)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{{\alpha_{\beta+1}}^{\beta+1}}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(5,2,0)(4,2,2)(5,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{{\alpha_{\beta+1}}^{\beta\cdot 2}})))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,2)(5,2,1) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{{\alpha_{\beta+1}}^{\Omega_{\beta+1}}}+$<br>$\psi_{\alpha_2}(\psi_\beta({\alpha_{\beta+1}}^{{\alpha_{\beta+1}}^{\Omega_{\beta+1}}})+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,2)(5,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{{\alpha_{\beta+1}}^{\alpha_{\beta+1}}}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,2,2)<br>(4,2,2)(5,2,2)(6,2,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{{\alpha_{\beta+1}}^{{\alpha_{\beta+1}}^{\alpha_{\beta+1}}}}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\varepsilon_{\alpha_{\beta+1}+1}))) = \psi(\psi_\alpha(\psi_\beta(\beta_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(1,1,1)(2,2,2)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2))\cdot 2)$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(2,2,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(2,2,1)(3,3,2)(4,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_{\alpha_3}(\psi_\beta(\beta_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(2,2,1)(3,3,2)(4,4,0)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\alpha_3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(2,2,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(2,2,2)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\alpha_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(2,2,2)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta(\alpha_{\beta+1}\cdot\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(2,2,2)<br>(3,2,2)(4,2,1)(5,3,2)(6,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta(\beta_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(2,2,2)<br>(3,2,2)(4,2,1)(5,3,2)(6,4,0)(3,2,1)<br>(4,3,2)(5,3,2)(6,3,0)(5,0,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta({\alpha_{\beta+1}}^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(2,2,2)<br>(3,2,2)(4,2,1)(5,3,2)(6,4,0)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_{\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1})}(\psi_\beta(\beta_2)+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(2,2,2)<br>(3,2,2)(4,2,1)(5,3,2)(6,4,0)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(2,2,2)(3,2,2)(4,2,1)(5,3,2)<br>(6,4,0)(5,3,1)(6,4,2)(7,5,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_{\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot 2)}(\psi_\beta(\beta_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(2,2,2)(3,2,2)(4,2,1)(5,3,2)(6,4,0)<br>(5,3,1)(6,4,2)(7,5,0)(6,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(2,2,2)(3,2,2)(4,2,1)<br>(5,3,2)(6,4,0)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(2,2,2)(3,2,2)(4,2,1)(5,3,2)<br>(6,4,0)(5,3,2)(6,3,0)(7,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega+\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(2,2,2)(3,2,2)(4,2,1)(5,3,2)<br>(6,4,0)(5,3,2)(6,3,1) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta({\alpha_{\beta+1}}^2+$<br>$\alpha_{\beta+1}\cdot\omega+\Omega_{\beta+1})+\psi_{\alpha_2}(\psi_\beta(\beta_2)+$<br>$\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega+\Omega_{\beta+1})+1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(2,2,2)(3,2,2)(4,2,1)(5,3,2)<br>(6,4,0)(5,3,2)(6,3,1)(7,4,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2) + \psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \omega \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(2,2,2)(3,2,2)(4,2,1)(5,3,2)(6,4,0)<br>(5,3,2)(6,3,2)(7,3,0)(6,0,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2) + \psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(2,2,2)(3,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2) + \psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(2,2,2)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2) \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(3,2,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2) \cdot \alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,0)(2,2,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2) \cdot \alpha_\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(3,2,0)<br>(2,2,2)(3,2,1)(4,3,2)(5,4,0)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2) \cdot \alpha_{\omega+1}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,0)(2,2,2)(3,2,1)(4,3,2)<br>(5,4,0)(5,3,0)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2) \cdot \alpha_{\omega \cdot 2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,0)(2,2,2)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2) \cdot \alpha_{\omega^2}))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,0)(2,2,2)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2) \cdot \psi_\beta(\alpha_{\beta+1} \cdot \beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,0)(2,2,2)(3,2,2)(4,2,1)<br>(5,3,2)(6,4,0)(6,2,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2) \cdot \psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta(\beta_2) \cdot \alpha_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,0)(2,2,2)(3,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2) \cdot \psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,0)(2,2,2)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,0)(2,2,2)(3,3,0)(2,2,2)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)^2 + \psi_\beta(\beta_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(3,2,0)<br>(2,2,2)(3,3,0)(3,2,0)(2,2,2)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)^2 \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,0)(3,2,0)(2,2,2)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)^3))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,0)(4,2,0)(2,2,2)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2)^{\psi}_{\beta}(\beta_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,0)(4,3,0)(3,2,0)(2,2,2)<br>(3,3,0)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \beta)^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,0)(4,3,0)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \beta)^{\psi}_{\beta}(\beta_2 + \beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,0)(4,3,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \beta \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,0)(4,3,1) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \Omega_{\beta+1}) + \Omega_{\alpha+1} \cdot \omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(3,2,1) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \Omega_{\beta+1}) +$<br>$\psi_{\alpha_2}(\psi_\beta(\beta_2 + \Omega_{\beta+1}) + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \alpha_{\beta+1} \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,1)(4,3,2)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \alpha_{\beta+1} \cdot \omega) \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,1)(4,3,2)(5,3,0)(6,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \alpha_{\beta+1} \cdot \omega + \beta)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,1)(4,3,2)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \alpha_{\beta+1} \cdot \omega^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,1)(4,3,2)(5,3,2)(6,3,0)(5,0,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \alpha_{\beta+1} \cdot \beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,1)(4,3,2)(5,3,2)(6,3,0)(7,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \alpha_{\beta+1} \cdot \varepsilon_{\beta+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,1)(4,3,2)(5,3,2)(6,3,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + {\alpha_{\beta+1}}^2 \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,1)(4,3,2)(5,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \psi_{\beta_2}(\beta_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,1)(4,3,2)(5,4,0)(4,3,2)(5,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \psi_{\beta_2}(\beta_2)) \cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,1)(4,3,2)(5,4,0)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \psi_{\beta_2}(\beta_2)) \cdot \psi_\beta(\beta_2 + \alpha_{\beta+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(3,2,1)<br>(4,3,2)(5,4,0)(5,3,0)(4,3,2)(5,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \psi_{\beta_2}(\beta_2))^2))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,1)(4,3,2)(5,4,0)(5,3,0)(6,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \psi_{\beta_2}(\beta_2) + \beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,1)(4,3,2)(5,4,0)(5,3,1)(6,4,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \psi_{\beta_2}(\beta_2) + \alpha_{\beta+1} \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \psi_{\beta_2}(\beta_2 + 1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,2)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \psi_{\beta_2}(\beta_2 + 2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \psi_{\beta_2}(\beta_2 + \beta))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,2)(4,2,0)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \psi_{\beta_2}(\beta_2 + \varepsilon_{\beta+1}))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \psi_{\beta_2}(\beta_2 + \alpha_{\beta+1} + 1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,2)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \psi_{\beta_2}(\beta_2 + \psi_{\beta_2}(\beta_2)))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(3,2,2)(4,3,0)(4,2,2)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 + \psi_{\beta_2}(\beta_2 + \psi_{\beta_2}(\beta_2 + \psi_{\beta_2}(\beta_2))))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(4,0,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(4,2,1) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \Omega_{\beta+1}) + \psi_{\alpha_2}(\psi_\beta(\beta_2 \cdot \Omega_{\beta+1}) + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(4,2,1)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \alpha_{\beta+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(4,2,1)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \alpha_{\beta+1} + \alpha_{\beta+1} \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,2,1)(5,3,2)(6,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \alpha_{\beta+1} + \psi_{\beta_2}(\beta_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,2,1)(5,3,2)(6,4,0)(6,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \alpha_{\beta+1} + \psi_{\beta_2}(\beta_2 \cdot 2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(4,2,1)<br>(5,3,2)(6,4,0)(7,3,1)(8,4,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \alpha_{\beta+1} + \psi_{\beta_2}(\beta_2 \cdot \alpha_{\beta+1}) + \alpha_{\beta+1} \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \alpha_{\beta+1} + \psi_{\beta_2}(\beta_2 \cdot \alpha_{\beta+1} + 1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,2,2)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \alpha_{\beta+1} + \psi_{\beta_2}(\beta_2 \cdot \alpha_{\beta+1} + 1) + \beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(4,2,2)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \alpha_{\beta+1} + \psi_{\beta_2}(\beta_2 \cdot \alpha_{\beta+1} + 2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,2,2)(3,2,2)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \alpha_{\beta+1} + \psi_{\beta_2}(\beta_2 \cdot \alpha_{\beta+1} + \psi_{\beta_2}(\beta_2)))))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(4,2,2)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot (\alpha_{\beta+1} + 1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,2,2)(3,3,0)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot (\alpha_{\beta+1} + \beta))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,2,2)(3,3,0)(4,2,0)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot (\alpha_{\beta+1} + \varepsilon_{\beta+1}))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,2,2)(3,3,0)(4,2,1) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot (\alpha_{\beta+1} + \Omega_{\beta+1}))+$<br>$\psi_{\alpha_2}(\psi_\beta(\beta_2 \cdot (\alpha_{\beta+1} + \Omega_{\beta+1})) + 1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,2,2)(3,3,0)(4,2,1)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \alpha_{\beta+1} \cdot 2 + \alpha_{\beta+1} \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(4,2,2)<br>(3,3,0)(4,2,1)(5,3,2)(6,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \alpha_{\beta+1} \cdot 2 + \psi_{\beta_2}(\beta_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,2,2)(3,3,0)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \alpha_{\beta+1} \cdot 2+$<br>$\psi_{\beta_2}(\beta_2 \cdot \alpha_{\beta+1} \cdot 2 + 1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,2,2)(3,3,0)(4,2,2)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \alpha_{\beta+1} \cdot 2 + \beta_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \alpha_{\beta+1} \cdot \beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(4,2,2)<br>(4,2,0)(3,3,0)(4,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \alpha_{\beta+1} \cdot \beta \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,2,2)(4,2,0)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \alpha_{\beta+1} \cdot \varepsilon_{\beta+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,2,2)(4,2,1)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot {\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(4,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot {\alpha_{\beta+1}}^2 + \psi_{\beta_2}(\beta_2 \cdot {\alpha_{\beta+1}}^2 + 1))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(4,2,2)(5,0,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot {\alpha_{\beta+1}}^\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,2,2)(5,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot {\alpha_{\beta+1}}^\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(4,2,2)(5,2,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot {\alpha_{\beta+1}}^{\alpha_{\beta+1}}+$<br>$\psi_{\beta_2}(\beta_2 \cdot {\alpha_{\beta+1}}^{\alpha_{\beta+1}} + 1))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(4,2,2)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \psi_{\beta_2}(\beta_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,2,2)(5,3,0)(3,3,0)(4,2,2)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \psi_{\beta_2}(\beta_2) \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(4,2,2)(5,3,0)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \psi_{\beta_2}(\beta_2 \cdot 2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,2,2)(5,3,0)(6,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \psi_{\beta_2}(\beta_2 \cdot \beta))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(4,3,0)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^2 + \beta_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,3,0)(3,3,0)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^2 + \beta_2 \cdot \alpha_{\beta+1}+$<br>$\psi_{\beta_2}({\beta_2}^2 + \beta_2 \cdot \alpha_{\beta+1} + 1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,3,0)(3,3,0)(4,2,2)(5,3,0)(6,3,0) | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^2 + \beta_2 \cdot \psi_{\beta_2}({\beta_2}^2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)<br>(4,3,0)(3,3,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^2 \cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(4,3,0)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^2 \cdot \alpha_{\beta+1} + \psi_{\beta_2}({\beta_2}^2 \cdot \alpha_{\beta+1} + 1))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(4,3,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^3)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(4,3,0)(5,0,0) | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,0)(4,3,0)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^{\beta_2})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(4,4,0) | $\psi(\psi_\alpha(\psi_\beta(\varepsilon_{\beta_2+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,0)(4,4,1) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1})+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1})+\psi_{\alpha_2}(\psi_\beta(\Omega_{\beta_2+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)(2,2,0) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1})+\alpha_2))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,1)(2,2,2)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1})+\psi_\beta(\beta_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,1)(2,2,2)(3,3,1) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1})\cdot 2+$<br>$\psi_{\alpha_2}(\psi_\beta(\Omega_{\beta_2+1})\cdot 2+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,1)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)(3,2,1) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\Omega_{\beta+1})+$<br>$\psi_{\alpha_2}(\psi_\beta(\Omega_{\beta_2+1}+\Omega_{\beta+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,1)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,1)(3,2,1)(4,3,2)(5,4,0) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\psi_{\beta_2}(\beta_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)<br>(3,2,2)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}+1)+\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)<br>(3,2,2)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}+2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)<br>(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}+\beta))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)<br>(3,2,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}+\alpha_{\beta+1}+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)<br>(3,2,2)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}+\psi_{\beta_2}(\beta_2)))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\beta_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)<br>(3,3,0)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\beta_2\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)<br>(3,3,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+{\beta_2}^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)<br>(3,3,0)(4,4,0) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\varepsilon_{\beta_2+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)(3,3,1) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}\cdot 2)+\psi_{\alpha_2}(\psi_\beta(\Omega_{\beta_2+1}\cdot 2)+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)(4,4,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)<br>(4,4,2)(5,4,0)(4,4,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\alpha_{\beta+1}\cdot\omega)^2))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,1)(4,4,2)(5,4,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\alpha_{\beta+1}\cdot\omega^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)<br>(4,4,2)(5,4,2)(6,4,0)(5,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\alpha_{\beta+1}\cdot\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,1)<br>(4,4,2)(5,4,2)(6,4,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+{\alpha_{\beta+1}}^2\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,1)(4,4,2)(5,5,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\beta_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,1)(4,4,2)(5,5,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}))+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}))+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\alpha_{\beta_2+1}+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(2,2,2)(3,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\alpha_{\beta_2+1}+1))\cdot 2))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\alpha_{\beta_2+1}+1)+\beta)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\alpha_{\beta_2+1}+2))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\alpha_{\beta_2+1}+\beta))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(3,2,2)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\alpha_{\beta_2+1}+\psi_{\beta_2}(\beta_2)))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\beta_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(3,3,0)(4,4,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\varepsilon_{\beta_2+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)(3,3,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\Omega_{\beta_2+1})+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta_2+1}+\Omega_{\beta_2+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(3,3,1)(4,4,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot 2+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)(3,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot 2+\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot 2+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)(4,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)<br>(4,2,0)(3,2,0)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,2,0)(3,2,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\Omega_{\beta+1})+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\Omega_{\beta+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)<br>(4,2,0)(3,2,1)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\varepsilon_{\Omega_{\beta+1}+1})))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,2,0)(3,2,1)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,2,0)(3,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot\beta+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)<br>(4,2,0)(3,2,2)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot\beta+\psi_{\beta_2}(\beta_2)))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)<br>(4,2,0)(3,2,2)(4,3,1)(5,4,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot\beta+$<br>$\psi_{\psi_\beta(\alpha_{\beta_2+1})}(\psi_\beta(\alpha_{\beta_2+1}+\alpha_{\beta+1}\cdot\omega))))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)<br>(4,2,0)(3,2,2)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+$<br>$\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot\beta+\psi_{\beta_2}(\alpha_{\beta_2+1})))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)<br>(4,2,0)(3,2,2)(4,3,2)(5,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+$<br>$\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot\beta+\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot\beta)))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,2,0)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\beta_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,2,0)(3,3,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\Omega_{\beta_2+1})+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\Omega_{\beta_2+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,2,0)(3,3,1)(4,4,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot(\beta+1)+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,2,0)(3,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot(\beta+1)+$<br>$\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot(\beta+1)+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)<br>(4,2,0)(3,3,2)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)<br>(4,2,0)(4,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)(4,2,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\Omega_{\beta+1})+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta_2+1}\cdot\Omega_{\beta+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,2,1)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\alpha_{\beta+1}+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\alpha_{\beta+1}+$<br>$\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot\alpha_{\beta+1}+1))))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)<br>(4,2,2)(3,3,2)(4,2,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\alpha_{\beta+1}\cdot 2+$<br>$\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot\alpha_{\beta+1}\cdot 2+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,2,2)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\psi_{\beta_2}(\beta_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,3,0)(3,3,2)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta_2\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)(4,3,1) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\Omega_{\beta_2+1})+$<br>$\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta_2+1}\cdot\Omega_{\beta_2+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,3,1)(5,4,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta_2+1}}^2+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta_2+1}}^2+\psi_{\beta_2}({\alpha_{\beta_2+1}}^2+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)<br>(4,3,2)(3,3,2)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta_2+1}}^2\cdot 2+\psi_{\beta_2}({\alpha_{\beta_2+1}}^2\cdot 2+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,3,2)(5,0,0) | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta_2+1}}^\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)(4,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_3)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)<br>(4,4,0)(3,3,2)(4,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_3+\psi_{\beta_3}(\beta_3))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,4,0)(4,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_3\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)<br>(4,4,0)(5,2,0)(3,0,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_3\cdot\beta)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,4,0)(5,2,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_3\cdot\alpha_{\beta+1}+\psi_{\beta_2}(\beta_3\cdot\alpha_{\beta+1}+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,4,0)(5,2,2)(6,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_3\cdot\psi_{\beta_2}(\beta_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)<br>(4,4,0)(5,2,2)(6,3,2)(7,4,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_3\cdot\psi_{\beta_2}(\beta_3))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,4,0)(5,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_3\cdot\beta_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,4,0)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta(\beta_3\cdot\alpha_{\beta_2+1}+\psi_{\beta_2}(\beta_3\cdot\alpha_{\beta_2+1}+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,4,0)(5,4,0) | $\psi(\psi_\alpha(\psi_\beta({\beta_3}^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,2)(4,4,1)(5,5,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_3+1}+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,2)(4,4,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_3+1}+\psi_{\beta_2}(\alpha_{\beta_3+1}+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,3) | $\psi(\psi_\alpha(\psi_\beta(\beta_\omega)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,3)<br>(3,2,1)(4,3,2)(5,4,3) | $\psi(\psi_\alpha(\psi_\beta(\beta_\omega+\psi_{\beta_2}(\beta_\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,3)(3,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_\omega+\beta_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,3)(3,3,3) | $\psi(\psi_\alpha(\psi_\beta(\beta_\omega\cdot 2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,3)(4,3,0) | $\psi(\psi_\alpha(\psi_\beta(\beta_\omega\cdot\beta_2)))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,3)(4,3,0)(3,3,3) | $\psi(\psi_\alpha(\psi_\beta({\beta_\omega}^2)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,3)(4,3,1) | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_\omega+1})+\psi_{\alpha_2}(\psi_\beta(\Omega_{\beta_\omega+1})+1)))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,3)(4,3,2) | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_\omega+1}+\psi_{\beta_2}(\alpha_{\beta_\omega+1}+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,3)(4,3,2)(5,4,3) | $\psi(\psi_\alpha(\psi_\beta(\beta_{\omega\cdot 2})))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,3)(4,3,3) | $\psi(\psi_\alpha(\psi_\beta(\beta_{\omega^2})))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,3)(4,3,3)(5,3,0)(4,0,0) | $\psi(\psi_\alpha(\psi_\beta(\psi_\gamma(\beta_{\gamma+1}\cdot\gamma))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,3)(4,3,3)(5,3,1) | $\psi(\psi_\alpha(\psi_\beta(\psi_\gamma(\beta_{\gamma+1}\cdot\Omega_{\gamma+1}))+$<br>$\psi_{\alpha_2}(\psi_\gamma(\beta_{\gamma+1}\cdot\Omega_{\gamma+1})+1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,3)(4,3,3)(5,3,2) | $\psi(\psi_\alpha(\psi_\beta(\psi_\gamma(\beta_{\gamma+1}\cdot\alpha_{\gamma+1})+$<br>$\psi_{\beta_2}(\psi_\gamma(\beta_{\gamma+1}\cdot\alpha_{\gamma+1})+1))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,3)(4,3,3)(5,3,3) | $\psi(\psi_\alpha(\psi_\beta(\psi_\gamma({\beta_{\gamma+1}}^2\cdot\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,3)(4,4,0) | $\psi(\psi_\alpha(\psi_\beta(\psi_\gamma(\gamma_2))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,3)(4,4,3) | $\psi(\psi_\alpha(\psi_\beta(\psi_\gamma(\beta_{\gamma_2+1}+\psi_{\gamma_2}(\beta_{\gamma_2+1}+1)))))$ |
| (0,0,0)(1,1,1)(2,2,2)<br>(3,3,3)(4,4,3)(5,0,0) | $\psi(\psi_\alpha(\psi_\beta(\psi_\gamma(\beta_{\gamma_2+1}\cdot\omega))))$ |
| (0,0,0)(1,1,1)(2,2,2)(3,3,3)(4,4,4) | $\psi(\psi_\alpha(\psi_\beta(\psi_\gamma(\gamma_\omega))))$ |
| (0,0,0,0)(1,1,1,1) | $\psi(\omega-P)=\psi(\psi_S(\sigma S\cdot\omega))$ |

## A.15 BMS vs Upward Projection (Zuicai Mengxin.ver)

The results in this section are primarily cited from the analysis by Zuicai Mengxin (最菜萌新).

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1) | $\psi(\omega-P)=\psi(\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+1)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,0,0,0)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(1,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,0,0,0)<br>(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega\cdot\psi(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega^2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,1,0,0)<br>(1,1,0,0)(2,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega^2\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,1,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega^\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_1(\Omega_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_1(\Omega_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,0)<br>(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_1(\Omega_\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,0)<br>(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_1(\Omega_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,0)<br>(3,2,1,0)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_1(OFP))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,0)<br>(3,2,1,0)(4,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_1(I_\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_1(\psi_{\alpha_2}(\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,0)<br>(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_1(\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_1(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,1)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_1(\psi_S(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,1)<br>(2,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_1(\psi_S(\sigma S\cdot\omega)+\Omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,1)<br>(2,1,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_1(\psi_S(\sigma S\cdot\omega)+\psi_1(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,1)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,1)<br>(2,2,0,0)(3,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_2(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_\omega\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_\omega\cdot\Omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,0,0)<br>(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+{\Omega_\omega}^2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\omega(\Omega_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,0,0)<br>(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,0,0)<br>(3,2,1,1)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,0,0)<br>(3,2,1,1)(2,1,0,0)<br>(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,0,0)<br>(3,2,1,1)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_{\omega+1})$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,0,0)<br>(3,2,1,1)(3,2,0,0)<br>(4,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\omega+1}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,0,0)<br>(3,2,1,1)(3,2,0,0)<br>(4,3,1,1)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_{\omega+2})$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,0,0)<br>(3,2,1,1)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_{\omega\cdot 2})$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_{\omega^2})$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_{\omega^3})$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(I))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\Omega_{I+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega))\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega))^2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)+1)+$<br>$\psi_{\psi_I(\psi_S(\sigma S\cdot\omega)+1)}(\psi_I(\psi_S(\sigma S\cdot\omega)+1)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)+1)\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)+2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(2,1,0,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)+\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(2,1,0,0)(3,2,1,0)<br>(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)+\omega)\cdot 2)$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(2,1,0,0)(3,2,1,0)<br>(4,2,0,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)+\omega)^2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(2,1,0,0)(3,2,1,0)<br>(4,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)+\omega^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(2,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+I)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(2,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(4,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+I\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(2,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\Omega_{I+1}}(\Omega_{I+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(2,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\Omega_{I+1}}(I_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(2,1,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\Omega_{I+1}}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\Omega_{I+1}}(\psi_S(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\Omega_{I+1}}(\psi_S(\sigma S\cdot\omega)+$<br>$\psi_{\Omega_{I+1}}(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_{I+1})$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(4,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_{I+\omega})$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(4,2,1,1)<br>(4,2,1,0)(5,2,1,0)<br>(6,2,0,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{I_2}(I_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+I_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,0,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_2\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,0,0)<br>(3,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_2\cdot\Omega_{\alpha+1}+\psi_{\alpha_2}(\alpha_2\cdot\Omega_{\alpha+1}+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha({\alpha_2}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,0,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\varepsilon_{\alpha_2+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,0,0)<br>(3,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\alpha(\Omega_{\alpha_2+1})}(\psi_\alpha(\Omega_{\alpha_2+1}+\Omega_{\alpha+1}\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,0,0)<br>(3,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\alpha(\Omega_{\alpha_2+1})}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,0,0)<br>(3,3,1,1)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,0,0)<br>(3,3,1,1)(2,2,0,0)<br>(3,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\alpha(\Omega_{\alpha_2+1}+\Omega_{\alpha+1})}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,0,0)<br>(3,3,1,1)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\Omega_{\alpha+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,0,0)<br>(3,3,1,1)(3,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,0,0)<br>(3,3,1,1)(3,3,1,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,0,0)<br>(3,3,1,1)(3,3,1,0)<br>(4,4,0,0)(5,5,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1})}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,1,0)<br>(1,1,1,0)(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}+1))\cdot 2)$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,1,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}+1)+\alpha))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,1,0)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}+\alpha)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,1,0)<br>(2,1,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\alpha_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,1,0)<br>(2,1,1,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1})+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,1,0)<br>(2,2,0,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\varepsilon_{\alpha_2+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,1,0)<br>(2,2,0,0)(3,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\alpha(\Omega_{\alpha_2+1}\cdot 2)}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,1,0)<br>(2,2,0,0)(3,3,1,1)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,1,0)<br>(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}\cdot 2+\psi_{\alpha_2}(\Omega_{\alpha_2+1}\cdot 2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,1,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}\cdot\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,1,0)<br>(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha({\Omega_{\alpha_2+1}}^2+\psi_{\alpha_2}({\Omega_{\alpha_2+1}}^2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,1,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_\omega+\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_{\omega+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,0)<br>(3,2,0,0)(4,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\alpha(\Omega_{\alpha_{\omega+1}+1})}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,0)<br>(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_{\omega+1}+1}+\psi_{\alpha_2}(\Omega_{\alpha_{\omega+1}+1}+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,0)<br>(3,2,1,0)(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_{\omega\cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,0)<br>(3,2,2,0)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_\beta(\beta_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,0)<br>(3,3,3,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\beta_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\alpha))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,1,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,1,1,0)(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,1,1,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+\alpha)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,1,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\alpha_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,1,1,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\alpha_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,1,1,0)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,1,1,0)(3,2,2,1)<br>(2,1,1,0)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega))\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,0,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\alpha_2\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,0,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\varepsilon_{\alpha_2+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,0,0)(3,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+$<br>$\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+$<br>$\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha_2+1}+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,1,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha_2+1}+\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,1,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha_2+1}\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha_2+1}\cdot\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,1,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_3}(\alpha_3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,1,0)(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_3}(\alpha_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,1,0)(3,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_3}(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,1,0)(3,3,2,1)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\alpha_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\alpha_\omega)$<br>$=\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\alpha_\omega+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(2,2,1,0)<br>(3,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\alpha_\omega+\psi_{\alpha_3}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\alpha_\omega\cdot 2)$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\alpha_\omega\cdot\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+{\alpha_\omega}^2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha_\omega+1}+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha_\omega+1}+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_{\omega+1}}(\alpha_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_{\omega+1}}(\alpha_{\omega\cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_{\omega+1}}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\alpha_{\omega+1})$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1)(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\alpha_{\omega\cdot 2})$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\alpha_{\omega^2})$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta+\beta))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,0)(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega))\cdot 2)$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,0)(5,3,0,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega)+\beta))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,0)(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega+1))}(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega+1))}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,1)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,1)(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,1)(4,3,2,0)<br>(5,3,2,0)(6,3,0,0)<br>(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\varepsilon_{\beta+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(5,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+$<br>$\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,0)(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\beta(\alpha_{\beta+1}{}^2)}(\psi_\beta(\alpha_{\beta+1}{}^2+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,0)(5,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\beta(\alpha_{\beta+1}{}^2)}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,0)(5,3,2,1)<br>(3,2,1,0)(4,3,2,1)<br>(4,3,2,0)(5,3,2,0)<br>(6,3,0,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}{}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,0)(5,3,2,1)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\beta(\alpha_{\beta+1}{}^2+\alpha_{\beta+1})}(\psi_S(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,0)(5,3,2,1)<br>(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}{}^2+\alpha_{\beta+1}\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}{}^2\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0)(3,2,2,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}{}^3\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\beta_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,0,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\beta_2\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,0,0)<br>(4,4,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\Omega_{\beta_2+1})+\Omega_{\alpha+1}\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\Omega_{\beta_2+1})+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\Omega_{\beta_2+1})+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,1,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\Omega_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,1,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\Omega_{\beta_2+1}+\beta_2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,1,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\varepsilon_{\Omega_{\beta_2+1}+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,1,0)<br>(4,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\beta(\alpha_{\beta_2+1})}(\psi_\beta(\alpha_{\beta_2+1}+\alpha_{\beta+1}\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,1,0)<br>(4,4,2,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\beta(\alpha_{\beta_2+1})}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,1,0)<br>(4,4,2,1)(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta_2+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,1,0)<br>(4,4,2,1)(4,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta_2+1}+\alpha_{\beta+1}\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\alpha_{\beta_2+1}+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,2,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta_2+1}+\beta_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,2,0)<br>(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta_2+1}\cdot 2+\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot 2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,2,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\beta_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\beta_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)<br>(2,2,2,0)(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\psi_S(\sigma S\cdot\omega))\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\psi_S(\sigma S\cdot\omega)+\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)<br>(3,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\psi_S(\sigma S\cdot\omega)+\alpha_\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)<br>(3,2,0,0)(2,2,2,0)<br>(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\beta)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\beta_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)<br>(3,3,3,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\beta_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)<br>(3,3,3,0)(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\gamma(\gamma_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)<br>(3,3,3,0)(4,4,4,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\gamma(\gamma_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)<br>(3,3,3,0)(4,4,4,1) | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\gamma(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)<br>(3,3,3,0)(4,4,4,1)<br>(4,3,0,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\gamma)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)<br>(3,3,3,0)(4,4,4,1)<br>(4,4,4,0) | $\psi(\psi_S(\sigma S\cdot\omega)+\gamma_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,1)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega)\cdot 2+\Omega_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,1)(1,1,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)\cdot 2+\psi_\alpha(\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,1)(1,1,1,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega)\cdot 2+\psi_\alpha(\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,1)(1,1,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega)\cdot 2+\psi_\alpha(\psi_S(\sigma S\cdot\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,1)(1,1,1,0)<br>(2,2,2,1)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega)\cdot 2+\psi_\alpha(\psi_S(\sigma S\cdot\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,1)(1,1,1,0)<br>(2,2,2,1)(2,2,2,1)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)\cdot 2+\alpha)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,1)(1,1,1,0)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)\cdot 2+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,1)(1,1,1,0)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,2,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)\cdot 2+\psi_\beta(\beta_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,1)(1,1,1,0)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)<br>(3,3,3,1)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega)\cdot 2+\beta)$ |
| (0,0,0,0)(1,1,1,1)<br>(1,1,1,1)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega)\cdot 3)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,0,0,0)(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_1(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_1(\psi_S(\sigma S\cdot\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega)+$<br>$\psi_1(\psi_S(\sigma S\cdot\omega+\Omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_2)$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,0,0)(3,3,1,1)<br>(4,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_2(\psi_S(\sigma S\cdot\omega+\Omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_\omega\cdot\Omega_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,2,0,0)<br>(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+{\Omega_\omega}^2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\omega(\Omega_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\omega^2})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\alpha(\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\alpha(\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\Omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\Omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,0,0)<br>(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\Omega)+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\Omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\alpha)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,0,0)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha+1}\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,0,0)<br>(4,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,0,0)<br>(4,3,1,1)(5,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,0,0)<br>(4,3,1,1)(5,1,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega))+$<br>$\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha+1}}(\psi_S(\sigma S\cdot\omega+\Omega)+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,0,0)<br>(4,3,1,1)(5,1,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega))+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,0,0)<br>(4,3,1,1)(5,1,0,0)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega))+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha+1}\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,0,0)<br>(4,3,1,1)(5,1,0,0)<br>(4,3,1,0)(5,4,2,1)<br>(6,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega))+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,0,0)<br>(4,3,1,1)(5,1,0,0)<br>(4,3,1,0)(5,4,2,1)<br>(6,1,0,0)(5,3,0,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega))+\alpha)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,0,0)<br>(4,3,1,1)(5,1,0,0)<br>(4,3,1,0)(5,4,2,1)<br>(6,1,0,0)(5,3,0,0)<br>(6,4,1,1)(7,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega))\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+1)+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+1)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+1)+\alpha)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(3,2,0,0)(4,3,1,1)<br>(5,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+1)+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\alpha))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,0,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\alpha+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\varepsilon_{\alpha+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,0,0)(5,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$<br>$\Omega_{\alpha+1})+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha+1}+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,1,0)(3,2,1,0)<br>(4,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha+1}\cdot 2+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\alpha_2\cdot 2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,0,0)(5,4,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1}))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1})+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,1,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1})+2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,1,0)(3,2,1,0)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1})\cdot 2+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,1,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1}+\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,1,0)(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$<br>$\psi_{\alpha_2}(\Omega_{\alpha_2+1}\cdot 2)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\alpha_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,2,1)(5,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,0,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\alpha_2\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,0,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\varepsilon_{\alpha_2+1})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,0,0)<br>(4,4,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha_2+1}+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha_2+1}+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,1,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha_2+1}+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,1,0)<br>(3,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha_2+1}\cdot 2+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha_2+1}\cdot 2+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,1,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_3}(\alpha_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,1,0)<br>(4,4,0,0)(5,5,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_3}(\Omega_{\alpha_3+1})+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,1,0)<br>(4,4,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_3}(\Omega_{\alpha_3+1})+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_3}(\Omega_{\alpha_3+1})+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,1,0)<br>(4,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_3}(\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,1,0)<br>(4,4,2,1)(5,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_3}(\psi_S(\sigma S\cdot\omega+\Omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,1,0)<br>(4,4,2,1)(5,1,0,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\alpha_3)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\alpha_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\alpha_\omega\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\alpha_\omega\cdot\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,0,0)(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+{\alpha_\omega}^2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,0,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\varepsilon_{\alpha_\omega+1})$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha_\omega+1}+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha_\omega+1}+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,1,0)(3,3,2,0)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha_\omega+1}\cdot 2+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha_\omega+1}\cdot 2+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,1,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\varepsilon_{\Omega_{\alpha_\omega+1}+1})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,1,0)(5,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_{\omega+1}}(\alpha_{\omega\cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,1,0)(5,4,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_{\omega+1}}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,1,0)(5,4,2,1)<br>(6,1,0,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\alpha_{\omega+1})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,1,0)(5,4,2,1)<br>(6,1,0,0)(5,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\alpha_{\omega\cdot 2})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\alpha_{\omega^2})$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,2,0)(5,3,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,2,0)(5,3,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,2,0)(5,3,0,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\beta(\alpha_{\beta+1}\cdot\varepsilon_{\beta+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,2,0)(5,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,2,0)(5,3,1,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\beta(\alpha_{\beta+1}\cdot\varepsilon_{\Omega_{\beta+1}+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,2,0)(5,3,1,0)<br>(6,4,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\psi_\beta(\alpha_{\beta+1}{}^2)}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,2,0)(5,3,1,0)<br>(6,4,2,1)(7,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\psi_\beta(\alpha_{\beta+1}{}^2)}(\psi_S(\sigma S\cdot\omega+\Omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,2,0)(5,3,1,0)<br>(6,4,2,1)(7,1,0,0)<br>(4,3,1,0)(5,4,2,0)<br>(6,4,2,0)(7,4,0,0)<br>(6,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\beta({\alpha_{\beta+1}}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,3,2,0)(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\beta(\beta_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,4,3,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\beta(\beta_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,4,3,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\beta(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,4,3,1)(5,1,0,0)<br>(4,3,0,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\beta)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,4,3,1)(5,1,0,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\beta_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,4,3,1)(5,1,0,0)<br>(4,4,3,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\beta_\omega)$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,1,0,0)(3,3,2,0)<br>(4,4,3,1)(5,1,0,0)<br>(4,4,3,0)(5,5,4,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_\gamma(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\Omega_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\Omega_{\omega^2})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\alpha)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,2,1,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+$<br>$\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$<br>$\psi_S(\sigma S\cdot\omega)+\alpha))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha_2+1}+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha_2+1}+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,1,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha_2+1}+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,1,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha_2+1}\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,1,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_3}(\alpha_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,1,0)<br>(4,4,2,1)(5,1,0,0)<br>(4,4,2,1)(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\alpha_3)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\alpha_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,2,0)<br>(4,3,0,0)(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+{\alpha_\omega}^2)$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,2,0)<br>(4,3,1,0)(5,4,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_{\omega+1}}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,2,0)<br>(4,3,1,0)(5,4,2,1)<br>(6,1,0,0)(5,4,2,1)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\alpha_{\omega+1})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,2,0)<br>(4,3,1,0)(5,4,2,1)<br>(6,1,0,0)(5,4,2,1)<br>(5,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\alpha_{\omega\cdot 2})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,2,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\alpha_{\omega^2})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,2,0)<br>(4,3,2,0)(5,3,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,2,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\psi_\beta(\beta_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,2,0)<br>(4,4,3,1)(5,1,0,0)<br>(4,4,3,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+$<br>$\psi_\beta(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,2,0)<br>(4,4,3,1)(5,1,0,0)<br>(4,4,3,1)(4,3,0,0)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\beta)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,2,0)<br>(4,4,3,1)(5,1,0,0)<br>(4,4,3,1)(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\beta_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1)(4,1,0,0)<br>(3,3,2,1)(3,3,2,0)<br>(4,4,3,1)(5,1,0,0)<br>(4,4,3,1)(4,4,3,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\beta_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(2,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(2,2,1,1)(3,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(3,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_1(\Omega_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)<br>(4,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_1(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,0,0)(3,3,1,1)<br>(4,2,0,0)(3,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2)+\Omega_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,0,0)(3,3,1,1)<br>(4,2,0,0)(3,3,1,0)<br>(4,4,2,1)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\Omega_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,0,0)(3,3,1,1)<br>(4,2,0,0)(3,3,1,0)<br>(4,4,2,1)(5,2,0,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2)+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,0,0)(3,3,1,1)<br>(4,2,0,0)(3,3,1,0)<br>(4,4,2,1)(5,2,0,0)<br>(4,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2)+\alpha_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,0,0)(3,3,1,1)<br>(4,2,0,0)(3,3,1,0)<br>(4,4,2,1)(5,2,0,0)<br>(4,4,2,0)(5,5,3,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2)+\psi_\beta(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,0,0)(3,3,1,1)<br>(4,2,0,0)(3,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2)+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,0,0)(3,3,1,1)<br>(4,2,0,0)(3,3,1,1)<br>(4,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2)+\psi_S(\sigma S\cdot\omega+\Omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,0,0)(3,3,1,1)<br>(4,2,0,0)(3,3,1,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2)\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,0,0)(3,3,1,1)<br>(4,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,0,0)(3,3,1,1)<br>(4,2,0,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,0,0)(3,3,1,1)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(1,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(1,1,0,0)(2,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_1(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(1,1,0,0)(2,2,1,1)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(1,1,0,0)(2,2,1,1)<br>(3,2,0,0)(2,2,0,0)<br>(3,3,1,1)(4,2,0,0)<br>(3,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega_2)+\Omega_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(1,1,0,0)(2,2,1,1)<br>(3,2,0,0)(2,2,0,0)<br>(3,3,1,1)(4,2,0,0)<br>(3,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega_2)+$<br>$\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(1,1,0,0)(2,2,1,1)<br>(3,2,0,0)(2,2,0,0)<br>(3,3,1,1)(4,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega_2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(1,1,0,0)(2,2,1,1)<br>(3,2,0,0)(2,2,0,0)<br>(3,3,1,1)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega_3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(1,1,0,0)(2,2,1,1)<br>(3,2,0,0)(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(1,1,0,0)(2,2,1,1)<br>(3,2,0,0)(2,2,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_\omega)$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_\omega\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_\omega\cdot\Omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_\omega\cdot\Omega+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,1,0)(3,1,0,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_\omega\cdot\Omega+\Omega_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,1,0)(3,1,0,0)<br>(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_\omega\cdot(\Omega+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_\omega\cdot\Omega_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+{\Omega_\omega}^2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+{\Omega_\omega}^2\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+{\Omega_\omega}^3)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\omega(\Omega_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\omega(\Omega_{\omega+1}\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\omega(\Omega_{\omega\cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega+\Omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,1,0)(3,2,0,0)<br>(4,3,1,1)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega+\Omega_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega+\Omega_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega+1})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega+1}\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_{\omega+1}(\Omega_{\omega+2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,0,0)<br>(4,3,1,1)(5,1,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega+2})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega\cdot 2})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,0,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega\cdot 2}{}^2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_{\omega\cdot 2}(\Omega_{\omega\cdot 2+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,0,0)(5,3,1,1)<br>(6,1,0,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_{\omega\cdot 2}(\psi_S(\sigma S\cdot\omega+\Omega_\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,0,0)(5,3,1,1)<br>(6,1,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega\cdot 2+1})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,0,0)(5,3,1,1)<br>(6,1,0,0)(5,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega\cdot 3})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega^2})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_I(I))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\alpha(\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,2,1)(5,1,0,0)<br>(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\Omega_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,2,1)(5,1,0,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\alpha)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,2,1)(5,1,0,0)<br>(4,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,2,1)(5,1,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\alpha_2)$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,2,1)(5,1,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\alpha_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,1)<br>(4,1,0,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(4,1,0,0)<br>(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_\omega(\Omega_{\omega+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,0,0)<br>(4,3,1,1)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega+2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega\cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,1,0)<br>(4,2,0,0)(5,3,1,1)<br>(6,3,0,0)(5,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega\cdot 3}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega^2})+\Omega_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(1,1,1,0)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega^2})+\Omega_{\omega^2})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega^2})+\psi_{\omega^2}(\Omega_{\omega^2+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(2,1,0,0)<br>(3,2,1,1)(4,1,0,0)<br>(1,1,1,0)(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega^2})+\psi_{\omega^2}(\psi_S(\sigma S\cdot\omega+\Omega_{\omega^2})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(2,1,0,0)<br>(3,2,1,1)(4,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega^2})+\Omega_{\omega^2+1})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(2,1,0,0)<br>(3,2,1,1)(4,1,0,0)<br>(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega^2})+\Omega_{\omega^2+\omega})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(2,1,0,0)<br>(3,2,1,1)(4,1,0,0)<br>(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega^2})+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(2,1,0,0)<br>(3,2,1,1)(4,1,0,0)<br>(4,1,0,0)(1,1,1,0)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega^2}\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(2,1,0,0)<br>(3,2,1,1)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega^2+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(2,1,0,0)<br>(3,2,1,1)(4,2,0,0)<br>(3,2,1,0)(4,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega^2\cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega^3}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\omega^\omega}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,0,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_I(I)\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_{\psi_I(I+1)}(\psi_I(I+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_{\psi_I(I+1)}(\psi_S(\sigma S\cdot\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+$<br>$\psi_{\psi_I(I+1)}(\psi_S(\sigma S\cdot\omega+\psi_I(I))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_I(I+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_I(I+\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_I(I\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_I(\Omega_{I+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(6,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_I(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(6,3,1,1)(7,1,0,0)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_I(\psi_S(\sigma S\cdot\omega+\psi_I(I))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(6,3,1,1)(7,1,0,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_I(\psi_S(\sigma S\cdot\omega+\psi_I(I))+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(6,3,1,1)(7,1,0,0)<br>(4,2,0,0)(5,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_I(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(6,3,1,1)(7,1,0,0)<br>(4,2,0,0)(5,3,1,0)<br>(6,3,1,0)(7,3,0,0)<br>(6,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+I)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(6,3,1,1)(7,1,0,0)<br>(4,2,0,0)(5,3,1,0)<br>(6,3,1,0)(7,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_{\Omega_{I+1}}(I_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(6,3,1,1)(7,1,0,0)<br>(4,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+$<br>$\psi_{\Omega_{I+1}}(\psi_S(\sigma S\cdot\omega+\psi_I(I))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(6,3,1,1)(7,1,0,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\Omega_{I+1})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(6,3,1,1)(7,1,0,0)<br>(6,3,1,0)(7,3,1,0)<br>(8,3,0,0)(7,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_{I_2}(I_2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+I_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_\alpha(\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_\alpha(\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,2,1)(5,1,0,0)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_I(I))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,2,1)(5,1,0,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\alpha)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,2,1)(5,1,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,2,1)(5,1,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\alpha_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,0)<br>(4,3,2,1)(5,1,0,0)<br>(4,3,2,0)(5,4,3,1)<br>(6,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_\beta(\psi_S(\sigma S\cdot\omega+\psi_I(I))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,1,0,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{\psi_I(I+1)}(\psi_I(I+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I+1))+\psi_I(I+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,0,0)<br>(4,3,1,1)(5,2,0,0)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I+1))+\psi_I(I+\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,0,0)<br>(4,3,1,1)(5,2,0,0)<br>(4,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I+1))+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,0,0)<br>(4,3,1,1)(5,2,0,0)<br>(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I+1)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,0,0)<br>(4,3,1,1)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I+\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I \cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,1,0)(2,1,1,0) | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I \cdot \omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(\Omega_{I+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1)(5,1,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(\psi_S(\sigma S \cdot \omega + \Omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1)(5,1,0,0)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(\psi_S(\sigma S \cdot \omega + \psi_I(I)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1)(5,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + I))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1)(5,1,0,0)<br>(2,0,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + I) + \psi_I(\psi_S(\sigma S \cdot \omega + I)) \cdot \omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1)(5,1,0,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + I) + \psi_I(\psi_S(\sigma S \cdot \omega + I) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1)(5,1,0,0)<br>(2,1,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + I) + I)$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1)(5,1,0,0)<br>(2,1,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+I)+\psi_{\Omega_{I+1}}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1)(5,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+I+\psi_I(\psi_S(\sigma S\cdot\omega+I)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1)(5,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+I\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1)(5,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(4,0,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+I\cdot 2)+\psi_I(\psi_S(\sigma S\cdot\omega+I)+I)\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1)(5,1,0,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0)<br>(6,3,1,1)(7,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+I\cdot 2+\psi_I(\psi_S(\sigma S\cdot\omega+I\cdot 2+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1)(5,1,0,0)<br>(2,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+I\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{I+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1)(5,2,0,0)<br>(4,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{I+\omega}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1)(5,2,0,0)<br>(4,2,1,0)(5,2,1,0)<br>(6,2,0,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{I_2}(I_2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,1,1)(5,2,0,0)<br>(4,2,1,0)(5,2,1,0)<br>(6,2,0,0)(7,3,1,1)<br>(8,2,0,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+I_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0)(3,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+I_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_\alpha(\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,0,0)(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_\alpha(\alpha_2+\psi_{\alpha_2}(\alpha_2+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,0,0)(2,1,1,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_\alpha(\alpha_2+\psi_{\alpha_2}(\alpha_2+\psi_{\alpha_2}(\alpha_2)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,0,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_\alpha(\alpha_2\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,0,0)(3,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_\alpha(\Omega_{\alpha_2+1}))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_\alpha(\Omega_{\alpha_2+1}))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{\alpha_2}(\Omega_{\alpha_2+1})+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,1,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_\alpha(\Omega_{\alpha_2+1}\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_\alpha(\alpha_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_\alpha(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\Omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)<br>(1,1,1,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_\alpha(\alpha_2)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)<br>(2,0,0,0)(1,1,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)<br>(2,0,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\alpha)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha)+\alpha)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\alpha)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha)+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha)+\alpha_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha)+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\alpha+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)<br>(4,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\alpha+1})+\Omega_{\alpha+1}\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\alpha+1})+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega_{\alpha+1})+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,1,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{\alpha_2}(\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,1,0)<br>(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_2)+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_2)+\Omega_{\alpha_2+1}+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\alpha_2)+\Omega_{\alpha_2+1}+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,0)(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_2)+\psi_{\alpha_3}(\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_2)+\alpha_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,2,0,0)(3,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_2)+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,2,0,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_2\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_\omega)+{\alpha_\omega}^2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_\omega)+\Omega_{\alpha_\omega+1})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_\omega)+\Omega_{\alpha_\omega+1}+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\alpha_\omega)+\Omega_{\alpha_\omega+1}+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_\omega)+\Omega_{\alpha_\omega+1}\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_\omega)+\psi_{\alpha_\omega+1}(\alpha_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_\omega)+\psi_{\alpha_\omega+1}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_\omega)+\psi_{\alpha_\omega+1}(\psi_S(\sigma S\cdot\omega+\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1)(5,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_\omega)+\psi_{\alpha_\omega+1}(\psi_S(\sigma S\cdot\omega+\alpha_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1)(5,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_\omega)+\alpha_{\omega+1})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1)(5,2,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_\omega)+\alpha_{\omega\cdot 2})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1)(5,2,0,0)<br>(4,3,2,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_\omega)+\psi_\beta(\beta_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1)(5,2,0,0)<br>(4,3,2,0)(5,4,3,1) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_\omega)+\psi_\beta(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1)(5,2,0,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_\omega)+\psi_S(\sigma S\cdot\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1)(5,2,0,0)<br>(5,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\omega \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1)(5,2,0,0)<br>(6,3,1,1) | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\alpha_\omega+1}) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1)(5,2,1,0) | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\alpha_\omega+1})+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega_{\alpha_\omega+1}) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1)(5,2,1,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{\alpha_{\omega+1}}(\alpha_{\omega+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1)(5,3,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\omega \cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0) | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,1,0)(4,3,2,1)<br>(5,2,0,0)(5,2,0,0)<br>(2,2,2,0)(3,2,2,0) | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\omega^2} \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,1,0)(4,3,2,1)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\omega^2+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,1,0)(4,3,2,1)<br>(5,3,0,0)(4,3,2,0)<br>(5,3,2,0) | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\omega^2 \cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(4,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\alpha))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\beta(\alpha_{\beta+1} \cdot \beta)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\beta(\beta_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1) | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\beta(\psi_S(\sigma S \cdot \omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,0,0)(2,2,2,0)<br>(3,3,3,1) | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\beta(\psi_S(\sigma S \cdot \omega + \psi_\beta(\psi_S(\sigma S \cdot \omega))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + \beta))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,0,0)(3,3,3,1) | $\psi(\psi_S(\sigma S \cdot \omega + \beta) + \psi_S(\sigma S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + \beta + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,0,0)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + \beta \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\beta+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,0,0)(5,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\beta+1})+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\beta+1})+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega_{\beta+1})+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,1,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\beta+1})+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,1,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\beta+1})+\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\Omega_{\beta+1})+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,1,0)(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_{\beta+1})+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,1,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_{\beta+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,1,0)(5,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_{\beta+1})+\psi_\beta(\psi_S(\sigma S\cdot\omega+\alpha_{\beta+1})+$<br>$\Omega_{\beta+1})+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\alpha_{\beta+1})+$<br>$\psi_\beta(\psi_S(\sigma S\cdot\omega+\alpha_{\beta+1})+\Omega_{\beta+1})+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,1,0)(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_{\beta+1})+\alpha_{\beta+1}\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,1,0)(5,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_{\beta+1})+\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,1,0)(5,3,2,1)<br>(3,3,3,1)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_{\beta+1})+\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega))+$<br>$\psi_{\psi_\beta(\psi_S(\sigma S\cdot\omega+\alpha_{\beta+1}))}(\psi_\beta(\psi_S(\sigma S\cdot\omega+\alpha_{\beta+1}))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\alpha_{\beta+1})+\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\alpha_{\beta+1})+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,2,2,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{\beta_2}(\beta_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\beta_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,0,0)(3,3,3,0) | $\psi(\psi_S(\sigma S\cdot\omega+\beta_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(1,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\Omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,1)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,1)<br>(3,2,0,0)(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,1)<br>(3,2,0,0)(2,2,1,0)<br>(3,3,2,1)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_1(\psi_S(\sigma S\cdot\omega+\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,1)<br>(3,2,0,0)(2,2,1,0)<br>(3,3,2,1)(4,3,0,0)<br>(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_1(\psi_S(\sigma S\cdot\omega+\alpha_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,1)<br>(3,2,0,0)(2,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_1(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(1,1,0,0)(2,2,1,1)<br>(3,2,0,0)(2,2,1,1)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\Omega_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\Omega_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(1,1,1,0)(2,1,1,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_I(I))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_\alpha(\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(2,2,2,1)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\alpha)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(2,2,2,1)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(2,2,2,1)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\alpha_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,1,1)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,1,0,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_\omega)+$<br>$\psi_\omega(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega+1})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,1,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_S(\sigma S\cdot\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,1,0,0)(4,1,0,0)<br>(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_{\omega\cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_\alpha(\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_\alpha(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\alpha))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,1,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_{\alpha+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_{\alpha+1})+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\Omega_{\alpha+1})+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,1,1,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{\alpha_2}(\alpha_2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))\cdot 3)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,1,0,0)<br>(3,2,1,1)(4,2,0,0)<br>(4,1,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_\omega)+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,1,0,0)<br>(3,2,1,1)(4,2,0,0)<br>(4,1,0,0)(3,2,1,1)<br>(4,2,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+$<br>$\Omega_\omega)+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,1,0,0)<br>(3,2,1,1)(4,2,0,0)<br>(4,1,0,0)(4,1,0,0)<br>(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,1,0,0)<br>(3,2,1,1)(4,2,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,1,0,0)<br>(1,1,1,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+$<br>$\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,1,0,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha)+\alpha)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,1,0,0)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha)+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,1,0,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha)+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,1,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha)+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,1,0,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,1,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha+1})+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha+1})+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,1,1,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha+1}+\alpha))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,1,1,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\alpha_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\psi_\beta(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,1)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)+\Omega_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,1)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,1)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)+\psi_S(\sigma S\cdot\omega+\Omega_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,2,0,0)<br>(2,2,2,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)+\psi_S(\sigma S\cdot\omega+\alpha))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)+\psi_S(\sigma S\cdot\omega+\alpha_\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(2,1,0,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(2,1,0,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2+\Omega_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(2,1,0,0)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,1,0,0)<br>(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega+\Omega_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(4,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega+\alpha)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(4,1,0,0)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega+\alpha)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(4,1,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega+\alpha+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega+\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,0,0)(4,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega+\alpha_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+S)+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+S)+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S)\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(2,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S+\Omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(2,1,0,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+S+\Omega_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(4,3,0,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S+\alpha))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+S+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S+\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(3,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S+\psi_S(\sigma S\cdot\omega+S+\Omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(3,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+S+\psi_S(\sigma S\cdot\omega+S+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(3,1,0,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S+\psi_S(\sigma S\cdot\omega+S+\psi_S(\sigma S\cdot\omega+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(4,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+S\cdot\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\Omega_{\alpha+1}\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+\Omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,0,0)<br>(2,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+\psi_1(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,0,0)<br>(2,2,1,1)(3,2,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+\Omega_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+\Omega_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,1,0,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+{\Omega_\omega}^2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+1)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+\alpha)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,1,0,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+\Omega_{\alpha+1}\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,1,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,0)(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_\beta(\psi_S(\sigma S\cdot\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))\cdot 2+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))\cdot 2+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,1,0,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\Omega_{\alpha+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\Omega_{\alpha+1})+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\Omega_{\alpha+1})+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,1,1,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_{\alpha_2}(\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+1)+\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}$<br>$(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+1)+\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}$<br>$(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(2,2,2,0)<br>(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+$<br>$\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_\beta(\psi_S(\sigma S\cdot\omega))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+$<br>$\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))))$ |

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| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(2,2,2,1)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+$<br>$\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+$<br>$\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+1)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+1)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\Omega_{\alpha+1}\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,3,0,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+\alpha)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+2))$ |

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| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,3,1,0)<br>(6,4,2,1)(7,4,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,4,2,1)<br>(6,4,0,0)(5,4,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,4,2,1)<br>(6,4,0,0)(7,5,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))\cdot 2)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,4,2,1)<br>(6,4,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))\cdot 2+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,4,2,1)<br>(6,4,1,0)(5,3,1,0)<br>(6,4,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))\cdot 2+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega))))$ |

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| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,4,2,1)<br>(6,4,1,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))\cdot 2+\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,4,2,1)<br>(6,4,1,0)(5,4,2,0)<br>(6,5,3,1)(7,5,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))\cdot 2+$<br>$\psi_\beta(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+$<br>$\psi_{\alpha_2}(\psi_\beta(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,4,2,1)<br>(6,4,1,0)(5,4,2,0)<br>(6,5,3,1)(7,5,1,0)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))\cdot 2+\psi_\beta(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,4,2,1)<br>(6,4,1,0)(5,4,2,0)<br>(6,5,3,1)(7,5,1,0)<br>(5,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))\cdot 2+\psi_\beta(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,4,2,1)<br>(6,4,1,0)(5,4,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))\cdot 2+\psi_S(\sigma S\cdot\omega)))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,4,2,1)<br>(6,4,1,0)(5,4,2,1)<br>(6,4,0,0)(7,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))\cdot 3)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,4,2,1)<br>(6,4,1,0)(5,4,2,1)<br>(6,4,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))\cdot 3+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,4,2,1)<br>(6,4,1,0)(6,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}$<br>$(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha_\omega))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,2,0,0)(5,4,2,1)<br>(6,4,1,0)(6,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,3,0,0)(5,4,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_S(\sigma S\cdot\omega))))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,3,0,0)(5,4,2,1)<br>(6,4,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,3,0,0)(5,4,2,1)<br>(6,4,1,0)(6,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))))))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,3,0,0)(5,4,2,1)<br>(6,4,1,0)(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,3,0,0)(6,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega)))))))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,3,0,0)(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))\cdot 2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,3,0,0)(7,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_\omega)))))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,3,0,0)(7,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega)))))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,3,0,0)(7,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\alpha)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\Omega_{\alpha+1}+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,3,1,0)(7,4,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,3,1,0)(7,4,2,1)<br>(8,4,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\alpha_2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,4,0,0)(5,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\alpha_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,1,1)<br>(5,3,1,0)(4,3,1,0)<br>(5,4,2,1)(6,4,1,0)<br>(6,4,0,0)(5,4,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(2,2,2,0)<br>(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_\beta(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(2,2,2,0)<br>(3,3,3,1)(4,3,1,0)<br>(4,3,0,0)(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_\beta(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(2,2,2,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+\alpha))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(2,2,2,1)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(2,2,2,1)<br>(3,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(2,2,2,1)<br>(3,2,0,0)(2,2,2,0)<br>(3,3,3,1)(4,3,1,0)<br>(4,3,0,0)(3,3,3,1)<br>(4,3,1,0)(3,3,3,1)<br>(4,3,0,0)(3,3,3,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+\beta_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(2,2,2,1)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(2,2,2,1)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(2,2,2,1)<br>(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))\cdot 2+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(2,2,2,1)<br>(3,2,1,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))\cdot 2+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(2,2,2,1)<br>(3,2,1,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))\cdot 2+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(3,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)+\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(3,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,1,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,1,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\Omega_{\alpha+1}))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\Omega_{\alpha+1}))+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,1,1,0)(5,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,2,0,0)(4,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,2,1,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S)+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,0,0)(2,2,2,1)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S)+\psi_S(\sigma S\cdot\omega+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,0,0)(2,2,2,1)<br>(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,0,0)(2,2,2,1)<br>(3,2,1,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S)\cdot 2)$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,0,0)(3,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,0,0)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,0,0)(5,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S\cdot\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,0,0)(5,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S\cdot\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$<br>$S\cdot\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,0,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,0,0)(5,4,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+\alpha_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,2,2,0)<br>(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+\psi_\beta(\psi_S(\sigma S\cdot\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,2,2,0)<br>(3,3,3,1)(4,3,1,0)<br>(5,4,0,0)(6,5,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+$<br>$\psi_\beta(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,2,2,0)<br>(3,3,3,1)(4,3,1,0)<br>(5,4,1,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+$<br>$\psi_\beta(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,2,2,0)<br>(3,3,3,1)(4,3,1,0)<br>(5,4,1,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+\beta)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,2,2,0)<br>(3,3,3,1)(4,3,1,0)<br>(5,4,1,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+\beta_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,2,2,0)<br>(3,3,3,1)(4,3,1,0)<br>(5,4,1,0)(3,3,3,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+\beta_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,2,2,1)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+\psi_S(\sigma S\cdot\omega+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,2,2,1)<br>(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)\cdot 2)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,2,2,1)<br>(3,2,1,0)(3,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,2,2,1)<br>(3,2,1,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+S)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,2,2,1)<br>(3,2,1,0)(4,3,0,0)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,2,2,1)<br>(3,2,1,0)(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)\cdot 2+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)\cdot 2+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(3,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(3,2,0,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(3,2,0,0)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(3,2,1,0)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2))+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(4,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(4,2,1,0)<br>(5,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)\cdot 2)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(4,3,0,0)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 3)+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 3)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,2,0,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_S(\sigma S\cdot\omega+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,2,1,0)<br>(6,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)\cdot 2)))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,0,0)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,0,0)<br>(3,2,1,0)(4,3,1,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,0,0)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S)+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S)+\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,0,0)<br>(4,3,1,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,0,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,0,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S)))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S)))+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,1,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))+$<br>$\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,1,0)<br>(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,1,0)<br>(4,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,1,0)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))+$<br>$\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))+\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,1,0)<br>(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,1,0)<br>(5,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,1,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,3,1,0)<br>(6,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,4,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_2)+S))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,4,0,0)<br>(4,3,1,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_2)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,4,0,0)<br>(5,3,1,0)(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_2+\psi_{S_2}(\sigma S+S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,4,0,0)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_2\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,4,0,0)<br>(6,5,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_3}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,4,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_3}(\sigma S)))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_3}(\sigma S)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,4,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_3}(\sigma S))+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,4,1,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_3}(\sigma S)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,4,1,0)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_3}(\sigma S)+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,4,1,0)<br>(5,4,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+$<br>$\psi_{S_3}(\sigma S)\cdot 2))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+$<br>$\psi_{S_3}(\sigma S)\cdot 2))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,4,1,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_3}(\sigma S+S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega)))$<br>$=\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\alpha_{\beta+1}\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega)+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(4,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(4,3,1,0)<br>(5,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(4,3,1,0)<br>(5,4,2,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(4,3,1,0)<br>(5,4,2,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega\cdot\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega\cdot\psi_S(\sigma S\cdot\omega))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,2,0,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega\cdot\psi_S(\sigma S\cdot\omega+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega\cdot S)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,0,0)<br>(4,3,1,0)(5,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega\cdot S)+\psi_{S_2}(\sigma S+S_\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,0,0)<br>(4,3,1,0)(5,4,2,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega\cdot S)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,0,0)<br>(4,3,1,0)(5,4,2,0)<br>(6,3,0,0)(5,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega\cdot(S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,0,0)<br>(4,3,1,0)(5,4,2,0)<br>(6,3,1,0)(7,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega\cdot\psi_{S_2}(\sigma S+S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,0,0)<br>(4,3,1,0)(5,4,2,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega\cdot S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+{S_\omega}^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,0,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_{\omega+1}}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_{\omega+1}}(\sigma S)))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+$<br>$\psi_{S_{\omega+1}}(\sigma S)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,1,0)<br>(6,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_{\omega\cdot 2})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,2,0)<br>(6,3,0,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\alpha_{\beta+1}\cdot\beta)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,2,0)<br>(6,3,0,0)(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\alpha_{\beta+1}\cdot\beta\cdot\omega)$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,2,0)<br>(6,3,0,0)(7,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\alpha_{\beta+1}\cdot\Omega_{\beta+1})$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,2,0)<br>(6,3,1,0)(7,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+{\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,2,0)<br>(6,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+{\alpha_{\beta+1}}^2\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\beta_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,4,0,0)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\beta_2\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,4,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\Omega_{\beta_2+1})+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+$<br>$\psi_{\beta_2}(\Omega_{\beta_2+1})+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,4,1,0)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\Omega_{\beta_2+1}+\beta_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,4,1,0)<br>(6,5,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\alpha_{\beta_2+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,4,1,0)<br>(6,5,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\alpha_{\beta_2+1})+\alpha_{\beta+1}\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\alpha_{\beta_2+1}+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,4,2,0)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\alpha_{\beta_2+1}+\beta_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,4,2,0)<br>(6,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}({\alpha_{\beta_2+1}}^2+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,4,2,0)<br>(6,5,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\beta_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,4,3,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\beta_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+$<br>$\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+$<br>$\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+1)+\alpha_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,0,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+$<br>$\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+1)+$<br>$\psi_\beta(\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+$<br>$\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+1)+$<br>$\psi_\beta(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+$<br>$\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,2,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\beta_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,2,0)(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,2,0)(3,3,3,1)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))\cdot 2+$<br>$\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))\cdot 2+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,2,0)(4,3,0,0)<br>(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2)+\psi_S(\sigma S\cdot\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,2,0)(4,3,1,0)<br>(5,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot 2)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,2,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,2,0)(5,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2)\cdot 2)+$<br>$\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2)\cdot 2)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,2,0)(5,4,2,0)<br>(6,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,2,0)(5,4,2,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,2,0)(5,4,2,0)<br>(6,5,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,2,0)(5,4,3,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 3))+\psi_{\gamma_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(2,2,2,1)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(2,2,2,1)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\alpha_\omega)$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\Omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\Omega_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,0)(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\Omega_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\alpha))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,1,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\alpha+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\Omega_{\alpha+1})+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\Omega_{\alpha+1})+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,1,1,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{\alpha_2}(\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,0,0)<br>(2,2,1,0)(3,3,2,1)<br>(4,3,2,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\alpha_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,0,0)<br>(2,2,2,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_\beta(\beta_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,0)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\beta))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,0)(4,3,0,0)<br>(3,3,3,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\beta_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,1,1)(4,2,1,0)<br>(4,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{\psi_{S_2}(\sigma S)}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,1,1)(4,2,1,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,1,1)(4,2,1,0)<br>(4,2,0,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\Omega_{\alpha+1}\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,1,1)(4,2,1,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\Omega_{\alpha+1}\cdot\omega^2))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,1,1)(4,2,1,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\alpha_2)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,1,1)(4,2,1,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\alpha_\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,1,1)(4,2,1,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)$<br>$+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,1,1)(4,2,1,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,1)(5,3,2,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,1,1)(4,2,1,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,1)(5,3,2,0)<br>(5,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\alpha)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,1,1)(4,2,1,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,1)(5,3,2,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,1,1)(4,2,1,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,1)(5,3,2,0)<br>(5,3,0,0)(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\alpha_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,1,1)(4,2,1,0)<br>(4,2,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,1,1)(4,2,1,0)<br>(4,2,0,0)(3,2,1,1)<br>(4,2,0,0)(5,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+$<br>$\Omega_{\alpha+1}\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,0,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,0,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,0,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,0,0)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)\cdot 2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)\cdot 2)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,1,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,1,0)<br>(4,3,1,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,1,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)$<br>$+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot 2)))+\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,1)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))+$<br>$\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,1)(2,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)))+\psi_S(\sigma S\cdot\omega+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,2,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(2,1,0,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\Omega_\omega))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(2,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)))\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(2,1,0,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\Omega_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,2,0)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(2,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(1,1,1,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,0,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega)))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_2)$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(2,0,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+1)+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(2,0,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\Omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(2,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,1,0)(1,1,1,1)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))+\psi_S(\sigma S\cdot\omega+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))\cdot 2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(2,1,1,0)(3,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(3,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(3,1,0,0)(2,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)\cdot 2)+$<br>$\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(3,1,0,0)(2,1,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)\cdot 2)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(3,1,0,0)(2,1,1,0)<br>(3,1,0,0)(3,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega)\cdot 2)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(3,1,0,0)(3,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)\cdot 3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(4,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))+$<br>$\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,0)<br>(4,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,0)<br>(4,2,1,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S))\cdot 2))+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,0)<br>(5,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))))+$<br>$\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,1,1,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S)+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S+\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(2,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(2,1,1,0)(3,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(2,1,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(3,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+S+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(3,1,1,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+S+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(4,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S\cdot\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(4,1,0,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S\cdot\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(4,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$S\cdot\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+S\cdot\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(4,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$S\cdot\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(4,1,1,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$S\cdot\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(4,2,0,0)(2,1,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$S^2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(4,2,0,0)(2,1,1,0)<br>(3,2,0,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^2+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(4,2,0,0)(3,1,1,0)<br>(4,2,0,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^2+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$S^2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(4,2,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^2+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(4,2,0,0)(3,2,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^2\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(4,2,0,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^S))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S))+\Omega_{\alpha+1}\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,0,0)<br>(4,3,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,3,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S))+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,3,1,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S))+\alpha_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,3,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,3,1,0)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,3,1,0)<br>(3,2,2,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,3,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,3,1,0)<br>(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,3,1,0)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,0)(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S\cdot 2))+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(2,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(2,1,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(2,1,1,0)(3,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(3,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(3,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(3,2,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+S\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+\psi_{S_2}(\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+$<br>$\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(3,2,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+1)))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,1,0,0)(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\Omega_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,1,0,0)(1,1,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_\alpha(\psi_S(\sigma S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,2,0)(5,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\alpha)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,2,0)(5,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\alpha_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,1,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,1,0,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_{S_2}(\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,1,0,0)(3,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,1,0,0)(3,2,1,0)<br>(4,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,1,0,0)(4,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,1,1,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S))))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,1,1,0)(5,2,1,0)<br>(6,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+S)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,2,0,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+S\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,2,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,2,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,2,1,0)(3,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega))+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,2,1,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,2,1,0)(4,2,0,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,2,1,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,2,1,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,2,1,0)(5,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,0,0)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_2+\psi_{S_2}(\sigma S\cdot\omega+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_2\cdot 2))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_3}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_3}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,1,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega)+$<br>$\psi_{S_2}(\sigma S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,1,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega)+$<br>$\psi_{S_2}(\sigma S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,1,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,1,0)(4,3,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,1,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,1,0)(5,2,1,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,1,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,1,0)(5,3,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,1,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega+\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(2,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(2,1,1,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega+\psi_S(\sigma S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega+S))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(3,2,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(3,2,1,0)(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(3,2,1,0)(4,3,2,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega\cdot\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega\cdot S+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega\cdot S+\psi_{S_2}(\sigma S\cdot\omega+S_\omega\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,0)(5,2,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega\cdot(S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,0)(5,2,1,0)<br>(6,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega\cdot\psi_{S_2}(\sigma S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_\omega\cdot S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+{S_\omega}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,0,0)(4,2,0,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+{S_\omega}^3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,0,0)(5,2,0,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+{S_\omega}^{S_\omega}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_{\omega+1}}(\sigma S)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_{\omega+1}}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,1,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,1,0)(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_{\omega\cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega+S_S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,0,0,0)(3,1,1,0)<br>(4,2,2,0)(5,2,2,0)<br>(6,2,0,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S))+$<br>$\psi_{\psi_S(\sigma S\cdot(\omega+1))}(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,0,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S))+\psi_S(\sigma S\cdot(\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,0,0,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S))+$<br>$\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,0,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,1,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\Omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,1,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_S(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,1,0,0)(4,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_S(\sigma S\cdot\omega)\cdot 2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,1,0,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,1,1,0)(5,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_S(\sigma S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_S(\sigma S\cdot(\omega+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,0)(5,3,2,0)<br>(6,3,0,0)(5,2,0,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_S(\sigma S\cdot(\omega+1)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,0)(5,3,2,0)<br>(6,3,0,0)(5,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,0,0)(3,2,1,0)<br>(4,3,2,0)(5,3,2,0)<br>(6,3,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,0,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,0,0)(3,2,2,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,0,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S))+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,0,0)(4,2,0,0)<br>(3,2,2,0)(4,2,2,0)<br>(5,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S))\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot(\omega+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(4,2,0,0)<br>(3,2,2,0)(4,2,2,0)<br>(5,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(4,2,0,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(4,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(5,3,0,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega)+S\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(5,3,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(5,3,1,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+S))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(5,3,1,0)<br>(6,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(5,3,1,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(5,3,1,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_{S_2}(\sigma S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(5,3,2,0)<br>(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+1))\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(5,3,2,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+1)+\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot(\omega+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(5,3,2,0)<br>(6,3,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(5,3,2,0)<br>(6,3,2,0)(7,2,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_S(\sigma S\cdot(\omega+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(5,3,2,0)<br>(6,3,2,0)(7,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot(\omega+1))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(5,3,2,0)<br>(6,3,2,0)(7,3,0,0)<br>(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot(\omega+1)+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(5,3,2,0)<br>(6,3,2,0)(7,3,0,0)<br>(5,3,2,0)(6,3,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot(\omega+1)+2)))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,1,0)(5,3,2,0)<br>(6,3,2,0)(7,3,0,0)<br>(6,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0)(4,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0)(4,1,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0)(4,1,1,0)<br>(5,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{\sigma S}(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+1))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$S+1)+\psi_S(\sigma S\cdot(\omega+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0)(4,2,0,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+$<br>$\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0)(4,2,0,0)<br>(3,2,2,0)(4,2,2,0)<br>(5,2,0,0)(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+$<br>$\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0)(4,2,0,0)<br>(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+$<br>$\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+$<br>$\psi_S(\sigma S\cdot(\omega+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0)(4,2,0,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+S))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0)(4,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+$<br>$\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0)(4,2,1,0)<br>(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+$<br>$\psi_S(\sigma S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0)(4,2,1,0)<br>(5,3,2,0)(6,3,2,0)<br>(7,3,0,0)(6,3,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0)(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0)(5,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$S+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+$<br>$\psi_S(\sigma S\cdot(\omega+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(5,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,1,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S)))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot\omega))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,1,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot\omega))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,1,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot(\omega+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,1,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot\omega))+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,1,0)<br>(4,2,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot\omega))+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,1,0)<br>(4,2,1,0)(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot\omega))+\psi_{S_2}(\sigma S\cdot(\omega+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,1,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,1,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot\omega)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,1,0)<br>(4,2,2,0)(5,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,1,0)<br>(5,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot\omega+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,1,0)<br>(5,2,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,1,0)<br>(6,3,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot\omega+S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,1,0)<br>(6,3,2,0)(7,3,2,0)<br>(8,3,0,0)(7,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)))+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1))+1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,2,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1))+1)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,2,0)<br>(4,2,0,0)(3,2,2,0)<br>(4,2,2,0)(5,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1))+1)+\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1))+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,2,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1))+1)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,2,0)<br>(4,2,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1))+1)+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,2,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1))+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1))+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1))+S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,2,0)<br>(4,2,2,0)(5,2,0,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1))+\psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,2,0)<br>(4,2,2,0)(5,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1))\cdot 2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,2,0)<br>(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_S(\sigma S\cdot(\omega+1))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,2,0)<br>(5,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,2,2,0)(5,2,2,0)<br>(5,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+S_2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,3,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+S_2\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,3,0,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_3}(\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,3,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_3}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,3,1,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_3}(\sigma S\cdot\omega)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,3,1,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_3}(\sigma S\cdot\omega+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,3,1,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_3}(\sigma S\cdot\omega+S_3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_3}(\sigma S\cdot(\omega+1))+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_3}(\sigma S\cdot(\omega+1))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,3,2,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_3}(\sigma S\cdot(\omega+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,3,2,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0)<br>(4,3,3,0) | $\psi(\psi_S(\sigma S\cdot(\omega+1)+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2)+\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(3,2,2,0)(4,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2)+\psi\psi_S(\sigma S\cdot(\omega+2))(\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(3,2,2,0)(4,3,3,1)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2)+\psi_S(\sigma S\cdot(\omega+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(3,2,2,0)(4,3,3,1)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2)+$<br>$\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(3,2,2,0)(4,3,3,1)<br>(4,3,3,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2)+$<br>$\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2)\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_{\psi_S(\sigma S\cdot(\omega+1))}(\psi_S(\sigma S\cdot\omega\cdot 2+$<br>$\psi_S(\sigma S\cdot\omega))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot(\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,0,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,0,0)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,0,0)(3,2,2,1)<br>(4,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega))\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,0,0)(4,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,0,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,0,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,1,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,1,0)(4,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,1,0)(4,1,0,0)<br>(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,1,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,1,0)(5,1,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,1,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,1,1,0)(5,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+S_\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot(\omega+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,0,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,0,0)(3,2,2,0)<br>(4,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{\psi_S(\sigma S\cdot(\omega+2))}(\psi_S(\sigma S\cdot\omega\cdot 2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,0,0)(3,2,2,0)<br>(4,3,3,1)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot(\omega+2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,0,0)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega\cdot 2)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,0,0)(4,2,0,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega\cdot 2)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,0,0)(5,2,0,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega\cdot 2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,1,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,1,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,1,0)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,1,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,1,0)(4,2,0,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,1,0)(4,2,0,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega\cdot 2+S)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,1,0)(4,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,1,0)(5,2,0,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega\cdot 2+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega\cdot 2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,1,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,1,0)(5,3,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,1,0)(5,3,1,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,1,0)(5,3,1,0)<br>(6,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,1,0)(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,1,0)(5,3,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)))+$<br>$\psi_{\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2)))}$<br>$(\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)))+$<br>$\psi_{\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2)))}$<br>$(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(4,3,3,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)))+$<br>$\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(4,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)))+\psi_S(\sigma S\cdot\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(5,3,0,0)(4,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(5,3,0,0)(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+$<br>$\psi_S(\sigma S\cdot\omega\cdot 2+S)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(5,3,1,0)(6,4,2,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+$<br>$\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega+S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(5,3,2,0)(4,3,3,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+$<br>$\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))))+$<br>$\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(6,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+$<br>$\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(6,3,0,0)(4,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+$<br>$\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+$<br>$\psi_S(\sigma S\cdot\omega\cdot 2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(6,4,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(6,4,2,0)(4,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))\cdot 2)+$<br>$\psi_S(\sigma S\cdot\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(6,4,2,0)(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))\cdot 2+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(6,4,2,0)(7,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(6,4,2,0)(7,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(6,4,2,0)(7,4,2,0)<br>(4,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1))))+\psi_S(\sigma S\cdot\omega\cdot 2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(6,4,2,0)(7,5,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(6,4,3,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,2,0)<br>(6,4,3,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+2)))+$<br>$\psi_{\psi_S(\sigma S\cdot(\omega+3)+\psi_{S_2}(\sigma S\cdot(\omega+3)))}$<br>$(\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,0)<br>(4,3,3,1)(5,3,3,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+2)))+$<br>$\psi_{\psi_S(\sigma S\cdot(\omega+3)+\psi_{S_2}(\sigma S\cdot(\omega+3)))}$<br>$(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+2)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,1)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2))+\psi_S(\sigma S\cdot\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,1)<br>(4,2,0,0)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2))+$<br>$\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,1)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2))+$<br>$\psi_S(\sigma S\cdot\omega\cdot 2+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,1)<br>(4,2,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2))+$<br>$\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,1)<br>(4,2,1,0)(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2))+$<br>$\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)))+$<br>$\psi_{\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2)))}$<br>$(\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,1)<br>(4,2,2,0)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2))\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(4,2,0,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2)+\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(4,2,1,0)<br>(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2)+$<br>$\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)))+$<br>$\psi_{\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2)))}$<br>$(\psi_S(\sigma S\cdot\omega\cdot 2))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(4,2,2,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2)+$<br>$\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(5,3,2,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(5,3,2,0)<br>(6,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(5,3,2,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(5,3,3,0) | $\psi(\psi_S(\sigma S\cdot(\omega\cdot 2+1)+\psi_{S_2}(\sigma S\cdot(\omega\cdot 2+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,0)(5,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega\cdot 3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(1,1,1,0)<br>(2,2,2,1)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_\alpha(\psi_S(\sigma S\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(1,1,1,0)<br>(2,2,2,1)(3,2,2,1)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2)+\alpha)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(1,1,1,0)<br>(2,2,2,1)(3,2,2,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_\beta(\psi_S(\sigma S\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(1,1,1,1)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(1,1,1,1)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_{\psi_S(\sigma S\cdot(\omega+1))}(\psi_S(\sigma S\cdot\omega^2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{\psi_S(\sigma S\cdot(\omega+1))}$<br>$(\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot(\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot(\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(3,2,2,0)<br>(4,3,3,1)(5,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_{\psi_S(\sigma S\cdot(\omega+2))}(\psi_S(\sigma S\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(3,2,2,0)<br>(4,3,3,1)(5,3,3,1)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot(\omega+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(1,1,1,1)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2)\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\Omega_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_\alpha(\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_\alpha(\psi_S(\sigma S\cdot\omega^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\alpha))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega)))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_{\psi_S(\sigma S\cdot(\omega+1))}(\psi_S(\sigma S\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_{\psi_S(\sigma S\cdot(\omega+1))}(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,1,0,0)(3,1,1,0)<br>(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_{\psi_S(\sigma S\cdot(\omega+1))}(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega))+$<br>$\psi_{\psi_S(\sigma S\cdot(\omega+1))}(\psi_S(\sigma S\cdot\omega\cdot 2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot(\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,1,0,0)(3,2,2,0)<br>(4,3,3,1)(5,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega))+\psi_{\psi_S(\sigma S\cdot(\omega+2))}(\psi_S(\sigma S\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,1,0,0)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,1,0,0)(3,2,2,1)<br>(4,2,2,0)(5,3,3,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega\cdot 2+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,1,0,0)(3,2,2,1)<br>(4,2,2,0)(5,3,3,1)<br>(6,3,3,1)(6,1,0,0)<br>(5,3,3,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega\cdot 3))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,1,0,0)(3,2,2,1)<br>(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,1,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,1,0,0)(4,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,1,0,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,1,1,0)(5,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,1,1,0)(5,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{\psi_S(\sigma S\cdot(\omega+1))}(\psi_S(\sigma S\cdot\omega\cdot 2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot(\omega+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,2,0,0)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,1)<br>(1,1,1,1)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega^2))+\psi_S(\sigma S\cdot\omega^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega^2)+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(4,2,0,0)<br>(4,1,0,0)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(4,2,0,0)<br>(4,1,0,0)(3,2,2,1)<br>(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(4,2,0,0)<br>(4,1,0,0)(4,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(4,2,0,0)<br>(4,1,0,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(4,2,0,0)<br>(4,2,0,0)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega^2)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_S(\sigma S\cdot\omega^2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(3,2,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S))+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(3,2,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(3,2,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(3,2,1,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega^2+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(3,2,1,0)<br>(3,2,1,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S)))+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(3,2,1,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega^2+$<br>$\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega^2+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S)\cdot 2)+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(3,2,1,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S+S_\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot 2))+$<br>$\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot 2))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,2,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega))+$<br>$\psi_{\psi_S(\sigma S\cdot(\omega+1))}(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,2,1,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,2,1,0)(3,2,2,1)<br>(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,2,1,0)(4,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,2,1,0)(4,2,0,0)<br>(3,2,2,1)(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,2,1,0)(4,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,2,1,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega)+S))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,2,1,0)(5,3,1,0)<br>(6,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,2,1,0)(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot(\omega+1)))+$<br>$\psi_{\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2)))}(\psi_S(\sigma S\cdot\omega^2+$<br>$\psi_{S_2}(\sigma S\cdot(\omega+1)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,2,2,0)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega^2+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)+$<br>$\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)+$<br>$\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,1,0)(1,1,1,1)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,1,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+S_2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot\omega^2+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,0)(4,2,2,0)<br>(5,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+\psi_{S_2}(\sigma S\cdot(\omega^2+1)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,0)(4,3,1,0)<br>(1,1,1,1)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+\psi_{S_3}(\sigma S\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,0)(4,3,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+\psi_{S_3}(\sigma S\cdot\omega^2)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,0)(4,3,1,0)<br>(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+\psi_{S_3}(\sigma S\cdot\omega^2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,0)(4,3,1,0)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+\psi_{S_3}(\sigma S\cdot\omega^2+S_3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,0)(4,3,1,0)<br>(5,4,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+\psi_{S_3}(\sigma S\cdot\omega^2+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,0)(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+\psi_{S_3}(\sigma S\cdot(\omega^2+1))+$<br>$\psi_{S_2}(\sigma S\cdot(\omega^2+1)+\psi_{S_3}(\sigma S\cdot(\omega^2+1))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,0)(4,3,3,0) | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot(\omega^2+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot(\omega^2+\omega)+\psi_{S_2}(\sigma S\cdot(\omega^2+\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,0)<br>(5,3,3,1) | $\psi(\psi_S(\sigma S\cdot(\omega^2+\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot(\omega^2\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega^3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega^3)+\psi_S(\sigma S\cdot\omega^2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(4,2,2,1)(4,2,0,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot(\omega^3)+\psi_S(\sigma S\cdot(\omega^2+\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,1)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega^3)+\psi_S(\sigma S\cdot(\omega^3))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,1)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega^3)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot(\omega^3)+\psi_{S_2}(\sigma S\cdot(\omega^3))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,1)<br>(2,1,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega^3)+\psi_{S_2}(\sigma S\cdot(\omega^3))+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(\omega^3)+\psi_{S_2}(\sigma S\cdot(\omega^3)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,1)<br>(2,1,1,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot(\omega^3)+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(2,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot\omega^3\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot\omega^\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0) | $\psi(\psi_S(\sigma S\cdot\Omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,0) | $\psi(\psi_S(\sigma S\cdot\Omega_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(4,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\alpha))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,2,2,1)(4,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,1)(5,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot\psi_\beta(\psi_S(\sigma S\cdot\alpha_\omega))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot \omega + S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot \omega + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega))+$<br>$\psi_{\psi_S(\sigma S \cdot (\omega+1))}(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot (\omega + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,1,0,0)(3,2,2,1) | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot \omega \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,1,0,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega)) \cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,1,0,0)(4,2,0,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega) + S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,1,0,0)(4,2,2,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot \omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,1,0,0)(4,2,2,0)<br>(5,3,3,1) | $\psi(\psi_S(\sigma S \cdot (\psi_S(\sigma S \cdot \omega) + \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,1,0,0)(4,2,2,1) | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega + 1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,1,0,0)(6,2,0,0) | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega + S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,1,1,0)(6,2,2,1)<br>(7,2,2,1) | $\psi(\psi_S(\sigma S \cdot \psi_{\psi_S(\sigma S \cdot (\omega+1))}(\psi_S(\sigma S \cdot \omega^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,0,0) | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot (\omega + 1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,0,0)(3,2,2,1) | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega \cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1)(2,1,1,1) | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,0,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S \cdot S) + \psi_S(\sigma S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,0,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,0,0) | $\psi(\psi_S(\sigma S \cdot S + \Omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S \cdot S + \psi_S(\sigma S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S \cdot S + \psi_S(\sigma S \cdot \omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S + \psi_S(\sigma S \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S \cdot S + \psi_S(\sigma S \cdot S + 1)))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot S + S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S)) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S)) + \psi_S(\sigma S \cdot S + S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,0,0,0) | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S))+$<br>$\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S)) + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S)) + S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,1,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) + S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,0) | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) + S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1) | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot (\psi_S(\sigma S \cdot S) + \omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot (\psi_S(\sigma S \cdot S) + \psi_S(\sigma S \cdot \omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) \cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) \cdot 2) + 1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S)\cdot 2)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(4,2,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S)\cdot 2)+$<br>$\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(4,2,1,0)(5,3,1,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S)\cdot 2)+$<br>$\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S)+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(4,2,1,0)(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S)\cdot 2)+$<br>$\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S)+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S)\cdot 2)+$<br>$\psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot S)+1)))+$<br>$\psi_{\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot S)+1))\cdot 2)}(\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S)\cdot 2)+$<br>$\psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot S)+1)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(4,2,2,0)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S)\cdot 2)+$<br>$\psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot S)+\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(4,2,2,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S))\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(4,2,2,0)(5,3,2,0)<br>(6,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S)\cdot 2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(4,2,2,0)(5,3,3,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S)\cdot 2+S_\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(4,2,2,0)(5,3,3,1)<br>(6,3,3,1)(7,1,0,0)<br>(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S)\cdot 3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(4,2,2,1)(5,1,0,0)<br>(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+\psi_S(\sigma S\cdot S)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,0,0)<br>(6,2,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,1,0)<br>(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,1,0)<br>(6,2,2,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S)+S_\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,1,1,0)<br>(6,2,2,1) | $\psi(\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot S)+1)))}$<br>$(\psi_S(\sigma S\cdot\psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot S)+\omega)))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot S)+1))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,0,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot S)+\omega))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,1)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+1)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+1)+$<br>$\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+1)+$<br>$\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+1))+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+1)+\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,0,0)(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S+1)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,1)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,1)(3,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+S)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+S)+$<br>$\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+S))+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+S)+\psi_{S_2}(\sigma S\cdot S)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,0,0)<br>(4,2,2,1)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+S)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+S\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S)))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,0,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,0,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+$<br>$\psi_{S_2}(\sigma S\cdot\psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,1,0)<br>(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,1,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+\psi_{S_2}(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))))))+$<br>$\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+$<br>$\psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))+\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,1,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))))+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,1,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))))+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,1,0)<br>(4,2,2,0)(5,3,3,1)<br>(6,3,3,1)(7,3,0,0)<br>(6,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot S))+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))))+\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,1,0)<br>(4,2,2,0)(5,3,3,1)<br>(6,3,3,1)(7,3,1,0)<br>(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))))\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,1,0)<br>(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,1,0)<br>(4,2,2,1)(5,2,0,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))))+\psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,1,0)<br>(4,2,2,1)(5,2,1,0)<br>(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))))\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,1,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,1,0)<br>(6,3,2,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))+S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,2,0)<br>(3,2,2,0)(4,3,3,1)<br>(5,3,3,1)(6,3,2,0)<br>(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))+1)))+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,2,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))+\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,2,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,1)(3,1,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)+\psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,1,1,1)(3,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot S+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_3}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot S+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+$<br>$\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot$<br>$\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot(S+\omega))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot$<br>$\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega))))+$<br>$\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}$<br>$(\psi_S(\sigma S\cdot S\cdot\omega))))+\psi_{\psi_S(\sigma S\cdot(S+1))}$<br>$(\psi_S(\sigma S\cdot S\cdot\omega)+\psi_S(\sigma S\cdot S+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}$<br>$(\psi_S(\sigma S\cdot S\cdot\omega))))+\psi_{\psi_S(\sigma S\cdot(S+1))}$<br>$(\psi_S(\sigma S\cdot S\cdot\omega)+\psi_S(\sigma S\cdot(S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega))))+$<br>$\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)+$<br>$\psi_{\psi_S(\sigma S\cdot(S+1)+\psi_{S_2}(\sigma S\cdot(S+1)))}$<br>$(\sigma\psi_S(\sigma S\cdot(S+1))\cdot\psi_S(\sigma S\cdot(S+1)))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,0)(6,3,3,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega))))+$<br>$\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)$<br>$+\psi_{\psi_S(\sigma S\cdot(S+1)+\psi_{S_2}(\sigma S\cdot(S+1)))}(\sigma\psi_S(\sigma S\cdot(S+1))+$<br>$\psi_S(\sigma S\cdot(S+1)+\psi_{S_2}(\sigma S\cdot(S+1)+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}$<br>$(\psi_S(\sigma S\cdot S\cdot\omega))))+\psi_{\psi_S(\sigma S\cdot(S+1))}$<br>$(\psi_S(\sigma S\cdot S\cdot\omega)+\psi_{\psi_S(\sigma S\cdot(S+1)+\psi_{S_2}(\sigma S\cdot(S+1)))}$<br>$(\psi_S(\sigma S\cdot S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1)(4,2,1,0)<br>(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(4,2,2,1)<br>(5,2,2,1)(6,2,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}$<br>$(\psi_S(\sigma S\cdot S\cdot\omega))))+\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)+$<br>$\psi_{\psi_S(\sigma S\cdot(S+1)+\psi_{S_2}(\sigma S\cdot(S+1)))}(\psi_S(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1)(4,2,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega))))+$<br>$\psi_S(\sigma S\cdot S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1)(4,2,1,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1)(4,2,1,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))+$<br>$\psi_S(\sigma S\cdot S\cdot\omega+S)))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1)(4,2,1,0)<br>(4,2,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))+$<br>$\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}$<br>$(\psi_S(\sigma S\cdot S\cdot\omega))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1)(4,2,1,0)<br>(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot$<br>$\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))+$<br>$\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}$<br>$(\psi_S(\sigma S\cdot S\cdot\omega)))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1)(4,2,1,0)<br>(5,2,0,0)(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot$<br>$\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))+$<br>$\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}$<br>$(\psi_S(\sigma S\cdot S\cdot\omega)))+\psi_S(\sigma S\cdot S\cdot\omega+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1)(4,2,1,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1)(4,2,1,0)<br>(5,3,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))+$<br>$\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1)(4,2,1,0)<br>(5,3,1,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot$<br>$\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))\cdot 2+S))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1)(4,2,1,0)<br>(5,3,1,0)(6,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_{\psi_S(\sigma S \cdot (S+1))}(\psi_S(\sigma S \cdot S \cdot \omega)) + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1)(4,2,1,0)<br>(5,3,1,0)(6,4,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_{\psi_S(\sigma S \cdot (S+1))}(\psi_S(\sigma S \cdot S \cdot \omega)) + S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1)(4,2,1,0)<br>(5,3,2,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_{\psi_S(\sigma S \cdot (S+1))}(\psi_S(\sigma S \cdot S \cdot \omega)) + S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1)(4,2,1,0)<br>(5,3,2,1) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot (\psi_{\psi_S(\sigma S \cdot (S+1))}(\psi_S(\sigma S \cdot S \cdot \omega)) + \omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1)(4,2,1,0)<br>(5,3,2,1)(6,3,2,1)<br>(7,3,0,0)(6,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot (S+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(2,1,1,1) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_{\psi_S(\sigma S \cdot (S+1) + \psi_{S_2}(\sigma S \cdot (S+1)))}(\psi_S(\sigma S \cdot S \cdot \omega) + 1))))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S+1)+\psi_{S_2}(\sigma S\cdot(S+1)+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1)(5,2,2,0)<br>(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S\cdot 2+1))}(\psi_S(\sigma S\cdot(S\cdot 2+\omega))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(1,1,1,1)(2,1,1,1)<br>(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,1)(2,1,1,1)<br>(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega))+$<br>$\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega))+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(3,2,1,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega)+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,2,0)<br>(6,3,3,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S+S_\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,2,1)<br>(4,2,2,0)(5,3,3,1)<br>(6,3,3,1)(7,3,0,0)<br>(6,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega)+\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+1)+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,1)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S+S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{\psi_{S_2}(\sigma S\cdot(S+1))}(\psi_{S_2}(\sigma S\cdot(S+\omega))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,1)(3,1,1,0)<br>(4,2,2,1)(5,2,2,1)<br>(6,2,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{\psi_{S_2}(\sigma S\cdot(S+1))}(\psi_{S_2}(\sigma S\cdot S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,1)(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega)+1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(3,1,0,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(3,1,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(3,1,1,0)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(3,1,1,0)(2,1,1,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S))+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(3,1,1,0)(2,1,1,1)<br>(3,1,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S))+\psi_{S_2}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(3,1,1,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(3,1,1,0)(3,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S)\cdot 2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(3,1,1,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S+S_2)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1)<br>(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2)+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,0,0)<br>(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_S(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,0,0)<br>(1,1,1,1)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_S(\sigma S\cdot S\cdot\omega+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_S(\sigma S\cdot S\cdot\omega+S_2+1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,0,0)<br>(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,0)<br>(1,1,1,1)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega+S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega+S_2))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega+S_2))+$<br>$\psi_S(\sigma S\cdot S\cdot\omega+S_2+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,0)<br>(2,1,1,0)(1,1,1,1)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega+S_2))+$<br>$\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega+S_2))+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,0)<br>(3,2,1,0)(1,1,1,1)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega+S_2))\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,0)<br>(3,2,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+$<br>$\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega+S_2))\cdot 2+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,0)<br>(3,2,1,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega+S_2)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+$<br>$\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega+S_2)+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot S\cdot\omega)+\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,0)<br>(3,2,2,1)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,1)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_2+2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,1)<br>(3,1,0,0)(2,1,1,0)<br>(3,2,2,1)(4,3,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + S)+$<br>$\psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,1)<br>(3,1,0,0)(2,1,1,0)<br>(3,2,2,1)(4,3,0,0)<br>(4,2,2,1)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + S) \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,1)<br>(3,1,0,0)(2,1,1,1) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,1)<br>(3,1,0,0)(2,1,1,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + S \cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,1)<br>(3,1,0,0)(4,2,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,1)<br>(3,1,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,1)<br>(3,1,1,0)(4,2,2,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2+$<br>$\psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S + S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,1)<br>(3,1,1,1) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2+$<br>$\psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,1,1,1)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2+$<br>$\psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(2,2,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 \cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,1,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 \cdot \psi_{S_2}(\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 \cdot \psi_{S_2}(\sigma S \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,1,1,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 \cdot \psi_{S_2}(\sigma S \cdot S + S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,1,1,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 \cdot \psi_{S_2}(\sigma S \cdot S + S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,1,1,0)<br>(4,2,2,1)(5,3,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 \cdot \psi_{\psi_{S_2}(\sigma S \cdot (S+1))}(\psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_2\cdot\psi_{S_2}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_2}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_2}^2+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_2}^2+\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_2}^2+\psi_{S_2}(\sigma S\cdot S\cdot\omega+{S_2}^2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(2,1,1,1)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_2}^2+\psi_{S_2}(\sigma S\cdot S\cdot\omega+{S_2}^2+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_2}^2+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(2,2,0,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_2}^2+S_2\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(2,2,0,0)(3,1,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_2}^2+S_2\cdot\psi_{S_2}(\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(2,2,0,0)(3,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_2}^2+S_2\cdot\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(2,2,0,0)(3,1,1,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_2}^2+S_2\cdot\psi_{S_2}(\sigma S\cdot S+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(2,2,0,0)(3,1,1,0)<br>(4,2,2,1)(5,3,0,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_2}^2+S_2\cdot$<br>$\psi_{\psi_{S_2}(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega+S^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(2,2,0,0)(3,1,1,0)<br>(4,2,2,1)(5,3,0,0)<br>(6,3,0,0)(5,3,0,0)<br>(6,2,2,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_2}^2+S_2\cdot\psi_{S_2}(\sigma S\cdot(S+1)))+$<br>$\psi_{\psi_S(\sigma S\cdot(S+2)+\psi_{S_2}(\sigma S\cdot(S+2)))}$<br>$(\psi_S(\sigma S\cdot S\cdot\omega+{S_2}^2+S_2\cdot\psi_{S_2}(\sigma S\cdot(S+1)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(2,2,0,0)(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_2}^2+S_2\cdot\psi_{S_2}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(2,2,0,0)(3,1,1,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_2}^2+S_2\cdot\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(2,2,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 \cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(3,1,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 \cdot \psi_{S_2}(\sigma S \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(3,1,1,1) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,2,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^{S_2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,3,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,3,0,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S))+$<br>$\psi_\alpha(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,3,1,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)) + \Omega_{\alpha+1} \cdot \omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,0,0)(3,3,1,1) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,0)<br>(2,2,2,1)(3,3,1,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)) + 1) + \alpha)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,0)<br>(2,2,2,1)(3,3,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)) + \alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,0)<br>(2,2,2,1)(3,3,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)) + \psi_S(\sigma S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,0)<br>(2,2,2,1)(3,3,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,0)<br>(2,2,2,1)(3,3,1,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S) + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,0)<br>(2,2,2,1)(3,3,1,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S + 1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,0)<br>(2,2,2,1)(3,3,1,0)<br>(4,4,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,0)<br>(2,2,2,1)(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot 2))+$<br>$\psi_{\beta_2}(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot 2))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega))+$<br>$\psi_{\psi_S(\sigma S\cdot(\omega+1))}(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,1,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,1,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,1,0)(4,3,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,1,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,1,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega+S_3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot(\omega+1)))+$<br>$\psi_{\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+1)))}$<br>$(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot(\omega+1)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,1)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,2,0)(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,1)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot\psi_S(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(1,1,1,1)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega+S_2))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,0,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,1,1,0)<br>(1,1,1,1)(2,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)+\psi_{S_2}$<br>$(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)+\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,1,1,0)<br>(3,2,2,1)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)+$<br>$\psi_{\psi_{S_2}(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega+S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,1,1,1)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)+\psi_{S_2}(\sigma S\cdot S\cdot\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,1,1,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,1,1,1)<br>(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,1,1,1)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,1,1,1)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,2,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)+{S_2}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,2,0,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)+\psi_{S_3}(\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)+\psi_{S_3}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_3}(\sigma S\cdot S)+\psi_{S_3}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)+\psi_{S_3}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(2,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,1,1,0)<br>(4,2,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+$<br>$\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,1,1,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,2,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+S_2\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+\psi_{S_3}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,2,1,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+\psi_{S_3}(\sigma S)))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_3}(\sigma S\cdot S+\psi_{S_3}(\sigma S)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+\psi_{S_3}(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+\psi_{S_3}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,2,1,0)<br>(2,2,1,0)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+\psi_{S_3}(\sigma S\cdot S))\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,2,1,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+\psi_{S_3}(\sigma S\cdot S)+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,2,1,0)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+\psi_{S_3}(\sigma S\cdot S)\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,2,1,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+\psi_{S_3}(\sigma S\cdot S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,2,1,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+\psi_{S_3}(\sigma S\cdot S+S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+S_3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,3,0,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+S_3\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,3,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+\psi_{S_4}(\sigma S\cdot S))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,3,1,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+S_4)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,3,2,0)<br>(4,3,0,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+\psi_{S_{\omega+1}}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,3,2,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+S_{\omega^2})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,3,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{\psi_{S_3}(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot(S+\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,3,2,1)<br>(4,4,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+$<br>$\psi_{\psi_{S_3}(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,3,2,1)<br>(4,4,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot(S+1)))+$<br>$\psi_{\psi_S(\sigma S\cdot(S+2)+\psi_{S_2}(\sigma S\cdot(S+2)))}$<br>$(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot(S+1)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,3,2,1)<br>(4,4,2,0)(3,3,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot(S+\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,0)(3,3,2,1)<br>(4,4,2,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,1,1,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+$<br>$\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega))+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,1,1,1)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,1,1,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,1,1,1)<br>(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,1,1,1)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_2))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,1,1,1)<br>(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,2,0,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+\psi_{S_3}(\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+\psi_{S_3}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,2,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+\psi_{S_3}(\sigma S\cdot S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,2,1,0)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+$<br>$\psi_{S_3}(\sigma S\cdot S+\psi_{S_3}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+\psi_{S_3}(\sigma S\cdot S+S_3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+$<br>$\psi_{S_3}(\sigma S\cdot S+\psi_{S_4}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+\psi_{S_3}(\sigma S\cdot S+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,2,1,0)<br>(3,3,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+$<br>$\psi_{\psi_{S_3}(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot(S+\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(2,2,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,0,0)<br>(2,2,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_2)+\psi_{S_3}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_2\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,0)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,0)<br>(4,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,0)<br>(4,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_3}(\sigma S\cdot S+\psi_{S_3}(\sigma S\cdot S)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+S_3))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,0)<br>(4,3,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_3}(\sigma S\cdot S+\psi_{S_4}(\sigma S\cdot S)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,0)<br>(4,3,1,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_3}(\sigma S\cdot S+\psi_{S_4}(\sigma S\cdot S+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{\psi_{S_3}(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot(S+\omega))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,1)<br>(2,2,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_3}(\sigma S\cdot S\cdot\omega))+\psi_{S_3}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,1)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,1)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,1)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_3}(\sigma S\cdot S\cdot\omega)+\psi_{S_3}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,1)<br>(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,2,1,1)<br>(4,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+$<br>$\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_3))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,3,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_4}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,1,1)(3,3,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_4}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,1,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,1,1,0)<br>(1,1,1,1)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega+S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,1,1,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega+S_\omega)+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,1,1,0)<br>(3,2,2,1)(4,3,3,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,1,1,1)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,1,1,1)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_3}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,2,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_3}(\sigma S\cdot S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,2,1,0)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_3}(\sigma S\cdot S+\psi_{S_3}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,2,1,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_3}(\sigma S\cdot S+S_3)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,2,1,0)<br>(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_3}(\sigma S\cdot S+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,2,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,2,1,1)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,2,1,1)<br>(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,2,1,1)<br>(3,3,2,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega\cdot\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega\cdot\psi_{S_2}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,1,1,1)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega\cdot\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega\cdot S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,0,0)<br>(2,2,1,1)(3,3,2,0)<br>(4,2,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega\cdot\psi_{S_3}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,0,0)<br>(2,2,1,1)(3,3,2,0)<br>(4,2,1,1)(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega\cdot\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,0,0)<br>(2,2,1,1)(3,3,2,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega\cdot S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_\omega}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_\omega}^2\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,0,0)<br>(3,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+{S_\omega}^3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_{\omega+1}}(S_{\omega+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_{\omega+1}}(\sigma S\cdot\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_{\omega+1}}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_{\omega+1}}(\sigma S\cdot S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,1,0)<br>(3,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_{\omega+1}}(\sigma S\cdot S+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,1,0)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_{\omega+1}}(\sigma S\cdot S+\psi_{S_{\omega+1}}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_{\omega+1}}(\sigma S\cdot S+S_{\omega+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_{\omega+1}}(\sigma S\cdot S+S_{\omega\cdot 2})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,1,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{\psi_{S_{\omega+1}}(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_{\omega+1}}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,1,1)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_{\omega+1}}(\sigma S\cdot S\cdot\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,1,1)<br>(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_{\omega+1}}(\sigma S\cdot S\cdot\omega+\psi_{S_{\omega+1}}(\sigma S\cdot S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_{\omega+2}}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_{\omega+2}}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_{\omega\cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_{\omega^2}+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(2,2,2,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_{\omega^2}+\psi_{S_{\omega+1}}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_{\omega^2}+\psi_{S_{\omega+1}}(\sigma S\cdot S\cdot\omega+S_{\omega\cdot 2})))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,0)(5,3,2,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} + \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega + S_{\omega^2})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,0)(5,3,2,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} + S_{\omega \cdot 2})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(2,2,2,0)(3,2,2,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0)(2,2,2,0)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0)(2,2,2,0)<br>(3,2,1,1) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0)(2,2,2,0)<br>(3,2,1,1)(4,3,2,0)<br>(5,3,2,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega + S_{\omega^2})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0)(2,2,2,0)<br>(3,2,1,1)(4,3,2,0)<br>(5,3,2,0)(5,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega +$<br>$\psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0)(2,2,2,0)<br>(3,2,1,1)(4,3,2,0)<br>(5,3,2,0)(5,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega + S_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0)(2,2,2,0)<br>(3,2,1,1)(4,3,2,0)<br>(5,3,2,0)(5,2,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega + S_{\omega \cdot 2}))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0)(2,2,2,0)<br>(3,2,1,1)(4,3,2,0)<br>(5,3,2,0)(5,2,0,0)<br>(4,3,2,0)(5,3,2,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot (S_\omega + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0)(2,2,2,0)<br>(3,2,1,1)(4,3,2,0)<br>(5,3,2,0)(5,2,0,0)<br>(5,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0)(2,2,2,0)<br>(3,2,1,1)(4,3,2,0)<br>(5,3,2,0)(5,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot \psi_{S_{\omega+1}}(\sigma S \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0)(2,2,2,0)<br>(3,2,1,1)(4,3,2,0)<br>(5,3,2,0)(5,2,1,1) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0)(2,2,2,0)<br>(3,2,1,1)(4,3,2,0)<br>(5,3,2,0)(5,3,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0)(2,2,2,0)<br>(3,2,1,1)(4,3,2,0)<br>(5,3,2,0)(5,3,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_{\omega \cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0)(2,2,2,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_{\omega^2}}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \varepsilon_{S_{\omega^2+1}}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega^2+1}}(\sigma S \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,1,1) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega^2+1}}(\sigma S \cdot S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^3}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,1,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_S \cdot \omega))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,1,0,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_{S\cdot\omega}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,1,0,0)(3,2,2,0)<br>(4,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_{S^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,1,0,0)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_{\varepsilon_{S+1}}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,1,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_{\psi_{S_2}(\sigma S\cdot S)}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,1,1,1) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_{\psi_{S_2}(\sigma S\cdot S\cdot\omega)}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_{S_2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S)+\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S)+$<br>$\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S)+\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S)+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S)+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,0)(5,3,2,0)<br>(6,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))))))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,0)(5,3,2,0)<br>(6,3,0,0)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0)<br>(4,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S$<br>$+\psi_S(\sigma S\cdot(S\cdot\omega+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot S))+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot S)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,0)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot S+S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,0)(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot S+S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,1)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,1)(5,3,0,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))))\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,1)(5,3,0,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))+S))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,1)(5,3,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))+\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,1)(5,3,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,1)(5,3,1,1)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,1)(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,1)(5,3,2,0)<br>(6,3,2,0)(7,3,0,0)<br>(6,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,1,1)(5,3,2,0)<br>(6,3,2,0)(7,3,1,1)<br>(8,4,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+1))+\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+1)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+1)+\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+1)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+1)+\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0)(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+1)+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0)(3,2,1,1)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0)(3,2,2,0)<br>(4,2,1,1)(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0)(3,2,2,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))\cdot 2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0)(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+S_2+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S_2)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,0,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+S_2+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S_2)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,0,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+S_2+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S_2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,0,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+S_2\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,0,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(S_3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,1,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,1,1)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1))+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,2,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1))+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,2,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1)+1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,2,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1)+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,2,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1)))+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1)))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,2,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(3,3,3,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega))+$<br>$\psi_S(\sigma S\cdot(S\cdot\omega+2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+2)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega))\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,0,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega)+\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,0)(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,0)(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+\omega))+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,0)(5,4,3,0)<br>(6,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+\omega)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,0)(5,4,3,0)<br>(6,5,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,0)(5,4,4,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega)+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,0)(5,4,4,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,1)(4,3,0,0)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega^2)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,1)(4,3,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega^3)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,1)(5,3,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,1)(5,3,0,0)<br>(4,3,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+S)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+S)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,1)(5,3,0,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+S)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+S)+\psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,1)(5,3,3,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+S)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+S)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+S)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,1)(5,3,3,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+S)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,1)(5,3,3,0)<br>(6,4,4,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+S)+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,1)(5,3,3,0)<br>(6,4,4,1)(7,4,4,1)<br>(8,4,0,0)(7,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+S\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,3,3,1)(5,3,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega\cdot 2)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega)+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,0)<br>(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1)))+$<br>$\psi_{\psi_S(\sigma S\cdot(S\cdot\omega+2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+2)))}$<br>$(\psi_S(\sigma S\cdot(S\cdot\omega+\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,0)<br>(3,3,3,1)(4,4,4,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1)))+$<br>$\psi_{\psi_S(\sigma S\cdot(S\cdot\omega+2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+2)))}$<br>$(\psi_S(\sigma S\cdot(S\cdot\omega^2))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,0)<br>(3,3,3,1)(4,4,4,1)<br>(4,1,0,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1)))+$<br>$\psi_S(\sigma S\cdot(S\cdot\omega+2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,0)<br>(3,3,3,1)(4,4,4,1)<br>(4,1,0,0)(3,3,3,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1)))+$<br>$\psi_S(\sigma S\cdot(S\cdot\omega+2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+2)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,0)<br>(3,3,3,1)(4,4,4,1)<br>(4,1,0,0)(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1)))+$<br>$\psi_S(\sigma S\cdot(S\cdot\omega+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,0)<br>(3,3,3,1)(4,4,4,1)<br>(4,1,0,0)(3,3,3,1)<br>(4,4,4,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1)))+$<br>$\psi_S(\sigma S\cdot(S\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,0)<br>(3,3,3,1)(4,4,4,1)<br>(4,1,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,0)<br>(3,3,3,1)(4,4,4,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,0)<br>(3,3,3,1)(4,4,4,1)<br>(4,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,0)<br>(3,3,3,1)(4,4,4,1)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+2)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+2)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,0)<br>(3,3,3,1)(4,4,4,1)<br>(4,3,0,0)(3,3,3,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+2)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+2)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,0)<br>(3,3,3,1)(4,4,4,1)<br>(4,3,0,0)(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,0)<br>(3,3,3,1)(4,4,4,1)<br>(4,3,0,0)(3,3,3,1)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot S\cdot\omega\cdot 2+S_2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,0)<br>(3,3,3,1)(4,4,4,1)<br>(4,3,0,0)(3,3,3,1)<br>(4,4,4,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot S\cdot\omega\cdot 2+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega^2))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega^2)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(1,1,1,1)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2)))+$<br>$\psi_S(\sigma S\cdot(S\cdot\omega^2)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(2,1,1,0)(1,1,1,1)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2)))+$<br>$\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2)))+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+$<br>$\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2)))+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,1,0)(1,1,1,1)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2)))\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,1,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,1,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2))+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,1,0)(4,2,1,0)<br>(1,1,1,1)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2))+$<br>$\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2))))))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,1,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2))+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2))+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,2,0)(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2))+S_{\omega^2})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,1)<br>(5,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,3,3,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,3,3,0)<br>(5,4,4,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,3,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,3,3,1)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)))+$<br>$\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,2,1)(4,3,3,1)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,0)<br>(3,2,2,1)(4,3,3,1)<br>(4,2,2,0)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+S)\cdot 2))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+S\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,1,0,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,1,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,1,1,0)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,1,1,0)(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S+S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,1,1,0)(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S+S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,1,1,1)(2,1,1,1)<br>(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,1,1,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$<br>$\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,2,2,0)(4,3,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega+\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,2,2,0)(4,3,3,1)<br>(5,4,4,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,2,2,0)(4,3,3,1)<br>(5,4,4,1)(5,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,2,2,0)(4,3,3,1)<br>(5,4,4,1)(5,3,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)))+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+1)))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,2,2,1)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2))+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,2,2,1)(2,1,1,1)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2))+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,2,2,1)(2,1,1,1)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2))\cdot 2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,2,2,1)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,2,2,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,1,1,1)<br>(3,2,2,1)(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,0,0)<br>(2,1,1,1)(3,2,2,1)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+S_2+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$<br>$S_2+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,0,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+S_2\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+{S_2}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,0,0)<br>(3,3,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,0)<br>(1,1,1,1)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,0)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S)+1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,0)<br>(2,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,0)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S+\psi_{S_3}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S+S_3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,0)<br>(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,0)<br>(3,3,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{\psi_{S_3}(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,0)<br>(3,3,2,1)(4,4,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{\psi_{S_3}(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot(S\cdot\omega^2)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,1)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,1)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,1)<br>(3,2,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,1)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,1)<br>(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,1)<br>(3,3,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,1,1)<br>(3,3,2,1)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+S_\omega\cdot S_2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+{S_\omega}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_{\omega+1}}(S_{\omega+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+S_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+S_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,2,2,0)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2+1)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,2,2,0)(4,2,0,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2+1)+S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,2,2,0)(4,2,0,0)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2+1)+\psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,2,2,0)(4,2,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2+1)+\psi_{S_2}(\sigma S\cdot S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,2,2,0)(4,2,1,1)<br>(5,3,2,1)(5,3,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,2,2,0)(4,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2+1))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2+1)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2+1)+$<br>$\psi_{S_3}(\sigma S\cdot(S\cdot\omega^2+1))+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2+1)+$<br>$\psi_{S_3}(\sigma S\cdot(S\cdot\omega^2+1))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,3,2,0)(4,4,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2+1)+S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,3,3,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2+1)+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,3,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,3,3,1)(4,3,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2+\omega^2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,3,3,1)(4,3,3,1)<br>(5,3,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,3,3,1)(4,3,3,1)<br>(5,3,3,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot(\omega^2+\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,0)<br>(3,3,3,1)(4,4,4,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^3)+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^3)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^3)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1)<br>(2,1,1,1)(3,2,2,1)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^3)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^3)+$<br>$\psi_{S_2}(\sigma S\cdot(S\cdot\omega^3)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^3)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^3)+\psi_{S_3}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,1,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^3)+\psi_{S_3}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,1,1)(3,3,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^3)+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,1,1)(3,3,2,1)<br>(3,3,2,1) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^3)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega^3))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,1,1)(3,3,2,1)<br>(3,3,2,1)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^3)+S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^3)+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^3)+S_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^3+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^3+1)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,2,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^3+1)+S_2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1) | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3 + \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,4,4,1)(4,4,4,1) | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3 \cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^4)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,0,0,0) | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S \cdot S \cdot \psi_S(\sigma S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot (S^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,0,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot (S^2) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_S(\sigma S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_S(\sigma S \cdot (S^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot (S^2) + S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S)) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot (S^2)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot (S^2)))+$<br>$\psi_S(\sigma S \cdot (S^2) + S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(2,1,1,0)<br>(1,1,1,1)(2,2,2,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot (S^2)))+$<br>$\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot (S^2))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot (S^2))) + S))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(3,2,1,0)<br>(1,1,1,1)(2,2,2,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S^2)))\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(3,2,1,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S^2))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(3,2,1,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S^2))+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S^2))+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S^2))+S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,3,1)(5,1,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot S\cdot\Omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,3,1)(5,1,0,0)<br>(1,1,1,1)(2,2,2,1)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot S\cdot\psi_S(\sigma S\cdot(S^2)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,3,1)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot(S^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,3,1)(5,2,0,0)<br>(4,0,0,0)(3,2,2,1)<br>(4,3,3,1)(5,1,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot(S^2)))+$<br>$\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot S\cdot\Omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,3,1)(5,2,0,0)<br>(4,0,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot(S^2))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot(S^2)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,3,3,1)<br>(5,2,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot(S^2)+1)+\psi_{S_2}(\sigma S\cdot(S^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,1)(2,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot(S^2)+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot(S^2)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,1)(3,1,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,1)(3,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,1)(3,1,1,1) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,1,1,1)(3,2,2,1)<br>(4,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot(S^2)+\psi_{S_2}(\sigma S\cdot(S^2)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,1,1) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_3}(\sigma S\cdot S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,1,1)(3,3,2,1) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+\psi_{S_3}(\sigma S\cdot(S^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,1,0,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S^2)+S_\omega))$ |

| BMS | Projection |
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| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2+1)+\psi_{S_2}(\sigma S\cdot(S^2+1)+S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,2,0)(3,3,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2+1)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,4,4,1)(5,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2+S\cdot\psi_S(\sigma S\cdot(S^2+1)+$<br>$\psi_{S_2}(\sigma S\cdot(S^2+1))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,4,4,1)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S^2\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,2,1)(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot(S^2\cdot\omega)+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,2,1)(2,2,2,0) | $\psi(\psi_S(\sigma S\cdot(S^2\cdot\omega)+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,2,1)(2,2,2,1) | $\psi(\psi_S(\sigma S\cdot(S^2\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(2,2,2,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S^3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S^\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot(S^S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S\cdot\psi_{S_2}(\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,1,0) | $\psi(\psi_S(\sigma S\cdot\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S\cdot\psi_{S_2}(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\psi_{S_2}(\sigma S\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,1,0)<br>(2,0,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S\cdot\psi_{S_2}(\sigma S\cdot S)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S\cdot\psi_{S_2}(\sigma S\cdot S)+S_2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,1,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,4,4,1)(5,3,3,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S) \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,1,0)<br>(2,2,2,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S + S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,1,0)<br>(2,2,2,1)(3,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,1,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S + S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,1,0)<br>(4,2,2,1) | $\psi(\psi_S(\sigma S \cdot \psi_{\psi_{S_2}(\sigma S \cdot (S+1))}(\psi_S(\sigma S \cdot \omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,1,1) | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,1,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,1,1)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,1,1)<br>(4,2,2,1)(5,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot (S^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,1,1,1)<br>(4,2,2,1)(5,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,1,0,0)(1,1,1,1) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_S(\sigma S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,1,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_S(\sigma S \cdot S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,1,1,0)(3,2,2,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S_2) + S_\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,3,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,3,1)(5,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot (S^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,1,1,0)(3,2,2,1)<br>(4,3,3,1)(5,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,1,1,1) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S_2 + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,1,1,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S_2 + S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,1,1,1)(3,2,2,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,1,1,1)(3,2,2,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot (S^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,1,0,0)(5,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot \psi_{S_2}(\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,1,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,1,1,1) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,1,1,1)(4,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega + 1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,1,1,1)(5,2,2,1)<br>(6,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(2,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2) + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S \cdot \omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,1)(5,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot (S^2)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,1)(5,1,0,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S^2 \cdot \omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,1)(5,1,0,0)<br>(6,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot \varepsilon_{S+1}))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,1)(5,1,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S)))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,1)(5,1,1,1)<br>(6,2,2,1)(7,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S_2)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2) + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2) + S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2) \cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + 1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(4,2,1,1)(5,3,2,1)<br>(6,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_4}(\sigma S \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,1,1)<br>(4,4,2,1)(5,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_4}(\sigma S \cdot S_2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,1,1)<br>(4,4,2,1)(5,2,0,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_4}(\sigma S \cdot S_2 + \psi_{S_4}(\sigma S \cdot S_2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,1,1)<br>(4,4,2,1)(5,2,0,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_4))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega + \psi_{S_3}(\sigma S \cdot S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega + \psi_{S_3}(\sigma S \cdot S_2 + S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega + \psi_{S_3}(\sigma S \cdot S_2 + S_\omega + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,2,0,0)(4,3,2,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega +$<br>$\psi_{S_3}(\sigma S \cdot S_2 + S_\omega + \psi_{S_3}(\sigma S \cdot S_2 + S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega + S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(3,3,2,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot 2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,0,0)(3,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot S_3 + S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,0,0)(3,3,1,1)<br>(4,4,2,1)(5,2,0,0)<br>(4,4,2,0)(5,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot S_3 + \psi_{S_4}(\sigma S \cdot S_2 + S_\omega \cdot S_3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,0,0)(3,3,1,1)<br>(4,4,2,1)(5,2,0,0)<br>(4,4,2,0)(5,3,0,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot S_3 + S_4))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,0,0)(3,3,1,1)<br>(4,4,2,1)(5,2,0,0)<br>(4,4,2,0)(5,3,0,0)<br>(4,4,2,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot (S_3 + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,0,0)(3,3,1,1)<br>(4,4,2,1)(5,2,0,0)<br>(4,4,2,0)(5,3,0,0)<br>(4,4,2,0)(5,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot S_3 \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,0,0)(3,3,1,1)<br>(4,4,2,1)(5,2,0,0)<br>(4,4,2,0)(5,3,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot \psi_{S_4}(\sigma S \cdot S)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,0,0)(3,3,1,1)<br>(4,4,2,1)(5,2,0,0)<br>(4,4,2,0)(5,4,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot S_4))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,0,0)(3,3,2,0) | $\psi(\psi_S(\sigma S \cdot S_2 + {S_\omega}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,0,0)(5,4,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_{\omega+1}}(S_{\omega+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_{\omega+1}}(\sigma S \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,1,1) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_{\omega+1}}(\sigma S \cdot S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,2,0,0)(5,4,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,2,0,0)(5,4,2,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_{\omega \cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S \cdot S_2 + S_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,2,0)(5,3,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S \cdot (S_2 + 1) + \psi_{S_2}(\sigma S \cdot (S_2 + 1) + S)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,3,2,0)(5,3,2,0) | $\psi(\psi_S(\sigma S \cdot (S_2+1) + \psi_{S_2}(\sigma S \cdot (S_2+1) + \psi_{S_2}(\sigma S \cdot (S_2+1)) + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma S \cdot (S_2+1) + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,4,3,0) | $\psi(\psi_S(\sigma S \cdot (S_2+1) + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,4,3,1) | $\psi(\psi_S(\sigma S \cdot (S_2+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,4,3,1)(5,5,4,1)<br>(6,4,0,0)(5,0,0,0) | $\psi(\psi_S(\sigma S \cdot (S_2+S^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0)<br>(4,4,3,1)(5,5,4,1)<br>(6,5,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,1) | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,1)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \omega + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,1)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \omega + \psi_{S_3}(\sigma S \cdot S_2 \cdot \omega + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,2,0,0)(4,3,2,1) | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \omega + \psi_{S_3}(\sigma S \cdot S_2 \cdot \omega + \psi_{S_3}(\sigma S \cdot S_2 \cdot \omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,1)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \omega + S_3))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,1)<br>(3,3,2,0) | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \omega + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,1)<br>(3,3,2,0)(4,4,3,1) | $\psi(\psi_S(\sigma S \cdot (S_2 \cdot \omega + \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,1)<br>(3,3,2,0)(4,4,3,1)<br>(5,5,4,1)(6,5,0,0)<br>(5,5,3,1)(6,6,4,2)<br>(7,5,0,0)(6,6,4,2) | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \omega \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,1)<br>(3,3,2,1) | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \omega^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,1)<br>(4,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 \cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,1)<br>(4,1,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \psi_{S_2}(\sigma S \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(3,3,2,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S \cdot {S_2}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(4,2,0,0) | $\psi(\psi_S(\sigma S \cdot {S_2}^{S_2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S \cdot \psi_{S_3}(S_3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot \psi_{S_3}(\sigma S \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,1,1) | $\psi(\psi_S(\sigma S \cdot \psi_{S_3}(\sigma S \cdot S \cdot \omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,2,1,1)(5,3,2,1)<br>(6,2,0,0)(7,3,0,0) | $\psi(\psi_S(\sigma S \cdot \psi_{S_3}(\sigma S \cdot \varepsilon_{(} S_2 + 1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,0,0)(2,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_3 + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,0,0)(3,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_3 + S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,0,0)(3,3,1,1)<br>(4,4,2,1)(5,3,0,0)<br>(4,4,2,0) | $\psi(\psi_S(\sigma S \cdot S_3 + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,0,0)(3,3,1,1)<br>(4,4,2,1)(5,3,0,0)<br>(4,4,2,1) | $\psi(\psi_S(\sigma S \cdot S_3 \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,0,0)(3,3,1,1)<br>(4,4,2,1)(5,3,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot \psi_{S_4}(\sigma S \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,0,0)(3,3,1,1)<br>(4,4,2,1)(5,4,0,0) | $\psi(\psi_S(\sigma S \cdot S_4))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S \cdot S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(2,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(2,2,1,1)<br>(3,3,2,1)(4,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + \psi_{S_3}(\sigma S \cdot S_3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(2,2,1,1)<br>(3,3,2,1)(4,3,0,0)<br>(3,3,2,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + \psi_{S_3}(\sigma S \cdot S_\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(2,2,1,1)<br>(3,3,2,1)(4,3,0,0)<br>(3,3,2,0)(3,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(2,2,2,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + S_\omega \cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,1,1,1)<br>(4,2,2,1)(5,2,0,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + S_\omega \cdot \psi_{S_2}(\sigma S \cdot S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + S_\omega \cdot S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + {S_\omega}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + \varepsilon_{S_\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S_\omega + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + S_{\omega+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + S_{\omega\cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(4,3,2,0)(5,3,2,0) | $\psi(\psi_S(\sigma S \cdot S_\omega + S_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(4,3,2,0)(5,4,0,0) | $\psi(\psi_S(\sigma S \cdot (S_\omega + 1) + \psi_{S_2}(\sigma S \cdot (S_\omega + 1) + S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(4,3,2,0)(5,4,3,0) | $\psi(\psi_S(\sigma S \cdot (S_\omega + 1) + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(4,3,2,0)(5,4,3,1)<br>(6,5,4,1)(7,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S \cdot S_\omega \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S \cdot S_\omega \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(4,3,2,1)(5,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S \cdot {S_\omega}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,2,1,1) | $\psi(\psi_S(\sigma S \cdot \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega+1} + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega+1} + \psi_{S_{\omega+1}}(\sigma S \cdot S_{\omega+1})))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega+1} + S_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,1,1) | $\psi(\psi_S(\sigma S \cdot S_{\omega+1} + \psi_{S_{\omega+2}}(\sigma S \cdot S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega+1} + \psi_{S_{\omega+2}}(\sigma S \cdot S_{\omega+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,3,0,0)(5,4,0,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega+1} + S_{\omega+2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,3,0,0)(5,4,2,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega+1} + S_{\omega\cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,3,0,0)(5,4,2,0)<br>(6,5,0,0) | $\psi(\psi_S(\sigma S \cdot (S_{\omega+1} + 1) + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,3,0,0)(5,4,2,1) | $\psi(\psi_S(\sigma S \cdot S_{\omega+1} \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,3,0,0)(5,4,2,1)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S \cdot {S_{\omega+1}}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega+2}))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega\cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,0,0)(3,3,2,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + \psi_{S_3}(\sigma S \cdot S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,0,0)(3,3,2,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + \psi_{S_3}(\sigma S \cdot S_{\omega^2})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,0,0)(3,3,2,0)<br>(4,3,2,0)(3,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + \psi_{S_{\omega+1}}(\sigma S \cdot S_{\omega\cdot 2})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,2,0)(5,3,2,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + S_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(2,2,2,0)(3,2,2,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + S_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + \varepsilon_{(} S_{\omega^2} + 1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,2,0,0)(2,2,2,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + \psi_{S_{\omega^2+1}}(\sigma S \cdot S_{\omega^2})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,2,0,0)(4,3,2,0)<br>(5,3,2,0)(5,3,2,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + S_{\omega^3}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,2,0,0)(4,3,2,1) | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega^2+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,0,0)(4,3,2,0)<br>(5,3,2,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega^2 \cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S \cdot S_{\omega^3}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S^2 + \psi_{S_2}(\sigma S^2 + S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0) | $\psi(\psi_S(\sigma S^2 + \psi_{S_2}(\sigma S^2 + S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma S^2 + \psi_{S_2}(\sigma S^2 + \varepsilon_{S+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,2,0) | $\psi(\psi_S(\sigma S^2 + \psi_{S_2}(\sigma S^2 + \psi_{S_2}(\sigma S^2) + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,0,0) | $\psi(\psi_S(\sigma S^2 + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,0) | $\psi(\psi_S(\sigma S^2 + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(1,1,1,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_S(\sigma S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(1,1,1,1)<br>(2,1,1,1)(3,1,1,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_S(\sigma S \cdot S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(1,1,1,1)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_S(\sigma S \cdot S \cdot \omega + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(1,1,1,1)<br>(2,2,2,0)(3,3,3,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_{\psi_S(\sigma S \cdot (S \cdot \omega + 1))}(\psi_S(\sigma S \cdot (S \cdot \omega + \omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(1,1,1,1)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_S(\sigma S \cdot S \cdot \omega^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_S(\sigma S \cdot S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_S(\sigma S \cdot S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,4,4,1)(5,4,0,0)<br>(4,4,4,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_{\psi_S(\sigma S^2)}(\psi_S(\sigma S^2 \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,4,4,1)(5,4,0,0)<br>(4,4,4,1)(2,2,2,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_{\psi_S(\sigma S^2)}(\psi_S(\sigma S^2 \cdot \omega)) +$<br>$S_\omega[\psi_{\psi_S(\sigma S^2)}(\psi_S(\sigma S^2 \cdot \omega))])$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,4,4,1)(5,4,0,0)<br>(4,4,4,1)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,2,0)(5,4,3,1)<br>(6,5,4,1)(7,4,0,0)<br>(6,5,4,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_{\psi_S(\sigma S^2)}(\psi_S(\sigma S^2 \cdot \omega) + 1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,4,4,1)(5,4,0,0)<br>(4,4,4,1)(3,2,2,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_{\psi_S(\sigma S^2 + \psi_{S_2}(\sigma S^2))}(\psi_S(\sigma S^2 \cdot \omega) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,4,4,1)(5,4,0,0)<br>(4,4,4,1)(3,3,0,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_S(\sigma S^2 + \sigma S + \psi_{S_2}(\sigma S^2 + \sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,4,4,1)(5,4,0,0)<br>(4,4,4,1)(3,3,3,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_S(\sigma S^2 + \sigma S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega) \cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,0,0,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,1,0,0)<br>(1,1,1,1)(2,2,2,1)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega + \psi_S(\sigma S^2 \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega + S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,1,1,0)<br>(1,1,1,1)(2,2,2,1)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S^2 \cdot \omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S^2 \cdot \omega)) + S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,1,0)(4,3,0,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S^2 \cdot \omega) + S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,1,1,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega + \psi_{S_2}(\sigma S \cdot (\psi_S(\sigma S^2 \cdot \omega) + \omega))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,1,1,1)<br>(3,2,2,1)(4,2,0,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega + \psi_{S_2}(\sigma S \cdot S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,1,1,1)<br>(3,2,2,1)(4,2,0,0)<br>(3,2,2,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega + \psi_{S_2}(\sigma S^2 \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,2,0,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,2,1,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,2,1,1)<br>(3,3,2,1)(4,3,0,0)<br>(3,3,2,1)(3,3,0,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega + S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,2,2,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,2,2,0)<br>(3,2,2,0)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega + \sigma S + \psi_{S_2}(\sigma S^2 \cdot \omega + \sigma S + S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,2,2,0)<br>(3,2,2,0)(4,2,2,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega + \sigma S + \psi_{S_2}(\sigma S^2 \cdot \omega + \sigma S +$<br>$\psi_{S_2}(\sigma S^2 \cdot \omega + \sigma S) + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,2,2,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S^2 \cdot \omega + \sigma S + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,2,2,0)<br>(3,3,3,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega + \sigma S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,2,2,0)<br>(3,3,3,1)(4,4,4,1)<br>(5,4,0,0)(4,4,4,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,2,2,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(2,2,2,1)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S^2 \cdot \omega^3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S^2 \cdot S))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,1,1,1)<br>(4,2,2,1)(5,2,0,0)<br>(4,2,2,1) | $\psi(\psi_S(\sigma S^2 \cdot \psi_{S_2}(\sigma S^2 \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0) | $\psi(\psi_S(\sigma S^2 \cdot S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,0,0)(3,3,2,1)<br>(4,2,0,0)(3,3,2,0) | $\psi(\psi_S(\sigma S^2 \cdot S_2 + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,0,0)(3,3,2,1)<br>(4,2,0,0)(3,3,2,1) | $\psi(\psi_S(\sigma S^2 \cdot S_2 \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,0,0)(3,3,2,1)<br>(4,2,0,0)(3,3,2,1)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S^2 \cdot {S_2}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,0,0)(3,3,2,1)<br>(4,2,1,1) | $\psi(\psi_S(\sigma S^2 \cdot \psi_{S_3}(\sigma S \cdot S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,0,0)(3,3,2,1)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S^2 \cdot S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S^2 \cdot S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(2,2,2,0) | $\psi(\psi_S(\sigma S^2 \cdot S_\omega + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S^2 \cdot S_\omega + \varepsilon_{S_\omega+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,2,1)(5,2,0,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S^2 \cdot S_\omega \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,2,1)(5,3,0,0) | $\psi(\psi_S(\sigma S^2 \cdot S_{\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S^2 \cdot S_{\omega\cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0) | $\psi(\psi_S(\sigma S^2 \cdot S_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma S^3 + \psi_{S_2}(\sigma S^3 + S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,0,0) | $\psi(\psi_S(\sigma S^3 + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1) | $\psi(\psi_S(\sigma S^3 + \sigma S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S^3 \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S^\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S^S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,1,0,0)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S^S + S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,1,0,0)(2,1,1,1)<br>(3,2,2,1)(4,2,0,0)<br>(4,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S^S + \psi_{S_2}(\sigma S^S)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,1,0,0)(2,2,0,0) | $\psi(\psi_S(\sigma S^S + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,1,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S^S + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,1,0,0)(2,2,2,0)<br>(3,3,3,1) | $\psi(\psi_S(\sigma S^S + \sigma S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,1,0,0)(2,2,2,0)<br>(3,3,3,1)(4,4,4,1)<br>(5,4,0,0)(5,3,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S^S \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,1,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S^S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,1,0,0)(2,2,2,1)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S^S \cdot S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,1,0,0)(2,2,2,1)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S^{S+1} \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,1,0,0)(2,2,2,1)<br>(3,2,0,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S^{S\cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,1,0,0)(4,2,0,0) | $\psi(\psi_S(\sigma S^{\varepsilon_{S+1}}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S^{S_2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,2,0,0)(2,2,0,0) | $\psi(\psi_S(\sigma S^{S_2} + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,2,0,0)(2,2,1,1)<br>(3,3,2,1)(4,3,0,0)<br>(4,2,0,0) | $\psi(\psi_S(\sigma S^{S_2} + \psi_{S_3}(\sigma S^{S_2})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,2,0,0)(2,2,1,1)<br>(3,3,2,1)(4,3,0,0)<br>(4,2,0,0)(3,3,2,1) | $\psi(\psi_S(\sigma S^{S_2} \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,2,0,0)(2,2,1,1)<br>(3,3,2,1)(4,3,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S^{S_3}))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S^{S_\omega}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,2,0,0)(2,2,2,0)<br>(3,2,2,0)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S^{\sigma S} + \psi_{S_2}(\sigma S^{\sigma S} + S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S^{\sigma S} \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(S(\sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S_2) + \Omega_{\alpha+1} \cdot \omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0) | $\psi(\psi_S(\sigma S_2) + \psi_{\alpha_2}(\psi_S(\sigma S_2) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0) | $\psi(\psi_S(\sigma S_2) + \psi_{\alpha_2}(\psi_S(\sigma S_2) + 1) + \Omega_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,0,0)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S_2) + \psi_{\alpha_2}(\psi_S(\sigma S_2) + 1) + \psi_\alpha(\psi_S(S(\sigma S + 1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0) | $\psi(\psi_S(\sigma S_2) + \psi_{\alpha_2}(\psi_S(\sigma S_2) + 1) + \psi_\alpha(\psi_S(\sigma S_2) +$<br>$\psi_{\alpha_2}(\psi_S(\sigma S_2) + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S_2) + \psi_{\alpha_2}(\psi_S(\sigma S_2) + 1) + \psi_\alpha(\psi_S(\sigma S_2) +$<br>$\psi_{\alpha_2}(\psi_S(\sigma S_2) + 1) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S_2) + \psi_{\alpha_2}(\psi_S(\sigma S_2) + 1) + \alpha)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(2,1,0,0)(3,2,1,1) | $\psi(\psi_S(\sigma S_2) + \psi_{\alpha_2}(\psi_S(\sigma S_2) + 1) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(2,1,1,0) | $\psi(\psi_S(\sigma S_2) + \psi_{\alpha_2}(\psi_S(\sigma S_2) + 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(2,1,1,0)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S_2) + \psi_{\alpha_2}(\psi_S(\sigma S_2) + \alpha))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(2,1,1,0)(3,1,1,0) | $\psi(\psi_S(\sigma S_2)+\psi_{\alpha_2}(\psi_S(\sigma S_2)+\Omega_{\alpha+1}+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(2,1,1,0)(3,2,2,1) | $\psi(\psi_S(\sigma S_2)+\psi_{\alpha_2}(\psi_S(\sigma S_2)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S_2)+\alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(2,2,1,0) | $\psi(\psi_S(\sigma S_2)+\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\psi_S(\sigma S_2)+\Omega_{\alpha_2+1}+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S_2)+\alpha_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S_2)+\psi_S(\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(2,2,2,1)(3,3,3,1)<br>(4,3,0,0)(5,4,0,0) | $\psi(\psi_S(\sigma S_2)+\psi_S(S(\sigma S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(2,2,2,1)(3,3,3,1)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S_2)\cdot 2+\psi_{\alpha_2}(\psi_S(\sigma S_2)\cdot 2+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S_2+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S_2+\alpha))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,1,1,0) | $\psi(\psi_S(\sigma S_2+\Omega_{\alpha+1})+\psi_{\alpha_2}(\psi_S(\sigma S_2+\Omega_{\alpha+1})+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,1,1,0)(4,2,0,0) | $\psi(\psi_S(\sigma S_2 + \psi_{\alpha_2}(\alpha_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,2,0,0) | $\psi(\psi_S(\sigma S_2 + \alpha_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S_2 + \alpha_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S_2 + \psi_S(\sigma S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0) | $\psi(\psi_S(\sigma S_2 + \psi_S(\sigma S_2)) + \psi_{\alpha_2}(\psi_S(\sigma S_2 + \psi_S(\sigma S_2)) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S_2 + S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,0,0) | $\psi(\psi_S(\sigma S_2 + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,1,1)(4,4,2,1)<br>(5,4,1,0)(4,4,0,0) | $\psi(\psi_S(\sigma S_2 + S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,0) | $\psi(\psi_S(\sigma S_2 + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,0)(4,3,3,0)<br>(5,3,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S_2 + \sigma S + \psi_{S_2}(\sigma S_2 + \sigma S + S)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,0)(4,4,4,1) | $\psi(\psi_S(\sigma S_2 + \sigma S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,0)(4,4,4,1)<br>(5,5,5,1)(6,5,0,0)<br>(7,6,0,0) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(S(\sigma S + 1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,0)(4,4,4,1)<br>(5,5,5,1)(6,5,1,0) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2)) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,0)(4,4,4,1)<br>(5,5,5,1)(6,5,1,0)<br>(4,4,4,1) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2)) + \psi_S(\sigma S_2 + \sigma S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(3,3,3,0) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + 1) + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(3,3,3,1) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(4,2,1,1) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(4,3,0,0) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + S_2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(4,3,0,0)<br>(3,3,3,0) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(4,3,0,0)<br>(3,3,3,0)(4,3,3,0)<br>(5,3,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S)+$<br>$\psi_{S_2}(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S) + S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(4,3,0,0)<br>(3,3,3,0)(4,4,0,0) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S) + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(4,3,0,0)<br>(3,3,3,0)(4,4,4,1) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S) + \sigma S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(4,3,0,0)<br>(3,3,3,1) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(4,3,0,0)<br>(3,3,3,1)(4,3,0,0) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S + S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(4,3,0,0)<br>(3,3,3,1)(4,3,0,0)<br>(3,3,3,1) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S \cdot 2 + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(4,3,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(4,3,0,0)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \varepsilon_{\sigma S+1})))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(4,3,1,0) | $\psi(\psi_S(\sigma S_2+\psi_{S(\sigma S+1)}(\sigma S_2+\psi_{S(\sigma S+1)}(\sigma S_2)))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S_2+\psi_{S(\sigma S+1)}(\sigma S_2+$<br>$\psi_{S(\sigma S+1)}(\sigma S_2)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S_2+\psi_{S(\sigma S+1)}(\sigma S_2+$<br>$\psi_{S(\sigma S+1)}(\sigma S_2+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(4,3,0,0)(3,3,3,1) | $\psi(\psi_S(\sigma S_2+\psi_{S(\sigma S+1)}(\sigma S_2+\psi_{S(\sigma S+1)}(\sigma S_2+\sigma S)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S_2+S(\sigma S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(5,4,1,0) | $\psi(\psi_S(\sigma S_2+\psi_{S(\sigma S+2)}(\sigma S_2))+$<br>$\psi_{\alpha_2}(\psi_S(\sigma S_2+\psi_{S(\sigma S+2)}(\sigma S_2))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(5,4,1,0)(3,3,3,0) | $\psi(\psi_S(\sigma S_2+\psi_{S(\sigma S+2)}(\sigma S_2)+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(5,4,1,0)(3,3,3,1) | $\psi(\psi_S(\sigma S_2+\psi_{S(\sigma S+2)}(\sigma S_2)+$<br>$\psi_{S(\sigma S+1)}(\sigma S_2+\psi_{S(\sigma S+2)}(\sigma S_2)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(5,4,1,0)(5,4,0,0) | $\psi(\psi_S(\sigma S_2+\psi_{S(\sigma S+2)}(\sigma S_2)+S(\sigma S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(5,4,2,0) | $\psi(\psi_S(\sigma S_2+S(\sigma S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,1,0)<br>(5,4,2,1) | $\psi(\psi_S(\sigma S_2\cdot 2)+\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,2,0) | $\psi(\psi_S(\sigma S_2\cdot 2)+\psi_{\beta_2}(\psi_S(\sigma S_2\cdot 2)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S_2\cdot\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,1)(2,1,1,1) | $\psi(\psi_S(\sigma S_2 \cdot \omega^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,1)(2,2,2,1)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot \psi_S(S(\sigma S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,1)(2,2,2,1)<br>(3,2,1,0) | $\psi(\psi_S(\sigma S_2 \cdot \psi_S(\sigma S_2)) + \psi_{\alpha_2}(\psi_S(\sigma S_2 + \psi_S(\sigma S_2)) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,0,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,0)(3,3,3,1) | $\psi(\psi_S(\sigma S_2 \cdot S+\sigma S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,4,4,1)(5,4,0,0)<br>(6,5,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S+\psi_{S(\sigma S+1)}(S(\sigma S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,4,4,1)(5,4,3,0)<br>(3,3,3,1) | $\psi(\psi_S(\sigma S_2 \cdot S+\psi_{S(\sigma S+1)}(\sigma S_2 \cdot \psi_S(\sigma S_2 \cdot S+\sigma S \cdot \omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,4,4,1)(5,4,3,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S+\psi_{S(\sigma S+1)}(\sigma S_2 \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S_2 \cdot S+\psi_{S(\sigma S+1)}(\sigma S_2 \cdot S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(2,2,2,1) | $\psi(\psi_S(\sigma S_2 \cdot S+\psi_{S(\sigma S+1)}(\sigma S_2 \cdot S+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S+\psi_{S(\sigma S+1)}(\sigma S_2 \cdot S+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1) | $\psi(\psi_S(\sigma S_2 \cdot S+\psi_{S(\sigma S+1)}(\sigma S_2 \cdot S+\sigma S)+\sigma S \cdot \omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0)(3,3,3,1)<br>(4,4,4,1)(5,4,3,0)<br>(4,4,4,1) | $\psi(\psi_S(\sigma S_2 \cdot S+ \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \sigma S) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(S(\sigma S + 1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,1,0) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S+ \psi_{S(\sigma S+1)}(\sigma S_2))) + \psi_{\alpha_2}(\psi_S(\sigma S_2 \cdot S+ \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2))) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,2,0)<br>(3,3,3,1)(4,3,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2))) + \alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,2,0)<br>(3,3,3,1)(4,3,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2)) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,2,0)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,0) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2)) + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,2,0)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2) + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,2,0)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(4,3,0,0)<br>(5,4,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S+ \psi_{S(\sigma S+1)}(\sigma S_2) + \varepsilon_{\sigma S+1})))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,2,0)<br>(3,3,3,1)(4,3,1,0)<br>(3,3,3,1)(4,3,1,0)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S+ \psi_{S(\sigma S+1)}(\sigma S_2) \cdot 2)) + \alpha_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)<br>(3,3,3,1)(4,3,2,0)<br>(3,3,3,1)(4,3,1,0)<br>(5,4,2,0) | $\psi(\psi_S(\sigma S_2 \cdot S+ \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 + S(\sigma S + \omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(1,1,1,1) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot \omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S) + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,2,2,1)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S+ \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S) + \varepsilon_{\sigma S+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + 1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S+ \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \sigma S) + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S+ \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S + S(\sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+2)}(\sigma S_2))+ \psi_{\alpha_2}(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+2)}(\sigma S_2)) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+2)}(\sigma S_2 \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(4,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+2)}(\sigma S_2 \cdot S) + S(\sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,1,0)(5,4,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S + S(\sigma S + 2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S_2 \cdot S + S(\sigma S + \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(4,3,2,0) | $\psi(\psi_S(\sigma S_2 \cdot S + S(\sigma S + \omega) \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,3,2,0)<br>(6,3,0,0)(5,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S_2}(\sigma S_2 \cdot (S+1) + S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,4,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,0)(5,4,3,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \sigma S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,1)(5,3,2,1)<br>(6,3,0,0)(5,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \sigma S \cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,1)(5,3,2,1)<br>(6,3,0,0)(7,4,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + \varepsilon_{S+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(4,3,2,1)(5,3,2,1)<br>(6,3,2,1) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \sigma S \cdot S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(2,2,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \sigma S \cdot S \cdot \omega + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(2,2,2,0)(3,3,3,1) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \sigma S \cdot (S \cdot \omega + \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(2,2,2,0)(3,3,3,1)<br>(4,4,4,1) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \sigma S \cdot S \cdot \omega^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(2,2,2,1)(3,2,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + S_\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(2,2,2,1)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + \varepsilon_{\sigma S+1}))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + S(\sigma S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + S(\sigma S+1) \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+2)}(\sigma S_2 \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,1,1) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+2)}(\sigma S_2 \cdot (S+1)) + \sigma S \cdot S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,1,1)(2,2,2,1) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+2)}(\sigma S_2 \cdot (S+1)) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+2)}(\sigma S_2 \cdot (S+1)) + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,1,1)(4,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+2)}(\sigma S_2 \cdot (S+1)) + S(\sigma S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,1,1)(5,4,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + S(\sigma S+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + S(\sigma S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,0)(5,3,2,0)<br>(6,3,0,0)(5,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + \psi_{S_2}(\sigma S_2 \cdot (S+2) + S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,0)(5,4,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,0)(5,4,3,1) | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + \sigma S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + \sigma S \cdot S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(2,2,2,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + \sigma S \cdot S \cdot \omega + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(2,2,2,1) | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+2)+1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(3,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+2) +$<br>$\psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+2)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(4,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + S(\sigma S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(4,3,2,0) | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + S(\sigma S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(4,3,2,1) | $\psi(\psi_S(\sigma S_2 \cdot (S+3) + \sigma S \cdot S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,1,0,0)<br>(2,0,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S \cdot 2+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,1,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,1,1)(5,4,2,1)<br>(6,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S \cdot 2 + \psi_{S_3}(\sigma S_2 \cdot S \cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,1,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S_2 \cdot S \cdot 2 + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,1,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S_2 \cdot S \cdot 2 + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S \cdot 2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,1,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S \cdot 2 + S(\sigma S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,1,0,0)<br>(4,3,2,1)(5,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S \cdot 3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,1,1,0) | $\psi(\psi_S(\sigma S_2 \cdot \psi_{S_2}(\sigma S)) + \psi_{\alpha_2}(\psi_S(\sigma S_2 \cdot \psi_{S_2}(\sigma S))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,1,1)(5,4,2,1)<br>(6,2,0,0)(3,3,2,0) | $\psi(\psi_S(\sigma S_2 \cdot S_2 + S_\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,1,1)(5,4,2,1)<br>(6,2,0,0)(5,4,2,0) | $\psi(\psi_S(\sigma S_2 \cdot S_2 + S(\sigma S + \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,1,1)(5,4,2,1)<br>(6,2,0,0)(5,4,2,1)<br>(6,2,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S_2 \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(2,2,1,1)(3,3,2,1)<br>(4,3,1,1)(5,4,2,1)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S_3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S_2 \cdot S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S_2 \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot \sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot \sigma S + S(\sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(4,3,2,1)(5,2,0,0) | $\psi(\psi_S(\sigma S_2 \cdot (\sigma S + S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot \varepsilon_{\sigma S+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S(\sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S(\sigma S + 1) + S(\sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S(\sigma S + 1) + \psi_{S(\sigma S+2)}(\sigma S_2 \cdot S)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,3,0,0)(5,4,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S(\sigma S+1)+S(\sigma S+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,3,0,0)(5,4,2,0) | $\psi(\psi_S(\sigma S_2 \cdot S(\sigma S+1)+S(\sigma S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,3,0,0)(5,4,2,1)<br>(6,3,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S(\sigma S+1)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S_2 \cdot S(\sigma S+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S_2 \cdot S(\sigma S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma {S_2}^2+\sigma S\cdot S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,2,1)(4,3,2,0) | $\psi(\psi_S(\sigma {S_2}^2+S(\sigma S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,2,1)(5,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma {S_2}^2+\sigma S_2\cdot S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(5,0,0,0) | $\psi(\psi_S(\sigma {S_2}^2\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(6,4,0,0) | $\psi(\psi_S(S(\sigma S_2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S_3\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0) | $\psi(\psi_S(\sigma S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(1,1,1,1)(2,2,2,1)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\sigma S_\omega)+\psi_S(\varepsilon_{\sigma S+1}))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S_\omega + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma S_\omega + S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S_\omega + S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,0)(3,3,3,1) | $\psi(\psi_S(\sigma S_\omega + \sigma S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_\omega + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_\omega + S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_\omega + \sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_\omega + \varepsilon_{\sigma S+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,1) | $\psi(\psi_S(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_\omega +$<br>$\psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1))) + \sigma S \cdot S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,1)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1)) + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,1)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + 1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_\omega +$<br>$\psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + S(\sigma S+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,0,0,0) | $\psi(\psi_S(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_\omega + \psi_{S(\sigma S+1)}(\varepsilon_{\sigma S_2+1}))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0) | $\psi(\psi_S(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_\omega + \psi_{S(\sigma S+1)}(\sigma S_\omega + 1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S_\omega + S(\sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S_\omega + S(\sigma S + \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0)<br>(4,3,2,0)(5,4,3,1) | $\psi(\psi_S(\sigma S_\omega + \sigma S_2 + \sigma S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0)<br>(4,3,2,1)(5,0,0,0) | $\psi(\psi_S(\sigma S_\omega + \sigma S_2 \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0)<br>(4,3,2,1)(5,3,0,0)<br>(6,4,0,0) | $\psi(\psi_S(\sigma S_\omega + \varepsilon_{\sigma S_2+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0)<br>(4,3,2,1)(5,3,1,1)<br>(6,4,2,1)(7,4,2,0) | $\psi(\psi_S(\sigma S_\omega + \psi_S(\sigma S_2 + 1)(\sigma S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(2,2,2,1)(3,2,2,0) | $\psi(\psi_S(\sigma S_\omega \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma S_\omega \cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S_\omega \cdot S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S + 2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S + \sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0) | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(5,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S +$<br>$\psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S) + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(5,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S +$<br>$\psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S + 1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(5,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + S(\sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(5,2,0,0)<br>(4,3,2,1)(5,0,0,0) | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + \sigma S_2 \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(5,2,0,0)<br>(4,3,2,1)(5,3,2,0) | $\psi(\psi_S(\sigma S_\omega \cdot (\sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(5,2,0,0)<br>(4,3,2,1)(5,3,2,0)<br>(5,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S \cdot 2 + \psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S \cdot 2 + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(5,2,0,0)<br>(5,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S \cdot S_\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(5,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S_\omega \cdot \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(5,3,0,0) | $\psi(\psi_S(\sigma S_\omega \cdot S(\sigma S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(5,3,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma S_\omega \cdot S(\sigma S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(5,3,0,0)<br>(4,3,2,1) | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S_2 + \sigma S \cdot S \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,2,0) | $\psi(\psi_S(\sigma {S_\omega}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,2,0)(2,2,2,1) | $\psi(\psi_S(\sigma {S_\omega}^2 + \psi_{S(\sigma S+1)}(\sigma {S_\omega}^2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,2,0)(2,2,2,1)<br>(3,2,2,0) | $\psi(\psi_S(\sigma {S_\omega}^2 + \sigma S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,2,0)(3,2,0,0) | $\psi(\psi_S(\sigma {S_\omega}^2 + \sigma S_\omega \cdot S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(3,2,0,0) | $\psi(\psi_S(\sigma {S_\omega}^2 \cdot S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(3,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\sigma {S_\omega}^2 \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma {S_\omega}^2 \cdot \sigma S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(3,2,0,0)<br>(2,2,2,1)(3,2,2,0) | $\psi(\psi_S(\sigma {S_\omega}^3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(S(\sigma S_\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,0) | $\psi(\psi_S(\sigma S_{\omega+1}) + \psi_{\alpha_2}(\psi_S(\sigma S_{\omega+1})+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,0)(1,1,1,1) | $\psi(\psi_S(\sigma S_{\omega+1}\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S_{\omega+1}\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,0)(2,2,2,1)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S_{\omega+1}\cdot S+\sigma S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,0)(2,2,2,1)<br>(3,2,2,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S_{\omega+1}\cdot S+\varepsilon_{\sigma S_\omega+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,0)(3,0,0,0) | $\psi(\psi_S(\sigma S_{\omega+1}\cdot S+\psi_S(\sigma S_\omega+1)(\sigma S_{\omega+1}\cdot S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\sigma S_{\omega+1}\cdot S+\psi_S(\sigma S_\omega+1)(\sigma S_{\omega+1}\cdot S+\varepsilon_{\sigma S_\omega+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,0)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\sigma S_{\omega+1}\cdot S+\psi_S(\sigma S_\omega+1)(\sigma S_{\omega+1}\cdot S+$<br>$\psi_S(\sigma S_\omega+1)(\sigma S_{\omega+1}\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,0)(4,3,0,0) | $\psi(\psi_S(\sigma S_{\omega+1}\cdot S+S(\sigma S_\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,0)(4,3,2,0) | $\psi(\psi_S(\sigma S_{\omega+1}\cdot S+S(\sigma S_\omega+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1) | $\psi(\psi_S(\sigma S_{\omega+1}\cdot(S+1)+\sigma S\cdot S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(2,2,2,1) | $\psi(\psi_S(\sigma S_{\omega+1}\cdot(S+1)+\psi_{S(\sigma S+1)}(\sigma S_{\omega+1}\cdot(S+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(2,2,2,1)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S_{\omega+1}\cdot(S+1)+\sigma S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(3,0,0,0) | $\psi(\psi_S(\sigma S_{\omega+1}\cdot(S+1)+$<br>$\psi_S(\sigma S_\omega+1)(\sigma S_{\omega+1}\cdot(S+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(3,2,0,0)<br>(2,2,2,1)(3,2,2,0) | $\psi(\psi_S(\sigma S_{\omega+1}\cdot(S+1)+$<br>$\psi_S(\sigma S_\omega+1)(\sigma S_{\omega+1}\cdot(S+1)+\sigma S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(4,0,0,0) | $\psi(\psi_S(\sigma S_{\omega+1}\cdot(S+1)+\psi_S(\sigma S_\omega+1)(\sigma S_{\omega+1}\cdot(S+1)+$<br>$\psi_S(\sigma S_\omega+1)(\sigma S_{\omega+1}\cdot(S+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(4,3,0,0) | $\psi(\psi_S(\sigma S_{\omega+1}\cdot(S+1)+S(\sigma S_\omega+1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(4,3,2,0) | $\psi(\psi_S(\sigma S_{\omega+1} \cdot (S+1) + S(\sigma S_\omega + \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,0,0,0) | $\psi(\psi_S(\sigma S_{\omega+1} \cdot (S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,1,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S_{\omega+1} \cdot \psi_{S_2}(\sigma S \cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma S_{\omega+1} \cdot S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S_{\omega+1} \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_{\omega+1} \cdot \sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,2,0,0)(2,2,2,1)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S_{\omega+1} \cdot \sigma S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,0,0) | $\psi(\psi_S(\sigma S_{\omega+1} \cdot S(\sigma S_\omega + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,0,0)(4,3,2,0) | $\psi(\psi_S(\sigma S_{\omega+1} \cdot S(\sigma S_\omega + \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,0,0)(4,3,2,1)<br>(5,0,0,0) | $\psi(\psi_S(\sigma {S_{\omega+1}}^2 + \sigma S_{\omega+1} \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0) | $\psi(\psi_S(\sigma S_{\omega \cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,2,0)(2,2,2,1)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S_{\omega^2} + \sigma S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,2,0)(2,2,2,1)<br>(3,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0)<br>(5,3,2,0)(4,3,2,0) | $\psi(\psi_S(\sigma S_{\omega^2} + S(\sigma S_\omega + \omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,2,0)(2,2,2,1)<br>(3,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0)<br>(5,3,2,0)(4,3,2,1)<br>(5,3,2,0) | $\psi(\psi_S(\sigma S_{\omega^2} + \sigma S_{\omega\cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,2,0)(2,2,2,1)<br>(3,2,2,0)(3,2,2,0) | $\psi(\psi_S(\sigma S_{\omega^2} \cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,2,0)(3,2,0,0)<br>(2,2,2,1)(3,2,2,0) | $\psi(\psi_S(\sigma S_{\omega^2} \cdot \sigma S_{\omega}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,2,0)(3,2,0,0)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(5,3,2,0)<br>(5,3,0,0)(4,3,2,1)<br>(5,3,2,0) | $\psi(\psi_S(\sigma S_{\omega^2} \cdot \sigma S_{\omega\cdot 2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,2,0)(3,2,0,0)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S_{\omega^2}{}^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,2,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(S(\sigma S_{\omega^2} + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(3,2,2,0)(3,2,2,0) | $\psi(\psi_S(\sigma S_{\omega^3}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma SS))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(2,2,2,0) | $\psi(\psi_S(\sigma SS_{\omega}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(2,2,2,1) | $\psi(\psi_S(\sigma S\sigma S + \psi_{S(\sigma S+1)}(\sigma S\sigma S + 1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(2,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(6,2,0,0)<br>(4,3,2,1)(5,0,0,0) | $\psi(\psi_S(\sigma S\sigma S + \sigma S_2 \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(2,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(6,2,0,0)<br>(5,3,2,0) | $\psi(\psi_S(\sigma S\sigma S \cdot \omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(2,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(6,3,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\sigma SS(\sigma S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(2,2,2,1)<br>(3,2,2,0) | $\psi(\psi_S(\sigma S\sigma S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S)+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0)<br>(2,2,2,1)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S)+\sigma S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S+S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,0,0)<br>(2,2,2,1)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S+\sigma S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,0,0)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S+\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S+S(\sigma\sigma S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot(\sigma\sigma S+1))\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,1)(5,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot(\sigma\sigma S+1))\cdot(S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(6,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot(\sigma\sigma S+1)+S(\sigma\sigma S+1))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot(\sigma\sigma S+\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0)<br>(6,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot(\sigma\sigma S+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S+S(\sigma\sigma S+1))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0)<br>(6,3,0,0)(5,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0)<br>(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot\omega+\sigma S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,2,0)(4,2,0,0)<br>(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot\omega+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot\omega+S(\sigma\sigma S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot(\sigma\sigma S\cdot\omega+1))\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot(\sigma\sigma S\cdot\omega+\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(6,3,0,0)<br>(5,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot(\omega+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0)<br>(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot\omega^2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0)<br>(4,2,0,0)(2,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S^2\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S^\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1)))+$<br>$\psi_{\alpha_2}(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(2,2,2,1)<br>(3,2,2,0)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(2,2,2,1)<br>(3,2,2,0)(4,2,0,0)<br>(5,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\varepsilon_{\sigma\sigma S+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}$<br>$(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1)))))+\psi_{\alpha_2}(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)$<br>$+\sigma S(\sigma\sigma S+1))))+1)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}$<br>$(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)\cdot\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)\cdot S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)\cdot S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,0,0)<br>(2,2,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S)\cdot\sigma S+$<br>$\psi_{S(\sigma S+1)}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S)\cdot\sigma S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,0,0)<br>(2,2,2,1)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)\cdot\sigma S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,0,0)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot$<br>$S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)\cdot\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}$<br>$(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,0,0)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S)\cdot\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}$<br>$(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S+$<br>$\psi_S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S)+1)(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}$<br>$(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S)+$<br>$S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}$<br>$(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S+$<br>$S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S)+1)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,0)<br>(4,3,0,0)(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S+$<br>$S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S)+1)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,0)<br>(4,3,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S+$<br>$\psi_S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S)+2)$<br>$(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,0)<br>(4,3,1,0)(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S+$<br>$\psi_S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S)+2)(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}$<br>$(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S))+$<br>$S(\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S+$<br>$S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S)+1)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,0)<br>(4,3,1,0)(5,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S+$<br>$\psi_S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S)+2)$<br>$(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,0)<br>(4,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S+$<br>$S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S)+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,0)<br>(4,3,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot(S+1)+\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot(S+1)+\sigma S\cdot S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,1)<br>(4,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot(S+1)+$<br>$S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)+\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,1)<br>(4,3,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot(S+2)+\sigma S\cdot S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,1)<br>(4,3,2,1)(5,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot(S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)$<br>$+\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\varepsilon_{\sigma\sigma S+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,1)<br>(4,3,2,1)(5,2,0,0)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}$<br>$(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)$<br>$\cdot S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}$<br>$(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(4,3,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)^2+\sigma S\cdot S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,1)<br>(4,3,2,1)(5,3,0,0)<br>(6,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$sigma_{S(\sigma\sigma S+1)})\cdot S+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0)<br>(6,3,0,0)(5,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0)<br>(6,3,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}$<br>$(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}$<br>$(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,2,0)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,0)(6,3,0,0)<br>(5,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}$<br>$(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+1)+\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,2,0)<br>(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}$<br>$(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,2,0)<br>(4,2,0,0)(2,2,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}$<br>$(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+\sigma S)+\psi_{S(\sigma S+1)}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+\sigma S)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}$<br>$(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,2,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma\sigma S))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(3,2,2,0)<br>(4,2,1,0)(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}$<br>$(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}$<br>$(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(5,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,0)(5,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot(S+1)+\sigma S\cdot S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,1)(5,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot(S+1)+\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,1)(5,3,2,1)<br>(6,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot(S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,1)(5,3,2,1)<br>(6,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,1)(5,3,2,1)<br>(6,3,0,0)(5,3,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))^2+\sigma S\cdot S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,1)(5,3,2,1)<br>(6,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,1,1)(5,3,2,1)<br>(6,3,2,0)(7,3,0,0)<br>(6,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(2,2,2,1)<br>(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+\sigma S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(2,2,2,1)<br>(3,2,2,0)(4,2,0,0)<br>(5,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\varepsilon_{\sigma\sigma S+1})))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,1)<br>(5,3,2,1)(6,3,2,0)<br>(7,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}$<br>$(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,1)<br>(5,3,2,1)(6,3,2,0)<br>(7,3,2,0)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)\cdot\omega)+1)\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,1)<br>(5,3,2,1)(6,3,2,0)<br>(7,3,2,0)(3,2,1,1)<br>(4,3,2,0)(5,3,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,1)<br>(5,3,2,1)(6,3,2,0)<br>(7,3,2,0)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)\cdot\omega)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,1)<br>(5,3,2,1)(6,3,2,0)<br>(7,3,2,0)(5,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,1)<br>(5,3,2,1)(6,3,2,0)<br>(7,3,2,0)(5,3,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)$<br>$+\sigma S(\sigma\sigma S+1))+\sigma S\cdot S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(2,2,2,1)<br>(3,2,2,0)(4,2,1,1)<br>(5,3,2,1)(6,3,2,0)<br>(7,3,2,0)(5,3,2,1)<br>(6,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(2,2,2,1)<br>(3,2,2,0)(4,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)\cdot 2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)\cdot S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(3,2,0,0)<br>(2,2,2,1)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)\cdot\sigma S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega+\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)$<br>$\cdot\omega+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)\cdot\omega+\sigma S(\sigma\sigma S+1))\cdot(S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0)<br>(6,3,0,0)(5,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot(\omega+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0)<br>(6,3,0,0)(7,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot(\omega+1)+\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(3,2,2,0)<br>(4,2,0,0)(2,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(3,2,2,0)<br>(4,2,0,0)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(3,2,2,0)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)^2+\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(3,2,2,0)<br>(4,2,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)^2+\sigma S(\sigma\sigma S+1))\cdot S))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(3,2,2,0)<br>(4,2,1,1)(5,3,2,1)<br>(6,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+1)^2+\sigma S(\sigma\sigma S+1))\cdot(S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(3,2,2,0)<br>(4,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)^2\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(3,2,2,0)<br>(4,2,2,0)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)^2\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(4,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)^\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)^\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,2,2,0)(4,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)^{S(\sigma\sigma S+1)}\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\varepsilon_{S(\sigma\sigma S+1)+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,1,0)(2,2,2,1)<br>(3,2,2,0)(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot\varepsilon_{S(\sigma\sigma S+1)+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,1,0)(2,2,2,1)<br>(3,2,2,0)(4,3,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,1,0)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,1,0)(3,2,1,1)<br>(4,3,2,1)(5,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\psi_S(\sigma\sigma S+2)(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1)))+\sigma S(\sigma\sigma S+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,1,0)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot\psi_S(\sigma\sigma S+2)(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,1,0)(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,1,0)(4,3,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,1,0)(5,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)+S(\sigma\sigma S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,1,0)(5,4,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)+S(\sigma\sigma S+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,1,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)+$<br>$\sigma S(\sigma\sigma S+1))\cdot(S+1)+\sigma S\cdot S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,1,0,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,4,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,2,0)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,2,0)(3,2,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)\cdot\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,2,0)(3,2,2,0)<br>(4,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot S(\sigma\sigma S+2)\cdot S(\sigma\sigma S+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,2,0)(3,2,2,0)<br>(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)\cdot\varepsilon_{S(\sigma\sigma S+1)+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,2,0)(3,2,2,0)<br>(4,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)^2\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,2,0)(4,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)^{S(\sigma\sigma S+1)}\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,2,0)(4,2,2,0)<br>(5,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)^2\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,2,0)(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\varepsilon_{S(\sigma\sigma S+2)+1})))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,2,0)(5,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,2,0)(5,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(S(\sigma\sigma S+1))\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,2,0)(5,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot\psi_{\sigma S(\sigma\sigma S+1)}(S(\sigma S(\sigma\sigma S+1)+1)^2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,2,0)(5,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$<br>$\cdot\psi_{\sigma S(\sigma\sigma S+1)}(S(\sigma S(\sigma\sigma S+1)+1)^2+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,2,0)(5,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\psi_{\sigma S(\sigma\sigma S+1)}(\sigma_{S(\sigma\sigma S+2)}))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\psi_{\sigma S(\sigma\sigma S+1)}(\sigma_{S(\sigma\sigma S+\omega)}))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,3,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)+\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(1,1,1,1)(2,2,2,1)<br>(3,2,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega)\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(2,2,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(2,2,2,1)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega+\sigma S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,3,1)(5,4,4,1)<br>(6,4,4,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega+$<br>$\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(2,2,2,1)(3,2,2,0)<br>(4,3,3,1)(5,4,4,1)<br>(6,4,4,1)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega+\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(2,2,2,1)(3,2,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,2,0,0)(2,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot S_\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2+\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2+\sigma S(\sigma\sigma S+1))\cdot(S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,0,0)(6,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2+\sigma S(\sigma\sigma S+1)+\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,3,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot 2)\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,2,2,0)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,2,2,0)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,2,2,0)(4,2,0,0)<br>(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot S(\sigma\sigma S+1))\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,2,2,0)(4,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot S(\sigma\sigma S+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,2,2,0)(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot S(\sigma\sigma S+2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,2,2,0)(4,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot S(\sigma\sigma S+2)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,2,2,0)(4,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot\psi_{\sigma S(\sigma\sigma S+1)}(\sigma_{S(\sigma\sigma S+\omega)}))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,2,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^3)\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(3,2,2,1)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^3\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(4,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(4,2,0,0)(2,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{S_\omega})))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{\sigma\sigma S})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(4,2,0,0)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{\sigma\sigma S}\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(4,2,0,0)(3,2,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{\sigma\sigma S+1})\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{\varepsilon_{\sigma\sigma S+1}})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(4,2,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{S(\sigma\sigma S+1)}+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(4,2,1,1)(5,3,2,1)<br>(6,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{S(\sigma\sigma S+1)}$<br>$+\sigma S(\sigma\sigma S+1))\cdot(S+\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(4,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{S(\sigma\sigma S+1)}\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(4,2,2,0)(3,2,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{S(\sigma\sigma S+1)+1})\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(4,2,2,0)(5,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{\psi_{\sigma S(\sigma\sigma S+1)}(\sigma_{S(\sigma\sigma S+\omega)})})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,2,2,1)<br>(4,2,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{\sigma S}(\sigma\sigma S+1))\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,0,0)<br>(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_2\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,1,0)<br>(2,2,2,1)(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)$<br>$+\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\sigma\sigma S}(\sigma\sigma S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,1,0)<br>(2,2,2,1)(3,3,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1))}(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,1,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\sigma\sigma S}(\psi_S(\sigma\sigma S_2+1)$<br>$(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S)+\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,1,0)<br>(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\sigma\sigma S}(\psi_S(\sigma\sigma S_2+1)$<br>$(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S+1))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,1,0)<br>(3,2,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\sigma\sigma S}(\psi_S(\sigma\sigma S_2+1)$<br>$(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\sigma S(\sigma\sigma S+1)))\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,1,0)<br>(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,1,0)<br>(3,3,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+S(\sigma\sigma S+1))}$<br>$(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,1,0)<br>(4,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+S(\sigma\sigma S+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,1,0)<br>(4,4,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S+$<br>$\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+S(\sigma\sigma S+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,1,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot(S+1)+\sigma S\cdot S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,1,1)<br>(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot(S+1)+$<br>$\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,1,1)<br>(4,4,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot(S+1)+$<br>$\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+S(\sigma\sigma S+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,1,1)<br>(4,4,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)$<br>$+\sigma S(\sigma\sigma S+1))\cdot(S+2)+\sigma S\cdot S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,1,1)<br>(4,4,2,1)(5,1,0,0)<br>(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)$<br>$+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(2,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)$<br>$+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1))+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(2,2,2,1)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)$<br>$+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1))+\sigma S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(2,2,2,1)(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$<br>$\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1))+\psi_{\sigma\sigma S}(\sigma\sigma S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(2,2,2,1)(3,3,1,1)<br>(4,4,2,1)(5,5,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1))+$<br>$\psi_{\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1))}(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$<br>$\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(2,2,2,1)(3,3,1,1)<br>(4,4,2,1)(5,5,2,0)<br>(4,4,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1))+$<br>$\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+S(\sigma\sigma S+1)\cdot\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(2,2,2,1)(3,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1))\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)$<br>$+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1)+\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$<br>$\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1)+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$<br>$\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(3,2,2,0)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$<br>$\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+\sigma\sigma S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(3,2,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$<br>$\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1)))\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(3,2,2,1)(4,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$<br>$\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1))+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma\sigma S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(3,3,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)\cdot 2+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(3,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)\cdot 2$<br>$+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)\cdot 2+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(4,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)\cdot\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)\cdot\sigma\sigma S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(4,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)^2+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)^2+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,0)<br>(4,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1))\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(2,1,0,0)(3,2,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1))\cdot\omega+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(2,2,2,1)(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1))\cdot\omega+\psi_{\sigma\sigma S}(\sigma\sigma S_2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1))\cdot\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(3,2,0,0)(2,2,2,1)<br>(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1))\cdot\psi_{\sigma\sigma S}(\sigma\sigma S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)+\psi_{\sigma\sigma S_2}(\sigma S(\sigma\sigma S_2+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)+\sigma\sigma S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(3,3,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)$<br>$+S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(3,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)+S(\sigma\sigma S_2+1)+$<br>$\psi_{\sigma\sigma S_2}(\sigma S(\sigma\sigma S_2+1)+S(\sigma\sigma S_2+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(3,3,2,0)(4,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)+S(\sigma\sigma S_2+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(3,3,2,0)(4,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)+S(\sigma\sigma S_2+2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(3,3,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot 2)\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(4,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(4,2,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot S(\sigma\sigma S+1)+$<br>$\sigma S(\sigma\sigma S+1))\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(4,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot S(\sigma\sigma S+1)+$<br>$\psi_{\sigma\sigma S_2}(\sigma S(\sigma\sigma S_2+1)\cdot S(\sigma\sigma S+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(4,2,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot\sigma S(\sigma\sigma S+1)+$<br>$\psi_{\sigma\sigma S_2}(\sigma S(\sigma\sigma S_2+1)\cdot\sigma S(\sigma\sigma S+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot\sigma\sigma S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(4,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot S(\sigma\sigma S_2+1)+$<br>$\psi_{\sigma\sigma S_2}(\sigma S(\sigma\sigma S_2+1)\cdot S(\sigma\sigma S_2+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(4,3,2,0)(5,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot S(\sigma\sigma S_2+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(4,3,2,0)(5,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot S(\sigma\sigma S_2+1)\cdot\sigma\sigma S_2)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(4,3,2,0)(5,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot S(\sigma\sigma S_2+2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(4,3,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)^2)\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(4,3,2,1)(3,3,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)^2+\sigma S(\sigma\sigma S_2+1))\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(4,3,2,1)(5,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)^\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(4,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma S(\sigma\sigma S_2+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(4,4,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+2))\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,2,1)<br>(4,4,2,1)(5,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(2,2,2,1)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega)+\sigma S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(2,2,2,1)(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega)+\psi_{\sigma\sigma S}(\sigma\sigma S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(2,2,2,1)(3,3,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega)+\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)+$<br>$\psi_{\sigma\sigma S_2}(\sigma S(\sigma\sigma S_2+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(2,2,2,1)(3,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega)\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(3,2,0,0)(2,2,2,1)<br>(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega+\sigma S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega+\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega+\psi_{\sigma\sigma S_2}(\sigma\sigma S_\omega+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega+\sigma\sigma S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(3,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,2,0,0)(2,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega\cdot S_\omega)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,2,0,0)(2,2,2,1)<br>(3,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega \cdot \psi_{\sigma\sigma S}(\sigma\sigma S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega \cdot \sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega \cdot S(\sigma\sigma S+1)+$<br>$\psi_{\sigma\sigma S_2}(\sigma\sigma S_\omega \cdot S(\sigma\sigma S+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,2,2,1)(5,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega \cdot \psi_{\sigma\sigma S_2}(\sigma\sigma S_\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega \cdot \sigma\sigma S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,0,0)(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega \cdot \sigma\sigma S_2 + \sigma\sigma S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,0,0)(3,3,2,1)<br>(4,4,3,0)(5,3,0,0)<br>(4,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega \cdot \sigma\sigma S_2 + \sigma\sigma S_3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,0,0)(3,3,2,1)<br>(4,4,3,0)(5,3,0,0)<br>(4,4,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega \cdot (\sigma\sigma S_2 + 1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,0,0)(3,3,2,1)<br>(4,4,3,0)(5,3,1,0)<br>(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega \cdot S(\sigma\sigma S_2 + 1) + \sigma S(\sigma\sigma S + 1)) \cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,0,0)(3,3,2,1)<br>(4,4,3,0)(5,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega \cdot \sigma\sigma S_3)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,0,0)(3,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma {S_\omega}^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,0,0)(5,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega + 1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega + 1) + \sigma S(\sigma\sigma S + 1)) \cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,1,0)(3,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega + 1) + \sigma S(\sigma\sigma S + 1)) \cdot S+$<br>$\psi_{\sigma\sigma S}(\psi_S(\sigma\sigma S_\omega + 1)(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega + 1)+$<br>$\sigma S(\sigma\sigma S + 1)) \cdot S) + \sigma\sigma S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,1,0)(5,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega + 1) + \sigma S(\sigma\sigma S + 1)) \cdot S+$<br>$\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega + 1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,1,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega + 1)+$<br>$\sigma S(\sigma\sigma S + 1)) \cdot (S + 1) + \sigma S \cdot S \cdot \omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,1,1)(5,4,2,1)<br>(6,4,0,0)(7,5,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+1)+\sigma S(\sigma\sigma S+1)+\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,2,0)(4,3,0,0)<br>(3,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+1)\cdot\sigma\sigma S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,2,0)(4,3,0,0)<br>(5,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+1)\cdot\varepsilon_{\sigma\sigma S_\omega+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,2,0)(5,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_\omega+1))\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,2,1)(3,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_\omega+1)+\sigma\sigma S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,2,1)(3,3,3,0)<br>(4,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_\omega+1)+S(\sigma\sigma S_\omega+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,2,1)(4,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_\omega+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,2,1)(5,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_\omega+1)\cdot S(\sigma\sigma S_\omega+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,2,1)(5,3,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_\omega+1)^2)\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,2,1)(5,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_{\omega+1})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_{\omega^2})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,3,0)(3,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_{\omega^2}+\sigma\sigma S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,3,0)(3,3,3,0)<br>(4,3,2,1)(5,4,3,0)<br>(6,4,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_{\omega^2}+\psi_{\sigma\sigma S_{\omega+1}}(\sigma\sigma S_{\omega^2}))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,3,0)(3,3,3,0)<br>(4,3,2,1)(5,4,3,0)<br>(6,4,3,0)(5,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_{\omega^2}+\sigma\sigma S_{\omega+1})))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,3,0)(3,3,3,0)<br>(4,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_{\omega^2}\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,3,0)(4,3,0,0)<br>(5,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_{\omega^2}+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,3,0)(4,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma_{S_{\omega^3}})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,3,3,0)(5,3,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\sigma\sigma S+\psi_{S_2}(\sigma\sigma S+S))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0)<br>(4,4,0,0) | $\psi(\psi_S(\sigma\sigma S+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1) | $\psi(\psi_S(\sigma\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(1,1,1,1)(2,2,2,1)<br>(3,3,3,0) | $\psi(\psi_S(\sigma\sigma S\cdot\omega)+\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(1,1,1,1)(2,2,2,1)<br>(3,3,3,0)(4,4,4,1)<br>(5,5,5,1)(6,6,6,1) | $\psi(\psi_S(\sigma\sigma S\cdot\omega)+\psi_{\psi_S(\sigma\sigma S)}(\psi_S(\sigma\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(1,1,1,1)(2,2,2,1)<br>(3,3,3,0)(4,4,4,1)<br>(5,5,5,1)(6,6,6,1)<br>(3,3,3,0) | $\psi(\psi_S(\sigma\sigma S\cdot\omega)+\psi_S(\sigma\sigma S+\psi_{S_2}(\sigma\sigma S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(1,1,1,1)(2,2,2,1)<br>(3,3,3,1) | $\psi(\psi_S(\sigma\sigma S\cdot\omega)\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(2,2,0,0) | $\psi(\psi_S(\sigma\sigma S\cdot\omega+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(2,2,2,0)(3,3,3,1) | $\psi(\psi_S(\sigma\sigma S\cdot\omega+\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(2,2,2,1) | $\psi(\psi_S(\sigma\sigma S\cdot\omega+\sigma S\cdot S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(2,2,2,1)(3,3,3,0) | $\psi(\psi_S(\sigma\sigma S\cdot\omega+\psi_{\sigma\sigma S}(\sigma\sigma S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(2,2,2,1)(3,3,3,1) | $\psi(\psi_S(\sigma\sigma S\cdot\omega^2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,2,0,0)(2,2,2,1)<br>(3,3,3,1) | $\psi(\psi_S(\sigma\sigma S^2)\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,2,0,0)(4,3,0,0) | $\psi(\psi_S(S(\sigma\sigma S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,2,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma S(\sigma\sigma S+1)\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,2,1,1)(4,3,2,1)<br>(5,4,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma S_2}(\sigma\sigma S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,2,2,0) | $\psi(\psi_S(\sigma\sigma S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,2,2,0)(4,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S+1)\cdot\sigma\sigma\sigma S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,2,2,0)(4,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S+1)\cdot S(\sigma\sigma\sigma S+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,2,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S+1)\cdot\sigma S(\sigma\sigma\sigma S+1))\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,2,2,1)(4,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S+1)\cdot$<br>$\psi_{\sigma\sigma S}(\sigma\sigma\sigma S+1)(\sigma\sigma S(\sigma\sigma\sigma S+\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,2,2,1)(4,3,3,1) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S+1)^2\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,2,2,1)(4,3,3,1)<br>(3,2,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S+1)^2\cdot\sigma S(\sigma\sigma\sigma S+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,2,2,1)(4,3,3,1)<br>(5,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S+1)^\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma\sigma S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,0,0)(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma\sigma S_2\cdot 2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,1,0)(2,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2+1)$<br>$+\psi_{\sigma\sigma\sigma S_2}(S(\sigma\sigma\sigma S_2+1)+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,2,0)(3,2,2,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2+1)$<br>$+\psi_{\sigma\sigma\sigma S_2}(S(\sigma\sigma\sigma S_2+1)+2))))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,2,0)(3,2,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2+1)+$<br>$\psi_{\sigma\sigma\sigma S_2}(S(\sigma\sigma\sigma S_2+1)+\sigma S(\sigma\sigma\sigma S+1)))\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,2,0)(3,2,2,1)<br>(4,3,3,1) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2+1)+$<br>$\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2+1)+\sigma\sigma S(\sigma\sigma\sigma S+1))\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,2,0)(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2+1)+\sigma\sigma\sigma S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,2,0)(4,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,2,0)(4,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2+2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,2,1) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma S(\sigma\sigma\sigma S_2+1))\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,2,1)(3,3,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma S(\sigma\sigma\sigma S_2+1)+\sigma\sigma\sigma S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,2,2,1)(4,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\psi_{\sigma\sigma S}(\sigma\sigma\sigma S_2+1)(\sigma\sigma S(\sigma\sigma\sigma S_2+\omega)))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,2,2,1)(4,3,3,1) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S_2+1))\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,2,1)(4,3,3,1)<br>(4,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S_2+1)\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma\sigma S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,3,0)(3,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma\sigma S_\omega\cdot 2))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,3,0)(4,3,0,0)<br>(5,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_\omega+1))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,3,0)(4,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma\sigma S_{\omega^2})))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,3,0)(4,4,0,0) | $\psi(\psi_S(\sigma\sigma\sigma\sigma S+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(3,3,3,1) | $\psi(\psi_S(\sigma\sigma\sigma\sigma S\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,0,0,0) | $\psi(\psi_S(\sigma^\omega S))=\psi(\psi_S(\sigma\theta S\cdot\omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,0,0,0)(3,3,0,0) | $\psi(\psi_S(\sigma\theta S\cdot\omega+\theta S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,0,0,0)(3,3,3,1) | $\psi(\psi_S(\sigma\theta S\cdot(\omega+1)+\psi_{\theta S}(\sigma\theta S\cdot(\omega+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,0,0,0)(4,0,0,0) | $\psi(\psi_S(\sigma\theta S\cdot\omega^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,1,0,0)(2,0,0,0) | $\psi(\psi_S(\sigma\theta S\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_S(\sigma\theta S\cdot\theta S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,2,0,0)(3,3,2,1)<br>(4,4,3,1)(5,2,0,0)<br>(3,0,0,0) | $\psi(\psi_S(\sigma\theta S\cdot\theta S+\psi_{\theta S_2}(\sigma\theta S\cdot\theta S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,2,0,0)(3,3,3,0) | $\psi(\psi_S(\sigma\theta S\cdot\theta S+\theta S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,2,0,0)(5,3,0,0) | $\psi(\psi_S(\sigma\theta S\cdot S(\theta S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,2,2,0) | $\psi(\psi_S(\sigma\theta S\cdot S(\theta S+1)+\psi_{\theta S_2}(\sigma\theta S\cdot S(\theta S+1)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,2,2,1) | $\psi(\psi_S(\sigma\theta S\cdot\psi_{\theta S_2}(\sigma\theta S)+\psi_{\theta S}(\sigma\theta S\cdot\psi_{\theta S_2}(\sigma\theta S)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,2,2,1)(5,3,3,1)<br>(6,0,0,0) | $\psi(\psi_S(\sigma\theta S\cdot\psi_{\theta S_2}(\sigma\theta S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,3,0,0) | $\psi(\psi_S(\sigma\theta S\cdot\theta S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,3,0,0)(3,3,2,1)<br>(4,4,3,1)(5,3,0,0)<br>(4,4,3,0) | $\psi(\psi_S(\sigma\theta S\cdot\theta S_2+\theta S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,3,0,0)(3,3,3,0) | $\psi(\psi_S(\sigma\theta S\cdot\theta S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,3,0,0)(3,3,3,1) | $\psi(\psi_S(\sigma\theta S^2+\psi_{\theta S}(\sigma\theta S^2+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,3,0,0)(5,4,0,0) | $\psi(\psi_S(S(\sigma\theta S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,3,1,0)(2,0,0,0) | $\psi(\psi_S(\sigma\theta S_2+\psi_{\theta S}(\sigma\theta S_2)\cdot S))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,3,2,1)(5,4,3,1)<br>(6,0,0,0) | $\psi(\psi_S(\sigma\theta S_2 \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,3,3,0) | $\psi(\psi_S(\sigma\theta S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,3,3,0)(4,3,3,0) | $\psi(\psi_S(\sigma\theta S_{\omega^2}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,3,3,0)(5,3,0,0)<br>(4,0,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma\theta S}(\sigma\theta S(\sigma\sigma\theta S+1)\cdot\sigma\sigma\theta S)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,3,3,0)(5,3,3,0) | $\psi(\psi_S(\psi_{\sigma\sigma\theta S}(\sigma\theta S(\sigma\sigma\theta S+1)\cdot\theta S(\sigma\sigma\theta S+1))+$<br>$\psi_{\theta S}(\psi_{\sigma\sigma\theta S}(\sigma\theta S(\sigma\sigma\theta S+1)\cdot\theta S(\sigma\sigma\theta S+1))+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,3,3,1) | $\psi(\psi_S(\psi_{\sigma\sigma\theta S}(\sigma\theta S(\sigma\sigma\theta S+1)^2)+$<br>$\psi_{\theta S}(\psi_{\sigma\sigma\theta S}(\sigma\theta S(\sigma\sigma\theta S+1)^2)+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,4,0,0) | $\psi(\psi_S(\psi_{\sigma\sigma\theta S}(\sigma\sigma\theta S_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,4,4,0) | $\psi(\psi_S(\psi_{\sigma\sigma\theta S}(\sigma\sigma\theta S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,4,4,1) | $\psi(\psi_S(\sigma\sigma\theta S+\psi_{\theta S}(\sigma\sigma\theta S+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2) | $\psi(\psi_S(\theta^\omega S))=\psi(\psi_X(\theta X\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(1,1,1,1)<br>(2,2,2,1)(3,3,3,0) | $\psi(\psi_X(\theta X\cdot\omega)+\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(1,1,1,1)<br>(2,2,2,1)(3,3,3,1)<br>(4,4,4,0) | $\psi(\psi_X(\theta X\cdot\omega)+\psi_S(\psi_{\sigma\sigma\theta S}(\sigma\sigma\theta S_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(1,1,1,1)<br>(2,2,2,2) | $\psi(\psi_X(\theta X\cdot\omega)+\psi_S(\psi_X(\theta X\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(2,1,0,0)<br>(1,1,1,1) | $\psi(\psi_X(\theta X\cdot\omega)+\psi_S(\psi_X(\theta X\cdot\omega)+\psi_S(\sigma S\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(2,1,0,0)<br>(1,1,1,1)(2,2,2,2) | $\psi(\psi_X(\theta X\cdot\omega)+\psi_S(\psi_X(\theta X\cdot\omega)+\psi_S(\psi_X(\theta X\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(2,1,0,0)<br>(3,2,0,0) | $\psi(\psi_X(\theta X\cdot\omega)+S)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(2,2,0,0) | $\psi(\psi_X(\theta X\cdot\omega)+S_2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(2,2,2,0) | $\psi(\psi_X(\theta X\cdot\omega)+S_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(2,2,2,1) | $\psi(\psi_X(\theta X\cdot\omega)+\sigma S+\psi_{\theta S}(\psi_X(\theta X\cdot\omega)+\sigma S+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(2,2,2,1)<br>(3,3,3,2) | $\psi(\psi_X(\theta X\cdot\omega)+\psi_{\theta S}(\psi_X(\theta X\cdot\omega)+\psi_{\theta S}(\psi_X(\theta X\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(2,2,2,1)<br>(3,3,3,2)(3,3,3,0) | $\psi(\psi_X(\theta X\cdot\omega)+\theta S_\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(2,2,2,2) | $\psi(\psi_X(\theta X\cdot\omega)\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,1,0,0)<br>(2,0,0,0) | $\psi(\psi_X(\theta X\cdot\omega+S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,0,0) | $\psi(\psi_X(\theta X\cdot\omega+S_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,0,0)<br>(2,2,2,0) | $\psi(\psi_X(\theta X\cdot\omega+S_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,0,0)<br>(2,2,2,1) | $\psi(\psi_X(\theta X\cdot\omega+\sigma S)+\psi_{\theta S_2}(\psi_X(\theta X\cdot\omega+\sigma S)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,0,0)<br>(2,2,2,1)(3,3,3,2) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{\theta S_2}(\psi_X(\theta X\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,0,0)<br>(2,2,2,2) | $\psi(\psi_X(\theta X\cdot\omega+\psi_X(\theta X\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,0,0)<br>(4,3,0,0) | $\psi(\psi_X(\theta X\cdot\omega+X))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,1,0) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\sigma X))+$<br>$\psi_{\alpha_2}(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\sigma X))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,1,0)<br>(2,0,0,0) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\sigma X))+$<br>$\psi_{\theta S}(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\sigma X))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,1,0)<br>(4,3,0,0) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\sigma X)+X))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,1,1) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\sigma X\cdot 2))+\sigma S\cdot S\cdot\omega)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,1,1)<br>(4,3,2,2) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X))+\psi_{\theta S_2}(\psi_X(\theta X\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X))+$<br>$\psi_{\theta S_2}(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(2,2,2,1)(3,3,3,2)<br>(4,2,2,0)(4,0,0,0) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(2,2,2,1)(3,3,3,2)<br>(4,2,2,0)(4,2,2,0) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X)\cdot 2)+$<br>$\psi_{\theta S_2}(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X)\cdot 2)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(2,2,2,1)(3,3,3,2)<br>(4,2,2,0)(5,2,0,0)<br>(3,0,0,0) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X+\theta S)))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(2,2,2,1)(3,3,3,2)<br>(4,2,2,0)(5,2,2,0) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X+\psi_{X_2}(\theta X)))+$<br>$\psi_{\theta S_2}(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X+\psi_{X_2}(\theta X)))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(2,2,2,1)(3,3,3,2)<br>(4,2,2,0)(5,3,0,0) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X+X_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(2,2,2,1)(3,3,3,2)<br>(4,2,2,1) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X+\sigma X))+$<br>$\psi_{\theta S}(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X+\sigma X))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(2,2,2,1)(3,3,3,2)<br>(4,2,2,1)(4,0,0,0) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X+\sigma X)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(2,2,2,1)(3,3,3,2)<br>(4,2,2,1)(5,0,0,0) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X+\sigma X+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(2,2,2,1)(3,3,3,2)<br>(4,2,2,1)(5,3,0,0) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X+\sigma X+X_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(2,2,2,1)(3,3,3,2)<br>(4,2,2,1)(5,3,3,2) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X\cdot 2))+\psi_{\theta\theta S}(\psi_X(\theta X\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(2,2,2,2) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(2,2,2,2)(3,2,2,0)<br>(2,2,2,1)(3,3,3,2)<br>(4,3,3,0)(3,3,3,2) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X\cdot\omega))+$<br>$\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X\cdot 2))+$<br>$\psi_{\theta\theta S}(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X\cdot\omega))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(3,0,0,0) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X\cdot\omega)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(3,2,2,0) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X\cdot\omega)+\psi_{X_2}(\theta X))+$<br>$\psi_{\theta S_2}(\psi_X(\theta X\cdot\omega+$<br>$\psi_{X_2}(\theta X\cdot\omega)+\psi_{X_2}(\theta X))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(4,0,0,0) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_2}(\theta X\cdot\omega+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(4,3,0,0) | $\psi(\psi_X(\theta X\cdot\omega+X_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(4,3,2,0)(2,2,2,2) | $\psi(\psi_X(\theta X\cdot\omega+\psi_{X_3}(\theta X\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,0)<br>(4,3,3,0) | $\psi(\psi_X(\theta X\cdot\omega+X_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,1) | $\psi(\psi_X(\theta X\cdot\omega+\sigma X)+\psi_{\theta S}(\psi_X(\theta X\cdot\omega+\sigma X)+1))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,1)<br>(3,0,0,0) | $\psi(\psi_X(\theta X\cdot\omega+\sigma X+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,1)<br>(3,2,2,0) | $\psi(\psi_X(\theta X\cdot\omega+\sigma X+\psi_{X_2}(\theta X))+$<br>$\psi_{\theta S_2}(\psi_X(\theta X\cdot\omega+\sigma X+\psi_{X_2}(\theta X))+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,1)<br>(3,2,2,0)(2,2,2,2) | $\psi(\psi_X(\theta X\cdot\omega+\sigma X+\psi_{X_2}(\theta X\cdot\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,1)<br>(4,0,0,0) | $\psi(\psi_X(\theta X\cdot\omega+\sigma X\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,1)<br>(4,3,0,0) | $\psi(\psi_X(\theta X\cdot\omega+\varepsilon_{\sigma X+1}))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,1)<br>(4,3,3,2) | $\psi(\psi_X(\theta X\cdot\omega\cdot 2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,2) | $\psi(\psi_X(\theta X\cdot\omega^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,2)<br>(2,2,2,2)(3,2,2,2) | $\psi(\psi_X(\theta X\cdot\omega^2)\cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,2)<br>(3,2,2,0)(2,2,2,2)<br>(3,2,2,2) | $\psi(\psi_X(\theta X\cdot\omega^2+\psi_{X_2}(\theta X\cdot\omega^2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,2)<br>(3,2,2,0)(4,3,0,0) | $\psi(\psi_X(\theta X\cdot\omega^2+X_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,2)<br>(3,2,2,2) | $\psi(\psi_X(\theta X\cdot\omega^3))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,2)<br>(4,1,0,0)(2,0,0,0) | $\psi(\psi_X(\theta X\cdot S))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,2)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_X(\theta X\cdot X))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,2)<br>(4,2,0,0)(3,2,2,2) | $\psi(\psi_X(\theta X\cdot X+\psi_{X_2}(\theta X\cdot X+1)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,2)<br>(4,2,2,0)(3,0,0,0) | $\psi(\psi_X(\theta X\cdot X+\psi_{X_2}(\theta X\cdot X+\psi_{X_2}(\theta X\cdot X))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,2)<br>(4,2,2,0)(5,3,0,0) | $\psi(\psi_X(\theta X\cdot X+X_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,2)<br>(4,2,2,1) | $\psi(\psi_X(\theta X\cdot X+\sigma X)+\psi_{\theta S}(\psi_X(\theta X\cdot X+\sigma X)+1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,2,2,2)<br>(4,2,2,2) | $\psi(\psi_X(\theta X\cdot X\cdot\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,0,0) | $\psi(\psi_X(\theta X\cdot X\cdot\omega+X_2))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,0) | $\psi(\psi_X(\theta X \cdot X \cdot \omega + X_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,1) | $\psi(\psi_X(\theta X \cdot X \cdot \omega + \sigma X)$<br>$+\psi_{\theta S}(\psi_X(\theta X \cdot X \cdot \omega + \sigma X) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,2) | $\psi(\psi_X(\theta X \cdot X \cdot \omega^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,2)<br>(4,2,0,0)(3,0,0,0) | $\psi(\psi_X(\theta X \cdot X^2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,2)<br>(4,3,0,0) | $\psi(\psi_X(\theta X \cdot X_2))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,2)<br>(4,3,0,0)(3,3,3,0) | $\psi(\psi_X(\theta X \cdot X_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,2)<br>(4,3,0,0)(3,3,3,1) | $\psi(\psi_X(\theta X \cdot \sigma X) + \psi_{\theta S}(\psi_X(\theta X \cdot \sigma X) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,2)<br>(4,3,0,0)(3,3,3,2) | $\psi(\psi_X(\theta X^2 \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,2)<br>(4,3,1,0)(2,0,0,0) | $\psi(\psi_X(\sigma X(\theta X + 1)) + \psi_{\theta S}(\psi_X(\sigma X(\theta X + 1)) + 1))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,2)<br>(4,3,3,0) | $\psi(\psi_X(\theta X_\omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,2)<br>(4,3,3,0)(5,3,0,0)<br>(4,0,0,0) | $\psi(\psi_X(\psi_{\theta\theta X}(\theta X(\theta\theta X + 1) \cdot \theta\theta X)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,2)<br>(4,3,3,2) | $\psi(\psi_X(\psi_{\theta\theta X}(\theta X(\theta\theta X + 1)^2 \cdot \omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,2)<br>(4,4,0,0) | $\psi(\psi_X(\psi_{\theta\theta X}(\theta\theta X_2)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,2)<br>(4,4,4,0) | $\psi(\psi_X(\psi_{\theta\theta X}(\theta\theta X_\omega)))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,2)<br>(4,4,4,2) | $\psi(\psi_X(\theta\theta X \cdot \omega))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,3) | $\psi(\psi_X(\theta^\omega X))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,3)<br>(3,3,3,3) | $\psi(\psi_A([1,1]A \cdot 2 + \psi_{[1,0]A}([1,1]A \cdot 2 + 1)) \cdot 2)$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,3)<br>(4,4,4,0) | $\psi(\psi_A([1,1]A \cdot 2 + \psi_{\psi_{[1,0]^2A}([1,1]A \cdot 2)}$<br>$(\psi_{[1,0]^2A}([1,1]A \cdot 2 + \psi_{[1,0]^3A}([1,0]^2A([1,0]^3A + 1) \cdot \omega))))))$ |
| (0,0,0,0)(1,1,1,1)<br>(2,2,2,2)(3,3,3,3)<br>(4,4,4,4) | $\psi(\psi_A([1,1]A \cdot 2 + [1,0]^\omega A))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1) | $\psi(\psi_A([1,1]A \cdot \omega))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,0,0,0,0) | $\psi(\psi_A([1,1]A\cdot\omega+1))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,1,0,0,0)<br>(3,2,0,0,0) | $\psi(\psi_A([1,1]A\cdot\omega+A))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,1,1,0,0)<br>(1,1,1,1,1) | $\psi(\psi_A([1,1]A\cdot\omega+\psi_{A_2}([1,1]A\cdot\omega)))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,1,1,0,0)<br>(3,2,2,0,0) | $\psi(\psi_A([1,1]A\cdot\omega+A_\omega))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,1,1,1,0) | $\psi(\psi_A([1,1]A\cdot\omega+\psi_{[1,0]A}([1,1]A))+$<br>$\psi_{\theta S}(\psi_A([1,1]A\cdot\omega+\psi_{[1,0]A}([1,1]A))+1))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,1,1,1,0)<br>(1,1,1,1,1) | $\psi(\psi_A([1,1]A\cdot\omega+\psi_{[1,0]A}([1,1]A\cdot\omega)))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,1,1,1,1) | $\psi(\psi_A([1,1]A\cdot\omega^2))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,1,1,1,1)<br>(3,1,0,0,0)<br>(2,0,0,0,0) | $\psi(\psi_A([1,1]A\cdot A))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,1,1,1,1)<br>(3,1,1,1,1) | $\psi(\psi_A([1,1]A\cdot A\cdot\omega))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,2,0,0,0) | $\psi(\psi_A([1,1]A\cdot A\cdot\omega+A_2))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,2,2,0,0) | $\psi(\psi_A([1,1]A\cdot A\cdot\omega+A_\omega))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,2,2,2,0) | $\psi(\psi_A([1,1]A\cdot A\cdot\omega+[1,0]^\omega A))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,2,2,2,1) | $\psi(\psi_A([1,1]A\cdot A\cdot\omega^2))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,2,2,2,1)<br>(3,2,0,0,0)<br>(2,2,2,0,0) | $\psi(\psi_A([1,1]A\cdot A_\omega))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,2,2,2,1)<br>(3,2,0,0,0)<br>(2,2,2,2,0) | $\psi(\psi_A([1,1]A\cdot[1,0]^\omega A))$ |

| BMS | Projection |
|---|---|
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,2,2,2,1)<br>(3,2,0,0,0)<br>(4,3,0,0,0) | $\psi(\psi_A(A([1,1]A+1)))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,2,2,2,1)<br>(3,2,2,0,0) | $\psi(\psi_A([1,1]A_\omega))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,2,2,2,1)<br>(3,2,2,2,0) | $\psi(\psi_A([1,1]^\omega A))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,2,2,2,1)<br>(3,3,3,3,1) | $\psi(\psi_A([1,2]A\cdot\omega))$ |
| (0,0,0,0,0)<br>(1,1,1,1,1)<br>(2,2,2,2,2) | $\psi(H^{H^{H\cdot\omega}})$ |
| (0,0,0,0,0,0)<br>(1,1,1,1,1,1) | $\psi(H^{H^{H^\omega}})$ |
| $(0)(1^\omega)=(0)(,,1)$ | $\psi(\varepsilon_{H+1})$ |

## A.16 0 − Y Sequence vs MOCF/Reflection/Stable

The results in this section are primarily cited from[32]. The reflection/stable collapsing notation used is the Madore-like version defined by Youshou Jiuxing.

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| ∅ | 0 |
| 1 | 1 |
| 1,1 | 2 |
| 1,2 | $\omega$ |
| 1,2,1 | $\omega+1$ |
| 1,2,1,2 | $\omega\cdot 2$ |
| 1,2,2 | $\omega^2$ |
| 1,2,2,2 | $\omega^3$ |
| 1,2,3 | $\omega^\omega$ |
| 1,2,3,2 | $\omega^{\omega+1}$ |
| 1,2,3,2,2 | $\omega^{\omega+2}$ |
| 1,2,3,2,3 | $\omega^{\omega\cdot 2}$ |
| 1,2,3,3 | $\omega^{\omega^2}$ |
| 1,2,3,3,3 | $\omega^{\omega^3}$ |
| 1,2,3,4 | $\omega^{\omega^\omega}$ |
| 1,2,3,4,5 | $\omega^{\omega^{\omega^\omega}}$ |
| 1,3 | $\varepsilon_0$ |
| 1,3,2 | $\varepsilon_0\cdot\omega$ |
| 1,3,2,3 | $\varepsilon_0\cdot\omega^\omega$ |
| 1,3,2,3,3 | $\varepsilon_0\cdot\omega^{\omega^2}$ |
| 1,3,2,3,4 | $\varepsilon_0\cdot\omega^{\omega^\omega}$ |
| 1,3,2,4 | $\varepsilon_0^2$ |
| 1,3,2,4,2,3,4 | $\varepsilon_0^2\cdot\omega^{\omega^\omega}$ |
| 1,3,2,4,2,4 | $\varepsilon_0^3$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,3,2,4,3 | $\varepsilon_0^{\omega}$ |
| 1,3,2,4,3,4 | $\varepsilon_0^{\omega^{\omega}}$ |
| 1,3,2,4,3,4,5 | $\varepsilon_0^{\omega^{\omega^{\omega}}}$ |
| 1,3,2,4,3,5 | $\varepsilon_0^{\varepsilon_0}$ |
| 1,3,3 | $\varepsilon_1$ |
| 1,3,3,2,4 | $\varepsilon_1 \cdot \varepsilon_0$ |
| 1,3,3,2,4,3 | $\varepsilon_1 \cdot \varepsilon_0^{\omega}$ |
| 1,3,3,2,4,3,4,5 | $\varepsilon_1 \cdot \varepsilon_0^{\varepsilon_0}$ |
| 1,3,3,2,4,4 | $\varepsilon_1^2$ |
| 1,3,3,2,4,4,3 | $\varepsilon_1^{\omega}$ |
| 1,3,3,2,4,4,3,5 | $\varepsilon_1^{\varepsilon_0}$ |
| 1,3,3,2,4,4,3,5,5 | $\varepsilon_1^{\varepsilon_1}$ |
| 1,3,3,3 | $\varepsilon_2$ |
| 1,3,4 | $\varepsilon_{\omega}$ |
| 1,3,4,2,4 | $\varepsilon_{\omega} \cdot \varepsilon_0$ |
| 1,3,4,2,4,4 | $\varepsilon_{\omega} \cdot \varepsilon_1$ |
| 1,3,4,2,4,5 | $\varepsilon_{\omega}^2$ |
| 1,3,4,3 | $\varepsilon_{\omega+1}$ |
| 1,3,4,3,3 | $\varepsilon_{\omega+2}$ |
| 1,3,4,3,4 | $\varepsilon_{\omega\cdot 2}$ |
| 1,3,4,4 | $\varepsilon_{\omega^2}$ |
| 1,3,4,5 | $\varepsilon_{\omega^{\omega}}$ |
| 1,3,4,6 | $\varepsilon_{\varepsilon_0}$ |
| 1,3,4,6,4 | $\varepsilon_{\varepsilon_0\cdot\omega}$ |
| 1,3,4,6,5 | $\varepsilon_{\varepsilon_0^{\omega}}$ |
| 1,3,4,6,6 | $\varepsilon_{\varepsilon_1}$ |
| 1,3,4,6,7 | $\varepsilon_{\varepsilon_{\omega}}$ |
| 1,3,4,6,7,9 | $\varepsilon_{\varepsilon_{\varepsilon_0}}$ |
| 1,3,5 | $\zeta_0$ |
| 1,3,5,3 | $\varepsilon_{\zeta_0+1}$ |
| 1,3,5,3,4,6 | $\varepsilon_{\zeta_0+\varepsilon_0}$ |
| 1,3,5,3,4,6,7,9 | $\varepsilon_{\zeta_0+\varepsilon_{\varepsilon_0}}$ |
| 1,3,5,3,4,6,8 | $\varepsilon_{\zeta_0\cdot 2}$ |
| 1,3,5,3,4,6,8,5,7,9 | $\varepsilon_{\zeta_0^2}$ |
| 1,3,5,3,4,6,8,6 | $\varepsilon_{\varepsilon_{\zeta_0+1}}$ |
| 1,3,5,3,4,6,8,6,7,9, 11,9 | $\varepsilon_{\varepsilon_{\varepsilon_{\zeta_0+1}}}$ |
| 1,3,5,3,5 | $\zeta_1$ |
| 1,3,5,3,5,3 | $\varepsilon_{\zeta_1+1}$ |
| 1,3,5,3,5,3,4,6,8,6, 8,6 | $\varepsilon_{\varepsilon_{\zeta_1+1}}$ |
| 1,3,5,3,5,3,5 | $\zeta_2$ |
| 1,3,5,4 | $\zeta_{\omega}$ |
| 1,3,5,4,3,5,4 | $\zeta_{\omega\cdot 2}$ |
| 1,3,5,4,4 | $\zeta_{\omega^2}$ |
| 1,3,5,4,5 | $\zeta_{\omega^{\omega}}$ |
| 1,3,5,4,6 | $\zeta_{\varepsilon_0}$ |
| 1,3,5,4,6,6 | $\zeta_{\varepsilon_1}$ |
| 1,3,5,4,6,7 | $\zeta_{\varepsilon_{\omega}}$ |
| 1,3,5,4,6,8 | $\zeta_{\zeta_0}$ |
| 1,3,5,4,6,8,7,9,11 | $\zeta_{\zeta_{\zeta_0}}$ |
| 1,3,5,5 | $\eta_0$ |
| 1,3,5,5,3,5,5 | $\eta_1$ |
| 1,3,5,5,4,5 | $\eta_{\omega^{\omega}}$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,3,5,5,4,6 | $\eta_{\varepsilon_0}$ |
| 1,3,5,5,4,6,8 | $\eta_{\zeta_0}$ |
| 1,3,5,5,4,6,8,8 | $\eta_{\eta_0}$ |
| 1,3,5,5,5 | $\varphi(4,0)$ |
| 1,3,5,6 | $\varphi(\omega,0)$ |
| 1,3,5,6,3 | $\varphi(1,\varphi(\omega,0)+1)$ |
| 1,3,5,6,3,5 | $\varphi(2,\varphi(\omega,0)+1)$ |
| 1,3,5,6,3,5,5 | $\varphi(3,\varphi(\omega,0)+1)$ |
| 1,3,5,6,3,5,6 | $\varphi(\omega,1)$ |
| 1,3,5,6,3,5,6,3 | $\varphi(1,\varphi(\omega,1)+1)$ |
| 1,3,5,6,3,5,6,3,5,6 | $\varphi(\omega,2)$ |
| 1,3,5,6,4 | $\varphi(\omega,\omega)$ |
| 1,3,5,6,4,3,5,6,4 | $\varphi(\omega,\omega\cdot 2)$ |
| 1,3,5,6,4,4 | $\varphi(\omega,\omega^2)$ |
| 1,3,5,6,4,5 | $\varphi(\omega,\omega^\omega)$ |
| 1,3,5,6,4,6 | $\varphi(\omega,\varphi(1,0))$ |
| 1,3,5,6,4,6,8 | $\varphi(\omega,\varphi(2,0))$ |
| 1,3,5,6,4,6,8,8 | $\varphi(\omega,\varphi(3,0))$ |
| 1,3,5,6,4,6,8,9 | $\varphi(\omega,\varphi(\omega,0))$ |
| 1,3,5,6,5 | $\varphi(\omega+1,0)$ |
| 1,3,5,6,5,4,6,8,9,8 | $\varphi(\omega+1,\varphi(\omega+1,0))$ |
| 1,3,5,6,5,5 | $\varphi(\omega+2,0)$ |
| 1,3,5,6,5,6 | $\varphi(\omega\cdot 2,0)$ |
| 1,3,5,6,5,6,5,6 | $\varphi(\omega\cdot 3,0)$ |
| 1,3,5,6,6 | $\varphi(\omega^2,0)$ |
| 1,3,5,6,7 | $\varphi(\omega^\omega,0)$ |
| 1,3,5,6,8 | $\varphi(\varphi(1,0),0)$ |
| 1,3,5,6,8,10 | $\varphi(\varphi(2,0),0)$ |
| 1,3,5,6,8,10,10 | $\varphi(\varphi(3,0),0)$ |
| 1,3,5,6,8,10,11 | $\varphi(\varphi(\omega,0),0)$ |
| 1,3,5,6,8,10,11,13 | $\varphi(\varphi(\varphi(1,0),0),0)$ |
| 1,3,5,6,8,10,11,13,15 | $\varphi(\varphi(\varphi(2,0),0),0)$ |
| 1,3,5,7 | $\varphi(1,0,0)$ |
| 1,3,5,7,3 | $\varphi(1,\varphi(1,0,0)+1)$ |
| 1,3,5,7,3,5 | $\varphi(2,\varphi(1,0,0)+1)$ |
| 1,3,5,7,3,5,6 | $\varphi(\omega,\varphi(1,0,0)+1)$ |
| 1,3,5,7,3,5,6,8 | $\varphi(\varphi(1,0),\varphi(1,0,0)+1)$ |
| 1,3,5,7,3,5,<br>6,8,10,12 | $\varphi(\varphi(1,0,0),1)$ |
| 1,3,5,7,3,5,6,8,10,12,3,<br>5,6,8,10,12 | $\varphi(\varphi(1,0,0),2)$ |
| 1,3,5,7,3,5,<br>6,8,10,12,4 | $\varphi(\varphi(1,0,0),\omega)$ |
| 1,3,5,7,3,5,6,<br>8,10,12,4,6 | $\varphi(\varphi(1,0,0),\varphi(1,0))$ |
| 1,3,5,7,3,5,6,8,<br>10,12,4,6,8,10 | $\varphi(\varphi(1,0,0),\varphi(1,0,0))$ |
| 1,3,5,7,3,5,6,<br>8,10,12,5 | $\varphi(\varphi(1,0,0)+1,0)$ |
| 1,3,5,7,3,5,6,<br>8,10,12,5,6 | $\varphi(\varphi(1,0,0)+\omega,0)$ |
| 1,3,5,7,3,5,6,8,<br>10,12,5,6,8 | $\varphi(\varphi(1,0,0)+\varphi(1,0),0)$ |
| 1,3,5,7,3,5,6,8,<br>10,12,5,6,8,10,12 | $\varphi(\varphi(1,0,0)\cdot 2,0)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,3,5,7,3,5,6,8, 10,12,6 | $\varphi(\varphi(1,0,0)\cdot\omega,0)$ |
| 1,3,5,7,3,5,6,8, 10,12,6,7 | $\varphi(\varphi(1,0,0)\cdot\omega^\omega,0)$ |
| 1,3,5,7,3,5,6,8, 10,12,6,8 | $\varphi(\varphi(1,0,0)\cdot\varphi(1,0),0)$ |
| 1,3,5,7,3,5,6,8, 10,12,6,8,10,12 | $\varphi(\varphi(1,0,0)^2,0)$ |
| 1,3,5,7,3,5,6,8, 10,12,7 | $\varphi(\varphi(\varphi(1,0,0)^\omega,0))$ |
| 1,3,5,7,3,5,6,8, 10,12,7,8 | $\varphi(\varphi(\varphi(1,0,0)^{\omega^\omega},0))$ |
| 1,3,5,7,3,5,6,8, 10,12,8 | $\varphi(\varphi(1,\varphi(1,0,0)+1),0)$ |
| 1,3,5,7,3,5,6,8, 10,12,8,10,11 | $\varphi(\varphi(\omega,\varphi(1,0,0)+1),0)$ |
| 1,3,5,7,3,5,7 | $\varphi(1,0,1)$ |
| 1,3,5,7,3,5,7,3,5,7 | $\varphi(1,0,2)$ |
| 1,3,5,7,4 | $\varphi(1,0,\omega)$ |
| 1,3,5,7,4,6 | $\varphi(1,0,\varphi(1,0))$ |
| 1,3,5,7,4,6,8,10 | $\varphi(1,0,\varphi(1,0,0))$ |
| 1,3,5,7,5 | $\varphi(1,1,0)$ |
| 1,3,5,7,5,4,6,8,10,8 | $\varphi(1,1,\varphi(1,1,0))$ |
| 1,3,5,7,5,5 | $\varphi(1,2,0)$ |
| 1,3,5,7,5,6 | $\varphi(1,\omega,0)$ |
| 1,3,5,7,5,6,8 | $\varphi(1,\varphi(1,0),0)$ |
| 1,3,5,7,5,6,8,10,12 | $\varphi(1,\varphi(1,0,0),0)$ |
| 1,3,5,7,5,7 | $\varphi(2,0,0)$ |
| 1,3,5,7,5,7,5,6, 8,10,12,10,12 | $\varphi(2,\varphi(2,0,0),0)$ |
| 1,3,5,7,5,7,5,7 | $\varphi(3,0,0)$ |
| 1,3,5,7,6 | $\varphi(\omega,0,0)$ |
| 1,3,5,7,6,8 | $\varphi(\varphi(1,0),0,0)$ |
| 1,3,5,7,6,8,10,12 | $\varphi(\varphi(1,0,0),0,0)$ |
| 1,3,5,7,7 | $\varphi(1,0,0,0)$ |
| 1,3,5,7,7,3,5,7,7 | $\varphi(1,0,0,1)$ |
| 1,3,5,7,7,5 | $\varphi(1,0,1,0)$ |
| 1,3,5,7,7,5,7 | $\varphi(1,1,0,0)$ |
| 1,3,5,7,7,5,7,7 | $\varphi(2,0,0,0)$ |
| 1,3,5,7,7,6 | $\varphi(\omega,0,0,0)$ |
| 1,3,5,7,7,6,8,10,12,12 | $\varphi(\varphi(1,0,0,0),0,0,0)$ |
| 1,3,5,7,7,7 | $\varphi(1,0,0,0,0)$ |
| 1,3,5,7,7,7,7 | $\varphi(1,0,0,0,0,0)$ |
| 1,3,5,7,8 | $\psi(\Omega^{\Omega^\omega})$ |
| 1,3,5,7,8,3 | $\psi(\Omega^{\Omega^\omega}+1)$ |
| 1,3,5,7,8,3,5,7,8 | $\psi(\Omega^{\Omega^\omega}\cdot 2)$ |
| 1,3,5,7,8,5 | $\psi(\Omega^{\Omega^\omega+1})$ |
| 1,3,5,7,8,5,7 | $\psi(\Omega^{\Omega^\omega+\Omega})$ |
| 1,3,5,7,8,5,7,8 | $\psi(\Omega^{\Omega^\omega\cdot 2})$ |
| 1,3,5,7,8,7 | $\psi(\Omega^{\Omega^{\omega+1}})$ |
| 1,3,5,7,8,7,8 | $\psi(\Omega^{\Omega^{\omega\cdot 2}})$ |
| 1,3,5,7,8,8 | $\psi(\Omega^{\Omega^{\omega^2}})$ |
| 1,3,5,7,8,10 | $\psi(\Omega^{\Omega^{\psi(0)}})$ |
| 1,3,5,7,8,10,12 | $\psi(\Omega^{\Omega^{\psi(\Omega)}})$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,3,5,7,8,10,12,14 | $\psi(\Omega^{\Omega^{\psi(\Omega^{\Omega})}})$ |
| 1,3,5,7,8,10,12,14,15 | $\psi(\Omega^{\Omega^{\psi(\Omega^{\Omega^{\omega}})}})$ |
| 1,3,5,7,9 | $\psi(\Omega^{\Omega^{\Omega}})$ |
| 1,3,5,7,9,3 | $\psi(\Omega^{\Omega^{\Omega}}+1)$ |
| 1,3,5,7,9,5 | $\psi(\Omega^{\Omega^{\Omega}+1})$ |
| 1,3,5,7,9,7 | $\psi(\Omega^{\Omega^{\Omega+1}})$ |
| 1,3,5,7,9,7,9 | $\psi(\Omega^{\Omega^{\Omega\cdot 2}})$ |
| 1,3,5,7,9,11 | $\psi(\Omega^{\Omega^{\Omega^{\Omega}}})$ |
| 1,3,6 | $\psi(\psi_1(0))$<br>BHO |
| 1,3,6,3 | $\psi(\psi_1(0)+1)$ |
| 1,3,6,3,4,6 | $\psi(\psi_1(0)+\psi(0))$ |
| 1,3,6,3,4,6,8 | $\psi(\psi_1(0)+\psi(\Omega))$ |
| 1,3,6,3,4,6,9 | $\psi(\psi_1(0)+\psi(\psi_1(0)))$ |
| 1,3,6,3,5 | $\psi(\psi_1(0)+\Omega)$ |
| 1,3,6,3,5,7 | $\psi(\psi_1(0)+\Omega^{\Omega})$ |
| 1,3,6,3,6 | $\psi(\psi_1(0)\cdot 2)$ |
| 1,3,6,3,6,3,6 | $\psi(\psi_1(0)\cdot 3)$ |
| 1,3,6,4 | $\psi(\psi_1(0)\cdot\omega)$ |
| 1,3,6,4,6 | $\psi(\psi_1(0)\cdot\psi(0))$ |
| 1,3,6,4,6,8 | $\psi(\psi_1(0)\cdot\psi(\Omega))$ |
| 1,3,6,4,6,8,10 | $\psi(\psi_1(0)\cdot\psi(\Omega^{\Omega}))$ |
| 1,3,6,4,6,9 | $\psi(\psi_1(0)\cdot\psi(\psi_1(0)))$ |
| 1,3,6,5 | $\psi(\psi_1(0)\cdot\Omega)$ |
| 1,3,6,5,6 | $\psi(\psi_1(0)\cdot\Omega^{\omega})$ |
| 1,3,6,5,6,8 | $\psi(\psi_1(0)\cdot\Omega^{\psi(0)})$ |
| 1,3,6,5,7 | $\psi(\psi_1(0)\cdot\Omega^{\Omega})$ |
| 1,3,6,5,8 | $\psi(\psi_1(0)^2)$ |
| 1,3,6,5,8,7 | $\psi(\psi_1(0)^{\Omega})$ |
| 1,3,6,5,8,7,10 | $\psi(\psi_1(0)^{\psi_1(0)})$ |
| 1,3,6,6 | $\psi(\psi_1(1))$ |
| 1,3,6,6,6 | $\psi(\psi_1(2))$ |
| 1,3,6,7 | $\psi(\psi_1(\omega))$ |
| 1,3,6,7,9 | $\psi(\psi_1(\psi(0)))$ |
| 1,3,6,7,9,12 | $\psi(\psi_1(\psi(\psi_1(0))))$ |
| 1,3,6,8 | $\psi(\psi_1(\Omega))$ |
| 1,3,6,8,6 | $\psi(\psi_1(\Omega+1))$ |
| 1,3,6,8,6,7,9,12,14,12 | $\psi(\psi_1(\Omega+\psi(\psi_1(\Omega))))$ |
| 1,3,6,8,6,8 | $\psi(\psi_1(\Omega\cdot 2))$ |
| 1,3,6,8,7 | $\psi(\psi_1(\Omega\cdot\omega))$ |
| 1,3,6,8,7,9 | $\psi(\psi_1(\Omega\cdot\psi(0)))$ |
| 1,3,6,8,7,9,12 | $\psi(\psi_1(\Omega\cdot\psi(\psi_1(0))))$ |
| 1,3,6,8,7,9,12,14 | $\psi(\psi_1(\Omega\cdot\psi(\psi_1(\Omega))))$ |
| 1,3,6,8,8 | $\psi(\psi_1(\Omega^2))$ |
| 1,3,6,8,8,7,9,12,14,14 | $\psi(\psi_1(\Omega^2\cdot\psi(\psi_1(\Omega^2))))$ |
| 1,3,6,8,8,8 | $\psi(\psi_1(\Omega^3))$ |
| 1,3,6,8,9 | $\psi(\psi_1(\Omega^{\omega}))$ |
| 1,3,6,8,9,11 | $\psi(\psi_1(\Omega^{\psi(0)}))$ |
| 1,3,6,8,9,11,14 | $\psi(\psi_1(\Omega^{\psi(\psi_1(0))}))$ |
| 1,3,6,8,9,11,14,16 | $\psi(\psi_1(\Omega^{\psi(\psi_1(\Omega))}))$ |
| 1,3,6,8,10 | $\psi(\psi_1(\Omega^{\Omega}))$ |
| 1,3,6,8,10,8 | $\psi(\psi_1(\Omega^{\Omega+1}))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,3,6,8,10,8,9, 11,14,16,18 | $\psi(\psi_1(\Omega^{\Omega+\psi(\psi_1(\Omega^{\Omega}))}))$ |
| 1,3,6,8,10,8,10 | $\psi(\psi_1(\Omega^{\Omega\cdot 2}))$ |
| 1,3,6,8,10,8,10,8,10 | $\psi(\psi_1(\Omega^{\Omega\cdot 3}))$ |
| 1,3,6,8,10,9 | $\psi(\psi_1(\Omega^{\Omega\cdot\omega}))$ |
| 1,3,6,8,10,9,11 | $\psi(\psi_1(\Omega^{\Omega\cdot\psi(0)}))$ |
| 1,3,6,8,10,9,11,14 | $\psi(\psi_1(\Omega^{\Omega\cdot\psi(\psi_1(0))}))$ |
| 1,3,6,8,10,9,11,14,16,18 | $\psi(\psi_1(\Omega^{\Omega\cdot\psi(\psi_1(\Omega^{\Omega}))}))$ |
| 1,3,6,8,10,10 | $\psi(\psi_1(\Omega^{\Omega^2}))$ |
| 1,3,6,8,10,10,9, 11,14,16,18,18 | $\psi(\psi_1(\Omega^{\Omega^2\cdot\psi(\psi_1(\Omega^{\Omega^2}))}))$ |
| 1,3,6,8,10,10,10 | $\psi(\psi_1(\Omega^{\Omega^3}))$ |
| 1,3,6,8,10,11 | $\psi(\psi_1(\Omega^{\Omega^{\omega}}))$ |
| 1,3,6,8,10,11,13,16 | $\psi(\psi_1(\Omega^{\Omega^{\psi(\psi_1(0))}}))$ |
| 1,3,6,8,10,11, 13,16,18,20,21 | $\psi(\psi_1(\Omega^{\Omega^{\psi(\psi_1(\Omega^{\Omega^{\omega}}))}}))$ |
| 1,3,6,8,10,12 | $\psi(\psi_1(\Omega^{\Omega^{\Omega}}))$ |
| 1,3,6,8,10,12,14 | $\psi(\psi_1(\Omega^{\Omega^{\Omega^{\Omega}}}))$ |
| 1,3,6,8,11 | $\psi(\psi_1(\psi_1(0)))$ |
| 1,3,6,8,11,6 | $\psi(\psi_1(\psi_1(0)+1))$ |
| 1,3,6,8,11,6,8,11 | $\psi(\psi_1(\psi_1(0)\cdot 2))$ |
| 1,3,6,8,11,7 | $\psi(\psi_1(\psi_1(0)\cdot\omega))$ |
| 1,3,6,8,11,8 | $\psi(\psi_1(\psi_1(0)\cdot\Omega))$ |
| 1,3,6,8,11,8,11 | $\psi(\psi_1(\psi_1(0)^2))$ |
| 1,3,6,8,11,9 | $\psi(\psi_1(\psi_1(0)^{\omega}))$ |
| 1,3,6,8,11,10 | $\psi(\psi_1(\psi_1(0)^{\Omega}))$ |
| 1,3,6,8,11,11 | $\psi(\psi_1(\psi_1(1)))$ |
| 1,3,6,8,11,12 | $\psi(\psi_1(\psi_1(\omega)))$ |
| 1,3,6,8,11,13 | $\psi(\psi_1(\psi_1(\Omega)))$ |
| 1,3,6,8,11,13,15 | $\psi(\psi_1(\psi_1(\Omega^{\Omega})))$ |
| 1,3,6,8,11,13,15,17 | $\psi(\psi_1(\psi_1(\Omega^{\Omega^{\Omega}})))$ |
| 1,3,6,8,11,13,16 | $\psi(\psi_1(\psi_1(\psi_1(0))))$ |
| 1,3,6,9 | $\psi(\Omega_2)$ |
| 1,3,6,9,3 | $\psi(\Omega_2+1)$ |
| 1,3,6,9,3,4,6 | $\psi(\Omega_2+\psi(0))$ |
| 1,3,6,9,3,4,6,9 | $\psi(\Omega_2+\psi(\psi_1(0)))$ |
| 1,3,6,9,3,4,6,9,12 | $\psi(\Omega_2+\psi(\Omega_2))$ |
| 1,3,6,9,3,5 | $\psi(\Omega_2+\Omega)$ |
| 1,3,6,9,3,5, 3,4,6,9,12,6,8 | $\psi(\Omega_2+\Omega+\psi(\Omega_2+\Omega))$ |
| 1,3,6,9,3,5,3,5 | $\psi(\Omega_2+\Omega\cdot 2)$ |
| 1,3,6,9,3,5,5 | $\psi(\Omega_2+\Omega^2)$ |
| 1,3,6,9,3,5,7 | $\psi(\Omega_2+\Omega^{\Omega})$ |
| 1,3,6,9,3,6 | $\psi(\Omega_2+\psi_1(0))$ |
| 1,3,6,9,3,6,8 | $\psi(\Omega_2+\psi_1(\Omega))$ |
| 1,3,6,9,3,6,8,11 | $\psi(\Omega_2+\psi_1(\psi_1(0)))$ |
| 1,3,6,9,3,6,9 | $\psi(\Omega_2+\psi_1(\Omega_2))$ |
| 1,3,6,9,3,6,9,3,6,9 | $\psi(\Omega_2+\psi_1(\Omega_2)\cdot 2)$ |
| 1,3,6,9,4 | $\psi(\Omega_2+\psi_1(\Omega_2)\cdot\omega)$ |
| 1,3,6,9,4,6,9 | $\psi(\Omega_2+\psi_1(\Omega_2)\cdot\psi(\psi_1(0)))$ |
| 1,3,6,9,4,6,9,12 | $\psi(\Omega_2+\psi_1(\Omega_2)\cdot\psi(\Omega_2))$ |
| 1,3,6,9,4,6,9,12,7 | $\psi(\Omega_2+\psi_1(\Omega_2)\cdot\psi(\Omega_2+\psi_1(\Omega_2)))$ |
| 1,3,6,9,5 | $\psi(\Omega_2+\psi_1(\Omega_2)\cdot\Omega)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,3,6,9,5,6,8 | $\psi(\Omega_2 + \psi_1(\Omega_2) \cdot \Omega^{\psi(0)})$ |
| 1,3,6,9,5,7 | $\psi(\Omega_2 + \psi_1(\Omega_2) \cdot \Omega^{\Omega})$ |
| 1,3,6,9,5,8 | $\psi(\Omega_2 + \psi_1(\Omega_2) \cdot \psi_1(0))$ |
| 1,3,6,9,5,8,9 | $\psi(\Omega_2 + \psi_1(\Omega_2) \cdot \psi_1(\omega))$ |
| 1,3,6,9,5,8,10 | $\psi(\Omega_2 + \psi_1(\Omega_2) \cdot \psi_1(\Omega))$ |
| 1,3,6,9,5,8,10,13 | $\psi(\Omega_2 + \psi_1(\Omega_2) \cdot \psi_1(\psi_1(0)))$ |
| 1,3,6,9,5,8,11 | $\psi(\Omega_2 + \psi_1(\Omega_2)^2)$ |
| 1,3,6,9,5,8,11,5,8,11 | $\psi(\Omega_2 + \psi_1(\Omega_2)^3)$ |
| 1,3,6,9,5,8,11,6 | $\psi(\Omega_2 + \psi_1(\Omega_2)^{\omega})$ |
| 1,3,6,9,5,8,11,<br>6,8,11,14,8,11,14 | $\psi(\Omega_2 + \psi_1(\Omega_2)^{\psi(\Omega_2+\psi_1(\Omega_2))})$ |
| 1,3,6,9,5,8,11,7 | $\psi(\Omega_2 + \psi_1(\Omega_2)^{\Omega})$ |
| 1,3,6,9,5,8,11,7,10,13 | $\psi(\Omega_2 + \psi_1(\Omega_2)^{\psi_1(\Omega_2)})$ |
| 1,3,6,9,6 | $\psi(\Omega_2 + \psi_1(\Omega_2 + 1))$ |
| 1,3,6,9,6,6 | $\psi(\Omega_2 + \psi_1(\Omega_2 + 2))$ |
| 1,3,6,9,6,7,9 | $\psi(\Omega_2 + \psi_1(\Omega_2 + \psi(0)))$ |
| 1,3,6,9,6,7,9,12 | $\psi(\Omega_2 + \psi_1(\Omega_2 + \psi(\psi_1(0))))$ |
| 1,3,6,9,6,7,9,12,15,12 | $\psi(\Omega_2 + \psi_1(\Omega_2 + \psi(\Omega_2 + \psi_1(\Omega_2 + 1))))$ |
| 1,3,6,9,6,8 | $\psi(\Omega_2 + \psi_1(\Omega_2 + \Omega))$ |
| 1,3,6,9,6,8,11 | $\psi(\Omega_2 + \psi_1(\Omega_2 + \psi_1(0)))$ |
| 1,3,6,9,6,8,11,14 | $\psi(\Omega_2 + \psi_1(\Omega_2 + \psi_1(\Omega_2)))$ |
| 1,3,6,9,6,9 | $\psi(\Omega_2 \cdot 2)$ |
| 1,3,6,9,6,9,6,9 | $\psi(\Omega_2 \cdot 3)$ |
| 1,3,6,9,7 | $\psi(\Omega_2 \cdot \omega)$ |
| 1,3,6,9,7,9 | $\psi(\Omega_2 \cdot \psi(0))$ |
| 1,3,6,9,7,9,12 | $\psi(\Omega_2 \cdot \psi(\psi_1(0)))$ |
| 1,3,6,9,7,9,12,15 | $\psi(\Omega_2 \cdot \psi(\Omega_2))$ |
| 1,3,6,9,8 | $\psi(\Omega_2 \cdot \Omega)$ |
| 1,3,6,9,8,9,11,14,17 | $\psi(\Omega_2 \cdot \Omega^{\psi(\Omega_2)})$ |
| 1,3,6,9,8,10 | $\psi(\Omega_2 \cdot \Omega^{\Omega})$ |
| 1,3,6,9,8,10,13 | $\psi(\Omega_2 \cdot \psi_1(0))$ |
| 1,3,6,9,8,10,13,16 | $\psi(\Omega_2 \cdot \psi_1(\Omega_2))$ |
| 1,3,6,9,9 | $\psi(\Omega_2^2)$ |
| 1,3,6,9,10 | $\psi(\Omega_2^{\omega})$ |
| 1,3,6,9,10,12,15,18 | $\psi(\Omega_2^{\psi(\Omega_2)})$ |
| 1,3,6,9,11 | $\psi(\Omega_2^{\Omega})$ |
| 1,3,6,9,11,14 | $\psi(\Omega_2^{\psi_1(0)})$ |
| 1,3,6,9,11,14,17 | $\psi(\Omega_2^{\psi_1(\Omega_2)})$ |
| 1,3,6,9,12 | $\psi(\Omega_2^{\Omega_2})$ |
| 1,3,6,9,12,15 | $\psi(\Omega_2^{\Omega_2^{\Omega_2}})$ |
| 1,3,6,10 | $\psi(\psi_2(0))$ |
| 1,3,6,10,13,17 | $\psi(\psi_2(\psi_2(0)))$ |
| 1,3,6,10,14 | $\psi(\Omega_3)$ |
| 1,3,6,10,14,18 | $\psi(\Omega_3^{\Omega_3})$ |
| 1,3,6,10,15 | $\psi(\psi_3(0))$ |
| 1,3,6,10,15,20 | $\psi(\Omega_4)$ |
| 1,3,6,10,15,21 | $\psi(\psi_4(0))$ |
| 1,4 | $\psi(\Omega_{\omega})$ |
| 1,4,3 | $\psi(\Omega_{\omega} + 1)$ |
| 1,4,3,4 | $\psi(\Omega_{\omega} + \omega)$ |
| 1,4,3,4,6 | $\psi(\Omega_{\omega} + \psi(0))$ |
| 1,4,3,4,6,9,12 | $\psi(\Omega_{\omega} + \psi(\Omega_2))$ |
| 1,4,3,4,7 | $\psi(\Omega_{\omega} + \psi(\Omega_{\omega}))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,3,4,7,6,7,10 | $\psi(\Omega_\omega + \psi(\Omega_\omega + \psi(\Omega_\omega)))$ |
| 1,4,3,5 | $\psi(\Omega_\omega + \Omega)$ |
| 1,4,3,5,3,5 | $\psi(\Omega_\omega + \Omega \cdot 2)$ |
| 1,4,3,5,4 | $\psi(\Omega_\omega + \Omega \cdot \omega)$ |
| 1,4,3,5,5 | $\psi(\Omega_\omega + \Omega^2)$ |
| 1,4,3,5,6 | $\psi(\Omega_\omega + \Omega^\omega)$ |
| 1,4,3,5,7 | $\psi(\Omega_\omega + \Omega^\Omega)$ |
| 1,4,3,6 | $\psi(\Omega_\omega + \psi_1(0))$ |
| 1,4,3,6,9 | $\psi(\Omega_\omega + \psi_1(\Omega_2))$ |
| 1,4,3,6,9,12 | $\psi(\Omega_\omega + \psi_1(\Omega_2^{\Omega_2}))$ |
| 1,4,3,6,10 | $\psi(\Omega_\omega + \psi_1(\psi_2(0)))$ |
| 1,4,3,6,10,14 | $\psi(\Omega_\omega + \psi_1(\Omega_3))$ |
| 1,4,3,6,10,15 | $\psi(\Omega_\omega + \psi_1(\psi_3(0)))$ |
| 1,4,3,7 | $\psi(\Omega_\omega + \psi_1(\Omega_\omega))$ |
| 1,4,3,7,4 | $\psi(\Omega_\omega + \psi_1(\Omega_\omega) \cdot \omega)$ |
| 1,4,3,7,5 | $\psi(\Omega_\omega + \psi_1(\Omega_\omega) \cdot \Omega)$ |
| 1,4,3,7,5,6,8,11 | $\psi(\Omega_\omega + \psi_1(\Omega_\omega) \cdot \psi_1(0))$ |
| 1,4,3,7,5,6,8,11,15 | $\psi(\Omega_\omega + \psi_1(\Omega_\omega) \cdot \psi_1(\psi_2(0)))$ |
| 1,4,3,7,5,6,9 | $\psi(\Omega_\omega + \psi_1(\Omega_\omega)^2)$ |
| 1,4,3,7,5,7 | $\psi(\Omega_\omega + \psi_1(\Omega_\omega)^{\psi_1(\Omega_\omega)})$ |
| 1,4,3,7,6 | $\psi(\Omega_\omega + \psi_1(\Omega_\omega + 1))$ |
| 1,4,3,7,6,7,9 | $\psi(\Omega_\omega + \psi_1(\Omega_\omega + \psi(0)))$ |
| 1,4,3,7,6,7,10 | $\psi(\Omega_\omega + \psi_1(\Omega_\omega + \psi(\Omega_\omega)))$ |
| 1,4,3,7,6,8 | $\psi(\Omega_\omega + \psi_1(\Omega_\omega + \Omega))$ |
| 1,4,3,7,6,8,11 | $\psi(\Omega_\omega + \psi_1(\Omega_\omega + \psi_1(0)))$ |
| 1,4,3,7,6,8,12 | $\psi(\Omega_\omega + \psi_1(\Omega_\omega + \psi_1(\Omega_\omega)))$ |
| 1,4,3,7,6,9 | $\psi(\Omega_\omega + \Omega_2)$ |
| 1,4,3,7,6,9,12 | $\psi(\Omega_\omega + \Omega_2^{\Omega_2})$ |
| 1,4,3,7,6,10 | $\psi(\Omega_\omega + \psi_2(0))$ |
| 1,4,3,7,6,10,12 | $\psi(\Omega_\omega + \psi_2(\Omega))$ |
| 1,4,3,7,6,11 | $\psi(\Omega_\omega + \psi_2(\Omega_\omega))$ |
| 1,4,3,7,6,11,10 | $\psi(\Omega_\omega + \psi_2(\Omega_\omega + 1))$ |
| 1,4,3,7,6,11,10,14 | $\psi(\Omega_\omega + \psi_2(\Omega_\omega + \psi_2(0)))$ |
| 1,4,3,7,6,11,10,15 | $\psi(\Omega_\omega + \Omega_3)$ |
| 1,4,3,7,6,11,10,16 | $\psi(\Omega_\omega + \psi_3(0))$ |
| 1,4,4 | $\psi(\Omega_\omega \cdot 2)$ |
| 1,4,4,4 | $\psi(\Omega_\omega \cdot 3)$ |
| 1,4,5 | $\psi(\Omega_\omega \cdot \omega)$ |
| 1,4,5,4 | $\psi(\Omega_\omega \cdot \omega + \Omega_\omega)$ |
| 1,4,5,5 | $\psi(\Omega_\omega \cdot \omega^2)$ |
| 1,4,5,7 | $\psi(\Omega_\omega \cdot \psi(0))$ |
| 1,4,5,8 | $\psi(\Omega_\omega \cdot \psi(\Omega_\omega))$ |
| 1,4,5,8,9,12 | $\psi(\Omega_\omega \cdot \psi(\Omega_\omega \cdot \psi(\Omega_\omega)))$ |
| 1,4,6 | $\psi(\Omega_\omega \cdot \Omega)$ |
| 1,4,6,3 | $\psi(\Omega_\omega \cdot \Omega + 1)$ |
| 1,4,6,3,4,7 | $\psi(\Omega_\omega \cdot \Omega + \psi(\Omega_\omega))$ |
| 1,4,6,3,5 | $\psi(\Omega_\omega \cdot \Omega + \Omega)$ |
| 1,4,6,3,6 | $\psi(\Omega_\omega \cdot \Omega + \psi_1(0))$ |
| 1,4,6,3,6,10 | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_2))$ |
| 1,4,6,3,7 | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega))$ |
| 1,4,6,3,7,9 | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \Omega))$ |
| 1,4,6,3,7,9,5 | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \Omega)^{\psi_1(\Omega_\omega \cdot \Omega)})$ |
| 1,4,6,3,7,9,6 | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \Omega + 1))$ |
| 1,4,6,3,7,9,6,7 | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \Omega + \omega))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,6,3,7,9,6,8 | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \Omega + \Omega))$ |
| 1,4,6,3,7,9,6,8,11 | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \Omega + \psi_1(0)))$ |
| 1,4,6,3,7,9,6,8,12 | $\psi(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \Omega + \psi_1(\Omega_\omega \cdot \Omega)))$ |
| 1,4,6,3,7,9,6,9 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2)$ |
| 1,4,6,3,7,9,6,9,3 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 + 1)$ |
| 1,4,6,3,7,9,6,9,3,7 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 + \psi_1(\Omega_\omega))$ |
| 1,4,6,3,7,<br>9,6,9,3,7,9 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega))$ |
| 1,4,6,3,7,9,<br>6,9,3,7,9,6,9 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega + \Omega_2))$ |
| 1,4,6,3,7,9,6,9,6 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega + \Omega_2 + 1))$ |
| 1,4,6,3,7,9,6,9,6,8 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega + \Omega_2 + \Omega))$ |
| 1,4,6,3,7,9,6,9,6,<br>8,12,14,11,14,11,13 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 +$<br>$\psi_1(\Omega_\omega \cdot \Omega + \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega + \Omega_2 + 1)))$ |
| 1,4,6,3,7,9,6,9,6,9 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 \cdot 2)$ |
| 1,4,6,3,7,9,6,9,7 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 \cdot \omega)$ |
| 1,4,6,3,7,9,6,9,7,9 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 \cdot \psi(0))$ |
| 1,4,6,3,7,9,6,9,7,10 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 \cdot \psi(\Omega_\omega))$ |
| 1,4,6,3,7,9,6,9,8 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 \cdot \Omega)$ |
| 1,4,6,3,7,9,<br>6,9,8,10,13 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 \cdot \psi_1(0))$ |
| 1,4,6,3,7,9,6,9,8,10,13,<br>15,12,16,18,15,18,17 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2 \cdot \psi_1(\Omega_\omega \cdot \Omega + \Omega_2 \cdot \Omega))$ |
| 1,4,6,3,7,9,6,9,9 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2^2)$ |
| 1,4,6,3,7,9,6,9,10 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2^\omega)$ |
| 1,4,6,3,7,9,6,9,12 | $\psi(\Omega_\omega \cdot \Omega + \Omega_2^{\Omega_2})$ |
| 1,4,6,3,7,9,6,10 | $\psi(\Omega_\omega \cdot \Omega + \psi_2(0))$ |
| 1,4,6,3,7,9,6,11 | $\psi(\Omega_\omega \cdot \Omega + \psi_2(\Omega_\omega))$ |
| 1,4,6,3,7,9,6,11,13 | $\psi(\Omega_\omega \cdot \Omega + \psi_2(\Omega_\omega \cdot \Omega))$ |
| 1,4,6,3,7,9,<br>6,11,13,10,14 | $\psi(\Omega_\omega \cdot \Omega + \Omega_3)$ |
| 1,4,6,3,7,9,<br>6,11,13,10,15 | $\psi(\Omega_\omega \cdot \Omega + \psi_3(0))$ |
| 1,4,6,3,7,9,<br>6,11,13,10,16 | $\psi(\Omega_\omega \cdot \Omega + \psi_3(\Omega_\omega))$ |
| 1,4,6,3,7,9,7 | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega)$ |
| 1,4,6,3,7,9,<br>7,3,7,9,7 | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_1(\Omega_\omega \cdot \Omega + \Omega_\omega))$ |
| 1,4,6,3,7,9,7,6 | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_1(\Omega_\omega \cdot \Omega + \Omega_\omega + 1))$ |
| 1,4,6,3,7,9,7,6,8,12 | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_1(\Omega_\omega \cdot \Omega + \Omega_\omega +$<br>$\psi_1(\Omega_\omega \cdot \Omega + \Omega_\omega)))$ |
| 1,4,6,3,7,9,7,6,9 | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \Omega_2)$ |
| 1,4,6,3,7,9,6,9,12 | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \Omega_2^{\Omega_2})$ |
| 1,4,6,3,7,9,7,6,10 | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_2(0))$ |
| 1,4,6,3,7,9,7,6,11 | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_2(\Omega_\omega))$ |
| 1,4,6,3,7,9,<br>7,6,11,13,11 | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_2(\Omega_\omega \cdot \Omega + \Omega_\omega))$ |
| 1,4,6,3,7,9,7,<br>6,11,13,11,10,14 | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \Omega_3)$ |
| 1,4,6,3,7,9,7,<br>6,11,13,11,10,15 | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega + \psi_3(0))$ |
| 1,4,6,3,7,9,7,7 | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega \cdot 2)$ |
| 1,4,6,3,7,9,7,8 | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega \cdot \omega)$ |
| 1,4,6,3,7,9,7,<br>8,11,13,10,14,16,14 | $\psi(\Omega_\omega \cdot \Omega + \Omega_\omega \cdot \psi(\Omega_\omega \cdot \Omega + \Omega_\omega))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,6,3,7,9,7,9 | $\psi(\Omega_\omega \cdot \Omega \cdot 2)$ |
| 1,4,6,3,7,9,7,9,7,9 | $\psi(\Omega_\omega \cdot \Omega \cdot 3)$ |
| 1,4,6,3,7,9,8 | $\psi(\Omega_\omega \cdot \Omega \cdot \omega)$ |
| 1,4,6,3,7,9,8,<br>11,13,10,14,16 | $\psi(\Omega_\omega \cdot \Omega \cdot \psi(\Omega_\omega \cdot \Omega))$ |
| 1,4,6,3,7,9,9 | $\psi(\Omega_\omega \cdot \Omega^2)$ |
| 1,4,6,3,7,9,10 | $\psi(\Omega_\omega \cdot \Omega^\omega)$ |
| 1,4,6,3,7,9,12 | $\psi(\Omega_\omega \cdot \psi_1(0))$ |
| 1,4,6,3,7,9,12,12 | $\psi(\Omega_\omega \cdot \psi_1(1))$ |
| 1,4,6,3,7,9,12,14 | $\psi(\Omega_\omega \cdot \psi_1(\Omega))$ |
| 1,4,6,3,7,9,12,14,17 | $\psi(\Omega_\omega \cdot \psi_1(\psi_1(0)))$ |
| 1,4,6,3,7,9,12,14,17,19 | $\psi(\Omega_\omega \cdot \psi_1(\psi_1(\Omega)))$ |
| 1,4,6,3,7,9,12,15 | $\psi(\Omega_\omega \cdot \psi_1(\Omega_2))$ |
| 1,4,6,3,7,9,12,16 | $\psi(\Omega_\omega \cdot \psi_1(\psi_2(0)))$ |
| 1,4,6,3,7,9,12,16,20 | $\psi(\Omega_\omega \cdot \psi_1(\Omega_3))$ |
| 1,4,6,3,7,9,13 | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega))$ |
| 1,4,6,3,7,9,13,15 | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \Omega))$ |
| 1,4,6,3,7,9,13,15,13 | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \Omega + \Omega_\omega))$ |
| 1,4,6,3,7,9,13,15,13,15 | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \Omega \cdot 2))$ |
| 1,4,6,3,7,9,13,15,15 | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \Omega^2))$ |
| 1,4,6,3,7,9,13,15,17 | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \Omega^\Omega))$ |
| 1,4,6,3,7,9,13,15,18 | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \psi_1(0)))$ |
| 1,4,6,3,7,9,13,15,19 | $\psi(\Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \psi_1(\Omega_\omega)))$ |
| 1,4,6,3,7,10 | $\psi(\Omega_\omega \cdot \Omega_2)$ |
| 1,4,6,3,7,10,3,7,10 | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega_2))$ |
| 1,4,6,3,7,10,6 | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega_2 + 1))$ |
| 1,4,6,3,7,10,6,8 | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega_2 + \Omega))$ |
| 1,4,6,3,7,10,6,8,11 | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_1(\Omega_\omega \cdot \Omega_2 + \psi_1(0)))$ |
| 1,4,6,3,7,10,6,9 | $\psi(\Omega_\omega \cdot \Omega_2 + \Omega_2)$ |
| 1,4,6,3,7,10,6,10 | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_2(0))$ |
| 1,4,6,3,7,10,6,11 | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_2(\Omega_\omega))$ |
| 1,4,6,3,7,10,6,11,14 | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_2(\Omega_\omega \cdot \Omega_2))$ |
| 1,4,6,3,7,10,<br>6,11,14,10,14 | $\psi(\Omega_\omega \cdot \Omega_2 + \Omega_3)$ |
| 1,4,6,3,7,10,<br>6,11,14,10,15 | $\psi(\Omega_\omega \cdot \Omega_2 + \psi_3(0))$ |
| 1,4,6,3,7,10,<br>6,11,14,10,15,20 | $\psi(\Omega_\omega \cdot \Omega_2 + \Omega_4)$ |
| 1,4,6,3,7,10,6,11,14,11 | $\psi(\Omega_\omega \cdot \Omega_2 + \Omega_\omega)$ |
| 1,4,6,3,7,10,<br>6,11,14,11,13 | $\psi(\Omega_\omega \cdot \Omega_2 + \Omega_\omega \cdot \Omega)$ |
| 1,4,6,3,7,10,<br>6,11,14,11,13,16 | $\psi(\Omega_\omega \cdot \Omega_2 + \Omega_\omega \cdot \psi_1(0))$ |
| 1,4,6,3,7,10,6,<br>11,14,11,13,16,19 | $\psi(\Omega_\omega \cdot \Omega_2 + \Omega_\omega \cdot \psi_1(\Omega_2))$ |
| 1,4,6,3,7,10,6,<br>11,14,11,13,17 | $\psi(\Omega_\omega \cdot \Omega_2 + \Omega_\omega \cdot \psi_1(\Omega_\omega))$ |
| 1,4,6,3,7,10,6,11,14,11,<br>13,17,20,16,21,24,21 | $\psi(\Omega_\omega \cdot \Omega_2 + \Omega_\omega \cdot \psi_1(\Omega_\omega \cdot \Omega_2 + \Omega_\omega))$ |
| 1,4,6,3,7,10,<br>6,11,14,11,14 | $\psi(\Omega_\omega \cdot \Omega_2 \cdot 2)$ |
| 1,4,6,3,7,10,6,11,14,12 | $\psi(\Omega_\omega \cdot \Omega_2 \cdot \omega)$ |
| 1,4,6,3,7,10,6,11,14,13 | $\psi(\Omega_\omega \cdot \Omega_2 \cdot \Omega)$ |
| 1,4,6,3,7,10,6,11,14,14 | $\psi(\Omega_\omega \cdot \Omega_2^2)$ |
| 1,4,6,3,7,10,6,11,14,15 | $\psi(\Omega_\omega \cdot \Omega_2^\omega)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,6,3,7,10,6,11,14,16 | $\psi(\Omega_\omega \cdot \Omega_2^\Omega)$ |
| 1,4,6,3,7,10,6,11,14,17 | $\psi(\Omega_\omega \cdot \Omega_2^{\Omega_2})$ |
| 1,4,6,3,7,10,6,11,14,18 | $\psi(\Omega_\omega \cdot \psi_2(0))$ |
| 1,4,6,3,7,10,6,11,14,19 | $\psi(\Omega_\omega \cdot \psi_2(\Omega_\omega))$ |
| 1,4,6,3,7,10,6,11,15 | $\psi(\Omega_\omega \cdot \Omega_3)$ |
| 1,4,6,3,7,10,<br>6,11,15,10,16,21 | $\psi(\Omega_\omega \cdot \Omega_4)$ |
| 1,4,6,4 | $\psi(\Omega_\omega^2)$ |
| 1,4,6,4,3 | $\psi(\Omega_\omega^2 + 1)$ |
| 1,4,6,4,3,5 | $\psi(\Omega_\omega^2 + \Omega)$ |
| 1,4,6,4,3,6 | $\psi(\Omega_\omega^2 + \psi_1(0))$ |
| 1,4,6,4,3,7 | $\psi(\Omega_\omega^2 + \psi_1(\Omega_\omega))$ |
| 1,4,6,4,3,7,10,7 | $\psi(\Omega_\omega^2 + \psi_1(\Omega_\omega^2))$ |
| 1,4,6,4,3,7,10,7,6,9 | $\psi(\Omega_\omega^2 + \Omega_2)$ |
| 1,4,6,4,3,7,<br>10,7,6,9,13 | $\psi(\Omega_\omega^2 + \psi_2(0))$ |
| 1,4,6,4,3,7,10,7,6,10 | $\psi(\Omega_\omega^2 + \psi_2(\Omega_\omega))$ |
| 1,4,6,4,3,7,10,7,6,11 | $\psi(\Omega_\omega^2 + \Omega_3)$ |
| 1,4,6,4,4 | $\psi(\Omega_\omega^2 + \Omega_\omega)$ |
| 1,4,6,4,5 | $\psi(\Omega_\omega^2 + \Omega_\omega \cdot \omega)$ |
| 1,4,6,4,6 | $\psi(\Omega_\omega^2 + \Omega_\omega \cdot \Omega)$ |
| 1,4,6,4,6,4 | $\psi(\Omega_\omega^2 \cdot 2)$ |
| 1,4,6,5 | $\psi(\Omega_\omega^2 \cdot \omega)$ |
| 1,4,6,6 | $\psi(\Omega_\omega^2 \cdot \Omega)$ |
| 1,4,6,6,4 | $\psi(\Omega_\omega^3)$ |
| 1,4,6,6,6 | $\psi(\Omega_\omega^3 \cdot \Omega)$ |
| 1,4,6,6,6,4 | $\psi(\Omega_\omega^4)$ |
| 1,4,6,7 | $\psi(\Omega_\omega^\omega)$ |
| 1,4,6,7,4 | $\psi(\Omega_\omega^{\omega+1})$ |
| 1,4,6,7,10 | $\psi(\Omega_\omega^{\psi(\Omega_\omega)})$ |
| 1,4,6,8 | $\psi(\Omega_\omega^\Omega)$ |
| 1,4,6,8,4 | $\psi(\Omega_\omega^{\Omega_\omega})$ |
| 1,4,6,8,8 | $\psi(\Omega_\omega^{\Omega_\omega \cdot \Omega})$ |
| 1,4,6,8,8,4 | $\psi(\Omega_\omega^{\Omega_\omega^2})$ |
| 1,4,6,8,9 | $\psi(\Omega_\omega^{\Omega_\omega^\omega})$ |
| 1,4,6,8,10 | $\psi(\Omega_\omega^{\Omega_\omega^\Omega})$ |
| 1,4,6,8,10,4 | $\psi(\Omega_\omega^{\Omega_\omega^{\Omega_\omega}})$ |
| 1,4,6,9 | $\psi(\psi_\omega(0))$ |
| 1,4,6,9,3 | $\psi(\psi_\omega(0) + 1)$ |
| 1,4,6,9,4 | $\psi(\psi_\omega(0) + \Omega_\omega)$ |
| 1,4,6,9,4,6 | $\psi(\psi_\omega(0) + \Omega_\omega \cdot \Omega)$ |
| 1,4,6,9,4,6,4 | $\psi(\psi_\omega(0) + \Omega_\omega^2)$ |
| 1,4,6,9,4,6,8 | $\psi(\psi_\omega(0) + \Omega_\omega^\Omega)$ |
| 1,4,6,9,4,6,8,4 | $\psi(\psi_\omega(0) + \Omega_\omega^{\Omega_\omega})$ |
| 1,4,6,9,4,6,9 | $\psi(\psi_\omega(0) \cdot 2)$ |
| 1,4,6,9,4,6,9,4,6,9 | $\psi(\psi_\omega(0) \cdot 3)$ |
| 1,4,6,9,5 | $\psi(\psi_\omega(0) \cdot \omega)$ |
| 1,4,6,9,5,8 | $\psi(\psi_\omega(0) \cdot \psi(\Omega_\omega))$ |
| 1,4,6,9,5,8,10,13 | $\psi(\psi_\omega(0) \cdot \psi(\psi_\omega(0)))$ |
| 1,4,6,9,6 | $\psi(\psi_\omega(0) \cdot \Omega)$ |
| 1,4,6,9,6,4 | $\psi(\psi_\omega(0) \cdot \Omega_\omega)$ |
| 1,4,6,9,6,8,4 | $\psi(\psi_\omega(0) \cdot \Omega_\omega^{\Omega_\omega})$ |
| 1,4,6,9,6,9 | $\psi(\psi_\omega(0)^2)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,6,9,7 | $\psi(\psi_\omega(0)^\omega)$ |
| 1,4,6,9,8 | $\psi(\psi_\omega(0)^\Omega)$ |
| 1,4,6,9,8,4 | $\psi(\psi_\omega(0)^{\Omega_\omega})$ |
| 1,4,6,9,8,11 | $\psi(\psi_\omega(0)^{\psi_\omega(0)})$ |
| 1,4,6,9,8,11,10,13 | $\psi(\psi_\omega(0)^{\psi_\omega(0)^{\psi_\omega(0)}})$ |
| 1,4,6,9,9 | $\psi(\psi_\omega(1))$ |
| 1,4,6,9,9,9 | $\psi(\psi_\omega(2))$ |
| 1,4,6,9,10 | $\psi(\psi_\omega(\omega))$ |
| 1,4,6,9,10,13 | $\psi(\psi_\omega(\psi(\Omega_\omega)))$ |
| 1,4,6,9,10,13,15,18 | $\psi(\psi_\omega(\psi(\psi_\omega(0))))$ |
| 1,4,6,9,11 | $\psi(\psi_\omega(\Omega))$ |
| 1,4,6,9,11,4 | $\psi(\psi_\omega(\Omega_\omega))$ |
| 1,4,6,9,11,4,4 | $\psi(\psi_\omega(\Omega_\omega)+\Omega_\omega)$ |
| 1,4,6,9,11,4,6,9 | $\psi(\psi_\omega(\Omega_\omega)+\psi_\omega(0))$ |
| 1,4,6,9,11,4,6,9,11,4 | $\psi(\psi_\omega(\Omega_\omega)\cdot 2)$ |
| 1,4,6,9,11,5 | $\psi(\psi_\omega(\Omega_\omega)\cdot\omega)$ |
| 1,4,6,9,11,6 | $\psi(\psi_\omega(\Omega_\omega)\cdot\Omega)$ |
| 1,4,6,9,11,6,4 | $\psi(\psi_\omega(\Omega_\omega)\cdot\Omega_\omega)$ |
| 1,4,6,9,11,6,8,4 | $\psi(\psi_\omega(\Omega_\omega)\cdot\Omega_\omega^{\Omega_\omega})$ |
| 1,4,6,9,11,6,9 | $\psi(\psi_\omega(\Omega_\omega)\cdot\psi_\omega(0))$ |
| 1,4,6,9,11,6,9,11 | $\psi(\psi_\omega(\Omega_\omega)\cdot\psi_\omega(\Omega))$ |
| 1,4,6,9,11,6,9,11,4 | $\psi(\psi_\omega(\Omega_\omega)^2)$ |
| 1,4,6,9,11,7 | $\psi(\psi_\omega(\Omega_\omega)^\omega)$ |
| 1,4,6,9,11,8,4 | $\psi(\psi_\omega(\Omega_\omega)^{\Omega_\omega})$ |
| 1,4,6,9,11,9 | $\psi(\psi_\omega(\Omega_\omega)^{\psi_\omega(0)})$ |
| 1,4,6,9,11,9,11 | $\psi(\psi_\omega(\Omega_\omega)^{\psi_\omega(\Omega)})$ |
| 1,4,6,9,11,10 | $\psi(\psi_\omega(\Omega_\omega+1))$ |
| 1,4,6,9,11,10,12 | $\psi(\psi_\omega(\Omega_\omega+\psi(0)))$ |
| 1,4,6,9,11,10,13,15,18,20 | $\psi(\psi_\omega(\Omega_\omega+\psi(\psi_\omega(\Omega))))$ |
| 1,4,6,9,11,11 | $\psi(\psi_\omega(\Omega_\omega+\Omega))$ |
| 1,4,6,9,11,11,4 | $\psi(\psi_\omega(\Omega_\omega\cdot 2))$ |
| 1,4,6,9,11,11,11 | $\psi(\psi_\omega(\Omega_\omega\cdot 2+\Omega))$ |
| 1,4,6,9,11,12 | $\psi(\psi_\omega(\Omega_\omega\cdot\omega))$ |
| 1,4,6,9,11,12,14 | $\psi(\psi_\omega(\Omega_\omega\cdot\psi(0)))$ |
| 1,4,6,9,11,12,<br>15,17,20,22,23 | $\psi(\psi_\omega(\Omega_\omega\cdot\psi(\psi_\omega(\Omega_\omega\cdot\omega))))$ |
| 1,4,6,9,11,13 | $\psi(\psi_\omega(\Omega_\omega\cdot\Omega))$ |
| 1,4,6,9,11,13,4 | $\psi(\psi_\omega(\Omega_\omega^2))$ |
| 1,4,6,9,11,13,5 | $\psi(\psi_\omega(\Omega_\omega^\omega))$ |
| 1,4,6,9,11,13,6 | $\psi(\psi_\omega(\Omega_\omega^\Omega))$ |
| 1,4,6,9,11,14 | $\psi(\psi_\omega(\psi_\omega(0)))$ |
| 1,4,6,9,12 | $\psi(\Omega_{\omega+1})$ |
| 1,4,6,9,12,4 | $\psi(\Omega_{\omega+1}+\Omega_\omega)$ |
| 1,4,6,9,12,4,6,9 | $\psi(\Omega_{\omega+1}+\psi_\omega(0))$ |
| 1,4,6,9,12,4,6,9,12 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1}))$ |
| 1,4,6,9,12,5 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})\cdot\omega)$ |
| 1,4,6,9,12,6 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})\cdot\Omega)$ |
| 1,4,6,9,12,6,4 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})\cdot\Omega_\omega)$ |
| 1,4,6,9,12,6,9 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})\cdot\psi_\omega(0))$ |
| 1,4,6,9,12,6,9,12 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})^2)$ |
| 1,4,6,9,12,7 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_\omega)^\omega)$ |
| 1,4,6,9,12,8 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})^\Omega)$ |
| 1,4,6,9,12,8,4 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})^{\Omega_\omega})$ |
| 1,4,6,9,12,8,11 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})^{\psi_\omega(0)})$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,6,9,12,8,11,14 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})^{\psi_\omega(\Omega_{\omega+1})})$ |
| 1,4,6,9,12,9 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1}+1))$ |
| 1,4,6,9,12,9,10 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1}+\omega))$ |
| 1,4,6,9,12,9,11 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1}+\Omega))$ |
| 1,4,6,9,12,9,11,4 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1}+\Omega_\omega))$ |
| 1,4,6,9,12,9,11,14 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1}+\psi_\omega(0)))$ |
| 1,4,6,9,12,9,11,14,17 | $\psi(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1}+\psi_\omega(\Omega_{\omega+1})))$ |
| 1,4,6,9,12,9,12 | $\psi(\Omega_{\omega+1}\cdot 2)$ |
| 1,4,6,9,12,10 | $\psi(\Omega_{\omega+1}\cdot\omega)$ |
| 1,4,6,9,12,11 | $\psi(\Omega_{\omega+1}\cdot\Omega)$ |
| 1,4,6,9,12,11,4 | $\psi(\Omega_{\omega+1}\cdot\Omega_\omega)$ |
| 1,4,6,9,12,11,14 | $\psi(\Omega_{\omega+1}\cdot\psi_\omega(0))$ |
| 1,4,6,9,12,11,14,17 | $\psi(\Omega_{\omega+1}\cdot\psi_\omega(\Omega_{\omega+1}))$ |
| 1,4,6,9,12,12 | $\psi(\Omega_{\omega+1}^2)$ |
| 1,4,6,9,12,13 | $\psi(\Omega_{\omega+1}^\omega)$ |
| 1,4,6,9,12,14 | $\psi(\Omega_{\omega+1}^\Omega)$ |
| 1,4,6,9,12,14,4 | $\psi(\Omega_{\omega+1}^{\Omega_\omega})$ |
| 1,4,6,9,12,14,17 | $\psi(\Omega_{\omega+1}^{\psi_\omega(0)})$ |
| 1,4,6,9,12,14,17,20 | $\psi(\Omega_{\omega+1}^{\psi_\omega(\Omega_{\omega+1})})$ |
| 1,4,6,9,12,15 | $\psi(\Omega_{\omega+1}^{\Omega_{\omega+1}})$ |
| 1,4,6,9,12,15,18 | $\psi(\Omega_{\omega+1}^{\Omega_{\omega+1}^{\Omega_{\omega+1}}})$ |
| 1,4,6,9,13 | $\psi(\psi_{\omega+1}(0))$ |
| 1,4,6,9,13,4,6,9,13 | $\psi(\psi_{\omega+1}(0)+\psi_\omega(\psi_{\omega+1}(0)))$ |
| 1,4,6,9,13,6 | $\psi(\psi_{\omega+1}(0)+\psi_\omega(\psi_{\omega+1}(0)+1))$ |
| 1,4,6,9,13,9 | $\psi(\psi_{\omega+1}(0)+\Omega_{\omega+1})$ |
| 1,4,6,9,13,9,12,15 | $\psi(\psi_{\omega+1}(0)+\Omega_{\omega+1}^{\Omega_{\omega+1}})$ |
| 1,4,6,9,13,9,13 | $\psi(\psi_{\omega+1}(0)\cdot 2)$ |
| 1,4,6,9,13,10 | $\psi(\psi_{\omega+1}(0)\cdot\omega)$ |
| 1,4,6,9,13,11 | $\psi(\psi_{\omega+1}(0)\cdot\Omega)$ |
| 1,4,6,9,13,12 | $\psi(\psi_{\omega+1}(0)^\omega)$ |
| 1,4,6,9,13,13 | $\psi(\psi_{\omega+1}(1))$ |
| 1,4,6,9,13,14 | $\psi(\psi_{\omega+1}(\omega))$ |
| 1,4,6,9,13,15 | $\psi(\psi_{\omega+1}(\Omega))$ |
| 1,4,6,9,13,15,4 | $\psi(\psi_{\omega+1}(\Omega_\omega))$ |
| 1,4,6,9,13,15,18 | $\psi(\psi_{\omega+1}(\psi_\omega(0)))$ |
| 1,4,6,9,13,15,18,21 | $\psi(\psi_{\omega+1}(\psi_\omega(\Omega_{\omega+1})))$ |
| 1,4,6,9,13,15,18,22 | $\psi(\psi_{\omega+1}(\psi_\omega(\psi_{\omega+1}(0))))$ |
| 1,4,6,9,13,16 | $\psi(\psi_{\omega+1}(\Omega_{\omega+1}))$ |
| 1,4,6,9,13,16,19 | $\psi(\psi_{\omega+1}(\psi_{\omega+1}(0)))$ |
| 1,4,6,9,13,16,19,22 | $\psi(\psi_{\omega+1}(\psi_{\omega+1}(\Omega_{\omega+1})))$ |
| 1,4,6,9,13,16,19,22,25 | $\psi(\psi_{\omega+1}(\psi_{\omega+1}(\psi_{\omega+1}(0))))$ |
| 1,4,6,9,13,17 | $\psi(\Omega_{\omega+2})$ |
| 1,4,6,9,13,18 | $\psi(\psi_{\omega+2}(0))$ |
| 1,4,6,9,13,18,23 | $\psi(\Omega_{\omega+3})$ |
| 1,4,6,9,13,18,24 | $\psi(\psi_{\omega+3}(0))$ |
| 1,4,6,10 | $\psi(\Omega_{\omega\cdot 2})$ |
| 1,4,6,10,4 | $\psi(\Omega_{\omega\cdot 2}+\Omega_\omega)$ |
| 1,4,6,10,4,6,9 | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(0))$ |
| 1,4,6,10,4,6,10 | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2}))$ |
| 1,4,6,10,5 | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2})\cdot\omega)$ |
| 1,4,6,10,6 | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2})\cdot\Omega)$ |
| 1,4,6,10,6,4 | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2})\cdot\Omega_\omega)$ |
| 1,4,6,10,6,9 | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2})\cdot\psi_\omega(0))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,6,10,6,10 | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2})^2)$ |
| 1,4,6,10,7 | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2})^\omega)$ |
| 1,4,6,10,8 | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2})^\Omega)$ |
| 1,4,6,10,8,11 | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2})^{\psi_\omega(0)})$ |
| 1,4,6,10,8,12 | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2})^{\psi_\omega(\Omega_{\omega\cdot 2})})$ |
| 1,4,6,10,9 | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2}+1))$ |
| 1,4,6,10,9,10 | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2}+\omega))$ |
| 1,4,6,10,9,11 | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2}+\Omega))$ |
| 1,4,6,10,9,11,14,13 | $\psi(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2}+\psi_\omega(\Omega_{\omega\cdot 2}+1)))$ |
| 1,4,6,10,9,12 | $\psi(\Omega_{\omega\cdot 2}+\Omega_{\omega+1})$ |
| 1,4,6,10,9,12,9,12 | $\psi(\Omega_{\omega\cdot 2}+\Omega_{\omega+1}\cdot 2)$ |
| 1,4,6,10,9,12,11 | $\psi(\Omega_{\omega\cdot 2}+\Omega_{\omega+1}\cdot\Omega)$ |
| 1,4,6,10,9,12,11,4 | $\psi(\Omega_{\omega\cdot 2}+\Omega_{\omega+1}\cdot\Omega_\omega)$ |
| 1,4,6,10,9,12,12 | $\psi(\Omega_{\omega\cdot 2}+\Omega_{\omega+1}^2)$ |
| 1,4,6,10,9,12,13 | $\psi(\Omega_{\omega\cdot 2}+\Omega_{\omega+1}^\omega)$ |
| 1,4,6,10,9,12,15 | $\psi(\Omega_{\omega\cdot 2}+\Omega_{\omega+1}^{\Omega_{\omega+1}})$ |
| 1,4,6,10,9,13 | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega+1}(0))$ |
| 1,4,6,10,9,14 | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega+1}(\Omega_{\omega\cdot 2}))$ |
| 1,4,6,10,9,14,13,17 | $\psi(\Omega_{\omega\cdot 2}+\Omega_{\omega+2})$ |
| 1,4,6,10,9,14,13,18 | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega+2}(0))$ |
| 1,4,6,10,9,14,13,19 | $\psi(\Omega_{\omega\cdot 2}+\psi_{\omega+2}(\Omega_{\omega\cdot 2}))$ |
| 1,4,6,10,10 | $\psi(\Omega_{\omega\cdot 2}\cdot 2)$ |
| 1,4,6,10,12 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega)$ |
| 1,4,6,10,12,4 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega)$ |
| 1,4,6,10,12,4,6,10,12 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\psi_\omega(\Omega_{\omega\cdot 2}\cdot\Omega_\omega))$ |
| 1,4,6,10,12,6,10,12 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\psi_\omega(\Omega_{\omega\cdot 2}\cdot\Omega_\omega)^2)$ |
| 1,4,6,10,12,9 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\psi_\omega(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+1))$ |
| 1,4,6,10,12,9,11,15,17 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\psi_\omega(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\psi_\omega(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+1)))$ |
| 1,4,6,10,12,9,12 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\Omega_{\omega+1})$ |
| 1,4,6,10,12,9,13 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\psi_{\omega+1}(0))$ |
| 1,4,6,10,12,9,14 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\psi_{\omega+1}(\Omega_{\omega\cdot 2}\cdot\Omega_\omega))$ |
| 1,4,6,10,12,9,14,13,17 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\Omega_{\omega+2})$ |
| 1,4,6,10,12,9,14,13,19 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\psi_{\omega+2}(\Omega_{\omega\cdot 2}\cdot\Omega_\omega))$ |
| 1,4,6,10,12,10 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\Omega_{\omega\cdot 2})$ |
| 1,4,6,10,12,10,12 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega+\Omega_{\omega\cdot 2}\cdot\Omega)$ |
| 1,4,6,10,12,10,12,4 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega\cdot 2)$ |
| 1,4,6,10,12,11 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega\cdot\omega)$ |
| 1,4,6,10,12,12 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega\cdot\Omega)$ |
| 1,4,6,10,12,12,4 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega^2)$ |
| 1,4,6,10,12,14,4 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_\omega^{\Omega_\omega})$ |
| 1,4,6,10,12,15 | $\psi(\Omega_{\omega\cdot 2}\cdot\psi_\omega(0))$ |
| 1,4,6,10,12,16 | $\psi(\Omega_{\omega\cdot 2}\cdot\psi_\omega(\Omega_{\omega\cdot 2}))$ |
| 1,4,6,10,12,16,18,22 | $\psi(\Omega_{\omega\cdot 2}\cdot\psi_\omega(\Omega_{\omega\cdot 2}\cdot\psi_\omega(\Omega_{\omega\cdot 2})))$ |
| 1,4,6,10,13 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1})$ |
| 1,4,6,10,13,9,12 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}+\Omega_{\omega+1})$ |
| 1,4,6,10,13,9,13 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}+\psi_{\omega+1}(0))$ |
| 1,4,6,10,13,9,14,17 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}+\psi_{\omega+1}(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}))$ |
| 1,4,6,10,13,9,14,17,13,17 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}+\Omega_{\omega+2})$ |
| 1,4,6,10,13,9,14,17,14 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}+\Omega_{\omega\cdot 2})$ |
| 1,4,6,10,13,9,14,17,14,17 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}\cdot 2)$ |
| 1,4,6,10,13,9,14,17,15 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}\cdot\omega)$ |
| 1,4,6,10,13,9,14,17,16,4 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}\cdot\Omega_\omega)$ |
| 1,4,6,10,13,9,14,17,17 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}^2)$ |
| 1,4,6,10,13,9,14,17,19 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1}^\Omega)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,6,10,13,9,14,17,21 | $\psi(\Omega_{\omega\cdot 2}\cdot\psi_{\omega+1}(0))$ |
| 1,4,6,10,13,9,14,17,22 | $\psi(\Omega_{\omega\cdot 2}\cdot\psi_{\omega+1}(\Omega_{\omega\cdot 2}))$ |
| 1,4,6,10,13,9,14,18 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+2})$ |
| 1,4,6,10,13,9,<br>14,18,13,19,24 | $\psi(\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+3})$ |
| 1,4,6,10,13,10 | $\psi(\Omega^2_{\omega\cdot 2})$ |
| 1,4,6,10,13,11,10,13,11 | $\psi(\Omega^{\omega\cdot 2}_{\omega\cdot 2})$ |
| 1,4,6,10,13,12 | $\psi(\Omega^{\Omega}_{\omega\cdot 2})$ |
| 1,4,6,10,13,12,4 | $\psi(\Omega^{\Omega_\omega}_{\omega\cdot 2})$ |
| 1,4,6,10,13,13,10 | $\psi(\Omega^{\Omega_{\omega\cdot 2}}_{\omega\cdot 2})$ |
| 1,4,6,10,13,17 | $\psi(\psi_{\omega\cdot 2}(0))$ |
| 1,4,6,10,13,17,17 | $\psi(\psi_{\omega\cdot 2}(1))$ |
| 1,4,6,10,13,17,19 | $\psi(\psi_{\omega\cdot 2}(\Omega))$ |
| 1,4,6,10,13,17,19,4 | $\psi(\psi_{\omega\cdot 2}(\Omega_\omega))$ |
| 1,4,6,10,13,17,20,10 | $\psi(\psi_{\omega\cdot 2}(\Omega_{\omega\cdot 2}))$ |
| 1,4,6,10,13,17,20,23 | $\psi(\psi_{\omega\cdot 2}(\psi_{\omega\cdot 2}(0)))$ |
| 1,4,6,10,13,17,21 | $\psi(\Omega_{\omega\cdot 2+1})$ |
| 1,4,6,10,13,17,22 | $\psi(\psi_{\omega\cdot 2+1}(0))$ |
| 1,4,6,10,13,18 | $\psi(\Omega_{\omega\cdot 3})$ |
| 1,4,6,10,13,18,18 | $\psi(\Omega_{\omega\cdot 3}\cdot 2)$ |
| 1,4,6,10,13,18,20 | $\psi(\Omega_{\omega\cdot 3}\cdot\Omega)$ |
| 1,4,6,10,13,18,20,4 | $\psi(\Omega_{\omega\cdot 3}\cdot\Omega_\omega)$ |
| 1,4,6,10,13,18,21,10 | $\psi(\Omega_{\omega\cdot 3}\cdot\Omega_{\omega\cdot 2})$ |
| 1,4,6,10,13,18,22,18 | $\psi(\Omega^2_{\omega\cdot 3})$ |
| 1,4,6,10,13,18,22,27 | $\psi(\psi_{\omega\cdot 3}(0))$ |
| 1,4,6,10,13,18,22,28 | $\psi(\Omega_{\omega\cdot 4})$ |
| 1,4,7 | $\psi(\Omega_{\omega^2})$ |
| 1,4,7,4 | $\psi(\Omega_{\omega^2}+\Omega_\omega)$ |
| 1,4,7,4,6,9 | $\psi(\Omega_{\omega^2}+\psi_\omega(0))$ |
| 1,4,7,4,6,10 | $\psi(\Omega_{\omega^2}+\psi_\omega(\Omega_{\omega\cdot 2}))$ |
| 1,4,7,4,6,10,14 | $\psi(\Omega_{\omega^2}+\psi_\omega(\Omega_{\omega^2}))$ |
| 1,4,7,4,6,10,14,9,12 | $\psi(\Omega_{\omega^2}+\Omega_{\omega+1})$ |
| 1,4,7,4,6,10,14,10 | $\psi(\Omega_{\omega^2}+\Omega_{\omega\cdot 2})$ |
| 1,4,7,4,6,10,<br>14,10,9,14,19,14 | $\psi(\Omega_{\omega^2}+\Omega_{\omega\cdot 3})$ |
| 1,4,7,4,7 | $\psi(\Omega_{\omega^2}\cdot 2)$ |
| 1,4,7,5 | $\psi(\Omega_{\omega^2}\cdot\omega)$ |
| 1,4,7,6 | $\psi(\Omega_{\omega^2}\cdot\Omega)$ |
| 1,4,7,6,4 | $\psi(\Omega_{\omega^2}\cdot\Omega_\omega)$ |
| 1,4,7,6,4,6,9 | $\psi(\Omega_{\omega^2}\cdot\psi_\omega(0))$ |
| 1,4,7,6,4,6,10 | $\psi(\Omega_{\omega^2}\cdot\psi_\omega(\Omega_{\omega\cdot 2}))$ |
| 1,4,7,6,4,6,10,14 | $\psi(\Omega_{\omega^2}\cdot\psi_\omega(\Omega_{\omega^2}))$ |
| 1,4,7,6,4,6,10,14,9,12 | $\psi(\Omega_{\omega^2}\cdot\Omega_{\omega+1})$ |
| 1,4,7,6,4,6,10,14,13,10 | $\psi(\Omega_{\omega^2}\cdot\Omega_{\omega\cdot 2})$ |
| 1,4,7,6,4,6,10,14,<br>13,10,13,18,23,22,18 | $\psi(\Omega_{\omega^2}\cdot\Omega_{\omega\cdot 3})$ |
| 1,4,7,6,4,7 | $\psi(\Omega^2_{\omega^2})$ |
| 1,4,7,6,5 | $\psi(\Omega^2_{\omega^2}\cdot\omega)$ |
| 1,4,7,6,6 | $\psi(\Omega^2_{\omega^2}\cdot\Omega)$ |
| 1,4,7,6,6,4 | $\psi(\Omega^2_{\omega^2}\cdot\Omega_\omega)$ |
| 1,4,7,6,6,4,7 | $\psi(\Omega^3_{\omega^2})$ |
| 1,4,7,6,7 | $\psi(\Omega^\omega_{\omega^2})$ |
| 1,4,7,6,8 | $\psi(\Omega^\Omega_{\omega^2})$ |
| 1,4,7,6,8,4,7 | $\psi(\Omega^{\Omega_{\omega^2}}_{\omega^2})$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,6,9 | $\psi(\psi_{\omega^2}(0))$ |
| 1,4,7,6,9,9 | $\psi(\psi_{\omega^2}(1))$ |
| 1,4,7,6,9,11 | $\psi(\psi_{\omega^2}(\Omega))$ |
| 1,4,7,6,9,11,4,7 | $\psi(\psi_{\omega^2}(\Omega_{\omega^2}))$ |
| 1,4,7,6,9,11,14 | $\psi(\psi_{\omega^2}(\psi_{\omega^2}(0)))$ |
| 1,4,7,6,9,12 | $\psi(\Omega_{\omega^2+1})$ |
| 1,4,7,6,9,13 | $\psi(\psi_{\omega^2+1}(0))$ |
| 1,4,7,6,10 | $\psi(\Omega_{\omega^2+\omega})$ |
| 1,4,7,6,10,14 | $\psi(\Omega_{\omega^2\cdot 2})$ |
| 1,4,7,7 | $\psi(\Omega_{\omega^3})$ |
| 1,4,7,8 | $\psi(\Omega_{\omega^\omega})$ |
| 1,4,7,8,7 | $\psi(\Omega_{\omega^{\omega+1}})$ |
| 1,4,7,8,8 | $\psi(\Omega_{\omega^{\omega\cdot 2}})$ |
| 1,4,7,8,9 | $\psi(\Omega_{\omega^{\omega^2}})$ |
| 1,4,7,8,10 | $\psi(\Omega_{\psi(0)})$ |
| 1,4,7,8,10,10 | $\psi(\Omega_{\psi(1)})$ |
| 1,4,7,8,10,11 | $\psi(\Omega_{\psi(\omega)})$ |
| 1,4,7,8,10,12 | $\psi(\Omega_{\psi(\Omega)})$ |
| 1,4,7,8,10,12,14 | $\psi(\Omega_{\psi(\Omega^\Omega)})$ |
| 1,4,7,8,10,13 | $\psi(\Omega_{\psi(\psi_1(0))})$ |
| 1,4,7,8,10,13,17 | $\psi(\Omega_{\psi(\psi_2(0))})$ |
| 1,4,7,8,11 | $\psi(\Omega_{\psi(\Omega_\omega)})$ |
| 1,4,7,8,11,14 | $\psi(\Omega_{\psi(\Omega_{\omega^2})})$ |
| 1,4,7,8,11,14,15,18,21 | $\psi(\Omega_{\psi(\Omega_{\psi(\Omega_{\omega^2})})})$ |
| 1,4,7,9 | $\psi(\Omega_\Omega)$ |
| 1,4,7,9,3 | $\psi(\Omega_\Omega+1)$ |
| 1,4,7,9,3,6 | $\psi(\Omega_\Omega+\psi_1(0))$ |
| 1,4,7,9,3,7 | $\psi(\Omega_\Omega+\psi_1(\Omega_\omega))$ |
| 1,4,7,9,3,7,11,13 | $\psi(\Omega_\Omega+\psi_1(\Omega_\Omega))$ |
| 1,4,7,9,3,7,11,13,6 | $\psi(\Omega_\Omega+\psi_1(\Omega_\Omega+1))$ |
| 1,4,7,9,3,7,11,<br>13,6,8,12,16,18 | $\psi(\Omega_\Omega+\psi_1(\Omega_\Omega+\psi_1(\Omega_\Omega)))$ |
| 1,4,7,9,3,7,11,13,6,9 | $\psi(\Omega_\Omega+\Omega_2)$ |
| 1,4,7,9,3,7,11,13,7 | $\psi(\Omega_\Omega+\Omega_\omega)$ |
| 1,4,7,9,3,7,11,13,7,7 | $\psi(\Omega_\Omega+\Omega_\omega\cdot 2)$ |
| 1,4,7,9,3,7,<br>11,13,7,10,7 | $\psi(\Omega_\Omega+\Omega_\omega^2)$ |
| 1,4,7,9,3,7,<br>11,13,7,10,14 | $\psi(\Omega_\Omega+\psi_\omega(0))$ |
| 1,4,7,9,3,7,11,<br>13,7,10,15,20,22 | $\psi(\Omega_\Omega+\psi_\omega(\Omega_\Omega))$ |
| 1,4,7,9,3,7,11,13,<br>7,10,15,20,22,14,18 | $\psi(\Omega_\Omega+\Omega_{\omega+1})$ |
| 1,4,7,9,3,7,11,13,<br>7,10,15,20,22,15 | $\psi(\Omega_\Omega+\Omega_{\omega\cdot 2})$ |
| 1,4,7,9,3,7,11,13,7,11 | $\psi(\Omega_\Omega+\Omega_{\omega^2})$ |
| 1,4,7,9,3,7,<br>11,13,7,11,7,11 | $\psi(\Omega_\Omega+\Omega_{\omega^2}\cdot 2)$ |
| 1,4,7,9,3,7,<br>11,13,7,11,10,7 | $\psi(\Omega_\Omega+\Omega_{\omega^2}^2)$ |
| 1,4,7,9,3,7,<br>11,13,7,11,10,14 | $\psi(\Omega_\Omega+\psi_{\omega^2}(0))$ |
| 1,4,7,9,3,7,<br>11,13,7,11,11 | $\psi(\Omega_\Omega+\Omega_{\omega^3})$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,9,3,7,<br>11,13,7,11,12 | $\psi(\Omega_\Omega + \Omega_{\omega^\omega})$ |
| 1,4,7,9,3,7,<br>11,13,7,11,12,15 | $\psi(\Omega_\Omega + \Omega_{\psi(\Omega_\omega)})$ |
| 1,4,7,9,3,7,11,13,7,11,12,<br>15,18,20,14,18,22,24,18,22 | $\psi(\Omega_\Omega + \Omega_{\psi(\Omega_\Omega+\Omega_2)})$ |
| 1,4,7,9,3,7,<br>11,13,7,11,13 | $\psi(\Omega_\Omega \cdot 2)$ |
| 1,4,7,9,3,7,11,13,8 | $\psi(\Omega_\Omega \cdot \omega)$ |
| 1,4,7,9,3,7,11,13,9 | $\psi(\Omega_\Omega \cdot \Omega)$ |
| 1,4,7,9,3,7,11,13,10,7 | $\psi(\Omega_\Omega \cdot \Omega_\omega)$ |
| 1,4,7,9,3,7,<br>11,13,10,7,11 | $\psi(\Omega_\Omega \cdot \Omega_{\omega^2})$ |
| 1,4,7,9,3,7,<br>11,13,10,7,11,13 | $\psi(\Omega_\Omega^2)$ |
| 1,4,7,9,3,7,11,13,10,8 | $\psi(\Omega_\Omega^\omega)$ |
| 1,4,7,9,3,7,<br>11,13,10,12,7 | $\psi(\Omega_\Omega^{\Omega_\omega})$ |
| 1,4,7,9,3,7,11,<br>13,10,13,7,11,13 | $\psi(\Omega_\Omega^{\Omega_\Omega})$ |
| 1,4,7,9,3,7,11,13,10,14 | $\psi(\psi_\Omega(0))$ |
| 1,4,7,9,3,7,<br>11,13,10,14,14 | $\psi(\psi_\Omega(1))$ |
| 1,4,7,9,3,7,<br>11,13,10,14,15 | $\psi(\psi_\Omega(\omega))$ |
| 1,4,7,9,3,7,<br>11,13,10,14,16 | $\psi(\psi_\Omega(\Omega))$ |
| 1,4,7,9,3,7,<br>11,13,10,14,17 | $\psi(\psi_\Omega(\Omega_2))$ |
| 1,4,7,9,3,7,11,<br>13,10,14,17,7,11,13 | $\psi(\psi_\Omega(\Omega_\Omega))$ |
| 1,4,7,9,3,7,11,<br>13,10,14,17,21 | $\psi(\psi_\Omega(\psi_\Omega(0)))$ |
| 1,4,7,9,3,7,<br>11,13,10,14,18 | $\psi(\Omega_{\Omega+1})$ |
| 1,4,7,9,3,7,11,13,10,15 | $\psi(\Omega_{\Omega+\omega})$ |
| 1,4,7,9,3,7,<br>11,13,10,15,19,25 | $\psi(\Omega_{\Omega+\omega\cdot 2})$ |
| 1,4,7,9,3,7,<br>11,13,10,15,20 | $\psi(\Omega_{\Omega+\omega^2})$ |
| 1,4,7,9,3,7,11,<br>13,10,15,20,21,23 | $\psi(\Omega_{\Omega+\psi(0)})$ |
| 1,4,7,9,3,7,11,<br>13,10,15,20,21,24 | $\psi(\Omega_{\Omega+\psi(\Omega_\omega)})$ |
| 1,4,7,9,3,7,11,<br>13,10,15,20,21,24,27,29 | $\psi(\Omega_{\Omega+\psi(\Omega_\Omega)})$ |
| 1,4,7,9,3,7,11,<br>13,10,15,20,22 | $\psi(\Omega_{\Omega\cdot 2})$ |
| 1,4,7,9,3,7,11,<br>13,10,15,20,22,7,11,13 | $\psi(\Omega_{\Omega\cdot 2} + \Omega_\Omega)$ |
| 1,4,7,9,3,7,11,13,10,15,<br>20,22,7,11,13,10,14 | $\psi(\Omega_{\Omega\cdot 2} + \psi_\Omega(0))$ |
| 1,4,7,9,3,7,11,13,10,15,<br>20,22,7,11,13,10,15,20,22 | $\psi(\Omega_{\Omega\cdot 2} + \psi_\Omega(\Omega_{\Omega\cdot 2}))$ |
| 1,4,7,9,3,7,11,13,<br>10,15,20,22,9 | $\psi(\Omega_{\Omega\cdot 2} + \psi_\Omega(\Omega_{\Omega\cdot 2}) \cdot \Omega)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,9,3,7,11,13,10, 15,20,22,10,7,11,13 | $\psi(\Omega_{\Omega\cdot 2}+\psi_\Omega(\Omega_{\Omega\cdot 2})\cdot\Omega_\Omega)$ |
| 1,4,7,9,3,7,11,13,10, 15,20,22,10,15,20,22 | $\psi(\Omega_{\Omega\cdot 2}+\psi_\Omega(\Omega_{\Omega\cdot 2})^2)$ |
| 1,4,7,9,3,7,11,13, 10,15,20,22,14 | $\psi(\Omega_{\Omega\cdot 2}+\psi_\Omega(\Omega_{\Omega\cdot 2}+1))$ |
| 1,4,7,9,3,7,11,13, 10,15,20,22,14,18 | $\psi(\Omega_{\Omega\cdot 2}+\Omega_{\Omega+1})$ |
| 1,4,7,9,3,7,11,13, 10,15,20,22,15,20,22 | $\psi(\Omega_{\Omega\cdot 2}\cdot 2)$ |
| 1,4,7,9,3,7,11,13, 10,15,20,22,18 | $\psi(\Omega_{\Omega\cdot 2}\cdot\Omega_2)$ |
| 1,4,7,9,3,7,11,13,10,15, 20,22,19,15,20,22 | $\psi(\Omega^2_{\Omega\cdot 2})$ |
| 1,4,7,9,3,7,11,13,10,15, 20,22,19,23,15,20,22 | $\psi(\Omega^{\Omega_{\Omega\cdot 2}}_{\Omega\cdot 2})$ |
| 1,4,7,9,3,7,11,13, 10,15,20,22,19,24 | $\psi(\psi_{\Omega\cdot 2}(0))$ |
| 1,4,7,9,3,7,11,13,10, 15,20,22,19,24,29 | $\psi(\Omega_{\Omega\cdot 2+1})$ |
| 1,4,7,9,3,7,11,13, 10,15,20,22,19,25 | $\psi(\Omega_{\Omega\cdot 2+\omega})$ |
| 1,4,7,9,3,7,11,13,10, 15,20,22,19,25,31,33 | $\psi(\Omega_{\Omega\cdot 3})$ |
| 1,4,7,9,3,7,11,13,11 | $\psi(\Omega_{\Omega\cdot\omega})$ |
| 1,4,7,9,3,7,11,13,11,11 | $\psi(\Omega_{\Omega\cdot\omega^2})$ |
| 1,4,7,9,3,7,11,13,11,13 | $\psi(\Omega_{\Omega^2})$ |
| 1,4,7,9,3,7,11,13,13 | $\psi(\Omega_{\Omega^\Omega})$ |
| 1,4,7,9,3,7,11,13, 13,10,13,7,11,13,13 | $\psi(\Omega^{\Omega_{\Omega^\Omega}}_{\Omega^\Omega})$ |
| 1,4,7,9,3,7,11,13,15 | $\psi(\Omega_{\Omega^{\Omega^\Omega}})$ |
| 1,4,7,9,3,7,11,13,16 | $\psi(\Omega_{\psi_1(0)})$ |
| 1,4,7,9,3,7,11,13,16,19 | $\psi(\Omega_{\psi_1(\Omega_2)})$ |
| 1,4,7,9,3,7,11,13,17 | $\psi(\Omega_{\psi_1(\Omega_\omega)})$ |
| 1,4,7,9,3,7, 11,13,17,21,23 | $\psi(\Omega_{\psi_1(\Omega_\Omega)})$ |
| 1,4,7,9,3,7,11, 13,17,21,23,26 | $\psi(\Omega_{\psi_1(\Omega_{\psi_1(0)})})$ |
| 1,4,7,9,3,7,11,14 | $\psi(\Omega_{\Omega_2})$ |
| 1,4,7,9,3,7,11,14, 6,11,16,19,15,20,25 | $\psi(\Omega_{\Omega_2+1})$ |
| 1,4,7,9,3,7,11, 14,6,11,16,19,19 | $\psi(\Omega_{\Omega_2^{\Omega_2}})$ |
| 1,4,7,9,3,7,11, 14,6,11,16,19,23 | $\psi(\Omega_{\psi_2(0)})$ |
| 1,4,7,9,3,7,11, 14,6,11,16,20 | $\psi(\Omega_{\Omega_3})$ |
| 1,4,7,9,4 | $\psi(\Omega_{\Omega_\omega})$ |
| 1,4,7,9,4,4 | $\psi(\Omega_{\Omega_\omega}+\Omega_\omega)$ |
| 1,4,7,9,4,6,9 | $\psi(\Omega_{\Omega_\omega}+\psi_\omega(0))$ |
| 1,4,7,9,4,6,10,14,16 | $\psi(\Omega_{\Omega_\omega}+\psi_\omega(\Omega_\Omega))$ |
| 1,4,7,9,4,6,10,14,16,4 | $\psi(\Omega_{\Omega_\omega}+\psi_\omega(\Omega_{\Omega_\omega}))$ |
| 1,4,7,9,4,6, 10,14,16,9,12 | $\psi(\Omega_{\Omega_\omega}+\Omega_{\omega+1})$ |
| 1,4,7,9,4,6,10,14,16,10 | $\psi(\Omega_{\Omega_\omega}+\Omega_{\omega\cdot 2})$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,9,4,6,<br>10,14,16,10,14,16 | $\psi(\Omega_{\Omega_\omega} + \Omega_\Omega)$ |
| 1,4,7,9,4,6,<br>10,14,16,10,14,16,4 | $\psi(\Omega_{\Omega_\omega} \cdot 2)$ |
| 1,4,7,9,4,6,10,14,16,11 | $\psi(\Omega_{\Omega_\omega} \cdot \omega)$ |
| 1,4,7,9,4,6,10,14,16,12 | $\psi(\Omega_{\Omega_\omega} \cdot \Omega)$ |
| 1,4,7,9,4,6,<br>10,13,15,12,4 | $\psi(\Omega_{\Omega_\omega} \cdot \Omega_\omega)$ |
| 1,4,7,9,4,6,<br>10,14,16,12,15 | $\psi(\Omega_{\Omega_\omega} \cdot \psi_\omega(0))$ |
| 1,4,7,9,4,6,10,14,16,13 | $\psi(\Omega_{\Omega_\omega} \cdot \Omega_{\omega+1})$ |
| 1,4,7,9,4,6,10,<br>14,16,13,10,14,16 | $\psi(\Omega_{\Omega_\omega} \cdot \Omega_\Omega)$ |
| 1,4,7,9,4,6,10,<br>14,16,13,10,14,16,4 | $\psi(\Omega^2_{\Omega_\omega})$ |
| 1,4,7,9,4,6,<br>10,14,16,13,12 | $\psi(\Omega^\Omega_{\Omega_\omega})$ |
| 1,4,7,9,4,6,<br>10,14,16,13,17 | $\psi(\psi_{\Omega_\omega}(0))$ |
| 1,4,7,9,4,6,10,<br>14,16,13,18,23,26 | $\psi(\Omega_{\Omega_\omega+1})$ |
| 1,4,7,9,4,6,10,<br>14,16,14,16,4 | $\psi(\Omega_{\Omega^2_\omega})$ |
| 1,4,7,9,4,6,<br>10,14,16,16,4 | $\psi(\Omega_{\Omega^{\Omega_\omega}_\omega})$ |
| 1,4,7,9,4,6,10,14,16,19 | $\psi(\Omega_{\psi_\omega(0)})$ |
| 1,4,7,9,4,6,10,14,17 | $\psi(\Omega_{\Omega_{\omega+1}})$ |
| 1,4,7,9,4,7 | $\psi(\Omega_{\Omega_{\omega^2}})$ |
| 1,4,7,9,4,7,8,10 | $\psi(\Omega_{\Omega_{\psi(0)}})$ |
| 1,4,7,9,4,7,8,11 | $\psi(\Omega_{\Omega_{\psi(\Omega_\omega)}})$ |
| 1,4,7,9,4,7,8,11,14,16 | $\psi(\Omega_{\Omega_{\psi(\Omega_\Omega)}})$ |
| 1,4,7,9,4,7,9 | $\psi(\Omega_{\Omega_\Omega})$ |
| 1,4,7,9,4,7,9,4,7,9 | $\psi(\Omega_{\Omega_{\Omega_\Omega}})$ |
| 1,4,7,9,5 | $\psi(\psi_I(0))$ |
| 1,4,7,9,5,3 | $\psi(\psi_I(0) + 1)$ |
| 1,4,7,9,5,3,<br>4,7,10,12,8 | $\psi(\psi_I(0) + \psi(\psi_I(0)))$ |
| 1,4,7,9,5,3,5 | $\psi(\psi_I(0) + \Omega)$ |
| 1,4,7,9,5,4,6,9 | $\psi(\psi_I(0) + \psi_{\Omega_{\omega+1}}(0))$ |
| 1,4,7,9,5,4,6,9,12 | $\psi(\psi_I(0) + \Omega_{\omega+1})$ |
| 1,4,7,9,5,4,7,9 | $\psi(\psi_I(0) + \Omega_\Omega)$ |
| 1,4,7,9,5,4,7,9,5 | $\psi(\psi_I(0) \cdot 2)$ |
| 1,4,7,9,5,5 | $\psi(\psi_I(0) \cdot \omega)$ |
| 1,4,7,9,5,8,11,13,9 | $\psi(\psi_I(0) \cdot \psi(\psi_I(0)))$ |
| 1,4,7,9,6 | $\psi(\psi_I(0) \cdot \Omega)$ |
| 1,4,7,9,6,4,7 | $\psi(\psi_I(0) \cdot \Omega_{\omega^2})$ |
| 1,4,7,9,6,4,7,9 | $\psi(\psi_I(0) \cdot \Omega_\Omega)$ |
| 1,4,7,9,6,4,7,9,6,4,7,9,5 | $\psi(\psi_I(0)^2 \cdot 2)$ |
| 1,4,7,9,6,5 | $\psi(\psi_I(0)^2 \cdot \omega)$ |
| 1,4,7,9,6,6 | $\psi(\psi_I(0)^2 \cdot \Omega)$ |
| 1,4,7,9,6,6,4,7,9,5 | $\psi(\psi_I(0)^3)$ |
| 1,4,7,9,6,7,10,13,15,11 | $\psi(\psi_I(0)^{\psi(\psi_I(0))})$ |
| 1,4,7,9,6,8 | $\psi(\psi_I(0)^\Omega)$ |
| 1,4,7,9,6,8,4,7,9,5 | $\psi(\psi_I(0)^{\psi_I(0)})$ |
| 1,4,7,9,6,9 | $\psi(\psi_{\Omega_{\psi_I(0)+1}}(0))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,9,6,9,9 | $\psi(\psi_{\Omega_{\psi_I(0)+1}}(1))$ |
| 1,4,7,9,6,9,<br>11,4,7,9,5 | $\psi(\psi_{\Omega_{\psi_I(0)+1}}(\psi_I(0)))$ |
| 1,4,7,9,6,9,11,14 | $\psi(\psi_{\Omega_{\psi_I(0)+1}}(\psi_{\Omega_{\psi_I(0)+1}}(0)))$ |
| 1,4,7,9,6,9,12 | $\psi(\Omega_{\psi_I(0)+1})$ |
| 1,4,7,9,6,10 | $\psi(\Omega_{\psi_I(0)+\omega})$ |
| 1,4,7,9,6,10,14,16 | $\psi(\Omega_{\psi_I(0)+\Omega})$ |
| 1,4,7,9,6,10,<br>14,16,4,7,9,5 | $\psi(\Omega_{\psi_I(0)\cdot 2})$ |
| 1,4,7,9,6,10,14,<br>16,4,7,9,5,4,7,9,5 | $\psi(\Omega_{\psi_I(0)\cdot 2}+\psi_I(0))$ |
| 1,4,7,9,6,10,14,<br>16,4,7,9,6,9 | $\psi(\Omega_{\psi_I(0)\cdot 2}+\psi_{\Omega_{\psi_I(0)+1}}(0))$ |
| 1,4,7,9,6,10,14,<br>16,4,7,9,6,9,12 | $\psi(\Omega_{\psi_I(0)\cdot 2}+\psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)+1}))$ |
| 1,4,7,9,6,10,14,16,4,7,<br>9,6,10,14,16,4,7,9,5 | $\psi(\Omega_{\psi_I(0)\cdot 2}+\psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2}))$ |
| 1,4,7,9,6,10,14,16,6 | $\psi(\Omega_{\psi_I(0)\cdot 2}+\psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2})\cdot\Omega)$ |
| 1,4,7,9,6,10,<br>14,16,6,4,7,9,5 | $\psi(\Omega_{\psi_I(0)\cdot 2}+\psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2})\cdot\psi_I(0))$ |
| 1,4,7,9,6,10,14,16,6,9 | $\psi(\Omega_{\psi_I(0)\cdot 2}+\psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2})\cdot\psi_{\Omega_{\psi_I(0)+1}}(0))$ |
| 1,4,7,9,6,10,14,16,<br>6,10,14,16,4,7,9,5 | $\psi(\Omega_{\psi_I(0)\cdot 2}+\psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2})^2)$ |
| 1,4,7,9,6,10,14,16,7 | $\psi(\Omega_{\psi_I(0)\cdot 2}+\psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2})^\omega)$ |
| 1,4,7,9,6,10,14,<br>16,8,4,7,9,5 | $\psi(\Omega_{\psi_I(0)\cdot 2}+\psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2})^{\psi_I(0)})$ |
| 1,4,7,9,6,10,14,16,9 | $\psi(\Omega_{\psi_I(0)\cdot 2}+\psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2}+1))$ |
| 1,4,7,9,6,10,14,16,9,12 | $\psi(\Omega_{\psi_I(0)\cdot 2}+\Omega_{\psi_I(0)+1})$ |
| 1,4,7,9,6,10,14,16,10 | $\psi(\Omega_{\psi_I(0)\cdot 2}+\Omega_{\psi_I(0)+\omega})$ |
| 1,4,7,9,6,10,14,16,<br>10,14,16,4,7,9,5 | $\psi(\Omega_{\psi_I(0)\cdot 2}\cdot 2)$ |
| 1,4,7,9,6,10,14,16,12 | $\psi(\Omega_{\psi_I(0)\cdot 2}\cdot\Omega)$ |
| 1,4,7,9,6,10,<br>14,16,12,4,7,9,5 | $\psi(\Omega_{\psi_I(0)\cdot 2}\cdot\psi_I(0))$ |
| 1,4,7,9,6,10,14,16,12,15 | $\psi(\Omega_{\psi_I(0)\cdot 2}\cdot\psi_{\Omega_{\psi_I(0)+1}}(0))$ |
| 1,4,7,9,6,10,14,<br>16,12,16,20,22,4,7,9,5 | $\psi(\Omega_{\psi_I(0)\cdot 2}\cdot\psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_I(0)\cdot 2}))$ |
| 1,4,7,9,6,10,14,16,13 | $\psi(\Omega_{\psi_I(0)\cdot 2}\cdot\Omega_{\psi_I(0)+1})$ |
| 1,4,7,9,6,10,14,16,13,<br>10,14,16,13,4,7,9,5 | $\psi(\Omega^2_{\psi_I(0)\cdot 2})$ |
| 1,4,7,9,6,10,14,<br>16,13,13,4,7,9,5 | $\psi(\Omega^{\Omega_{\psi_I(0)\cdot 2}}_{\psi_I(0)\cdot 2})$ |
| 1,4,7,9,6,10,14,16,13,17 | $\psi(\psi_{\Omega_{\psi_I(0)\cdot 2+1}}(0))$ |
| 1,4,7,9,6,10,<br>14,16,13,17,21 | $\psi(\Omega_{\psi_I(0)\cdot 2+1})$ |
| 1,4,7,9,6,10,14,16,<br>13,18,23,25,4,7,9,5 | $\psi(\Omega_{\psi_I(0)\cdot 3})$ |
| 1,4,7,9,6,10,14,16,14 | $\psi(\Omega_{\psi_I(0)\cdot\omega})$ |
| 1,4,7,9,6,10,14,<br>16,14,10,14,16,14 | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot 2)$ |
| 1,4,7,9,6,10,14,16,14,12 | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\Omega)$ |
| 1,4,7,9,6,10,14,<br>16,14,12,4,7,9,5 | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\psi_I(0))$ |
| 1,4,7,9,6,10,14,<br>16,14,12,9,12 | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\psi_I(0)+\Omega_{\psi_I(0)+1})$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,9,6,10,14,16,14,12, 10,14,16,14,12,4,7,9,5 | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\psi_I(0)\cdot 2)$ |
| 1,4,7,9,6,10, 14,16,14,12,12 | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\psi_I(0)\cdot\Omega)$ |
| 1,4,7,9,6,10,14, 16,14,12,12,4,7,9,5 | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\psi_I(0)^2)$ |
| 1,4,7,9,6,10,14, 16,14,12,14,4,7,9,5 | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\psi_I(0)^{\psi_I(0)})$ |
| 1,4,7,9,6,10, 14,16,14,12,15 | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\psi_{\Omega_{\psi_I(0)+1}}(0))$ |
| 1,4,7,9,6,10,14,16,14,13 | $\psi(\Omega_{\psi_I(0)\cdot\omega}\cdot\Omega_{\psi_I(0)+1})$ |
| 1,4,7,9,6,10,14, 16,14,13,10,14,16,14 | $\psi(\Omega^2_{\psi_I(0)\cdot\omega})$ |
| 1,4,7,9,6,10, 14,16,14,13,15 | $\psi(\Omega^{\Omega}_{\psi_I(0)\cdot\omega})$ |
| 1,4,7,9,6,10, 14,16,14,13,17 | $\psi(\psi_{\Omega_{\psi_I(0)\cdot\omega+1}}(0))$ |
| 1,4,7,9,6,10, 14,16,14,13,17,21 | $\psi(\Omega_{\psi_I(0)\cdot\omega+1})$ |
| 1,4,7,9,6,10,14, 16,14,13,18,23,25,23 | $\psi(\Omega_{\psi_I(0)\cdot\omega\cdot 2})$ |
| 1,4,7,9,6,10,14,16,14,14 | $\psi(\Omega_{\psi_I(0)\cdot\omega^2})$ |
| 1,4,7,9,6,10,14,16,14,16 | $\psi(\Omega_{\psi_I(0)\cdot\Omega})$ |
| 1,4,7,9,6,10,14, 16,14,16,4,7,9,5 | $\psi(\Omega_{\psi_I(0)^2})$ |
| 1,4,7,9,6,10,14, 16,16,4,7,9,5 | $\psi(\Omega_{\psi_I(0)^{\psi_I(0)}})$ |
| 1,4,7,9,6,10,14,16,19 | $\psi(\Omega_{\psi_{\Omega_{\psi_I(0)+1}}(0)})$ |
| 1,4,7,9,6,10,14,16,19,19 | $\psi(\Omega_{\psi_{\Omega_{\psi_I(0)+1}}(1)})$ |
| 1,4,7,9,6,10,14, 16,20,24,26,29 | $\psi(\Omega_{\psi_{\Omega_{\psi_I(0)+1}}(\Omega_{\psi_{\Omega_{\psi_I(0)+1}}(0)})})$ |
| 1,4,7,9,6,10,14,17 | $\psi(\Omega_{\Omega_{\psi_I(0)+1}})$ |
| 1,4,7,9,6,10, 14,17,10,14,17 | $\psi(\Omega_{\Omega_{\Omega_{\psi_I(0)+1}}})$ |
| 1,4,7,9,6,10,14,17,11 | $\psi(\psi_I(1))$ |
| 1,4,7,9,6,10, 14,17,11,4,7,9,5 | $\psi(\psi_I(1)+\psi_I(0))$ |
| 1,4,7,9,6,10,14, 17,11,4,7,9,6,9 | $\psi(\psi_I(1)+\psi_{\Omega_{\psi_I(0)+1}}(0))$ |
| 1,4,7,9,6,10,14,17, 11,4,7,9,6,10,14,17 | $\psi(\psi_I(1)+\psi_{\Omega_{\psi_I(0)+1}}(\psi_I(1)))$ |
| 1,4,7,9,6,10,14, 17,11,6,4,7,9,5 | $\psi(\psi_I(1)+\psi_{\Omega_{\psi_I(0)+1}}(\psi_I(1))\cdot\psi_I(0))$ |
| 1,4,7,9,6,10,14,17,11,9 | $\psi(\psi_I(1)+\psi_{\Omega_{\psi_I(0)+1}}(\psi_I(1)+1))$ |
| 1,4,7,9,6,10, 14,17,11,9,12 | $\psi(\psi_I(1)+\Omega_{\psi_I(0)+1})$ |
| 1,4,7,9,6,10,14,17,11,10 | $\psi(\psi_I(1)+\Omega_{\psi_I(0)+\omega})$ |
| 1,4,7,9,6,10,14, 17,11,10,14,17,11 | $\psi(\psi_I(1)\cdot 2)$ |
| 1,4,7,9,6,10,14, 17,12,4,7,9,5 | $\psi(\psi_I(1)\cdot\psi_I(0))$ |
| 1,4,7,9,6,10,14, 17,12,6,10,14,17,11 | $\psi(\psi_I(1)\cdot\psi_{\Omega_{\psi_I(0)+1}}(\psi_I(1)))$ |
| 1,4,7,9,6,10, 14,17,12,9,12 | $\psi(\psi_I(1)\cdot\Omega_{\psi_I(0)+1})$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,9,6,10,14,17,12,10 | $\psi(\psi_I(1)\cdot\Omega_{\psi_I(0)+\omega})$ |
| 1,4,7,9,6,10,14,<br>17,12,10,14,17,11 | $\psi(\psi_I(1)^2)$ |
| 1,4,7,9,6,10,14,17,12,11 | $\psi(\psi_I(1)^\omega)$ |
| 1,4,7,9,6,10,14,17,12,12 | $\psi(\psi_I(1)^\Omega)$ |
| 1,4,7,9,6,10,14,<br>17,12,12,4,7,9,5 | $\psi(\psi_I(1)^{\psi_I(0)})$ |
| 1,4,7,9,6,10,14,17,13 | $\psi(\psi_I(1)^{\Omega_{\psi_I(0)+1}})$ |
| 1,4,7,9,6,10,14,<br>17,13,10,14,17,11 | $\psi(\psi_I(1)^{\psi_I(1)})$ |
| 1,4,7,9,6,10,14,17,13,17 | $\psi(\psi_{\Omega_{\psi_I(1)+1}}(0))$ |
| 1,4,7,9,6,10,<br>14,17,13,17,21 | $\psi(\Omega_{\psi_I(1)+1})$ |
| 1,4,7,9,6,10,14,17,13,18 | $\psi(\Omega_{\psi_I(1)+\omega})$ |
| 1,4,7,9,6,10,<br>14,17,13,18,23 | $\psi(\Omega_{\psi_I(1)+\Omega})$ |
| 1,4,7,9,6,10,14,17,<br>13,18,23,25,4,7,9,5 | $\psi(\Omega_{\psi_I(1)+\psi_I(0)})$ |
| 1,4,7,9,6,10,14,17,13,<br>18,23,26,10,14,17,11 | $\psi(\Omega_{\psi_I(1)\cdot 2})$ |
| 1,4,7,9,6,10,14,<br>17,13,18,23,26,22,28 | $\psi(\Omega_{\psi_{\Omega_{\psi_I(1)+1}}(0)})$ |
| 1,4,7,9,6,10,14,<br>17,13,18,23,27 | $\psi(\Omega_{\Omega_{\psi_I(1)+1}})$ |
| 1,4,7,9,6,10,14,<br>17,13,18,23,27,19 | $\psi(\psi_I(2))$ |
| 1,4,7,9,7 | $\psi(\psi_I(\omega))$ |
| 1,4,7,9,7,3,5 | $\psi(\psi_I(\omega)+\Omega)$ |
| 1,4,7,9,7,4,7,9,7 | $\psi(\psi_I(\omega)\cdot 2)$ |
| 1,4,7,9,7,6 | $\psi(\psi_I(\omega)\cdot\Omega)$ |
| 1,4,7,9,7,6,4,7,9,7 | $\psi(\psi_I(\omega)^2)$ |
| 1,4,7,9,7,6,9 | $\psi(\psi_{\Omega_{\psi_I(\omega)+1}}(0))$ |
| 1,4,7,9,7,6,9,9 | $\psi(\psi_{\Omega_{\psi_I(\omega)+1}}(1))$ |
| 1,4,7,9,7,6,9,12 | $\psi(\Omega_{\psi_I(\omega)+1})$ |
| 1,4,7,9,7,6,10,<br>14,16,4,7,9,7 | $\psi(\Omega_{\psi_I(\omega)\cdot 2})$ |
| 1,4,7,9,7,6,10,<br>14,16,16,4,7,9,7 | $\psi(\Omega_{\psi_I(\omega)^{\psi_I(\omega)}})$ |
| 1,4,7,9,7,6,10,14,16,19 | $\psi(\Omega_{\psi_{\Omega_{\psi_I(\omega)+1}}(0)})$ |
| 1,4,7,9,7,6,10,<br>14,16,20,24,26,29 | $\psi(\Omega_{\psi_{\Omega_{\psi_I(\omega)+1}}(\Omega_{\psi_{\Omega_{\psi_I(\omega)+1}}(0)})})$ |
| 1,4,7,9,7,6,10,14,17 | $\psi(\Omega_{\Omega_{\psi_I(\omega)+1}})$ |
| 1,4,7,9,7,6,10,14,17,11 | $\psi(\psi_I(\omega+1))$ |
| 1,4,7,9,7,6,10,14,17,14 | $\psi(\psi_I(\omega\cdot 2))$ |
| 1,4,7,9,7,7 | $\psi(\psi_I(\omega^2))$ |
| 1,4,7,9,7,8,10 | $\psi(\psi_I(\psi(0)))$ |
| 1,4,7,9,7,8,11 | $\psi(\psi_I(\psi(\Omega_\omega)))$ |
| 1,4,7,9,7,8,11,14,16,12 | $\psi(\psi_I(\psi(\psi_I(0))))$ |
| 1,4,7,9,7,9 | $\psi(\psi_I(\Omega))$ |
| 1,4,7,9,7,9,4,7,9,5 | $\psi(\psi_I(\psi_I(0)))$ |
| 1,4,7,9,7,9,4,7,9,6 | $\psi(\psi_I(\psi_I(0))+\psi_I(0)\cdot\Omega)$ |
| 1,4,7,9,7,9,4,<br>7,9,6,10,14,17,11 | $\psi(\psi_I(\psi_I(0))+\psi_I(1))$ |
| 1,4,7,9,7,9,4,<br>7,9,6,10,14,17,14 | $\psi(\psi_I(\psi_I(0))+\psi_I(\omega))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,9,7,9,4,7,<br>9,6,10,14,17,14,16 | $\psi(\psi_I(\psi_I(0)) + \psi_I(\Omega))$ |
| 1,4,7,9,7,9,4,7,9,6,<br>10,14,17,14,16,4,7,9,5 | $\psi(\psi_I(\psi_I(0)) \cdot 2)$ |
| 1,4,7,9,7,9,4,7,9,<br>6,10,14,17,14,16,6 | $\psi(\psi_I(\psi_I(0)) \cdot \Omega)$ |
| 1,4,7,9,7,9,4,7,9,<br>6,10,14,17,14,16,6,9 | $\psi(\psi_{\Omega_{\psi_I(\psi_I(0))+1}}(0))$ |
| 1,4,7,9,7,9,4,7,9,6,<br>10,14,17,14,16,6,9,12 | $\psi(\Omega_{\psi_I(\psi_I(0))+1})$ |
| 1,4,7,9,7,9,4,7,9,6,10,<br>14,17,14,16,6,10,14,17,11 | $\psi(\psi_I(\psi_I(0) + 1))$ |
| 1,4,7,9,7,9,4,7,9,6,10,<br>14,17,14,16,6,10,14,17,14 | $\psi(\psi_I(\psi_I(0) + \omega))$ |
| 1,4,7,9,7,9,4,7,9,6,10,<br>14,17,14,16,6,10,14,17,14,16 | $\psi(\psi_I(\psi_I(0) + \Omega))$ |
| 1,4,7,9,7,9,4,7,9,<br>6,10,14,17,14,16,6,10,<br>14,17,14,16,4,7,9,5 | $\psi(\psi_I(\psi_I(0) \cdot 2))$ |
| 1,4,7,9,7,9,4,7,<br>9,6,10,14,17,14,16,7 | $\psi(\psi_I(\psi_I(0) \cdot \omega))$ |
| 1,4,7,9,7,9,4,7,<br>9,6,10,14,17,14,16,19 | $\psi(\psi_I(\psi_{\Omega_{\psi_I(0)+1}}(0)))$ |
| 1,4,7,9,7,9,4,7,9,6,<br>10,14,17,14,16,20,24,27,21 | $\psi(\psi_I(\psi_I(1)))$ |
| 1,4,7,9,7,9,4,7,9,7 | $\psi(\psi_I(\psi_I(\omega)))$ |
| 1,4,7,9,7,9,4,7,9,7,9 | $\psi(\psi_I(\psi_I(\Omega)))$ |
| 1,4,7,9,7,9,4,<br>7,9,7,9,4,7,9,5 | $\psi(\psi_I(\psi_I(\psi_I(0))))$ |
| 1,4,7,9,7,9,5 | $\psi(I)$<br>$\psi(2\ 1-2)$ |
| 1,4,7,9,7,9,5,3 | $\psi(I + 1)$ |
| 1,4,7,9,7,9,5,<br>3,4,7,10,12,10,12,8 | $\psi(I + \psi(I))$ |
| 1,4,7,9,7,9,5,3,5 | $\psi(I + \Omega)$ |
| 1,4,7,9,7,9,<br>5,4,7,9,5 | $\psi(I + \psi_I(0))$ |
| 1,4,7,9,7,9,<br>5,4,7,9,7,9,5 | $\psi(I + \psi_I(I))$ |
| 1,4,7,9,7,9,6 | $\psi(I + \psi_I(I) \cdot \Omega)$ |
| 1,4,7,9,7,9,<br>6,4,7,9,7,9,5 | $\psi(I + \psi_I(I)^2)$ |
| 1,4,7,9,7,9,6,9 | $\psi(I + \psi_{\Omega_{\psi_I(I)+1}}(0))$ |
| 1,4,7,9,7,9,6,9,12 | $\psi(I + \Omega_{\psi_I(I)+1})$ |
| 1,4,7,9,7,9,<br>6,10,14,17,11 | $\psi(I + \psi_I(I + 1))$ |
| 1,4,7,9,7,9,<br>6,10,14,17,14 | $\psi(I + \psi_I(I + \omega))$ |
| 1,4,7,9,7,9,6,10,14,<br>17,14,16,4,7,9,7,9,5 | $\psi(I + \psi_I(I + \psi_I(I)))$ |
| 1,4,7,9,7,9,6,10,<br>14,17,14,17,11 | $\psi(I \cdot 2)$ |
| 1,4,7,9,7,9,7 | $\psi(I \cdot \omega)$ |
| 1,4,7,9,7,9,7,7 | $\psi(I \cdot \omega^2)$ |
| 1,4,7,9,7,9,7,8,<br>11,14,16,14,16,12 | $\psi(I \cdot \psi(I))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,9,7,9,7,9 | $\psi(I \cdot \Omega)$ |
| 1,4,7,9,7,9,7,<br>9,4,7,9,7,9,5 | $\psi(I \cdot \psi_I(I))$ |
| 1,4,7,9,7,9,7,<br>9,4,7,9,7,9,7,9 | $\psi(I \cdot \psi_I(I \cdot \Omega))$ |
| 1,4,7,9,7,9,7,9,5 | $\psi(I^2)$ |
| 1,4,7,9,7,9,<br>7,9,7,9,5 | $\psi(I^3)$ |
| 1,4,7,9,8 | $\psi(I^\omega)$ |
| 1,4,7,9,8,7,9,5 | $\psi(I^{\omega+1})$ |
| 1,4,7,9,8,7,9,8 | $\psi(I^{\omega \cdot 2})$ |
| 1,4,7,9,9 | $\psi(I^\Omega)$ |
| 1,4,7,9,9,4,7,9,9 | $\psi(I^{\psi_I(I^\Omega)})$ |
| 1,4,7,9,9,5 | $\psi(I^I)$ |
| 1,4,7,9,9,7,9,5 | $\psi(I^{I+1})$ |
| 1,4,7,9,9,7,9,9,5 | $\psi(I^{I \cdot 2})$ |
| 1,4,7,9,9,8 | $\psi(I^{I \cdot \omega})$ |
| 1,4,7,9,9,9 | $\psi(I^{I \cdot \Omega})$ |
| 1,4,7,9,9,9,5 | $\psi(I^{I^2})$ |
| 1,4,7,9,10 | $\psi(I^{I^\omega})$ |
| 1,4,7,9,11 | $\psi(I^{I^\Omega})$ |
| 1,4,7,9,11,5 | $\psi(I^{I^I})$ |
| 1,4,7,9,11,13,5 | $\psi(I^{I^{I^I}})$ |
| 1,4,7,9,12 | $\psi(\psi_{\Omega_{I+1}}(0))$ |
| 1,4,7,9,12,3 | $\psi(\psi_{\Omega_{I+1}}(0) + 1)$ |
| 1,4,7,9,12,4,7,9,12 | $\psi(\psi_{\Omega_{I+1}}(0) + \psi_I(\psi_{\Omega_{I+1}}(0)))$ |
| 1,4,7,9,12,6,9 | $\psi(\psi_{\Omega_{I+1}}(0) + \psi_{\psi_I(\psi_{\Omega_{I+1}}(0))+1}(0))$ |
| 1,4,7,9,12,6,9,12 | $\psi(\psi_{\Omega_{I+1}}(0) + \Omega_{\psi_I(\psi_{\Omega_{I+1}}(0))+1})$ |
| 1,4,7,9,12,6,10,14,17,11 | $\psi(\psi_{\Omega_{I+1}}(0) + \psi_I(\psi_{\Omega_{I+1}}(0) + 1))$ |
| 1,4,7,9,12,6,<br>10,14,17,14,16 | $\psi(\psi_{\Omega_{I+1}}(0) + \psi_I(\psi_{\Omega_{I+1}}(0) + \Omega))$ |
| 1,4,7,9,12,6,10,14,<br>17,14,16,4,7,9,12 | $\psi(\psi_{\Omega_{I+1}}(0) + \psi_I(\psi_{\Omega_{I+1}}(0) + \psi_I(\psi_{\Omega_{I+1}}(0))))$ |
| 1,4,7,9,12,6,10,<br>14,17,14,17,11 | $\psi(\psi_{\Omega_{I+1}}(0) + I)$ |
| 1,4,7,9,12,6,10,<br>14,17,14,17,14 | $\psi(\psi_{\Omega_{I+1}}(0) + I \cdot \omega)$ |
| 1,4,7,9,12,6,10,14,<br>17,14,17,14,17,11 | $\psi(\psi_{\Omega_{I+1}}(0) + I^2)$ |
| 1,4,7,9,12,6,10,14,17,15 | $\psi(\psi_{\Omega_{I+1}}(0) + I^\omega)$ |
| 1,4,7,9,12,6,<br>10,14,17,17,11 | $\psi(\psi_{\Omega_{I+1}}(0) + I^I)$ |
| 1,4,7,9,12,6,10,14,17,21 | $\psi(\psi_{\Omega_{I+1}}(0) \cdot 2)$ |
| 1,4,7,9,12,7 | $\psi(\psi_{\Omega_{I+1}}(0) \cdot \omega)$ |
| 1,4,7,9,12,7,9 | $\psi(\psi_{\Omega_{I+1}}(0) \cdot \Omega)$ |
| 1,4,7,9,12,7,<br>9,4,7,9,12 | $\psi(\psi_{\Omega_{I+1}}(0) \cdot \psi_I(\psi_{\Omega_{I+1}}(0)))$ |
| 1,4,7,9,12,7,9,5 | $\psi(\psi_{\Omega_{I+1}}(0) \cdot I)$ |
| 1,4,7,9,12,7,9,12 | $\psi(\psi_{\Omega_{I+1}}(0)^2)$ |
| 1,4,7,9,12,9 | $\psi(\psi_{\Omega_{I+1}}(0)^\Omega)$ |
| 1,4,7,9,12,9,5 | $\psi(\psi_{\Omega_{I+1}}(0)^I)$ |
| 1,4,7,9,12,9,12 | $\psi(\psi_{\Omega_{I+1}}(0)^{\psi_{\Omega_{I+1}}(0)})$ |
| 1,4,7,9,12,11 | $\psi(\psi_{\Omega_{I+1}}(0)^{\psi_{\Omega_{I+1}}(0)^\Omega})$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,9,12,12 | $\psi(\psi_{\Omega_{I+1}}(1))$ |
| 1,4,7,9,12,14 | $\psi(\psi_{\Omega_{I+1}}(\Omega))$ |
| 1,4,7,9,12,14,5 | $\psi(\psi_{\Omega_{I+1}}(I))$ |
| 1,4,7,9,12,14,12,14,5 | $\psi(\psi_{\Omega_{I+1}}(I\cdot 2))$ |
| 1,4,7,9,12,14,14,5 | $\psi(\psi_{\Omega_{I+1}}(I^2))$ |
| 1,4,7,9,12,14,17 | $\psi(\psi_{\Omega_{I+1}}(\psi_{\Omega_{I+1}}(0)))$ |
| 1,4,7,9,12,15 | $\psi(\Omega_{I+1})$ |
| 1,4,7,9,12,16 | $\psi(\psi_{\Omega_{I+2}}(0))$ |
| 1,4,7,9,12,16,20 | $\psi(\Omega_{I+2})$ |
| 1,4,7,9,13 | $\psi(\Omega_{I+\omega})$ |
| 1,4,7,9,13,17,19 | $\psi(\Omega_{I+\Omega})$ |
| 1,4,7,9,13,17,<br>19,4,7,9,13,17,19 | $\psi(\Omega_{I+\psi_I(\Omega_{I+\Omega})})$ |
| 1,4,7,9,13,17,19,5 | $\psi(\Omega_{I\cdot 2})$ |
| 1,4,7,9,13,17,19,6 | $\psi(\Omega_{I\cdot 2}+\psi_I(\Omega_{I\cdot 2})\cdot\Omega)$ |
| 1,4,7,9,13,17,<br>19,6,10,14,17,11 | $\psi(\Omega_{I\cdot 2}+\psi_I(\Omega_{I\cdot 2}+1))$ |
| 1,4,7,9,13,17,19,<br>6,10,14,17,14,17,11 | $\psi(\Omega_{I\cdot 2}+I)$ |
| 1,4,7,9,13,17,<br>19,6,10,14,17,21 | $\psi(\Omega_{I\cdot 2}+\psi_{\Omega_{I+1}}(0))$ |
| 1,4,7,9,13,17,19,<br>6,10,14,17,22,27,30 | $\psi(\Omega_{I\cdot 2}\cdot 2)$ |
| 1,4,7,9,13,17,19,7 | $\psi(\Omega_{I\cdot 2}\cdot\omega)$ |
| 1,4,7,9,13,17,19,7,9 | $\psi(\Omega_{I\cdot 2}\cdot\Omega)$ |
| 1,4,7,9,13,17,19,7,9,5 | $\psi(\Omega_{I\cdot 2}\cdot I)$ |
| 1,4,7,9,13,17,19,9 | $\psi(\Omega^{\Omega}_{I\cdot 2})$ |
| 1,4,7,9,13,17,19,9,5 | $\psi(\Omega^{I}_{I\cdot 2})$ |
| 1,4,7,9,13,17,<br>19,9,13,17,19 | $\psi(\Omega^{\psi_{\Omega_{I+1}}(\Omega_{I\cdot 2})}_{I\cdot 2})$ |
| 1,4,7,9,13,17,19,11 | $\psi(\Omega^{\psi_{\Omega_{I+1}}(\Omega_{I\cdot 2})\cdot\Omega}_{I\cdot 2})$ |
| 1,4,7,9,13,17,19,12 | $\psi(\Omega^{\psi_{\Omega_{I+1}}(\Omega_{I\cdot 2}+1)}_{I\cdot 2})$ |
| 1,4,7,9,13,17,19,12,16,20 | $\psi(\Omega^{\Omega_{I+1}}_{I\cdot 2})$ |
| 1,4,7,9,13,17,19,13 | $\psi(\Omega^{\Omega_{I+\omega}}_{I\cdot 2})$ |
| 1,4,7,9,13,17,<br>19,13,17,19,5 | $\psi(\Omega^{\Omega_{I\cdot 2}}_{I\cdot 2})$ |
| 1,4,7,9,13,17,19,16,20 | $\psi(\psi_{\Omega_{I\cdot 2+1}}(0))$ |
| 1,4,7,9,13,17,19,16,20,24 | $\psi(\Omega_{I\cdot 2+1})$ |
| 1,4,7,9,13,17,<br>19,16,21,26,28,5 | $\psi(\Omega_{I\cdot 3})$ |
| 1,4,7,9,13,17,19,17 | $\psi(\Omega_{I\cdot\omega})$ |
| 1,4,7,9,13,17,19,17,19,5 | $\psi(\Omega_{I^2})$ |
| 1,4,7,9,13,17,19,19,5 | $\psi(\Omega_{I^I})$ |
| 1,4,7,9,13,17,19,22 | $\psi(\Omega_{\psi_{\Omega_{I+1}}(0)})$ |
| 1,4,7,9,13,17,20 | $\psi(\Omega_{\Omega_{I+1}})$ |
| 1,4,7,9,13,17,20,13,17,20 | $\psi(\Omega_{\Omega_{\Omega_{I+1}}})$ |
| 1,4,7,9,13,17,20,14 | $\psi(\psi_{I_2}(0))$ |
| 1,4,7,9,13,17,20,14,6 | $\psi(\psi_{I_2}(0)+\psi_I(\psi_{I_2}(0))\cdot\Omega)$ |
| 1,4,7,9,13,17,20,14,6,<br>10,14,17,14,17,11 | $\psi(\psi_{I_2}(0)+I)$ |
| 1,4,7,9,13,17,20,<br>14,6,10,14,17,21 | $\psi(\psi_{I_2}(0)+\psi_{\Omega_{I+1}}(0))$ |
| 1,4,7,9,13,17,20,14,6,<br>10,14,17,22,27,31,23 | $\psi(\psi_{I_2}(0)\cdot 2)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,9,13,17,20,14,7 | $\psi(\psi_{I_2}(0)\cdot\omega)$ |
| 1,4,7,9,13,17,<br>20,14,7,9,5 | $\psi(\psi_{I_2}(0)\cdot I)$ |
| 1,4,7,9,13,17,20,<br>14,7,9,13,17,20,14 | $\psi(\psi_{I_2}(0)^2)$ |
| 1,4,7,9,13,17,20,14,9 | $\psi(\psi_{I_2}(0)^\Omega)$ |
| 1,4,7,9,13,17,20,14,9,5 | $\psi(\psi_{I_2}(0)^I)$ |
| 1,4,7,9,13,17,20,<br>14,9,13,17,20,14 | $\psi(\psi_{I_2}(0)^{\psi_{\Omega_{I+1}}(\psi_{I_2}(0))})$ |
| 1,4,7,9,13,17,20,<br>14,13,17,20,14 | $\psi(\psi_{I_2}(0)^{\psi_{I_2}(0)})$ |
| 1,4,7,9,13,17,20,16,20 | $\psi(\psi_{\Omega_{\psi_{I_2}(0)+1}}(0))$ |
| 1,4,7,9,13,17,20,16,20,20 | $\psi(\psi_{\Omega_{\psi_{I_2}(0)+1}}(1))$ |
| 1,4,7,9,13,17,<br>20,14,16,20,21 | $\psi(\psi_{\Omega_{\psi_{I_2}(0)+1}}(\omega))$ |
| 1,4,7,9,13,17,20,16,20,22 | $\psi(\psi_{\Omega_{\psi_{I_2}(0)+1}}(\Omega))$ |
| 1,4,7,9,13,17,20,16,20,24 | $\psi(\Omega_{\psi_{I_2}(0)+1})$ |
| 1,4,7,9,13,17,<br>20,16,21,26,30 | $\psi(\Omega_{\Omega_{\psi_{I_2}(0)+1}})$ |
| 1,4,7,9,13,17,<br>20,16,21,26,30,22 | $\psi(\psi_{I_2}(1))$ |
| 1,4,7,9,13,17,20,17 | $\psi(\psi_{I_2}(\omega))$ |
| 1,4,7,9,13,17,20,17,19 | $\psi(\psi_{I_2}(\Omega))$ |
| 1,4,7,9,13,17,20,17,19 | $\psi(\psi_{I_2}(\Omega))$ |
| 1,4,7,9,13,17,20,17,19,5 | $\psi(\psi_{I_2}(I))$ |
| 1,4,7,9,13,17,20,17,19,22 | $\psi(\psi_{I_2}(\psi_{\Omega_{I+1}}(0)))$ |
| 1,4,7,9,13,17,20,17,20 | $\psi(\psi_{I_2}(\Omega_{I+1}))$ |
| 1,4,7,9,13,17,<br>20,17,20,13,17,20 | $\psi(\psi_{I_2}(\Omega_{\Omega_{I+1}}))$ |
| 1,4,7,9,13,17,<br>20,17,20,13,17,20,14 | $\psi(\psi_{I_2}(\psi_{I_2}(0)))$ |
| 1,4,7,9,13,17,20,<br>17,20,13,17,20,17,20 | $\psi(\psi_{I_2}(\psi_{I_2}(\Omega_{I+1})))$ |
| 1,4,7,9,13,17,20,17,20,14 | $\psi(I_2)$ |
| 1,4,7,9,13,17,20,17,20,<br>16,21,26,30,35,30,35,31 | $\psi(I_2\cdot 2)$ |
| 1,4,7,9,13,17,20,17,20,17 | $\psi(I_2\cdot\omega)$ |
| 1,4,7,9,13,17,<br>20,17,20,17,20,14 | $\psi(I_2^2)$ |
| 1,4,7,9,13,17,20,19 | $\psi(I_2^\Omega)$ |
| 1,4,7,9,13,17,20,19,5 | $\psi(I_2^I)$ |
| 1,4,7,9,13,17,20,20,14 | $\psi(I_2^{I_2})$ |
| 1,4,7,9,13,17,20,24 | $\psi(\psi_{\Omega_{I_2+1}}(0))$ |
| 1,4,7,9,13,17,20,24,24 | $\psi(\psi_{\Omega_{I_2+1}}(1))$ |
| 1,4,7,9,13,17,20,24,28 | $\psi(\Omega_{I_2+1})$ |
| 1,4,7,9,13,17,20,24,29 | $\psi(\psi_{\Omega_{I_2+2}}(0))$ |
| 1,4,7,9,13,17,20,25,30,34 | $\psi(\Omega_{\Omega_{I_2+1}})$ |
| 1,4,7,9,13,17,<br>20,25,30,34,26 | $\psi(\psi_{I_3}(0))$ |
| 1,4,7,9,13,17,20,<br>25,30,34,30,34,26 | $\psi(I_3)$ |
| 1,4,7,10 | $\psi(I_\omega)$ |
| 1,4,7,10,3 | $\psi(I_\omega+1)$ |
| 1,4,7,10,4 | $\psi(I_\omega+\Omega_\omega)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,4,7,9,5 | $\psi(I_\omega + \psi_I(0))$ |
| 1,4,7,10,4,7,9,7,9,5 | $\psi(I_\omega + \psi_I(I))$ |
| 1,4,7,10,4,7,9,12 | $\psi(I_\omega + \psi_I(\psi_{\Omega_{I+1}}(0)))$ |
| 1,4,7,10,4,7,9,12,15 | $\psi(I_\omega + \psi_I(\Omega_{I+1}))$ |
| 1,4,7,10,4,7,<br>9,13,17,20,14 | $\psi(I_\omega + \psi_I(\psi_{I_2}(0)))$ |
| 1,4,7,10,4,7,9,13,17,21 | $\psi(I_\omega + \psi_I(I_\omega))$ |
| 1,4,7,10,4,7,<br>9,13,17,21,6,9 | $\psi(I_\omega + \psi_{\Omega_{\psi_I(I_\omega)+1}}(0))$ |
| 1,4,7,10,4,7,<br>9,13,17,21,6,9,12 | $\psi(I_\omega + \Omega_{\psi_I(I_\omega)+1})$ |
| 1,4,7,10,4,7,9,13,<br>17,21,6,10,14,17,11 | $\psi(I_\omega + \psi_I(I_\omega + 1))$ |
| 1,4,7,10,4,7,9,13,17,<br>21,6,10,14,17,14,17,11 | $\psi(I_\omega + I)$ |
| 1,4,7,10,4,7,9,13,<br>17,21,6,10,14,17,21 | $\psi(I_\omega + \psi_{\Omega_{I+1}}(0))$ |
| 1,4,7,10,4,7,9,13,17,<br>21,6,10,14,17,21,21 | $\psi(I_\omega + \psi_{\Omega_{I+1}}(1))$ |
| 1,4,7,10,4,7,9,13,17,<br>21,6,10,14,17,21,25 | $\psi(I_\omega + \psi_{\Omega_{I+1}}(\Omega_{I+1}))$ |
| 1,4,7,10,4,7,9,13,<br>17,21,6,10,14,18 | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega))$ |
| 1,4,7,10,4,7,9,13,17,21,6,<br>10,14,18,10,14,17,22,27,32 | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) \cdot 2)$ |
| 1,4,7,10,4,7,9,13,17,21,7 | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega) \cdot \omega)$ |
| 1,4,7,10,4,7,9,13,<br>17,21,7,9,13,17,21 | $\psi(I_\omega + \psi_{\Omega_{I+1}}(I_\omega)^2)$ |
| 1,4,7,10,4,7,9,<br>13,17,21,12,15 | $\psi(I_\omega + \Omega_{I+1})$ |
| 1,4,7,10,4,7,9,<br>13,17,21,13,17,20 | $\psi(I_\omega + \Omega_{\Omega_{I+1}})$ |
| 1,4,7,10,4,7,9,<br>13,17,21,13,17,20,14 | $\psi(I_\omega + \psi_{I_2}(0))$ |
| 1,4,7,10,4,7,9,13,<br>17,21,13,17,20,25,<br>30,35,25,30,34,26 | $\psi(I_\omega + \psi_{I_3}(0))$ |
| 1,4,7,10,4,7,10 | $\psi(I_\omega \cdot 2)$ |
| 1,4,7,10,5 | $\psi(I_\omega \cdot \omega)$ |
| 1,4,7,10,6 | $\psi(I_\omega \cdot \Omega)$ |
| 1,4,7,10,6,4,7,9,5 | $\psi(I_\omega \cdot \psi_I(0))$ |
| 1,4,7,10,6,4,<br>7,9,13,17,21 | $\psi(I_\omega \cdot \psi_I(I_\omega))$ |
| 1,4,7,10,6,4,<br>7,9,13,17,21,16 | $\psi(I_\omega \cdot \psi_I(I_\omega \cdot \Omega))$ |
| 1,4,7,10,6,4,7,9,13,17,<br>21,16,4,7,9,13,17,21 | $\psi(I_\omega \cdot \psi_I(I_\omega \cdot \psi_I(I_\omega)))$ |
| 1,4,7,10,6,4,7,<br>9,13,17,21,16,5 | $\psi(I_\omega \cdot I)$ |
| 1,4,7,10,6,4,7,9,<br>13,17,21,16,12 | $\psi(I_\omega \cdot \psi_{\Omega_{I+1}}(0))$ |
| 1,4,7,10,6,4,7,9,<br>13,17,21,16,12,15 | $\psi(I_\omega \cdot \Omega_{I+1})$ |
| 1,4,7,10,6,4,7,9,<br>13,17,21,16,13,17,20 | $\psi(I_\omega \cdot \Omega_{\Omega_{I+1}})$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,6,4,7,9,13, 17,21,16,13,17,20,14 | $\psi(I_\omega \cdot \psi_{I_2}(0))$ |
| 1,4,7,10,6,4,7,9,13, 17,21,16,13,17,20,25,30, 35,29,25,30,34,26 | $\psi(I_\omega \cdot \psi_{I_3}(0))$ |
| 1,4,7,10,6,4,7,10 | $\psi(I_\omega^2)$ |
| 1,4,7,10,6,4, 7,10,4,7,10 | $\psi(I_\omega^2 + I_\omega)$ |
| 1,4,7,10,6,4, 7,10,6,4,7,10 | $\psi(I_\omega^2 \cdot 2)$ |
| 1,4,7,10,6,5 | $\psi(I_\omega^2 \cdot \omega)$ |
| 1,4,7,10,6,6 | $\psi(I_\omega^2 \cdot \Omega)$ |
| 1,4,7,10,6,6,4,7,10 | $\psi(I_\omega^3)$ |
| 1,4,7,10,6,7 | $\psi(I_\omega^\omega)$ |
| 1,4,7,10,6,8 | $\psi(I_\omega^\Omega)$ |
| 1,4,7,10,6,8,4,7,10 | $\psi(I_\omega^{I_\omega})$ |
| 1,4,7,10,6,9 | $\psi(\psi_{\Omega_{I_\omega+1}}(0))$ $\psi((1-)^{1,0}$ aft $1-2\ 1-2)$ |
| 1,4,7,10,6,9,9 | $\psi(\psi_{\Omega_{I_\omega+1}}(1))$ |
| 1,4,7,10,6,9,12 | $\psi(\Omega_{I_\omega+1})$ |
| 1,4,7,10,6,10 | $\psi(\Omega_{I_\omega+\omega})$ |
| 1,4,7,10,6,10,14,16,19 | $\psi(\Omega_{\psi_{\Omega_{I_\omega+1}}(0)})$ |
| 1,4,7,10,6,10,14,17 | $\psi(\psi_{I_{\omega+1}}(0))$ |
| 1,4,7,10,6,10, 14,17,14,17,11 | $\psi(I_{\omega+1})$ |
| 1,4,7,10,6,10,14,18 | $\psi(I_{\omega\cdot 2})$ |
| 1,4,7,10,7 | $\psi(I_{\omega^2})$ |
| 1,4,7,10,7,7 | $\psi(I_{\omega^3})$ |
| 1,4,7,10,7,9 | $\psi(I_\Omega)$ |
| 1,4,7,10,7,9,4,7,9,5 | $\psi(I_{\psi_I(0)})$ |
| 1,4,7,10,7,9, 4,7,9,13,17,21 | $\psi(I_{\psi_I(I_\omega)})$ |
| 1,4,7,10,7,9,4, 7,9,13,17,21,17,19 | $\psi(I_{\psi_I(I_\Omega)})$ |
| 1,4,7,10,7,9,4,7,9, 13,17,21,17,19,4,7,9, 13,17,21,17,19 | $\psi(I_{\psi_I(I_{\psi_I(I_\Omega)})})$ |
| 1,4,7,10,7,9,4,7, 9,13,17,21,17,19,5 | $\psi(I_I)$ |
| 1,4,7,10,7,9,4,7,10 | $\psi(I_{I_\omega})$ |
| 1,4,7,10,7,9, 4,7,10,7,9 | $\psi(I_{I_\Omega})$ |
| 1,4,7,10,7,9, 4,7,10,7,9,4,7,10 | $\psi(I_{I_{I_\omega}})$ |
| 1,4,7,10,7,9,5 | $\psi(\psi_{I(1,0)}(0))$ |
| 1,4,7,10,7,9,5,3 | $\psi(\psi_{I(1,0)}(0)+1)$ |
| 1,4,7,10,7,9,5,4,7,10 | $\psi(\psi_{I(1,0)}(0)+I_\omega)$ |
| 1,4,7,10,7,9, 5,4,7,10,7,9 | $\psi(\psi_{I(1,0)}(0)+I_\Omega)$ |
| 1,4,7,10,7,9, 5,4,7,10,7,9,5 | $\psi(\psi_{I(1,0)}(0)\cdot 2)$ |
| 1,4,7,10,7,9,5,5 | $\psi(\psi_{I(1,0)}(0)\cdot \omega)$ |
| 1,4,7,10,7,9,6 | $\psi(\psi_{I(1,0)}(0)\cdot \Omega)$ |
| 1,4,7,10,7,9,6,4,7,10 | $\psi(\psi_{I(1,0)}(0)\cdot I_\omega)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,7,9,<br>6,4,7,10,7,9,5 | $\psi(\psi_{I(1,0)}(0)^2)$ |
| 1,4,7,10,7,9,6,<br>4,7,10,7,9,6 | $\psi(\psi_{I(1,0)}(0)^2 + \psi_{I(1,0)}(0) \cdot \Omega)$ |
| 1,4,7,10,7,9,6,4,7,<br>10,7,9,6,4,7,10,7,9,5 | $\psi(\psi_{I(1,0)}(0)^2 \cdot 2)$ |
| 1,4,7,10,7,9,6,5 | $\psi(\psi_{I(1,0)}(0)^2 \cdot \omega)$ |
| 1,4,7,10,7,9,6,6 | $\psi(\psi_{I(1,0)}(0)^2 \cdot \Omega)$ |
| 1,4,7,10,7,9,<br>6,6,4,7,10,7,9,5 | $\psi(\psi_{I(1,0)}(0)^3)$ |
| 1,4,7,10,7,9,6,<br>6,6,4,7,10,7,9,5 | $\psi(\psi_{I(1,0)}(0)^4)$ |
| 1,4,7,10,7,9,6,7 | $\psi(\psi_{I(1,0)}(0)^\omega)$ |
| 1,4,7,10,7,9,6,8 | $\psi(\psi_{I(1,0)}(0)^\Omega)$ |
| 1,4,7,10,7,9,6,<br>8,4,7,10,7,9,5 | $\psi(\psi_{I(1,0)}(0)^{\psi_{I(1,0)}(0)})$ |
| 1,4,7,10,7,9,6,9 | $\psi(\psi_{\Omega_{\psi_{I(1,0)}(0)+1}}(0))$ |
| 1,4,7,10,7,9,6,9,9 | $\psi(\psi_{\Omega_{\psi_{I(1,0)}(0)+1}}(1))$ |
| 1,4,7,10,7,9,6,9,11 | $\psi(\psi_{\Omega_{\psi_{I(1,0)}(0)+1}}(\Omega))$ |
| 1,4,7,10,7,9,6,9,12 | $\psi(\Omega_{\psi_{I(1,0)}(0)+1})$ |
| 1,4,7,10,7,9,6,10,<br>14,16,4,7,10,7,9,5 | $\psi(\Omega_{\psi_{I(1,0)}(0)\cdot 2})$ |
| 1,4,7,10,7,9,6,10,14,17 | $\psi(\Omega_{\Omega_{\psi_{I(1,0)}(0)+1}})$ |
| 1,4,7,10,7,9,<br>6,10,14,17,11 | $\psi(\psi_{I_{\psi_{I(1,0)}(0)+1}}(0))$ |
| 1,4,7,10,7,9,<br>6,10,14,17,14 | $\psi(\psi_{I_{\psi_{I(1,0)}(0)+1}}(\omega))$ |
| 1,4,7,10,7,9,<br>6,10,14,17,14,17,11 | $\psi(I_{\psi_{I(1,0)}(0)+1})$ |
| 1,4,7,10,7,9,<br>6,10,14,17,21,25 | $\psi(\Omega_{I_{\psi_{I(1,0)}(0)+1}+1})$ |
| 1,4,7,10,7,9,6,<br>10,14,17,22,27,31,23 | $\psi(\psi_{I_{\psi_{I(1,0)}(0)+2}}(0))$ |
| 1,4,7,10,7,9,6,10,14,18 | $\psi(I_{\psi_{I(1,0)}(0)+\omega})$ |
| 1,4,7,10,7,9,6,10,14,<br>18,14,16,4,7,10,7,9,5 | $\psi(I_{\psi_{I(1,0)}(0)\cdot 2})$ |
| 1,4,7,10,7,9,6,<br>10,14,18,14,17 | $\psi(I_{\Omega_{\psi_{I(1,0)}(0)+1}})$ |
| 1,4,7,10,7,9,6,<br>10,14,18,14,17,11 | $\psi(\psi_{I(1,0)}(1))$ |
| 1,4,7,10,7,9,6,10,14,18,<br>14,17,13,18,23,27,23,26,19 | $\psi(\psi_{I(1,0)}(2))$ |
| 1,4,7,10,7,9,7 | $\psi(\psi_{I(1,0)}(\omega))$ |
| 1,4,7,10,7,9,7,7 | $\psi(\psi_{I(1,0)}(\omega^2))$ |
| 1,4,7,10,7,9,7,9 | $\psi(\psi_{I(1,0)}(\Omega))$ |
| 1,4,7,10,7,9,7,<br>9,4,7,10,7,9,7,9 | $\psi(\psi_{I(1,0)}(\psi_{I(1,0)}(\Omega)))$ |
| 1,4,7,10,7,9,7,9,5 | $\psi(I(1,0))$<br>$\psi(2\ 1-2\ 1-2)$ |
| 1,4,7,10,7,9,7,9,<br>5,4,7,10,7,9,7,9,5 | $\psi(I(1,0) + \psi_{I(1,0)}(I(1,0)))$ |
| 1,4,7,10,7,9,7,9,6 | $\psi(I(1,0) + \psi_{I(1,0)}(I(1,0)) \cdot \Omega)$ |
| 1,4,7,10,7,9,7,9,6,9 | $\psi(I(1,0) + \psi_{\Omega_{\psi_{I(1,0)}(I(1,0))+1}}(0))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,7,9,<br>7,9,6,9,12 | $\psi(I(1,0)+\Omega_{\psi_{I(1,0)}(I(1,0))+1})$ |
| 1,4,7,10,7,9,<br>7,9,6,10,14,17 | $\psi(I(1,0)+\Omega_{\Omega_{\psi_{I(1,0)}(I(1,0))+1}})$ |
| 1,4,7,10,7,9,7,<br>9,6,10,14,17,11 | $\psi(I(1,0)+\psi_{I_{\psi_{I(1,0)}(I(1,0))+1}}(0))$ |
| 1,4,7,10,7,9,7,<br>9,6,10,14,17,14,17,11 | $\psi(I(1,0)+I_{\psi_{I(1,0)}(I(1,0))+1})$ |
| 1,4,7,10,7,9,7,<br>9,6,10,14,18 | $\psi(I(1,0)+I_{\psi_{I(1,0)}(I(1,0))+\omega})$ |
| 1,4,7,10,7,9,7,<br>9,6,10,14,18,14,17,11 | $\psi(I(1,0)+\psi_{I(1,0)}(I(1,0)+1))$ |
| 1,4,7,10,7,9,7,<br>9,6,10,14,18,14,17,<br>14,16,4,7,10,7,9,7,9,5 | $\psi(I(1,0)+\psi_{I(1,0)}(I(1,0)+\psi_{I(1,0)}(I(1,0))))$ |
| 1,4,7,10,7,9,7,9,6,10,<br>14,18,14,17,14,17,11 | $\psi(I(1,0)\cdot 2)$ |
| 1,4,7,10,7,9,7,9,7 | $\psi(I(1,0)\cdot\omega)$ |
| 1,4,7,10,7,9,7,9,7,9 | $\psi(I(1,0)\cdot\Omega)$ |
| 1,4,7,10,7,9,7,9,7,<br>9,4,7,10,7,9,7,9,5 | $\psi(I(1,0)\cdot\psi_{I(1,0)}(I(1,0)))$ |
| 1,4,7,10,7,<br>9,7,9,7,9,5 | $\psi(I(1,0)^2)$ |
| 1,4,7,10,7,9,7,9,7,9,<br>7,9,4,7,10,7,9,7,9,5 | $\psi(I(1,0)^2\cdot\psi_{I(1,0)}(I(1,0)))$ |
| 1,4,7,10,7,9,<br>7,9,7,9,7,9,5 | $\psi(I(1,0)^3)$ |
| 1,4,7,10,7,9,8 | $\psi(I(1,0)^\omega)$ |
| 1,4,7,10,7,9,9,5 | $\psi(I(1,0)^{I(1,0)})$ |
| 1,4,7,10,7,9,12 | $\psi(\psi_{\Omega_{I(1,0)}+1}(0))$ |
| 1,4,7,10,7,9,12,15 | $\psi(\Omega_{I(1,0)+1})$ |
| 1,4,7,10,7,9,13 | $\psi(\Omega_{I(1,0)+\omega})$ |
| 1,4,7,10,7,9,13,17,19 | $\psi(\Omega_{I(1,0)+\Omega})$ |
| 1,4,7,10,7,9,13,17,<br>19,4,7,10,7,9,7,9,5 | $\psi(\Omega_{I(1,0)\cdot 2})$ |
| 1,4,7,10,7,9,13,17,19,17 | $\psi(\Omega_{I(1,0)\cdot\omega})$ |
| 1,4,7,10,7,9,13,17,19,22 | $\psi(\Omega_{\psi_{\Omega_{I(1,0)+1}}(0)})$ |
| 1,4,7,10,7,9,13,17,20 | $\psi(\Omega_{\Omega_{I(1,0)+1}})$ |
| 1,4,7,10,7,9,13,17,20,14 | $\psi(\psi_{I_{I(1,0)+1}}(0))$ |
| 1,4,7,10,7,9,<br>13,17,20,17,20,14 | $\psi(I_{I(1,0)+1})$ |
| 1,4,7,10,7,9,13,17,21 | $\psi(I_{I(1,0)+\omega})$ |
| 1,4,7,10,7,9,13,17,21,<br>17,19,4,7,10,7,9,5 | $\psi(I_{I(1,0)+\psi_{I(1,0)}(0)})$ |
| 1,4,7,10,7,9,13,<br>17,21,17,19,5 | $\psi(I_{I(1,0)\cdot 2})$ |
| 1,4,7,10,7,9,13,<br>17,21,17,19,17 | $\psi(I_{I(1,0)\cdot\omega})$ |
| 1,4,7,10,7,9,13,<br>17,21,17,19,18 | $\psi(I_{I(1,0)^\omega})$ |
| 1,4,7,10,7,9,13,<br>17,21,17,19,22 | $\psi(I_{\psi_{\Omega_{I(1,0)+1}}(0)})$ |
| 1,4,7,10,7,9,<br>13,17,21,17,20 | $\psi(I_{\Omega_{I(1,0)+1}})$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,7,9,13,17,<br>21,17,20,13,17,20,14 | $\psi(I_{\psi_{I_{I(1,0)+1}}(0)})$ |
| 1,4,7,10,7,9,13,17,21,<br>17,20,13,17,20,17,20,14 | $\psi(I_{I_{I(1,0)+1}})$ |
| 1,4,7,10,7,9,13,<br>17,21,17,20,13,17,21 | $\psi(I_{I_{I(1,0)+\omega}})$ |
| 1,4,7,10,7,9,13,17,21,17,<br>20,13,17,21,17,19,22 | $\psi(I_{I_{\psi_{\Omega_{I(1,0)+1}}(0)}})$ |
| 1,4,7,10,7,9,13,17,21,<br>17,20,13,17,21,17,20 | $\psi(I_{I_{\Omega_{I(1,0)+1}}})$ |
| 1,4,7,10,7,9,13,17,<br>21,17,20,13,17,21,17,<br>20,13,17,21,17,20 | $\psi(I_{I_{I_{\Omega_{I(1,0)+1}}}})$ |
| 1,4,7,10,7,9,13,<br>17,21,17,20,14 | $\psi(\psi_{I(1,1)}(0))$<br>$\psi((1-)^{1,0}\ 2\ 1-2 \text{ aft } (2\ 1-)^2\ 2)$ |
| 1,4,7,10,7,9,13,<br>17,21,17,20,16,20 | $\psi(\psi_{\Omega_{\psi_{I(1,1)}(0)+1}}(0))$ |
| 1,4,7,10,7,9,13,17,21,<br>17,20,16,21,26,30,22 | $\psi(\psi_{I_{\psi_{I(1,1)}(0)+1}}(0))$ |
| 1,4,7,10,7,9,13,17,21,<br>17,20,16,21,26,30,26,30,22 | $\psi(I_{\psi_{I(1,1)}(0)+1})$ |
| 1,4,7,10,7,9,13,17,<br>21,17,20,16,21,26,31 | $\psi(I_{\psi_{I(1,1)}(0)+\omega})$ |
| 1,4,7,10,7,9,13,17,21,<br>17,20,16,21,26,31,26,30,22 | $\psi(\psi_{I(1,1)}(1))$ |
| 1,4,7,10,7,9,13,<br>17,21,17,20,17 | $\psi(\psi_{I(1,1)}(\omega))$ |
| 1,4,7,10,7,9,13,<br>17,21,17,20,17,20 | $\psi(\psi_{I(1,1)}(\Omega_{\psi_{I(1,1)}(0)+1}))$ |
| 1,4,7,10,7,9,13,<br>17,21,17,20,17,20,14 | $\psi(I(1,1))$ |
| 1,4,7,10,7,9,<br>13,17,21,17,20,24 | $\psi(\psi_{\Omega_{I(1,1)+1}}(0))$ |
| 1,4,7,10,7,9,<br>13,17,21,17,20,25 | $\psi(\Omega_{I(1,1)+\omega})$ |
| 1,4,7,10,7,9,13,17,<br>21,17,20,25,30,34,26 | $\psi(\psi_{I_{I(1,1)+1}}(0))$ |
| 1,4,7,10,7,9,13,17,21,<br>17,20,25,30,34,30,34,26 | $\psi(I_{I(1,1)+1})$ |
| 1,4,7,10,7,9,13,17,21,<br>17,20,25,30,35,30,34 | $\psi(I_{\Omega_{I(1,1)+1}})$ |
| 1,4,7,10,7,9,13,17,21,<br>17,20,25,30,35,30,34,26 | $\psi(\psi_{I(1,2)}(0))$ |
| 1,4,7,10,7,9,13,17,<br>21,17,20,25,30,35,<br>30,34,30,34,26 | $\psi(I(1,2))$ |
| 1,4,7,10,7,10 | $\psi(I(1,\omega))$ |
| 1,4,7,10,7,10,7 | $\psi(I(1,\omega^2))$ |
| 1,4,7,10,7,10,7,9 | $\psi(I(1,\Omega))$ |
| 1,4,7,10,7,<br>10,7,9,4,7,9,5 | $\psi(I(1,\psi_I(0)))$ |
| 1,4,7,10,7,10,7,9,4,<br>7,9,13,17,21,17,<br>21,17,19,5 | $\psi(I(1,I))$ |
| 1,4,7,10,7,10,<br>7,9,4,7,10 | $\psi(I(1,I_\omega))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,7,10,<br>7,9,4,7,10,7,9,5 | $\psi(I(1,\psi_{I(1,0)}(0)))$ |
| 1,4,7,10,7,10,7,9,4,<br>7,10,7,9,13,17,21,<br>17,21,17,19,5 | $\psi(I(1,I(1,0)))$ |
| 1,4,7,10,7,10,7,<br>9,4,7,10,7,10,7,9 | $\psi(I(1,I(1,\Omega)))$ |
| 1,4,7,10,7,10,7,9,5 | $\psi(\psi_{I(2,0)}(0))$ |
| 1,4,7,10,7,10,7,9,7 | $\psi(\psi_{I(2,0)}(\omega))$ |
| 1,4,7,10,7,10,7,9,7,9,5 | $\psi(I(2,0))$ |
| 1,4,7,10,7,10,7,9,12 | $\psi(\psi_{\Omega_{I(2,0)+1}}(0))$ |
| 1,4,7,10,7,10,7,9,12,15 | $\psi(\Omega_{I(2,0)+1})$ |
| 1,4,7,10,7,10,<br>7,9,13,17,20,14 | $\psi(\psi_{I_{I(2,0)+1}}(0))$ |
| 1,4,7,10,7,10,7,<br>9,13,17,20,17,20,14 | $\psi(I_{I(2,0)+1})$ |
| 1,4,7,10,7,10,7,<br>9,13,17,21,17,20,14 | $\psi(\psi_{I(1,I(2,0)+1)}(0))$ |
| 1,4,7,10,7,10,7,9,13,<br>17,21,17,20,17,20,14 | $\psi(I(1,I(2,0)+1))$ |
| 1,4,7,10,7,10,7,9,13,<br>17,21,17,21,17,20,14 | $\psi(\psi_{I(2,1)}(0))$ |
| 1,4,7,10,7,10,7,9,13,<br>17,21,17,21,17,20,17,20,14 | $\psi(I(2,1))$ |
| 1,4,7,10,7,10,7,10 | $\psi(I(2,\omega))$ |
| 1,4,7,10,7,10,7,10,7,9 | $\psi(I(2,\Omega))$ |
| 1,4,7,10,7,10,7,10,7,<br>9,4,7,10,7,10,7,9,5 | $\psi(I(2,\psi_{I(2,0)}(0)))$ |
| 1,4,7,10,7,10,7,10,<br>7,9,4,7,10,7,10,7,9,13,<br>17,21,17,21,17,21,17,19,5 | $\psi(I(2,I(2,0)))$ |
| 1,4,7,10,7,10,<br>7,10,7,9,5 | $\psi(\psi_{I(3,0)}(0))$ |
| 1,4,7,10,7,10,7,10,7,10 | $\psi(I(3,\omega))$ |
| 1,4,7,10,7,10,<br>7,10,7,10,7,10 | $\psi(I(4,\omega))$ |
| 1,4,7,10,8 | $\psi(I(\omega,0))$ |
| 1,4,7,10,8,3 | $\psi(I(\omega,0)+1)$ |
| 1,4,7,10,8,4,7,10,8 | $\psi(I(\omega,0)\cdot 2)$ |
| 1,4,7,10,8,6 | $\psi(I(\omega,0)\cdot\Omega)$ |
| 1,4,7,10,8,6,9 | $\psi(\psi_{\Omega_{I(\omega,0)+1}}(0))$ |
| 1,4,7,10,8,6,10,14,17,11 | $\psi(\psi_{I_{I(\omega,0)+1}}(0))$ |
| 1,4,7,10,8,6,10,14,18 | $\psi(I_{I(\omega,0)+\omega})$ |
| 1,4,7,10,8,6,10,<br>14,18,14,17,14,17,11 | $\psi(I(1,I(\omega,0)+1))$ |
| 1,4,7,10,8,6,10,<br>14,18,14,18,14,17,11 | $\psi(I(2,I(\omega,0)+1))$ |
| 1,4,7,10,8,6,10,14,18,15 | $\psi(I(\omega,1))$ |
| 1,4,7,10,8,6,10,14,<br>18,15,13,18,23,28,24 | $\psi(I(\omega,2))$ |
| 1,4,7,10,8,7 | $\psi(I(\omega,\omega))$ |
| 1,4,7,10,8,7,9 | $\psi(I(\omega,\Omega))$ |
| 1,4,7,10,8,<br>7,9,4,7,10,8 | $\psi(I(\omega,I(\omega,0)))$ |
| 1,4,7,10,8,7,9,5 | $\psi(\psi_{I(\omega+1,0)}(0))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,8,7,9,7,9,5 | $\psi(I(\omega+1,0))$<br>$\psi(\text{real.}(2\ 1-)^{\omega}\ 2)$ |
| 1,4,7,10,8,7,10 | $\psi(I(\omega+1,\omega))$ |
| 1,4,7,10,8,7,<br>10,7,9,7,9,5 | $\psi(I(\omega+2,0))$ |
| 1,4,7,10,8,7,10,8 | $\psi(I(\omega\cdot 2,0))$ |
| 1,4,7,10,8,8 | $\psi(I(\omega^2,0))$ |
| 1,4,7,10,8,9 | $\psi(I(\omega^{\omega},0))$ |
| 1,4,7,10,9 | $\psi(I(\Omega,0))$ |
| 1,4,7,10,9,4,<br>7,9,13,17,21,19,5 | $\psi(I(I,0))$ |
| 1,4,7,10,9,4,7,10,9 | $\psi(I(I(\Omega,0),0))$ |
| 1,4,7,10,9,5 | $\psi(\psi_{I(1,0,0)}(0))$<br>$\psi((2\ 1-)^{1,0}\ 2)$<br>$\psi(\psi_{\psi_M(M^M)}(0))$<br>TBO |
| 1,4,7,10,9,6 | $\psi(\psi_{I(1,0,0)}(0)\cdot\Omega)$ |
| 1,4,7,10,9,6,9 | $\psi(\psi_{\Omega_{\psi_{I(1,0,0)}(0)+1}}(0))$ |
| 1,4,7,10,9,6,9,12 | $\psi(\Omega_{\psi_{I(1,0,0)}(0)+1})$ |
| 1,4,7,10,9,6,10,14,17,11 | $\psi(\psi_{I_{I(1,0,0)+1}}(0))$ |
| 1,4,7,10,9,6,10,<br>14,18,14,17,14,17,11 | $\psi(I(1,\psi_{I(1,0,0)}(0)+1))$ |
| 1,4,7,10,9,6,10,14,18,15 | $\psi(I(\omega,\psi_{I(1,0,0)}(0)+1))$ |
| 1,4,7,10,9,6,10,14,18,16 | $\psi(I(\Omega,\psi_{I(1,0,0)}(0)+1))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0),1))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,4,7,10,9,6 | $\psi(I(\psi_{I(1,0,0)}(0),1)+\psi_{I(1,0,0)}(0)\cdot\Omega)$ |
| 1,4,7,10,9,6,10,<br>14,18,16,4,7,10,9,6,9 | $\psi(I(\psi_{I(1,0,0)}(0),1)+\psi_{\Omega_{\psi_{I(1,0,0)}(0)+1}}(0))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,4,7,10,9,6,10,<br>14,18,16,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0),1)+$<br>$\psi_{\Omega_{\psi_{I(1,0,0)}(0)+1}}(I(\psi_{I(1,0,0)}(0),1)))$ |
| 1,4,7,10,9,6,<br>10,14,18,16,6 | $\psi(I(\psi_{I(1,0,0)}(0),1)+$<br>$\psi_{\Omega_{\psi_{I(1,0,0)}(0)+1}}(I(\psi_{I(1,0,0)}(0),1))\cdot\Omega)$ |
| 1,4,7,10,9,6,<br>10,14,18,16,9 | $\psi(I(\psi_{I(1,0,0)}(0),1)+$<br>$\psi_{\Omega_{\psi_{I(1,0,0)}(0)+1}}(I(\psi_{I(1,0,0)}(0),1)+1))$ |
| 1,4,7,10,9,6,<br>10,14,18,16,9,12 | $\psi(I(\psi_{I(1,0,0)}(0),1)+\Omega_{\psi_{I(1,0,0)}(0)+1})$ |
| 1,4,7,10,9,6,<br>10,14,18,16,10 | $\psi(I(\psi_{I(1,0,0)}(0),1)+\Omega_{\psi_{I(1,0,0)}(0)+\omega})$ |
| 1,4,7,10,9,6,10,14,18,<br>16,10,14,17,14,17,11 | $\psi(I(\psi_{I(1,0,0)}(0),1)+I_{\psi_{I(1,0,0)}(0)+1})$ |
| 1,4,7,10,9,6,10,<br>14,18,16,10,14,18 | $\psi(I(\psi_{I(1,0,0)}(0),1)+I_{\psi_{I(1,0,0)}(0)+\omega})$ |
| 1,4,7,10,9,6,10,14,18,<br>16,10,14,18,14,17,14,17,11 | $\psi(I(\psi_{I(1,0,0)}(0),1)+I(1,\psi_{I(1,0,0)}(0)+1))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,10,14,18,15 | $\psi(I(\psi_{I(1,0,0)}(0),1)+I(\omega,\psi_{I(1,0,0)}(0)+1))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,10,14,18,16 | $\psi(I(\psi_{I(1,0,0)}(0),1)+I(\Omega,\psi_{I(1,0,0)}(0)+1))$ |
| 1,4,7,10,9,6,10,14,18,<br>16,10,14,18,16,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0),1)\cdot 2)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,9,6,<br>10,14,18,16,11 | $\psi(I(\psi_{I(1,0,0)}(0),1)\cdot\omega)$ |
| 1,4,7,10,9,6,<br>10,14,18,16,12 | $\psi(I(\psi_{I(1,0,0)}(0),1)\cdot\Omega)$ |
| 1,4,7,10,9,6,10,14,<br>18,16,12,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0),1)\cdot\psi_{I(1,0,0)}(0))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,12,10,14,<br>18,16,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0),1)\cdot\psi_{I(1,0,0)}(0)+I(\psi_{I(1,0,0)}(0),1))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,12,10,14,18,<br>16,12,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0),1)\cdot\psi_{I(1,0,0)}(0)\cdot 2)$ |
| 1,4,7,10,9,6,10,14,18,<br>16,12,12,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0),1)\cdot\psi_{I(1,0,0)}(0)^2)$ |
| 1,4,7,10,9,6,10,<br>14,18,16,12,15 | $\psi(I(\psi_{I(1,0,0)}(0),1)\cdot\psi_{\Omega_{\psi_{I(1,0,0)}(0)+1}}(0))$ |
| 1,4,7,10,9,6,10,14,<br>18,16,12,16,20,24 | $\psi(I(\psi_{I(1,0,0)}(0),1)\cdot I_{\psi_{I(1,0,0)}(0)+\omega})$ |
| 1,4,7,10,9,6,10,14,18,16,<br>12,16,20,24,22,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0),1)^2)$ |
| 1,4,7,10,9,6,10,14,<br>18,16,12,16,20,24,22,14,18,16,<br>12,16,20,24,22,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0),1)^3)$ |
| 1,4,7,10,9,6,<br>10,14,18,16,13 | $\psi(\psi_{\Omega_{I(\psi_{I(1,0,0)}(0),1)+1}}(0))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,13,18 | $\psi(\Omega_{I(\psi_{I(1,0,0)}(0),1)+\omega})$ |
| 1,4,7,10,9,6,10,<br>14,18,16,13,18,23,28 | $\psi(I_{I(\psi_{I(1,0,0)}(0),1)+\omega})$ |
| 1,4,7,10,9,6,10,14,<br>18,16,13,18,23,28,<br>23,27,23,27,14 | $\psi(I(1,I(\psi_{I(1,0,0)}(0),1)+1))$ |
| 1,4,7,10,9,6,10,14,<br>18,16,13,18,23,28,24 | $\psi(I(\omega,I(\psi_{I(1,0,0)}(0),1)+1))$ |
| 1,4,7,10,9,6,10,14,<br>18,16,13,18,23,28,25 | $\psi(I(\Omega,I(\psi_{I(1,0,0)}(0),1)+1))$ |
| 1,4,7,10,9,6,10,14,<br>18,16,13,18,23,28,<br>25,4,7,10,8 | $\psi(I(I(\omega,0),I(\psi_{I(1,0,0)}(0),1)+1))$ |
| 1,4,7,10,9,6,10,14,<br>18,16,13,18,23,28,25,<br>4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0),2))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,13,18,23,28,25,22,<br>28,34,40,36,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0),3))$ |
| 1,4,7,10,9,6,<br>10,14,18,16,14 | $\psi(I(\psi_{I(1,0,0)}(0),\omega))$ |
| 1,4,7,10,9,6,<br>10,14,18,16,14,14 | $\psi(I(\psi_{I(1,0,0)}(0),\omega^2))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,14,16 | $\psi(I(\psi_{I(1,0,0)}(0),\Omega))$ |
| 1,4,7,10,9,6,10,14,18,<br>16,14,16,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0),\psi_{I(1,0,0)}(0)))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,14,16,14 | $\psi(I(\psi_{I(1,0,0)}(0),\psi_{I(1,0,0)}(0)+\omega))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,9,6,10,<br>14,18,16,14,16,14,16 | $\psi(I(\psi_{I(1,0,0)}(0), \psi_{I(1,0,0)}(0) + \Omega))$ |
| 1,4,7,10,9,6,10,14,<br>18,16,14,16,14,16,<br>4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0), \psi_{I(1,0,0)}(0) \cdot 2))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,14,16,15 | $\psi(I(\psi_{I(1,0,0)}(0), \psi_{I(1,0,0)}(0) \cdot \omega))$ |
| 1,4,7,10,9,6,10,14,18,<br>16,14,16,16,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0), \psi_{I(1,0,0)}(0)^2))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,14,16,17 | $\psi(I(\psi_{I(1,0,0)}(0), \psi_{I(1,0,0)}(0)^\omega))$ |
| 1,4,7,10,9,6,10,14,18,<br>16,14,16,18,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0), \psi_{I(1,0,0)}(0)^{\psi_{I(1,0,0)}(0)}))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,14,16,19 | $\psi(I(\psi_{I(1,0,0)}(0), \psi_{\Omega_{\psi_{I(1,0,0)}(0)+1}}(0)))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,14,17 | $\psi(I(\psi_{I(1,0,0)}(0), \Omega_{\psi_{I(1,0,0)}(0)+1}))$ |
| 1,4,7,10,9,6,10,14,18,<br>16,14,17,10,14,17,11 | $\psi(I(\psi_{I(1,0,0)}(0), \psi_{I_{\psi_{I(1,0,0)}(0)+1}}(0)))$ |
| 1,4,7,10,9,6,10,14,18,<br>16,14,17,10,14,18,<br>14,17,14,17,11 | $\psi(I(\psi_{I(1,0,0)}(0), I(1, \psi_{I(1,0,0)}(0) + 1)))$ |
| 1,4,7,10,9,6,10,14,<br>18,16,14,17,10,14,<br>18,16,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0), I(\psi_{I(1,0,0)}(0), 1)))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,14,17,11 | $\psi(\psi_{I(\psi_{I(1,0,0)}(0)+1,0)}(0))$ |
| 1,4,7,10,9,6,10,14,<br>18,16,14,17,14,17,11 | $\psi(I(\psi_{I(1,0,0)}(0) + 1, 0))$ |
| 1,4,7,10,9,6,10,14,<br>18,16,14,17,14,17,11,10,<br>14,18,16,14,17,14,17,11 | $\psi(I(\psi_{I(1,0,0)}(0) + 1, 0) \cdot 2)$ |
| 1,4,7,10,9,6,10,14,<br>18,16,14,17,14,17,13 | $\psi(\psi_{\Omega_{I(\psi_{I(1,0,0)}(0)+1,0)}+1}(0))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,14,17,14,17,13,<br>18,23,28,25,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0), I(\psi_{I(1,0,0)}(0) + 1, 0) + 1))$ |
| 1,4,7,10,9,6,10,14,<br>18,16,14,17,14,17,13,<br>18,23,28,25,23,26,19 | $\psi(\psi_{I(\psi_{I(1,0,0)}(0)+1,1)}(0))$ |
| 1,4,7,10,9,6,10,14,<br>18,16,14,17,14,17,14 | $\psi(I(\psi_{I(1,0,0)}(0) + 1, \omega))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,14,17,14,17,<br>14,16,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0) + 1, \psi_{I(1,0,0)}(0)))$ |
| 1,4,7,10,9,6,10,14,18,<br>16,14,17,14,17,14,16,10,14,<br>18,16,14,17,14,17,14,16 | $\psi(I(\psi_{I(1,0,0)}(0) + 1, I(\psi_{I(1,0,0)}(0) + 1, \Omega)))$ |
| 1,4,7,10,9,6,10,14,18,<br>16,14,17,14,17,14,17,11 | $\psi(\psi_{I(\psi_{I(1,0,0)}(0)+2,0)}(0))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,14,17,15 | $\psi(I(\psi_{I(1,0,0)}(0) + \omega, 0))$ |
| 1,4,7,10,9,6,10,<br>14,18,16,14,17,16 | $\psi(I(\psi_{I(1,0,0)}(0) + \Omega, 0))$ |
| 1,4,7,10,9,6,10,14,18,<br>16,14,17,16,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0) \cdot 2, 0))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,9,6,<br>10,14,18,16,15 | $\psi(I(\psi_{I(1,0,0)}(0)\cdot\omega,0))$ |
| 1,4,7,10,9,6,10,14,<br>18,16,16,4,7,10,9,5 | $\psi(I(\psi_{I(1,0,0)}(0)^2,0))$ |
| 1,4,7,10,9,6,<br>10,14,18,16,19 | $\psi(I(\psi_{\Omega_{\psi_{I(1,0,0)}(0)+1}}(0),0))$ |
| 1,4,7,10,9,6,10,14,18,17 | $\psi(I(\Omega_{\psi_{I(1,0,0)}(0)+1},0))$ |
| 1,4,7,10,9,6,10,<br>14,18,17,10,14,17,11 | $\psi(I(\psi_{I_{\psi_{I(1,0,0)}(0)+1}}(0),0))$ |
| 1,4,7,10,9,6,10,14,18,<br>17,10,14,18,14,17,14,17,11 | $\psi(I(I(1,\psi_{I(1,0,0)}(0)+1),0))$ |
| 1,4,7,10,9,6,10,14,18,<br>17,10,14,18,16,4,7,10,9,5 | $\psi(I(I(\psi_{I(1,0,0)}(0),1),0))$ |
| 1,4,7,10,9,6,10,14,<br>18,17,10,14,18,16,<br>14,17,14,17,11 | $\psi(I(I(\psi_{I(1,0,0)}(0)+1,0),0))$ |
| 1,4,7,10,9,6,<br>10,14,18,17,11 | $\psi(\psi_{I(1,0,0)}(1))$ |
| 1,4,7,10,9,6,10,14,<br>18,17,13,18,23,28,14 | $\psi(\psi_{I(1,0,0)}(2))$ |
| 1,4,7,10,9,7 | $\psi(\psi_{I(1,0,0)}(\omega))$ |
| 1,4,7,10,9,7,9 | $\psi(\psi_{I(1,0,0)}(\Omega))$ |
| 1,4,7,10,9,7,<br>9,4,7,10,9,7,9 | $\psi(\psi_{I(1,0,0)}(\psi_{I(1,0,0)}(\Omega)))$ |
| 1,4,7,10,9,7,9,5 | $\psi(I(1,0,0))$ |
| 1,4,7,10,9,7,9,5,3 | $\psi(I(1,0,0)+1)$ |
| 1,4,7,10,9,7,9,<br>5,4,7,10,7,9,7,9,5 | $\psi(I(1,0,0)+I(1,0))$ |
| 1,4,7,10,9,7,9,<br>5,4,7,10,9,7,9,5 | $\psi(I(1,0,0)+\psi_{I(1,0,0)}(I(1,0,0)))$ |
| 1,4,7,10,9,7,9,6 | $\psi(I(1,0,0)+\psi_{I(1,0,0)}(I(1,0,0))\cdot\Omega)$ |
| 1,4,7,10,9,7,9,6,9 | $\psi(I(1,0,0)+\psi_{\Omega_{\psi_{I(1,0,0)}(I(1,0,0))+1}}(0))$ |
| 1,4,7,10,9,7,9,6,9,12 | $\psi(I(1,0,0)+\Omega_{\psi_{I(1,0,0)}(I(1,0,0))+1})$ |
| 1,4,7,10,9,7,<br>9,6,10,14,17,11 | $\psi(I(1,0,0)+\psi_{I_{\psi_{I(1,0,0)}(I(1,0,0))+1}}(0))$ |
| 1,4,7,10,9,7,9,6,10,<br>14,18,14,17,14,17,11 | $\psi(I(1,0,0)+I(1,\psi_{I(1,0,0)}(I(1,0,0))+1))$ |
| 1,4,7,10,9,7,<br>9,6,10,14,18,15 | $\psi(I(1,0,0)+I(\omega,\psi_{I(1,0,0)}(I(1,0,0))+1))$ |
| 1,4,7,10,9,7,9,6,10,14,<br>18,16,4,7,10,9,7,9,5 | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)}(I(1,0,0)),1))$ |
| 1,4,7,10,9,7,9,6,10,<br>14,18,16,10,14,18,<br>16,4,7,10,9,5 | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)}(I(1,0,0)),1)\cdot 2)$ |
| 1,4,7,10,9,7,9,<br>6,10,14,18,16,13 | $\psi(I(1,0,0)+\psi_{\Omega_{I(\psi_{I(1,0,0)}(I(1,0,0)),1)+1}}(0))$ |
| 1,4,7,10,9,7,9,6,10,<br>14,18,16,13,18,23,28 | $\psi(I(1,0,0)+I_{I(\psi_{I(1,0,0)}(I(1,0,0)),1)+\omega})$ |
| 1,4,7,10,9,7,9,<br>6,10,14,18,16,13,18,<br>23,28,23,27,23,27,14 | $\psi(I(1,0,0)+I(1,I(\psi_{I(1,0,0)}(I(1,0,0)),1)+1))$ |
| 1,4,7,10,9,7,9,6,10,<br>14,18,16,13,18,23,28,24 | $\psi(I(1,0,0)+I(\omega,I(\psi_{I(1,0,0)}(I(1,0,0)),1)+1))$ |
| 1,4,7,10,9,7,9,6,<br>10,14,18,16,13,18,23,<br>28,25,4,7,10,9,7,9,5 | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)}(I(1,0,0)),2))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,9,7,9, 6,10,14,18,16,14 | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)}(I(1,0,0)),\omega))$ |
| 1,4,7,10,9,7,9,6,10, 14,18,16,14,16,4, 7,10,9,7,9,5 | $\psi(I(1,0,0)+ I(\psi_{I(1,0,0)}(I(1,0,0)),\psi_{I(1,0,0)}(I(1,0,0))))$ |
| 1,4,7,10,9,7,9,6, 10,14,18,16,14,16,19 | $\psi(I(1,0,0)+ I(\psi_{I(1,0,0)}(I(1,0,0)),\psi_{\Omega_{\psi_{I(1,0,0)}(I(1,0,0))+1}}(0)))$ |
| 1,4,7,10,9,7,9,6, 10,14,18,16,14,17 | $\psi(I(1,0,0)+ I(\psi_{I(1,0,0)}(I(1,0,0)),\Omega_{\psi_{I(1,0,0)}(I(1,0,0))+1}))$ |
| 1,4,7,10,9,7,9,6, 10,14,18,16,14,17,10, 14,18,14,17,14,17,11 | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)} (I(1,0,0)),I(1,\psi_{I(1,0,0)}(I(1,0,0))+1)))$ |
| 1,4,7,10,9,7,9,6,10,14, 18,16,14,17,10,14,18,16 | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)} (I(1,0,0)),I(\Omega,\psi_{I(1,0,0)}(I(1,0,0))+1)))$ |
| 1,4,7,10,9,7,9,6, 10,14,18,16,14,17,10,14, 18,16,4,7,10,9,7,9,5 | $\psi(I(1,0,0)+ I(\psi_{I(1,0,0)}(I(1,0,0)),I(\psi_{I(1,0,0)}(I(1,0,0)),1)))$ |
| 1,4,7,10,9,7,9,6, 10,14,18,16,14,17,11 | $\psi(I(1,0,0)+\psi_{I(\psi_{I(1,0,0)}(I(1,0,0))+1,0)}(0))$ |
| 1,4,7,10,9,7,9,6, 10,14,18,16,14,17,15 | $\psi(I(1,0,0)+I(\psi_{I(1,0,0)}(I(1,0,0))+\omega,0))$ |
| 1,4,7,10,9,7,9, 6,10,14,18,17 | $\psi(I(1,0,0)+I(\Omega_{\psi_{I(1,0,0)}(I(1,0,0))+1},0))$ |
| 1,4,7,10,9,7,9,6,10, 14,18,17,10,14,18,15 | $\psi(I(1,0,0)+I(I(\omega,\psi_{I(1,0,0)}(I(1,0,0))+1),0))$ |
| 1,4,7,10,9,7,9,6,10, 14,18,17,10,14,18,17 | $\psi(I(1,0,0)+I(I(\Omega_{\psi_{I(1,0,0)}(I(1,0,0))+1},0),0))$ |
| 1,4,7,10,9,7,9, 6,10,14,18,17,11 | $\psi(I(1,0,0)+\psi_{I(1,0,0)}(I(1,0,0)+1))$ |
| 1,4,7,10,9,7,9, 6,10,14,18,17,14 | $\psi(I(1,0,0)+\psi_{I(1,0,0)}(I(1,0,0)+\omega))$ |
| 1,4,7,10,9,7,9,6, 10,14,18,17,14,16, 4,7,10,9,7,9,5 | $\psi(I(1,0,0)+ \psi_{I(1,0,0)}(I(1,0,0)+\psi_{I(1,0,0)}(I(1,0,0))))$ |
| 1,4,7,10,9,7,9,6, 10,14,18,17,14,17,11 | $\psi(I(1,0,0)\cdot 2)$ |
| 1,4,7,10,9,7,9,7 | $\psi(I(1,0,0)\cdot\omega)$ |
| 1,4,7,10,9,7,9,7,7 | $\psi(I(1,0,0)\cdot\omega^2)$ |
| 1,4,7,10,9,7,9, 7,9,4,7,10,9,7,9,5 | $\psi(I(1,0,0)\cdot\psi_{I(1,0,0)}(I(1,0,0)))$ |
| 1,4,7,10,9,7,9,7,9,5 | $\psi(I(1,0,0)^2)$ |
| 1,4,7,10,9,7,9,8 | $\psi(I(1,0,0)^\omega)$ |
| 1,4,7,10,9,7,9,12 | $\psi(\psi_{\Omega_{I(1,0,0)+1}}(0))$ |
| 1,4,7,10,9,7,9,12,15 | $\psi(\Omega_{I(1,0,0)+1})$ |
| 1,4,7,10,9,7, 9,13,17,20,14 | $\psi(\psi_{I_{I(1,0,0)+1}}(0))$ |
| 1,4,7,10,9,7,9,13,17,21 | $\psi(I_{I(1,0,0)+\omega})$ |
| 1,4,7,10,9,7,9,13, 17,21,17,20,17,20,14 | $\psi(I(1,I(1,0,0)+1))$ |
| 1,4,7,10,9,7, 9,13,17,21,18 | $\psi(I(\omega,I(1,0,0)+1))$ |
| 1,4,7,10,9,7,9,13,17, 21,19,4,7,10,9,7,9,5 | $\psi(I(\psi_{I(1,0,0)}(I(1,0,0)),I(1,0,0)+1))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,9,7,9,13,<br>17,21,19,4,7,10,9,7,9,<br>13,17,21,17,20,17,20,14 | $\psi(I(I(1, I(1,0,0)+1), I(1,0,0)+1))$ |
| 1,4,7,10,9,7,9,13,<br>17,21,19,4,7,10,9,<br>7,9,13,17,21,19 | $\psi(I(\psi_{I(1,0,0)}(I(\Omega, I(1,0,0)+1)), I(1,0,0)+1))$ |
| 1,4,7,10,9,7,<br>9,13,17,21,19,5 | $\psi(I(I(1,0,0),1))$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,5,4,7,10,<br>9,7,9,13,17,21,19,5 | $\psi(I(I(1,0,0),1) + \psi_{I(1,0,0)}(I(I(1,0,0),1)))$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,6 | $\psi(I(I(1,0,0),1) + \psi_{I(1,0,0)}(I(I(1,0,0),1)) \cdot \Omega)$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,6,9 | $\psi(I(I(1,0,0),1) + \psi_{\Omega_{\psi_{I(1,0,0)}(I(I(1,0,0),1))+1}}(0))$ |
| 1,4,7,10,9,7,9,13,<br>17,21,19,6,10,14,18 | $\psi(I(I(1,0,0),1) + I_{I(I(1,0,0),1)+\omega})$ |
| 1,4,7,10,9,7,9,13,17,21,<br>19,6,10,14,18,17,11 | $\psi(I(I(1,0,0),1) + \psi_{I(1,0,0)}(I(I(1,0,0),1)+1))$ |
| 1,4,7,10,9,7,9,13,17,21,<br>19,6,10,14,18,17,14,17,11 | $\psi(I(I(1,0,0),1) + I(1,0,0))$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,6,10,14,18,<br>17,14,17,22,27,32,30,11 | $\psi(I(I(1,0,0),1) \cdot 2)$ |
| 1,4,7,10,9,7,<br>9,13,17,21,19,7 | $\psi(I(I(1,0,0),1) \cdot \omega)$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,7,9,5 | $\psi(I(I(1,0,0),1) \cdot I(1,0,0))$ |
| 1,4,7,10,9,7,9,13,17,21,<br>19,7,9,13,17,21,19,5 | $\psi(I(I(1,0,0),1)^2)$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,8 | $\psi(I(I(1,0,0),1)^\omega)$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,12 | $\psi(\psi_{\Omega_{I(I(1,0,0),1)+1}}(0))$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,12,15 | $\psi(\Omega_{I(I(1,0,0),1)+1})$ |
| 1,4,7,10,9,7,9,13,<br>17,21,19,13,17,20,14 | $\psi(\psi_{I_{I(I(1,0,0),1)+1}}(0))$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,13,17,21 | $\psi(I_{I(I(1,0,0),1)+\omega})$ |
| 1,4,7,10,9,7,9,13,17,<br>21,19,13,17,21,17,<br>19,17,19,5 | $\psi(I(1, I(I(1,0,0),1)+1))$ |
| 1,4,7,10,9,7,9,13,<br>17,21,19,13,17,21,19 | $\psi(I(\Omega, I(I(1,0,0),1)+1))$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,13,17,21,19,4,7,<br>10,9,7,9,13,17,<br>21,19,13,17,21,19 | $\psi(I(\psi_{I(1,0,0)}(I(\Omega, I(I(1,0,0),1)+1)),$<br>$I(I(1,0,0),1)+1))$ |
| 1,4,7,10,9,7,9,13,17,<br>21,19,13,17,21,19,5 | $\psi(I(I(1,0,0),2))$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,14 | $\psi(I(I(1,0,0),\omega))$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,15 | $\psi(I(I(1,0,0),\Omega))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,9,7,9,<br>13,17,21,19,15,5 | $\psi(\psi_{I(I(1,0,0)+1,0)}(0))$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,17 | $\psi(\psi_{I(I(1,0,0)+1,0)}(\omega))$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,17,19,5 | $\psi(I(I(1,0,0)+1,0))$ |
| 1,4,7,10,9,7,9,13,<br>17,21,19,17,21,17,19,5 | $\psi(I(I(1,0,0)+2,0))$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,17,21,18 | $\psi(I(I(1,0,0)+\omega,0))$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,17,21,19,5 | $\psi(I(I(1,0,0)\cdot 2,0))$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,19,5 | $\psi(I(I(1,0,0)^2,0))$ |
| 1,4,7,10,9,7,9,<br>13,17,21,19,22 | $\psi(I(\psi_{\Omega_{I(1,0,0)+1}}(0),1))$ |
| 1,4,7,10,9,7,<br>9,13,17,21,20 | $\psi(I(\Omega_{I(1,0,0)+1},0))$ |
| 1,4,7,10,9,7,9,13,<br>17,21,20,13,17,21,20 | $\psi(I(I(\Omega_{I(1,0,0)+1},0),0))$ |
| 1,4,7,10,9,7,<br>9,13,17,21,20,14 | $\psi(\psi_{I(1,0,1)}(0))$ |
| 1,4,7,10,9,7,9,13,17,<br>21,20,17,20,25,30,35,34,26 | $\psi(\psi_{I(1,0,2)}(0))$ |
| 1,4,7,10,9,7,10 | $\psi(I(1,0,\omega))$ |
| 1,4,7,10,9,7,10,7,9,5 | $\psi(\psi_{I(1,1,0)}(0))$ |
| 1,4,7,10,9,<br>7,10,7,9,7,9,5 | $\psi(I(1,1,0))$ |
| 1,4,7,10,9,7,10,7,10 | $\psi(I(1,1,\omega))$ |
| 1,4,7,10,9,7,10,8 | $\psi(I(1,\omega,0))$ |
| 1,4,7,10,9,7,10,9 | $\psi(I(1,\Omega,0))$ |
| 1,4,7,10,9,7,10,9,5 | $\psi(\psi_{I(2,0,0)}(0))$ |
| 1,4,7,10,9,<br>7,10,9,7,9,5 | $\psi(I(2,0,0))$ |
| 1,4,7,10,9,7,10,9,7,<br>9,13,17,21,20,17,21,20,14 | $\psi(\psi_{I(2,0,1)}(0))$ |
| 1,4,7,10,9,7,10,9,7,10 | $\psi(I(2,0,\omega))$ |
| 1,4,7,10,9,7,<br>10,9,7,10,7,10 | $\psi(I(2,1,\omega))$ |
| 1,4,7,10,9,7,<br>10,9,7,10,8 | $\psi(I(2,\omega,0))$ |
| 1,4,7,10,9,7,<br>10,9,7,10,9 | $\psi(I(2,\Omega,0))$ |
| 1,4,7,10,9,7,<br>10,9,7,10,9,7,9,5 | $\psi(I(3,0,0))$ |
| 1,4,7,10,9,8 | $\psi(I(\omega,0,0))$ |
| 1,4,7,10,9,9 | $\psi(I(\Omega,0,0))$ |
| 1,4,7,10,9,9,7,9,5 | $\psi(I(1,0,0,0))$ |
| 1,4,7,10,9,9,7,10 | $\psi(I(1,0,0,\omega))$ |
| 1,4,7,10,9,9,<br>7,10,7,9,5 | $\psi(I(1,0,1,0))$ |
| 1,4,7,10,9,9,7,10,8 | $\psi(I(1,0,\omega,0))$ |
| 1,4,7,10,9,<br>9,7,10,9,7,9,5 | $\psi(I(1,1,0,0))$ |
| 1,4,7,10,9,9,<br>7,10,9,9,7,9,5 | $\psi(I(2,0,0,0))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,9,9,8 | $\psi(I(\omega,0,0,0))$ |
| 1,4,7,10,9,9,9 | $\psi(I(\Omega,0,0,0))$ |
| 1,4,7,10,9,9,9,7,9,5 | $\psi(I(1,0,0,0,0))$ |
| 1,4,7,10,9,<br>9,9,9,7,9,5 | $\psi(I(1,0,0,0,0,0))$ |
| 1,4,7,10,9,10 | $\psi(I(1@\omega))$<br>$\psi(M^{M^{\omega}})$ |
| 1,4,7,10,9,10,7 | $\psi(I(1@\omega,\omega@0))$<br>$\psi(M^{M^{\omega}}\cdot\omega)$ |
| 1,4,7,10,9,10,7,9,5 | $\psi(\psi_{I(1@\omega,1@1)}(0))$ |
| 1,4,7,10,9,10,7,10 | $\psi(I(1@\omega,1@1,\omega@0))$ |
| 1,4,7,10,9,10,7,10,9,10 | $\psi(I(2@\omega))$ |
| 1,4,7,10,9,10,9 | $\psi(I(\Omega@\omega))$ |
| 1,4,7,10,9,10,9,7,9,5 | $\psi(I(1@\omega+1))$ |
| 1,4,7,10,9,11 | $\psi(I(1@\Omega))$ |
| 1,4,7,10,9,11,5 | $\psi(\psi_{I(1@(1,0))}(0))$<br>$\psi(M^{M^{\psi_{\psi_M(M^{M^M})}(0)}})$ |
| 1,4,7,10,9,11,7,9,5 | $\psi(I(1@(1,0)))$<br>$\psi(M^{M^M})$ |
| 1,4,7,10,9,11,7,10 | $\psi(I(1@(1,0),\omega@0))$<br>$\psi(M^{M^M}\cdot\omega)$ |
| 1,4,7,10,9,<br>11,7,10,7,9,5 | $\psi(\psi_{I(1@(1,0),1@1)}(0))$<br>$\psi(M^{M^M}\cdot\psi_{\psi_M(M^{M^M+1})}(0))$ |
| 1,4,7,10,9,11,<br>7,10,7,9,7,9,5 | $\psi(I(1@(1,0),1@1))$<br>$\psi(M^{M^M+1})$ |
| 1,4,7,10,9,11,7,10,8 | $\psi(I(1@(1,0),\omega@1))$<br>$\psi(M^{M^M+\omega})$ |
| 1,4,7,10,9,11,7,10,9,10 | $\psi(I(1@(1,0),1@\omega))$<br>$\psi(M^{M^M+M^{\omega}})$ |
| 1,4,7,10,9,11,7,10,9,11 | $\psi(I(1@(1,0),1@\Omega))$<br>$\psi(M^{M^M+M^{\Omega}})$ |
| 1,4,7,10,9,11,<br>7,10,9,11,7,9,5 | $\psi(I(2@(1,0)))$<br>$\psi(M^{M^M\cdot 2})$ |
| 1,4,7,10,9,11,8 | $\psi(I(\omega@(1,0)))$<br>$\psi(M^{M^M\cdot\omega})$ |
| 1,4,7,10,9,11,9 | $\psi(I(\Omega@(1,0)))$<br>$\psi(M^{M^M\cdot\Omega})$ |
| 1,4,7,10,9,11,9,7,9,5 | $\psi(I(1@(1,1)))$<br>$\psi(M^{M^{M+1}})$ |
| 1,4,7,10,9,<br>11,9,9,7,9,5 | $\psi(I(1@(1,2)))$ |
| 1,4,7,10,9,11,9,10 | $\psi(I(1@(1,\omega)))$ |
| 1,4,7,10,9,11,9,11,7,9,5 | $\psi(I(1@(2,0)))$<br>$\psi(M^{M^{M\cdot 2}})$ |
| 1,4,7,10,9,11,10 | $\psi(I(1@(\omega,0)))$<br>$\psi(M^{M^{M\cdot\omega}})$ |
| 1,4,7,10,9,11,11 | $\psi(I(1@(\Omega,0)))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,9,11,11,7,9,5 | $\psi(I(1@(1,0,0)))$<br>$\psi(M^{M^{M^2}})$ |
| 1,4,7,10,9,11,<br>11,9,7,9,5 | $\psi(I(1@(1,0,1)))$ |
| 1,4,7,10,9,11,<br>11,9,11,7,9,5 | $\psi(I(1@(1,1,0)))$ |
| 1,4,7,10,9,11,<br>11,9,11,11,7,9,5 | $\psi(I(1@(2,0,0)))$ |
| 1,4,7,10,9,11,11,11 | $\psi(I(1@(\Omega,0,0)))$ |
| 1,4,7,10,9,<br>11,11,11,7,9,5 | $\psi(I(1@(1,0,0,0)))$ |
| 1,4,7,10,9,11,12 | $\psi(I(1@(1@\omega)))$<br>$\psi(M^{M^{M^{\omega}}})$ |
| 1,4,7,10,9,<br>11,13,7,9,5 | $\psi(I(1@(1@(1,0))))$<br>$\psi(M^{M^{M^{M}}})$ |
| 1,4,7,10,9,11,<br>13,15,7,9,5 | $\psi(I(1@(1@(1@(1,0)))))$<br>$\psi(M^{M^{M^{M^{M}}}})$ |
| 1,4,7,10,9,12 | $\psi(\psi_{\Omega_{M+1}}(0))$ |
| 1,4,7,10,9,12,3 | $\psi(\psi_{\Omega_{M+1}}(0)+1)$ |
| 1,4,7,10,9,12,4 | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\omega))$ |
| 1,4,7,10,9,12,4,7,10 | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(M\cdot\omega))$<br>$\psi(\psi_{\Omega_{M+1}}(0)+I_\omega)$ |
| 1,4,7,10,9,<br>12,4,7,10,9,12 | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0)))$ |
| 1,4,7,10,9,12,6 | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0))\cdot\Omega)$ |
| 1,4,7,10,9,<br>12,6,4,7,10,9,12 | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0))^2)$ |
| 1,4,7,10,9,12,6,9 | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_{\Omega_{\psi_M(\psi_{\Omega_{M+1}}(0))+1}}(0))$ |
| 1,4,7,10,9,12,6,9,12 | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0)+1))$<br>$\psi(\psi_{\Omega_{M+1}}(0)+\Omega_{\psi_M(\psi_{\Omega_{M+1}}(0))+1})$ |
| 1,4,7,10,9,12,6,10 | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0)+\omega))$ |
| 1,4,7,10,9,<br>12,6,10,14,16 | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0)+\Omega))$ |
| 1,4,7,10,9,12,<br>6,10,14,16,4,7,10,9,12 | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0)+$<br>$\psi_M(\psi_{\Omega_{M+1}}(0))))$<br>$\psi(\psi_{\Omega_{M+1}}(0)+\Omega_{\psi_M(\psi_{\Omega_{M+1}}(0))\cdot 2})$ |
| 1,4,7,10,9,12,6,10,14,17 | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_M(\psi_{\Omega_{M+1}}(0)+$<br>$\psi_M(\psi_{\Omega_{M+1}}(0)+1)))$<br>$\psi(\psi_{\Omega_{M+1}}(0)+\Omega_{\Omega_{\psi_M(\psi_{\Omega_{M+1}}(0))+1}})$ |
| 1,4,7,10,9,<br>12,6,10,14,17,11 | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_{\psi_M(\psi_{\Omega_{M+1}}(0)+M)}(0))$<br>$\psi(\psi_{\Omega_{M+1}}(0)+\psi_{I_{\psi_M(\psi_{\Omega_{M+1}}(0))+1}}(0))$ |
| 1,4,7,10,9,12,6,10,<br>14,17,11,4,7,10,9,12 | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_{\psi_M(\psi_{\Omega_{M+1}}(0)+M)}(0)+\psi_M(\psi_{\Omega_{M+1}}(0)))$ |
| 1,4,7,10,9,12,6,<br>10,14,17,11,9,12 | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_{\psi_M(\psi_{\Omega_{M+1}}(0)+M)}(0)+$<br>$\psi_M(\psi_{\Omega_{M+1}}(0)+\psi_{\psi_M(\psi_{\Omega_{M+1}}(0)+M)}(0)+1))$<br>$\psi(\psi_{\Omega_{M+1}}(0)+\Omega_{\psi_{I_{\psi_M(\psi_{\Omega_{M+1}}(0))+1}}(0)+1})$ |
| 1,4,7,10,9,12,6,10,<br>14,17,13,18,23,27,19 | $\psi(\psi_{\Omega_{M+1}}(0)+\psi_{\psi_M(\psi_{\Omega_{M+1}}(0)+M)}(1))$<br>$\psi(\psi_{\Omega_{M+1}}(0)+\psi_{I_{\psi_M(\psi_{\Omega_{M+1}}(0))+1}}(1))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,9,12,6,<br>10,14,17,14,17,11 | $\psi(\psi_{\Omega_{M+1}}(0)+M)$<br>$\psi(\psi_{\Omega_{M+1}}(0)+I_{\psi_M(\psi_{\Omega_{M+1}}(0))+1})$ |
| 1,4,7,10,9,12,6,10,14,18 | $\psi(\psi_{\Omega_{M+1}}(0)+M\cdot\omega)$ |
| 1,4,7,10,9,12,6,10,<br>14,18,14,17,14,17,11 | $\psi(\psi_{\Omega_{M+1}}(0)+I(1,\psi_M(\psi_{\Omega_{M+1}}(0))+1))$ |
| 1,4,7,10,9,12,6,10,<br>14,18,17,14,17,11 | $\psi(\psi_{\Omega_{M+1}}(0)+M^M)$<br>$\psi(\psi_{\Omega_{M+1}}(0)+I(1,0,\psi_M(\psi_{\Omega_{M+1}}(0))+1))$ |
| 1,4,7,10,9,12,6,<br>10,14,18,17,18 | $\psi(\psi_{\Omega_{M+1}}(0)+M^{M^\omega})$ |
| 1,4,7,10,9,12,6,10,<br>14,18,17,19,14,17,11 | $\psi(\psi_{\Omega_{M+1}}(0)+M^{M^M})$ |
| 1,4,7,10,9,12,6,<br>10,14,18,17,21 | $\psi(\psi_{\Omega_{M+1}}(0)\cdot 2)$ |
| 1,4,7,10,9,12,7 | $\psi(\psi_{\Omega_{M+1}}(0)\cdot\omega)$ |
| 1,4,7,10,9,12,7,9 | $\psi(\psi_{\Omega_{M+1}}(0)\cdot\Omega)$ |
| 1,4,7,10,9,12,7,<br>9,4,7,10,9,12 | $\psi(\psi_{\Omega_{M+1}}(0)\cdot\psi_M(\psi_{\Omega_{M+1}}(0)))$ |
| 1,4,7,10,9,12,7,9,5 | $\psi(\psi_{\Omega_{M+1}}(0)\cdot\psi_{\psi_M(\psi_{\Omega_{M+1}}(0)\cdot M)}(0))$ |
| 1,4,7,10,9,12,<br>7,9,7,9,5 | $\psi(\psi_{\Omega_{M+1}}(0)\cdot M)$ |
| 1,4,7,10,9,12,7,10 | $\psi(\psi_{\Omega_{M+1}}(0)\cdot M\cdot\omega)$ |
| 1,4,7,10,9,12,<br>7,10,7,9,7,9,5 | $\psi(\psi_{\Omega_{M+1}}(0)\cdot M^2)$ |
| 1,4,7,10,9,12,<br>7,10,9,7,9,5 | $\psi(\psi_{\Omega_{M+1}}(0)\cdot M^M)$ |
| 1,4,7,10,9,12,7,10,9,12 | $\psi(\psi_{\Omega_{M+1}}(0)^2)$ |
| 1,4,7,10,9,12,8 | $\psi(\psi_{\Omega_{M+1}}(0)^\omega)$ |
| 1,4,7,10,9,12,9 | $\psi(\psi_{\Omega_{M+1}}(0)^\Omega)$ |
| 1,4,7,10,9,12,9,5 | $\psi(\psi_{\Omega_{M+1}}(0)^{\psi_{\psi_M(\psi_{\Omega_{M+1}}(0)M)}(0)})$ |
| 1,4,7,10,9,12,9,7,9,5 | $\psi(\psi_{\Omega_{M+1}}(0)^M)$ |
| 1,4,7,10,9,12,9,12 | $\psi(\psi_{\Omega_{M+1}}(0)^{\psi_{\Omega_{M+1}}(0)})$ |
| 1,4,7,10,9,12,10 | $\psi(\psi_{\Omega_{M+1}}(0)^{\psi_{\Omega_{M+1}}(0)^\omega})$ |
| 1,4,7,10,9,12,11 | $\psi(\psi_{\Omega_{M+1}}(0)^{\psi_{\Omega_{M+1}}(0)^\Omega})$ |
| 1,4,7,10,9,12,11,14 | $\psi(\psi_{\Omega_{M+1}}(0)^{\psi_{\Omega_{M+1}}(0)^{\psi_{\Omega_{M+1}}(0)}})$ |
| 1,4,7,10,9,12,12 | $\psi(\psi_{\Omega_{M+1}}(1))$ |
| 1,4,7,10,9,12,13 | $\psi(\psi_{\Omega_{M+1}}(\omega))$ |
| 1,4,7,10,9,12,14 | $\psi(\psi_{\Omega_{M+1}}(\Omega))$ |
| 1,4,7,10,9,<br>12,14,4,7,10,9,12 | $\psi(\psi_{\Omega_{M+1}}(\psi_M(\psi_{\Omega_{M+1}}(0))))$ |
| 1,4,7,10,9,12,<br>14,4,7,10,9,12,14 | $\psi(\psi_{\Omega_{M+1}}(\psi_M(\psi_{\Omega_{M+1}}(\Omega))))$ |
| 1,4,7,10,9,12,14,5 | $\psi(\psi_{\Omega_{M+1}}(\psi_{\psi_M(\psi_{\Omega_{M+1}}(M))}(0)))$ |
| 1,4,7,10,9,12,14,7,9,5 | $\psi(\psi_{\Omega_{M+1}}(M))$ |
| 1,4,7,10,9,12,14,17 | $\psi(\psi_{\Omega_{M+1}}(\psi_{\Omega_{M+1}}(0)))$ |
| 1,4,7,10,9,12,15 | $\psi(\Omega_{M+1})$<br>$\psi(\psi_{M_2}(0))$<br>$\psi(2 \text{ aft } 2-2)$ |
| 1,4,7,10,9,13 | $\psi(\Omega_{M+\omega})$<br>$\psi(\psi_{M_2}(\omega))$ |
| 1,4,7,10,9,13,17,19 | $\psi(\Omega_{M+\Omega})$<br>$\psi(\psi_{M_2}(\Omega))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,9,13,<br>17,19,4,7,10,9,12 | $\psi(\Omega_{M+\psi_M(\psi_{\Omega_{M+1}}(0))})$ |
| 1,4,7,10,9,13,<br>17,19,4,7,10,9,12,15 | $\psi(\Omega_{M+\psi_M(\Omega_{M+1})})$<br>$\psi(\psi_{M_2}(\psi_M(\psi_{M_2}(0))))$ |
| 1,4,7,10,9,13,17,19,5 | $\psi(\Omega_{M+\psi_{\psi_M(\Omega_{M\cdot 2})}(0)})$<br>$\psi(\psi_{M_2}(\psi_{\psi_M(\psi_{M_2}(M))}(0)))$ |
| 1,4,7,10,9,13,<br>17,19,7,9,5 | $\psi(\Omega_{M\cdot 2})$<br>$\psi(\psi_{M_2}(M))$ |
| 1,4,7,10,9,13,17,19,22 | $\psi(\Omega_{\psi_{\Omega_{M+1}}(0)})$<br>$\psi(\psi_{M_2}(\psi_{\Omega_{M+1}}(0)))$ |
| 1,4,7,10,9,13,17,20 | $\psi(\Omega_{\Omega_{M+1}})$<br>$\psi(\psi_{M_2}(\psi_{M_2}(0)))$ |
| 1,4,7,10,9,13,17,20,14 | $\psi(\psi_{I_{M+1}}(0))$<br>$\psi(\psi_{\psi_{M_2}(M_2)}(0))$ |
| 1,4,7,10,9,13,<br>17,20,17,20,14 | $\psi(I_{M+1})$<br>$\psi(\psi_{M_2}(M_2))$<br>$\psi(M_2)$ |
| 1,4,7,10,9,13,17,21 | $\psi(I_{M+\omega})$<br>$\psi(M_2\cdot\omega)$ |
| 1,4,7,10,9,13,17,21,17,19 | $\psi(I_{M+\Omega})$ |
| 1,4,7,10,9,13,<br>17,21,17,19,7,9,5 | $\psi(I_{M\cdot 2})$<br>$\psi(M_2\cdot M)$ |
| 1,4,7,10,9,13,17,21,17,20 | $\psi(I_{\Omega_{M+1}})$<br>$\psi(M_2\cdot\psi_{M_2}(0))$ |
| 1,4,7,10,9,13,<br>17,21,17,20,14 | $\psi(\psi_{I(1,M+1)}(0))$<br>$\psi(M_2\cdot\psi_{\psi_{M_2}(M_2^2)}(0))$ |
| 1,4,7,10,9,13,17,<br>21,17,20,17,20,14 | $\psi(I(1,M+1))$<br>$\psi(M_2^2)$ |
| 1,4,7,10,9,13,17,21,18 | $\psi(I(\omega,M+1))$<br>$\psi(M_2^\omega)$ |
| 1,4,7,10,9,13,17,21,19 | $\psi(I(\Omega,M+1))$<br>$\psi(M_2^\Omega)$ |
| 1,4,7,10,9,13,17,21,20 | $\psi(I(\Omega_{M+1},M+1))$<br>$\psi(M_2^{\psi_{M_2}(0)})$ |
| 1,4,7,10,9,13,17,21,20,14 | $\psi(\psi_{I(1,0,M+1)}(0))$<br>$\psi(M_2^{\psi_{\psi_{M_2}(M_2^{M_2})}(0)})$ |
| 1,4,7,10,9,13,17,<br>21,20,17,20,14 | $\psi(I(1,0,M+1))$<br>$\psi(M_2^{M_2})$ |
| 1,4,7,10,9,13,17,21,20,21 | $\psi(I(1@\omega,M+1@0))$<br>$\psi(M_2^{M_2^\omega})$ |
| 1,4,7,10,9,13,17,<br>21,20,22,14 | $\psi(I(1@(1,0),M+1@0))$<br>$\psi(M_2^{M_2^{M_2}})$ |
| 1,4,7,10,9,13,17,21,20,24 | $\psi(\psi_{\Omega_{M_2+1}}(0))$ |
| 1,4,7,10,9,13,<br>17,21,20,24,24 | $\psi(\psi_{\Omega_{M_2+1}}(1))$ |
| 1,4,7,10,9,13,<br>17,21,20,24,28 | $\psi(\Omega_{M_2+1})$<br>$\psi(\psi_{M_3}(0))$ |
| 1,4,7,10,9,13,17,21,20,25 | $\psi(\Omega_{M_2+\omega})$<br>$\psi(\psi_{M_3}(\omega))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,9,13,<br>17,21,20,25,30,34,26 | $\psi(\psi_{I_{M_2+1}}(0))$<br>$\psi(\psi_{\psi_{M_3}(M_3)}(0))$ |
| 1,4,7,10,9,13,17,21,<br>20,25,30,34,30,34,26 | $\psi(I_{M_2+1})$<br>$\psi(M_3)$ |
| 1,4,7,10,9,13,<br>17,21,20,25,30,35 | $\psi(I_{M_2+\omega})$<br>$\psi(M_3 \cdot \omega)$ |
| 1,4,7,10,9,13,17,21,20,<br>25,30,35,30,34,30,34,26 | $\psi(I(1, M_2+1))$<br>$\psi(M_3^2)$ |
| 1,4,7,10,9,13,17,21,20,<br>25,30,35,34,30,34,26 | $\psi(I(1, 0, M_2+1))$<br>$\psi(M_3^{M_3})$ |
| 1,4,7,10,9,13,17,21,20,<br>25,30,35,34,36,30,34,26 | $\psi(I(1@(1,0), M_2+1@0))$<br>$\psi(M_3^{M_3^{M_3}})$ |
| 1,4,7,10,9,13,17,21,<br>20,25,30,35,34,39 | $\psi(\psi_{\Omega_{M_3+1}}(0))$ |
| 1,4,7,10,9,13,17,21,20,<br>25,30,35,34,39,44 | $\psi(\Omega_{M_3+1})$<br>$\psi(M_4)$ |
| 1,4,7,10,9,13,17,21,<br>20,25,30,35,34,40,<br>46,52,51,57,63 | $\psi(\Omega_{M_4+1})$<br>$\psi(M_5)$ |
| 1,4,7,10,10 | $\psi(M_\omega)$<br>$\psi(1-2-2)$<br>SMO |
| 1,4,7,10,10,4,7,10,4 | $\psi(M_\omega + I_\omega + \Omega_\omega)$ |
| 1,4,7,10,10,4,<br>7,10,9,13,17,21,21 | $\psi(M_\omega + \psi_M(M_\omega))$ |
| 1,4,7,10,10,4,7,10,10 | $\psi(M_\omega \cdot 2)$ |
| 1,4,7,10,10,6 | $\psi(M_\omega \cdot \Omega)$ |
| 1,4,7,10,10,6,4,7,10,10 | $\psi(M_\omega^2)$ |
| 1,4,7,10,10,6,7 | $\psi(M_\omega^\omega)$ |
| 1,4,7,10,10,<br>6,8,4,7,10,10 | $\psi(M_\omega^{M_\omega})$ |
| 1,4,7,10,10,6,9 | $\psi(\psi_{\Omega_{M_\omega+1}}(0))$ |
| 1,4,7,10,10,6,9,12 | $\psi(\Omega_{M_\omega+1})$<br>$\psi(\psi_{M_{\omega+1}}(0))$ |
| 1,4,7,10,10,6,10,14,17,11 | $\psi(\psi_{I_{M_\omega+1}}(0))$<br>$\psi(\psi_{\psi_{M_{\omega+1}}(M_{\omega+1})}(0))$ |
| 1,4,7,10,10,6,<br>10,14,17,14,17,11 | $\psi(I_{M_\omega+1})$<br>$\psi(M_{\omega+1})$ |
| 1,4,7,10,10,6,10,14,18 | $\psi(I_{M_\omega+\omega})$<br>$\psi(M_{\omega+1} \cdot \omega)$ |
| 1,4,7,10,10,6,<br>10,14,18,17,14,17,11 | $\psi(I(1, 0, M_\omega+1))$<br>$\psi(M_{\omega+1}^{M_{\omega+1}})$ |
| 1,4,7,10,10,6,<br>10,14,18,17,21 | $\psi(\psi_{\Omega_{M_{\omega+1}+1}}(0))$ |
| 1,4,7,10,10,6,<br>10,14,18,17,21,25 | $\psi(\Omega_{M_{\omega+1}+1})$<br>$\psi(\psi_{M_{\omega+2}}(0))$ |
| 1,4,7,10,10,6,10,<br>14,18,17,22,27,31,23 | $\psi(\psi_{I_{M_{\omega+1}+1}}(0))$<br>$\psi(\psi_{\psi_{M_{\omega+2}}(M_{\omega+2})}(0))$ |
| 1,4,7,10,10,6,10,14,18,<br>17,22,27,31,27,31,23 | $\psi(I_{M_{\omega+1}+1})$<br>$\psi(M_{\omega+2})$ |
| 1,4,7,10,10,6,10,14,18,18 | $\psi(M_{\omega \cdot 2})$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,10,6,10,14,18, 18,13,18,23,28,28 | $\psi(M_{\omega\cdot 3})$ |
| 1,4,7,10,10,7 | $\psi(M_{\omega^2})$ |
| 1,4,7,10,10,7,7 | $\psi(M_{\omega^3})$ |
| 1,4,7,10,10,7,8 | $\psi(M_{\omega^\omega})$ |
| 1,4,7,10,10,7,9 | $\psi(M_\Omega)$ |
| 1,4,7,10,10,7,9,4,7, 9,13,17,21,21,17,19,5 | $\psi(M_I)$ |
| 1,4,7,10,10,7,9,4,7,10 | $\psi(M_{I_\omega})$ |
| 1,4,7,10,10,7,9,4, 7,10,9,13,17,21,21 | $\psi(M_{\psi_M(M_\omega)})$ |
| 1,4,7,10,10,7,9,4,7,10, 9,13,17,21,21,17,19,5 | $\psi(M_{\psi_{\psi_M(M_M)}(0)})$ |
| 1,4,7,10,10,7,9,4,7,10, 9,13,17,21,21,17,19,7,9,5 | $\psi(M_M)$ |
| 1,4,7,10,10,7, 9,4,7,10,10 | $\psi(M_{M_\omega})$ |
| 1,4,7,10,10,7, 9,4,7,10,10,7,9 | $\psi(M_{M_\Omega})$ |
| 1,4,7,10,10,7,9,5 | $\psi(\psi_{M(1,0)}(0))$ |
| 1,4,7,10,10,7,9,7,9,5 | $\psi(M(1,0))$<br>$\psi(\psi_{M(1;0)}(0))$ |
| 1,4,7,10,10,7,9,7,9,6 | $\psi(M(1,0)+\psi_{M(1,0)}(M(1,0))\cdot\Omega)$ |
| 1,4,7,10,10,7,9,7,9,6,9 | $\psi(M(1,0)+\psi_{\Omega_{\psi_{M(1,0)}(M(1,0))+1}}(0))$ |
| 1,4,7,10,10,7,9, 7,9,6,10,14,17,11 | $\psi(M(1,0)+\psi_{I_{\psi_{M(1,0)}(M(1,0))+1}}(0))$<br>$\psi(M(1,0)+\psi_{\psi_{M_{\psi_{M(1,0)}(M(1,0))+1}}(M_{\psi_{M(1,0)}(M(1,0))+1})}(0))$ |
| 1,4,7,10,10,7,9, 7,9,6,10,14,18 | $\psi(M(1,0)+I_{\psi_{M(1,0)}(M(1,0))+\omega})$<br>$\psi(M(1,0)+M_{\psi_{M(1,0)}(M(1,0))+1}\cdot\omega)$ |
| 1,4,7,10,10,7,9,7,9,6, 10,14,18,17,14,17,11 | $\psi(M(1,0)+I(1,0,\psi_{M(1,0)}(M(1,0))+1))$<br>$\psi(M(1,0)+M^{M_{\psi_{M(1,0)}(M(1,0))+1}}_{\psi_{M(1,0)}(M(1,0))+1})$ |
| 1,4,7,10,10,7,9, 7,9,6,10,14,18,17,21 | $\psi(M(1,0)+\psi_{\Omega_{M_{\psi_{M(1,0)}(M(1,0))+1}+1}}(0))$ |
| 1,4,7,10,10,7,9,7,9,6, 10,14,18,17,22,27,31,23 | $\psi(M(1,0)+\psi_{I_{M_{\psi_{M(1,0)}(M(1,0))+1}+1}}(0))$<br>$\psi(M(1,0)+\psi_{\psi_{M_{\psi_{M(1,0)}(M(1,0))+2}}(M_{\psi_{M(1,0)}(M(1,0))+2})}(0))$ |
| 1,4,7,10,10,7,9, 7,9,6,10,14,18,18 | $\psi(M(1,0)+M_{\psi_{M(1,0)}(M(1,0))+\omega})$ |
| 1,4,7,10,10,7,9,7, 9,6,10,14,18,18,14 | $\psi(M(1,0)+M_{\psi_{M(1,0)}(M(1,0))+\omega^2})$ |
| 1,4,7,10,10,7,9, 7,9,6,10,14,18,18,14, 16,4,7,10,10,7,9,5 | $\psi(M(1,0)+M_{\psi_{M(1,0)}(M(1,0))\cdot 2})$ |
| 1,4,7,10,10,7,9,7,9,6, 10,14,18,18,14,17,11 | $\psi(M(1,0)+\psi_{M(1,0)}(M(1,0)+1))$ |
| 1,4,7,10,10,7,9,7, 9,6,10,14,18,18,14, 17,14,17,11 | $\psi(M(1,0)\cdot 2)$ |
| 1,4,7,10,10,7,9,7,9,7 | $\psi(M(1,0)\cdot\omega)$ |
| 1,4,7,10,10, 7,9,7,9,7,9 | $\psi(M(1,0)\cdot\Omega)$ |
| 1,4,7,10,10,7,9,7,9,7, 9,4,7,10,10,7,9,7,9,5 | $\psi(M(1,0)\cdot\psi_{M(1,0)}(M(1,0)))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,10,<br>7,9,7,9,7,9,5 | $\psi(M(1,0)^2)$ |
| 1,4,7,10,10,7,9,8 | $\psi(M(1,0)^\omega)$ |
| 1,4,7,10,10,7,9,12 | $\psi(\psi_{\Omega_{M(1,0)+1}}(0))$ |
| 1,4,7,10,10,7,9,12,15 | $\psi(\Omega_{M(1,0)+1})$<br>$\psi(\psi_{M_{M(1,0)+1}}(0))$ |
| 1,4,7,10,10,<br>7,9,13,17,20,14 | $\psi(\psi_{I_{M(1,0)+1}}(0))$<br>$\psi(\psi_{\psi_{M_{M(1,0)+1}}(M_{M(1,0)+1})}(0))$ |
| 1,4,7,10,10,7,9,<br>13,17,20,17,20,14 | $\psi(I_{M(1,0)+1})$<br>$\psi(M_{M(1,0)+1})$ |
| 1,4,7,10,10,7,<br>9,13,17,21,21 | $\psi(M_{M(1,0)+\omega})$ |
| 1,4,7,10,10,7,9,<br>13,17,21,21,17,20,14 | $\psi(\psi_{M(1,1)}(0))$<br>$\psi((1-)^{1,0}\ 2-2 \text{ aft } 2\ 1-2-2)$ |
| 1,4,7,10,10,7,9,13,17,<br>21,21,17,20,17,20,14 | $\psi(M(1,1))$<br>$\psi(\psi_{M(1;0)}(1))$ |
| 1,4,7,10,10,7,9,<br>13,17,21,21,17,20,25,<br>30,35,35,30,34,30,34,26 | $\psi(M(1,2))$<br>$\psi(\psi_{M(1;0)}(2))$ |
| 1,4,7,10,10,7,10 | $\psi(M(1,\omega))$<br>$\psi(\psi_{M(1;0)}(\omega))$ |
| 1,4,7,10,10,7,10,7 | $\psi(M(1,\omega^2))$<br>$\psi(\psi_{M(1;0)}(\omega^2))$ |
| 1,4,7,10,10,7,10,7,9 | $\psi(M(1,\Omega))$<br>$\psi(\psi_{M(1;0)}(\Omega))$ |
| 1,4,7,10,10,7,10,7,9,5 | $\psi(\psi_{M(2,0)}(0))$<br>$\psi(\psi_{\psi_{M(1;0)}(M(1;0))}(0))$ |
| 1,4,7,10,10,7,10,7,9,7 | $\psi(\psi_{M(2,0)}(\omega))$<br>$\psi(\psi_{\psi_{M(1;0)}(M(1;0))}(\omega))$ |
| 1,4,7,10,10,7,10,7,9,7,9 | $\psi(\psi_{M(2,0)}(\Omega))$<br>$\psi(\psi_{\psi_{M(1;0)}(M(1;0))}(\Omega))$ |
| 1,4,7,10,10,<br>7,10,7,9,7,9,5 | $\psi(M(2,0))$ |
| 1,4,7,10,10,<br>7,10,7,9,7,9,5,3 | $\psi(M(2,0)+1)$ |
| 1,4,7,10,10,7,<br>10,7,9,7,9,7,9,5 | $\psi(M(2,0)^2)$ |
| 1,4,7,10,10,7,10,7,9,8 | $\psi(M(2,0)^\omega)$ |
| 1,4,7,10,10,7,<br>10,7,9,9,5 | $\psi(M(2,0)^{M(2,0)})$ |
| 1,4,7,10,10,7,10,7,9,12 | $\psi(\psi_{\Omega_{M(2,0)+1}}(0))$ |
| 1,4,7,10,10,7,<br>10,7,9,12,12 | $\psi(\psi_{\Omega_{M(2,0)+1}}(1))$ |
| 1,4,7,10,10,7,<br>10,7,9,12,15 | $\psi(\Omega_{M(2,0)+1})$ |
| 1,4,7,10,10,7,<br>10,7,9,13,17,20,14 | $\psi(\psi_{I_{M(2,0)+1}}(0))$<br>$\psi(\psi_{\psi_{M_{M(2,0)+1}}(M_{M(2,0)+1})}(0))$ |
| 1,4,7,10,10,7,10,<br>7,9,13,17,20,17,20,14 | $\psi(I_{M(2,0)+1})$<br>$\psi(M_{M(2,0)+1})$ |
| 1,4,7,10,10,7,<br>10,7,9,13,17,21,21 | $\psi(M_{M(2,0)+\omega})$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,10,7,10,7,9,<br>13,17,21,21,17,20,14 | $\psi(\psi_{M(1,M(2,0)+1)}(0))$ |
| 1,4,7,10,10,7,10,<br>7,9,13,17,21,21,17,21 | $\psi(M(1,M(2,0)+\omega))$ |
| 1,4,7,10,10,7,10,7,<br>9,13,17,21,21,17,21,<br>17,20,17,20,14 | $\psi(M(2,1))$ |
| 1,4,7,10,10,7,10,7,10 | $\psi(M(2,\omega))$ |
| 1,4,7,10,10,7,<br>10,7,10,7,9,5 | $\psi(\psi_{M(3,0)}(0))$ |
| 1,4,7,10,10,7,10,<br>7,10,7,9,7,9,5 | $\psi(M(3,0))$<br>$\psi(M(1;0)^2)$ |
| 1,4,7,10,10,7,<br>10,7,10,7,10 | $\psi(M(3,\omega))$ |
| 1,4,7,10,10,7,10,8 | $\psi(M(\omega,0))$<br>$\psi(M(1;0)^\omega)$ |
| 1,4,7,10,10,7,10,9 | $\psi(M(\Omega,0))$ |
| 1,4,7,10,10,7,10,9,5 | $\psi(\psi_{M(1,0,0)}(0))$<br>$\psi(\psi_{\psi_{M(1;0)}(M(1;0)^{M(1;0)})}(0))$ |
| 1,4,7,10,10,7,<br>10,9,7,9,5 | $\psi(M(1,0,0))$<br>$\psi(M(1;0)^{M(1;0)})$ |
| 1,4,7,10,10,7,10,9,7,<br>9,13,17,21,21,17,<br>21,20,17,20,14 | $\psi(M(1,0,1))$<br>$\psi(M(1;0)^{M(1;0)}\cdot 2)$ |
| 1,4,7,10,10,7,10,9,7,10 | $\psi(M(1,0,\omega))$<br>$\psi(M(1;0)^{M(1;0)}\cdot\omega)$ |
| 1,4,7,10,10,7,10,<br>9,7,10,7,9,7,9,5 | $\psi(M(1,1,0))$ |
| 1,4,7,10,10,7,<br>10,9,7,10,7,10 | $\psi(M(1,1,\omega))$ |
| 1,4,7,10,10,7,10,9,<br>7,10,7,10,7,9,7,9,5 | $\psi(M(1,2,0))$ |
| 1,4,7,10,10,7,<br>10,9,7,10,8 | $\psi(M(1,\omega,0))$ |
| 1,4,7,10,10,7,<br>10,9,7,10,9 | $\psi(M(1,\Omega,0))$ |
| 1,4,7,10,10,7,<br>10,9,7,10,9,7,9,5 | $\psi(M(2,0,0))$ |
| 1,4,7,10,10,7,10,9,8 | $\psi(M(\omega,0,0))$ |
| 1,4,7,10,10,<br>7,10,9,9,7,9,5 | $\psi(M(1,0,0,0))$<br>$\psi(M(1;0)^{M(1;0)^2})$ |
| 1,4,7,10,10,7,<br>10,9,9,9,7,9,5 | $\psi(M(1,0,0,0,0))$ |
| 1,4,7,10,10,7,<br>10,9,9,9,9,7,9,5 | $\psi(M(1,0,0,0,0,0))$ |
| 1,4,7,10,10,7,10,9,10 | $\psi(M(1@\omega))$<br>$\psi(M(1;0)^{M(1;0)^\omega})$ |
| 1,4,7,10,10,7,10,9,11 | $\psi(M(1@\Omega))$ |
| 1,4,7,10,10,7,10,9,11,5 | $\psi(\psi_{M(1@(1,0))}(0))$ |
| 1,4,7,10,10,7,<br>10,9,11,7,9,5 | $\psi(M(1@(1,0)))$<br>$\psi(M(1;0)^{M(1;0)^{M(1;0)}})$ |
| 1,4,7,10,10,7,10,9,12 | $\psi(\psi_{\Omega_{M(1;0)+1}}(0))$ |
| 1,4,7,10,10,7,10,9,12,12 | $\psi(\psi_{\Omega_{M(1;0)+1}}(1))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,10,7,10,9,12,15 | $\psi(\Omega_{M(1;0)+1})$<br>$\psi(\psi_{M_{M(1;0)+1}}(0))$ |
| 1,4,7,10,10,7,10,9,13 | $\psi(\Omega_{M(1;0)+\omega})$<br>$\psi(\psi_{M_{M(1;0)+1}}(\omega))$ |
| 1,4,7,10,10,7,<br>10,9,13,17,20,14 | $\psi(\psi_{I_{M(1;0)+1}}(0))$<br>$\psi(\psi_{\psi_{M_{M(1;0)+1}}(M_{M(1;0)+1})}(0))$ |
| 1,4,7,10,10,7,10,<br>9,13,17,20,17,20,14 | $\psi(I_{M(1;0)+1})$<br>$\psi(M_{M(1;0)+1})$ |
| 1,4,7,10,10,7,10,<br>9,13,17,21,21 | $\psi(M_{M(1;0)+\omega})$ |
| 1,4,7,10,10,7,10,9,<br>13,17,21,21,17,20,14 | $\psi(\psi_{M(1,M(1;0)+1)}(0))$ |
| 1,4,7,10,10,7,10,9,13,17,<br>21,21,17,20,17,20,14 | $\psi(M(1,M(1;0)+1))$<br>$\psi(\psi_{M(1;1)}(0))$ |
| 1,4,7,10,10,7,10,9,13,<br>17,21,21,17,21,17,20,14 | $\psi(\psi_{M(2,M(1;0)+1)}(0))$<br>$\psi(\psi_{\psi_{M(1;1)}(M(1;1))}(0))$ |
| 1,4,7,10,10,7,10,9,13,17,<br>21,21,17,21,17,20,17,20,14 | $\psi(M(2,M(1;0)+1))$<br>$\psi(M(1;1))$ |
| 1,4,7,10,10,7,10,9,<br>13,17,21,21,17,21,18 | $\psi(M(\omega,M(1;0)+1))$<br>$\psi(M(1;1)^{\omega})$ |
| 1,4,7,10,10,7,10,9,13,<br>17,21,21,17,21,20,14 | $\psi(\psi_{M(1,0,M(1;0)+1)})$<br>$\psi(\psi_{\psi_{M(1;1)}(M(1;1)^{M(1;1)})}(0))$ |
| 1,4,7,10,10,7,10,9,13,17,<br>21,21,17,21,20,17,20,14 | $\psi(M(1,0,M(1;0)+1))$<br>$\psi(M(1;1)^{M(1;1)})$ |
| 1,4,7,10,10,7,10,9,13,17,<br>21,21,17,21,20,20,17,20,14 | $\psi(M(1,0,0,M(1;0)+1))$<br>$\psi(M(1;1)^{M(1;1)^2})$ |
| 1,4,7,10,10,7,10,9,13,<br>17,21,21,17,21,20,21 | $\psi(M(1@\omega,M(1;0)+1@0))$<br>$\psi(M(1;1)^{M(1;1)^{\omega}})$ |
| 1,4,7,10,10,7,10,9,13,<br>17,21,21,17,21,20,23,14 | $\psi(M(1@(1,0),M(1;0)+1@0))$<br>$\psi(M(1;1)^{M(1;1)^{M(1;1)}})$ |
| 1,4,7,10,10,7,10,9,13,<br>17,21,21,17,21,20,24 | $\psi(\psi_{\Omega_{M(1;1)+1}}(0))$ |
| 1,4,7,10,10,7,10,9,13,<br>17,21,21,17,21,20,24,28 | $\psi(\Omega_{M(1;1)+1})$<br>$\psi(\psi_{M_{M(1;1)+1}}(0))$ |
| 1,4,7,10,10,7,10,9,<br>13,17,21,21,17,21,<br>20,25,30,34,26 | $\psi(\psi_{I_{M(1;1)+1}}(0))$ |
| 1,4,7,10,10,7,10,<br>9,13,17,21,21,17,21,<br>20,25,30,34,30,34,26 | $\psi(I_{M(1;1)+1})$<br>$\psi(M_{M(1;1)+1})$ |
| 1,4,7,10,10,7,10,9,13,<br>17,21,21,17,21,20,<br>25,30,35,35 | $\psi(M_{M(1;1)+\omega})$ |
| 1,4,7,10,10,7,10,<br>9,13,17,21,21,17,21,20,<br>25,30,35,35,30,34,30,34,26 | $\psi(M(1,M(1;1)+1))$<br>$\psi(\psi_{M(1;2)}(0))$ |
| 1,4,7,10,10,7,10,9,<br>13,17,21,21,17,21,20,25,<br>30,35,35,30,35,30,34,26 | $\psi(\psi_{M(2,M(1;1)+1)}(0))$<br>$\psi(\psi_{\psi_{M(1;2)}(M(1;2))}(0))$ |
| 1,4,7,10,10,7,10,9,<br>13,17,21,21,17,21,20,25,30,35,<br>35,30,35,30,34,30,34,26 | $\psi(M(2,M(1;1)+1))$<br>$\psi(M(1;2))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,10,7,10,9,<br>13,17,21,21,17,21,20,25,<br>30,35,35,30,35,34,30,34,26 | $\psi(M(1,0,M(1;1)+1))$<br>$\psi(M(1;2)^{M(1;2)})$ |
| 1,4,7,10,10,7,10,9,<br>13,17,21,21,17,21,20,25,30,<br>35,35,30,35,34,39,30,34,26 | $\psi(M(1@(1,0),M(1;1)+1@0))$<br>$\psi(M(1;2)^{M(1;2)^{M(1;2)}})$ |
| 1,4,7,10,10,7,10,9,<br>13,17,21,21,17,21,20,<br>25,30,35,35,30,35,34,39 | $\psi(\psi_{\Omega_{M(1;2)+1}}(0))$ |
| 1,4,7,10,10,7,10,9,<br>13,17,21,21,17,21,20,25,<br>30,35,35,30,35,34,40,46,52,<br>52,46,52,46,51,46,51,41 | $\psi(M(2,M(1;2)+1))$<br>$\psi(M(1;3))$ |
| 1,4,7,10,10,7,10,10 | $\psi(M(1;\omega))$ |
| 1,4,7,10,10,7,10,10,6 | $\psi(M(1;\omega)\cdot\Omega)$ |
| 1,4,7,10,10,7,10,10,6,9 | $\psi(\psi_{\Omega_{M(1;\omega)+1}}(0))$ |
| 1,4,7,10,10,7,10,10,<br>6,10,14,17,14,17,11 | $\psi(I_{M(1;\omega)+1})$<br>$\psi(M_{M(1;\omega)+1})$ |
| 1,4,7,10,10,7,10,10,6,<br>10,14,18,18,14,17,14,17,11 | $\psi(M(1,M(1;\omega)+1))$<br>$\psi(\psi_{M(1;\omega+1)}(0))$ |
| 1,4,7,10,10,7,10,<br>10,6,10,14,18,18,14,<br>18,14,17,14,17,11 | $\psi(M(2,M(1;\omega)+1))$<br>$\psi(M(1;\omega+1))$ |
| 1,4,7,10,10,7,10,10,6,<br>10,14,18,18,14,18,18 | $\psi(M(1;\omega\cdot 2))$ |
| 1,4,7,10,10,7,10,10,7 | $\psi(M(1;\omega^2))$ |
| 1,4,7,10,10,7,<br>10,10,7,9,5 | $\psi(\psi_{M(1;1,0)}(0))$ |
| 1,4,7,10,10,7,<br>10,10,7,9,7,9,5 | $\psi(M(1;1,0))$<br>$\psi(\psi_{M(2;0)}(0))$<br>$\psi(2\ 1-2-2\ 1-2-2)$ |
| 1,4,7,10,10,7,10,10,7,10 | $\psi(M(1;1,\omega))$<br>$\psi(\psi_{M(2;0)}(\omega))$ |
| 1,4,7,10,10,7,<br>10,10,7,10,7,9,5 | $\psi(\psi_{M(1;2,0)}(0))$<br>$\psi(\psi_{\psi_{M(2;0)}(M(2;0))}(0))$ |
| 1,4,7,10,10,7,10,<br>10,7,10,7,9,7,9,5 | $\psi(M(1;2,0))$<br>$\psi(M(2;0))$ |
| 1,4,7,10,10,7,<br>10,10,7,10,7,10 | $\psi(M(1;2,\omega))$<br>$\psi(M(2;0)\cdot\omega)$ |
| 1,4,7,10,10,7,<br>10,10,7,10,8 | $\psi(M(1;\omega,0))$<br>$\psi(M(2;0)^{\omega})$ |
| 1,4,7,10,10,7,<br>10,10,7,10,9,5 | $\psi(\psi_{M(1;1,0,0)}(0))$ |
| 1,4,7,10,10,7,10,<br>10,7,10,9,7,9,5 | $\psi(M(1;1,0,0))$<br>$\psi(M(2;0)^{M(2;0)})$ |
| 1,4,7,10,10,7,<br>10,10,7,10,9,10 | $\psi(M(1;1@\omega))$<br>$\psi(M(2;0)^{M(2;0)^{\omega}})$ |
| 1,4,7,10,10,7,<br>10,10,7,10,9,12 | $\psi(\psi_{\Omega_{M(2;0)+1}}(0))$ |
| 1,4,7,10,10,7,10,10,7,<br>10,9,13,17,20,17,20,14 | $\psi(I_{M(2;0)+1})$<br>$\psi(M_{M(2;0)+1})$ |
| 1,4,7,10,10,7,10,<br>10,7,10,9,13,17,21,21 | $\psi(M_{M(2;0)+\omega})$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,10,7,10,<br>10,7,10,9,13,17,21,21,<br>17,21,17,20,17,20,14 | $\psi(M(2, M(2;0)+1))$<br>$\psi(M(2;1))$ |
| 1,4,7,10,10,7,<br>10,10,7,10,10 | $\psi(M(2;\omega))$ |
| 1,4,7,10,10,8 | $\psi(M(\omega;0))$<br>$\psi(\psi_N(\omega))$<br>$\psi((2-2\ 1-)^\omega\ 2-2)$ |
| 1,4,7,10,10,8,7 | $\psi(M(\omega;\omega))$ |
| 1,4,7,10,10,8,<br>7,10,7,9,7,9,5 | $\psi(M(\omega+1;0))$<br>$\psi(\psi_N(\omega+1))$ |
| 1,4,7,10,10,8,7,10,10 | $\psi(M(\omega+1,\omega))$ |
| 1,4,7,10,10,8,7,<br>10,10,7,10,7,9,7,9,5 | $\psi(M(\omega+2;0))$ |
| 1,4,7,10,10,8,7,10,10,8 | $\psi(M(\omega\cdot 2;0))$ |
| 1,4,7,10,10,8,8 | $\psi(M(\omega^2;0))$ |
| 1,4,7,10,10,9 | $\psi(M(\Omega;0))$<br>$\psi(\psi_N(\Omega))$ |
| 1,4,7,10,10,9,5 | $\psi(\psi_{M(1,0;0)}(0))$<br>$\psi(\psi_{\psi_{\psi_N(N)}(0)}(0))$ |
| 1,4,7,10,10,9,7,9,5 | $\psi(M(1,0;0))$<br>$\psi(\psi_{\psi_N(N)}(0))$ |
| 1,4,7,10,10,9,7,9,7 | $\psi(M(1,0;0)\cdot\omega)$ |
| 1,4,7,10,10,<br>9,7,9,7,9,5 | $\psi(M(1,0;0)^2)$ |
| 1,4,7,10,10,9,7,9,12 | $\psi(\psi_{\Omega_{M(1,0;0)+1}}(0))$ |
| 1,4,7,10,10,9,7,<br>9,13,17,20,17,20,14 | $\psi(I_{M(1,0;0)+1})$<br>$\psi(M_{M(1,0;0)+1})$ |
| 1,4,7,10,10,9,7,9,13,<br>17,21,21,17,20,17,20,14 | $\psi(M(1, M(1,0;0)+1))$ |
| 1,4,7,10,10,9,7,9,<br>13,17,21,21,17,21,<br>17,20,17,20,14 | $\psi(M(2, M(1,0;0)+1))$<br>$\psi(M(1,0;1))$ |
| 1,4,7,10,10,9,7,10 | $\psi(M(1,0;\omega))$ |
| 1,4,7,10,10,9,<br>7,10,7,9,7,9,5 | $\psi(M(1,0;1,0))$<br>$\psi(\psi_{M(1,1;0)}(0))$ |
| 1,4,7,10,10,9,7,<br>10,7,10,7,9,7,9,5 | $\psi(M(1,0;2,0))$<br>$\psi(M(1,1;0))$ |
| 1,4,7,10,10,9,7,10,8 | $\psi(M(1,0;\omega,0))$ |
| 1,4,7,10,10,9,<br>7,10,9,7,9,5 | $\psi(M(1,0;1,0,0))$<br>$\psi(M(1,1;0)^{M(1,1;0)})$ |
| 1,4,7,10,10,9,7,10,9,12 | $\psi(\psi_{\Omega_{M(1,1;0)+1}}(0))$ |
| 1,4,7,10,10,9,7,10,<br>9,13,17,20,17,20,14 | $\psi(M_{M(1,1;0)+1})$ |
| 1,4,7,10,10,9,7,10,9,13,<br>17,21,21,17,20,17,20,14 | $\psi(M(1, M(1,1;0)+1))$ |
| 1,4,7,10,10,9,<br>7,10,9,13,17,21,21,17,<br>21,17,20,17,20,14 | $\psi(M(2, M(1,1;0)+1))$<br>$\psi(M(1,1;1))$ |
| 1,4,7,10,10,9,7,10,10 | $\psi(M(1,1;\omega))$ |
| 1,4,7,10,10,9,7,10,10,8 | $\psi(M(1,\omega;0))$<br>$\psi(\psi_N(\psi_{\psi_N(N)}(0)+\omega))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,10,9,7,10,10,9 | $\psi(M(1,\Omega;0))$<br>$\psi(\psi_N(\psi_{\psi_N(N)}(0)+\Omega))$ |
| 1,4,7,10,10,9,<br>7,10,10,9,5 | $\psi(\psi_{M(2,0;0)}(0))$<br>$\psi(\psi_{\psi_{\psi_N(N)}(1)}(0))$ |
| 1,4,7,10,10,9,<br>7,10,10,9,7,9,5 | $\psi(M(2,0;0))$<br>$\psi(\psi_{\psi_N(N)}(1))$ |
| 1,4,7,10,10,9,7,<br>10,10,9,7,10,10,9,7,9,5 | $\psi(M(3,0;0))$<br>$\psi(\psi_{\psi_N(N)}(2))$ |
| 1,4,7,10,10,9,8 | $\psi(M(\omega,0;0))$<br>$\psi(\psi_{\psi_N(N)}(\omega))$ |
| 1,4,7,10,10,9,9 | $\psi(M(\Omega,0;0))$<br>$\psi(\psi_{\psi_N(N)}(\Omega))$ |
| 1,4,7,10,10,9,9,5 | $\psi(\psi_{M(1,0,0;0)}(0))$<br>$\psi(\psi_N(N))$<br>$\psi(N)$<br>$\psi(2-2-2)$ |
| 1,4,7,10,10,9,9,7,9,5 | $\psi(M(1,0,0;0))$<br>$\psi(N+\psi_{\psi_N(N\cdot 2)}(0))$ |
| 1,4,7,10,10,9,<br>9,7,10,7,9,5 | $\psi(\psi_{M(1,0,1;0)}(0))$<br>$\psi(N+\psi_{\psi_N(N\cdot 2)}(N))$ |
| 1,4,7,10,10,9,<br>9,7,10,7,9,7,9,5 | $\psi(M(1,0,1;0))$<br>$\psi(N\cdot 2)$ |
| 1,4,7,10,10,9,9,7,10,10 | $\psi(M(1,0,1;\omega))$ |
| 1,4,7,10,10,9,<br>9,7,10,10,8 | $\psi(M(1,0,\omega;0))$<br>$\psi(N\cdot\omega)$ |
| 1,4,7,10,10,9,<br>9,7,10,10,9 | $\psi(M(1,0,\Omega;0))$<br>$\psi(N\cdot\Omega)$ |
| 1,4,7,10,10,9,9,<br>7,10,10,9,7,9,5 | $\psi(M(1,1,0;0))$<br>$\psi(N\cdot\psi_{\psi_N(N^2)}(0))$ |
| 1,4,7,10,10,9,<br>9,7,10,10,9,8 | $\psi(M(1,\omega,0;0))$<br>$\psi(N\cdot\psi_{\psi_N(N^2)}(\omega))$ |
| 1,4,7,10,10,9,<br>9,7,10,10,9,9,5 | $\psi(\psi_{M(2,0,0;0)}(0))$<br>$\psi(N^2)$ |
| 1,4,7,10,10,9,9,7,10,<br>10,9,9,7,10,10,9,9,5 | $\psi(\psi_{M(3,0,0;0)}(0))$<br>$\psi(N^3)$ |
| 1,4,7,10,10,9,9,8 | $\psi(M(\omega,0,0;0))$<br>$\psi(N^\omega)$ |
| 1,4,7,10,10,9,9,9,5 | $\psi(\psi_{M(1,0,0,0;0)}(0))$<br>$\psi(N^N)$ |
| 1,4,7,10,10,9,10 | $\psi(M(1@\omega;0))$<br>$\psi(N^{N^\omega})$ |
| 1,4,7,10,10,9,11,7,9,5 | $\psi(M(1@(1,0);0))$<br>$\psi(N^{N^N})$ |
| 1,4,7,10,10,9,12 | $\psi(\psi_{\Omega_{N+1}}(0))$<br>$\psi((1-)^{1,0}\ \text{aft}\ \ 2-2-2)$ |
| 1,4,7,10,10,9,12,3 | $\psi(\psi_{\Omega_{N+1}}(0)+1)$ |
| 1,4,7,10,10,9,12,12 | $\psi(\psi_{\Omega_{N+1}}(1))$ |
| 1,4,7,10,10,9,12,14 | $\psi(\psi_{\Omega_{N+1}}(\Omega))$ |
| 1,4,7,10,10,9,12,14,5 | $\psi(\psi_{\Omega_{N+1}}(N))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,10,9,12,15 | $\psi(\Omega_{N+1})$<br>$\psi(\psi_{M_{N+1}}(0))$<br>$\psi(2 \text{ aft } 2-2-2)$ |
| 1,4,7,10,10,9,13,17,20,14 | $\psi(\psi_{I_{N+1}}(0))$<br>$\psi(\alpha \to \psi_{M_{N+1}}(\alpha) \text{ FP})$<br>$\psi((1-)^{1,0}\ 2 \text{ aft } 2-2-2)$ |
| 1,4,7,10,10,9,13,17,<br>21,21,17,20,17,20,14 | $\psi(M(1,N+1))$<br>$\psi(\psi_{M(1;N+1)}(0))$ |
| 1,4,7,10,10,9,13,17,21,<br>21,17,21,17,20,17,20,14 | $\psi(M(2,N+1))$<br>$\psi(M(1;N+1))$<br>$\psi(\psi_{N_2}(0))$ |
| 1,4,7,10,10,9,13,<br>17,21,21,17,21,21 | $\psi(M(2;N+\omega))$ |
| 1,4,7,10,10,9,<br>13,17,21,21,18 | $\psi(M(\omega;N+1))$<br>$\psi(\psi_{N_2}(\omega))$ |
| 1,4,7,10,10,9,<br>13,17,21,21,19 | $\psi(M(\Omega;N+1))$<br>$\psi(\psi_{N_2}(\Omega))$ |
| 1,4,7,10,10,9,<br>13,17,21,21,19,5 | $\psi(M(N;1))$<br>$\psi(\psi_{N_2}(N))$ |
| 1,4,7,10,10,9,<br>13,17,21,21,20,14 | $\psi(\psi_{M(1,0;N+1)}(0))$<br>$\psi(\alpha \to \psi_{N_2}(\alpha) \text{ FP})$ |
| 1,4,7,10,10,9,13,<br>17,21,21,20,17,20,14 | $\psi(M(1,0;N+1))$<br>$\psi(\alpha \to \psi_{N_2}(\alpha) \text{ AP})$<br>$\psi(\psi_{\psi_{N_2}(N_2)}(0))$ |
| 1,4,7,10,10,9,13,17,21,21,<br>20,17,21,21,20,17,20,14 | $\psi(M(2,0;N+1))$<br>$\psi(\text{2nd } \alpha \to \psi_{N_2}(\alpha) \text{ AP})$<br>$\psi(\psi_{\psi_{N_2}(N_2)}(1))$ |
| 1,4,7,10,10,9,<br>13,17,21,21,20,18 | $\psi(M(\omega,0;N+1))$<br>$\psi(1-2\ 1-2-2\ 1-2-2 \text{ aft } 2-2-2)$<br>$\psi(\psi_{\psi_{N_2}(N_2)}(\omega))$ |
| 1,4,7,10,10,9,<br>13,17,21,21,20,19 | $\psi(M(\Omega,0;N+1))$<br>$\psi(\psi_{\psi_{N_2}(N_2)}(\Omega))$ |
| 1,4,7,10,10,9,13,<br>17,21,21,20,20,14 | $\psi(\psi_{M(1,0,0;N+1)}(0))$<br>$\psi(N_2)$<br>$\psi(\text{2nd } 2-2-2)$ |
| 1,4,7,10,10,9,13,17,21,<br>21,20,20,17,20,14 | $\psi(M(1,0,0;N+1))$<br>$\psi(N_2+\psi_{\psi_{N_2}(N_2\cdot 2)}(0))$ |
| 1,4,7,10,10,9,13,17,21,21,<br>20,20,17,21,17,20,17,20,14 | $\psi(M(1,0,1;N+1))$<br>$\psi(N_2\cdot 2)$ |
| 1,4,7,10,10,9,13,17,21,<br>21,20,20,17,21,21,18 | $\psi(M(1,0,\omega;N+1))$<br>$\psi(N_2\cdot\omega)$ |
| 1,4,7,10,10,9,13,17,21,<br>21,20,20,17,21,21,19 | $\psi(M(1,0,\Omega;N+1))$<br>$\psi(N_2\cdot\Omega)$ |
| 1,4,7,10,10,9,13,17,21,<br>21,20,20,17,21,21,19,5 | $\psi(M(1,0,N;1))$<br>$\psi(N_2\cdot N)$ |
| 1,4,7,10,10,9,13,17,21,<br>21,20,20,17,21,21,20,14 | $\psi(\psi_{M(1,1,0;N+1)}(0))$<br>$\psi(\alpha \to N_2\cdot\psi_{N_2}(\alpha) \text{ FP})$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,10,9,13,17,<br>21,21,20,20,17,21,<br>21,20,17,20,14 | $\psi(M(1,1,0;N+1))$<br>$\psi(\alpha \to N_2 \cdot \psi_{N_2}(\alpha)$ AP) |
| 1,4,7,10,10,9,13,<br>17,21,21,20,20,17,21,21,<br>20,17,21,21,20,17,20,14 | $\psi(M(1,2,0;N+1))$<br>$\psi(2\text{nd } \alpha \to N_2 \cdot \psi_{N_2}(\alpha)$ AP) |
| 1,4,7,10,10,9,13,17,21,<br>21,20,20,17,21,21,20,18 | $\psi(M(1,\omega,0;N+1))$<br>$\psi(N_2 \cdot \psi_{\psi_{N_2}(N_2^2)}(\omega))$ |
| 1,4,7,10,10,9,13,17,21,21,<br>20,20,17,21,21,20,19,5 | $\psi(M(1,N,0;1))$<br>$\psi(N_2 \cdot \psi_{\psi_{N_2}(N_2^2)}(N))$ |
| 1,4,7,10,10,9,13,17,21,21,<br>20,20,17,21,21,20,20,14 | $\psi(\psi_{M(2,0,0;N+1)}(0))$<br>$\psi(N_2^2)$ |
| 1,4,7,10,10,9,13,<br>17,21,21,20,20,18 | $\psi(M(\omega,0,0;N+1))$<br>$\psi(N_2^\omega)$ |
| 1,4,7,10,10,9,13,<br>17,21,21,20,20,19 | $\psi(M(\Omega,0,0;N+1))$<br>$\psi(N_2^\Omega)$ |
| 1 ,4,7,10,10,9,13,<br>17,21,21,20,20,20,14 | $\psi(\psi_{M(1,0,0,0;N+1)}(0))$<br>$\psi(N_2^{N_2})$ |
| 1,4,7,10,10,9,13,<br>17,21,21,20,21 | $\psi(M(1@\omega;N+1))$<br>$\psi(N_2^{N_2^\omega})$ |
| 1,4,7,10,10,9,13,<br>17,21,21,20,25 | $\psi(\psi_{\Omega_{N_2+1}}(0))$<br>$\psi((1-)^{1,0}$ aft 2nd $2-2-2)$ |
| 1,4,7,10,10,9,13,17,21,<br>21,20,25,30,35,35,34,40 | $\psi(\psi_{\Omega_{N_3+1}}(0))$<br>$\psi((1-)^{1,0}$ aft 3rd $2-2-2)$ |
| 1,4,7,10,10,10 | $\psi(N_\omega)$<br>$\psi(1-2-2-2)$<br>SNO |
| 1,4,7,10,10,10,7 | $\psi(N_{\omega^2})$<br>$\psi(1-1-2-2-2)$ |
| 1,4,7,10,10,10,7,9 | $\psi(N_\Omega)$ |
| 1,4,7,10,10,10,<br>7,9,4,7,10,10,9,13,17,21,<br>21,21,17,21,21,17,19,7,9,5 | $\psi(N_N)$ |
| 1,4,7,10,10,10,7,9,5 | $\psi(\psi_{N(1,0)}(0))$<br>$\psi((1-)^{1,0}\ 2-2-2)$ |
| 1,4,7,10,10,<br>10,7,9,7,9,5 | $\psi(N(1,0))$<br>$\psi(2\ 1-2-2-2)$ |
| 1,4,7,10,10,10,7,10 | $\psi(N(1,\omega))$<br>$\psi(1-2\ 1-2-2-2)$ |
| 1,4,7,10,10,10,<br>7,10,7,9,7,9,5 | $\psi(N(2,0))$<br>$\psi(2\ 1-2\ 1-2-2-2)$ |
| 1,4,7,10,10,10,7,10,7,10 | $\psi(N(2,\omega))$<br>$\psi(1-2\ 1-2\ 1-2-2-2)$ |
| 1,4,7,10,10,10,7,10,8 | $\psi(N(\omega,0))$<br>$\psi((2\ 1-)^\omega\ 2-2-2)$ |
| 1,4,7,10,10,10,<br>7,10,9,7,9,5 | $\psi(N(1,0,0))$<br>$\psi((2\ 1-)^{1,1}\ 2-2-2)$ |
| 1,4,7,10,10,10,7,10,10 | $\psi(1-2-2\ 1-2-2-2)$ |
| 1,4,7,10,10,10,<br>7,10,10,7,10,10 | $\psi(1-2-2\ 1-2-2\ 1-2-2-2)$ |
| 1,4,7,10,10,10,7,10,10,8 | $\psi((2-2\ 1-)^\omega\ 2-2-2)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,10,10,7,10,10,10 | $\psi(1-2-2-2\ 1-2-2-2)$ |
| 1,4,7,10,10,10,8 | $\psi((2-2-2\ 1-)^{\omega}\ 2-2-2)$ |
| 1,4,7,10,10,10,9 | $\psi((2-2-2\ 1-)^{(2)}\ 2-2-2)$ |
| 1,4,7,10,10,10,9,5 | $\psi((2-2-2\ 1-)^{1,0}\ 2-2-2)$ |
| 1,4,7,10,10,10,9,9,9,5 | $\psi(2-2-2-2)$ |
| 1,4,7,10,10,10,10 | $\psi(1-2-2-2-2)$ |
| 1,4,7,10,10,10,<br>10,7,9,7,9,5 | $\psi(2\ 1-2-2-2-2)$ |
| 1,4,7,10,10,10,<br>10,7,10,7,9,7,9,5 | $\psi(2-2\ 1-2-2-2-2)$ |
| 1,4,7,10,10,10,10,7,10,<br>10,7,10,7,9,7,9,5 | $\psi(2-2-2\ 1-2-2-2-2)$ |
| 1,4,7,10,10,10,<br>10,7,10,10,10,10 | $\psi(1-2-2-2-2\ 1-2-2-2-2)$ |
| 1,4,7,10,10,10,10,8 | $\psi((2-2-2-2\ 1-)^{\omega}\ 2-2-2-2)$ |
| 1,4,7,10,10,10,10,10 | $\psi(1-2-2-2-2-2)$ |
| 1,4,7,10,10,10,10,10,10 | $\psi(1-2-2-2-2-2-2)$ |
| 1,4,7,10,11 | $\psi((2-)^{\omega})$ |
| 1,4,7,10,12 | $\psi((2-)^{(2)})$ |
| 1,4,7,10,12,5 | $\psi((2-)^{1,0})$<br>$\psi(\psi_K(0))$ |
| 1,4,7,10,12,6,<br>10,14,18,21,11 | $\psi(2\text{nd}\ (2-)^{1,0})$ |
| 1,4,7,10,12,7 | $\psi(1-(2-)^{1,0})$ |
| 1,4,7,10,12,7,10 | $\psi(1-2\ 1-(2-)^{1,0})$ |
| 1,4,7,10,12,7,10,10 | $\psi(1-2-2\ 1-(2-)^{1,0})$ |
| 1,4,7,10,12,7,10,11 | $\psi((2-)^{\omega}\ 1-(2-)^{1,0})$ |
| 1,4,7,10,12,7,10,12 | $\psi((2-)^{(2)}\ 1-(2-)^{1,0})$ |
| 1,4,7,10,12,7,10,12,5 | $\psi((2-)^{1,0}\ 1-(2-)^{1,0})$ |
| 1,4,7,10,12,7,10,12,7 | $\psi(1-(2-)^{1,0}\ 1-(2-)^{1,0})$ |
| 1,4,7,10,12,7,<br>10,12,7,10,12,5 | $\psi(((2-)^{1,0}\ 1-)^{2}\ (2-)^{1,0})$ |
| 1,4,7,10,12,8 | $\psi(((2-)^{1,0}\ 1-)^{\omega}\ (2-)^{1,0})$ |
| 1,4,7,10,12,9 | $\psi(((2-)^{1,0}\ 1-)^{(2)}\ (2-)^{1,0})$ |
| 1,4,7,10,12,9,5 | $\psi(((2-)^{1,0}\ 1-)^{1,0}\ (2-)^{1,0})$ |
| 1,4,7,10,12,9,7,9,5 | $\psi(((2-)^{1,0}\ 1-)^{1,1}\ (2-)^{1,0})$ |
| 1,4,7,10,12,9,7,10,12,5 | $\psi(((2-)^{1,0}\ 1-)^{1,2}\ (2-)^{1,0})$ |
| 1,4,7,10,12,9,7,10,12,8 | $\psi(((2-)^{1,0}\ 1-)^{1,\omega}\ (2-)^{1,0})$ |
| 1,4,7,10,12,9,<br>7,10,12,9,5 | $\psi(((2-)^{1,0}\ 1-)^{2,0}\ (2-)^{1,0})$ |
| 1,4,7,10,12,9,8 | $\psi(((2-)^{1,0}\ 1-)^{\omega,0}\ (2-)^{1,0})$ |
| 1,4,7,10,12,9,11,5 | $\psi(((2-)^{1,0}\ 1-)^{1@(1,0)}\ (2-)^{1,0})$ |
| 1,4,7,10,12,10 | $\psi(1-(2-)^{1,1})$ |
| 1,4,7,10,12,10,11 | $\psi((2-)^{1,\omega})$<br>$\psi(\psi_K(\omega))$ |
| 1,4,7,10,12,10,12 | $\psi((2-)^{1,\Omega})$<br>$\psi(\psi_K(\Omega))$ |
| 1,4,7,10,12,10,<br>12,4,7,10,12,10,12 | $\psi((2-)^{1,(2-)^{1,\Omega}})$<br>$\psi(\psi_K(\psi_K(\Omega)))$ |
| 1,4,7,10,12,10,12,5 | $\psi((2-)^{2,0})$<br>$\psi(\psi_K(K))$<br>$\psi(K)$<br>$\psi(\Pi_3)$ |

| $0-\mathrm{Y}$ sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,12,10,12,<br>5,4,7,10,12,10,12,5 | $\psi(K+\psi_K(K))$ |
| 1,4,7,10,12,10,12,6,9 | $\psi(K+\psi_{\Omega_{\psi_K(K)+1}}(0))$ |
| 1,4,7,10,12,10,<br>12,6,10,14,18 | $\psi(K+I_{\psi_K(K)+\omega})$<br>$\psi(K+(1-2\ 1-2 \text{ aft } (2-)^{2,0}))$ |
| 1,4,7,10,12,10,<br>12,6,10,14,18,18 | $\psi(K+M_{\psi_K(K)+\omega})$<br>$\psi(K+(1-2-2 \text{ aft } (2-)^{2,0}))$ |
| 1,4,7,10,12,10,<br>12,6,10,14,18,19 | $\psi(K+((2-)^{\omega} \text{ aft } (2-)^{2,0}))$ |
| 1,4,7,10,12,10,<br>12,6,10,14,18,21,11 | $\psi(K+((2-)^{1,0} \text{ aft } (2-)^{2,0}))$<br>$\psi(K+\psi_K(K+1))$ |
| 1,4,7,10,12,10,12,6,10,<br>14,18,21,18,21,11 | $\psi(K\cdot 2)$<br>$\psi(\text{2nd } (2-)^{2,0})$ |
| 1,4,7,10,12,10,12,7 | $\psi(K\cdot\omega)$<br>$\psi(1-(2-)^{2,0})$ |
| 1,4,7,10,12,10,12,10 | $\psi(1-(2-)^{2,1})$ |
| 1,4,7,10,12,<br>10,12,10,12,5 | $\psi(K^2)$<br>$\psi((2-)^{3,0})$ |
| 1,4,7,10,12,11 | $\psi(K^{\omega})$<br>$\psi((2-)^{\omega,0})$ |
| 1,4,7,10,12,12,5 | $\psi(K^K)$<br>$\psi((2-)^{1,0,0})$ |
| 1,4,7,10,12,14,15 | $\psi(K^{K^{\omega}})$<br>$\psi((2-)^{1@\omega})$ |
| 1,4,7,10,12,14,16,5 | $\psi(K^{K^K})$<br>$\psi((2-)^{1@(1,0)})$ |
| 1,4,7,10,12,15 | $\psi(\psi_{\Omega_{K+1}}(0))$<br>$\psi(\varepsilon_{K+1})$<br>$\psi((1-)^{1,0} \text{ aft } 3)$<br>$RO$ |
| 1,4,7,10,12,15,18 | $\psi(\Omega_{K+1})$<br>$\psi(2 \text{ aft } 3)$ |
| 1,4,7,10,12,16 | $\psi(\Omega_{K+\omega})$<br>$\psi(1-2 \text{ aft } 3)$ |
| 1,4,7,10,12,16,20,24 | $\psi(I_{K+\omega})$<br>$\psi(1-2\ 1-2 \text{ aft } 3)$ |
| 1,4,7,10,12,16,20,24,24 | $\psi(M_{K+\omega})$<br>$\psi(1-2-2 \text{ aft } 3)$ |
| 1,4,7,10,12,16,20,24,24,24 | $\psi(N_{K+\omega})$<br>$\psi(1-2-2-2 \text{ aft } 3)$ |
| 1,4,7,10,12,16,20,24,25 | $\psi((2-)^{\omega} \text{ aft } 3)$ |
| 1,4,7,10,12,16,20,24,26 | $\psi((2-)^{(2)} \text{ aft } 3)$ |
| 1,4,7,10,12,16,20,<br>24,26,4,7,10,12,5 | $\psi((2-)^{(2-)^{1,0}} \text{ aft } 3)$ |
| 1,4,7,10,12,16,20,24,<br>26,4,7,10,12,10,12,5 | $\psi((2-)^{(2-)^{2,0}} \text{ aft } 3)$ |
| 1,4,7,10,12,16,20,<br>24,26,4,7,10,12,15 | $\psi((2-)^{\psi_K(\psi_{\Omega_{K+1}}(0))} \text{ aft } 3)$ |
| 1,4,7,10,12,16,20,24,26,5 | $\psi((2-)^{(3)} \text{ aft } 3)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,12,16,20,24,27,17 | $\psi((2-)^{1,0}$ aft 3) <br> $\psi(\psi_{K_2}(0))$ |
| 1,4,7,10,12,16, <br> 20,24,27,24,27,17 | $\psi(K_2)$ <br> $\psi(2\text{nd } 3)$ |
| 1,4,7,10,12,16,20,24,27,31 | $\psi(\psi_{\Omega_{K_2+1}}(0))$ <br> $\psi((1-)^{1,0}$ aft 2nd 3) |
| 1,4,7,10,12,16,20, <br> 24,27,32,37,42,46,28 | $\psi(\psi_{K_3}(0))$ <br> $\psi((2-)^{1,0}$ aft 2nd 3) |
| 1,4,7,10,13 | $\psi(K_\omega)$ <br> $\psi(1-3)$ |
| 1,4,7,10,13,3 | $\psi(K_\omega+1)$ |
| 1,4,7,10,13,4,7,10,13 | $\psi(K_\omega \cdot 2)$ |
| 1,4,7,10,13,6,10, <br> 14,18,21,18,21,11 | $\psi(K_{\omega+1})$ <br> $\psi(3 \text{ aft } 1-3)$ |
| 1,4,7,10,13,6,10,14,18,22 | $\psi(K_{\omega\cdot 2})$ |
| 1,4,7,10,13,7 | $\psi(K_{\omega^2})$ <br> $\psi(1-1-3)$ |
| 1,4,7,10,13,7,9 | $\psi(K_\Omega)$ |
| 1,4,7,10,13,7,9,4,7,10, <br> 12,16,20,24,28,20,22,5 | $\psi(K_K)$ |
| 1,4,7,10,13,7,9,5 | $\psi(\psi_{K(1,0)}(0))$ <br> $\psi((1-)^{1,0}\ 3)$ |
| 1,4,7,10,13,7,10,8 | $\psi((2\ 1-)^{\omega}\ 3)$ |
| 1,4,7,10,13,7,10,10 | $\psi(1-2-2\ 1-3)$ |
| 1,4,7,10,13,7,10,11 | $\psi((2-)^{\omega}\ 1-3)$ |
| 1,4,7,10,13,7,10,12 | $\psi((2-)^{\Omega}\ 1-3)$ |
| 1,4,7,10,13,7,10,12,5 | $\psi((2-)^{1,0}\ 1-3)$ |
| 1,4,7,10,13,7,10,13 | $\psi(1-3\ 1-3)$ |
| 1,4,7,10,13, <br> 7,10,13,7,10,13 | $\psi(1-3\ 1-3\ 1-3)$ |
| 1,4,7,10,13,8 | $\psi((3\ 1-)^{\omega}\ 3)$ |
| 1,4,7,10,13,9 | $\psi((3\ 1-)^{\Omega}\ 3)$ |
| 1,4,7,10,13,9,5 | $\psi((3\ 1-)^{1,0}\ 3)$ |
| 1,4,7,10,13,10 | $\psi(1-2-3)$ |
| 1,4,7,10,13, <br> 10,4,7,10,13,10 | $\psi(1-2-3\ 1-2-3)$ |
| 1,4,7,10,13,10,10 | $\psi(1-2-2-3)$ |
| 1,4,7,10,13,10,11 | $\psi((2-)^{\omega}\ 3)$ |
| 1,4,7,10,13,10,12,5 | $\psi((2-)^{1,0}\ 3)$ |
| 1,4,7,10,13,10,13 | $\psi(1-3\ 2-3)$ |
| 1,4,7,10,13,10, <br> 13,7,10,13,10,13 | $\psi(1-3\ 2-3\ 1-3\ 2-3)$ |
| 1,4,7,10,13,10,13,10,13 | $\psi(1-3\ 2-3\ 2-3)$ |
| 1,4,7,10,13,11 | $\psi((3\ 2-)^{\omega}\ 3)$ |
| 1,4,7,10,13,12,5 | $\psi((3\ 2-)^{1,0}\ 3)$ |
| 1,4,7,10,13,13 | $\psi(1-3-3)$ |
| 1,4,7,10,13,13,7,10,13 | $\psi(1-3\ 1-3-3)$ |
| 1,4,7,10,13,13,7,10,13,13 | $\psi(1-3-3\ 1-3-3)$ |
| 1,4,7,10,13,13,10 | $\psi(1-2-3-3)$ |
| 1,4,7,10,13,13,10,13 | $\psi(1-3\ 2-3-3)$ |
| 1,4,7,10,13,13,10,13,13 | $\psi(1-3-3\ 2-3-3)$ |
| 1,4,7,10,13,13,13 | $\psi(1-3-3-3)$ |
| 1,4,7,10,13,14 | $\psi((3-)^{\omega})$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,7,10,13,15,5 | $\psi((3-)^{1,0})$ |
| 1,4,7,10,13,15,13,15,5 | $\psi((3-)^{2,0})$<br>$\psi(\Pi_4)$<br>$\psi(\kappa)$ |
| 1,4,7,10,13,16 | $\psi(\kappa_\omega)$<br>$\psi(1-4)$ |
| 1,4,7,10,13,16,7 | $\psi(\kappa_{\omega^2})$<br>$\psi(1-1-4)$ |
| 1,4,7,10,13,16,7,10 | $\psi(1-2\ 1-4)$ |
| 1,4,7,10,13,16,7,10,10 | $\psi(1-2-2\ 1-4)$ |
| 1,4,7,10,13,16,7,10,12,5 | $\psi((2-)^{1,0}\ 1-4)$ |
| 1,4,7,10,13,16,7,10,13 | $\psi(1-3\ 1-4)$ |
| 1,4,7,10,13,16,7,10,13,10 | $\psi(1-2-3\ 1-4)$ |
| 1,4,7,10,13,16,<br>7,10,13,10,13 | $\psi(1-3\ 2-3\ 1-4)$ |
| 1,4,7,10,13,16,7,10,13,13 | $\psi(1-3-3\ 1-4)$ |
| 1,4,7,10,13,16,7,10,13,14 | $\psi((3-)^\omega\ 1-4)$ |
| 1,4,7,10,13,16,7,10,13,16 | $\psi(1-4\ 1-4)$ |
| 1,4,7,10,13,16,8 | $\psi((4\ 1-)^\omega\ 4)$ |
| 1,4,7,10,13,16,10 | $\psi(1-2-4)$ |
| 1,4,7,10,13,16,10,13 | $\psi(1-3\ 2-4)$ |
| 1,4,7,10,13,16,10,13,16 | $\psi(1-4\ 2-4)$ |
| 1,4,7,10,13,16,11 | $\psi((4\ 2-)^\omega\ 4)$ |
| 1,4,7,10,13,16,13 | $\psi(1-3-4)$ |
| 1,4,7,10,13,16,13,14 | $\psi((3-)^\omega\ 4)$ |
| 1,4,7,10,13,16,13,16 | $\psi(1-4\ 3-4)$ |
| 1,4,7,10,13,16,14 | $\psi((4\ 3-)^\omega\ 4)$ |
| 1,4,7,10,13,16,16 | $\psi(1-4-4)$ |
| 1,4,7,10,13,16,16,13,16 | $\psi(1-4\ 3-4-4)$ |
| 1,4,7,10,13,16,16,13,16,16 | $\psi(1-4-4\ 3-4-4)$ |
| 1,4,7,10,13,16,16,16 | $\psi(1-4-4-4)$ |
| 1,4,7,10,13,16,17 | $\psi((4-)^\omega)$ |
| 1,4,7,10,13,16,18 | $\psi((4-)^{(2)})$ |
| 1,4,7,10,13,16,19 | $\psi(1-5)$ |
| 1,4,7,10,13,16,19,10 | $\psi(1-2-5)$ |
| 1,4,7,10,13,16,19,13 | $\psi(1-3-5)$ |
| 1,4,7,10,13,16,19,16 | $\psi(1-4-5)$ |
| 1,4,7,10,13,16,19,19 | $\psi(1-5-5)$ |
| 1,4,7,10,13,16,19,20 | $\psi((5-)^\omega)$ |
| 1,4,7,10,13,16,19,22 | $\psi(1-6)$ |
| 1,4,7,10,13,16,19,22,25 | $\psi(1-7)$ |
| 1,4,7,10,13,16,19,22,25,28 | $\psi(1-8)$ |
| 1,4,7,10,13,16,<br>19,22,25,28,31 | $\psi(1-9)$ |
| 1,4,7,10,13,16,<br>19,22,25,28,31,34 | $\psi(1-10)$ |
| 1,4,8 | $\psi(\lambda\alpha.(\alpha+1)-\Pi_0)$<br>$\psi(\psi_a(a_2))$ |
| 1,4,8,5 | $\psi(\psi_a(\psi_{a_2}(a_2)+1))$ |
| 1,4,8,6 | $\psi(\psi_a(\psi_{a_2}(a_2)+\psi_a(0)))$ |
| 1,4,8,6,4,8 | $\psi(\psi_a(\psi_{a_2}(a_2)+\psi_a(\psi_{a_2}(a_2))))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,8,6,9 | $\psi((1-)^{1,0}\text{ aft }\omega)$<br>$\psi(\psi_{\Omega_{\lambda\alpha.(\alpha+1)-\Pi_0+1}}(0))$<br>$\psi(\psi_a(\psi_{a_2}(a_2)+a))$ |
| 1,4,8,6,9,12 | $\psi(2\text{ aft }\omega)$<br>$\psi(\psi_a(\psi_{a_2}(a_2)+\Omega_{a+1}))$ |
| 1,4,8,6,10,14,17,14,17,11 | $\psi(2-2\text{ aft }\omega)$<br>$\psi(\psi_a(\psi_{a_2}(a_2)+\Omega_{a+1}^{\Omega_{a+1}}))$ |
| 1,4,8,6,10,14,18,18,18 | $\psi(1-2-2-2\text{ aft }\omega)$<br>$\psi(\psi_a(\psi_{a_2}(a_2)+\Omega_{a+1}^{\Omega_{a+1}^2}\cdot\omega))$ |
| 1,4,8,6,10,14,18,19 | $\psi((2-)^{\omega}\text{ aft }\omega)$<br>$\psi(\psi_a(\psi_{a_2}(a_2)+\Omega_{a+1}^{\Omega_{a+1}^{\omega}}))$ |
| 1,4,7,10,6,10,14,18,22 | $\psi(1-3\text{ aft }\omega)$<br>$\psi(\psi_a(\psi_{a_2}(a_2)+\Omega_{a+1}^{\Omega_{a+1}^{\Omega_{a+1}}}\cdot\omega))$ |
| 1,4,8,6,10,14,18,22,26 | $\psi(1-4\text{ aft }\omega)$<br>$\psi(\psi_a(\psi_{a_2}(a_2)+\Omega_{a+1}^{\Omega_{a+1}^{\Omega_{a+1}^{\Omega_{a+1}}}}\cdot\omega))$ |
| 1,4,8,6,10,15 | $\psi(2\text{nd }\omega)$<br>$\psi(\psi_a(\psi_{a_2}(a_2)\cdot 2))$ |
| 1,4,8,6,10,15,13,18,24 | $\psi(3\text{rd }\omega)$<br>$\psi(\psi_a(\psi_{a_2}(a_2)\cdot 3))$ |
| 1,4,8,7 | $\psi(1-\omega)$<br>$\psi(\psi_a(\psi_{a_2}(a_2+1)))$ |
| 1,4,8,7,7 | $\psi(1-1-\omega)$<br>$\psi(\psi_a(\psi_{a_2}(a_2+2))))$ |
| 1,4,8,7,9 | $\psi((1-)^{\Omega}\ \omega)$<br>$\psi(\psi_a(\psi_{a_2}(a_2+\psi_a(0))))$ |
| 1,4,8,7,9,5 | $\psi((1-)^{1,0}\ \omega)$<br>$\psi(\psi_a(\psi_{a_2}(a_2+a)))$ |
| 1,4,8,7,10 | $\psi(1-2\ 1-\omega)$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}+1}\cdot\Omega_{a+1}\cdot\omega))$<br>$\psi(\psi_a(\psi_{a_2}(a_2+\psi_{a_2}(0)+1)))$ |
| 1,4,8,7,10,10 | $\psi(1-2-2\ 1-\omega)$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}+1}\cdot\Omega_{a+1}^{\Omega_{a+1}}\cdot\omega))$ |
| 1,4,8,7,10,11 | $\psi((2-)^{\omega}\ 1-\omega)$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}+1}\cdot\Omega_{a+1}^{\Omega_{a+1}^{\omega}}))$ |
| 1,4,8,7,10,13 | $\psi(1-3\ 1-\omega)$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}+1}\cdot\Omega_{a+1}^{\Omega_{a+1}^{\Omega_{a+1}}}\cdot\omega))$ |
| 1,4,8,7,10,13,16 | $\psi(1-4\ 1-\omega)$ |
| 1,4,8,7,11 | $\psi(\omega\ 1-\omega)$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}+1}^2))$<br>$\psi(\psi_a(\psi_{a_2}(a_2+\psi_{a_2}(a_2))))$ |
| 1,4,8,7,11,7 | $\psi(1-\omega\ 1-\omega)$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}+1}^2\cdot\omega))$ |
| 1,4,8,7,11,7,10 | $\psi(1-2\ 1-\omega\ 1-\omega)$ |
| 1,4,8,7,11,7,10,13 | $\psi(1-3\ 1-\omega\ 1-\omega)$ |
| 1,4,8,7,11,7,11 | $\psi(\omega\ 1-\omega\ 1-\omega)$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}+1}^3))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,8,7,11,8 | $\psi((\omega\ 1-)^{\omega}\ \omega)$<br>$\psi(\psi_a(\varepsilon^{\omega}_{\Omega_{a+1}+1}))$ |
| 1,4,8,7,11,9 | $\psi((\omega\ 1-)^{(2)}\ \omega)$<br>$\psi(\psi_a(\varepsilon^{\psi_a(0)}_{\Omega_{a+1}+1}))$ |
| 1,4,8,7,11,9,5 | $\psi((\omega\ 1-)^{1,0}\ \omega)$<br>$\psi(\psi_a(\varepsilon^{a}_{\Omega_{a+1}+1}))$<br>$\psi(\psi_a(\psi_{a_2}(a_2+\psi_{a_2}(a_2+a))))$ |
| 1,4,8,7,11,10 | $\psi(1-2-\omega)$<br>$\psi(\psi_a(\varepsilon^{\Omega_{a+1}}_{\Omega_{a+1}+1}\cdot\omega))$ |
| 1,4,8,7,11,10,13 | $\psi(1-3\ 2-\omega)$<br>$\psi(\psi_a(\varepsilon^{\Omega^2_{a+1}}_{\Omega_{a+1}+1}\cdot\omega))$ |
| 1,4,8,7,11,10,14 | $\psi(\omega\ 2-\omega)$<br>$\psi(\psi_a(\varepsilon^{\varepsilon_{\Omega_{a+1}+1}}_{\Omega_{a+1}+1}))$ |
| 1,4,8,7,11,10,14,13,17 | $\psi(\omega\ 3-\omega)$ |
| 1,4,8,8 | $\psi(\omega-\omega)$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\alpha+1)-\Pi_0)$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}+2}))$<br>$\psi(\psi_a(\psi_{a_2}(a_2\cdot 2)))$ |
| 1,4,8,8,8 | $\psi(\omega-\omega-\omega)$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}+3}))$ |
| 1,4,8,9 | $\psi((\omega-)^{\omega})$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}+\omega}))$ |
| 1,4,8,10 | $\psi((\omega-)^{(2)})$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}+\psi_a(0)}))$ |
| 1,4,8,10,5 | $\psi((\omega-)^{1,0})$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}+a}))$ |
| 1,4,8,10,8 | $\psi((\omega-)^{1,1})$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}+a+1}))$ |
| 1,4,8,10,8,10,5 | $\psi((\omega-)^{2,0})$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}+a\cdot 2}))$ |
| 1,4,8,10,13 | $\psi((1-)^{1,0}\ \text{aft}\ (\omega+1))$ |
| 1,4,8,11 | $\psi(1-(\omega+1))$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}\cdot 2}\cdot\omega))$ |
| 1,4,8,11,7 | $\psi(1-1-(\omega+1))$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}\cdot 2}\cdot\omega^2))$ |
| 1,4,8,11,7,10 | $\psi(1-2\ 1-(\omega+1))$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}\cdot 2}\cdot\Omega_{a+1}\cdot\omega))$ |
| 1,4,8,11,7,10,13 | $\psi(1-3\ 1-(\omega+1))$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}\cdot 2}\cdot\Omega^{\Omega_{a+1}}_{a+1}\cdot\omega))$ |
| 1,4,8,11,7,11 | $\psi(\omega\ 1-(\omega+1))$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}\cdot 2}\cdot\varepsilon_{\Omega_{a+1}+1}))$ |
| 1,4,8,11,7,11,11 | $\psi(\omega-\omega\ 1-(\omega+1))$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}\cdot 2}\cdot\varepsilon_{\Omega_{a+1}+2}))$ |
| 1,4,8,11,7,11,14 | $\psi(1-(\omega+1)\ 1-(\omega+1))$<br>$\psi(\psi_a(\varepsilon^{2}_{\Omega_{a+1}\cdot 2}))$ |
| 1,4,8,11,7,11,14,10 | $\psi(1-2-(\omega+1))$<br>$\psi(\psi_a(\varepsilon^{\Omega_{a+1}}_{\Omega_{a+1}\cdot 2}\cdot\omega))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,8,11,7,11,14,10,14 | $\psi(\omega\ 2-(\omega+1))$<br>$\psi(\psi_a(\varepsilon^{\varepsilon_{\Omega_{a+1}+1}+1}_{\Omega_{a+1}\cdot 2}))$ |
| 1,4,8,11,7,11,14,10,14,17 | $\psi(1-(\omega+1)\ 2-(\omega+1))$<br>$\psi(\psi_a(\varepsilon^{\varepsilon_{\Omega_{a+1}\cdot 2}}_{\Omega_{a+1}\cdot 2}\cdot\omega))$ |
| 1,4,8,11,7,11,<br>14,10,14,17,13 | $\psi(1-3-(\omega+1))$ |
| 1,4,8,11,7,11,<br>14,10,14,17,13,17,20 | $\psi(1-(\omega+1)\ 3-(\omega+1))$ |
| 1,4,8,11,8 | $\psi(\omega-(\omega+1))$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}\cdot 2+1}))$ |
| 1,4,8,11,8,8 | $\psi(\omega-\omega-(\omega+1))$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}\cdot 2+2}))$ |
| 1,4,8,11,8,9 | $\psi((\omega-)^{\omega}\ (\omega+1))$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}\cdot 2+\omega}))$ |
| 1,4,8,11,8,10,5 | $\psi((\omega-)^{1,0}\ (\omega+1))$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}\cdot 2+a}))$ |
| 1,4,8,11,8,11 | $\psi(1-(\omega+1)\ \omega-(\omega+1))$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}\cdot 3}))$ |
| 1,4,8,11,9 | $\psi(((\omega+1)\ \omega-)^{\omega}\ (\omega+1))$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}\cdot\omega}))$ |
| 1,4,8,11,10,5 | $\psi(((\omega+1)\ \omega-)^{1,0}\ (\omega+1))$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}\cdot a}))$ |
| 1,4,8,11,11 | $\psi(1-(\omega+1)-(\omega+1))$<br>$\psi(\psi_a(\varepsilon_{\Omega^2_{a+1}}\cdot\omega))$ |
| 1,4,8,11,12 | $\psi(((\omega+1)-)^{\omega})$<br>$\psi(\psi_a(\varepsilon_{\Omega^{\omega}_{a+1}}))$ |
| 1,4,8,11,13,5 | $\psi(((\omega+1)-)^{1,0})$<br>$\psi(\psi_a(\varepsilon_{\Omega^{a}_{a+1}}))$ |
| 1,4,8,11,14 | $\psi(1-(\omega+2))$<br>$\psi(\psi_a(\varepsilon_{\Omega^{\Omega_{a+1}}_{a+1}}))$ |
| 1,4,8,11,14,8 | $\psi(\omega-(\omega+2))$<br>$\psi(\psi_a(\varepsilon_{\Omega^{\Omega_{a+1}}_{a+1}+1}))$ |
| 1,4,8,11,14,8,11 | $\psi(1-(\omega+1)\ \ \omega-(\omega+2))$<br>$\psi(\psi_a(\varepsilon_{\Omega^{\Omega_{a+1}}_{a+1}+\Omega_{a+1}}\cdot\omega))$ |
| 1,4,8,11,14,8,11,14 | $\psi(1-(\omega+2)\ \ \omega-(\omega+2))$<br>$\psi(\psi_a(\varepsilon_{\Omega^{\Omega_{a+1}}_{a+1}\cdot 2}\cdot\omega))$ |
| 1,4,8,11,14,10,5 | $\psi(((\omega+2)\ \ \omega-)^{1,0}\ \ (\omega+2))$<br>$\psi(\psi_a(\varepsilon_{\Omega^{\Omega_{a+1}}_{a+1}\cdot a}))$ |
| 1,4,8,11,14,11 | $\psi(1-(\omega+1)-(\omega+2))$<br>$\psi(\psi_a(\varepsilon_{\Omega^{\Omega_{a+1}+1}_{a+1}}\cdot\omega))$ |
| 1,4,8,11,14,11,14 | $\psi(1-(\omega+2)\ \ (\omega+1)-(\omega+2))$<br>$\psi(\psi_a(\varepsilon_{\Omega^{\Omega_{a+1}\cdot 2}_{a+1}}\cdot\omega))$ |
| 1,4,8,11,14,13,5 | $\psi(((\omega+2)\ \ (\omega+1)-)^{1,0}\ (\omega+2))$<br>$\psi(\psi_a(\varepsilon_{\Omega^{\Omega_{a+1}\cdot a}_{a+1}}))$ |
| 1,4,8,11,14,14 | $\psi(1-(\omega+2)-(\omega+2))$<br>$\psi(\psi_a(\varepsilon_{\Omega^{\Omega^2_{a+1}}_{a+1}}\cdot\omega))$ |

| $0-\mathrm{Y}$ sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,8,11,14,14,11,14 | $\psi(1-(\omega+2)\ \ (\omega+1)-(\omega+2)-(\omega+2))$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}^{\Omega_{a+1}^2+\Omega_{a+1}}}\cdot\omega))$ |
| 1,4,8,11,14,14,14 | $\psi(1-(\omega+2)-(\omega+2)-(\omega+2))$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}^{\Omega_{a+1}^3}}\cdot\omega))$ |
| 1,4,8,11,14,15 | $\psi(((\omega+2)-)^{\omega})$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}^{\Omega_{a+1}^{\omega}}}))$ |
| 1,4,8,11,14,17 | $\psi(1-(\omega+3))$<br>$\psi(\psi_a(\varepsilon_{\Omega_{a+1}^{\Omega_{a+1}^{\Omega_{a+1}}}}\cdot\omega))$ |
| 1,4,8,11,14,17,20 | $\psi(1-(\omega+4))$ |
| 1,4,8,11,15 | $\psi(\Pi_{\omega\cdot 2})$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\Omega_{a+1}+1}+1}))$<br>$\psi(\psi_a(\psi_{a_2}(a_2\cdot\psi_{a_2}(a_2))))$<br>$\psi(\lambda\alpha.(\alpha+2)-\Pi_0)$ |
| 1,4,8,11,15,8 | $\psi(\omega-(\omega\cdot 2))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\Omega_{a+1}+1}+1}))$ |
| 1,4,8,11,15,8,11 | $\psi(1-(\omega+1)\ \ \omega-(\omega\cdot 2))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\Omega_{a+1}+1}+\Omega_{a+1}}\cdot\omega))$ |
| 1,4,8,11,15,8,11,15 | $\psi((\omega\cdot 2)\ \ \omega-(\omega\cdot 2))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\Omega_{a+1}+1}\cdot 2}))$ |
| 1,4,8,11,15,11 | $\psi(1-(\omega+1)-(\omega\cdot 2))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\Omega_{a+1}+1}\cdot\Omega_{a+1}}\cdot\omega))$ |
| 1,4,8,11,15,11,15 | $\psi((\omega\cdot 2)\ \ (\omega+1)-(\omega\cdot 2))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon^2_{\Omega_{a+1}+1}}))$ |
| 1,4,8,11,15,14 | $\psi(1-(\omega+2)-(\omega\cdot 2))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\Omega_{a+1}+1}^{\Omega_{a+1}}}\cdot\omega))$ |
| 1,4,8,11,15,14,18 | $\psi((\omega\cdot 2)\ \ (\omega+2)-(\omega\cdot 2))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\Omega_{a+1}+1}^{\varepsilon_{\Omega_{a+1}+1}}}))$ |
| 1,4,8,11,15,15 | $\psi((\omega\cdot 2)-(\omega\cdot 2))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\Omega_{a+1}+2}}))$ |
| 1,4,8,11,15,16 | $\psi(((\omega\cdot 2)-)^{\omega})$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\Omega_{a+1}+\omega}}))$ |
| 1,4,8,11,15,18 | $\psi(1-(\omega\cdot 2+1))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\Omega_{a+1}\cdot 2}}\cdot\omega))$ |
| 1,4,8,11,15,18,15 | $\psi((\omega\cdot 2)-(\omega\cdot 2+1))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\Omega_{a+1}\cdot 2+1}}))$ |
| 1,4,8,11,15,18,15,18 | $\psi(1-(\omega\cdot 2+1)\ \ (\omega\cdot 2)-(\omega\cdot 2+1))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\Omega_{a+1}\cdot 3}}))$ |
| 1,4,8,11,15,18,18 | $\psi(1-(\omega\cdot 2+1)-(\omega\cdot 2+1))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\Omega^2_{a+1}}}\cdot\omega))$ |
| 1,4,8,11,15,18,20,5 | $\psi(((\omega\cdot 2+1)-)^{1,0})$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\Omega^a_{a+1}}}))$ |
| 1,4,8,11,15,18,21 | $\psi(1-(\omega\cdot 2+2))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\Omega_{a+1}^{\Omega_{a+1}}}}\cdot\omega))$ |
| 1,4,8,11,15,18,21,24 | $\psi(1-(\omega\cdot 2+3))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,8,11,15,18,22 | $\psi(\Pi_{\omega\cdot 3})$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\varepsilon_{\Omega_{a+1}+1}}}))$<br>$\psi(\psi_a(\psi_{a_2}(a_2\cdot\psi_{a_2}(a_2\cdot\psi_{a_2}(a_2)))))$<br>$\psi(\lambda\alpha.(\alpha+3)-\Pi_0)$ |
| 1,4,8,11,15,18,22,25,29 | $\psi(\Pi_{\omega\cdot 4})$<br>$\psi(\lambda\alpha.(\alpha+4)-\Pi_0)$ |
| 1,4,8,12 | $\psi(\Pi_{\omega^2})$<br>$\psi(\psi_a(\zeta_{\Omega_{a+1}+1}))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2)))$<br>$\psi(\lambda\alpha.(\alpha+\omega)-\Pi_0)$ |
| 1,4,8,12,8 | $\psi(\omega-(\omega^2))$<br>$\psi(\psi_a(\varepsilon_{\zeta_{\Omega_{a+1}+1}+1}))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2+a_2)))$ |
| 1,4,8,12,8,11 | $\psi(1-(\omega+1)\ \ \omega-(\omega^2))$<br>$\psi(\psi_a(\varepsilon_{\zeta_{\Omega_{a+1}+1}+\Omega_{a+1}}\cdot\omega))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2+a_2\cdot\psi_{a_2}(0)+1)))$ |
| 1,4,8,12,8,11,14 | $\psi(1-(\omega+2)\ \ \omega-(\omega^2))$ |
| 1,4,8,12,8,11,15 | $\psi((\omega\cdot 2)\ \ \omega-(\omega^2))$<br>$\psi(\psi_a(\varepsilon_{\zeta_{\Omega_{a+1}+1}+\varepsilon_{\Omega_{a+1}}}))$ |
| 1,4,8,12,8,11,15,19 | $\psi((\omega^2)\ \ \omega-(\omega^2))$<br>$\psi(\psi_a(\varepsilon_{\zeta_{\Omega_{a+1}+1}\cdot 2}))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2+a_2\cdot\psi_{a_2}(a_2^2))))$ |
| 1,4,8,12,8,11,15,19,9 | $\psi(((\omega^2)\ \ \omega-)^{\omega}\ \ (\omega^2))$ |
| 1,4,8,12,8,11,15,19,11 | $\psi(1-(\omega+1)-(\omega^2))$<br>$\psi(\psi_a(\varepsilon_{\zeta_{\Omega_{a+1}+1}\cdot\Omega_{a+1}}\cdot\omega))$ |
| 1,4,8,12,8,11,15,19,11,14 | $\psi(1-(\omega+2)\ \ (\omega+1)-(\omega^2))$<br>$\psi(\psi_a(\varepsilon_{\zeta_{\Omega_{a+1}+1}\cdot\Omega_{a+1}^{\Omega_{a+1}}}\cdot\omega))$ |
| 1,4,8,12,8,11,15,19,11,15 | $\psi((\omega\cdot 2)\ \ (\omega+1)-(\omega^2))$ |
| 1,4,8,12,8,11,<br>15,19,11,15,19 | $\psi((\omega^2)\ \ (\omega+1)-(\omega^2))$<br>$\psi(\psi_a(\varepsilon_{\zeta^2_{\Omega_{a+1}+1}}))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2+a_2\cdot\psi_{a_2}(a_2^2+\psi_{a_2}(a_2^2)))))$ |
| 1,4,8,12,8,11,15,19,14 | $\psi(1-(\omega+2)-(\omega^2))$<br>$\psi(\psi_a(\varepsilon_{\zeta_{\Omega_{a+1}+1}^{\Omega_{a+1}}}\cdot\omega))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2+a_2\cdot\psi_{a_2}(a_2^2+\psi_{a_2}(a_2^2+\psi_{a_2}(0)))+1)))$ |
| 1,4,8,12,8,11,15,19,14,17 | $\psi(1-(\omega+3)\ \ (\omega+2)-(\omega^2))$ |
| 1,4,8,12,8,11,<br>15,19,14,18,22 | $\psi((\omega^2)\ \ (\omega+2)-(\omega^2))$<br>$\psi(\psi_a(\varepsilon_{\zeta_{\Omega_{a+1}+1}^{\zeta_{\Omega_{a+1}+1}}}))$ |
| 1,4,8,12,8,11,<br>15,19,14,18,22,15 | $\psi(((\omega^2)\ \ (\omega+2)-)^{\omega}\ (\omega^2))$ |
| 1,4,8,12,8,11,<br>15,19,14,18,22,17 | $\psi(1-(\omega+3)-(\omega^2))$ |
| 1,4,8,12,8,11,15,19,15 | $\psi((\omega\cdot 2)-(\omega^2))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\zeta_{\Omega_{a+1}+1}+1}}))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2+a_2\cdot\psi_{a_2}(a_2^2+a_2))))$ |
| 1,4,8,12,8,11,15,19,15,18 | $\psi(1-(\omega\cdot 2+1)\ \ (\omega\cdot 2)-(\omega^2))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\zeta_{\Omega_{a+1}+1}+\Omega_{a+1}\cdot 2}}\cdot\omega))$ |

| $0-\mathrm{Y}$ sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,8,12,8,11,<br>15,19,15,18,22,26 | $\psi((\omega^2)\ \ (\omega\cdot 2)-(\omega^2))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\zeta_{\Omega_{a+1}+1}\cdot 2}}))$ |
| 1,4,8,12,8,11,15,<br>19,15,18,22,26,21 | $\psi(1-(\omega\cdot 2+1)-(\omega^2))$ |
| 1,4,8,12,8,11,15,<br>19,15,18,22,26,22 | $\psi((\omega\cdot 3)-(\omega^2))$<br>$\psi(\psi_a(\varepsilon_{\varepsilon_{\varepsilon_{\zeta_{\Omega_{a+1}+1}+1}+1}}))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2+a_2\cdot\psi_{a_2}(a_2^2+a_2\cdot\psi_{a_2}(a_2^2+a_2)))))$ |
| 1,4,8,12,8,12 | $\psi((\omega^2)-(\omega^2))$<br>$\psi(\psi_a(\zeta_{\Omega_{a+1}+2}))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot 2)))$ |
| 1,4,8,12,9 | $\psi(((\omega^2)-)^\omega)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\omega)))$ |
| 1,4,8,12,10,5 | $\psi(((\omega^2)-)^{1,0})$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot a)))$ |
| 1,4,8,12,11 | $\psi(1-(\omega^2+1))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(0)+1)))$ |
| 1,4,8,12,11,8,12 | $\psi((\omega^2)-(\omega^2+1))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(0)+a_2^2)))$ |
| 1,4,8,12,11,8,12,11 | $\psi(1-(\omega^2+1)\ \ (\omega^2)-(\omega^2+1))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(0)\cdot 2+1)))$ |
| 1,4,8,12,11,10,5 | $\psi(((\omega^2+1)\ \ (\omega^2)-)^{1,0}\ (\omega^2+1))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(a))))$ |
| 1,4,8,12,11,11 | $\psi(1-(\omega^2+1)-(\omega^2+1))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(\psi_{a_2}(0)))))$ |
| 1,4,8,12,11,12 | $\psi(((\omega^2+1)-)^\omega)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(\psi_{a_2}(\psi_{a_2}(1))))))$ |
| 1,4,8,12,11,14 | $\psi(1-(\omega^2+2))$ |
| 1,4,8,12,11,15 | $\psi(\Pi_{\omega^2+\omega})$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(a_2))))$ |
| 1,4,8,12,11,15,8,12 | $\psi((\omega^2)-(\omega^2+\omega))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(a_2)+a_2^2)))$ |
| 1,4,8,12,11,15,8,12,11 | $\psi(1-(\omega^2+1)\ \ (\omega^2)-(\omega^2+\omega))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(a_2)+a_2^2\cdot\psi_{a_2}(0)+1)))$ |
| 1,4,8,12,11,15,8,12,11,15 | $\psi((\omega^2+\omega)\ \ (\omega^2)-(\omega^2+\omega))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(a_2)\cdot 2)))$ |
| 1,4,8,12,11,15,11 | $\psi(1-(\omega^2+1)-(\omega^2+\omega))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(a_2+\psi_{a_2}(0))+1)))$ |
| 1,4,8,12,11,15,11,12 | $\psi(((\omega^2+1)-)^\omega\ \ (\omega^2+\omega))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(a_2+\psi_{a_2}(1)))))$ |
| 1,4,8,12,11,15,11,15 | $\psi((\omega^2+\omega)\ \ (\omega^2+1)-(\omega^2+\omega))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(a_2+\psi_{a_2}(a_2)))))$ |
| 1,4,8,12,11,15,14 | $\psi(1-(\omega^2+2)-(\omega^2+\omega))$ |
| 1,4,8,12,11,15,15 | $\psi((\omega^2+\omega)-(\omega^2+\omega))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(a_2\cdot 2))))$ |
| 1,4,8,12,11,15,18 | $\psi((1-(\omega^2+\omega+1)))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(a_2\cdot\psi_{a_2}(0)))))$ |
| 1,4,8,12,11,15,18,22 | $\psi(\Pi_{\omega^2+\omega\cdot 2})$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(a_2\cdot\psi_{a_2}(a_2)))))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,8,12,11,15,19 | $\psi(\Pi_{\omega^2\cdot 2})$<br>$\psi(\psi_a(\psi_{a_2}(a_2^2\cdot\psi_{a_2}(a_2^2))))$ |
| 1,4,8,12,12 | $\psi(\Pi_{\omega^3})$<br>$\psi(\psi_a(\psi_{a_2}(a_2^3)))$<br>$\psi(\lambda\alpha.(\alpha+\omega^2)-\Pi_0)$ |
| 1,4,8,12,12,12 | $\psi(\Pi_{\omega^4})$<br>$\psi(\psi_a(\psi_{a_2}(a_2^4)))$<br>$\psi(\lambda\alpha.(\alpha+\omega^3)-\Pi_0)$ |
| 1,4,8,12,13 | $\psi(\Pi_{\omega^\omega})$<br>$\psi(\psi_a(\psi_{a_2}(a_2^\omega)))$ |
| 1,4,8,12,13,15 | $\psi(\Pi_{\psi(0)})$<br>$\psi(\psi_a(\psi_{a_2}(a_2^{\psi(0)})))$ |
| 1,4,8,12,13,16,20 | $\psi(\Pi_{\psi(\Pi_\omega)})$<br>$\psi(\psi_a(\psi_{a_2}(a_2^{\psi(\psi_a(a_2))})))$ |
| 1,4,8,12,14 | $\psi(\Pi_\Omega)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^{\psi_a(0)})))$ |
| 1,4,8,12,14,4 | $\psi(\Pi_{\Omega_\omega})$ |
| 1,4,8,12,14,4,8 | $\psi(\Pi_{\Pi_\omega})$ |
| 1,4,8,12,14,4,8,12,14 | $\psi(\Pi_{\Pi_\Omega})$ |
| 1,4,8,12,14,5 | $\psi(\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^a)))$ |
| 1,4,8,12,14,5,3 | $\psi(\lambda\alpha.(\alpha\cdot 2)-\Pi_0+1)$<br>$\psi(\psi_a(a_2^a)+1)$ |
| 1,4,8,12,14,7 | $\psi(1-\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^a+1)))$ |
| 1,4,8,12,14,8 | $\psi(\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^a+a_2)))$ |
| 1,4,8,12,14,8,12 | $\psi((\omega^2)-(\lambda\alpha.(\alpha\cdot 2)-\Pi_0))$<br>$\psi(\psi_a(\psi_{a_2}(a_2^a+a_2^2)))$ |
| 1,4,8,12,14,8,12,14 | $\psi(\lambda\alpha.(\alpha+\Omega)-\Pi_0-\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^a+a_2^{\psi_a(0)})))$ |
| 1,4,8,12,14,8,<br>12,14,4,8,12,14,5 | $\psi(\lambda\alpha.(\alpha+\lambda\alpha.(\alpha\cdot 2)-\Pi_0)-\Pi_0-\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^a+a_2^{\psi_a(\psi_{a_2}(a_2^a))})))$ |
| 1,4,8,12,14,8,12,14,5 | $\psi(\lambda\alpha.(\alpha\cdot 2)-\Pi_0-\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^a\cdot 2)))$ |
| 1,4,8,12,14,8,<br>12,14,8,12,14,5 | $\psi((\lambda\alpha.(\alpha\cdot 2)-\Pi_0-)^3)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^a\cdot 3)))$ |
| 1,4,8,12,14,9 | $\psi((\lambda\alpha.(\alpha\cdot 2)-\Pi_0-)^\omega)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^a\cdot\omega)))$ |
| 1,4,8,12,14,10,5 | $\psi((\lambda\alpha.(\alpha\cdot 2)-\Pi_0-)^{1,0})$<br>$\psi(\psi_a(\psi_{a_2}(a_2^a\cdot a)))$ |
| 1,4,8,12,14,11 | $\psi(1-\lambda\alpha.(\alpha\cdot 2)-\Pi_1)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^a\cdot\psi_{a_2}(0)+1)))$ |
| 1,4,8,12,14,11,8,12,14,5 | $\psi(\lambda\alpha.(\alpha\cdot 2)-\Pi_0-\lambda\alpha.(\alpha\cdot 2)-\Pi_1)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^a\cdot\psi_{a_2}(0)+a_2^a)))$ |
| 1,4,8,12,14,11,8,12,14,11 | $\psi(1-\lambda\alpha.(\alpha\cdot 2)-\Pi_1\quad\lambda\alpha.(\alpha\cdot 2)$<br>$-\Pi_0-\lambda\alpha.(\alpha\cdot 2)-\Pi_1)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^a\cdot\psi_{a_2}(0)\cdot 2+1)))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,8,12,14,11,11 | $\psi(1-\lambda\alpha.(\alpha\cdot 2)-\Pi_1-\lambda\alpha.(\alpha\cdot 2)-\Pi_1)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^a\cdot\psi_{a_2}(\psi_{a_2}(0))+1)))$ |
| 1,4,8,12,14,11,14 | $\psi(1-\lambda\alpha.(\alpha\cdot 2)-\Pi_2)$ |
| 1,4,8,12,14,11,15 | $\psi(\lambda\alpha.(\alpha\cdot 2+1)-\Pi_0)$<br>$\psi(\psi_a(a_2^a\cdot\psi_{a_2}(a_2)))$ |
| 1,4,8,12,14,11,15,19,21 | $\psi(\lambda\alpha.(\alpha\cdot 2+\Omega)-\Pi_0)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^a\cdot\psi_{a_2}(a_2^{\psi_a(0)}))))$ |
| 1,4,8,12,14,11,15,19,21,5 | $\psi(\lambda\alpha.(\alpha\cdot 3)-\Pi_0)$<br>$\psi(\psi_a(\psi_{a_2}(a_2^a\cdot\psi_{a_2}(a_2^a))))$ |
| 1,4,8,12,14,11,<br>15,19,21,15,19,21,5 | $\psi(\lambda\alpha.(\alpha\cdot 3)-\Pi_0-\lambda\alpha.(\alpha\cdot 3)-\Pi_0)$<br>$\psi(\psi_a(a_2^a\cdot\psi_{a_2}(a_2^a\cdot 2)))$ |
| 1,4,8,12,14,11,<br>15,19,21,18,22 | $\psi(\lambda\alpha.(\alpha\cdot 3+1)-\Pi_0)$<br>$\psi(\psi_a(a_2^a\cdot\psi_{a_2}(a_2^a\cdot\psi_{a_2}(a_2))))$ |
| 1,4,8,12,14,11,<br>15,19,21,18,22,26,28,5 | $\psi(\lambda\alpha.(\alpha\cdot 4)-\Pi_0)$<br>$\psi(\psi_a(a_2^a\cdot\psi_{a_2}(a_2^a\cdot\psi_{a_2}(a_2^a))))$ |
| 1,4,8,12,14,12 | $\psi(\lambda\alpha.(\alpha\cdot\omega)-\Pi_0)$<br>$\psi(\psi_a(a_2^{a+1}))$ |
| 1,4,8,12,14,12,<br>11,15,19,21,5 | $\psi(\lambda\alpha.(\alpha\cdot\omega+\alpha)-\Pi_0)$<br>$\psi(\psi_a(a_2^{a+1}\cdot\psi_{a_2}(a_2^a)))$ |
| 1,4,8,12,14,12,<br>11,15,19,21,19 | $\psi(\lambda\alpha.(\alpha\cdot\omega\cdot 2)-\Pi_0)$<br>$\psi(\psi_a(a_2^{a+1}\cdot\psi_{a_2}(a_2^{a+1})))$ |
| 1,4,8,12,14,12,12 | $\psi(\lambda\alpha.(\alpha\cdot\omega^2)-\Pi_0)$<br>$\psi(\psi_a(a_2^{a+2}))$ |
| 1,4,8,12,14,12,14 | $\psi(\lambda\alpha.(\alpha\cdot\Omega)-\Pi_0)$<br>$\psi(\psi_a(a_2^{a+\psi_a(0)}))$ |
| 1,4,8,12,14,12,<br>14,4,8,12,14,12 | $\psi(\lambda\alpha.(\alpha\cdot\lambda\alpha.(\alpha\cdot\omega)-\Pi_0)-\Pi_0)$<br>$\psi(\psi_a(a_2^{a+\psi_a(a_2^{a+1})}))$ |
| 1,4,8,12,14,12,14,5 | $\psi(\lambda\alpha.(\alpha^2)-\Pi_0)$<br>$\psi(\psi_a(a_2^{a\cdot 2}))$ |
| 1,4,8,12,14,12,<br>14,11,15,19,21,5 | $\psi(\lambda\alpha.(\alpha^2+\alpha)-\Pi_0)$<br>$\psi(\psi_a(a_2^{a\cdot 2}\cdot\psi_{a_2}(a_2^a)))$ |
| 1,4,8,12,14,12,14,<br>11,15,19,21,19,21,5 | $\psi(\lambda\alpha.(\alpha^2\cdot 2)-\Pi_0)$<br>$\psi(\psi_a(a_2^{a\cdot 2}\cdot\psi_{a_2}(a_2^{a\cdot 2})))$ |
| 1,4,8,12,14,12,14,12 | $\psi(\lambda\alpha.(\alpha^2\cdot\omega)-\Pi_0)$<br>$\psi(\psi_a(a_2^{a\cdot 2+1}))$ |
| 1,4,8,12,14,12,14,12,14,5 | $\psi(\lambda\alpha.(\alpha^3)-\Pi_0)$<br>$\psi(\psi_a(a_2^{a\cdot 3}))$ |
| 1,4,8,12,14,13 | $\psi(\lambda\alpha.(\alpha^\omega)-\Pi_0)$<br>$\psi(\psi_a(a_2^{a\cdot\omega}))$ |
| 1,4,8,12,14,14 | $\psi(\lambda\alpha.(\alpha^\Omega)-\Pi_0)$<br>$\psi(\psi_a(a_2^{a\cdot\psi_a(0)}))$ |
| 1,4,8,12,14,14,5 | $\psi(\lambda\alpha.(\alpha^\alpha)-\Pi_0)$<br>$\psi(\psi_a(a_2^{a^2}))$ |
| 1,4,8,12,14,14,12,14,5 | $\psi(\lambda\alpha.(\alpha^{\alpha+1})-\Pi_0)$<br>$\psi(\psi_a(a_2^{a^2+a}))$ |
| 1,4,8,12,14,14,12,14,14,5 | $\psi(\lambda\alpha.(\alpha^{\alpha\cdot 2})-\Pi_0)$<br>$\psi(\psi_a(a_2^{a^2\cdot 2}))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,8,12,14,14,14,5 | $\psi(\lambda\alpha.(\alpha^{\alpha^2}) - \Pi_0)$<br>$\psi(\psi_a(a_2^{a^3}))$ |
| 1,4,8,12,14,16,5 | $\psi(\lambda\alpha.(\alpha^{\alpha^\alpha}) - \Pi_0)$<br>$\psi(\psi_a(a_2^{a^a}))$ |
| 1,4,8,12,14,17 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0)) - \Pi_0)$<br>$\psi(\lambda\alpha.(\varepsilon_{\alpha+1}) - \Pi_0)$<br>$\psi(\psi_a(a_2^{\psi_{\Omega_{a+1}}(0)}))$ |
| 1,4,8,12,14,17,<br>11,15,19,21,5 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0) + \alpha) - \Pi_0)$<br>$\psi(\psi_a(a_2^{\psi_{\Omega_{a+1}}(0)} \cdot \psi_{a_2}(a_2^a)))$ |
| 1,4,8,12,14,17,<br>11,15,19,21,23,23,5 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0) + \alpha^\alpha) - \Pi_0)$<br>$\psi(\psi_a(a_2^{\psi_{\Omega_{a+1}}(0)} \cdot \psi_{a_2}(a_2^{a^2})))$ |
| 1,4,8,12,14,17,<br>11,15,19,21,24 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0) \cdot 2) - \Pi_0)$<br>$\psi(\psi_a(a_2^{\psi_{\Omega_{a+1}}(0)} \cdot \psi_{a_2}(a_2^{\psi_{\Omega_{a+1}}(0)})))$ |
| 1,4,8,12,14,17,12 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0) \cdot \omega) - \Pi_0)$<br>$\psi(\psi_a(a_2^{\psi_{\Omega_{a+1}}(0)+1}))$ |
| 1,4,8,12,14,17,12,14,17 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0)^2) - \Pi_0)$<br>$\psi(\psi_a(a_2^{\psi_{\Omega_{a+1}}(0)\cdot 2}))$ |
| 1,4,8,12,14,17,13 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0)^\omega) - \Pi_0)$<br>$\psi(\psi_a(a_2^{\psi_{\Omega_{a+1}}(0)\cdot\omega}))$ |
| 1,4,8,12,14,17,14,5 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0)^\alpha) - \Pi_0)$<br>$\psi(\psi_a(a_2^{\psi_{\Omega_{a+1}}(0)\cdot a}))$ |
| 1,4,8,12,14,17,14,17 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(0)^{\psi_{\Omega_{\alpha+1}}(0)}) - \Pi_0)$ |
| 1,4,8,12,14,17,17 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(1)) - \Pi_0)$<br>$\psi(\psi_a(a_2^{\psi_{\Omega_{a+1}}(1)}))$ |
| 1,4,8,12,14,17,18 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\omega)) - \Pi_0)$<br>$\psi(\psi_a(a_2^{\psi_{\Omega_{a+1}}(\omega)}))$ |
| 1,4,8,12,14,17,19,5 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\alpha)) - \Pi_0)$<br>$\psi(\psi_a(a_2^{\psi_{\Omega_{a+1}}(a)}))$ |
| 1,4,8,12,14,17,19,22 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+1}}(0))) - \Pi_0)$<br>$\psi(\psi_a(a_2^{\psi_{\Omega_{a+1}}(\psi_{\Omega_{a+1}}(0))}))$ |
| 1,4,8,12,14,17,20 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1})) - \Pi_0)$<br>$\psi(\psi_a(a_2^{\psi_{\Omega_{a+1}}(\Omega_{a+1})}))$ |
| 1,4,8,12,14,17,20,23 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}^{\Omega_{\alpha+1}})) - \Pi_0)$ |
| 1,4,8,12,14,17,21 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\psi_{\Omega_{\alpha+2}}(0))) - \Pi_0)$ |
| 1,4,8,12,14,18 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+\omega})) - \Pi_0)$ |
| 1,4,8,12,14,18,22,26 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(I_{\alpha+\omega})) - \Pi_0)$ |
| 1,4,8,12,14,18,22,26,26 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(M_{\alpha+\omega})) - \Pi_0)$ |
| 1,4,8,12,14,18,22,26,30 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(K_{\alpha+\omega})) - \Pi_0)$ |
| 1,4,8,12,14,18,23 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'+1) - \Pi_0)) - \Pi_0)$ |
| 1,4,8,12,14,18,23,28,30,5 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'+\alpha) - \Pi_0)) - \Pi_0)$ |
| 1,4,8,12,14,18,23,28,30,33 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'+\psi_{\Omega_{\alpha+1}}(0)) - \Pi_0)) - \Pi_0)$ |
| 1,4,8,12,14,18,<br>23,28,30,33,36 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'+\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1})) - \Pi_0)) - \Pi_0)$ |
| 1,4,8,12,14,18,23,28,31 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'+\Omega_{\alpha+1}) - \Pi_0)) - \Pi_0)$ |
| 1,4,8,12,14,18,23,28,31,18 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'+\Omega_{\alpha+\omega}) - \Pi_0)) - \Pi_0)$ |
| 1,4,8,12,14,18,23,28,31,19 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'\cdot 2) - \Pi_0)) - \Pi_0)$ |
| 1,4,8,12,14,18,23,28,31,28 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'\cdot \omega) - \Pi_0)) - \Pi_0)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,8,12,14,18, 23,28,31,28,31,19 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'^2)-\Pi_0))-\Pi_0)$ |
| 1,4,8,12,14,18, 23,28,31,31,19 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'^{\alpha'})-\Pi_0))-\Pi_0)$ |
| 1,4,8,12,14,18,23,28,31,35 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+1}}(0))-\Pi_0))-\Pi_0)$ |
| 1,4,8,12,14,18, 23,28,31,35,39 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+1}}(\Omega_{\alpha'+1}))-\Pi_0))-\Pi_0)$ |
| 1,4,8,12,14,18,23,28,31,36 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+1}}(\Omega_{\alpha'+\omega}))-\Pi_0))-\Pi_0)$ |
| 1,4,8,12,14,18, 23,28,31,36,42 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.$ $(\psi_{\Omega_{\alpha'+1}}(\lambda\alpha''.(\alpha''+1)-\Pi_0))-\Pi_0))-\Pi_0)$ |
| 1,4,8,12,14,18,23, 28,31,36,42,48,52,57 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+1}}(\lambda\alpha''.$ $(\psi_{\Omega_{\alpha''+1}}(0))-\Pi_0))-\Pi_0))-\Pi_0)$ |
| 1,4,8,12,15 | $\psi(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1)$ $\psi(\psi_a(a_2^{\psi_{a_2}(0)}))$ |
| 1,4,8,12,15,3 | $\psi(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1+1)$ |
| 1,4,8,12,15,7 | $\psi(1-\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1)$ $\psi(\psi_a(a_2^{\psi_{a_2}(0)}+\psi_{a_2}(a_2^{\psi_{a_2}(0)}+1)))$ |
| 1,4,8,12,15,8 | $\psi(\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1)$ |
| 1,4,8,12,15,8,12 | $\psi(\lambda\alpha.(\alpha+\omega)-\Pi_0-\lambda\alpha.(\Omega_{\alpha+1})-\Pi_0)$ |
| 1,4,8,12,15,8,12,14,5 | $\psi(\lambda\alpha.(\alpha\cdot 2)-\Pi_0-\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1)$ |
| 1,4,8,12,15,8,12,15 | $\psi(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1)$ |
| 1,4,8,12,15,9 | $\psi((\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1-)^{\omega})$ |
| 1,4,8,12,15,10,5 | $\psi((\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1-)^{1,0})$ |
| 1,4,8,12,15,11 | $\psi(1-\lambda\alpha.(\Omega_{\alpha+1})-\Pi_2)$ |
| 1,4,8,12,15,11,8,12,15,11 | $\psi(1-\lambda\alpha.(\Omega_{\alpha+1})-\Pi_2\quad\lambda\alpha.$ $(\Omega_{\alpha+1})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+1})-\Pi_2)$ |
| 1,4,8,12,15,11,11 | $\psi(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_2-\lambda\alpha.(\Omega_{\alpha+1})-\Pi_2)$ |
| 1,4,8,12,15,11,12 | $\psi((\lambda\alpha.(\Omega_{\alpha+1})-\Pi_2-)^{\omega})$ |
| 1,4,8,12,15,11,14 | $\psi(1-\lambda\alpha.(\Omega_{\alpha+1})-\Pi_3)$ |
| 1,4,8,12,15,11,15 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+1)-\Pi_0)$ $\psi(\psi_a(a_2^{\psi_{a_2}(0)}\cdot\psi_{a_2}(a_2)))$ |
| 1,4,8,12,15,11,15,19 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\omega)-\Pi_0)$ $\psi(\psi_a(a_2^{\psi_{a_2}(0)}\cdot\psi_{a_2}(a_2^2)))$ |
| 1,4,8,12,15,11,15,19,21,5 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\alpha)-\Pi_0)$ |
| 1,4,8,12,15,11, 15,19,21,18,22,26,28,5 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\alpha\cdot 2)-\Pi_0)$ |
| 1,4,8,12,15,11,15,19,21,19 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\alpha\cdot\omega)-\Pi_0)$ |
| 1,4,8,12,15,11, 15,19,21,19,21,5 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\alpha^2)-\Pi_0)$ |
| 1,4,8,12,15,11,15,19,21,24 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(0))-\Pi_0)$ |
| 1,4,8,12,15,11, 15,19,21,24,24 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(1))-\Pi_0)$ |
| 1,4,8,12,15,11, 15,19,21,24,27 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}))-\Pi_0)$ |
| 1,4,8,12,15,11,15,19,21,25 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+\omega}))-\Pi_0)$ |
| 1,4,8,12,15,11, 15,19,21,25,30 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'+1)-\Pi_0))-\Pi_0)$ |
| 1,4,8,12,15,11,15, 19,21,25,30,35,37,5 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'+\alpha)-\Pi_0))-\Pi_0)$ |
| 1,4,8,12,15,11,15,19,21, 25,30,35,38,35,38,26 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'\cdot 2)-\Pi_0))-\Pi_0)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,8,12,15,11,15, 19,21,25,30,35,38,42 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+1}}(0))-\Pi_0))-\Pi_0)$ |
| 1,4,8,12,15,11,15, 19,21,25,30,35,39 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}+\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\alpha'+1})-\Pi_1))-\Pi_0)$ |
| 1,4,8,12,15,11,15,19,22 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot 2)-\Pi_0)$ |
| 1,4,8,12,15,11,15, 19,22,18,22,26,29 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot 3)-\Pi_0)$ |
| 1,4,8,12,15,12 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,8,12,15,12,14,5 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\alpha)-\Pi_0)$ $\psi(\psi_a(a_2^{\psi_{a_2}(0)+a}))$ |
| 1,4,8,12,15,12,14,17 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\psi_{\Omega_{\alpha+1}}(0))-\Pi_0)$ |
| 1,4,8,12,15,12,14,17,20 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1}))-\Pi_0)$ |
| 1,4,8,12,15,12,14,18 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+\omega}))-\Pi_0)$ |
| 1,4,8,12,15,12,14,18,23 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'+1)-\Pi_0))-\Pi_0)$ |
| 1,4,8,12,15,12, 14,18,23,28,32 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\alpha'+1})-\Pi_1))-\Pi_0)$ |
| 1,4,8,12,15,12, 14,18,23,28,32,28 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\alpha'+1}\cdot\omega)-\Pi_0))-\Pi_0)$ |
| 1,4,8,12,15,12, 14,18,23,28,32,28,30,19 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\alpha'+1}\cdot\alpha')-\Pi_0))-\Pi_0)$ |
| 1,4,8,12,15,12,14, 18,23,28,32,28,31,35 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}\cdot\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\alpha'+1}\cdot\psi_{\Omega_{\alpha'+1}}(0))-\Pi_0))-\Pi_0)$ |
| 1,4,8,12,15,12,15 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}^2)-\Pi_0)$ $\psi(\psi_a(a_2^{\psi_{a_2}(0)\cdot 2}))$ |
| 1,4,8,12,15,12,15,12,15 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}^3)-\Pi_0)$ |
| 1,4,8,12,15,13 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}^\omega)-\Pi_0)$ $\psi(\psi_a(a_2^{\psi_{a_2}(1)}))$ |
| 1,4,8,12,15,14,5 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}^\alpha)-\Pi_0)$ |
| 1,4,8,12,15,14,17 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}^{\psi_{\Omega_{\alpha+1}}(0)})-\Pi_0)$ |
| 1,4,8,12,15,14,17,20 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}^{\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1})})-\Pi_0)$ |
| 1,4,8,12,15,14,18,23,28,32 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}^{\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\alpha'+1})-\Pi_1)})-\Pi_0)$ |
| 1,4,8,12,15,14, 18,23,28,32,31,19 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}^{\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\alpha'+1}^{\alpha'})-\Pi_0)})-\Pi_0)$ |
| 1,4,8,12,15,15 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}^{\Omega_{\alpha+1}})-\Pi_0)$ $\psi(\psi_a(a_2^{\psi_{a_2}(\psi_{a_2}(0))}))$ |
| 1,4,8,12,15,16 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}^{\Omega_{\alpha+1}^\omega})-\Pi_0)$ |
| 1,4,8,12,15,18 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}^{\Omega_{\alpha+1}^{\Omega_{\alpha+1}}})-\Pi_0)$ |
| 1,4,8,12,15,19 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(0))-\Pi_0)$ $\psi(\psi_a(a_2^{\psi_{a_2}(a_2)}))$ |
| 1,4,8,12,15,19,19 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(1))-\Pi_0)$ $\psi(\psi_a(a_2^{\psi_{a_2}(a_2\cdot 2)}))$ |
| 1,4,8,12,15,19,23 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}))-\Pi_0)$ $\psi(\psi_a(a_2^{\psi_{a_2}(a_2^2)}))$ |
| 1,4,8,12,15,19,23,19,23 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}\cdot 2))-\Pi_0)$ |
| 1,4,8,12,15,19,23,20 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}\cdot\omega))-\Pi_0)$ |
| 1,4,8,12,15,19,23,21,5 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}\cdot\alpha))-\Pi_0)$ |
| 1,4,8,12,15,19,23,22 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}\cdot\Omega_{\alpha+1}))-\Pi_0)$ |
| 1,4,8,12,15,19,23,23 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}^2))-\Pi_0)$ |
| 1,4,8,12,15,19,23,26 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}^{\Omega_{\alpha+1}}))-\Pi_0)$ |

| $0-\mathrm{Y}$ sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,8,12,15,19,23,26,30 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}^{\psi_{\Omega_{\alpha+2}}(0)})) - \Pi_0)$ |
| 1,4,8,12,15,19, 23,26,30,34,37 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}^{\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}^{\Omega_{\alpha+1}})})) - \Pi_0)$ |
| 1,4,8,12,16 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}^{\Omega_{\alpha+2}})) - \Pi_0)$ |
| 1,4,8,12,16,20 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}^{\Omega_{\alpha+2}^{\Omega_{\alpha+2}}})) - \Pi_0)$ |
| 1,4,8,13 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(0))) - \Pi_0)$ <br> $\psi(\psi_a(a_2^{\psi_{a_2}(a_2^3)}))$ |
| 1,4,8,13,18 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+3})) - \Pi_0)$ |
| 1,4,8,14 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+\omega})) - \Pi_0)$ |
| 1,4,8,14,20,26 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(I_{\alpha+\omega})) - \Pi_0)$ |
| 1,4,8,14,21 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'+1) - \Pi_0)) - \Pi_0)$ |
| 1,4,8,14,21,28,30,5 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'+\alpha) - \Pi_0)) - \Pi_0)$ |
| 1,4,8,14,21,28,33,15 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'\cdot 2) - \Pi_0)) - \Pi_0)$ |
| 1,4,8,14,21,28,33,39 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+1}}(0)) - \Pi_0)) - \Pi_0)$ |
| 1,4,8,14,21,28,34 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\Omega_{\alpha'+1}) - \Pi_1)) - \Pi_0)$ |
| 1,4,8,14,21,28,35 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\Omega_{\alpha'+1}^{\Omega_{\alpha'+1}}) - \Pi_0)) - \Pi_0)$ |
| 1,4,8,14,21,29 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+2}}(0)) - \Pi_0)) - \Pi_0)$ |
| 1,4,8,14,21,29,37 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+2}}(\Omega_{\alpha'+2})) - \Pi_0)) - \Pi_0)$ |
| 1,4,8,14,21,30 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\psi_{\Omega_{\alpha'+2}}(\Omega_{\alpha'+\omega})) - \Pi_0)) - \Pi_0)$ |
| 1,4,8,14,21,30,40 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.$ <br> $(\psi_{\Omega_{\alpha'+2}}(\lambda\alpha''.(\alpha''+1) - \Pi_0)) - \Pi_0)) - \Pi_0)$ |
| 1,4,9 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ <br> $\psi(\psi_a(\Omega_{a_2+1}))$ <br> BGO |
| 1,4,9,3 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1 + 1)$ |
| 1,4,9,8 | $\psi(\lambda\alpha.(\alpha+1) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ |
| 1,4,9,8,12,14,5 | $\psi(\lambda\alpha.(\alpha\cdot 2) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ |
| 1,4,9,8,12,15 | $\psi(\lambda\alpha.(\Omega_{\alpha+1}) - \Pi_1 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ |
| 1,4,9,8,12,15,19 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(0)) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ |
| 1,4,9,8,12,15,19,23 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2})) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ |
| 1,4,9,8,12,15,19,25 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+\omega})) - \Pi_0 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_0)$ |
| 1,4,9,8,12,15,20 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ |
| 1,4,9,8,12,15, 20,8,12,15,20 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1 - \lambda\alpha.$ <br> $(\Omega_{\alpha+2}) - \Pi_1 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)$ |
| 1,4,9,8,12,15,20,9 | $\psi((\lambda\alpha.(\Omega_{\alpha+2}) - \Pi_1)^{\omega})$ |
| 1,4,9,8,12,15,20,11 | $\psi(1 - \lambda\alpha.(\Omega_{\alpha+2}) - \Pi_2)$ |
| 1,4,9,8,12,15,20,11,15 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+1) - \Pi_0)$ |
| 1,4,9,8,12,15,20,11,15,19 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\omega) - \Pi_0)$ |
| 1,4,9,8,12,15, 20,11,15,19,21 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\Omega) - \Pi_0)$ |
| 1,4,9,8,12,15, 20,11,15,19,21,5 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\alpha) - \Pi_0)$ |
| 1,4,9,8,12,15, 20,11,15,19,21,24 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+1}}(0)) - \Pi_0)$ |
| 1,4,9,8,12,15, 20,11,15,19,22 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\Omega_{\alpha+1}) - \Pi_0)$ |
| 1,4,9,8,12,15, 20,11,15,19,22,26 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(0)) - \Pi_0)$ |
| 1,4,9,8,12,15, 20,11,15,19,22,26,30 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2})) - \Pi_0)$ |
| 1,4,9,8,12,15, 20,11,15,19,23 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2}^{\Omega_{\alpha+2}})) - \Pi_0)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,9,8,12,15,20,11,15,20 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\psi_{\Omega_{\alpha+3}}(0)))-\Pi_0)$ |
| 1,4,9,8,12,15,20,11,15,21 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+\omega}))-\Pi_0)$ |
| 1,4,9,8,12,15,<br>20,11,15,21,28 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'+1)-\Pi_0))-\Pi_0)$ |
| 1,4,9,8,12,15,<br>20,11,15,21,28,35,41 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\Omega_{\alpha'+1})-\Pi_1))-\Pi_0)$ |
| 1,4,9,8,12,15,<br>20,11,15,21,29 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\Omega_{\alpha'+2})-\Pi_1))-\Pi_0)$ |
| 1,4,9,8,12,15,20,11,15,<br>21,29,28,35,41,49,34,41 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}+\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.$<br>$(\Omega_{\alpha'+2}+1)-\Pi_0))-\Pi_0)$ |
| 1,4,9,8,12,15,20,11,16 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot 2)-\Pi_0)$ |
| 1,4,9,8,12,15,20,11,16,<br>15,19,22,27,18,23 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot 3)-\Pi_0)$ |
| 1,4,9,8,12,15,20,12 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega)-\Pi_0)$ |
| 1,4,9,8,12,15,20,12,12 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\omega^2)-\Pi_0)$ |
| 1,4,9,8,12,15,20,12,14,5 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\alpha)-\Pi_0)$ |
| 1,4,9,8,12,15,20,12,14,17 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\psi_{\Omega_{\alpha+1}}(0))-\Pi_0)$ |
| 1,4,9,8,12,15,20,12,15 | $\psi(\lambda\alpha.(\Omega_{\alpha+2}\cdot\Omega_{\alpha+1})-\Pi_0)$ |
| 1,4,9,8,12,15,20,12,15,20 | $\psi(\lambda\alpha.(\Omega^2_{\alpha+2})-\Pi_0)$ |
| 1,4,9,8,12,15,20,13 | $\psi(\lambda\alpha.(\Omega^{\omega}_{\alpha+2})-\Pi_0)$ |
| 1,4,9,8,12,15,20,14,5 | $\psi(\lambda\alpha.(\Omega^{\alpha}_{\alpha+2})-\Pi_0)$ |
| 1,4,9,8,12,15,20,15 | $\psi(\lambda\alpha.(\Omega^{\Omega_{\alpha+1}}_{\alpha+2})-\Pi_0)$ |
| 1,4,9,8,12,15,20,15,20 | $\psi(\lambda\alpha.(\Omega^{\Omega_{\alpha+2}}_{\alpha+2})-\Pi_0)$ |
| 1,4,9,8,12,15,20,19 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(0))-\Pi_0)$ |
| 1,4,9,8,12,15,20,19,23 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3}))-\Pi_0)$ |
| 1,4,9,8,12,15,<br>20,19,23,19,23 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+3}\cdot 2))-\Pi_0)$ |
| 1,4,9,8,12,15,20,19,23,23 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega^2_{\alpha+3}))-\Pi_0)$ |
| 1,4,9,8,12,16 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega^{\Omega_{\alpha+3}}_{\alpha+3}))-\Pi_0)$ |
| 1,4,9,8,13 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\psi_{\Omega_{\alpha+4}}(0)))-\Pi_0)$ |
| 1,4,9,8,14 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\Omega_{\alpha+\omega}))-\Pi_0)$ |
| 1,4,9,8,14,21 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\lambda\alpha'.(\alpha'+1)-\Pi_0))-\Pi_0)$ |
| 1,4,9,8,14,21,28,34 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\lambda\alpha'.(\Omega_{\alpha'+1})-\Pi_1))-\Pi_0)$ |
| 1,4,9,8,14,22 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\lambda\alpha'.(\Omega_{\alpha'+2})-\Pi_1))-\Pi_0)$ |
| 1,4,9,8,14,22,21,28,35 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+3}}(\lambda\alpha'.(\psi_{\Omega_{\alpha+3}}(0))-\Pi_0))-\Pi_0)$ |
| 1,4,9,9 | $\psi(\lambda\alpha.(\Omega_{\alpha+3})-\Pi_1)$<br>$\psi(\psi_a(\Omega_{a_2+1}\cdot 2))$ |
| 1,4,9,9,9 | $\psi(\lambda\alpha.(\Omega_{\alpha+4})-\Pi_1)$<br>$\psi(\psi_a(\Omega_{a_2+1}\cdot 3))$ |
| 1,4,9,10 | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_0)$<br>$\psi(\psi_a(\Omega_{a_2+1}\cdot\omega))$ |
| 1,4,9,10,3 | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_0+1)$<br>$\psi(\psi_a(\Omega_{a_2+1}\cdot\omega)+1)$ |
| 1,4,9,10,7 | $\psi(1-\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_0)$ |
| 1,4,9,10,8 | $\psi(\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_0)$ |
| 1,4,9,10,8,12,15 | $\psi(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_0)$ |
| 1,4,9,10,8,12,15,20 | $\psi(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_0)$ |
| 1,4,9,10,8,12,15,20,20 | $\psi(\lambda\alpha.(\Omega_{\alpha+3})-\Pi_1-\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_0)$ |
| 1,4,9,10,8,12,15,20,21 | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_0-\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_0)$ |
| 1,4,9,10,8,12,15,20,21,11 | $\psi(1-\lambda\alpha.(\Omega_{\alpha+\omega})-\Pi_2)$ |
| 1,4,9,10,8,12,<br>15,20,21,11,15 | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}+1)-\Pi_0)$ |
| 1,4,9,10,8,12,15,<br>20,21,11,15,19,21,5 | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}+\alpha)-\Pi_0)$ |

| $0-\mathrm{Y}$ sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,9,10,8,12,15, 20,21,11,15,19,22 | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}+\Omega_{\alpha+1})-\Pi_0)$ |
| 1,4,9,10,8,12,15, 20,21,11,15,20 | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}+\Omega_{\alpha+2})-\Pi_0)$ |
| 1,4,9,10,8,12,15, 20,21,11,15,20,21 | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}\cdot 2)-\Pi_0)$ |
| 1,4,9,10,8,12,15,20,21,12 | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}\cdot\omega)-\Pi_0)$ |
| 1,4,9,10,8,12, 15,20,21,12,14,5 | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}\cdot\alpha)-\Pi_0)$ |
| 1,4,9,10,8,12,15,20,21,12,15 | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}\cdot\Omega_{\alpha+1})-\Pi_0)$ |
| 1,4,9,10,8,12, 15,20,21,12,15,20 | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega}\cdot\Omega_{\alpha+2})-\Pi_0)$ |
| 1,4,9,10,8,12,15, 20,21,12,15,20,21 | $\psi(\lambda\alpha.(\Omega^2_{\alpha+\omega})-\Pi_0)$ |
| 1,4,9,10,8,12, 15,20,21,14,5 | $\psi(\lambda\alpha.(\Omega^{\alpha}_{\alpha+\omega})-\Pi_0)$ |
| 1,4,9,10,8,12,15,20,21,15 | $\psi(\lambda\alpha.(\Omega^{\Omega_{\alpha+1}}_{\alpha+\omega})-\Pi_0)$ |
| 1,4,9,10,8,12,15, 20,21,15,20,21 | $\psi(\lambda\alpha.(\Omega^{\Omega_{\alpha+\omega}}_{\alpha+\omega})-\Pi_0)$ |
| 1,4,9,10,8,12,15,20,21,19 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+\omega+1}}(0))-\Pi_0)$ |
| 1,4,9,10,8,12, 15,20,21,19,23 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+\omega+1}}(\Omega_{\alpha+\omega+1}))-\Pi_0)$ |
| 1,4,9,10,8,12,16 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+\omega+1}}(\Omega^{\Omega_{\alpha+\omega+1}}_{\alpha+\omega+1}))-\Pi_0)$ |
| 1,4,9,10,8,14 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+\omega+1}}(\Omega_{\alpha+\omega\cdot 2}))-\Pi_0)$ |
| 1,4,9,10,8,14,21 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha+\omega+1}}(\lambda\alpha'.(\alpha'+1)-\Pi_0))-\Pi_0)$ |
| 1,4,9,10,9 | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega+1})-\Pi_1)$ |
| 1,4,9,10,9,10 | $\psi(\lambda\alpha.(\Omega_{\alpha+\omega\cdot 2})-\Pi_0)$ |
| 1,4,9,11 | $\psi(\lambda\alpha.(\Omega_{\alpha+\Omega})-\Pi_0)$ |
| 1,4,9,11,5 | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot 2})-\Pi_0)$<br>$\psi(\psi_a(\Omega_{a_2+1}\cdot a))$ |
| 1,4,9,11,8 | $\psi(\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\Omega_{\alpha\cdot 2})-\Pi_0)$ |
| 1,4,9,11,8,12,15, 20,22,12,15,20,22,5 | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot 2}\cdot 2)-\Pi_0)$ |
| 1,4,9,11,8,14,21 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}(\lambda\alpha'.(\alpha'+1)-\Pi_0))-\Pi_0)$ |
| 1,4,9,11,8,14,22 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}(\lambda\alpha'.(\Omega_{\alpha'+2})-\Pi_1))-\Pi_0)$ |
| 1,4,9,11,8,14,22,24,5 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}(\lambda\alpha'.(\Omega_{\alpha'+\alpha})-\Pi_0))-\Pi_0)$ |
| 1,4,9,11,8,14,22,25 | $\psi(\lambda\alpha.(\psi_{\Omega_{\alpha\cdot 2+1}}(\lambda\alpha'.(\Omega_{\alpha'+\Omega_{\alpha+1}})-\Pi_0))-\Pi_0)$ |
| 1,4,9,11,9 | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot 2+1})-\Pi_1)$ |
| 1,4,9,11,9,11,5 | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot 3})-\Pi_0)$ |
| 1,4,9,11,10 | $\psi(\lambda\alpha.(\Omega_{\alpha\cdot\omega})-\Pi_0)$ |
| 1,4,9,11,11,5 | $\psi(\lambda\alpha.(\Omega_{\alpha^2})-\Pi_0)$ |
| 1,4,9,11,13,5 | $\psi(\lambda\alpha.(\Omega_{\alpha^\alpha})-\Pi_0)$ |
| 1,4,9,11,14 | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(0)})-\Pi_0)$ |
| 1,4,9,11,14,17 | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\Omega_{\alpha+1})})-\Pi_0)$ |
| 1,4,9,11,15,20 | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\alpha'+1)-\Pi_0)})-\Pi_0)$ |
| 1,4,9,11,15,21 | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\alpha'+1})-\Pi_1)})-\Pi_0)$ |
| 1,4,9,11,15,21,23,5 | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\alpha'+\alpha})-\Pi_0)})-\Pi_0)$ |
| 1,4,9,11,15,21,24,16 | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\alpha'\cdot 2})-\Pi_0)})-\Pi_0)$ |
| 1,4,9,11,15,21,24,28 | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+1}}(\lambda\alpha'.(\Omega_{\psi_{\Omega_{\alpha'+1}}(0)})-\Pi_0)})-\Pi_0)$ |
| 1,4,9,12 | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}})-\Pi_0)$<br>$\psi(\psi_a(\Omega_{a_2+1}\cdot\psi_{a_2}(0)))$ |
| 1,4,9,12,9 | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}+1})-\Pi_1)$ |
| 1,4,9,12,9,11,5 | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}+\alpha})-\Pi_0)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,9,12,9,12 | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}\cdot 2}) - \Pi_0)$ |
| 1,4,9,12,11,5 | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+1}\cdot\alpha}) - \Pi_0)$ |
| 1,4,9,12,12 | $\psi(\lambda\alpha.(\Omega_{\Omega^2_{\alpha+1}}) - \Pi_0)$ |
| 1,4,9,12,14,5 | $\psi(\lambda\alpha.(\Omega_{\Omega^\alpha_{\alpha+1}}) - \Pi_0)$ |
| 1,4,9,12,15 | $\psi(\lambda\alpha.(\Omega_{\Omega^{\Omega_{\alpha+1}}_{\alpha+1}}) - \Pi_0)$ |
| 1,4,9,12,16 | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+2}}(0)}) - \Pi_0)$<br>$\psi(\psi_a(\Omega_{a_2+1}\cdot\psi_{a_2}(a_2)))$ |
| 1,4,9,12,16,20 | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+2}}(\Omega_{\alpha+2})}) - \Pi_0)$ |
| 1,4,9,12,16,22,29 | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\alpha'+1)-\Pi_0)}) - \Pi_0)$ |
| 1,4,9,12,16,22,30 | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\Omega_{\alpha'+1})-\Pi_1)}) - \Pi_0)$ |
| 1,4,9,12,16,22,30,36 | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\alpha+2}}(\lambda\alpha'.(\Omega_{\Omega_{\alpha'+1}})-\Pi_0)}) - \Pi_0)$ |
| 1,4,9,12,17 | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+2}}) - \Pi_0)$ |
| 1,4,9,12,17,17 | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+3}}) - \Pi_0)$ |
| 1,4,9,12,17,18 | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha+\omega}}) - \Pi_0)$ |
| 1,4,9,12,17,19,5 | $\psi(\lambda\alpha.(\Omega_{\Omega_{\alpha\cdot 2}}) - \Pi_0)$ |
| 1,4,9,12,17,20 | $\psi(\lambda\alpha.(\Omega_{\Omega_{\Omega_{\alpha+1}}}) - \Pi_0)$ |
| 1,4,9,13 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(0)) - \Pi_0)$<br>$\psi(\psi_a(\Omega_{a_2+1}\cdot a_2))$ |
| 1,4,9,13,9 | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)+1}) - \Pi_1)$ |
| 1,4,9,13,9,11,5 | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)+\alpha}) - \Pi_0)$ |
| 1,4,9,13,9,12 | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)+\Omega_{\alpha+1}}) - \Pi_0)$ |
| 1,4,9,13,9,12,17 | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)+\Omega_{\alpha+2}}) - \Pi_0)$ |
| 1,4,9,13,9,12,17,19,5 | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)+\Omega_{\alpha\cdot 2}}) - \Pi_0)$ |
| 1,4,9,13,9,12,17,20 | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)+\Omega_{\Omega_{\alpha+1}}}) - \Pi_0)$ |
| 1,4,9,13,9,12,17,21 | $\psi(\lambda\alpha.(\Omega_{\psi_{I_{\alpha+1}}(0)\cdot 2}) - \Pi_0)$ |
| 1,4,9,13,9,12,<br>17,21,9,12,17,21 | $\psi(\Omega_{\psi_{I_{\alpha+1}}(0)\cdot 3})$ |
| 1,4,9,13,9,12,17,21,16 | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{\psi_{I_{\alpha+1}}(0)+1}}(0)}) - \Pi_0)$ |
| 1,4,9,13,9,12,17,21,17 | $\psi(\lambda\alpha.(\Omega_{\Omega_{\psi_{I_{\alpha+1}}(0)+1}}) - \Pi_0)$ |
| 1,4,9,13,9,13 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(1)) - \Pi_0)$ |
| 1,4,9,13,10 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\omega)) - \Pi_0)$ |
| 1,4,9,13,11,5 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\alpha)) - \Pi_0)$ |
| 1,4,9,13,12 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\Omega_{\alpha+1})) - \Pi_0)$ |
| 1,4,9,13,12,17,21 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\psi_{I_{\alpha+1}}(0))) - \Pi_0)$ |
| 1,4,9,13,13 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1})) - \Pi_0)$ |
| 1,4,9,13,13,9,13,13 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I_{\alpha+1}\cdot 2)) - \Pi_0)$ |
| 1,4,9,13,13,13 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I^2_{\alpha+1})) - \Pi_0)$ |
| 1,4,9,13,15,5 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I^\alpha_{\alpha+1})))$ |
| 1,4,9,13,16 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I^{\Omega_{\alpha+1}}_{\alpha+1})) - \Pi_0)$ |
| 1,4,9,13,17 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(I^{I_{\alpha+1}}_{\alpha+1})) - \Pi_0)$ |
| 1,4,9,13,18 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\psi_{\Omega_{I_{\alpha+1}+1}}(0))) - \Pi_0)$ |
| 1,4,9,13,19 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\Omega_{I_{\alpha+1}+\omega})) - \Pi_0)$ |
| 1,4,9,13,19,26 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\lambda\alpha'.(\alpha'+1) - \Pi_0)) - \Pi_0)$ |
| 1,4,9,13,19,27,34 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+1}}(\lambda\alpha'.(\psi_{I_{\alpha'+1}}(0)) - \Pi_0)) - \Pi_0)$ |
| 1,4,9,14 | $\psi(\lambda\alpha.(I_{\alpha+1}) - \Pi_1)$<br>$\psi(\psi_a(\Omega^2_{a_2+1}))$ |
| 1,4,9,14,9 | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}+1}) - \Pi_1)$ |
| 1,4,9,14,9,12,17,21 | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}+\psi_{I_{\alpha+1}}(0)}) - \Pi_0)$ |
| 1,4,9,14,9,12,17,22 | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}\cdot 2}) - \Pi_0)$ |
| 1,4,9,14,9,12,<br>17,22,9,12,17,22 | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}\cdot 3}) - \Pi_0)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,9,14,9,12,17,22,12 | $\psi(\lambda\alpha.(\Omega_{I_{\alpha+1}\cdot\Omega_{\alpha+1}}) - \Pi_0)$ |
| 1,4,9,14,9,12,<br>17,22,12,17,22 | $\psi(\lambda\alpha.(\Omega_{I^2_{\alpha+1}}) - \Pi_0)$ |
| 1,4,9,14,9,12,17,22,16 | $\psi(\lambda\alpha.(\Omega_{\psi_{\Omega_{I_{\alpha+1}+1}}(0)}) - \Pi_0)$ |
| 1,4,9,14,9,12,17,22,17 | $\psi(\lambda\alpha.(\Omega_{\Omega_{I_{\alpha+1}}}) - \Pi_0)$ |
| 1,4,9,14,9,13 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+2}}(0)) - \Pi_0)$ |
| 1,4,9,14,9,13,13 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+2}}(I_{\alpha+2})) - \Pi_0)$ |
| 1,4,9,14,9,13,18 | $\psi(\lambda\alpha.(\psi_{I_{\alpha+2}}(\psi_{\Omega_{I_{\alpha+2}+1}}(0))) - \Pi_0)$ |
| 1,4,9,14,9,14 | $\psi(\lambda\alpha.(I_{\alpha+2}) - \Pi_1)$ |
| 1,4,9,14,10 | $\psi(\lambda\alpha.(I_{\alpha+\omega}) - \Pi_0)$ |
| 1,4,9,14,11,5 | $\psi(\lambda\alpha.(I_{\alpha\cdot 2}) - \Pi_0)$ |
| 1,4,9,14,11,14 | $\psi(\lambda\alpha.(I_{\psi_{\Omega_{\alpha+1}}(0)}) - \Pi_0)$ |
| 1,4,9,14,12 | $\psi(\lambda\alpha.(I_{\Omega_{\alpha+1}}) - \Pi_0)$ |
| 1,4,9,14,12,17 | $\psi(\lambda\alpha.(I_{\Omega_{\alpha+2}}) - \Pi_0)$ |
| 1,4,9,14,12,17,21 | $\psi(\lambda\alpha.(I_{\psi_{I_{\alpha+1}}(0)}) - \Pi_0)$ |
| 1,4,9,14,12,17,22 | $\psi(\lambda\alpha.(I_{I_{\alpha+1}}) - \Pi_0)$ |
| 1,4,9,14,12,17,22,20,25,29 | $\psi(\lambda\alpha.(I_{I_{I_{\alpha+1}}}) - \Pi_0)$ |
| 1,4,9,14,13 | $\psi(\lambda\alpha.(\psi_{I(1,\alpha+1)}(0)) - \Pi_0)$<br>$\psi(\psi_a(\Omega^2_{a_2+1}\cdot a))$ |
| 1,4,9,14,13,9 | $\psi(\lambda\alpha.(\Omega_{\psi_{I(1,\alpha+1)}(0)+1}) - \Pi_1)$ |
| 1,4,9,14,13,9,14 | $\psi(\lambda\alpha.(I_{\psi_{I(1,\alpha+1)}(0)+1}) - \Pi_1)$ |
| 1,4,9,14,13,9,14,13 | $\psi(\lambda\alpha.(\psi_{I(1,\alpha+1)}(1)) - \Pi_0)$ |
| 1,4,9,14,13,13 | $\psi(\lambda\alpha.(\psi_{I(1,\alpha+1)}(I(1,\alpha+1))) - \Pi_0)$ |
| 1,4,9,14,13,18 | $\psi(\lambda\alpha.(\psi_{I(1,\alpha+1)}(\psi_{\Omega_{I(1,\alpha+1)+1}}(0))) - \Pi_0)$ |
| 1,4,9,14,14 | $\psi(\lambda\alpha.(I(1,\alpha+1)) - \Pi_0)$ |
| 1,4,9,14,14,9,14,14 | $\psi(\lambda\alpha.(I(1,\alpha+2)) - \Pi_0)$ |
| 1,4,9,14,14,11,5 | $\psi(\lambda\alpha.(I(1,\alpha\cdot 2)) - \Pi_0)$ |
| 1,4,9,14,14,11,14 | $\psi(\lambda\alpha.(I(1,\psi_{\Omega_{\alpha+1}}(0))) - \Pi_0)$ |
| 1,4,9,14,14,12 | $\psi(\lambda\alpha.(I(1,\Omega_{\alpha+1})) - \Pi_0)$ |
| 1,4,9,14,14,12,17,22 | $\psi(\lambda\alpha.(I(1,I_{\alpha+1})) - \Pi_0)$ |
| 1,4,9,14,14,12,17,22,22 | $\psi(\lambda\alpha.(I(1,I(1,\alpha+1))) - \Pi_0)$ |
| 1,4,9,14,14,13 | $\psi(\lambda\alpha.(\psi_{I(2,\alpha+1)}(0)) - \Pi_0)$<br>$\psi(\psi_a(\Omega^3_{a_2+1}\cdot a))$ |
| 1,4,9,14,14,14 | $\psi(\lambda\alpha.(I(2,\alpha+1)) - \Pi_1)$<br>$\psi(\psi_a(\Omega^4_{a_2+1}))$ |
| 1,4,9,14,15 | $\psi(\lambda\alpha.(I(\omega,\alpha+1)) - \Pi_0)$<br>$\psi(\psi_a(\Omega^\omega_{a_2+1}))$ |
| 1,4,9,14,16,5 | $\psi(\lambda\alpha.(I(\alpha,1)) - \Pi_0)$<br>$\psi(\psi_a(\Omega^a_{a_2+1}))$ |
| 1,4,9,14,16,14 | $\psi(\lambda\alpha.(I(\alpha+1,0)) - \Pi_1)$ |
| 1,4,9,14,16,14,16,5 | $\psi(\lambda\alpha.(I(\alpha\cdot 2,0)) - \Pi_0)$ |
| 1,4,9,14,16,19 | $\psi(\lambda\alpha.(I(\psi_{\Omega_{\alpha+1}}(0),0)) - \Pi_0)$ |
| 1,4,9,14,17 | $\psi(\lambda\alpha.(I(\Omega_{\alpha+1},0)) - \Pi_0)$ |
| 1,4,9,14,17,22,27,30 | $\psi(\lambda\alpha.(I(I(\Omega_{\alpha+1},0),0)) - \Pi_0)$ |
| 1,4,9,14,18 | $\psi(\lambda\alpha.(\psi_{I(1,0,\alpha+1)}(0)) - \Pi_0)$<br>$\psi(\psi_a(\Omega^{a_2}_{a_2+1}))$ |
| 1,4,9,14,18,9,14,18 | $\psi(\lambda\alpha.(\psi_{I(1,0,\alpha+1)}(1)) - \Pi_0)$ |
| 1,4,9,14,18,13 | $\psi(\lambda\alpha.(\psi_{I(1,0,\alpha+1)}(I(1,0,\alpha+1))) - \Pi_0)$ |
| 1,4,9,14,18,14 | $\psi(\lambda\alpha.(I(1,0,\alpha+1)) - \Pi_1)$<br>$\psi(\psi_a(\Omega^{a_2+1}_{a_2+1}))$ |
| 1,4,9,14,18,14,13 | $\psi(\lambda\alpha.(\psi_{I(1,1,\alpha+1)}(0)) - \Pi_0)$ |
| 1,4,9,14,18,14,14 | $\psi(\lambda\alpha.(I(1,1,\alpha+1)) - \Pi_1)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,9,14,18,14,15 | $\psi(\lambda\alpha.(I(1,\omega,\alpha+1))-\Pi_0)$ |
| 1,4,9,14,18,14,16,5 | $\psi(\lambda\alpha.(I(1,\alpha,1))-\Pi_0)$ |
| 1,4,9,14,18,14,18 | $\psi(\lambda\alpha.(\psi_{I(2,0,\alpha+1)}(0))-\Pi_0)$ |
| 1,4,9,14,18,14,18,14 | $\psi(\lambda\alpha.(I(2,0,\alpha+1))-\Pi_0)$ |
| 1,4,9,14,18,15 | $\psi(\lambda\alpha.(I(\omega,0,\alpha+1))-\Pi_0)$ |
| 1,4,9,14,18,16,5 | $\psi(\lambda\alpha.(\psi_{I(1,0,0,\alpha+1)}(0))-\Pi_0)$ |
| 1,4,9,14,18,16,14 | $\psi(\lambda\alpha.(I(1,0,0,\alpha+1))-\Pi_1)$ |
| 1,4,9,14,18,16,14,18,14 | $\psi(\lambda\alpha.(I(1,0,1,\alpha+1))-\Pi_1)$ |
| 1,4,9,14,18,16,14,18,16,14 | $\psi(\lambda\alpha.(I(2,0,0,\alpha+1))-\Pi_1)$ |
| 1,4,9,14,18,18 | $\psi(\lambda\alpha.(\psi_{I(1,0,0,0,\alpha+1)}(0))-\Pi_0)$ |
| 1,4,9,14,18,18,14 | $\psi(\lambda\alpha.(I(1,0,0,0,\alpha+1))-\Pi_1)$ |
| 1,4,9,14,18,18,18,14 | $\psi(\lambda\alpha.(I(1,0,0,0,0,\alpha+1))-\Pi_1)$ |
| 1,4,9,14,18,19 | $\psi(\lambda\alpha.(I(1@\omega,\alpha+1@0))-\Pi_0)$ |
| 1,4,9,14,18,20,5 | $\psi(\lambda\alpha.(I(1@\alpha,1@0))-\Pi_0)$ |
| 1,4,9,14,18,22,14 | $\psi(\lambda\alpha.(I(1@(1,0),\alpha+1@0))-\Pi_1)$ |
| 1,4,9,14,18,22,14 | $\psi(\lambda\alpha.(I(1@(1,0),\alpha+1@0))-\Pi_1)$ |
| 1,4,9,14,18,23 | $\psi(\lambda\alpha.(\psi_{M_{\alpha+1}}(\psi_{\Omega_{M_{\alpha+1}+1}}(0)))-\Pi_0)$ |
| 1,4,9,14,18,24 | $\psi(\lambda\alpha.(\psi_{M_{\alpha+1}}(\Omega_{M_{\alpha+1}+\omega}))-\Pi_0)$ |
| 1,4,9,14,19 | $\psi(\lambda\alpha.(M_{\alpha+1})-\Pi_1)$<br>$\psi(\psi_a(\Omega_{a_2+1}^{\Omega_{a_2+1}}))$ |
| 1,4,9,14,19,9 | $\psi(\lambda\alpha.(\Omega_{M_{\alpha+1}+1})-\Pi_1)$ |
| 1,4,9,14,19,9,14 | $\psi(\lambda\alpha.(I_{M_{\alpha+1}+1})-\Pi_1)$ |
| 1,4,9,14,19,9,14,19 | $\psi(\lambda\alpha.(M_{\alpha+2})-\Pi_1)$<br>$\psi(\psi_a(\Omega_{a_2+1}^{\Omega_{a_2+1}}\cdot 2))$ |
| 1,4,9,14,19,10 | $\psi(\lambda\alpha.(M_{\alpha+\omega})-\Pi_0)$<br>$\psi(\psi_a(\Omega_{a_2+1}^{\Omega_{a_2+1}}\cdot\omega))$ |
| 1,4,9,14,19,11,5 | $\psi(\lambda\alpha.(M_{\alpha\cdot 2})-\Pi_0)$ |
| 1,4,9,14,19,12 | $\psi(\lambda\alpha.(M_{\Omega_{\alpha+1}})-\Pi_0)$ |
| 1,4,9,14,19,12,17,22 | $\psi(\lambda\alpha.(M_{I_{\alpha+1}})-\Pi_0)$ |
| 1,4,9,14,19,12,17,22,27 | $\psi(\lambda\alpha.(M_{M_{\alpha+1}})-\Pi_0)$ |
| 1,4,9,14,19,13 | $\psi(\lambda\alpha.(\psi_{M(1,\alpha+1)}(0))-\Pi_0)$ |
| 1,4,9,14,19,14 | $\psi(\lambda\alpha.(M(1,\alpha+1))-\Pi_1)$<br>$\psi(\lambda\alpha.(2\ 1-2-2\ \text{aft}\ \alpha)-\Pi_1)$<br>$\psi(\psi_a(\Omega_{a_2+1}^{\Omega_{a_2+1}+1}))$ |
| 1,4,9,14,19,14,13 | $\psi(\lambda\alpha.(\psi_{M(2,\alpha+1)}(0))-\Pi_0)$ |
| 1,4,9,14,19,14,14 | $\psi(\lambda\alpha.(M(2,\alpha+1))-\Pi_1)$ |
| 1,4,9,14,19,14,15 | $\psi(\lambda\alpha.(M(\omega,\alpha+1))-\Pi_0)$ |
| 1,4,9,14,19,14,16,5 | $\psi(\lambda\alpha.(M(\alpha,1))-\Pi_0)$<br>$\psi(\psi_a(\Omega_{a_2+1}^{\Omega_{a_2+1}+a}))$ |
| 1,4,9,14,19,14,18 | $\psi(\lambda\alpha.(\psi_{M(1,0,\alpha+1)}(0))-\Pi_0)$ |
| 1,4,9,14,19,14,18,14 | $\psi(\lambda\alpha.(M(1,0,\alpha+1))-\Pi_1)$ |
| 1,4,9,14,19,14,19 | $\psi(\lambda\alpha.(M(1;\alpha+1))-\Pi_1)$ |
| 1,4,9,14,19,14,19,14,19 | $\psi(\lambda\alpha.(M(2;\alpha+1))-\Pi_1)$ |
| 1,4,9,14,19,16,5 | $\psi(\lambda\alpha.(M(\alpha;1))-\Pi_0)$ |
| 1,4,9,14,19,18,14 | $\psi(\lambda\alpha.(M(1,0;\alpha+1))-\Pi_1)$ |
| 1,4,9,14,19,18,23 | $\psi(\lambda\alpha.(\psi_{N_{\alpha+1}}(\psi_{\Omega_{N_{\alpha+1}+1}}(0)))-\Pi_0)$ |
| 1,4,9,14,19,19 | $\psi(\lambda\alpha.(N_{\alpha+1})-\Pi_1)$<br>$\psi(\psi_a(\Omega_{a_2+1}^{\Omega_{a_2+1}\cdot 2}))$ |
| 1,4,9,14,19,19,9,14,19,19 | $\psi(\lambda\alpha.(N_{\alpha+2})-\Pi_1)$<br>$\psi(\psi_a(\Omega_{a_2+1}^{\Omega_{a_2+1}\cdot 2}\cdot 2))$ |
| 1,4,9,14,19,19,10 | $\psi(\lambda\alpha.(N_{\alpha+\omega})-\Pi_0)$ |
| 1,4,9,14,19,19,11,5 | $\psi(\lambda\alpha.(N_{\alpha\cdot 2})-\Pi_0)$ |

| $0-\mathrm{Y}$ sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,9,14,19,19,12 | $\psi(\lambda\alpha.(N_{\Omega_{\alpha+1}})-\Pi_0)$ |
| 1,4,9,14,19,19,13 | $\psi(\lambda\alpha.(\psi_{N(1,\alpha+1)}(0))-\Pi_0)$ |
| 1,4,9,14,19,19,14 | $\psi(\lambda\alpha.(N(1,\alpha+1))-\Pi_1)$<br>$\psi(\lambda\alpha.(2\ 1-2-2-2 \text{ aft } \alpha)-{}_0)$ |
| 1,4,9,14,19,19,14,19 | $\psi(\lambda\alpha.(2-2\ 1-2-2-2 \text{ aft } \alpha)-{}_1)$ |
| 1,4,9,14,19,19,14,19,19 | $\psi(\lambda\alpha.(2-2-2\ 1-2-2-2 \text{ aft } \alpha)-{}_1)$ |
| 1,4,9,14,19,19,16,5 | $\psi(\lambda\alpha.((2-2-2\ 1-)^{\alpha}\ 2-2-2 \text{ aft } \alpha)-{}_0)$ |
| 1,4,9,14,19,19,18,14 | $\psi(\lambda\alpha.((2-2-2\ 1-)^{1,0}\ 2-2-2 \text{ aft } \alpha)-{}_0)$ |
| 1,4,9,14,19,19,19 | $\psi(\lambda\alpha.(2-2-2-2 \text{ aft } \alpha)-{}_1)$ |
| 1,4,9,14,19,20 | $\psi(\lambda\alpha.((2-)^{\omega} \text{ aft } \alpha)-{}_0)$ |
| 1,4,9,14,19,21,5 | $\psi(\lambda\alpha.((2-)^{\alpha} \text{ aft } \alpha)-{}_0)$ |
| 1,4,9,14,19,23,14 | $\psi(\lambda\alpha.((2-)^{1,0} \text{ aft } \alpha)-\Pi_1)$<br>$\psi(\lambda\alpha.(\psi_{K_{\alpha+1}}(0))-\Pi_1)$ |
| 1,4,9,14,19,24 | $\psi(\lambda\alpha.(K_{\alpha+1})-\Pi_2)$<br>$\psi(\lambda\alpha.(3 \text{ aft } \alpha)-\Pi_2)$<br>$\psi(\psi_a(\Omega_{a_2+1}^{\Omega_{a_2+1}}))$ |
| 1,4,9,14,19,24,9,14,19,24 | $\psi(\lambda\alpha.(K_{\alpha+2})-\Pi_2)$ |
| 1,4,9,14,19,24,11,5 | $\psi(\lambda\alpha.(K_{\alpha\cdot 2})-\Pi_0)$ |
| 1,4,9,14,19,24,13 | $\psi(\lambda\alpha.((1-)^{1,0}\ 3 \text{ aft } \alpha)-{}_0)$ |
| 1,4,9,14,19,24,14 | $\psi(\lambda\alpha.(2\ 1-3 \text{ aft } \alpha)-{}_1)$ |
| 1,4,9,14,19,24,14,19 | $\psi(\lambda\alpha.(2-2\ 1-3 \text{ aft } \alpha)-{}_1)$ |
| 1,4,9,14,19,24,14,19,24 | $\psi(\lambda\alpha.(3\ 1-3 \text{ aft } \alpha)-{}_2)$ |
| 1,4,9,14,19,24,19 | $\psi(\lambda\alpha.(2-3 \text{ aft } \alpha)-{}_1)$ |
| 1,4,9,14,19,24,19,19 | $\psi(\lambda\alpha.(2-2-3 \text{ aft } \alpha)-{}_1)$ |
| 1,4,9,14,19,24,19,24 | $\psi(\lambda\alpha.(3\ 2-3 \text{ aft } \alpha)-{}_2)$ |
| 1,4,9,14,19,24,19,24,19,24 | $\psi(\lambda\alpha.(3\ 2-3\ 2-3 \text{ aft } \alpha)-{}_2)$ |
| 1,4,9,14,19,24,23 | $\psi(\lambda\alpha.((3\ 2-)^{1,0}\ 3 \text{ aft } \alpha)-{}_0)$ |
| 1,4,9,14,19,24,24 | $\psi(\lambda\alpha.(3-3 \text{ aft } \alpha)-{}_2)$ |
| 1,4,9,14,19,24,24,24 | $\psi(\lambda\alpha.(3-3-3 \text{ aft } \alpha)-{}_2)$ |
| 1,4,9,14,19,24,25 | $\psi(\lambda\alpha.((3-)^{\omega} \text{ aft } \alpha)-{}_0)$ |
| 1,4,9,14,19,24,26,5 | $\psi(\lambda\alpha.((3-)^{\alpha} \text{ aft } \alpha)-{}_0)$ |
| 1,4,9,14,19,24,28 | $\psi(\lambda\alpha.((3-)^{1,0} \text{ aft } \alpha)-{}_0)$ |
| 1,4,9,14,19,24,29 | $\psi(\lambda\alpha.(\kappa_{\alpha+1})-\Pi_3)$<br>$\psi(\lambda\alpha.(4 \text{ aft } \alpha)-{}_3)$ |
| 1,4,9,14,19,24,29,34 | $\psi(\lambda\alpha.(5 \text{ aft } \alpha)-{}_4)$ |
| 1,4,9,15 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$<br>$\psi(\psi_a(a_3))$ |
| 1,4,9,15,3 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0+1)$<br>$\psi(\psi_a(a_3)+1)$ |
| 1,4,9,15,7 | $\psi(1-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$<br>$\psi(\psi_a(a_3+\psi_{a_3}(a_3+1)))$ |
| 1,4,9,15,8 | $\psi(\lambda\alpha.(\alpha+1)-\Pi_0-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$<br>$\psi(\psi_a(a_3+a_2))$ |
| 1,4,9,15,8,12,15 | $\psi(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$<br>$\psi(\psi_a(a_3+a_2^{\psi_{a_2}(0)}))$ |
| 1,4,9,15,8,12,15,20,26 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0-$<br>$\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$<br>$\psi(\psi_a(a_3\cdot 2))$ |
| 1,4,9,15,8,12,15,20,26,11 | $\psi(1-\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0)-\Pi_1)$ |
| 1,4,9,15,8,12,<br>15,20,26,11,15 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0+1)-\Pi_0)$<br>$\psi(\psi_a(a_3\cdot\psi_{a_2}(a_2)))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,9,15,8,12,15, 20,26,11,15,19,21,5 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0+\alpha)-\Pi_0)$ $\psi(\psi_a(a_3\cdot\psi_{a_2}(a_2^a)))$ |
| 1,4,9,15,8,12,15,20,26,12 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0\cdot\omega)-\Pi_0)$ |
| 1,4,9,15,8,12, 15,20,26,12,14,5 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0\cdot\alpha)-\Pi_0)$ |
| 1,4,9,15,8,12,15, 20,26,12,15,20,26 | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1)-\Pi_0)^2)-\Pi_0)$ |
| 1,4,9,15,8,12,15,20,26,19 | $\psi((\lambda\alpha.(\psi_{\Omega_{\lambda\beta.(\beta+1)-\Pi_0+1}}(0))-\Pi_0)$ |
| 1,4,9,15,8,12, 15,20,26,19,23 | $\psi(\lambda\alpha.(\psi_{\Omega_{\lambda\beta.(\beta+1)-\Pi_0+1}}(\Omega_{\lambda\beta.(\beta+1)-\Pi_0+1})))-\Pi_0)$ |
| 1,4,9,15,9 | $\psi(\lambda\alpha.(\Omega_{\lambda\beta.(\beta+1)-\Pi_0+1})-\Pi_1)$ |
| 1,4,9,15,9,14,19,24 | $\psi((\lambda\alpha.(K_{\lambda\beta.(\beta+1)-\Pi_0+1})-\Pi_2)$ |
| 1,4,9,15,9,15 | $\psi((\lambda\alpha.(2\text{nd }\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,10 | $\psi((\lambda\alpha.(1-\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,10,10 | $\psi((\lambda\alpha.(1-1-\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,11,5 | $\psi((\lambda\alpha.((1-)^\alpha\ \lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,13 | $\psi((\lambda\alpha.((1-)^{1,0}\ \lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,14 | $\psi((\lambda\alpha.(2\ 1-\lambda\beta.(\beta+1)-\Pi_0)-\Pi_1)$ |
| 1,4,9,15,14,20 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0\ 1-\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,14,20,19 | $\psi(\lambda\alpha.(2-\lambda\beta.(\beta+1)-\Pi_0)-\Pi_1)$ |
| 1,4,9,15,14,20,19,25 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0\ 2-\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,14,20,19,25,24 | $\psi(\lambda\alpha.(3-\lambda\beta.(\beta+1)-\Pi_0)-\Pi_2)$ |
| 1,4,9,15,15 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0-\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,17,5 | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1)-\Pi_0-)^\alpha)-\Pi_0)$ |
| 1,4,9,15,19 | $\psi(\lambda\alpha.((\lambda\beta.(\beta+1)-\Pi_0-)^{1,0})-\Pi_0)$ $\psi(\lambda\alpha.(\psi_{\lambda\beta.(\beta+1)-\Pi_1}(0))-\Pi_0)$ |
| 1,4,9,15,19,19 | $\psi(\lambda\alpha.(\psi_{\lambda\beta.(\beta+1)-\Pi_1}(1))-\Pi_0)$ |
| 1,4,9,15,19,23 | $\psi(\lambda\alpha.(\psi_{\lambda\beta.(\beta+1)-\Pi_1}(\lambda\beta.(\beta+1)-\Pi_1))-\Pi_0)$ |
| 1,4,9,15,19,23,27 | $\psi(\lambda\alpha.(\psi_{\lambda\beta.(\beta+1)-\Pi_1}(\Omega_{\lambda\beta.(\beta+1)-\Pi_1+1})))-\Pi_0)$ |
| 1,4,9,15,19,25,32 | $\psi(\lambda\alpha.(\psi_{\lambda\beta.(\beta+1)-\Pi_1}(\lambda\alpha'.(\alpha'+1)-\Pi_0))-\Pi_0)$ |
| 1,4,9,15,19,25,33,42 | $\psi(\lambda\alpha.(\psi_{\lambda\beta.(\beta+1)-\Pi_1}(\lambda\alpha'.$ $(\lambda\beta'.(\beta'+1)-\Pi_0)-\Pi_0))-\Pi_0)$ |
| 1,4,9,15,20 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_1)-\Pi_1)$ $\psi(\psi_a(a_3\cdot\psi_{a_3}(0)))$ |
| 1,4,9,15,20,15 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_0-\lambda\beta.(\beta+1)-\Pi_1)-\Pi_0)$ |
| 1,4,9,15,20,15,20 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_1\ \lambda\beta.(\beta+1)-$ $\Pi_0-\lambda\beta.(\beta+1)-\Pi_1)-\Pi_1)$ $\psi(\psi_a(a_3\cdot\psi_{a_3}(0)\cdot 2))$ |
| 1,4,9,15,20,20 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-$ $\Pi_1-\lambda\beta.(\beta+1)-\Pi_1)-\Pi_1)$ |
| 1,4,9,15,20,25 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_2)-\Pi_2)$ |
| 1,4,9,15,20,26 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+2)-\Pi_0)-\Pi_0)$ $\psi(\psi_a(a_3\cdot\psi_{a_3}(a_3)))$ |
| 1,4,9,15,21 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\omega)-\Pi_0)-\Pi_0)$ $\psi(\psi_a(a_3^2))$ |
| 1,4,9,15,21,21 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\omega^2)-\Pi_0)-\Pi_0)$ $\psi(\psi_a(a_3^3))$ |
| 1,4,9,15,21,23,5 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\alpha)-\Pi_0)-\Pi_0)$ $\psi(\psi_a(a_3^a))$ |
| 1,4,9,15,21,24 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\Omega_{\alpha+1})-\Pi_0)-\Pi_0)$ $\psi(\psi_a(a_3^{\psi_{a_2}(0)}))$ |
| 1,4,9,15,21,24,28 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\psi_{\Omega_{\alpha+2}}(0))-\Pi_0)-\Pi_0)$ |

| $0-\mathrm{Y}$ sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,9,15,21,24,29 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\Omega_{\alpha+2})-\Pi_0)-\Pi_0)$ $\psi(\psi_a(a_3^{\psi_{a_2}(\psi_{a_3}(0))}))$ |
| 1,4,9,15,21,24,29,34 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+I_{\alpha+1})-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,21,24,29,35 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\lambda\beta.(\beta+1)-\Pi_0)-\Pi_0)-\Pi_0)$ $\psi(\psi_a(a_3^{\psi_{a_2}(a_3)}))$ |
| 1,4,9,15,21,24, 29,35,41,44,49,55 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+\lambda\beta.(\beta+\lambda\beta.(\beta+1)$ $-\Pi_0)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,21,25 | $\psi(\lambda\alpha.(\lambda\beta.(\beta\cdot 2)-\Pi_0)-\Pi_0)$ $\psi(\psi_a(a_3^{a_2}))$ |
| 1,4,9,15,21,25,20 | $\psi(\lambda\alpha.(\lambda\beta.(\beta\cdot 2)-\Pi_1)-\Pi_1)$ $\psi(\psi_a(a_3^{a_2}\cdot\psi_{a_3}(0)))$ |
| 1,4,9,15,21,25,21 | $\psi(\lambda\alpha.(\lambda\beta.(\beta\cdot\omega)-\Pi_0)-\Pi_0)$ $\psi(\psi_a(a_3^{a_2+1}))$ |
| 1,4,9,15,21,25,21,25 | $\psi(\lambda\alpha.(\lambda\beta.(\beta^2)-\Pi_0)-\Pi_0)$ $\psi(\psi_a(a_3^{a_2\cdot 2}))$ |
| 1,4,9,15,21,25,25 | $\psi(\lambda\alpha.(\lambda\beta.(\beta^\beta)-\Pi_0)-\Pi_0)$ $\psi(\psi_a(a_3^{a_2^2}))$ |
| 1,4,9,15,21,25,30 | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(0))-\Pi_0)-\Pi_0)$ $\psi(\psi_a(a_3^{\psi_{\Omega_{a_2+1}}(0)}))$ |
| 1,4,9,15,21,25,30,35 | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\Omega_{\beta+1}))-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,21,25,31 | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\Omega_{\beta+\omega}))-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,21,25,31,38 | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\alpha'.$ $(\alpha'+1)-\Pi_0))-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,21,25,31,39,48 | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\alpha'.(\lambda\beta'.(\beta'+1)$ $-\Pi_0)-\Pi_0))-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,21,25,31,39,48,57 | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\alpha'.(\lambda\beta'.(\beta'+\omega)$ $-\Pi_0)-\Pi_0))-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,21,25, 31,39,48,57,64,72 | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+1}}(\lambda\alpha'.(\lambda\beta'.(\psi_{\Omega_{\beta'+1}}(0))$ $-\Pi_0)-\Pi_0))-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,21,26 | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+1})-\Pi_1)-\Pi_1)$ $\psi(\psi_a(a_3^{\psi_{a_3}(0)}))$ |
| 1,4,9,15,21,26,21 | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+1}\cdot\omega)-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,21,26,21,26 | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+1}^2)-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,21,26,26 | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+1}^{\Omega_{\beta+1}})-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,21,26,32 | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(0))-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,21,26,32,38 | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\Omega_{\beta+2}))-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,21,27 | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\Omega_{\beta+2}^{\Omega_{\beta+2}}))-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,22 | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\psi_{\Omega_{\beta+3}}(0)))-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,23 | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\Omega_{\beta+\omega}))-\Pi_0)-\Pi_0)$ |
| 1,4,9,15,23,32 | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\lambda\alpha'.(\alpha'+1)-\Pi_0)))-\Pi_0)-\Pi_0)))$ |
| 1,4,9,15,23,33,44 | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{\Omega_{\beta+2}}(\lambda\alpha'.$ $(\lambda\beta'.(\beta'+1)-\Pi_0)-\Pi_0))-\Pi_0)-\Pi_0)$ |
| 1,4,9,16 | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+2})-\Pi_1)-\Pi_1)$ $\psi(\psi_a(\Omega_{a_3+1}))$ |
| 1,4,9,16,16 | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+3})-\Pi_1)-\Pi_1)$ $\psi(\psi_a(\Omega_{a_3+1}\cdot 2))$ |
| 1,4,9,16,17 | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+\omega})-\Pi_0)-\Pi_0)$ $\psi(\psi_a(\Omega_{a_3+1}\cdot\omega))$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,9,16,18,5 | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta+\alpha}) - \Pi_0) - \Pi_0)$<br>$\psi(\psi_a(\Omega_{a_3+1} \cdot a))$ |
| 1,4,9,16,20 | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta\cdot 2}) - \Pi_0) - \Pi_0)$<br>$\psi(\psi_a(\Omega_{a_3+1} \cdot a_2))$ |
| 1,4,9,16,20,16,20 | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta\cdot 3}) - \Pi_0) - \Pi_0)$ |
| 1,4,9,16,20,20 | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\beta^2}) - \Pi_0) - \Pi_0)$ |
| 1,4,9,16,20,25 | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\psi_{\Omega_{\beta+1}}(0)}) - \Pi_0) - \Pi_0)$<br>$\psi(\psi_a(\Omega_{a_3+1} \cdot \psi_{\Omega_{a_2+1}}(0)))$ |
| 1,4,9,16,21 | $\psi(\lambda\alpha.(\lambda\beta.(\Omega_{\Omega_{\beta+1}}) - \Pi_0) - \Pi_0)$<br>$\psi(\psi_a(\Omega_{a_3+1} \cdot \psi_{a_3}(0)))$ |
| 1,4,9,16,22 | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{I_{\beta+1}}(0)) - \Pi_0) - \Pi_0)$<br>$\psi(\psi_a(\Omega_{a_3+1} \cdot a_3))$ |
| 1,4,9,16,23 | $\psi(\lambda\alpha.(\lambda\beta.(I_{\beta+1}) - \Pi_1) - \Pi_1)$<br>$\psi(\psi_a(\Omega^2_{a_3+1}))$ |
| 1,4,9,16,23,23 | $\psi(\lambda\alpha.(\lambda\beta.(I(1,\beta+1)) - \Pi_1) - \Pi_1)$<br>$\psi(\psi_a(\Omega^3_{a_3+1}))$ |
| 1,4,9,16,23,24 | $\psi(\lambda\alpha.(\lambda\beta.(I(\omega,\beta+1)) - \Pi_0) - \Pi_0)$<br>$\psi(\psi_a(\Omega^\omega_{a_3+1}))$ |
| 1,4,9,16,23,25,5 | $\psi(\lambda\alpha.(\lambda\beta.(I(\alpha,\beta+1)) - \Pi_0) - \Pi_0)$<br>$\psi(\psi_a(\Omega^a_{a_3+1}))$ |
| 1,4,9,16,23,27 | $\psi(\lambda\alpha.(\lambda\beta.(I(\beta,1)) - \Pi_0) - \Pi_0)$<br>$\psi(\psi_a(\Omega^{a_2}_{a_3+1}))$ |
| 1,4,9,16,23,29 | $\psi(\lambda\alpha.(\lambda\beta.(\psi_{I(1,0,\beta+1)}(0)) - \Pi_0) - \Pi_0)$<br>$\psi(\psi_a(\Omega^{a_3}_{a_3+1}))$ |
| 1,4,9,16,23,29,23 | $\psi(\lambda\alpha.(\lambda\beta.(I(1,0,\beta+1)) - \Pi_1) - \Pi_1)$<br>$\psi(\psi_a(\Omega^{a_3+1}_{a_3+1}))$ |
| 1,4,9,16,23,30 | $\psi(\lambda\alpha.(\lambda\beta.(M_{\beta+1}) - \Pi_1) - \Pi_1)$<br>$\psi(\psi_a(\Omega^{\Omega_{a_3+1}}_{a_3+1}))$ |
| 1,4,9,16,23,30,30 | $\psi(\lambda\alpha.(\lambda\beta.(N_{\beta+1}) - \Pi_1) - \Pi_1)$<br>$\psi(\psi_a(\Omega^{\Omega^2_{a_3+1}}_{a_3+1}))$ |
| 1,4,9,16,23,30,37 | $\psi(\lambda\alpha.(\lambda\beta.(K_{\beta+1}) - \Pi_2) - \Pi_2)$<br>$\psi(\psi_a(\Omega^{\Omega^{\Omega_{a_3+1}}_{a_3+1}}_{a_3+1}))$ |
| 1,4,9,16,24 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_0)$<br>$\psi(3 - \pi - (+1) - \Pi_0)$<br>$\psi(\psi_a(a_4))$<br>TSO |
| 1,4,9,16,24,9,16,24 | $\psi(\lambda\alpha.(2\text{nd } \lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| 1,4,9,16,24,15 | $\psi(\lambda\alpha.(\lambda\beta.(\beta+1) - \Pi_0 - \lambda\beta.$<br>$(\lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| 1,4,9,16,24,15,<br>21,25,31,39,48 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0 -$<br>$\lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| 1,4,9,16,24,16 | $\psi(\lambda\alpha.(\lambda\beta.(2 \text{ aft } \lambda\gamma.(\gamma+1) - {}_0) - {}_1) - {}_1)$ |
| 1,4,9,16,24,16,24 | $\psi(\lambda\alpha.(\lambda\beta.(2\text{nd } \lambda\gamma.(\gamma+1) - {}_0) - {}_0) - {}_0)$ |
| 1,4,9,16,24,24 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_0 -$<br>$\lambda\gamma.(\gamma+1) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| 1,4,9,16,24,31 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+1) - \Pi_1) - \Pi_1) - \Pi_1)$ |
| 1,4,9,16,24,31,39 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+2) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| 1,4,9,16,24,32 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+\omega) - \Pi_0) - \Pi_0) - \Pi_0)$ |
| 1,4,9,16,24,32,34,5 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+\alpha) - \Pi_0) - \Pi_0) - \Pi_0)$ |

| 0 − Y sequence | MOCF/Reflection/Stable |
|---|---|
| 1,4,9,16,24,32,36 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+\beta)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| 1,4,9,16,24,32,38 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma\cdot 2)-\Pi_0)-\Pi_0)-\Pi_0)$ |
| 1,4,9,16,24,32,38,32 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma\cdot\omega)-\Pi_0)-\Pi_0)-\Pi_0)$<br>$\psi(\psi_a(a_4^{a_3+1}))$ |
| 1,4,9,16,24,32,38,32,38 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma^2)-\Pi_0)-\Pi_0)-\Pi_0)$<br>$\psi(\psi_a(a_4^{a_3\cdot 2}))$ |
| 1,4,9,16,24,32,38,45 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\psi_{\Omega_{\gamma+1}}(0))-\Pi_0)-\Pi_0)-\Pi_0)$<br>$\psi(\psi_a(a_4^{\psi_{\Omega_{a_3+1}}(\Omega_{a_3+1})}))$ |
| 1,4,9,16,24,32,39 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Omega_{\gamma+1})-\Pi_1)-\Pi_1)-\Pi_1)$<br>$\psi(\psi_a(a_4^{\psi_{a_4}(0)}))$ |
| 1,4,9,16,25 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Omega_{\gamma+2})-\Pi_1)-\Pi_1)-\Pi_1)$<br>$\psi(\psi_a(\Omega_{a_4+1}))$ |
| 1,4,9,16,25,25 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Omega_{\gamma+3})-\Pi_1)-\Pi_1)-\Pi_1)$<br>$\psi(\psi_a(\Omega_{a_4+1}\cdot 2))$ |
| 1,4,9,16,25,26 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Omega_{\gamma+\omega})-\Pi_0)-\Pi_0)-\Pi_0)$<br>$\psi(\psi_a(\Omega_{a_4+1}\cdot\omega))$ |
| 1,4,9,16,25,27,5 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Omega_{\gamma+\alpha})-\Pi_0)-\Pi_0)-\Pi_0)$<br>$\psi(\psi_a(\Omega_{a_4+1}\cdot a))$ |
| 1,4,9,16,25,29 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Omega_{\gamma+\beta})-\Pi_0)-\Pi_0)-\Pi_0)$<br>$\psi(\psi_a(\Omega_{a_4+1}\cdot a_2))$ |
| 1,4,9,16,25,31 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\Omega_{\gamma\cdot 2})-\Pi_0)-\Pi_0)-\Pi_0)$<br>$\psi(\psi_a(\Omega_{a_4+1}\cdot a_3))$ |
| 1,4,9,16,25,33 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\psi_{I_{\gamma+1}}(0))-\Pi_0)-\Pi_0)-\Pi_0)$<br>$\psi(\psi_a(\Omega_{a_4+1}\cdot a_4))$ |
| 1,4,9,16,25,34 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(I_{\gamma+1})-\Pi_1)-\Pi_1)-\Pi_1)$<br>$\psi(\psi_a(\Omega^2_{a_4+1}))$ |
| 1,4,9,16,25,34,43 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(M_{\gamma+1})-\Pi_1)-\Pi_1)-\Pi_1)$<br>$\psi(\psi_a(\Omega_{a_4+1}^{\Omega_{a_4+1}}))$ |
| 1,4,9,16,25,34,43,52 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(K_{\gamma+1})-\Pi_2)-\Pi_2)-\Pi_2)$<br>$\psi(\psi_a(\Omega_{a_4+1}^{\Omega_{a_4+1}^{\Omega_{a_4+1}}}))$ |
| 1,4,9,16,25,35 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\lambda\delta.(\delta+1)-\Pi_0)-\Pi_0)-\Pi_0)-\Pi_0)$<br>$\psi(4-\pi-(+1)-\Pi_0)$<br>$\psi(\psi_a(a_5))$ |
| 1,4,9,16,25,36,48 | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\lambda\delta.(\lambda\epsilon.(\epsilon+1)$<br>$-\Pi_0)-\Pi_0)-\Pi_0)-\Pi_0)-\Pi_0)$<br>$\psi(\psi_a(a_6))$ |
| 1,4,10 | $\psi(\omega-\pi-\Pi_0)$<br>$\psi(\psi_a(a_\omega))$<br>$\psi(\psi_a(\psi_b(a_{b+1}\cdot\omega)))$<br>p.f.e.c.LRO |

## A.17 0 − Y vs Projection

| 0 − Y | Projection |
|---|---|
| 1,4,7,10,10,7,9,<br>5 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha}))$ |
| 1,4,7,10,10,7,9,<br>5,5 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha}+1))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,7,10,10,7,9, 6,9 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\alpha}+\Omega_{\alpha+1}))$ |
| 1,4,7,10,10,7,9, 6,10 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\alpha}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,7,10,10,7,9, 6,10,14,18,15 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\alpha}+{\Omega_{\alpha+1}}^{\omega}))$ |
| 1,4,7,10,10,7,9, 6,10,14,18,17,11 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\alpha+{\Omega_{\alpha+1}}^{\alpha}}))$ |
| 1,4,7,10,10,7,9, 6,10,14,18,17,21 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\alpha}+{\Omega_{\alpha+1}}^{\varepsilon_{\alpha+1}}))$ $=\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot(\alpha+1)}+\Omega_{\alpha+1}))$ |
| 1,4,7,10,10,7,9, 6,10,14,18,18 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot(\alpha+\omega)}))$ |
| 1,4,7,10,10,7,9, 6,10,14,18,18,14 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot(\alpha+\omega^2)}))$ |
| 1,4,7,10,10,7,9, 6,10,14,18,18,14,17 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot(\alpha+\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\alpha}+\Omega_{\alpha+1}))}))$ |
| 1,4,7,10,10,7,9, 6,10,14,18,18,14,17, 11 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\alpha\cdot 2}))$ |
| 1,4,7,10,10,7,9, 6,10,14,18,18,14,17, 13,18,23,28,28 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot(\alpha\cdot 2+\omega)}))$ |
| 1,4,7,10,10,7,9, 7 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\alpha\cdot\omega}))$ |
| 1,4,7,10,10,7,9, 7,6,10,14,18,18,14, 17,14 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\alpha\cdot\omega\cdot 2}))$ |
| 1,4,7,10,10,7,9, 7,7 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\alpha\cdot\omega^2}))$ |
| 1,4,7,10,10,7,9, 7,9 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\alpha\cdot\Omega}))$ |
| 1,4,7,10,10,7,9, 7,9,5 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\alpha^2}))$ |
| 1,4,7,10,10,7,9, 12 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\varepsilon_{\alpha+1}}))$ $=\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+1}+\Omega_{\alpha+1}))$ |
| 1,4,7,10,10,7,9, 13,17,21,18 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+1}+{\Omega_{\alpha+1}}^{\omega}))$ |
| 1,4,7,10,10,7,9, 13,17,21,21 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+1}+{\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\omega}))$ |
| 1,4,7,10,10,7,9, 13,17,21,21,17,20,14 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+1}\cdot 2))$ |
| 1,4,7,10,10,7,10 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+1}\cdot\omega))$ |
| 1,4,7,10,10,7,10, 7 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+1}\cdot\omega^2))$ |
| 1,4,7,10,10,7,10, 7,9,5 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+2}))$ $=\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+1}\cdot\alpha))$ |
| 1,4,7,10,10,7,10, 8 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\omega}))$ |
| 1,4,7,10,10,7,10, 9,5 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\alpha}))$ |
| 1,4,7,10,10,7,10, 9,5,5 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\alpha})\cdot\omega)$ |
| 1,4,7,10,10,7,10, 9,6,10,14,18,18,14, 18,17,11 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\alpha}\cdot 2))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,7,10,10,7,10, 9,7 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}+\alpha}\cdot\omega))$ |
| 1,4,7,10,10,7,10, 9,7,9,12 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}+\alpha}\cdot\varepsilon_{\alpha+1}))$ $=\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}+\alpha+1}+\Omega_{\alpha+1}))$ |
| 1,4,7,10,10,7,10, 9,7,10 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}+\alpha+1}\cdot\omega))$ |
| 1,4,7,10,10,7,10, 9,7,10,8 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}+\alpha+\omega}))$ |
| 1,4,7,10,10,7,10, 9,7,10,9,5 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}+\alpha\cdot 2}))$ |
| 1,4,7,10,10,7,10, 9,8 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}+\alpha\cdot\omega}))$ |
| 1,4,7,10,10,7,10, 9,12 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}+\varepsilon_{\alpha+1}}))$ $=\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2}+\Omega_{\alpha+1}))$ |
| 1,4,7,10,10,7,10, 9,13 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,7,10,10,7,10, 9,13,17,21,21 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2}+\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}\cdot\omega))$ |
| 1,4,7,10,10,7,10, 9,13,17,21,21,17,21, 20,14 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2}+\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}+\alpha}))$ |
| 1,4,7,10,10,7,10, 9,13,17,21,21,17,21, 20,24 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2}\cdot 2+\Omega_{\alpha+1}))$ |
| 1,4,7,10,10,7,10, 10 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2}\cdot\omega))$ |
| 1,4,7,10,10,7,10, 10,7 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2}\cdot\omega^2))$ |
| 1,4,7,10,10,7,10, 10,7,9,5 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2}\cdot\alpha))$ |
| 1,4,7,10,10,7,10, 10,7,10 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2+1}\cdot\omega))$ |
| 1,4,7,10,10,7,10, 10,7,10,9,5 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2+\alpha}))$ |
| 1,4,7,10,10,7,10, 10,7,10,9,12 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 2+\varepsilon_{\alpha+1}}))$ |
| 1,4,7,10,10,7,10, 10,7,10,10 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot 3}\cdot\omega))$ |
| 1,4,7,10,10,8 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\omega}))$ |
| 1,4,7,10,10,8,7, 10,10 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot(\omega+1)}\cdot\omega))$ |
| 1,4,7,10,10,8,8 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\omega^2}))$ |
| 1,4,7,10,10,9,5 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha}))$ |
| 1,4,7,10,10,9,6, 10,14,18,18,17,11 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha}\cdot 2))$ |
| 1,4,7,10,10,9,7, 10 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha+1}\cdot\omega))$ |
| 1,4,7,10,10,9,7, 10,10 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot(\alpha+1)}\cdot\omega))$ |
| 1,4,7,10,10,9,7, 10,10,9,5 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha\cdot 2}))$ |
| 1,4,7,10,10,9,8 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha\cdot\omega}))$ |
| 1,4,7,10,10,9,9, 5 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\alpha^2}))$ |
| 1,4,7,10,10,9,12 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}\cdot\varepsilon_{\alpha+1}}))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,7,10,10,9,13 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2} + \Omega_{\alpha+1} \cdot \omega))$ |
| 1,4,7,10,10,9,13, 17,21,21,20,24 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2} \cdot 2 + \Omega_{\alpha+1}))$ |
| 1,4,7,10,10,10 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2} \cdot \omega))$ |
| 1,4,7,10,10,10,7, 9,5 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2} \cdot \alpha))$ |
| 1,4,7,10,10,10,7, 9,7 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2} \cdot \alpha \cdot \omega))$ |
| 1,4,7,10,10,10,7, 9,12 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2+1} + \Omega_{\alpha+1}))$ |
| 1,4,7,10,10,10,7, 9,13,17,21,21,21 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2+1} + \Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2} \cdot \omega))$ |
| 1,4,7,10,10,10,7, 10 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2+1} \cdot \omega))$ |
| 1,4,7,10,10,10,7, 10,9,12 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2+\Omega_{\alpha+1}} + \Omega_{\alpha+1}))$ |
| 1,4,7,10,10,10,7, 10,10 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2+\Omega_{\alpha+1}} \cdot \omega))$ |
| 1,4,7,10,10,10,7, 10,10,10 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2 \cdot 2} \cdot \omega))$ |
| 1,4,7,10,10,10,9, 5 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2 \cdot \alpha}))$ |
| 1,4,7,10,10,10,9, 12 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2 \cdot \varepsilon_{\alpha+1}}))$ |
| 1,4,7,10,10,10,9, 13,17,21,21,21 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^3} + \Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^2} \cdot \omega))$ |
| 1,4,7,10,10,10,10 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^3} \cdot \omega))$ |
| 1,4,7,10,11 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^\omega}))$ |
| 1,4,7,10,12 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^\Omega}))$ |
| 1,4,7,10,12,4,7, 10,11 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^\omega})}}))$ |
| 1,4,7,10,12,5 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^\alpha}))$ |
| 1,4,7,10,12,7 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^\alpha} \cdot \omega))$ |
| 1,4,7,10,12,7,10, 11 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^\alpha+\Omega_{\alpha+1}{}^\omega}))$ |
| 1,4,7,10,12,9,5 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^\alpha \cdot \alpha}))$ |
| 1,4,7,10,12,10 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\alpha+1}} \cdot \omega))$ |
| 1,4,7,10,12,10,12, 5 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\alpha \cdot 2}}))$ |
| 1,4,7,10,12,12,5 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\alpha^2}}))$ |
| 1,4,7,10,12,15 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\varepsilon_{\alpha+1}}}))$ $= \psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} + \Omega_{\alpha+1}))$ |
| 1,4,7,10,12,16 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} + \Omega_{\alpha+1} \cdot \omega))$ |
| 1,4,7,10,12,16,20, 23,17 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} + \Omega_{\alpha+1} \cdot \alpha))$ |
| 1,4,7,10,12,16,20, 24,25 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} + \Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^\omega}))$ |
| 1,4,7,10,12,16,20, 24,27,31 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} \cdot 2 + \Omega_{\alpha+1}))$ |
| 1,4,7,10,13 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} \cdot \omega))$ |
| 1,4,7,10,13,6,10, 14,18,22 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} \cdot \omega \cdot 2))$ |
| 1,4,7,10,13,7 | $\psi(\psi_\alpha(\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} \cdot \omega^2))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,7,10,13,7,9, 5 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{\Omega_{\alpha+1}}} \cdot \alpha))$ |
| 1,4,7,10,13,7,10 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+1}} \cdot \omega))$ |
| 1,4,7,10,13,7,10, 12,5 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+{\Omega_{\alpha+1}}^{\alpha}}}))$ |
| 1,4,7,10,13,7,10, 13 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot 2}} \cdot \omega))$ |
| 1,4,7,10,13,10 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+1}} \cdot \omega))$ |
| 1,4,7,10,13,10,12, 5 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{\Omega_{\alpha+1}+\alpha}}))$ |
| 1,4,7,10,13,10,13 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot 2}} \cdot \omega))$ |
| 1,4,7,10,13,11 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\omega}}))$ |
| 1,4,7,10,13,12,5 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{\Omega_{\alpha+1}\cdot\alpha}}))$ |
| 1,4,7,10,13,13 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{2}}}))$ |
| 1,4,7,10,13,15,5 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{\alpha}}}))$ |
| 1,4,7,10,13,16 | $\psi(\psi_\alpha({\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{\Omega_{\alpha+1}}}} \cdot \omega))$ |
| 1,4,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}))$ $= \psi(\Pi_\omega)$ |
| 1,4,8,4 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) + \Omega_\omega)$ |
| 1,4,8,4,7,10 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) + I_\omega)$ |
| 1,4,8,4,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot 2)$ |
| 1,4,8,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \omega)$ |
| 1,4,8,6 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega)$ |
| 1,4,8,6,3,7,12, 9,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega \cdot \omega)$ |
| 1,4,8,6,3,7,12, 10 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_2)$ |
| 1,4,8,6,4 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_\omega)$ |
| 1,4,8,6,4,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_\omega + \Omega_\omega \cdot \omega)$ |
| 1,4,8,6,4,6,9 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_\omega + \psi_\omega(\Omega_{\omega+1}))$ |
| 1,4,8,6,4,6,10, 15,12,4 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_\omega + \psi_\omega(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_\omega))$ |
| 1,4,8,6,4,6,10, 15,12,9 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_\omega + \Omega_{\omega+1})$ |
| 1,4,8,6,4,6,10, 15,12,10,15 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot (\Omega_\omega + 1))$ |
| 1,4,8,6,4,6,10, 15,12,10,15,12,4 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_\omega \cdot 2)$ |
| 1,4,8,6,4,6,10, 15,12,11 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_\omega \cdot \omega)$ |
| 1,4,8,6,4,6,10, 15,12,12,4 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot {\Omega_\omega}^2)$ |
| 1,4,8,6,4,6,10, 15,12,15 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \psi_\omega(\Omega_{\omega+1}))$ |
| 1,4,8,6,4,6,10, 15,13 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_{\omega+1})$ |
| 1,4,8,6,4,6,10, 15,13,10 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_{\omega\cdot 2})$ |
| 1,4,8,6,4,7 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \Omega_{\omega^2})$ |
| 1,4,8,6,4,7,9, 5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot \psi_I(I))$ |
| 1,4,8,6,4,7,9, 13,18,15,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}) \cdot I)$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,8,6,4,7,9, 13,18,16 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot\Omega_{I+1}))$ |
| 1,4,8,6,4,7,10 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1})\cdot I_\omega)$ |
| 1,4,8,6,4,7,10, 13 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1})\cdot\psi_\alpha({\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{\Omega_{\alpha+1}}}\cdot\omega))$ |
| 1,4,8,6,4,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1})^2)$ |
| 1,4,8,6,4,8,4, 8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1})^2+\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,8,6,4,8,6, 4,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1})^2\cdot 2)$ |
| 1,4,8,6,6,4,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1})^3)$ |
| 1,4,8,6,7 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1})^\omega)$ |
| 1,4,8,6,8,4,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1})^{\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1})})$ |
| 1,4,8,6,9 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}+\Omega_{\alpha+1}))$ |
| 1,4,8,6,9,9 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}+\Omega_{\alpha+1})\cdot 2)$ |
| 1,4,8,6,10 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,8,6,10,14,18 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}+{\Omega_{\alpha+1}}^2\cdot\omega))$ |
| 1,4,8,6,10,15 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot 2))$ |
| 1,4,8,6,10,15,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot 2)+$ $\psi_{\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}+\Omega_{\alpha+1})}(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot 2)+1))$ |
| 1,4,8,6,10,15,9 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot 2)+\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}+\Omega_{\alpha+1}))$ |
| 1,4,8,6,10,15,10, 15 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot 2)\cdot 2)$ |
| 1,4,8,6,10,15,11 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot 2)\cdot\omega)$ |
| 1,4,8,6,10,15,13 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot 2)\cdot\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}+\Omega_{\alpha+1}))$ |
| 1,4,8,6,10,15,13, 10,15 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot 2)^2)$ |
| 1,4,8,6,10,15,13, 17 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot 2+\Omega_{\alpha+1}))$ |
| 1,4,8,6,10,15,13, 18,24 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot 3))$ |
| 1,4,8,7 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot\omega))$ |
| 1,4,8,7,4,8,7 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot\omega)\cdot 2)$ |
| 1,4,8,7,6,9 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot\omega+\Omega_{\alpha+1}))$ |
| 1,4,8,7,6,10,15 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot(\omega+1)))$ |
| 1,4,8,7,7 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot\omega^2))$ |
| 1,4,8,7,9 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot\Omega))$ |
| 1,4,8,7,9,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot\alpha))$ |
| 1,4,8,7,9,6,10, 15,14,17,11 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot\alpha\cdot 2))$ |
| 1,4,8,7,9,7 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot\alpha\cdot\omega))$ |
| 1,4,8,7,9,12 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot\varepsilon_{\alpha+1}))$ |
| 1,4,8,7,9,13 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot\Omega_{\alpha+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,8,7,9,13,18 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot\Omega_{\alpha+1}+\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,8,7,9,13,18, 17,20,14 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot(\Omega_{\alpha+1}+\alpha)))$ |
| 1,4,8,7,9,13,18, 17,20,24 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot\Omega_{\alpha+1}\cdot 2+\Omega_{\alpha+1}))$ |
| 1,4,8,7,10 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,8,7,10,7 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot\Omega_{\alpha+1}\cdot\omega^2))$ |
| 1,4,8,7,10,7,9, 5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot\Omega_{\alpha+1}\cdot\alpha))$ |
| 1,4,8,7,10,7,10 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot{\Omega_{\alpha+1}}^2\cdot\omega))$ |
| 1,4,8,7,10,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot{\Omega_{\alpha+1}}^\omega))$ |
| 1,4,8,7,10,9,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}\cdot{\Omega_{\alpha+1}}^\alpha))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,8,7,10,9,12 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot {\Omega_{\alpha+1}}^{\varepsilon_{\alpha+1}}))$ |
| 1,4,8,7,10,11 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1} \cdot {\Omega_{\alpha+1}}^{{\Omega_{\alpha+1}}^{\omega}}))$ |
| 1,4,8,7,11 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^2))$ |
| 1,4,8,7,11,6,9 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^2 + \Omega_{\alpha+1}))$ |
| 1,4,8,7,11,6,10, 15,14,19 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^2 \cdot 2))$ |
| 1,4,8,7,11,7 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^2 \cdot \omega))$ |
| 1,4,8,7,11,7,9, 5 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^2 \cdot \alpha))$ |
| 1,4,8,7,11,7,10 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^2 \cdot \Omega_{\alpha+1} \cdot \omega))$ |
| 1,4,8,7,11,7,11 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^3))$ |
| 1,4,8,7,11,9,5 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^\alpha))$ |
| 1,4,8,7,11,9,5, 5 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^\alpha) \cdot \omega)$ |
| 1,4,8,7,11,9,6, 9 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^\alpha + \Omega_{\alpha+1}))$ |
| 1,4,8,7,11,9,6, 10,15 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^\alpha + \varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,8,7,11,9,6, 10,15,14,19,16,4,8, 7,11,9,5 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^\alpha + \varepsilon_{\Omega_{\alpha+1}+1}{}^{\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^\alpha)}))$ |
| 1,4,8,7,11,9,6, 10,15,14,19,17 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^\alpha + \varepsilon_{\Omega_{\alpha+1}+1}{}^{\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^\alpha+\Omega_{\alpha+1})}))$ |
| 1,4,8,7,11,9,6, 10,15,14,19,17,11 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^\alpha \cdot 2))$ |
| 1,4,8,7,11,9,7 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^\alpha \cdot \omega))$ |
| 1,4,8,7,11,9,7, 11,9,5 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^\alpha \cdot 2))$ |
| 1,4,8,7,11,9,9, 5 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{\alpha^2}))$ |
| 1,4,8,7,11,9,12 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{\varepsilon_{\alpha+1}}))$ |
| 1,4,8,7,11,9,13 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{\Omega_{\alpha+1}} + \Omega_{\alpha+1} \cdot \omega))$ |
| 1,4,8,7,11,10 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{\Omega_{\alpha+1}} \cdot \omega))$ |
| 1,4,8,7,11,10,7 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{\Omega_{\alpha+1}} \cdot \omega^2))$ |
| 1,4,8,7,11,10,7, 11 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{\Omega_{\alpha+1}+1}))$ |
| 1,4,8,7,11,10,7, 11,9,5 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{\Omega_{\alpha+1}+\alpha}))$ |
| 1,4,8,7,11,10,7, 11,10 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{\Omega_{\alpha+1}\cdot 2} \cdot \omega))$ |
| 1,4,8,7,11,10,8 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{\Omega_{\alpha+1}\cdot\omega}))$ |
| 1,4,8,7,11,10,9, 5 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{\Omega_{\alpha+1}\cdot\alpha}))$ |
| 1,4,8,7,11,10,9, 12 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{\Omega_{\alpha+1}\cdot\varepsilon_{\alpha+1}}))$ |
| 1,4,8,7,11,10,10 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{{\Omega_{\alpha+1}}^2} \cdot \omega))$ |
| 1,4,8,7,11,10,10, 7,11,10,10 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{{\Omega_{\alpha+1}}^2\cdot 2} \cdot \omega))$ |
| 1,4,8,7,11,10,10, 9,5 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{{\Omega_{\alpha+1}}^2\cdot\alpha}))$ |
| 1,4,8,7,11,10,10, 10 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{{\Omega_{\alpha+1}}^3} \cdot \omega))$ |
| 1,4,8,7,11,10,11 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{{\Omega_{\alpha+1}}^{\omega}}))$ |
| 1,4,8,7,11,10,12, 5 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+1}+1}}^{{\Omega_{\alpha+1}}^{\alpha}}))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,8,7,11,10,12, 10,12,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}{}^{\alpha\cdot 2}}))$ |
| 1,4,8,7,11,10,12, 15 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}{}^{\varepsilon_{\alpha+1}}}))$ |
| 1,4,8,7,11,10,13 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}\cdot\omega}))$ |
| 1,4,8,7,11,10,14 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\varepsilon_{\Omega_{\alpha+1}+1}}))$ |
| 1,4,8,7,11,10,14, 7,11 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\varepsilon_{\Omega_{\alpha+1}+1}+1}))$ |
| 1,4,8,7,11,10,14, 7,11,10,14 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\varepsilon_{\Omega_{\alpha+1}+1}\cdot 2}))$ |
| 1,4,8,7,11,10,14, 10 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\varepsilon_{\Omega_{\alpha+1}+1}\cdot\Omega_{\alpha+1}}\cdot\omega))$ |
| 1,4,8,7,11,10,14, 10,14 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\varepsilon_{\Omega_{\alpha+1}+1}{}^2}))$ |
| 1,4,8,7,11,10,14, 12,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\varepsilon_{\Omega_{\alpha+1}+1}{}^\alpha}))$ |
| 1,4,8,7,11,10,14, 13 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\varepsilon_{\Omega_{\alpha+1}+1}{}^{\Omega_{\alpha+1}}}\cdot\omega))$ |
| 1,4,8,7,11,10,14, 13,17 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}{}^{\varepsilon_{\Omega_{\alpha+1}+1}{}^{\varepsilon_{\Omega_{\alpha+1}+1}}}))$ |
| 1,4,8,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+2}))$ |
| 1,4,8,8,6,9 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+2}+\Omega_{\alpha+1}))$ |
| 1,4,8,8,7 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+2}\cdot\omega))$ |
| 1,4,8,8,7,11 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+2}\cdot\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,8,8,7,11,11 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+2}{}^2))$ |
| 1,4,8,8,7,11,11, 9,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+2}{}^\alpha))$ |
| 1,4,8,8,7,11,11, 10,14 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+2}{}^{\varepsilon_{\Omega_{\alpha+1}+1}}))$ |
| 1,4,8,8,7,11,11, 10,14,14 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+2}{}^{\varepsilon_{\Omega_{\alpha+1}+2}}))$ |
| 1,4,8,8,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+3}))$ |
| 1,4,8,9 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+\omega}))$ |
| 1,4,8,10,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+\alpha}))$ |
| 1,4,8,10,7 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+\alpha}\cdot\omega))$ |
| 1,4,8,10,7,11,13, 5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+\alpha}{}^2))$ |
| 1,4,8,10,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+\alpha+1}))$ |
| 1,4,8,10,8,10,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+\alpha\cdot 2}))$ |
| 1,4,8,10,10,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+\alpha^2}))$ |
| 1,4,8,10,13 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}+\Omega_{\alpha+1}))$ |
| 1,4,8,10,14,19 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}+\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,8,10,14,19,20 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}+\varepsilon_{\Omega_{\alpha+1}+\omega}))$ |
| 1,4,8,10,14,19,22, 15 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}+\varepsilon_{\Omega_{\alpha+1}+\alpha}))$ |
| 1,4,8,11 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}\cdot\omega))$ |
| 1,4,8,11,7 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}\cdot\omega^2))$ |
| 1,4,8,11,7,9,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}\cdot\alpha))$ |
| 1,4,8,11,7,11 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}\cdot\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,8,11,7,11,14 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2}{}^2\cdot\omega))$ |
| 1,4,8,11,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2+1}))$ |
| 1,4,8,11,8,10,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 2+\alpha}))$ |
| 1,4,8,11,8,11 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot 3}\cdot\omega))$ |
| 1,4,8,11,10,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}\cdot\alpha}))$ |
| 1,4,8,11,10,13 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}{}^2}+\Omega_{\alpha+1}))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,8,11,11 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}{}^2} \cdot \omega))$ |
| 1,4,8,11,11,8,11, 11 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}{}^2 \cdot 2} \cdot \omega))$ |
| 1,4,8,11,11,11 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}{}^3} \cdot \omega))$ |
| 1,4,8,11,13,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}{}^\alpha}))$ |
| 1,4,8,11,14 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}{}^{\Omega_{\alpha+1}}} \cdot \omega))$ |
| 1,4,8,11,15 | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\Omega_{\alpha+1}+1}}))$ |
| 1,4,8,11,15,11,15 | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\Omega_{\alpha+1}+1} \cdot 2}))$ |
| 1,4,8,11,15,15 | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\Omega_{\alpha+1}+2}}))$ |
| 1,4,8,11,15,18 | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\Omega_{\alpha+1} \cdot 2}} \cdot \omega))$ |
| 1,4,8,11,15,18,22 | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\varepsilon_{\Omega_{\alpha+1}+1}}}))$ |
| 1,4,8,12 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+1}))$ |
| 1,4,8,12,6,9 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+1} + \Omega_{\alpha+1}))$ |
| 1,4,8,12,6,10,15, 20 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+1} \cdot 2))$ |
| 1,4,8,12,7 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+1} \cdot \omega))$ |
| 1,4,8,12,7,11 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+1} \cdot \varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,8,12,7,11,15 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+1}{}^2))$ |
| 1,4,8,12,8 | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+1}+1}))$ |
| 1,4,8,12,8,10,5 | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+1}+\alpha}))$ |
| 1,4,8,12,8,11,15, 19 | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+1} \cdot 2}))$ |
| 1,4,8,12,8,11,15, 19,10,5 | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+1} \cdot \alpha}))$ |
| 1,4,8,12,8,11,15, 19,11 | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+1} \cdot \Omega_{\alpha+1}} \cdot \omega))$ |
| 1,4,8,12,8,11,15, 19,11,15,19 | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+1}{}^2}))$ |
| 1,4,8,12,8,11,15, 19,15 | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\zeta_{\Omega_{\alpha+1}+1}+1}+1}))$ |
| 1,4,8,12,8,12 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+2}))$ |
| 1,4,8,12,8,12,8 | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+2}+1}))$ |
| 1,4,8,12,8,12,8, 11,15,19,15,19 | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+2} \cdot 2}))$ |
| 1,4,8,12,8,12,8, 12 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+3}))$ |
| 1,4,8,12,10,5 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+\alpha}))$ |
| 1,4,8,12,10,7 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+\alpha} \cdot \omega))$ |
| 1,4,8,12,10,8 | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+1}+\alpha}+1}))$ |
| 1,4,8,12,10,8,12 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+\alpha+1}))$ |
| 1,4,8,12,10,8,12, 10 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}+\alpha \cdot 2}))$ |
| 1,4,8,12,10,13 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1} \cdot 2} + \Omega_{\alpha+1}))$ |
| 1,4,8,12,11 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1} \cdot 2} \cdot \omega))$ |
| 1,4,8,12,11,9 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1} \cdot \omega}))$ |
| 1,4,8,12,11,11 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+1}{}^2} \cdot \omega))$ |
| 1,4,8,12,11,15 | $\psi(\psi_\alpha(\zeta_{\varepsilon_{\Omega_{\alpha+1}+1}+1}))$ |
| 1,4,8,12,11,15,19 | $\psi(\psi_\alpha(\zeta_{\zeta_{\Omega_{\alpha+1}+1}+1}))$ |
| 1,4,8,12,12 | $\psi(\psi_\alpha(\eta_{\Omega_{\alpha+1}+1}))$ |
| 1,4,8,12,13 | $\psi(\psi_\alpha(\varphi(\omega, \Omega_{\alpha+1} + 1)))$ |
| 1,4,8,12,14,5 | $\psi(\psi_\alpha(\varphi(\alpha, \Omega_{\alpha+1} + 1)))$ |
| 1,4,8,12,14,8,12, 14,5 | $\psi(\psi_\alpha(\varphi(\alpha, \Omega_{\alpha+1} + 2)))$ |
| 1,4,8,12,14,12 | $\psi(\psi_\alpha(\varphi(\alpha + 1, \Omega_{\alpha+1} + 1)))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,8,12,14,17 | $\psi(\psi_\alpha(\varphi(\Omega_{\alpha+1},1)+\Omega_{\alpha+1}))$ |
| 1,4,8,12,15 | $\psi(\psi_\alpha(\varphi(\Omega_{\alpha+1},1)\cdot\omega))$ |
| 1,4,8,12,15,19 | $\psi(\psi_\alpha(\varphi(\varepsilon_{\Omega_{\alpha+1}+1},0)))$ |
| 1,4,8,12,15,19,23, 24 | $\psi(\psi_\alpha(\varphi(\varphi(\omega,\Omega_{\alpha+1}+1),0)))$ |
| 1,4,8,12,16 | $\psi(\psi_\alpha(\Gamma\Omega_{\alpha+1}+1))$ |
| 1,4,8,13 | $\psi(\psi_\alpha(\Omega_{\alpha+2}+\Omega_{\alpha+1}))$ |
| 1,4,8,14 | $\psi(\psi_\alpha(\Omega_{\alpha+2}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,8,14,8 | $\psi(\psi_\alpha(\Omega_{\alpha+2}+\Omega_{\alpha+1}\cdot\omega)+\psi_\alpha(\Omega_{\alpha+2}+\Omega_{\alpha+1}))$ |
| 1,4,8,14,14 | $\psi(\psi_\alpha(\Omega_{\alpha+2}+\Omega_{\alpha+1}\cdot\omega)\cdot 2)$ |
| 1,4,8,14,21 | $\psi(\psi_\alpha(\Omega_{\alpha+2}+\varepsilon_{\Omega_{\alpha+1}+1}))$ $=\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot 2+\Omega_{\alpha+1}))$ |
| 1,4,9 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega))$ |
| 1,4,9,4,8 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega)+\psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,9,4,8,14,21 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega)+$ $\psi_{\psi_\alpha(\Omega_{\alpha+2})}(\psi_\alpha(\Omega_{\alpha+2}\cdot 2+\Omega_{\alpha+1})))$ |
| 1,4,9,4,8,14,22 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega)+\psi_{\psi_\alpha(\Omega_{\alpha+2})}(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega)))$ |
| 1,4,9,4,8,14,22, 8 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega)+\psi_\alpha(\Omega_{\alpha+2}))$ |
| 1,4,9,4,8,14,22, 14 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega)+\psi_\alpha(\Omega_{\alpha+2}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,4,9 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega)\cdot 2)$ |
| 1,4,9,6,4,9 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega)^2)$ |
| 1,4,9,6,9 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega+\Omega_{\alpha+1}))$ |
| 1,4,9,6,10,15 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot(\omega+1)))$ $=\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega+\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,9,6,10,16 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega\cdot 2))$ |
| 1,4,9,7 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\omega^2))$ |
| 1,4,9,7,9,5 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\alpha))$ |
| 1,4,9,7,9,12 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\Omega_{\alpha+1}+\Omega_{\alpha+1}))$ |
| 1,4,9,7,9,13,19 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot(\Omega_{\alpha+1}+\omega)))$ |
| 1,4,9,7,10 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,7,11 | $\psi(\psi_\alpha(\Omega_{\alpha+2}\cdot\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,9,7,11,17,25 | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^2+\Omega_{\alpha+2}\cdot\omega))$ |
| 1,4,9,7,11,17,25, 23 | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^2+\Omega_{\alpha+2}\cdot\omega^2))$ |
| 1,4,9,7,11,17,25, 23,30 | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^2\cdot 2+\Omega_{\alpha+1}))$ |
| 1,4,9,7,12 | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^2\cdot\omega))$ |
| 1,4,9,7,12,6,9 | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^2\cdot\omega+\Omega_{\alpha+1}))$ |
| 1,4,9,7,12,7 | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^2\cdot\omega^2))$ |
| 1,4,9,7,12,7,11 | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^2\cdot\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,9,7,12,7,12 | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^3\cdot\omega))$ |
| 1,4,9,7,12,8 | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^\omega))$ |
| 1,4,9,7,12,9,5 | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^\alpha))$ |
| 1,4,9,7,12,9,12 | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^{\Omega_{\alpha+1}}+\Omega_{\alpha+1}))$ |
| 1,4,9,7,12,10 | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^{\Omega_{\alpha+1}}\cdot\omega))$ |
| 1,4,9,7,12,10,15 | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^{\Omega_{\alpha+2}}\cdot\omega))$ |
| 1,4,9,7,12,10,15, 13,18 | $\psi(\psi_\alpha({\Omega_{\alpha+2}}^{{\Omega_{\alpha+2}}^{\Omega_{\alpha+2}}}\cdot\omega))$ |
| 1,4,9,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+1}))$ |
| 1,4,9,8,7 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+1}\cdot\omega))$ |
| 1,4,9,8,7,12 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+1}\cdot\Omega_{\alpha+2}\cdot\omega))$ |
| 1,4,9,8,7,12,11 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha+2}+1}}^2))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,9,8,7,12,11, 8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+1}{}^{\omega}))$ |
| 1,4,9,8,7,12,11, 10,15,14 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+1}{}^{\varepsilon_{\Omega_{\alpha+2}+1}}))$ |
| 1,4,9,8,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+2}))$ |
| 1,4,9,8,9 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+\omega}))$ |
| 1,4,9,8,10,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+\alpha}))$ |
| 1,4,9,8,10,13 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+\Omega_{\alpha+1}} + \Omega_{\alpha+1}))$ |
| 1,4,9,8,11 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}+\Omega_{\alpha+1}} \cdot \omega))$ |
| 1,4,9,8,11,15 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}\cdot 2}))$ |
| 1,4,9,8,11,16 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}\cdot 2} \cdot \omega))$ |
| 1,4,9,8,11,16,8, 11,16 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}\cdot 3} \cdot \omega))$ |
| 1,4,9,8,11,16,11 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+2}\cdot\Omega_{\alpha+1}} \cdot \omega))$ |
| 1,4,9,8,11,16,15 | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\Omega_{\alpha+2}+1}}))$ |
| 1,4,9,8,12 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+2}+1}))$ |
| 1,4,9,8,12,8 | $\psi(\psi_\alpha(\varepsilon_{\zeta_{\Omega_{\alpha+2}+1}} + 1)))))$ |
| 1,4,9,8,12,8,11, 16,15,19,15 | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\zeta_{\Omega_{\alpha+2}+1}}}))$ |
| 1,4,9,8,12,8,12 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+2}+2}))$ |
| 1,4,9,8,12,10,5 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+2}+\alpha}))$ |
| 1,4,9,8,12,11 | $\psi(\psi_\alpha(\zeta_{\Omega_{\alpha+2}+\Omega_{\alpha+1}} \cdot \omega))$ |
| 1,4,9,8,12,11,16, 15,19 | $\psi(\psi_\alpha(\zeta_{\zeta_{\Omega_{\alpha+2}+1}}))$ |
| 1,4,9,8,12,12 | $\psi(\psi_\alpha(\eta_{\Omega_{\alpha+2}+1}))$ |
| 1,4,9,8,13 | $\psi(\psi_\alpha(\Omega_{\alpha+3} + \Omega_{\alpha+1}))$ |
| 1,4,9,8,14,22 | $\psi(\psi_\alpha(\Omega_{\alpha+3} + \Omega_{\alpha+2} \cdot \omega))$ |
| 1,4,9,8,14,22,21 | $\psi(\psi_\alpha(\Omega_{\alpha+3} + \varepsilon_{\Omega_{\alpha+2}+1}))$ $= \psi(\psi_\alpha(\Omega_{\alpha+3} \cdot 2))$ |
| 1,4,9,9 | $\psi(\psi_\alpha(\Omega_{\alpha+3} \cdot \omega))$ |
| 1,4,9,9,7 | $\psi(\psi_\alpha(\Omega_{\alpha+3} \cdot \omega^2))$ |
| 1,4,9,9,7,12 | $\psi(\psi_\alpha(\Omega_{\alpha+3} \cdot \Omega_{\alpha+2} \cdot \omega))$ |
| 1,4,9,9,7,12,12 | $\psi(\psi_\alpha({\Omega_{\alpha+3}}^2 \cdot \omega))$ |
| 1,4,9,9,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+3}+1}))$ |
| 1,4,9,9,8,14,22, 22,21 | $\psi(\psi_\alpha(\Omega_{\alpha+4} \cdot 2))$ |
| 1,4,9,9,9 | $\psi(\psi_\alpha(\Omega_{\alpha+4} \cdot \omega))$ |
| 1,4,9,10 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega}))$ |
| 1,4,9,10,5 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega}) \cdot \omega)$ |
| 1,4,9,10,6,9 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega} + \Omega_{\alpha+1}))$ |
| 1,4,9,10,6,10,15 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega} + \varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,9,10,6,10,16 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega} + \Omega_{\alpha+2} \cdot \omega))$ |
| 1,4,9,10,6,10,16, 17 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega} \cdot 2))$ |
| 1,4,9,10,7 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega} \cdot \omega))$ |
| 1,4,9,10,7,9,5 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega} \cdot \alpha))$ |
| 1,4,9,10,7,9,7 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega} \cdot \alpha \cdot \omega))$ |
| 1,4,9,10,7,9,12 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega} \cdot \varepsilon_{\alpha+1}))$ |
| 1,4,9,10,7,10 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega} \cdot \Omega_{\alpha+1} \cdot \omega))$ |
| 1,4,9,10,7,12 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega} \cdot \Omega_{\alpha+2} \cdot \omega))$ |
| 1,4,9,10,7,12,13 | $\psi(\psi_\alpha({\Omega_{\alpha+\omega}}^2))$ |
| 1,4,9,10,8 | $\psi(\psi_\alpha(\varepsilon_(\Omega_{\alpha+\omega} + 1)))$ |
| 1,4,9,10,8,12 | $\psi(\psi_\alpha(\zeta_(\Omega_{\alpha+\omega} + 1)))$ |
| 1,4,9,10,8,13 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega+1}))$ |
| 1,4,9,10,8,14 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega+1} + \Omega_{\alpha+1} \cdot \omega))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,9,10,8,14,22, 23 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega+1}+\Omega_{\alpha+\omega}))$ |
| 1,4,9,10,9 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega+1}\cdot\omega))$ |
| 1,4,9,10,9,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha+\omega+1}+1}))$ |
| 1,4,9,10,9,9 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega+2}\cdot\omega))$ |
| 1,4,9,10,9,10 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega\cdot 2}))$ |
| 1,4,9,10,10 | $\psi(\psi_\alpha(\Omega_{\alpha+\omega^2}))$ |
| 1,4,9,11 | $\psi(\psi_\alpha(\Omega_{\alpha+\Omega}))$ |
| 1,4,9,11,4,9,10 | $\psi(\psi_\alpha(\Omega_(\alpha+\psi_\alpha(\Omega_{\alpha+\omega))))$ |
| 1,4,9,11,5 | $\psi(\psi_\alpha(\Omega_{\alpha\cdot 2}))$ |
| 1,4,9,11,9 | $\psi(\psi_\alpha(\Omega_{\alpha\cdot 2+1}\cdot\omega))$ |
| 1,4,9,11,9,11,5 | $\psi(\psi_\alpha(\Omega_{\alpha\cdot 3}))$ |
| 1,4,9,11,10 | $\psi(\psi_\alpha(\Omega_{\alpha\cdot\omega}))$ |
| 1,4,9,11,11,5 | $\psi(\psi_\alpha(\Omega_{\alpha^2}))$ |
| 1,4,9,11,14 | $\psi(\psi_\alpha(\Omega_{\varepsilon_{\alpha+1}}))$ |
| 1,4,9,11,15 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,11,15,21,22 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}}+\Omega_{\alpha+\omega}))$ |
| 1,4,9,12 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}}\cdot\omega))$ |
| 1,4,9,12,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\Omega_{\alpha+1}}+1}))$ |
| 1,4,9,12,9 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}+1}\cdot\omega))$ |
| 1,4,9,12,9,11,14 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}+\varepsilon_{\alpha+1}}))$ |
| 1,4,9,12,9,12 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}\cdot 2}\cdot\omega))$ |
| 1,4,9,12,11,5 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}\cdot\alpha}))$ |
| 1,4,9,12,11,14 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}{}^2}+\Omega_{\alpha+1}))$ |
| 1,4,9,12,12 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}{}^2}\cdot\omega))$ |
| 1,4,9,12,14,5 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+1}{}^\alpha}))$ |
| 1,4,9,12,16 | $\psi(\psi_\alpha(\Omega_{\varepsilon_{\Omega_{\alpha+1}+1}}))$ |
| 1,4,9,12,16,22,30, 31 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+2}}+\Omega_{\alpha+\omega}))$ |
| 1,4,9,12,17 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+2}}\cdot\omega))$ |
| 1,4,9,12,17,9,12, 17 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+2}\cdot 2}\cdot\omega))$ |
| 1,4,9,12,17,10 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+2}\cdot\omega}))$ |
| 1,4,9,12,17,12,17 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+2}{}^2}\cdot\omega))$ |
| 1,4,9,12,17,16 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+3}}))$ |
| 1,4,9,12,17,18 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha+\omega}}))$ |
| 1,4,9,13 | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1)))$ $=\psi(\psi_\alpha(I_{\alpha+1}))$ |
| 1,4,9,13,5 | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1))\cdot\omega)$ |
| 1,4,9,13,6,9 | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1)+\Omega_{\alpha+1}))$ |
| 1,4,9,13,6,10,16, 17 | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1)+\Omega_{\alpha+\omega}))$ |
| 1,4,9,13,7 | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1)\cdot\omega))$ |
| 1,4,9,13,7,7 | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1)\cdot\omega^2))$ |
| 1,4,9,13,7,9,5 | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1)\cdot\alpha))$ |
| 1,4,9,13,7,12,13 | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1)\cdot\Omega_{\alpha+\omega}))$ |
| 1,4,9,13,7,12,16 | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+1)^2))$ |
| 1,4,9,13,8 | $\psi(\psi_\alpha(\varepsilon_{\mathrm{OFP}(\alpha+1)+1}))$ |
| 1,4,9,13,8,14 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,13,8,14,22, 23 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)+1}+\Omega_{\alpha+\omega}))$ |
| 1,4,9,13,9 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)+1}\cdot\omega))$ |
| 1,4,9,13,9,11,5 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)}+\alpha))$ |
| 1,4,9,13,9,11,15 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)+\Omega_{\alpha+1}}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,13,9,12 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)+\Omega_{\alpha+1}}\cdot\omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,9,13,9,12,16 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)+\varepsilon_{\Omega_{\alpha+1}+1}}))$ |
| 1,4,9,13,9,12,17, 18 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)+\Omega_{\alpha+\omega}}))$ |
| 1,4,9,13,9,12,17, 20 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)+\Omega_{\Omega_{\alpha+1}}} \cdot \omega))$ |
| 1,4,9,13,9,12,17, 21 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)\cdot 2}))$ |
| 1,4,9,13,9,12,17, 21,9,12,17,21 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)\cdot 3}))$ |
| 1,4,9,13,9,12,17, 21,10 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)\cdot\omega}))$ |
| 1,4,9,13,9,12,17, 21,12 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)\cdot\Omega_{\alpha+1}} \cdot \omega))$ |
| 1,4,9,13,9,12,17, 21,12,17,18 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)\cdot\Omega_{\alpha+\omega}}))$ |
| 1,4,9,13,9,12,17, 21,12,17,21 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)^2}))$ |
| 1,4,9,13,9,12,17, 21,15,20,24 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+1)^{\mathrm{OFP}(\alpha+1)}}))$ |
| 1,4,9,13,9,12,17, 21,16 | $\psi(\psi_\alpha(\Omega_{\varepsilon_{\mathrm{OFP}(\alpha+1)+1}}))$ |
| 1,4,9,13,9,12,17, 21,17 | $\psi(\psi_\alpha(\Omega_{\Omega_{\mathrm{OFP}(\alpha+1)+1}} \cdot \omega))$ |
| 1,4,9,13,9,12,17, 21,17,18 | $\psi(\psi_\alpha(\Omega_{\Omega_{\mathrm{OFP}(\alpha+1)+\omega}}))$ |
| 1,4,9,13,9,13 | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+2)))$ |
| 1,4,9,13,9,13,9 | $\psi(\psi_\alpha(\Omega_{(}\mathrm{OFP}(\alpha+2)+1)\cdot\omega))$ |
| 1,4,9,13,9,13,9, 12,17,21 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+2)+\mathrm{OFP}(\alpha+1)}))$ |
| 1,4,9,13,9,13,9, 12,17,21,17,21 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(\alpha+2)\cdot 2}))$ |
| 1,4,9,13,9,13,9, 12,17,21,17,21,17 | $\psi(\psi_\alpha(\Omega_{\Omega_{\mathrm{OFP}(\alpha+2)+1}} \cdot \omega))$ |
| 1,4,9,13,9,13,9, 13 | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+3)))$ |
| 1,4,9,13,10 | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha+\omega)))$ |
| 1,4,9,13,11,5 | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha\cdot 2)))$ |
| 1,4,9,13,11,9,13, 11,5 | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha\cdot 3)))$ |
| 1,4,9,13,11,14 | $\psi(\psi_\alpha(\mathrm{OFP}(\varepsilon_{\alpha+1})))$ |
| 1,4,9,13,11,15 | $\psi(\psi_\alpha(\mathrm{OFP}(\Omega_{\alpha+1})+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,13,12 | $\psi(\psi_\alpha(\mathrm{OFP}(\Omega_{\alpha+1})\cdot\omega))$ |
| 1,4,9,13,12,9,13, 10 | $\psi(\psi_\alpha(\mathrm{OFP}(\Omega_{\alpha+1}+\omega)))$ |
| 1,4,9,13,12,9,13, 12 | $\psi(\psi_\alpha(\mathrm{OFP}(\Omega_{\alpha+1}\cdot 2)\cdot\omega))$ |
| 1,4,9,13,12,10 | $\psi(\psi_\alpha(\mathrm{OFP}(\Omega_{\alpha+1}\cdot\omega)))$ |
| 1,4,9,13,12,12 | $\psi(\psi_\alpha(\mathrm{OFP}({\Omega_{\alpha+1}}^2)\cdot\omega))$ |
| 1,4,9,13,12,16 | $\psi(\psi_\alpha(\mathrm{OFP}(\varepsilon_{\Omega_{\alpha+1}+1})))$ |
| 1,4,9,13,12,17,18 | $\psi(\psi_\alpha(\mathrm{OFP}(\Omega_{\alpha+\omega})))$ |
| 1,4,9,13,12,17,21 | $\psi(\psi_\alpha(\mathrm{OFP}(\mathrm{OFP}(\alpha+1))))$ |
| 1,4,9,13,13 | $\psi(\psi_\alpha(I_{\alpha+1})^2)$ $= \psi(\psi_\alpha(\psi_{I_{\alpha+1}}({I_{\alpha+1}}^2)))$ |
| 1,4,9,13,13,8 | $\psi(\psi_\alpha(\varepsilon_{(}\Phi(2,\alpha+1)+1)))$ |
| 1,4,9,13,13,9 | $\psi(\psi_\alpha(\Omega_{(}\Phi(2,\alpha+1)+1)\cdot\omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,9,13,13,9,13 | $\psi(\psi_\alpha(\mathrm{OFP}(\Phi(2,\alpha+1)+1)))$ |
| 1,4,9,13,13,9,13, 12 | $\psi(\psi_\alpha(\mathrm{OFP}(\Phi(2,\alpha+1)+\Omega_{\alpha+1})\cdot\omega))$ |
| 1,4,9,13,13,9,13, 12,17,21,21 | $\psi(\psi_\alpha(\mathrm{OFP}(\Phi(2,\alpha+1)\cdot 2)))$ |
| 1,4,9,13,13,9,13, 13 | $\psi(\psi_\alpha(\Phi(2,\alpha+2)))$ |
| 1,4,9,13,13,10 | $\psi(\psi_\alpha(\Phi(2,\alpha+\omega)))$ |
| 1,4,9,13,13,11,5 | $\psi(\psi_\alpha(\Phi(2,\alpha\cdot 2)))$ |
| 1,4,9,13,13,11,14 | $\psi(\psi_\alpha(\Phi(2,\Omega_{\alpha+1})+\Omega_{\alpha+1}))$ |
| 1,4,9,13,13,12 | $\psi(\psi_\alpha(\Phi(2,\Omega_{\alpha+1})\cdot\omega))$ |
| 1,4,9,13,13,12,17, 21,21 | $\psi(\psi_\alpha(\Phi(2,\Phi(2,\alpha+1))))$ |
| 1,4,9,13,13,13 | $\psi(\psi_\alpha(\Phi(3,\alpha+1)))$ |
| 1,4,9,13,14 | $\psi(\psi_\alpha(\Phi(\omega,\alpha+1)))$ $=\psi(\psi_\alpha(I_{\alpha+1})^\omega)$ |
| 1,4,9,13,16 | $\psi(\psi_\alpha(\psi_{I_{\alpha+1}}(I_{\alpha+1}{}^{\Omega_{\alpha+1}})\cdot\omega))$ |
| 1,4,9,13,16,21,25, 26 | $\psi(\psi_\alpha(\psi_{I_{\alpha+1}}(I_{\alpha+1}{}^{\psi_{I_{\alpha+1}}(I_{\alpha+1}{}^\omega)})))$ |
| 1,4,9,13,17 | $\psi(\psi_\alpha(I_{\alpha+1})^{\psi_\alpha(I_{\alpha+1})})$ |
| 1,4,9,13,18 | $\psi(\psi_\alpha(I_{\alpha+1}+\Omega_{\alpha+1}))$ |
| 1,4,9,13,19 | $\psi(\psi_\alpha(I_{\alpha+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,13,19,27,28 | $\psi(\psi_\alpha(I_{\alpha+1}+\Omega_{\alpha+\omega}))$ |
| 1,4,9,13,19,27,34, 42 | $\psi(\psi_\alpha(I_{\alpha+1}\cdot 2+\Omega_{\alpha+1}))$ |
| 1,4,9,14 | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\omega))$ |
| 1,4,9,14,5 | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\omega)\cdot\omega)$ |
| 1,4,9,14,6,9 | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\omega+\Omega_{\alpha+1}))$ |
| 1,4,9,14,7 | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\omega^2))$ |
| 1,4,9,14,7,9,5 | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\alpha))$ |
| 1,4,9,14,7,10 | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,14,7,11 | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\Omega_{\alpha+2}))$ |
| 1,4,9,14,7,12,13 | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\Omega_{\alpha+\omega}))$ |
| 1,4,9,14,7,12,16 | $\psi(\psi_\alpha(I_{\alpha+1}\cdot\mathrm{OFP}(\alpha+1)))$ |
| 1,4,9,14,7,12,16, 17 | $\psi(\psi_\alpha(I_{\alpha+1}{}^2)^\omega)$ |
| 1,4,9,14,7,12,16, 21 | $\psi(\psi_\alpha(I_{\alpha+1}{}^2+\Omega_{\alpha+1}))$ |
| 1,4,9,14,7,12,16, 22,30,38 | $\psi(\psi_\alpha(I_{\alpha+1}{}^2+I_{\alpha+1}\cdot\omega))$ |
| 1,4,9,14,7,12,17 | $\psi(\psi_\alpha(I_{\alpha+1}{}^2\cdot\omega))$ |
| 1,4,9,14,7,12,17, 7,9,5 | $\psi(\psi_\alpha(I_{\alpha+1}{}^2\cdot\alpha))$ |
| 1,4,9,14,7,12,17, 7,12,16,21 | $\psi(\psi_\alpha(I_{\alpha+1}{}^3+\Omega_{\alpha+1}))$ |
| 1,4,9,14,7,12,17, 7,12,16,22,30,38,28, 36,44 | $\psi(\psi_\alpha(I_{\alpha+1}{}^3+I_{\alpha+1}\cdot\omega))$ |
| 1,4,9,14,7,12,17, 7,12,17 | $\psi(\psi_\alpha(I_{\alpha+1}{}^3\cdot\omega))$ |
| 1,4,9,14,7,12,17, 8 | $\psi(\psi_\alpha(I_{\alpha+1}{}^\omega))$ |
| 1,4,9,14,7,12,17, 10 | $\psi(\psi_\alpha(I_{\alpha+1}{}^{\Omega_{\alpha+1}}\cdot\omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,9,14,7,12,17, 10,15,20 | $\psi(\psi_\alpha({I_{\alpha+1}}^{I_{\alpha+1}} \cdot \omega))$ |
| 1,4,9,14,8 | $\psi(\psi_\alpha(\varepsilon_{I_{\alpha+1}+1}))$ |
| 1,4,9,14,8,10,5 | $\psi(\psi_\alpha(\varepsilon_{I_{\alpha+1}+\alpha}))$ |
| 1,4,9,14,8,11,16, 21,15 | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{I_{\alpha+1}+1}}))$ |
| 1,4,9,14,8,12 | $\psi(\psi_\alpha(\zeta_{I_{\alpha+1}+1}))$ |
| 1,4,9,14,8,13 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+1} + \Omega_{\alpha+1}))$ |
| 1,4,9,14,8,14,22, 30 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+1} + I_{\alpha+1} \cdot \omega))$ |
| 1,4,9,14,9 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+1} \cdot \omega))$ |
| 1,4,9,14,9,10 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+\omega}))$ |
| 1,4,9,14,9,11,5 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+\alpha}))$ |
| 1,4,9,14,9,11,14 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+\Omega_{\alpha+1}} + \Omega_{\alpha+1}))$ |
| 1,4,9,14,9,12 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+\Omega_{\alpha+1}} \cdot \omega))$ |
| 1,4,9,14,9,12,17, 18 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+\Omega_{\alpha+\omega}}))$ |
| 1,4,9,14,9,12,17, 21 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}+\mathrm{OFP}(\alpha+1)}))$ |
| 1,4,9,14,9,12,17, 21,26 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 2} + \Omega_{\alpha+1}))$ |
| 1,4,9,14,9,12,17, 22 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 2} \cdot \omega))$ |
| 1,4,9,14,9,12,17, 22,5 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 2} \cdot \omega) \cdot \omega)$ |
| 1,4,9,14,9,12,17, 22,7 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 2} \cdot \omega^2))$ |
| 1,4,9,14,9,12,17, 22,7,12,17 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 2} \cdot I_{\alpha+1} \cdot \omega))$ |
| 1,4,9,14,9,12,17, 22,7,12,17,12,15,20, 25 | $\psi(\psi_\alpha({\Omega_{I_{\alpha+1}\cdot 2}}^2 \cdot \omega))$ |
| 1,4,9,14,9,12,17, 22,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{I_{\alpha+1}\cdot 2}+1}))$ |
| 1,4,9,14,9,12,17, 22,9 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 2+1} \cdot \omega))$ |
| 1,4,9,14,9,12,17, 22,9,11,5 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 2+\alpha}))$ |
| 1,4,9,14,9,12,17, 22,9,12,17,22 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot 3} \cdot \omega))$ |
| 1,4,9,14,9,12,17, 22,11,5 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot \alpha}))$ |
| 1,4,9,14,9,12,17, 22,12 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot \Omega_{\alpha+1}} \cdot \omega))$ |
| 1,4,9,14,9,12,17, 22,12,17,18 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+1}\cdot \Omega_{\alpha+\omega}}))$ |
| 1,4,9,14,9,12,17, 22,12,17,22 | $\psi(\psi_\alpha(\Omega_{{I_{\alpha+1}}^2} \cdot \omega))$ |
| 1,4,9,14,9,12,17, 22,13 | $\psi(\psi_\alpha(\Omega_{{I_{\alpha+1}}^\omega}))$ |
| 1,4,9,14,9,12,17, 22,15,20,25 | $\psi(\psi_\alpha(\Omega_{{I_{\alpha+1}}^{I_{\alpha+1}}} \cdot \omega))$ |
| 1,4,9,14,9,12,17, 22,16 | $\psi(\psi_\alpha(\Omega_{\varepsilon_{I_{\alpha+1}+1}}))$ |
| 1,4,9,14,9,12,17, 22,16,21 | $\psi(\psi_\alpha(\Omega_{\Omega_{I_{\alpha+1}+1}} + \Omega_{\alpha+1}))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,9,14,9,12,17, 22,17 | $\psi(\psi_\alpha(\Omega_{\Omega_{I_{\alpha+1}+1}} \cdot \omega))$ |
| 1,4,9,14,9,12,17, 22,17,9,12,17,22 | $\psi(\psi_\alpha(\Omega_{\Omega_{I_{\alpha+1}+1}+I_{\alpha+1}} \cdot \omega))$ |
| 1,4,9,14,9,12,17, 22,17,9,12,17,22,17 | $\psi(\psi_\alpha(\Omega_{\Omega_{I_{\alpha+1}+1}\cdot 2} \cdot \omega))$ |
| 1,4,9,14,9,12,17, 22,17,16 | $\psi(\psi_\alpha(\Omega_{(\varepsilon_{\Omega_{I_{\alpha+1}+1}+1}+1)}))$ |
| 1,4,9,14,9,12,17, 22,17,17 | $\psi(\psi_\alpha(\Omega_{\Omega_{I_{\alpha+1}+2}} \cdot \omega))$ |
| 1,4,9,14,9,12,17, 22,17,20,25,30 | $\psi(\psi_\alpha(\Omega_{\Omega_{I_{\alpha+1}\cdot 2}} \cdot \omega))$ |
| 1,4,9,14,9,12,17, 22,17,20,25,30,25 | $\psi(\psi_\alpha(\Omega_{\Omega_{\Omega_{I_{\alpha+1}+1}+1}}))$ |
| 1,4,9,14,9,13 | $\psi(\psi_\alpha(I_{\alpha+2}))$ $= \psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha+1}+1)))$ |
| 1,4,9,14,9,13,7 | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha+1}+1)\cdot\omega))$ |
| 1,4,9,14,9,13,8 | $\psi(\psi_\alpha(\varepsilon_{(}\mathrm{OFP}(I_{\alpha+1}+1)+1)))$ |
| 1,4,9,14,9,13,9 | $\psi(\psi_\alpha(\Omega_{(}\mathrm{OFP}(I_{\alpha+1}+1)+1)\cdot\omega))$ |
| 1,4,9,14,9,13,9, 11,5 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(I_{\alpha+1}+1)+\alpha}))$ |
| 1,4,9,14,9,13,9, 12,17,22,17,21 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(I_{\alpha+1}+1)\cdot 2}))$ |
| 1,4,9,14,9,13,9, 12,17,22,17,21,17 | $\psi(\psi_\alpha(\Omega_{\Omega_{\mathrm{OFP}(I_{\alpha+1}+1)+1}} \cdot \omega))$ |
| 1,4,9,14,9,13,9, 13 | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha+1}+2)))$ |
| 1,4,9,14,9,13,11, 5 | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha+1}+\alpha)))$ |
| 1,4,9,14,9,13,12 | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha+1}+\Omega_{\alpha+1})\cdot\omega))$ |
| 1,4,9,14,9,13,12, 17,22 | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha+1}\cdot 2)\cdot\omega))$ |
| 1,4,9,14,9,13,12, 17,22,12,17,22 | $\psi(\psi_\alpha(\mathrm{OFP}({I_{\alpha+1}}^2)\cdot\omega))$ |
| 1,4,9,14,9,13,12, 17,22,16 | $\psi(\psi_\alpha(\mathrm{OFP}(\varepsilon_{I_{\alpha+1}+1})))$ |
| 1,4,9,14,9,13,12, 17,22,17 | $\psi(\psi_\alpha(\mathrm{OFP}(\Omega_{I_{\alpha+1}+1})\cdot\omega))$ |
| 1,4,9,14,9,13,12, 17,22,17,20,25,30 | $\psi(\psi_\alpha(\mathrm{OFP}(\Omega_{\Omega_{I_{\alpha+1}+1}})\cdot\omega))$ |
| 1,4,9,14,9,13,12, 17,22,17,21 | $\psi(\psi_\alpha(\mathrm{OFP}(\mathrm{OFP}(I_{\alpha+1}+1))))$ |
| 1,4,9,14,9,13,13 | $\psi(\psi_\alpha(\Phi(2, I_{\alpha+1}+1)))$ |
| 1,4,9,14,9,13,14 | $\psi(\psi_\alpha(I_{\alpha+2})^\omega)$ |
| 1,4,9,14,9,13,18 | $\psi(\psi_\alpha(I_{\alpha+2}+\Omega_{\alpha+1}))$ |
| 1,4,9,14,9,13,19 | $\psi(\psi_\alpha(I_{\alpha+2}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,14,9,13,19, 27,28 | $\psi(\psi_\alpha(I_{\alpha+2}+\Omega_{\alpha+\omega}))$ |
| 1,4,9,14,9,13,19, 27,35 | $\psi(\psi_\alpha(I_{\alpha+2}+I_{\alpha+1}\cdot\omega))$ |
| 1,4,9,14,9,13,19, 27,35,27,34 | $\psi(\psi_\alpha(I_{\alpha+2}\cdot 2))$ |
| 1,4,9,14,9,14 | $\psi(\psi_\alpha(I_{\alpha+2}\cdot\omega))$ |
| 1,4,9,14,9,14,7, 12,17,12,17 | $\psi(\psi_\alpha({I_{\alpha+2}}^2\cdot\omega))$ |
| 1,4,9,14,9,14,8 | $\psi(\psi_\alpha(\varepsilon_{(}I_{\alpha+2}+1)))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,9,14,9,14,9 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+2}+1} \cdot \omega))$ |
| 1,4,9,14,9,14,9, 10 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+2}+\omega}))$ |
| 1,4,9,14,9,14,9, 12,17,22 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+2}+I_{\alpha+1}} \cdot \omega))$ |
| 1,4,9,14,9,14,9, 12,17,22,17,22 | $\psi(\psi_\alpha(\Omega_{I_{\alpha+2}\cdot 2} \cdot \omega))$ |
| 1,4,9,14,9,14,9, 12,17,22,17,22,16 | $\psi(\psi_\alpha(\Omega_{\varepsilon_{I_{\alpha+2}+1}}))$ |
| 1,4,9,14,9,14,9, 12,17,22,17,22,17 | $\psi(\psi_\alpha(\Omega_{\Omega_{I_{\alpha+2}+1}}))$ |
| 1,4,9,14,9,14,9, 13 | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha+2}+1)))$ |
| 1,4,9,14,9,14,9, 13,9,13 | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha+2}+2)))$ |
| 1,4,9,14,9,14,9, 13,12,17,22,17,22,17, 21 | $\psi(\psi_\alpha(\mathrm{OFP}(\mathrm{OFP}(I_{\alpha+2}+1))))$ |
| 1,4,9,14,9,14,9, 13,13 | $\psi(\psi_\alpha(I_{\alpha+3})^2)$ |
| 1,4,9,14,9,14,9, 14 | $\psi(\psi_\alpha(I_{\alpha+3} \cdot \omega))$ |
| 1,4,9,14,10 | $\psi(\psi_\alpha(I_{\alpha+\omega}))$ |
| 1,4,9,14,11,5 | $\psi(\psi_\alpha(I_{\alpha\cdot 2}))$ |
| 1,4,9,14,11,5,5 | $\psi(\psi_\alpha(I_{\alpha\cdot 2}) \cdot \omega)$ |
| 1,4,9,14,11,6,4, 9,14,11,5 | $\psi(\psi_\alpha(I_{\alpha\cdot 2})^2)$ |
| 1,4,9,14,11,6,9 | $\psi(\psi_\alpha(I_{\alpha\cdot 2}+\Omega_{\alpha+1}))$ |
| 1,4,9,14,11,6,10, 16,22,18,4,9,14,11, 5 | $\psi(\psi_\alpha(I_{\alpha\cdot 2}+I(\alpha+\psi_\alpha(I_{\alpha\cdot 2}))))$ |
| 1,4,9,14,11,6,10, 16,22,18,5 | $\psi(\psi_\alpha(I_{\alpha\cdot 2}+I(\alpha+\psi_\alpha(I_{\alpha\cdot 2})) \cdot \omega))$ |
| 1,4,9,14,11,6,10, 16,22,19 | $\psi(\psi_\alpha(I_{\alpha\cdot 2}+I(\alpha+\psi_\alpha(I_{\alpha\cdot 2}+\Omega_{\alpha+1}))))$ |
| 1,4,9,14,11,6,10, 16,22,19,10 | $\psi(\psi_\alpha(I_{\alpha\cdot 2}+I(\alpha+\psi_\alpha(I_{\alpha+1}+\Omega_{\alpha+1} \cdot \omega))))$ |
| 1,4,9,14,11,6,10, 16,22,19,11 | $\psi(\psi_\alpha(I_{\alpha\cdot 2} \cdot 2))$ |
| 1,4,9,14,11,7 | $\psi(\psi_\alpha(I_{\alpha\cdot 2} \cdot \omega))$ |
| 1,4,9,14,11,7,9, 5 | $\psi(\psi_\alpha(I_{\alpha\cdot 2} \cdot \alpha))$ |
| 1,4,9,14,11,7,9, 12 | $\psi(\psi_\alpha(I_{\alpha\cdot 2} \cdot \varepsilon_{\alpha+1}))$ |
| 1,4,9,14,11,7,10 | $\psi(\psi_\alpha(I_{\alpha\cdot 2} \cdot \Omega_{\alpha+1} \cdot \omega))$ |
| 1,4,9,14,11,7,11 | $\psi(\psi_\alpha(I_{\alpha\cdot 2} \cdot \varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,9,14,11,7,12, 13 | $\psi(\psi_\alpha(I_{\alpha\cdot 2} \cdot \Omega_{\alpha+\omega}))$ |
| 1,4,9,14,11,7,12, 17,18 | $\psi(\psi_\alpha(I_{\alpha\cdot 2} \cdot I(\alpha+\omega)))$ |
| 1,4,9,14,11,7,12, 17,19,5 | $\psi(\psi_\alpha({I_{\alpha\cdot 2}}^2))$ |
| 1,4,9,14,11,7,12, 17,19,8 | $\psi(\psi_\alpha({I_{\alpha\cdot 2}}^\omega))$ |
| 1,4,9,14,11,7,12, 17,19,10,15,20,22,5 | $\psi(\psi_\alpha({I_{\alpha\cdot 2}}^{I_{\alpha\cdot 2}}))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,9,14,11,8 | $\psi(\psi_\alpha(\varepsilon_{I_{\alpha\cdot 2}+1}))$ |
| 1,4,9,14,11,8,8 | $\psi(\psi_\alpha(\varepsilon_{I_{\alpha\cdot 2}+2}))$ |
| 1,4,9,14,11,8,14 | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,14,11,8,14, 22,30,23 | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}+1}+I(\alpha+\omega)))$ |
| 1,4,9,14,11,9 | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}+1}\cdot\omega))$ |
| 1,4,9,14,11,9,7 | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}+1}\cdot\omega^2))$ |
| 1,4,9,14,11,9,7, 12,17,19,12 | $\psi(\psi_\alpha({\Omega_{I_{\alpha\cdot 2}+1}}^2\cdot\omega))$ |
| 1,4,9,14,11,9,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{I_{\alpha\cdot 2}+1}+1}))$ |
| 1,4,9,14,11,9,9 | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}+2}\cdot\omega))$ |
| 1,4,9,14,11,9,11, 5 | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}+\alpha}))$ |
| 1,4,9,14,11,9,12 | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}+\Omega_{\alpha+1}}\cdot\omega))$ |
| 1,4,9,14,11,9,12, 17,18 | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}+\Omega_{\alpha+\omega}}))$ |
| 1,4,9,14,11,9,12, 17,22,19,5 | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}\cdot 2}))$ |
| 1,4,9,14,11,9,12, 17,22,19,9,10 | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}\cdot 2+\omega}))$ |
| 1,4,9,14,11,9,12, 17,22,19,10 | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}\cdot\omega}))$ |
| 1,4,9,14,11,9,12, 17,22,19,12 | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}\cdot\Omega_{\alpha+1}}\cdot\omega))$ |
| 1,4,9,14,11,9,12, 17,22,19,12,17,22,19, 5 | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2}{}^2}))$ |
| 1,4,9,14,11,9,12, 17,22,19,16 | $\psi(\psi_\alpha(\Omega_{\varepsilon_{I_{\alpha\cdot 2}+1}}))$ |
| 1,4,9,14,11,9,12, 17,22,19,17 | $\psi(\psi_\alpha(\Omega_{\Omega_{I_{\alpha\cdot 2}+1}}\cdot\omega))$ |
| 1,4,9,14,11,9,13 | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha\cdot 2}+1)))$ |
| 1,4,9,14,11,9,13, 8 | $\psi(\psi_\alpha(\varepsilon_{\mathrm{OFP}(I_{\alpha\cdot 2}+1)+1}))$ |
| 1,4,9,14,11,9,13, 9 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(I_{\alpha\cdot 2}+1)+1}\cdot\omega))$ |
| 1,4,9,14,11,9,13, 9,12,17,22,19,17,21 | $\psi(\psi_\alpha(\Omega_{\mathrm{OFP}(I_{\alpha\cdot 2}+1)\cdot 2}))$ |
| 1,4,9,14,11,9,13, 9,12,17,22,19,17,21, 17 | $\psi(\psi_\alpha(\Omega_{\Omega_{\mathrm{OFP}(I_{\alpha\cdot 2}+1)}}\cdot\omega))$ |
| 1,4,9,14,11,9,13, 9,13 | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha\cdot 2}+2)))$ |
| 1,4,9,14,11,9,13, 10 | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha\cdot 2}+\omega)))$ |
| 1,4,9,14,11,9,13, 12 | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha\cdot 2}+\Omega_{\alpha+1})\cdot\omega))$ |
| 1,4,9,14,11,9,13, 12,17,22,19,5 | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha\cdot 2}\cdot 2)))$ |
| 1,4,9,14,11,9,13, 12,17,22,19,9,13,12, 17,22,19,5 | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha\cdot 2}\cdot 3)))$ |
| 1,4,9,14,11,9,13, 12,17,22,19,16 | $\psi(\psi_\alpha(\mathrm{OFP}(\varepsilon_{I_{\alpha\cdot 2}+1})))$ |
| 1,4,9,14,11,9,13, 12,17,22,19,17,21 | $\psi(\psi_\alpha(\mathrm{OFP}(\mathrm{OFP}(I_{\alpha\cdot 2}+1))))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,9,14,11,9,13, 13 | $\psi(\psi_\alpha(I_{\alpha\cdot 2+1})^2)$ |
| 1,4,9,14,11,9,13, 18 | $\psi(\psi_\alpha(I_{\alpha\cdot 2+1}+\Omega_{\alpha+1}))$ |
| 1,4,9,14,11,9,13, 19,27,35,32,20 | $\psi(\psi_\alpha(I_{\alpha\cdot 2+1}+I_{\alpha\cdot 2}))$ |
| 1,4,9,14,11,9,14 | $\psi(\psi_\alpha(I_{\alpha\cdot 2+1}\cdot\omega))$ |
| 1,4,9,14,11,9,14, 8 | $\psi(\psi_\alpha(\varepsilon_{(}I_{\alpha\cdot 2+1}+1)))$ |
| 1,4,9,14,11,9,14, 9 | $\psi(\psi_\alpha(\Omega_{I_{\alpha\cdot 2+1}+1}\cdot\omega))$ |
| 1,4,9,14,11,9,14, 9,13 | $\psi(\psi_\alpha(\mathrm{OFP}(I_{\alpha\cdot 2+1}+1)))$ |
| 1,4,9,14,11,9,14, 9,14 | $\psi(\psi_\alpha(I_{\alpha\cdot 2+2}\cdot\omega))$ |
| 1,4,9,14,11,9,14, 10 | $\psi(\psi_\alpha(I_{\alpha\cdot 2+\omega}))$ |
| 1,4,9,14,11,9,14, 11,5 | $\psi(\psi_\alpha(I_{\alpha\cdot 3}))$ |
| 1,4,9,14,11,10 | $\psi(\psi_\alpha(I_{\alpha\cdot\omega}))$ |
| 1,4,9,14,11,11,5 | $\psi(\psi_\alpha(I_{\alpha^2}))$ |
| 1,4,9,14,11,14 | $\psi(\psi_\alpha(I_{\varepsilon_{\alpha+1}}))$ |
| 1,4,9,14,11,15 | $\psi(\psi_\alpha(I_{\Omega_{\alpha+1}}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,14,12 | $\psi(\psi_\alpha(I_{\Omega_{\alpha+1}}\cdot\omega))$ |
| 1,4,9,14,12,9,14 | $\psi(\psi_\alpha(I_{\Omega_{\alpha+1}+1}\cdot\omega))$ |
| 1,4,9,14,12,9,14, 11,5 | $\psi(\psi_\alpha(I_{\Omega_{\alpha+1}+\alpha}))$ |
| 1,4,9,14,12,9,14, 12 | $\psi(\psi_\alpha(I_{\Omega_{\alpha+1}\cdot 2}\cdot\omega))$ |
| 1,4,9,14,12,10 | $\psi(\psi_\alpha(I_{\Omega_{\alpha+1}\cdot\omega}))$ |
| 1,4,9,14,12,12 | $\psi(\psi_\alpha(I_{\Omega_{\alpha+1}{}^2}\cdot\omega))$ |
| 1,4,9,14,12,16 | $\psi(\psi_\alpha(I_{\varepsilon_{\Omega_{\alpha+1}+1}}))$ |
| 1,4,9,14,12,17 | $\psi(\psi_\alpha(I_{\Omega_{\alpha+2}}\cdot\omega))$ |
| 1,4,9,14,12,17,18 | $\psi(\psi_\alpha(I_{\Omega_{\alpha+\omega}}))$ |
| 1,4,9,14,12,17,22, 18 | $\psi(\psi_\alpha(I_{I_{\alpha+\omega}}))$ |
| 1,4,9,14,13 | $\psi(\psi_\alpha(\psi_{I(1,\alpha+1)}(I(1,\alpha+1))))$ |
| 1,4,9,14,13,9 | $\psi(\psi_\alpha(\Omega_{\mathrm{IFP}(\alpha+1)+1}\cdot\omega))$ |
| 1,4,9,14,13,9,14 | $\psi(\psi_\alpha(I_{\mathrm{IFP}(\alpha+1)+1}\cdot\omega))$ |
| 1,4,9,14,13,9,14, 11,5 | $\psi(\psi_\alpha(I_{\mathrm{IFP}(\alpha+1)+\alpha}))$ |
| 1,4,9,14,13,9,14, 12,17,22,18 | $\psi(\psi_\alpha(I_{\mathrm{IFP}(\alpha+1)+I_{\alpha+1}}))$ |
| 1,4,9,14,13,9,14, 12,17,22,21 | $\psi(\psi_\alpha(I_{\mathrm{IFP}(\alpha+1)\cdot 2}))$ |
| 1,4,9,14,13,9,14, 12,17,22,21,17 | $\psi(\psi_\alpha(I_{\Omega_{\mathrm{IFP}(\alpha+1)+1}}\cdot\omega))$ |
| 1,4,9,14,13,9,14, 12,17,22,21,17,22 | $\psi(\psi_\alpha(I_{I_{\mathrm{IFP}(\alpha+1)+1}}\cdot\omega))$ |
| 1,4,9,14,13,9,14, 13 | $\psi(\psi_\alpha(\mathrm{IFP}(\alpha+2)))$ |
| 1,4,9,14,13,10 | $\psi(\psi_\alpha(\mathrm{IFP}(\alpha+\omega)))$ |
| 1,4,9,14,13,12,17, 22,21 | $\psi(\psi_\alpha(\mathrm{IFP}(\mathrm{IFP}(\alpha+1))))$ |
| 1,4,9,14,13,13 | $\psi(\psi_\alpha(\psi_{I(1,\alpha+1)}(I(1,\alpha+1)^2)))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,9,14,13,13,9, 14,13,13 | $\psi(\psi_\alpha(I(1,\alpha+1))^2\cdot 2)$ |
| 1,4,9,14,13,13,10 | $\psi(\psi_\alpha(I(1,\alpha+1))^2\cdot\omega)$ |
| 1,4,9,14,13,13,12, 17,22,21,21 | $\psi(\psi_\alpha(I(1,\alpha+1))^2\cdot$ $\psi_{\psi_\alpha(I(1,\alpha+1))}(\psi_\alpha(I(1,\alpha+1))^2))$ |
| 1,4,9,14,13,13,13 | $\psi(\psi_\alpha(I(1,\alpha+1))^3)$ |
| 1,4,9,14,13,18 | $\psi(\psi_\alpha(I(1,\alpha+1)+\Omega_{\alpha+1}))$ |
| 1,4,9,14,13,19,27, 35,34 | $\psi(\psi_\alpha(I(1,\alpha+1)\cdot 2))$ |
| 1,4,9,14,14 | $\psi(\psi_\alpha(I(1,\alpha+1)\cdot\omega))$ |
| 1,4,9,14,14,6,9 | $\psi(\psi_\alpha(I(1,\alpha+1)\cdot\omega+\Omega_{\alpha+1}))$ |
| 1,4,9,14,14,6,10, 16,22,22 | $\psi(\psi_\alpha(I(1,\alpha+1)\cdot\omega\cdot 2))$ |
| 1,4,9,14,14,7 | $\psi(\psi_\alpha(I(1,\alpha+1)\cdot\omega^2))$ |
| 1,4,9,14,14,7,9, 5 | $\psi(\psi_\alpha(I(1,\alpha+1)\cdot\alpha))$ |
| 1,4,9,14,14,7,10 | $\psi(\psi_\alpha(I(1,\alpha+1)\cdot\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,14,14,7,11 | $\psi(\psi_\alpha(I(1,\alpha+1)\cdot\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,9,14,14,7,12, 17,16 | $\psi(\psi_\alpha(I(1,\alpha+1)^2))$ $=\psi(\psi_\alpha(I(1,\alpha+1)\cdot\mathrm{IFP}(\alpha+1)))$ |
| 1,4,9,14,14,7,12, 17,17 | $\psi(\psi_\alpha(I(1,\alpha+1)^2\cdot\omega))$ |
| 1,4,9,14,14,8 | $\psi(\psi_\alpha(\varepsilon_{I(1,\alpha+1)+1}))$ |
| 1,4,9,14,14,8,14 | $\psi(\psi_\alpha(\Omega_(I(1,\alpha+1)+1)+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,14,14,9 | $\psi(\psi_\alpha(\Omega_{I(1,\alpha+1)+1}\cdot\omega))$ |
| 1,4,9,14,14,9,12, 17,22,22 | $\psi(\psi_\alpha(\Omega_{I(1,\alpha+1)\cdot 2}\cdot\omega))$ |
| 1,4,9,14,14,9,13 | $\psi(\psi_\alpha(\mathrm{OFP}(I(1,\alpha+1)+1)))$ |
| 1,4,9,14,14,9,14 | $\psi(\psi_\alpha(I_{I(1,\alpha+1)+1}\cdot\omega))$ |
| 1,4,9,14,14,9,14, 13 | $\psi(\psi_\alpha(\mathrm{IFP}(I(1,\alpha+1)+1)))$ |
| 1,4,9,14,14,9,14, 14 | $\psi(\psi_\alpha(I(1,\alpha+2)\cdot\omega))$ |
| 1,4,9,14,14,10 | $\psi(\psi_\alpha(I(1,\alpha+\omega)))$ |
| 1,4,9,14,14,11,5 | $\psi(\psi_\alpha(I(1,\alpha\cdot 2)))$ |
| 1,4,9,14,14,11,14 | $\psi(\psi_\alpha(I(1,\varepsilon_{\alpha+1})))$ |
| 1,4,9,14,14,12 | $\psi(\psi_\alpha(I(1,\Omega_{\alpha+1})\cdot\omega))$ |
| 1,4,9,14,14,12,17, 22,22 | $\psi(\psi_\alpha(I(1,I(1,\alpha+1))\cdot\omega))$ |
| 1,4,9,14,14,13 | $\psi(\psi_\alpha(I(2,\alpha+1)))$ |
| 1,4,9,14,14,13,9, 14,14 | $\psi(\psi_\alpha(I(1,\psi_{I(2,\alpha+1)}(I(2,\alpha+1))+1)\cdot\omega))$ |
| 1,4,9,14,14,13,9, 14,14,12,17,22,22,21, 10 | $\psi(\psi_\alpha(I(1,\psi_{I(2,\alpha+1)}(I(2,\alpha+1))\cdot\omega)))$ |
| 1,4,9,14,14,13,9, 14,14,12,17,22,22,21, 12,17,22,22,21 | $\psi(\psi_\alpha(I(1,\psi_{I(2,\alpha+1)}(I(2,\alpha+1))^2)))$ |
| 1,4,9,14,14,13,9, 14,14,12,17,22,22,21, 17,22,22 | $\psi(\psi_\alpha(I(1,I(1,\psi_{I(2,\alpha+1)}$ $(I(2,\alpha+1))+1))\cdot\omega))$ |
| 1,4,9,14,14,13,9, 14,14,13 | $\psi(\psi_\alpha(I(2,\alpha+1))\cdot 2)$ |
| 1,4,9,14,14,13,11, 5 | $\psi(\psi_\alpha(\psi_{I(2,\alpha+1)}(I(2,\alpha+1)\cdot\alpha)))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,9,14,14,13,13 | $\psi(\psi_\alpha(I(2,\alpha+1))^2)$ |
| 1,4,9,14,14,13,18 | $\psi(\psi_\alpha(I(2,\alpha+1)+\Omega_{\alpha+1}))$ |
| 1,4,9,14,14,14 | $\psi(\psi_\alpha(I(2,\alpha+1)\cdot\omega))$ |
| 1,4,9,14,14,14,9, 14,14,14 | $\psi(\psi_\alpha(I(2,\alpha+2)\cdot\omega))$ |
| 1,4,9,14,14,14,13 | $\psi(\psi_\alpha(I(3,\alpha+1)))$ |
| 1,4,9,14,14,14,14 | $\psi(\psi_\alpha(I(3,\alpha+1)\cdot\omega))$ |
| 1,4,9,14,15 | $\psi(\psi_\alpha(I(\omega,\alpha+1)))$ |
| 1,4,9,14,16,5 | $\psi(\psi_\alpha(I(\alpha,1)))$ |
| 1,4,9,14,16,6,9 | $\psi(\psi_\alpha(I(\alpha,1)+\Omega_{\alpha+1}))$ |
| 1,4,9,14,16,6,10, 16,22,25,11 | $\psi(\psi_\alpha(I(\alpha,1)\cdot 2))$ |
| 1,4,9,14,16,7 | $\psi(\psi_\alpha(I(\alpha,1)\cdot\omega))$ |
| 1,4,9,14,16,8 | $\psi(\psi_\alpha(\varepsilon_(I(\alpha,1)+1)))$ |
| 1,4,9,14,16,9 | $\psi(\psi_\alpha(\Omega_(I(\alpha,1)+1)\cdot\omega))$ |
| 1,4,9,14,16,9,14, 16,5 | $\psi(\psi_\alpha(I(\alpha,2)))$ |
| 1,4,9,14,16,10 | $\psi(\psi_\alpha(I(\alpha,\omega)))$ |
| 1,4,9,14,16,11,5 | $\psi(\psi_\alpha(I(\alpha,\alpha)))$ |
| 1,4,9,14,16,12,17, 22,24,5 | $\psi(\psi_\alpha(I(\alpha,I(\alpha,1))))$ |
| 1,4,9,14,16,13 | $\psi(\psi_\alpha(I(\alpha+1,0)))$ |
| 1,4,9,14,16,14 | $\psi(\psi_\alpha(I(\alpha+1,0)\cdot\omega))$ |
| 1,4,9,14,16,14,16, 5 | $\psi(\psi_\alpha(I(\alpha\cdot 2,0)))$ |
| 1,4,9,14,16,16,5 | $\psi(\psi_\alpha(I(\alpha^2,0)))$ |
| 1,4,9,14,16,19 | $\psi(\psi_\alpha(I(\varepsilon_{\alpha+1},0)))$ |
| 1,4,9,14,17 | $\psi(\psi_\alpha(I(\Omega_{\alpha+1},0)\cdot\omega))$ |
| 1,4,9,14,17,22,27, 29,5 | $\psi(\psi_\alpha(I(I(\alpha,1),0)))$ |
| 1,4,9,14,18 | $\psi(\psi_\alpha(I(1,0,\alpha+1)))$ |
| 1,4,9,14,18,9,14, 15 | $\psi(\psi_\alpha(I(\omega,\psi_{I(1,0,\alpha+1)}(I(1,0,\alpha+1))+1)))$ |
| 1,4,9,14,18,9,14, 17,22,27,31 | $\psi(\psi_\alpha(I(\psi_{I(1,0,\alpha+1)}(I(1,0,\alpha+1)),1)))$ |
| 1,4,9,14,18,9,14, 18 | $\psi(\psi_\alpha(I(1,0,\alpha+1))\cdot 2)$ |
| 1,4,9,14,18,12 | $\psi(\psi_\alpha(\psi_{I(1,0,\alpha+1)}(I(1,0,\alpha+1)\cdot\Omega_{\alpha+1})\cdot\omega))$ |
| 1,4,9,14,18,12,17, 22,26 | $\psi(\psi_\alpha(\psi_{I(1,0,\alpha+1)}(I(1,0,\alpha+1)\cdot$ $\psi_{I(1,0,\alpha+1)}(I(1,0,\alpha+1)))))$ |
| 1,4,9,14,18,13 | $\psi(\psi_\alpha(I(1,0,\alpha+1))^2)$ |
| 1,4,9,14,18,13,18 | $\psi(\psi_\alpha(I(1,0,\alpha+1)+\Omega_{\alpha+1}))$ |
| 1,4,9,14,18,14 | $\psi(\psi_\alpha(I(1,0,\alpha+1)\cdot\omega))$ |
| 1,4,9,14,18,14,7 | $\psi(\psi_\alpha(I(1,0,\alpha+1)\cdot\omega^2))$ |
| 1,4,9,14,18,14,7, 12,17,21 | $\psi(\psi_\alpha(I(1,0,\alpha+1)\cdot$ $\psi_{I(1,0,\alpha+1)}(I(1,0,\alpha+1))))$ |
| 1,4,9,14,18,14,7, 12,17,21,17 | $\psi(\psi_\alpha(I(1,0,\alpha+1)^2\cdot\omega))$ |
| 1,4,9,14,18,14,8 | $\psi(\psi_\alpha(\varepsilon_{I(1,0,\alpha+1)+1}))$ |
| 1,4,9,14,18,14,9 | $\psi(\psi_\alpha(\Omega_{I(1,0,\alpha+1)+1}\cdot\omega))$ |
| 1,4,9,14,18,14,9, 14,15 | $\psi(\psi_\alpha(I(\omega,I(1,0,\alpha+1)+1)))$ |
| 1,4,9,14,18,14,9, 14,17,22,27,31 | $\psi(\psi_\alpha(I(I(1,0,\alpha+1),1)))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,9,14,18,14,9, 14,17,22,27,31,27,21 | $\psi(\psi_\alpha(I(\varepsilon_{I(1,0,\alpha+1)+1},0)))$ |
| 1,4,9,14,18,14,9, 14,18 | $\psi(\psi_\alpha(I(1,0,\alpha+2)))$ |
| 1,4,9,14,18,14,9, 14,18,14 | $\psi(\psi_\alpha(I(1,0,\alpha+2)\cdot\omega))$ |
| 1,4,9,14,18,14,10 | $\psi(\psi_\alpha(I(1,0,\alpha+\omega)))$ |
| 1,4,9,14,18,14,11, 5 | $\psi(\psi_\alpha(I(1,0,\alpha\cdot 2)))$ |
| 1,4,9,14,18,14,12 | $\psi(\psi_\alpha(I(1,0,\Omega_{\alpha+1})\cdot\omega))$ |
| 1,4,9,14,18,14,12, 17,22,26 | $\psi(\psi_\alpha(I(1,0,I(1,0,\alpha+1))))$ |
| 1,4,9,14,18,14,13 | $\psi(\psi_\alpha(I(1,1,\alpha+1)))$ |
| 1,4,9,14,18,14,13, 18 | $\psi(\psi_\alpha(I(1,1,\alpha+1)+\Omega_{\alpha+1}))$ |
| 1,4,9,14,18,14,14 | $\psi(\psi_\alpha(I(1,1,\alpha+1)\cdot\omega))$ |
| 1,4,9,14,18,14,15 | $\psi(\psi_\alpha(I(1,\omega,\alpha+1)))$ |
| 1,4,9,14,18,14,16, 5 | $\psi(\psi_\alpha(I(1,\alpha,1)))$ |
| 1,4,9,14,18,14,16, 19 | $\psi(\psi_\alpha(I(1,\varepsilon_{\alpha+1},0)))$ |
| 1,4,9,14,18,14,17 | $\psi(\psi_\alpha(I(1,\Omega_{\alpha+1},0)\cdot\omega))$ |
| 1,4,9,14,18,14,18 | $\psi(\psi_\alpha(I(2,0,\alpha+1)))$ |
| 1,4,9,14,18,14,18, 8 | $\psi(\psi_\alpha(\varepsilon_{\psi_{I(2,0,\alpha+1)}(I(2,0,\alpha+1))+1}))$ |
| 1,4,9,14,18,14,18, 14 | $\psi(\psi_\alpha(I(2,0,\alpha+1)\cdot\omega))$ |
| 1,4,9,14,18,14,18, 14,14 | $\psi(\psi_\alpha(I(2,1,\alpha+1)\cdot\omega))$ |
| 1,4,9,14,18,14,18, 14,18 | $\psi(\psi_\alpha(I(3,0,\alpha+1)))$ |
| 1,4,9,14,18,15 | $\psi(\psi_\alpha(I(\omega,0,\alpha+1)))$ |
| 1,4,9,14,18,16,5 | $\psi(\psi_\alpha(I(\alpha,0,1)))$ |
| 1,4,9,14,18,16,19 | $\psi(\psi_\alpha(I(\varepsilon_{\alpha+1},0,0)))$ |
| 1,4,9,14,18,17 | $\psi(\psi_\alpha(I(\Omega_{\alpha+1},0,0)\cdot\omega))$ |
| 1,4,9,14,18,17,22, 27,31,28 | $\psi(\psi_\alpha(I(I(\omega,0,\alpha+1),0,0)))$ |
| 1,4,9,14,18,18 | $\psi(\psi_\alpha(I(1,0,0,\alpha+1)))$ |
| 1,4,9,14,18,18,14, 18,18 | $\psi(\psi_\alpha(I(2,0,0,\alpha+1)))$ |
| 1,4,9,14,18,18,18 | $\psi(\psi_\alpha(I(1,0,0,0,\alpha+1)))$ |
| 1,4,9,14,18,19 | $\psi(\psi_\alpha(I(1@\omega,\alpha+1)))$ |
| 1,4,9,14,18,20,5 | $\psi(\psi_\alpha(I(1@\alpha,1)))$ |
| 1,4,9,14,18,20,9, 14,18,20,5 | $\psi(\psi_\alpha(I(1@\alpha,2)))$ |
| 1,4,9,14,18,20,14 | $\psi(\psi_\alpha(I(1@\alpha,1,0)\cdot\omega))$ |
| 1,4,9,14,18,20,14, 18,20,5 | $\psi(\psi_\alpha(I(2@\alpha)))$ |
| 1,4,9,14,18,20,15 | $\psi(\psi_\alpha(I(\omega@\alpha)))$ |
| 1,4,9,14,18,20,18 | $\psi(\psi_\alpha(I(1@(\alpha+1))))$ |
| 1,4,9,14,18,20,18, 20,5 | $\psi(\psi_\alpha(I(1@(\alpha\cdot 2))))$ |
| 1,4,9,14,18,20,20, 5 | $\psi(\psi_\alpha(I(1@(\alpha^2))))$ |
| 1,4,9,14,18,20,23 | $\psi(\psi_\alpha(I(1@\varepsilon_{\alpha+1})))$ |
| 1,4,9,14,18,21 | $\psi(\psi_\alpha(I(1@\Omega_{\alpha+1})\cdot\omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,9,14,18,22 | $\psi(\psi_\alpha(M_{\alpha+1})^{\psi_\alpha(M_{\alpha+1})^{\psi_\alpha(M_{\alpha+1})}})$ |
| 1,4,9,14,18,23 | $\psi(\psi_\alpha(M_{\alpha+1}+\Omega_{\alpha+1}))$ |
| 1,4,9,14,18,24 | $\psi(\psi_\alpha(M_{\alpha+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,14,18,24,30, 30 | $\psi(\psi_\alpha(\psi_{M_{\alpha+1}}(M_{\alpha+\omega})))$ $=\psi(\psi_\alpha(M_{\alpha+1}+{\Omega_{\alpha+1}}^{\Omega_{\alpha+1}}\cdot\omega))$ |
| 1,4,9,14,18,24,31 | $\psi(\psi_\alpha(M_{\alpha+1}+\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,9,14,18,24,32, 33 | $\psi(\psi_\alpha(M_{\alpha+1}+\Omega_{\alpha+\omega}))$ |
| 1,4,9,14,18,24,32, 40,47,56 | $\psi(\psi_\alpha(M_{\alpha+1}\cdot 2+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,14,19 | $\psi(\psi_\alpha(M_{\alpha+1}\cdot\omega))$ |
| 1,4,9,14,19,6,9 | $\psi(\psi_\alpha(M_{\alpha+1}\cdot\omega+\Omega_{\alpha+1}))$ |
| 1,4,9,14,19,7 | $\psi(\psi_\alpha(M_{\alpha+1}\cdot\omega^2))$ |
| 1,4,9,14,19,7,9, 5 | $\psi(\psi_\alpha(M_{\alpha+1}\cdot\alpha))$ |
| 1,4,9,14,19,7,11 | $\psi(\psi_\alpha(M_{\alpha+1}\cdot\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,9,14,19,7,12, 13 | $\psi(\psi_\alpha(M_{\alpha+1}\cdot\Omega_{\alpha+\omega}))$ |
| 1,4,9,14,19,7,12, 17,22 | $\psi(\psi_\alpha({M_{\alpha+1}}^2\cdot\omega))$ |
| 1,4,9,14,19,8 | $\psi(\psi_\alpha(\varepsilon_{M_{\alpha+1}+1}))$ |
| 1,4,9,14,19,8,14 | $\psi(\psi_\alpha(\Omega_{M_{\alpha+1}+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,14,19,9 | $\psi(\psi_\alpha(\Omega_{M_{\alpha+1}+1}\cdot\omega))$ |
| 1,4,9,14,19,9,10 | $\psi(\psi_\alpha(\Omega_{M_{\alpha+1}+\omega}))$ |
| 1,4,9,14,19,9,11, 5 | $\psi(\psi_\alpha(\Omega_{M_{\alpha+1}+\alpha}))$ |
| 1,4,9,14,19,9,12, 16 | $\psi(\psi_\alpha(\Omega_{M_{\alpha+1}+\varepsilon_{\Omega_{\alpha+1}+1}+1}))$ |
| 1,4,9,14,19,9,12, 17,22,27 | $\psi(\psi_\alpha(\Omega_{M_{\alpha+1}\cdot 2}\cdot\omega))$ |
| 1,4,9,14,19,9,12, 17,22,27,16 | $\psi(\psi_\alpha(\Omega_{\varepsilon_{M_{\alpha+1}+1}+1}))$ |
| 1,4,9,14,19,9,12, 17,22,27,17 | $\psi(\psi_\alpha(\Omega_{\Omega_{M_{\alpha+1}+1}+1}\cdot\omega))$ |
| 1,4,9,14,19,9,13 | $\psi(\psi_\alpha(\mathrm{OFP}(M_{\alpha+1})))$ |
| 1,4,9,14,19,9,13, 19 | $\psi(\psi_\alpha(I(M_{\alpha+1}+1)+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,14,19,9,14 | $\psi(\psi_\alpha(I(M_{\alpha+1}+1)\cdot\omega))$ |
| 1,4,9,14,19,9,14, 13 | $\psi(\psi_\alpha(\mathrm{IFP}(M_{\alpha+1}+1)))$ |
| 1,4,9,14,19,9,14, 14 | $\psi(\psi_\alpha(I(1,M_{\alpha+1}+1)\cdot\omega))$ |
| 1,4,9,14,19,9,14, 15,9,14,14 | $\psi(\psi_\alpha(I(\omega,M_{\alpha+1}+1)))$ |
| 1,4,9,14,19,9,14, 16,5 | $\psi(\psi_\alpha(I(\alpha,M_{\alpha+1}+1)))$ |
| 1,4,9,14,19,9,14, 17,22,27,32 | $\psi(\psi_\alpha(I(M_{\alpha+1},1)\cdot\omega))$ |
| 1,4,9,14,19,9,14, 17,22,27,32,14 | $\psi(\psi_\alpha(I(M_{\alpha+1}+1,0)\cdot\omega))$ |
| 1,4,9,14,19,9,14, 17,22,27,32,22,23 | $\psi(\psi_\alpha(I(\Omega_{M_{\alpha+1}+\omega},0)))$ |
| 1,4,9,14,19,9,14, 18 | $\psi(\psi_\alpha(I(1,0,M_{\alpha+1}+1)))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,9,14,19,9,14, 18,10 | $\psi(\psi_\alpha(I(1,0,M_{\alpha+1}+1))\cdot\omega)$ |
| 1,4,9,14,19,9,14, 18,13,18 | $\psi(\psi_\alpha(I(1,0,M_{\alpha+1}+1)+\Omega_{\alpha+1}))$ |
| 1,4,9,14,19,9,14, 18,14 | $\psi(\psi_\alpha(I(1,0,M_{\alpha+1}+1)\cdot\omega))$ |
| 1,4,9,14,19,9,14, 18,14,8 | $\psi(\psi_\alpha(\varepsilon_{I(1,0,M_{\alpha+1}+1)+1}))$ |
| 1,4,9,14,19,9,14, 18,14,9 | $\psi(\psi_\alpha(\Omega_{I(1,0,M_{\alpha+1}+1)\cdot\omega})$ |
| 1,4,9,14,19,9,14, 18,14,9,14,17,22,27, 32,22,27,31 | $\psi(\psi_\alpha(I(I(1,0,M_{\alpha+1}+1),1)$ |
| 1,4,9,14,19,9,14, 18,14,9,14,18 | $\psi(\psi_\alpha(I(1,0,M_{\alpha+1}+2)))$ |
| 1,4,9,14,19,9,14, 18,14,9,14,18,14 | $\psi(\psi_\alpha(I(1,0,M_{\alpha+1}+2)\cdot\omega))$ |
| 1,4,9,14,19,9,14, 18,14,11,5 | $\psi(\psi_\alpha(I(1,0,M_{\alpha+1}+\alpha)))$ |
| 1,4,9,14,19,9,14, 18,14,12,17,22,27,17, 22,26 | $\psi(\psi_\alpha(I(1,0,I(1,0,M_{\alpha+1}+1))))$ |
| 1,4,9,14,19,9,14, 18,14,13 | $\psi(\psi_\alpha(I(1,1,M_{\alpha+1}+1)))$ |
| 1,4,9,14,19,9,14, 18,14,18 | $\psi(\psi_\alpha(I(2,0,M_{\alpha+1}+1)))$ |
| 1,4,9,14,19,9,14, 18,18 | $\psi(\psi_\alpha(I(1,0,0,M_{\alpha+1}+1)))$ |
| 1,4,9,14,19,9,14, 18,23 | $\psi(\psi_\alpha(M_{\alpha+2}+\Omega_{\alpha+1}))$ |
| 1,4,9,14,19,9,14, 19 | $\psi(\psi_\alpha(M_{\alpha+2}\cdot\omega))$ |
| 1,4,9,14,19,9,14, 19,9 | $\psi(\psi_\alpha(\Omega_{M_{\alpha+2}+1}\cdot\omega))$ |
| 1,4,9,14,19,9,14, 19,9,14,15 | $\psi(\psi_\alpha(I(\omega,M_{\alpha+2}+1)))$ |
| 1,4,9,14,19,9,14, 19,9,14,18 | $\psi(\psi_\alpha(I(1,0,M_{\alpha+2}+1)))$ |
| 1,4,9,14,19,9,14, 19,9,14,18,23 | $\psi(\psi_\alpha(M_{\alpha+3}+\Omega_{\alpha+1}))$ |
| 1,4,9,14,19,9,14, 19,9,14,19 | $\psi(\psi_\alpha(M_{\alpha+3}\cdot\omega))$ |
| 1,4,9,14,19,10 | $\psi(\psi_\alpha(M_{\alpha+\omega}))$ |
| 1,4,9,14,19,11,5 | $\psi(\psi_\alpha(M_{\alpha\cdot 2}))$ |
| 1,4,9,14,19,11,5, 5 | $\psi(\psi_\alpha(M_{\alpha\cdot 2})\cdot\omega)$ |
| 1,4,9,14,19,11,6, 9 | $\psi(\psi_\alpha(M_{\alpha\cdot 2}+\Omega_{\alpha+1}))$ |
| 1,4,9,14,19,11,6, 10,16,22,28,19,11 | $\psi(\psi_\alpha(M_{\alpha\cdot 2}\cdot 2))$ |
| 1,4,9,14,19,11,7 | $\psi(\psi_\alpha(M_{\alpha\cdot 2}\cdot\omega))$ |
| 1,4,9,14,19,11,7, 11 | $\psi(\psi_\alpha(M_{\alpha\cdot 2}\cdot\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,9,14,19,11,7, 12,17,22,14,5 | $\psi(\psi_\alpha({M_{\alpha\cdot 2}}^2))$ |
| 1,4,9,14,19,11,8 | $\psi(\psi_\alpha(\varepsilon_{M_{\alpha\cdot 2}+1}))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,9,14,19,11,9 | $\psi(\psi_\alpha(\Omega_{M_{\alpha\cdot 2}+1}\cdot\omega))$ |
| 1,4,9,14,19,11,9, 14,18 | $\psi(\psi_\alpha(I(1,0,M_{\alpha\cdot 2}+1)))$ |
| 1,4,9,14,19,11,9, 14,19 | $\psi(\psi_\alpha(M_{\alpha\cdot 2+1}\cdot\omega))$ |
| 1,4,9,14,19,11,9, 14,19,11,5 | $\psi(\psi_\alpha(M_{\alpha\cdot 3}))$ |
| 1,4,9,14,19,11,11 | $\psi(\psi_\alpha(M_{\alpha^2}))$ |
| 1,4,9,14,19,11,14 | $\psi(\psi_\alpha(M_{\varepsilon_{\alpha+1}}))$ |
| 1,4,9,14,19,12 | $\psi(\psi_\alpha(M_{\Omega_{\alpha+1}}\cdot\omega))$ |
| 1,4,9,14,19,12,9, 14,19 | $\psi(\psi_\alpha(M_{\Omega_{\alpha+1}+1}\cdot\omega))$ |
| 1,4,9,14,19,12,10 | $\psi(\psi_\alpha(M_{\Omega_{\alpha+1}\cdot\omega}))$ |
| 1,4,9,14,19,12,16 | $\psi(\psi_\alpha(M_{\varepsilon_{\Omega_{\alpha+1}+1}}))$ |
| 1,4,9,14,19,12,17, 18 | $\psi(\psi_\alpha(M_{\Omega_{\alpha+\omega}}))$ |
| 1,4,9,14,19,12,17, 22,27 | $\psi(\psi_\alpha(M_{M_{\alpha+1}}\cdot\omega))$ |
| 1,4,9,14,19,12,17, 22,27,20 | $\psi(\psi_\alpha(M_{M_{\Omega_{\alpha+1}}}\cdot\omega))$ |
| 1,4,9,14,19,13 | $\psi(\psi_\alpha(\mathrm{MFP}(\alpha+1)))$ $=\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2))$ |
| 1,4,9,14,19,13,7 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+$ $\psi_{\alpha_2}({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+1)))$ |
| 1,4,9,14,19,13,8 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+\alpha_2))$ |
| 1,4,9,14,19,13,8, 13 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+\varepsilon_{\alpha_2+1}))$ |
| 1,4,9,14,19,13,9 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+\Omega_{\alpha_2+1}+$ $\psi_{\alpha_2}({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+\Omega_{\alpha_2+1}+1)))$ |
| 1,4,9,14,19,13,9, 10 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+\Omega_{\alpha_2+1}\cdot\omega))$ |
| 1,4,9,14,19,13,9, 14,15 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+{\Omega_{\alpha_2+1}}^{\omega}))$ |
| 1,4,9,14,19,13,9, 14,18 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2+{\Omega_{\alpha_2+1}}^{\alpha_2}))$ |
| 1,4,9,14,19,13,9, 14,19 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot(\alpha_2+1)+$ $\psi_{\alpha_2}({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot(\alpha_2+1)+1)))$ |
| 1,4,9,14,19,13,9, 14,19,10 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot(\alpha_2+\omega)))$ |
| 1,4,9,14,19,13,9, 14,19,11,5 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot(\alpha_2+\alpha)))$ |
| 1,4,9,14,19,13,9, 14,19,12,17,22,27,21 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot(\alpha_2+$ $\psi_{\alpha_2}({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2))))$ |
| 1,4,9,14,19,13,9, 14,19,13 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2\cdot 2))$ |
| 1,4,9,14,19,13,11, 5 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2\cdot\alpha))$ |
| 1,4,9,14,19,13,12, 17,22,27,21,18 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2\cdot$ $\psi_{\alpha_2}({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2\cdot\omega)))$ |
| 1,4,9,14,19,13,13 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot({\alpha_2}^2)))$ |
| 1,4,9,14,19,13,18 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\varepsilon_{\alpha_2+1}))$ |
| 1,4,9,14,19,13,19, 27,35,43,34 | $\psi(\psi_\alpha({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}+1}+$ $\psi_{\alpha_2}({\Omega_{\alpha_2+1}}^{\Omega_{\alpha_2+1}}\cdot\alpha_2)))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,9,14,19,14 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}+ \psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}+1)))$ |
| 1,4,9,14,19,14,8 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}+\alpha_2))$ |
| 1,4,9,14,19,14,8 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}+\Omega_{\alpha_2+1} +\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}+\Omega_{\alpha_2+1}+1)))$ |
| 1,4,9,14,19,14,9, 10 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}+\Omega_{\alpha_2+1}\cdot\omega))$ |
| 1,4,9,14,19,14,9, 14,19,10 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1} +\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}}\cdot\omega))$ |
| 1,4,9,14,19,14,9, 14,19,14 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}\cdot 2+ \psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}\cdot 2+1)))$ |
| 1,4,9,14,19,14,10 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}\cdot\omega))$ |
| 1,4,9,14,19,14,11, 5 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}\cdot\alpha))$ |
| 1,4,9,14,19,14,11, 9,14,19,14,11,5 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}\cdot\alpha\cdot 2))$ |
| 1,4,9,14,19,14,11, 14 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}\cdot\varepsilon_{\alpha+1}))$ |
| 1,4,9,14,19,14,12 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}\cdot\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,9,14,19,14,12, 17,18 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}\cdot\Omega_{\alpha+\omega}))$ |
| 1,4,9,14,19,14,13 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}\cdot\alpha_2))$ |
| 1,4,9,14,19,14,13, 9 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}\cdot\alpha_2+ \psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}\cdot\alpha_2+\Omega_{\alpha_2+1}+1)))$ |
| 1,4,9,14,19,14,13, 9,14,19,14,11,5 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}\cdot(\alpha_2+\alpha)))$ |
| 1,4,9,14,19,14,13, 9,14,19,14,12,17,18 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}\cdot(\alpha_2+\Omega_{\alpha+\omega})))$ |
| 1,4,9,14,19,14,13, 9,14,19,14,13 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}\cdot\alpha_2\cdot 2))$ |
| 1,4,9,14,19,14,13, 18 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}\cdot\varepsilon_{\alpha_2+1}))$ |
| 1,4,9,14,19,14,13, 19,27,35,43,35,34 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+2}+ \psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+1}\cdot\alpha_2)))$ |
| 1,4,9,14,19,14,14 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+2}+ \psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+2}+1)))$ |
| 1,4,9,14,19,14,15 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\omega}))$ |
| 1,4,9,14,19,14,16, 5 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha}))$ |
| 1,4,9,14,19,14,16, 14 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha+1}+ \psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha+1}+1)))$ |
| 1,4,9,14,19,14,16, 19 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\varepsilon_{\alpha+1}}))$ |
| 1,4,9,14,19,14,17, 22,27,32,27,28 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\omega})}))$ |
| 1,4,9,14,19,14,18 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2}))$ |
| 1,4,9,14,19,14,18, 8 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2}+\alpha_2))$ |
| 1,4,9,14,19,14,18, 9,14,19,14,15 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2}+\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\omega}))$ |
| 1,4,9,14,19,14,18, 9,14,19,14,18 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2}\cdot 2))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,9,14,19,14,18, 13 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2}\cdot\alpha_2))$ |
| 1,4,9,14,19,14,18, 13,18 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2}\cdot\varepsilon_{\alpha_2+1}))$ |
| 1,4,9,14,19,14,18, 14 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2+1}+$ $\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2+1}+1)))$ |
| 1,4,9,14,19,14,18, 14,18 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\alpha_2\cdot 2}))$ |
| 1,4,9,14,19,14,18, 23 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}+\varepsilon_{\alpha_2+1}}))$ |
| 1,4,9,14,19,14,19 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}\cdot 2}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}\cdot 2}+1)))$ |
| 1,4,9,14,19,14,19, 10 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}\cdot 2}\cdot\omega))$ |
| 1,4,9,14,19,14,19, 14,15 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}\cdot 2+\omega}))$ |
| 1,4,9,14,19,16,5 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}\cdot\alpha}))$ |
| 1,4,9,14,19,18 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}\cdot\alpha_2}))$ |
| 1,4,9,14,19,18,9, 14,19,18 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}\cdot\alpha_2}\cdot 2))$ |
| 1,4,9,14,19,18,13 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}\cdot\alpha_2}\cdot\alpha_2))$ |
| 1,4,9,14,19,18,14, 19 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}\cdot(\alpha_2+1)}+$ $\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}\cdot(\alpha_2+1)}+1)))$ |
| 1,4,9,14,19,18,14, 19,18 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}\cdot\alpha_2\cdot 2}))$ |
| 1,4,9,14,19,18,23 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}\cdot\varepsilon_{\alpha_2+1}}))$ |
| 1,4,9,14,19,19 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^2}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^2}+1)))$ |
| 1,4,9,14,19,19,10 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^2}\cdot\omega))$ |
| 1,4,9,14,19,19,14, 19,19 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^2\cdot 2}+$ $\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^2\cdot 2}+1)))$ |
| 1,4,9,14,19,19,19 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^3}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^3}+1)))$ |
| 1,4,9,14,19,21,5 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^\alpha}))$ |
| 1,4,9,14,19,22,27, 32,37,38 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^{\psi_{\alpha_2}(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^\omega})}}))$ |
| 1,4,9,14,19,23 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^{\alpha_2}}))$ |
| 1,4,9,14,19,24,10 | $\psi(\psi_\alpha(\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}{}^{\Omega_{\alpha_2+1}}}\cdot\omega))$ |
| 1,4,9,15 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}))$ |
| 1,4,9,15,6,9 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}+\Omega_{\alpha+1}))$ |
| 1,4,9,15,7 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}+$ $\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}+1}+1)))$ |
| 1,4,9,15,7,12,18 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}+$ $\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}+1}+\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}+1}))))$ |
| 1,4,9,15,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}+\alpha_2))$ |
| 1,4,9,15,9 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}+\Omega_{\alpha_2+1}+$ $\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}+1}+\Omega_{\alpha_2+1}+1)))$ |
| 1,4,9,15,9,15 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}\cdot 2))$ |
| 1,4,9,15,11,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}\cdot\alpha))$ |
| 1,4,9,15,12,17,23 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}\cdot\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}+1})))$ |
| 1,4,9,15,13 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}\cdot\alpha_2))$ |
| 1,4,9,15,14 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}\cdot\Omega_{\alpha_2+1}+$ $\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}+1}\cdot\Omega_{\alpha_2+1}+1)))$ |
| 1,4,9,15,14,20 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}{}^2))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,9,15,14,20,9, 15,14,20 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha_2+1}+1}}^2 \cdot 2))$ |
| 1,4,9,15,14,20,14 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha_2+1}+1}}^2 \cdot \Omega_{\alpha_2+1}+ \psi_{\alpha_2}({\varepsilon_{\Omega_{\alpha_2+1}+1}}^2 \cdot \Omega_{\alpha_2+1}+1)))$ |
| 1,4,9,15,14,20,15 | $\psi(\psi_\alpha({\varepsilon_{\Omega_{\alpha_2+1}+1}}^\omega))$ |
| 1,4,9,15,14,20,19, 25 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+1}{}^{\varepsilon_{\Omega_{\alpha_2+1}+1}+1}))$ |
| 1,4,9,15,15 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+2}))$ |
| 1,4,9,15,17,5 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+\alpha}))$ |
| 1,4,9,15,19 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}+\alpha_2}))$ |
| 1,4,9,15,20 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2}+\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2}+1)))$ |
| 1,4,9,15,20,7 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2}+\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2}+\omega)))$ |
| 1,4,9,15,20,7,12, 18 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2}+ \psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2}+\psi_{\alpha_2}(\varepsilon_{\Omega_{\alpha+1}+1}))))$ |
| 1,4,9,15,20,8 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2}+\alpha_2))$ |
| 1,4,9,15,20,9,15 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2}+\varepsilon_{\Omega_{\alpha_2+1}+1}))$ |
| 1,4,9,15,20,15 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}\cdot 2+1}))$ |
| 1,4,9,15,20,16 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_2+1}\cdot\omega}))$ |
| 1,4,9,15,20,26 | $\psi(\psi_\alpha(\varepsilon_{\varepsilon_{\Omega_{\alpha_2+1}+1}+1}))$ |
| 1,4,9,15,21 | $\psi(\psi_\alpha({\alpha_3}^2))$ |
| 1,4,9,15,22 | $\psi(\psi_\alpha(\varepsilon_{\alpha_3+1}))$ |
| 1,4,9,15,23,33,44 | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\psi_{\alpha_2}(\alpha_3)))$ |
| 1,4,9,16 | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\psi_{\alpha_2}(\Omega_{\alpha_3+1}+1)))$ |
| 1,4,9,16,8 | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\alpha_2))$ |
| 1,4,9,16,9,16 | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\psi_{\alpha_3}(\Omega_{\alpha_3+1})+ \psi_{\alpha_2}(\Omega_{\alpha_3+1}+\psi_{\alpha_3}(\Omega_{\alpha_3+1})+1)))$ |
| 1,4,9,16,10 | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\psi_{\alpha_3}(\Omega_{\alpha_3+1}+1)))$ |
| 1,4,9,16,13 | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\psi_{\alpha_3}(\Omega_{\alpha_3+1}+\alpha_2)))$ |
| 1,4,9,16,14,20 | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\psi_{\alpha_3}(\Omega_{\alpha_3+1}+\psi_{\alpha_3}(\alpha_3))))$ |
| 1,4,9,16,15 | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\alpha_3))$ |
| 1,4,9,16,15,22 | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}+\varepsilon_{\alpha_3+1}))$ |
| 1,4,9,16,16 | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}\cdot 2+\psi_{\alpha_2}(\Omega_{\alpha_3+1}\cdot 2+1)))$ |
| 1,4,9,16,17 | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}\cdot\omega))$ |
| 1,4,9,16,20 | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}\cdot\alpha_2))$ |
| 1,4,9,16,21,27 | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}\cdot\psi_{\alpha_3}(\alpha_3)))$ |
| 1,4,9,16,22 | $\psi(\psi_\alpha(\Omega_{\alpha_3+1}\cdot\alpha_3))$ |
| 1,4,9,16,23 | $\psi(\psi_\alpha({\Omega_{\alpha_3+1}}^2+\psi_{\alpha_2}({\Omega_{\alpha_3+1}}^2+1)))$ |
| 1,4,9,16,23,8 | $\psi(\psi_\alpha({\Omega_{\alpha_3+1}}^2+\alpha_2))$ |
| 1,4,9,16,23,10 | $\psi(\psi_\alpha({\Omega_{\alpha_3+1}}^2+\psi_{\alpha_3}({\Omega_{\alpha_3+1}}^2+1)))$ |
| 1,4,9,16,23,15 | $\psi(\psi_\alpha({\Omega_{\alpha_3+1}}^2+\alpha_3))$ |
| 1,4,9,16,23,16,23 | $\psi(\psi_\alpha({\Omega_{\alpha_3+1}}^2\cdot 2+\psi_{\alpha_2}({\Omega_{\alpha_3+1}}^2\cdot 2+1)))$ |
| 1,4,9,16,23,17 | $\psi(\psi_\alpha({\Omega_{\alpha_3+1}}^2\cdot\omega))$ |
| 1,4,9,16,23,23 | $\psi(\psi_\alpha({\Omega_{\alpha_3+1}}^3+\psi_{\alpha_2}({\Omega_{\alpha_3+1}}^3+1)))$ |
| 1,4,9,16,23,24 | $\psi(\psi_\alpha({\Omega_{\alpha_3+1}}^\omega))$ |
| 1,4,9,16,24 | $\psi(\psi_\alpha(\alpha_4))$ |
| 1,4,9,16,24,16,24 | $\psi(\psi_\alpha(\alpha_4+\psi_{\alpha_4}(\alpha_4)))$ |
| 1,4,9,16,24,24 | $\psi(\psi_\alpha(\alpha_4\cdot 2))$ |
| 1,4,9,16,25 | $\psi(\psi_\alpha(\Omega_{\alpha_4+1}+\psi_{\alpha_2}(\Omega_{\alpha_4+1}+1)))$ |
| 1,4,9,16,25,35 | $\psi(\psi_\alpha(\alpha_5))$ |
| 1,4,10 | $\psi(\psi_\alpha(\alpha_\omega))$ |
| 1,4,10,3 | $\psi(\psi_\alpha(\alpha_\omega)+\Omega)$ |
| 1,4,10,4 | $\psi(\psi_\alpha(\alpha_\omega)+\Omega_\omega)$ |
| 1,4,10,4,8 | $\psi(\psi_\alpha(\alpha_\omega)+\psi_\alpha(\alpha_2))$ |
| 1,4,10,4,9 | $\psi(\psi_\alpha(\alpha_\omega)+\psi_\alpha(\Omega_{\alpha+2}\cdot\omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,4,10 | $\psi(\psi_\alpha(\alpha_\omega)\cdot 2)$ |
| 1,4,10,6 | $\psi(\psi_\alpha(\alpha_\omega)\cdot \Omega)$ |
| 1,4,10,6,3,7,14, 10 | $\psi(\psi_\alpha(\alpha_\omega)\cdot \Omega_2)$ |
| 1,4,10,6,4 | $\psi(\psi_\alpha(\alpha_\omega)\cdot \Omega_\omega)$ |
| 1,4,10,6,4,8 | $\psi(\psi_\alpha(\alpha_\omega)\cdot \psi_\alpha(\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,10,6,4,8,14, 23,19 | $\psi(\psi_\alpha(\alpha_\omega)\cdot \psi_\alpha(\Omega_{\alpha+2}+\Omega_{\alpha+1}))$ |
| 1,4,10,6,4,9 | $\psi(\psi_\alpha(\alpha_\omega)\cdot \psi_\alpha(\Omega_{\alpha+2}\cdot\omega))$ |
| 1,4,10,6,4,9,15, 23,34,30 | $\psi(\psi_\alpha(\alpha_\omega)\cdot \psi_\alpha(\Omega_{\alpha_2+2}+\Omega_{\alpha+1}))$ |
| 1,4,10,6,4,9,16 | $\psi(\psi_\alpha(\alpha_\omega)\cdot \psi_\alpha(\Omega_{\alpha_2+2}+\psi_{\alpha_2}(\Omega_{\alpha_2+2}+1)))$ |
| 1,4,10,6,4,10 | $\psi(\psi_\alpha(\alpha_\omega)^2)$ |
| 1,4,10,6,4,10,4, 10 | $\psi(\psi_\alpha(\alpha_\omega)^2+\psi_\alpha(\alpha_\omega))$ |
| 1,4,10,6,5 | $\psi(\psi_\alpha(\alpha_\omega)^2\cdot\omega)$ |
| 1,4,10,6,7 | $\psi(\psi_\alpha(\alpha_\omega)^\omega)$ |
| 1,4,10,6,9 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha+1}))$ |
| 1,4,10,6,9,5 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha+1})+$ $\psi_{\psi_\alpha(\alpha_\omega+\Omega_{\alpha+1})}(\psi_\alpha(\alpha_\omega+\Omega_{\alpha+1})+1))$ |
| 1,4,10,6,9,9 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha+1})\cdot 2)$ |
| 1,4,10,6,10 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,10,6,10,15 | $\psi(\psi_\alpha(\alpha_\omega+\varepsilon_{\Omega_{\alpha+1}+1}))$ |
| 1,4,10,6,10,17 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega)))$ |
| 1,4,10,7 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+1)))$ |
| 1,4,10,7,4,10 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+1))+\psi_\alpha(\alpha_\omega))$ |
| 1,4,10,7,4,10,6, 10,17,14 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+1))+$ $\psi_{\psi_\alpha(\alpha_\omega+\Omega_{\alpha+1})}(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+1))))$ |
| 1,4,10,7,4,10,6, 10,17,14,9 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+1))+\psi_\alpha(\alpha_\omega+\Omega_{\alpha+1}))$ |
| 1,4,10,7,4,10,6, 10,17,14,10,17 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+1))+\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega)))$ |
| 1,4,10,7,4,10,7 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+1))\cdot 2)$ |
| 1,4,10,7,6,4 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+1))\cdot \Omega_\omega)$ |
| 1,4,10,7,6,4,10 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+1))\cdot \psi_\alpha(\alpha_\omega))$ |
| 1,4,10,7,6,4,10, 6,10,17,14,13 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+1))\cdot \psi_\alpha(\alpha_\omega+\Omega_{\alpha+1}))$ |
| 1,4,10,7,6,4,10, 7 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+1))^2)$ |
| 1,4,10,7,6,9 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+1)+\Omega_{\alpha+1}))$ |
| 1,4,10,7,6,10,17, 14 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+1)\cdot 2))$ |
| 1,4,10,7,7 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+2)))$ |
| 1,4,10,7,8 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\omega)))$ |
| 1,4,10,7,9,5 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\alpha)))$ |
| 1,4,10,7,9,12 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\varepsilon_{\alpha+1})))$ |
| 1,4,10,7,10 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\Omega_{\alpha+1}+1)))$ |
| 1,4,10,7,10,6,9 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\Omega_{\alpha+1}+1)+\Omega_{\alpha+1}))$ |
| 1,4,10,7,10,7,9, 5 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\Omega_{\alpha+1}+\alpha)))$ |
| 1,4,10,7,10,7,10 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\Omega_{\alpha+1}\cdot 2+1)))$ |
| 1,4,10,7,10,9,5 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\Omega_{\alpha+1}\cdot\alpha)))$ |
| 1,4,10,7,10,9,7, 10,9,5 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\Omega_{\alpha+1}\cdot\alpha\cdot 2)))$ |
| 1,4,10,7,10,10 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+{\Omega_{\alpha+1}}^2+1)))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,10,7,10,11 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+{\Omega_{\alpha+1}}^\omega)))$ |
| 1,4,10,7,11 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\psi_{\alpha_2}(\alpha_2))))$ |
| 1,4,10,7,11,7,11 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\psi_{\alpha_2}(\alpha_2)\cdot 2)))$ |
| 1,4,10,7,11,10,14 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\psi_{\alpha_2}(\alpha_2+\psi_{\alpha_2}(\alpha_2)))))$ |
| 1,4,10,7,11,11 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\psi_{\alpha_2}(\alpha_2\cdot 2))))$ |
| 1,4,10,7,11,17 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\psi_{\alpha_2}(\Omega_{\alpha_2+1}))+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,10,7,12 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\psi_{\alpha_2}(\Omega_{\alpha_2+1})+1)))$ |
| 1,4,10,7,12,7,12 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\psi_{\alpha_2}(\Omega_{\alpha_2+1})\cdot 2+1)))$ |
| 1,4,10,7,12,11 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\psi_{\alpha_2}(\Omega_{\alpha_2+1}+\alpha_2))))$ |
| 1,4,10,7,12,18 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\psi_{\alpha_2}(\alpha_3))))$ |
| 1,4,10,7,13 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega))))$ |
| 1,4,10,7,13,7,13 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega)\cdot 2)))$ |
| 1,4,10,7,13,10,16 | $\psi(\psi_\alpha(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega+\psi_{\alpha_2}(\alpha_\omega)))))$ |
| 1,4,10,8 | $\psi(\psi_\alpha(\alpha_\omega+\alpha_2))$ |
| 1,4,10,8,6,9 | $\psi(\psi_\alpha(\alpha_\omega+\alpha_2+\Omega_{\alpha+1}))$ |
| 1,4,10,8,6,10,17 | $\psi(\psi_\alpha(\alpha_\omega+\alpha_2+\psi_{\alpha_2}(\alpha_\omega)))$ |
| 1,4,10,8,6,10,17, 15 | $\psi(\psi_\alpha(\alpha_\omega+\alpha_2+\psi_{\alpha_2}(\alpha_\omega+\alpha_2)))$ |
| 1,4,10,8,7 | $\psi(\psi_\alpha(\alpha_\omega+\alpha_2+\psi_{\alpha_2}(\alpha_\omega+\alpha_2+1)))$ |
| 1,4,10,8,7,10 | $\psi(\psi_\alpha(\alpha_\omega+\alpha_2+\psi_{\alpha_2}(\alpha_\omega+\alpha_2+\Omega_{\alpha+1}+1)))$ |
| 1,4,10,8,7,11 | $\psi(\psi_\alpha(\alpha_\omega+\alpha_2+\psi_{\alpha_2}(\alpha_\omega+\alpha_2+\psi_{\alpha_2}(\alpha_2))))$ |
| 1,4,10,8,7,13,11 | $\psi(\psi_\alpha(\alpha_\omega+\alpha_2+\psi_{\alpha_2}(\alpha_\omega+\alpha_2+\psi_{\alpha_2}(\alpha_\omega+\alpha_2))))$ |
| 1,4,10,8,8 | $\psi(\psi_\alpha(\alpha_\omega+\alpha_2\cdot 2))$ |
| 1,4,10,8,10,5 | $\psi(\psi_\alpha(\alpha_\omega+\alpha_2\cdot\alpha))$ |
| 1,4,10,8,10,14,21 | $\psi(\psi_\alpha(\alpha_\omega+\alpha_2\cdot\Omega_{\alpha+1}+\psi_{\alpha_2}(\alpha_\omega)))$ |
| 1,4,10,8,11 | $\psi(\psi_\alpha(\alpha_\omega+\alpha_2\cdot\Omega_{\alpha+1}+$ $\psi_{\alpha_2}(\alpha_\omega+\alpha_2\cdot\Omega_{\alpha+1}+1)))$ |
| 1,4,10,8,11,8 | $\psi(\psi_\alpha(\alpha_\omega+\alpha_2\cdot(\Omega_{\alpha+1}+1)))$ |
| 1,4,10,8,11,8,11 | $\psi(\psi_\alpha(\alpha_\omega+\alpha_2\cdot\Omega_{\alpha+1}\cdot 2+$ $\psi_{\alpha_2}(\alpha_\omega+\alpha_2\cdot\Omega_{\alpha+1}\cdot 2+1)))$ |
| 1,4,10,8,11,15 | $\psi(\psi_\alpha(\alpha_\omega+\alpha_2\cdot\psi_{\alpha_2}(\alpha_2)))$ |
| 1,4,10,8,11,17 | $\psi(\psi_\alpha(\alpha_\omega+\alpha_2\cdot\psi_{\alpha_2}(\alpha_\omega)))$ |
| 1,4,10,8,12 | $\psi(\psi_\alpha(\alpha_\omega+{\alpha_2}^2))$ |
| 1,4,10,8,12,8,12 | $\psi(\psi_\alpha(\alpha_\omega+{\alpha_2}^2\cdot 2))$ |
| 1,4,10,8,13 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha_2+1}))$ |
| 1,4,10,8,14 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha_2+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,10,8,14,23 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\alpha_\omega)))$ |
| 1,4,10,9 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\alpha_\omega+\Omega_{\alpha_2+1}+1)))$ |
| 1,4,10,9,7 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\alpha_\omega+\Omega_{\alpha_2+1}+2)))$ |
| 1,4,10,9,7,13 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha_2+1}+$ $\psi_{\alpha_2}(\alpha_\omega+\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\alpha_\omega))))$ |
| 1,4,10,9,8 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha_2+1}+\alpha_2))$ |
| 1,4,10,9,8,13 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha_2+1}\cdot 2))$ |
| 1,4,10,9,8,14,23 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha_2+1}\cdot 2+\psi_{\alpha_2}(\alpha_\omega)))$ |
| 1,4,10,9,9 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha_2+1}\cdot 2+$ $\psi_{\alpha_2}(\alpha_\omega+\Omega_{\alpha_2+1}\cdot 2+1)))$ |
| 1,4,10,9,10 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha_2+1}\cdot\omega))$ |
| 1,4,10,9,11,5 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha_2+1}\cdot\alpha))$ |
| 1,4,10,9,12 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha_2+1}\cdot\Omega_{\alpha+1}+$ $\psi_{\alpha_2}(\alpha_\omega+\Omega_{\alpha_2+1}\cdot\Omega_{\alpha+1}+1)))$ |
| 1,4,10,9,12,16 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha_2+1}\cdot\psi_{\alpha_2}(\alpha_2)))$ |
| 1,4,10,9,12,18 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha_2+1}\cdot\psi_{\alpha_2}(\alpha_\omega)))$ |
| 1,4,10,9,13 | $\psi(\psi_\alpha(\alpha_\omega+\Omega_{\alpha_2+1}\cdot\alpha_2))$ |
| 1,4,10,9,13,18 | $\psi(\psi_\alpha(\alpha_\omega+{\Omega_{\alpha_2+1}}^2))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,9,14 | $\psi(\psi_\alpha(\alpha_\omega + {\Omega_{\alpha_2+1}}^2 + \psi_{\alpha_2}(\alpha_\omega + {\Omega_{\alpha_2+1}}^2 + 1)))$ |
| 1,4,10,9,15 | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_3}(\alpha_3)))$ |
| 1,4,10,9,17,16 | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_3+1} + \psi_{\alpha_2}(\alpha_\omega + \Omega_{\alpha_3+1} + 1)))$ |
| 1,4,10,9,17,16,22 | $\psi(\psi_\alpha(\alpha_\omega + \Omega_{\alpha_3+1} \cdot \alpha_3))$ |
| 1,4,10,9,17,16,24 | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_4}(\alpha_4)))$ |
| 1,4,10,10 | $\psi(\psi_\alpha(\alpha_\omega \cdot 2))$ |
| 1,4,10,10,7 | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \psi_{\alpha_2}(\alpha_\omega \cdot 2 + 1)))$ |
| 1,4,10,10,7,13,13 | $\psi(\psi_\alpha(\alpha_\omega + \psi_{\alpha_2}(\alpha_\omega \cdot 2 + \psi_{\alpha_2}(\alpha_\omega \cdot 2))))$ |
| 1,4,10,10,8 | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \alpha_2))$ |
| 1,4,10,10,8,14 | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \Omega_{\alpha_2+1} + \Omega_{\alpha+1} \cdot \omega))$ |
| 1,4,10,10,9 | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \Omega_{\alpha_2+1} + \psi_{\alpha_2}(\alpha_\omega \cdot 2 + \Omega_{\alpha_2+1} + 1)))$ |
| 1,4,10,10,9,10 | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \Omega_{\alpha_2+1} \cdot \omega))$ |
| 1,4,10,10,9,13 | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \Omega_{\alpha_2+1} \cdot \alpha_2))$ |
| 1,4,10,10,9,15 | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \psi_{\alpha_3}(\alpha_3)))$ |
| 1,4,10,10,9,16 | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \psi_{\alpha_3}(\Omega_{\alpha_3+1}) + \psi_{\alpha_2}(\alpha_\omega \cdot 2 + \psi_{\alpha_3}(\Omega_{\alpha_3+1}) + 1)))$ |
| 1,4,10,10,9,17 | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \psi_{\alpha_3}(\alpha_\omega)))$ |
| 1,4,10,10,9,17,17 | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \psi_{\alpha_3}(\alpha_\omega \cdot 2)))$ |
| 1,4,10,10,9,17,17, 14,22,22 | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \psi_{\alpha_3}(\alpha_\omega \cdot 2 + \psi_{\alpha_3}(\alpha_\omega \cdot 2))))$ |
| 1,4,10,10,9,17,17, 15 | $\psi(\psi_\alpha(\alpha_\omega \cdot 2 + \alpha_3))$ |
| 1,4,10,10,10 | $\psi(\psi_\alpha(\alpha_\omega \cdot 3))$ |
| 1,4,10,11 | $\psi(\psi_\alpha(\alpha_\omega \cdot \omega))$ |
| 1,4,10,12,5 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha))$ |
| 1,4,10,12,6,9 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha+1}))$ |
| 1,4,10,12,6,10,17, 19 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega)))$ |
| 1,4,10,12,6,10,17, 19,4,10,12,5 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha))))$ |
| 1,4,10,12,6,10,17, 19,4,10,12,6,10,17, 19 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha))) + \psi_{\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha+1})}(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega))))$ |
| 1,4,10,12,6,10,17, 19,5 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha))) + \psi_{\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha+1})}(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha))) + 1))$ |
| 1,4,10,12,6,10,17, 19,7 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha))) + \psi_{\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha+1})}(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha))) + \psi_{\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha+1})}(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha))) + 1)))$ |
| 1,4,10,12,6,10,17, 19,9 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha))) + \psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha+1}))$ |
| 1,4,10,12,6,10,17, 19,10,17,19 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha))) + \psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega)))$ |
| 1,4,10,12,6,10,17, 19,11 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha))) \cdot \omega)$ |
| 1,4,10,12,6,10,17, 19,13,10,17,19,4,10, 12,5 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha)))^2)$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,10,12,6,10,17, 19,13,17 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha)) + \Omega_{\alpha+1}))$ |
| 1,4,10,12,6,10,17, 19,15 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha) + \alpha_2)))$ |
| 1,4,10,12,6,10,17, 19,17 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot (\psi_\alpha(\alpha_\omega \cdot \alpha) + 1))))$ |
| 1,4,10,12,6,10,17, 20 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha+1}))))$ |
| 1,4,10,12,6,10,17, 20,10,17 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha +$ $\psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega)))))$ |
| 1,4,10,12,6,10,17, 20,11 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha)))$ |
| 1,4,10,12,7 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha + 1)))$ |
| 1,4,10,12,7,11 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_2))))$ |
| 1,4,10,12,7,13,15, 5 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha))))$ |
| 1,4,10,12,8 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \alpha_2))$ |
| 1,4,10,12,8,14 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha_2+1} + \Omega_{\alpha+1} \cdot \omega))$ |
| 1,4,10,12,9 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha_2+1} +$ $\psi_{\alpha_2}(\alpha_\omega \cdot \alpha + \Omega_{\alpha_2+1} + 1)))$ |
| 1,4,10,12,9,13 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \Omega_{\alpha_2+1} \cdot \alpha_2))$ |
| 1,4,10,12,9,15 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_3}(\alpha_3)))$ |
| 1,4,10,12,9,17,21, 10 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha)))$ |
| 1,4,10,12,9,17,21, 15 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha + \alpha_3))$ |
| 1,4,10,12,10 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha + 1)))$ |
| 1,4,10,12,10,12,5 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha \cdot 2))$ |
| 1,4,10,12,11 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha \cdot \omega))$ |
| 1,4,10,12,12,5 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha^2))$ |
| 1,4,10,12,15 | $\psi(\psi_\alpha(\alpha_\omega \cdot \varepsilon_{\alpha+1}))$ |
| 1,4,10,12,16,23 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_\omega)))$ |
| 1,4,10,12,16,23,26, 17 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha)))$ |
| 1,4,10,13 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} + 1)))$ |
| 1,4,10,13,6,9 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} +$ $\psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} + 1) + \Omega_{\alpha+1})))))$ |
| 1,4,10,13,7 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} + 2)))$ |
| 1,4,10,13,7,10 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} +$ $\psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} + \Omega_{\alpha+1} + 1)))$ |
| 1,4,10,13,7,11 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} +$ $\psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_2))))$ |
| 1,4,10,13,7,13,15, 5 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} +$ $\psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha))))$ |
| 1,4,10,13,7,13,16 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} +$ $\psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1}) + 1)))$ |
| 1,4,10,13,8 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \alpha_2))$ |
| 1,4,10,13,8,13 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \varepsilon_{\alpha_2+1}))$ |
| 1,4,10,13,9 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \Omega_{\alpha_2+1} +$ $\psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} + \Omega_{\alpha_2+1} + 1)))$ |
| 1,4,10,13,9,17 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} + \psi_{\alpha_3}(\alpha_\omega)))$ |
| 1,4,10,13,10 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\Omega_{\alpha+1} + 1)))$ |
| 1,4,10,13,10,12,5 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\Omega_{\alpha+1} + \alpha)))$ |
| 1,4,10,13,10,13 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} \cdot 2 + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} \cdot 2 + 1)))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,13,11 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha+1} \cdot \omega))$ |
| 1,4,10,13,13 | $\psi(\psi_\alpha(\alpha_\omega \cdot ({\Omega_{\alpha+1}}^2) + \psi_{\alpha_2}(\alpha_\omega \cdot ({\Omega_{\alpha+1}}^2) + 1)))$ |
| 1,4,10,13,17 | $\psi(\psi_\alpha(\alpha_\omega \cdot \psi_{\alpha_2}(\alpha_2)))$ |
| 1,4,10,13,19 | $\psi(\psi_\alpha(\alpha_\omega \cdot \psi_{\alpha_2}(\alpha_\omega)))$ |
| 1,4,10,13,19,21,5 | $\psi(\psi_\alpha(\alpha_\omega \cdot \psi_{\alpha_2}(\alpha_\omega \cdot \alpha)))$ |
| 1,4,10,14 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2))$ |
| 1,4,10,14,6,10 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha+1} \cdot \omega))$ |
| 1,4,10,14,6,10,17, 18 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \omega)))$ |
| 1,4,10,14,6,10,17, 20 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 +$ $\psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha+1}))))$ |
| 1,4,10,14,6,10,17, 20,9,14,22,26 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 +$ $\psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha+1} \cdot 2))))$ |
| 1,4,10,14,6,10,17, 20,10,17 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 +$ $\psi_{\alpha_2}(\alpha_\omega \cdot \psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega)))))$ |
| 1,4,10,14,6,10,17, 20,11 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha)))$ |
| 1,4,10,14,6,10,17, 20,17 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot (\alpha + 1))))$ |
| 1,4,10,14,6,10,17, 20,24 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \varepsilon_{\alpha+1})))$ |
| 1,4,10,14,6,10,17, 21 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha+1} + 1)))$ |
| 1,4,10,14,6,10,17, 22 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2)))$ |
| 1,4,10,14,7 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + 1)))$ |
| 1,4,10,14,7,9,5 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \alpha)))$ |
| 1,4,10,14,7,11 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 +$ $\psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_2))))$ |
| 1,4,10,14,7,13,17 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 +$ $\psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2))))$ |
| 1,4,10,14,8 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \alpha_2))$ |
| 1,4,10,14,8,14 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_2+1} + \Omega_{\alpha+1} \cdot \omega))$ |
| 1,4,10,14,9 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_2+1} +$ $\psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_2+1} + 1)))$ |
| 1,4,10,14,9,8 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_2+1} + \alpha_2))$ |
| 1,4,10,14,9,10 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_2+1} \cdot \omega))$ |
| 1,4,10,14,9,15 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_3)))$ |
| 1,4,10,14,9,17,18 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \omega)))$ |
| 1,4,10,14,9,17,19, 5 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha)))$ |
| 1,4,10,14,9,17,19, 22 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_\omega \cdot \varepsilon_{\alpha+1})))$ |
| 1,4,10,14,9,17,20 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \Omega_{\alpha+1}) +$ $\psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \Omega_{\alpha+1}) + 1)))$ |
| 1,4,10,14,9,17,20, 24 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \psi_{\alpha_2}(\alpha_2))))$ |
| 1,4,10,14,9,17,21 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha_2)))$ |
| 1,4,10,14,9,17,21, 10 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha_2 + 1)))$ |
| 1,4,10,14,9,17,21, 13 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha_2 + \alpha_2)))$ |
| 1,4,10,14,9,17,21, 14 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_2+1}) +$ $\psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_2+1}) + 1)))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,10,14,9,17,21, 14,22 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_\omega))))$ |
| 1,4,10,14,9,17,21, 15 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \alpha_3))$ |
| 1,4,10,14,9,17,21, 15,22 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \varepsilon_{\alpha_3+1}))$ |
| 1,4,10,14,9,17,21, 16 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_3+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \Omega_{\alpha_3+1} + 1)))$ |
| 1,4,10,14,9,17,21, 16,26 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_4}(\alpha_\omega)))$ |
| 1,4,10,14,9,17,21, 16,26,30 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_4}(\alpha_\omega \cdot \alpha_2)))$ |
| 1,4,10,14,9,17,21, 16,26,30,24 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 + \alpha_4))$ |
| 1,4,10,14,9,17,21, 17 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 1)))$ |
| 1,4,10,14,9,17,21, 17,7 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 1) + \psi_{\alpha_2}(\alpha_\omega \cdot (\alpha_2 + 1) + 1)))$ |
| 1,4,10,14,9,17,21, 17,7,13 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 1) + \psi_{\alpha_2}(\alpha_\omega \cdot (\alpha_2 + 1) + \psi_{\alpha_2}(\alpha_\omega))))$ |
| 1,4,10,14,9,17,21, 17,8 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 1) + \alpha_2))$ |
| 1,4,10,14,9,17,21, 17,9 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 1) + \Omega_{\alpha_2+1} + \psi_{\alpha_2}(\alpha_\omega \cdot (\alpha_2 + 1) + \Omega_{\alpha_2+1} + 1)))$ |
| 1,4,10,14,9,17,21, 17,9,17,21,17 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 1) + \psi_{\alpha_3}(\alpha_\omega \cdot (\alpha_2 + 1))))$ |
| 1,4,10,14,9,17,21, 17,14,22 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 1) + \psi_{\alpha_3}(\alpha_\omega \cdot (\alpha_2 + 1) + \psi_{\alpha_3}(\alpha_\omega))))$ |
| 1,4,10,14,9,17,21, 17,15 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 1) + \alpha_3))$ |
| 1,4,10,14,9,17,21, 17,17 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + 2)))$ |
| 1,4,10,14,9,17,21, 17,19,5 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + \alpha)))$ |
| 1,4,10,14,9,17,21, 17,19,8 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + \alpha) + \alpha_2))$ |
| 1,4,10,14,9,17,21, 17,19,15 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + \alpha) + \alpha_3))$ |
| 1,4,10,14,9,17,21, 17,19,17 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + \alpha + 1)))$ |
| 1,4,10,14,9,17,21, 17,19,17,19,5 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + \alpha \cdot 2)))$ |
| 1,4,10,14,9,17,21, 17,19,22 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + \varepsilon_{\alpha+1})))$ |
| 1,4,10,14,9,17,21, 17,20 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + \Omega_{\alpha+1}) + \psi_{\alpha_2}(\alpha_\omega \cdot (\alpha_2 + \Omega_{\alpha+1}) + 1)))$ |
| 1,4,10,14,9,17,21, 17,20,24 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_2 + \psi_{\alpha_2}(\alpha_2))))$ |
| 1,4,10,14,9,17,21, 17,21 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 \cdot 2))$ |
| 1,4,10,14,9,17,21, 19,5 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 \cdot \alpha))$ |
| 1,4,10,14,9,17,21, 20 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_2 \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 \cdot \Omega_{\alpha+1} + 1)))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,14,9,17,21, 21 | $\psi(\psi_\alpha(\alpha_\omega \cdot {\alpha_2}^2))$ |
| 1,4,10,14,9,17,21, 26 | $\psi(\psi_\alpha(\alpha_\omega \cdot \varepsilon_{\alpha_2+1}))$ |
| 1,4,10,14,9,17,22 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha_2+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha_2+1} + 1)))$ |
| 1,4,10,14,9,17,22, 8 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha_2+1} + \alpha_2))$ |
| 1,4,10,14,9,17,22, 17 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\Omega_{\alpha_2+1} + 1)))$ |
| 1,4,10,14,9,17,22, 17,21 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\Omega_{\alpha_2+1} + \alpha_2)))$ |
| 1,4,10,14,9,17,22, 17,22 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha_2+1} \cdot 2 + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha_2+1} \cdot 2 + 1)))$ |
| 1,4,10,14,9,17,22, 28 | $\psi(\psi_\alpha(\alpha_\omega \cdot \psi_{\alpha_3}(\alpha_3)))$ |
| 1,4,10,14,9,17,22, 30 | $\psi(\psi_\alpha(\alpha_\omega \cdot \psi_{\alpha_3}(\alpha_\omega)))$ |
| 1,4,10,14,9,17,23 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_3))$ |
| 1,4,10,14,9,17,23, 8 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_3 + \alpha_2))$ |
| 1,4,10,14,9,17,23, 15 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_3 + \alpha_3))$ |
| 1,4,10,14,9,17,23, 16 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_3 + \Omega_{\alpha_3+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_3 + \Omega_{\alpha_3+1} + 1)))$ |
| 1,4,10,14,9,17,23, 16,26 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_3 + \psi_{\alpha_4}(\alpha_\omega)))$ |
| 1,4,10,14,9,17,23, 16,26,32 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_3 + \psi_{\alpha_4}(\alpha_\omega \cdot \alpha_3)))$ |
| 1,4,10,14,9,17,23, 16,26,32,24 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_3 + \alpha_4))$ |
| 1,4,10,14,9,17,23, 16,26,32,26 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_3 + 1)))$ |
| 1,4,10,14,9,17,23, 16,26,32,26,30 | $\psi(\psi_\alpha(\alpha_\omega \cdot (\alpha_3 + \alpha_2)))$ |
| 1,4,10,14,9,17,23, 16,26,32,26,32 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_3 \cdot 2))$ |
| 1,4,10,14,9,17,23, 16,26,32,32 | $\psi(\psi_\alpha(\alpha_\omega \cdot {\alpha_3}^2))$ |
| 1,4,10,14,9,17,23, 16,26,32,39 | $\psi(\psi_\alpha(\alpha_\omega \cdot \varepsilon_{\alpha_3+1}))$ |
| 1,4,10,14,9,17,23, 16,26,33 | $\psi(\psi_\alpha(\alpha_\omega \cdot \Omega_{\alpha_3+1} + \psi_{\alpha_2}(\alpha_\omega \cdot \Omega_{\alpha_3+1} + 1)))$ |
| 1,4,10,14,9,17,23, 16,26,33,43 | $\psi(\psi_\alpha(\alpha_\omega \cdot \psi_{\alpha_4}(\alpha_\omega)))$ |
| 1,4,10,14,9,17,23, 16,26,34 | $\psi(\psi_\alpha(\alpha_\omega \cdot \alpha_4))$ |
| 1,4,10,14,10 | $\psi(\psi_\alpha({\alpha_\omega}^2))$ |
| 1,4,10,14,10,6,9 | $\psi(\psi_\alpha({\alpha_\omega}^2 + \Omega_{\alpha+1}))$ |
| 1,4,10,14,10,6,10, 17,22 | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2)))$ |
| 1,4,10,14,10,6,10, 17,22,14 | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + 1)))$ |
| 1,4,10,14,10,6,10, 17,22,14,19 | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \psi_{\alpha_2}(\alpha_2))))$ |
| 1,4,10,14,10,6,10, 17,22,15 | $\psi(\psi_\alpha({\alpha_\omega}^2 + \psi_{\alpha_2}(\alpha_\omega \cdot \alpha_2 + \alpha_2)))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,10,14,10,6,10, 17,22,16,25 | $\psi(\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_2}(\alpha_\omega\cdot\alpha_2+\psi_{\alpha_3}(\alpha_\omega))))$ |
| 1,4,10,14,10,6,10, 17,22,16,25,30 | $\psi(\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_2}(\alpha_\omega\cdot\alpha_2+\psi_{\alpha_3}(\alpha_\omega\cdot\alpha_2))))$ |
| 1,4,10,14,10,6,10, 17,22,16,25,30,25 | $\psi(\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_2}(\alpha_\omega\cdot(\alpha_2+1))))$ |
| 1,4,10,14,10,6,10, 17,22,16,25,30,25,30 | $\psi(\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_2}(\alpha_\omega\cdot\alpha_2\cdot 2)))$ |
| 1,4,10,14,10,6,10, 17,22,16,25,32 | $\psi(\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_2}(\alpha_\omega\cdot\alpha_3)))$ |
| 1,4,10,14,10,6,10, 17,22,17 | $\psi(\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_2}({\alpha_\omega}^2)))$ |
| 1,4,10,14,10,6,10, 17,22,17,11 | $\psi(\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_2}({\alpha_\omega}^2)+1))$ |
| 1,4,10,14,10,6,10, 17,22,17,13 | $\psi(\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_2}({\alpha_\omega}^2)+\psi_\alpha({\alpha_\omega}^2+\Omega_{\alpha+1})))$ |
| 1,4,10,14,10,6,10, 17,22,17,13,10,17,22, 17 | $\psi(\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_2}({\alpha_\omega}^2)+\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_2}({\alpha_\omega}^2))))$ |
| 1,4,10,14,10,6,10, 17,22,17,13,17 | $\psi(\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_2}({\alpha_\omega}^2)+\alpha))$ |
| 1,4,10,14,10,7 | $\psi(\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_2}({\alpha_\omega}^2+1)))$ |
| 1,4,10,14,10,7,9, 5 | $\psi(\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_2}({\alpha_\omega}^2+\alpha)))$ |
| 1,4,10,14,10,7,10 | $\psi(\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_2}({\alpha_\omega}^2+\Omega_{\alpha+1}+1)))$ |
| 1,4,10,14,10,7,11 | $\psi(\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_2}({\alpha_\omega}^2+\psi_{\alpha_2}(\alpha_2))))$ |
| 1,4,10,14,10,7,13, 17,13 | $\psi(\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_2}({\alpha_\omega}^2+\psi_{\alpha_2}({\alpha_\omega}^2))))$ |
| 1,4,10,14,10,8 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_2))$ |
| 1,4,10,14,10,8,13 | $\psi(\psi_\alpha({\alpha_\omega}^2+\varepsilon_{\alpha_2+1}))$ |
| 1,4,10,14,10,9 | $\psi(\psi_\alpha({\alpha_\omega}^2+\Omega_{\alpha_2+1}+\psi_{\alpha_2}({\alpha_\omega}^2+\Omega_{\alpha_2+1}+1)))$ |
| 1,4,10,14,10,9,17 | $\psi(\psi_\alpha({\alpha_\omega}^2+\psi_{\alpha_3}(\alpha_\omega)))$ |
| 1,4,10,14,10,10 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_\omega))$ |
| 1,4,10,14,10,11 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_\omega\cdot\omega))$ |
| 1,4,10,14,10,12,5 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_\omega\cdot\alpha))$ |
| 1,4,10,14,10,12,10 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_\omega\cdot(\alpha+1)))$ |
| 1,4,10,14,10,12,15 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_\omega\cdot\varepsilon_{\alpha+1}))$ |
| 1,4,10,14,10,12,16, 23 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_\omega\cdot\Omega_{\alpha+1}+\psi_{\alpha_2}(\alpha_\omega)))$ |
| 1,4,10,14,10,13 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_\omega\cdot\Omega_{\alpha+1}+$ $\psi_{\alpha_2}({\alpha_\omega}^2+\alpha_\omega\cdot\Omega_{\alpha+1}+1)))$ |
| 1,4,10,14,10,13,7 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_\omega\cdot\Omega_{\alpha+1}+$ $\psi_{\alpha_2}({\alpha_\omega}^2+\alpha_\omega\cdot\Omega_{\alpha+1}+2)))$ |
| 1,4,10,14,10,13,8 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_\omega\cdot\Omega_{\alpha+1}+\alpha_2))$ |
| 1,4,10,14,10,13,10 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_\omega\cdot(\Omega_{\alpha+1}+1)))$ |
| 1,4,10,14,10,13,10, 13 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_\omega\cdot\Omega_{\alpha+1}\cdot 2+$ $\psi_{\alpha_2}({\alpha_\omega}^2+\alpha_\omega\cdot\Omega_{\alpha+1}\cdot 2+1)))$ |
| 1,4,10,14,10,13,17 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_\omega\cdot\psi_{\alpha_2}(\alpha_2)))$ |
| 1,4,10,14,10,14 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_\omega\cdot\alpha_2))$ |
| 1,4,10,14,10,14,7 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_\omega\cdot\alpha_2+$ $\psi_{\alpha_2}({\alpha_\omega}^2+\alpha_\omega\cdot\alpha_2+1)))$ |
| 1,4,10,14,10,14,8 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_\omega\cdot\alpha_2+\alpha_2))$ |
| 1,4,10,14,10,14,9, 17,23,17 | $\psi(\psi_\alpha({\alpha_\omega}^2+\alpha_\omega\cdot\alpha_2+\psi_{\alpha_3}({\alpha_\omega}^2)))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,14,10,14,9, 17,23,17,21 | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \alpha_2 + \psi_{\alpha_3}({\alpha_\omega}^2 + \alpha_\omega \cdot \alpha_2)))$ |
| 1,4,10,14,10,14,9, 17,23,17,21,15 | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \alpha_2 + \alpha_3))$ |
| 1,4,10,14,10,14,9, 17,23,17,21,17 | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot (\alpha_2 + 1)))$ |
| 1,4,10,14,10,14,9, 17,23,17,21,17,21 | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \alpha_2 \cdot 2))$ |
| 1,4,10,14,10,14,9, 17,23,17,21,26 | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \varepsilon_{\alpha_2+1}))$ |
| 1,4,10,14,10,14,9, 17,23,17,23 | $\psi(\psi_\alpha({\alpha_\omega}^2 + \alpha_\omega \cdot \alpha_3))$ |
| 1,4,10,14,10,14,10 | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot 2))$ |
| 1,4,10,14,12,5 | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha))$ |
| 1,4,10,14,12,10 | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha + \alpha_\omega))$ |
| 1,4,10,14,12,10,14, 10 | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot (\alpha + 1)))$ |
| 1,4,10,14,13 | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}({\alpha_\omega}^2 \cdot \Omega_{\alpha+1} + 1)))$ |
| 1,4,10,14,13,17 | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \psi_{\alpha_2}(\alpha_2)))$ |
| 1,4,10,14,14 | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha_2))$ |
| 1,4,10,14,14,8 | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha_2 + \alpha_2))$ |
| 1,4,10,14,14,9,17, 23,21 | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha_2 + \psi_{\alpha_3}({\alpha_\omega}^2 \cdot \alpha_2)))$ |
| 1,4,10,14,14,9,17, 23,21,17 | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha_2 + \alpha_\omega))$ |
| 1,4,10,14,14,9,17, 23,21,17,21 | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha_2 + \alpha_\omega \cdot \alpha_2))$ |
| 1,4,10,14,14,9,17, 23,21,17,23 | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha_2 + \alpha_\omega \cdot \alpha_3))$ |
| 1,4,10,14,14,9,17, 23,21,17,23,17 | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot (\alpha_2 + 1)))$ |
| 1,4,10,14,14,9,17, 23,21,17,23,21 | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha_2 \cdot 2))$ |
| 1,4,10,14,14,9,17, 23,21,26 | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \varepsilon_{\alpha_2+1}))$ |
| 1,4,10,14,14,9,17, 23,23 | $\psi(\psi_\alpha({\alpha_\omega}^2 \cdot \alpha_3))$ |
| 1,4,10,14,14,10 | $\psi(\psi_\alpha({\alpha_\omega}^3))$ |
| 1,4,10,14,14,10,10 | $\psi(\psi_\alpha({\alpha_\omega}^3 + \alpha_\omega))$ |
| 1,4,10,14,14,10,14, 10 | $\psi(\psi_\alpha({\alpha_\omega}^3 + {\alpha_\omega}^2))$ |
| 1,4,10,14,14,14,10 | $\psi(\psi_\alpha({\alpha_\omega}^4))$ |
| 1,4,10,14,16,5 | $\psi(\psi_\alpha({\alpha_\omega}^\alpha))$ |
| 1,4,10,14,16,10 | $\psi(\psi_\alpha({\alpha_\omega}^\alpha + \alpha_\omega))$ |
| 1,4,10,14,16,10,14, 16,5 | $\psi(\psi_\alpha({\alpha_\omega}^\alpha \cdot 2))$ |
| 1,4,10,14,17 | $\psi(\psi_\alpha({\alpha_\omega}^{\Omega_{\alpha+1}} + \psi_{\alpha_2}({\alpha_\omega}^{\Omega_{\alpha+1}} + 1)))$ |
| 1,4,10,14,18 | $\psi(\psi_\alpha({\alpha_\omega}^{\alpha_2}))$ |
| 1,4,10,14,18,9,17, 23,27,23,17 | $\psi(\psi_\alpha({\alpha_\omega}^{(}\alpha_2 + 1)))$ |
| 1,4,10,14,18,9,17, 23,27,23,27 | $\psi(\psi_\alpha({\alpha_\omega}^{(}\alpha_2 \cdot 2)))$ |
| 1,4,10,14,18,9,17, 23,29 | $\psi(\psi_\alpha({\alpha_\omega}^{\alpha_3}))$ |
| 1,4,10,14,18,10 | $\psi(\psi_\alpha({\alpha_\omega}^{\alpha_\omega}))$ |
| 1,4,10,14,19 | $\psi(\psi_\alpha(\varepsilon_{\alpha_\omega+1})) = \psi(\psi_\alpha(\Omega_{\alpha_\omega+1}))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,10,14,19,19 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1})\cdot 2)$ |
| 1,4,10,14,20,29 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\alpha_\omega)))$ |
| 1,4,10,15 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}+1)))$ |
| 1,4,10,15,6,10,17, 23 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}+1)\cdot 2))$ |
| 1,4,10,15,7 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}+2)))$ |
| 1,4,10,15,7,10 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}+\Omega_{\alpha+1}+1)))$ |
| 1,4,10,15,7,11 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\alpha_2))))$ |
| 1,4,10,15,8 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\alpha_2))$ |
| 1,4,10,15,9 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\Omega_{\alpha_2+1}+$ $\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}+\Omega_{\alpha_2+1}+1)))$ |
| 1,4,10,15,9,17,23, 17 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\psi_{\alpha_3}({\alpha_\omega}^2)))$ |
| 1,4,10,15,9,17,24 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\psi_{\alpha_3}(\Omega_{\alpha_\omega+1})+$ $\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}+\psi_{\alpha_3}(\Omega_{\alpha_\omega+1})+1)))$ |
| 1,4,10,15,10 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\alpha_\omega))$ |
| 1,4,10,15,10,14 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\alpha_\omega\cdot\alpha_2))$ |
| 1,4,10,15,10,14,10 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+{\alpha_\omega}^2))$ |
| 1,4,10,15,10,14,19 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}+\varepsilon_{\alpha_\omega+1}))$ |
| 1,4,10,15,10,15 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot 2+\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}\cdot 2+1)))$ |
| 1,4,10,15,12,5 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha))$ |
| 1,4,10,15,13 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\Omega_{\alpha+1}+$ $\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}\cdot\Omega_{\alpha+1}+1)))$ |
| 1,4,10,15,13,8 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\Omega_{\alpha+1}+\alpha_2))$ |
| 1,4,10,15,13,10 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\Omega_{\alpha+1}+\alpha_\omega))$ |
| 1,4,10,15,13,10,15 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot(\Omega_{\alpha+1}+1)+$ $\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}\cdot(\Omega_{\alpha+1}+1)+1)))$ |
| 1,4,10,15,13,11 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,10,15,13,12,5 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\Omega_{\alpha+1}\cdot\alpha))$ |
| 1,4,10,15,13,13 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot({\Omega_{\alpha+1}}^2)+$ $\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}\cdot({\Omega_{\alpha+1}}^2)+1)))$ |
| 1,4,10,15,13,15,5 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot({\Omega_{\alpha+1}}^\alpha)))$ |
| 1,4,10,15,13,17 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\psi_{\alpha_2}(\alpha_2)))$ |
| 1,4,10,15,14 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_2))$ |
| 1,4,10,15,14,9,17, 24,21 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_2+\psi_{\alpha_3}(\Omega_{\alpha_\omega+1}\cdot\alpha_2)))$ |
| 1,4,10,15,14,9,17, 24,21,15 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_2+\alpha_3))$ |
| 1,4,10,15,14,9,17, 24,21,17 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_2+\alpha_\omega))$ |
| 1,4,10,15,14,9,17, 24,21,17,23,17 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_2+{\alpha_\omega}^2))$ |
| 1,4,10,15,14,9,17, 24,21,17,23,30 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_2+\varepsilon_{\alpha_\omega+1}))$ |
| 1,4,10,15,14,9,17, 24,21,17,24 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot(\alpha_2+1)+$ $\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}\cdot(\alpha_2+1)+1)))$ |
| 1,4,10,15,14,9,17, 24,21,17,24,21 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_2\cdot 2))$ |
| 1,4,10,15,14,9,17, 24,23 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_3))$ |
| 1,4,10,15,14,10 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_\omega))$ |
| 1,4,10,15,14,10,10 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_\omega+\alpha_\omega))$ |
| 1,4,10,15,14,10,14, 10 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_\omega+{\alpha_\omega}^2))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,15,14,10,15 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot(\alpha_\omega+1)+$ $\psi_{\alpha_2}(\Omega_{\alpha_\omega+1}\cdot(\alpha_\omega+1)+1)))$ |
| 1,4,10,15,14,10,15, 14 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot(\alpha_\omega+\alpha_2)))$ |
| 1,4,10,15,14,10,15, 14,10 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_\omega\cdot 2))$ |
| 1,4,10,15,14,14 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\alpha_\omega\cdot\alpha_2))$ |
| 1,4,10,15,14,14,10 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot{\alpha_\omega}^2))$ |
| 1,4,10,15,14,19 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+1}\cdot\varepsilon_{\alpha_\omega+1}))$ |
| 1,4,10,15,15 | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^2+\psi_{\alpha_2}({\Omega_{\alpha_\omega+1}}^2+1)))$ |
| 1,4,10,15,15,10,15 | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^2+\Omega_{\alpha_\omega+1}+$ $\psi_{\alpha_2}({\Omega_{\alpha_\omega+1}}^2+\Omega_{\alpha_\omega+1}+1)))$ |
| 1,4,10,15,15,14 | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^2\cdot\alpha_2))$ |
| 1,4,10,15,15,14,9, 17,24,24,23 | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^2\cdot\alpha_3))$ |
| 1,4,10,15,15,14,10 | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^2\cdot\alpha_\omega))$ |
| 1,4,10,15,15,15 | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^3+\psi_{\alpha_2}({\Omega_{\alpha_\omega+1}}^3+1)))$ |
| 1,4,10,15,17,5 | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^\alpha))$ |
| 1,4,10,15,18 | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^{\Omega_{\alpha+1}}+\psi_{\alpha_2}({\Omega_{\alpha_\omega+1}}^{\Omega_{\alpha+1}}+1)))$ |
| 1,4,10,15,18,22 | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^{\psi_{\alpha_2}(\alpha_2)}))$ |
| 1,4,10,15,19 | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^{\alpha_2}))$ |
| 1,4,10,15,19,10 | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^{\alpha_\omega}))$ |
| 1,4,10,15,19,24 | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^{\varepsilon_{\alpha_\omega+1}}))$ |
| 1,4,10,15,20 | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+1}}^{\Omega_{\alpha_\omega+1}}+\psi_{\alpha_2}({\Omega_{\alpha_\omega+1}}^{\Omega_{\alpha_\omega+1}}+1)))$ |
| 1,4,10,15,21 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_\omega+1}+1}))$ |
| 1,4,10,15,21,21 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_\omega+1}+2}))$ |
| 1,4,10,15,21,29 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,10,15,21,29,40 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}+\psi_{\alpha_2}(\alpha_\omega)))$ |
| 1,4,10,15,22 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}+\psi_{\alpha_2}(\Omega_{\alpha_\omega+2}+1)))$ |
| 1,4,10,15,22,8 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}+\alpha_2))$ |
| 1,4,10,15,22,10 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}+\alpha_\omega))$ |
| 1,4,10,15,22,10,15 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}+\Omega_{\alpha_\omega+1}+$ $\psi_{\alpha_2}(\Omega_{\alpha_\omega+2}+\Omega_{\alpha_\omega+1}+1)))$ |
| 1,4,10,15,22,10,15, 21 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}+\varepsilon_{\Omega_{\alpha_\omega+1}+1}))$ |
| 1,4,10,15,22,10,15, 22 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}\cdot 2+\psi_{\alpha_2}(\Omega_{\alpha_\omega+2}\cdot 2+1)))$ |
| 1,4,10,15,22,12,5 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}\cdot\alpha))$ |
| 1,4,10,15,22,14 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}\cdot\alpha_2))$ |
| 1,4,10,15,22,14,10 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}\cdot\alpha_\omega))$ |
| 1,4,10,15,22,15 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+2}\cdot\Omega_{\alpha_\omega+1}+$ $\psi_{\alpha_2}(\Omega_{\alpha_\omega+2}\cdot\Omega_{\alpha_\omega+1}+1)))$ |
| 1,4,10,15,22,15,22 | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+2}}^2+\psi_{\alpha_2}({\Omega_{\alpha_\omega+2}}^2+1)))$ |
| 1,4,10,15,22,19,10 | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+2}}^{\alpha_\omega}))$ |
| 1,4,10,15,22,20,27 | $\psi(\psi_\alpha({\Omega_{\alpha_\omega+2}}^{\Omega_{\alpha_\omega+2}}+\psi_{\alpha_2}({\Omega_{\alpha_\omega+2}}^{\Omega_{\alpha_\omega+2}}+1)))$ |
| 1,4,10,15,22,21 | $\psi(\psi_\alpha(\varepsilon_{\Omega_{\alpha_\omega+2}+1}))$ |
| 1,4,10,15,22,22 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+3}+\psi_{\alpha_2}(\Omega_{\alpha_\omega+3}+1)))$ |
| 1,4,10,15,22,23 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+\omega}))$ |
| 1,4,10,15,22,26 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega+\alpha_2}))$ |
| 1,4,10,15,22,26,10 | $\psi(\psi_\alpha(\Omega_{\alpha_\omega\cdot 2}))$ |
| 1,4,10,15,22,27 | $\psi(\psi_\alpha(\Omega_{\Omega_{\alpha_\omega+1}}+\psi_{\alpha_2}(\Omega_{\Omega_{\alpha_\omega+1}}+1)))$ |
| 1,4,10,15,22,28 | $\psi(\psi_\alpha(\mathrm{OFP}(\alpha_\omega+1)))$ $=\psi(\psi_\alpha(\Omega_{\alpha_{\omega+1}+1}\cdot\alpha_{\omega+1}))$ |
| 1,4,10,15,22,29 | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2+\psi_{\alpha_2}({\Omega_{\alpha_{\omega+1}+1}}^2+1)))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,10,15,22,29,10 | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2+\alpha_\omega))$ |
| 1,4,10,15,22,29,10, 15 | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2+\Omega_{\alpha_\omega+1}+$ $\psi_{\alpha_2}({\Omega_{\alpha_{\omega+1}+1}}^2+\Omega_{\alpha_\omega+1}+1)))$ |
| 1,4,10,15,22,29,10, 15,22,23 | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2+\psi_{\alpha_{\omega+1}}(\Omega_{\alpha_{\omega+1}+1}\cdot\omega)))$ |
| 1,4,10,15,22,29,11 | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2+\psi_{\alpha_{\omega+1}}({\Omega_{\alpha_{\omega+1}+1}}^2+1)))$ |
| 1,4,10,15,22,29,14, 10 | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2+\psi_{\alpha_{\omega+1}}({\Omega_{\alpha_{\omega+1}+1}}^2+\alpha_\omega)))$ |
| 1,4,10,15,22,29,15, 22,29 | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2+\psi_{\alpha_{\omega+1}}({\Omega_{\alpha_{\omega+1}+1}}^2+$ $\psi_{\alpha_{\omega+1}}({\Omega_{\alpha_{\omega+1}+1}}^2))+\psi_{\alpha_2}({\Omega_{\alpha_{\omega+1}+1}}^2+$ $\psi_{\alpha_{\omega+1}}({\Omega_{\alpha_{\omega+1}+1}}^2+\psi_{\alpha_{\omega+1}}({\Omega_{\alpha_{\omega+1}+1}}^2)))+1)))$ |
| 1,4,10,15,22,29,21 | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2+\alpha_{\omega+1}))$ |
| 1,4,10,15,22,29,22 | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2+\Omega_{\alpha_{\omega+1}+1}+$ $\psi_{\alpha_2}({\Omega_{\alpha_{\omega+1}+1}}^2+\Omega_{\alpha_{\omega+1}+1}+1)))$ |
| 1,4,10,15,22,29,22, 29 | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2\cdot 2+\psi_{\alpha_2}({\Omega_{\alpha_{\omega+1}+1}}^2\cdot 2+1)))$ |
| 1,4,10,15,22,29,28 | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^2\cdot\alpha_{\omega+1}))$ |
| 1,4,10,15,22,29,30 | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^\omega))$ |
| 1,4,10,15,22,29,33, 10 | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^{\alpha_\omega}))$ |
| 1,4,10,15,22,29,36 | $\psi(\psi_\alpha({\Omega_{\alpha_{\omega+1}+1}}^{\Omega_{\alpha_{\omega+1}+1}}+$ $\psi_{\alpha_2}({\Omega_{\alpha_{\omega+1}+1}}^{\Omega_{\alpha_{\omega+1}+1}}+1)))$ |
| 1,4,10,15,22,30 | $\psi(\psi_\alpha(\alpha_{\omega+2}))$ |
| 1,4,10,15,22,31 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega+2}+1}+\psi_{\alpha_2}(\Omega_{\alpha_{\omega+2}+1}+1)))$ |
| 1,4,10,15,23 | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2}))$ |
| 1,4,10,15,23,10,15, 23 | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2}+\psi_{\alpha_{\omega+1}}(\alpha_{\omega\cdot 2})))$ |
| 1,4,10,15,23,21 | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2}+\alpha_{\omega+1}))$ |
| 1,4,10,15,23,23 | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2}\cdot 2))$ |
| 1,4,10,15,23,27 | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2}\cdot\alpha_2))$ |
| 1,4,10,15,23,27,10 | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2}\cdot\alpha_\omega))$ |
| 1,4,10,15,23,27,23 | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2}\cdot(\alpha_\omega+1)))$ |
| 1,4,10,15,23,28 | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2}\cdot\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\alpha_{\omega\cdot 2}\cdot\Omega_{\alpha_\omega+1}+1)))$ |
| 1,4,10,15,23,29 | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2}\cdot\alpha_{\omega+1}))$ |
| 1,4,10,15,23,29,23 | $\psi(\psi_\alpha({\alpha_{\omega\cdot 2}}^2))$ |
| 1,4,10,15,23,29,30 | $\psi(\psi_\alpha({\alpha_{\omega\cdot 2}}^\omega))$ |
| 1,4,10,15,23,29,36 | $\psi(\psi_\alpha(\varepsilon_{\alpha_{\omega\cdot 2}+1}))$ |
| 1,4,10,15,23,30 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega\cdot 2}+1}+\psi_{\alpha_2}(\Omega_{\alpha_{\omega\cdot 2}+1}+1)))$ |
| 1,4,10,15,23,30,23 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega\cdot 2}+1}+\alpha_{\omega\cdot 2}))$ |
| 1,4,10,15,23,30,23, 30 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega\cdot 2}+1}\cdot 2+\psi_{\alpha_2}(\Omega_{\alpha_{\omega\cdot 2}+1}\cdot 2+1)))$ |
| 1,4,10,15,23,30,38 | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2+1}))$ |
| 1,4,10,15,23,30,39 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega\cdot 2+1}+1}+\psi_{\alpha_2}(\Omega_{\alpha_{\omega\cdot 2+1}+1}+1)))$ |
| 1,4,10,15,23,30,39, 49 | $\psi(\psi_\alpha(\alpha_{\omega\cdot 2+2}))$ |
| 1,4,10,15,23,30,40 | $\psi(\psi_\alpha(\alpha_{\omega\cdot 3}))$ |
| 1,4,10,16 | $\psi(\psi_\alpha(\alpha_{\omega^2}))$ |
| 1,4,10,16,7 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_2}(\alpha_{\omega^2}+1)))$ |
| 1,4,10,16,7,13,19 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_2}(\alpha_{\omega^2}+\psi_{\alpha_2}(\alpha_{\omega^2}))))$ |
| 1,4,10,16,8 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_2))$ |
| 1,4,10,16,9,17,25 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_3}(\alpha_{\omega^2})))$ |
| 1,4,10,16,9,17,25, 15 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_3))$ |
| 1,4,10,16,10 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_\omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,16,10,12,5 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \alpha_\omega \cdot \alpha))$ |
| 1,4,10,16,10,14,10 | $\psi(\psi_\alpha(\alpha_{\omega^2} + {\alpha_\omega}^2))$ |
| 1,4,10,16,10,14,20 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + \Omega_{\alpha+1} \cdot \omega))$ |
| 1,4,10,16,10,14,20, 29,38 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + \psi_{\alpha_2}(\alpha_{\omega^2})))$ |
| 1,4,10,16,10,15 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + \psi_{\alpha_2}(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + 1)))$ |
| 1,4,10,16,10,15,7 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + \psi_{\alpha_2}(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + 2)))$ |
| 1,4,10,16,10,15,7, 13 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} +$ $\psi_{\alpha_2}(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + \psi_{\alpha_2}(\alpha_\omega))))$ |
| 1,4,10,16,10,15,8 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + \alpha_2))$ |
| 1,4,10,16,10,15,9 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + \Omega_{\alpha_2+1} +$ $\psi_{\alpha_2}(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + \Omega_{\alpha_2+1} + 1)))$ |
| 1,4,10,16,10,15,9, 15 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + \psi_{\alpha_3}(\alpha_3)))$ |
| 1,4,10,16,10,15,9, 17,25,17,24 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + \psi_{\alpha_3}(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1}) +$ $\psi_{\alpha_2}(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + \psi_{\alpha_3}(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + 1))))$ |
| 1,4,10,16,10,15,9, 17,25,17,24,15 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + \alpha_3))$ |
| 1,4,10,16,10,15,10 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + \alpha_\omega))$ |
| 1,4,10,16,10,15,10, 14 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} + \alpha_\omega \cdot \alpha_2))$ |
| 1,4,10,16,10,15,10, 15 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot 2 +$ $\psi_{\alpha_2}(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot 2 + 1)))$ |
| 1,4,10,16,10,15,10, 15,8 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot 2 + \alpha_2))$ |
| 1,4,10,16,10,15,10, 15,10 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot 2 + \alpha_\omega))$ |
| 1,4,10,16,10,15,11 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot \omega))$ |
| 1,4,10,16,10,15,12, 5 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot \alpha))$ |
| 1,4,10,16,10,15,13 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot \Omega_{\alpha+1} +$ $\psi_{\alpha_2}(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot \Omega_{\alpha+1} + 1)))$ |
| 1,4,10,16,10,15,13, 17 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot \psi_{\alpha_2}(\alpha_2)))$ |
| 1,4,10,16,10,15,14 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot \alpha_2))$ |
| 1,4,10,16,10,15,14, 9,17,25,17,24,21,15 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot \alpha_2 + \alpha_3))$ |
| 1,4,10,16,10,15,14, 9,17,25,17,24,21,17 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot \alpha_2 + \alpha_\omega))$ |
| 1,4,10,16,10,15,14, 9,17,25,17,24,21,17, 24 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot (\alpha_2 + 1) +$ $\psi_{\alpha_2}(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot (\alpha_2 + 1) + 1)))$ |
| 1,4,10,16,10,15,14, 9,17,25,17,24,21,17, 24,21 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot \alpha_2 \cdot 2))$ |
| 1,4,10,16,10,15,14, 9,17,25,17,24,21,21 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot {\alpha_2}^2))$ |
| 1,4,10,16,10,15,14, 9,17,25,17,24,23 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot \alpha_3))$ |
| 1,4,10,16,10,15,14, 10 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot \alpha_\omega))$ |
| 1,4,10,16,10,15,14, 10,15 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot (\alpha_\omega + 1) +$ $\psi_{\alpha_2}(\alpha_{\omega^2} + \Omega_{\alpha_\omega+1} \cdot (\alpha_\omega + 1) + 1)))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,10,16,10,15,14, 10,15,14,10 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot\alpha_\omega\cdot 2))$ |
| 1,4,10,16,10,15,14, 14,10 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot{\alpha_\omega}^2))$ |
| 1,4,10,16,10,15,14, 19 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_\omega+1}\cdot\varepsilon_{\alpha_\omega+1}))$ |
| 1,4,10,16,10,15,15 | $\psi(\psi_\alpha(\alpha_{\omega^2}+{\Omega_{\alpha_\omega+1}}^2+$ $\psi_{\alpha_2}(\alpha_{\omega^2}+{\Omega_{\alpha_\omega+1}}^2+1)))$ |
| 1,4,10,16,10,15,19, 10 | $\psi(\psi_\alpha(\alpha_{\omega^2}+{\Omega_{\alpha_\omega+1}}^{\alpha_\omega}))$ |
| 1,4,10,16,10,15,20 | $\psi(\psi_\alpha(\alpha_{\omega^2}+{\Omega_{\alpha_\omega+1}}^{\Omega_{\alpha_\omega+1}}+$ $\psi_{\alpha_2}(\alpha_{\omega^2}+{\Omega_{\alpha_\omega+1}}^{\Omega_{\alpha_\omega+1}}+1)))$ |
| 1,4,10,16,10,15,21 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\alpha_{\omega+1})))$ |
| 1,4,10,16,10,15,21, 21 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\alpha_{\omega+1}\cdot 2)))$ |
| 1,4,10,16,10,15,22 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\Omega_{\alpha_{\omega+1}+1})+$ $\psi_{\alpha_2}(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\Omega_{\alpha_{\omega+1}+1})+1)))$ |
| 1,4,10,16,10,15,23 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\alpha_{\omega\cdot 2})))$ |
| 1,4,10,16,10,15,23, 31 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\alpha_{\omega^2})))$ |
| 1,4,10,16,10,15,23, 31,11 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\alpha_{\omega^2}+1)))$ |
| 1,4,10,16,10,15,23, 31,15,23,31 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\alpha_{\omega^2}+\psi_{\alpha_{\omega+1}}(\alpha_{\omega^2}))))$ |
| 1,4,10,16,10,15,23, 31,21 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega+1}))$ |
| 1,4,10,16,10,15,23, 31,21,29 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_{\omega+1}+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,10,16,10,15,23, 31,22 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_{\omega+1}+1}+$ $\psi_{\alpha_2}(\alpha_{\omega^2}+\Omega_{\alpha_{\omega+1}+1}+1)))$ |
| 1,4,10,16,10,15,23, 31,22,21 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_{\omega+1}+1}+\alpha_{\omega+1}))$ |
| 1,4,10,16,10,15,23, 31,22,23 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\Omega_{\alpha_{\omega+1}+1}\cdot\omega))$ |
| 1,4,10,16,10,15,23, 31,22,30 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+2}}(\alpha_{\omega+2})))$ |
| 1,4,10,16,10,15,23, 31,22,32,42 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\psi_{\alpha_{\omega+2}}(\alpha_{\omega^2})))$ |
| 1,4,10,16,10,15,23, 31,22,32,42,30 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega+2}))$ |
| 1,4,10,16,10,15,23, 31,23 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega\cdot 2}))$ |
| 1,4,10,16,10,15,23, 31,23,21 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega\cdot 2}+\alpha_{\omega+1}))$ |
| 1,4,10,16,10,15,23, 31,23,23 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega\cdot 2}\cdot 2))$ |
| 1,4,10,16,10,15,23, 31,23,27 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega\cdot 2}\cdot\alpha_2))$ |
| 1,4,10,16,10,15,23, 31,23,27,10 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega\cdot 2}\cdot\alpha_\omega))$ |
| 1,4,10,16,10,15,23, 31,23,29 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega\cdot 2}\cdot\alpha_{\omega+1}))$ |
| 1,4,10,16,10,15,23, 31,23,29,22,32,42,32, 40 | $\psi(\psi_\alpha(\alpha_{\omega^2}+\alpha_{\omega\cdot 2}\cdot\alpha_{\omega+2}))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,16,10,15,23, 31,23,29,23 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \alpha_{\omega\cdot 2}{}^2))$ |
| 1,4,10,16,10,15,23, 31,23,29,36 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \varepsilon_{\alpha_{\omega\cdot 2}+1}))$ |
| 1,4,10,16,10,15,23, 31,23,30 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_{\omega\cdot 2}+1} + \psi_{\alpha_2}(\alpha_{\omega^2} + \Omega_{\alpha_{\omega\cdot 2}+1} + 1)))$ |
| 1,4,10,16,10,15,23, 31,23,30,23 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_{\omega\cdot 2}+1} + \alpha_{\omega\cdot 2}))$ |
| 1,4,10,16,10,15,23, 31,23,30,24 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \Omega_{\alpha_{\omega\cdot 2}+1} \cdot \omega))$ |
| 1,4,10,16,10,15,23, 31,23,30,38 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \psi_{\alpha_{\omega\cdot 2+1}}(\alpha_{\omega\cdot 2+1})))$ |
| 1,4,10,16,10,15,23, 31,23,30,40,50 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \psi_{\alpha_{\omega\cdot 2+1}}(\alpha_{\omega^2})))$ |
| 1,4,10,16,10,15,23, 31,23,30,40,50,38 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \alpha_{\omega\cdot 2+1}))$ |
| 1,4,10,16,10,15,23, 31,23,30,40,50,40 | $\psi(\psi_\alpha(\alpha_{\omega^2} + \alpha_{\omega\cdot 3}))$ |
| 1,4,10,16,10,16 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 2))$ |
| 1,4,10,16,10,16,7 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 2 + \psi_{\alpha_2}(\alpha_{\omega^2} \cdot 2 + 1)))$ |
| 1,4,10,16,10,16,8 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 2 + \alpha_2))$ |
| 1,4,10,16,10,16,10 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 2 + \alpha_\omega))$ |
| 1,4,10,16,10,16,10, 15,21 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 2 + \psi_{\alpha_{\omega+1}}(\alpha_{\omega+1})))$ |
| 1,4,10,16,10,16,10, 15,23,31,23,31 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 2 + \psi_{\alpha_{\omega+1}}(\alpha_{\omega^2} \cdot 2)))$ |
| 1,4,10,16,10,16,10, 15,23,31,23,31,21 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 2 + \alpha_{\omega+1}))$ |
| 1,4,10,16,10,16,10, 15,23,31,23,31,23 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 2 + \alpha_{\omega\cdot 2}))$ |
| 1,4,10,16,10,16,10, 16 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot 3))$ |
| 1,4,10,16,11 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \omega))$ |
| 1,4,10,16,12,5 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha))$ |
| 1,4,10,16,13 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \Omega_{\alpha+1} + \psi_{\alpha_2}(\alpha_{\omega^2} \cdot \Omega_{\alpha+1} + 1)))$ |
| 1,4,10,16,14 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_2))$ |
| 1,4,10,16,14,8 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_2 + \alpha_2))$ |
| 1,4,10,16,14,9,17, 25,21 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_2 + \psi_{\alpha_3}(\alpha_{\omega^2} \cdot \alpha_2)))$ |
| 1,4,10,16,14,9,17, 25,21,15 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_2 + \alpha_3))$ |
| 1,4,10,16,14,9,17, 25,21,17 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_2 + \alpha_\omega))$ |
| 1,4,10,16,14,9,17, 25,21,17,25 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot (\alpha_2 + 1)))$ |
| 1,4,10,16,14,9,17, 25,21,17,25,21 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_2 \cdot 2))$ |
| 1,4,10,16,14,9,17, 25,22 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \Omega_{\alpha_2+1} + \psi_{\alpha_2}(\alpha_{\omega^2} \cdot \Omega_{\alpha_2+1} + 1)))$ |
| 1,4,10,16,14,9,17, 25,23 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_3))$ |
| 1,4,10,16,14,10 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_\omega))$ |
| 1,4,10,16,14,10,10 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_\omega + \alpha_\omega))$ |
| 1,4,10,16,14,10,15 | $\psi(\psi_\alpha(\alpha_{\omega^2} \cdot \alpha_\omega + \Omega_{\alpha_\omega+1} + \psi_{\alpha_2}(\alpha_{\omega^2} \cdot \alpha_\omega + \Omega_{\alpha_\omega+1} + 1)))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,10,16,14,10,15, 23,31 | $\psi(\psi_\alpha(\alpha_{\omega^2}\cdot\alpha_\omega+\psi_{\alpha_{\omega+1}}(\alpha_{\omega^2})))$ |
| 1,4,10,16,14,10,15, 23,31,27,10 | $\psi(\psi_\alpha(\alpha_{\omega^2}\cdot\alpha_\omega+\psi_{\alpha_{\omega+1}}(\alpha_{\omega^2}\cdot\alpha_\omega)))$ |
| 1,4,10,16,14,10,15, 23,31,27,21 | $\psi(\psi_\alpha(\alpha_{\omega^2}\cdot\alpha_\omega+\alpha_{\omega+1}))$ |
| 1,4,10,16,14,10,15, 23,31,27,23 | $\psi(\psi_\alpha(\alpha_{\omega^2}\cdot\alpha_\omega+\alpha_{\omega\cdot 2}))$ |
| 1,4,10,16,14,10,15, 23,31,27,23,31 | $\psi(\psi_\alpha(\alpha_{\omega^2}\cdot(\alpha_\omega+1)))$ |
| 1,4,10,16,14,10,15, 23,31,27,23,31,27,10 | $\psi(\psi_\alpha(\alpha_{\omega^2}\cdot\alpha_\omega\cdot 2))$ |
| 1,4,10,16,14,10,15, 23,31,27,32 | $\psi(\psi_\alpha(\alpha_{\omega^2}\cdot\varepsilon_{\alpha_\omega+1}))$ |
| 1,4,10,16,14,10,15, 23,31,28 | $\psi(\psi_\alpha(\alpha_{\omega^2}\cdot\Omega_{\alpha_\omega+1}+\psi_{\alpha_2}(\alpha_{\omega^2}\cdot\Omega_{\alpha_\omega+1}+1)))$ |
| 1,4,10,16,14,10,15, 23,31,28,36,44 | $\psi(\psi_\alpha(\alpha_{\omega^2}\cdot\psi_{\alpha_{\omega+1}}(\alpha_{\omega^2})))$ |
| 1,4,10,16,14,10,15, 23,31,29 | $\psi(\psi_\alpha(\alpha_{\omega^2}\cdot\alpha_{\omega+1}))$ |
| 1,4,10,16,14,10,15, 23,31,29,23 | $\psi(\psi_\alpha(\alpha_{\omega^2}\cdot\alpha_{\omega\cdot 2}))$ |
| 1,4,10,16,14,10,16 | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2))$ |
| 1,4,10,16,14,10,16, 10,16 | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2+\alpha_{\omega^2}))$ |
| 1,4,10,16,14,10,16, 14 | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2+\alpha_{\omega^2}\cdot\alpha_2))$ |
| 1,4,10,16,14,10,16, 14,10 | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2+\alpha_{\omega^2}\cdot\alpha_\omega))$ |
| 1,4,10,16,14,10,16, 14,10,15,23,31,29,23, 31,29 | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2+\alpha_{\omega^2}\cdot\alpha_{\omega+1}))$ |
| 1,4,10,16,14,10,16, 14,10,15,23,31,29,23, 31,29,23 | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2+\alpha_{\omega^2}\cdot\alpha_{\omega\cdot 2}))$ |
| 1,4,10,16,14,10,16, 14,10,16 | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2\cdot 2))$ |
| 1,4,10,16,14,12,5 | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2\cdot\alpha))$ |
| 1,4,10,16,14,14 | $\psi(\psi_\alpha({\alpha_{\omega^2}}^2\cdot\alpha_2))$ |
| 1,4,10,16,14,14,10, 16 | $\psi(\psi_\alpha({\alpha_{\omega^2}}^3))$ |
| 1,4,10,16,14,15 | $\psi(\psi_\alpha({\alpha_{\omega^2}}^\omega))$ |
| 1,4,10,16,14,18 | $\psi(\psi_\alpha({\alpha_{\omega^2}}^{\alpha_2}))$ |
| 1,4,10,16,14,18,10 | $\psi(\psi_\alpha({\alpha_{\omega^2}}^{\alpha_\omega}))$ |
| 1,4,10,16,14,18,10, 15,23,31,29,35,23 | $\psi(\psi_\alpha({\alpha_{\omega^2}}^{\alpha_{\omega\cdot 2}}))$ |
| 1,4,10,16,14,18,10, 16 | $\psi(\psi_\alpha({\alpha_{\omega^2}}^{\alpha_{\omega^2}}))$ |
| 1,4,10,16,14,19 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}))$ |
| 1,4,10,16,15 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}+\psi_{\alpha_2}(\Omega_{\alpha_{\omega^2}+1}+1)))$ |
| 1,4,10,16,15,8 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}+\alpha_2))$ |
| 1,4,10,16,15,10 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}+\alpha_\omega))$ |
| 1,4,10,16,15,10,16 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}+\alpha_{\omega^2}))$ |
| 1,4,10,16,15,10,16, 14,10,16 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2}+1}+{\alpha_{\omega^2}}^2))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,16,15,10,16, 14,20 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}} \cdot 2 + \Omega_{\alpha+1} \cdot \omega))$ |
| 1,4,10,16,15,10,16, 15 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}} \cdot 2 + \psi_{\alpha_2}(\Omega_{\alpha_{\omega^2+1}} \cdot 2 + 1)))$ |
| 1,4,10,16,15,14 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}} \cdot \alpha_2))$ |
| 1,4,10,16,15,14,10 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}} \cdot \alpha_\omega))$ |
| 1,4,10,16,15,14,10, 16 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}} \cdot \alpha_{\omega^2}))$ |
| 1,4,10,16,15,14,10, 16,15,14 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}} \cdot (\alpha_{\omega^2} + \alpha_2)))$ |
| 1,4,10,16,15,14,19 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}}{}^2))$ |
| 1,4,10,16,15,15 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}}{}^2 + \psi_{\alpha_2}(\Omega_{\alpha_{\omega^2+1}}{}^2 + 1)))$ |
| 1,4,10,16,15,15,10, 16 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}}{}^2 + \alpha_{\omega^2}))$ |
| 1,4,10,16,15,15,14 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}}{}^2 \cdot \alpha_2))$ |
| 1,4,10,16,15,15,15 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}}{}^3 + \psi_{\alpha_2}(\Omega_{\alpha_{\omega^2+1}}{}^3 + 1)))$ |
| 1,4,10,16,15,19 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}}{}^{\alpha_2}))$ |
| 1,4,10,16,15,19,10 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}}{}^{\alpha_\omega}))$ |
| 1,4,10,16,15,19,10, 15,23,31,30,36,23 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}}{}^{\alpha_{\omega\cdot 2}}))$ |
| 1,4,10,16,15,19,10, 16 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}}{}^{\alpha_{\omega^2}}))$ |
| 1,4,10,16,15,19,15, 19,10,16 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}}{}^{(}\alpha_{\omega^2} \cdot 2)))$ |
| 1,4,10,16,15,19,24 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}}{}^{\varepsilon_{\alpha_{\omega^2+1}}}))$ |
| 1,4,10,16,15,20 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}}{}^{\Omega_{\alpha_{\omega^2+1}}} + \psi_{\alpha_2}(\Omega_{\alpha_{\omega^2+1}}{}^{\Omega_{\alpha_{\omega^2+1}}} + 1)))$ |
| 1,4,10,16,15,21 | $\psi(\psi_\alpha(\alpha_{\omega^2+1}))$ |
| 1,4,10,16,15,22 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+1}+1} + \psi_{\alpha_2}(\Omega_{\alpha_{\omega^2+1}+1} + 1)))$ |
| 1,4,10,16,15,23 | $\psi(\psi_\alpha(\alpha_{\omega^2+\omega}))$ |
| 1,4,10,16,15,23,23 | $\psi(\psi_\alpha(\alpha_{\omega^2+\omega} \cdot 2))$ |
| 1,4,10,16,15,23,29 | $\psi(\psi_\alpha(\alpha_{\omega^2+\omega} \cdot \alpha_{\omega^2+1}))$ |
| 1,4,10,16,15,23,29, 23 | $\psi(\psi_\alpha(\alpha_{\omega^2+\omega}{}^2))$ |
| 1,4,10,16,15,23,29, 36 | $\psi(\psi_\alpha(\varepsilon_{\alpha_{\omega^2+\omega}+1}))$ |
| 1,4,10,16,15,23,30 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^2+\omega}+1} + \psi_{\alpha_2}(\Omega_{\alpha_{\omega^2+\omega}+1} + 1)))$ |
| 1,4,10,16,15,23,30, 40 | $\psi(\psi_\alpha(\alpha_{\omega^2+\omega\cdot 2}))$ |
| 1,4,10,16,15,23,31 | $\psi(\psi_\alpha(\alpha_{\omega^2\cdot 2}))$ |
| 1,4,10,16,16 | $\psi(\psi_\alpha(\alpha_{\omega^3}))$ |
| 1,4,10,16,16,10,16 | $\psi(\psi_\alpha(\alpha_{\omega^3} + \alpha_{\omega^2}))$ |
| 1,4,10,16,16,10,16, 15,23,31,31,23 | $\psi(\psi_\alpha(\alpha_{\omega^3} + \alpha_{\omega^2+\omega}))$ |
| 1,4,10,16,16,10,16, 15,23,31,31,23,31 | $\psi(\psi_\alpha(\alpha_{\omega^3} + \alpha_{\omega^2\cdot 2}))$ |
| 1,4,10,16,16,10,16, 16 | $\psi(\psi_\alpha(\alpha_{\omega^3} \cdot 2))$ |
| 1,4,10,16,16,14,10, 16 | $\psi(\psi_\alpha(\alpha_{\omega^3} \cdot \alpha_{\omega^2}))$ |
| 1,4,10,16,16,14,10, 16,15,23,31,31,29 | $\psi(\psi_\alpha(\alpha_{\omega^3} \cdot \alpha_{\omega^2+1}))$ |
| 1,4,10,16,16,14,10, 16,15,23,31,31,29,23, 31 | $\psi(\psi_\alpha(\alpha_{\omega^3} \cdot \alpha_{\omega^2\cdot 2}))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,10,16,16,14,10, 16,16 | $\psi(\psi_\alpha({\alpha_{\omega^3}}^2))$ |
| 1,4,10,16,16,14,19 | $\psi(\psi_\alpha(\varepsilon_{\alpha_{\omega^3+1}}))$ |
| 1,4,10,16,16,15 | $\psi(\psi_\alpha(\Omega_{\alpha_{\omega^3+1}}+\psi_{\alpha_2}(\Omega_{\alpha_{\omega^3+1}}+1)))$ |
| 1,4,10,16,16,15,23, 31,31 | $\psi(\psi_\alpha(\alpha_{\omega^3\cdot 2}))$ |
| 1,4,10,16,16,16 | $\psi(\psi_\alpha(\alpha_{\omega^4}))$ |
| 1,4,10,16,18,5 | $\psi(\psi_\alpha(\alpha_\alpha))$ |
| 1,4,10,16,18,7,13, 19,21,5 | $\psi(\psi_\alpha(\alpha_\alpha+\psi_{\alpha_2}(\alpha_\alpha)))$ |
| 1,4,10,16,18,8 | $\psi(\psi_\alpha(\alpha_\alpha+\alpha_2))$ |
| 1,4,10,16,18,10 | $\psi(\psi_\alpha(\alpha_\alpha+\alpha_\omega))$ |
| 1,4,10,16,18,10,16, 18,5 | $\psi(\psi_\alpha(\alpha_\alpha\cdot 2))$ |
| 1,4,10,16,18,15 | $\psi(\psi_\alpha(\Omega_{\alpha_\alpha+1}+\psi_{\alpha_2}(\Omega_{\alpha_\alpha+1}+1)))$ |
| 1,4,10,16,18,15,23 | $\psi(\psi_\alpha(\alpha_{\alpha+\omega}))$ |
| 1,4,10,16,18,15,23, 31,33,5 | $\psi(\psi_\alpha(\alpha_{\alpha\cdot 2}))$ |
| 1,4,10,16,18,16 | $\psi(\psi_\alpha(\alpha_{\alpha\cdot\omega}))$ |
| 1,4,10,16,18,16,18, 5 | $\psi(\psi_\alpha(\alpha_{\alpha^2}))$ |
| 1,4,10,16,18,21 | $\psi(\psi_\alpha(\alpha_{\varepsilon_{\alpha+1}}))$ |
| 1,4,10,16,19 | $\psi(\psi_\alpha(\alpha_{\Omega_{\alpha+1}}+\psi_{\alpha_2}(\alpha_{\Omega_{\alpha+1}}+1)))$ |
| 1,4,10,16,19,25 | $\psi(\psi_\alpha(\alpha_{\psi_{\alpha_2}(\alpha_\omega)}))$ |
| 1,4,10,16,20 | $\psi(\psi_\alpha(\alpha_{\alpha_2}))$ |
| 1,4,10,16,20,9,17, 25,29,25 | $\psi(\psi_\alpha(\alpha_{\alpha_2\cdot\omega}))$ |
| 1,4,10,16,20,9,17, 25,31 | $\psi(\psi_\alpha(\alpha_{\alpha_3}))$ |
| 1,4,10,16,20,10 | $\psi(\psi_\alpha(\alpha_{\alpha_\omega}))$ |
| 1,4,10,16,20,10,10 | $\psi(\psi_\alpha(\alpha_{\alpha_\omega}+\alpha_\omega))$ |
| 1,4,10,16,20,10,15, 23 | $\psi(\psi_\alpha(\alpha_{\alpha_\omega}+\psi_{\alpha_{\omega+1}}(\alpha_{\omega\cdot 2})))$ |
| 1,4,10,16,20,10,15, 23,31,35 | $\psi(\psi_\alpha(\alpha_{\alpha_\omega}+\psi_{\alpha_{\omega+1}}(\alpha_{\alpha_2})))$ |
| 1,4,10,16,20,10,15, 23,31,35,10 | $\psi(\psi_\alpha(\alpha_{\alpha_\omega}+\psi_{\alpha_{\omega+1}}(\alpha_{\alpha_\omega})))$ |
| 1,4,10,16,20,10,15, 23,31,35,21 | $\psi(\psi_\alpha(\alpha_{\alpha_\omega}+\alpha_{\omega+1}))$ |
| 1,4,10,16,20,10,15, 23,31,35,23,31,35 | $\psi(\psi_\alpha(\alpha_{\alpha_\omega}+\alpha_{\alpha_2}))$ |
| 1,4,10,16,20,10,15, 23,31,35,23,31,35,10 | $\psi(\psi_\alpha(\alpha_{\alpha_\omega}\cdot 2))$ |
| 1,4,10,16,20,10,15, 23,31,37 | $\psi(\psi_\alpha(\alpha_{\alpha_{\omega+1}}))$ |
| 1,4,10,16,20,10,15, 23,31,37,23 | $\psi(\psi_\alpha(\alpha_{\alpha_{\omega\cdot 2}}))$ |
| 1,4,10,16,20,10,16 | $\psi(\psi_\alpha(\alpha_{\alpha_{\omega^2}}))$ |
| 1,4,10,16,20,10,16, 20 | $\psi(\psi_\alpha(\alpha_{\alpha_{\alpha_2}}))$ |
| 1,4,10,16,20,11 | $\psi(\psi_\alpha(\alpha\mathrm{fp}))=\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)))$ |
| 1,4,10,16,20,11,10 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)+\alpha_\omega))$ |
| 1,4,10,16,20,11,10, 16 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)+\alpha_{\omega^2}))$ |
| 1,4,10,16,20,11,10, 16,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)\cdot 2))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,16,20,11,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)\cdot\omega))$ |
| 1,4,10,16,20,12,5 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)\cdot\alpha))$ |
| 1,4,10,16,20,13 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)\cdot\Omega_{\alpha+1}+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\beta)\cdot\Omega_{\alpha+1}+1)))$ |
| 1,4,10,16,20,13,17 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)\cdot\psi_{\alpha_2}(\alpha_2)))$ |
| 1,4,10,16,20,14 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)\cdot\alpha_2))$ |
| 1,4,10,16,20,14,10 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)\cdot\alpha_\omega))$ |
| 1,4,10,16,20,14,10, 16,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)^2))$ |
| 1,4,10,16,20,14,14 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)^2\cdot\alpha_2))$ |
| 1,4,10,16,20,14,14, 10,16,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)^3))$ |
| 1,4,10,16,20,14,18, 10,16,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)^{\psi_\beta(\alpha_{\beta+1}\cdot\beta)}))$ |
| 1,4,10,16,20,14,19 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta+\beta)))$ |
| 1,4,10,16,20,15 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta+\Omega_{\beta+1})+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\beta+\Omega_{\beta+1})+1)))$ |
| 1,4,10,16,20,15,10, 16,20,15 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta+\Omega_{\beta+1})\cdot 2+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\beta+\Omega_{\beta+1})\cdot 2+1)))$ |
| 1,4,10,16,20,15,15 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta+\Omega_{\beta+1})^2+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\beta+\Omega_{\beta+1})^2+1)))$ |
| 1,4,10,16,20,15,21 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta+\Omega_{\beta+1}+\beta)))$ |
| 1,4,10,16,20,15,23 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega))))$ |
| 1,4,10,16,20,15,23, 23 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega))\cdot 2))$ |
| 1,4,10,16,20,15,23, 29,36 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega)+\beta)))$ |
| 1,4,10,16,20,15,23, 31 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega^2))))$ |
| 1,4,10,16,20,15,23, 31,33,5 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\alpha))))$ |
| 1,4,10,16,20,15,23, 31,35 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\alpha_2))))$ |
| 1,4,10,16,20,15,23, 31,35,10 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\alpha_\omega))))$ |
| 1,4,10,16,20,15,23, 31,35,10,16,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\psi_\beta(\alpha_{\beta+1}\cdot\beta)))))$ |
| 1,4,10,16,20,15,23, 31,37 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\psi_\beta(\alpha_{\beta+1}\cdot(\beta+1))))))$ |
| 1,4,10,16,20,15,23, 31,37,23 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega))))))$ |
| 1,4,10,16,20,15,23, 31,37,24 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot 2)))$ |
| 1,4,10,16,20,16 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega)))$ |
| 1,4,10,16,20,16,10, 16,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta)))$ |
| 1,4,10,16,20,16,10, 16,20,15,23,31,37,31, 23,31,37,24 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot 2)))$ |
| 1,4,10,16,20,16,10, 16,20,16 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega)\cdot 2))$ |
| 1,4,10,16,20,16,14, 10,16,20,16 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega)^2))$ |
| 1,4,10,16,20,16,14, 19 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega+\beta)))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,10,16,20,16,15 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega+\Omega_{\beta+1})+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega+\Omega_{\beta+1})+1)))$ |
| 1,4,10,16,20,16,15, 23 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta\cdot\omega+\omega))))$ |
| 1,4,10,16,20,16,15, 23,31,37,24 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot(\omega+1))))$ |
| 1,4,10,16,20,16,16 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega^2)))$ |
| 1,4,10,16,20,16,18, 5 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\alpha)))$ |
| 1,4,10,16,20,16,18, 16 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\alpha\cdot\omega)))$ |
| 1,4,10,16,20,16,20 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\alpha_2)))$ |
| 1,4,10,16,20,16,20, 10 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\alpha_\omega)))$ |
| 1,4,10,16,20,16,20, 10,16,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\psi_\beta(\alpha_{\beta+1}\cdot\beta))))$ |
| 1,4,10,16,20,16,20, 10,16,20,16 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega))))$ |
| 1,4,10,16,20,16,20, 11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta^2)))$ |
| 1,4,10,16,20,16,20, 14,19 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta^2+\beta)))$ |
| 1,4,10,16,20,16,20, 15,23 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\beta^2+\omega))))$ |
| 1,4,10,16,20,16,20, 16 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta^2\cdot\omega)))$ |
| 1,4,10,16,20,16,20, 16,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta^3)))$ |
| 1,4,10,16,20,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta^\beta)))$ |
| 1,4,10,16,20,25 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\varepsilon_{\beta+1})))$ |
| 1,4,10,16,21 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+1)))$ |
| 1,4,10,16,21,8 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+\alpha_2))$ |
| 1,4,10,16,21,10 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+\alpha_\omega))$ |
| 1,4,10,16,21,10,16, 20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+\psi_\beta(\alpha_{\beta+1}\cdot\beta)))$ |
| 1,4,10,16,21,10,16, 21 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})\cdot 2+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})\cdot 2+1)))$ |
| 1,4,10,16,21,14,19 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1}+\beta)))$ |
| 1,4,10,16,21,15 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1}+\Omega_{\beta+1})+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1}+\Omega_{\beta+1})+1)))$ |
| 1,4,10,16,21,15,23 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot(\Omega_{\beta+1}+\omega))))$ |
| 1,4,10,16,21,15,23, 31,38 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1}\cdot 2)+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1}\cdot 2)+1)))$ |
| 1,4,10,16,21,16 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1}\cdot\omega)))$ |
| 1,4,10,16,21,16,20, 11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1}\cdot\beta)))$ |
| 1,4,10,16,21,16,21 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot{\Omega_{\beta+1}}^2)+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}\cdot{\Omega_{\beta+1}}^2)+1)))$ |
| 1,4,10,16,21,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot{\Omega_{\beta+1}}^\beta)))$ |
| 1,4,10,16,21,27 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\varepsilon_{\Omega_{\beta+1}+1})))$ |
| 1,4,10,16,21,28,29 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+\omega})))$ $=\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2+\Omega_{\beta+1}\cdot\omega)))$ |
| 1,4,10,16,21,29 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,16,21,29,29 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \omega) \cdot 2))$ |
| 1,4,10,16,21,29,37 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \omega^2)))$ |
| 1,4,10,16,21,29,37, 41 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \alpha_2)))$ |
| 1,4,10,16,21,29,37, 41,10,16,21,29,37,41 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \alpha_2)))))$ |
| 1,4,10,16,21,29,37, 41,11 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \psi_\beta({\alpha_{\beta+1}}^2))))$ |
| 1,4,10,16,21,29,37, 43 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1}))))$ |
| 1,4,10,16,21,29,37, 43,29 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \omega))))$ |
| 1,4,10,16,21,29,37, 43,30 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \beta)))$ |
| 1,4,10,16,21,29,37, 43,37 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \beta \cdot \omega)))$ |
| 1,4,10,16,21,29,37, 43,50 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \varepsilon_{\beta+1})))$ |
| 1,4,10,16,21,29,37, 44,54 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot 2 + \alpha_{\beta+1} \cdot \omega)))$ |
| 1,4,10,16,22 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega)))$ |
| 1,4,10,16,22,8 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \alpha_2))$ |
| 1,4,10,16,22,10 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \alpha_\omega))$ |
| 1,4,10,16,22,10,16 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \alpha_{\omega^2}))$ |
| 1,4,10,16,22,10,16, 20,11 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \psi_\beta(\alpha_{\beta+1} \cdot \beta)))$ |
| 1,4,10,16,22,10,16, 20,25 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \psi_\beta(\alpha_{\beta+1} \cdot \varepsilon_{\beta+1})))$ |
| 1,4,10,16,22,10,16, 21 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \psi_\beta(\alpha_{\beta+1} \cdot \Omega_{\beta+1}) + \psi_{\alpha_2}(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \psi_\beta(\alpha_{\beta+1} \cdot \Omega_{\beta+1}) + 1)))$ |
| 1,4,10,16,22,10,16, 21,29 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \omega))))$ |
| 1,4,10,16,22,10,16, 21,29,37,45 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega))))$ |
| 1,4,10,16,22,10,16, 21,29,37,45,11 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega)) \cdot \omega))$ |
| 1,4,10,16,22,10,16, 21,29,37,45,15,23,31, 35,10,16,21,29,37,45 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega)))))$ |
| 1,4,10,16,22,10,16, 21,29,37,45,15,23,31, 37,24 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \psi_\beta({\alpha_{\beta+1}}^2)))$ |
| 1,4,10,16,22,10,16, 21,29,37,45,16 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \psi_\beta({\alpha_{\beta+1}}^2) \cdot \omega))$ |
| 1,4,10,16,22,10,16, 21,29,37,45,27 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1})))$ |
| 1,4,10,16,22,10,16, 21,29,37,45,29 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \omega)))$ |
| 1,4,10,16,22,10,16, 21,29,37,45,29,37,44, 54,64,74 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) + \psi_{\psi_\beta({\alpha_{\beta+1}}^2 \cdot 2)}(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega))))$ |
| 1,4,10,16,22,10,16, 22 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot 2))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,16,22,14 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot \alpha_2))$ |
| 1,4,10,16,22,14,10, 16 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot \alpha_{\omega^2}))$ |
| 1,4,10,16,22,14,10, 16,20,11 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot \psi_\beta(\alpha_{\beta+1} \cdot \beta)))$ |
| 1,4,10,16,22,14,10, 16,20,16 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot \psi_\beta(\alpha_{\beta+1} \cdot \beta \cdot \omega)))$ |
| 1,4,10,16,22,14,10, 16,21,29,37,45,33 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot$ $\psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \alpha_2))))$ |
| 1,4,10,16,22,14,10, 16,21,29,37,45,33,11 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot \psi_\beta({\alpha_{\beta+1}}^2)))$ |
| 1,4,10,16,22,14,10, 16,21,29,37,45,35 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot \psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1})))$ |
| 1,4,10,16,22,14,10, 16,21,29,37,45,35,29 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega) \cdot \psi_\beta({\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \omega)))$ |
| 1,4,10,16,22,14,10, 16,22 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega)^2))$ |
| 1,4,10,16,22,14,14 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega)^2 \cdot \alpha_2))$ |
| 1,4,10,16,22,14,14, 10,16,22 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega)^3))$ |
| 1,4,10,16,22,14,19 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega + \beta)))$ |
| 1,4,10,16,22,15 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega + \Omega_{\beta+1}) +$ $\psi_{\alpha_2}(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega + \Omega_{\beta+1}) + 1)))$ |
| 1,4,10,16,22,15,21 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega + \varepsilon_{\Omega_{\beta+1}+1})))$ |
| 1,4,10,16,22,15,23 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega + \alpha_{\beta+1} \cdot \omega)))$ |
| 1,4,10,16,22,15,23, 31,37,24 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega + \alpha_{\beta+1} \cdot \beta)))$ |
| 1,4,10,16,22,15,23, 31,38,48 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot (\omega + 1) + \alpha_{\beta+1} \cdot \omega)))$ |
| 1,4,10,16,22,15,23, 31,39 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega \cdot 2)))$ |
| 1,4,10,16,22,16 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega^2)))$ |
| 1,4,10,16,22,16,20 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \alpha_2)))$ |
| 1,4,10,16,22,16,20, 11 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \beta)))$ |
| 1,4,10,16,22,16,20, 14,19 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \beta + \beta)))$ |
| 1,4,10,16,22,16,20, 15,23 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \beta + \alpha_{\beta+1} \cdot \omega)))$ |
| 1,4,10,16,22,16,20, 15,23,31,39 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot (\beta + \omega))))$ |
| 1,4,10,16,22,16,20, 16 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \beta \cdot \omega)))$ |
| 1,4,10,16,22,16,20, 16,20,11 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \beta^2)))$ |
| 1,4,10,16,22,16,20, 25 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \varepsilon_{\beta+1})))$ |
| 1,4,10,16,22,16,21 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \Omega_{\beta+1}) +$ $\psi_{\alpha_2}(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \Omega_{\beta+1}) + 1)))$ |
| 1,4,10,16,22,16,21, 11 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \Omega_{\beta+1} + 1)))$ |
| 1,4,10,16,22,16,21, 15,23 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \Omega_{\beta+1} + \alpha_{\beta+1} \cdot \omega)))$ |
| 1,4,10,16,22,16,21, 16 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \Omega_{\beta+1} \cdot \omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,16,22,16,21, 16,21 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot {\Omega_{\beta+1}}^2)+ \psi_{\alpha_2}(\psi_\beta({\alpha_{\beta+1}}^2 \cdot {\Omega_{\beta+1}}^2)+1)))$ |
| 1,4,10,16,22,16,21, 27 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^2 \cdot \varepsilon_{\Omega_{\beta+1}+1})))$ |
| 1,4,10,16,22,16,21, 29 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^3 + \alpha_{\beta+1} \cdot \omega)))$ |
| 1,4,10,16,22,16,21, 29,37,45 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^3 + {\alpha_{\beta+1}}^2 \cdot \omega)))$ |
| 1,4,10,16,22,16,22 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^3 \cdot \omega)))$ |
| 1,4,10,16,22,17 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\omega)))$ |
| 1,4,10,16,22,18,5 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\alpha)))$ |
| 1,4,10,16,22,18,14, 19 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\alpha + \beta)))$ |
| 1,4,10,16,22,18,16 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\alpha \cdot \omega)))$ |
| 1,4,10,16,22,18,16, 20 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\alpha \cdot \alpha_2)))$ |
| 1,4,10,16,22,18,16, 20,11 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\alpha \cdot \beta)))$ |
| 1,4,10,16,22,18,16, 22 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^(\alpha + 1) \cdot \omega)))$ |
| 1,4,10,16,22,18,16, 22,18,5 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^(\alpha \cdot 2))))$ |
| 1,4,10,16,22,18,21 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\varepsilon_{\alpha+1}})))$ |
| 1,4,10,16,22,20 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\alpha_2})))$ |
| 1,4,10,16,22,20,10 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\alpha_\omega})))$ |
| 1,4,10,16,22,20,10, 16,20,11 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\psi_\beta(\alpha_{\beta+1} \cdot \beta)})))$ |
| 1,4,10,16,22,20,10, 16,22,20 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\psi_\beta({\alpha_{\beta+1}}^{\alpha_2})})))$ |
| 1,4,10,16,22,20,11 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\beta)))$ |
| 1,4,10,16,22,20,14, 19 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\beta + \beta)))$ |
| 1,4,10,16,22,20,16 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\beta \cdot \omega)))$ |
| 1,4,10,16,22,20,16, 20,11 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\beta \cdot \beta)))$ |
| 1,4,10,16,22,20,16, 20,25 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^\beta \cdot \varepsilon_{\beta+1})))$ |
| 1,4,10,16,22,20,16, 21,29 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^(\beta + 1) + \alpha_{\beta+1} \cdot \omega)))$ |
| 1,4,10,16,22,20,16, 22 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^(\beta + 1) \cdot \omega)))$ |
| 1,4,10,16,22,20,16, 22,18,5 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^(\beta + \alpha))))$ |
| 1,4,10,16,22,20,16, 22,20 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^(\beta + \alpha_2))))$ |
| 1,4,10,16,22,20,16, 22,20,11 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^(\beta \cdot 2))))$ |
| 1,4,10,16,22,20,20, 11 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\beta^2})))$ |
| 1,4,10,16,22,20,25 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\varepsilon_{\beta+1}})))$ |
| 1,4,10,16,22,21 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\Omega_{\beta+1}})+ \psi_{\alpha_2}(\psi_\beta({\alpha_{\beta+1}}^{\Omega_{\beta+1})} + 1)))$ |
| 1,4,10,16,22,21,15, 23 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\Omega_{\beta+1}} + \alpha_{\beta+1} \cdot \omega)))$ |
| 1,4,10,16,22,21,16 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta+1}}^{\Omega_{\beta+1}} \cdot \omega)))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,10,16,22,21,16, 22 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\Omega_{\beta+1}+1)\cdot\omega)))$ |
| 1,4,10,16,22,21,16, 22,21 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\Omega_{\beta+1}\cdot 2))+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}{}^{(}\Omega_{\beta+1}\cdot 2))+1)))$ |
| 1,4,10,16,22,21,20, 11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\Omega_{\beta+1}\cdot\beta))))$ |
| 1,4,10,16,22,21,27 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\varepsilon_{\Omega_{\beta+1}+1}})))$ |
| 1,4,10,16,22,21,29 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,4,10,16,22,21,29, 37,45,38 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}+\alpha_{\beta+1}{}^{\omega})))$ |
| 1,4,10,16,22,21,29, 37,45,41 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}+\alpha_{\beta+1}{}^{\alpha_2})))$ |
| 1,4,10,16,22,21,29, 37,45,41,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}+\alpha_{\beta+1}{}^{\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}})})))$ |
| 1,4,10,16,22,21,29, 37,45,43 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}+$ $\alpha_{\beta+1}{}^{\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}+\alpha_{\beta+1})})))$ |
| 1,4,10,16,22,21,29, 37,45,43,30 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}+\alpha_{\beta+1}{}^{\beta})))$ |
| 1,4,10,16,22,21,29, 37,45,44,54 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}\cdot 2+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,4,10,16,22,22 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}\cdot\omega)))$ |
| 1,4,10,16,22,22,10, 16,22,17 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}\cdot\omega)+\psi_\beta(\alpha_{\beta+1}{}^{\omega})))$ |
| 1,4,10,16,22,22,10, 16,22,21,29,37,45,45 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}\cdot\omega)+$ $\psi_{\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}})}(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}\cdot\omega))))$ |
| 1,4,10,16,22,22,10, 16,22,21,29,37,45,45, 16,22,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}\cdot\omega)+\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}})))$ |
| 1,4,10,16,22,22,10, 16,22,22 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}\cdot\omega)\cdot 2))$ |
| 1,4,10,16,22,22,14, 10,16,22,22 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}\cdot\omega)^2))$ |
| 1,4,10,16,22,22,14, 19 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}\cdot\omega+\beta)))$ |
| 1,4,10,16,22,22,15, 23 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}\cdot\omega+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,4,10,16,22,22,16 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}\cdot\omega^2)))$ |
| 1,4,10,16,22,22,16, 20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}}\cdot\beta)))$ |
| 1,4,10,16,22,22,16, 21,29 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}+1)+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,4,10,16,22,22,16, 22 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}+1)\cdot\omega)))$ |
| 1,4,10,16,22,22,16, 22,17 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}+\omega))))$ |
| 1,4,10,16,22,22,16, 22,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}+\beta))))$ |
| 1,4,10,16,22,22,16, 22,20,25 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}+\varepsilon_{\beta+1}))))$ |
| 1,4,10,16,22,22,16, 22,21 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}+\Omega_{\beta+1})+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}+\Omega_{\beta+1}))+1))))$ |
| 1,4,10,16,22,22,16, 22,21,29 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}\cdot 2)+\alpha_{\beta+1}\cdot\omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,16,22,22,16, 22,22 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}\cdot 2)\cdot\omega)))$ |
| 1,4,10,16,22,22,17 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}\cdot\omega))))$ |
| 1,4,10,16,22,22,18, 5 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}\cdot\alpha))))$ |
| 1,4,10,16,22,22,20, 11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}\cdot\beta))))$ |
| 1,4,10,16,22,22,20, 16 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}\cdot\beta)\cdot\omega)))$ |
| 1,4,10,16,22,22,20, 16,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}\cdot\beta)\cdot\beta)))$ |
| 1,4,10,16,22,22,20, 16,22 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}\cdot\beta+1)\cdot\omega)))$ |
| 1,4,10,16,22,22,20, 16,22,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}\cdot\beta+\beta))))$ |
| 1,4,10,16,22,22,20, 16,22,22,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}\cdot\beta\cdot 2))))$ |
| 1,4,10,16,22,22,22 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}{}^{2}}\cdot\omega)))$ |
| 1,4,10,16,22,23 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}{}^{\omega}})))$ |
| 1,4,10,16,22,26,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}{}^{\beta}})))$ |
| 1,4,10,16,22,26,16, 22,26,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}{}^{\beta}\cdot 2))))$ |
| 1,4,10,16,22,26,20, 11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{(}\alpha_{\beta+1}{}^{\beta}\cdot\beta))))$ |
| 1,4,10,16,22,26,22 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}{}^{\beta+1}}\cdot\omega)))$ |
| 1,4,10,16,22,26,22, 26,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}{}^{\beta\cdot 2}})))$ |
| 1,4,10,16,22,27 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}{}^{\Omega_{\beta+1}}}+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}{}^{\Omega_{\beta+1}}})+1))))$ |
| 1,4,10,16,22,28 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}{}^{\alpha_{\beta+1}}}\cdot\omega)))$ |
| 1,4,10,16,22,28,34 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta+1}{}^{\alpha_{\beta+1}{}^{\alpha_{\beta+1}{}^{\alpha_{\beta+1}}}}\cdot\omega)))$ |
| 1,4,10,17 | $\psi(\psi_\alpha(\psi_\beta(\varepsilon_{\alpha_{\beta+1}+1}))) = \psi(\psi_\alpha(\psi_\beta(\beta_2)))$ |
| 1,4,10,17,4,10,17 | $\psi(\psi_\alpha(\psi_\beta(\beta_2))\cdot 2)$ |
| 1,4,10,17,8 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\alpha_2))$ |
| 1,4,10,17,9,17,26 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_{\alpha_3}(\psi_\beta(\beta_2))))$ |
| 1,4,10,17,9,17,26, 15 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\alpha_3))$ |
| 1,4,10,17,10 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\alpha_\omega))$ |
| 1,4,10,17,10,16 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\alpha_{\omega^2}))$ |
| 1,4,10,17,10,16,20, 11 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta(\alpha_{\beta+1}\cdot\beta)))$ |
| 1,4,10,17,10,16,21, 29,38 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_{\psi_\beta(\alpha_{\beta+1}{}^2)}(\psi_\beta(\beta_2))))$ |
| 1,4,10,17,10,16,21, 29,38,15,23,31,37,24 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta(\alpha_{\beta+1}{}^2)))$ |
| 1,4,10,17,10,16,21, 29,38,16 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_{\psi_\beta(\alpha_{\beta+1}{}^2+\alpha_{\beta+1})}(\psi_\beta(\beta_2)+1)))$ |
| 1,4,10,17,10,16,21, 29,38,27 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta(\alpha_{\beta+1}{}^2+\alpha_{\beta+1})))$ |
| 1,4,10,17,10,16,21, 29,38,28,38,49 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_{\psi_\beta(\alpha_{\beta+1}{}^2+\alpha_{\beta+1}\cdot 2)}(\psi_\beta(\beta_2))))$ |
| 1,4,10,17,10,16,21, 29,38,28,38,49 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta(\alpha_{\beta+1}{}^2+\alpha_{\beta+1}\cdot 2)))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,17,10,16,21, 29,38,29 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,4,10,17,10,16,21, 29,38,29,35,4 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega+\beta)))$ |
| 1,4,10,17,10,16,21, 29,38,29,36 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta({\alpha_{\beta+1}}^2+$ $\alpha_{\beta+1}\cdot\omega+\Omega_{\beta+1})+\psi_{\alpha_2}(\psi_\beta(\beta_2)+$ $\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega+\Omega_{\beta+1})+1)))$ |
| 1,4,10,17,10,16,21, 29,38,29,36,46 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega\cdot 2)))$ |
| 1,4,10,17,10,16,21, 29,38,29,37,43,30 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\beta)))$ |
| 1,4,10,17,10,16,22 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)+\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)))$ |
| 1,4,10,17,10,17 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)\cdot 2))$ |
| 1,4,10,17,14 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)\cdot\alpha_2))$ |
| 1,4,10,17,14,10 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)\cdot\alpha_\omega))$ |
| 1,4,10,17,14,10,15, 23,32,29 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)\cdot\alpha_{\omega+1}))$ |
| 1,4,10,17,14,10,15, 23,32,29,23 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)\cdot\alpha_{\omega\cdot 2}))$ |
| 1,4,10,17,14,10,16 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)\cdot\alpha_{\omega^2}))$ |
| 1,4,10,17,14,10,16, 20,11 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)\cdot\psi_\beta(\alpha_{\beta+1}\cdot\beta)))$ |
| 1,4,10,17,14,10,16, 21,29,38,33 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)\cdot\psi_{\psi_\beta({\alpha_{\beta+1}}^2)}(\psi_\beta(\beta_2)\cdot\alpha_2)))$ |
| 1,4,10,17,14,10,16, 22 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)\cdot\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega)))$ |
| 1,4,10,17,14,10,17 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)^2))$ |
| 1,4,10,17,14,10,17, 10,17 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)^2+\psi_\beta(\beta_2)))$ |
| 1,4,10,17,14,10,17, 14,10,17 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)^2\cdot 2))$ |
| 1,4,10,17,14,18,10, 17 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)^3))$ |
| 1,4,10,17,14,18,10, 17 | $\psi(\psi_\alpha(\psi_\beta(\beta_2)^{\psi}_{\beta}(\beta_2)))$ |
| 1,4,10,17,14,19 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\beta)))$ |
| 1,4,10,17,14,19,14, 10,17,14,19 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\beta)^2))$ |
| 1,4,10,17,14,19,14, 19 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\beta)^{\psi}_{\beta}(\beta_2+\beta)))$ |
| 1,4,10,17,14,19,19 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\beta\cdot 2)))$ |
| 1,4,10,17,14,20 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\Omega_{\beta+1})+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,4,10,17,15 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\Omega_{\beta+1})+$ $\psi_{\alpha_2}(\psi_\beta(\beta_2+\Omega_{\beta+1})+1)))$ |
| 1,4,10,17,15,23 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,4,10,17,15,23,23 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\alpha_{\beta+1}\cdot\omega)\cdot 2))$ |
| 1,4,10,17,15,23,29, 36 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\alpha_{\beta+1}\cdot\omega+\beta)))$ |
| 1,4,10,17,15,23,31 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\alpha_{\beta+1}\cdot\omega^2)))$ |
| 1,4,10,17,15,23,31, 37,24 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\alpha_{\beta+1}\cdot\beta)))$ |
| 1,4,10,17,15,23,31, 37,44 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\alpha_{\beta+1}\cdot\varepsilon_{\beta+1})))$ |
| 1,4,10,17,15,23,31, 39 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+{\alpha_{\beta+1}}^2\cdot\omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,17,15,23,32 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\psi_{\beta_2}(\beta_2))))$ |
| 1,4,10,17,15,23,32, 23,32 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\psi_{\beta_2}(\beta_2))\cdot 2))$ |
| 1,4,10,17,15,23,32, 29 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\psi_{\beta_2}(\beta_2))\cdot\psi_\beta(\beta_2+\alpha_{\beta+1})))$ |
| 1,4,10,17,15,23,32, 29,23,32 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\psi_{\beta_2}(\beta_2))^2))$ |
| 1,4,10,17,15,23,32, 29,36 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\psi_{\beta_2}(\beta_2)+\beta)))$ |
| 1,4,10,17,15,23,32, 30,40 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\psi_{\beta_2}(\beta_2)+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,4,10,17,16 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\psi_{\beta_2}(\beta_2+1))))$ |
| 1,4,10,17,16,16 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\psi_{\beta_2}(\beta_2+2))))$ |
| 1,4,10,17,16,20,11 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\psi_{\beta_2}(\beta_2+\beta))))$ |
| 1,4,10,17,16,20,25 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\psi_{\beta_2}(\beta_2+\varepsilon_{\beta+1}))))$ |
| 1,4,10,17,16,22 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\psi_{\beta_2}(\beta_2+\alpha_{\beta+1}+1))))$ |
| 1,4,10,17,16,23 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\psi_{\beta_2}(\beta_2+\psi_{\beta_2}(\beta_2)))))$ |
| 1,4,10,17,16,23,22, 29 | $\psi(\psi_\alpha(\psi_\beta(\beta_2+\psi_{\beta_2}(\beta_2+\psi_{\beta_2}(\beta_2+\psi_{\beta_2}(\beta_2))))))$ |
| 1,4,10,17,17 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot 2)))$ |
| 1,4,10,17,18 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\omega)))$ |
| 1,4,10,17,21,11 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\beta)))$ |
| 1,4,10,17,22 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\Omega_{\beta+1})+\psi_{\alpha_2}(\psi_\beta(\beta_2\cdot\Omega_{\beta+1})+1)))$ |
| 1,4,10,17,22,28 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\alpha_{\beta+1})))$ |
| 1,4,10,17,22,30 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\alpha_{\beta+1}+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,4,10,17,22,30,39 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\alpha_{\beta+1}+\psi_{\beta_2}(\beta_2))))$ |
| 1,4,10,17,22,30,39, 39 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\alpha_{\beta+1}+\psi_{\beta_2}(\beta_2\cdot 2))))$ |
| 1,4,10,17,22,30,39, 46,56 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\alpha_{\beta+1}+\psi_{\beta_2}(\beta_2\cdot\alpha_{\beta+1})+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,4,10,17,23 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\alpha_{\beta+1}+\psi_{\beta_2}(\beta_2\cdot\alpha_{\beta+1}+1))))$ |
| 1,4,10,17,23,14,19 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\alpha_{\beta+1}+\psi_{\beta_2}(\beta_2\cdot\alpha_{\beta+1}+1)+\beta)))$ |
| 1,4,10,17,23,16 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\alpha_{\beta+1}+\psi_{\beta_2}(\beta_2\cdot\alpha_{\beta+1}+2))))$ |
| 1,4,10,17,23,16,23 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\alpha_{\beta+1}+\psi_{\beta_2}(\beta_2\cdot\alpha_{\beta+1}+\psi_{\beta_2}(\beta_2)))))$ |
| 1,4,10,17,23,17 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot(\alpha_{\beta+1}+1))))$ |
| 1,4,10,17,23,17,21, 11 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot(\alpha_{\beta+1}+\beta))))$ |
| 1,4,10,17,23,17,21, 26 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot(\alpha_{\beta+1}+\varepsilon_{\beta+1}))))$ |
| 1,4,10,17,23,17,22 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot(\alpha_{\beta+1}+\Omega_{\beta+1}))+$ $\psi_{\alpha_2}(\psi_\beta(\beta_2\cdot(\alpha_{\beta+1}+\Omega_{\beta+1}))+1)))$ |
| 1,4,10,17,23,17,22, 30 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\alpha_{\beta+1}\cdot 2+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,4,10,17,23,17,22, 30,39 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\alpha_{\beta+1}\cdot 2+\psi_{\beta_2}(\beta_2))))$ |
| 1,4,10,17,23,17,23 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\alpha_{\beta+1}\cdot 2+$ $\psi_{\beta_2}(\beta_2\cdot\alpha_{\beta+1}\cdot 2+1))))$ |
| 1,4,10,17,23,17,23, 17 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\alpha_{\beta+1}\cdot 2+\beta_2)))$ |
| 1,4,10,17,23,21,11 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\alpha_{\beta+1}\cdot\beta)))$ |
| 1,4,10,17,23,21,17, 23,21,11 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\alpha_{\beta+1}\cdot\beta\cdot 2)))$ |
| 1,4,10,17,23,21,26 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot\alpha_{\beta+1}\cdot\varepsilon_{\beta+1})))$ |
| 1,4,10,17,23,22,30 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot{\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,4,10,17,23,23 | $\psi(\psi_\alpha(\psi_\beta(\beta_2\cdot{\alpha_{\beta+1}}^2+\psi_{\beta_2}(\beta_2\cdot{\alpha_{\beta+1}}^2+1))))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,4,10,17,23,24 | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot {\alpha_{\beta+1}}^\omega)))$ |
| 1,4,10,17,23,27,11 | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot {\alpha_{\beta+1}}^\beta)))$ |
| 1,4,10,17,23,29 | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot {\alpha_{\beta+1}}^{\alpha_{\beta+1}}+$ $\psi_{\beta_2}(\beta_2 \cdot {\alpha_{\beta+1}}^{\alpha_{\beta+1}}+1))))$ |
| 1,4,10,17,23,30 | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \psi_{\beta_2}(\beta_2))))$ |
| 1,4,10,17,23,30,17, 23,30 | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \psi_{\beta_2}(\beta_2) \cdot 2)))$ |
| 1,4,10,17,23,30,30 | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \psi_{\beta_2}(\beta_2 \cdot 2))))$ |
| 1,4,10,17,23,30,34, 11 | $\psi(\psi_\alpha(\psi_\beta(\beta_2 \cdot \psi_{\beta_2}(\beta_2 \cdot \beta))))$ |
| 1,4,10,17,24 | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^2)))$ |
| 1,4,10,17,24,17 | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^2+\beta_2)))$ |
| 1,4,10,17,24,17,23 | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^2+\beta_2 \cdot \alpha_{\beta+1}+$ $\psi_{\beta_2}({\beta_2}^2+\beta_2 \cdot \alpha_{\beta+1}+1))))$ |
| 1,4,10,17,24,17,23, 30,37 | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^2+\beta_2 \cdot \psi_{\beta_2}({\beta_2}^2))))$ |
| 1,4,10,17,24,17,24 | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^2 \cdot 2)))$ |
| 1,4,10,17,24,23 | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^2 \cdot \alpha_{\beta+1}+\psi_{\beta_2}({\beta_2}^2 \cdot \alpha_{\beta+1}+1))))$ |
| 1,4,10,17,24,24 | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^3)))$ |
| 1,4,10,17,24,25 | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^\omega)))$ |
| 1,4,10,17,24,31 | $\psi(\psi_\alpha(\psi_\beta({\beta_2}^{\beta_2})))$ |
| 1,4,10,17,25 | $\psi(\psi_\alpha(\psi_\beta(\varepsilon_{\beta_2+1})))$ |
| 1,4,10,17,26 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1})+\Omega_{\alpha+1} \cdot \omega))$ |
| 1,4,10,18 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1})+\psi_{\alpha_2}(\psi_\beta(\Omega_{\beta_2+1})+1)))$ |
| 1,4,10,18,8 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1})+\alpha_2))$ |
| 1,4,10,18,10,17 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1})+\psi_\beta(\beta_2)))$ |
| 1,4,10,18,10,18 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}) \cdot 2+$ $\psi_{\alpha_2}(\psi_\beta(\Omega_{\beta_2+1}) \cdot 2+1)))$ |
| 1,4,10,18,14,19 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\beta)))$ |
| 1,4,10,18,15 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\Omega_{\beta+1})+$ $\psi_{\alpha_2}(\psi_\beta(\Omega_{\beta_2+1}+\Omega_{\beta+1})+1)))$ |
| 1,4,10,18,15,23 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\alpha_{\beta+1} \cdot \omega)))$ |
| 1,4,10,18,15,23,32 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\psi_{\beta_2}(\beta_2))))$ |
| 1,4,10,18,16 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}+1))))$ |
| 1,4,10,18,16,14,19 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}+1)+\beta)))$ |
| 1,4,10,18,16,16 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}+2))))$ |
| 1,4,10,18,16,20,11 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}+\beta))))$ |
| 1,4,10,18,16,22 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}+\alpha_{\beta+1}+1))))$ |
| 1,4,10,18,16,23 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}+\psi_{\beta_2}(\beta_2)))))$ |
| 1,4,10,18,17 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\beta_2)))$ |
| 1,4,10,18,17,17 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\beta_2 \cdot 2)))$ |
| 1,4,10,18,17,24 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+{\beta_2}^2)))$ |
| 1,4,10,18,17,25 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1}+\varepsilon_{\beta_2+1})))$ |
| 1,4,10,18,18 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_2+1} \cdot 2)+\psi_{\alpha_2}(\psi_\beta(\Omega_{\beta_2+1} \cdot 2)+1)))$ |
| 1,4,10,18,29 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\alpha_{\beta+1} \cdot \omega)))$ |
| 1,4,10,18,29,38,29 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\alpha_{\beta+1} \cdot \omega)^2))$ |
| 1,4,10,18,29,40 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\alpha_{\beta+1} \cdot \omega^2)))$ |
| 1,4,10,18,29,40,49, 30 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\alpha_{\beta+1} \cdot \beta)))$ |
| 1,4,10,18,29,40,51 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+{\alpha_{\beta+1}}^2 \cdot \omega)))$ |
| 1,4,10,18,29,41 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\beta_2))))$ |
| 1,4,10,18,29,42 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}))+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}))+1)))$ |
| 1,4,10,19 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\alpha_{\beta_2+1}+1))))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,19,10,19 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\alpha_{\beta_2+1}+1))\cdot 2))$ |
| 1,4,10,19,14,19 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\alpha_{\beta_2+1}+1)+\beta)))$ |
| 1,4,10,19,16 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\alpha_{\beta_2+1}+2))))$ |
| 1,4,10,19,16,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\alpha_{\beta_2+1}+\beta))))$ |
| 1,4,10,19,16,23 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\alpha_{\beta_2+1}+\psi_{\beta_2}(\beta_2)))))$ |
| 1,4,10,19,17 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\beta_2)))$ |
| 1,4,10,19,17,25 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\varepsilon_{\beta_2+1})))$ |
| 1,4,10,19,18 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}+\Omega_{\beta_2+1})+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta_2+1}+\Omega_{\beta_2+1})+1)))$ |
| 1,4,10,19,18,29 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot 2+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,4,10,19,19 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot 2+\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot 2+1))))$ |
| 1,4,10,19,20 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\omega)))$ |
| 1,4,10,16,20,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta)))$ |
| 1,4,10,19,23,14,19 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\beta)))$ |
| 1,4,10,19,23,15 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\Omega_{\beta+1})+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\Omega_{\beta+1})+1)))$ |
| 1,4,10,19,23,15,21 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\varepsilon_{\Omega_{\beta+1}+1})))$ |
| 1,4,10,19,23,15,23 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,4,10,19,23,16 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot\beta+1))))$ |
| 1,4,10,19,23,16,23 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot\beta+\psi_{\beta_2}(\beta_2)))))$ |
| 1,4,10,19,23,16,24, 35 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot\beta+$ $\psi_{\psi_\beta(\alpha_{\beta_2+1})}(\psi_\beta(\alpha_{\beta_2+1}+\alpha_{\beta+1}\cdot\omega))))))$ |
| 1,4,10,19,23,16,25 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+$ $\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot\beta+\psi_{\beta_2}(\alpha_{\beta_2+1}))))))$ |
| 1,4,10,19,23,16,25, 29,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+$ $\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot\beta+\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot\beta)))))$ |
| 1,4,10,19,23,17 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\beta_2)))$ |
| 1,4,10,19,23,18 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\Omega_{\beta_2+1})+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta+\Omega_{\beta_2+1})+1)))$ |
| 1,4,10,19,23,18,29 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot(\beta+1)+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,4,10,19,23,19 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot(\beta+1)+$ $\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot(\beta+1)+1))))$ |
| 1,4,10,19,23,19,23, 11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta\cdot 2)))$ |
| 1,4,10,19,23,23,11 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta^2)))$ |
| 1,4,10,19,24 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\Omega_{\beta+1})+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta_2+1}\cdot\Omega_{\beta+1})+1)))$ |
| 1,4,10,19,24,32 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\alpha_{\beta+1}+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,4,10,19,25 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\alpha_{\beta+1}+$ $\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot\alpha_{\beta+1}+1))))$ |
| 1,4,10,19,25,19,25 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\alpha_{\beta+1}\cdot 2+$ $\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot\alpha_{\beta+1}\cdot 2+1))))$ |
| 1,4,10,19,25,32 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\psi_{\beta_2}(\beta_2))))$ |
| 1,4,10,19,26 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta_2)))$ |
| 1,4,10,19,26,19,26 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\beta_2\cdot 2)))$ |
| 1,4,10,19,27 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_2+1}\cdot\Omega_{\beta_2+1})+$ $\psi_{\alpha_2}(\psi_\beta(\alpha_{\beta_2+1}\cdot\Omega_{\beta_2+1})+1)))$ |
| 1,4,10,19,27,38 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta_2+1}}^2+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,4,10,19,28 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta_2+1}}^2+\psi_{\beta_2}({\alpha_{\beta_2+1}}^2+1))))$ |
| 1,4,10,19,28,19,28 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta_2+1}}^2\cdot 2+\psi_{\beta_2}({\alpha_{\beta_2+1}}^2\cdot 2+1))))$ |
| . 1,4,10,19,28,29 | $\psi(\psi_\alpha(\psi_\beta({\alpha_{\beta_2+1}}^\omega)))$ |
| 1,4,10,19,29 | $\psi(\psi_\alpha(\psi_\beta(\beta_3)))$ |
| 1,4,10,19,29,19,29 | $\psi(\psi_\alpha(\psi_\beta(\beta_3+\psi_{\beta_3}(\beta_3))))$ |

| 0 − Y | Projection |
|---|---|
| 1,4,10,19,29,29 | $\psi(\psi_\alpha(\psi_\beta(\beta_3 \cdot 2)))$ |
| 1,4,10,19,29,33,11 | $\psi(\psi_\alpha(\psi_\beta(\beta_3 \cdot \beta)))$ |
| 1,4,10,19,29,35 | $\psi(\psi_\alpha(\psi_\beta(\beta_3 \cdot \alpha_{\beta+1} + \psi_{\beta_2}(\beta_3 \cdot \alpha_{\beta+1} + 1))))$ |
| 1,4,10,19,29,35,42 | $\psi(\psi_\alpha(\psi_\beta(\beta_3 \cdot \psi_{\beta_2}(\beta_2))))$ |
| 1,4,10,19,29,35,44, 54 | $\psi(\psi_\alpha(\psi_\beta(\beta_3 \cdot \psi_{\beta_2}(\beta_3))))$ |
| 1,4,10,19,29,36 | $\psi(\psi_\alpha(\psi_\beta(\beta_3 \cdot \beta_2)))$ |
| 1,4,10,19,29,38 | $\psi(\psi_\alpha(\psi_\beta(\beta_3 \cdot \alpha_{\beta_2+1} + \psi_{\beta_2}(\beta_3 \cdot \alpha_{\beta_2+1} + 1))))$ |
| 1,4,10,19,29,39 | $\psi(\psi_\alpha(\psi_\beta({\beta_3}^2)))$ |
| 1,4,10,19,30,44 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_3+1} + \alpha_{\beta+1} \cdot \omega)))$ |
| 1,4,10,19,31 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_3+1} + \psi_{\beta_2}(\alpha_{\beta_3+1} + 1))))$ |
| 1,4,10,20 | $\psi(\psi_\alpha(\psi_\beta(\beta_\omega)))$ |
| 1,4,10,20,15,23,35 | $\psi(\psi_\alpha(\psi_\beta(\beta_\omega + \psi_{\beta_2}(\beta_\omega))))$ |
| 1,4,10,20,17 | $\psi(\psi_\alpha(\psi_\beta(\beta_\omega + \beta_2)))$ |
| 1,4,10,20,20 | $\psi(\psi_\alpha(\psi_\beta(\beta_\omega \cdot 2)))$ |
| 1,4,10,20,27 | $\psi(\psi_\alpha(\psi_\beta(\beta_\omega \cdot \beta_2)))$ |
| 1,4,10,20,27,20 | $\psi(\psi_\alpha(\psi_\beta({\beta_\omega}^2)))$ |
| 1,4,10,20,28 | $\psi(\psi_\alpha(\psi_\beta(\Omega_{\beta_\omega+1}) + \psi_{\alpha_2}(\psi_\beta(\Omega_{\beta_\omega+1}) + 1)))$ |
| 1,4,10,20,29 | $\psi(\psi_\alpha(\psi_\beta(\alpha_{\beta_\omega+1} + \psi_{\beta_2}(\alpha_{\beta_\omega+1} + 1))))$ |
| 1,4,10,20,29,42 | $\psi(\psi_\alpha(\psi_\beta(\beta_{\omega \cdot 2})))$ |
| 1,4,10,20,30 | $\psi(\psi_\alpha(\psi_\beta(\beta_{\omega^2})))$ |
| 1,4,10,20,30,37,21 | $\psi(\psi_\alpha(\psi_\beta(\psi_\gamma(\beta_{\gamma+1} \cdot \gamma))))$ |
| 1,4,10,20,30,38 | $\psi(\psi_\alpha(\psi_\beta(\psi_\gamma(\beta_{\gamma+1} \cdot \Omega_{\gamma+1}))+$ $\psi_{\alpha_2}(\psi_\gamma(\beta_{\gamma+1} \cdot \Omega_{\gamma+1}) + 1)))$ |
| 1,4,10,20,30,39 | $\psi(\psi_\alpha(\psi_\beta(\psi_\gamma(\beta_{\gamma+1} \cdot \alpha_{\gamma+1})+$ $\psi_{\beta_2}(\psi_\gamma(\beta_{\gamma+1} \cdot \alpha_{\gamma+1}) + 1))))$ |
| 1,4,10,20,30,40 | $\psi(\psi_\alpha(\psi_\beta(\psi_\gamma({\beta_{\gamma+1}}^2 \cdot \omega))))$ |
| 1,4,10,20,31 | $\psi(\psi_\alpha(\psi_\beta(\psi_\gamma(\gamma_2))))$ |
| 1,4,10,20,34 | $\psi(\psi_\alpha(\psi_\beta(\psi_\gamma(\beta_{\gamma_2+1} + \psi_{\gamma_2}(\beta_{\gamma_2+1} + 1)))))$ |
| 1,4,10,20,34,35 | $\psi(\psi_\alpha(\psi_\beta(\psi_\gamma(\beta_{\gamma_2+1} \cdot \omega))))$ |
| 1,4,10,20,35 | $\psi(\psi_\alpha(\psi_\beta(\psi_\gamma(\gamma_\omega))))$ |
| 1,5 | $\psi(\omega - P) = \psi(\psi_S(\sigma S \cdot \omega))$ |

## A.18 0 − Y vs Upward Projection

| 0 − Y | Projection |
|---|---|
| 1,5 | $\psi(\omega - P) = \psi(\psi_S(\sigma S \cdot \omega))$ |
| 1,5,2 | $\psi(\psi_S(\sigma S \cdot \omega) + 1)$ |
| 1,5,2,6 | $\psi(\psi_S(\sigma S \cdot \omega) + \psi(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,3 | $\psi(\psi_S(\sigma S \cdot \omega) + \Omega)$ |
| 1,5,3,3 | $\psi(\psi_S(\sigma S \cdot \omega) + \Omega \cdot 2)$ |
| 1,5,3,4,8 | $\psi(\psi_S(\sigma S \cdot \omega) + \Omega \cdot \psi(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,3,5 | $\psi(\psi_S(\sigma S \cdot \omega) + \Omega^2)$ |
| 1,5,3,5,3,5 | $\psi(\psi_S(\sigma S \cdot \omega) + \Omega^2 \cdot 2)$ |
| 1,5,3,5,6 | $\psi(\psi_S(\sigma S \cdot \omega) + \Omega^\omega)$ |
| 1,5,3,6 | $\psi(\psi_S(\sigma S \cdot \omega) + \psi_1(\Omega_2))$ |
| 1,5,3,7 | $\psi(\psi_S(\sigma S \cdot \omega) + \psi_1(\Omega_\omega))$ |
| 1,5,3,7,7 | $\psi(\psi_S(\sigma S \cdot \omega) + \psi_1(\Omega_\omega \cdot 2))$ |
| 1,5,3,7,11 | $\psi(\psi_S(\sigma S \cdot \omega) + \psi_1(\Omega_{\omega^2}))$ |
| 1,5,3,7,11,14,8 | $\psi(\psi_S(\sigma S \cdot \omega) + \psi_1(OFP))$ |
| 1,5,3,7,11,15 | $\psi(\psi_S(\sigma S \cdot \omega) + \psi_1(I_\omega))$ |
| 1,5,3,7,12 | $\psi(\psi_S(\sigma S \cdot \omega) + \psi_1(\psi_{\alpha_2}(\alpha_2)))$ |
| 1,5,3,7,14 | $\psi(\psi_S(\sigma S \cdot \omega) + \psi_1(\alpha_\omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,3,8 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_1(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,3,8,4 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_1(\psi_S(\sigma S\cdot\omega)+1))$ |
| 1,5,3,8,5 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_1(\psi_S(\sigma S\cdot\omega)+\Omega))$ |
| 1,5,3,8,5,10 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_1(\psi_S(\sigma S\cdot\omega)+\psi_1(\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,3,8,6 | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_2)$ |
| 1,5,3,8,6,12 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_2(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4 | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_\omega)$ |
| 1,5,4,4 | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_\omega\cdot 2)$ |
| 1,5,4,6 | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_\omega\cdot\Omega)$ |
| 1,5,4,6,4 | $\psi(\psi_S(\sigma S\cdot\omega)+{\Omega_\omega}^2)$ |
| 1,5,4,6,9 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\omega(\Omega_{\omega+1}))$ |
| 1,5,4,6,11 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,6,11,5 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega)+1))$ |
| 1,5,4,6,11,6,11 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,4,6,11,9 | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_{\omega+1})$ |
| 1,5,4,6,11,9,15 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\omega+1}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,6,11,9,15, 13 | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_{\omega+2})$ |
| 1,5,4,6,11,10 | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_{\omega\cdot 2})$ |
| 1,5,4,7 | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_{\omega^2})$ |
| 1,5,4,7,7 | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_{\omega^3})$ |
| 1,5,4,7,9,5 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(I))$ |
| 1,5,4,7,9,12 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\Omega_{I+1}))$ |
| 1,5,4,7,9,14 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,7,9,14,4, 7,9,14 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega))\cdot 2)$ |
| 1,5,4,7,9,14,6, 4,7,9,14 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega))^2)$ |
| 1,5,4,7,9,14,6, 9 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)+1))$ |
| 1,5,4,7,9,14,6, 9,5 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)+1)+$ $\psi_{\psi_I(\psi_S(\sigma S\cdot\omega)+1)}(\psi_I(\psi_S(\sigma S\cdot\omega)+1)+1))$ |
| 1,5,4,7,9,14,6, 9,9 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)+1)\cdot 2)$ |
| 1,5,4,7,9,14,6, 9,13 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)+2))$ |
| 1,5,4,7,9,14,6, 10 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)+\omega))$ |
| 1,5,4,7,9,14,6, 10,10 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)+\omega)\cdot 2)$ |
| 1,5,4,7,9,14,6, 10,13,10 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)+\omega)^2)$ |
| 1,5,4,7,9,14,6, 10,14 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_I(\psi_S(\sigma S\cdot\omega)+\omega^2))$ |
| 1,5,4,7,9,14,6, 10,14,17,11 | $\psi(\psi_S(\sigma S\cdot\omega)+I)$ |
| 1,5,4,7,9,14,6, 10,14,17,14 | $\psi(\psi_S(\sigma S\cdot\omega)+I\cdot\omega)$ |
| 1,5,4,7,9,14,6, 10,14,17,21 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\Omega_{I+1}}(\Omega_{I+1}))$ |
| 1,5,4,7,9,14,6, 10,14,18 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\Omega_{I+1}}(I_\omega))$ |
| 1,5,4,7,9,14,6, 11 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\Omega_{I+1}}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,7,9,14,7 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\Omega_{I+1}}(\psi_S(\sigma S\cdot\omega)+1))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,4,7,9,14,7, 9,14 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\Omega_{I+1}}(\psi_S(\sigma S\cdot\omega)+\psi_{\Omega_{I+1}}(\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,4,7,9,14,12 | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_{I+1})$ |
| 1,5,4,7,9,14,13 | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_{I+\omega})$ |
| 1,5,4,7,9,14,13, 17,20,14 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{I_2}(I_2))$ |
| 1,5,4,7,10 | $\psi(\psi_S(\sigma S\cdot\omega)+I_\omega)$ |
| 1,5,4,8 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_2))$ |
| 1,5,4,8,8 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_2\cdot 2))$ |
| 1,5,4,8,11 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_2\cdot\Omega_{\alpha+1}+\psi_{\alpha_2}(\alpha_2\cdot\Omega_{\alpha+1}+1)))$ |
| 1,5,4,8,12 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha({\alpha_2}^2))$ |
| 1,5,4,8,13 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\varepsilon_{\alpha_2+1}))$ |
| 1,5,4,8,14 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\alpha(\Omega_{\alpha_2+1})}(\psi_\alpha(\Omega_{\alpha_2+1}+\Omega_{\alpha+1}\cdot\omega)))$ |
| 1,5,4,8,15 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\alpha(\Omega_{\alpha_2+1})}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,8,15,8 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}))$ |
| 1,5,4,8,15,8,15 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\alpha(\Omega_{\alpha_2+1}+\Omega_{\alpha+1})}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,8,15,13 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\Omega_{\alpha+1}))$ |
| 1,5,4,8,15,14 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\Omega_{\alpha+1}\cdot\omega))$ |
| 1,5,4,8,15,14,21 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\alpha_2)))$ |
| 1,5,4,8,15,14,21, 31 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1})}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,9 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}+1)))$ |
| 1,5,4,9,4,9 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}+1))\cdot 2)$ |
| 1,5,4,9,6,9 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}+1)+\alpha))$ |
| 1,5,4,9,7 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}+2)))$ |
| 1,5,4,9,7,9,5 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}+\alpha)))$ |
| 1,5,4,9,7,11 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\alpha_2))))$ |
| 1,5,4,9,7,12 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\Omega_{\alpha_2+1})+1)))$ |
| 1,5,4,9,8 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\alpha_2))$ |
| 1,5,4,9,8,13 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}+\varepsilon_{\alpha_2+1}))$ |
| 1,5,4,9,8,15 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\alpha(\Omega_{\alpha_2+1}\cdot 2)}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,9,8,15,8 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}\cdot 2))$ |
| 1,5,4,9,9 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}\cdot 2+\psi_{\alpha_2}(\Omega_{\alpha_2+1}\cdot 2+1)))$ |
| 1,5,4,9,10 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}\cdot\omega))$ |
| 1,5,4,9,13 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_2+1}\cdot\alpha_2))$ |
| 1,5,4,9,14 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha({\Omega_{\alpha_2+1}}^2+\psi_{\alpha_2}({\Omega_{\alpha_2+1}}^2+1)))$ |
| 1,5,4,9,15 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_3))$ |
| 1,5,4,10 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_\omega))$ |
| 1,5,4,10,8 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_\omega+\alpha_2))$ |
| 1,5,4,10,10 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_\omega\cdot 2))$ |
| 1,5,4,10,14,19 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_{\omega+1}))$ |
| 1,5,4,10,14,21 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\alpha(\Omega_{\alpha_{\omega+1}+1})}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,10,15 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\Omega_{\alpha_{\omega+1}+1}+\psi_{\alpha_2}(\Omega_{\alpha_{\omega+1}+1}+1)))$ |
| 1,5,4,10,15,23 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_{\omega\cdot 2}))$ |
| 1,5,4,10,16 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_{\omega^2}))$ |
| 1,5,4,10,16,20,11 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_\beta(\alpha_{\beta+1}\cdot\beta)))$ |
| 1,5,4,10,17 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_\beta(\beta_2)))$ |
| 1,5,4,10,20 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\beta_\omega))$ |
| 1,5,4,11 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,11,5 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+1))$ |
| 1,5,4,11,6,4,11 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,4,11,6,9 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\alpha))$ |
| 1,5,4,11,6,11 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega))))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,4,11,7 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+1)))$ |
| 1,5,4,11,7,7 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+2)))$ |
| 1,5,4,11,7,9,5 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+\alpha)))$ |
| 1,5,4,11,7,11 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\alpha_2))))$ |
| 1,5,4,11,7,13 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\alpha_\omega))))$ |
| 1,5,4,11,7,14 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))))$ |
| 1,5,4,11,7,14,7, 14 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega))\cdot 2)))$ |
| 1,5,4,11,8 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\alpha_2))$ |
| 1,5,4,11,8,8 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\alpha_2\cdot 2))$ |
| 1,5,4,11,8,13 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\varepsilon_{\alpha_2+1}))$ |
| 1,5,4,11,8,15 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+$ $\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,4,11,9 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+$ $\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha_2+1}+1)))$ |
| 1,5,4,11,9,8 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha_2+1}+\alpha_2))$ |
| 1,5,4,11,9,10 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha_2+1}\cdot\omega))$ |
| 1,5,4,11,9,13 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha_2+1}\cdot\alpha_2))$ |
| 1,5,4,11,9,15 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_3}(\alpha_3)))$ |
| 1,5,4,11,9,17 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_3}(\alpha_\omega)))$ |
| 1,5,4,11,9,18 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_3}(\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,4,11,9,18,15 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\alpha_3))$ |
| 1,5,4,11,10 | $\psi(\psi_S(\sigma S\cdot\omega)+\alpha_\omega)$ $=\psi(\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)+\alpha_\omega))$ |
| 1,5,4,11,10,8 | $\psi(\psi_S(\sigma S\cdot\omega)+\alpha_\omega+\alpha_2)$ |
| 1,5,4,11,10,9,18 | $\psi(\psi_S(\sigma S\cdot\omega)+\alpha_\omega+\psi_{\alpha_3}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,11,10,10 | $\psi(\psi_S(\sigma S\cdot\omega)+\alpha_\omega\cdot 2)$ |
| 1,5,4,11,10,14 | $\psi(\psi_S(\sigma S\cdot\omega)+\alpha_\omega\cdot\alpha_2)$ |
| 1,5,4,11,10,14,10 | $\psi(\psi_S(\sigma S\cdot\omega)+{\alpha_\omega}^2)$ |
| 1,5,4,11,10,15 | $\psi(\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha_\omega+1}+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha_\omega+1}+1))$ |
| 1,5,4,11,10,15,21 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_{\omega+1}}(\alpha_{\omega+1}))$ |
| 1,5,4,11,10,15,23 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_{\omega+1}}(\alpha_{\omega\cdot 2}))$ |
| 1,5,4,11,10,15,24 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_{\omega+1}}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,11,10,15,24, 21 | $\psi(\psi_S(\sigma S\cdot\omega)+\alpha_{\omega+1})$ |
| 1,5,4,11,10,15,24, 23 | $\psi(\psi_S(\sigma S\cdot\omega)+\alpha_{\omega\cdot 2})$ |
| 1,5,4,11,10,16 | $\psi(\psi_S(\sigma S\cdot\omega)+\alpha_{\omega^2})$ |
| 1,5,4,11,10,16,20, 11 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta))$ |
| 1,5,4,11,10,16,20, 14,19 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta+\beta))$ |
| 1,5,4,11,10,16,20, 15,23 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega)))$ |
| 1,5,4,11,10,16,20, 15,23,23 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega))\cdot 2)$ |
| 1,5,4,11,10,16,20, 15,23,29,36 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega)+\beta))$ |
| 1,5,4,11,10,16,20, 15,23,31 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega+1))}(\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega^2))))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,4,11,10,16,20, 15,24 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega+1))}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,11,10,16,20, 15,24,21 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega+1)))$ |
| 1,5,4,11,10,16,20, 15,24,23 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot(\beta+\omega\cdot 2)))$ |
| 1,5,4,11,10,16,20, 15,24,23,31,37,24 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot 2))$ |
| 1,5,4,11,10,16,20, 16 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta\cdot\omega))$ |
| 1,5,4,11,10,16,20, 16,20,11 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta^2))$ |
| 1,5,4,11,10,16,20, 25 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\varepsilon_{\beta+1}))$ |
| 1,5,4,11,10,16,20, 27 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,11,10,16,21 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+$ $\psi_\beta(\alpha_{\beta+1}\cdot\Omega_{\beta+1})+1))$ |
| 1,5,4,11,10,16,21, 29 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\beta(\alpha_{\beta+1}{}^2)}(\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,5,4,11,10,16,21, 30 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\beta(\alpha_{\beta+1}{}^2)}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,11,10,16,21, 30,15,24,23,31,37,24 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta({\alpha_{\beta+1}}^2))$ |
| 1,5,4,11,10,16,21, 30,16 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\beta(\alpha_{\beta+1}{}^2+\alpha_{\beta+1})}(\psi_S(\sigma S\cdot\omega)+1))$ |
| 1,5,4,11,10,16,21, 30,29 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta({\alpha_{\beta+1}}^2+\alpha_{\beta+1}\cdot\omega))$ |
| 1,5,4,11,10,16,22 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta({\alpha_{\beta+1}}^2\cdot\omega))$ |
| 1,5,4,11,10,16,22, 16,22 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta({\alpha_{\beta+1}}^3\cdot\omega))$ |
| 1,5,4,11,10,17 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\beta_2))$ |
| 1,5,4,11,10,17,17 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\beta_2\cdot 2))$ |
| 1,5,4,11,10,17,26 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\Omega_{\beta_2+1})+\Omega_{\alpha+1}\cdot\omega)$ |
| 1,5,4,11,10,18 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\Omega_{\beta_2+1})+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\Omega_{\beta_2+1})+1))$ |
| 1,5,4,11,10,18,16 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\Omega_{\beta_2+1}+\psi_{\beta_2}(\Omega_{\beta_2+1}+1)))$ |
| 1,5,4,11,10,18,17 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\Omega_{\beta_2+1}+\beta_2))$ |
| 1,5,4,11,10,18,27 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\varepsilon_{\Omega_{\beta_2+1}+1}))$ |
| 1,5,4,11,10,18,29 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\beta(\alpha_{\beta_2+1})}(\psi_\beta(\alpha_{\beta_2+1}+\alpha_{\beta+1}\cdot\omega)))$ |
| 1,5,4,11,10,18,30 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_{\psi_\beta(\alpha_{\beta_2+1})}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,11,10,18,30, 27 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta_2+1}))$ |
| 1,5,4,11,10,18,30, 29 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta_2+1}+\alpha_{\beta+1}\cdot\omega))$ |
| 1,5,4,11,10,19 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta_2+1}+\psi_{\beta_2}(\alpha_{\beta_2+1}+1)))$ |
| 1,5,4,11,10,19,17 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta_2+1}+\beta_2))$ |
| 1,5,4,11,10,19,19 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta_2+1}\cdot 2+\psi_{\beta_2}(\alpha_{\beta_2+1}\cdot 2+1)))$ |
| 1,5,4,11,10,19,29 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\beta_3))$ |
| 1,5,4,11,10,20 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\beta_\omega))$ |
| 1,5,4,11,10,21 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,4,11,10,21,10, 21 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\psi_S(\sigma S\cdot\omega))\cdot 2)$ |
| 1,5,4,11,10,21,14 | $\psi(\psi_S(\sigma S\cdot\omega)+\psi_\beta(\psi_S(\sigma S\cdot\omega)+\alpha_2))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,4,11,10,21,14, 10 | $\psi(\psi_S(\sigma S \cdot \omega) + \psi_\beta(\psi_S(\sigma S \cdot \omega) + \alpha_\omega))$ |
| 1,5,4,11,10,21,14, 10,21 | $\psi(\psi_S(\sigma S \cdot \omega) + \psi_\beta(\psi_S(\sigma S \cdot \omega) + \psi_\beta(\psi_S(\sigma S \cdot \omega))))$ |
| 1,5,4,11,10,21,14, 19 | $\psi(\psi_S(\sigma S \cdot \omega) + \beta)$ |
| 1,5,4,11,10,21,16 | $\psi(\psi_S(\sigma S \cdot \omega) + \psi_{\beta_2}(\psi_S(\sigma S \cdot \omega) + 1))$ |
| 1,5,4,11,10,21,17 | $\psi(\psi_S(\sigma S \cdot \omega) + \beta_2)$ |
| 1,5,4,11,10,21,20 | $\psi(\psi_S(\sigma S \cdot \omega) + \beta_\omega)$ |
| 1,5,4,11,10,21,20, 31 | $\psi(\psi_S(\sigma S \cdot \omega) + \psi_\gamma(\gamma_2))$ |
| 1,5,4,11,10,21,20, 35 | $\psi(\psi_S(\sigma S \cdot \omega) + \psi_\gamma(\gamma_\omega))$ |
| 1,5,4,11,10,21,20, 36 | $\psi(\psi_S(\sigma S \cdot \omega) + \psi_\gamma(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,4,11,10,21,20, 36,27,35 | $\psi(\psi_S(\sigma S \cdot \omega) + \gamma)$ |
| 1,5,4,11,10,21,20, 36,35 | $\psi(\psi_S(\sigma S \cdot \omega) + \gamma_\omega)$ |
| 1,5,5 | $\psi(\psi_S(\sigma S \cdot \omega) \cdot 2)$ |
| 1,5,5,4 | $\psi(\psi_S(\sigma S \cdot \omega) \cdot 2 + \Omega_\omega)$ |
| 1,5,5,4,8 | $\psi(\psi_S(\sigma S \cdot \omega) \cdot 2 + \psi_\alpha(\alpha_2))$ |
| 1,5,5,4,10 | $\psi(\psi_S(\sigma S \cdot \omega) \cdot 2 + \psi_\alpha(\alpha_\omega))$ |
| 1,5,5,4,11 | $\psi(\psi_S(\sigma S \cdot \omega) \cdot 2 + \psi_\alpha(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,5,4,11,11 | $\psi(\psi_S(\sigma S \cdot \omega) \cdot 2 + \psi_\alpha(\psi_S(\sigma S \cdot \omega) \cdot 2))$ |
| 1,5,5,4,11,11,6, 9 | $\psi(\psi_S(\sigma S \cdot \omega) \cdot 2 + \alpha)$ |
| 1,5,5,4,11,11,8 | $\psi(\psi_S(\sigma S \cdot \omega) \cdot 2 + \alpha_2)$ |
| 1,5,5,4,11,11,10, 17 | $\psi(\psi_S(\sigma S \cdot \omega) \cdot 2 + \psi_\beta(\beta_2))$ |
| 1,5,5,4,11,11,10, 21,21,14,19 | $\psi(\psi_S(\sigma S \cdot \omega) \cdot 2 + \beta)$ |
| 1,5,5,5 | $\psi(\psi_S(\sigma S \cdot \omega) \cdot 3)$ |
| 1,5,6 | $\psi(\psi_S(\sigma S \cdot \omega + 1))$ |
| 1,5,6,10 | $\psi(\psi_S(\sigma S \cdot \omega + \psi(\psi_S(\sigma S \cdot \omega))))$ |
| 1,5,7 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega))$ |
| 1,5,7,3 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \Omega)$ |
| 1,5,7,3,8 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_1(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,3,8,9 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_1(\psi_S(\sigma S \cdot \omega + 1)))$ |
| 1,5,7,3,8,10 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_1(\psi_S(\sigma S \cdot \omega + \Omega)))$ |
| 1,5,7,3,8,10,4 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_1(\psi_S(\sigma S \cdot \omega + \Omega) + 1))$ |
| 1,5,7,3,8,10,5, 10,12 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_1(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_1(\psi_S(\sigma S \cdot \omega + \Omega))))$ |
| 1,5,7,3,8,10,6 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \Omega_2)$ |
| 1,5,7,3,8,10,6, 12,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_2(\psi_S(\sigma S \cdot \omega + \Omega)))$ |
| 1,5,7,3,8,10,7 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \Omega_\omega)$ |
| 1,5,7,3,8,10,7, 10 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \Omega_\omega \cdot \Omega_2)$ |
| 1,5,7,3,8,10,7, 10,7 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + {\Omega_\omega}^2)$ |
| 1,5,7,3,8,10,7, 10,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\omega(\Omega_{\omega+1}))$ |
| 1,5,7,3,8,10,7, 11 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \Omega_{\omega^2})$ |

| 0 − Y | Projection |
|---|---|
| 1,5,7,3,8,10,7, 12 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\alpha(\alpha_2))$ |
| 1,5,7,3,8,10,7, 14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\alpha(\alpha_\omega))$ |
| 1,5,7,3,8,10,7, 15 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\alpha(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,3,8,10,7, 15,17 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\alpha(\psi_S(\sigma S \cdot \omega + \Omega)))$ |
| 1,5,7,3,8,10,7, 15,17,8 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\alpha(\psi_S(\sigma S \cdot \omega + \Omega) + 1))$ |
| 1,5,7,3,8,10,7, 15,17,10 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\alpha(\psi_S(\sigma S \cdot \omega + \Omega) + \Omega_2))$ |
| 1,5,7,3,8,10,7, 15,17,10,7 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\alpha(\psi_S(\sigma S \cdot \omega + \Omega) + \Omega_\omega))$ |
| 1,5,7,3,8,10,7, 15,17,10,7,15,17 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\alpha(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\alpha(\psi_S(\sigma S \cdot \omega + \Omega))))$ |
| 1,5,7,3,8,10,7, 15,17,10,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \alpha)$ |
| 1,5,7,3,8,10,7, 15,17,10,15 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \Omega_{\alpha+1} \cdot \omega)$ |
| 1,5,7,3,8,10,7, 15,17,10,16 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,3,8,10,7, 15,17,10,16,18 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega)))$ |
| 1,5,7,3,8,10,7, 15,17,10,16,18,8 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega)) + \psi_{\psi_\alpha(\psi_S(\sigma S \cdot \omega + \Omega) + \Omega_{\alpha+1}}(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega)) + 1))$ |
| 1,5,7,3,8,10,7, 15,17,10,16,18,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega)) + \psi_\alpha(\psi_S(\sigma S \cdot \omega + \Omega) + \Omega_{\alpha+1}))$ |
| 1,5,7,3,8,10,7, 15,17,10,16,18,15 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega)) + \psi_\alpha(\psi_S(\sigma S \cdot \omega + \Omega) + \Omega_{\alpha+1} \cdot \omega))$ |
| 1,5,7,3,8,10,7, 15,17,10,16,18,15,24, 26 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega)) + \psi_\alpha(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega))))$ |
| 1,5,7,3,8,10,7, 15,17,10,16,18,15,24, 26,19,24 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega)) + \alpha)$ |
| 1,5,7,3,8,10,7, 15,17,10,16,18,15,24, 26,19,26,28 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega)) \cdot 2)$ |
| 1,5,7,3,8,10,7, 15,17,11 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega) + 1))$ |
| 1,5,7,3,8,10,7, 15,17,11,8 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega) + 1) + \psi_\alpha(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega) + 1) + 1))$ |
| 1,5,7,3,8,10,7, 15,17,11,10,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega) + 1) + \alpha)$ |
| 1,5,7,3,8,10,7, 15,17,11,10,16,18 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega) + 1) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega)))$ |
| 1,5,7,3,8,10,7, 15,17,11,11 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega) + 2))$ |
| 1,5,7,3,8,10,7, 15,17,11,14,8 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega) + \alpha))$ |
| 1,5,7,3,8,10,7, 15,17,11,14,11 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega) + \alpha + 1))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,7,3,8,10,7, 15,17,11,14,18 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\varepsilon_{\alpha+1}))$ |
| 1,5,7,3,8,10,7, 15,17,11,14,20 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$ $\Omega_{\alpha+1})+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,3,8,10,7, 15,17,11,15 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha+1}+1))$ |
| 1,5,7,3,8,10,7, 15,17,11,15,11,15 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha+1}\cdot 2+1))$ |
| 1,5,7,3,8,10,7, 15,17,11,16 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\alpha_2)))$ |
| 1,5,7,3,8,10,7, 15,17,11,16,16 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\alpha_2\cdot 2)))$ |
| 1,5,7,3,8,10,7, 15,17,11,16,24 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$ $\psi_{\alpha_2}(\Omega_{\alpha_2+1}))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,3,8,10,7, 15,17,11,17 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$ $\psi_{\alpha_2}(\Omega_{\alpha_2+1})+1))$ |
| 1,5,7,3,8,10,7, 15,17,11,17,11 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$ $\psi_{\alpha_2}(\Omega_{\alpha_2+1})+2))$ |
| 1,5,7,3,8,10,7, 15,17,11,17,11,17 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$ $\psi_{\alpha_2}(\Omega_{\alpha_2+1})\cdot 2+1))$ |
| 1,5,7,3,8,10,7, 15,17,11,17,16 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$ $\psi_{\alpha_2}(\Omega_{\alpha_2+1}+\alpha_2)))$ |
| 1,5,7,3,8,10,7, 15,17,11,17,17 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$ $\psi_{\alpha_2}(\Omega_{\alpha_2+1}\cdot 2)+1))$ |
| 1,5,7,3,8,10,7, 15,17,11,18 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\alpha_\omega)))$ |
| 1,5,7,3,8,10,7, 15,17,11,19 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,7,3,8,10,7, 15,17,11,19,21 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega))))$ |
| 1,5,7,3,8,10,7, 15,17,12 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\alpha_2)$ |
| 1,5,7,3,8,10,7, 15,17,12,12 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\alpha_2\cdot 2)$ |
| 1,5,7,3,8,10,7, 15,17,12,18 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\varepsilon_{\alpha_2+1})$ |
| 1,5,7,3,8,10,7, 15,17,12,20 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,3,8,10,7, 15,17,13 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha_2+1}+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha_2+1}+1))$ |
| 1,5,7,3,8,10,7, 15,17,13,12 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha_2+1}+\alpha_2)$ |
| 1,5,7,3,8,10,7, 15,17,13,13 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha_2+1}\cdot 2+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\Omega_{\alpha_2+1}\cdot 2+1))$ |
| 1,5,7,3,8,10,7, 15,17,13,20 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_3}(\alpha_3))$ |
| 1,5,7,3,8,10,7, 15,17,13,20,30 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_3}(\Omega_{\alpha_3+1})+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,3,8,10,7, 15,17,13,21 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_3}(\Omega_{\alpha_3+1})+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_3}(\Omega_{\alpha_3+1})+1))$ |
| 1,5,7,3,8,10,7, 15,17,13,22 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_{\alpha_3}(\alpha_\omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,7,3,8,10,7, 15,17,13,23,25 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_3}(\psi_S(\sigma S \cdot \omega + \Omega)))$ |
| 1,5,7,3,8,10,7, 15,17,13,23,25,20 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \alpha_3)$ |
| 1,5,7,3,8,10,7, 15,17,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \alpha_\omega)$ |
| 1,5,7,3,8,10,7, 15,17,14,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \alpha_\omega \cdot 2)$ |
| 1,5,7,3,8,10,7, 15,17,14,19 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \alpha_\omega \cdot \alpha_2)$ |
| 1,5,7,3,8,10,7, 15,17,14,19,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + {\alpha_\omega}^2)$ |
| 1,5,7,3,8,10,7, 15,17,14,19,25 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \varepsilon_{\alpha_\omega+1})$ |
| 1,5,7,3,8,10,7, 15,17,14,20 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \Omega_{\alpha_\omega+1}+$ $\psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega) + \Omega_{\alpha_\omega+1} + 1))$ |
| 1,5,7,3,8,10,7, 15,17,14,20,14,20 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \Omega_{\alpha_\omega+1} \cdot 2+$ $\psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega) + \Omega_{\alpha_\omega+1} \cdot 2 + 1))$ |
| 1,5,7,3,8,10,7, 15,17,14,20,27 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \varepsilon_{\Omega_{\alpha_\omega+1}+1})$ |
| 1,5,7,3,8,10,7, 15,17,14,20,29 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_{\omega+1}}(\alpha_{\omega\cdot 2}))$ |
| 1,5,7,3,8,10,7, 15,17,14,20,30 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\alpha_{\omega+1}}(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,3,8,10,7, 15,17,14,20,30,32,27 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \alpha_{\omega+1})$ |
| 1,5,7,3,8,10,7, 15,17,14,20,30,32,29 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \alpha_{\omega\cdot 2})$ |
| 1,5,7,3,8,10,7, 15,17,14,21 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \alpha_{\omega^2})$ |
| 1,5,7,3,8,10,7, 15,17,14,21,26,15 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\beta(\alpha_{\beta+1} \cdot \beta))$ |
| 1,5,7,3,8,10,7, 15,17,14,21,26,21 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\beta(\alpha_{\beta+1} \cdot \beta \cdot \omega))$ |
| 1,5,7,3,8,10,7, 15,17,14,21,26,32 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\beta(\alpha_{\beta+1} \cdot \varepsilon_{\beta+1}))$ |
| 1,5,7,3,8,10,7, 15,17,14,21,27 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\beta(\alpha_{\beta+1} \cdot \Omega_{\beta+1})+$ $\psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\beta(\alpha_{\beta+1} \cdot \Omega_{\beta+1}) + 1))$ |
| 1,5,7,3,8,10,7, 15,17,14,21,27,34 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\beta(\alpha_{\beta+1} \cdot \varepsilon_{\Omega_{\beta+1}+1}))$ |
| 1,5,7,3,8,10,7, 15,17,14,21,27,37 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\psi_\beta(\alpha_{\beta+1}{}^2)}(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,3,8,10,7, 15,17,14,21,27,37,39 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_{\psi_\beta(\alpha_{\beta+1}{}^2)}(\psi_S(\sigma S \cdot \omega + \Omega)))$ |
| 1,5,7,3,8,10,7, 15,17,14,21,27,37,39, 20,29,38,45,30 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\beta({\alpha_{\beta+1}}^2))$ |
| 1,5,7,3,8,10,7, 15,17,14,21,28 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\beta({\alpha_{\beta+1}}^2 \cdot \omega))$ |
| 1,5,7,3,8,10,7, 15,17,14,22 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\beta(\beta_2))$ |
| 1,5,7,3,8,10,7, 15,17,14,25 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\beta(\beta_\omega))$ |
| 1,5,7,3,8,10,7, 15,17,14,26 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\beta(\psi_S(\sigma S \cdot \omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,7,3,8,10,7, 15,17,14,26,28,19,25 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \beta)$ |
| 1,5,7,3,8,10,7, 15,17,14,26,28,22 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \beta_2)$ |
| 1,5,7,3,8,10,7, 15,17,14,26,28,25 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \beta_\omega)$ |
| 1,5,7,3,8,10,7, 15,17,14,26,28,25,42 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_\gamma(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,3,8,10,8 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega))$ |
| 1,5,7,3,8,10,8, 7 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \Omega_\omega)$ |
| 1,5,7,3,8,10,8, 7,11 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \Omega_{\omega^2})$ |
| 1,5,7,3,8,10,8, 7,12 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \psi_\alpha(\alpha_2))$ |
| 1,5,7,3,8,10,8, 7,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \psi_\alpha(\alpha_\omega))$ |
| 1,5,7,3,8,10,8, 7,15 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \psi_\alpha(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,3,8,10,8, 7,15,17,15,10,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \alpha)$ |
| 1,5,7,3,8,10,8, 7,15,17,15,11 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) +$ $\psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + 1))$ |
| 1,5,7,3,8,10,8, 7,15,17,15,11,14,8 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) +$ $\psi_S(\sigma S \cdot \omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega) +$ $\psi_S(\sigma S \cdot \omega) + \alpha))$ |
| 1,5,7,3,8,10,8, 7,15,17,15,12 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \alpha_2)$ |
| 1,5,7,3,8,10,8, 7,15,17,15,13 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \Omega_{\alpha_2+1} +$ $\psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \Omega_{\alpha_2+1} + 1))$ |
| 1,5,7,3,8,10,8, 7,15,17,15,13,12 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \Omega_{\alpha_2+1} + \alpha_2)$ |
| 1,5,7,3,8,10,8, 7,15,17,15,13,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \Omega_{\alpha_2+1} \cdot \omega)$ |
| 1,5,7,3,8,10,8, 7,15,17,15,13,20 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \psi_{\alpha_3}(\alpha_3))$ |
| 1,5,7,3,8,10,8, 7,15,17,15,13,23,25, 23,20 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \alpha_3)$ |
| 1,5,7,3,8,10,8, 7,15,17,15,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \alpha_\omega)$ |
| 1,5,7,3,8,10,8, 7,15,17,15,14,19,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + {\alpha_\omega}^2)$ |
| 1,5,7,3,8,10,8, 7,15,17,15,14,20,30 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \psi_{\alpha_{\omega+1}}(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,3,8,10,8, 7,15,17,15,14,20,30, 32,30,27 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \alpha_{\omega+1})$ |
| 1,5,7,3,8,10,8, 7,15,17,15,14,20,30, 32,30,29 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \alpha_{\omega \cdot 2})$ |
| 1,5,7,3,8,10,8, 7,15,17,15,14,21 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega) + \psi_S(\sigma S \cdot \omega) + \alpha_{\omega^2})$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,5,7,3,8,10,8, 7,15,17,15,14,21,26, 15 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\psi_\beta(\alpha_{\beta+1}\cdot\beta))$ |
| 1,5,7,3,8,10,8, 7,15,17,15,14,22 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\psi_\beta(\beta_2))$ |
| 1,5,7,3,8,10,8, 7,15,17,15,14,26,28, 26 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+$ $\psi_\beta(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,3,8,10,8, 7,15,17,15,14,26,28, 26,19,25 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\beta)$ |
| 1,5,7,3,8,10,8, 7,15,17,15,14,26,28, 26,22 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\beta_2)$ |
| 1,5,7,3,8,10,8, 7,15,17,15,14,26,28, 26,25 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)+\beta_\omega)$ |
| 1,5,7,3,8,10,8, 8 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega)\cdot 2)$ |
| 1,5,7,3,8,10,8, 9 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)+\psi_S(\sigma S\cdot\omega+1))$ |
| 1,5,7,3,8,10,8, 10 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega)\cdot 2)$ |
| 1,5,7,3,8,10,9 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega+1))$ |
| 1,5,7,3,8,10,10 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega\cdot 2))$ |
| 1,5,7,3,8,10,13 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_1(\Omega_2)))$ |
| 1,5,7,3,8,10,15 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_1(\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,7,3,8,11 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2))$ |
| 1,5,7,3,8,11,6, 12,15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2)+\Omega_\omega)$ |
| 1,5,7,3,8,11,6, 12,15,11,20,23 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\Omega_2)))$ |
| 1,5,7,3,8,11,6, 12,15,11,20,23,17 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2)+\alpha_2)$ |
| 1,5,7,3,8,11,6, 12,15,11,20,23,19 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2)+\alpha_\omega)$ |
| 1,5,7,3,8,11,6, 12,15,11,20,23,19,32 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2)+\psi_\beta(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,3,8,11,6, 12,15,12 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2)+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,7,3,8,11,6, 12,15,12,14 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2)+\psi_S(\sigma S\cdot\omega+\Omega))$ |
| 1,5,7,3,8,11,6, 12,15,12,15 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2)\cdot 2)$ |
| 1,5,7,3,8,11,6, 12,15,13 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2+1))$ |
| 1,5,7,3,8,11,6, 12,15,15 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_2\cdot 2))$ |
| 1,5,7,3,8,11,6, 12,16 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_3))$ |
| 1,5,7,4 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega))$ |
| 1,5,7,4,3 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega)$ |
| 1,5,7,4,3,8 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_1(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,4,3,8,11 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega_2)))$ |
| 1,5,7,4,3,8,11, 6,12,15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega_2)+\Omega_\omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,7,4,3,8,11, 6,12,15,12 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega_2)+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,4,3,8,11, 6,12,15,13 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega_2+1)))$ |
| 1,5,7,4,3,8,11, 6,12,16 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega_3)))$ |
| 1,5,7,4,3,8,11, 7 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega_\omega)))$ |
| 1,5,7,4,3,8,11, 7,6 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_2)$ |
| 1,5,7,4,4 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_\omega)$ |
| 1,5,7,4,5 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_\omega\cdot\omega)$ |
| 1,5,7,4,6 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_\omega\cdot\Omega)$ |
| 1,5,7,4,6,3,8, 11,7 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_\omega\cdot\Omega+\psi_1(\psi_S(\sigma S\cdot\omega+\Omega_\omega)))$ |
| 1,5,7,4,6,3,8, 11,7,9,6 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_\omega\cdot\Omega+\Omega_2)$ |
| 1,5,7,4,6,3,8, 11,7,9,7 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_\omega\cdot(\Omega+1))$ |
| 1,5,7,4,6,3,8, 11,7,10 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_\omega\cdot\Omega_2)$ |
| 1,5,7,4,6,4 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+{\Omega_\omega}^2)$ |
| 1,5,7,4,6,4,6, 4 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+{\Omega_\omega}^2\cdot 2)$ |
| 1,5,7,4,6,6,4 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+{\Omega_\omega}^3)$ |
| 1,5,7,4,6,9 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\omega(\Omega_{\omega+1}))$ |
| 1,5,7,4,6,9,9 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\omega(\Omega_{\omega+1}\cdot 2))$ |
| 1,5,7,4,6,10 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\omega(\Omega_{\omega\cdot 2}))$ |
| 1,5,7,4,6,11 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,4,6,11,13 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega+\Omega)))$ |
| 1,5,7,4,6,11,13, 3,8,11,7,10,16,19 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega+\Omega_2)))$ |
| 1,5,7,4,6,11,13, 4 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega+\Omega_\omega)))$ |
| 1,5,7,4,6,11,13, 9 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega+1})$ |
| 1,5,7,4,6,11,13, 9,9 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega+1}\cdot 2)$ |
| 1,5,7,4,6,11,13, 9,13 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_{\omega+1}(\Omega_{\omega+2}))$ |
| 1,5,7,4,6,11,13, 9,15,17,13 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega+2})$ |
| 1,5,7,4,6,11,13, 10 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega\cdot 2})$ |
| 1,5,7,4,6,11,13, 10,13 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega\cdot 2}\cdot\Omega_{\omega+1})$ |
| 1,5,7,4,6,11,13, 10,13,10 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+{\Omega_{\omega\cdot 2}}^2)$ |
| 1,5,7,4,6,11,13, 10,13,17 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_{\omega\cdot 2}(\Omega_{\omega\cdot 2+1}))$ |
| 1,5,7,4,6,11,13, 10,13,19,21,4 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_{\omega\cdot 2}(\psi_S(\sigma S\cdot\omega+\Omega_\omega)))$ |
| 1,5,7,4,6,11,13, 10,13,19,21,17 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega\cdot 2+1})$ |
| 1,5,7,4,6,11,13, 10,13,19,21,18 | $\psi(\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega\cdot 3})$ |

| 0 − Y | Projection |
|---|---|
| 1,5,7,4,6,11,13, 10,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_\omega) + \Omega_{\omega^2})$ |
| 1,5,7,4,6,11,13, 10,14,17,7 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_\omega) + \psi_I(I))$ |
| 1,5,7,4,6,11,13, 10,15 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_\omega) + \psi_\alpha(\alpha_2))$ |
| 1,5,7,4,6,11,13, 10,18 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_\omega) + \psi_\alpha(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,4,6,11,13, 10,18,20,4 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_\omega) + \psi_\alpha(\psi_S(\sigma S \cdot \omega + \Omega_\omega)))$ |
| 1,5,7,4,6,11,13, 10,18,20,13,17 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_\omega) + \alpha)$ |
| 1,5,7,4,6,11,13, 10,18,20,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_\omega) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega_\omega) + 1))$ |
| 1,5,7,4,6,11,13, 10,18,20,15 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_\omega) + \alpha_2)$ |
| 1,5,7,4,6,11,13, 10,18,20,17 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_\omega) + \alpha_\omega)$ |
| 1,5,7,4,6,11,13, 11 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_\omega) + \psi_S(\sigma S \cdot \omega))$ |
| 1,5,7,4,6,11,13, 11,13,4 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_\omega) \cdot 2)$ |
| 1,5,7,4,6,11,13, 13,4 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_\omega \cdot 2))$ |
| 1,5,7,4,6,11,13, 16 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\omega(\Omega_{\omega+1})))$ |
| 1,5,7,4,6,11,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega+1}))$ |
| 1,5,7,4,6,11,14, 9,15,19 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega+2}))$ |
| 1,5,7,4,6,11,14, 10 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega \cdot 2}))$ |
| 1,5,7,4,6,11,14, 10,13,19,23,18 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega \cdot 3}))$ |
| 1,5,7,4,7 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega^2}))$ |
| 1,5,7,4,7,4 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega^2}) + \Omega_\omega)$ |
| 1,5,7,4,7,4,7 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega^2}) + \Omega_{\omega^2})$ |
| 1,5,7,4,7,6,9 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega^2}) + \psi_{\omega^2}(\Omega_{\omega^2+1}))$ |
| 1,5,7,4,7,6,11, 13,4,7 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega^2}) + \psi_{\omega^2}(\psi_S(\sigma S \cdot \omega + \Omega_{\omega^2})))$ |
| 1,5,7,4,7,6,11, 13,9 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega^2}) + \Omega_{\omega^2+1})$ |
| 1,5,7,4,7,6,11, 13,10 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega^2}) + \Omega_{\omega^2+\omega})$ |
| 1,5,7,4,7,6,11, 13,11 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega^2}) + \psi_S(\sigma S \cdot \omega))$ |
| 1,5,7,4,7,6,11, 13,13,4,7 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega^2} \cdot 2))$ |
| 1,5,7,4,7,6,11, 14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega^2+1}))$ |
| 1,5,7,4,7,6,11, 14,10,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega^2 \cdot 2}))$ |
| 1,5,7,4,7,7 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega^3}))$ |
| 1,5,7,4,7,8 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\omega^\omega}))$ |
| 1,5,7,4,7,9,5 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)))$ |
| 1,5,7,4,7,9,5, 5 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \psi_I(I) \cdot \omega)$ |

| 0 − Y | Projection |
|---|---|
| 1,5,7,4,7,9,6, 9 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \psi_{\psi_I(I+1)}(\psi_I(I+1)))$ |
| 1,5,7,4,7,9,6, 11 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \psi_{\psi_I(I+1)}(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,4,7,9,6, 11,13,4,7,9,5 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I))+$ $\psi_{\psi_I(I+1)}(\psi_S(\sigma S \cdot \omega + \psi_I(I))))$ |
| 1,5,7,4,7,9,6, 11,13,9 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \psi_I(I+1))$ |
| 1,5,7,4,7,9,6, 11,13,10 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \psi_I(I+\omega))$ |
| 1,5,7,4,7,9,6, 11,13,10,14,17,11 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \psi_I(I \cdot 2))$ |
| 1,5,7,4,7,9,6, 11,13,10,14,17,21 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \psi_I(\Omega_{I+1}))$ |
| 1,5,7,4,7,9,6, 11,13,10,14,17,23 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \psi_I(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,4,7,9,6, 11,13,10,14,17,23,25, 4,7,9,5 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \psi_I(\psi_S(\sigma S \cdot \omega + \psi_I(I))))$ |
| 1,5,7,4,7,9,6, 11,13,10,14,17,23,25, 13,17 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \psi_I(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + 1))$ |
| 1,5,7,4,7,9,6, 11,13,10,14,17,23,25, 13,18 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \psi_I(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \omega))$ |
| 1,5,7,4,7,9,6, 11,13,10,14,17,23,25, 13,18,23,27,19 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + I)$ |
| 1,5,7,4,7,9,6, 11,13,10,14,17,23,25, 13,18,23,28 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \psi_{\Omega_{I+1}}(I_\omega))$ |
| 1,5,7,4,7,9,6, 11,13,10,14,17,23,25, 14 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I))+$ $\psi_{\Omega_{I+1}}(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + 1))$ |
| 1,5,7,4,7,9,6, 11,13,10,14,17,23,25, 21 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \Omega_{I+1})$ |
| 1,5,7,4,7,9,6, 11,13,10,14,17,23,25, 22,27,31,23 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \psi_{I_2}(I_2))$ |
| 1,5,7,4,7,9,6, 11,13,10,14,18 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + I_\omega)$ |
| 1,5,7,4,7,9,6, 11,13,10,15 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \psi_\alpha(\alpha_2))$ |
| 1,5,7,4,7,9,6, 11,13,10,17 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \psi_\alpha(\alpha_\omega))$ |
| 1,5,7,4,7,9,6, 11,13,10,18,20,4,7, 9,5 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \psi_\alpha(\psi_S(\sigma S \cdot \omega + \psi_I(I))))$ |
| 1,5,7,4,7,9,6, 11,13,10,18,20,13,17 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \alpha)$ |
| 1,5,7,4,7,9,6, 11,13,10,18,20,15 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \alpha_2)$ |
| 1,5,7,4,7,9,6, 11,13,10,18,20,17 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_I(I)) + \alpha_\omega)$ |

| 0 − Y | Projection |
|---|---|
| 1,5,7,4,7,9,6, 11,13,10,18,20,17,29, 31,4,7,9,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_\beta(\psi_S(\sigma S\cdot\omega+\psi_I(I))))$ |
| 1,5,7,4,7,9,6, 11,13,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I))+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,7,4,7,9,6, 11,13,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I)+1))$ |
| 1,5,7,4,7,9,6, 11,13,16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{\psi_I(I+1)}(\psi_I(I+1))))$ |
| 1,5,7,4,7,9,6, 11,14 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I+1)))$ |
| 1,5,7,4,7,9,6, 11,14,9 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I+1))+\psi_I(I+1))$ |
| 1,5,7,4,7,9,6, 11,14,9,15,18,14 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I+1))+\psi_I(I+\omega))$ |
| 1,5,7,4,7,9,6, 11,14,9,15,18,15 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I+1))+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,7,4,7,9,6, 11,14,9,15,18,18 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I+1)\cdot 2))$ |
| 1,5,7,4,7,9,6, 11,14,9,15,19 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I+2)))$ |
| 1,5,7,4,7,9,6, 11,14,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I+\omega)))$ |
| 1,5,7,4,7,9,6, 11,14,10,14,17,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I\cdot 2)))$ |
| 1,5,7,4,7,9,7 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I\cdot\omega)))$ |
| 1,5,7,4,7,9,7, 7 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I\cdot\omega^2)))$ |
| 1,5,7,4,7,9,7, 9,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(I^2)))$ |
| 1,5,7,4,7,9,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(\Omega_{I+1})))$ |
| 1,5,7,4,7,9,14, 16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(\psi_S(\sigma S\cdot\omega+\Omega))))$ |
| 1,5,7,4,7,9,14, 16,4,7,9,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_I(\psi_S(\sigma S\cdot\omega+\psi_I(I)))))$ |
| 1,5,7,4,7,9,14, 16,5 | $\psi(\psi_S(\sigma S\cdot\omega+I))$ |
| 1,5,7,4,7,9,14, 16,5,5 | $\psi(\psi_S(\sigma S\cdot\omega+I)+\psi_I(\psi_S(\sigma S\cdot\omega+I))\cdot\omega)$ |
| 1,5,7,4,7,9,14, 16,6,9 | $\psi(\psi_S(\sigma S\cdot\omega+I)+\psi_I(\psi_S(\sigma S\cdot\omega+I)+1))$ |
| 1,5,7,4,7,9,14, 16,6,10,14,17,11 | $\psi(\psi_S(\sigma S\cdot\omega+I)+I)$ |
| 1,5,7,4,7,9,14, 16,6,11 | $\psi(\psi_S(\sigma S\cdot\omega+I)+\psi_{\Omega_{I+1}}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,4,7,9,14, 16,6,11,14 | $\psi(\psi_S(\sigma S\cdot\omega+I+\psi_I(\psi_S(\sigma S\cdot\omega+I)+1)))$ |
| 1,5,7,4,7,9,14, 16,6,11,14,10,14,17, 11 | $\psi(\psi_S(\sigma S\cdot\omega+I\cdot 2))$ |
| 1,5,7,4,7,9,14, 16,6,11,14,10,14,17, 11,11 | $\psi(\psi_S(\sigma S\cdot\omega+I\cdot 2)+\psi_I(\psi_S(\sigma S\cdot\omega+I)+I)\cdot\omega)$ |
| 1,5,7,4,7,9,14, 16,6,11,14,10,14,17, 23,27 | $\psi(\psi_S(\sigma S\cdot\omega+I\cdot 2+\psi_I(\psi_S(\sigma S\cdot\omega+I\cdot 2+1))))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,7,4,7,9,14, 16,7 | $\psi(\psi_S(\sigma S \cdot \omega + I \cdot \omega))$ |
| 1,5,7,4,7,9,14, 17 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{I+1}))$ |
| 1,5,7,4,7,9,14, 17,13 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{I+\omega}))$ |
| 1,5,7,4,7,9,14, 17,13,17,20,14 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{I_2}(I_2)))$ |
| 1,5,7,4,7,9,14, 17,13,17,20,26,29,14 | $\psi(\psi_S(\sigma S \cdot \omega + I_2))$ |
| 1,5,7,4,7,10 | $\psi(\psi_S(\sigma S \cdot \omega + I_\omega))$ |
| 1,5,7,4,8 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\alpha(\alpha_2)))$ |
| 1,5,7,4,8,7 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\alpha(\alpha_2 + \psi_{\alpha_2}(\alpha_2 + 1))))$ |
| 1,5,7,4,8,7,11 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\alpha(\alpha_2 + \psi_{\alpha_2}(\alpha_2 + \psi_{\alpha_2}(\alpha_2)))))$ |
| 1,5,7,4,8,8 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\alpha(\alpha_2 \cdot 2)))$ |
| 1,5,7,4,8,15 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\alpha(\Omega_{\alpha_2+1})) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,4,9 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\alpha(\Omega_{\alpha_2+1}))+$ $\psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \psi_{\alpha_2}(\Omega_{\alpha_2+1}) + 1)))$ |
| 1,5,7,4,9,10 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\alpha(\Omega_{\alpha_2+1} \cdot \omega)))$ |
| 1,5,7,4,10 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\alpha(\alpha_\omega)))$ |
| 1,5,7,4,11 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\alpha(\psi_S(\sigma S \cdot \omega))))$ |
| 1,5,7,4,11,13 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\alpha(\psi_S(\sigma S \cdot \omega + \Omega))))$ |
| 1,5,7,4,11,13,4, 8 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\alpha(\psi_S(\sigma S \cdot \omega + \psi_\alpha(\alpha_2)))))$ |
| 1,5,7,4,11,13,5 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha))$ |
| 1,5,7,4,11,13,5, 4,11 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha) + \psi_\alpha(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,4,11,13,5, 5 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha) + \psi_\alpha(\psi_S(\sigma S \cdot \omega + \alpha) + 1))$ |
| 1,5,7,4,11,13,6, 9 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha) + \alpha)$ |
| 1,5,7,4,11,13,6, 11 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,4,11,13,7 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \alpha) + 1))$ |
| 1,5,7,4,11,13,8 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha) + \alpha_2)$ |
| 1,5,7,4,11,13,10 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha) + \alpha_\omega)$ |
| 1,5,7,4,11,13,11 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha) + \psi_S(\sigma S \cdot \omega))$ |
| 1,5,7,4,11,13,12 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha + 1))$ |
| 1,5,7,4,11,13,13, 5 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha \cdot 2))$ |
| 1,5,7,4,11,13,16 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\alpha+1}))$ |
| 1,5,7,4,11,13,17 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\alpha+1}) + \Omega_{\alpha+1} \cdot \omega)$ |
| 1,5,7,4,11,14 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\alpha+1}) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega_{\alpha+1}) + 1))$ |
| 1,5,7,4,11,14,18 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{\alpha_2}(\alpha_2)))$ |
| 1,5,7,4,11,14,21 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega))))$ |
| 1,5,7,4,11,15 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_2))$ |
| 1,5,7,4,11,15,8 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_2) + \alpha_2)$ |
| 1,5,7,4,11,15,9 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_2) + \Omega_{\alpha_2+1}+$ $\psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \alpha_2) + \Omega_{\alpha_2+1} + 1))$ |
| 1,5,7,4,11,15,9, 17 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_2) + \psi_{\alpha_3}(\alpha_\omega))$ |
| 1,5,7,4,11,15,9, 18,22,17 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_2) + \alpha_\omega)$ |
| 1,5,7,4,11,15,9, 18,22,18 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_2) + \psi_S(\sigma S \cdot \omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,7,4,11,15,9, 18,22,22 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_2 \cdot 2))$ |
| 1,5,7,4,11,15,9, 18,24 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_3))$ |
| 1,5,7,4,11,15,10 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\omega))$ |
| 1,5,7,4,11,15,10, 14,10 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\omega) + {\alpha_\omega}^2)$ |
| 1,5,7,4,11,15,10, 14,19 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\omega) + \Omega_{\alpha_\omega+1})$ |
| 1,5,7,4,11,15,10, 15 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\omega) + \Omega_{\alpha_\omega+1} +$ $\psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \alpha_\omega) + \Omega_{\alpha_\omega+1} + 1))$ |
| 1,5,7,4,11,15,10, 15,11 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\omega) + \Omega_{\alpha_\omega+1} \cdot \omega)$ |
| 1,5,7,4,11,15,10, 15,21 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\omega) + \psi_{\alpha_{\omega+1}}(\alpha_{\omega+1}))$ |
| 1,5,7,4,11,15,10, 15,24 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\omega) + \psi_{\alpha_{\omega+1}}(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,4,11,15,10, 15,24,28 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\omega) + \psi_{\alpha_{\omega+1}}(\psi_S(\sigma S \cdot \omega + \alpha_2)))$ |
| 1,5,7,4,11,15,10, 15,24,28,10 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\omega) + \psi_{\alpha_{\omega+1}}(\psi_S(\sigma S \cdot \omega + \alpha_\omega)))$ |
| 1,5,7,4,11,15,10, 15,24,28,21 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\omega) + \alpha_{\omega+1})$ |
| 1,5,7,4,11,15,10, 15,24,28,23 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\omega) + \alpha_{\omega \cdot 2})$ |
| 1,5,7,4,11,15,10, 15,24,28,23,32 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\omega) + \psi_\beta(\beta_2))$ |
| 1,5,7,4,11,15,10, 15,24,28,23,36 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\omega) + \psi_\beta(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,4,11,15,10, 15,24,28,24 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\omega) + \psi_S(\sigma S \cdot \omega))$ |
| 1,5,7,4,11,15,10, 15,24,28,28,10 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\omega \cdot 2))$ |
| 1,5,7,4,11,15,10, 15,24,28,35 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\alpha_\omega+1}) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,4,11,15,10, 15,24,29 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\alpha_\omega+1}) +$ $\psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega_{\alpha_\omega+1}) + 1))$ |
| 1,5,7,4,11,15,10, 15,24,29,35 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{\alpha_{\omega+1}}(\alpha_{\omega+1})))$ |
| 1,5,7,4,11,15,10, 15,24,30 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\omega+1}))$ |
| 1,5,7,4,11,15,10, 15,24,30,23 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\omega \cdot 2}))$ |
| 1,5,7,4,11,15,10, 16 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\omega^2}))$ |
| 1,5,7,4,11,15,10, 16,15,24,28,28,10,16 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\omega^2} \cdot 2))$ |
| 1,5,7,4,11,15,10, 16,15,24,30 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\omega^2+1}))$ |
| 1,5,7,4,11,15,10, 16,15,24,30,23,31 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\omega^2 \cdot 2}))$ |
| 1,5,7,4,11,15,10, 16,18,5 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_\alpha))$ |
| 1,5,7,4,11,15,10, 16,20,11 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\beta(\alpha_{\beta+1} \cdot \beta)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,7,4,11,15,10, 17 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\beta(\beta_2)))$ |
| 1,5,7,4,11,15,10, 21 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\beta(\psi_S(\sigma S \cdot \omega))))$ |
| 1,5,7,4,11,15,10, 21,25,10,21 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_\beta(\psi_S(\sigma S \cdot \omega + \psi_\beta(\psi_S(\sigma S \cdot \omega))))))$ |
| 1,5,7,4,11,15,10, 21,25,11 | $\psi(\psi_S(\sigma S \cdot \omega + \beta))$ |
| 1,5,7,4,11,15,10, 21,25,21 | $\psi(\psi_S(\sigma S \cdot \omega + \beta) + \psi_S(\sigma S \cdot \omega))$ |
| 1,5,7,4,11,15,10, 21,25,22 | $\psi(\psi_S(\sigma S \cdot \omega + \beta + 1))$ |
| 1,5,7,4,11,15,10, 21,25,25,11 | $\psi(\psi_S(\sigma S \cdot \omega + \beta \cdot 2))$ |
| 1,5,7,4,11,15,10, 21,25,30 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\beta+1}))$ |
| 1,5,7,4,11,15,10, 21,25,32 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\beta+1}) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,4,11,15,10, 21,26 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\beta+1}) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \Omega_{\beta+1}) + 1))$ |
| 1,5,7,4,11,15,10, 21,26,8 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\beta+1}) + \alpha_2)$ |
| 1,5,7,4,11,15,10, 21,26,16 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\beta+1}) + \psi_{\beta_2}(\psi_S(\sigma S \cdot \omega + \Omega_{\beta+1}) + 1))$ |
| 1,5,7,4,11,15,10, 21,26,21 | $\psi(\psi_S(\sigma S \cdot \omega + \Omega_{\beta+1}) + \psi_S(\sigma S \cdot \omega))$ |
| 1,5,7,4,11,15,10, 21,26,32 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\beta+1}))$ |
| 1,5,7,4,11,15,10, 21,26,33 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\beta+1}) + \psi_\beta(\psi_S(\sigma S \cdot \omega + \alpha_{\beta+1}) +$ $\Omega_{\beta+1}) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \alpha_{\beta+1}) +$ $\psi_\beta(\psi_S(\sigma S \cdot \omega + \alpha_{\beta+1}) + \Omega_{\beta+1}) + 1))$ |
| 1,5,7,4,11,15,10, 21,26,34 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\beta+1}) + \alpha_{\beta+1} \cdot \omega)$ |
| 1,5,7,4,11,15,10, 21,26,35 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\beta+1}) + \psi_{\beta_2}(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,4,11,15,10, 21,26,35,21,25,11 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\beta+1}) + \psi_{\beta_2}(\psi_S(\sigma S \cdot \omega)) +$ $\psi_{\psi_\beta(\psi_S(\sigma S \cdot \omega + \alpha_{\beta+1}))}(\psi_\beta(\psi_S(\sigma S \cdot \omega + \alpha_{\beta+1}))))$ |
| 1,5,7,4,11,15,10, 21,27 | $\psi(\psi_S(\sigma S \cdot \omega + \alpha_{\beta+1}) + \psi_{\beta_2}(\psi_S(\sigma S \cdot \omega + \alpha_{\beta+1}) + 1))$ |
| 1,5,7,4,11,15,10, 21,27,34 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{\beta_2}(\beta_2)))$ |
| 1,5,7,4,11,15,10, 21,28 | $\psi(\psi_S(\sigma S \cdot \omega + \beta_2))$ |
| 1,5,7,4,11,15,10, 21,28,20 | $\psi(\psi_S(\sigma S \cdot \omega + \beta_\omega))$ |
| 1,5,7,5 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_S(\sigma S \cdot \omega)))$ |
| 1,5,7,5,3 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_S(\sigma S \cdot \omega)) + \Omega)$ |
| 1,5,7,5,3,8,11 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_S(\sigma S \cdot \omega)) + \psi_1(\psi_S(\sigma S \cdot \omega + \Omega_2)))$ |
| 1,5,7,5,3,8,11, 7 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_S(\sigma S \cdot \omega)) + \psi_1(\psi_S(\sigma S \cdot \omega + \Omega_\omega)))$ |
| 1,5,7,5,3,8,11, 7,15,20 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_S(\sigma S \cdot \omega)) + \psi_1(\psi_S(\sigma S \cdot \omega + \alpha_2)))$ |
| 1,5,7,5,3,8,11, 7,15,20,14 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_S(\sigma S \cdot \omega)) + \psi_1(\psi_S(\sigma S \cdot \omega + \alpha_\omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,7,5,3,8,11, 8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+$ $\psi_1(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,7,5,3,8,11, 8,6 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\Omega_2)$ |
| 1,5,7,5,4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\Omega_\omega)$ |
| 1,5,7,5,4,7,9, 5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_I(I))$ |
| 1,5,7,5,4,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_\alpha(\alpha_2))$ |
| 1,5,7,5,4,11,15, 11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+$ $\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,7,5,4,11,15, 11,6,9 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\alpha)$ |
| 1,5,7,5,4,11,15, 11,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\alpha_2)$ |
| 1,5,7,5,4,11,15, 11,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\alpha_\omega)$ |
| 1,5,7,5,7 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega))$ |
| 1,5,7,5,7,3,8, 11,8,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_2))$ |
| 1,5,7,5,7,4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_\omega))$ |
| 1,5,7,5,7,4,6, 11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_\omega(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,5,7,4,6, 11,14,11,13,4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_\omega)+$ $\psi_\omega(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_\omega)))$ |
| 1,5,7,5,7,4,6, 11,14,11,13,9 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\Omega_{\omega+1})$ |
| 1,5,7,5,7,4,6, 11,14,11,13,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\Omega_\omega)+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,7,5,7,4,6, 11,14,11,13,13,4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_\omega\cdot 2))$ |
| 1,5,7,5,7,4,6, 11,14,11,14 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_{\omega+1}))$ |
| 1,5,7,5,7,4,6, 11,14,11,14,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_{\omega\cdot 2}))$ |
| 1,5,7,5,7,4,7 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_{\omega^2}))$ |
| 1,5,7,5,7,4,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_\alpha(\alpha_2)))$ |
| 1,5,7,5,7,4,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_\alpha(\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,7,5,7,4,11, 15,11,13,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\alpha))$ |
| 1,5,7,5,7,4,11, 15,11,13,16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_{\alpha+1}))$ |
| 1,5,7,5,7,4,11, 15,11,14 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega_{\alpha+1})+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\Omega_{\alpha+1})+1))$ |
| 1,5,7,5,7,4,11, 15,11,14,18 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{\alpha_2}(\alpha_2)))$ |
| 1,5,7,5,7,4,11, 15,11,15 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\alpha_2))$ |
| 1,5,7,5,7,4,11, 15,11,15,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\alpha_\omega))$ |
| 1,5,7,5,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))\cdot 2)$ |
| 1,5,7,5,7,5,7, 5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))\cdot 3)$ |

| 0 − Y | Projection |
|---|---|
| 1,5,7,6 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+1))$ |
| 1,5,7,7 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega))$ |
| 1,5,7,7,4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_\omega))$ |
| 1,5,7,7,4,6,11, 14,13,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_\omega)+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,7,7,4,6,11, 14,13,11,14,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+$ $\Omega_\omega)+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,7,4,6,11, 14,13,13,4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_\omega\cdot 2))$ |
| 1,5,7,7,4,6,11, 14,14 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_{\omega+1}))$ |
| 1,5,7,7,4,7 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_{\omega^2}))$ |
| 1,5,7,7,4,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\alpha_2)))$ |
| 1,5,7,7,4,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,7,7,4,11,15, 13 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+$ $\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega))))$ |
| 1,5,7,7,4,11,15, 13,4,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+$ $\psi_S(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)))))))$ |
| 1,5,7,7,4,11,15, 13,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha))$ |
| 1,5,7,7,4,11,15, 13,6,9 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha)+\alpha)$ |
| 1,5,7,7,4,11,15, 13,7 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha)+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha)+1))$ |
| 1,5,7,7,4,11,15, 13,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha)+\alpha_2)$ |
| 1,5,7,7,4,11,15, 13,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha)+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,7,7,4,11,15, 13,13,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha\cdot 2))$ |
| 1,5,7,7,4,11,15, 13,16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha+1}))$ |
| 1,5,7,7,4,11,15, 14 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha+1})+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha+1})+1))$ |
| 1,5,7,7,4,11,15, 14,13,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_{\alpha+1}+\alpha))$ |
| 1,5,7,7,4,11,15, 14,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\psi_{\alpha_2}(\alpha_\omega)))$ |
| 1,5,7,7,4,11,15, 15 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha_2))$ |
| 1,5,7,7,4,11,15, 15,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha_\omega))$ |
| 1,5,7,7,4,11,15, 15,10,16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\alpha_{\omega^2}))$ |
| 1,5,7,7,4,11,15, 15,10,21 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\psi_\beta(\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,7,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2))$ |
| 1,5,7,7,5,4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)+\Omega_\omega)$ |
| 1,5,7,7,5,4,11, 15,15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)+$ $\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)))$ |
| 1,5,7,7,5,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,7,7,5,7,4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)+\psi_S(\sigma S\cdot\omega+\Omega_\omega))$ |
| 1,5,7,7,5,7,4, 11,15,15,11,13,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)+\psi_S(\sigma S\cdot\omega+\alpha))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,7,7,5,7,4, 11,15,15,11,15,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)+\psi_S(\sigma S\cdot\omega+\alpha_\omega))$ |
| 1,5,7,7,5,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)+$ $\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,7,5,7,7, 4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)+$ $\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)+\Omega_\omega))$ |
| 1,5,7,7,5,7,7, 5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)\cdot 2)$ |
| 1,5,7,7,6 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2+1))$ |
| 1,5,7,7,7,4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2+\Omega_\omega))$ |
| 1,5,7,7,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 3))$ |
| 1,5,7,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega+1)))$ |
| 1,5,7,9,4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega+\Omega_\omega)))$ |
| 1,5,7,9,4,11,15, 17,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega+\alpha)))$ |
| 1,5,7,9,4,11,15, 17,15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega+\alpha)+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,9,4,11,15, 17,16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega+\alpha+1)))$ |
| 1,5,7,9,4,11,15, 19 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega+\alpha_2)))$ |
| 1,5,7,9,4,11,15, 19,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega+\alpha_\omega)))$ |
| 1,5,7,9,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,7,10 | $\psi(\psi_S(\sigma S\cdot\omega+S))$ |
| 1,5,7,10,5 | $\psi(\psi_S(\sigma S\cdot\omega+S)+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,7,10,5,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+S)+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,10,5,7,10 | $\psi(\psi_S(\sigma S\cdot\omega+S)\cdot 2)$ |
| 1,5,7,10,7 | $\psi(\psi_S(\sigma S\cdot\omega+S+\Omega))$ |
| 1,5,7,10,7,4 | $\psi(\psi_S(\sigma S\cdot\omega+S+\Omega_\omega))$ |
| 1,5,7,10,7,4,11, 15,20,13,5 | $\psi(\psi_S(\sigma S\cdot\omega+S+\alpha))$ |
| 1,5,7,10,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+S+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,10,7,10 | $\psi(\psi_S(\sigma S\cdot\omega+S+\psi_S(\sigma S\cdot\omega+S)))$ |
| 1,5,7,10,9 | $\psi(\psi_S(\sigma S\cdot\omega+S+\psi_S(\sigma S\cdot\omega+S+\Omega)))$ |
| 1,5,7,10,9,5 | $\psi(\psi_S(\sigma S\cdot\omega+S+\psi_S(\sigma S\cdot\omega+S+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,7,10,9,12 | $\psi(\psi_S(\sigma S\cdot\omega+S+\psi_S(\sigma S\cdot\omega+S+\psi_S(\sigma S\cdot\omega+S))))$ |
| 1,5,7,10,10 | $\psi(\psi_S(\sigma S\cdot\omega+S\cdot 2))$ |
| 1,5,7,10,12,5 | $\psi(\psi_S(\sigma S\cdot\omega+S\cdot\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,7,10,13 | $\psi(\psi_S(\sigma S\cdot\omega+S^2))$ |
| 1,5,7,10,14 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(S_2)))$ |
| 1,5,7,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\Omega_{\alpha+1}\cdot\omega)$ |
| 1,5,7,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |
| 1,5,8,3 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+\Omega)$ |
| 1,5,8,3,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+\psi_1(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,3,8,12,6 | $\psi(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+\Omega_2)$ |
| 1,5,8,4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+\Omega_\omega)$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,4,6,4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+{\Omega_\omega}^2)$ |
| 1,5,8,4,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+\psi_\alpha(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+1)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)))$ |
| 1,5,8,4,11,16,6, 9 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+\alpha)$ |
| 1,5,8,4,11,16,6, 10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+\Omega_{\alpha+1}\cdot\omega)$ |
| 1,5,8,4,11,16,7 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+2))$ |
| 1,5,8,4,11,16,7, 9,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha))$ |
| 1,5,8,4,11,16,7, 11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\alpha_2)))$ |
| 1,5,8,4,11,16,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_2)$ |
| 1,5,8,4,11,16,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_\omega)$ |
| 1,5,8,4,11,16,10, 21 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_\beta(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,4,11,16,11, 15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16,11, 15,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+S))$ |
| 1,5,8,4,11,16,11, 16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))\cdot 2+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))\cdot 2+1))$ |
| 1,5,8,4,11,16,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+1))$ |
| 1,5,8,4,11,16,13, 5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha))$ |
| 1,5,8,4,11,16,13, 16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\Omega_{\alpha+1}))$ |
| 1,5,8,4,11,16,14 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\Omega_{\alpha+1})+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\Omega_{\alpha+1})+1))$ |
| 1,5,8,4,11,16,14, 18 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_{\alpha_2}(\alpha_2)))$ |
| 1,5,8,4,11,16,15 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha_2))$ |
| 1,5,8,4,11,16,15, 10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha_\omega))$ |
| 1,5,8,4,11,16,15, 11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16,15, 11,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,4,11,16,15, 11,15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16,15, 11,15,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S))$ |
| 1,5,8,4,11,16,15, 11,15,22 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16,15, 11,15,22,28 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+1)+\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}$ $(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha_2)))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+1)+\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}$ $(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha_\omega)))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,10,21 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+$ $\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_\beta(\psi_S(\sigma S\cdot\omega))))))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+$ $\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,11,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+$ $\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,11,15,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+$ $\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S)))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+1)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,21 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+1)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\Omega_{\alpha+1}\cdot\omega)))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,26,32 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)+\alpha)$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,27 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+2))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,27,37,45 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,28 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_2))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,31,38,31 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,31,38,48 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))\cdot 2)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,31,39 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))\cdot 2+1))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,31,39,27,37 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))\cdot 2+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,31,39,28 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))\cdot 2+\alpha_2))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,31,39,30,44,55 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))\cdot 2+$ $\psi_\beta(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+$ $\psi_{\alpha_2}(\psi_\beta(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1)))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,31,39,30,44,55,28 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))\cdot 2+\psi_\beta(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\alpha_2))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,31,39,30,44,55,30 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))\cdot 2+\psi_\beta(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\alpha_\omega))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,31,39, | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))\cdot 2+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,31,39,31,38,45 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))\cdot 3)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,31,39,31,39 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))\cdot 3+1))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,31,39,35,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}$ $(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha_\omega))))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 35,31,39,35,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 36 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 36,31 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 36,31,39 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 36,31,39,35,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))))))))))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 36,31,39,36 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))+$ $\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 36,35,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega)))))))))))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 36,36 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))\cdot 2))))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 36,40,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_\omega)))))))))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 36,40,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega)))))))))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 36,42 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\alpha)))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 37 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\Omega_{\alpha+1}+1)))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 37,47 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 37,47,55 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))+1))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 38 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\alpha_2)))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 38,30 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\alpha_\omega)))$ |
| 1,5,8,4,11,16,15, 11,15,22,28,21,31,39, 38,30 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,4,11,16,15, 11,16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |
| 1,5,8,4,11,16,15, 11,16,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_2)$ |
| 1,5,8,4,11,16,15, 11,16,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_\omega)$ |
| 1,5,8,4,11,16,15, 11,16,10,21 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_\beta(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16,15, 11,16,10,21,29,28,21 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_\beta(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,4,11,16,15, 11,16,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,4,11,16,15, 11,16,11,13,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+\alpha))$ |
| 1,5,8,4,11,16,15, 11,16,11,15 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+\alpha_2))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,4,11,16,15, 11,16,11,15,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+\alpha_\omega))$ |
| 1,5,8,4,11,16,15, 11,16,11,15,10,21,29, 28,21,29,21,28,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+\beta_\omega))$ |
| 1,5,8,4,11,16,15, 11,16,11,15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16,15, 11,16,11,15,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+S))$ |
| 1,5,8,4,11,16,15, 11,16,11,16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))\cdot 2+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |
| 1,5,8,4,11,16,15, 11,16,11,16,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))\cdot 2+\alpha_2)$ |
| 1,5,8,4,11,16,15, 11,16,11,16,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))\cdot 2+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,4,11,16,15, 11,16,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+1))$ |
| 1,5,8,4,11,16,15, 11,16,15 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha_2))$ |
| 1,5,8,4,11,16,15, 11,16,15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))\cdot 2)$ |
| 1,5,8,4,11,16,15, 12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)+1))$ |
| 1,5,8,4,11,16,15, 15,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)+\alpha_\omega))$ |
| 1,5,8,4,11,16,15, 15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)\cdot 2))$ |
| 1,5,8,4,11,16,15, 16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+1)))$ |
| 1,5,8,4,11,16,15, 20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+S)))$ |
| 1,5,8,4,11,16,16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |
| 1,5,8,4,11,16,16, 8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\alpha_2)$ |
| 1,5,8,4,11,16,16, 12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |
| 1,5,8,4,11,16,17 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+1)))$ |
| 1,5,8,4,11,16,18, 5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\alpha)))$ |
| 1,5,8,4,11,16,19 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\Omega_{\alpha+1}))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\Omega_{\alpha+1}))+1))$ |
| 1,5,8,4,11,16,19, 26 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,4,11,16,20 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + \alpha_2)))$ |
| 1,5,8,4,11,16,20, 11 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega))))$ |
| 1,5,8,4,11,16,20, 12 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega)) + 1))$ |
| 1,5,8,4,11,16,20, 16,20,11 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega)) \cdot 2))$ |
| 1,5,8,4,11,16,20, 20,11 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega) \cdot 2)))$ |
| 1,5,8,4,11,16,20, 25 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega + S))))$ |
| 1,5,8,4,11,16,21, 8 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S)))) + \alpha_2)$ |
| 1,5,8,4,11,16,22 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + S))$ |
| 1,5,8,4,11,16,22, 11 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + S) + \psi_S(\sigma S \cdot \omega))$ |
| 1,5,8,4,11,16,22, 11,15,20 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + S) + \psi_S(\sigma S \cdot \omega + S))$ |
| 1,5,8,4,11,16,22, 11,16 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + S) + \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S)) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + S) + \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S)) + 1))$ |
| 1,5,8,4,11,16,22, 11,16,22 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + S) \cdot 2)$ |
| 1,5,8,4,11,16,22, 15,11 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + S + \psi_S(\sigma S \cdot \omega)))$ |
| 1,5,8,4,11,16,22, 16,22 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + S + \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + S)))$ |
| 1,5,8,4,11,16,22, 22 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + S \cdot 2))$ |
| 1,5,8,4,11,16,22, 26,11 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + S \cdot \psi_S(\sigma S \cdot \omega)))$ |
| 1,5,8,4,11,16,22, 27 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + S \cdot \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S))) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + S \cdot \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S))) + 1))$ |
| 1,5,8,4,11,16,22, 28 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) + S^2))$ |
| 1,5,8,4,11,16,22, 29 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) \cdot 2))$ |
| 1,5,8,4,11,16,22, 31 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) \cdot 2) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,8,4,11,16,23 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) \cdot 2) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) \cdot 2) + 1))$ |
| 1,5,8,4,11,16,23, 8 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) \cdot 2) + \alpha_2)$ |
| 1,5,8,4,11,16,23, 10 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) \cdot 2) + \alpha_\omega)$ |
| 1,5,8,4,11,16,23, 10,21 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) \cdot 2) + \psi_\beta(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,8,4,11,16,23, 10,21,29,38,48 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) \cdot 2) + \psi_\beta(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S) \cdot 2)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,4,11,16,23, 10,21,29,39,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+ \psi_\beta(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+1))$ |
| 1,5,8,4,11,16,23, 10,21,29,39,14,19 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+\beta)$ |
| 1,5,8,4,11,16,23, 10,21,29,39,17 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+\beta_2)$ |
| 1,5,8,4,11,16,23, 10,21,29,39,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+\beta_\omega)$ |
| 1,5,8,4,11,16,23, 11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,4,11,16,23, 11,15,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+\psi_S(\sigma S\cdot\omega+S))$ |
| 1,5,8,4,11,16,23, 11,16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+\psi_S(\sigma S\cdot\omega+ \psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+ \psi_{S_2}(\sigma S)\cdot 2)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |
| 1,5,8,4,11,16,23, 11,16,15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+ \psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16,23, 11,16,15,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+ \psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega+S)))$ |
| 1,5,8,4,11,16,23, 11,16,22,29 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+ \psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2))$ |
| 1,5,8,4,11,16,23, 11,16,23 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)\cdot 2+ \psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)\cdot 2+1))$ |
| 1,5,8,4,11,16,23, 15,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\alpha_\omega))$ |
| 1,5,8,4,11,16,23, 15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16,23, 15,11,16,23 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+ \psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+ \psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega))+ \psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)+1))$ |
| 1,5,8,4,11,16,23, 15,15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega)\cdot 2))$ |
| 1,5,8,4,11,16,23, 15,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega+S)))$ |
| 1,5,8,4,11,16,23, 16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega+ \psi_{S_2}(\sigma S)))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+ \psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |
| 1,5,8,4,11,16,23, 16,15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+ \psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16,23, 16,22 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+ \psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S)))$ |
| 1,5,8,4,11,16,23, 16,23 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+ \psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+ \psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2))+1))$ |
| 1,5,8,4,11,16,23, 17 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+ \psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+1)))$ |
| 1,5,8,4,11,16,23, 20,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+ \psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega))))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,4,11,16,23, 21,28 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega+ \psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)))+ \psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+ \psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+\psi_S(\sigma S\cdot\omega+ \psi_{S_2}(\sigma S)\cdot 2)))+1))$ |
| 1,5,8,4,11,16,23, 22 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2+S))$ |
| 1,5,8,4,11,16,23, 22,29 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 3))$ |
| 1,5,8,4,11,16,23, 23 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 3)+ \psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 3)+1))$ |
| 1,5,8,4,11,16,23, 24 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+1)))$ |
| 1,5,8,4,11,16,23, 27,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,4,11,16,23, 27,32 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_S(\sigma S\cdot\omega+S))))$ |
| 1,5,8,4,11,16,23, 28 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_S(\sigma S\cdot\omega+ \psi_{S_2}(\sigma S))))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+ \psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))+1))$ |
| 1,5,8,4,11,16,23, 28,35 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_S(\sigma S\cdot\omega+ \psi_{S_2}(\sigma S)\cdot 2)))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+ \psi_{S_2}(\sigma S+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)\cdot 2)))+1))$ |
| 1,5,8,4,11,16,23, 29 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S)))$ |
| 1,5,8,4,11,16,23, 29,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S)+1))$ |
| 1,5,8,4,11,16,23, 29,15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S)+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16,23, 29,16,23,29 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S)+ \psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S))))$ |
| 1,5,8,4,11,16,23, 29,22 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S)+S))$ |
| 1,5,8,4,11,16,23, 29,23 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S)+\psi_{S_2}(\sigma S))+ \psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S)+\psi_{S_2}(\sigma S))+1))$ |
| 1,5,8,4,11,16,23, 29,23,29 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S)\cdot 2))$ |
| 1,5,8,4,11,16,23, 29,29 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S\cdot 2)))$ |
| 1,5,8,4,11,16,23, 29,36 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))))$ |
| 1,5,8,4,11,16,23, 30 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S)))+ \psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S)))+1))$ |
| 1,5,8,4,11,16,23, 30,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S)))+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,4,11,16,23, 30,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))+1))$ |
| 1,5,8,4,11,16,23, 30,15,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))+ \psi_S(\sigma S\cdot\omega+S)))$ |
| 1,5,8,4,11,16,23, 30,16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))+ \psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+ \psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,4,11,16,23, 30,20,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16,23, 30,22 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))+S))$ |
| 1,5,8,4,11,16,23, 30,23 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))+$ $\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S))+\psi_{S_2}(\sigma S))+1))$ |
| 1,5,8,4,11,16,23, 30,24 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S)+1)))$ |
| 1,5,8,4,11,16,23, 30,27,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,4,11,16,23, 30,29 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S)+S)))$ |
| 1,5,8,4,11,16,23, 30,31 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_2}(\sigma S+1))))$ |
| 1,5,8,4,11,16,23, 31 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_2)))$ |
| 1,5,8,4,11,16,23, 31,22 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_2)+S))$ |
| 1,5,8,4,11,16,23, 31,23,31 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_2)\cdot 2))$ |
| 1,5,8,4,11,16,23, 31,30,38 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_2+\psi_{S_2}(\sigma S+S_2))))$ |
| 1,5,8,4,11,16,23, 31,31 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_2\cdot 2)))$ |
| 1,5,8,4,11,16,23, 31,40 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_3}(\sigma S))))$ |
| 1,5,8,4,11,16,23, 32 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_3}(\sigma S)))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_3}(\sigma S)))+1))$ |
| 1,5,8,4,11,16,23, 32,22 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_3}(\sigma S))+S))$ |
| 1,5,8,4,11,16,23, 32,29 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_3}(\sigma S)+S)))$ |
| 1,5,8,4,11,16,23, 32,31 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_3}(\sigma S)+S_2)))$ |
| 1,5,8,4,11,16,23, 32,32 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+$ $\psi_{S_3}(\sigma S)\cdot 2))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+$ $\psi_{S_3}(\sigma S)\cdot 2))+1))$ |
| 1,5,8,4,11,16,23, 32,40 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+\psi_{S_3}(\sigma S+S_2))))$ |
| 1,5,8,4,11,16,24 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega)))$ $=\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\alpha_{\beta+1}\cdot\omega)$ |
| 1,5,8,4,11,16,24, 12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega)+1))$ |
| 1,5,8,4,11,16,24, 20,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega)+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,16,24, 22 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega)+S))$ |
| 1,5,8,4,11,16,24, 23,33 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega)\cdot 2))$ |
| 1,5,8,4,11,16,24, 23,33,29 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega+S)))$ |
| 1,5,8,4,11,16,24, 23,33,31 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega+S_2)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,4,11,16,24, 24 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S + S_\omega \cdot 2)))$ |
| 1,5,8,4,11,16,24, 28,11 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S + S_\omega \cdot \psi_S(\sigma S \cdot \omega))))$ |
| 1,5,8,4,11,16,24, 28,12 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S + S_\omega \cdot \psi_S(\sigma S \cdot \omega)) + 1))$ |
| 1,5,8,4,11,16,24, 28,33 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S + S_\omega \cdot \psi_S(\sigma S \cdot \omega + S))))$ |
| 1,5,8,4,11,16,24, 30 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S + S_\omega \cdot S)))$ |
| 1,5,8,4,11,16,24, 30,22 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S + S_\omega \cdot S) + S))$ |
| 1,5,8,4,11,16,24, 30,23,33 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S + S_\omega \cdot S) + \psi_{S_2}(\sigma S + S_\omega)))$ |
| 1,5,8,4,11,16,24, 30,23,33,39 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S + S_\omega \cdot S) \cdot 2))$ |
| 1,5,8,4,11,16,24, 30,23,33,39,33 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S + S_\omega \cdot (S + 1))))$ |
| 1,5,8,4,11,16,24, 30,23,33,40,50 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S + S_\omega \cdot \psi_{S_2}(\sigma S + S_\omega))))$ |
| 1,5,8,4,11,16,24, 30,23,33,41 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S + S_\omega \cdot S_2)))$ |
| 1,5,8,4,11,16,24, 30,24 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S + {S_\omega}^2)))$ |
| 1,5,8,4,11,16,24, 30,37 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S + \psi_{S_{\omega+1}}(\sigma S))))$ |
| 1,5,8,4,11,16,24, 31 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S + \psi_{S_{\omega+1}}(\sigma S)))+$ $\psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S+$ $\psi_{S_{\omega+1}}(\sigma S))) + 1))$ |
| 1,5,8,4,11,16,24, 31,41 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S + S_{\omega \cdot 2})))$ |
| 1,5,8,4,11,16,24, 32,38,25 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot 2)) + \alpha_{\beta+1} \cdot \beta)$ |
| 1,5,8,4,11,16,24, 32,38,32 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot 2)) + \alpha_{\beta+1} \cdot \beta \cdot \omega)$ |
| 1,5,8,4,11,16,24, 32,38,45 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot 2)) + \alpha_{\beta+1} \cdot \Omega_{\beta+1})$ |
| 1,5,8,4,11,16,24, 32,39,49 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot 2)) + {\alpha_{\beta+1}}^2 + \alpha_{\beta+1} \cdot \omega)$ |
| 1,5,8,4,11,16,24, 32,40 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot 2)) + {\alpha_{\beta+1}}^2 \cdot \omega)$ |
| 1,5,8,4,11,16,24, 33 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot 2)) + \psi_{\beta_2}(\beta_2))$ |
| 1,5,8,4,11,16,24, 33,33 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot 2)) + \psi_{\beta_2}(\beta_2 \cdot 2))$ |
| 1,5,8,4,11,16,24, 34 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot 2)) + \psi_{\beta_2}(\Omega_{\beta_2+1})+$ $\psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot 2))+$ $\psi_{\beta_2}(\Omega_{\beta_2+1}) + 1))$ |
| 1,5,8,4,11,16,24, 34,33 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot 2)) + \psi_{\beta_2}(\Omega_{\beta_2+1} + \beta_2))$ |
| 1,5,8,4,11,16,24, 34,45 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot 2)) + \psi_{\beta_2}(\alpha_{\beta_2+1}))$ |
| 1,5,8,4,11,16,24, 34,47 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot 2)) + \psi_{\beta_2}(\alpha_{\beta_2+1}) + \alpha_{\beta+1} \cdot \omega)$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,4,11,16,24, 35 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\alpha_{\beta_2+1}+1))$ |
| 1,5,8,4,11,16,24, 35,33 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\alpha_{\beta_2+1}+\beta_2))$ |
| 1,5,8,4,11,16,24, 35,46 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}({\alpha_{\beta_2+1}}^2+1))$ |
| 1,5,8,4,11,16,24, 35,47 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\beta_3))$ |
| 1,5,8,4,11,16,24, 36 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\beta_\omega))$ |
| 1,5,8,4,11,16,25 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,17 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+$ $\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+1))$ |
| 1,5,8,4,11,17,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+$ $\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+1)+\alpha_\omega)$ |
| 1,5,8,4,11,17,10, 21,28,36 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+$ $\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+1)+$ $\psi_\beta(\psi_S(\sigma S\cdot\omega+S)))$ |
| 1,5,8,4,11,17,10, 21,30 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+$ $\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+1)+$ $\psi_\beta(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+$ $\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+1)))$ |
| 1,5,8,4,11,16,10, 21,30,17 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\beta_2)$ |
| 1,5,8,4,11,17,10, 21,30,21 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,4,11,17,10, 21,30,21,30 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))\cdot 2+$ $\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))\cdot 2+1))$ |
| 1,5,8,4,11,17,10, 21,30,28,21 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2)+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,4,11,17,10, 21,30,29,40 | $\psi(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot 2)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega))))$ |
| 1,5,8,4,11,17,10, 21,30,40 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2)+S))$ |
| 1,5,8,4,11,17,10, 21,30,42 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2)\cdot 2)+$ $\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2)\cdot 2)+1))$ |
| 1,5,8,4,11,17,10, 21,30,42,43 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2+1)))$ |
| 1,5,8,4,11,17,10, 21,30,42,52 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2+S)))$ |
| 1,5,8,4,11,17,10, 21,30,42,55 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2+S_2)))$ |
| 1,5,8,4,11,17,10, 21,30,44 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 3))+\psi_{\gamma_2}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,8,5,4,11,17, 11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+$ $\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| 1,5,8,5,4,11,17, 11,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\alpha_2)$ |
| 1,5,8,5,4,11,17, 11,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\alpha_\omega)$ |
| 1,5,8,5,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,5,7 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\Omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,5,7,4,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_\alpha(\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,5,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,5,8,4,11, 17,11,16,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_2)$ |
| 1,5,8,5,8,4,11, 17,11,16,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,5,8,4,11, 17,11,16,15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,5,8,4,11, 17,11,16,22 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S))$ |
| 1,5,8,5,8,4,11, 17,11,16,24 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega)))$ |
| 1,5,8,5,8,4,11, 17,11,17 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))+1))$ |
| 1,5,8,5,8,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))\cdot 2)$ |
| 1,5,8,6 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+1))$ |
| 1,5,8,7 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\Omega))$ |
| 1,5,8,7,4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\Omega_\omega))$ |
| 1,5,8,7,4,7 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\Omega_{\omega^2}))$ |
| 1,5,8,7,4,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,7,4,11,17, 13,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\alpha))$ |
| 1,5,8,7,4,11,17, 13,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\alpha+1))$ |
| 1,5,8,7,4,11,17, 14 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\Omega_{\alpha+1})+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\Omega_{\alpha+1})+1))$ |
| 1,5,8,7,4,11,17, 14,18 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{\alpha_2}(\alpha_2)))$ |
| 1,5,8,7,4,11,17, 15 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\alpha_2))$ |
| 1,5,8,7,4,11,17, 15,9,18,26,24 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\alpha_3))$ |
| 1,5,8,7,4,11,17, 15,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\alpha_\omega))$ |
| 1,5,8,7,4,11,17, 15,10,17 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_\beta(\beta_2)))$ |
| 1,5,8,7,4,11,17, 15,10,21,31,25,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\beta))$ |
| 1,5,8,7,4,11,17, 15,10,21,31,28,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\beta_\omega))$ |
| 1,5,8,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,7,5,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,7,5,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,7,5,7,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S))$ |
| 1,5,8,7,5,7,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,7,5,7,12, 16,14,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{\psi_{S_2}(\sigma S)}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))))))))$ |
| 1,5,8,7,5,7,12, 16,15 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))))$ |
| 1,5,8,7,5,7,12, 16,15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\Omega_{\alpha+1}\cdot\omega))))$ |
| 1,5,8,7,5,7,12, 16,15,11,15 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\Omega_{\alpha+1}\cdot\omega^2))))$ |
| 1,5,8,7,5,7,12, 16,15,11,16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_\alpha(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\alpha_2)))))$ |
| 1,5,8,7,5,7,12, 16,15,11,18 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\alpha_\omega)))))$ |
| 1,5,8,7,5,7,12, 16,15,11,19 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)$ $+\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega))))))$ |
| 1,5,8,7,5,7,12, 16,15,11,19,26,19 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_\alpha(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))))))$ |
| 1,5,8,7,5,7,12, 16,15,11,19,26,22,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\alpha)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,7,5,7,12, 16,15,11,19,26,24 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\alpha_2)))$ |
| 1,5,8,7,5,7,12, 16,15,11,19,26,24,18 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\alpha_\omega)))$ |
| 1,5,8,7,5,7,12, 16,15,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,7,5,7,12, 16,15,12,15,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\Omega_{\alpha+1}\cdot\omega)$ |
| 1,5,8,7,5,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |
| 1,5,8,7,5,8,4, 11,17,15,11,16,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,7,5,8,4, 11,17,15,11,16,11,15, 11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,7,5,8,4, 11,17,15,11,16,11,15, 20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+S))$ |
| 1,5,8,7,5,8,4, 11,17,15,11,16,24 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega)))$ |
| 1,5,8,7,5,8,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,8,7,5,8,6 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+1))$ |
| 1,5,8,7,5,8,7, 5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))\cdot 2)$ |
| 1,5,8,7,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)\cdot 2))$ |
| 1,5,8,7,7,5,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)\cdot 2)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |
| 1,5,8,7,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+1)))$ |
| 1,5,8,7,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+S)))$ |
| 1,5,8,7,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,8,4,11,17, 16,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\alpha_2)$ |
| 1,5,8,8,4,11,17, 16,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,8,4,11,17, 16,11,17,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,8,8,4,11,17, 16,11,17,15,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\alpha_\omega))$ |
| 1,5,8,8,4,11,17, 16,11,17,15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,8,4,11,17, 16,11,17,15,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+S)))$ |
| 1,5,8,8,4,11,17, 16,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1))$ |
| 1,5,8,8,4,11,17, 16,17 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+1)))$ |
| 1,5,8,8,4,11,17, 16,22 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S)))$ |
| 1,5,8,8,4,11,17, 16,23,24 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+1))))$ |
| 1,5,8,8,4,11,17, 16,24 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)$ $+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S+S_\omega))))$ |
| 1,5,8,8,4,11,17, 17 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot 2)))+\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2)))+1))$ |
| 1,5,8,8,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| 1,5,8,8,5,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))+$ $\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,8,5,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))+$ $\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,8,5,7,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)))+\psi_S(\sigma S\cdot\omega+S))$ |
| 1,5,8,8,5,8,4, 11,17,17,11,16,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_2)$ |
| 1,5,8,8,5,8,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,8,8,5,8,7, 4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\Omega_\omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,8,5,8,7,<br>5 | $\psi(\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,8,5,8,7,<br>10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+S)))$ |
| 1,5,8,8,5,8,8,<br>5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)))\cdot 2)$ |
| 1,5,8,8,6 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+1))$ |
| 1,5,8,8,7,4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\Omega_\omega))$ |
| 1,5,8,8,7,4,11,<br>17,17,15,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\alpha_\omega))$ |
| 1,5,8,8,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,8,7,5,8,<br>5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,8,8,7,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega)\cdot 2))$ |
| 1,5,8,8,7,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S)))$ |
| 1,5,8,8,8,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))\cdot 2))$ |
| 1,5,8,9 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$<br>$\psi_{S_2}(\sigma S\cdot\omega)+1)))$ |
| 1,5,8,10,4,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_\alpha(\psi_S(\sigma S\cdot\omega)))))$ |
| 1,5,8,10,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,10,5,8,4,<br>11,17,15,11,16,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega)))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+\alpha_2)$ |
| 1,5,8,10,6 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+1))$ |
| 1,5,8,10,6,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+1)+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,10,6,6 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+2))$ |
| 1,5,8,10,7 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\Omega))$ |
| 1,5,8,10,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$<br>$\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,10,7,6 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega)+1))$ |
| 1,5,8,10,7,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega)\cdot 2))$ |
| 1,5,8,10,7,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S)))$ |
| 1,5,8,10,8,4,11, 17,21,16,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\alpha_2)$ |
| 1,5,8,10,8,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| 1,5,8,10,8,5,6 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))+\psi_S(\sigma S\cdot\omega+1))$ |
| 1,5,8,10,8,6 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+1))$ |
| 1,5,8,10,8,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,10,8,8,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))\cdot 2))$ |
| 1,5,8,10,8,10,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))\cdot 2))$ |
| 1,5,8,10,9 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)+1)))$ |
| 1,5,8,10,10,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)\cdot 2)))$ |
| 1,5,8,10,10,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)\cdot 2)+$ $\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,10,10,8,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)\cdot 2)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| 1,5,8,10,10,8,10, 10,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega)\cdot 2)\cdot 2))$ |
| 1,5,8,10,10,10,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)\cdot 3)))$ |
| 1,5,8,10,12,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))))$ |
| 1,5,8,10,13 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+S))))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,5,8,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))+1))$ |
| 1,5,8,11,4,11,17, 22,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))+$ $\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,11,4,11,17, 22,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |
| 1,5,8,11,4,11,17, 22,18 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))+1)))$ |
| 1,5,8,11,4,11,17, 22,21,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,11,4,11,17, 22,21,26 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega+S))))$ |
| 1,5,8,11,4,11,17, 22,22,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S))\cdot 2))+\alpha_2)$ |
| 1,5,8,11,4,11,17, 22,26,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))))$ |
| 1,5,8,11,4,11,17, 22,28 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)+S))))$ |
| 1,5,8,11,4,11,17, 23 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))))+$ $\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot 2))))+1))$ |
| 1,5,8,11,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)))))$ |
| 1,5,8,11,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+1))))$ |
| 1,5,8,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S))$ |
| 1,5,8,12,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S)+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,12,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,12,7,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S+\psi_S(\sigma S\cdot\omega+S)))$ |
| 1,5,8,12,8,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| 1,5,8,12,8,10,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,12,8,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+S)))$ |
| 1,5,8,12,10,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+S+\psi_S(\sigma S\cdot\omega))))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,12,11,15 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S))))$ |
| 1,5,8,12,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S\cdot 2))$ |
| 1,5,8,12,13 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S\cdot\omega))$ |
| 1,5,8,12,14,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S\cdot\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,12,14,17 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S\cdot\psi_S(\sigma S\cdot\omega+S)))$ |
| 1,5,8,12,15 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S\cdot\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S))))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S\cdot\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |
| 1,5,8,12,15,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S\cdot\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| 1,5,8,12,15,19 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S\cdot\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S)))$ |
| 1,5,8,12,16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^2))$ |
| 1,5,8,12,16,8,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| 1,5,8,12,16,8,12, 16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^2)))$ |
| 1,5,8,12,16,11,15, 19 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^2))))$ |
| 1,5,8,12,16,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^2+S))$ |
| 1,5,8,12,16,12,16 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^2\cdot 2))$ |
| 1,5,8,12,16,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S^S))$ |
| 1,5,8,12,17 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S)))$ |
| 1,5,8,12,18 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S))+\Omega_{\alpha+1}\cdot\omega)$ |
| 1,5,8,12,19 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,13 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S))+1))$ |
| 1,5,8,13,4,11,17, 25,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S))+\alpha_2)$ |
| 1,5,8,13,4,11,17, 25,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S))+\alpha_\omega)$ |
| 1,5,8,13,4,11,17, 25,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S))+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,8,13,4,11,17, 25,15,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,13,4,11,17, 25,17,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S)+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,13,4,11,17, 25,24 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S)+S))$ |
| 1,5,8,13,4,11,17, 25,26 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S+1)))$ |
| 1,5,8,13,4,11,17, 25,34 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S+S_2)))$ |
| 1,5,8,13,4,11,17, 26 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S\cdot 2))+\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+\psi_{S_2}(\sigma S\cdot 2))+1))$ |
| 1,5,8,13,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2))$ |
| 1,5,8,13,6 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+1))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,13,7,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,13,7,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,13,8 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S)))+1))$ |
| 1,5,8,13,8,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| 1,5,8,13,8,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S)))$ |
| 1,5,8,13,8,13,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2)))$ |
| 1,5,8,13,9 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+1)))$ |
| 1,5,8,13,10,5 | $\psi(\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,13,11,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))))$ |
| 1,5,8,13,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+S))$ |
| 1,5,8,13,12,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+S\cdot 2))$ |
| 1,5,8,13,12,17 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+\psi_{S_2}(\sigma S)))$ |
| 1,5,8,13,13 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+$ $\psi_{S_2}(\sigma S))+1))$ |
| 1,5,8,13,13,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)\cdot 3))$ |
| 1,5,8,13,14 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+1)))$ |
| 1,5,8,13,15,4 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\Omega_\omega)))$ |
| 1,5,8,13,15,4,11 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_\alpha(\psi_S(\sigma S\cdot\omega)))))$ |
| 1,5,8,13,15,4,11, 17,26,28,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\alpha)))$ |
| 1,5,8,13,15,4,11, 17,26,30,10 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\alpha_\omega)))$ |
| 1,5,8,13,15,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,13,15,6 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+1))$ |
| 1,5,8,13,15,9 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+$ $\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))))$ |
| 1,5,8,13,15,12 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+S))$ |
| 1,5,8,13,15,12,17 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))+\psi_{S_2}(\sigma S)))$ |
| 1,5,8,13,15,13,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+$ $\psi_S(\sigma S\cdot\omega))+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,8,13,15,13,15, 5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega))\cdot 2))$ |
| 1,5,8,13,15,15,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)\cdot 2)))$ |
| 1,5,8,13,16,5 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))))$ |
| 1,5,8,13,16,20 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S))))$ |
| 1,5,8,13,16,21,22 | $\psi(\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega+1)))))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,13,17 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega + S)))$ |
| 1,5,8,13,17,13,17 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega + S) \cdot 2))$ |
| 1,5,8,13,17,17 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega + S \cdot 2)))$ |
| 1,5,8,13,17,22 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega + \psi_{S_2}(\sigma S))))$ |
| 1,5,8,13,18,5 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega))))$ |
| 1,5,8,13,18,12 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega)) + S))$ |
| 1,5,8,13,18,13,5 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega +$ $\psi_{S_2}(\sigma S \cdot \omega)) + \psi_{S_2}(\sigma S \cdot \omega)))$ |
| 1,5,8,13,18,14 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega) + 1)))$ |
| 1,5,8,13,18,17,22 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega) + \psi_{S_2}(\sigma S))))$ |
| 1,5,8,13,18,19 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega + 1))))$ |
| 1,5,8,13,18,22 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega + S))))$ |
| 1,5,8,13,18,23,5 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega +$ $\psi_{S_2}(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega)))))$ |
| 1,5,8,13,19 | $\psi(\psi_S(\sigma S \cdot \omega + S_2))$ |
| 1,5,8,13,19,13,19 | $\psi(\psi_S(\sigma S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot \omega + S_2)))$ |
| 1,5,8,13,19,19 | $\psi(\psi_S(\sigma S \cdot \omega + S_2 \cdot 2))$ |
| 1,5,8,13,20 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_3}(\sigma S)) +$ $\psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega + \psi_{S_3}(\sigma S)) + 1))$ |
| 1,5,8,13,20,5 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega)))$ |
| 1,5,8,13,20,12 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega) + S))$ |
| 1,5,8,13,20,14 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega) +$ $\psi_{S_2}(\sigma S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega) + 1)))$ |
| 1,5,8,13,20,17 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega) +$ $\psi_{S_2}(\sigma S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega) + S)))$ |
| 1,5,8,13,20,19 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega) + S_2))$ |
| 1,5,8,13,20,20,5 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega) \cdot 2))$ |
| 1,5,8,13,20,24 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega + S)))$ |
| 1,5,8,13,20,25,31 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega + S_2))))$ |
| 1,5,8,13,20,26 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega + S_2)))$ |
| 1,5,8,13,20,27,5 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega))))$ |
| 1,5,8,13,20,28 | $\psi(\psi_S(\sigma S \cdot \omega + S_3))$ |
| 1,5,8,14 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega))$ |
| 1,5,8,14,6 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega + 1))$ |
| 1,5,8,14,7,10 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega + \psi_S(\sigma S \cdot \omega + S)))$ |
| 1,5,8,14,8,5 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega + \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega))))$ |
| 1,5,8,14,8,14 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega + \psi_S(\sigma S \cdot \omega + S_\omega)))$ |
| 1,5,8,14,12 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega + S))$ |
| 1,5,8,14,13,5 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega + \psi_{S_2}(\sigma S \cdot \omega)))$ |
| 1,5,8,14,13,21 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega + \psi_{S_2}(\sigma S \cdot \omega + S_\omega)))$ |
| 1,5,8,14,13,21,19 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega + S_2))$ |
| 1,5,8,14,14 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega \cdot 2))$ |
| 1,5,8,14,16,5 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega \cdot \psi_S(\sigma S \cdot \omega)))$ |
| 1,5,8,14,18 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega \cdot S))$ |
| 1,5,8,14,18,12 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega \cdot S + S))$ |
| 1,5,8,14,18,13,21, 25 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega \cdot S + \psi_{S_2}(\sigma S \cdot \omega + S_\omega \cdot S)))$ |
| 1,5,8,14,18,13,21, 25,21 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega \cdot (S + 1)))$ |
| 1,5,8,14,18,13,21, 26,34 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega \cdot \psi_{S_2}(\sigma S \cdot \omega + S_\omega)))$ |
| 1,5,8,14,18,13,21, 27 | $\psi(\psi_S(\sigma S \cdot \omega + S_\omega \cdot S_2))$ |
| 1,5,8,14,18,14 | $\psi(\psi_S(\sigma S \cdot \omega + {S_\omega}^2))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,14,18,18,14 | $\psi(\psi_S(\sigma S \cdot \omega + {S_\omega}^3))$ |
| 1,5,8,14,18,22,14 | $\psi(\psi_S(\sigma S \cdot \omega + {S_\omega}^{S_\omega}))$ |
| 1,5,8,14,18,23 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_{\omega+1}}(\sigma S)))$ |
| 1,5,8,14,19,5 | $\psi(\psi_S(\sigma S \cdot \omega + \psi_{S_{\omega+1}}(\sigma S \cdot \omega)))$ |
| 1,5,8,14,19,25 | $\psi(\psi_S(\sigma S \cdot \omega + S_{\omega+1}))$ |
| 1,5,8,14,19,27 | $\psi(\psi_S(\sigma S \cdot \omega + S_{\omega \cdot 2}))$ |
| 1,5,8,14,20 | $\psi(\psi_S(\sigma S \cdot \omega + S_{\omega^2}))$ |
| 1,5,8,14,20,24 | $\psi(\psi_S(\sigma S \cdot \omega + S_S))$ |
| 1,5,8,14,20,24,15 | $\psi(\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S)))$ |
| 1,5,8,14,20,24,15, 11,17,23,27,18 | $\psi(\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S))+$ $\psi_{\psi_S(\sigma S \cdot (\omega+1))}(\psi_S(\sigma S \cdot (\omega+1)+$ $\psi_{S_2}(\sigma S \cdot (\omega+1) + S))))$ |
| 1,5,8,14,20,24,15, 12 | $\psi(\psi_S(\sigma S \cdot (\omega+1)+$ $\psi_{S_2}(\sigma S \cdot (\omega+1) + S)) + \psi_S(\sigma S \cdot (\omega+1)))$ |
| 1,5,8,14,20,24,15, 14 | $\psi(\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S))+$ $\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + 1)))$ |
| 1,5,8,14,20,24,15, 15 | $\psi(\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S) + 1))$ |
| 1,5,8,14,20,24,16 | $\psi(\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S) + \Omega))$ |
| 1,5,8,14,20,24,16, 5 | $\psi(\psi_S(\sigma S \cdot (\omega+1)+$ $\psi_{S_2}(\sigma S \cdot (\omega+1) + S) + \psi_S(\sigma S \cdot \omega)))$ |
| 1,5,8,14,20,24,16, 15 | $\psi(\psi_S(\sigma S \cdot (\omega+1)+$ $\psi_{S_2}(\sigma S \cdot (\omega+1) + S) + \psi_S(\sigma S \cdot \omega) + 1))$ |
| 1,5,8,14,20,24,16, 16,5 | $\psi(\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S)+$ $\psi_S(\sigma S \cdot \omega) \cdot 2))$ |
| 1,5,8,14,20,24,16, 19 | $\psi(\psi_S(\sigma S \cdot (\omega+1)+$ $\psi_{S_2}(\sigma S \cdot (\omega+1) + S) + \psi_S(\sigma S \cdot \omega + S)))$ |
| 1,5,8,14,20,24,17, 5 | $\psi(\psi_S(\sigma S \cdot (\omega+1)+$ $\psi_{S_2}(\sigma S \cdot (\omega+1) + S) + \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega))))$ |
| 1,5,8,14,20,24,17, 23 | $\psi(\psi_S(\sigma S \cdot (\omega+1)+$ $\psi_{S_2}(\sigma S \cdot (\omega+1) + S) + \psi_S(\sigma S \cdot \omega + S_\omega)))$ |
| 1,5,8,14,20,24,18 | $\psi(\psi_S(\sigma S \cdot (\omega+1)+$ $\psi_{S_2}(\sigma S \cdot (\omega+1) + S) + \psi_S(\sigma S \cdot (\omega+1))))$ |
| 1,5,8,14,20,24,18, 13,21,29,35,25,30 | $\psi(\psi_S(\sigma S \cdot (\omega+1)+$ $\psi_{S_2}(\sigma S \cdot (\omega+1) + S) + \psi_S(\sigma S \cdot (\omega+1) + S)))$ |
| 1,5,8,14,20,24,18, 13,21,29,35,26,5 | $\psi(\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S)+$ $\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot \omega))))$ |
| 1,5,8,14,20,24,18, 13,21,29,35,27 | $\psi(\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S)+$ $\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1)))))$ |
| 1,5,8,14,20,24,18, 14 | $\psi(\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S)+$ $\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + 1))))$ |
| 1,5,8,14,20,24,18, 14,20 | $\psi(\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S)+$ $\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + 2))))$ |
| 1,5,8,14,20,24,18, 14,20,24,15 | $\psi(\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S)+$ $\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S))))$ |
| 1,5,8,14,20,24,18, 15 | $\psi(\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S)+$ $\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S)) + 1))$ |
| 1,5,8,14,20,24,18, 18,14,20,24,15 | $\psi(\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S)+$ $\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S)) \cdot 2))$ |
| 1,5,8,14,20,24,18, 23 | $\psi(\psi_S(\sigma S \cdot (\omega+1) + \psi_{S_2}(\sigma S \cdot (\omega+1) + S) + S))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,14,20,24,19, 5 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,8,14,20,24,19, 18 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+ \psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot(\omega+1))))$ |
| 1,5,8,14,20,24,19, 18,14,20,24,15 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+ \psi_{S_2}(\sigma S\cdot\omega)+ \psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S))))$ |
| 1,5,8,14,20,24,19, 18,23 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+ \psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+S)+S)))$ |
| 1,5,8,14,20,24,19, 19,5 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+ \psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| 1,5,8,14,20,24,19, 25 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega)+S))$ |
| 1,5,8,14,20,24,19, 25,25 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega)+S\cdot 2))$ |
| 1,5,8,14,20,24,19, 26,5 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2))$ |
| 1,5,8,14,20,24,19, 26,25 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+S))$ |
| 1,5,8,14,20,24,19, 26,27 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega+1)))$ |
| 1,5,8,14,20,24,19, 26,32 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega+S)))$ |
| 1,5,8,14,20,24,19, 26,34 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot\omega+S_2)))$ |
| 1,5,8,14,20,24,19, 27 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+ \psi_{S_2}(\sigma S\cdot\omega+S_\omega)))$ |
| 1,5,8,14,20,24,19, 27,27 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+1))\cdot 2)$ |
| 1,5,8,14,20,24,19, 27,33 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+1)+\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot(\omega+1)))))$ |
| 1,5,8,14,20,24,19, 27,35 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+2)))$ |
| 1,5,8,14,20,24,19, 27,35,39 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_S(\sigma S\cdot(\omega+1)))))$ |
| 1,5,8,14,20,24,19, 27,35,41 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot(\omega+1))))))$ |
| 1,5,8,14,20,24,19, 27,35,41,27 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot(\omega+1)+1)))))$ |
| 1,5,8,14,20,24,19, 27,35,41,27,35 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+S)+\psi_{S_2}(\sigma S\cdot(\omega+1)+2)))))$ |
| 1,5,8,14,20,24,19, 27,35,41,28 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)\cdot 2))$ |
| 1,5,8,14,20,24,20 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,14,20,24,20, 15 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+1))$ |
| 1,5,8,14,20,24,20, 16,5 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,14,20,24,20, 17,5 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| 1,5,8,14,20,24,20, 17,23 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+$ $\psi_S(\sigma S\cdot\omega+\psi_{\sigma S}(\psi_S(\sigma S\cdot(\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(\omega+1)+1))))))$ |
| 1,5,8,14,20,24,20, 18 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$ $S+1)+\psi_S(\sigma S\cdot(\omega+1))))$ |
| 1,5,8,14,20,24,20, 18,14 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+$ $\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+1))))$ |
| 1,5,8,14,20,24,20, 18,14,20,24,20 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+$ $\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1))))$ |
| 1,5,8,14,20,24,20, 18,22 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+$ $\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+$ $\psi_S(\sigma S\cdot(\omega+1)))))$ |
| 1,5,8,14,20,24,20, 18,23 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+S))$ |
| 1,5,8,14,20,24,20, 19,5 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+$ $\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,8,14,20,24,20, 19,27 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)+$ $\psi_S(\sigma S\cdot\omega+S_\omega)))$ |
| 1,5,8,14,20,24,20, 19,27,35,41,35 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+1)\cdot 2))$ |
| 1,5,8,14,20,24,20, 20 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+2)))$ |
| 1,5,8,14,20,24,20, 22,5 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$ $S+\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,8,14,20,24,20, 24 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S+$ $\psi_S(\sigma S\cdot(\omega+1)))))$ |
| 1,5,8,14,20,24,20, 24,15 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S\cdot 2)))$ |
| 1,5,8,14,20,24,21 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S\cdot\omega)))$ |
| 1,5,8,14,20,24,24, 15 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+S^2)))$ |
| 1,5,8,14,20,24,29 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S))))$ |
| 1,5,8,14,20,25 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$ $\psi_{S_2}(\sigma S)))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot(\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S)))+1))$ |
| 1,5,8,14,20,25,5 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| 1,5,8,14,20,25,15 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$ $\psi_{S_2}(\sigma S\cdot\omega))+1))$ |
| 1,5,8,14,20,25,18 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$ $\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot(\omega+1))))$ |
| 1,5,8,14,20,25,18, 23 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$ $\psi_{S_2}(\sigma S\cdot\omega))+S))$ |
| 1,5,8,14,20,25,19, 5 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+$ $\psi_{S_2}(\sigma S\cdot\omega))+\psi_{S_2}(\sigma S\cdot\omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,14,20,25,19, 27 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot\omega))+\psi_{S_2}(\sigma S\cdot(\omega+1)+1)))$ |
| 1,5,8,14,20,25,20 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot\omega)+1)))$ |
| 1,5,8,14,20,25,20, 24,15 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot\omega)+S)))$ |
| 1,5,8,14,20,25,20, 25,5 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2)))$ |
| 1,5,8,14,20,25,24, 15 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot\omega+S))))$ |
| 1,5,8,14,20,25,25, 5 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))))$ |
| 1,5,8,14,20,25,33 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot\omega+S_\omega))))$ |
| 1,5,8,14,20,25,33, 41,47,34 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)))+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)))$ |
| 1,5,8,14,20,26 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1))+1)))$ |
| 1,5,8,14,20,26,15 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1))+1)+1))$ |
| 1,5,8,14,20,26,18, 14,20,26 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1))+1)+\psi_S(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1))+1))))$ |
| 1,5,8,14,20,26,18, 23 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1))+1)+S))$ |
| 1,5,8,14,20,26,19, 5 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1))+1)+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,8,14,20,26,20 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1))+2)))$ |
| 1,5,8,14,20,26,20, 24,15 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1))+S)))$ |
| 1,5,8,14,20,26,20, 24,20 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1))+S+1)))$ |
| 1,5,8,14,20,26,20, 24,29 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1))+\psi_{S_2}(\sigma S))))$ |
| 1,5,8,14,20,26,20, 26 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1))\cdot 2+1)))$ |
| 1,5,8,14,20,26,24 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_S(\sigma S\cdot(\omega+1))))))$ |
| 1,5,8,14,20,26,24, 15 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+S))))$ |
| 1,5,8,14,20,26,26 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+ \psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)))+1)))$ |
| 1,5,8,14,21 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+S_2))$ |
| 1,5,8,14,21,21 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+S_2\cdot 2))$ |
| 1,5,8,14,21,29 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_3}(\sigma S)))$ |
| 1,5,8,14,22,5 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_3}(\sigma S\cdot\omega)))$ |
| 1,5,8,14,22,21 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_3}(\sigma S\cdot\omega)+S_2))$ |
| 1,5,8,14,22,29 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_3}(\sigma S\cdot\omega+S_2)))$ |
| 1,5,8,14,22,31 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_3}(\sigma S\cdot\omega+S_3)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,14,23 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_3}(\sigma S\cdot(\omega+1))+$ $\psi_{S_2}(\sigma S\cdot(\omega+1)+\psi_{S_3}(\sigma S\cdot(\omega+1))+1)))$ |
| 1,5,8,14,23,24 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_3}(\sigma S\cdot(\omega+1)+1)))$ |
| 1,5,8,14,23,33 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+S_3))$ |
| 1,5,8,14,24 | $\psi(\psi_S(\sigma S\cdot(\omega+1)+S_\omega))$ |
| 1,5,8,15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2))$ |
| 1,5,8,15,14 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2)+\psi_S(\sigma S\cdot(\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(\omega+1)+1)))$ |
| 1,5,8,15,14,25 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2)+\psi\psi_S(\sigma S\cdot(\omega+2))(\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| 1,5,8,15,14,25,18, 23 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2)+\psi_S(\sigma S\cdot(\omega+2)))$ |
| 1,5,8,15,14,25,21 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2)+$ $\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2))))$ |
| 1,5,8,15,14,25,24 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2)+$ $\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2)+1)))$ |
| 1,5,8,15,15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2)\cdot 2)$ |
| 1,5,8,15,16 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+1))$ |
| 1,5,8,15,17,5 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,15,17,6 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega))+$ $\psi_{\psi_S(\sigma S\cdot(\omega+1))}(\psi_S(\sigma S\cdot\omega\cdot 2+$ $\psi_S(\sigma S\cdot\omega))+1))$ |
| 1,5,8,15,17,12 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot(\omega+1)))$ |
| 1,5,8,15,17,14 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+1)))$ |
| 1,5,8,15,17,15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega\cdot 2))$ |
| 1,5,8,15,17,15,17, 5 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega))\cdot 2)$ |
| 1,5,8,15,17,17,5 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega)\cdot 2))$ |
| 1,5,8,15,17,18 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+1)))$ |
| 1,5,8,15,17,20 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+S)))$ |
| 1,5,8,15,18,5 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| 1,5,8,15,18,5 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega))+1))$ |
| 1,5,8,15,18,17,5 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,8,15,18,17,20 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S)))$ |
| 1,5,8,15,18,19 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+1)))$ |
| 1,5,8,15,18,21,5 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+$ $\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)))))$ |
| 1,5,8,15,18,22 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+\psi_{S_2}(\sigma S\cdot\omega)+S)))$ |
| 1,5,8,15,18,24 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega+S_\omega)))$ |
| 1,5,8,15,19 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot(\omega+1))))$ |
| 1,5,8,15,19,14 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot(\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(\omega+1)+1))))$ |
| 1,5,8,15,19,14,25 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{\psi_S(\sigma S\cdot(\omega+2))}(\psi_S(\sigma S\cdot\omega\cdot 2))))$ |
| 1,5,8,15,19,14,25, 29,15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot(\omega+2))))$ |
| 1,5,8,15,19,15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| 1,5,8,15,19,16 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega\cdot 2)+1))$ |
| 1,5,8,15,19,19,15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega\cdot 2)\cdot 2))$ |
| 1,5,8,15,19,23,15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega\cdot 2))))$ |
| 1,5,8,15,19,24 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+S))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,15,20 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S))+1))$ |
| 1,5,8,15,20,5 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,8,15,20,14 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+1)))$ |
| 1,5,8,15,20,15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega\cdot 2))$ |
| 1,5,8,15,20,16 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)+1))$ |
| 1,5,8,15,20,19,15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| 1,5,8,15,20,19,24 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega\cdot 2+S)))$ |
| 1,5,8,15,20,20,5 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| 1,5,8,15,20,24,15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)+$ $\psi_S(\sigma S\cdot\omega\cdot 2+$ $\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega\cdot 2))))$ |
| 1,5,8,15,20,26 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)+S))$ |
| 1,5,8,15,20,27,5 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2))$ |
| 1,5,8,15,20,27,26 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)\cdot 2+S))$ |
| 1,5,8,15,20,27,28 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega+1)))$ |
| 1,5,8,15,20,28 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega+S_\omega)))$ |
| 1,5,8,15,20,29 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)))+$ $\psi_{\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2)))}$ $(\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| 1,5,8,15,21 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)))+$ $\psi_{\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2)))}$ $(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)))+1))$ |
| 1,5,8,15,21,14,25, 34,24 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)))+$ $\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2)+1)))$ |
| 1,5,8,15,21,14,25, 34,25 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)))+\psi_S(\sigma S\cdot\omega\cdot 2))$ |
| 1,5,8,15,21,14,25, 34,32,25 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| 1,5,8,15,21,14,25, 34,32,40 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+$ $\psi_S(\sigma S\cdot\omega\cdot 2+S)))$ |
| 1,5,8,15,21,14,25, 34,33,44 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+$ $\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega+S_\omega))))$ |
| 1,5,8,15,21,14,25, 34,34,24 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+$ $\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))))+$ $\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2)+1)))$ |
| 1,5,8,15,21,14,25, 34,35 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+$ $\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+1)))$ |
| 1,5,8,15,21,14,25, 34,41,25 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+$ $\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+$ $\psi_S(\sigma S\cdot\omega\cdot 2))))$ |
| 1,5,8,15,21,14,25, 34,44 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+S))$ |
| 1,5,8,15,21,14,25, 34,45,5 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,8,15,21,14,25, 34,46,25 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))\cdot 2)+$ $\psi_S(\sigma S\cdot\omega\cdot 2))$ |
| 1,5,8,15,21,14,25, 34,46,44 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1))\cdot 2+S))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,8,15,21,14,25, 34,46,47 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)+1)))$ |
| 1,5,8,15,21,14,25, 34,46,56 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)+S)))$ |
| 1,5,8,15,21,14,25, 34,46,58,25 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(\omega+1))))+\psi_S(\sigma S\cdot\omega\cdot 2))$ |
| 1,5,8,15,21,14,25, 34,46,59 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)+S_2)))$ |
| 1,5,8,15,21,14,25, 34,47 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)+S_\omega)))$ |
| 1,5,8,15,21,14,25, 34,48 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+2)))+$ $\psi_{\psi_S(\sigma S\cdot(\omega+3)+\psi_{S_2}(\sigma S\cdot(\omega+3)))}$ $(\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| 1,5,8,15,21,14,25, 35 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+2)))+$ $\psi_{\psi_S(\sigma S\cdot(\omega+3)+\psi_{S_2}(\sigma S\cdot(\omega+3)))}$ $(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+2)))+1))$ |
| 1,5,8,15,21,15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2)))$ |
| 1,5,8,15,21,15,15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2))+\psi_S(\sigma S\cdot\omega\cdot 2))$ |
| 1,5,8,15,21,15,19, 15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2))+$ $\psi_S(\sigma S\cdot\omega\cdot 2+\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| 1,5,8,15,21,15,19, 24 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2))+$ $\psi_S(\sigma S\cdot\omega\cdot 2+S))$ |
| 1,5,8,15,21,15,20, 5 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2))+$ $\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,8,15,21,15,20, 28 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2))+$ $\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)))+$ $\psi_{\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2)))}$ $(\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| 1,5,8,15,21,15,21, 15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2))\cdot 2)$ |
| 1,5,8,15,21,19,15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2)+\psi_S(\sigma S\cdot\omega\cdot 2)))$ |
| 1,5,8,15,21,20,28 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2)+$ $\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot(\omega+1)))+$ $\psi_{\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2)))}$ $(\psi_S(\sigma S\cdot\omega\cdot 2))))$ |
| 1,5,8,15,21,21,15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2)+$ $\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2))))$ |
| 1,5,8,15,21,28 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2)+S))$ |
| 1,5,8,15,21,30,15 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2)\cdot 2))$ |
| 1,5,8,15,21,30,31 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2+1)))$ |
| 1,5,8,15,21,30,40 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 2+S_2))$ |
| 1,5,8,15,21,31 | $\psi(\psi_S(\sigma S\cdot(\omega\cdot 2+1)+\psi_{S_2}(\sigma S\cdot(\omega\cdot 2+1)+1)))$ |
| 1,5,8,15,21,32 | $\psi(\psi_S(\sigma S\cdot\omega\cdot 3))$ |
| 1,5,9 | $\psi(\psi_S(\sigma S\cdot\omega^2))$ |
| 1,5,9,4,11,18 | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_\alpha(\psi_S(\sigma S\cdot\omega^2)))$ |
| 1,5,9,4,11,18,6, 9 | $\psi(\psi_S(\sigma S\cdot\omega^2)+\alpha)$ |
| 1,5,9,4,11,18,10, 21,32 | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_\beta(\psi_S(\sigma S\cdot\omega^2)))$ |
| 1,5,9,5 | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,9,5,7,5 | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,9,5,7,10 | $\psi(\psi_S(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega+S))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,9,5,8,14 | $\psi(\psi_S(\sigma S \cdot \omega^2) + \psi_S(\sigma S \cdot \omega + S_\omega))$ |
| 1,5,9,5,8,15,22 | $\psi(\psi_S(\sigma S \cdot \omega^2) + \psi_{\psi_S(\sigma S \cdot (\omega+1))}(\psi_S(\sigma S \cdot \omega^2)))$ |
| 1,5,9,5,8,15,22, 7,10 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_{\psi_S(\sigma S \cdot (\omega+1))}$ $(\psi_S(\sigma S \cdot \omega^2) + \psi_S(\sigma S \cdot \omega + S))))$ |
| 1,5,9,5,8,15,22, 12 | $\psi(\psi_S(\sigma S \cdot \omega^2) + \psi_S(\sigma S \cdot (\omega + 1)))$ |
| 1,5,9,5,8,15,22, 14 | $\psi(\psi_S(\sigma S \cdot \omega^2) + \psi_S(\sigma S \cdot (\omega + 1) +$ $\psi_{S_2}(\sigma S \cdot (\omega + 1) + 1)))$ |
| 1,5,9,5,8,15,22, 14,25,36 | $\psi(\psi_S(\sigma S \cdot \omega^2) + \psi_{\psi_S(\sigma S \cdot (\omega+2))}(\psi_S(\sigma S \cdot \omega^2)))$ |
| 1,5,9,5,8,15,22, 14,25,36,21 | $\psi(\psi_S(\sigma S \cdot \omega^2) + \psi_S(\sigma S \cdot (\omega + 2)))$ |
| 1,5,9,5,8,15,22, 15 | $\psi(\psi_S(\sigma S \cdot \omega^2) + \psi_S(\sigma S \cdot \omega \cdot 2))$ |
| 1,5,9,5,9 | $\psi(\psi_S(\sigma S \cdot \omega^2) \cdot 2)$ |
| 1,5,9,6 | $\psi(\psi_S(\sigma S \cdot \omega^2 + 1))$ |
| 1,5,9,7,4 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \Omega_\omega))$ |
| 1,5,9,7,4,8 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_\alpha(\alpha_2)))$ |
| 1,5,9,7,4,11,18 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_\alpha(\psi_S(\sigma S \cdot \omega^2))))$ |
| 1,5,9,7,4,11,18, 13,5 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \alpha))$ |
| 1,5,9,7,4,11,18, 15,10 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \alpha_\omega))$ |
| 1,5,9,7,5 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega)))$ |
| 1,5,9,7,5,5 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot \omega))$ |
| 1,5,9,7,5,7,10 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot \omega + S))$ |
| 1,5,9,7,5,8,14 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot \omega + S_\omega))$ |
| 1,5,9,7,5,8,15, 22 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega))+$ $\psi_{\psi_S(\sigma S \cdot (\omega+1))}(\psi_S(\sigma S \cdot \omega^2)))$ |
| 1,5,9,7,5,8,15, 22,17,5 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega))+$ $\psi_{\psi_S(\sigma S \cdot (\omega+1))}(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega))))$ |
| 1,5,9,7,5,8,15, 22,17,11,18 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega))+$ $\psi_{\psi_S(\sigma S \cdot (\omega+1))}(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega))+$ $\psi_{\psi_S(\sigma S \cdot (\omega+1))}(\psi_S(\sigma S \cdot \omega \cdot 2))))$ |
| 1,5,9,7,5,8,15, 22,17,12 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot (\omega + 1)))$ |
| 1,5,9,7,5,8,15, 22,17,14,25,36 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega)) + \psi_{\psi_S(\sigma S \cdot (\omega+2))}(\psi_S(\sigma S \cdot \omega^2)))$ |
| 1,5,9,7,5,8,15, 22,17,15 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot \omega \cdot 2))$ |
| 1,5,9,7,5,8,15, 22,17,15,21,31 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot \omega \cdot 2 + S_\omega))$ |
| 1,5,9,7,5,8,15, 22,17,15,21,32,43,34, 32 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot \omega \cdot 3))$ |
| 1,5,9,7,5,8,15, 22,17,15,22 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot \omega^2))$ |
| 1,5,9,7,5,8,15, 22,17,16 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega) + 1))$ |
| 1,5,9,7,5,8,15, 22,17,17,5 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega) \cdot 2))$ |
| 1,5,9,7,5,8,15, 22,17,20 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega + S)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,9,7,5,8,15, 22,18,5 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega))))$ |
| 1,5,9,7,5,8,15, 22,18,24 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega + S_\omega)))$ |
| 1,5,9,7,5,8,15, 22,18,25 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_{\psi_S(\sigma S \cdot (\omega+1))}(\psi_S(\sigma S \cdot \omega \cdot 2))))$ |
| 1,5,9,7,5,8,15, 22,19 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot (\omega + 1))))$ |
| 1,5,9,7,5,8,15, 22,19,15 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega \cdot 2)))$ |
| 1,5,9,7,5,9 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega^2)))$ |
| 1,5,9,7,5,9,5, 9 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega^2)) + \psi_S(\sigma S \cdot \omega^2))$ |
| 1,5,9,7,6 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega^2) + 1))$ |
| 1,5,9,7,7,5 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega^2) + \psi_S(\sigma S \cdot \omega)))$ |
| 1,5,9,7,7,5,8, 15,22,19,17,15 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega^2) + \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot \omega \cdot 2))$ |
| 1,5,9,7,7,5,8, 15,22,19,17,15,22 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega^2) + \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot \omega^2))$ |
| 1,5,9,7,7,5,8, 15,22,19,17,17,5 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega^2) + \psi_S(\sigma S \cdot \omega) \cdot 2))$ |
| 1,5,9,7,7,5,8, 15,22,19,17,20 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega^2) + \psi_S(\sigma S \cdot \omega + S)))$ |
| 1,5,9,7,7,5,8, 15,22,19,19,15 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega^2) + \psi_S(\sigma S \cdot \omega \cdot 2)))$ |
| 1,5,9,7,7,5,9 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega^2) \cdot 2))$ |
| 1,5,9,7,8 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_S(\sigma S \cdot \omega^2 + 1)))$ |
| 1,5,9,7,10 | $\psi(\psi_S(\sigma S \cdot \omega^2 + S))$ |
| 1,5,9,8 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S)) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S)) + 1))$ |
| 1,5,9,8,4,11,18, 16,8 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S)) + \alpha_2)$ |
| 1,5,9,8,4,11,18, 16,11 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S)) + \psi_S(\sigma S \cdot \omega))$ |
| 1,5,9,8,4,11,18, 16,12 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S) + 1))$ |
| 1,5,9,8,4,11,18, 16,15,11 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega)))$ |
| 1,5,9,8,4,11,18, 16,15,11,18 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega^2)))$ |
| 1,5,9,8,4,11,18, 16,15,20 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega^2 + S)))$ |
| 1,5,9,8,4,11,18, 16,16,8 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S))) + \alpha_2)$ |
| 1,5,9,8,4,11,18, 16,20,25 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S) + \psi_S(\sigma S \cdot \omega^2 + S))))$ |
| 1,5,9,8,4,11,18, 16,22 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S) + S))$ |
| 1,5,9,8,4,11,18, 16,23,8 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S) \cdot 2) + \alpha_2)$ |
| 1,5,9,8,4,11,18, 16,23,24 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S + 1)))$ |
| 1,5,9,8,4,11,18, 16,23,31 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S + S_2)))$ |
| 1,5,9,8,4,11,18, 16,24 | $\psi(\psi_S(\sigma S \cdot \omega^2 + \psi_{S_2}(\sigma S + S_\omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,9,8,4,11,18, 17 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot 2))+$ $\psi_{\beta_2}(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot 2))+1))$ |
| 1,5,9,8,5 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,9,8,5,7,10 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S))$ |
| 1,5,9,8,5,8,14 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega+S_\omega))$ |
| 1,5,9,8,5,8,15, 22,20,5 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega))+$ $\psi_{\psi_S(\sigma S\cdot(\omega+1))}(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| 1,5,9,8,5,8,15, 22,20,14 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega))+$ $\psi_S(\sigma S\cdot(\omega+1)+\psi_{S_2}(\sigma S\cdot(\omega+1)+1)))$ |
| 1,5,9,8,5,8,15, 22,20,15,22 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega))+\psi_S(\sigma S\cdot\omega^2))$ |
| 1,5,9,8,5,8,15, 22,20,17,5 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,9,8,5,8,15, 22,20,19,15,22 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega^2)))$ |
| 1,5,9,8,5,8,15, 22,20,20,5 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega)+\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega))))$ |
| 1,5,9,8,5,8,15, 22,20,26 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega)+S))$ |
| 1,5,9,8,5,8,15, 22,20,27,28 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega+1)))$ |
| 1,5,9,8,5,8,15, 22,20,28 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega+S_\omega)))$ |
| 1,5,9,8,5,8,15, 22,21 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot(\omega+1)))+$ $\psi_{\psi_S(\sigma S\cdot(\omega+2)+\psi_{S_2}(\sigma S\cdot(\omega+2)))}(\psi_S(\sigma S\cdot\omega^2+$ $\psi_{S_2}(\sigma S\cdot(\omega+1)))+1))$ |
| 1,5,9,8,5,8,15, 22,21,15 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega\cdot 2)))$ |
| 1,5,9,8,5,9 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)))$ |
| 1,5,9,8,6 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)+1))$ |
| 1,5,9,8,7,5,9 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega^2)))$ |
| 1,5,9,8,7,10 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega^2+S)))$ |
| 1,5,9,8,8,5,9 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)+$ $\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2))))$ |
| 1,5,9,8,9 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)+$ $\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)+1)))$ |
| 1,5,9,8,12 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)+S))$ |
| 1,5,9,8,13,5 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,9,8,13,5,9 | $\psi(\psi_S(\sigma S\cdot\omega^2+\psi_{S_2}(\sigma S\cdot\omega^2)\cdot 2))$ |
| 1,5,9,8,13,19 | $\psi(\psi_S(\sigma S\cdot\omega^2+S_2))$ |
| 1,5,9,8,14 | $\psi(\psi_S(\sigma S\cdot\omega^2+S_\omega))$ |
| 1,5,9,8,14,20,24, 15 | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+\psi_{S_2}(\sigma S\cdot(\omega^2+1)+S)))$ |
| 1,5,9,8,14,21 | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+S_2))$ |
| 1,5,9,8,14,22,5, 9 | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+\psi_{S_3}(\sigma S\cdot\omega^2)))$ |
| 1,5,9,8,14,22,21 | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+\psi_{S_3}(\sigma S\cdot\omega^2)+S_2))$ |
| 1,5,9,8,14,22,23 | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+\psi_{S_3}(\sigma S\cdot\omega^2+1)))$ |
| 1,5,9,8,14,22,31 | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+\psi_{S_3}(\sigma S\cdot\omega^2+S_3)))$ |
| 1,5,9,8,14,22,33 | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+\psi_{S_3}(\sigma S\cdot\omega^2+S_\omega)))$ |
| 1,5,9,8,14,23 | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+\psi_{S_3}(\sigma S\cdot(\omega^2+1))+$ $\psi_{S_2}(\sigma S\cdot(\omega^2+1)+\psi_{S_3}(\sigma S\cdot(\omega^2+1))+1)))$ |
| 1,5,9,8,14,24 | $\psi(\psi_S(\sigma S\cdot(\omega^2+1)+S_\omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,9,8,15 | $\psi(\psi_S(\sigma S \cdot (\omega^2 + \omega)))$ |
| 1,5,9,8,15,21,15 | $\psi(\psi_S(\sigma S \cdot (\omega^2 + \omega) + \psi_{S_2}(\sigma S \cdot (\omega^2 + \omega))))$ |
| 1,5,9,8,15,21,32 | $\psi(\psi_S(\sigma S \cdot (\omega^2 + \omega \cdot 2)))$ |
| 1,5,9,8,15,22 | $\psi(\psi_S(\sigma S \cdot (\omega^2 \cdot 2)))$ |
| 1,5,9,9 | $\psi(\psi_S(\sigma S \cdot (\omega^3)))$ |
| 1,5,9,9,7,5,9 | $\psi(\psi_S(\sigma S \cdot (\omega^3) + \psi_S(\sigma S \cdot \omega^2)))$ |
| 1,5,9,9,7,5,9, 8,15,22,22,19,15 | $\psi(\psi_S(\sigma S \cdot (\omega^3) + \psi_S(\sigma S \cdot (\omega^2 + \omega))))$ |
| 1,5,9,9,7,5,9, 9 | $\psi(\psi_S(\sigma S \cdot (\omega^3) + \psi_S(\sigma S \cdot (\omega^3))))$ |
| 1,5,9,9,7,10 | $\psi(\psi_S(\sigma S \cdot (\omega^3) + S))$ |
| 1,5,9,9,8,5,9, 9 | $\psi(\psi_S(\sigma S \cdot (\omega^3) + \psi_{S_2}(\sigma S \cdot (\omega^3))))$ |
| 1,5,9,9,8,12 | $\psi(\psi_S(\sigma S \cdot (\omega^3) + \psi_{S_2}(\sigma S \cdot (\omega^3)) + S))$ |
| 1,5,9,9,8,13,14 | $\psi(\psi_S(\sigma S \cdot (\omega^3) + \psi_{S_2}(\sigma S \cdot (\omega^3) + 1)))$ |
| 1,5,9,9,8,14 | $\psi(\psi_S(\sigma S \cdot (\omega^3) + S_\omega))$ |
| 1,5,9,9,8,15,22, 22 | $\psi(\psi_S(\sigma S \cdot \omega^3 \cdot 2))$ |
| 1,5,9,10 | $\psi(\psi_S(\sigma S \cdot \omega^\omega))$ |
| 1,5,9,11 | $\psi(\psi_S(\sigma S \cdot \Omega))$ |
| 1,5,9,11,4 | $\psi(\psi_S(\sigma S \cdot \Omega_\omega))$ |
| 1,5,9,11,4,11,18, 20,5 | $\psi(\psi_S(\sigma S \cdot \alpha))$ |
| 1,5,9,11,4,11,18, 22 | $\psi(\psi_S(\sigma S \cdot \alpha_2))$ |
| 1,5,9,11,4,11,18, 22,10,21,32,36,10 | $\psi(\psi_S(\sigma S \cdot \psi_\beta(\psi_S(\sigma S \cdot \alpha_\omega))))$ |
| 1,5,9,11,5 | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega)))$ |
| 1,5,9,11,5,7,10 | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot \omega + S))$ |
| 1,5,9,11,5,8,14 | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot \omega + S_\omega))$ |
| 1,5,9,11,5,8,15, 22,24,5 | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega)) + \psi_{\psi_S(\sigma S \cdot (\omega+1))}(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega))))$ |
| 1,5,9,11,5,8,15, 22,24,12 | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot (\omega + 1)))$ |
| 1,5,9,11,5,8,15, 22,24,15 | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot \omega \cdot 2))$ |
| 1,5,9,11,5,8,15, 22,24,15,22,24,5 | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega)) \cdot 2)$ |
| 1,5,9,11,5,8,15, 22,24,19,24 | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega) + S))$ |
| 1,5,9,11,5,8,15, 22,24,21,15,22,24,5 | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot \omega))))$ |
| 1,5,9,11,5,8,15, 22,24,21,32 | $\psi(\psi_S(\sigma S \cdot (\psi_S(\sigma S \cdot \omega) + \omega)))$ |
| 1,5,9,11,5,8,15, 22,24,22 | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega + 1)))$ |
| 1,5,9,11,5,8,15, 22,24,27 | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega + S)))$ |
| 1,5,9,11,5,8,15, 22,25,5 | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega))))$ |
| 1,5,9,11,5,8,15, 22,25,32,39 | $\psi(\psi_S(\sigma S \cdot \psi_{\psi_S(\sigma S \cdot (\omega+1))}(\psi_S(\sigma S \cdot \omega^2))))$ |
| 1,5,9,11,5,8,15, 22,26 | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot (\omega + 1))))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,9,11,5,8,15, 22,26,15 | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega \cdot 2)))$ |
| 1,5,9,11,5,9 | $\psi(\psi_S(\sigma S \cdot \psi_S(\sigma S \cdot \omega^2)))$ |
| 1,5,9,11,6 | $\psi(\psi_S(\sigma S \cdot S))$ |
| 1,5,9,11,6,5 | $\psi(\psi_S(\sigma S \cdot S) + \psi_S(\sigma S \cdot \omega))$ |
| 1,5,9,11,6,6 | $\psi(\psi_S(\sigma S \cdot S + 1))$ |
| 1,5,9,11,7 | $\psi(\psi_S(\sigma S \cdot S + \Omega))$ |
| 1,5,9,11,7,5 | $\psi(\psi_S(\sigma S \cdot S + \psi_S(\sigma S \cdot \omega)))$ |
| 1,5,9,11,7,5,9 | $\psi(\psi_S(\sigma S \cdot S + \psi_S(\sigma S \cdot \omega^2)))$ |
| 1,5,9,11,7,5,9, 11,6 | $\psi(\psi_S(\sigma S \cdot S + \psi_S(\sigma S \cdot S)))$ |
| 1,5,9,11,7,8 | $\psi(\psi_S(\sigma S \cdot S + \psi_S(\sigma S \cdot S + 1)))$ |
| 1,5,9,11,7,10 | $\psi(\psi_S(\sigma S \cdot S + S))$ |
| 1,5,9,11,8,5 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \omega)))$ |
| 1,5,9,11,8,5,9, 11,6 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S))))$ |
| 1,5,9,11,8,6 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S)) + 1))$ |
| 1,5,9,11,8,7,10 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S)) + \psi_S(\sigma S \cdot S + S)))$ |
| 1,5,9,11,8,9 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S))+$ $\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S)) + 1)))$ |
| 1,5,9,11,8,12 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S)) + S))$ |
| 1,5,9,11,8,13,14 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) + 1)))$ |
| 1,5,9,11,8,13,19 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) + S_2)))$ |
| 1,5,9,11,8,14 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) + S_\omega)))$ |
| 1,5,9,11,8,15 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot (\psi_S(\sigma S \cdot S) + \omega))))$ |
| 1,5,9,11,8,15,22, 24,5 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot (\psi_S(\sigma S \cdot S) + \psi_S(\sigma S \cdot \omega)))))$ |
| 1,5,9,11,8,15,22, 24,5,9,11,6 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) \cdot 2)))$ |
| 1,5,9,11,8,15,22, 24,16 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) \cdot 2) + 1))$ |
| 1,5,9,11,8,15,22, 24,19,24 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) \cdot 2) + S))$ |
| 1,5,9,11,8,15,22, 24,20,5,9,11,6 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) \cdot 2)+$ $\psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S))))$ |
| 1,5,9,11,8,15,22, 24,20,27,35 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) \cdot 2)+$ $\psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) + S_2)))$ |
| 1,5,9,11,8,15,22, 24,20,28 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) \cdot 2)+$ $\psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) + S_\omega)))$ |
| 1,5,9,11,8,15,22, 24,21 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) \cdot 2)+$ $\psi_{S_2}(\sigma S \cdot (\psi_S(\sigma S \cdot S) + 1)))+$ $\psi_{\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot (\psi_S(\sigma S \cdot S)+1)) \cdot 2)}(\psi_S(\sigma S \cdot S+$ $\psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) \cdot 2)+$ $\psi_{S_2}(\sigma S \cdot (\psi_S(\sigma S \cdot S) + 1))) + 1))$ |
| 1,5,9,11,8,15,22, 24,21,15 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) \cdot 2)+$ $\psi_{S_2}(\sigma S \cdot (\psi_S(\sigma S \cdot S) + \omega))))$ |
| 1,5,9,11,8,15,22, 24,21,15,22,24,5,9, 11,6 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S)) \cdot 2))$ |
| 1,5,9,11,8,15,22, 24,21,30,31 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) \cdot 2 + 1)))$ |
| 1,5,9,11,8,15,22, 24,21,31 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) \cdot 2 + S_\omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,9,11,8,15,22, 24,21,32,43,45,5,9, 11,6 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) \cdot 3)))$ |
| 1,5,9,11,8,15,22, 24,22 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S + 1))))$ |
| 1,5,9,11,8,15,22, 24,22,24,5,9,11,6 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S + \psi_S(\sigma S \cdot S)))))$ |
| 1,5,9,11,8,15,22, 24,27 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S + S))))$ |
| 1,5,9,11,8,15,22, 25,5,9,11,6 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S+ \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S))))))$ |
| 1,5,9,11,8,15,22, 25,31 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S+ \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S) + S_\omega)))))$ |
| 1,5,9,11,8,15,22, 25,32 | $\psi(\psi_S(\sigma S \cdot S+ \psi_{S_2}(\sigma S \cdot \psi_{\psi_S(\sigma S \cdot S+\psi_{S_2}(\sigma S \cdot (\psi_S(\sigma S \cdot S)+1)))} (\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot (\psi_S(\sigma S \cdot S) + \omega)))))))$ |
| 1,5,9,11,8,15,22, 26 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S+ \psi_{S_2}(\sigma S \cdot (\psi_S(\sigma S \cdot S) + 1))))))$ |
| 1,5,9,11,8,15,22, 26,15 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S+ \psi_{S_2}(\sigma S \cdot (\psi_S(\sigma S \cdot S) + \omega))))))$ |
| 1,5,9,11,8,15,22, 26,16 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,9,11,9 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + 1)))$ |
| 1,5,9,11,9,7,10 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + 1) + S))$ |
| 1,5,9,11,9,8,5, 9,11,9 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + 1)+ \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + 1)))))$ |
| 1,5,9,11,9,8,14 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + 1)+ \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + 1)) + S_\omega)))$ |
| 1,5,9,11,9,8,15, 22,26,16 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + 1) + \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,9,11,9,8,15, 22,26,22 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S + 1) \cdot 2))$ |
| 1,5,9,11,9,9 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + 2)))$ |
| 1,5,9,11,9,11,5 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + \psi_S(\sigma S \cdot \omega))))$ |
| 1,5,9,11,9,11,6 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + S)))$ |
| 1,5,9,11,9,11,7, 10 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + S) + S))$ |
| 1,5,9,11,9,11,8, 14 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + S)+ \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + S)) + S_\omega)))$ |
| 1,5,9,11,9,11,8, 15,22,26,16 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + S) + \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,9,11,9,11,8, 15,22,26,22,26,16 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + S) \cdot 2))$ |
| 1,5,9,11,9,11,9 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + S + 1)))$ |
| 1,5,9,11,11,6 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + S \cdot 2)))$ |
| 1,5,9,11,14 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + \psi_{S_2}(\sigma S))))$ |
| 1,5,9,12 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + \psi_{S_2}(\sigma S)))+ \psi_{\alpha_2}(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + \psi_{S_2}(\sigma S))) + 1))$ |
| 1,5,9,12,5 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot \omega))))$ |
| 1,5,9,12,6 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S))))$ |
| 1,5,9,12,6,6 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S)) + 1))$ |
| 1,5,9,12,7,10 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S)) + S))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,9,12,8,15,22, 26,31 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+ \psi_{S_2}(\sigma S\cdot\psi_{S_2}(\sigma S))))$ |
| 1,5,9,12,8,15,22, 27,5,9,12,6 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))))))$ |
| 1,5,9,12,8,15,22, 27,15 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+\psi_{S_2}(\sigma S\cdot S+ \psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))))))+ \psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+ \psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))+\omega))))$ |
| 1,5,9,12,8,15,22, 27,16 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))))+1))$ |
| 1,5,9,12,8,15,22, 27,19,24 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))))+S))$ |
| 1,5,9,12,8,15,22, 27,21,32,43,50,33 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+ \psi_{S_2}(\sigma S\cdot S))+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))))+\psi_{S_2}(\sigma S\cdot S)))$ |
| 1,5,9,12,8,15,22, 27,21,32,43,51,5,9, 12,6 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))))\cdot 2))$ |
| 1,5,9,12,8,15,22, 27,22 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))))+1)))$ |
| 1,5,9,12,8,15,22, 27,22,26,31 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))))+\psi_{S_2}(\sigma S))))$ |
| 1,5,9,12,8,15,22, 27,22,27,5,9,12,6 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))))\cdot 2)))$ |
| 1,5,9,12,8,15,22, 27,33 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))+1))))$ |
| 1,5,9,12,8,15,22, 27,35 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))+S_\omega))))$ |
| 1,5,9,12,8,15,22, 28,14,25,36,45,26 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot S+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))+1)))+1))$ |
| 1,5,9,12,8,15,22, 28,15 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot S+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)))+\omega)))))$ |
| 1,5,9,12,8,15,22, 28,16 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))+ \psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S))))$ |
| 1,5,9,12,9 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)+1)))$ |
| 1,5,9,12,9,11,6 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)+S)))$ |
| 1,5,9,12,9,11,14 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)+\psi_{S_2}(\sigma S))))$ |
| 1,5,9,12,9,12,6 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S)\cdot 2)))$ |
| 1,5,9,12,11,6 | $\psi(\psi_S(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+\psi_{S_2}(\sigma S\cdot S+S))))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,5,9,12,16 | $\psi(\psi_S(\sigma S \cdot S + S_2))$ |
| 1,5,9,12,17,6 | $\psi(\psi_S(\sigma S \cdot S + \psi_{S_3}(\sigma S \cdot S)))$ |
| 1,5,9,12,18 | $\psi(\psi_S(\sigma S \cdot S + S_\omega))$ |
| 1,5,9,12,19 | $\psi(\psi_S(\sigma S \cdot (S + \omega)))$ |
| 1,5,9,12,19,26,32, 20 | $\psi(\psi_S(\sigma S \cdot S \cdot 2))$ |
| 1,5,9,13 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega))$ |
| 1,5,9,13,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + 1))$ |
| 1,5,9,13,7,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S))$ |
| 1,5,9,13,8,5 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \omega)))$ |
| 1,5,9,13,8,5,9, 11,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S))))$ |
| 1,5,9,13,8,5,9, 12,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S+ \psi_{S_2}(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S))))))$ |
| 1,5,9,13,8,5,9, 12,18 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S + S_\omega))))$ |
| 1,5,9,13,8,5,9, 12,19 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S\cdot \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S \cdot (S + \omega))))))$ |
| 1,5,9,13,8,5,9, 12,19,26,33 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S \cdot S \cdot \omega)))))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S\cdot \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S \cdot S \cdot \omega))))+ \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S \cdot S \cdot \omega) + 1))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,7,5,9, 12,18 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_{\psi_S(\sigma S\cdot(S+1))} (\psi_S(\sigma S \cdot S \cdot \omega)))) + \psi_{\psi_S(\sigma S\cdot(S+1))} (\psi_S(\sigma S \cdot S \cdot \omega) + \psi_S(\sigma S \cdot S + S_\omega)))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,7,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_{\psi_S(\sigma S\cdot(S+1))} (\psi_S(\sigma S \cdot S \cdot \omega)))) + \psi_{\psi_S(\sigma S\cdot(S+1))} (\psi_S(\sigma S \cdot S \cdot \omega) + \psi_S(\sigma S \cdot (S + 1))))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 26,16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega+ \psi_{S_2}(\sigma S \cdot \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S \cdot S \cdot \omega))))+ \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S \cdot S \cdot \omega)+ \psi_{\psi_S(\sigma S\cdot(S+1)+\psi_{S_2}(\sigma S\cdot(S+1)))} (\sigma\psi_S(\sigma S \cdot (S + 1)) \cdot \psi_S(\sigma S \cdot (S + 1)))))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 28,38 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S \cdot S \cdot \omega))))+ \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S \cdot S \cdot \omega) +\psi_{\psi_S(\sigma S\cdot(S+1)+\psi_{S_2}(\sigma S\cdot(S+1)))}(\sigma\psi_S(\sigma S \cdot (S + 1))+ \psi_S(\sigma S \cdot (S + 1) + \psi_{S_2}(\sigma S \cdot (S + 1) + 1)))))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 29 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_{\psi_S(\sigma S\cdot(S+1))} (\psi_S(\sigma S \cdot S \cdot \omega)))) + \psi_{\psi_S(\sigma S\cdot(S+1))} (\psi_S(\sigma S \cdot S \cdot \omega) + \psi_{\psi_S(\sigma S\cdot(S+1)+\psi_{S_2}(\sigma S\cdot(S+1)))} (\psi_S(\sigma S \cdot S \cdot \omega))))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 29,20,5,9,12,19,26, 33 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_{\psi_S(\sigma S\cdot(S+1))} (\psi_S(\sigma S \cdot S \cdot \omega)))) + \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S \cdot S \cdot \omega)+ \psi_{\psi_S(\sigma S\cdot(S+1)+\psi_{S_2}(\sigma S\cdot(S+1)))}(\psi_S(\sigma S \cdot S \cdot \omega+ \psi_{S_2}(\sigma S \cdot \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S \cdot S \cdot \omega)))))))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 29,20,15,22,29 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega+ \psi_{S_2}(\sigma S \cdot \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S \cdot S \cdot \omega))))+ \psi_S(\sigma S \cdot S \cdot \omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 29,20,16 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+ \psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))+1))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 29,20,19,24 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+ \psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))+ \psi_S(\sigma S\cdot S\cdot\omega+S)))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 29,20,20,5,9,12,19, 26,33 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+ \psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))+ \psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))} (\psi_S(\sigma S\cdot S\cdot\omega))))))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 29,20,21 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))+ \psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))} (\psi_S(\sigma S\cdot S\cdot\omega)))+1)))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 29,20,24,29 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))+ \psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))} (\psi_S(\sigma S\cdot S\cdot\omega)))+\psi_S(\sigma S\cdot S\cdot\omega+S))))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 29,20,26 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+ \psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))+S))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 29,20,27,5 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+ \psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))+ \psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 29,20,27,26 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega)))\cdot 2+S))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 29,20,27,28 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega))+1)))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 29,20,27,35 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega))+S_2)))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 29,20,28 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot \psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega))+S_\omega)))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 29,20,29 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot (\psi_{\psi_S(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot S\cdot\omega))+\omega))))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,8,15,22, 29,20,29,38,44,30 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S+1)))))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,9 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+ \psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S+1)+\psi_{S_2}(\sigma S\cdot(S+1)))} (\psi_S(\sigma S\cdot S\cdot\omega)+1))))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,18 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+ \psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S+1)+\psi_{S_2}(\sigma S\cdot(S+1)+1)))))$ |
| 1,5,9,13,8,5,9, 12,19,26,33,25,19 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+ \psi_{S_2}(\sigma S\cdot\psi_{\psi_S(\sigma S\cdot(S\cdot 2+1))}(\psi_S(\sigma S\cdot(S\cdot 2+\omega))))))$ |
| 1,5,9,13,8,5,9, 13 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega))))$ |
| 1,5,9,13,8,6 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega))+1))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,9,13,8,7,5, 9,13 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega))+ \psi_S(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,9,13,8,7,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega))+ \psi_S(\sigma S \cdot S \cdot \omega + S)))$ |
| 1,5,9,13,8,8,5, 9,13 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega))+ \psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega)))))$ |
| 1,5,9,13,8,12 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega)) + S))$ |
| 1,5,9,13,8,13,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega) + 1)))$ |
| 1,5,9,13,8,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega) + S_\omega)))$ |
| 1,5,9,13,8,15,22, 26,16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,9,13,8,15,22, 28,38 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S + S_\omega)))$ |
| 1,5,9,13,8,15,22, 29 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,9,13,8,15,22, 29,21,32,43,50,33 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega) + \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,9,13,9 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + 1)))$ |
| 1,5,9,13,9,8,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + 1) + S_\omega))$ |
| 1,5,9,13,9,9 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + 2)))$ |
| 1,5,9,13,9,11,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S)))$ |
| 1,5,9,13,9,11,9 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S + 1)))$ |
| 1,5,9,13,9,11,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S))))$ |
| 1,5,9,13,9,12,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S))))$ |
| 1,5,9,13,9,12,16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S + S_2))))$ |
| 1,5,9,13,9,12,19 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega+ \psi_{\psi_{S_2}(\sigma S \cdot (S+1))}(\psi_{S_2}(\sigma S \cdot (S + \omega))))))$ |
| 1,5,9,13,9,12,19, 26,33 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega+ \psi_{\psi_{S_2}(\sigma S \cdot (S+1))}(\psi_{S_2}(\sigma S \cdot S \cdot \omega)))))$ |
| 1,5,9,13,9,13 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega))))$ |
| 1,5,9,13,9,13,9 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega) + 1)))$ |
| 1,5,9,13,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + 1))))$ |
| 1,5,9,13,11,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S))))$ |
| 1,5,9,13,11,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega+ \psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S)))))$ |
| 1,5,9,13,12,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega+ \psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S)))))$ |
| 1,5,9,13,12,9 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega+ \psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S)) + 1)))$ |
| 1,5,9,13,12,9,11, 6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega+ \psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S)) + S)))$ |
| 1,5,9,13,12,9,12, 6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega+ \psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S)) + \psi_{S_2}(\sigma S \cdot S))))$ |
| 1,5,9,13,12,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega+ \psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S) + 1))))$ |
| 1,5,9,13,12,12,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega+ \psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S) \cdot 2))))$ |
| 1,5,9,13,12,16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega+ \psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S + S_2)))))$ |
| 1,5,9,13,13 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega+ \psi_{S_2}(\sigma S \cdot S \cdot \omega)))))$ |
| 1,5,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,10,5 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2) + \psi_S(\sigma S \cdot \omega))$ |
| 1,5,10,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + 1))$ |
| 1,5,10,7,5 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_S(\sigma S \cdot \omega)))$ |
| 1,5,10,7,5,9,11, 6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_S(\sigma S \cdot S)))$ |
| 1,5,10,7,5,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_S(\sigma S \cdot S \cdot \omega + S_2)))$ |
| 1,5,10,7,8 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_S(\sigma S \cdot S \cdot \omega + S_2 + 1)))$ |
| 1,5,10,7,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + S))$ |
| 1,5,10,7,12 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S)) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,10,8 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S)) + 1))$ |
| 1,5,10,8,5,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega + S_2))))$ |
| 1,5,10,8,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega + S_2)) + 1))$ |
| 1,5,10,8,7,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega + S_2))+$ $\psi_S(\sigma S \cdot S \cdot \omega + S_2 + S)))$ |
| 1,5,10,8,8,5,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega + S_2))+$ $\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega)))))$ |
| 1,5,10,8,12 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega + S_2)) + S))$ |
| 1,5,10,8,13,5,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega + S_2)) \cdot 2))$ |
| 1,5,10,8,13,12 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2+$ $\psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega + S_2)) \cdot 2 + S))$ |
| 1,5,10,8,13,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega + S_2) + 1)))$ |
| 1,5,10,8,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2+$ $\psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega + S_2) + S_\omega)))$ |
| 1,5,10,8,15 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot (\psi_S(\sigma S \cdot S \cdot \omega) + \omega))))$ |
| 1,5,10,8,15,22,26, 16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,10,8,15,22,29 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,10,8,15,23 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2)))$ |
| 1,5,10,9 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + 1)))$ |
| 1,5,10,9,9 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + 2)))$ |
| 1,5,10,9,11,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + S)))$ |
| 1,5,10,9,11,8,15, 23 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + S)+$ $\psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2)))$ |
| 1,5,10,9,11,8,15, 23,22,26,16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + S) \cdot 2))$ |
| 1,5,10,9,11,9 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + S + 1)))$ |
| 1,5,10,9,11,9,11, 6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + S \cdot 2)))$ |
| 1,5,10,9,11,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S))))$ |
| 1,5,10,9,12,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S))))$ |
| 1,5,10,9,12,18 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2+$ $\psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S + S_\omega))))$ |
| 1,5,10,9,13 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2+$ $\psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega))))$ |
| 1,5,10,9,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2+$ $\psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2))))$ |
| 1,5,10,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 \cdot 2))$ |
| 1,5,10,12,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 \cdot S))$ |
| 1,5,10,12,15 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 \cdot \psi_{S_2}(\sigma S)))$ |
| 1,5,10,13,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 \cdot \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,10,13,17 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 \cdot \psi_{S_2}(\sigma S \cdot S + S_2)))$ |
| 1,5,10,13,19 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 \cdot \psi_{S_2}(\sigma S \cdot S + S_\omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,10,13,20,28 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 \cdot \psi_{\psi_{S_2}(\sigma S \cdot (S+1))}(\psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2))))$ |
| 1,5,10,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_2 \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,10,15 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2))$ |
| 1,5,10,15,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 + 1))$ |
| 1,5,10,15,8,15,22, 26,16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 + \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,10,15,9 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + {S_2}^2 + 1)))$ |
| 1,5,10,15,9,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + {S_2}^2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2))))$ |
| 1,5,10,15,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 + S_2))$ |
| 1,5,10,15,10,12,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 + S_2 \cdot S))$ |
| 1,5,10,15,10,12,15 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 + S_2 \cdot \psi_{S_2}(\sigma S)))$ |
| 1,5,10,15,10,13,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 + S_2 \cdot \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,10,15,10,13,19 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 + S_2 \cdot \psi_{S_2}(\sigma S \cdot S + S_\omega)))$ |
| 1,5,10,15,10,13,20, 28,36 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 + S_2 \cdot \psi_{\psi_{S_2}(\sigma S \cdot (S+1))}(\psi_S(\sigma S \cdot S \cdot \omega + S^2))))$ |
| 1,5,10,15,10,13,20, 28,36,28,34,21 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 + S_2 \cdot \psi_{S_2}(\sigma S \cdot (S+1))) + \psi_{\psi_S(\sigma S \cdot (S+2) + \psi_{S_2}(\sigma S \cdot (S+2)))}(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 + S_2 \cdot \psi_{S_2}(\sigma S \cdot (S+1))) + 1))$ |
| 1,5,10,15,10,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 + S_2 \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,10,15,10,14,19 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 + S_2 \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2)))$ |
| 1,5,10,15,10,15 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 \cdot 2))$ |
| 1,5,10,15,11 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 \cdot \omega))$ |
| 1,5,10,15,12,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 \cdot S))$ |
| 1,5,10,15,13,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 \cdot \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,10,15,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^2 \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,10,15,15 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^3))$ |
| 1,5,10,15,20 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_2}^{S_2}))$ |
| 1,5,10,16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)))$ |
| 1,5,10,16,16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)) + \psi_\alpha(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S))))$ |
| 1,5,10,17 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)) + \Omega_{\alpha+1} \cdot \omega)$ |
| 1,5,10,18 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,11 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)) + 1))$ |
| 1,5,11,4,11,20,6, 9 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)) + 1) + \alpha)$ |
| 1,5,11,4,11,20,8 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)) + \alpha_2)$ |
| 1,5,11,4,11,20,11 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S)) + \psi_S(\sigma S \cdot \omega))$ |
| 1,5,11,4,11,20,12 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S) + 1))$ |
| 1,5,11,4,11,20,19 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S) + S_2))$ |
| 1,5,11,4,11,20,21 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S + 1)))$ |
| 1,5,11,4,11,20,32 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S + S_\omega)))$ |
| 1,5,11,4,11,20,32 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot 2)) + \psi_{\beta_2}(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot 2)) + 1))$ |
| 1,5,11,5 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega)))$ |
| 1,5,11,5,8,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega)) + \psi_S(\sigma S \cdot \omega + S_\omega))$ |
| 1,5,11,5,8,15,24, 6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega)) + \psi_{\psi_S(\sigma S \cdot (\omega+1))}(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega)) + 1))$ |
| 1,5,11,5,8,15,24, 16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega) + 1))$ |
| 1,5,11,5,8,15,24, 23 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega) + S_2))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,11,5,8,15,24, 24,5 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega) \cdot 2))$ |
| 1,5,11,5,8,15,24, 32 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega + S_2)))$ |
| 1,5,11,5,8,15,24, 34 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega + S_3)))$ |
| 1,5,11,5,8,15,25 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot (\omega + 1)))+$ $\psi_{\psi_S(\sigma S \cdot (\omega+2)+\psi_{S_2}(\sigma S \cdot (\omega+1)))}$ $(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot (\omega + 1))) + 1))$ |
| 1,5,11,5,8,15,25, 15 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega \cdot 2)))$ |
| 1,5,11,5,9 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot \omega^2)))$ |
| 1,5,11,5,9,11,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot \psi_S(\sigma S \cdot S))))$ |
| 1,5,11,5,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega + S_2))))$ |
| 1,5,11,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S)))$ |
| 1,5,11,6,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S) + 1))$ |
| 1,5,11,7,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S) + S))$ |
| 1,5,11,8,5,11,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S) + \psi_{S_2}$ $(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S)))))$ |
| 1,5,11,8,15,22,26, 16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S) + \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,11,8,15,23 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S)+$ $\psi_{\psi_{S_2}(\sigma S \cdot (S+1))}(\psi_S(\sigma S \cdot S \cdot \omega + S_2))))$ |
| 1,5,11,9 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S) + \psi_{S_2}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,11,9,9 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S) + \psi_{S_2}(\sigma S \cdot S \cdot \omega + 1)))$ |
| 1,5,11,9,11,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S) + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S)))$ |
| 1,5,11,9,13 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S)+$ $\psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega))))$ |
| 1,5,11,9,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S) + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2)))$ |
| 1,5,11,9,15,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S)+$ $\psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S))))$ |
| 1,5,11,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S) + S_2))$ |
| 1,5,11,10,15 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S) + {S_2}^2))$ |
| 1,5,11,10,16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S) + \psi_{S_3}(\sigma S)))$ |
| 1,5,11,11 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S) + \psi_{S_3}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S \cdot S \cdot \omega+$ $\psi_{S_3}(\sigma S \cdot S) + \psi_{S_3}(\sigma S)) + 1))$ |
| 1,5,11,11,5 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S) + \psi_{S_3}(\sigma S \cdot \omega)))$ |
| 1,5,11,11,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S) \cdot 2))$ |
| 1,5,11,12 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + 1)))$ |
| 1,5,11,13,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + S)))$ |
| 1,5,11,14,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S))))$ |
| 1,5,11,14,21 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S+$ $\psi_{\psi_S(\sigma S \cdot (S+1))}(\psi_S(\sigma S \cdot \omega)))))$ |
| 1,5,11,15 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S \cdot \omega))))$ |
| 1,5,11,15,20 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2))))$ |
| 1,5,11,16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + S_2)))$ |
| 1,5,11,16,16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + S_2 \cdot 2)))$ |
| 1,5,11,16,22 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + \psi_{S_3}(\sigma S))))$ |
| 1,5,11,17 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + \psi_{S_3}(\sigma S)))+$ $\psi_{\alpha_2}(\psi_S(\sigma S \cdot S \cdot \omega+$ $\psi_{S_3}(\sigma S \cdot S + \psi_{S_3}(\sigma S))) + 1))$ |
| 1,5,11,17,5 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + \psi_{S_3}(\sigma S \cdot \omega))))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,11,17,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + \psi_{S_3}(\sigma S \cdot S))))$ |
| 1,5,11,17,11,17,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + \psi_{S_3}(\sigma S \cdot S)) \cdot 2))$ |
| 1,5,11,17,16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + \psi_{S_3}(\sigma S \cdot S) + S_2)))$ |
| 1,5,11,17,17,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + \psi_{S_3}(\sigma S \cdot S) \cdot 2)))$ |
| 1,5,11,17,18 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + \psi_{S_3}(\sigma S \cdot S + 1))))$ |
| 1,5,11,17,22 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + \psi_{S_3}(\sigma S \cdot S + S_2))))$ |
| 1,5,11,18 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + S_3)))$ |
| 1,5,11,18,18 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + S_3 \cdot 2)))$ |
| 1,5,11,19,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + \psi_{S_4}(\sigma S \cdot S))))$ |
| 1,5,11,19,28 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + S_4)))$ |
| 1,5,11,20 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + S_\omega)))$ |
| 1,5,11,20,27,35 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + \psi_{S_{\omega+1}}(\sigma S))))$ |
| 1,5,11,20,29 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S + S_{\omega^2})))$ |
| 1,5,11,21 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{\psi_{S_3}(\sigma S \cdot (S+1))}(\psi_S(\sigma S \cdot (S + \omega)))))$ |
| 1,5,11,21,33,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega +$ $\psi_{\psi_{S_3}(\sigma S \cdot (S+1))}(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S)))))$ |
| 1,5,11,21,34 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot (S + 1)))+$ $\psi_{\psi_S(\sigma S \cdot (S+2) + \psi_{S_2}(\sigma S \cdot (S+2)))}$ $(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot (S + 1))) + 1))$ |
| 1,5,11,21,34,21 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot (S + \omega))))$ |
| 1,5,11,21,34,22 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot 2)))$ |
| 1,5,12 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,12,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega) + 1))$ |
| 1,5,12,7,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega) + S))$ |
| 1,5,12,8,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)+$ $\psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)) + S_\omega)))$ |
| 1,5,12,8,15,22,26, 16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega) + \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,12,9 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)+$ $\psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega) + 1)))$ |
| 1,5,12,9,9 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)+$ $\psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega) + 2)))$ |
| 1,5,12,9,11,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)+$ $\psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega) + S)))$ |
| 1,5,12,9,13 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)+$ $\psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)+$ $\psi_{S_2}(\sigma S \cdot S \cdot \omega))))$ |
| 1,5,12,9,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)+$ $\psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)+$ $\psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2))))$ |
| 1,5,12,9,16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)+$ $\psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)+$ $\psi_{S_2}(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)))))$ |
| 1,5,12,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega) + S_2))$ |
| 1,5,12,10,16 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega) + \psi_{S_3}(\sigma S)))$ |
| 1,5,12,11,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega) + \psi_{S_3}(\sigma S \cdot S)))$ |
| 1,5,12,11,12 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega) + \psi_{S_3}(\sigma S \cdot S + 1)))$ |
| 1,5,12,11,17,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)+$ $\psi_{S_3}(\sigma S \cdot S + \psi_{S_3}(\sigma S \cdot S))))$ |
| 1,5,12,11,18 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega) + \psi_{S_3}(\sigma S \cdot S + S_3)))$ |
| 1,5,12,11,19,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)+$ $\psi_{S_3}(\sigma S \cdot S + \psi_{S_4}(\sigma S \cdot S))))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,12,11,20 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+\psi_{S_3}(\sigma S\cdot S+S_\omega)))$ |
| 1,5,12,11,21 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+$ $\psi_{\psi_{S_3}(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot(S+\omega)))))$ |
| 1,5,12,12 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)\cdot 2))$ |
| 1,5,12,13 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+1)))$ |
| 1,5,12,14,6 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S)))$ |
| 1,5,12,15,6 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S))))$ |
| 1,5,12,17 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_2)))$ |
| 1,5,12,17,12 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_2)+\psi_{S_3}(\sigma S\cdot S\cdot\omega)))$ |
| 1,5,12,17,13 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_2+1)))$ |
| 1,5,12,17,17 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_2\cdot 2)))$ |
| 1,5,12,17,23 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S))))$ |
| 1,5,12,18,5 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot\omega))))$ |
| 1,5,12,18,6 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S))))$ |
| 1,5,12,18,18,6 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)\cdot 2)))$ |
| 1,5,12,18,20,6 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+S))))$ |
| 1,5,12,18,23 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+S_2))))$ |
| 1,5,12,18,24,6 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+$ $\psi_{S_3}(\sigma S\cdot S+\psi_{S_3}(\sigma S\cdot S)))))$ |
| 1,5,12,18,25 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+S_3))))$ |
| 1,5,12,18,26,6 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+$ $\psi_{S_3}(\sigma S\cdot S+\psi_{S_4}(\sigma S\cdot S)))))$ |
| 1,5,12,18,26,27 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+$ $\psi_{S_3}(\sigma S\cdot S+\psi_{S_4}(\sigma S\cdot S+1)))))$ |
| 1,5,12,18,27 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S+S_\omega))))$ |
| 1,5,12,18,28 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+$ $\psi_{\psi_{S_3}(\sigma S\cdot(S+1))}(\psi_S(\sigma S\cdot(S+\omega))))))$ |
| 1,5,12,19 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega))))$ |
| 1,5,12,19,12 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+$ $\psi_{S_3}(\sigma S\cdot S\cdot\omega))+\psi_{S_3}(\sigma S\cdot S\cdot\omega)))$ |
| 1,5,12,19,13 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+1)))$ |
| 1,5,12,19,17 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)+S_2)))$ |
| 1,5,12,19,18,6 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+$ $\psi_{S_3}(\sigma S\cdot S\cdot\omega)+\psi_{S_3}(\sigma S\cdot S))))$ |
| 1,5,12,19,19 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega)\cdot 2)))$ |
| 1,5,12,19,24 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+S_2))))$ |
| 1,5,12,19,25,6 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S\cdot\omega+$ $\psi_{S_3}(\sigma S\cdot S\cdot\omega+\psi_{S_3}(\sigma S\cdot S)))))$ |
| 1,5,12,20 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_3))$ |
| 1,5,12,21,6 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_4}(\sigma S\cdot S)))$ |
| 1,5,12,22 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+\psi_{S_4}(\sigma S\cdot S\cdot\omega)))$ |
| 1,5,13 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega))$ |
| 1,5,13,6 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+1))$ |
| 1,5,13,7,10 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+S))$ |
| 1,5,13,8 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S))+1))$ |
| 1,5,13,8,5 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,13,8,5,13 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega+S_\omega))))$ |
| 1,5,13,8,14 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot S\cdot\omega+S_\omega)+S_\omega)))$ |
| 1,5,13,8,15,22,26, 16 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot S)))$ |
| 1,5,13,8,15,22,29 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))$ |
| 1,5,13,8,15,27 | $\psi(\psi_S(\sigma S\cdot S\cdot\omega+S_\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_\omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,13,9 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_\omega + 1)))$ |
| 1,5,13,9,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_\omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2))))$ |
| 1,5,13,9,17 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_\omega + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_\omega))))$ |
| 1,5,13,10 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega + S_2))$ |
| 1,5,13,11,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega + \psi_{S_3}(\sigma S \cdot S)))$ |
| 1,5,13,11,12 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega + \psi_{S_3}(\sigma S \cdot S + 1)))$ |
| 1,5,13,11,17,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega + \psi_{S_3}(\sigma S \cdot S + \psi_{S_3}(\sigma S \cdot S))))$ |
| 1,5,13,11,18 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega + \psi_{S_3}(\sigma S \cdot S + S_3)))$ |
| 1,5,13,11,20 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega + \psi_{S_3}(\sigma S \cdot S + S_\omega)))$ |
| 1,5,13,12 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,13,12,20 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega + S_3)))$ |
| 1,5,13,12,23 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega + S_\omega)))$ |
| 1,5,13,12,23,20 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega + S_3))$ |
| 1,5,13,13 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega \cdot 2))$ |
| 1,5,13,15,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega \cdot S))$ |
| 1,5,13,16,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega \cdot \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,13,17 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,13,17,25 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_\omega)))$ |
| 1,5,13,18 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega \cdot S_2))$ |
| 1,5,13,18,12,23,30 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega \cdot \psi_{S_3}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,13,18,12,23,30, 41 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega \cdot \psi_{S_3}(\sigma S \cdot S \cdot \omega + S_\omega)))$ |
| 1,5,13,18,12,23,31 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_\omega \cdot S_3))$ |
| 1,5,13,18,13 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_\omega}^2))$ |
| 1,5,13,18,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_\omega}^2 \cdot \omega))$ |
| 1,5,13,18,18,13 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_\omega}^3))$ |
| 1,5,13,18,24 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega+1}}(S_{\omega+1})))$ |
| 1,5,13,19,5 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega+1}}(\sigma S \cdot \omega)))$ |
| 1,5,13,19,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega+1}}(\sigma S \cdot S)))$ |
| 1,5,13,19,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega+1}}(\sigma S \cdot S + 1)))$ |
| 1,5,13,19,18,13 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega+1}}(\sigma S \cdot S + S_\omega)))$ |
| 1,5,13,19,19,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega+1}}(\sigma S \cdot S + \psi_{S_{\omega+1}}(\sigma S \cdot S))))$ |
| 1,5,13,19,26 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega+1}}(\sigma S \cdot S + S_{\omega+1})))$ |
| 1,5,13,19,28 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega+1}}(\sigma S \cdot S + S_{\omega \cdot 2})))$ |
| 1,5,13,19,29 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{\psi_{S_{\omega+1}}(\sigma S \cdot (S+1))}(\psi_S(\sigma S \cdot \omega))))$ |
| 1,5,13,20 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,13,20,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega + 1)))$ |
| 1,5,13,20,20 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega))))$ |
| 1,5,13,20,28 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega+1}))$ |
| 1,5,13,20,29,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega+2}}(\sigma S \cdot S)))$ |
| 1,5,13,20,30 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega+2}}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,13,20,31 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega \cdot 2}))$ |
| 1,5,13,21 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2}))$ |
| 1,5,13,21,13 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} + S_\omega))$ |
| 1,5,13,21,13,20 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} + \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,13,21,13,20,31 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} + \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega + S_{\omega \cdot 2})))$ |
| 1,5,13,21,13,20,31, 42 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} + \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega + S_{\omega^2})))$ |
| 1,5,13,21,13,20,31, 42,31 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} + S_{\omega \cdot 2})))$ |
| 1,5,13,21,13,21 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot 2))$ |
| 1,5,13,21,15,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,13,21,18 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_2))$ |
| 1,5,13,21,18,13 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega))$ |
| 1,5,13,21,18,13,19, 6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S)))$ |
| 1,5,13,21,18,13,20 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,13,21,18,13,20, 31,42 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega + S_{\omega^2})))$ |
| 1,5,13,21,18,13,20, 31,42,36,13 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega +$ $\psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega)))$ |
| 1,5,13,21,18,13,20, 31,42,36,28 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega + S_{\omega+1}))$ |
| 1,5,13,21,18,13,20, 31,42,36,31 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega + S_{\omega \cdot 2}))$ |
| 1,5,13,21,18,13,20, 31,42,36,31,42 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot (S_\omega + 1)))$ |
| 1,5,13,21,18,13,20, 31,42,36,36,13 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_\omega \cdot 2))$ |
| 1,5,13,21,18,13,20, 31,42,37,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot \psi_{S_{\omega+1}}(\sigma S \cdot S)))$ |
| 1,5,13,21,18,13,20, 31,42,38 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,13,21,18,13,20, 31,42,39 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_{\omega+1}))$ |
| 1,5,13,21,18,13,20, 31,42,39,31 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^2} \cdot S_{\omega \cdot 2}))$ |
| 1,5,13,21,18,13,21 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + {S_{\omega^2}}^2))$ |
| 1,5,13,21,18,24 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \varepsilon_{S_{\omega^2}+1}))$ |
| 1,5,13,21,19,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega^2+1}}(\sigma S \cdot S)))$ |
| 1,5,13,21,20 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + \psi_{S_{\omega^2+1}}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,13,21,21 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\omega^3}))$ |
| 1,5,13,21,23,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_S))$ |
| 1,5,13,21,23,14 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_S \cdot \omega))$ |
| 1,5,13,21,23,21 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{S \cdot \omega}))$ |
| 1,5,13,21,23,21,23, 6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{S^2}))$ |
| 1,5,13,21,23,26 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\varepsilon_{S+1}}))$ |
| 1,5,13,21,24,6 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\psi_{S_2}(\sigma S \cdot S)}))$ |
| 1,5,13,21,25 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{\psi_{S_2}(\sigma S \cdot S \cdot \omega)}))$ |
| 1,5,13,21,26 | $\psi(\psi_S(\sigma S \cdot S \cdot \omega + S_{S_2}))$ |
| 1,5,13,21,26,14 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega + 1) + \psi_{S_2}(\sigma S \cdot (S \cdot \omega + 1) + S)))$ |
| 1,5,13,21,26,14,14 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega + 1) + \psi_{S_2}(\sigma S \cdot (S \cdot \omega + 1) + S) + 1))$ |
| 1,5,13,21,26,18 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega + 1) +$ $\psi_{S_2}(\sigma S \cdot (S \cdot \omega + 1) + S) + \psi_S(\sigma S \cdot (S \cdot \omega + 1) +$ $\psi_{S_2}(\sigma S \cdot (S \cdot \omega + 1)))))$ |
| 1,5,13,21,26,18,13, 21,26,14 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega + 1) + \psi_{S_2}(\sigma S \cdot (S \cdot \omega + 1) + S) +$ $\psi_S(\sigma S \cdot (S \cdot \omega + 1) + \psi_{S_2}(\sigma S \cdot (S \cdot \omega + 1) + S))))$ |
| 1,5,13,21,26,18,24 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega + 1) + \psi_{S_2}(\sigma S \cdot (S \cdot \omega + 1) + S) + S))$ |
| 1,5,13,21,26,19,6 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega + 1) +$ $\psi_{S_2}(\sigma S \cdot (S \cdot \omega + 1) + S) + \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,13,21,26,20 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega + 1) +$ $\psi_{S_2}(\sigma S \cdot (S \cdot \omega + 1) + S) + \psi_{S_2}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,13,21,26,20,31 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega + 1) +$ $\psi_{S_2}(\sigma S \cdot (S \cdot \omega + 1) + S) + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_\omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,13,21,26,20,31, 42,47 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_S(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))))))$ |
| 1,5,13,21,26,20,31, 42,50,32 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S)\cdot 2))$ |
| 1,5,13,21,26,21 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S+1)))$ |
| 1,5,13,21,26,21,23, 6 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S$ $+\psi_S(\sigma S\cdot(S\cdot\omega+1)))))$ |
| 1,5,13,21,26,21,26, 14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S\cdot 2)))$ |
| 1,5,13,21,26,26,14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S^2)))$ |
| 1,5,13,21,26,32 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(S_2))))$ |
| 1,5,13,21,27,6 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot S))))$ |
| 1,5,13,21,27,18,24 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot S))+S))$ |
| 1,5,13,21,27,21 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot S)+1)))$ |
| 1,5,13,21,27,22 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot S+1))))$ |
| 1,5,13,21,27,34 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot S+S_2))))$ |
| 1,5,13,21,27,36 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot S+S_\omega))))$ |
| 1,5,13,21,28 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot S\cdot\omega))))$ |
| 1,5,13,21,28,36 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))))$ |
| 1,5,13,21,28,36,36 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))))\cdot 2)$ |
| 1,5,13,21,28,36,45 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))+S))$ |
| 1,5,13,21,28,37,6 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))+\psi_{S_2}(\sigma S\cdot S)))$ |
| 1,5,13,21,28,38 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))$ |
| 1,5,13,21,28,38,49 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))))$ |
| 1,5,13,21,28,39 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+1)))$ |
| 1,5,13,21,28,39,50, 58,40 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S)))$ |
| 1,5,13,21,28,39,50, 60,71 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))\cdot 2))$ |
| 1,5,13,21,29 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+1)))$ |
| 1,5,13,21,29,13 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+1))+\psi_S(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+1)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,13,21,29,14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+1)+1))$ |
| 1,5,13,21,29,18,13 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+1)+\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+1))))$ |
| 1,5,13,21,29,18,24 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+1)+S))$ |
| 1,5,13,21,29,19,6 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+1)+\psi_{S_2}(\sigma S\cdot S)))$ |
| 1,5,13,21,29,20 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+1)+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))$ |
| 1,5,13,21,29,20,31 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+1)))$ |
| 1,5,13,21,29,21 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+2)))$ |
| 1,5,13,21,29,21,26,14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))+S)))$ |
| 1,5,13,21,29,21,28,36 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))\cdot 2)))$ |
| 1,5,13,21,29,21,29 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1))\cdot 2+1)))$ |
| 1,5,13,21,29,26,14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S))))$ |
| 1,5,13,21,29,29 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))+1)))$ |
| 1,5,13,22 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+S_2))$ |
| 1,5,13,22,14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+S_2+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S_2)+1))$ |
| 1,5,13,22,18,24 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+S_2+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S_2)+S))$ |
| 1,5,13,22,21 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+S_2+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+S_2+1)))$ |
| 1,5,13,22,22 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+S_2\cdot 2))$ |
| 1,5,13,22,32 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(S_3)))$ |
| 1,5,13,23 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S))+1))$ |
| 1,5,13,23,6 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot S)))$ |
| 1,5,13,24 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot S\cdot\omega)))$ |
| 1,5,13,24,36 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1))))$ |
| 1,5,13,25 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1))+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1))+1)))$ |
| 1,5,13,25,22 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1))+S_2))$ |
| 1,5,13,25,26 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1)+1)))$ |
| 1,5,13,25,34 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1)+S_2)))$ |
| 1,5,13,25,37 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1)))+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1)+\psi_{S_3}(\sigma S\cdot(S\cdot\omega+1)))+1)))$ |
| 1,5,13,25,38 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+S_3))$ |
| 1,5,13,26 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+1)+S_\omega))$ |
| 1,5,13,27 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,13,27,26 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega))+$ $\psi_S(\sigma S\cdot(S\cdot\omega+2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+2)+1)))$ |
| 1,5,13,27,27 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega))\cdot 2)$ |
| 1,5,13,27,36,46 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega)+S))$ |
| 1,5,13,27,37,6 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega)+\psi_{S_2}(\sigma S\cdot S)))$ |
| 1,5,13,27,40,27 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+\omega))))$ |
| 1,5,13,27,40,54 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+\omega))+S))$ |
| 1,5,13,27,40,58,59 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+\omega)+1)))$ |
| 1,5,13,27,40,58,77 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega)+S_2))$ |
| 1,5,13,27,40,59 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega)+S_\omega))$ |
| 1,5,13,27,40,60 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega\cdot 2)))$ |
| 1,5,13,27,41 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega^2)))$ |
| 1,5,13,27,41,36,46 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega^2)+S))$ |
| 1,5,13,27,41,41 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+\omega^3)))$ |
| 1,5,13,27,41,50,28 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+S)))$ |
| 1,5,13,27,41,50,41 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+S)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+S)+1)))$ |
| 1,5,13,27,41,50,60 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+S)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+S)+\psi_{S_2}(\sigma S))))$ |
| 1,5,13,27,41,54,28 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+S)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+S)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+S)))))$ |
| 1,5,13,27,41,54,68 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+S)+S_2))$ |
| 1,5,13,27,41,54,73 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+S)+S_\omega))$ |
| 1,5,13,27,41,54,74, 94,108,75 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega+S\cdot 2)))$ |
| 1,5,13,27,41,55 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega\cdot 2)))$ |
| 1,5,13,27,42 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega\cdot 2)+S_2))$ |
| 1,5,14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)))$ |
| 1,5,14,6 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+1))$ |
| 1,5,14,7,5 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,14,7,5,13 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega)+S_\omega)))$ |
| 1,5,14,7,5,13,27 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1)))+$ $\psi_{\psi_S(\sigma S\cdot(S\cdot\omega+2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+2)))}$ $(\psi_S(\sigma S\cdot(S\cdot\omega+\omega))))$ |
| 1,5,14,7,5,13,27, 49 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1)))+$ $\psi_{\psi_S(\sigma S\cdot(S\cdot\omega+2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+2)))}$ $(\psi_S(\sigma S\cdot(S\cdot\omega^2))))$ |
| 1,5,14,7,5,13,27, 49,29,22 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1)))+$ $\psi_S(\sigma S\cdot(S\cdot\omega+2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+2))))$ |
| 1,5,14,7,5,13,27, 49,29,26 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1)))+$ $\psi_S(\sigma S\cdot(S\cdot\omega+2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+2)+1)))$ |
| 1,5,14,7,5,13,27, 49,29,27 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1)))+$ $\psi_S(\sigma S\cdot(S\cdot\omega+\omega)))$ |
| 1,5,14,7,5,13,27, 49,29,27,49 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1)))+$ $\psi_S(\sigma S\cdot(S\cdot\omega^2)))$ |
| 1,5,14,7,5,13,27, 49,29,28 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1))+1))$ |
| 1,5,14,7,5,13,27, 49,32 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)))))$ |
| 1,5,14,7,5,13,27, 49,32,13 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+1)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+1)+1))))$ |
| 1,5,14,7,5,13,27, 49,36 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+2)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+2)))))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,7,5,13,27, 49,36,26 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+2)+ \psi_{S_2}(\sigma S\cdot(S\cdot\omega+2)+1))))$ |
| 1,5,14,7,5,13,27, 49,36,27 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega+\omega))))$ |
| 1,5,14,7,5,13,27, 49,36,27,42 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot S\cdot\omega\cdot 2+S_2)))$ |
| 1,5,14,7,5,13,27, 49,36,27,48 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot S\cdot\omega\cdot 2+S_\omega)))$ |
| 1,5,14,7,5,14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega^2))))$ |
| 1,5,14,7,6 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega^2))+1))$ |
| 1,5,14,7,8 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_S(\sigma S\cdot(S\cdot\omega^2)+1)))$ |
| 1,5,14,7,10 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+S))$ |
| 1,5,14,8 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S))+ \psi_{\alpha_2}(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S))+1))$ |
| 1,5,14,8,5 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,14,8,5,14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2)))))$ |
| 1,5,14,8,6 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2)))+1))$ |
| 1,5,14,8,7,10 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2)))+ \psi_S(\sigma S\cdot(S\cdot\omega^2)+S)))$ |
| 1,5,14,8,8,5,14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2)))+ \psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2))))))$ |
| 1,5,14,8,12 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2)))+S))$ |
| 1,5,14,8,13,5 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+ \psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2)))+\psi_{S_2}(\sigma S\cdot\omega)))$ |
| 1,5,14,8,13,5,14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2)))\cdot 2))$ |
| 1,5,14,8,13,14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2))+1)))$ |
| 1,5,14,8,13,17 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2))+S)))$ |
| 1,5,14,8,13,18,5, 14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2))+ \psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2))))))$ |
| 1,5,14,8,13,19 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2))+S_2)))$ |
| 1,5,14,8,14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2))+S_\omega)))$ |
| 1,5,14,8,14,20 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2))+S_{\omega^2})))$ |
| 1,5,14,8,15 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\omega)))))$ |
| 1,5,14,8,15,22,26, 16 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S)))$ |
| 1,5,14,8,15,22,29 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))$ |
| 1,5,14,8,15,23 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_2)))$ |
| 1,5,14,8,15,27 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_\omega)))$ |
| 1,5,14,8,15,27,46 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+\omega))))$ |
| 1,5,14,8,15,28 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2))))$ |
| 1,5,14,8,15,28,15 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)))+ \psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega+\omega))))$ |
| 1,5,14,8,15,28,16 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2))+1))$ |
| 1,5,14,9 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+1)))$ |
| 1,5,14,9,9 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+2)))$ |
| 1,5,14,9,11,6 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+S)))$ |
| 1,5,14,9,11,8,15, 28,21,25,16 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+S)\cdot 2))$ |
| 1,5,14,9,11,9 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+S+1)))$ |
| 1,5,14,9,11,9,11, 6 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+S\cdot 2)))$ |
| 1,5,14,9,11,14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S))))$ |
| 1,5,14,9,12,6 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S))))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,5,14,9,12,9 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S)+1)))$ |
| 1,5,14,9,12,16 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S+S_2))))$ |
| 1,5,14,9,12,18 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S+S_\omega))))$ |
| 1,5,14,9,13 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S\cdot\omega))))$ |
| 1,5,14,9,13,9,13 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$ $\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))))$ |
| 1,5,14,9,13,11,6 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$ $\psi_{S_2}(\sigma S\cdot S\cdot\omega+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S)))))$ |
| 1,5,14,9,14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$ $\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_2))))$ |
| 1,5,14,9,17 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$ $\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_\omega))))$ |
| 1,5,14,9,17,31 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega+\omega)))))$ |
| 1,5,14,9,17,31,53 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)))))$ |
| 1,5,14,9,17,31,53, 32 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)))+1))$ |
| 1,5,14,9,17,31,53, 45 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)))+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+1)))$ |
| 1,5,14,9,18 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2))+1)))$ |
| 1,5,14,9,18,9 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2))+2)))$ |
| 1,5,14,9,18,9,17 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2))+\psi_{S_2}(\sigma S\cdot S\cdot\omega+S_\omega))))$ |
| 1,5,14,9,18,9,18 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2))\cdot 2+1)))$ |
| 1,5,14,9,18,10 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+1))))$ |
| 1,5,14,9,18,11,6 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+S))))$ |
| 1,5,14,9,18,13 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_2}(\sigma S\cdot S\cdot\omega)))))$ |
| 1,5,14,10 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+S_2))$ |
| 1,5,14,10,9,18,14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+S_2+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+$ $S_2+\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+S_2))))$ |
| 1,5,14,10,10 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+S_2\cdot 2))$ |
| 1,5,14,10,15 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+S_2{}^2))$ |
| 1,5,14,10,18 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S))+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,14,11 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S))+1))$ |
| 1,5,14,11,5 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot\omega)))$ |
| 1,5,14,11,5,14 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot\psi_S(\sigma S\cdot(S\cdot\omega^2)))))$ |
| 1,5,14,11,6 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S)))$ |
| 1,5,14,11,7,10 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S)+S))$ |
| 1,5,14,11,9 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S)+$ $\psi_{S_2}(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S)+1)))$ |
| 1,5,14,11,10 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S)+S_2))$ |
| 1,5,14,11,11,6 | $\psi(\psi_S(\sigma S\cdot(S\cdot\omega^2)+\psi_{S_3}(\sigma S\cdot S)\cdot 2))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,11,16 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + \psi_{S_3}(\sigma S \cdot S + S_2)))$ |
| 1,5,14,11,17,6 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + \psi_{S_3}(\sigma S \cdot S + \psi_{S_3}(\sigma S \cdot S))))$ |
| 1,5,14,11,18 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + \psi_{S_3}(\sigma S \cdot S + S_3)))$ |
| 1,5,14,11,20 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + \psi_{S_3}(\sigma S \cdot S + S_\omega)))$ |
| 1,5,14,11,21 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + \psi_{\psi_{S_3}(\sigma S \cdot (S+1))}(\psi_S(\sigma S \cdot \omega))))$ |
| 1,5,14,11,21,37 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + \psi_{\psi_{S_3}(\sigma S \cdot (S+1))}(\psi_S(\sigma S \cdot (S \cdot \omega^2)))))$ |
| 1,5,14,12 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + \psi_{S_3}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,14,12,14,6 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + \psi_{S_3}(\sigma S \cdot S \cdot \omega + S)))$ |
| 1,5,14,12,17 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + \psi_{S_3}(\sigma S \cdot S \cdot \omega + S_2)))$ |
| 1,5,14,12,18,6 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + \psi_{S_3}(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S))))$ |
| 1,5,14,12,19 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + \psi_{S_3}(\sigma S \cdot S \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega))))$ |
| 1,5,14,12,20 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + \psi_{S_3}(\sigma S \cdot S \cdot \omega + S_3)))$ |
| 1,5,14,12,23 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + \psi_{S_3}(\sigma S \cdot S \cdot \omega + S_\omega)))$ |
| 1,5,14,12,24 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + \psi_{S_3}(\sigma S \cdot (S \cdot \omega^2))))$ |
| 1,5,14,12,24,20 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + S_3))$ |
| 1,5,14,13 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + S_\omega))$ |
| 1,5,14,13,18 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + S_\omega \cdot S_2))$ |
| 1,5,14,13,18,13 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + {S_\omega}^2))$ |
| 1,5,14,13,18,24 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + \psi_{S_{\omega+1}}(S_{\omega+1})))$ |
| 1,5,14,13,20,32,28 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + S_{\omega+1}))$ |
| 1,5,14,13,21 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2) + S_{\omega^2}))$ |
| 1,5,14,13,21,26,14 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2 + 1) + \psi_{S_2}(\sigma S \cdot (S \cdot \omega^2 + 1) + S)))$ |
| 1,5,14,13,21,26,21 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2 + 1) + \psi_{S_2}(\sigma S \cdot (S \cdot \omega^2 + 1) + S + 1)))$ |
| 1,5,14,13,21,26,32 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2 + 1) + \psi_{S_2}(\sigma S \cdot (S \cdot \omega^2 + 1) + \psi_{S_2}(\sigma S))))$ |
| 1,5,14,13,21,28 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2 + 1) +$ $\psi_{S_2}(\sigma S \cdot (S \cdot \omega^2 + 1) + \psi_{S_2}(\sigma S \cdot S \cdot \omega))))$ |
| 1,5,14,13,21,28,40, 39 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2 + 1) + \psi_{S_2}(\sigma S \cdot (S \cdot \omega^2 + 1) +$ $\psi_{S_2}(\sigma S \cdot (S \cdot \omega^2) + S_\omega))))$ |
| 1,5,14,13,21,29 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2 + 1) + \psi_{S_2}(\sigma S \cdot (S \cdot \omega^2 + 1) +$ $\psi_{S_2}(\sigma S \cdot (S \cdot \omega^2 + 1)) + 1)))$ |
| 1,5,14,13,22 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2 + 1) + S_2))$ |
| 1,5,14,13,25 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2 + 1) +$ $\psi_{S_3}(\sigma S \cdot (S \cdot \omega^2 + 1)) + \psi_{S_2}(\sigma S \cdot (S \cdot \omega^2 + 1) +$ $\psi_{S_3}(\sigma S \cdot (S \cdot \omega^2 + 1)) + 1)))$ |
| 1,5,14,13,25,38 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2 + 1) + S_3))$ |
| 1,5,14,13,26 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2 + 1) + S_\omega))$ |
| 1,5,14,13,27 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2 + \omega)))$ |
| 1,5,14,13,27,41 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2 + \omega^2)))$ |
| 1,5,14,13,27,41,50, 28 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2 + S)))$ |
| 1,5,14,13,27,41,55 | $\psi(\psi_S(\sigma S \cdot (S \cdot (\omega^2 + \omega))))$ |
| 1,5,14,13,27,49 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^2 \cdot 2)))$ |
| 1,5,14,14 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3)))$ |
| 1,5,14,14,7,10 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3) + S))$ |
| 1,5,14,14,9 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3) + \psi_{S_2}(\sigma S \cdot (S \cdot \omega^3) + 1)))$ |
| 1,5,14,14,9,18,18 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3) + \psi_{S_2}(\sigma S \cdot (S \cdot \omega^3) +$ $\psi_{S_2}(\sigma S \cdot (S \cdot \omega^3)))))$ |
| 1,5,14,14,10 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3) + S_2))$ |
| 1,5,14,14,11,6 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3) + \psi_{S_3}(\sigma S \cdot S)))$ |
| 1,5,14,14,12 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3) + \psi_{S_3}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,14,14,12,23 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3) + \psi_{S_3}(\sigma S \cdot S \cdot \omega + S_\omega)))$ |
| 1,5,14,14,12,24,24 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3) + \psi_{S_3}(\sigma S \cdot (S \cdot \omega^3))))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,14,12,24,24,20 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3) + S_3))$ |
| 1,5,14,14,13 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3) + S_\omega))$ |
| 1,5,14,14,13,21 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3) + S_{\omega^2}))$ |
| 1,5,14,14,13,21,26,14 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3 + 1) + \psi_{S_2}(\sigma S \cdot (S \cdot \omega^3 + 1) + S)))$ |
| 1,5,14,14,13,22 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3 + 1) + S_2))$ |
| 1,5,14,14,13,27 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3 + \omega)))$ |
| 1,5,14,14,13,27,49,49 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^3 \cdot 2)))$ |
| 1,5,14,14,14 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^4)))$ |
| 1,5,14,15 | $\psi(\psi_S(\sigma S \cdot (S \cdot \omega^\omega)))$ |
| 1,5,14,16,5 | $\psi(\psi_S(\sigma S \cdot S \cdot \psi_S(\sigma S \cdot \omega)))$ |
| 1,5,14,16,6 | $\psi(\psi_S(\sigma S \cdot (S^2)))$ |
| 1,5,14,16,6,6 | $\psi(\psi_S(\sigma S \cdot (S^2) + 1))$ |
| 1,5,14,16,7,5 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_S(\sigma S \cdot \omega)))$ |
| 1,5,14,16,7,5,14,16,6 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_S(\sigma S \cdot (S^2))))$ |
| 1,5,14,16,7,10 | $\psi(\psi_S(\sigma S \cdot (S^2) + S))$ |
| 1,5,14,16,8 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S))+$ $\psi_{\alpha_2}(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S)) + 1))$ |
| 1,5,14,16,8,5 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \omega)))$ |
| 1,5,14,16,8,5,14,16,6 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot (S^2)))))$ |
| 1,5,14,16,8,7,10 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot (S^2)))+$ $\psi_S(\sigma S \cdot (S^2) + S)))$ |
| 1,5,14,16,8,8,5,14,16,6 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot (S^2)))+$ $\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot (S^2))))))$ |
| 1,5,14,16,8,12 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot (S^2))) + S))$ |
| 1,5,14,16,8,13,5,14,16,6 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot (S^2))) \cdot 2))$ |
| 1,5,14,16,8,13,14 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot (S^2)) + 1)))$ |
| 1,5,14,16,8,13,17 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot (S^2)) + S)))$ |
| 1,5,14,16,8,13,19 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot (S^2)) + S_2)))$ |
| 1,5,14,16,8,14 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot (S^2)) + S_\omega)))$ |
| 1,5,14,16,8,15,22,26,16 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,14,16,8,15,22,29 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,14,16,8,15,23 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2)))$ |
| 1,5,14,16,8,15,28,30 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot S \cdot \Omega)))$ |
| 1,5,14,16,8,15,28,30,5,14,16,6 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot S \cdot \psi_S(\sigma S \cdot (S^2)))))$ |
| 1,5,14,16,8,15,28,32,16 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot (S^2))))$ |
| 1,5,14,16,8,15,28,32,16,15,28,30 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot (S^2)))+$ $\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot S \cdot \Omega)))$ |
| 1,5,14,16,8,15,28,32,16,16 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot (S^2)) + 1))$ |
| 1,5,14,16,9 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot (S^2) + 1)))$ |
| 1,5,14,16,9,8,15,28,32,16 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot (S^2) + 1) + \psi_{S_2}(\sigma S \cdot (S^2))))$ |
| 1,5,14,16,9,9 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot (S^2) + 2)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,16,9,11,6 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot (S^2) + S)))$ |
| 1,5,14,16,9,11,14 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S))))$ |
| 1,5,14,16,9,12,6 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot S))))$ |
| 1,5,14,16,9,13 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot S \cdot \omega))))$ |
| 1,5,14,16,9,18,20, 6 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot (S^2) + \psi_{S_2}(\sigma S \cdot (S^2)))))$ |
| 1,5,14,16,10 | $\psi(\psi_S(\sigma S \cdot (S^2) + S_2))$ |
| 1,5,14,16,12 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_3}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,14,16,12,24 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_3}(\sigma S \cdot (S \cdot \omega^2))))$ |
| 1,5,14,16,12,24,26, 6 | $\psi(\psi_S(\sigma S \cdot (S^2) + \psi_{S_3}(\sigma S \cdot (S^2))))$ |
| 1,5,14,16,12,24,26, 20 | $\psi(\psi_S(\sigma S \cdot (S^2) + S_3))$ |
| 1,5,14,16,13 | $\psi(\psi_S(\sigma S \cdot (S^2) + S_\omega))$ |
| 1,5,14,16,13,21,26, 14 | $\psi(\psi_S(\sigma S \cdot (S^2 + 1) + \psi_{S_2}(\sigma S \cdot (S^2 + 1) + S)))$ |
| 1,5,14,16,13,22 | $\psi(\psi_S(\sigma S \cdot (S^2 + 1) + S_2))$ |
| 1,5,14,16,13,27,49, 54 | $\psi(\psi_S(\sigma S \cdot (S^2 + S \cdot \psi_S(\sigma S \cdot (S^2 + 1) + \psi_{S_2}(\sigma S \cdot (S^2 + 1))))))$ |
| 1,5,14,16,13,27,49, 54,28 | $\psi(\psi_S(\sigma S \cdot (S^2 \cdot 2)))$ |
| 1,5,14,16,14 | $\psi(\psi_S(\sigma S \cdot (S^2 \cdot \omega)))$ |
| 1,5,14,16,14,10 | $\psi(\psi_S(\sigma S \cdot (S^2 \cdot \omega) + S_2))$ |
| 1,5,14,16,14,13 | $\psi(\psi_S(\sigma S \cdot (S^2 \cdot \omega) + S_\omega))$ |
| 1,5,14,16,14,14 | $\psi(\psi_S(\sigma S \cdot (S^2 \cdot \omega^2)))$ |
| 1,5,14,16,14,16,6 | $\psi(\psi_S(\sigma S \cdot (S^3)))$ |
| 1,5,14,16,15 | $\psi(\psi_S(\sigma S \cdot (S^\omega)))$ |
| 1,5,14,16,16,6 | $\psi(\psi_S(\sigma S \cdot (S^S)))$ |
| 1,5,14,16,19 | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S)))$ |
| 1,5,14,17 | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S)) + \psi_{\alpha_2}(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S)) + 1))$ |
| 1,5,14,17,5 | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot \omega)))$ |
| 1,5,14,17,6 | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,14,17,6,6 | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S) + 1))$ |
| 1,5,14,17,10 | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S) + S_2))$ |
| 1,5,14,17,13,27,49, 62,28 | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S) \cdot 2))$ |
| 1,5,14,17,14 | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S + 1)))$ |
| 1,5,14,17,14,16,6 | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S + S)))$ |
| 1,5,14,17,14,17,6 | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S))))$ |
| 1,5,14,17,21 | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S + S_2)))$ |
| 1,5,14,17,24 | $\psi(\psi_S(\sigma S \cdot \psi_{\psi_{S_2}(\sigma S \cdot (S+1))}(\psi_S(\sigma S \cdot \omega))))$ |
| 1,5,14,18 | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,14,18,23 | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_2)))$ |
| 1,5,14,18,26 | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_\omega)))$ |
| 1,5,14,18,27,29,6 | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot (S^2))))$ |
| 1,5,14,18,27,30,6 | $\psi(\psi_S(\sigma S \cdot \psi_{S_2}(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S))))$ |
| 1,5,14,19 | $\psi(\psi_S(\sigma S \cdot S_2))$ |
| 1,5,14,19,6 | $\psi(\psi_S(\sigma S \cdot S_2 + 1))$ |
| 1,5,14,19,7,5 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_S(\sigma S \cdot \omega)))$ |
| 1,5,14,19,7,5,14, 19 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_S(\sigma S \cdot S_2)))$ |
| 1,5,14,19,7,10 | $\psi(\psi_S(\sigma S \cdot S_2 + S))$ |
| 1,5,14,19,8,14 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S \cdot S_2) + S_\omega)))$ |
| 1,5,14,19,8,15,22, 26,16 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,19,8,15,27 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_\omega)))$ |
| 1,5,14,19,8,15,28, 32 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot (S^2))))$ |
| 1,5,14,19,8,15,28, 36 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S_2)))$ |
| 1,5,14,19,9 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S_2 + 1)))$ |
| 1,5,14,19,9,11,6 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S_2 + S)))$ |
| 1,5,14,19,9,17 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega + S_\omega))))$ |
| 1,5,14,19,9,18,23 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S_2 + \psi_{S_2}(\sigma S \cdot S_2))))$ |
| 1,5,14,19,10 | $\psi(\psi_S(\sigma S \cdot S_2 + S_2))$ |
| 1,5,14,19,11,6 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S)))$ |
| 1,5,14,19,12,24,26, 6 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot (S^2))))$ |
| 1,5,14,19,12,24,26, 29 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot \psi_{S_2}(\sigma S))))$ |
| 1,5,14,19,12,24,27, 6 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S))))$ |
| 1,5,14,19,12,24,28 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega))))$ |
| 1,5,14,19,12,24,28, 25 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S \cdot \omega + 1))))$ |
| 1,5,14,19,12,24,28, 37,42 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S_2))))$ |
| 1,5,14,19,12,24,29 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2)))$ |
| 1,5,14,19,12,24,29, 10 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2) + S_2))$ |
| 1,5,14,19,12,24,29, 13 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + 1)))$ |
| 1,5,14,19,12,24,29, 17 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + S_2)))$ |
| 1,5,14,19,12,24,29, 19 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S \cdot \omega))))$ |
| 1,5,14,19,12,24,29, 19,31,33,6 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot (S^2)))))$ |
| 1,5,14,19,12,24,29, 19,31,33,31 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S^2 \cdot \omega))))$ |
| 1,5,14,19,12,24,29, 19,31,33,36 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot \varepsilon_{S+1}))))$ |
| 1,5,14,19,12,24,29, 19,31,34,6 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S)))))$ |
| 1,5,14,19,12,24,29, 19,31,35,44,49 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot \psi_{S_2}(\sigma S \cdot S_2)))))$ |
| 1,5,14,19,12,24,29, 19,31,36 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2))))$ |
| 1,5,14,19,12,24,29, 19,31,36,13 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2) + 1)))$ |
| 1,5,14,19,12,24,29, 19,31,36,17 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2) + S_2)))$ |
| 1,5,14,19,12,24,29, 19,31,36,19,31,36 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2) \cdot 2)))$ |
| 1,5,14,19,12,24,29, 19,31,36,20 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + 1))))$ |
| 1,5,14,19,12,24,29, 19,31,36,26,38,43 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2 + \psi_{S_3}(\sigma S \cdot S_2)))))$ |
| 1,5,14,19,12,24,29, 20 | $\psi(\psi_S(\sigma S \cdot S_2 + S_3))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,19,12,24,29, 21,6 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_4}(\sigma S \cdot S)))$ |
| 1,5,14,19,12,24,29, 22,37,42 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_4}(\sigma S \cdot S_2)))$ |
| 1,5,14,19,12,24,29, 22,37,42,32,47,52 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_4}(\sigma S \cdot S_2 + \psi_{S_4}(\sigma S \cdot S_2))))$ |
| 1,5,14,19,12,24,29, 22,37,42,33 | $\psi(\psi_S(\sigma S \cdot S_2 + S_4))$ |
| 1,5,14,19,12,24,29, 23 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega))$ |
| 1,5,14,19,12,24,29, 23,10 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega + S_2))$ |
| 1,5,14,19,12,24,29, 23,12,24,29 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega + \psi_{S_3}(\sigma S \cdot S_2)))$ |
| 1,5,14,19,12,24,29, 23,12,24,29,23 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega + \psi_{S_3}(\sigma S \cdot S_2 + S_\omega)))$ |
| 1,5,14,19,12,24,29, 23,13 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega + \psi_{S_3}(\sigma S \cdot S_2 + S_\omega + 1)))$ |
| 1,5,14,19,12,24,29, 23,19,31,36,30 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega +$ $\psi_{S_3}(\sigma S \cdot S_2 + S_\omega + \psi_{S_3}(\sigma S \cdot S_2 + S_\omega))))$ |
| 1,5,14,19,12,24,29, 23,20 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega + S_3))$ |
| 1,5,14,19,12,24,29, 23,23 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot 2))$ |
| 1,5,14,19,12,24,29, 23,28 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot S_2))$ |
| 1,5,14,19,12,24,29, 23,31 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot S_3))$ |
| 1,5,14,19,12,24,29, 23,31,20 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot S_3 + S_3))$ |
| 1,5,14,19,12,24,29, 23,31,22,37,42,36,44 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot S_3 + \psi_{S_4}(\sigma S \cdot S_2 + S_\omega \cdot S_3)))$ |
| 1,5,14,19,12,24,29, 23,31,22,37,42,36,44, 33 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot S_3 + S_4))$ |
| 1,5,14,19,12,24,29, 23,31,22,37,42,36,44, 36 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot (S_3 + 1)))$ |
| 1,5,14,19,12,24,29, 23,31,22,37,42,36,44, 36,44 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot S_3 \cdot 2))$ |
| 1,5,14,19,12,24,29, 23,31,22,37,42,36,45, 6 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot \psi_{S_4}(\sigma S \cdot S)))$ |
| 1,5,14,19,12,24,29, 23,31,22,37,42,36,47 | $\psi(\psi_S(\sigma S \cdot S_2 + S_\omega \cdot S_4))$ |
| 1,5,14,19,12,24,29, 23,31,23 | $\psi(\psi_S(\sigma S \cdot S_2 + {S_\omega}^2))$ |
| 1,5,14,19,12,24,29, 23,31,40 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_{\omega+1}}(S_{\omega+1})))$ |
| 1,5,14,19,12,24,29, 23,32,6 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_{\omega+1}}(\sigma S \cdot S)))$ |
| 1,5,14,19,12,24,29, 23,33 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,14,19,12,24,29, 23,33,48,53 | $\psi(\psi_S(\sigma S \cdot S_2 + \psi_{S_{\omega+1}}(\sigma S \cdot S_2)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,19,12,24,29, 23,33,48,53,44 | $\psi(\psi_S(\sigma S \cdot S_2 + S_{\omega+1}))$ |
| 1,5,14,19,12,24,29, 23,33,48,53,47 | $\psi(\psi_S(\sigma S \cdot S_2 + S_{\omega \cdot 2}))$ |
| 1,5,14,19,12,24,29, 23,34 | $\psi(\psi_S(\sigma S \cdot S_2 + S_{\omega^2}))$ |
| 1,5,14,19,12,24,29, 23,34,42,24 | $\psi(\psi_S(\sigma S \cdot (S_2 + 1) + \psi_{S_2}(\sigma S \cdot (S_2 + 1) + S)))$ |
| 1,5,14,19,12,24,29, 23,34,45 | $\psi(\psi_S(\sigma S \cdot (S_2 + 1) + \psi_{S_2}(\sigma S \cdot (S_2 + 1) + \psi_{S_2}(\sigma S \cdot (S_2 + 1)) + 1)))$ |
| 1,5,14,19,12,24,29, 23,35 | $\psi(\psi_S(\sigma S \cdot (S_2 + 1) + S_2))$ |
| 1,5,14,19,12,24,29, 23,39 | $\psi(\psi_S(\sigma S \cdot (S_2 + 1) + S_\omega))$ |
| 1,5,14,19,12,24,29, 23,40 | $\psi(\psi_S(\sigma S \cdot (S_2 + \omega)))$ |
| 1,5,14,19,12,24,29, 23,40,65,77,41 | $\psi(\psi_S(\sigma S \cdot (S_2 + S^2)))$ |
| 1,5,14,19,12,24,29, 23,40,65,83 | $\psi(\psi_S(\sigma S \cdot S_2 \cdot 2))$ |
| 1,5,14,19,12,24,29, 24 | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \omega))$ |
| 1,5,14,19,12,24,29, 24,10 | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \omega + S_2))$ |
| 1,5,14,19,12,24,29, 24,13 | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \omega + \psi_{S_3}(\sigma S \cdot S_2 \cdot \omega + 1)))$ |
| 1,5,14,19,12,24,29, 24,19,31,36,31 | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \omega + \psi_{S_3}(\sigma S \cdot S_2 \cdot \omega + \psi_{S_3}(\sigma S \cdot S_2 \cdot \omega))))$ |
| 1,5,14,19,12,24,29, 24,20 | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \omega + S_3))$ |
| 1,5,14,19,12,24,29, 24,23 | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \omega + S_\omega))$ |
| 1,5,14,19,12,24,29, 24,23,40 | $\psi(\psi_S(\sigma S \cdot (S_2 \cdot \omega + \omega)))$ |
| 1,5,14,19,12,24,29, 24,23,40,65,83,63,95, 113,95 | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \omega \cdot 2))$ |
| 1,5,14,19,12,24,29, 24,24 | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \omega^2))$ |
| 1,5,14,19,12,24,29, 24,26,6 | $\psi(\psi_S(\sigma S \cdot S_2 \cdot S))$ |
| 1,5,14,19,12,24,29, 24,27,6 | $\psi(\psi_S(\sigma S \cdot S_2 \cdot \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,14,19,12,24,29, 24,29 | $\psi(\psi_S(\sigma S \cdot {S_2}^2))$ |
| 1,5,14,19,12,24,29, 29 | $\psi(\psi_S(\sigma S \cdot {S_2}^{S_2}))$ |
| 1,5,14,19,12,24,29, 35 | $\psi(\psi_S(\sigma S \cdot \psi_{S_3}(S_3)))$ |
| 1,5,14,19,12,24,30, 6 | $\psi(\psi_S(\sigma S \cdot \psi_{S_3}(\sigma S \cdot S)))$ |
| 1,5,14,19,12,24,31 | $\psi(\psi_S(\sigma S \cdot \psi_{S_3}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,14,19,12,24,31, 43,48,54 | $\psi(\psi_S(\sigma S \cdot \psi_{S_3}(\sigma S \cdot \varepsilon_(S_2 + 1))))$ |
| 1,5,14,19,12,24,32 | $\psi(\psi_S(\sigma S \cdot S_3))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,19,12,24,32, 10 | $\psi(\psi_S(\sigma S \cdot S_3 + S_2))$ |
| 1,5,14,19,12,24,32, 20 | $\psi(\psi_S(\sigma S \cdot S_3 + S_3))$ |
| 1,5,14,19,12,24,32, 22,37,45,36 | $\psi(\psi_S(\sigma S \cdot S_3 + S_\omega))$ |
| 1,5,14,19,12,24,32, 22,37,45,37 | $\psi(\psi_S(\sigma S \cdot S_3 \cdot \omega))$ |
| 1,5,14,19,12,24,32, 22,37,46,6 | $\psi(\psi_S(\sigma S \cdot \psi_{S_4}(\sigma S \cdot S)))$ |
| 1,5,14,19,12,24,32, 22,37,48 | $\psi(\psi_S(\sigma S \cdot S_4))$ |
| 1,5,14,19,13 | $\psi(\psi_S(\sigma S \cdot S_\omega))$ |
| 1,5,14,19,13,10 | $\psi(\psi_S(\sigma S \cdot S_\omega + S_2))$ |
| 1,5,14,19,13,12,24, 32 | $\psi(\psi_S(\sigma S \cdot S_\omega + \psi_{S_3}(\sigma S \cdot S_3)))$ |
| 1,5,14,19,13,12,24, 32,23 | $\psi(\psi_S(\sigma S \cdot S_\omega + \psi_{S_3}(\sigma S \cdot S_\omega)))$ |
| 1,5,14,19,13,12,24, 32,23,20 | $\psi(\psi_S(\sigma S \cdot S_\omega + S_3))$ |
| 1,5,14,19,13,13 | $\psi(\psi_S(\sigma S \cdot S_\omega + S_\omega))$ |
| 1,5,14,19,13,15,6 | $\psi(\psi_S(\sigma S \cdot S_\omega + S_\omega \cdot S))$ |
| 1,5,14,19,13,17,26, 31,25 | $\psi(\psi_S(\sigma S \cdot S_\omega + S_\omega \cdot \psi_{S_2}(\sigma S \cdot S_\omega)))$ |
| 1,5,14,19,13,18 | $\psi(\psi_S(\sigma S \cdot S_\omega + S_\omega \cdot S_2))$ |
| 1,5,14,19,13,18,13 | $\psi(\psi_S(\sigma S \cdot S_\omega + {S_\omega}^2))$ |
| 1,5,14,19,13,18,24 | $\psi(\psi_S(\sigma S \cdot S_\omega + \varepsilon_{S_\omega+1}))$ |
| 1,5,14,19,13,19,6 | $\psi(\psi_S(\sigma S \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S)))$ |
| 1,5,14,19,13,20,32, 37 | $\psi(\psi_S(\sigma S \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S_2)))$ |
| 1,5,14,19,13,20,32, 37,13 | $\psi(\psi_S(\sigma S \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S_\omega)))$ |
| 1,5,14,19,13,20,32, 37,14 | $\psi(\psi_S(\sigma S \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S_\omega + 1)))$ |
| 1,5,14,19,13,20,32, 37,20,32,37,13 | $\psi(\psi_S(\sigma S \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S_\omega + \psi_{S_{\omega+1}}(\sigma S \cdot S_\omega))))$ |
| 1,5,14,19,13,20,32, 37,28 | $\psi(\psi_S(\sigma S \cdot S_\omega + S_{\omega+1}))$ |
| 1,5,14,19,13,20,32, 37,31 | $\psi(\psi_S(\sigma S \cdot S_\omega + S_{\omega\cdot 2}))$ |
| 1,5,14,19,13,20,32, 37,31,42 | $\psi(\psi_S(\sigma S \cdot S_\omega + S_{\omega^2}))$ |
| 1,5,14,19,13,20,32, 37,31,43 | $\psi(\psi_S(\sigma S \cdot (S_\omega + 1) + \psi_{S_2}(\sigma S \cdot (S_\omega + 1) + S)))$ |
| 1,5,14,19,13,20,32, 37,31,47 | $\psi(\psi_S(\sigma S \cdot (S_\omega + 1) + S_\omega))$ |
| 1,5,14,19,13,20,32, 37,31,48,73,78,13 | $\psi(\psi_S(\sigma S \cdot S_\omega \cdot 2))$ |
| 1,5,14,19,13,20,32, 37,32 | $\psi(\psi_S(\sigma S \cdot S_\omega \cdot \omega))$ |
| 1,5,14,19,13,20,32, 37,32,37,13 | $\psi(\psi_S(\sigma S \cdot {S_\omega}^2))$ |
| 1,5,14,19,13,20,32, 39 | $\psi(\psi_S(\sigma S \cdot \psi_{S_{\omega+1}}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,14,19,13,20,32, 40 | $\psi(\psi_S(\sigma S \cdot S_{\omega+1}))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,19,13,20,32, 40,13 | $\psi(\psi_S(\sigma S \cdot S_{\omega+1} + S_\omega))$ |
| 1,5,14,19,13,20,32, 40,13,20,32,40 | $\psi(\psi_S(\sigma S \cdot S_{\omega+1} + \psi_{S_{\omega+1}}(\sigma S \cdot S_{\omega+1})))$ |
| 1,5,14,19,13,20,32, 40,28 | $\psi(\psi_S(\sigma S \cdot S_{\omega+1} + S_{\omega+1}))$ |
| 1,5,14,19,13,20,32, 40,30 | $\psi(\psi_S(\sigma S \cdot S_{\omega+1} + \psi_{S_{\omega+2}}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,14,19,13,20,32, 40,30,45,53 | $\psi(\psi_S(\sigma S \cdot S_{\omega+1} + \psi_{S_{\omega+2}}(\sigma S \cdot S_{\omega+1})))$ |
| 1,5,14,19,13,20,32, 40,30,45,53,41 | $\psi(\psi_S(\sigma S \cdot S_{\omega+1} + S_{\omega+2}))$ |
| 1,5,14,19,13,20,32, 40,30,45,53,44 | $\psi(\psi_S(\sigma S \cdot S_{\omega+1} + S_{\omega\cdot 2}))$ |
| 1,5,14,19,13,20,32, 40,30,45,53,44,59 | $\psi(\psi_S(\sigma S \cdot (S_{\omega+1} + 1) + S_2))$ |
| 1,5,14,19,13,20,32, 40,30,45,53,45 | $\psi(\psi_S(\sigma S \cdot S_{\omega+1} \cdot \omega))$ |
| 1,5,14,19,13,20,32, 40,30,45,53,45,53 | $\psi(\psi_S(\sigma S \cdot {S_{\omega+1}}^2))$ |
| 1,5,14,19,13,20,32, 40,30,45,56 | $\psi(\psi_S(\sigma S \cdot S_{\omega+2}))$ |
| 1,5,14,19,13,20,32, 40,31 | $\psi(\psi_S(\sigma S \cdot S_{\omega\cdot 2}))$ |
| 1,5,14,19,13,21 | $\psi(\psi_S(\sigma S \cdot S_{\omega^2}))$ |
| 1,5,14,19,13,21,10 | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + S_2))$ |
| 1,5,14,19,13,21,12, 24,32,23 | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + \psi_{S_3}(\sigma S \cdot S_\omega)))$ |
| 1,5,14,19,13,21,12, 24,32,23,34 | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + \psi_{S_3}(\sigma S \cdot S_{\omega^2})))$ |
| 1,5,14,19,13,21,12, 24,32,23,34,20 | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + S_3))$ |
| 1,5,14,19,13,21,13 | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + S_\omega))$ |
| 1,5,14,19,13,21,13, 20,32,40,31 | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + \psi_{S_{\omega+1}}(\sigma S \cdot S_{\omega\cdot 2})))$ |
| 1,5,14,19,13,21,13, 20,32,40,31,42,28 | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + S_{\omega+1}))$ |
| 1,5,14,19,13,21,13, 21 | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + S_{\omega^2}))$ |
| 1,5,14,19,13,21,18, 24 | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + \varepsilon_(S_{\omega^2} + 1)))$ |
| 1,5,14,19,13,21,20, 32,37,13,21 | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + \psi_{S_{\omega^2+1}}(\sigma S \cdot S_{\omega^2})))$ |
| 1,5,14,19,13,21,20, 32,37,31,42,42 | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} + S_{\omega^3}))$ |
| 1,5,14,19,13,21,20, 32,37,32 | $\psi(\psi_S(\sigma S \cdot S_{\omega^2} \cdot \omega))$ |
| 1,5,14,19,13,21,20, 32,40 | $\psi(\psi_S(\sigma S \cdot S_{\omega^2+1}))$ |
| 1,5,14,19,13,21,20, 32,40,31,42 | $\psi(\psi_S(\sigma S \cdot S_{\omega^2\cdot 2}))$ |
| 1,5,14,19,13,21,21 | $\psi(\psi_S(\sigma S \cdot S_{\omega^3}))$ |
| 1,5,14,19,13,21,26, 14 | $\psi(\psi_S(\sigma S^2 + \psi_{S_2}(\sigma S^2 + S)))$ |
| 1,5,14,19,13,21,26, 21 | $\psi(\psi_S(\sigma S^2 + \psi_{S_2}(\sigma S^2 + S + 1)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,19,13,21,26, 32 | $\psi(\psi_S(\sigma S^2 + \psi_{S_2}(\sigma S^2 + \varepsilon_{S+1})))$ |
| 1,5,14,19,13,21,29 | $\psi(\psi_S(\sigma S^2 + \psi_{S_2}(\sigma S^2 + \psi_{S_2}(\sigma S^2) + 1)))$ |
| 1,5,14,19,13,22 | $\psi(\psi_S(\sigma S^2 + S_2))$ |
| 1,5,14,19,13,26 | $\psi(\psi_S(\sigma S^2 + S_\omega))$ |
| 1,5,14,19,14 | $\psi(\psi_S(\sigma S^2 \cdot \omega))$ |
| 1,5,14,19,14,5 | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_S(\sigma S \cdot \omega))$ |
| 1,5,14,19,14,5,9, 13 | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_S(\sigma S \cdot S \cdot \omega))$ |
| 1,5,14,19,14,5,10 | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_S(\sigma S \cdot S \cdot \omega + S_2))$ |
| 1,5,14,19,14,5,13, 27 | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_{\psi_S(\sigma S \cdot (S \cdot \omega + 1))}(\psi_S(\sigma S \cdot (S \cdot \omega + \omega))))$ |
| 1,5,14,19,14,5,14 | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_S(\sigma S \cdot S \cdot \omega^2))$ |
| 1,5,14,19,14,5,14, 19 | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_S(\sigma S \cdot S_2))$ |
| 1,5,14,19,14,5,14, 19,13 | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_S(\sigma S \cdot S_\omega))$ |
| 1,5,14,19,14,5,14, 19,13,27,49,64,49 | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_{\psi_S(\sigma S^2)}(\psi_S(\sigma S^2 \cdot \omega)))$ |
| 1,5,14,19,14,5,14, 19,13,27,49,64,49,13 | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_{\psi_S(\sigma S^2)}(\psi_S(\sigma S^2 \cdot \omega)) + S_\omega[\psi_{\psi_S(\sigma S^2)}(\psi_S(\sigma S^2 \cdot \omega))])$ |
| 1,5,14,19,14,5,14, 19,13,27,49,64,49,20, 32,40,31,48,73,85,73 | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_{\psi_S(\sigma S^2)}(\psi_S(\sigma S^2 \cdot \omega) + 1))$ |
| 1,5,14,19,14,5,14, 19,13,27,49,64,49,21 | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_{\psi_S(\sigma S^2 + \psi_{S_2}(\sigma S^2))}(\psi_S(\sigma S^2 \cdot \omega) + 1))$ |
| 1,5,14,19,14,5,14, 19,13,27,49,64,49,22 | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_S(\sigma S^2 + \sigma S + \psi_{S_2}(\sigma S^2 + \sigma S)))$ |
| 1,5,14,19,14,5,14, 19,13,27,49,64,49,27 | $\psi(\psi_S(\sigma S^2 \cdot \omega) + \psi_S(\sigma S^2 + \sigma S \cdot \omega))$ |
| 1,5,14,19,14,5,14, 19,14 | $\psi(\psi_S(\sigma S^2 \cdot \omega) \cdot 2)$ |
| 1,5,14,19,14,6 | $\psi(\psi_S(\sigma S^2 \cdot \omega + 1))$ |
| 1,5,14,19,14,7,5, 14,19,14 | $\psi(\psi_S(\sigma S^2 \cdot \omega + \psi_S(\sigma S^2 \cdot \omega)))$ |
| 1,5,14,19,14,7,10 | $\psi(\psi_S(\sigma S^2 \cdot \omega + S))$ |
| 1,5,14,19,14,8,5, 14,19,14 | $\psi(\psi_S(\sigma S^2 \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S^2 \cdot \omega))))$ |
| 1,5,14,19,14,8,12 | $\psi(\psi_S(\sigma S^2 \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S^2 \cdot \omega)) + S))$ |
| 1,5,14,19,14,8,13, 19 | $\psi(\psi_S(\sigma S^2 \cdot \omega + \psi_{S_2}(\sigma S \cdot \psi_S(\sigma S^2 \cdot \omega) + S_2)))$ |
| 1,5,14,19,14,8,15 | $\psi(\psi_S(\sigma S^2 \cdot \omega + \psi_{S_2}(\sigma S \cdot (\psi_S(\sigma S^2 \cdot \omega) + \omega))))$ |
| 1,5,14,19,14,9,18, 23,17 | $\psi(\psi_S(\sigma S^2 \cdot \omega + \psi_{S_2}(\sigma S \cdot S_\omega)))$ |
| 1,5,14,19,14,9,18, 23,18 | $\psi(\psi_S(\sigma S^2 \cdot \omega + \psi_{S_2}(\sigma S^2 \cdot \omega)))$ |
| 1,5,14,19,14,10 | $\psi(\psi_S(\sigma S^2 \cdot \omega + S_2))$ |
| 1,5,14,19,14,12 | $\psi(\psi_S(\sigma S^2 \cdot \omega + \psi_{S_3}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,14,19,14,12,24, 32,24,20 | $\psi(\psi_S(\sigma S^2 \cdot \omega + S_3))$ |
| 1,5,14,19,14,13 | $\psi(\psi_S(\sigma S^2 \cdot \omega + S_\omega))$ |
| 1,5,14,19,14,13,21, 26,14 | $\psi(\psi_S(\sigma S^2 \cdot \omega + \sigma S + \psi_{S_2}(\sigma S^2 \cdot \omega + \sigma S + S)))$ |
| 1,5,14,19,14,13,21, 29 | $\psi(\psi_S(\sigma S^2 \cdot \omega + \sigma S + \psi_{S_2}(\sigma S^2 \cdot \omega + \sigma S + \psi_{S_2}(\sigma S^2 \cdot \omega + \sigma S) + 1)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,19,14,13,22 | $\psi(\psi_S(\sigma S^2 \cdot \omega + \sigma S + S_2))$ |
| 1,5,14,19,14,13,27 | $\psi(\psi_S(\sigma S^2 \cdot \omega + \sigma S \cdot \omega))$ |
| 1,5,14,19,14,13,27, 49,64,49 | $\psi(\psi_S(\sigma S^2 \cdot \omega \cdot 2))$ |
| 1,5,14,19,14,14 | $\psi(\psi_S(\sigma S^2 \cdot \omega^2))$ |
| 1,5,14,19,14,14,14 | $\psi(\psi_S(\sigma S^2 \cdot \omega^3))$ |
| 1,5,14,19,14,16,6 | $\psi(\psi_S(\sigma S^2 \cdot S))$ |
| 1,5,14,19,14,18,27, 32,27 | $\psi(\psi_S(\sigma S^2 \cdot \psi_{S_2}(\sigma S^2 \cdot \omega)))$ |
| 1,5,14,19,14,19 | $\psi(\psi_S(\sigma S^2 \cdot S_2))$ |
| 1,5,14,19,14,19,12, 24,32,24,29,23 | $\psi(\psi_S(\sigma S^2 \cdot S_2 + S_\omega))$ |
| 1,5,14,19,14,19,12, 24,32,24,29,24 | $\psi(\psi_S(\sigma S^2 \cdot S_2 \cdot \omega))$ |
| 1,5,14,19,14,19,12, 24,32,24,29,24,29 | $\psi(\psi_S(\sigma S^2 \cdot {S_2}^2))$ |
| 1,5,14,19,14,19,12, 24,32,24,31 | $\psi(\psi_S(\sigma S^2 \cdot \psi_{S_3}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,14,19,14,19,12, 24,32,24,32 | $\psi(\psi_S(\sigma S^2 \cdot S_3))$ |
| 1,5,14,19,14,19,13 | $\psi(\psi_S(\sigma S^2 \cdot S_\omega))$ |
| 1,5,14,19,14,19,13, 13 | $\psi(\psi_S(\sigma S^2 \cdot S_\omega + S_\omega))$ |
| 1,5,14,19,14,19,13, 18,24 | $\psi(\psi_S(\sigma S^2 \cdot S_\omega + \varepsilon_{S_\omega+1}))$ |
| 1,5,14,19,14,19,13, 20,32,40,32,37,32 | $\psi(\psi_S(\sigma S^2 \cdot S_\omega \cdot \omega))$ |
| 1,5,14,19,14,19,13, 20,32,40,32,40 | $\psi(\psi_S(\sigma S^2 \cdot S_{\omega+1}))$ |
| 1,5,14,19,14,19,13, 20,32,40,32,40,31 | $\psi(\psi_S(\sigma S^2 \cdot S_{\omega\cdot 2}))$ |
| 1,5,14,19,14,19,13, 21 | $\psi(\psi_S(\sigma S^2 \cdot S_{\omega^2}))$ |
| 1,5,14,19,14,19,13, 21,26,14 | $\psi(\psi_S(\sigma S^3 + \psi_{S_2}(\sigma S^3 + S)))$ |
| 1,5,14,19,14,19,13, 22 | $\psi(\psi_S(\sigma S^3 + S_2))$ |
| 1,5,14,19,14,19,13, 27 | $\psi(\psi_S(\sigma S^3 + \sigma S \cdot \omega))$ |
| 1,5,14,19,14,19,14 | $\psi(\psi_S(\sigma S^3 \cdot \omega))$ |
| 1,5,14,19,15 | $\psi(\psi_S(\sigma S^\omega))$ |
| 1,5,14,19,16,6 | $\psi(\psi_S(\sigma S^S))$ |
| 1,5,14,19,16,7,10 | $\psi(\psi_S(\sigma S^S + S))$ |
| 1,5,14,19,16,9,18, 23,20,6 | $\psi(\psi_S(\sigma S^S + \psi_{S_2}(\sigma S^S)))$ |
| 1,5,14,19,16,10 | $\psi(\psi_S(\sigma S^S + S_2))$ |
| 1,5,14,19,16,13 | $\psi(\psi_S(\sigma S^S + S_\omega))$ |
| 1,5,14,19,16,13,27 | $\psi(\psi_S(\sigma S^S + \sigma S \cdot \omega))$ |
| 1,5,14,19,16,13,27, 49,64,58,28 | $\psi(\psi_S(\sigma S^S \cdot 2))$ |
| 1,5,14,19,16,14 | $\psi(\psi_S(\sigma S^S \cdot \omega))$ |
| 1,5,14,19,16,14,19 | $\psi(\psi_S(\sigma S^S \cdot S_2))$ |
| 1,5,14,19,16,14,19, 14 | $\psi(\psi_S(\sigma S^{S+1} \cdot \omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,19,16,14,19, 16,6 | $\psi(\psi_S(\sigma S^{S\cdot 2}))$ |
| 1,5,14,19,16,19 | $\psi(\psi_S(\sigma S^{\varepsilon_{S+1}}))$ |
| 1,5,14,19,19 | $\psi(\psi_S(\sigma S^{S_2}))$ |
| 1,5,14,19,19,10 | $\psi(\psi_S(\sigma S^{S_2}+S_2))$ |
| 1,5,14,19,19,12,24, 32,29 | $\psi(\psi_S(\sigma S^{S_2}+\psi_{S_3}(\sigma S^{S_2})))$ |
| 1,5,14,19,19,12,24, 32,29,24 | $\psi(\psi_S(\sigma S^{S_2}\cdot\omega))$ |
| 1,5,14,19,19,12,24, 32,32 | $\psi(\psi_S(\sigma S^{S_3}))$ |
| 1,5,14,19,19,13 | $\psi(\psi_S(\sigma S^{S_\omega}))$ |
| 1,5,14,19,19,13,21, 26,14 | $\psi(\psi_S(\sigma S^{\sigma S}+\psi_{S_2}(\sigma S^{\sigma S}+S)))$ |
| 1,5,14,19,19,14 | $\psi(\psi_S(\sigma S^{\sigma S}\cdot\omega))$ |
| 1,5,14,19,25 | $\psi(\psi_S(S(\sigma S+1)))$ |
| 1,5,14,19,26 | $\psi(\psi_S(\sigma S_2)+\Omega_{\alpha+1}\cdot\omega)$ |
| 1,5,14,20 | $\psi(\psi_S(\sigma S_2)+\psi_{\alpha_2}(\psi_S(\sigma S_2)+1))$ |
| 1,5,14,20,4 | $\psi(\psi_S(\sigma S_2)+\psi_{\alpha_2}(\psi_S(\sigma S_2)+1)+\Omega_\omega)$ |
| 1,5,14,20,4,11,24, 32,41 | $\psi(\psi_S(\sigma S_2)+\psi_{\alpha_2}(\psi_S(\sigma S_2)+1)+\psi_\alpha(\psi_S(S(\sigma S+1))))$ |
| 1,5,14,20,4,11,24, 33 | $\psi(\psi_S(\sigma S_2)+\psi_{\alpha_2}(\psi_S(\sigma S_2)+1)+\psi_\alpha(\psi_S(\sigma S_2)+$ $\psi_{\alpha_2}(\psi_S(\sigma S_2)+1)))$ |
| 1,5,14,20,4,11,24, 33,5 | $\psi(\psi_S(\sigma S_2)+\psi_{\alpha_2}(\psi_S(\sigma S_2)+1)+\psi_\alpha(\psi_S(\sigma S_2)+$ $\psi_{\alpha_2}(\psi_S(\sigma S_2)+1)+1))$ |
| 1,5,14,20,4,11,24, 33,6,9 | $\psi(\psi_S(\sigma S_2)+\psi_{\alpha_2}(\psi_S(\sigma S_2)+1)+\alpha)$ |
| 1,5,14,20,4,11,24, 33,6,11 | $\psi(\psi_S(\sigma S_2)+\psi_{\alpha_2}(\psi_S(\sigma S_2)+1)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega)))$ |
| 1,5,14,20,4,11,24, 33,7 | $\psi(\psi_S(\sigma S_2)+\psi_{\alpha_2}(\psi_S(\sigma S_2)+2))$ |
| 1,5,14,20,4,11,24, 33,7,9,5 | $\psi(\psi_S(\sigma S_2)+\psi_{\alpha_2}(\psi_S(\sigma S_2)+\alpha))$ |
| 1,5,14,20,4,11,24, 33,7,10 | $\psi(\psi_S(\sigma S_2)+\psi_{\alpha_2}(\psi_S(\sigma S_2)+\Omega_{\alpha+1}+1))$ |
| 1,5,14,20,4,11,24, 33,7,14 | $\psi(\psi_S(\sigma S_2)+\psi_{\alpha_2}(\psi_S(\sigma S_2)+\psi_{\alpha_2}(\psi_S(\sigma S\cdot\omega))))$ |
| 1,5,14,20,4,11,24, 33,8 | $\psi(\psi_S(\sigma S_2)+\alpha_2)$ |
| 1,5,14,20,4,11,24, 33,9 | $\psi(\psi_S(\sigma S_2)+\Omega_{\alpha_2+1}+\psi_{\alpha_2}(\psi_S(\sigma S_2)+\Omega_{\alpha_2+1}+1))$ |
| 1,5,14,20,4,11,24, 33,10 | $\psi(\psi_S(\sigma S_2)+\alpha_\omega)$ |
| 1,5,14,20,4,11,24, 33,11 | $\psi(\psi_S(\sigma S_2)+\psi_S(\sigma S\cdot\omega))$ |
| 1,5,14,20,4,11,24, 33,11,24,32,41 | $\psi(\psi_S(\sigma S_2)+\psi_S(S(\sigma S+1)))$ |
| 1,5,14,20,4,11,24, 33,11,24,33 | $\psi(\psi_S(\sigma S_2)\cdot 2+\psi_{\alpha_2}(\psi_S(\sigma S_2)\cdot 2+1))$ |
| 1,5,14,20,4,11,24, 33,12 | $\psi(\psi_S(\sigma S_2+1))$ |
| 1,5,14,20,4,11,24, 33,13,5 | $\psi(\psi_S(\sigma S_2+\alpha))$ |
| 1,5,14,20,4,11,24, 33,14 | $\psi(\psi_S(\sigma S_2+\Omega_{\alpha+1})+\psi_{\alpha_2}(\psi_S(\sigma S_2+\Omega_{\alpha+1})+1))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,20,4,11,24, 33,14,18 | $\psi(\psi_S(\sigma S_2 + \psi_{\alpha_2}(\alpha_2)))$ |
| 1,5,14,20,4,11,24, 33,15 | $\psi(\psi_S(\sigma S_2 + \alpha_2))$ |
| 1,5,14,20,4,11,24, 33,15,10 | $\psi(\psi_S(\sigma S_2 + \alpha_\omega))$ |
| 1,5,14,20,4,11,24, 33,15,11 | $\psi(\psi_S(\sigma S_2 + \psi_S(\sigma S \cdot \omega)))$ |
| 1,5,14,20,4,11,24, 33,15,11,24,33 | $\psi(\psi_S(\sigma S_2 + \psi_S(\sigma S_2)) + \psi_{\alpha_2}(\psi_S(\sigma S_2 + \psi_S(\sigma S_2)) + 1))$ |
| 1,5,14,20,4,11,24, 33,15,20 | $\psi(\psi_S(\sigma S_2 + S))$ |
| 1,5,14,20,4,11,24, 33,19 | $\psi(\psi_S(\sigma S_2 + S_2))$ |
| 1,5,14,20,4,11,24, 33,21,36,48,32 | $\psi(\psi_S(\sigma S_2 + S_3))$ |
| 1,5,14,20,4,11,24, 33,23 | $\psi(\psi_S(\sigma S_2 + S_\omega))$ |
| 1,5,14,20,4,11,24, 33,23,35,43,24 | $\psi(\psi_S(\sigma S_2 + \sigma S + \psi_{S_2}(\sigma S_2 + \sigma S + S)))$ |
| 1,5,14,20,4,11,24, 33,23,42 | $\psi(\psi_S(\sigma S_2 + \sigma S \cdot \omega))$ |
| 1,5,14,20,4,11,24, 33,23,42,70,90,111 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(S(\sigma S + 1))))$ |
| 1,5,14,20,4,11,24, 33,23,42,70,91 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2))+$ $\psi_{\alpha_2}(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2)) + 1))$ |
| 1,5,14,20,4,11,24, 33,23,42,70,91,42 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2)) + \psi_S(\sigma S_2 + \sigma S \cdot \omega))$ |
| 1,5,14,20,4,11,24, 33,24 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + 1)))$ |
| 1,5,14,20,4,11,24, 33,24,23 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + 1) + S_\omega))$ |
| 1,5,14,20,4,11,24, 33,24,24 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + 2)))$ |
| 1,5,14,20,4,11,24, 33,24,28,12 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + S)))$ |
| 1,5,14,20,4,11,24, 33,24,30 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \psi_{S_2}(\sigma S \cdot S \cdot \omega))))$ |
| 1,5,14,20,4,11,24, 33,24,32 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + S_2)))$ |
| 1,5,14,20,4,11,24, 33,24,32,23 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + S_\omega)))$ |
| 1,5,14,20,4,11,24, 33,24,32,23,35,43,24 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S)+$ $\psi_{S_2}(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S) + S)))$ |
| 1,5,14,20,4,11,24, 33,24,32,23,36 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S) + S_2))$ |
| 1,5,14,20,4,11,24, 33,24,32,23,42 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S) + \sigma S \cdot \omega))$ |
| 1,5,14,20,4,11,24, 33,24,32,24 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S + 1)))$ |
| 1,5,14,20,4,11,24, 33,24,32,24,32 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S + S_2)))$ |
| 1,5,14,20,4,11,24, 33,24,32,24,32,24 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S \cdot 2 + 1)))$ |
| 1,5,14,20,4,11,24, 33,24,32,25 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S \cdot \omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,20,4,11,24, 33,24,32,41 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \varepsilon_{\sigma S+1})))$ |
| 1,5,14,20,4,11,24, 33,24,33 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2)))+$ $\psi_{\alpha_2}(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2+$ $\psi_{S(\sigma S+1)}(\sigma S_2))) + 1))$ |
| 1,5,14,20,4,11,24, 33,25 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2+$ $\psi_{S(\sigma S+1)}(\sigma S_2 + 1))))$ |
| 1,5,14,20,4,11,24, 33,32,24 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \psi_{S(\sigma S+1)}(\sigma S_2 + \sigma S) + 1)))$ |
| 1,5,14,20,4,11,24, 33,43 | $\psi(\psi_S(\sigma S_2 + S(\sigma S + 1)))$ |
| 1,5,14,20,4,11,24, 33,44 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+2)}(\sigma S_2))+$ $\psi_{\alpha_2}(\psi_S(\sigma S_2 + \psi_{S(\sigma S+2)}(\sigma S_2)) + 1))$ |
| 1,5,14,20,4,11,24, 33,44,23 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+2)}(\sigma S_2) + S_\omega))$ |
| 1,5,14,20,4,11,24, 33,44,24 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+2)}(\sigma S_2)+$ $\psi_{S(\sigma S+1)}(\sigma S_2 + \psi_{S(\sigma S+2)}(\sigma S_2) + 1)))$ |
| 1,5,14,20,4,11,24, 33,44,43 | $\psi(\psi_S(\sigma S_2 + \psi_{S(\sigma S+2)}(\sigma S_2) + S(\sigma S + 1)))$ |
| 1,5,14,20,4,11,24, 33,45 | $\psi(\psi_S(\sigma S_2 + S(\sigma S + \omega)))$ |
| 1,5,14,20,4,11,24, 33,46 | $\psi(\psi_S(\sigma S_2 \cdot 2) + \psi_{\beta_2}(\psi_S(\sigma S \cdot \omega)))$ |
| 1,5,14,20,4,11,24, 34 | $\psi(\psi_S(\sigma S_2 \cdot 2) + \psi_{\beta_2}(\psi_S(\sigma S_2 \cdot 2) + 1))$ |
| 1,5,14,20,5 | $\psi(\psi_S(\sigma S_2 \cdot \omega))$ |
| 1,5,14,20,5,9 | $\psi(\psi_S(\sigma S_2 \cdot \omega^2))$ |
| 1,5,14,20,5,14,19, 25 | $\psi(\psi_S(\sigma S_2 \cdot \psi_S(S(\sigma S + 1))))$ |
| 1,5,14,20,5,14,20 | $\psi(\psi_S(\sigma S_2 \cdot \psi_S(\sigma S_2)) + \psi_{\alpha_2}(\psi_S(\sigma S_2 + \psi_S(\sigma S_2)) + 1))$ |
| 1,5,14,20,6 | $\psi(\psi_S(\sigma S_2 \cdot S))$ |
| 1,5,14,20,6,6 | $\psi(\psi_S(\sigma S_2 \cdot S + 1))$ |
| 1,5,14,20,7,10 | $\psi(\psi_S(\sigma S_2 \cdot S + S))$ |
| 1,5,14,20,10 | $\psi(\psi_S(\sigma S_2 \cdot S + S_2))$ |
| 1,5,14,20,13,27 | $\psi(\psi_S(\sigma S_2 \cdot S + \sigma S \cdot \omega))$ |
| 1,5,14,20,13,27,49, 64,80 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(S(\sigma S + 1))))$ |
| 1,5,14,20,13,27,49, 68,27 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot \psi_S(\sigma S_2 \cdot S + \sigma S \cdot \omega))))$ |
| 1,5,14,20,13,27,49, 68,28 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S)))$ |
| 1,5,14,20,14 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + 1)))$ |
| 1,5,14,20,14,14 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + 2)))$ |
| 1,5,14,20,14,19 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + S_2)))$ |
| 1,5,14,20,14,19,13, 27 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \sigma S) + \sigma S \cdot \omega))$ |
| 1,5,14,20,14,19,13, 27,49,68,49 | $\psi(\psi_S(\sigma S_2 \cdot S+$ $\psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \sigma S) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + 1)))$ |
| 1,5,14,20,14,19,14 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \sigma S + 1)))$ |
| 1,5,14,20,14,19,25 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(S(\sigma S + 1)))))$ |
| 1,5,14,20,14,20 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S+$ $\psi_{S(\sigma S+1)}(\sigma S_2))) + \psi_{\alpha_2}(\psi_S(\sigma S_2 \cdot S+$ $\psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2))) + 1))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,20,14,20,4, 11,24,34,24,33,8 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2))) + \alpha_2)$ |
| 1,5,14,20,14,20,4, 11,24,34,24,33,12 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2)) + 1))$ |
| 1,5,14,20,14,20,4, 11,24,34,24,33,23 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2)) + S_\omega))$ |
| 1,5,14,20,14,20,4, 11,24,34,24,33,24 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2) + 1)))$ |
| 1,5,14,20,14,20,4, 11,24,34,24,33,24,32, 41 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S+ \psi_{S(\sigma S+1)}(\sigma S_2) + \varepsilon_{\sigma S+1})))$ |
| 1,5,14,20,14,20,4, 11,24,34,24,33,24,33, 8 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S+ \psi_{S(\sigma S+1)}(\sigma S_2) \cdot 2)) + \alpha_2)$ |
| 1,5,14,20,14,20,4, 11,24,34,24,33,45 | $\psi(\psi_S(\sigma S_2 \cdot S+ \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 + S(\sigma S + \omega)))))$ |
| 1,5,14,20,14,20,5 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot \omega))))$ |
| 1,5,14,20,14,20,6 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S))))$ |
| 1,5,14,20,14,20,14 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S) + 1)))$ |
| 1,5,14,20,14,20,14, 19,25 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S+ \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S) + \varepsilon_{\sigma S+1})))$ |
| 1,5,14,20,15 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + 1))))$ |
| 1,5,14,20,19,14 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S+ \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \sigma S) + 1)))$ |
| 1,5,14,20,20,6 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S+ \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S)))))$ |
| 1,5,14,20,27 | $\psi(\psi_S(\sigma S_2 \cdot S + S(\sigma S + 1)))$ |
| 1,5,14,20,28 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+2)}(\sigma S_2))+ \psi_{\alpha_2}(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+2)}(\sigma S_2)) + 1))$ |
| 1,5,14,20,28,6 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+2)}(\sigma S_2 \cdot S)))$ |
| 1,5,14,20,28,27 | $\psi(\psi_S(\sigma S_2 \cdot S + \psi_{S(\sigma S+2)}(\sigma S_2 \cdot S) + S(\sigma S + 1)))$ |
| 1,5,14,20,28,37 | $\psi(\psi_S(\sigma S_2 \cdot S + S(\sigma S + 2)))$ |
| 1,5,14,20,29 | $\psi(\psi_S(\sigma S_2 \cdot S + S(\sigma S + \omega)))$ |
| 1,5,14,20,29,29 | $\psi(\psi_S(\sigma S_2 \cdot S + S(\sigma S + \omega) \cdot 2))$ |
| 1,5,14,20,29,38,45, 30 | $\psi(\psi_S(\sigma S_2 \cdot (S + 1) + \psi_{S_2}(\sigma S_2 \cdot (S + 1) + S)))$ |
| 1,5,14,20,29,39 | $\psi(\psi_S(\sigma S_2 \cdot (S + 1) + S_2))$ |
| 1,5,14,20,29,42 | $\psi(\psi_S(\sigma S_2 \cdot (S + 1) + S_\omega))$ |
| 1,5,14,20,30 | $\psi(\psi_S(\sigma S_2 \cdot (S + 1) + \sigma S \cdot \omega))$ |
| 1,5,14,20,30,40,47, 31 | $\psi(\psi_S(\sigma S_2 \cdot (S + 1) + \sigma S \cdot S))$ |
| 1,5,14,20,30,40,47, 55 | $\psi(\psi_S(\sigma S_2 \cdot (S + 1) + \sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S + \varepsilon_{S+1})))$ |
| 1,5,14,20,30,40,50 | $\psi(\psi_S(\sigma S_2 \cdot (S + 1) + \sigma S \cdot S + \psi_{S_2}(\sigma S \cdot S \cdot \omega)))$ |
| 1,5,14,21 | $\psi(\psi_S(\sigma S_2 \cdot (S + 1) + \sigma S \cdot S \cdot \omega))$ |
| 1,5,14,21,10 | $\psi(\psi_S(\sigma S_2 \cdot (S + 1) + \sigma S \cdot S \cdot \omega + S_2))$ |
| 1,5,14,21,13,27 | $\psi(\psi_S(\sigma S_2 \cdot (S + 1) + \sigma S \cdot (S \cdot \omega + \omega)))$ |
| 1,5,14,21,13,27,49 | $\psi(\psi_S(\sigma S_2 \cdot (S + 1) + \sigma S \cdot S \cdot \omega^2))$ |
| 1,5,14,21,14 | $\psi(\psi_S(\sigma S_2 \cdot (S + 1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S + 1) + 1)))$ |
| 1,5,14,21,14,19 | $\psi(\psi_S(\sigma S_2 \cdot (S + 1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S + 1) + S_2)))$ |
| 1,5,14,21,14,19,13 | $\psi(\psi_S(\sigma S_2 \cdot (S + 1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S + 1) + S_\omega)))$ |
| 1,5,14,21,14,20,6 | $\psi(\psi_S(\sigma S_2 \cdot (S + 1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S + 1)+ \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S))))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,21,15 | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1)+1))))$ |
| 1,5,14,21,19,25 | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + \varepsilon_{\sigma S+1}))))$ |
| 1,5,14,21,20,6 | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S)))))$ |
| 1,5,14,21,29 | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + S(\sigma S+1)))$ |
| 1,5,14,21,29,29 | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + S(\sigma S+1) \cdot 2))$ |
| 1,5,14,21,30,6 | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+2)}(\sigma S_2 \cdot S)))$ |
| 1,5,14,21,31 | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+2)}(\sigma S_2 \cdot (S+1)) + \sigma S \cdot S \cdot \omega))$ |
| 1,5,14,21,31,14 | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+2)}(\sigma S_2 \cdot (S+1)) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+2)}(\sigma S_2 \cdot (S+1)) + 1)))$ |
| 1,5,14,21,31,29 | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + \psi_{S(\sigma S+2)}(\sigma S_2 \cdot (S+1)) + S(\sigma S+1)))$ |
| 1,5,14,21,31,42 | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + S(\sigma S+2)))$ |
| 1,5,14,21,32 | $\psi(\psi_S(\sigma S_2 \cdot (S+1) + S(\sigma S+\omega)))$ |
| 1,5,14,21,32,43,51,33 | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + \psi_{S_2}(\sigma S_2 \cdot (S+2) + S)))$ |
| 1,5,14,21,32,44 | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + S_2))$ |
| 1,5,14,21,32,49 | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + \sigma S \cdot \omega))$ |
| 1,5,14,21,33 | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + \sigma S \cdot S \cdot \omega))$ |
| 1,5,14,21,33,13 | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + \sigma S \cdot S \cdot \omega + S_\omega))$ |
| 1,5,14,21,33,14 | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+2) + 1)))$ |
| 1,5,14,21,33,15 | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+2) + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot (S+2) + 1))))$ |
| 1,5,14,21,33,29 | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + S(\sigma S+1)))$ |
| 1,5,14,21,33,32 | $\psi(\psi_S(\sigma S_2 \cdot (S+2) + S(\sigma S+\omega)))$ |
| 1,5,14,21,33,33 | $\psi(\psi_S(\sigma S_2 \cdot (S+3) + \sigma S \cdot S \cdot \omega))$ |
| 1,5,14,21,33,35,6 | $\psi(\psi_S(\sigma S_2 \cdot S \cdot 2))$ |
| 1,5,14,21,33,35,6,6 | $\psi(\psi_S(\sigma S_2 \cdot S \cdot 2 + 1))$ |
| 1,5,14,21,33,35,12,24,34,49,51,6 | $\psi(\psi_S(\sigma S_2 \cdot S \cdot 2 + \psi_{S_3}(\sigma S_2 \cdot S \cdot 2)))$ |
| 1,5,14,21,33,35,13 | $\psi(\psi_S(\sigma S_2 \cdot S \cdot 2 + S_\omega))$ |
| 1,5,14,21,33,35,14 | $\psi(\psi_S(\sigma S_2 \cdot S \cdot 2 + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S \cdot 2 + 1)))$ |
| 1,5,14,21,33,35,29 | $\psi(\psi_S(\sigma S_2 \cdot S \cdot 2 + S(\sigma S+1)))$ |
| 1,5,14,21,33,35,33,35,6 | $\psi(\psi_S(\sigma S_2 \cdot S \cdot 3))$ |
| 1,5,14,21,33,36 | $\psi(\psi_S(\sigma S_2 \cdot \psi_{S_2}(\sigma S)) + \psi_{\alpha_2}(\psi_S(\sigma S_2 \cdot \psi_{S_2}(\sigma S)) + 1))$ |
| 1,5,14,21,33,38 | $\psi(\psi_S(\sigma S_2 \cdot S_2))$ |
| 1,5,14,21,33,38,12,24,34,49,54,23 | $\psi(\psi_S(\sigma S_2 \cdot S_2 + S_\omega))$ |
| 1,5,14,21,33,38,12,24,34,49,54,48 | $\psi(\psi_S(\sigma S_2 \cdot S_2 + S(\sigma S+\omega)))$ |
| 1,5,14,21,33,38,12,24,34,49,54,49,54 | $\psi(\psi_S(\sigma S_2 \cdot S_2 \cdot 2))$ |
| 1,5,14,21,33,38,12,24,34,49,57 | $\psi(\psi_S(\sigma S_2 \cdot S_3))$ |
| 1,5,14,21,33,38,13 | $\psi(\psi_S(\sigma S_2 \cdot S_\omega))$ |
| 1,5,14,21,33,38,14 | $\psi(\psi_S(\sigma S_2 \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot \sigma S + 1)))$ |
| 1,5,14,21,33,38,29 | $\psi(\psi_S(\sigma S_2 \cdot \sigma S + S(\sigma S+1)))$ |
| 1,5,14,21,33,38,33,38 | $\psi(\psi_S(\sigma S_2 \cdot (\sigma S + S_2)))$ |
| 1,5,14,21,33,38,44 | $\psi(\psi_S(\sigma S_2 \cdot \varepsilon_{\sigma S+1}))$ |
| 1,5,14,21,33,39,6 | $\psi(\psi_S(\sigma S_2 \cdot \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,21,33,41 | $\psi(\psi_S(\sigma S_2 \cdot S(\sigma S+1)))$ |
| 1,5,14,21,33,41,29 | $\psi(\psi_S(\sigma S_2 \cdot S(\sigma S+1)+S(\sigma S+1)))$ |
| 1,5,14,21,33,41,30, 6 | $\psi(\psi_S(\sigma S_2 \cdot S(\sigma S+1)+\psi_{S(\sigma S+2)}(\sigma S_2 \cdot S)))$ |
| 1,5,14,21,33,41,31, 46,54,42 | $\psi(\psi_S(\sigma S_2 \cdot S(\sigma S+1)+S(\sigma S+2)))$ |
| 1,5,14,21,33,41,31, 46,54,45 | $\psi(\psi_S(\sigma S_2 \cdot S(\sigma S+1)+S(\sigma S+\omega)))$ |
| 1,5,14,21,33,41,31, 46,54,46,54 | $\psi(\psi_S(\sigma S_2 \cdot S(\sigma S+1)\cdot 2))$ |
| 1,5,14,21,33,41,31, 46,57 | $\psi(\psi_S(\sigma S_2 \cdot S(\sigma S+2)))$ |
| 1,5,14,21,33,41,32 | $\psi(\psi_S(\sigma S_2 \cdot S(\sigma S+\omega)))$ |
| 1,5,14,21,33,41,33 | $\psi(\psi_S(\sigma {S_2}^2+\sigma S\cdot S\cdot\omega))$ |
| 1,5,14,21,33,41,33, 32 | $\psi(\psi_S(\sigma {S_2}^2+S(\sigma S+\omega)))$ |
| 1,5,14,21,33,41,33, 38,13 | $\psi(\psi_S(\sigma {S_2}^2+\sigma S_2\cdot S_\omega))$ |
| 1,5,14,21,33,41,34 | $\psi(\psi_S(\sigma {S_2}^2\cdot\omega))$ |
| 1,5,14,21,33,41,50 | $\psi(\psi_S(S(\sigma S_2+1)))$ |
| 1,5,14,21,33,42,6 | $\psi(\psi_S(\sigma S_3\cdot S))$ |
| 1,5,14,22 | $\psi(\psi_S(\sigma S_\omega))$ |
| 1,5,14,22,5,14,19, 25 | $\psi(\psi_S(\sigma S_\omega)+\psi_S(\varepsilon_{\sigma S+1}))$ |
| 1,5,14,22,6 | $\psi(\psi_S(\sigma S_\omega+1))$ |
| 1,5,14,22,7,10 | $\psi(\psi_S(\sigma S_\omega+S))$ |
| 1,5,14,22,13 | $\psi(\psi_S(\sigma S_\omega+S_\omega))$ |
| 1,5,14,22,13,27 | $\psi(\psi_S(\sigma S_\omega+\sigma S\cdot\omega))$ |
| 1,5,14,22,14 | $\psi(\psi_S(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_\omega+1)))$ |
| 1,5,14,22,14,19,13 | $\psi(\psi_S(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_\omega+S_\omega)))$ |
| 1,5,14,22,14,19,14 | $\psi(\psi_S(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_\omega+\sigma S+1)))$ |
| 1,5,14,22,14,19,25 | $\psi(\psi_S(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_\omega+\varepsilon_{\sigma S+1})))$ |
| 1,5,14,22,14,20,6 | $\psi(\psi_S(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_2\cdot S))))$ |
| 1,5,14,22,14,21 | $\psi(\psi_S(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_\omega+$ $\psi_{S(\sigma S+1)}(\sigma S_2\cdot(S+1)))+\sigma S\cdot S\cdot\omega))$ |
| 1,5,14,22,14,21,14 | $\psi(\psi_S(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_2\cdot(S+1))+1)))$ |
| 1,5,14,22,14,21,15 | $\psi(\psi_S(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_2\cdot(S+1)+1))))$ |
| 1,5,14,22,14,21,29 | $\psi(\psi_S(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_\omega+$ $\psi_{S(\sigma S+1)}(\sigma S_2\cdot(S+1)+S(\sigma S+1)))))$ |
| 1,5,14,22,14,21,33, 34 | $\psi(\psi_S(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_2\cdot(S+\omega)))))$ |
| 1,5,14,22,14,21,33, 41,50 | $\psi(\psi_S(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_\omega+\psi_{S(\sigma S+1)}(\varepsilon_{\sigma S_2+1}))))$ |
| 1,5,14,22,14,21,33, 44 | $\psi(\psi_S(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_\omega))))$ |
| 1,5,14,22,14,21,33, 44,15 | $\psi(\psi_S(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_\omega+\psi_{S(\sigma S+1)}(\sigma S_\omega+1))))$ |
| 1,5,14,22,14,21,33, 44,29 | $\psi(\psi_S(\sigma S_\omega+S(\sigma S+1)))$ |
| 1,5,14,22,14,21,33, 44,32 | $\psi(\psi_S(\sigma S_\omega+S(\sigma S+\omega)))$ |
| 1,5,14,22,14,21,33, 44,32,49 | $\psi(\psi_S(\sigma S_\omega+\sigma S_2+\sigma S\cdot\omega))$ |
| 1,5,14,22,14,21,33, 44,33,34 | $\psi(\psi_S(\sigma S_\omega+\sigma S_2\cdot\omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,22,14,21,33, 44,33,41,50 | $\psi(\psi_S(\sigma S_\omega + \varepsilon_{\sigma S_2+1}))$ |
| 1,5,14,22,14,21,33, 44,33,43,58,72 | $\psi(\psi_S(\sigma S_\omega + \psi_S(\sigma S_2 + 1)(\sigma S_\omega)))$ |
| 1,5,14,22,14,22 | $\psi(\psi_S(\sigma S_\omega \cdot 2))$ |
| 1,5,14,22,16,6 | $\psi(\psi_S(\sigma S_\omega \cdot S))$ |
| 1,5,14,22,19,13 | $\psi(\psi_S(\sigma S_\omega \cdot S_\omega))$ |
| 1,5,14,22,19,14 | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S + 1)))$ |
| 1,5,14,22,19,14,14 | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S + 2)))$ |
| 1,5,14,22,19,14,19, 14 | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S + \sigma S + 1)))$ |
| 1,5,14,22,19,14,20, 6 | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S))))$ |
| 1,5,14,22,19,14,21, 33,44 | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_\omega))))$ |
| 1,5,14,22,19,14,21, 33,44,38,14 | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S +$ $\psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S) + 1)))$ |
| 1,5,14,22,19,14,21, 33,44,38,15 | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S +$ $\psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S + 1))))$ |
| 1,5,14,22,19,14,21, 33,44,38,29 | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + S(\sigma S + 1)))$ |
| 1,5,14,22,19,14,21, 33,44,38,33,34 | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S + \sigma S_2 \cdot \omega))$ |
| 1,5,14,22,19,14,21, 33,44,38,33,44 | $\psi(\psi_S(\sigma S_\omega \cdot (\sigma S + 1)))$ |
| 1,5,14,22,19,14,21, 33,44,38,33,44,38,14 | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S \cdot 2 + \psi_{S(\sigma S+1)}(\sigma S_\omega \cdot \sigma S \cdot 2 + 1)))$ |
| 1,5,14,22,19,14,21, 33,44,38,38,13 | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S \cdot S_\omega))$ |
| 1,5,14,22,19,14,21, 33,44,39,6 | $\psi(\psi_S(\sigma S_\omega \cdot \psi_{S(\sigma S+1)}(\sigma S_2 \cdot S)))$ |
| 1,5,14,22,19,14,21, 33,44,41 | $\psi(\psi_S(\sigma S_\omega \cdot S(\sigma S + 1)))$ |
| 1,5,14,22,19,14,21, 33,44,41,32 | $\psi(\psi_S(\sigma S_\omega \cdot S(\sigma S + \omega)))$ |
| 1,5,14,22,19,14,21, 33,44,41,33 | $\psi(\psi_S(\sigma S_\omega \cdot \sigma S_2 + \sigma S \cdot S \cdot \omega))$ |
| 1,5,14,22,19,14,22 | $\psi(\psi_S(\sigma {S_\omega}^2))$ |
| 1,5,14,22,19,14,22, 14 | $\psi(\psi_S(\sigma {S_\omega}^2 + \psi_{S(\sigma S+1)}(\sigma {S_\omega}^2 + 1)))$ |
| 1,5,14,22,19,14,22, 14,22 | $\psi(\psi_S(\sigma {S_\omega}^2 + \sigma S_\omega))$ |
| 1,5,14,22,19,14,22, 19 | $\psi(\psi_S(\sigma {S_\omega}^2 + \sigma S_\omega \cdot S_2))$ |
| 1,5,14,22,19,19 | $\psi(\psi_S(\sigma {S_\omega}^2 \cdot S_2))$ |
| 1,5,14,22,19,19,14 | $\psi(\psi_S(\sigma {S_\omega}^2 \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma {S_\omega}^2 \cdot \sigma S + 1)))$ |
| 1,5,14,22,19,19,14, 22 | $\psi(\psi_S(\sigma {S_\omega}^3))$ |
| 1,5,14,22,19,25 | $\psi(\psi_S(S(\sigma S_\omega + 1)))$ |
| 1,5,14,22,20 | $\psi(\psi_S(\sigma S_{\omega+1}) + \psi_{\alpha_2}(\psi_S(\sigma S_{\omega+1}) + 1))$ |
| 1,5,14,22,20,5 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot \omega))$ |
| 1,5,14,22,20,6 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot S))$ |
| 1,5,14,22,20,14,22 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot S + \sigma S_\omega))$ |
| 1,5,14,22,20,14,22, 19,25 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot S + \varepsilon_{\sigma S_\omega+1}))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,22,20,15 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot S + \psi_S(\sigma S_\omega + 1)(\sigma S_{\omega+1} \cdot S + 1)))$ |
| 1,5,14,22,20,19,25 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot S + \psi_S(\sigma S_\omega + 1)(\sigma S_{\omega+1} \cdot S + \varepsilon_{\sigma S_\omega+1})))$ |
| 1,5,14,22,20,20,6 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot S + \psi_S(\sigma S_\omega + 1)(\sigma S_{\omega+1} \cdot S +$ $\psi_S(\sigma S_\omega + 1)(\sigma S_{\omega+1} \cdot S))))$ |
| 1,5,14,22,20,27 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot S + S(\sigma S_\omega + 1)))$ |
| 1,5,14,22,20,29 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot S + S(\sigma S_\omega + \omega)))$ |
| 1,5,14,22,21 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot (S+1) + \sigma S \cdot S \cdot \omega))$ |
| 1,5,14,22,21,14 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot (S+1) + \psi_{S(\sigma S+1)}(\sigma S_{\omega+1} \cdot (S+1) + 1)))$ |
| 1,5,14,22,21,14,22 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot (S+1) + \sigma S_\omega))$ |
| 1,5,14,22,21,15 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot (S+1) +$ $\psi_S(\sigma S_\omega + 1)(\sigma S_{\omega+1} \cdot (S+1) + 1)))$ |
| 1,5,14,22,21,19,14, 22 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot (S+1) +$ $\psi_S(\sigma S_\omega + 1)(\sigma S_{\omega+1} \cdot (S+1) + \sigma S_\omega)))$ |
| 1,5,14,22,21,22 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot (S+1) + \psi_S(\sigma S_\omega + 1)(\sigma S_{\omega+1} \cdot (S+1) +$ $\psi_S(\sigma S_\omega + 1)(\sigma S_{\omega+1} \cdot (S+1) + 1))))$ |
| 1,5,14,22,21,29 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot (S+1) + S(\sigma S_\omega + 1)))$ |
| 1,5,14,22,21,32 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot (S+1) + S(\sigma S_\omega + \omega)))$ |
| 1,5,14,22,21,33,34 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot (S + \omega)))$ |
| 1,5,14,22,21,33,36, 6 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot \psi_{S_2}(\sigma S \cdot S)))$ |
| 1,5,14,22,21,33,38, 13 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot S_\omega))$ |
| 1,5,14,22,21,33,38, 14 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot \sigma S + \psi_{S(\sigma S+1)}(\sigma S_{\omega+1} \cdot \sigma S + 1)))$ |
| 1,5,14,22,21,33,38, 14,22 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot \sigma S_\omega))$ |
| 1,5,14,22,21,33,41 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot S(\sigma S_\omega + 1)))$ |
| 1,5,14,22,21,33,41, 32 | $\psi(\psi_S(\sigma S_{\omega+1} \cdot S(\sigma S_\omega + \omega)))$ |
| 1,5,14,22,21,33,41, 33,34 | $\psi(\psi_S(\sigma {S_{\omega+1}}^2 + \sigma S_{\omega+1} \cdot \omega))$ |
| 1,5,14,22,21,33,44 | $\psi(\psi_S(\sigma S_{\omega \cdot 2}))$ |
| 1,5,14,22,22 | $\psi(\psi_S(\sigma S_{\omega^2}))$ |
| 1,5,14,22,22,14,22 | $\psi(\psi_S(\sigma S_{\omega^2} + \sigma S_\omega))$ |
| 1,5,14,22,22,14,22, 21,33,44,44,32 | $\psi(\psi_S(\sigma S_{\omega^2} + S(\sigma S_\omega + \omega)))$ |
| 1,5,14,22,22,14,22, 21,33,44,44,33,44 | $\psi(\psi_S(\sigma S_{\omega^2} + \sigma S_{\omega \cdot 2}))$ |
| 1,5,14,22,22,14,22, 22 | $\psi(\psi_S(\sigma S_{\omega^2} \cdot 2))$ |
| 1,5,14,22,22,19,14, 22 | $\psi(\psi_S(\sigma S_{\omega^2} \cdot \sigma S_\omega))$ |
| 1,5,14,22,22,19,14, 22,21,33,44,44,41,33, 44 | $\psi(\psi_S(\sigma S_{\omega^2} \cdot \sigma S_{\omega \cdot 2}))$ |
| 1,5,14,22,22,19,14, 22,22 | $\psi(\psi_S(\sigma {S_{\omega^2}}^2))$ |
| 1,5,14,22,22,19,25 | $\psi(\psi_S(S(\sigma S_{\omega^2} + 1)))$ |
| 1,5,14,22,22,22 | $\psi(\psi_S(\sigma S_{\omega^3}))$ |
| 1,5,14,22,24,6 | $\psi(\psi_S(\sigma S S))$ |
| 1,5,14,22,27,13 | $\psi(\psi_S(\sigma S S_\omega))$ |
| 1,5,14,22,27,14 | $\psi(\psi_S(\sigma S \sigma S + \psi_{S(\sigma S+1)}(\sigma S \sigma S + 1)))$ |
| 1,5,14,22,27,14,21, 33,44,49,33,34 | $\psi(\psi_S(\sigma S \sigma S + \sigma S_2 \cdot \omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,22,27,14,21, 33,44,49,44 | $\psi(\psi_S(\sigma S\sigma S\cdot\omega))$ |
| 1,5,14,22,27,14,21, 33,44,52,32 | $\psi(\psi_S(\sigma SS(\sigma S+\omega)))$ |
| 1,5,14,22,27,14,22 | $\psi(\psi_S(\sigma S\sigma S_\omega))$ |
| 1,5,14,22,27,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S)))$ |
| 1,5,14,22,27,15,13 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S)+S_\omega))$ |
| 1,5,14,22,27,15,14, 22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S)+\sigma S_\omega))$ |
| 1,5,14,22,27,15,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S+1)))$ |
| 1,5,14,22,27,19 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S+S_2)))$ |
| 1,5,14,22,27,19,14, 22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S+\sigma S_\omega)))$ |
| 1,5,14,22,27,19,14, 22,27,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S+\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S))))$ |
| 1,5,14,22,27,19,25 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S+S(\sigma\sigma S+1))))$ |
| 1,5,14,22,27,20,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot(\sigma\sigma S+1))\cdot S))$ |
| 1,5,14,22,27,21,33, 34 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot(\sigma\sigma S+1))\cdot(S+\omega)))$ |
| 1,5,14,22,27,21,33, 41,50 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot(\sigma\sigma S+1)+S(\sigma\sigma S+1))))$ |
| 1,5,14,22,27,21,33, 44 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot(\sigma\sigma S+\omega))))$ |
| 1,5,14,22,27,21,33, 44,52 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot(\sigma\sigma S+$ $\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S+S(\sigma\sigma S+1))))))$ |
| 1,5,14,22,27,21,33, 44,52,34 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot 2)))$ |
| 1,5,14,22,27,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot\omega)))$ |
| 1,5,14,22,27,22,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot\omega+1)))$ |
| 1,5,14,22,27,22,19, 14,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot\omega+\sigma S_\omega)))$ |
| 1,5,14,22,27,22,19, 14,22,27,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot\omega+$ $\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot\omega))))$ |
| 1,5,14,22,27,22,19, 25 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot\omega+S(\sigma\sigma S+1))))$ |
| 1,5,14,22,27,22,20, 6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot(\sigma\sigma S\cdot\omega+1))\cdot S))$ |
| 1,5,14,22,27,22,21, 33,44 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot(\sigma\sigma S\cdot\omega+\omega))))$ |
| 1,5,14,22,27,22,21, 33,44,52,34 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot(\omega+1))))$ |
| 1,5,14,22,27,22,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot\omega^2)))$ |
| 1,5,14,22,27,22,27, 13 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot S_\omega)))$ |
| 1,5,14,22,27,22,27, 15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S^2)))$ |
| 1,5,14,22,27,22,27, 22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S^2\cdot\omega)))$ |
| 1,5,14,22,27,27,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S^\sigma\sigma S)))$ |
| 1,5,14,22,27,33 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))))$ |
| 1,5,14,22,28 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1)))+$ $\psi_{\alpha_2}(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1)))+1))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,5,14,22,28,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+ \sigma S(\sigma\sigma S+1))\cdot S))$ |
| 1,5,14,22,28,14,22, 27,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+ \sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\sigma\sigma S)))$ |
| 1,5,14,22,28,14,22, 27,33 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+ \sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\varepsilon_{\sigma\sigma S+1})))$ |
| 1,5,14,22,28,14,22, 28 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))} (\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+ \sigma S(\sigma\sigma S+1))))) + \psi_{\alpha_2}(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+ \sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1) +\sigma S(\sigma\sigma S+1)))) + 1)$ |
| 1,5,14,22,28,14,22, 28,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+ \sigma S(\sigma\sigma S+1))\cdot S+\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))} (\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| 1,5,14,22,28,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)\cdot\omega))$ |
| 1,5,14,22,28,19 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)\cdot S_2))$ |
| 1,5,14,22,28,19,13 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)\cdot S_\omega))$ |
| 1,5,14,22,28,19,14 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+ \sigma S(\sigma\sigma S+1))\cdot S)\cdot\sigma S+ \psi_{S(\sigma S+1)}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+ \sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+ \sigma S(\sigma\sigma S+1))\cdot S)\cdot\sigma S+1)))$ |
| 1,5,14,22,28,19,14, 22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)\cdot\sigma S_\omega))$ |
| 1,5,14,22,28,19,14, 22,27,33 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)\cdot\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))} (\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma\sigma S))))$ |
| 1,5,14,22,28,19,14, 22,28,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+ \sigma S(\sigma\sigma S+1))\cdot S)\cdot\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))} (\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| 1,5,14,22,28,19,25 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+ S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)+1)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,22,28,20,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S))$ |
| 1,5,14,22,28,20,19, 25 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S+$ $\psi_S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S)+1)(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}$ $(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S)+$ $S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}$ $(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)+1))))$ |
| 1,5,14,22,28,20,27 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S+$ $S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S)+1)))$ |
| 1,5,14,22,28,20,27, 27 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S+$ $S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S)+1)\cdot 2))$ |
| 1,5,14,22,28,20,28, 6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S+$ $\psi_S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S)+2)$ $(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S)))$ |
| 1,5,14,22,28,20,28, 27 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S+$ $\psi_S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S)+2)(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}$ $(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S))+$ $S(\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S+$ $S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S)+1)))+1))$ |
| 1,5,14,22,28,20,28, 29 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S+$ $\psi_S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S)+2)$ $(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S)+1))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,5,14,22,28,20,29 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S+$ $S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S)+\omega)))$ |
| 1,5,14,22,28,20,30 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot(S+1)+\sigma S\cdot\omega))$ |
| 1,5,14,22,28,21 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot(S+1)+\sigma S\cdot S\cdot\omega))$ |
| 1,5,14,22,28,21,32 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot(S+1)+$ $S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)+\omega)))$ |
| 1,5,14,22,28,21,33 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot(S+2)+\sigma S\cdot S\cdot\omega))$ |
| 1,5,14,22,28,21,33, 34 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot(S+\omega)))$ |
| 1,5,14,22,28,21,33, 38,13 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S_\omega))$ |
| 1,5,14,22,28,21,33, 38,14,22,27,33 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)$ $+\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\varepsilon_{\sigma\sigma S+1})))$ |
| 1,5,14,22,28,21,33, 38,14,22,28,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}$ $(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| 1,5,14,22,28,21,33, 41 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)$ $\cdot S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S)+1)))$ |
| 1,5,14,22,28,21,33, 41,32 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)\cdot S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}$ $(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)+\omega)))$ |
| 1,5,14,22,28,21,33, 41,33 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)^2+\sigma S\cdot S\cdot\omega))$ |
| 1,5,14,22,28,21,33, 41,50 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $sigma_{S(\sigma\sigma S+1)})\cdot S+1)+1)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,22,28,21,33, 44 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+\omega)))$ |
| 1,5,14,22,28,21,33, 44,52,34 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))))$ |
| 1,5,14,22,28,21,33, 44,53,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}$ $(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| 1,5,14,22,28,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}$ $(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+1)))$ |
| 1,5,14,22,28,22,21, 33,44,52,34 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}$ $(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+1)+\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))))$ |
| 1,5,14,22,28,22,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}$ $(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+2)))$ |
| 1,5,14,22,28,22,27, 14 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}$ $(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+\sigma S)+\psi_{S(\sigma S+1)}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+\sigma S)+1)))$ |
| 1,5,14,22,28,22,27, 15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)))))$ |
| 1,5,14,22,28,22,27, 22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))+1)))$ |
| 1,5,14,22,28,22,27, 22,27,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}$ $(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))\cdot 2)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,22,28,22,27, 33 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma\sigma S))))$ |
| 1,5,14,22,28,22,28, 15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}$ $(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}$ $(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S))))$ |
| 1,5,14,22,28,35 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))))$ |
| 1,5,14,22,28,37 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1)\cdot\omega)))$ |
| 1,5,14,22,29 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot(S+1)+\sigma S\cdot S\cdot\omega))$ |
| 1,5,14,22,29,37 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot(S+1)+\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))))$ |
| 1,5,14,22,29,41,42 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot(S+\omega)))$ |
| 1,5,14,22,29,41,49 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))\cdot\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))))$ |
| 1,5,14,22,29,41,49, 41 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1))^2+\sigma S\cdot S\cdot\omega))$ |
| 1,5,14,22,29,41,52 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+$ $\sigma S(\sigma\sigma S+1)\cdot\omega)))$ |
| 1,5,14,22,29,41,52, 60,42 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot 2)))$ |
| 1,5,14,22,30 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)))$ |
| 1,5,14,22,30,14,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+\sigma S_\omega))$ |
| 1,5,14,22,30,14,22, 27,33 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+$ $\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\varepsilon_{\sigma\sigma S+1})))$ |
| 1,5,14,22,30,14,22, 28,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| 1,5,14,22,30,14,22, 29,41,52,63 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}$ $(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)))))$ |
| 1,5,14,22,30,14,22, 29,41,52,63,20,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot S(\sigma\sigma S+1)\cdot\omega)+1)\cdot S))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,22,30,14,22, 29,41,52,63,21,32,40, 22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+ \psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1))))$ |
| 1,5,14,22,30,14,22, 29,41,52,63,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+ \psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+1)\cdot\omega)+1)))$ |
| 1,5,14,22,30,14,22, 29,41,52,63,37 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+ \psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+S(\sigma\sigma S+1))))$ |
| 1,5,14,22,30,14,22, 29,41,52,63,41 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+ \psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1) +\sigma S(\sigma\sigma S+1))+\sigma S\cdot S\cdot\omega))$ |
| 1,5,14,22,30,14,22, 29,41,52,63,41,52 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)+ \psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)+\sigma S(\sigma\sigma S+1)\cdot\omega)))$ |
| 1,5,14,22,30,14,22, 30 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)\cdot 2))$ |
| 1,5,14,22,30,19,13 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)\cdot S_\omega))$ |
| 1,5,14,22,30,19,14, 22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega)\cdot\sigma S_\omega))$ |
| 1,5,14,22,30,19,25 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega+\sigma\sigma S)))$ |
| 1,5,14,22,30,20,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1) \cdot\omega+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| 1,5,14,22,30,21,33, 34 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+1)\cdot\omega+\sigma S(\sigma\sigma S+1))\cdot(S+\omega)))$ |
| 1,5,14,22,30,21,33, 44,52,34 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot(\omega+1))))$ |
| 1,5,14,22,30,21,33, 44,52,61 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot(\omega+1)+\sigma\sigma S)))$ |
| 1,5,14,22,30,21,33, 44,52,61 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\omega^2)))$ |
| 1,5,14,22,30,22,27, 13 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot S_\omega)))$ |
| 1,5,14,22,30,22,27, 15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\sigma\sigma S)))$ |
| 1,5,14,22,30,22,27, 22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)\cdot\sigma\sigma S\cdot\omega)))$ |
| 1,5,14,22,30,22,27, 33 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)^2+\sigma\sigma S)))$ |
| 1,5,14,22,30,22,28, 6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+1)^2+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| 1,5,14,22,30,22,29, 41,42 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+1)^2+\sigma S(\sigma\sigma S+1))\cdot(S+\omega)))$ |
| 1,5,14,22,30,22,30 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)^2\cdot\omega)))$ |
| 1,5,14,22,30,22,30, 22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)^2\cdot\omega^2)))$ |
| 1,5,14,22,30,23 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)^\omega)))$ |
| 1,5,14,22,30,27,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)^\sigma\sigma S)))$ |
| 1,5,14,22,30,30 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+1)^{S(\sigma\sigma S+1)}\cdot\omega)))$ |
| 1,5,14,22,31 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\varepsilon_{S(\sigma\sigma S+1)+1})))$ |
| 1,5,14,22,32,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| 1,5,14,22,32,14,22, 31 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)+ \sigma S(\sigma\sigma S+1))\cdot S+\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot\varepsilon_{S(\sigma\sigma S+1)+1})))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,22,32,14,22, 32,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)+ \sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| 1,5,14,22,32,19,25 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S+ S(\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S)+1)))$ |
| 1,5,14,22,32,21,33, 44 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\psi_S(\sigma\sigma S+2)(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1)))+\sigma S(\sigma\sigma S+1)\cdot\omega)))$ |
| 1,5,14,22,32,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot\psi_S(\sigma\sigma S+2)(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S+1))))$ |
| 1,5,14,22,32,31 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2))))$ |
| 1,5,14,22,32,32,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)+S(\sigma\sigma S+1))}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| 1,5,14,22,32,43 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)+S(\sigma\sigma S+1))))$ |
| 1,5,14,22,32,45 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S+ \psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)+S(\sigma\sigma S+1)\cdot\omega)))$ |
| 1,5,14,22,33 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)+ \sigma S(\sigma\sigma S+1))\cdot(S+1)+\sigma S\cdot S\cdot\omega))$ |
| 1,5,14,22,33,49,51, 6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1))\cdot S\cdot 2))$ |
| 1,5,14,22,33,49,64 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+2)+\sigma S(\sigma\sigma S+1)\cdot\omega)))$ |
| 1,5,14,22,34 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)\cdot\omega)))$ |
| 1,5,14,22,34,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)\cdot\omega^2)))$ |
| 1,5,14,22,34,22,27, 15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)\cdot\sigma\sigma S)))$ |
| 1,5,14,22,34,22,30 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1) \cdot S(\sigma\sigma S+2)\cdot S(\sigma\sigma S+1)\cdot\omega)))$ |
| 1,5,14,22,34,22,31 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)\cdot\varepsilon_{S(\sigma\sigma S+1)+1})))$ |
| 1,5,14,22,34,22,34 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)^2\cdot\omega)))$ |
| 1,5,14,22,34,30 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)^{S(\sigma\sigma S+1)}\cdot\omega)))$ |
| 1,5,14,22,34,30,42 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+2)^2\cdot\omega)))$ |
| 1,5,14,22,34,31 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\varepsilon_{S(\sigma\sigma S+2)+1})))$ |
| 1,5,14,22,34,35 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(\sigma\sigma S+\omega))))$ |
| 1,5,14,22,34,42 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot S(S(\sigma\sigma S+1))\cdot\omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,22,34,43 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot\psi_{\sigma S(\sigma\sigma S+1)}(S(\sigma S(\sigma\sigma S+1)+1)^2))))$ |
| 1,5,14,22,34,46 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)$ $\cdot\psi_{\sigma S(\sigma\sigma S+1)}(S(\sigma S(\sigma\sigma S+1)+1)^2+1))))$ |
| 1,5,14,22,34,47 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\psi_{\sigma S(\sigma\sigma S+1)}(\sigma_{S(\sigma\sigma S+2)}))))$ |
| 1,5,14,22,35 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)\cdot\psi_{\sigma S(\sigma\sigma S+1)}(\sigma_{S(\sigma\sigma S+\omega)}))))$ |
| 1,5,14,22,36 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)+\sigma S\cdot\omega))$ |
| 1,5,14,23 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega))$ |
| 1,5,14,23,5,14,23 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega)\cdot 2)$ |
| 1,5,14,23,7,10 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega+S))$ |
| 1,5,14,23,10 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega+S_2))$ |
| 1,5,14,23,14,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega+\sigma S_\omega))$ |
| 1,5,14,23,14,22,36, 58,80 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega+$ $\psi_{\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)}(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega)))$ |
| 1,5,14,23,14,22,36, 58,80,27,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega+\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)))$ |
| 1,5,14,23,14,23 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega\cdot 2))$ |
| 1,5,14,23,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot\omega^2))$ |
| 1,5,14,23,19,13 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2)\cdot S_\omega))$ |
| 1,5,14,23,19,25 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2+\sigma\sigma S)))$ |
| 1,5,14,23,20,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| 1,5,14,23,21,33,34 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2+\sigma S(\sigma\sigma S+1))\cdot(S+\omega)))$ |
| 1,5,14,23,21,33,41, 50 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2+\sigma S(\sigma\sigma S+1)+\sigma\sigma S)))$ |
| 1,5,14,23,21,33,45 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot 2)\cdot\omega))$ |
| 1,5,14,23,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot\omega)))$ |
| 1,5,14,23,22,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot\omega^2)))$ |
| 1,5,14,23,22,27,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot\sigma\sigma S)))$ |
| 1,5,14,23,22,27,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot S(\sigma\sigma S+1))\cdot\omega))$ |
| 1,5,14,23,22,30 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot S(\sigma\sigma S+1)\cdot\omega)))$ |
| 1,5,14,23,22,31 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot S(\sigma\sigma S+2))))$ |
| 1,5,14,23,22,34 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot S(\sigma\sigma S+2)\cdot\omega)))$ |
| 1,5,14,23,22,35 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^2\cdot\psi_{\sigma S(\sigma\sigma S+1)}(\sigma_{S(\sigma\sigma S+\omega)}))))$ |
| 1,5,14,23,23 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^3)\cdot\omega))$ |
| 1,5,14,23,23,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^3\cdot\omega)))$ |
| 1,5,14,23,24 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^\omega)))$ |
| 1,5,14,23,28,13 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{S_\omega})))$ |
| 1,5,14,23,28,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{\sigma\sigma S})))$ |
| 1,5,14,23,28,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{\sigma\sigma S}\cdot\omega)))$ |
| 1,5,14,23,28,23 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{\sigma\sigma S+1})\cdot\omega))$ |
| 1,5,14,23,28,34 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{\varepsilon_{\sigma\sigma S+1}})))$ |
| 1,5,14,23,29,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{S(\sigma\sigma S+1)}+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| 1,5,14,23,30,42,43 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{S(\sigma\sigma S+1)}$ $+\sigma S(\sigma\sigma S+1))\cdot(S+\omega)))$ |
| 1,5,14,23,31 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{S(\sigma\sigma S+1)}\cdot\omega)))$ |
| 1,5,14,23,31,23 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{S(\sigma\sigma S+1)+1})\cdot\omega))$ |
| 1,5,14,23,31,44 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{\psi_{\sigma S(\sigma\sigma S+1)}(\sigma_{S(\sigma\sigma S+\omega)})}))))$ |
| 1,5,14,23,32 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S+1)^{\sigma S}(\sigma\sigma S+1))\cdot\omega))$ |
| 1,5,14,24 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_2)))$ |
| 1,5,14,24,24 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_2\cdot 2)))$ |
| 1,5,14,25,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| 1,5,14,25,14,24 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)$ $+\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\sigma\sigma S}(\sigma\sigma S_2)))$ |

| $0-\mathrm{Y}$ | Projection |
|---|---|
| 1,5,14,25,14,25,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1))}(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| 1,5,14,25,19,25 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\sigma\sigma S}(\psi_S(\sigma\sigma S_2+1)$ $(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S)+\sigma\sigma S)))$ |
| 1,5,14,25,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\sigma\sigma S}(\psi_S(\sigma\sigma S_2+1)$ $(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S+1))))$ |
| 1,5,14,25,23 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\sigma\sigma S}(\psi_S(\sigma\sigma S_2+1)$ $(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\sigma S(\sigma\sigma S+1)))\cdot\omega))$ |
| 1,5,14,25,24 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1))))$ |
| 1,5,14,25,25,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S+\psi_{\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+S(\sigma\sigma S+1))}$ $(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S)))$ |
| 1,5,14,25,37 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+S(\sigma\sigma S+1))))$ |
| 1,5,14,25,39 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+S(\sigma\sigma S+1)\cdot\omega)))$ |
| 1,5,14,26 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$ $\sigma S(\sigma\sigma S+1))\cdot(S+1)+\sigma S\cdot S\cdot\omega))$ |
| 1,5,14,26,24 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot(S+1)+$ $\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1))))$ |
| 1,5,14,26,42 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot(S+1)+$ $\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+S(\sigma\sigma S+1)\cdot\omega)))$ |
| 1,5,14,26,43 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)$ $+\sigma S(\sigma\sigma S+1))\cdot(S+2)+\sigma S\cdot S\cdot\omega))$ |
| 1,5,14,26,43,45,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S\cdot 2))$ |
| 1,5,14,27 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1))))$ |
| 1,5,14,27,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)$ $+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1))+1))$ |
| 1,5,14,27,13 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)$ $+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1))+S_\omega))$ |
| 1,5,14,27,14,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)$ $+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1))+\sigma S_\omega))$ |
| 1,5,14,27,14,24 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$ $\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1))+\psi_{\sigma\sigma S}(\sigma\sigma S_2)))$ |
| 1,5,14,27,14,26,43, 64 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1))+$ $\psi_{\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1))}(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+$ $\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1)))))$ |
| 1,5,14,27,14,26,43, 64,42 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1))+$ $\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+S(\sigma\sigma S+1)\cdot\omega)))$ |
| 1,5,14,27,14,27 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1))\cdot 2))$ |
| 1,5,14,27,19,25 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)$ $+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1)+\sigma\sigma S)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,27,20,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+ \psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+1)+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| 1,5,14,27,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+ \psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+2))))$ |
| 1,5,14,27,22,27,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+ \psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+\sigma\sigma S))))$ |
| 1,5,14,27,23 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+ \psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1)))\cdot\omega))$ |
| 1,5,14,27,23,36 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+ \psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1))+1))))$ |
| 1,5,14,27,24 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)+\sigma\sigma S_2)))$ |
| 1,5,14,27,25,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)\cdot 2+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| 1,5,14,27,27 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)\cdot 2 +\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)\cdot 2+1))))$ |
| 1,5,14,27,28 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)\cdot\omega)))$ |
| 1,5,14,27,32,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)\cdot\sigma\sigma S)))$ |
| 1,5,14,27,37 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)\cdot\sigma\sigma S_2)))$ |
| 1,5,14,27,40 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+1)^2+\psi_{\sigma\sigma S_2}(S(\sigma\sigma S_2+1)^2+1))))$ |
| 1,5,14,27,41 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_2+2))))$ |
| 1,5,14,28 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1))\cdot\omega))$ |
| 1,5,14,28,7,10 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1))\cdot\omega+S))$ |
| 1,5,14,28,14,24 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1))\cdot\omega+\psi_{\sigma\sigma S}(\sigma\sigma S_2)))$ |
| 1,5,14,28,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1))\cdot\omega\cdot 2))$ |
| 1,5,14,28,19,14,24 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1))\cdot\psi_{\sigma\sigma S}(\sigma\sigma S_2)))$ |
| 1,5,14,28,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)+\psi_{\sigma\sigma S_2}(\sigma S(\sigma\sigma S_2+1)+1))))$ |
| 1,5,14,28,24 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)+\sigma\sigma S_2)))$ |
| 1,5,14,28,25,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1) +S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| 1,5,14,28,27 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)+S(\sigma\sigma S_2+1)+ \psi_{\sigma\sigma S_2}(\sigma S(\sigma\sigma S_2+1)+S(\sigma\sigma S_2+1)+1))))$ |
| 1,5,14,28,27,28 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)+S(\sigma\sigma S_2+1)\cdot\omega)))$ |
| 1,5,14,28,27,41 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)+S(\sigma\sigma S_2+2))))$ |
| 1,5,14,28,28 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot 2)\cdot\omega))$ |
| 1,5,14,28,29 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot\omega)))$ |
| 1,5,14,28,33,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot\sigma\sigma S)))$ |
| 1,5,14,28,34,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot S(\sigma\sigma S+1)+ \sigma S(\sigma\sigma S+1))\cdot S))$ |
| 1,5,14,28,36 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot S(\sigma\sigma S+1)+ \psi_{\sigma\sigma S_2}(\sigma S(\sigma\sigma S_2+1)\cdot S(\sigma\sigma S+1)+1))))$ |
| 1,5,14,28,37 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot\sigma S(\sigma\sigma S+1)+ \psi_{\sigma\sigma S_2}(\sigma S(\sigma\sigma S_2+1)\cdot\sigma S(\sigma\sigma S+1)+1))))$ |
| 1,5,14,28,38 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot\sigma\sigma S_2)))$ |
| 1,5,14,28,41 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot S(\sigma\sigma S_2+1)+ \psi_{\sigma\sigma S_2}(\sigma S(\sigma\sigma S_2+1)\cdot S(\sigma\sigma S_2+1)+1))))$ |
| 1,5,14,28,41,42 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot S(\sigma\sigma S_2+1)\cdot\omega)))$ |
| 1,5,14,28,41,51 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot S(\sigma\sigma S_2+1)\cdot\sigma\sigma S_2)))$ |
| 1,5,14,28,41,55 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)\cdot S(\sigma\sigma S_2+2))))$ |
| 1,5,14,28,42 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)^2)\cdot\omega))$ |
| 1,5,14,28,42,28 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)^2+\sigma S(\sigma\sigma S_2+1))\cdot\omega))$ |
| 1,5,14,28,42,43 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)^\omega)))$ |
| 1,5,14,28,43 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma S(\sigma\sigma S_2+1)+1))))$ |
| 1,5,14,28,47 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+2))\cdot\omega))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,28,47,48 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+\omega))))$ |
| 1,5,14,29 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega)))$ |
| 1,5,14,29,14,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega)+\sigma S_\omega))$ |
| 1,5,14,29,14,24 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega)+\psi_{\sigma\sigma S}(\sigma\sigma S_2)))$ |
| 1,5,14,29,14,28 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega)+\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_2+1)+$ $\psi_{\sigma\sigma S_2}(\sigma S(\sigma\sigma S_2+1)+1))))$ |
| 1,5,14,29,14,29 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega)\cdot 2))$ |
| 1,5,14,29,19,14,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega+\sigma S_\omega)))$ |
| 1,5,14,29,19,25 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega+\sigma\sigma S)))$ |
| 1,5,14,29,22 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega+\psi_{\sigma\sigma S_2}(\sigma\sigma S_\omega+1))))$ |
| 1,5,14,29,24 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega+\sigma\sigma S_2)))$ |
| 1,5,14,29,29 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega\cdot 2)))$ |
| 1,5,14,29,34,13 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega\cdot S_\omega)))$ |
| 1,5,14,29,34,14,29 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega\cdot\psi_{\sigma\sigma S}(\sigma\sigma S_\omega))))$ |
| 1,5,14,29,34,15 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega\cdot\sigma\sigma S)))$ |
| 1,5,14,29,37 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega\cdot S(\sigma\sigma S+1)+$ $\psi_{\sigma\sigma S_2}(\sigma\sigma S_\omega\cdot S(\sigma\sigma S+1)+1))))$ |
| 1,5,14,29,38,53 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega\cdot\psi_{\sigma\sigma S_2}(\sigma\sigma S_\omega))))$ |
| 1,5,14,29,39 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega\cdot\sigma\sigma S_2)))$ |
| 1,5,14,29,39,24 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega\cdot\sigma\sigma S_2+\sigma\sigma S_2)))$ |
| 1,5,14,29,39,28,48, 58,43 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega\cdot\sigma\sigma S_2+\sigma\sigma S_3)))$ |
| 1,5,14,29,39,28,48, 58,48 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega\cdot(\sigma\sigma S_2+1))))$ |
| 1,5,14,29,39,28,48, 59,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega\cdot S(\sigma\sigma S_2+1)+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| 1,5,14,29,39,28,48, 63 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega\cdot\sigma\sigma S_3)))$ |
| 1,5,14,29,39,29 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma {S_\omega}^2)))$ |
| 1,5,14,29,39,50 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+1))))$ |
| 1,5,14,29,40,6 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+1)+\sigma S(\sigma\sigma S+1))\cdot S))$ |
| 1,5,14,29,40,29 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\sigma\sigma S}(\psi_S(\sigma\sigma S_\omega+1)(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+1)+$ $\sigma S(\sigma\sigma S+1))\cdot S)+\sigma\sigma S_\omega)))$ |
| 1,5,14,29,40,52 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+1)+\sigma S(\sigma\sigma S+1))\cdot S+$ $\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+1))))$ |
| 1,5,14,29,41 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+1)+$ $\sigma S(\sigma\sigma S+1))\cdot(S+1)+\sigma S\cdot S\cdot\omega))$ |
| 1,5,14,29,41,58,71, 85 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+1)+\sigma S(\sigma\sigma S+1)+\sigma\sigma S)))$ |
| 1,5,14,29,42 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+1)\cdot\omega)))$ |
| 1,5,14,29,42,39,29 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+1)\cdot\sigma\sigma S_\omega)))$ |
| 1,5,14,29,42,39,50 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+1)\cdot\varepsilon_{\sigma\sigma S_\omega+1})))$ |
| 1,5,14,29,42,56 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_\omega+2))))$ |
| 1,5,14,29,43 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_\omega+1))\cdot\omega))$ |
| 1,5,14,29,43,29 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_\omega+1)+\sigma\sigma S_\omega)))$ |
| 1,5,14,29,43,29,42 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_\omega+1)+S(\sigma\sigma S_\omega+1)\cdot\omega)))$ |
| 1,5,14,29,43,30 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_\omega+1)\cdot\omega)))$ |
| 1,5,14,29,43,56 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_\omega+1)\cdot S(\sigma\sigma S_\omega+1)\cdot\omega)))$ |
| 1,5,14,29,43,57 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma S(\sigma\sigma S_\omega+1)^2)\cdot\omega))$ |
| 1,5,14,29,43,58 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_{\omega+1})))$ |
| 1,5,14,29,44 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_{\omega^2})))$ |
| 1,5,14,29,44,29 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_{\omega^2}+\sigma\sigma S_\omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,29,44,29,43, 63,83 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_{\omega^2} + \psi_{\sigma\sigma S_{\omega+1}}(\sigma\sigma S_{\omega^2}))))$ |
| 1,5,14,29,44,29,43, 63,83,58 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_{\omega^2} + \sigma\sigma S_{\omega+1})))$ |
| 1,5,14,29,44,29,44 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_{\omega^2} \cdot 2)))$ |
| 1,5,14,29,44,39,50 | $\psi(\psi_S(\psi_{\sigma\sigma S}(S(\sigma\sigma S_{\omega^2} + 1))))$ |
| 1,5,14,29,44,44 | $\psi(\psi_S(\psi_{\sigma\sigma S}(\sigma\sigma_{S_{\omega^3}})))$ |
| 1,5,14,29,44,54,30 | $\psi(\psi_S(\sigma\sigma S + \psi_{S_2}(\sigma\sigma S + S)))$ |
| 1,5,14,29,45 | $\psi(\psi_S(\sigma\sigma S + S_2))$ |
| 1,5,14,30 | $\psi(\psi_S(\sigma\sigma S \cdot \omega))$ |
| 1,5,14,30,5,14,29 | $\psi(\psi_S(\sigma\sigma S \cdot \omega) + \psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega)))$ |
| 1,5,14,30,5,14,29, 52,85,130 | $\psi(\psi_S(\sigma\sigma S \cdot \omega) + \psi_{\psi_S(\sigma\sigma S)}(\psi_S(\sigma\sigma S \cdot \omega)))$ |
| 1,5,14,30,5,14,29, 52,85,130,29 | $\psi(\psi_S(\sigma\sigma S \cdot \omega) + \psi_S(\sigma\sigma S + \psi_{S_2}(\sigma\sigma S + 1)))$ |
| 1,5,14,30,5,14,30 | $\psi(\psi_S(\sigma\sigma S \cdot \omega) \cdot 2)$ |
| 1,5,14,30,10 | $\psi(\psi_S(\sigma\sigma S \cdot \omega + S_2))$ |
| 1,5,14,30,13,27 | $\psi(\psi_S(\sigma\sigma S \cdot \omega + \sigma S \cdot \omega))$ |
| 1,5,14,30,14 | $\psi(\psi_S(\sigma\sigma S \cdot \omega + \sigma S \cdot S \cdot \omega))$ |
| 1,5,14,30,14,29 | $\psi(\psi_S(\sigma\sigma S \cdot \omega + \psi_{\sigma\sigma S}(\sigma\sigma S_\omega)))$ |
| 1,5,14,30,14,30 | $\psi(\psi_S(\sigma\sigma S \cdot \omega^2))$ |
| 1,5,14,30,19,14,30 | $\psi(\psi_S(\sigma\sigma S^2) \cdot \omega)$ |
| 1,5,14,30,19,25 | $\psi(\psi_S(S(\sigma\sigma S + 1)))$ |
| 1,5,14,30,20,6 | $\psi(\psi_S(\sigma S(\sigma\sigma S + 1) \cdot S))$ |
| 1,5,14,30,21,33,51 | $\psi(\psi_S(\psi_{\sigma\sigma S_2}(\sigma\sigma S_\omega)))$ |
| 1,5,14,30,22 | $\psi(\psi_S(\sigma\sigma S_\omega))$ |
| 1,5,14,30,22,27,15 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S + 1) \cdot \sigma\sigma\sigma S)))$ |
| 1,5,14,30,22,30 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S + 1) \cdot S(\sigma\sigma\sigma S + 1) \cdot \omega)))$ |
| 1,5,14,30,23 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S + 1) \cdot \sigma S(\sigma\sigma\sigma S + 1)) \cdot \omega))$ |
| 1,5,14,30,23,38 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S + 1) \cdot$ $\psi_{\sigma\sigma S}(\sigma\sigma\sigma S + 1)(\sigma\sigma S(\sigma\sigma\sigma S + \omega)))))$ |
| 1,5,14,30,23,39 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S + 1)^2 \cdot \omega)))$ |
| 1,5,14,30,23,39,23 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S + 1)^2 \cdot \sigma S(\sigma\sigma\sigma S + 1) \cdot \omega)))$ |
| 1,5,14,30,23,39,40 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S + 1)^\omega)))$ |
| 1,5,14,30,24 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma\sigma S_2)))$ |
| 1,5,14,30,24,24 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma\sigma S_2 \cdot 2)))$ |
| 1,5,14,30,25,6 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2 + 1) + \sigma S(\sigma\sigma S + 1)) \cdot S))$ |
| 1,5,14,30,27 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2 + 1)$ $+\psi_{\sigma\sigma\sigma S_2}(S(\sigma\sigma\sigma S_2 + 1) + 1))))$ |
| 1,5,14,30,27,22 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2 + 1)$ $+\psi_{\sigma\sigma\sigma S_2}(S(\sigma\sigma\sigma S_2 + 1) + 2))))$ |
| 1,5,14,30,27,23 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2 + 1)+$ $\psi_{\sigma\sigma\sigma S_2}(S(\sigma\sigma\sigma S_2 + 1) + \sigma S(\sigma\sigma\sigma S + 1))) \cdot \omega))$ |
| 1,5,14,30,27,23,39 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2 + 1)+$ $\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2 + 1) + \sigma\sigma S(\sigma\sigma\sigma S + 1)) \cdot \omega)))$ |
| 1,5,14,30,27,24 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2 + 1) + \sigma\sigma\sigma S_2)))$ |
| 1,5,14,30,27,28 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2 + 1) \cdot \omega)))$ |
| 1,5,14,30,27,41 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_2 + 2))))$ |
| 1,5,14,30,28 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma S(\sigma\sigma\sigma S_2 + 1)) \cdot \omega))$ |
| 1,5,14,30,28,24 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma S(\sigma\sigma\sigma S_2 + 1) + \sigma\sigma\sigma S_2)))$ |
| 1,5,14,30,23,38 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\psi_{\sigma\sigma S}(\sigma\sigma\sigma S_2 + 1)(\sigma\sigma S(\sigma\sigma\sigma S_2 + \omega)))))$ |
| 1,5,14,30,23,39 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S_2 + 1)) \cdot \omega))$ |
| 1,5,14,30,28,44,29 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma S(\sigma\sigma\sigma S_2 + 1) \cdot \omega)))$ |
| 1,5,14,30,29 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma\sigma S_\omega)))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,14,30,29,29 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma\sigma S_\omega \cdot 2)))$ |
| 1,5,14,30,29,39,50 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(S(\sigma\sigma\sigma S_\omega + 1))))$ |
| 1,5,14,30,29,44 | $\psi(\psi_S(\psi_{\sigma\sigma\sigma S}(\sigma\sigma\sigma S_{\omega^2})))$ |
| 1,5,14,30,29,45 | $\psi(\psi_S(\sigma\sigma\sigma\sigma S + S_2))$ |
| 1,5,14,30,30 | $\psi(\psi_S(\sigma\sigma\sigma\sigma S \cdot \omega))$ |
| 1,5,14,30,31 | $\psi(\psi_S(\sigma^\omega S)) = \psi(\psi_S(\sigma\theta S \cdot \omega))$ |
| 1,5,14,30,31,24 | $\psi(\psi_S(\sigma\theta S \cdot \omega + \theta S_2))$ |
| 1,5,14,30,31,30 | $\psi(\psi_S(\sigma\theta S \cdot (\omega + 1) + \psi_{\theta S}(\sigma\theta S \cdot (\omega + 1) + 1)))$ |
| 1,5,14,30,31,31 | $\psi(\psi_S(\sigma\theta S \cdot \omega^2))$ |
| 1,5,14,30,32,6 | $\psi(\psi_S(\sigma\theta S \cdot S))$ |
| 1,5,14,30,35,15 | $\psi(\psi_S(\sigma\theta S \cdot \theta S))$ |
| 1,5,14,30,35,28,49, 54,15 | $\psi(\psi_S(\sigma\theta S \cdot \theta S + \psi_{\theta S_2}(\sigma\theta S \cdot \theta S)))$ |
| 1,5,14,30,35,29 | $\psi(\psi_S(\sigma\theta S \cdot \theta S + \theta S_\omega))$ |
| 1,5,14,30,35,41 | $\psi(\psi_S(\sigma\theta S \cdot S(\theta S + 1)))$ |
| 1,5,14,30,38 | $\psi(\psi_S(\sigma\theta S \cdot S(\theta S + 1) + \psi_{\theta S_2}(\sigma\theta S \cdot S(\theta S + 1) + 1)))$ |
| 1,5,14,30,39 | $\psi(\psi_S(\sigma\theta S \cdot \psi_{\theta S_2}(\sigma\theta S) + \psi_{\theta S}(\sigma\theta S \cdot \psi_{\theta S_2}(\sigma\theta S) + 1)))$ |
| 1,5,14,30,39,55,56 | $\psi(\psi_S(\sigma\theta S \cdot \psi_{\theta S_2}(\sigma\theta S \cdot \omega)))$ |
| 1,5,14,30,40 | $\psi(\psi_S(\sigma\theta S \cdot \theta S_2))$ |
| 1,5,14,30,40,28,49, 59,48 | $\psi(\psi_S(\sigma\theta S \cdot \theta S_2 + \theta S_\omega))$ |
| 1,5,14,30,40,29 | $\psi(\psi_S(\sigma\theta S \cdot \theta S_\omega))$ |
| 1,5,14,30,40,30 | $\psi(\psi_S(\sigma\theta S^2 + \psi_{\theta S}(\sigma\theta S^2 + 1)))$ |
| 1,5,14,30,40,51 | $\psi(\psi_S(S(\sigma\theta S + 1)))$ |
| 1,5,14,30,41,6 | $\psi(\psi_S(\sigma\theta S_2 + \psi_{\theta S}(\sigma\theta S_2) \cdot S))$ |
| 1,5,14,30,44,65,66 | $\psi(\psi_S(\sigma\theta S_2 \cdot \omega))$ |
| 1,5,14,30,45 | $\psi(\psi_S(\sigma\theta S_\omega))$ |
| 1,5,14,30,45,45 | $\psi(\psi_S(\sigma\theta S_{\omega^2}))$ |
| 1,5,14,30,45,55,31 | $\psi(\psi_S(\psi_{\sigma\sigma\theta S}(\sigma\theta S(\sigma\sigma\theta S + 1) \cdot \sigma\sigma\theta S)))$ |
| 1,5,14,30,45,60 | $\psi(\psi_S(\psi_{\sigma\sigma\theta S}(\sigma\theta S(\sigma\sigma\theta S + 1) \cdot \theta S(\sigma\sigma\theta S + 1))+$ $\psi_{\theta S}(\psi_{\sigma\sigma\theta S}(\sigma\theta S(\sigma\sigma\theta S + 1) \cdot \theta S(\sigma\sigma\theta S + 1)) + 1)))$ |
| 1,5,14,30,46 | $\psi(\psi_S(\psi_{\sigma\sigma\theta S}(\sigma\theta S(\sigma\sigma\theta S + 1)^2)+$ $\psi_{\theta S}(\psi_{\sigma\sigma\theta S}(\sigma\theta S(\sigma\sigma\theta S + 1)^2) + 1)))$ |
| 1,5,14,30,47 | $\psi(\psi_S(\psi_{\sigma\sigma\theta S}(\sigma\sigma\theta S_2)))$ |
| 1,5,14,30,54 | $\psi(\psi_S(\psi_{\sigma\sigma\theta S}(\sigma\sigma\theta S_\omega)))$ |
| 1,5,14,30,55 | $\psi(\psi_S(\sigma\sigma\theta S + \psi_{\theta S}(\sigma\sigma\theta S + 1)))$ |
| 1,5,15 | $\psi(\psi_S(\theta^\omega S)) = \psi(\psi_X(\theta X \cdot \omega))$ |
| 1,5,15,5,14,29 | $\psi(\psi_X(\theta X \cdot \omega) + \psi_S(\psi_{\sigma\sigma S}(\sigma\sigma S_\omega)))$ |
| 1,5,15,5,14,30,54 | $\psi(\psi_X(\theta X \cdot \omega) + \psi_S(\psi_{\sigma\sigma\theta S}(\sigma\sigma\theta S_\omega)))$ |
| 1,5,15,5,15 | $\psi(\psi_X(\theta X \cdot \omega) + \psi_S(\psi_X(\theta X \cdot \omega)))$ |
| 1,5,15,7,5 | $\psi(\psi_X(\theta X \cdot \omega) + \psi_S(\psi_X(\theta X \cdot \omega) + \psi_S(\sigma S \cdot \omega)))$ |
| 1,5,15,7,5,15 | $\psi(\psi_X(\theta X \cdot \omega) + \psi_S(\psi_X(\theta X \cdot \omega) + \psi_S(\psi_X(\theta X \cdot \omega))))$ |
| 1,5,15,7,10 | $\psi(\psi_X(\theta X \cdot \omega) + S)$ |
| 1,5,15,10 | $\psi(\psi_X(\theta X \cdot \omega) + S_2)$ |
| 1,5,15,13 | $\psi(\psi_X(\theta X \cdot \omega) + S_\omega)$ |
| 1,5,15,14 | $\psi(\psi_X(\theta X \cdot \omega) + \sigma S + \psi_{\theta S}(\psi_X(\theta X \cdot \omega) + \sigma S + 1))$ |
| 1,5,15,14,31 | $\psi(\psi_X(\theta X \cdot \omega) + \psi_{\theta S}(\psi_X(\theta X \cdot \omega) + \psi_{\theta S}(\psi_X(\theta X \cdot \omega))))$ |
| 1,5,15,14,31,29 | $\psi(\psi_X(\theta X \cdot \omega) + \theta S_\omega)$ |
| 1,5,15,15 | $\psi(\psi_X(\theta X \cdot \omega) \cdot 2)$ |
| 1,5,15,17,6 | $\psi(\psi_X(\theta X \cdot \omega + S))$ |
| 1,5,15,20 | $\psi(\psi_X(\theta X \cdot \omega + S_2))$ |
| 1,5,15,20,13 | $\psi(\psi_X(\theta X \cdot \omega + S_\omega))$ |
| 1,5,15,20,14 | $\psi(\psi_X(\theta X \cdot \omega + \sigma S) + \psi_{\theta S_2}(\psi_X(\theta X \cdot \omega + \sigma S) + 1))$ |
| 1,5,15,20,14,31 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{\theta S_2}(\psi_X(\theta X \cdot \omega))))$ |

| 0 − Y | Projection |
|---|---|
| 1,5,15,20,15 | $\psi(\psi_X(\theta X \cdot \omega + \psi_X(\theta X \cdot \omega)))$ |
| 1,5,15,20,26 | $\psi(\psi_X(\theta X \cdot \omega + X))$ |
| 1,5,15,21 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\sigma X))+$ $\psi_{\alpha_2}(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\sigma X)) + 1))$ |
| 1,5,15,21,6 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\sigma X))+$ $\psi_{\theta S}(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\sigma X)) + 1))$ |
| 1,5,15,21,28 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\sigma X) + X))$ |
| 1,5,15,22 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\sigma X \cdot 2)) + \sigma S \cdot S \cdot \omega)$ |
| 1,5,15,22,35 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X)) + \psi_{\theta S_2}(\psi_X(\theta X \cdot \omega)))$ |
| 1,5,15,23 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X))+$ $\psi_{\theta S_2}(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X)) + 1))$ |
| 1,5,15,23,14,31,39, 32 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X) + 1))$ |
| 1,5,15,23,14,31,39, 39 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X) \cdot 2)+$ $\psi_{\theta S_2}(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X) \cdot 2) + 1))$ |
| 1,5,15,23,14,31,39, 44,15 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X + \theta S)))$ |
| 1,5,15,23,14,31,39, 47 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X + \psi_{X_2}(\theta X)))+$ $\psi_{\theta S_2}(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X + \psi_{X_2}(\theta X))) + 1))$ |
| 1,5,15,23,14,31,39, 48 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X + X_2)))$ |
| 1,5,15,23,14,31,40 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X + \sigma X))+$ $\psi_{\theta S}(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X + \sigma X)) + 1))$ |
| 1,5,15,23,14,31,40, 32 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X + \sigma X) + 1))$ |
| 1,5,15,23,14,31,40, 41 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X + \sigma X + 1)))$ |
| 1,5,15,23,14,31,40, 50 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X + \sigma X + X_2)))$ |
| 1,5,15,23,14,31,40, 57 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X \cdot 2)) + \psi_{\theta\theta S}(\psi_X(\theta X \cdot \omega)))$ |
| 1,5,15,23,15 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X \cdot \omega)))$ |
| 1,5,15,23,15,23,14, 31,46,31 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X \cdot \omega))+$ $\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X \cdot 2))+$ $\psi_{\theta\theta S}(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X \cdot \omega))))$ |
| 1,5,15,23,16 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X \cdot \omega) + 1))$ |
| 1,5,15,23,23 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X \cdot \omega) + \psi_{X_2}(\theta X))+$ $\psi_{\theta S_2}(\psi_X(\theta X \cdot \omega+$ $\psi_{X_2}(\theta X \cdot \omega) + \psi_{X_2}(\theta X)) + 1))$ |
| 1,5,15,23,24 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_2}(\theta X \cdot \omega + 1)))$ |
| 1,5,15,23,32 | $\psi(\psi_X(\theta X \cdot \omega + X_2))$ |
| 1,5,15,23,35,15 | $\psi(\psi_X(\theta X \cdot \omega + \psi_{X_3}(\theta X \cdot \omega)))$ |
| 1,5,15,23,36 | $\psi(\psi_X(\theta X \cdot \omega + X_\omega))$ |
| 1,5,15,24 | $\psi(\psi_X(\theta X \cdot \omega + \sigma X) + \psi_{\theta S}(\psi_X(\theta X \cdot \omega + \sigma X) + 1))$ |
| 1,5,15,24,16 | $\psi(\psi_X(\theta X \cdot \omega + \sigma X + 1))$ |
| 1,5,15,24,23 | $\psi(\psi_X(\theta X \cdot \omega + \sigma X + \psi_{X_2}(\theta X))+$ $\psi_{\theta S_2}(\psi_X(\theta X \cdot \omega + \sigma X + \psi_{X_2}(\theta X)) + 1))$ |
| 1,5,15,24,23,15 | $\psi(\psi_X(\theta X \cdot \omega + \sigma X + \psi_{X_2}(\theta X \cdot \omega)))$ |
| 1,5,15,24,25 | $\psi(\psi_X(\theta X \cdot \omega + \sigma X \cdot \omega))$ |
| 1,5,15,24,34 | $\psi(\psi_X(\theta X \cdot \omega + \varepsilon_{\sigma X+1}))$ |
| 1,5,15,24,41 | $\psi(\psi_X(\theta X \cdot \omega \cdot 2))$ |
| 1,5,15,25 | $\psi(\psi_X(\theta X \cdot \omega^2))$ |
| 1,5,15,25,15,25 | $\psi(\psi_X(\theta X \cdot \omega^2) \cdot 2)$ |

| 0 − Y | Projection |
|---|---|
| 1,5,15,25,23,15,25 | $\psi(\psi_X(\theta X \cdot \omega^2 + \psi_{X_2}(\theta X \cdot \omega^2)))$ |
| 1,5,15,25,23,32 | $\psi(\psi_X(\theta X \cdot \omega^2 + X_2))$ |
| 1,5,15,25,25 | $\psi(\psi_X(\theta X \cdot \omega^3))$ |
| 1,5,15,25,27,6 | $\psi(\psi_X(\theta X \cdot S))$ |
| 1,5,15,25,30,16 | $\psi(\psi_X(\theta X \cdot X))$ |
| 1,5,15,25,30,25 | $\psi(\psi_X(\theta X \cdot X + \psi_{X_2}(\theta X \cdot X + 1)))$ |
| 1,5,15,25,33,16 | $\psi(\psi_X(\theta X \cdot X + \psi_{X_2}(\theta X \cdot X + \psi_{X_2}(\theta X \cdot X))))$ |
| 1,5,15,25,33,42 | $\psi(\psi_X(\theta X \cdot X + X_2))$ |
| 1,5,15,25,34 | $\psi(\psi_X(\theta X \cdot X + \sigma X) + \psi_{\theta S}(\psi_X(\theta X \cdot X + \sigma X) + 1))$ |
| 1,5,15,25,35 | $\psi(\psi_X(\theta X \cdot X \cdot \omega))$ |
| 1,5,15,26 | $\psi(\psi_X(\theta X \cdot X \cdot \omega + X_2))$ |
| 1,5,15,32 | $\psi(\psi_X(\theta X \cdot X \cdot \omega + X_\omega))$ |
| 1,5,15,33 | $\psi(\psi_X(\theta X \cdot X \cdot \omega + \sigma X)$ $+\psi_{\theta S}(\psi_X(\theta X \cdot X \cdot \omega + \sigma X) + 1))$ |
| 1,5,15,34 | $\psi(\psi_X(\theta X \cdot X \cdot \omega^2))$ |
| 1,5,15,34,39,16 | $\psi(\psi_X(\theta X \cdot X^2))$ |
| 1,5,15,34,45 | $\psi(\psi_X(\theta X \cdot X_2))$ |
| 1,5,15,34,45,32 | $\psi(\psi_X(\theta X \cdot X_\omega))$ |
| 1,5,15,34,45,33 | $\psi(\psi_X(\theta X \cdot \sigma X) + \psi_{\theta S}(\psi_X(\theta X \cdot \sigma X) + 1))$ |
| 1,5,15,34,45,34 | $\psi(\psi_X(\theta X^2 \cdot \omega))$ |
| 1,5,15,34,46,6 | $\psi(\psi_X(\sigma X(\theta X + 1)) + \psi_{\theta S}(\psi_X(\sigma X(\theta X + 1)) + 1))$ |
| 1,5,15,34,51 | $\psi(\psi_X(\theta X_\omega))$ |
| 1,5,15,34,51,62,35 | $\psi(\psi_X(\psi_{\theta\theta X}(\theta X(\theta\theta X + 1) \cdot \theta\theta X)))$ |
| 1,5,15,34,53 | $\psi(\psi_X(\psi_{\theta\theta X}(\theta X(\theta\theta X + 1)^2 \cdot \omega)))$ |
| 1,5,15,34,54 | $\psi(\psi_X(\psi_{\theta\theta X}(\theta\theta X_2)))$ |
| 1,5,15,34,63 | $\psi(\psi_X(\psi_{\theta\theta X}(\theta\theta X_\omega)))$ |
| 1,5,15,34,65 | $\psi(\psi_X(\theta\theta X \cdot \omega))$ |
| 1,5,15,35 | $\psi(\psi_X(\theta^\omega X))$ |
| 1,5,15,35,35 | $\psi(\psi_A([1,1]A \cdot 2 + \psi_{[1,0]A}([1,1]A \cdot 2 + 1)) \cdot 2)$ |
| 1,5,15,35,66 | $\psi(\psi_A([1,1]A \cdot 2 + \psi_{\psi_{[1,0]^2A}([1,1]A \cdot 2)}$ $(\psi_{[1,0]^2A}([1,1]A \cdot 2 + \psi_{[1,0]^3A}([1,0]^2A([1,0]^3A + 1) \cdot \omega))))))$ |
| 1,5,15,35,70 | $\psi(\psi_A([1,1]A \cdot 2 + [1,0]^\omega A))$ |
| 1,6 | $\psi(\psi_A([1,1]A \cdot \omega))$ |
| 1,6,7 | $\psi(\psi_A([1,1]A \cdot \omega + 1))$ |
| 1,6,8,11 | $\psi(\psi_A([1,1]A \cdot \omega + A))$ |
| 1,6,9,6 | $\psi(\psi_A([1,1]A \cdot \omega + \psi_{A_2}([1,1]A \cdot \omega)))$ |
| 1,6,9,15 | $\psi(\psi_A([1,1]A \cdot \omega + A_\omega))$ |
| 1,6,10 | $\psi(\psi_A([1,1]A \cdot \omega + \psi_{[1,0]A}([1,1]A))+$ $\psi_{\theta S}(\psi_A([1,1]A \cdot \omega + \psi_{[1,0]A}([1,1]A)) + 1))$ |
| 1,6,10,6 | $\psi(\psi_A([1,1]A \cdot \omega + \psi_{[1,0]A}([1,1]A \cdot \omega)))$ |
| 1,6,10,6 | $\psi(\psi_A([1,1]A \cdot \omega^2))$ |
| 1,6,11,13,7 | $\psi(\psi_A([1,1]A \cdot A))$ |
| 1,6,11,16 | $\psi(\psi_A([1,1]A \cdot A \cdot \omega))$ |
| 1,6,12 | $\psi(\psi_A([1,1]A \cdot A \cdot \omega + A_2))$ |
| 1,6,16 | $\psi(\psi_A([1,1]A \cdot A \cdot \omega + A_\omega))$ |
| 1,6,19 | $\psi(\psi_A([1,1]A \cdot A \cdot \omega + [1,0]^\omega A))$ |
| 1,6,20 | $\psi(\psi_A([1,1]A \cdot A \cdot \omega^2))$ |
| 1,6,20,26,16 | $\psi(\psi_A([1,1]A \cdot A_\omega))$ |
| 1,6,20,26,19 | $\psi(\psi_A([1,1]A \cdot [1,0]^\omega A))$ |
| 1,6,20,26,33 | $\psi(\psi_A(A([1,1]A + 1)))$ |
| 1,6,20,30 | $\psi(\psi_A([1,1]A_\omega))$ |
| 1,6,20,33 | $\psi(\psi_A([1,1]^\omega A))$ |
| 1,6,20,50 | $\psi(\psi_A([1,2]A \cdot \omega))$ |
| 1,6,21 | $\psi(H^{H^{H \cdot \omega}})$ |

| 0 − Y | Projection |
|---|---|
| 1,7 | $\psi(H^{H^{H^{\omega}}})$ |
| limit | $\psi(\varepsilon_{H+1})$ |

## A.19 0 − Y vs BMS

The results in this section are primarily cited from[2].

| 0 − Y sequence | BMS |
|---|---|
| 1 | (0) |
| 1,1 | (0)(0) |
| 1,1,1 | (0)(0)(0) |
| 1,1,1,1 | (0)(0)(0)(0) |
| 1,2 | (0)(1) |
| 1,2,1 | (0)(1)(0) |
| 1,2,1,1 | (0)(1)(0)(0) |
| 1,2,1,2 | (0)(1)(0)(1) |
| 1,2,1,2,1 | (0)(1)(0)(1)(0) |
| 1,2,1,2,1,2 | (0)(1)(0)(1)(0)(1) |
| 1,2,2 | (0)(1)(1) |
| 1,2,2,1 | (0)(1)(1)(0) |
| 1,2,2,1,2 | (0)(1)(1)(0)(1) |
| 1,2,2,1,2,2 | (0)(1)(1)(0)(1)(1) |
| 1,2,2,2 | (0)(1)(1)(1) |
| 1,2,2,2,2 | (0)(1)(1)(1)(1) |
| 1,2,3 | (0)(1)(2) |
| 1,2,3,1 | (0)(1)(2)(0) |
| 1,2,3,1,2 | (0)(1)(2)(0)(1) |
| 1,2,3,1,2,3 | (0)(1)(2)(0)(1)(2) |
| 1,2,3,2 | (0)(1)(2)(1) |
| 1,2,3,2,2 | (0)(1)(2)(1)(1) |
| 1,2,3,2,3 | (0)(1)(2)(1)(2) |
| 1,2,3,2,3,2 | (0)(1)(2)(1)(2)(1) |
| 1,2,3,2,3,2,3 | (0)(1)(2)(1)(2)(1)(2) |
| 1,2,3,3 | (0)(1)(2)(2) |
| 1,2,3,3,2 | (0)(1)(2)(2)(1) |
| 1,2,3,3,2,3 | (0)(1)(2)(2)(1)(2) |
| 1,2,3,3,2,3,3 | (0)(1)(2)(2)(1)(2)(2) |
| 1,2,3,3,3 | (0)(1)(2)(2)(2) |
| 1,2,3,3,3,3 | (0)(1)(2)(2)(2)(2) |
| 1,2,3,4 | (0)(1)(2)(3) |
| 1,2,3,4,2 | (0)(1)(2)(3)(1) |
| 1,2,3,4,2,3,4 | (0)(1)(2)(3)(1)(2)(3) |
| 1,2,3,4,3 | (0)(1)(2)(3)(2) |
| 1,2,3,4,3,4 | (0)(1)(2)(3)(2)(3) |
| 1,2,3,4,4 | (0)(1)(2)(3)(3) |
| 1,2,3,4,5 | (0)(1)(2)(3)(4) |
| 1,2,3,4,5,4 | (0)(1)(2)(3)(4)(3) |
| 1,2,3,4,5,4,5 | (0)(1)(2)(3)(4)(3)(4) |
| 1,2,3,4,5,5 | (0)(1)(2)(3)(4)(4) |
| 1,2,3,4,5,6 | (0)(1)(2)(3)(4)(5) |
| 1,2,3,4,5,6,7 | (0)(1)(2)(3)(4)(5)(6) |
| 1,3 | (0,0)(1,1) |
| 1,3,1 | (0,0)(1,1)(0,0) |
| 1,3,1,2 | (0,0)(1,1)(0,0)(1,0) |

| 0 − Y sequence | BMS |
|---|---|
| 1,3,1,2,3 | (0,0)(1,1)(0,0)(1,0)(2,0) |
| 1,3,1,3 | (0,0)(1,1)(0,0)(1,1) |
| 1,3,2 | (0,0)(1,1)(1,0) |
| 1,3,2,2 | (0,0)(1,1)(1,0)(1,0) |
| 1,3,2,3 | (0,0)(1,1)(1,0)(2,0) |
| 1,3,2,4 | (0,0)(1,1)(1,0)(2,1) |
| 1,3,2,4,3 | (0,0)(1,1)(1,0)(2,1)(2,0) |
| 1,3,2,4,3,5 | (0,0)(1,1)(1,0)(2,1)(2,0)(3,1) |
| 1,3,2,4,3,5,4,6 | (0,0)(1,1)(1,0)(2,1)<br>(2,0)(3,1)(3,0)(4,1) |
| 1,3,3 | (0,0)(1,1)(1,1) |
| 1,3,3,1,3,3 | (0,0)(1,1)(1,1)(0,0)(1,1)(1,1) |
| 1,3,3,2 | (0,0)(1,1)(1,1)(1,0) |
| 1,3,3,2,4,4 | (0,0)(1,1)(1,1)(1,0)(2,1)(2,1) |
| 1,3,3,2,4,4,3 | (0,0)(1,1)(1,1)(1,0)(2,1)(2,1)(2,0) |
| 1,3,3,2,4,4,3,5,5 | (0,0)(1,1)(1,1)(1,0)(2,1)<br>(2,1)(2,0)(3,1)(3,1) |
| 1,3,3,3 | (0,0)(1,1)(1,1)(1,1) |
| 1,3,3,3,3 | (0,0)(1,1)(1,1)(1,1)(1,1) |
| 1,3,4 | (0,0)(1,1)(2,0) |
| 1,3,4,2 | (0,0)(1,1)(2,0)(1,0) |
| 1,3,4,2,4,5 | (0,0)(1,1)(2,0)(1,0)(2,1)(3,0) |
| 1,3,4,2,4,5,3 | (0,0)(1,1)(2,0)(1,0)(2,1)(3,0)(2,0) |
| 1,3,4,2,4,5,3,5,6 | (0,0)(1,1)(2,0)(1,0)(2,1)<br>(3,0)(2,0)(3,1)(4,0) |
| 1,3,4,3 | (0,0)(1,1)(2,0)(1,1) |
| 1,3,4,3,3 | (0,0)(1,1)(2,0)(1,1)(1,1) |
| 1,3,4,3,4 | (0,0)(1,1)(2,0)(1,1)(2,0) |
| 1,3,4,3,4,3,4 | (0,0)(1,1)(2,0)(1,1)(2,0)(1,1)(2,0) |
| 1,3,4,4 | (0,0)(1,1)(2,0)(2,0) |
| 1,3,4,4,3,4 | (0,0)(1,1)(2,0)(2,0)(1,1)(2,0) |
| 1,3,4,4,3,4,4 | (0,0)(1,1)(2,0)(2,0)(1,1)(2,0)(2,0) |
| 1,3,4,4,4 | (0,0)(1,1)(2,0)(2,0)(2,0) |
| 1,3,4,4,4,4 | (0,0)(1,1)(2,0)(2,0)(2,0)(2,0) |
| 1,3,4,5 | (0,0)(1,1)(2,0)(3,0) |
| 1,3,4,5,6 | (0,0)(1,1)(2,0)(3,0)(4,0) |
| 1,3,4,6 | (0,0)(1,1)(2,0)(3,1) |
| 1,3,4,6,6 | (0,0)(1,1)(2,0)(3,1)(3,1) |
| 1,3,4,6,7 | (0,0)(1,1)(2,0)(3,1)(4,0) |
| 1,3,4,6,7,9 | (0,0)(1,1)(2,0)(3,1)(4,0)(5,1) |
| 1,3,5 | (0,0)(1,1)(2,1) |
| 1,3,5,2 | (0,0)(1,1)(2,1)(1,0) |
| 1,3,5,2,4,6 | (0,0)(1,1)(2,1)(1,0)(2,1)(3,1) |
| 1,3,5,2,4,6,3 | (0,0)(1,1)(2,1)(1,0)(2,1)(3,1)(2,0) |
| 1,3,5,2,4,6,3,5,7 | (0,0)(1,1)(2,1)(1,0)<br>(2,1)(3,1)(2,0)(3,1)(4,1) |
| 1,3,5,3 | (0,0)(1,1)(2,1)(1,1) |
| 1,3,5,3,4 | (0,0)(1,1)(2,1)(1,1)(2,0) |
| 1,3,5,3,4,3 | (0,0)(1,1)(2,1)(1,1)(2,0)(1,1) |
| 1,3,5,3,4,3,4 | (0,0)(1,1)(2,1)(1,1)(2,0)(1,1)(2,0) |
| 1,3,5,3,4,4 | (0,0)(1,1)(2,1)(1,1)(2,0)(2,0) |
| 1,3,5,3,4,5 | (0,0)(1,1)(2,1)(1,1)(2,0)(3,0) |
| 1,3,5,3,4,6 | (0,0)(1,1)(2,1)(1,1)(2,0)(3,1) |
| 1,3,5,3,4,6,7 | (0,0)(1,1)(2,1)(1,1)(2,0)(3,1)(4,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,3,5,3,4,6,7,9 | (0,0)(1,1)(2,1)(1,1)<br>(2,0)(3,1)(4,0)(5,1) |
| 1,3,5,3,4,6,8 | (0,0)(1,1)(2,1)(1,1)(2,0)(3,1)(4,1) |
| 1,3,5,3,4,6,8,6 | (0,0)(1,1)(2,1)(1,1)<br>(2,0)(3,1)(4,1)(3,1) |
| 1,3,5,3,5 | (0,0)(1,1)(2,1)(1,1)(2,1) |
| 1,3,5,3,5,3,5 | (0,0)(1,1)(2,1)(1,1)(2,1)(1,1)(2,1) |
| 1,3,5,4 | (0,0)(1,1)(2,1)(2,0) |
| 1,3,5,4,4 | (0,0)(1,1)(2,1)(2,0)(2,0) |
| 1,3,5,4,5 | (0,0)(1,1)(2,1)(2,0)(3,0) |
| 1,3,5,4,6 | (0,0)(1,1)(2,1)(2,0)(3,1) |
| 1,3,5,4,6,8 | (0,0)(1,1)(2,1)(2,0)(3,1)(4,1) |
| 1,3,5,4,6,8,7,9,11 | (0,0)(1,1)(2,1)(2,0)<br>(3,1)(4,1)(4,0)(5,1)(6,1) |
| 1,3,5,5 | (0,0)(1,1)(2,1)(2,1) |
| 1,3,5,5,3 | (0,0)(1,1)(2,1)(2,1)(1,1) |
| 1,3,5,5,3,4,6,8,8 | (0,0)(1,1)(2,1)(2,1)(1,1)<br>(2,0)(3,1)(4,1)(4,1) |
| 1,3,5,5,3,4,6,8,8,6 | (0,0)(1,1)(2,1)(2,1)(1,1)<br>(2,0)(3,1)(4,1)(4,1)(3,1) |
| 1,3,5,5,3,5 | (0,0)(1,1)(2,1)(2,1)(1,1)(2,1) |
| 1,3,5,5,3,5,4,6,8,8 | (0,0)(1,1)(2,1)(2,1)(1,1)<br>(2,1)(2,0)(3,1)(4,1)(4,1) |
| 1,3,5,5,3,5,<br>4,6,8,8,6,8 | (0,0)(1,1)(2,1)(2,1)(1,1)(2,1)<br>(2,0)(3,1)(4,1)(4,1)(3,1)(4,1) |
| 1,3,5,5,3,5,5 | (0,0)(1,1)(2,1)(2,1)(1,1)(2,1)(2,1) |
| 1,3,5,5,4 | (0,0)(1,1)(2,1)(2,1)(2,0) |
| 1,3,5,5,4,6 | (0,0)(1,1)(2,1)(2,1)(2,0)(3,1) |
| 1,3,5,5,4,6,7 | (0,0)(1,1)(2,1)(2,1)(2,0)(3,1)(4,0) |
| 1,3,5,5,4,6,8 | (0,0)(1,1)(2,1)(2,1)(2,0)(3,1)(4,1) |
| 1,3,5,5,4,6,8,8 | (0,0)(1,1)(2,1)(2,1)<br>(2,0)(3,1)(4,1)(4,1) |
| 1,3,5,5,5 | (0,0)(1,1)(2,1)(2,1)(2,1) |
| 1,3,5,5,5,3,5,5,5 | (0,0)(1,1)(2,1)(2,1)(2,1)<br>(1,1)(2,1)(2,1)(2,1) |
| 1,3,5,5,5,4 | (0,0)(1,1)(2,1)(2,1)(2,1)(2,0) |
| 1,3,5,5,5,5 | (0,0)(1,1)(2,1)(2,1)(2,1)(2,1) |
| 1,3,5,6 | (0,0)(1,1)(2,1)(3,0) |
| 1,3,5,6,2 | (0,0)(1,1)(2,1)(3,0)(1,0) |
| 1,3,5,6,2,4,6,7,3 | (0,0)(1,1)(2,1)(3,0)(1,0)<br>(2,1)(3,1)(4,0)(2,0) |
| 1,3,5,6,3 | (0,0)(1,1)(2,1)(3,0)(1,1) |
| 1,3,5,6,3,5,6 | (0,0)(1,1)(2,1)(3,0)(1,1)(2,1)(3,0) |
| 1,3,5,6,4 | (0,0)(1,1)(2,1)(3,0)(2,0) |
| 1,3,5,6,4,6,8,9 | (0,0)(1,1)(2,1)(3,0)<br>(2,0)(3,1)(4,1)(5,0) |
| 1,3,5,6,5 | (0,0)(1,1)(2,1)(3,0)(2,1) |
| 1,3,5,6,5,3,5,6 | (0,0)(1,1)(2,1)(3,0)<br>(2,1)(1,1)(2,1)(3,0) |
| 1,3,5,6,5,3,5,6,5 | (0,0)(1,1)(2,1)(3,0)(2,1)<br>(1,1)(2,1)(3,0)(2,1) |
| 1,3,5,6,5,5 | (0,0)(1,1)(2,1)(3,0)(2,1)(2,1) |
| 1,3,5,6,5,6 | (0,0)(1,1)(2,1)(3,0)(2,1)(3,0) |
| 1,3,5,6,6 | (0,0)(1,1)(2,1)(3,0)(3,0) |
| 1,3,5,6,7 | (0,0)(1,1)(2,1)(3,0)(4,0) |

| 0 − Y sequence | BMS |
|---|---|
| 1,3,5,6,7,4 | (0,0)(1,1)(2,1)(3,0)(4,0)(2,0) |
| 1,3,5,6,8 | (0,0)(1,1)(2,1)(3,0)(4,1) |
| 1,3,5,6,8,5 | (0,0)(1,1)(2,1)(3,0)(4,1)(2,1) |
| 1,3,5,6,8,5,2,4 | (0,0)(1,1)(2,1)(3,0)<br>(4,1)(2,1)(1,0)(2,1) |
| 1,3,5,6,8,5,3,5,6,8 | (0,0)(1,1)(2,1)(3,0)(4,1)<br>(2,1)(1,1)(2,1)(3,0)(4,1) |
| 1,3,5,6,8,6,3,5,6,8 | (0,0)(1,1)(2,1)(3,0)(4,1)<br>(3,0)(1,1)(2,1)(3,0)(4,1) |
| 1,3,5,6,8,8 | (0,0)(1,1)(2,1)(3,0)(4,1)(4,1) |
| 1,3,5,6,8,8,8 | (0,0)(1,1)(2,1)(3,0)(4,1)(4,1)(4,1) |
| 1,3,5,6,8,9 | (0,0)(1,1)(2,1)(3,0)(4,1)(5,0) |
| 1,3,5,6,8,9,3,5,6,8 | (0,0)(1,1)(2,1)(3,0)(4,1)<br>(5,0)(1,1)(2,1)(3,0)(4,1) |
| 1,3,5,6,8,10 | (0,0)(1,1)(2,1)(3,0)(4,1)(5,1) |
| 1,3,5,6,8,10,8,9,11,13 | (0,0)(1,1)(2,1)(3,0)(4,1)<br>(5,1)(4,1)(5,0)(6,1)(7,1) |
| 1,3,5,6,8,10,10 | (0,0)(1,1)(2,1)(3,0)(4,1)(5,1)(5,1) |
| 1,3,5,6,8,10,11 | (0,0)(1,1)(2,1)(3,0)(4,1)(5,1)(6,0) |
| 1,3,5,6,8,10,11,11 | (0,0)(1,1)(2,1)(3,0)<br>(4,1)(5,1)(6,0)(6,0) |
| 1,3,5,6,8,10,11,13 | (0,0)(1,1)(2,1)(3,0)<br>(4,1)(5,1)(6,0)(7,1) |
| 1,3,5,6,8,10,11,13,15 | (0,0)(1,1)(2,1)(3,0)(4,1)<br>(5,1)(6,0)(7,1)(8,1) |
| 1,3,5,7 | (0,0)(1,1)(2,1)(3,1) |
| 1,3,5,7,2 | (0,0)(1,1)(2,1)(3,1)(1,0) |
| 1,3,5,7,2,2 | (0,0)(1,1)(2,1)(3,1)(1,0)(1,0) |
| 1,3,5,7,2,3 | (0,0)(1,1)(2,1)(3,1)(1,0)(2,0) |
| 1,3,5,7,2,4 | (0,0)(1,1)(2,1)(3,1)(1,0)(2,1) |
| 1,3,5,7,2,4,5 | (0,0)(1,1)(2,1)(3,1)(1,0)(2,1)(3,0) |
| 1,3,5,7,2,4,5,7 | (0,0)(1,1)(2,1)(3,1)<br>(1,0)(2,1)(3,0)(4,1) |
| 1,3,5,7,2,4,6 | (0,0)(1,1)(2,1)(3,1)(1,0)(2,1)(3,1) |
| 1,3,5,7,2,4,6,7 | (0,0)(1,1)(2,1)(3,1)<br>(1,0)(2,1)(3,1)(4,0) |
| 1,3,5,7,2,4,6,7,9 | (0,0)(1,1)(2,1)(3,1)(1,0)<br>(2,1)(3,1)(4,0)(5,1) |
| 1,3,5,7,2,4,6,7,9,11 | (0,0)(1,1)(2,1)(3,1)(1,0)<br>(2,1)(3,1)(4,0)(5,1)(6,1) |
| 1,3,5,7,2,4,6,8 | (0,0)(1,1)(2,1)(3,1)<br>(1,0)(2,1)(3,1)(4,1) |
| 1,3,5,7,2,4,6,8,2 | (0,0)(1,1)(2,1)(3,1)(1,0)<br>(2,1)(3,1)(4,1)(1,0) |
| 1,3,5,7,2,4,<br>6,8,2,4,6,8 | (0,0)(1,1)(2,1)(3,1)(1,0)(2,1)<br>(3,1)(4,1)(1,0)(2,1)(3,1)(4,1) |
| 1,3,5,7,2,4,6,8,3 | (0,0)(1,1)(2,1)(3,1)(1,0)<br>(2,1)(3,1)(4,1)(2,0) |
| 1,3,5,7,2,4,<br>6,8,3,5,7,9 | (0,0)(1,1)(2,1)(3,1)(1,0)(2,1)<br>(3,1)(4,1)(2,0)(3,1)(4,1)(5,1) |
| 1,3,5,7,3 | (0,0)(1,1)(2,1)(3,1)(1,1) |
| 1,3,5,7,3,2 | (0,0)(1,1)(2,1)(3,1)(1,1)(1,0) |
| 1,3,5,7,3,2,4 | (0,0)(1,1)(2,1)(3,1)(1,1)(1,0)(2,1) |
| 1,3,5,7,3,2,4,6 | (0,0)(1,1)(2,1)(3,1)<br>(1,1)(1,0)(2,1)(3,1) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,3,5,7,3,2,4,6,8 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(1,0)(2,1)(3,1)(4,1) |
| 1,3,5,7,3,2,4,6,8,3 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(1,0)(2,1)(3,1)(4,1)<br>(2,0)(3,1)(4,1)(5,1) |
| 1,3,5,7,3,2,4,6,8,4 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(1,0)(2,1)(3,1)(4,1)(2,1) |
| 1,3,5,7,3,3 | (0,0)(1,1)(2,1)(3,1)(1,1)(1,1) |
| 1,3,5,7,3,4 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,0) |
| 1,3,5,7,3,4,6,8,10 | (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,0)(3,1)(4,1)(5,1) |
| 1,3,5,7,3,4,6,8,10,6 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,0)(3,1)(4,1)(5,1)(3,1) |
| 1,3,5,7,3,5 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1) |
| 1,3,5,7,3,5,6 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)(3,0) |
| 1,3,5,7,3,5,8 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)(3,0)(4,1) |
| 1,3,5,7,3,5,6,8,10 | (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(3,0)(4,1)(5,1) |
| 1,3,5,7,3,5,6,8,10,12 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1) |
| 1,3,5,7,3,5,<br>6,8,10,12,4 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1)(2,0) |
| 1,3,5,7,3,5,6,8,10,<br>12,4,3,5,6,8,10,12 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(2,0)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1) |
| 1,3,5,7,3,5,<br>6,8,10,12,4,4 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(2,0)(2,0) |
| 1,3,5,7,3,5,<br>6,8,10,12,4,5 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(2,0)(3,0) |
| 1,3,5,7,3,5,<br>6,8,10,12,4,6 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(2,0)(3,1) |
| 1,3,5,7,3,5,6,<br>8,10,12,4,6,8,10 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)(3,0)<br>(4,1)(5,1)(6,1)(2,0)(3,1)(4,1)(5,1) |
| 1,3,5,7,3,5,<br>6,8,10,12,5 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1)(2,1) |
| 1,3,5,7,3,5,6,8,<br>10,12,5,6,8,10,12 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1) |
| 1,3,5,7,3,5,<br>6,8,10,12,6 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(3,0) |
| 1,3,5,7,3,5,6,8,<br>10,12,6,8,10,12 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)(3,0)<br>(4,1)(5,1)(6,1)(3,0)(4,1)(5,1)(6,1) |
| 1,3,5,7,3,5,<br>6,8,10,12,8 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1)(4,1) |
| 1,3,5,7,3,5,6,8,10,<br>12,8,10,11,13,15,17 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1)(4,1)<br>(5,1)(6,0)(7,1)(8,1)(9,1) |
| 1,3,5,7,3,5,7 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)(3,1) |
| 1,3,5,7,4 | (0,0)(1,1)(2,1)(3,1)(2,0) |
| 1,3,5,7,4,6,8,10 | (0,0)(1,1)(2,1)(3,1)(2,0)(3,1)(4,1)(5,1) |
| 1,3,5,7,5 | (0,0)(1,1)(2,1)(3,1)(2,1) |
| 1,3,5,7,5,6 | (0,0)(1,1)(2,1)(3,1)(2,1)(3,0) |
| 1,3,5,7,5,6,8 | (0,0)(1,1)(2,1)(3,1)(2,1)(3,0)(4,1) |
| 1,3,5,7,5,6,8,10,12 | (0,0)(1,1)(2,1)(3,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1) |
| 1,3,5,7,5,7 | (0,0)(1,1)(2,1)(3,1)(2,1)(3,1) |

| 0 − Y sequence | BMS |
|---|---|
| 1,3,5,7,5,7,5,7 | (0,0)(1,1)(2,1)(3,1)<br>(2,1)(3,1)(2,1)(3,1) |
| 1,3,5,7,6 | (0,0)(1,1)(2,1)(3,1)(3,0) |
| 1,3,5,7,6,4 | (0,0)(1,1)(2,1)(3,1)(3,0)(2,0) |
| 1,3,5,7,6,5 | (0,0)(1,1)(2,1)(3,1)(3,0)(2,1) |
| 1,3,5,7,6,5,6 | (0,0)(1,1)(2,1)(3,1)(3,0)(2,1)(3,0) |
| 1,3,5,7,6,4 | (0,0)(1,1)(2,1)(3,1)(3,0)(2,1)(3,1) |
| 1,3,5,7,6,5,7 | (0,0)(1,1)(2,1)(3,1)<br>(3,0)(2,1)(3,1)(3,0) |
| 1,3,5,7,6,4 | (0,0)(1,1)(2,1)(3,1)(3,0)(3,0) |
| 1,3,5,7,6,6 | (0,0)(1,1)(2,1)(3,1)(3,0)(4,0) |
| 1,3,5,7,6,8 | (0,0)(1,1)(2,1)(3,1)(3,0)(4,1) |
| 1,3,5,7,6,8,10,12 | (0,0)(1,1)(2,1)(3,1)<br>(3,0)(4,1)(5,1)(6,1) |
| 1,3,5,7,7 | (0,0)(1,1)(2,1)(3,1)(3,1) |
| 1,3,5,7,7,5 | (0,0)(1,1)(2,1)(3,1)(3,1)(2,1) |
| 1,3,5,7,7,5,7 | (0,0)(1,1)(2,1)(3,1)(3,1)(2,1)(3,1) |
| 1,3,5,7,7,5,7,6 | (0,0)(1,1)(2,1)(3,1)<br>(3,1)(2,1)(3,1)(3,0) |
| 1,3,5,7,7,5,7,7 | (0,0)(1,1)(2,1)(3,1)<br>(3,1)(2,1)(3,1)(3,1) |
| 1,3,5,7,7,6 | (0,0)(1,1)(2,1)(3,1)(3,1)(3,0) |
| 1,3,5,7,7,7 | (0,0)(1,1)(2,1)(3,1)(3,1)(3,1) |
| 1,3,5,7,7,7,5,7,7,7 | (0,0)(1,1)(2,1)(3,1)(3,1)<br>(3,1)(2,1)(3,1)(3,1)(3,1) |
| 1,3,5,7,7,7,6 | (0,0)(1,1)(2,1)(3,1)(3,1)(3,1)(3,0) |
| 1,3,5,7,7,7,7 | (0,0)(1,1)(2,1)(3,1)(3,1)(3,1)(3,1) |
| 1,3,5,7,8 | (0,0)(1,1)(2,1)(3,1)(4,0) |
| 1,3,5,7,8,7 | (0,0)(1,1)(2,1)(3,1)(4,0)(3,1) |
| 1,3,5,7,8,7,8 | (0,0)(1,1)(2,1)(3,1)(4,0)(3,1)(4,0) |
| 1,3,5,7,8,8 | (0,0)(1,1)(2,1)(3,1)(4,0)(4,0) |
| 1,3,5,7,8,10 | (0,0)(1,1)(2,1)(3,1)(4,0)(5,1) |
| 1,3,5,7,8,10,12,14 | (0,0)(1,1)(2,1)(3,1)<br>(4,0)(5,1)(6,1)(7,1) |
| 1,3,5,7,8,10,12,14,15 | (0,0)(1,1)(2,1)(3,1)(4,0)<br>(5,1)(6,1)(7,1)(8,0) |
| 1,3,5,7,8,10,12,14,15,17 | (0,0)(1,1)(2,1)(3,1)(4,0)<br>(5,1)(6,1)(7,1)(8,0)(9,1) |
| 1,3,5,7,9 | (0,0)(1,1)(2,1)(3,1)(4,1) |
| 1,3,5,7,9,5 | (0,0)(1,1)(2,1)(3,1)(4,1)(2,1) |
| 1,3,5,7,9,7 | (0,0)(1,1)(2,1)(3,1)(4,1)(3,1) |
| 1,3,5,7,9,7,9 | (0,0)(1,1)(2,1)(3,1)(4,1)(3,1)(4,1) |
| 1,3,5,7,9,8 | (0,0)(1,1)(2,1)(3,1)(4,1)(4,0) |
| 1,3,5,7,9,9 | (0,0)(1,1)(2,1)(3,1)(4,1)(4,1) |
| 1,3,5,7,9,9,9 | (0,0)(1,1)(2,1)(3,1)(4,1)(4,1)(4,1) |
| 1,3,5,7,9,10 | (0,0)(1,1)(2,1)(3,1)(4,1)(5,0) |
| 1,3,5,7,9,11 | (0,0)(1,1)(2,1)(3,1)(4,1)(5,1) |
| 1,3,5,7,9,11,13 | (0,0)(1,1)(2,1)(3,1)(4,1)(5,1)(6,1) |
| 1,3,6 | (0,0)(1,1)(2,2) |
| 1,3,6,2 | (0,0)(1,1)(2,2)(1,0) |
| 1,3,6,2,4,7 | (0,0)(1,1)(2,2)(1,0)(2,1)(3,2) |
| 1,3,6,3 | (0,0)(1,1)(2,2)(1,1) |
| 1,3,6,3,5 | (0,0)(1,1)(2,2)(1,1)(2,1) |
| 1,3,6,3,5,7 | (0,0)(1,1)(2,2)(1,1)(2,1)(3,1) |
| 1,3,6,3,6 | (0,0)(1,1)(2,2)(1,1)(2,2) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,3,6,3,6,3,6 | (0,0)(1,1)(2,2)(1,1)(2,2)(1,1)(2,2) |
| 1,3,6,4 | (0,0)(1,1)(2,2)(2,0) |
| 1,3,6,4,4 | (0,0)(1,1)(2,2)(2,0)(2,0) |
| 1,3,6,4,5 | (0,0)(1,1)(2,2)(2,0)(3,0) |
| 1,3,6,4,6 | (0,0)(1,1)(2,2)(2,0)(3,1) |
| 1,3,6,4,6,9 | (0,0)(1,1)(2,2)(2,0)(3,1)(4,2) |
| 1,3,6,5 | (0,0)(1,1)(2,2)(2,1) |
| 1,3,6,5,6 | (0,0)(1,1)(2,2)(2,1)(3,0) |
| 1,3,6,5,7 | (0,0)(1,1)(2,2)(2,1)(3,1) |
| 1,3,6,5,8 | (0,0)(1,1)(2,2)(2,1)(3,2) |
| 1,3,6,5,8,7 | (0,0)(1,1)(2,2)(2,1)(3,2)(3,1) |
| 1,3,6,5,8,7,10 | (0,0)(1,1)(2,2)(2,1)(3,2)(3,1)(4,2) |
| 1,3,6,6 | (0,0)(1,1)(2,2)(2,2) |
| 1,3,6,6,3,6,6 | (0,0)(1,1)(2,2)(2,2)(1,1)(2,2)(2,2) |
| 1,3,6,6,4 | (0,0)(1,1)(2,2)(2,2)(2,0) |
| 1,3,6,6,5 | (0,0)(1,1)(2,2)(2,2)(2,1) |
| 1,3,6,6,5,8,8 | (0,0)(1,1)(2,2)(2,2)(2,1)(3,2)(3,2) |
| 1,3,6,6,6 | (0,0)(1,1)(2,2)(2,2)(2,2) |
| 1,3,6,7 | (0,0)(1,1)(2,2)(3,0) |
| 1,3,6,7,9 | (0,0)(1,1)(2,2)(3,0)(4,1) |
| 1,3,6,7,9,12 | (0,0)(1,1)(2,2)(3,0)(4,1)(5,2) |
| 1,3,6,8 | (0,0)(1,1)(2,2)(3,1) |
| 1,3,6,8,6,8 | (0,0)(1,1)(2,2)(3,1)(2,2)(3,1) |
| 1,3,6,8,7 | (0,0)(1,1)(2,2)(3,1)(3,0) |
| 1,3,6,8,8 | (0,0)(1,1)(2,2)(3,1)(3,1) |
| 1,3,6,8,11 | (0,0)(1,1)(2,2)(3,1)(4,2) |
| 1,3,6,9 | (0,0)(1,1)(2,2)(3,2) |
| 1,3,6,9,9 | (0,0)(1,1)(2,2)(3,2)(3,2) |
| 1,3,6,9,10 | (0,0)(1,1)(2,2)(3,2)(4,0) |
| 1,3,6,9,11 | (0,0)(1,1)(2,2)(3,2)(4,1) |
| 1,3,6,9,11,6,9,11 | (0,0)(1,1)(2,2)(3,2)<br>(4,1)(2,2)(3,2)(4,1) |
| 1,3,6,9,11,7 | (0,0)(1,1)(2,2)(3,2)(4,1)(3,0) |
| 1,3,6,9,11,7,9 | (0,0)(1,1)(2,2)(3,2)(4,1)(3,0)(4,1) |
| 1,3,6,9,11,8 | (0,0)(1,1)(2,2)(3,2)(4,1)(3,1) |
| 1,3,6,9,11,8,11,14,16,13 | (0,0)(1,1)(2,2)(3,2)(4,1)<br>(3,1)(4,2)(5,2)(6,1)(5,1) |
| 1,3,6,9,11,9 | (0,0)(1,1)(2,2)(3,2)(4,1)(3,2) |
| 1,3,6,9,11,9,11 | (0,0)(1,1)(2,2)(3,2)(4,1)(3,2)(4,1) |
| 1,3,6,9,11,10 | (0,0)(1,1)(2,2)(3,2)(4,1)(4,0) |
| 1,3,6,9,11,11 | (0,0)(1,1)(2,2)(3,2)(4,1)(4,1) |
| 1,3,6,9,11,12 | (0,0)(1,1)(2,2)(3,2)(4,1)(5,0) |
| 1,3,6,9,11,13 | (0,0)(1,1)(2,2)(3,2)(4,1)(5,1) |
| 1,3,6,9,11,14 | (0,0)(1,1)(2,2)(3,2)(4,1)(5,2) |
| 1,3,6,9,11,14,17 | (0,0)(1,1)(2,2)(3,2)(4,1)(5,2)(6,2) |
| 1,3,6,9,11,14,17,19 | (0,0)(1,1)(2,2)(3,2)<br>(4,1)(5,2)(6,2)(7,1) |
| 1,3,6,9,12 | (0,0)(1,1)(2,2)(3,2)(4,2) |
| 1,3,6,9,12,1,3,6,9,12 | (0,0)(1,1)(2,2)(3,2)(4,2)<br>(0,0)(1,1)(2,2)(3,2)(4,2) |
| 1,3,6,9,12,3,6,9,12 | (0,0)(1,1)(2,2)(3,2)(4,2)<br>(1,1)(2,2)(3,2)(4,2) |
| 1,3,6,9,12,5,8,11,14 | (0,0)(1,1)(2,2)(3,2)(4,2)<br>(2,1)(3,2)(4,2)(5,2) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,3,6,9,12,6,9,12 | (0,0)(1,1)(2,2)(3,2)<br>(4,2)(2,2)(3,2)(4,2) |
| 1,3,6,9,12,9 | (0,0)(1,1)(2,2)(3,2)(4,2)(3,2) |
| 1,3,6,9,12,15 | (0,0)(1,1)(2,2)(3,2)(4,2)(5,2) |
| 1,3,6,10 | (0,0)(1,1)(2,2)(3,3) |
| 1,3,6,10,3,6,10 | (0,0)(1,1)(2,2)(3,3)(1,1)(2,2)(3,3) |
| 1,3,6,10,5,8,12 | (0,0)(1,1)(2,2)(3,3)(2,1)(3,2)(4,3) |
| 1,3,6,10,6 | (0,0)(1,1)(2,2)(3,3)(2,2) |
| 1,3,6,10,6,9,12 | (0,0)(1,1)(2,2)(3,3)(2,2)(3,2)(4,2) |
| 1,3,6,10,6,10 | (0,0)(1,1)(2,2)(3,3)(2,2)(3,3) |
| 1,3,6,10,9,13 | (0,0)(1,1)(2,2)(3,3)(3,2)(4,3) |
| 1,3,6,10,10 | (0,0)(1,1)(2,2)(3,3)(3,3) |
| 1,3,6,10,11 | (0,0)(1,1)(2,2)(3,3)(4,0) |
| 1,3,6,10,14 | (0,0)(1,1)(2,2)(3,3)(4,3) |
| 1,3,6,10,14,18 | (0,0)(1,1)(2,2)(3,3)(4,3)(5,3) |
| 1,3,6,10,15 | (0,0)(1,1)(2,2)(3,3)(4,4) |
| 1,3,6,10,15,20,25 | (0,0)(1,1)(2,2)(3,3)(4,4)(5,4)(6,4) |
| 1,3,6,10,15,21 | (0,0)(1,1)(2,2)(3,3)(4,4)(5,5) |
| 1,4 | (0,0,0)(1,1,1) |
| 1,4,1 | (0,0,0)(1,1,1)(0,0,0) |
| 1,4,2 | (0,0,0)(1,1,1)(1,0,0) |
| 1,4,2,5 | (0,0,0)(1,1,1)(1,0,0)(2,1,1) |
| 1,4,2,5,3 | (0,0,0)(1,1,1)(1,0,0)(2,1,1)(2,0,0) |
| 1,4,3 | (0,0,0)(1,1,1)(1,1,0) |
| 1,4,3,5,7 | (0,0,0)(1,1,1)(1,1,0)(2,1,0)(3,1,0) |
| 1,4,3,6 | (0,0,0)(1,1,1)(1,1,0)(2,2,0) |
| 1,4,3,6,10 | (0,0,0)(1,1,1)(1,1,0)(2,2,0)(3,3,0) |
| 1,4,3,7 | (0,0,0)(1,1,1)(1,1,0)(2,2,1) |
| 1,4,3,7,4 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)(2,0,0) |
| 1,4,3,7,4,3,7,4 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,0,0)(1,1,0)(2,2,1)(2,0,0) |
| 1,4,3,7,4,4 | (0,0,0)(1,1,1)(1,1,0)<br>(2,2,1)(2,0,0)(2,0,0) |
| 1,4,3,7,4,4,3,7,4 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)(2,0,0)<br>(2,0,0)(1,1,0)(2,2,1)(2,0,0) |
| 1,4,3,7,5 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)(2,1,0) |
| 1,4,3,7,5,8 | (0,0,0)(1,1,1)(1,1,0)<br>(2,2,1)(2,1,0)(3,2,0) |
| 1,4,3,7,5,9 | (0,0,0)(1,1,1)(1,1,0)<br>(2,2,1)(2,1,0)(3,2,1) |
| 1,4,3,7,5,9,6 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,1,0)(3,2,1)(3,0,0) |
| 1,4,3,7,5,9,7 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,1,0)(3,2,1)(3,1,0) |
| 1,4,3,7,5,9,7,11 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,1,0)(3,2,1)(3,1,0)(4,2,1) |
| 1,4,3,7,6 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)(2,2,0) |
| 1,4,3,7,6,11 | (0,0,0)(1,1,1)(1,1,0)<br>(2,2,1)(2,2,0)(3,3,1) |
| 1,4,3,7,6,11,7 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,2,0)(3,3,1)(3,0,0) |
| 1,4,3,7,6,11,10 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,2,0)(3,3,1)(3,3,0) |
| 1,4,4 | (0,0,0)(1,1,1)(1,1,1) |
| 1,4,4,3 | (0,0,0)(1,1,1)(1,1,1)(1,1,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,4,3,7,7,6 | (0,0,0)(1,1,1)(1,1,1)(1,1,0)<br>(2,2,1)(2,2,1)(2,2,0) |
| 1,4,4,4 | (0,0,0)(1,1,1)(1,1,1)(1,1,1) |
| 1,4,5 | (0,0,0)(1,1,1)(2,0,0) |
| 1,4,5,3,7,8 | (0,0,0)(1,1,1)(2,0,0)<br>(1,1,0)(2,2,1)(3,0,0) |
| 1,4,5,4 | (0,0,0)(1,1,1)(2,0,0)(1,1,1) |
| 1,4,5,4,5 | (0,0,0)(1,1,1)(2,0,0)(1,1,1)(2,0,0) |
| 1,4,5,5 | (0,0,0)(1,1,1)(2,0,0)(2,0,0) |
| 1,4,5,7 | (0,0,0)(1,1,1)(2,0,0)(3,1,0) |
| 1,4,5,7,10 | (0,0,0)(1,1,1)(2,0,0)(3,1,0)(4,2,0) |
| 1,4,5,8 | (0,0,0)(1,1,1)(2,0,0)(3,1,1) |
| 1,4,6 | (0,0,0)(1,1,1)(2,1,0) |
| 1,4,6,3,7,9 | (0,0,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,2,1)(3,1,0) |
| 1,4,6,3,7,9,6 | (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0) |
| 1,4,6,3,7,9,6,11,13 | (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,0)(3,3,1)(4,1,0) |
| 1,4,6,3,7,9,7 | (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1) |
| 1,4,6,3,7,9,9 | (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(3,1,0) |
| 1,4,6,3,7,9,11 | (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,1,0) |
| 1,4,6,3,7,9,12 | (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,0) |
| 1,4,6,3,7,9,13 | (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,1) |
| 1,4,6,3,7,10 | (0,0,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,2,1)(3,2,0) |
| 1,4,6,3,7,10,6,11,15 | (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,1)(4,3,0) |
| 1,4,6,4 | (0,0,0)(1,1,1)(2,1,0)(1,1,1) |
| 1,4,6,4,3 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0) |
| 1,4,6,4,3,7 | (0,0,0)(1,1,1)(2,1,0)<br>(1,1,1)(1,1,0)(2,2,1) |
| 1,4,6,4,3,7,7 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(2,2,1) |
| 1,4,6,4,3,7,9 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,1,0) |
| 1,4,6,4,3,7,9,7 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,1,0)(2,2,1) |
| 1,4,6,4,3,7,10 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0) |
| 1,4,6,4,3,7,<br>10,6,11,14,11 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,0)<br>(3,3,1)(4,2,0)(3,3,1) |
| 1,4,6,4,3,7,10,7 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1) |
| 1,4,6,4,3,7,10,7,6 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,1)(2,2,0) |
| 1,4,6,4,4 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,1) |
| 1,4,6,4,6 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)(2,1,0) |
| 1,4,6,4,6,3,7,10,7,10 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1)(3,2,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,6,4,6,4 | (0,0,0)(1,1,1)(2,1,0)<br>(1,1,1)(2,1,0)(1,1,1) |
| 1,4,6,5 | (0,0,0)(1,1,1)(2,1,0)(2,0,0) |
| 1,4,6,5,4 | (0,0,0)(1,1,1)(2,1,0)(2,0,0)(1,1,1) |
| 1,4,6,6 | (0,0,0)(1,1,1)(2,1,0)(2,1,0) |
| 1,4,6,6,3,7,10,10 | (0,0,0)(1,1,1)(2,1,0)(2,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(3,2,0) |
| 1,4,6,6,4 | (0,0,0)(1,1,1)(2,1,0)(2,1,0)(1,1,1) |
| 1,4,6,6,4,6,4 | (0,0,0)(1,1,1)(2,1,0)(2,1,0)<br>(1,1,1)(2,1,0)(1,1,1) |
| 1,4,6,6,4,6,5 | (0,0,0)(1,1,1)(2,1,0)(2,1,0)<br>(1,1,1)(2,1,0)(2,0,0) |
| 1,4,6,6,4,6,6,4 | (0,0,0)(1,1,1)(2,1,0)(2,1,0)<br>(1,1,1)(2,1,0)(2,1,0)(1,1,1) |
| 1,4,6,6,5 | (0,0,0)(1,1,1)(2,1,0)(2,1,0)(2,0,0) |
| 1,4,6,6,6,4 | (0,0,0)(1,1,1)(2,1,0)<br>(2,1,0)(2,1,0)(1,1,1) |
| 1,4,6,7 | (0,0,0)(1,1,1)(2,1,0)(3,0,0) |
| 1,4,6,8 | (0,0,0)(1,1,1)(2,1,0)(3,1,0) |
| 1,4,6,8,3,7,10,13 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,2,0) |
| 1,4,6,8,4 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)(1,1,1) |
| 1,4,6,8,5 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)(2,0,0) |
| 1,4,6,8,6,4 | (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(2,1,0)(1,1,1) |
| 1,4,6,8,6,8,4 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(2,1,0)(3,1,0)(1,1,1) |
| 1,4,6,8,7 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)(3,0,0) |
| 1,4,6,8,8,4 | (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(3,1,0)(1,1,1) |
| 1,4,6,8,8,8,4 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(3,1,0)(3,1,0)(1,1,1) |
| 1,4,6,8,9 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)(4,0,0) |
| 1,4,6,8,10,4 | (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(4,1,0)(1,1,1) |
| 1,4,6,8,10,8,4 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(4,1,0)(3,1,0)(1,1,1) |
| 1,4,6,8,10,8,10,4 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(4,1,0)(3,1,0)(4,1,0)(1,1,1) |
| 1,4,6,8,10,8,10,7 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(4,1,0)(3,1,0)(4,1,0)(3,0,0) |
| 1,4,6,8,10,10,4 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(4,1,0)(4,1,0)(1,1,1) |
| 1,4,6,8,10,11 | (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(4,1,0)(5,0,0) |
| 1,4,6,8,10,12,4 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(4,1,0)(5,1,0)(1,1,1) |
| 1,4,6,9 | (0,0,0)(1,1,1)(2,1,0)(3,2,0) |
| 1,4,6,9,3,7 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(1,1,0)(2,2,1) |
| 1,4,6,9,3,7,8,10 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,0,0)(4,1,0) |
| 1,4,6,9,3,7,8,<br>10,5,9,10,12 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,0,0)(4,1,0)<br>(2,1,0)(3,2,1)(4,0,0)(5,1,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,6,9,3,7,8,10,6 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)(1,1,0)<br>(2,2,1)(3,0,0)(4,1,0)(2,2,0) |
| 1,4,6,9,3,7,8,10,7 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)(1,1,0)<br>(2,2,1)(3,0,0)(4,1,0)(2,2,1) |
| 1,4,6,9,3,7,8,11 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,0,0)(4,1,1) |
| 1,4,6,9,3,7,9 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,1,0) |
| 1,4,6,9,3,7,10,7 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1) |
| 1,4,6,9,3,7,10,14 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0) |
| 1,4,6,9,3,7,10,<br>14,6,11,15,20 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(2,2,0)(3,3,1)(4,3,0)(5,4,0) |
| 1,4,6,9,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)(1,1,1) |
| 1,4,6,9,4,6,9 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,1)(2,1,0)(3,2,0) |
| 1,4,6,9,6,9 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(2,1,0)(3,2,0) |
| 1,4,6,9,8,11 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(3,1,0)(4,2,0) |
| 1,4,6,9,9 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)(3,2,0) |
| 1,4,6,9,10 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,0,0) |
| 1,4,6,9,11,4 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,1,0)(1,1,1) |
| 1,4,6,9,12 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,2,0) |
| 1,4,6,9,12,13 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(5,0,0) |
| 1,4,6,9,12,14,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(5,1,0)(1,1,1) |
| 1,4,6,9,12,15 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(5,2,0) |
| 1,4,6,9,13 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,3,0) |
| 1,4,6,9,13,13 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(4,3,0) |
| 1,4,6,9,13,14 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,0,0) |
| 1,4,6,9,13,17 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,3,0) |
| 1,4,6,9,13,17,21 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,3,0)(5,3,0)(6,3,0) |
| 1,4,6,9,13,18 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,4,0) |
| 1,4,6,10 | (0,0,0)(1,1,1)(2,1,0)(3,2,1) |
| 1,4,6,10,3,7 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(1,1,0)(2,2,1) |
| 1,4,6,10,3,7,10,7 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1) |
| 1,4,6,10,3,7,10,10,7 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(3,2,0)(2,2,1) |
| 1,4,6,10,3,7,10,14 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0) |
| 1,4,6,10,3,7,10,14,19 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(1,1,0)<br>(2,2,1)(3,2,0)(4,3,0)(5,4,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,6,10,3,7,10,15 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,1) |
| 1,4,6,10,3,7,<br>10,15,6,11,14 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,1)<br>(2,2,0)(3,3,1)(4,2,0) |
| 1,4,6,10,3,7,10,<br>15,6,11,15,20 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,1)<br>(2,2,0)(3,3,1)(4,3,0)(5,4,0) |
| 1,4,6,10,3,7,10,<br>15,6,11,15,21 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,1)<br>(2,2,0)(3,3,1)(4,3,0)(5,4,1) |
| 1,4,6,10,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(1,1,1) |
| 1,4,6,10,4,4 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(1,1,1)(1,1,1) |
| 1,4,6,10,4,6 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(1,1,1)(2,1,0) |
| 1,4,6,10,4,6,3,<br>7,10,15,7,10 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(4,3,1)(2,2,1)(3,2,0) |
| 1,4,6,10,4,6,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(1,1,1) |
| 1,4,6,10,4,6,6,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(2,1,0)(1,1,1) |
| 1,4,6,10,4,6,7 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(3,0,0) |
| 1,4,6,10,4,6,9 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(3,2,0) |
| 1,4,6,10,4,6,10 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(3,2,1) |
| 1,4,6,10,4,6,10,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(3,2,1)(1,1,1) |
| 1,4,6,10,4,6,10,4,6 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(1,1,1)<br>(2,1,0)(3,2,1)(1,1,1)(2,1,0) |
| 1,4,6,10,4,6,10,4,6,9 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(1,1,1)<br>(2,1,0)(3,2,1)(1,1,1)(2,1,0)(3,2,0) |
| 1,4,6,10,4,6,10,4,6,10 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(1,1,1)<br>(2,1,0)(3,2,1)(1,1,1)(2,1,0)(3,2,1) |
| 1,4,6,10,5 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(2,0,0) |
| 1,4,6,10,6 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(2,1,0) |
| 1,4,6,10,6,4 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(2,1,0)(1,1,1) |
| 1,4,6,10,6,9 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(2,1,0)(3,2,0) |
| 1,4,6,10,6,9,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(2,1,0)(3,2,0)(1,1,1) |
| 1,4,6,10,6,10 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(2,1,0)(3,2,1) |
| 1,4,6,10,8,12 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(3,1,0)(4,2,1) |
| 1,4,6,10,9 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(3,2,0) |
| 1,4,6,10,9,12,15 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(3,2,0)(4,2,0)(5,2,0) |
| 1,4,6,10,9,14 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(3,2,0)(4,3,1) |
| 1,4,6,10,9,14,13 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(3,2,0)(4,3,1)(4,3,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,6,10,10 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(3,2,1) |
| 1,4,6,10,11 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,0,0) |
| 1,4,6,10,11,9,14,15 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,0,0)(3,2,0)(4,3,1)(5,0,0) |
| 1,4,6,10,11,10 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,0,0)(3,2,1) |
| 1,4,6,10,11,12 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,0,0)(5,0,0) |
| 1,4,6,10,11,13 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,0,0)(5,1,0) |
| 1,4,6,10,11,14 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,0,0)(5,1,1) |
| 1,4,6,10,12 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0) |
| 1,4,6,10,12,4 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(1,1,1) |
| 1,4,6,10,12,4,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(1,1,1)(1,1,1) |
| 1,4,6,10,12,4,5,8,10,14 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0)<br>(1,1,1)(2,0,0)(3,1,1)(4,1,0)(5,2,1) |
| 1,4,6,10,12,4,6 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(1,1,1)(2,1,0) |
| 1,4,6,10,12,4,6,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(1,1,1)(2,1,0)(1,1,1) |
| 1,4,6,10,12,4,6,6,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0)<br>(1,1,1)(2,1,0)(2,1,0)(1,1,1) |
| 1,4,6,10,12,4,6,9 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(1,1,1)(2,1,0)(3,2,0) |
| 1,4,6,10,12,4,6,10 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(1,1,1)(2,1,0)(3,2,1) |
| 1,4,6,10,12,4,6,10,12,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,1,0)(1,1,1) |
| 1,4,6,10,12,5 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(2,0,0) |
| 1,4,6,10,12,6 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(2,1,0) |
| 1,4,6,10,12,6,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(2,1,0)(1,1,1) |
| 1,4,6,10,12,6,4,6,10 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0)<br>(2,1,0)(1,1,1)(2,1,0)(3,2,1) |
| 1,4,6,10,12,6,5 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(2,1,0)(2,0,0) |
| 1,4,6,10,12,6,6,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(2,1,0)(2,1,0)(1,1,1) |
| 1,4,6,10,12,6,8,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(2,1,0)(3,1,0)(1,1,1) |
| 1,4,6,10,12,6,9 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(2,1,0)(3,2,0) |
| 1,4,6,10,12,6,10 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(2,1,0)(3,2,1) |
| 1,4,6,10,12,6,10,12,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0)<br>(2,1,0)(3,2,1)(4,1,0)(1,1,1) |
| 1,4,6,10,12,7 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(3,0,0) |
| 1,4,6,10,12,8,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,1,0)(1,1,1) |

| 0 − Y sequence | BMS |
|---|---|
| 1,4,6,10,12,8,10,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,1,0)(4,1,0)(1,1,1) |
| 1,4,6,10,12,8,11 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,1,0)(4,2,0) |
| 1,4,6,10,12,8,12 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,1,0)(4,2,1) |
| 1,4,6,10,12,9 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(3,2,0) |
| 1,4,6,10,12,9,14 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,2,0)(4,3,1) |
| 1,4,6,10,12,9,14,16,13 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,2,0)(4,3,1)(5,1,0)(4,3,0) |
| 1,4,6,10,12,10 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(3,2,1) |
| 1,4,6,10,12,11 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(4,0,0) |
| 1,4,6,10,12,12,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(4,1,0)(1,1,1) |
| 1,4,6,10,12,14,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(5,1,0)(1,1,1) |
| 1,4,6,10,12,15 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(5,2,0) |
| 1,4,6,10,12,16 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(5,2,1) |
| 1,4,6,10,12,16,18,4 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(5,2,1)(6,1,0)(1,1,1) |
| 1,4,6,10,13 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,2,0) |
| 1,4,6,10,13,9,14,18 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(3,2,0)(4,3,1)(5,3,0) |
| 1,4,6,10,13,10 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(3,2,1) |
| 1,4,6,10,13,16,10 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,2,0)(3,2,1) |
| 1,4,6,10,13,17 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,0) |
| 1,4,6,10,13,17,22 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,0)(6,4,0) |
| 1,4,6,10,13,18 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,1) |
| 1,4,6,10,13,18,22,28 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,1)(6,3,0)(7,4,1) |
| 1,4,7 | (0,0,0)(1,1,1)(2,1,1) |
| 1,4,7,3,7 | (0,0,0)(1,1,1)(2,1,1)(1,1,0)(2,2,1) |
| 1,4,7,3,7,9 | (0,0,0)(1,1,1)(2,1,1)<br>(1,1,0)(2,2,1)(3,1,0) |
| 1,4,7,3,7,10 | (0,0,0)(1,1,1)(2,1,1)<br>(1,1,0)(2,2,1)(3,2,0) |
| 1,4,7,3,7,10,15 | (0,0,0)(1,1,1)(2,1,1)(1,1,0)<br>(2,2,1)(3,2,0)(4,3,1) |
| 1,4,7,3,7,11 | (0,0,0)(1,1,1)(2,1,1)<br>(1,1,0)(2,2,1)(3,2,1) |
| 1,4,7,4 | (0,0,0)(1,1,1)(2,1,1)(1,1,1) |
| 1,4,7,4,6,10 | (0,0,0)(1,1,1)(2,1,1)<br>(1,1,1)(2,1,0)(3,2,1) |
| 1,4,7,4,6,10,14 | (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,4,6,10,14,9 | (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(3,2,0) |
| 1,4,7,4,6,10,14,10 | (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(3,2,1) |
| 1,4,7,4,7 | (0,0,0)(1,1,1)(2,1,1)(1,1,1)(2,1,1) |
| 1,4,7,4,7,4,7 | (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,1)(1,1,1)(2,1,1) |
| 1,4,7,5 | (0,0,0)(1,1,1)(2,1,1)(2,0,0) |
| 1,4,7,5,3,7,11,8 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)<br>(1,1,0)(2,2,1)(3,2,1)(3,0,0) |
| 1,4,7,5,4 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)(1,1,1) |
| 1,4,7,5,4,6,10 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)<br>(1,1,1)(2,1,0)(3,2,1) |
| 1,4,7,5,4,6,10,14 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1) |
| 1,4,7,5,4,6,10,14,11 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)(4,0,0) |
| 1,4,7,5,4,6,10,<br>14,11,6,10,14,11 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)(4,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(4,0,0) |
| 1,4,7,5,4,6,10,<br>14,11,8,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(4,0,0)(3,1,0)(4,2,1)(5,2,1)(5,0,0) |
| 1,4,7,5,4,6,10,14,11,9 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,0,0)(3,2,0) |
| 1,4,7,5,4,6,10,14,11,10 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,0,0)(3,2,1) |
| 1,4,7,5,4,6,10,14,<br>11,10,13,18,23,19 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,0,0)(3,2,1)<br>(4,2,0)(5,3,1)(6,3,1)(6,0,0) |
| 1,4,7,5,4,7 | (0,0,0)(1,1,1)(2,1,1)<br>(2,0,0)(1,1,1)(2,1,1) |
| 1,4,7,5,4,7,5 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)<br>(1,1,1)(2,1,1)(2,0,0) |
| 1,4,7,5,5 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)(2,0,0) |
| 1,4,7,6 | (0,0,0)(1,1,1)(2,1,1)(2,1,0) |
| 1,4,7,6,4 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1) |
| 1,4,7,6,4,6,10,14,13 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,2,0) |
| 1,4,7,6,4,6,10,14,13,10 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,2,0)(3,2,1) |
| 1,4,7,6,4,7 | (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1) |
| 1,4,7,6,5 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)(2,0,0) |
| 1,4,7,6,6,4,7 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(2,1,0)(1,1,1)(2,1,1) |
| 1,4,7,6,7 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)(3,0,0) |
| 1,4,7,6,8,4,7 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(3,1,0)(1,1,1)(2,1,1) |
| 1,4,7,6,8,10,4,7 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(3,1,0)(4,1,0)(1,1,1)(2,1,1) |
| 1,4,7,6,9 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)(3,2,0) |
| 1,4,7,6,9,13 | (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(3,2,0)(4,3,0) |
| 1,4,7,6,10 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)(3,2,1) |

| 0 − Y sequence | BMS |
|---|---|
| 1,4,7,6,10,13,10 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(3,2,1) |
| 1,4,7,6,10,13,18 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,1) |
| 1,4,7,6,10,14 | (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1) |
| 1,4,7,7 | (0,0,0)(1,1,1)(2,1,1)(2,1,1) |
| 1,4,7,7,4,7 | (0,0,0)(1,1,1)(2,1,1)<br>(2,1,1)(1,1,1)(2,1,1) |
| 1,4,7,7,4,7,5 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(1,1,1)(2,1,1)(2,0,0) |
| 1,4,7,7,4,7,6 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(1,1,1)(2,1,1)(2,1,0) |
| 1,4,7,7,4,7,6,4,7 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(1,1,1)(2,1,1)(2,1,0)(1,1,1)(2,1,1) |
| 1,4,7,7,4,7,6,10,14,14 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)(1,1,1)<br>(2,1,1)(2,1,0)(3,2,1)(4,2,1)(4,2,1) |
| 1,4,7,7,4,7,7 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(1,1,1)(2,1,1)(2,1,1) |
| 1,4,7,7,6,3,<br>7,11,11,9,7,11 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(2,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(3,2,1)(3,1,0)(2,2,1)(3,2,1) |
| 1,4,7,7,6,4 | (0,0,0)(1,1,1)(2,1,1)<br>(2,1,1)(2,1,0)(1,1,1) |
| 1,4,7,7,6,4,7 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1) |
| 1,4,7,7,6,4,7,6 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(2,1,0) |
| 1,4,7,7,6,4,<br>7,6,10,14,14 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(4,2,1) |
| 1,4,7,7,6,4,7,6,<br>10,14,14,12,16,20,20 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)(2,1,0)<br>(1,1,1)(2,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(4,2,1)(4,1,0)(5,2,1)(6,2,1)(6,2,1) |
| 1,4,7,7,6,4,7,7 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(2,1,1) |
| 1,4,7,7,6,8,4,7,7 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)(2,1,0)<br>(3,1,0)(1,1,1)(2,1,1)(2,1,1) |
| 1,4,7,7,6,9 | (0,0,0)(1,1,1)(2,1,1)<br>(2,1,1)(2,1,0)(3,2,0) |
| 1,4,7,7,6,10,14,14 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,2,1) |
| 1,4,7,7,7 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)(2,1,1) |
| 1,4,7,7,7,7 | (0,0,0)(1,1,1)(2,1,1)<br>(2,1,1)(2,1,1)(2,1,1) |
| 1,4,7,8 | (0,0,0)(1,1,1)(2,1,1)(3,0,0) |
| 1,4,7,8,10 | (0,0,0)(1,1,1)(2,1,1)(3,0,0)(4,1,0) |
| 1,4,7,8,10,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,0,0)<br>(4,1,0)(5,1,0)(6,1,0) |
| 1,4,7,8,10,13 | (0,0,0)(1,1,1)(2,1,1)<br>(3,0,0)(4,1,0)(5,2,0) |
| 1,4,7,8,11 | (0,0,0)(1,1,1)(2,1,1)(3,0,0)(4,1,1) |
| 1,4,7,8,11,14 | (0,0,0)(1,1,1)(2,1,1)<br>(3,0,0)(4,1,1)(5,1,1) |
| 1,4,7,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0) |
| 1,4,7,9,3 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,9,3,7 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1) |
| 1,4,7,9,3,7,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1) |
| 1,4,7,9,3,7,11,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,0,0) |
| 1,4,7,9,3,7,11,12,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,0,0)(5,1,1) |
| 1,4,7,9,3,7,11,12,15,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,0,0)(5,1,1)(6,1,1) |
| 1,4,7,9,3,7,<br>11,12,15,18,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,0,0)<br>(5,1,1)(6,1,1)(7,1,0) |
| 1,4,7,9,3,7,11,<br>12,15,18,20,6 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,0,0)<br>(5,1,1)(6,1,1)(7,1,0)(2,2,0) |
| 1,4,7,9,3,7,11,<br>12,15,18,20,10,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,0,0)<br>(5,1,1)(6,1,1)(7,1,0)(3,2,0)(4,3,0) |
| 1,4,7,9,3,7,11,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0) |
| 1,4,7,9,3,7,11,13,6 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)(2,2,0) |
| 1,4,7,9,3,7,11,13,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1) |
| 1,4,7,9,3,7,11,13,7,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)(3,2,1) |
| 1,4,7,9,3,7,11,<br>13,7,11,12,15,18,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)(3,2,1)<br>(4,0,0)(5,1,1)(6,1,1)(7,1,0) |
| 1,4,7,9,3,7,11,13,<br>7,11,12,15,18,20,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)(3,2,1)<br>(4,0,0)(5,1,1)(6,1,1)(7,1,0)(3,0,0) |
| 1,4,7,9,3,7,11,13,<br>7,11,12,15,18,20,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)(3,2,1)<br>(4,0,0)(5,1,1)(6,1,1)(7,1,0)(3,1,0) |
| 1,4,7,9,3,7,<br>11,13,7,11,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,1)(3,2,1)(4,1,0) |
| 1,4,7,9,3,7,11,13,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,0,0) |
| 1,4,7,9,3,7,11,13,10,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(2,2,1) |
| 1,4,7,9,3,7,11,<br>13,10,7,11,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(2,2,1)(3,2,1)(4,1,0) |
| 1,4,7,9,3,7,11,<br>13,13,10,7,11,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(4,2,0)(2,2,1)(3,2,1)(4,1,0) |
| 1,4,7,9,3,7,11,13,10,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,0) |
| 1,4,7,9,3,7,11,13,10,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,1) |
| 1,4,7,9,3,7,<br>11,13,10,15,20,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,0)(4,3,1)(5,3,1)(6,1,0) |

| 0 − Y sequence | BMS |
|---|---|
| 1,4,7,9,3,7,11,13,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,1) |
| 1,4,7,9,3,7,11,13,11,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,1)(4,1,0) |
| 1,4,7,9,3,7,11,13,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(4,1,0) |
| 1,4,7,9,3,7,11,13,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(5,1,0) |
| 1,4,7,9,3,7,11,13,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(5,2,0) |
| 1,4,7,9,3,7,11,13,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(5,2,1) |
| 1,4,7,9,3,7,11,13,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(5,2,1)(6,2,1) |
| 1,4,7,9,3,7,<br>11,13,17,21,23 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,1,0) |
| 1,4,7,9,3,7,11,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,0) |
| 1,4,7,9,3,7,11,<br>14,3,7,11,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0) |
| 1,4,7,9,3,7,11,14,<br>6,11,16,19,11,16,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,0)<br>(2,2,0)(3,3,1)(4,3,1)(5,2,0)<br>(3,3,1)(4,3,1)(5,1,0) |
| 1,4,7,9,3,7,11,14,6,11,<br>16,19,11,16,18,22,26,29 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(2,2,0)(3,3,1)<br>(4,3,1)(5,2,0)(3,3,1)(4,3,1)<br>(5,1,0)(6,2,1)(7,2,1)(8,2,0) |
| 1,4,7,9,3,7,11,<br>14,6,11,16,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,0)<br>(2,2,0)(3,3,1)(4,3,1)(5,3,0) |
| 1,4,7,9,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1) |
| 1,4,7,9,4,6,10,14,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0) |
| 1,4,7,9,4,7 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1) |
| 1,4,7,9,4,7,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0) |
| 1,4,7,9,4,7,9,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(1,1,1) |
| 1,4,7,9,4,7,<br>9,4,7,9,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(1,1,1) |
| 1,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,9,5,3 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,0) |
| 1,4,7,9,5,4 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1) |
| 1,4,7,9,5,4,7,9,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(1,1,1) |
| 1,4,7,9,5,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,9,5,5 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(2,0,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,9,6 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0) |
| 1,4,7,9,6,4 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(1,1,1) |
| 1,4,7,9,6,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,9,6,4,7,9,5,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0)(2,0,0) |
| 1,4,7,9,6,4,7,9,6 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0) |
| 1,4,7,9,6,5 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(2,0,0) |
| 1,4,7,9,6,7 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,0,0) |
| 1,4,7,9,6,8,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,9,6,9 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,0) |
| 1,4,7,9,6,10 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1) |
| 1,4,7,9,6,10,14,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0) |
| 1,4,7,9,6,10,<br>14,16,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,9,6,10,14,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0) |
| 1,4,7,9,6,10,14,17,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(3,2,1) |
| 1,4,7,9,6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,9,6,10,14,<br>17,13,18,23,27,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,0)(6,0,0) |
| 1,4,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,9,7,4 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(1,1,1) |
| 1,4,7,9,7,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,9,7,4,7,9,5,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0)(2,0,0) |
| 1,4,7,9,7,4,7,9,6 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0) |
| 1,4,7,9,7,4,7,<br>9,6,10,14,17,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1) |
| 1,4,7,9,7,4,7,<br>9,6,10,14,17,14,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(3,2,0) |
| 1,4,7,9,7,4,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,9,7,5 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(2,0,0) |
| 1,4,7,9,7,6,4,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) |

| 0 − Y sequence | BMS |
|---|---|
| 1,4,7,9,7,<br>6,8,4,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,0)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,9,7,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,0)(3,2,0) |
| 1,4,7,9,7,6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,9,7,6,10,14,17,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1) |
| 1,4,7,9,7,7 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(2,1,1) |
| 1,4,7,9,7,7,<br>6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,9,7,7,6,<br>10,14,17,14,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(4,2,1) |
| 1,4,7,9,7,7,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,1)(2,1,1) |
| 1,4,7,9,7,8 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,0,0) |
| 1,4,7,9,7,9 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0) |
| 1,4,7,9,7,9,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(1,1,1) |
| 1,4,7,9,7,9,4,7,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0) |
| 1,4,7,9,7,9,4,7,9,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1) |
| 1,4,7,9,7,9,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,9,7,9,4,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,9,7,9,<br>4,7,9,7,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,0,0) |
| 1,4,7,9,7,9,<br>4,7,9,7,9,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1) |
| 1,4,7,9,7,9,4,<br>7,9,7,9,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,9,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,9,7,9,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,0) |
| 1,4,7,9,7,9,<br>6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,9,7,9,6,<br>10,14,17,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,9,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,9,7,9,7,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1)(3,0,0) |
| 1,4,7,9,7,9,7,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0) |
| 1,4,7,9,7,9,7,9,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1) |
| 1,4,7,9,7,9,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,9,7,9,<br>7,9,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,9,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,0,0) |
| 1,4,7,9,9,4,7,9,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(3,0,0) |
| 1,4,7,9,9,5 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,9,9,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(2,1,0)(3,2,0) |
| 1,4,7,9,9,6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,9,9,6,<br>10,14,17,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(5,2,0)(4,0,0) |
| 1,4,7,9,9,7 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(2,1,1) |
| 1,4,7,9,9,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,9,9,7,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,1,0)<br>(2,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,9,9,8 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(3,0,0) |
| 1,4,7,9,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,9,9,<br>9,7,9,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,0)<br>(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,9,9,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,9,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,0,0) |
| 1,4,7,9,11,5 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(2,0,0) |
| 1,4,7,9,11,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(2,1,0)(3,2,0) |
| 1,4,7,9,11,7 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(2,1,1) |
| 1,4,7,9,11,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,9,11,7,9,11,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,1,0)<br>(2,1,1)(3,1,0)(4,1,0)(2,0,0) |
| 1,4,7,9,11,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(3,1,0)(2,0,0) |
| 1,4,7,9,11,9,11,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(3,1,0)(4,1,0)(2,0,0) |
| 1,4,7,9,11,10 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(4,0,0) |

| 0 − Y sequence | BMS |
|---|---|
| 1,4,7,9,11,11,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(4,1,0)(2,0,0) |
| 1,4,7,9,11,11,9,11,11,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,1,0)<br>(4,1,0)(3,1,0)(4,1,0)(4,1,0)(2,0,0) |
| 1,4,7,9,11,11,11,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(4,1,0)(4,1,0)(2,0,0) |
| 1,4,7,9,11,12 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(5,0,0) |
| 1,4,7,9,11,13,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(5,1,0)(2,0,0) |
| 1,4,7,9,11,13,13,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(5,1,0)(5,1,0)(2,0,0) |
| 1,4,7,9,11,13,15,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(5,1,0)(6,1,0)(2,0,0) |
| 1,4,7,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,4,7,9,12,6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,9,12,6,10,14,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(3,1,0)(4,2,0) |
| 1,4,7,9,12,12 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(4,2,0) |
| 1,4,7,9,12,15,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,2,0)(6,2,0) |
| 1,4,7,9,12,16 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(5,3,0) |
| 1,4,7,9,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1) |
| 1,4,7,9,13,<br>17,19,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,9,13,17,19,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(2,0,0) |
| 1,4,7,9,13,17,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0) |
| 1,4,7,9,13,17,20,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(4,2,1) |
| 1,4,7,9,13,17,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) |
| 1,4,7,9,13,17,20,<br>14,6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,9,13,17,<br>20,25,30,34,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,1)<br>(8,3,1)(9,3,0)(8,0,0) |
| 1,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,4,7,9,13,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1) |
| 1,4,7,10,4,7,9,13,17,<br>21,13,17,20,25,30,35 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(4,2,1)(5,2,1)(6,2,0)<br>(7,3,1)(8,3,1)(9,3,1) |
| 1,4,7,10,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,0,0) |
| 1,4,7,10,6,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,1) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,6,9 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,0)(3,2,0) |
| 1,4,7,10,6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,6,10,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,4,7,10,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1) |
| 1,4,7,10,7,6,10,14,18,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(4,2,1) |
| 1,4,7,10,7,7 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(2,1,1) |
| 1,4,7,10,7,8 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,0,0) |
| 1,4,7,10,7,9 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0) |
| 1,4,7,10,7,9,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1) |
| 1,4,7,10,7,9,4,7,<br>9,13,17,21,17,19,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(2,0,0) |
| 1,4,7,10,7,9,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,7,9,5,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1) |
| 1,4,7,10,7,<br>9,5,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,7,9,<br>5,4,7,9,6,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1) |
| 1,4,7,10,7,9,5,<br>4,7,9,6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,7,9,5,4,<br>7,9,6,10,14,17,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1) |
| 1,4,7,10,7,9,5,4,<br>7,9,6,10,14,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(6,3,0) |
| 1,4,7,10,7,9,5,4,<br>7,9,6,10,14,17,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(6,3,1) |
| 1,4,7,10,7,9,5,4,<br>7,9,6,10,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,4,7,10,7,9,5,4,<br>7,9,6,10,14,18,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(4,2,1) |
| 1,4,7,10,7,9,<br>5,4,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,7,9,5,<br>4,7,9,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,7,9,<br>5,4,7,9,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,0)(3,1,0)(2,1,1) |
| 1,4,7,10,7,9,<br>5,4,7,9,11,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,0)(4,1,0)(2,1,1) |
| 1,4,7,10,7,9,<br>5,4,7,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,7,9,5,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,7,9,<br>5,4,7,10,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,1)(2,1,1) |
| 1,4,7,10,7,9,<br>5,4,7,10,7,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0) |
| 1,4,7,10,7,9,5,<br>4,7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,7,9,6 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0) |
| 1,4,7,10,7,9,<br>6,4,7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,7,9,6,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(2,0,0) |
| 1,4,7,10,7,9,6,6 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(2,1,0) |
| 1,4,7,10,7,9,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,0) |
| 1,4,7,10,7,9,6,<br>10,14,18,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,7,9,7,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(2,1,1) |
| 1,4,7,10,7,9,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,7,9,<br>7,9,6,10,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,4,7,10,7,9,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,7,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,10,7,9,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(3,1,0)(2,1,1) |
| 1,4,7,10,7,9,9,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,7,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,7,9,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,7,9,13,17,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,0)(5,0,0) |
| 1,4,7,10,7,9,13,17,20,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,0)(5,2,1) |
| 1,4,7,10,7,9,13,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) |
| 1,4,7,10,7,9,13,17,21,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1) |
| 1,4,7,10,7,9,<br>13,17,21,17,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0) |
| 1,4,7,10,7,9,13,<br>17,21,17,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(5,0,0) |
| 1,4,7,10,7,10 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,7,10,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1) |
| 1,4,7,10,7,10,7,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0) |
| 1,4,7,10,7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,7,10,7,9,6,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,0)(2,0,0) |
| 1,4,7,10,7,10,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,7,<br>10,7,9,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,7,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,7,10,<br>7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,7,10,7,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,8,4 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(1,1,1) |
| 1,4,7,10,8,4,7,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0) |
| 1,4,7,10,8,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,8,4,7,<br>9,5,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,8,4,7,<br>9,6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,8,4,7,<br>9,6,10,14,17,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1) |
| 1,4,7,10,8,4,7,<br>9,6,10,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,8,4,7,<br>9,6,10,14,18,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(4,2,1)(5,2,1) |
| 1,4,7,10,8,4,7,<br>9,6,10,14,18,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,0,0) |
| 1,4,7,10,8,4,7,<br>9,6,10,14,18,15,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0)(2,1,0)(3,2,0) |
| 1,4,7,10,8,4,7,<br>9,6,10,14,18,15,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0)(3,1,0) |
| 1,4,7,10,8,4,7,9,<br>6,10,14,18,15,8,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0)(3,1,0)(4,2,0) |
| 1,4,7,10,8,4,7,9,<br>6,10,14,18,15,8,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0)(3,1,0)(4,2,1) |
| 1,4,7,10,8,4,7,9,<br>6,10,14,18,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0)(3,2,0) |
| 1,4,7,10,8,4,7,9,<br>6,10,14,18,15,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0)(3,2,1) |
| 1,4,7,10,8,4,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,8,<br>4,7,9,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,8,4,7,9,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(3,1,0)(2,1,1) |
| 1,4,7,10,8,4,7,9,11,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,1,0)(2,1,1) |
| 1,4,7,10,8,4,7,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,8,4,7,9,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1) |
| 1,4,7,10,8,4,<br>7,9,13,17,21,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,0,0) |
| 1,4,7,10,8,4,7,<br>9,13,17,21,18,6,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,0,0)(2,1,0)(3,2,1) |
| 1,4,7,10,8,4,7,<br>9,13,17,21,18,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0)(2,1,1) |
| 1,4,7,10,8,4,7,<br>9,13,17,21,18,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0)(3,1,0) |
| 1,4,7,10,8,4,7,9,<br>13,17,21,18,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,0,0)(3,1,0)(4,2,0) |
| 1,4,7,10,8,4,7,<br>9,13,17,21,18,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0)(4,1,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,8,4,7,9,<br>13,17,21,18,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0)(4,2,0) |
| 1,4,7,10,8,4,7,9,<br>13,17,21,18,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0)(4,2,1) |
| 1,4,7,10,8,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,8,4,7,10,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(2,1,1) |
| 1,4,7,10,8,4,7,10,7,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,0) |
| 1,4,7,10,8,4,<br>7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,8,4,<br>7,10,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,8,4,<br>7,10,7,9,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,8,4,<br>7,10,7,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,8,4,7,<br>10,7,9,13,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) |
| 1,4,7,10,8,4,7,10,<br>7,9,13,17,21,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,0,0) |
| 1,4,7,10,8,4,7,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,8,4,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,8,5 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(2,0,0) |
| 1,4,7,10,8,6,4,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,8,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,0)(3,2,0) |
| 1,4,7,10,8,6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,8,6,10,14,18,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,0,0) |
| 1,4,7,10,8,7 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(2,1,1) |
| 1,4,7,10,8,7,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(2,1,0)(3,2,0) |
| 1,4,7,10,8,7,<br>6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,8,7,6,10,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1) | |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,8,7,6,<br>10,14,18,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,8,7,6,<br>10,14,18,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,1) |
| 1,4,7,10,8,7,<br>6,10,14,18,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0) |
| 1,4,7,10,8,7,6,<br>10,14,18,15,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0)(4,2,1) |
| 1,4,7,10,8,7,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(2,1,1) |
| 1,4,7,10,8,7,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,0,0) |
| 1,4,7,10,8,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,8,7,9,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,0) |
| 1,4,7,10,8,7,9,<br>6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,8,7,<br>9,6,10,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1) |
| 1,4,7,10,8,7,9,<br>6,10,14,18,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,0,0) |
| 1,4,7,10,8,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,8,7,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,8,7,<br>9,13,17,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) |
| 1,4,7,10,8,<br>7,9,13,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1) |
| 1,4,7,10,8,7,<br>9,13,17,21,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0) |
| 1,4,7,10,8,7,9,13,<br>17,21,18,17,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,0,0)(5,2,1)(6,2,0)(5,0,0) |
| 1,4,7,10,8,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,1) |
| 1,4,7,10,8,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,8,8 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(3,0,0) |
| 1,4,7,10,8,9 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(4,0,0) |
| 1,4,7,10,8,10 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(4,1,0) |
| 1,4,7,10,8,11,14,17,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(4,1,1)(5,1,1)(6,1,1)(6,0,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0) |
| 1,4,7,10,9,2 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(1,0,0) |
| 1,4,7,10,9,3 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(1,1,0) |
| 1,4,7,10,9,3,7,11,<br>15,13,7,11,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,1)(4,1,0)<br>(2,2,1)(3,2,1)(4,2,1)(4,0,0) |
| 1,4,7,10,9,3,7,11,<br>15,13,7,11,15,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(4,1,0)(2,2,1)(3,2,1)(4,2,1)(4,1,0) |
| 1,4,7,10,9,3,<br>7,11,15,13,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(4,2,1)(4,1,0)(3,0,0) |
| 1,4,7,10,9,3,<br>7,11,15,13,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(4,2,1)(4,1,0)(3,1,0) |
| 1,4,7,10,9,3,<br>7,11,15,13,9,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(4,2,1)(4,1,0)(3,1,0)(3,0,0) |
| 1,4,7,10,9,3,7,<br>11,15,13,9,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(4,2,1)(4,1,0)(3,1,0)(4,0,0) |
| 1,4,7,10,9,3,7,<br>11,15,13,9,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(4,2,1)(4,1,0)(3,1,0)(4,1,0) |
| 1,4,7,10,9,3,7,11,<br>15,13,9,13,17,21,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,1)(4,1,0)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,0,0) |
| 1,4,7,10,9,3,7,11,<br>15,13,10,7,11,15,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,1)(4,1,0)<br>(3,2,0)(2,2,1)(3,2,1)(4,2,1)(4,1,0) |
| 1,4,7,10,9,<br>3,7,11,15,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,1)(4,2,0) |
| 1,4,7,10,9,4 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(1,1,1) |
| 1,4,7,10,9,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,4,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,9,4,7,10,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0) |
| 1,4,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,5,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,0,0)(1,1,1) |
| 1,4,7,10,9,5,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,0,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,5,4,7,10,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0) |
| 1,4,7,10,9,5,4,<br>7,10,9,4,7,10,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,9,<br>5,4,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,5,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,0,0)(2,0,0) |
| 1,4,7,10,9,6 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,0) |
| 1,4,7,10,9,6,4,7,10,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0) |
| 1,4,7,10,9,6,<br>4,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,6,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,1,0) |
| 1,4,7,10,9,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,0) |
| 1,4,7,10,9,6,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1) |
| 1,4,7,10,9,6,<br>10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,9,6,10,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,4,7,10,9,6,10,<br>14,18,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,9,6,<br>10,14,18,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,1) |
| 1,4,7,10,9,<br>6,10,14,18,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,0,0) |
| 1,4,7,10,9,6,10,<br>14,18,16,4,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,6,<br>10,14,18,16,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(2,0,0) |
| 1,4,7,10,9,6,<br>10,14,18,16,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(3,2,0) |
| 1,4,7,10,9,6,<br>10,14,18,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,0) |
| 1,4,7,10,9,6,<br>10,14,18,16,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,1) |
| 1,4,7,10,9,6,10,<br>14,18,16,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,9,6,10,<br>14,18,16,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,1)(5,2,1) |
| 1,4,7,10,9,6,10,<br>14,18,16,14,18,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,2,1)(5,0,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,9,6,<br>10,14,18,16,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(6,2,0) |
| 1,4,7,10,9,6,<br>10,14,18,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,2,0) |
| 1,4,7,10,9,6,<br>10,14,18,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,9,7 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,7,<br>6,10,14,18,17,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(4,2,1) |
| 1,4,7,10,9,7,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(2,1,1) |
| 1,4,7,10,9,7,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,0,0) |
| 1,4,7,10,9,7,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0) |
| 1,4,7,10,9,7,9,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1) |
| 1,4,7,10,9,7,<br>9,4,7,10,9,7,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,0) |
| 1,4,7,10,9,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,7,9,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(3,1,0)(2,1,1) |
| 1,4,7,10,9,7,9,11,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,1,0)(2,1,1) |
| 1,4,7,10,9,7,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,9,7,9,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1) |
| 1,4,7,10,9,7,9,13,17,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,0) |
| 1,4,7,10,9,7,<br>9,13,17,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) |
| 1,4,7,10,9,7,<br>9,13,17,20,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1) |
| 1,4,7,10,9,7,9,13,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1) |
| 1,4,7,10,9,7,9,<br>13,17,21,17,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(5,0,0) |
| 1,4,7,10,9,7,<br>9,13,17,21,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,2,1) |
| 1,4,7,10,9,7,<br>9,13,17,21,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0) |

| 0 − Y sequence | BMS |
|---|---|
| 1,4,7,10,9,7,9,<br>13,17,21,19,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,1,0)(2,0,0) |
| 1,4,7,10,9,7,<br>9,13,17,21,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0) |
| 1,4,7,10,9,7,<br>9,13,17,21,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(5,0,0) |
| 1,4,7,10,9,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1) |
| 1,4,7,10,9,7,10,<br>6,10,14,18,17,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,2,1)(5,2,1) |
| 1,4,7,10,9,7,10,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1) |
| 1,4,7,10,9,7,10,7,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0) |
| 1,4,7,10,9,7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,7,10,7,9,6 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,0) |
| 1,4,7,10,9,7,10,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,7,10,7,9,7,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0) |
| 1,4,7,10,9,7,<br>10,7,9,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,7,10,7,9,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(3,0,0) |
| 1,4,7,10,9,7,<br>10,7,9,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(2,1,1) |
| 1,4,7,10,9,7,<br>10,7,9,11,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(2,1,1) |
| 1,4,7,10,9,7,10,7,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,9,7,10,7,<br>9,13,17,21,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(5,0,0) |
| 1,4,7,10,9,7,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,7,<br>10,7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,9,7,10,8,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0)(2,0,0) |
| 1,4,7,10,9,7,10,8,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0)(2,1,1) |
| 1,4,7,10,9,7,10,8,7,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0)(2,1,1)(3,1,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,9,7,<br>10,8,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,7,10,8,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0)(2,1,1)(3,1,1) |
| 1,4,7,10,9,7,10,<br>8,7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,7,<br>10,8,7,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,7,<br>10,8,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,9,7,10,8,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0)(3,0,0) |
| 1,4,7,10,9,7,10,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0) |
| 1,4,7,10,9,7,10,9,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(1,1,1) |
| 1,4,7,10,9,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,<br>7,10,9,7,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,9,7,10,9,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,1) |
| 1,4,7,10,9,7,<br>10,9,7,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,7,<br>10,9,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,9,7,<br>10,9,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,7,10,<br>9,7,10,9,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,8 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,0,0) |
| 1,4,7,10,9,8,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1) |
| 1,4,7,10,9,8,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,8,4,7,<br>9,6,10,14,18,17,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,0)(5,0,0) |
| 1,4,7,10,9,8,4,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,8,4,<br>7,9,13,17,21,20,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(6,0,0) |
| 1,4,7,10,9,8,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,9,8,<br>4,7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,8,<br>4,7,10,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,8,<br>4,7,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,8,4,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,9,8,4,7,10,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0) |
| 1,4,7,10,9,8,<br>4,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,8,<br>4,7,10,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,8,<br>4,7,10,9,7,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,0) |
| 1,4,7,10,9,8,<br>4,7,10,9,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1) |
| 1,4,7,10,9,8,4,<br>7,10,9,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,9,8,4,7,<br>10,9,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,<br>8,4,7,10,9,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,0,0) |
| 1,4,7,10,9,8,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,0,0) |
| 1,4,7,10,9,8,6 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,0) |
| 1,4,7,10,9,8,<br>6,4,7,10,9,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(3,0,0) |
| 1,4,7,10,9,8,<br>6,10,14,18,17,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(5,0,0) |
| 1,4,7,10,9,8,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,1) |
| 1,4,7,10,9,8,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,8,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,1)(3,1,1) |
| 1,4,7,10,9,8,7,10,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,1)(2,1,1) |
| 1,4,7,10,9,8,<br>7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,9,8,7,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,8,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,9,8,7,10,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,1)(3,1,0) |
| 1,4,7,10,9,8,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,8,7,10,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,8,<br>7,10,9,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1) |
| 1,4,7,10,9,8,<br>7,10,9,7,10,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0) |
| 1,4,7,10,9,8,7,<br>10,9,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,8,7,10,9,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,1)(3,1,0)(3,0,0) |
| 1,4,7,10,9,8,8 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,0,0)(3,0,0) |
| 1,4,7,10,9,9 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,1,0) |
| 1,4,7,10,9,9,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(1,1,1) |
| 1,4,7,10,9,9,<br>4,7,10,9,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,1,0) |
| 1,4,7,10,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,10,9,9,<br>5,4,7,10,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,10,9,9,6 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,0) |
| 1,4,7,10,9,9,<br>6,4,7,10,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,10,9,9,6,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,0)(3,2,1) |
| 1,4,7,10,9,9,6,10,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,4,7,10,9,9,6,<br>10,14,18,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,9,9,6,<br>10,14,18,17,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(5,2,0)(4,0,0) |
| 1,4,7,10,9,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1) |
| 1,4,7,10,9,9,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,9,7,9,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1) |

| 0 − Y sequence | BMS |
|---|---|
| 1,4,7,10,9,9,<br>7,9,13,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1) |
| 1,4,7,10,9,9,7,<br>9,13,17,21,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)(5,0,0) |
| 1,4,7,10,9,9,7,9,<br>13,17,21,20,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(6,2,0)(5,0,0) |
| 1,4,7,10,9,9,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1) |
| 1,4,7,10,9,9,7,10,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1) |
| 1,4,7,10,9,9,<br>7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,9,7,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,9,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,9,<br>9,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,<br>9,7,10,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,9,7,<br>10,9,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,9,7,10,9,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(3,0,0) |
| 1,4,7,10,9,<br>9,7,10,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,10,9,9,7,<br>10,9,9,7,10,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,10,9,9,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(3,0,0) |
| 1,4,7,10,9,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,10,9,10 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,0,0) |
| 1,4,7,10,9,10,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1) |
| 1,4,7,10,9,10,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,10,4,7,<br>9,6,10,14,18,17,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,0)(6,0,0) |
| 1,4,7,10,9,10,4,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,10,4,7,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,9,10,4,<br>7,9,13,17,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,9,10,<br>4,7,9,13,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) |
| 1,4,7,10,9,10,4,<br>7,9,13,17,21,20,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)(7,0,0) |
| 1,4,7,10,9,10,4,7,<br>9,13,17,21,20,21,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(7,0,0)(4,2,0) |
| 1,4,7,10,9,10,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,10,4,<br>7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,10,<br>4,7,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,10,4,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,9,10,<br>4,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,10,<br>4,7,10,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,10,<br>4,7,10,9,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1) |
| 1,4,7,10,9,10,<br>4,7,10,9,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,10,<br>4,7,10,9,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,0,0) |
| 1,4,7,10,9,10,<br>4,7,10,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,10,9,<br>10,4,7,10,9,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,0,0) |
| 1,4,7,10,9,10,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,0,0) |
| 1,4,7,10,9,10,<br>6,4,7,10,9,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,0,0) |
| 1,4,7,10,9,10,6,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,0)(3,2,1) |
| 1,4,7,10,9,10,6,10,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,4,7,10,9,10,6,<br>10,14,18,17,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(6,0,0) |
| 1,4,7,10,9,10,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,1) |
| 1,4,7,10,9,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,1)(3,1,0)(2,0,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,9,10,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,10,7,9,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,0)(4,2,1) |
| 1,4,7,10,9,10,<br>7,9,13,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1) |
| 1,4,7,10,9,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,1)(3,1,1) |
| 1,4,7,10,9,10,7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,10,7,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,10,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,9,10,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,10,7,10,9,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,1)(3,1,0)(4,0,0) |
| 1,4,7,10,9,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(3,0,0) |
| 1,4,7,10,9,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(3,1,0)(2,0,0) |
| 1,4,7,10,9,10,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(3,1,0)(2,1,1) |
| 1,4,7,10,9,10,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,10,9,10,9,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(3,1,0)(4,0,0) |
| 1,4,7,10,9,10,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(4,0,0) |
| 1,4,7,10,9,11 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,1,0) |
| 1,4,7,10,9,11,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,0,0) |
| 1,4,7,10,9,11,5,<br>4,7,10,9,11,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,1,0)(2,0,0) |
| 1,4,7,10,9,11,6 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,1,0) |
| 1,4,7,10,9,11,6,<br>10,14,18,17,20,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,2,0)(4,0,0) |
| 1,4,7,10,9,11,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,1,1) |
| 1,4,7,10,9,11,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,11,7,9,<br>13,17,21,20,23,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(7,2,0)(5,0,0) |
| 1,4,7,10,9,11,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,1,1)(3,1,1) |
| 1,4,7,10,9,11,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,1)(3,0,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,9,11,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,11,7,10,9,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,1)(3,1,0)(4,0,0) |
| 1,4,7,10,9,11,<br>7,10,9,11,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,0,0) |
| 1,4,7,10,9,11,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(3,0,0) |
| 1,4,7,10,9,11,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(3,1,0)(2,0,0) |
| 1,4,7,10,9,11,9,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(3,1,0)(4,0,0) |
| 1,4,7,10,9,11,9,11,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(3,1,0)(4,1,0)(2,0,0) |
| 1,4,7,10,9,11,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(4,0,0) |
| 1,4,7,10,9,11,11,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(4,1,0)(2,0,0) |
| 1,4,7,10,9,11,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(5,0,0) |
| 1,4,7,10,9,11,13,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(5,1,0)(2,0,0) |
| 1,4,7,10,9,11,13,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(5,1,0)(6,0,0) |
| 1,4,7,10,9,11,13,15,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(5,1,0)(6,1,0)(2,0,0) |
| 1,4,7,10,9,12 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,9,12,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1) |
| 1,4,7,10,9,12,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,4,<br>7,9,6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,9,12,<br>4,7,9,6,10,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,4,7,10,9,12,4,<br>7,9,6,10,14,18,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(4,2,1) |
| 1,4,7,10,9,12,4,7,<br>9,6,10,14,18,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(4,2,1)(5,2,1) |
| 1,4,7,10,9,12,4,7,<br>9,6,10,14,18,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,0,0) |
| 1,4,7,10,9,12,4,7,<br>9,6,10,14,18,15,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,0,0)(4,2,1) |
| 1,4,7,10,9,12,4,7,<br>9,6,10,14,18,15,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,0,0)(5,0,0) |

| 0 − Y sequence | BMS |
|---|---|
| 1,4,7,10,9,12,4,7,<br>9,6,10,14,18,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,9,12,4,7,<br>9,6,10,14,18,17,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,0)(5,0,0) |
| 1,4,7,10,9,12,4,7,<br>9,6,10,14,18,17,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,0)(6,0,0) |
| 1,4,7,10,9,12,4,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,<br>12,4,7,9,7,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(2,1,1) |
| 1,4,7,10,9,<br>12,4,7,9,7,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0) |
| 1,4,7,10,9,12,<br>4,7,9,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,4,7,9,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(3,0,0) |
| 1,4,7,10,9,12,4,7,9,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(3,1,0) |
| 1,4,7,10,9,12,<br>4,7,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,4,7,9,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,0,0) |
| 1,4,7,10,9,12,<br>4,7,9,11,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(2,0,0) |
| 1,4,7,10,9,12,4,7,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,9,12,4,7,9,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1) |
| 1,4,7,10,9,12,4,<br>7,9,13,17,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) |
| 1,4,7,10,9,12,<br>4,7,9,13,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) |
| 1,4,7,10,9,12,4,<br>7,9,13,17,21,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1) |
| 1,4,7,10,9,12,4,7,<br>9,13,17,21,17,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,2,0)(5,0,0) |
| 1,4,7,10,9,12,4,7,<br>9,13,17,21,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(5,0,0) |
| 1,4,7,10,9,12,4,7,<br>9,13,17,21,20,24 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(7,3,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,9,12,4,7,<br>9,13,17,21,20,24,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(7,3,0)(3,0,0) |
| 1,4,7,10,9,12,4,7,<br>9,13,17,21,20,24,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(7,3,0)(4,0,0) |
| 1,4,7,10,9,12,4,7,<br>9,13,17,21,20,24,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(7,3,0)(4,2,0) |
| 1,4,7,10,9,12,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,12,<br>4,7,10,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,<br>4,7,10,4,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,12,<br>4,7,10,4,7,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,9,12,<br>4,7,10,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,12,4,7,10,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(2,0,0) |
| 1,4,7,10,9,<br>12,4,7,10,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,0)(3,2,0) |
| 1,4,7,10,9,12,4,7,<br>10,6,10,14,18,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,0) |
| 1,4,7,10,9,12,4,7,10,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1) |
| 1,4,7,10,9,12,<br>4,7,10,7,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,<br>4,7,10,7,4,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,12,<br>4,7,10,7,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(1,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,12,4,<br>7,10,7,4,7,10,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(1,1,1)(2,1,1)(3,1,1)(2,1,1) |
| 1,4,7,10,9,12,<br>4,7,10,7,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(2,1,1) |
| 1,4,7,10,9,12,<br>4,7,10,7,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,0,0) |
| 1,4,7,10,9,12,<br>4,7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,9,12,4,<br>7,10,7,9,5,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,12,4,<br>7,10,7,9,5,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(2,0,0) |
| 1,4,7,10,9,12,<br>4,7,10,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,12,4,<br>7,10,7,9,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,4,<br>7,10,7,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,4,<br>7,10,7,9,11,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,1,0)(2,0,0) |
| 1,4,7,10,9,12,<br>4,7,10,7,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,9,12,<br>4,7,10,7,9,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1) |
| 1,4,7,10,9,12,<br>4,7,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,12,4,<br>7,10,7,10,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,4,7,<br>10,7,10,4,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,12,4,<br>7,10,7,10,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(1,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,12,<br>4,7,10,7,10,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(2,1,1) |
| 1,4,7,10,9,12,4,<br>7,10,7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,4,<br>7,10,7,10,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,12,4,<br>7,10,7,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,12,4,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,9,12,<br>4,7,10,8,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,9,12,4,<br>7,10,8,4,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,12,<br>4,7,10,8,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,12,4,<br>7,10,8,4,7,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,12,4,<br>7,10,8,4,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,9,12,<br>4,7,10,8,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(2,1,0)(3,2,0) |
| 1,4,7,10,9,12,4,7,<br>10,8,6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,9,12,4,7,<br>10,8,6,10,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,4,7,10,9,12,<br>4,7,10,8,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(2,1,1) |
| 1,4,7,10,9,12,4,<br>7,10,8,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,<br>4,7,10,8,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(2,1,1)(3,1,1) |
| 1,4,7,10,9,12,<br>4,7,10,8,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(3,0,0) |
| 1,4,7,10,9,12,4,7,10,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0) |
| 1,4,7,10,9,12,<br>4,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,<br>4,7,10,9,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,0)(3,2,0) |
| 1,4,7,10,9,12,<br>4,7,10,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,12,4,<br>7,10,9,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,4,<br>7,10,9,7,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,9,12,<br>4,7,10,9,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1) |
| 1,4,7,10,9,12,4,<br>7,10,9,7,10,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,0)(3,2,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,9,12,4,<br>7,10,9,7,10,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1)(2,1,1) |
| 1,4,7,10,9,12,4,<br>7,10,9,7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,4,<br>7,10,9,7,10,7,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,4,7,10,9,12,4,7,<br>10,9,7,10,7,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,9,12,4,<br>7,10,9,7,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,9,12,4,<br>7,10,9,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,9,12,4,<br>7,10,9,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,<br>4,7,10,9,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,0,0) |
| 1,4,7,10,9,12,<br>4,7,10,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,<br>4,7,10,9,9,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(2,1,1) |
| 1,4,7,10,9,12,<br>4,7,10,9,9,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(3,0,0) |
| 1,4,7,10,9,12,<br>4,7,10,9,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,<br>4,7,10,9,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,0,0) |
| 1,4,7,10,9,12,<br>4,7,10,9,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,1,0) |
| 1,4,7,10,9,12,<br>4,7,10,9,11,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,1,0)(2,0,0) |
| 1,4,7,10,9,12,<br>4,7,10,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,9,12,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,0) |
| 1,4,7,10,9,12,<br>6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,9,12,<br>6,10,14,18,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(4,0,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,9,12,6,<br>10,14,18,17,20,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1<br>(4,2,1)(5,2,1)(5,2,0)(6,2,0)(4,0,0) |
| 1,4,7,10,9,12,6,<br>10,14,18,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(6,3,0) |
| 1,4,7,10,9,12,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1) |
| 1,4,7,10,9,12,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1) |
| 1,4,7,10,9,<br>12,7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,9,12,<br>7,10,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,9,12,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(3,1,0)(4,2,0) |
| 1,4,7,10,9,12,11,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(4,1,0)(5,2,0) |
| 1,4,7,10,9,12,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(4,2,0) |
| 1,4,7,10,9,12,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,0,0) |
| 1,4,7,10,9,12,<br>14,4,7,10,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,9,12,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,2,0) |
| 1,4,7,10,9,12,15,17,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(5,2,0)(6,1,0)(2,0,0) |
| 1,4,7,10,9,12,15,17,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(5,2,0)(6,1,0)(5,0,0) |
| 1,4,7,10,9,12,15,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,2,0)(6,2,0) |
| 1,4,7,10,9,12,<br>15,18,12,15,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,2,0)(6,2,0)<br>(4,2,0)(5,2,0)(6,2,0) |
| 1,4,7,10,9,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,3,0) |
| 1,4,7,10,9,13 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,1) |
| 1,4,7,10,9,13,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1) |
| 1,4,7,10,9,13,17,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,1,0) |
| 1,4,7,10,9,13,17,19,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,1,0)(2,0,0) |
| 1,4,7,10,9,13,17,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,0) |
| 1,4,7,10,9,13,17,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) |

| 0 − Y sequence | BMS |
|---|---|
| 1,4,7,10,9,13,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) |
| 1,4,7,10,9,13,<br>17,21,20,24 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)(7,3,0) |
| 1,4,7,10,9,13,17,<br>21,20,24,28,32 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(7,3,0)(8,3,0)(9,3,0) |
| 1,4,7,10,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1) |
| 1,4,7,10,10,4,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,10,4,<br>7,9,13,17,21,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1) |
| 1,4,7,10,10,4,7,<br>9,13,17,21,21,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1)(4,2,0) |
| 1,4,7,10,10,4,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,10,4,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,10,4,7,10,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,10,4,7,<br>10,9,13,17,21,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1) |
| 1,4,7,10,10,4,7,<br>10,9,13,17,21,21,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,0)(3,2,0) |
| 1,4,7,10,10,4,7,<br>10,9,13,17,21,21,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1)(4,2,0) |
| 1,4,7,10,10,4,7,10,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,1)(3,1,1) |
| 1,4,7,10,10,5 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(2,0,0) |
| 1,4,7,10,10,6,4,7,10,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1) |
| 1,4,7,10,10,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,0)(3,2,0) |
| 1,4,7,10,10,6,9,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,0)(3,2,0)(4,3,0) |
| 1,4,7,10,10,6,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,0)(3,2,1) |
| 1,4,7,10,10,6,10,14,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0) |
| 1,4,7,10,10,<br>6,10,14,17,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,4,7,10,10,<br>6,10,14,18,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,2,1) |
| 1,4,7,10,10,7 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(2,1,1) |
| 1,4,7,10,10,7,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0) |
| 1,4,7,10,10,<br>7,9,4,7,10,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,1) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,7,10,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,10,7,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,10,7,<br>9,13,17,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) |
| 1,4,7,10,10,7,9,<br>13,17,21,21,17,20,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,0)(5,0,0) |
| 1,4,7,10,10,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1) |
| 1,4,7,10,10,7,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,10,7,10,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,10,7,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,10,7,10,9,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,4,7,10,10,7,10,9,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,0,0) |
| 1,4,7,10,10,7,10,9,11,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,1,0)(2,0,0) |
| 1,4,7,10,10,7,10,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,10,7,10,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1) |
| 1,4,7,10,10,<br>7,10,10,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,10,8 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,0,0) |
| 1,4,7,10,10,8,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,10,9 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,1,0) |
| 1,4,7,10,10,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,10,9,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(3,0,0) |
| 1,4,7,10,10,9,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0) |
| 1,4,7,10,10,10 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,1,1) |
| 1,4,7,10,10,10,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,1) |
| 1,4,7,10,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,0,0) |
| 1,4,7,10,11,14,17,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,0,0)(5,1,1)(6,1,1)(7,1,1) |
| 1,4,7,10,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0) |
| 1,4,7,10,12,5 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(2,0,0) |
| 1,4,7,10,12,6 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(2,1,0) |

| 0 − Y sequence | BMS |
|---|---|
| 1,4,7,10,12,6,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,0)(3,2,0) |
| 1,4,7,10,12,<br>6,10,14,18,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,2,1) |
| 1,4,7,10,12,<br>6,10,14,18,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(6,0,0) |
| 1,4,7,10,12,<br>6,10,14,18,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(6,1,0) |
| 1,4,7,10,12,6,10,<br>14,18,20,4,7,10,12,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(6,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(4,1,0)(2,0,0) |
| 1,4,7,10,12,6,<br>10,14,18,21,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(6,2,0)(4,0,0) |
| 1,4,7,10,12,7 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(2,1,1) |
| 1,4,7,10,12,7,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,4,7,10,12,7,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,1)(3,1,1) |
| 1,4,7,10,12,7,10,12,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,1)(3,1,1)(4,1,0)(2,0,0) |
| 1,4,7,10,12,8 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(3,0,0) |
| 1,4,7,10,12,9,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(3,1,0)(2,0,0) |
| 1,4,7,10,12,10 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(3,1,1) |
| 1,4,7,10,12,10,12,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(3,1,1)(4,1,0)(2,0,0) |
| 1,4,7,10,12,11 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(4,0,0) |
| 1,4,7,10,12,12,5 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(4,1,0)(2,0,0) |
| 1,4,7,10,12,15 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(5,2,0) |
| 1,4,7,10,12,16,20,24 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,1)(6,2,1)(7,2,1) |
| 1,4,7,10,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1) |
| 1,4,7,10,13,16 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(5,1,1) |
| 1,4,7,10,13,16,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,1) |
| 1,4,8 | (0,0,0)(1,1,1)(2,2,0) |
| 1,4,8,4 | (0,0,0)(1,1,1)(2,2,0)(1,1,1) |
| 1,4,8,4,7 | (0,0,0)(1,1,1)(2,2,0)(1,1,1)(2,1,1) |
| 1,4,8,4,7,10 | (0,0,0)(1,1,1)(2,2,0)<br>(1,1,1)(2,1,1)(3,1,1) |
| 1,4,8,4,8 | (0,0,0)(1,1,1)(2,2,0)(1,1,1)(2,2,0) |
| 1,4,8,4,8,4,8 | (0,0,0)(1,1,1)(2,2,0)(1,1,1)<br>(2,2,0)(1,1,1)(2,2,0) |
| 1,4,8,5 | (0,0,0)(1,1,1)(2,2,0)(2,0,0) |
| 1,4,8,5,8,12 | (0,0,0)(1,1,1)(2,2,0)<br>(2,0,0)(3,1,1)(4,2,0) |
| 1,4,8,6 | (0,0,0)(1,1,1)(2,2,0)(2,1,0) |
| 1,4,8,6,9 | (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,0) |

| 0 − Y sequence | BMS |
|---|---|
| 1,4,8,6,10,15 | (0,0,0)(1,1,1)(2,2,0)<br>(2,1,0)(3,2,1)(4,3,0) |
| 1,4,8,7 | (0,0,0)(1,1,1)(2,2,0)(2,1,1) |
| 1,4,8,7,10 | (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,1) |
| 1,4,8,7,11 | (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0) |
| 1,4,8,7,11,10,14 | (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(3,1,1)(4,2,0) |
| 1,4,8,8 | (0,0,0)(1,1,1)(2,2,0)(2,2,0) |
| 1,4,8,8,8 | (0,0,0)(1,1,1)(2,2,0)(2,2,0)(2,2,0) |
| 1,4,8,9 | (0,0,0)(1,1,1)(2,2,0)(3,0,0) |
| 1,4,8,10 | (0,0,0)(1,1,1)(2,2,0)(3,1,0) |
| 1,4,8,10,3,7,12,15 | (0,0,0)(1,1,1)(2,2,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,3,0)(4,2,0) |
| 1,4,8,11 | (0,0,0)(1,1,1)(2,2,0)(3,1,1) |
| 1,4,8,12 | (0,0,0)(1,1,1)(2,2,0)(3,2,0) |
| 1,4,8,12,14,5 | (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(2,0,0) |
| 1,4,8,12,15 | (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1) |
| 1,4,8,12,16 | (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,2,0) |
| 1,4,8,13 | (0,0,0)(1,1,1)(2,2,0)(3,3,0) |
| 1,4,8,13,19 | (0,0,0)(1,1,1)(2,2,0)(3,3,0)(4,4,0) |
| 1,4,8,14 | (0,0,0)(1,1,1)(2,2,0)(3,3,1) |
| 1,4,8,14,19,14 | (0,0,0)(1,1,1)(2,2,0)<br>(3,3,1)(4,3,0)(3,3,1) |
| 1,4,8,14,20 | (0,0,0)(1,1,1)(2,2,0)(3,3,1)(4,3,1) |
| 1,4,8,14,20,25,15 | (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,3,1)(5,3,0)(4,0,0) |
| 1,4,8,14,21 | (0,0,0)(1,1,1)(2,2,0)(3,3,1)(4,4,0) |
| 1,4,8,14,21,30 | (0,0,0)(1,1,1)(2,2,0)<br>(3,3,1)(4,4,0)(5,5,1) |
| 1,4,8,14,21,30,40,52 | (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,5,1)(6,6,0)(7,7,1) |
| 1,4,9 | (0,0,0)(1,1,1)(2,2,1) |
| 1,4,9,4,9 | (0,0,0)(1,1,1)(2,2,1)(1,1,1)(2,2,1) |
| 1,4,9,6,4,8,14,22 | (0,0,0)(1,1,1)(2,2,1)(2,1,0)<br>(1,1,1)(2,2,0)(3,3,1)(4,4,1) |
| 1,4,9,6,4,8,14,22,16,5 | (0,0,0)(1,1,1)(2,2,1)(2,1,0)(1,1,1)<br>(2,2,0)(3,3,1)(4,4,1)(4,1,0)(2,0,0) |
| 1,4,9,7 | (0,0,0)(1,1,1)(2,2,1)(2,1,1) |
| 1,4,9,7,11 | (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,0) |
| 1,4,9,7,11,17 | (0,0,0)(1,1,1)(2,2,1)<br>(2,1,1)(3,2,0)(4,3,1) |
| 1,4,9,7,11,17,25 | (0,0,0)(1,1,1)(2,2,1)(2,1,1)<br>(3,2,0)(4,3,1)(5,4,1) |
| 1,4,9,7,12 | (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,1) |
| 1,4,9,7,12,5 | (0,0,0)(1,1,1)(2,2,1)<br>(2,1,1)(3,2,1)(2,0,0) |
| 1,4,9,7,12,7 | (0,0,0)(1,1,1)(2,2,1)<br>(2,1,1)(3,2,1)(2,1,1) |
| 1,4,9,7,12,7,12 | (0,0,0)(1,1,1)(2,2,1)(2,1,1)<br>(3,2,1)(2,1,1)(3,2,1) |
| 1,4,9,7,12,8 | (0,0,0)(1,1,1)(2,2,1)<br>(2,1,1)(3,2,1)(3,0,0) |
| 1,4,9,7,12,9,5 | (0,0,0)(1,1,1)(2,2,1)(2,1,1)<br>(3,2,1)(3,1,0)(2,0,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,9,7,12,10 | (0,0,0)(1,1,1)(2,2,1)<br>(2,1,1)(3,2,1)(3,1,1) |
| 1,4,9,7,12,10,15 | (0,0,0)(1,1,1)(2,2,1)(2,1,1)<br>(3,2,1)(3,1,1)(4,2,1) |
| 1,4,9,7,12,10,15,13,18 | (0,0,0)(1,1,1)(2,2,1)(2,1,1)<br>(3,2,1)(3,1,1)(4,2,1)(4,1,1)(5,2,1) |
| 1,4,9,8 | (0,0,0)(1,1,1)(2,2,1)(2,2,0) |
| 1,4,9,8,12 | (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0) |
| 1,4,9,8,12,16 | (0,0,0)(1,1,1)(2,2,1)<br>(2,2,0)(3,2,0)(4,2,0) |
| 1,4,9,8,14 | (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,3,1) |
| 1,4,9,8,14,17 | (0,0,0)(1,1,1)(2,2,1)<br>(2,2,0)(3,3,1)(4,1,1) |
| 1,4,9,8,14,18 | (0,0,0)(1,1,1)(2,2,1)<br>(2,2,0)(3,3,1)(4,2,0) |
| 1,4,9,8,14,19,14 | (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,3,1)(4,3,0)(3,3,1) |
| 1,4,9,8,14,22 | (0,0,0)(1,1,1)(2,2,1)<br>(2,2,0)(3,3,1)(4,4,1) |
| 1,4,9,8,14,22,14 | (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,3,1)(4,4,1)(3,3,1) |
| 1,4,9,9 | (0,0,0)(1,1,1)(2,2,1)(2,2,1) |
| 1,4,9,9,7,9,5 | (0,0,0)(1,1,1)(2,2,1)(2,2,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,4,9,9,9 | (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,1) |
| 1,4,9,10 | (0,0,0)(1,1,1)(2,2,1)(3,0,0) |
| 1,4,9,10,8 | (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0) |
| 1,4,9,10,9,8 | (0,0,0)(1,1,1)(2,2,1)<br>(3,0,0)(2,2,1)(2,2,0) |
| 1,4,9,10,9,10 | (0,0,0)(1,1,1)(2,2,1)<br>(3,0,0)(2,2,1)(3,0,0) |
| 1,4,9,10,10 | (0,0,0)(1,1,1)(2,2,1)(3,0,0)(3,0,0) |
| 1,4,9,11 | (0,0,0)(1,1,1)(2,2,1)(3,1,0) |
| 1,4,9,11,3,7,13,16 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,3,1)(4,2,0) |
| 1,4,9,11,4 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)(1,1,1) |
| 1,4,9,11,4,7,10 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,4,9,11,4,7,10,10 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,1) |
| 1,4,9,11,4,9 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,0)(1,1,1)(2,2,1) |
| 1,4,9,11,4,9,9 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(1,1,1)(2,2,1)(2,2,1) |
| 1,4,9,11,4,9,10 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(1,1,1)(2,2,1)(3,0,0) |
| 1,4,9,11,5 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,0,0) |
| 1,4,9,11,6,9 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,0)(2,1,0)(3,2,0) |
| 1,4,9,11,8 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,0) |
| 1,4,9,11,9 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,1) |
| 1,4,9,11,14 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)(4,2,0) |
| 1,4,9,11,15,21,24 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(4,2,1)(5,3,1)(6,2,0) |
| 1,4,9,12 | (0,0,0)(1,1,1)(2,2,1)(3,1,1) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,9,12,7,12,15 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(2,1,1)(3,2,1)(4,1,1) |
| 1,4,9,12,8 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)(2,2,0) |
| 1,4,9,12,9 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)(2,2,1) |
| 1,4,9,12,9,12 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(2,2,1)(3,1,1) |
| 1,4,9,12,12 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)(3,1,1) |
| 1,4,9,12,15 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,1,1) |
| 1,4,9,12,16 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,0) |
| 1,4,9,12,16,22 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,0)(5,3,1) |
| 1,4,9,12,17 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,1) |
| 1,4,9,12,17,9 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,1)(2,2,1) |
| 1,4,9,12,17,9,12 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(2,2,1)(3,1,1) |
| 1,4,9,12,17,9,12,17 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(2,2,1)(3,1,1)(4,2,1) |
| 1,4,9,12,17,12 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,1)(3,1,1) |
| 1,4,9,12,17,12,17 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(3,1,1)(4,2,1) |
| 1,4,9,12,17,15,20 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(4,1,1)(5,2,1) |
| 1,4,9,12,17,16 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,1)(4,2,0) |
| 1,4,9,12,17,17 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,1)(4,2,1) |
| 1,4,9,12,17,18 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,0,0) |
| 1,4,9,12,17,19,5 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,1,0)(2,0,0) |
| 1,4,9,12,17,20 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,1,1) |
| 1,4,9,12,17,20,24 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,1,1)(6,2,0) |
| 1,4,9,12,17,20,25 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,1,1)(6,2,1) |
| 1,4,9,13 | (0,0,0)(1,1,1)(2,2,1)(3,2,0) |
| 1,4,9,13,4,9,13 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(1,1,1)(2,2,1)(3,2,0) |
| 1,4,9,13,6,4,9,13 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,1,0)(1,1,1)(2,2,1)(3,2,0) |
| 1,4,9,13,6,9 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(2,1,0)(3,2,0) |
| 1,4,9,13,6,10 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(2,1,0)(3,2,1) |
| 1,4,9,13,6,10,16 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,1,0)(3,2,1)(4,3,1) |
| 1,4,9,13,6,10,16,17 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,1,0)(3,2,1)(4,3,1)(5,0,0) |
| 1,4,9,13,6,10,16,19,11 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,1,0)(3,2,1)(4,3,1)(5,2,0)(4,0,0) |
| 1,4,9,13,6,10,<br>16,20,26,29,11 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,1,0)(3,2,1)(4,3,1)(5,2,1)<br>(6,3,1)(7,2,0)(4,0,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,9,13,6,10,16,21 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,1,0)(3,2,1)(4,3,1)(5,3,0) |
| 1,4,9,13,6,10,16,<br>21,13,18,25,31 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,1,0)(3,2,1)(4,3,1)(5,3,0)<br>(4,2,0)(5,3,1)(6,4,1)(7,4,0) |
| 1,4,9,13,7 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,1,1) |
| 1,4,9,13,8 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,0) |
| 1,4,9,13,9,8 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(2,2,1)(2,2,0) |
| 1,4,9,13,9,10 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(2,2,1)(3,0,0) |
| 1,4,9,13,9,11,5 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,0)(2,0,0) |
| 1,4,9,13,9,12,17,21 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0) |
| 1,4,9,13,9,12,<br>17,21,17,19,5 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,1,0)(2,0,0) |
| 1,4,9,13,9,12,17,<br>21,17,20,25,29 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,1,1)(6,2,1)(7,2,0) |
| 1,4,9,13,9,13 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(2,2,1)(3,2,0) |
| 1,4,9,13,9,13,6,9 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,2,0)(2,1,0)(3,2,0) |
| 1,4,9,13,9,13,6,10,16,21 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,2,0)(2,1,0)(3,2,1)(4,3,1)(5,3,0) |
| 1,4,9,13,9,13,<br>6,10,16,21,16,21 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,2,0)(2,1,0)(3,2,1)<br>(4,3,1)(5,3,0)(4,3,1)(5,3,0) |
| 1,4,9,13,9,13,9,11,5 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,2,0)(2,2,1)(3,1,0)(2,0,0) |
| 1,4,9,13,9,13,9,13 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,2,0)(2,2,1)(3,2,0) |
| 1,4,9,13,10 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,0,0) |
| 1,4,9,13,11,5 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(3,1,0)(2,0,0) |
| 1,4,9,13,11,9,13 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,1,0)(2,2,1)(3,2,0) |
| 1,4,9,13,12 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,1) |
| 1,4,9,13,12,17,18 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,1,1)(4,2,1)(5,0,0) |
| 1,4,9,13,12,17,21 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,1,1)(4,2,1)(5,2,0) |
| 1,4,9,13,12,17,21,23,5 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,1,1)(4,2,1)(5,2,0)(5,1,0)(2,0,0) |
| 1,4,9,13,13 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0) |
| 1,4,9,13,15,5 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(4,1,0)(2,0,0) |
| 1,4,9,13,15,9,13,15,5 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,1,0)<br>(2,2,1)(3,2,0)(4,1,0)(2,0,0) |
| 1,4,9,13,15,13,15,5 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0)(2,0,0) |
| 1,4,9,13,15,17,5 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,1,0)(5,1,0)(2,0,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,9,13,15,18 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(4,1,0)(5,2,0) |
| 1,4,9,13,16,21,25 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,1,1)(5,2,1)(6,2,0) |
| 1,4,9,13,16,21,25,27,5 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,1,1)(5,2,1)(6,2,0)(7,1,0)(2,0,0) |
| 1,4,9,13,17 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,2,0) |
| 1,4,9,13,18 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,3,0) |
| 1,4,9,13,18,23,28 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,0)(5,3,0)(6,3,0) |
| 1,4,9,13,19 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,3,1) |
| 1,4,9,13,19,20 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(4,3,1)(5,0,0) |
| 1,4,9,13,19,21,24 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,1,0)(6,2,0) |
| 1,4,9,13,19,21,25 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,1,0)(6,2,1) |
| 1,4,9,13,19,21,25,29 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,1,0)(6,2,1)(7,2,1) |
| 1,4,9,13,19,21,25,31 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,1,0)(6,2,1)(7,3,1) |
| 1,4,9,13,19,21,25,31,36 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,1,0)(6,2,1)(7,3,1)(8,3,0) |
| 1,4,9,13,19,22 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(4,3,1)(5,1,1) |
| 1,4,9,13,19,22,27,31 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,1,1)(6,2,1)(7,2,0) |
| 1,4,9,13,19,23 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(4,3,1)(5,2,0) |
| 1,4,9,13,19,24,19 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,3,0)(4,3,1) |
| 1,4,9,13,19,27 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(4,3,1)(5,4,1) |
| 1,4,9,13,19,27,28 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,4,1)(6,0,0) |
| 1,4,9,13,19,27,32,20 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,4,1)(6,3,0)(5,0,0) |
| 1,4,9,13,19,27,34 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,4,1)(6,4,0) |
| 1,4,9,14 | (0,0,0)(1,1,1)(2,2,1)(3,2,1) |
| 1,4,9,14,4,7 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(1,1,1)(2,1,1) |
| 1,4,9,14,4,9 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(1,1,1)(2,2,1) |
| 1,4,9,14,4,9,13 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(1,1,1)(2,2,1)(3,2,0) |
| 1,4,9,14,4,9,14 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(1,1,1)(2,2,1)(3,2,1) |
| 1,4,9,14,8 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,0) |
| 1,4,9,14,8,12,16 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,0)(3,2,0)(4,2,0) |
| 1,4,9,14,9 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(2,2,1) |
| 1,4,9,14,9,10 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,0,0) |

| 0 − Y sequence | BMS |
|---|---|
| 1,4,9,14,9,11,5 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,0)(2,0,0) |
| 1,4,9,14,9,12 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,1,1) |
| 1,4,9,14,9,12,17 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1) |
| 1,4,9,14,9,12,17,21 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0) |
| 1,4,9,14,9,12,17,21,26 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0)(6,3,0) |
| 1,4,9,14,9,12,17,21,27 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(6,3,1) |
| 1,4,9,14,9,12,17,22 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,1) |
| 1,4,9,14,9,<br>12,17,22,17,19,5 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,1,0)(2,0,0) |
| 1,4,9,14,9,13 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,2,0) |
| 1,4,9,14,9,13,18,23,28 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,0)(4,3,0)(5,3,0)(6,3,0) |
| 1,4,9,14,9,14 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,2,1) |
| 1,4,9,14,9,14,8,12,16 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,1)(2,2,0)(3,2,0)(4,2,0) |
| 1,4,9,14,9,14,9,11,5 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,1)(2,2,1)(3,1,0)(2,0,0) |
| 1,4,9,14,9,14,9,12,17,21 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,1,1)(4,2,1)(5,2,0) |
| 1,4,9,14,9,14,9,<br>12,17,22,17,21 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0) |
| 1,4,9,14,9,14,9,13 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,1)(2,2,1)(3,2,0) |
| 1,4,9,14,9,14,<br>9,13,18,23,28 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,1)(2,2,1)(3,2,0)<br>(4,3,0)(5,3,0)(6,3,0) |
| 1,4,9,14,9,14,9,14,9,13 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,2,1)(2,2,1)(3,2,0) |
| 1,4,9,14,9,14,9,<br>14,9,13,18,23,28 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,0)(4,3,0)(5,3,0)(6,3,0) |
| 1,4,9,14,10 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,0,0) |
| 1,4,9,14,11,5 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,1,0)(2,0,0) |
| 1,4,9,14,11,15,21,26 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,1,0)(4,2,1)(5,3,1)(6,3,0) |
| 1,4,9,14,11,<br>15,21,27,21,26 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(4,2,1)(5,3,1)(6,3,1)(5,3,1)(6,3,0) |
| 1,4,9,14,11,15,21,27,22 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(4,2,1)(5,3,1)(6,3,1)(6,0,0) |
| 1,4,9,14,12 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1) |
| 1,4,9,14,12,17,22 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,1,1)(4,2,1)(5,2,1) |
| 1,4,9,14,13 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,4,9,14,13,17,21 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,0)(4,2,0)(5,2,0) |
| 1,4,9,14,13,19 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,2,0)(4,3,1) |
| 1,4,9,14,14 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1) |
| 1,4,9,14,14,9,13 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,1)(2,2,1)(3,2,0) |
| 1,4,9,14,14,9,14,13 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,1)(2,2,1)(3,2,1)(3,2,0) |
| 1,4,9,14,14,13 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,2,1)(3,2,0) |
| 1,4,9,14,18 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0) |
| 1,4,9,14,19,13 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,1)(3,2,0) |
| 1,4,9,14,19,14,18 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,2,0) |
| 1,4,9,14,19,18 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,1)(4,2,0) |
| 1,4,9,14,19,23 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,1)(5,2,0) |
| 1,4,9,15 | (0,0,0)(1,1,1)(2,2,1)(3,3,0) |
| 1,4,9,15,21,27 | (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,3,0)(5,3,0) |
| 1,4,9,15,23 | (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,4,1) |
| 1,4,9,15,23,33,42 | (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,4,1)(5,5,1)(6,5,0) |
| 1,4,9,15,23,33,43,34 | (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,4,1)(5,5,1)(6,5,1)(6,0,0) |
| 1,4,9,15,23,33,44 | (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,4,1)(5,5,1)(6,6,0) |
| 1,4,9,16 | (0,0,0)(1,1,1)(2,2,1)(3,3,1) |
| 1,4,9,16,9 | (0,0,0)(1,1,1)(2,2,1)(3,3,1)(2,2,1) |
| 1,4,9,16,9,14 | (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(2,2,1)(3,2,1) |
| 1,4,9,16,9,15 | (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(2,2,1)(3,3,0) |
| 1,4,9,16,9,16 | (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(2,2,1)(3,3,1) |
| 1,4,9,16,14 | (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,2,1) |
| 1,4,9,16,14,21 | (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(3,2,1)(4,3,1) |
| 1,4,9,16,15 | (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,3,0) |
| 1,4,9,16,16 | (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,3,1) |
| 1,4,9,16,17 | (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,0,0) |
| 1,4,9,16,22 | (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,3,0) |
| 1,4,9,16,25 | (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,4,1) |
| 1,4,9,16,25,36 | (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,4,1)(5,5,1) |
| 1,4,10 | (0,0,0)(1,1,1)(2,2,2) |
| 1,4,10,16 | (0,0,0)(1,1,1)(2,2,2)(3,2,2) |
| 1,4,10,16,17 | (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,0,0) |
| 1,4,10,16,18 | (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,1,0) |
| 1,4,10,16,18,5 | (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,1,0)(2,0,0) |
| 1,4,10,16,19 | (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,1,1) |
| 1,4,10,16,20 | (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0) |

| 0 − Y sequence | BMS |
|---|---|
| 1,4,10,16,20,10 | (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,0)(2,2,2) |
| 1,4,10,16,20,10,16,20,10 | (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(2,2,2)(3,2,2)(4,2,0)(2,2,2) |
| 1,4,10,16,20,11 | (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,0)(3,0,0) |
| 1,4,10,16,21 | (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,1) |
| 1,4,10,16,22 | (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2) |
| 1,4,10,16,22,22 | (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,2)(4,2,2) |
| 1,4,10,16,22,28 | (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,2)(5,2,2) |
| 1,4,10,17 | (0,0,0)(1,1,1)(2,2,2)(3,3,0) |
| 1,4,10,18 | (0,0,0)(1,1,1)(2,2,2)(3,3,1) |
| 1,4,10,18,28 | (0,0,0)(1,1,1)(2,2,2)(3,3,1)(4,4,1) |
| 1,4,10,18,29 | (0,0,0)(1,1,1)(2,2,2)(3,3,1)(4,4,2) |
| 1,4,10,19 | (0,0,0)(1,1,1)(2,2,2)(3,3,2) |
| 1,4,10,19,20 | (0,0,0)(1,1,1)(2,2,2)(3,3,2)(4,0,0) |
| 1,4,10,19,31 | (0,0,0)(1,1,1)(2,2,2)(3,3,2)(4,4,2) |
| 1,4,10,20 | (0,0,0)(1,1,1)(2,2,2)(3,3,3) |
| 1,4,10,20,35 | (0,0,0)(1,1,1)(2,2,2)(3,3,3)(4,4,4) |
| 1,4,10,20,35,56 | (0,0,0)(1,1,1)(2,2,2)<br>(3,3,3)(4,4,4)(5,5,5) |
| 1,5 | (0,0,0,0)(1,1,1,1) |
| 1,5,4 | (0,0,0,0)(1,1,1,1)(1,1,1,0) |
| 1,5,4,9 | (0,0,0,0)(1,1,1,1)(1,1,1,0)(2,2,1,0) |
| 1,5,4,9,16 | (0,0,0,0)(1,1,1,1)(1,1,1,0)<br>(2,2,1,0)(3,3,1,0) |
| 1,5,4,10 | (0,0,0,0)(1,1,1,1)(1,1,1,0)(2,2,2,0) |
| 1,5,4,10,20 | (0,0,0,0)(1,1,1,1)(1,1,1,0)<br>(2,2,2,0)(3,3,3,0) |
| 1,5,4,11 | (0,0,0,0)(1,1,1,1)(1,1,1,0)(2,2,2,1) |
| 1,5,4,11,9 | (0,0,0,0)(1,1,1,1)(1,1,1,0)<br>(2,2,2,1)(2,2,1,0) |
| 1,5,4,11,9,18,16 | (0,0,0,0)(1,1,1,1)(1,1,1,0)(2,2,2,1)<br>(2,2,1,0)(3,3,2,1)(3,3,1,0) |
| 1,5,4,11,10 | (0,0,0,0)(1,1,1,1)(1,1,1,0)<br>(2,2,2,1)(2,2,2,0) |
| 1,5,4,11,10,21 | (0,0,0,0)(1,1,1,1)(1,1,1,0)<br>(2,2,2,1)(2,2,2,0)(3,3,3,1) |
| 1,5,4,11,10,21,20 | (0,0,0,0)(1,1,1,1)(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)(3,3,3,0) |
| 1,5,5 | (0,0,0,0)(1,1,1,1)(1,1,1,1) |
| 1,5,5,5 | (0,0,0,0)(1,1,1,1)(1,1,1,1)(1,1,1,1) |
| 1,5,6 | (0,0,0,0)(1,1,1,1)(2,0,0,0) |
| 1,5,7 | (0,0,0,0)(1,1,1,1)(2,1,0,0) |
| 1,5,7,3,8 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,0,0)(2,2,1,1) |
| 1,5,7,3,8,10 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,0,0)(2,2,1,1)(3,1,0,0) |
| 1,5,7,3,8,10,7 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)(2,2,1,0) |
| 1,5,7,3,8,10,15,17 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,0,0)(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)(4,1,0,0) |

| $0-\mathrm{Y}$ sequence | BMS |
|---|---|
| 1,5,7,3,8,10,8 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)(2,2,1,1) |
| 1,5,7,3,8,11 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,0,0)(2,2,1,1)(3,1,1,0) |
| 1,5,7,4 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0) |
| 1,5,7,4,11,13 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)(3,1,0,0) |
| 1,5,7,4,11,13,4 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(1,1,1,0) |
| 1,5,7,4,11,13,5 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(2,0,0,0) |
| 1,5,7,4,11,13,11 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(2,2,2,1) |
| 1,5,7,4,11,13,11,13,11 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)(3,1,0,0)<br>(2,2,2,1)(3,1,0,0)(2,2,2,1) |
| 1,5,7,4,11,13,12 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(3,0,0,0) |
| 1,5,7,4,11,13,13 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(3,1,0,0) |
| 1,5,7,4,11,13,13,4 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(3,1,0,0)(1,1,1,0) |
| 1,5,7,4,11,13,13,11 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(3,1,0,0)(2,2,2,1) |
| 1,5,7,4,11,13,16 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(4,2,0,0) |
| 1,5,7,4,11,13,17 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(4,2,1,0) |
| 1,5,7,4,11,13,18 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(4,2,1,1) |
| 1,5,7,4,11,14 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)(3,1,1,0) |
| 1,5,7,4,11,14,19 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,1,0)(4,2,1,0) |
| 1,5,7,4,11,14,20 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,1,0)(4,2,2,0) |
| 1,5,7,4,11,14,21 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,1,0)(4,2,2,1) |
| 1,5,7,4,11,14,21,24,31 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)(3,1,1,0)<br>(4,2,2,1)(5,1,1,0)(6,2,2,1) |
| 1,5,7,4,11,15 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)(3,2,0,0) |
| 1,5,7,4,11,15,10 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)(2,2,2,0) |
| 1,5,7,5 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,1) |
| 1,5,7,5,7,5 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)(1,1,1,1) |
| 1,5,7,7,5 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(2,1,0,0)(1,1,1,1) |
| 1,5,7,10 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(3,2,0,0) |
| 1,5,8 | (0,0,0,0)(1,1,1,1)(2,1,1,0) |
| 1,5,8,4,11 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1) |
| 1,5,8,4,11,14 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,1,1,0) |

| 0 − Y sequence | BMS |
|---|---|
| 1,5,8,4,11,14,21 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,1,1,0)(4,2,2,1) |
| 1,5,8,4,11,15 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,0,0) |
| 1,5,8,4,11,16 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,1,0) |
| 1,5,8,4,11,16,9,18,25 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,1,0)<br>(2,2,1,0)(3,3,2,1)(4,3,1,0) |
| 1,5,8,4,11,16,10,21,29 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,1,0)<br>(2,2,2,0)(3,3,3,1)(4,3,1,0) |
| 1,5,8,4,11,16,11 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)(2,2,2,1) |
| 1,5,8,4,11,16,16,11 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)(3,2,1,0)(2,2,2,1) |
| 1,5,8,4,11,16,17 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)(4,0,0,0) |
| 1,5,8,4,11,16,21,11 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)(4,2,1,0)(2,2,2,1) |
| 1,5,8,4,11,16,23 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)(4,3,1,0) |
| 1,5,8,4,11,16,24 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)(4,3,2,0) |
| 1,5,8,4,11,16,25 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)(4,3,2,1) |
| 1,5,8,4,11,16,25,36,51 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,1,0)<br>(4,3,2,1)(5,4,1,0)(6,5,2,1) |
| 1,5,8,4,11,17 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,2,0) |
| 1,5,8,4,11,17,9,18,26 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,2,0)<br>(2,2,1,0)(3,3,2,1)(4,3,2,0) |
| 1,5,8,4,11,17,10 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)(2,2,2,0) |
| 1,5,8,4,11,17,10,21,31 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,2,0)<br>(2,2,2,0)(3,3,3,1)(4,3,3,0) |
| 1,5,8,5 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,1) |
| 1,5,8,5,8 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0) |
| 1,5,8,5,8,5 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)(1,1,1,1) |
| 1,5,8,8 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(2,1,1,0) |
| 1,5,8,8,5 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(2,1,1,0)(1,1,1,1) |
| 1,5,8,11,5 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,1,1,0)(1,1,1,1) |
| 1,5,8,13 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(3,2,1,0) |
| 1,5,8,13,4,11,17,26 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,2,0)(4,3,2,0) |
| 1,5,8,13,5 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,2,1,0)(1,1,1,1) |
| 1,5,8,13,20,5 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,2,1,0)(4,3,1,0)(1,1,1,1) |

| 0 − Y sequence | BMS |
|---|---|
| 1,5,8,14 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(3,2,2,0) |
| 1,5,8,14,24 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,2,2,0)(4,3,3,0) |
| 1,5,8,15 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(3,2,2,1) |
| 1,5,8,15,15 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(3,2,2,1) |
| 1,5,8,15,18,5 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,1,1,0)(1,1,1,1) |
| 1,5,8,15,20,5 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,1,0)(1,1,1,1) |
| 1,5,8,15,21 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,0) |
| 1,5,8,15,21,32 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,0)(5,3,3,1) |
| 1,5,9 | (0,0,0,0)(1,1,1,1)(2,1,1,1) |
| 1,5,9,5,9 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(1,1,1,1)(2,1,1,1) |
| 1,5,9,8 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(2,1,1,0) |
| 1,5,9,8,5,9 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(2,1,1,0)(1,1,1,1)(2,1,1,1) |
| 1,5,9,8,11,5,9 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(2,1,1,0)<br>(3,1,1,0)(1,1,1,1)(2,1,1,1) |
| 1,5,9,8,13,5,9 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(2,1,1,0)<br>(3,2,1,0)(1,1,1,1)(2,1,1,1) |
| 1,5,9,8,14 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(2,1,1,0)(3,2,2,0) |
| 1,5,9,8,15 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(2,1,1,0)(3,2,2,1) |
| 1,5,9,8,15,22 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(2,1,1,0)(3,2,2,1)(4,2,2,1) |
| 1,5,9,9 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(2,1,1,1) |
| 1,5,9,10 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(3,0,0,0) |
| 1,5,9,11 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(3,1,0,0) |
| 1,5,9,11,5 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(1,1,1,1) |
| 1,5,9,11,5,9 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(1,1,1,1)(2,1,1,1) |
| 1,5,9,11,5,9,11 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(1,1,1,1)(2,1,1,1)(3,1,0,0) |
| 1,5,9,11,6 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,0,0,0) |
| 1,5,9,11,7 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,0,0) |
| 1,5,9,11,8 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,0) |
| 1,5,9,11,8,5,9,11 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(1,1,1,1)(2,1,1,1)(3,1,0,0) |
| 1,5,9,11,8,6 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,0)(2,0,0,0) |
| 1,5,9,11,8,13,5,9,11 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,0)(3,2,1,0)<br>(1,1,1,1)(2,1,1,1)(3,1,0,0) |
| 1,5,9,11,8,14 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,0)(3,2,2,0) |
| 1,5,9,11,8,15 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,0)(3,2,2,1) |

| 0 − Y sequence | BMS |
|---|---|
| 1,5,9,11,8,15,22,26,16 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,0,0)(4,0,0,0) |
| 1,5,9,11,9 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,1) |
| 1,5,9,11,14 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(4,2,0,0) |
| 1,5,9,12 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(3,1,1,0) |
| 1,5,9,12,4,11,18,24 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(3,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,2,1)(4,2,2,0) |
| 1,5,9,12,5 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(1,1,1,1) |
| 1,5,9,12,5,9 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(1,1,1,1)(2,1,1,1) |
| 1,5,9,12,5,9,12 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(3,1,1,0)<br>(1,1,1,1)(2,1,1,1)(3,1,1,0) |
| 1,5,9,12,6 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,0,0,0) |
| 1,5,9,12,7 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,1,0,0) |
| 1,5,9,12,8 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,1,1,0) |
| 1,5,9,12,8,5 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,1,1,0)(1,1,1,1) |
| 1,5,9,12,8,5,9,12,6 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)(2,0,0,0) |
| 1,5,9,12,8,12 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,1,1,0)(3,2,0,0) |
| 1,5,9,12,8,13,5,9,12,6 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)(2,0,0,0) |
| 1,5,9,12,8,14 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,1,1,0)(3,2,2,0) |
| 1,5,9,12,9 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,1,1,1) |
| 1,5,9,13 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(3,1,1,1) |
| 1,5,11 | (0,0,0,0)(1,1,1,1)(2,2,1,0) |
| 1,5,11,4,11,21 | (0,0,0,0)(1,1,1,1)(2,2,1,0)<br>(1,1,1,0)(2,2,2,1)(3,3,2,0) |
| 1,5,11,5 | (0,0,0,0)(1,1,1,1)(2,2,1,0)(1,1,1,1) |
| 1,5,12 | (0,0,0,0)(1,1,1,1)(2,2,1,1) |
| 1,5,12,22 | (0,0,0,0)(1,1,1,1)(2,2,1,1)(3,3,1,1) |
| 1,5,12,22,35 | (0,0,0,0)(1,1,1,1)(2,2,1,1)<br>(3,3,1,1)(4,4,1,1) |
| 1,5,13 | (0,0,0,0)(1,1,1,1)(2,2,2,0) |
| 1,5,15 | (0,0,0,0)(1,1,1,1)(2,2,2,2) |
| 1,5,15,35 | (0,0,0,0)(1,1,1,1)(2,2,2,2)(3,3,3,3) |
| 1,6 | (0,0,0,0,0)(1,1,1,1,1) |
| 1,6,11 | (0,0,0,0,0)(1,1,1,1,1)(2,1,1,1,1) |
| 1,6,15 | (0,0,0,0,0)(1,1,1,1,1)(2,2,1,1,1) |
| 1,6,18 | (0,0,0,0,0)(1,1,1,1,1)(2,2,2,1,1) |
| 1,6,19 | (0,0,0,0,0)(1,1,1,1,1)(2,2,2,2,0) |
| 1,6,20 | (0,0,0,0,0)(1,1,1,1,1)(2,2,2,2,1) |
| 1,6,21 | (0,0,0,0,0)(1,1,1,1,1)(2,2,2,2,2) |
| 1,7 | (0,0,0,0,0,0)(1,1,1,1,1,1) |

| 0 − Y sequence | BMS |
|---|---|
| 1,8 | (0,0,0,0,0,0,0)(1,1,1,1,1,1,1) |
| 1,9 | (0,0,0,0,0,0,0,0)(1,1,1,1,1,1,1,1) |

## A.20 Y sequence vs BMS

The results in this section are primarily cited from[2].

| Y sequence | BMS |
|---|---|
| 1 | (0) |
| 1,1 | (0)(0) |
| 1,1,1 | (0)(0)(0) |
| 1,1,1,1 | (0)(0)(0)(0) |
| 1,2 | (0)(1) |
| 1,2,1 | (0)(1)(0) |
| 1,2,1,1 | (0)(1)(0)(0) |
| 1,2,1,2 | (0)(1)(0)(1) |
| 1,2,1,2,1 | (0)(1)(0)(1)(0) |
| 1,2,1,2,1,2 | (0)(1)(0)(1)(0)(1) |
| 1,2,2 | (0)(1)(1) |
| 1,2,2,1 | (0)(1)(1)(0) |
| 1,2,2,1,2 | (0)(1)(1)(0)(1) |
| 1,2,2,1,2,2 | (0)(1)(1)(0)(1)(1) |
| 1,2,2,2 | (0)(1)(1)(1) |
| 1,2,2,2,2 | (0)(1)(1)(1)(1) |
| 1,2,3 | (0)(1)(2) |
| 1,2,3,1 | (0)(1)(2)(0) |
| 1,2,3,1,2 | (0)(1)(2)(0)(1) |
| 1,2,3,1,2,3 | (0)(1)(2)(0)(1)(2) |
| 1,2,3,2 | (0)(1)(2)(1) |
| 1,2,3,2,2 | (0)(1)(2)(1)(1) |
| 1,2,3,2,3 | (0)(1)(2)(1)(2) |
| 1,2,3,2,3,2 | (0)(1)(2)(1)(2)(1) |
| 1,2,3,2,3,2,3 | (0)(1)(2)(1)(2)(1)(2) |
| 1,2,3,3 | (0)(1)(2)(2) |
| 1,2,3,3,2 | (0)(1)(2)(2)(1) |
| 1,2,3,3,2,3 | (0)(1)(2)(2)(1)(2) |
| 1,2,3,3,2,3,3 | (0)(1)(2)(2)(1)(2)(2) |
| 1,2,3,3,3 | (0)(1)(2)(2)(2) |
| 1,2,3,3,3,3 | (0)(1)(2)(2)(2)(2) |
| 1,2,3,4 | (0)(1)(2)(3) |
| 1,2,3,4,2 | (0)(1)(2)(3)(1) |
| 1,2,3,4,2,3,4 | (0)(1)(2)(3)(1)(2)(3) |
| 1,2,3,4,3 | (0)(1)(2)(3)(2) |
| 1,2,3,4,3,4 | (0)(1)(2)(3)(2)(3) |
| 1,2,3,4,4 | (0)(1)(2)(3)(3) |
| 1,2,3,4,5 | (0)(1)(2)(3)(4) |
| 1,2,3,4,5,4 | (0)(1)(2)(3)(4)(3) |
| 1,2,3,4,5,4,5 | (0)(1)(2)(3)(4)(3)(4) |
| 1,2,3,4,5,5 | (0)(1)(2)(3)(4)(4) |
| 1,2,3,4,5,6 | (0)(1)(2)(3)(4)(5) |
| 1,2,3,4,5,6,7 | (0)(1)(2)(3)(4)(5)(6) |
| 1,2,4 | (0,0)(1,1) |
| 1,2,4,1 | (0,0)(1,1)(0,0) |
| 1,2,4,1,2 | (0,0)(1,1)(0,0)(1,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,1,2,3 | (0,0)(1,1)(0,0)(1,0)(2,0) |
| 1,2,4,1,2,4 | (0,0)(1,1)(0,0)(1,1) |
| 1,2,4,2 | (0,0)(1,1)(1,0) |
| 1,2,4,2,2 | (0,0)(1,1)(1,0)(1,0) |
| 1,2,4,2,3 | (0,0)(1,1)(1,0)(2,0) |
| 1,2,4,2,4 | (0,0)(1,1)(1,0)(2,1) |
| 1,2,4,3 | (0,0)(1,1)(1,0)(2,1)(2,0) |
| 1,2,4,3,5 | (0,0)(1,1)(1,0)(2,1)(2,0)(3,1) |
| 1,2,4,3,5,4,6 | (0,0)(1,1)(1,0)(2,1)<br>(2,0)(3,1)(3,0)(4,1) |
| 1,2,4,4 | (0,0)(1,1)(1,1) |
| 1,2,4,4,1,2,4,4 | (0,0)(1,1)(1,1)(0,0)(1,1)(1,1) |
| 1,2,4,4,2 | (0,0)(1,1)(1,1)(1,0) |
| 1,2,4,4,2,4,4 | (0,0)(1,1)(1,1)(1,0)(2,1)(2,1) |
| 1,2,4,4,3 | (0,0)(1,1)(1,1)(1,0)(2,1)(2,1)(2,0) |
| 1,2,4,4,3,5,5 | (0,0)(1,1)(1,1)(1,0)(2,1)<br>(2,1)(2,0)(3,1)(3,1) |
| 1,2,4,4,4 | (0,0)(1,1)(1,1)(1,1) |
| 1,2,4,4,4,4 | (0,0)(1,1)(1,1)(1,1)(1,1) |
| 1,2,4,5 | (0,0)(1,1)(2,0) |
| 1,2,4,5,2 | (0,0)(1,1)(2,0)(1,0) |
| 1,2,4,5,2,4,5 | (0,0)(1,1)(2,0)(1,0)(2,1)(3,0) |
| 1,2,4,5,3 | (0,0)(1,1)(2,0)(1,0)(2,1)(3,0)(2,0) |
| 1,2,4,5,3,5,6 | (0,0)(1,1)(2,0)(1,0)(2,1)<br>(3,0)(2,0)(3,1)(4,0) |
| 1,2,4,5,4 | (0,0)(1,1)(2,0)(1,1) |
| 1,2,4,5,4,4 | (0,0)(1,1)(2,0)(1,1)(1,1) |
| 1,2,4,5,4,5 | (0,0)(1,1)(2,0)(1,1)(2,0) |
| 1,2,4,5,4,5,4,5 | (0,0)(1,1)(2,0)(1,1)(2,0)(1,1)(2,0) |
| 1,2,4,5,5 | (0,0)(1,1)(2,0)(2,0) |
| 1,2,4,5,5,4,5 | (0,0)(1,1)(2,0)(2,0)(1,1)(2,0) |
| 1,2,4,5,5,4,5,5 | (0,0)(1,1)(2,0)(2,0)(1,1)(2,0)(2,0) |
| 1,2,4,5,5,5 | (0,0)(1,1)(2,0)(2,0)(2,0) |
| 1,2,4,5,5,5,5 | (0,0)(1,1)(2,0)(2,0)(2,0)(2,0) |
| 1,2,4,5,6 | (0,0)(1,1)(2,0)(3,0) |
| 1,2,4,5,6,7 | (0,0)(1,1)(2,0)(3,0)(4,0) |
| 1,2,4,5,7 | (0,0)(1,1)(2,0)(3,1) |
| 1,2,4,5,7,7 | (0,0)(1,1)(2,0)(3,1)(3,1) |
| 1,2,4,5,7,8 | (0,0)(1,1)(2,0)(3,1)(4,0) |
| 1,2,4,5,7,8,10 | (0,0)(1,1)(2,0)(3,1)(4,0)(5,1) |
| 1,2,4,6 | (0,0)(1,1)(2,1) |
| 1,2,4,6,2 | (0,0)(1,1)(2,1)(1,0) |
| 1,2,4,6,2,4,6 | (0,0)(1,1)(2,1)(1,0)(2,1)(3,1) |
| 1,2,4,6,3 | (0,0)(1,1)(2,1)(1,0)(2,1)(3,1)(2,0) |
| 1,2,4,6,3,5,7 | (0,0)(1,1)(2,1)(1,0)<br>(2,1)(3,1)(2,0)(3,1)(4,1) |
| 1,2,4,6,4 | (0,0)(1,1)(2,1)(1,1) |
| 1,2,4,6,4,5 | (0,0)(1,1)(2,1)(1,1)(2,0) |
| 1,2,4,6,4,5,4 | (0,0)(1,1)(2,1)(1,1)(2,0)(1,1) |
| 1,2,4,6,4,5,4,5 | (0,0)(1,1)(2,1)(1,1)(2,0)(1,1)(2,0) |
| 1,2,4,6,4,5,5 | (0,0)(1,1)(2,1)(1,1)(2,0)(2,0) |
| 1,2,4,6,4,5,6 | (0,0)(1,1)(2,1)(1,1)(2,0)(3,0) |
| 1,2,4,6,4,5,7 | (0,0)(1,1)(2,1)(1,1)(2,0)(3,1) |
| 1,2,4,6,4,5,7,8 | (0,0)(1,1)(2,1)(1,1)(2,0)(3,1)(4,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,6,4,5,7,8,10 | (0,0)(1,1)(2,1)(1,1)<br>(2,0)(3,1)(4,0)(5,1) |
| 1,2,4,6,4,5,7,9 | (0,0)(1,1)(2,1)(1,1)(2,0)(3,1)(4,1) |
| 1,2,4,6,4,5,7,9,7 | (0,0)(1,1)(2,1)(1,1)<br>(2,0)(3,1)(4,1)(3,1) |
| 1,2,4,6,4,6 | (0,0)(1,1)(2,1)(1,1)(2,1) |
| 1,2,4,6,4,6,4,6 | (0,0)(1,1)(2,1)(1,1)(2,1)(1,1)(2,1) |
| 1,2,4,6,5 | (0,0)(1,1)(2,1)(2,0) |
| 1,2,4,6,5,5 | (0,0)(1,1)(2,1)(2,0)(2,0) |
| 1,2,4,6,5,6 | (0,0)(1,1)(2,1)(2,0)(3,0) |
| 1,2,4,6,5,7 | (0,0)(1,1)(2,1)(2,0)(3,1) |
| 1,2,4,6,5,7,9 | (0,0)(1,1)(2,1)(2,0)(3,1)(4,1) |
| 1,2,4,6,5,7,9,8,10,12 | (0,0)(1,1)(2,1)(2,0)<br>(3,1)(4,1)(4,0)(5,1)(6,1) |
| 1,2,4,6,6 | (0,0)(1,1)(2,1)(2,1) |
| 1,2,4,6,6,4 | (0,0)(1,1)(2,1)(2,1)(1,1) |
| 1,2,4,6,6,4,5,7,9,9 | (0,0)(1,1)(2,1)(2,1)(1,1)<br>(2,0)(3,1)(4,1)(4,1) |
| 1,2,4,6,6,4,<br>5,7,9,9,7 | (0,0)(1,1)(2,1)(2,1)(1,1)<br>(2,0)(3,1)(4,1)(4,1)(3,1) |
| 1,2,4,6,6,4,6 | (0,0)(1,1)(2,1)(2,1)(1,1)(2,1) |
| 1,2,4,6,6,4,<br>6,5,7,9,9 | (0,0)(1,1)(2,1)(2,1)(1,1)<br>(2,1)(2,0)(3,1)(4,1)(4,1) |
| 1,2,4,6,6,4,<br>6,5,7,9,9,7,9 | (0,0)(1,1)(2,1)(2,1)(1,1)(2,1)<br>(2,0)(3,1)(4,1)(4,1)(3,1)(4,1) |
| 1,2,4,6,6,4,6,6 | (0,0)(1,1)(2,1)(2,1)(1,1)(2,1)(2,1) |
| 1,2,4,6,6,5 | (0,0)(1,1)(2,1)(2,1)(2,0) |
| 1,2,4,6,6,5,7 | (0,0)(1,1)(2,1)(2,1)(2,0)(3,1) |
| 1,2,4,6,6,5,7,8 | (0,0)(1,1)(2,1)(2,1)(2,0)(3,1)(4,0) |
| 1,2,4,6,6,5,7,9 | (0,0)(1,1)(2,1)(2,1)(2,0)(3,1)(4,1) |
| 1,2,4,6,6,5,7,9,9 | (0,0)(1,1)(2,1)(2,1)<br>(2,0)(3,1)(4,1)(4,1) |
| 1,2,4,6,6,6 | (0,0)(1,1)(2,1)(2,1)(2,1) |
| 1,2,4,6,6,6,4,6,6,6 | (0,0)(1,1)(2,1)(2,1)(2,1)<br>(1,1)(2,1)(2,1)(2,1) |
| 1,2,4,6,6,6,5 | (0,0)(1,1)(2,1)(2,1)(2,1)(2,0) |
| 1,2,4,6,6,6,6 | (0,0)(1,1)(2,1)(2,1)(2,1)(2,1) |
| 1,2,4,6,7 | (0,0)(1,1)(2,1)(3,0) |
| 1,2,4,6,7,2 | (0,0)(1,1)(2,1)(3,0)(1,0) |
| 1,2,4,6,7,3 | (0,0)(1,1)(2,1)(3,0)(1,0)<br>(2,1)(3,1)(4,0)(2,0) |
| 1,2,4,6,7,4 | (0,0)(1,1)(2,1)(3,0)(1,1) |
| 1,2,4,6,7,4,6,7 | (0,0)(1,1)(2,1)(3,0)(1,1)(2,1)(3,0) |
| 1,2,4,6,7,5 | (0,0)(1,1)(2,1)(3,0)(2,0) |
| 1,2,4,6,7,5,7,9,10 | (0,0)(1,1)(2,1)(3,0)<br>(2,0)(3,1)(4,1)(5,0) |
| 1,2,4,6,7,6 | (0,0)(1,1)(2,1)(3,0)(2,1) |
| 1,2,4,6,7,6,4,6,7 | (0,0)(1,1)(2,1)(3,0)<br>(2,1)(1,1)(2,1)(3,0) |
| 1,2,4,6,7,6,4,6,7,6 | (0,0)(1,1)(2,1)(3,0)(2,1)<br>(1,1)(2,1)(3,0)(2,1) |
| 1,2,4,6,7,6,6 | (0,0)(1,1)(2,1)(3,0)(2,1)(2,1) |
| 1,2,4,6,7,6,7 | (0,0)(1,1)(2,1)(3,0)(2,1)(3,0) |
| 1,2,4,6,7,7 | (0,0)(1,1)(2,1)(3,0)(3,0) |
| 1,2,4,6,7,8 | (0,0)(1,1)(2,1)(3,0)(4,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,6,7,8,5 | (0,0)(1,1)(2,1)(3,0)(4,0)(2,0) |
| 1,2,4,6,7,9 | (0,0)(1,1)(2,1)(3,0)(4,1) |
| 1,2,4,6,7,9,6 | (0,0)(1,1)(2,1)(3,0)(4,1)(2,1) |
| 1,2,4,6,7,9,6,2,4 | (0,0)(1,1)(2,1)(3,0)<br>(4,1)(2,1)(1,0)(2,1) |
| 1,2,4,6,7,9,<br>6,4,6,7,9 | (0,0)(1,1)(2,1)(3,0)(4,1)<br>(2,1)(1,1)(2,1)(3,0)(4,1) |
| 1,2,4,6,7,9,<br>7,4,6,7,9 | (0,0)(1,1)(2,1)(3,0)(4,1)<br>(3,0)(1,1)(2,1)(3,0)(4,1) |
| 1,2,4,6,7,9,9 | (0,0)(1,1)(2,1)(3,0)(4,1)(4,1) |
| 1,2,4,6,7,9,9,9 | (0,0)(1,1)(2,1)(3,0)(4,1)(4,1)(4,1) |
| 1,2,4,6,7,9,10 | (0,0)(1,1)(2,1)(3,0)(4,1)(5,0) |
| 1,2,4,6,7,9,<br>10,4,6,7,9 | (0,0)(1,1)(2,1)(3,0)(4,1)<br>(5,0)(1,1)(2,1)(3,0)(4,1) |
| 1,2,4,6,7,9,11 | (0,0)(1,1)(2,1)(3,0)(4,1)(5,1) |
| 1,2,4,6,7,9,<br>11,9,10,12,14 | (0,0)(1,1)(2,1)(3,0)(4,1)<br>(5,1)(4,1)(5,0)(6,1)(7,1) |
| 1,2,4,6,7,9,11,11 | (0,0)(1,1)(2,1)(3,0)(4,1)(5,1)(5,1) |
| 1,2,4,6,7,9,11,12 | (0,0)(1,1)(2,1)(3,0)(4,1)(5,1)(6,0) |
| 1,2,4,6,7,9,11,12,12 | (0,0)(1,1)(2,1)(3,0)<br>(4,1)(5,1)(6,0)(6,0) |
| 1,2,4,6,7,9,11,12,14 | (0,0)(1,1)(2,1)(3,0)<br>(4,1)(5,1)(6,0)(7,1) |
| 1,2,4,6,7,9,11,12,14,16 | (0,0)(1,1)(2,1)(3,0)(4,1)<br>(5,1)(6,0)(7,1)(8,1) |
| 1,2,4,6,8 | (0,0)(1,1)(2,1)(3,1) |
| 1,2,4,6,8,2 | (0,0)(1,1)(2,1)(3,1)(1,0) |
| 1,2,4,6,8,2,2 | (0,0)(1,1)(2,1)(3,1)(1,0)(1,0) |
| 1,2,4,6,8,2,3 | (0,0)(1,1)(2,1)(3,1)(1,0)(2,0) |
| 1,2,4,6,8,2,4 | (0,0)(1,1)(2,1)(3,1)(1,0)(2,1) |
| 1,2,4,6,8,2,4,5 | (0,0)(1,1)(2,1)(3,1)(1,0)(2,1)(3,0) |
| 1,2,4,6,8,2,4,5,7 | (0,0)(1,1)(2,1)(3,1)<br>(1,0)(2,1)(3,0)(4,1) |
| 1,2,4,6,8,2,4,6 | (0,0)(1,1)(2,1)(3,1)(1,0)(2,1)(3,1) |
| 1,2,4,6,8,2,4,6,7 | (0,0)(1,1)(2,1)(3,1)<br>(1,0)(2,1)(3,1)(4,0) |
| 1,2,4,6,8,2,4,6,7,9 | (0,0)(1,1)(2,1)(3,1)(1,0)<br>(2,1)(3,1)(4,0)(5,1) |
| 1,2,4,6,8,2,<br>4,6,7,9,11 | (0,0)(1,1)(2,1)(3,1)(1,0)<br>(2,1)(3,1)(4,0)(5,1)(6,1) |
| 1,2,4,6,8,2,4,6,8 | (0,0)(1,1)(2,1)(3,1)<br>(1,0)(2,1)(3,1)(4,1) |
| 1,2,4,6,8,2,4,6,8,2 | (0,0)(1,1)(2,1)(3,1)(1,0)<br>(2,1)(3,1)(4,1)(1,0) |
| 1,2,4,6,8,2,<br>4,6,8,2,4,6,8 | (0,0)(1,1)(2,1)(3,1)(1,0)(2,1)<br>(3,1)(4,1)(1,0)(2,1)(3,1)(4,1) |
| 1,2,4,6,8,3 | (0,0)(1,1)(2,1)(3,1)(1,0)<br>(2,1)(3,1)(4,1)(2,0) |
| 1,2,4,6,8,3,5,7,9 | (0,0)(1,1)(2,1)(3,1)(1,0)(2,1)<br>(3,1)(4,1)(2,0)(3,1)(4,1)(5,1) |
| 1,2,4,6,8,4 | (0,0)(1,1)(2,1)(3,1)(1,1) |
| 1,2,4,6,8,4,2 | (0,0)(1,1)(2,1)(3,1)(1,1)(1,0) |
| 1,2,4,6,8,4,2,4 | (0,0)(1,1)(2,1)(3,1)(1,1)(1,0)(2,1) |
| 1,2,4,6,8,4,2,4,6 | (0,0)(1,1)(2,1)(3,1)<br>(1,1)(1,0)(2,1)(3,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,6,8,4,2,4,6,8 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(1,0)(2,1)(3,1)(4,1) |
| 1,2,4,6,8,4,<br>2,4,6,8,3 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(1,0)(2,1)(3,1)(4,1)<br>(2,0)(3,1)(4,1)(5,1) |
| 1,2,4,6,8,4,<br>2,4,6,8,4 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(1,0)(2,1)(3,1)(4,1)(2,1) |
| 1,2,4,6,8,4,4 | (0,0)(1,1)(2,1)(3,1)(1,1)(1,1) |
| 1,2,4,6,8,4,5 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,0) |
| 1,2,4,6,8,4,5,7,9,11 | (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,0)(3,1)(4,1)(5,1) |
| 1,2,4,6,8,4,<br>5,7,9,11,7 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,0)(3,1)(4,1)(5,1)(3,1) |
| 1,2,4,6,8,4,6 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1) |
| 1,2,4,6,8,4,6,7 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)(3,0) |
| 1,2,4,6,8,4,6,7,9 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)(3,0)(4,1) |
| 1,2,4,6,8,4,6,7,9,11 | (0,0)(1,1)(2,1)(3,1)<br>(1,1)(2,1)(3,0)(4,1)(5,1) |
| 1,2,4,6,8,4,<br>6,7,9,11,13 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1) |
| 1,2,4,6,8,4,<br>6,7,9,11,13,5 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1)(2,0) |
| 1,2,4,6,8,4,6,7,9,<br>11,13,5,4,6,7,9,11,13 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(2,0)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1) |
| 1,2,4,6,8,4,<br>6,7,9,11,13,5,5 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(2,0)(2,0) |
| 1,2,4,6,8,4,<br>6,7,9,11,13,5,6 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(2,0)(3,0) |
| 1,2,4,6,8,4,6,<br>7,9,11,13,5,7 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(2,0)(3,1) |
| 1,2,4,6,8,4,6,7,<br>9,11,13,5,7,9,11 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)(3,0)<br>(4,1)(5,1)(6,1)(2,0)(3,1)(4,1)(5,1) |
| 1,2,4,6,8,4,<br>6,7,9,11,13,6 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1)(2,1) |
| 1,2,4,6,8,4,6,7,<br>9,11,13,6,7,9,11,13 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1) |
| 1,2,4,6,8,4,<br>6,7,9,11,13,7 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1)(3,0) |
| 1,2,4,6,8,4,6,7,<br>9,11,13,7,9,11,13 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)(3,0)<br>(4,1)(5,1)(6,1)(3,0)(4,1)(5,1)(6,1) |
| 1,2,4,6,8,4,<br>6,7,9,11,13,9 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1)(4,1) |
| 1,2,4,6,8,4,6,7,<br>9,11,13,9,10,12,14,16 | (0,0)(1,1)(2,1)(3,1)(1,1)<br>(2,1)(3,0)(4,1)(5,1)(6,1)(4,1)<br>(5,1)(6,0)(7,1)(8,1)(9,1) |
| 1,2,4,6,8,4,6,8 | (0,0)(1,1)(2,1)(3,1)(1,1)(2,1)(3,1) |
| 1,2,4,6,8,5 | (0,0)(1,1)(2,1)(3,1)(2,0) |
| 1,2,4,6,8,5,7,9,11 | (0,0)(1,1)(2,1)(3,1)(2,0)(3,1)(4,1)(5,1) |
| 1,2,4,6,8,6 | (0,0)(1,1)(2,1)(3,1)(2,1) |
| 1,2,4,6,8,6,7 | (0,0)(1,1)(2,1)(3,1)(2,1)(3,0) |
| 1,2,4,6,8,6,7,9 | (0,0)(1,1)(2,1)(3,1)(2,1)(3,0)(4,1) |
| 1,2,4,6,8,6,7,9,11,13 | (0,0)(1,1)(2,1)(3,1)(2,1)<br>(3,0)(4,1)(5,1)(6,1) |
| 1,2,4,6,8,6,8 | (0,0)(1,1)(2,1)(3,1)(2,1)(3,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,6,8,6,8,6,8 | (0,0)(1,1)(2,1)(3,1)<br>(2,1)(3,1)(2,1)(3,1) |
| 1,2,4,6,8,7 | (0,0)(1,1)(2,1)(3,1)(3,0) |
| 1,2,4,6,8,7,5 | (0,0)(1,1)(2,1)(3,1)(3,0)(2,0) |
| 1,2,4,6,8,7,6 | (0,0)(1,1)(2,1)(3,1)(3,0)(2,1) |
| 1,2,4,6,8,7,6,7 | (0,0)(1,1)(2,1)(3,1)(3,0)(2,1)(3,0) |
| 1,2,4,6,8,7,6,8 | (0,0)(1,1)(2,1)(3,1)(3,0)(2,1)(3,1) |
| 1,2,4,6,8,7,6,8,7 | (0,0)(1,1)(2,1)(3,1)<br>(3,0)(2,1)(3,1)(3,0) |
| 1,2,4,6,8,7,7 | (0,0)(1,1)(2,1)(3,1)(3,0)(3,0) |
| 1,2,4,6,8,7,8 | (0,0)(1,1)(2,1)(3,1)(3,0)(4,0) |
| 1,2,4,6,8,7,9 | (0,0)(1,1)(2,1)(3,1)(3,0)(4,1) |
| 1,2,4,6,8,7,9,11,13 | (0,0)(1,1)(2,1)(3,1)<br>(3,0)(4,1)(5,1)(6,1) |
| 1,2,4,6,8,8 | (0,0)(1,1)(2,1)(3,1)(3,1) |
| 1,2,4,6,8,8,6 | (0,0)(1,1)(2,1)(3,1)(3,1)(2,1) |
| 1,2,4,6,8,8,6,8 | (0,0)(1,1)(2,1)(3,1)(3,1)(2,1)(3,1) |
| 1,2,4,6,8,8,6,8,7 | (0,0)(1,1)(2,1)(3,1)<br>(3,1)(2,1)(3,1)(3,0) |
| 1,2,4,6,8,8,6,8,8 | (0,0)(1,1)(2,1)(3,1)<br>(3,1)(2,1)(3,1)(3,1) |
| 1,2,4,6,8,8,7 | (0,0)(1,1)(2,1)(3,1)(3,1)(3,0) |
| 1,2,4,6,8,8,8 | (0,0)(1,1)(2,1)(3,1)(3,1)(3,1) |
| 1,2,4,6,8,8,8,6,8,8,8 | (0,0)(1,1)(2,1)(3,1)(3,1)<br>(3,1)(2,1)(3,1)(3,1)(3,1) |
| 1,2,4,6,8,8,8,7 | (0,0)(1,1)(2,1)(3,1)(3,1)(3,1)(3,0) |
| 1,2,4,6,8,8,8,8 | (0,0)(1,1)(2,1)(3,1)(3,1)(3,1)(3,1) |
| 1,2,4,6,8,9 | (0,0)(1,1)(2,1)(3,1)(4,0) |
| 1,2,4,6,8,9,8 | (0,0)(1,1)(2,1)(3,1)(4,0)(3,1) |
| 1,2,4,6,8,9,8,9 | (0,0)(1,1)(2,1)(3,1)(4,0)(3,1)(4,0) |
| 1,2,4,6,8,9,9 | (0,0)(1,1)(2,1)(3,1)(4,0)(4,0) |
| 1,2,4,6,8,9,11 | (0,0)(1,1)(2,1)(3,1)(4,0)(5,1) |
| 1,2,4,6,8,9,11,13,15 | (0,0)(1,1)(2,1)(3,1)<br>(4,0)(5,1)(6,1)(7,1) |
| 1,2,4,6,8,9,11,13,15,16 | (0,0)(1,1)(2,1)(3,1)(4,0)<br>(5,1)(6,1)(7,1)(8,0) |
| 1,2,4,6,8,9,<br>11,13,15,16,18 | (0,0)(1,1)(2,1)(3,1)(4,0)<br>(5,1)(6,1)(7,1)(8,0)(9,1) |
| 1,2,4,6,8,10 | (0,0)(1,1)(2,1)(3,1)(4,1) |
| 1,2,4,6,8,10,6 | (0,0)(1,1)(2,1)(3,1)(4,1)(2,1) |
| 1,2,4,6,8,10,8 | (0,0)(1,1)(2,1)(3,1)(4,1)(3,1) |
| 1,2,4,6,8,10,8,10 | (0,0)(1,1)(2,1)(3,1)(4,1)(3,1)(4,1) |
| 1,2,4,6,8,10,9 | (0,0)(1,1)(2,1)(3,1)(4,1)(4,0) |
| 1,2,4,6,8,10,10 | (0,0)(1,1)(2,1)(3,1)(4,1)(4,1) |
| 1,2,4,6,8,10,10,10 | (0,0)(1,1)(2,1)(3,1)(4,1)(4,1)(4,1) |
| 1,2,4,6,8,10,11 | (0,0)(1,1)(2,1)(3,1)(4,1)(5,0) |
| 1,2,4,6,8,10,12 | (0,0)(1,1)(2,1)(3,1)(4,1)(5,1) |
| 1,2,4,6,8,10,12,14 | (0,0)(1,1)(2,1)(3,1)(4,1)(5,1)(6,1) |
| 1,2,4,7 | (0,0)(1,1)(2,2) |
| 1,2,4,7,2 | (0,0)(1,1)(2,2)(1,0) |
| 1,2,4,7,2,4,7 | (0,0)(1,1)(2,2)(1,0)(2,1)(3,2) |
| 1,2,4,7,4 | (0,0)(1,1)(2,2)(1,1) |
| 1,2,4,7,4,6 | (0,0)(1,1)(2,2)(1,1)(2,1) |
| 1,2,4,7,4,6,8 | (0,0)(1,1)(2,2)(1,1)(2,1)(3,1) |
| 1,2,4,7,4,7 | (0,0)(1,1)(2,2)(1,1)(2,2) |

| Y sequence | BMS |
|---|---|
| 1,2,4,7,4,7,4,7 | (0,0)(1,1)(2,2)(1,1)(2,2)(1,1)(2,2) |
| 1,2,4,7,5 | (0,0)(1,1)(2,2)(2,0) |
| 1,2,4,7,5,5 | (0,0)(1,1)(2,2)(2,0)(2,0) |
| 1,2,4,7,5,6 | (0,0)(1,1)(2,2)(2,0)(3,0) |
| 1,2,4,7,5,7 | (0,0)(1,1)(2,2)(2,0)(3,1) |
| 1,2,4,7,5,7,10 | (0,0)(1,1)(2,2)(2,0)(3,1)(4,2) |
| 1,2,4,7,6 | (0,0)(1,1)(2,2)(2,1) |
| 1,2,4,7,6,7 | (0,0)(1,1)(2,2)(2,1)(3,0) |
| 1,2,4,7,6,8 | (0,0)(1,1)(2,2)(2,1)(3,1) |
| 1,2,4,7,6,9 | (0,0)(1,1)(2,2)(2,1)(3,2) |
| 1,2,4,7,6,9,8 | (0,0)(1,1)(2,2)(2,1)(3,2)(3,1) |
| 1,2,4,7,6,9,8,11 | (0,0)(1,1)(2,2)(2,1)(3,2)(3,1)(4,2) |
| 1,2,4,7,7 | (0,0)(1,1)(2,2)(2,2) |
| 1,2,4,7,7,4,7,7 | (0,0)(1,1)(2,2)(2,2)(1,1)(2,2)(2,2) |
| 1,2,4,7,7,5 | (0,0)(1,1)(2,2)(2,2)(2,0) |
| 1,2,4,7,7,6 | (0,0)(1,1)(2,2)(2,2)(2,1) |
| 1,2,4,7,7,6,9,9 | (0,0)(1,1)(2,2)(2,2)(2,1)(3,2)(3,2) |
| 1,2,4,7,7,7 | (0,0)(1,1)(2,2)(2,2)(2,2) |
| 1,2,4,7,8 | (0,0)(1,1)(2,2)(3,0) |
| 1,2,4,7,8,10 | (0,0)(1,1)(2,2)(3,0)(4,1) |
| 1,2,4,7,8,10,13 | (0,0)(1,1)(2,2)(3,0)(4,1)(5,2) |
| 1,2,4,7,9 | (0,0)(1,1)(2,2)(3,1) |
| 1,2,4,7,9,7,9 | (0,0)(1,1)(2,2)(3,1)(2,2)(3,1) |
| 1,2,4,7,9,8 | (0,0)(1,1)(2,2)(3,1)(3,0) |
| 1,2,4,7,9,9 | (0,0)(1,1)(2,2)(3,1)(3,1) |
| 1,2,4,7,9,12 | (0,0)(1,1)(2,2)(3,1)(4,2) |
| 1,2,4,7,10 | (0,0)(1,1)(2,2)(3,2) |
| 1,2,4,7,10,10 | (0,0)(1,1)(2,2)(3,2)(3,2) |
| 1,2,4,7,10,11 | (0,0)(1,1)(2,2)(3,2)(4,0) |
| 1,2,4,7,10,12 | (0,0)(1,1)(2,2)(3,2)(4,1) |
| 1,2,4,7,10,12,7,10,12 | (0,0)(1,1)(2,2)(3,2)<br>(4,1)(2,2)(3,2)(4,1) |
| 1,2,4,7,10,12,8 | (0,0)(1,1)(2,2)(3,2)(4,1)(3,0) |
| 1,2,4,7,10,12,8,10 | (0,0)(1,1)(2,2)(3,2)(4,1)(3,0)(4,1) |
| 1,2,4,7,10,12,9 | (0,0)(1,1)(2,2)(3,2)(4,1)(3,1) |
| 1,2,4,7,10,12,9,12,15,17 | (0,0)(1,1)(2,2)(3,2)(4,1)<br>(3,1)(4,2)(5,2)(6,1)(5,1) |
| 1,2,4,7,10,12,10 | (0,0)(1,1)(2,2)(3,2)(4,1)(3,2) |
| 1,2,4,7,10,12,10,12 | (0,0)(1,1)(2,2)(3,2)(4,1)(3,2)(4,1) |
| 1,2,4,7,10,12,11 | (0,0)(1,1)(2,2)(3,2)(4,1)(4,0) |
| 1,2,4,7,10,12,12 | (0,0)(1,1)(2,2)(3,2)(4,1)(4,1) |
| 1,2,4,7,10,12,13 | (0,0)(1,1)(2,2)(3,2)(4,1)(5,0) |
| 1,2,4,7,10,12,14 | (0,0)(1,1)(2,2)(3,2)(4,1)(5,1) |
| 1,2,4,7,10,12,15 | (0,0)(1,1)(2,2)(3,2)(4,1)(5,2) |
| 1,2,4,7,10,12,15,18 | (0,0)(1,1)(2,2)(3,2)(4,1)(5,2)(6,2) |
| 1,2,4,7,10,12,15,18,20 | (0,0)(1,1)(2,2)(3,2)<br>(4,1)(5,2)(6,2)(7,1) |
| 1,2,4,7,10,13 | (0,0)(1,1)(2,2)(3,2)(4,2) |
| 1,2,4,7,10,13<br>,1,2,4,7,10,13 | (0,0)(1,1)(2,2)(3,2)(4,2)<br>(0,0)(1,1)(2,2)(3,2)(4,2) |
| 1,2,4,7,10,13,4,7,10,13 | (0,0)(1,1)(2,2)(3,2)(4,2)<br>(1,1)(2,2)(3,2)(4,2) |
| 1,2,4,7,10,13,6,9,12,15 | (0,0)(1,1)(2,2)(3,2)(4,2)<br>(2,1)(3,2)(4,2)(5,2) |

| Y sequence | BMS |
|---|---|
| 1,2,4,7,10,13,7,10,13 | (0,0)(1,1)(2,2)(3,2)<br>(4,2)(2,2)(3,2)(4,2) |
| 1,2,4,7,10,13,10 | (0,0)(1,1)(2,2)(3,2)(4,2)(3,2) |
| 1,2,4,7,10,13,16 | (0,0)(1,1)(2,2)(3,2)(4,2)(5,2) |
| 1,2,4,7,11 | (0,0)(1,1)(2,2)(3,3) |
| 1,2,4,7,11,4,7,11 | (0,0)(1,1)(2,2)(3,3)(1,1)(2,2)(3,3) |
| 1,2,4,7,11,6,9,13 | (0,0)(1,1)(2,2)(3,3)(2,1)(3,2)(4,3) |
| 1,2,4,7,11,7 | (0,0)(1,1)(2,2)(3,3)(2,2) |
| 1,2,4,7,11,7,10,13 | (0,0)(1,1)(2,2)(3,3)(2,2)(3,2)(4,2) |
| 1,2,4,7,11,7,11 | (0,0)(1,1)(2,2)(3,3)(2,2)(3,3) |
| 1,2,4,7,11,10,14 | (0,0)(1,1)(2,2)(3,3)(3,2)(4,3) |
| 1,2,4,7,11,11 | (0,0)(1,1)(2,2)(3,3)(3,3) |
| 1,2,4,7,11,12 | (0,0)(1,1)(2,2)(3,3)(4,0) |
| 1,2,4,7,11,15 | (0,0)(1,1)(2,2)(3,3)(4,3) |
| 1,2,4,7,11,15,19 | (0,0)(1,1)(2,2)(3,3)(4,3)(5,3) |
| 1,2,4,7,11,16 | (0,0)(1,1)(2,2)(3,3)(4,4) |
| 1,2,4,7,11,16,21,26 | (0,0)(1,1)(2,2)(3,3)(4,4)(5,4)(6,4) |
| 1,2,4,7,11,16,22 | (0,0)(1,1)(2,2)(3,3)(4,4)(5,5) |
| 1,2,4,8 | (0,0,0)(1,1,1) |
| 1,2,4,8,1 | (0,0,0)(1,1,1)(0,0,0) |
| 1,2,4,8,2 | (0,0,0)(1,1,1)(1,0,0) |
| 1,2,4,8,2,4,8 | (0,0,0)(1,1,1)(1,0,0)(2,1,1) |
| 1,2,4,8,3 | (0,0,0)(1,1,1)(1,0,0)(2,1,1)(2,0,0) |
| 1,2,4,8,4 | (0,0,0)(1,1,1)(1,1,0) |
| 1,2,4,8,4,6,8 | (0,0,0)(1,1,1)(1,1,0)(2,1,0)(3,1,0) |
| 1,2,4,8,4,7 | (0,0,0)(1,1,1)(1,1,0)(2,2,0) |
| 1,2,4,8,4,7,11 | (0,0,0)(1,1,1)(1,1,0)(2,2,0)(3,3,0) |
| 1,2,4,8,4,8 | (0,0,0)(1,1,1)(1,1,0)(2,2,1) |
| 1,2,4,8,5 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)(2,0,0) |
| 1,2,4,8,5,4,8,5 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,0,0)(1,1,0)(2,2,1)(2,0,0) |
| 1,2,4,8,5,5 | (0,0,0)(1,1,1)(1,1,0)<br>(2,2,1)(2,0,0)(2,0,0) |
| 1,2,4,8,5,5,4,8,5 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)(2,0,0)<br>(2,0,0)(1,1,0)(2,2,1)(2,0,0) |
| 1,2,4,8,6 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)(2,1,0) |
| 1,2,4,8,6,9 | (0,0,0)(1,1,1)(1,1,0)<br>(2,2,1)(2,1,0)(3,2,0) |
| 1,2,4,8,6,10 | (0,0,0)(1,1,1)(1,1,0)<br>(2,2,1)(2,1,0)(3,2,1) |
| 1,2,4,8,6,10,7 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,1,0)(3,2,1)(3,0,0) |
| 1,2,4,8,6,10,8 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,1,0)(3,2,1)(3,1,0) |
| 1,2,4,8,6,10,8,12 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,1,0)(3,2,1)(3,1,0)(4,2,1) |
| 1,2,4,8,7 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)(2,2,0) |
| 1,2,4,8,7,12 | (0,0,0)(1,1,1)(1,1,0)<br>(2,2,1)(2,2,0)(3,3,1) |
| 1,2,4,8,7,12,8 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,2,0)(3,3,1)(3,0,0) |
| 1,2,4,8,7,12,11 | (0,0,0)(1,1,1)(1,1,0)(2,2,1)<br>(2,2,0)(3,3,1)(3,3,0) |
| 1,2,4,8,8 | (0,0,0)(1,1,1)(1,1,1) |
| 1,2,4,8,8,4 | (0,0,0)(1,1,1)(1,1,1)(1,1,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,8,7 | (0,0,0)(1,1,1)(1,1,1)(1,1,0)<br>(2,2,1)(2,2,1)(2,2,0) |
| 1,2,4,8,8,8 | (0,0,0)(1,1,1)(1,1,1)(1,1,1) |
| 1,2,4,8,9 | (0,0,0)(1,1,1)(2,0,0) |
| 1,2,4,8,9,4,8,9 | (0,0,0)(1,1,1)(2,0,0)<br>(1,1,0)(2,2,1)(3,0,0) |
| 1,2,4,8,9,8 | (0,0,0)(1,1,1)(2,0,0)(1,1,1) |
| 1,2,4,8,9,8,9 | (0,0,0)(1,1,1)(2,0,0)(1,1,1)(2,0,0) |
| 1,2,4,8,9,9 | (0,0,0)(1,1,1)(2,0,0)(2,0,0) |
| 1,2,4,8,9,11 | (0,0,0)(1,1,1)(2,0,0)(3,1,0) |
| 1,2,4,8,9,11,14 | (0,0,0)(1,1,1)(2,0,0)(3,1,0)(4,2,0) |
| 1,2,4,8,9,11,15 | (0,0,0)(1,1,1)(2,0,0)(3,1,1) |
| 1,2,4,8,10 | (0,0,0)(1,1,1)(2,1,0) |
| 1,2,4,8,10,4,8,10 | (0,0,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,2,1)(3,1,0) |
| 1,2,4,8,10,7 | (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,0) |
| 1,2,4,8,10,7,12,14 | (0,0,0)(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,1,0)(2,2,0)(3,3,1)(4,1,0) |
| 1,2,4,8,10,8 | (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(2,2,1) |
| 1,2,4,8,10,10 | (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(3,1,0) |
| 1,2,4,8,10,12 | (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,1,0) |
| 1,2,4,8,10,13 | (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,0) |
| 1,2,4,8,10,14 | (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,1,0)(4,2,1) |
| 1,2,4,8,11 | (0,0,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,2,1)(3,2,0) |
| 1,2,4,8,11,7,12,16 | (0,0,0)(1,1,1)(2,1,0)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,0)(3,3,1)(4,3,0) |
| 1,2,4,8,11,8 | (0,0,0)(1,1,1)(2,1,0)(1,1,1) |
| 1,2,4,8,11,8,4 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0) |
| 1,2,4,8,11,8,4,8 | (0,0,0)(1,1,1)(2,1,0)<br>(1,1,1)(1,1,0)(2,2,1) |
| 1,2,4,8,11,8,4,8,8 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(2,2,1) |
| 1,2,4,8,11,8,4,8,10 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,1,0) |
| 1,2,4,8,11,8,4,8,10,8 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,1,0)(2,2,1) |
| 1,2,4,8,11,8,4,8,11 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0) |
| 1,2,4,8,11,8,<br>4,8,11,7,12,15,12 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,0)<br>(3,3,1)(4,2,0)(3,3,1) |
| 1,2,4,8,11,8,4,8,11,8 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1) |
| 1,2,4,8,11,8,7 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,0)<br>(2,2,1)(3,2,0)(2,2,1)(2,2,0) |
| 1,2,4,8,11,8,8 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)(1,1,1) |
| 1,2,4,8,11,8,10 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)(2,1,0) |
| 1,2,4,8,11,8,11 | (0,0,0)(1,1,1)(2,1,0)(1,1,1)(2,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1)(3,2,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,11,8,11,8 | (0,0,0)(1,1,1)(2,1,0)<br>(1,1,1)(2,1,0)(1,1,1) |
| 1,2,4,8,11,9 | (0,0,0)(1,1,1)(2,1,0)(2,0,0) |
| 1,2,4,8,11,9,8 | (0,0,0)(1,1,1)(2,1,0)(2,0,0)(1,1,1) |
| 1,2,4,8,11,10 | (0,0,0)(1,1,1)(2,1,0)(2,1,0) |
| 1,2,4,8,11,11 | (0,0,0)(1,1,1)(2,1,0)(2,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(3,2,0) |
| 1,2,4,8,11,11,8 | (0,0,0)(1,1,1)(2,1,0)(2,1,0)(1,1,1) |
| 1,2,4,8,11,11,8,11,8 | (0,0,0)(1,1,1)(2,1,0)(2,1,0)<br>(1,1,1)(2,1,0)(1,1,1) |
| 1,2,4,8,11,11,8,11,9 | (0,0,0)(1,1,1)(2,1,0)(2,1,0)<br>(1,1,1)(2,1,0)(2,0,0) |
| 1,2,4,8,11,11,8,11,11,8 | (0,0,0)(1,1,1)(2,1,0)(2,1,0)<br>(1,1,1)(2,1,0)(2,1,0)(1,1,1) |
| 1,2,4,8,11,11,9 | (0,0,0)(1,1,1)(2,1,0)(2,1,0)(2,0,0) |
| 1,2,4,8,11,11,11,8 | (0,0,0)(1,1,1)(2,1,0)<br>(2,1,0)(2,1,0)(1,1,1) |
| 1,2,4,8,11,12 | (0,0,0)(1,1,1)(2,1,0)(3,0,0) |
| 1,2,4,8,11,13 | (0,0,0)(1,1,1)(2,1,0)(3,1,0) |
| 1,2,4,8,11,14 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,2,0) |
| 1,2,4,8,11,14,8 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)(1,1,1) |
| 1,2,4,8,11,14,9 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,11,14,11,8 | (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(2,1,0)(1,1,1) |
| 1,2,4,8,11,14,11,14,8 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(2,1,0)(3,1,0)(1,1,1) |
| 1,2,4,8,11,14,12 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)(3,0,0) |
| 1,2,4,8,11,14,14,8 | (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(3,1,0)(1,1,1) |
| 1,2,4,8,11,14,14,14,8 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(3,1,0)(3,1,0)(1,1,1) |
| 1,2,4,8,11,14,15 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)(4,0,0) |
| 1,2,4,8,11,14,17,8 | (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(4,1,0)(1,1,1) |
| 1,2,4,8,11,14,17,14,8 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(4,1,0)(3,1,0)(1,1,1) |
| 1,2,4,8,11,14,17,14,17,8 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(4,1,0)(3,1,0)(4,1,0)(1,1,1) |
| 1,2,4,8,11,14,17,14,17,12 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(4,1,0)(3,1,0)(4,1,0)(3,0,0) |
| 1,2,4,8,11,14,17,17,8 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(4,1,0)(4,1,0)(1,1,1) |
| 1,2,4,8,11,14,17,18 | (0,0,0)(1,1,1)(2,1,0)<br>(3,1,0)(4,1,0)(5,0,0) |
| 1,2,4,8,11,14,17,20,8 | (0,0,0)(1,1,1)(2,1,0)(3,1,0)<br>(4,1,0)(5,1,0)(1,1,1) |
| 1,2,4,8,11,15 | (0,0,0)(1,1,1)(2,1,0)(3,2,0) |
| 1,2,4,8,11,15,4,8 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(1,1,0)(2,2,1) |
| 1,2,4,8,11,15,4,8,9,11 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,0,0)(4,1,0) |
| 1,2,4,8,11,15,4,<br>8,9,11,6,10,11,13 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,0,0)(4,1,0)<br>(2,1,0)(3,2,1)(4,0,0)(5,1,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,11,15,<br>4,8,9,11,7 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)(1,1,0)<br>(2,2,1)(3,0,0)(4,1,0)(2,2,0) |
| 1,2,4,8,11,15,<br>4,8,9,11,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)(1,1,0)<br>(2,2,1)(3,0,0)(4,1,0)(2,2,1) |
| 1,2,4,8,11,15,<br>4,8,9,11,15 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,0,0)(4,1,1) |
| 1,2,4,8,11,15,4,8,10 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,1,0) |
| 1,2,4,8,11,15,4,8,11,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1) |
| 1,2,4,8,11,15,4,8,11,15 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0) |
| 1,2,4,8,11,15,7,12,16,21 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0)<br>(2,2,0)(3,3,1)(4,3,0)(5,4,0) |
| 1,2,4,8,11,15,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)(1,1,1) |
| 1,2,4,8,11,15,8,11,15 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(1,1,1)(2,1,0)(3,2,0) |
| 1,2,4,8,11,15,11,15 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(2,1,0)(3,2,0) |
| 1,2,4,8,11,15,13,17 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(3,1,0)(4,2,0) |
| 1,2,4,8,11,15,15 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)(3,2,0) |
| 1,2,4,8,11,15,16 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,0,0) |
| 1,2,4,8,11,15,18,8 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,1,0)(1,1,1) |
| 1,2,4,8,11,15,19 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,2,0) |
| 1,2,4,8,11,15,19,20 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(5,0,0) |
| 1,2,4,8,11,15,19,22,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,2,0)(5,1,0)(1,1,1) |
| 1,2,4,8,11,15,19,23 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,2,0)(5,2,0) |
| 1,2,4,8,11,15,20 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)(4,3,0) |
| 1,2,4,8,11,15,20,20 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(4,3,0) |
| 1,2,4,8,11,15,20,21 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,0,0) |
| 1,2,4,8,11,15,20,25 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,3,0) |
| 1,2,4,8,11,15,20,25,30 | (0,0,0)(1,1,1)(2,1,0)(3,2,0)<br>(4,3,0)(5,3,0)(6,3,0) |
| 1,2,4,8,11,15,20,26 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,0)(4,3,0)(5,4,0) |
| 1,2,4,8,11,16 | (0,0,0)(1,1,1)(2,1,0)(3,2,1) |
| 1,2,4,8,11,16,4,8 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(1,1,0)(2,2,1) |
| 1,2,4,8,11,16,4,8,11,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(2,2,1) |
| 1,2,4,8,11,16,<br>4,8,11,11,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(3,2,0)(2,2,1) |
| 1,2,4,8,11,16,4,8,11,15 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,0) |
| 1,2,4,8,11,16,<br>4,8,11,15,20 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(1,1,0)<br>(2,2,1)(3,2,0)(4,3,0)(5,4,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,11,16,4,8,11,16 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,1) |
| 1,2,4,8,11,16,7,12,15 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,1)<br>(2,2,0)(3,3,1)(4,2,0) |
| 1,2,4,8,11,16,7,12,16,21 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,1)<br>(2,2,0)(3,3,1)(4,3,0)(5,4,0) |
| 1,2,4,8,11,16,7,12,16,22 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,0)(2,2,1)(3,2,0)(4,3,1)<br>(2,2,0)(3,3,1)(4,3,0)(5,4,1) |
| 1,2,4,8,11,16,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(1,1,1) |
| 1,2,4,8,11,16,8,8 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(1,1,1)(1,1,1) |
| 1,2,4,8,11,16,8,10 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(1,1,1)(2,1,0) |
| 1,2,4,8,11,16,8,11 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(1,1,0)(2,2,1)<br>(3,2,0)(4,3,1)(2,2,1)(3,2,0) |
| 1,2,4,8,11,16,8,11,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(1,1,1) |
| 1,2,4,8,11,16,8,11,11,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(2,1,0)(1,1,1) |
| 1,2,4,8,11,16,8,11,12 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(3,0,0) |
| 1,2,4,8,11,16,8,11,15 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(3,2,0) |
| 1,2,4,8,11,16,8,11,16 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(3,2,1) |
| 1,2,4,8,11,16,8,11,16,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(1,1,1)(2,1,0)(3,2,1)(1,1,1) |
| 1,2,4,8,11,16,<br>8,11,16,8,10 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(1,1,1)<br>(2,1,0)(3,2,1)(1,1,1)(2,1,0) |
| 1,2,4,8,11,16,<br>8,11,16,8,11,15 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(1,1,1)<br>(2,1,0)(3,2,1)(1,1,1)(2,1,0)(3,2,0) |
| 1,2,4,8,11,16,<br>8,11,16,8,11,16 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(1,1,1)<br>(2,1,0)(3,2,1)(1,1,1)(2,1,0)(3,2,1) |
| 1,2,4,8,11,16,9 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(2,0,0) |
| 1,2,4,8,11,16,10 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(2,1,0) |
| 1,2,4,8,11,16,11,8 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(2,1,0)(1,1,1) |
| 1,2,4,8,11,16,11,15 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(2,1,0)(3,2,0) |
| 1,2,4,8,11,16,11,15,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(2,1,0)(3,2,0)(1,1,1) |
| 1,2,4,8,11,16,11,16 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(2,1,0)(3,2,1) |
| 1,2,4,8,11,16,14,19 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(3,1,0)(4,2,1) |
| 1,2,4,8,11,16,15 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(3,2,0) |
| 1,2,4,8,11,16,15,19,23 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(3,2,0)(4,2,0)(5,2,0) |
| 1,2,4,8,11,16,15,21 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(3,2,0)(4,3,1) |
| 1,2,4,8,11,16,15,21,20 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(3,2,0)(4,3,1)(4,3,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,11,16,16 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(3,2,1) |
| 1,2,4,8,11,16,17 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,0,0) |
| 1,2,4,8,11,16,17,15,21,22 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,0,0)(3,2,0)(4,3,1)(5,0,0) |
| 1,2,4,8,11,16,17,16 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,0,0)(3,2,1) |
| 1,2,4,8,11,16,17,18 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,0,0)(5,0,0) |
| 1,2,4,8,11,16,17,19 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,0,0)(5,1,0) |
| 1,2,4,8,11,16,17,19,23 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,0,0)(5,1,1) |
| 1,2,4,8,11,16,18 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0) |
| 1,2,4,8,11,16,19,8 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(1,1,1) |
| 1,2,4,8,11,16,19,8,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(1,1,1)(1,1,1) |
| 1,2,4,8,11,16,<br>19,8,9,11,15,18,23 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0)<br>(1,1,1)(2,0,0)(3,1,1)(4,1,0)(5,2,1) |
| 1,2,4,8,11,16,19,8,10 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(1,1,1)(2,1,0) |
| 1,2,4,8,11,16,19,8,11,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(1,1,1)(2,1,0)(1,1,1) |
| 1,2,4,8,11,16,19,8,11,11,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0)<br>(1,1,1)(2,1,0)(2,1,0)(1,1,1) |
| 1,2,4,8,11,16,19,8,11,15 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(1,1,1)(2,1,0)(3,2,0) |
| 1,2,4,8,11,16,19,8,11,16 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(1,1,1)(2,1,0)(3,2,1) |
| 1,2,4,8,11,16,<br>19,8,11,16,19,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,1,0)(1,1,1) |
| 1,2,4,8,11,16,19,9 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(2,0,0) |
| 1,2,4,8,11,16,19,10 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(2,1,0) |
| 1,2,4,8,11,16,19,11,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(2,1,0)(1,1,1) |
| 1,2,4,8,11,16,<br>19,11,8,11,16 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0)<br>(2,1,0)(1,1,1)(2,1,0)(3,2,1) |
| 1,2,4,8,11,16,19,11,9 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(2,1,0)(2,0,0) |
| 1,2,4,8,11,16,19,11,11,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(2,1,0)(2,1,0)(1,1,1) |
| 1,2,4,8,11,16,19,11,14,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(2,1,0)(3,1,0)(1,1,1) |
| 1,2,4,8,11,16,19,11,15 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(2,1,0)(3,2,0) |
| 1,2,4,8,11,16,19,11,16 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(2,1,0)(3,2,1) |
| 1,2,4,8,11,16,<br>19,11,16,19,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,1,0)<br>(2,1,0)(3,2,1)(4,1,0)(1,1,1) |
| 1,2,4,8,11,16,19,12 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(3,0,0) |
| 1,2,4,8,11,16,19,14,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,1,0)(1,1,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,11,16,19,14,17,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,1,0)(4,1,0)(1,1,1) |
| 1,2,4,8,11,16,19,14,18 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,1,0)(4,2,0) |
| 1,2,4,8,11,16,19,14,19 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,1,0)(4,2,1) |
| 1,2,4,8,11,16,19,15 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(3,2,0) |
| 1,2,4,8,11,16,19,15,21 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,2,0)(4,3,1) |
| 1,2,4,8,11,16,<br>19,15,21,24,20 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(3,2,0)(4,3,1)(5,1,0)(4,3,0) |
| 1,2,4,8,11,16,19,16 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(3,2,1) |
| 1,2,4,8,11,16,19,17 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(4,0,0) |
| 1,2,4,8,11,16,19,19,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(4,1,0)(1,1,1) |
| 1,2,4,8,11,16,19,22,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(5,1,0)(1,1,1) |
| 1,2,4,8,11,16,19,23 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(5,2,0) |
| 1,2,4,8,11,16,19,24 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,1,0)(5,2,1) |
| 1,2,4,8,11,16,19,24,27,8 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,1,0)(5,2,1)(6,1,0)(1,1,1) |
| 1,2,4,8,11,16,20 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)(4,2,0) |
| 1,2,4,8,11,16,20,15,21,26 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(3,2,0)(4,3,1)(5,3,0) |
| 1,2,4,8,11,16,20,16 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(3,2,1) |
| 1,2,4,8,11,16,20,24,16 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,2,0)(3,2,1) |
| 1,2,4,8,11,16,20,25 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,0) |
| 1,2,4,8,11,16,20,25,31 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,0)(6,4,0) |
| 1,2,4,8,11,16,20,26 | (0,0,0)(1,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,1) |
| 1,2,4,8,11,16,20,26,31,38 | (0,0,0)(1,1,1)(2,1,0)(3,2,1)<br>(4,2,0)(5,3,1)(6,3,0)(7,4,1) |
| 1,2,4,8,12 | (0,0,0)(1,1,1)(2,1,1) |
| 1,2,4,8,12,4,8 | (0,0,0)(1,1,1)(2,1,1)(1,1,0)(2,2,1) |
| 1,2,4,8,12,4,8,10 | (0,0,0)(1,1,1)(2,1,1)<br>(1,1,0)(2,2,1)(3,1,0) |
| 1,2,4,8,12,4,8,11 | (0,0,0)(1,1,1)(2,1,1)<br>(1,1,0)(2,2,1)(3,2,0) |
| 1,2,4,8,12,4,8,11,16 | (0,0,0)(1,1,1)(2,1,1)(1,1,0)<br>(2,2,1)(3,2,0)(4,3,1) |
| 1,2,4,8,12,4,8,12 | (0,0,0)(1,1,1)(2,1,1)<br>(1,1,0)(2,2,1)(3,2,1) |
| 1,2,4,8,12,8 | (0,0,0)(1,1,1)(2,1,1)(1,1,1) |
| 1,2,4,8,12,8,11,16 | (0,0,0)(1,1,1)(2,1,1)<br>(1,1,1)(2,1,0)(3,2,1) |
| 1,2,4,8,12,8,11,16,21 | (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,8,11,16,21,15 | (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(3,2,0) |
| 1,2,4,8,12,8,11,16,21,16 | (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(3,2,1) |
| 1,2,4,8,12,8,12 | (0,0,0)(1,1,1)(2,1,1)(1,1,1)(2,1,1) |
| 1,2,4,8,12,8,12,8,12 | (0,0,0)(1,1,1)(2,1,1)(1,1,1)<br>(2,1,1)(1,1,1)(2,1,1) |
| 1,2,4,8,12,9 | (0,0,0)(1,1,1)(2,1,1)(2,0,0) |
| 1,2,4,8,12,9,4,8,12,9 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)<br>(1,1,0)(2,2,1)(3,2,1)(3,0,0) |
| 1,2,4,8,12,9,8 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)(1,1,1) |
| 1,2,4,8,12,9,8,11,16 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)<br>(1,1,1)(2,1,0)(3,2,1) |
| 1,2,4,8,12,9,8,11,16,21 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1) |
| 1,2,4,8,12,9,<br>8,11,16,21,17 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)(4,0,0) |
| 1,2,4,8,12,9,8,11,<br>16,21,17,11,16,21,17 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)(4,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(4,0,0) |
| 1,2,4,8,12,9,8,11,<br>16,21,17,14,19,24,20 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)<br>(1,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(4,0,0)(3,1,0)(4,2,1)(5,2,1)(5,0,0) |
| 1,2,4,8,12,9,8,<br>11,16,21,17,15 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,0,0)(3,2,0) |
| 1,2,4,8,12,9,8,<br>11,16,21,17,16 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,0,0)(3,2,1) |
| 1,2,4,8,12,9,8,11,<br>16,21,17,16,20,26,32,27 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,0,0)(3,2,1)<br>(4,2,0)(5,3,1)(6,3,1)(6,0,0) |
| 1,2,4,8,12,9,8,12 | (0,0,0)(1,1,1)(2,1,1)<br>(2,0,0)(1,1,1)(2,1,1) |
| 1,2,4,8,12,9,8,12,9 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)<br>(1,1,1)(2,1,1)(2,0,0) |
| 1,2,4,8,12,9,9 | (0,0,0)(1,1,1)(2,1,1)(2,0,0)(2,0,0) |
| 1,2,4,8,12,10 | (0,0,0)(1,1,1)(2,1,1)(2,1,0) |
| 1,2,4,8,12,11,8 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1) |
| 1,2,4,8,12,11,<br>8,11,16,21,20 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,2,0) |
| 1,2,4,8,12,11,<br>8,11,16,21,20,16 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,2,0)(3,2,1) |
| 1,2,4,8,12,11,8,12 | (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1) |
| 1,2,4,8,12,11,9 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)(2,0,0) |
| 1,2,4,8,12,11,11,8,12 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(2,1,0)(1,1,1)(2,1,1) |
| 1,2,4,8,12,11,12 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)(3,0,0) |
| 1,2,4,8,12,11,14,8,12 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(3,1,0)(1,1,1)(2,1,1) |
| 1,2,4,8,12,11,14,17,8,12 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(3,1,0)(4,1,0)(1,1,1)(2,1,1) |
| 1,2,4,8,12,11,15 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)(3,2,0) |
| 1,2,4,8,12,11,15,20 | (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(3,2,0)(4,3,0) |
| 1,2,4,8,12,11,16 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)(3,2,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,11,16,20,16 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(3,2,1) |
| 1,2,4,8,12,11,16,20,26 | (0,0,0)(1,1,1)(2,1,1)(2,1,0)<br>(3,2,1)(4,2,0)(5,3,1) |
| 1,2,4,8,12,11,16,21 | (0,0,0)(1,1,1)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1) |
| 1,2,4,8,12,12 | (0,0,0)(1,1,1)(2,1,1)(2,1,1) |
| 1,2,4,8,12,12,8,12 | (0,0,0)(1,1,1)(2,1,1)<br>(2,1,1)(1,1,1)(2,1,1) |
| 1,2,4,8,12,12,8,12,9 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(1,1,1)(2,1,1)(2,0,0) |
| 1,2,4,8,12,12,8,12,10 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(1,1,1)(2,1,1)(2,1,0) |
| 1,2,4,8,12,12,<br>8,12,11,8,12 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(1,1,1)(2,1,1)(2,1,0)(1,1,1)(2,1,1) |
| 1,2,4,8,12,12,<br>8,12,11,16,21,21 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)(1,1,1)<br>(2,1,1)(2,1,0)(3,2,1)(4,2,1)(4,2,1) |
| 1,2,4,8,12,12,8,12,12 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(1,1,1)(2,1,1)(2,1,1) |
| 1,2,4,8,12,12,10,8,12 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(2,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(3,2,1)(3,1,0)(2,2,1)(3,2,1) |
| 1,2,4,8,12,12,11,8 | (0,0,0)(1,1,1)(2,1,1)<br>(2,1,1)(2,1,0)(1,1,1) |
| 1,2,4,8,12,12,11,8,12 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1) |
| 1,2,4,8,12,12,11,8,12,10 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(2,1,0) |
| 1,2,4,8,12,12,<br>11,8,12,11,16,21,21 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(4,2,1) |
| 1,2,4,8,12,12,11,8,12,<br>11,16,21,21,19,24,29,29 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)(2,1,0)<br>(1,1,1)(2,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(4,2,1)(4,1,0)(5,2,1)(6,2,1)(6,2,1) |
| 1,2,4,8,12,12,11,8,12,12 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(2,1,1) |
| 1,2,4,8,12,12,<br>11,14,8,12,12 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)(2,1,0)<br>(3,1,0)(1,1,1)(2,1,1)(2,1,1) |
| 1,2,4,8,12,12,11,15 | (0,0,0)(1,1,1)(2,1,1)<br>(2,1,1)(2,1,0)(3,2,0) |
| 1,2,4,8,12,12,11,16,21,21 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(4,2,1) |
| 1,2,4,8,12,12,12 | (0,0,0)(1,1,1)(2,1,1)(2,1,1)(2,1,1) |
| 1,2,4,8,12,12,12,12 | (0,0,0)(1,1,1)(2,1,1)<br>(2,1,1)(2,1,1)(2,1,1) |
| 1,2,4,8,12,13 | (0,0,0)(1,1,1)(2,1,1)(3,0,0) |
| 1,2,4,8,12,13,15 | (0,0,0)(1,1,1)(2,1,1)(3,0,0)(4,1,0) |
| 1,2,4,8,12,13,15,17,19 | (0,0,0)(1,1,1)(2,1,1)(3,0,0)<br>(4,1,0)(5,1,0)(6,1,0) |
| 1,2,4,8,12,13,15,18 | (0,0,0)(1,1,1)(2,1,1)<br>(3,0,0)(4,1,0)(5,2,0) |
| 1,2,4,8,12,13,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,0,0)(4,1,1) |
| 1,2,4,8,12,13,15,19,23 | (0,0,0)(1,1,1)(2,1,1)<br>(3,0,0)(4,1,1)(5,1,1) |
| 1,2,4,8,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0) |
| 1,2,4,8,12,14,4 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,14,4,8 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(1,1,0)(2,2,1) |
| 1,2,4,8,12,14,4,8,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1) |
| 1,2,4,8,12,14,4,8,12,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,0,0) |
| 1,2,4,8,12,14,<br>4,8,12,13,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,0,0)(5,1,1) |
| 1,2,4,8,12,14,4,<br>8,12,13,15,19,23 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,0,0)(5,1,1)(6,1,1) |
| 1,2,4,8,12,14,4,<br>8,12,13,15,19,23,25 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,0,0)<br>(5,1,1)(6,1,1)(7,1,0) |
| 1,2,4,8,12,14,4,8,<br>12,13,15,19,23,25,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,0,0)<br>(5,1,1)(6,1,1)(7,1,0)(2,2,0) |
| 1,2,4,8,12,14,4,8,12,<br>13,15,19,23,25,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,0,0)<br>(5,1,1)(6,1,1)(7,1,0)(3,2,0)(4,3,0) |
| 1,2,4,8,12,14,4,8,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0) |
| 1,2,4,8,12,14,7 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)(2,2,0) |
| 1,2,4,8,12,14,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1) |
| 1,2,4,8,12,14,8,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)(3,2,1) |
| 1,2,4,8,12,14,8,<br>12,13,15,19,23,25 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)(3,2,1)<br>(4,0,0)(5,1,1)(6,1,1)(7,1,0) |
| 1,2,4,8,12,14,8,<br>12,13,15,19,23,25,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)(3,2,1)<br>(4,0,0)(5,1,1)(6,1,1)(7,1,0)(3,0,0) |
| 1,2,4,8,12,14,8,<br>12,13,15,19,23,25,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(2,2,1)(3,2,1)<br>(4,0,0)(5,1,1)(6,1,1)(7,1,0)(3,1,0) |
| 1,2,4,8,12,14,8,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(2,2,1)(3,2,1)(4,1,0) |
| 1,2,4,8,12,14,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,0,0) |
| 1,2,4,8,12,14,11,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(2,2,1) |
| 1,2,4,8,12,14,11,8,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(2,2,1)(3,2,1)(4,1,0) |
| 1,2,4,8,12,14,<br>11,14,8,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)(3,2,0)<br>(4,2,0)(2,2,1)(3,2,1)(4,1,0) |
| 1,2,4,8,12,14,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,0) |
| 1,2,4,8,12,14,11,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,0)(4,3,1) |
| 1,2,4,8,12,14,11,16,21,23 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(3,2,0)(4,3,1)(5,3,1)(6,1,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,14,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,1) |
| 1,2,4,8,12,14,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(3,2,1)(4,1,0) |
| 1,2,4,8,12,14,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(4,1,0) |
| 1,2,4,8,12,14,16,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(5,1,0) |
| 1,2,4,8,12,14,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(5,2,0) |
| 1,2,4,8,12,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(5,2,1) |
| 1,2,4,8,12,14,18,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,1,0)(5,2,1)(6,2,1) |
| 1,2,4,8,12,14,18,22,24 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,1,0)<br>(5,2,1)(6,2,1)(7,1,0) |
| 1,2,4,8,12,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,0) |
| 1,2,4,8,12,15,4,8,12,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0) |
| 1,2,4,8,12,15,7,<br>12,17,20,12,17,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,0)<br>(2,2,0)(3,3,1)(4,3,1)(5,2,0)<br>(3,3,1)(4,3,1)(5,1,0) |
| 1,2,4,8,12,15,7,12,<br>17,20,12,17,19,23,27,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,0)<br>(2,2,1)(3,2,1)(4,2,0)(2,2,0)(3,3,1)<br>(4,3,1)(5,2,0)(3,3,1)(4,3,1)<br>(5,1,0)(6,2,1)(7,2,1)(8,2,0) |
| 1,2,4,8,12,15,7,12,17,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,0)<br>(2,2,0)(3,3,1)(4,3,1)(5,3,0) |
| 1,2,4,8,12,15,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1) |
| 1,2,4,8,12,15,<br>8,11,16,21,25 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0) |
| 1,2,4,8,12,15,8,12 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1) |
| 1,2,4,8,12,15,8,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0) |
| 1,2,4,8,12,15,8,12,15,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(1,1,1) |
| 1,2,4,8,12,15,8,<br>12,15,8,12,15,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(1,1,1) |
| 1,2,4,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,9,4 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,0) |
| 1,2,4,8,12,15,9,8 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1) |
| 1,2,4,8,12,15,<br>9,8,12,15,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(1,1,1) |
| 1,2,4,8,12,15,<br>9,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,9,9 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(2,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,15,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0) |
| 1,2,4,8,12,15,11,8 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(1,1,1) |
| 1,2,4,8,12,15,<br>11,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,<br>11,8,12,15,9,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0)(2,0,0) |
| 1,2,4,8,12,15,<br>11,8,12,15,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0) |
| 1,2,4,8,12,15,11,9 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(2,0,0) |
| 1,2,4,8,12,15,11,12 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,0,0) |
| 1,2,4,8,12,15,<br>11,14,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,11,15 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,0) |
| 1,2,4,8,12,15,11,16 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,0)(3,2,1) |
| 1,2,4,8,12,15,11,16,21,24 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0) |
| 1,2,4,8,12,15,11,16,<br>21,24,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,11,16,21,25 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0) |
| 1,2,4,8,12,15,<br>11,16,21,25,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(3,2,1) |
| 1,2,4,8,12,15,<br>11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,15,11,<br>16,21,25,20,26,32,37,27 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,0)<br>(5,3,1)(6,3,1)(7,3,0)(6,0,0) |
| 1,2,4,8,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,15,12,8 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(1,1,1) |
| 1,2,4,8,12,15,<br>12,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,<br>12,8,12,15,9,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0)(2,0,0) |
| 1,2,4,8,12,15,<br>12,8,12,15,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0) |
| 1,2,4,8,12,15,12,8,<br>12,15,11,16,21,25,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1) |
| 1,2,4,8,12,15,12,8,<br>12,15,11,16,21,25,21,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1)(3,2,0) |
| 1,2,4,8,12,15,<br>12,8,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,15,12,9 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(2,0,0) |
| 1,2,4,8,12,15,<br>12,11,8,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,15,12,<br>11,14,8,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,0)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,15,12,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,0)(3,2,0) |
| 1,2,4,8,12,15,<br>12,11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,15,<br>12,11,16,21,25,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1) |
| 1,2,4,8,12,15,12,12 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(2,1,1) |
| 1,2,4,8,12,15,12,<br>12,11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,15,12,<br>12,11,16,21,25,21,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,2,1)(4,2,1) |
| 1,2,4,8,12,15,12,12,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(2,1,1)(2,1,1) |
| 1,2,4,8,12,15,12,13 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,0,0) |
| 1,2,4,8,12,15,12,14 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0) |
| 1,2,4,8,12,15,12,15,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(1,1,1) |
| 1,2,4,8,12,15,<br>12,15,8,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0) |
| 1,2,4,8,12,15,<br>12,15,8,12,15,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(1,1,1) |
| 1,2,4,8,12,15,<br>12,15,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,<br>12,15,8,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,15,<br>12,15,8,12,15,12,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,0,0) |
| 1,2,4,8,12,15,12,<br>15,8,12,15,12,15,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1) |
| 1,2,4,8,12,15,12,15,<br>8,12,15,12,15,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,12,15,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,0) |
| 1,2,4,8,12,15,12,<br>15,11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,15,12,<br>15,11,16,21,25,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,15,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,15,12,15,12,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1)(3,0,0) |
| 1,2,4,8,12,15,12,15,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0) |
| 1,2,4,8,12,15,<br>12,15,12,15,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1) |
| 1,2,4,8,12,15,<br>12,15,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,12,<br>15,12,15,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,0,0) |
| 1,2,4,8,12,15,<br>15,8,12,15,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(3,0,0) |
| 1,2,4,8,12,15,15,9 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,15,10,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(2,1,0)(3,2,0) |
| 1,2,4,8,12,15,<br>15,11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,15,<br>15,11,16,21,25,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,0)(5,2,0)(4,0,0) |
| 1,2,4,8,12,15,15,12 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(2,1,1) |
| 1,2,4,8,12,15,15,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,<br>15,12,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(3,1,0)<br>(2,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,15,13 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(3,0,0) |
| 1,2,4,8,12,15,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,15,<br>15,12,15,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,0)<br>(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,15,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(3,1,0)(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,0,0) |
| 1,2,4,8,12,15,18,9 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(2,0,0) |
| 1,2,4,8,12,15,18,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(2,1,0)(3,2,0) |
| 1,2,4,8,12,15,18,12 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(2,1,1) |
| 1,2,4,8,12,15,18,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,<br>18,12,15,18,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,1,0)<br>(2,1,1)(3,1,0)(4,1,0)(2,0,0) |
| 1,2,4,8,12,15,18,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,18,15,18,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(3,1,0)(4,1,0)(2,0,0) |
| 1,2,4,8,12,15,18,16 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(4,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,15,18,18,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(4,1,0)(2,0,0) |
| 1,2,4,8,12,15,<br>18,18,15,18,18,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,1,0)<br>(4,1,0)(3,1,0)(4,1,0)(4,1,0)(2,0,0) |
| 1,2,4,8,12,15,18,18,18,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(4,1,0)(4,1,0)(2,0,0) |
| 1,2,4,8,12,15,18,19 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(5,0,0) |
| 1,2,4,8,12,15,18,21,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(5,1,0)(2,0,0) |
| 1,2,4,8,12,15,18,21,21,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(5,1,0)(5,1,0)(2,0,0) |
| 1,2,4,8,12,15,18,21,24,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,1,0)(5,1,0)(6,1,0)(2,0,0) |
| 1,2,4,8,12,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,15,<br>19,11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,15,<br>19,11,16,21,25,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(3,1,0)(4,2,0) |
| 1,2,4,8,12,15,19,19 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(4,2,0) |
| 1,2,4,8,12,15,19,23,27 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,0)(5,2,0)(6,2,0) |
| 1,2,4,8,12,15,19,24 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,0)(5,3,0) |
| 1,2,4,8,12,15,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1) |
| 1,2,4,8,12,15,20,<br>25,28,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,15,20,25,28,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,1,0)(2,0,0) |
| 1,2,4,8,12,15,20,25,29 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0) |
| 1,2,4,8,12,15,20,25,29,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(4,2,1) |
| 1,2,4,8,12,15,20,25,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) |
| 1,2,4,8,12,15,20,<br>25,29,21,11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,15,<br>20,25,29,35,41,46,36 | (0,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(7,3,1)<br>(8,3,1)(9,3,0)(8,0,0) |
| 1,2,4,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>8,12,15,20,25,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1) |
| 1,2,4,8,12,16,8,12,15,20,<br>25,30,20,25,29,35,41,47 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(4,2,1)(5,2,1)(6,2,0)<br>(7,3,1)(8,3,1)(9,3,1) |
| 1,2,4,8,12,16,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,0,0) |
| 1,2,4,8,12,16,11,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,11,15 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,0)(3,2,0) |
| 1,2,4,8,12,16,<br>11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,11,16,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1) |
| 1,2,4,8,12,16,<br>12,11,16,21,26,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(4,2,1) |
| 1,2,4,8,12,16,12,12 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(2,1,1) |
| 1,2,4,8,12,16,12,13 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,0,0) |
| 1,2,4,8,12,16,12,14 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0) |
| 1,2,4,8,12,16,12,15,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1) |
| 1,2,4,8,12,16,12,15,<br>8,12,15,20,25,30,25,28,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>12,15,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,12,15,9,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1) |
| 1,2,4,8,12,16,12,<br>15,9,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,12,<br>15,9,8,12,15,11,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1) |
| 1,2,4,8,12,16,12,15,<br>9,8,12,15,11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,12,15,<br>9,8,12,15,11,16,21,25,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,2,1) |
| 1,2,4,8,12,16,12,15,<br>9,8,12,15,11,16,21,25,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(6,3,0) |
| 1,2,4,8,12,16,12,15,<br>9,8,12,15,11,16,21,25,31 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(6,3,1) |
| 1,2,4,8,12,16,12,15,<br>9,8,12,15,11,16,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,12,15,<br>9,8,12,15,11,16,21,26,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(4,2,1) |
| 1,2,4,8,12,16,12,<br>15,9,8,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,12,15,<br>9,8,12,15,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,12,<br>15,9,8,12,15,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,0)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,12,<br>15,9,8,12,15,18,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,0)(4,1,0)(2,1,1) |
| 1,2,4,8,12,16,12,<br>15,9,8,12,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,<br>12,15,9,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,12,<br>15,9,8,12,16,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,1)(2,1,1) |
| 1,2,4,8,12,16,12,<br>15,9,8,12,16,12,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0) |
| 1,2,4,8,12,16,<br>12,15,9,8,12,16,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,12,15,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0) |
| 1,2,4,8,12,16,12,<br>15,11,8,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,12,15,11,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(2,0,0) |
| 1,2,4,8,12,16,12,15,11,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(2,1,0) |
| 1,2,4,8,12,16,12,15,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,0) |
| 1,2,4,8,12,16,12,<br>15,11,16,21,26,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,12,15,12,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(2,1,1) |
| 1,2,4,8,12,16,<br>12,15,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,12,<br>15,12,15,11,16,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,<br>12,15,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,12,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,12,15,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,<br>12,15,15,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,12,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,12,15,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,<br>12,15,20,25,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,0)(5,0,0) |
| 1,2,4,8,12,16,<br>12,15,20,25,29,25 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,0)(5,2,1) |
| 1,2,4,8,12,16,<br>12,15,20,25,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) |
| 1,2,4,8,12,16,<br>12,15,20,25,30,25 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1) |
| 1,2,4,8,12,16,12,<br>15,20,25,30,25,29 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0) |
| 1,2,4,8,12,16,12,<br>15,20,25,30,25,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(5,0,0) |
| 1,2,4,8,12,16,12,16 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,12,16,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1) |
| 1,2,4,8,12,16,12,16,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0) |
| 1,2,4,8,12,16,<br>12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>12,16,12,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>12,16,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,12,<br>16,12,15,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,12,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>12,16,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>12,16,12,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,13,8 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(1,1,1) |
| 1,2,4,8,12,16,13,8,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0) |
| 1,2,4,8,12,16,<br>13,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,13,<br>8,12,15,9,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,13,<br>8,12,15,11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,13,<br>8,12,15,11,16,21,25,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,2,1) |
| 1,2,4,8,12,16,13,<br>8,12,15,11,16,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,13,8,<br>12,15,11,16,21,26,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,13,<br>8,12,15,11,16,21,26,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,0,0) |
| 1,2,4,8,12,16,13,8,12,<br>15,11,16,21,26,22,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0)(2,1,0)(3,2,0) |
| 1,2,4,8,12,16,13,8,<br>12,15,11,16,21,26,22,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0)(3,1,0) |
| 1,2,4,8,12,16,13,8,12,<br>15,11,16,21,26,22,14,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,13,8,12,<br>15,11,16,21,26,22,14,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0)(3,1,0)(4,2,1) |
| 1,2,4,8,12,16,13,8,<br>12,15,11,16,21,26,22,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0)(3,2,0) |
| 1,2,4,8,12,16,13,8,<br>12,15,11,16,21,26,22,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0)(3,2,1) |
| 1,2,4,8,12,16,<br>13,8,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,<br>13,8,12,15,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,<br>13,8,12,15,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,<br>13,8,12,15,18,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,1,0)(2,1,1) |
| 1,2,4,8,12,16,<br>13,8,12,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,<br>13,8,12,15,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1) |
| 1,2,4,8,12,16,13,<br>8,12,15,20,25,30,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,0,0) |
| 1,2,4,8,12,16,13,8,<br>12,15,20,25,30,26,11,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,0,0)(2,1,0)(3,2,1) |
| 1,2,4,8,12,16,13,8,<br>12,15,20,25,30,26,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0)(2,1,1) |
| 1,2,4,8,12,16,13,8,<br>12,15,20,25,30,26,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0)(3,1,0) |
| 1,2,4,8,12,16,13,8,<br>12,15,20,25,30,26,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,0,0)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,13,<br>8,12,15,20,25,30,26,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0)(4,1,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,13,<br>8,12,15,20,25,30,26,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0)(4,2,0) |
| 1,2,4,8,12,16,13,<br>8,12,15,20,25,30,26,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0)(4,2,1) |
| 1,2,4,8,12,16,13,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>13,8,12,16,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(2,1,1) |
| 1,2,4,8,12,16,<br>13,8,12,16,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,0) |
| 1,2,4,8,12,16,<br>13,8,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>13,8,12,16,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,13,<br>8,12,16,12,15,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,13,<br>8,12,16,12,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,13,8,<br>12,16,12,15,20,25,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) |
| 1,2,4,8,12,16,13,8,<br>12,16,12,15,20,25,30,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,0,0) |
| 1,2,4,8,12,16,<br>13,8,12,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>13,8,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,13,9 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(2,0,0) |
| 1,2,4,8,12,16,<br>13,11,8,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,13,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,0)(3,2,0) |
| 1,2,4,8,12,16,<br>13,11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,<br>13,11,16,21,26,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,0,0) |
| 1,2,4,8,12,16,13,12 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(2,1,1) |
| 1,2,4,8,12,16,13,12,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(2,1,0)(3,2,0) |
| 1,2,4,8,12,16,<br>13,12,11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,<br>13,12,11,16,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,<br>13,12,11,16,21,26,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,13,<br>12,11,16,21,26,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,13,<br>12,11,16,21,26,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,13,<br>12,11,16,21,26,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0) |
| 1,2,4,8,12,16,13,<br>12,11,16,21,26,22,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,0,0)(4,2,1) |
| 1,2,4,8,12,16,13,12,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(2,1,1) |
| 1,2,4,8,12,16,13,12,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,0,0) |
| 1,2,4,8,12,16,13,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>13,12,15,11,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,0)(2,1,0)(3,2,0) |
| 1,2,4,8,12,16,<br>13,12,15,11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,<br>13,12,15,11,16,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,13,<br>12,15,11,16,21,26,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,0,0) |
| 1,2,4,8,12,16,13,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,13,12,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,<br>13,12,15,20,25,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) |
| 1,2,4,8,12,16,<br>13,12,15,20,25,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1) |
| 1,2,4,8,12,16,<br>13,12,15,20,25,30,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0) |
| 1,2,4,8,12,16,13,<br>12,15,20,25,30,26,25,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,0,0)(5,2,1)(6,2,0)(5,0,0) |
| 1,2,4,8,12,16,13,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,13,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,13,13 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(3,0,0) |
| 1,2,4,8,12,16,13,14 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(4,0,0) |
| 1,2,4,8,12,16,13,15 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(4,1,0) |
| 1,2,4,8,12,16,<br>13,15,19,23,27,24 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(4,1,1)(5,1,1)(6,1,1)(6,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0) |
| 1,2,4,8,12,16,14,2 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(1,0,0) |
| 1,2,4,8,12,16,14,4 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(1,1,0) |
| 1,2,4,8,12,16,<br>14,8,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,1)(4,1,0)<br>(2,2,1)(3,2,1)(4,2,1)(4,0,0) |
| 1,2,4,8,12,16,<br>14,8,12,16,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)(4,2,1)<br>(4,1,0)(2,2,1)(3,2,1)(4,2,1)(4,1,0) |
| 1,2,4,8,12,16,14,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(4,2,1)(4,1,0)(3,0,0) |
| 1,2,4,8,12,16,14,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(4,2,1)(4,1,0)(3,1,0) |
| 1,2,4,8,12,16,14,10,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(4,2,1)(4,1,0)(3,1,0)(3,0,0) |
| 1,2,4,8,12,16,14,10,11 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(4,2,1)(4,1,0)(3,1,0)(4,0,0) |
| 1,2,4,8,12,16,14,10,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,0)(2,2,1)(3,2,1)<br>(4,2,1)(4,1,0)(3,1,0)(4,1,0) |
| 1,2,4,8,12,16,<br>14,10,14,18,22,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,1)(4,1,0)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,0,0) |
| 1,2,4,8,12,16,<br>14,11,8,12,16,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,1)(4,1,0)<br>(3,2,0)(2,2,1)(3,2,1)(4,2,1)(4,1,0) |
| 1,2,4,8,12,16,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,2,1)(4,2,1)(4,2,0) |
| 1,2,4,8,12,16,15,8 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(1,1,1) |
| 1,2,4,8,12,16,<br>15,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>15,8,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,<br>15,8,12,16,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0) |
| 1,2,4,8,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,9,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,0,0)(1,1,1) |
| 1,2,4,8,12,16,<br>15,9,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,0,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>15,9,8,12,16,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0) |
| 1,2,4,8,12,16,15,9,<br>8,12,16,15,8,12,16,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,<br>15,9,8,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,9,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,0,0)(2,0,0) |
| 1,2,4,8,12,16,15,10 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,0) |
| 1,2,4,8,12,16,<br>15,11,8,12,16,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0) |
| 1,2,4,8,12,16,<br>15,11,8,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,11,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,1,0) |
| 1,2,4,8,12,16,15,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,0) |
| 1,2,4,8,12,16,15,11,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1) |
| 1,2,4,8,12,16,<br>15,11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,<br>15,11,16,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,<br>15,11,16,21,26,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,<br>15,11,16,21,26,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,<br>15,11,16,21,26,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,0,0) |
| 1,2,4,8,12,16,15,11,<br>16,21,26,24,8,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>11,16,21,26,24,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>11,16,21,26,24,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(3,2,0) |
| 1,2,4,8,12,16,15,<br>11,16,21,26,24,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,0) |
| 1,2,4,8,12,16,15,<br>11,16,21,26,24,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,1) |
| 1,2,4,8,12,16,15,<br>11,16,21,26,24,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,15,<br>11,16,21,26,24,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,15,<br>11,16,21,26,24,21,26,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1)<br>(5,1,0)(4,2,1)(5,2,1)(5,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,15,<br>11,16,21,26,24,28 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,1,0)(6,2,0) |
| 1,2,4,8,12,16,15,<br>11,16,21,26,25 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,2,0) |
| 1,2,4,8,12,16,15,<br>11,16,21,26,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,15,12 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,<br>12,11,16,21,26,25,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(4,2,1) |
| 1,2,4,8,12,16,15,12,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(2,1,1) |
| 1,2,4,8,12,16,15,12,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,0,0) |
| 1,2,4,8,12,16,15,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0) |
| 1,2,4,8,12,16,15,12,15,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1) |
| 1,2,4,8,12,16,15,<br>12,15,8,12,16,15,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,0) |
| 1,2,4,8,12,16,15,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,<br>15,12,15,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,<br>15,12,15,18,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,12,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,15,12,15,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1) |
| 1,2,4,8,12,16,<br>15,12,15,20,25,29 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,0) |
| 1,2,4,8,12,16,15,<br>12,15,20,25,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) |
| 1,2,4,8,12,16,15,<br>12,15,20,25,29,25 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,2,1) |
| 1,2,4,8,12,16,<br>15,12,15,20,25,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)(6,2,1) |
| 1,2,4,8,12,16,15,<br>12,15,20,25,30,25,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(5,2,1)(6,2,0)(5,0,0) |
| 1,2,4,8,12,16,15,<br>12,15,20,25,30,25,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,2,1) |
| 1,2,4,8,12,16,15,<br>12,15,20,25,30,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,15,<br>12,15,20,25,30,28,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>12,15,20,25,30,29 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0) |
| 1,2,4,8,12,16,15,<br>12,15,20,25,30,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(5,0,0) |
| 1,2,4,8,12,16,15,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,12,<br>16,11,16,21,26,25,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,15,12,16,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1) |
| 1,2,4,8,12,16,<br>15,12,16,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0) |
| 1,2,4,8,12,16,<br>15,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>15,12,16,12,15,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,0) |
| 1,2,4,8,12,16,<br>15,12,16,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,<br>12,16,12,15,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0) |
| 1,2,4,8,12,16,15,<br>12,16,12,15,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>12,16,12,15,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(3,0,0) |
| 1,2,4,8,12,16,15,<br>12,16,12,15,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,<br>12,16,12,15,18,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,<br>12,16,12,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,15,12,<br>16,12,15,20,25,30,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(5,0,0) |
| 1,2,4,8,12,16,<br>15,12,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,<br>12,16,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,<br>15,12,16,13,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0)(2,0,0) |
| 1,2,4,8,12,16,<br>15,12,16,13,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0)(2,1,1) |
| 1,2,4,8,12,16,<br>15,12,16,13,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0)(2,1,1)(3,1,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,<br>15,12,16,13,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>15,12,16,13,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,<br>12,16,13,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>12,16,13,12,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,<br>12,16,13,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,<br>15,12,16,13,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0)(3,0,0) |
| 1,2,4,8,12,16,15,12,16,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0) |
| 1,2,4,8,12,16,<br>15,12,16,15,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(1,1,1) |
| 1,2,4,8,12,16,<br>15,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>12,16,15,12,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,<br>15,12,16,15,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,<br>12,16,15,12,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,<br>12,16,15,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,15,<br>12,16,15,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,12,<br>16,15,12,16,15,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,13 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,0,0) |
| 1,2,4,8,12,16,15,13,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1) |
| 1,2,4,8,12,16,<br>15,13,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,13,<br>8,12,15,11,16,21,26,25,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,0)(5,0,0) |
| 1,2,4,8,12,16,<br>15,13,8,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,<br>13,8,12,15,20,25,30,29,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(6,0,0) |
| 1,2,4,8,12,16,<br>15,13,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,15,<br>13,8,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>13,8,12,16,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,<br>13,8,12,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>15,13,8,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,<br>15,13,8,12,16,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0) |
| 1,2,4,8,12,16,<br>15,13,8,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>13,8,12,16,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,<br>13,8,12,16,15,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,0) |
| 1,2,4,8,12,16,15,<br>13,8,12,16,15,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,<br>13,8,12,16,15,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,15,<br>13,8,12,16,15,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>13,8,12,16,15,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,0,0) |
| 1,2,4,8,12,16,15,13,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,0,0) |
| 1,2,4,8,12,16,15,13,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,0) |
| 1,2,4,8,12,16,15,<br>13,11,8,12,16,15,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(3,0,0) |
| 1,2,4,8,12,16,15,<br>13,11,16,21,26,25,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(5,0,0) |
| 1,2,4,8,12,16,15,13,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,1) |
| 1,2,4,8,12,16,<br>15,13,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,13,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>15,13,12,16,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,1)(2,1,1) |
| 1,2,4,8,12,16,15,<br>13,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,<br>15,13,12,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>15,13,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,<br>15,13,12,16,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,1)(3,1,0) |
| 1,2,4,8,12,16,<br>15,13,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>15,13,12,16,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,<br>15,13,12,16,15,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,<br>13,12,16,15,12,16,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0) |
| 1,2,4,8,12,16,15,<br>13,12,16,15,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,0,0)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>15,13,12,16,15,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,0,0)(2,1,1)(3,1,1)(3,1,0)(3,0,0) |
| 1,2,4,8,12,16,15,13,13 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,0,0)(3,0,0) |
| 1,2,4,8,12,16,15,14 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,1,0) |
| 1,2,4,8,12,16,15,15,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(1,1,1) |
| 1,2,4,8,12,16,<br>15,15,8,12,16,15,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,1,0) |
| 1,2,4,8,12,16,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>15,9,8,12,16,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,15,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,0) |
| 1,2,4,8,12,16,15,<br>15,11,8,12,16,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,15,11,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,0)(3,2,1) |
| 1,2,4,8,12,16,<br>15,15,11,16,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,<br>15,15,11,16,21,26,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,15,<br>15,11,16,21,26,25,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,15,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,<br>15,15,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,<br>15,15,12,15,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,15,<br>15,12,15,20,25,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1) |
| 1,2,4,8,12,16,15,<br>15,12,15,20,25,30,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)(5,0,0) |
| 1,2,4,8,12,16,15,15,<br>12,15,20,25,30,29,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(6,2,0)(5,0,0) |
| 1,2,4,8,12,16,15,15,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>15,15,12,16,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1) |
| 1,2,4,8,12,16,15,<br>15,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>15,12,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>15,15,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,<br>15,15,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>15,15,12,16,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,<br>15,12,16,15,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>15,12,16,15,13,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(3,0,0) |
| 1,2,4,8,12,16,15,<br>15,12,16,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,15,12,<br>16,15,15,12,16,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,15,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(3,0,0) |
| 1,2,4,8,12,16,15,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,16 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,0,0) |
| 1,2,4,8,12,16,15,16,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1) |
| 1,2,4,8,12,16,<br>15,16,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,16,8,<br>12,15,11,16,21,26,25,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,0)(6,0,0) |
| 1,2,4,8,12,16,<br>15,16,8,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,<br>15,16,8,12,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,15,<br>16,8,12,15,20,25,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,15,<br>16,8,12,15,20,25,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) |
| 1,2,4,8,12,16,15,<br>16,8,12,15,20,25,30,29,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)(7,0,0) |
| 1,2,4,8,12,16,15,16,8,<br>12,15,20,25,30,29,30,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(7,0,0)(4,2,0) |
| 1,2,4,8,12,16,<br>15,16,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,<br>16,8,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>16,8,12,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,<br>16,8,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,15,<br>16,8,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>16,8,12,16,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,<br>16,8,12,16,15,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,16,<br>8,12,16,15,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>16,8,12,16,15,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,0,0) |
| 1,2,4,8,12,16,15,<br>16,8,12,16,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>16,8,12,16,15,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,0,0) |
| 1,2,4,8,12,16,15,16,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>16,11,8,12,16,15,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,0,0) |
| 1,2,4,8,12,16,15,16,11,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,0)(3,2,1) |
| 1,2,4,8,12,16,<br>15,16,11,16,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,15,<br>16,11,16,21,26,25,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(6,0,0) |
| 1,2,4,8,12,16,15,16,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,1) |
| 1,2,4,8,12,16,<br>15,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,1)(3,1,0)(2,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,<br>15,16,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,<br>15,16,12,15,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,0)(4,2,1) |
| 1,2,4,8,12,16,15,<br>16,12,15,20,25,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1) |
| 1,2,4,8,12,16,15,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>15,16,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>15,16,12,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>15,16,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,<br>15,16,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>15,16,12,16,15,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,0,0)(2,1,1)(3,1,1)(3,1,0)(4,0,0) |
| 1,2,4,8,12,16,15,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(3,0,0) |
| 1,2,4,8,12,16,15,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,16,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,<br>15,16,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,16,15,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(3,1,0)(4,0,0) |
| 1,2,4,8,12,16,15,16,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,0,0)(4,0,0) |
| 1,2,4,8,12,16,15,17 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,1,0) |
| 1,2,4,8,12,16,15,18,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>18,9,8,12,16,15,18,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,0,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,18,10 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,1,0) |
| 1,2,4,8,12,16,15,<br>18,11,16,21,26,25,29,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,2,0)(4,0,0) |
| 1,2,4,8,12,16,15,18,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,1,1) |
| 1,2,4,8,12,16,<br>15,18,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,18,<br>12,15,20,25,30,29,33,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,0)(7,2,0)(5,0,0) |
| 1,2,4,8,12,16,15,18,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>15,18,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,1)(3,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,<br>15,18,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>15,18,12,16,15,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(2,1,1)(3,1,1)(3,1,0)(4,0,0) |
| 1,2,4,8,12,16,<br>15,18,12,16,15,18,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,18,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(3,0,0) |
| 1,2,4,8,12,16,15,18,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,18,15,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(3,1,0)(4,0,0) |
| 1,2,4,8,12,16,<br>15,18,15,18,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(3,1,0)(4,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,18,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(4,0,0) |
| 1,2,4,8,12,16,15,18,18,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(4,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,18,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(5,0,0) |
| 1,2,4,8,12,16,15,18,21,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(5,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,18,21,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,1,0)(5,1,0)(6,0,0) |
| 1,2,4,8,12,16,<br>15,18,21,24,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,1,0)(5,1,0)(6,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,19 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,15,19,8 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1) |
| 1,2,4,8,12,16,<br>15,19,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,15,11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,15,11,16,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,15,11,16,21,26,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(4,2,1) |
| 1,2,4,8,12,16,15,19,<br>8,12,15,11,16,21,26,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,15,11,16,21,26,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,0,0) |
| 1,2,4,8,12,16,15,19,<br>8,12,15,11,16,21,26,22,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,0,0)(4,2,1) |
| 1,2,4,8,12,16,15,19,<br>8,12,15,11,16,21,26,22,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,0,0)(5,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,15,19,<br>8,12,15,11,16,21,26,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,15,19,<br>8,12,15,11,16,21,26,25,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,0)(5,0,0) |
| 1,2,4,8,12,16,15,19,<br>8,12,15,11,16,21,26,25,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,0)<br>(3,2,1)(4,2,1)(5,2,1)(5,2,0)(6,0,0) |
| 1,2,4,8,12,16,<br>15,19,8,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,<br>15,19,8,12,15,12,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(2,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,15,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,15,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>15,19,8,12,15,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(3,0,0) |
| 1,2,4,8,12,16,<br>15,19,8,12,15,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(3,1,0) |
| 1,2,4,8,12,16,<br>15,19,8,12,15,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>15,19,8,12,15,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,0,0) |
| 1,2,4,8,12,16,<br>15,19,8,12,15,18,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>15,19,8,12,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,<br>15,19,8,12,15,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,15,20,25,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) |
| 1,2,4,8,12,16,<br>15,19,8,12,15,20,25,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,15,20,25,30,25 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(5,2,1) |
| 1,2,4,8,12,16,15,19,<br>8,12,15,20,25,30,25,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(5,2,1)(6,2,0)(5,0,0) |
| 1,2,4,8,12,16,15,19,<br>8,12,15,20,25,30,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(5,0,0) |
| 1,2,4,8,12,16,15,19,<br>8,12,15,20,25,30,29,35 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(7,3,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,15,19,<br>8,12,15,20,25,30,29,35,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(7,3,0)(3,0,0) |
| 1,2,4,8,12,16,15,19,<br>8,12,15,20,25,30,29,35,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(7,3,0)(4,0,0) |
| 1,2,4,8,12,16,15,19,<br>8,12,15,20,25,30,29,35,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,0)(7,3,0)(4,2,0) |
| 1,2,4,8,12,16,<br>15,19,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,8,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,8,12,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>15,19,8,12,16,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(2,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,0)(3,2,0) |
| 1,2,4,8,12,16,15,19,8,<br>12,16,11,16,21,26,25,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(5,2,0)(6,3,0) |
| 1,2,4,8,12,16,<br>15,19,8,12,16,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1) |
| 1,2,4,8,12,16,15,19,<br>8,12,16,12,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,19,<br>8,12,16,12,8,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,8,12,16,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(1,1,1)(2,1,1)(3,1,1)(2,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(2,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,15,19,<br>8,12,16,12,15,9,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,0)(2,0,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,15,9,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,15,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,15,18,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(4,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,15,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,19,<br>8,12,16,12,16,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(1,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,19,<br>8,12,16,12,16,8,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(2,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,19,<br>8,12,16,12,16,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,16,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(2,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,19,<br>8,12,16,12,16,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,19,<br>8,12,16,12,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>15,19,8,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,13,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,15,<br>19,8,12,16,13,8,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,13,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,19,<br>8,12,16,13,8,12,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,19,<br>8,12,16,13,8,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,13,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(2,1,0)(3,2,0) |
| 1,2,4,8,12,16,15,19,8,<br>12,16,13,11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,15,19,<br>8,12,16,13,11,16,21,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,0)(3,2,1)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,13,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(2,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,13,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,13,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,13,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,0,0)(3,0,0) |
| 1,2,4,8,12,16,<br>15,19,8,12,16,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0) |
| 1,2,4,8,12,16,<br>15,19,8,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,0)(3,2,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,12,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,19,<br>8,12,16,15,12,16,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(2,1,0)(3,2,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,12,16,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(2,1,1)(3,1,1)(2,1,1) |
| 1,2,4,8,12,16,15,19,8,<br>12,16,15,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,19,8,<br>12,16,15,12,16,12,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,19,8,<br>12,16,15,12,16,12,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,15,19,<br>8,12,16,15,12,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,15,19,<br>8,12,16,15,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,15,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(2,1,1) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,15,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(3,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,1,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,18,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,15,19,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,0) |
| 1,2,4,8,12,16,15,<br>19,11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,15,<br>19,11,16,21,26,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(4,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,15,<br>19,11,16,21,26,25,29,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1<br>(4,2,1)(5,2,1)(5,2,0)(6,2,0)(4,0,0) |
| 1,2,4,8,12,16,15,<br>19,11,16,21,26,25,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,0)(3,2,1)<br>(4,2,1)(5,2,1)(5,2,0)(6,3,0) |
| 1,2,4,8,12,16,15,19,12 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1) |
| 1,2,4,8,12,16,<br>15,19,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,19,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>15,19,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(2,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>15,19,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>15,19,12,16,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(2,1,1)(3,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,15,19,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,15,19,18,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(4,1,0)(5,2,0) |
| 1,2,4,8,12,16,15,19,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(4,2,0) |
| 1,2,4,8,12,16,15,19,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,0,0) |
| 1,2,4,8,12,16,15,<br>19,22,8,12,16,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,15,19,23 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,2,0) |
| 1,2,4,8,12,16,<br>15,19,23,26,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(5,2,0)(6,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>15,19,23,26,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,0)(5,2,0)(6,1,0)(5,0,0) |
| 1,2,4,8,12,16,15,19,23,27 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,2,0)(6,2,0) |
| 1,2,4,8,12,16,15,<br>19,23,27,19,23,27 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,2,0)(6,2,0)<br>(4,2,0)(5,2,0)(6,2,0) |
| 1,2,4,8,12,16,15,19,24 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,0)(5,3,0) |
| 1,2,4,8,12,16,15,20 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,0)(4,2,1) |
| 1,2,4,8,12,16,15,20,25 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1) |
| 1,2,4,8,12,16,15,20,25,27 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,1,0) |
| 1,2,4,8,12,16,<br>15,20,25,28,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,1,0)(2,0,0) |
| 1,2,4,8,12,16,15,20,25,29 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,0) |
| 1,2,4,8,12,16,<br>15,20,25,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,0)(5,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,15,20,25,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1) |
| 1,2,4,8,12,16,<br>15,20,25,30,29,34 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,0)(7,3,0) |
| 1,2,4,8,12,16,15,<br>20,25,30,29,34,39,44 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)<br>(6,2,0)(7,3,0)(8,3,0)(9,3,0) |
| 1,2,4,8,12,16,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>16,8,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>16,8,12,15,20,25,30,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1) |
| 1,2,4,8,12,16,16,<br>8,12,15,20,25,30,30,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1)(4,2,0) |
| 1,2,4,8,12,16,16,8,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>16,8,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,<br>16,8,12,16,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,16,<br>8,12,16,15,20,25,30,30 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,0)(4,2,1)(5,2,1)(6,2,1)(6,2,1) |
| 1,2,4,8,12,16,16,8,12,<br>16,15,20,25,30,30,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,1)(6,2,1)(2,1,0)(3,2,0) |
| 1,2,4,8,12,16,16,8,<br>12,16,15,20,25,30,30,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,0)<br>(4,2,1)(5,2,1)(6,2,1)(6,2,1)(4,2,0) |
| 1,2,4,8,12,16,<br>16,8,12,16,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(1,1,1)(2,1,1)(3,1,1)(3,1,1) |
| 1,2,4,8,12,16,16,9 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(2,0,0) |
| 1,2,4,8,12,16,<br>16,11,8,12,16,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1) |
| 1,2,4,8,12,16,16,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,0)(3,2,0) |
| 1,2,4,8,12,16,16,11,15,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,0)(3,2,0)(4,3,0) |
| 1,2,4,8,12,16,16,11,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,0)(3,2,1) |
| 1,2,4,8,12,16,<br>16,11,16,21,25 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0) |
| 1,2,4,8,12,16,<br>16,11,16,21,25,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,0)(4,0,0) |
| 1,2,4,8,12,16,<br>16,11,16,21,26,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,2,1) |
| 1,2,4,8,12,16,16,12 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(2,1,1) |
| 1,2,4,8,12,16,16,12,14 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0) |
| 1,2,4,8,12,16,16,<br>12,15,8,12,16,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(1,1,1)<br>(2,1,1)(3,1,1)(3,1,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,16,12,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,16,<br>12,15,20,25,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)<br>(5,2,1)(6,2,0)(5,0,0) |
| 1,2,4,8,12,16,16,12,<br>15,20,25,30,30,25,29,21 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,0)(4,2,1)(5,2,1)<br>(6,2,1)(6,2,1)(5,2,1)(6,2,0)(5,0,0) |
| 1,2,4,8,12,16,16,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>16,12,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,16,12,16,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,<br>16,12,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>16,12,16,15,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>16,12,16,15,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,0,0) |
| 1,2,4,8,12,16,<br>16,12,16,15,18,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,1,0)(2,0,0) |
| 1,2,4,8,12,16,<br>16,12,16,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,16,12,16,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>16,12,16,16,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,16,13 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,0,0) |
| 1,2,4,8,12,16,<br>16,13,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)<br>(3,0,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,16,14 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,1,0) |
| 1,2,4,8,12,16,16,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,16,15,13 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(3,0,0) |
| 1,2,4,8,12,16,16,15,19 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,0)(4,2,0) |
| 1,2,4,8,12,16,16,16 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(3,1,1)(3,1,1) |
| 1,2,4,8,12,16,16,16,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(3,1,1)(3,1,1)(3,1,1) |
| 1,2,4,8,12,16,17 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,0,0) |
| 1,2,4,8,12,16,<br>17,19,23,27,31 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,0,0)(5,1,1)(6,1,1)(7,1,1) |
| 1,2,4,8,12,16,18 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0) |
| 1,2,4,8,12,16,19,9 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(2,0,0) |
| 1,2,4,8,12,16,19,10 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(2,1,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,12,16,19,11,15 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,0)(3,2,0) |
| 1,2,4,8,12,16,<br>19,11,16,21,26,26 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(5,2,1) |
| 1,2,4,8,12,16,<br>19,11,16,21,26,27 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(6,0,0) |
| 1,2,4,8,12,16,<br>19,11,16,21,26,28 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(6,1,0) |
| 1,2,4,8,12,16,19,11,<br>16,21,26,29,8,12,16,19,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)<br>(2,1,0)(3,2,1)(4,2,1)(5,2,1)(6,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(4,1,0)(2,0,0) |
| 1,2,4,8,12,16,19,<br>11,16,21,26,30,22 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,0)(3,2,1)(4,2,1)<br>(5,2,1)(6,2,0)(4,0,0) |
| 1,2,4,8,12,16,19,12 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(2,1,1) |
| 1,2,4,8,12,16,19,12,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,19,12,16 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,1)(3,1,1) |
| 1,2,4,8,12,16,<br>19,12,16,19,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(2,1,1)(3,1,1)(4,1,0)(2,0,0) |
| 1,2,4,8,12,16,19,13 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(3,0,0) |
| 1,2,4,8,12,16,19,15,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(3,1,0)(2,0,0) |
| 1,2,4,8,12,16,19,16 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(3,1,1) |
| 1,2,4,8,12,16,19,16,19,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(3,1,1)(4,1,0)(2,0,0) |
| 1,2,4,8,12,16,19,17 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(4,0,0) |
| 1,2,4,8,12,16,19,19,9 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(4,1,0)(2,0,0) |
| 1,2,4,8,12,16,19,23 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,0)(5,2,0) |
| 1,2,4,8,12,16,19,24,29,34 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,0)(5,2,1)(6,2,1)(7,2,1) |
| 1,2,4,8,12,16,20 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1) |
| 1,2,4,8,12,16,20,24 | (0,0,0)(1,1,1)(2,1,1)<br>(3,1,1)(4,1,1)(5,1,1) |
| 1,2,4,8,12,16,20,24,28 | (0,0,0)(1,1,1)(2,1,1)(3,1,1)<br>(4,1,1)(5,1,1)(6,1,1) |
| 1,2,4,8,13 | (0,0,0)(1,1,1)(2,2,0) |
| 1,2,4,8,13,8 | (0,0,0)(1,1,1)(2,2,0)(1,1,1) |
| 1,2,4,8,13,8,12 | (0,0,0)(1,1,1)(2,2,0)(1,1,1)(2,1,1) |
| 1,2,4,8,13,8,12,16 | (0,0,0)(1,1,1)(2,2,0)<br>(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,13,8,13 | (0,0,0)(1,1,1)(2,2,0)(1,1,1)(2,2,0) |
| 1,2,4,8,13,8,13,8,13 | (0,0,0)(1,1,1)(2,2,0)(1,1,1)<br>(2,2,0)(1,1,1)(2,2,0) |
| 1,2,4,8,13,9 | (0,0,0)(1,1,1)(2,2,0)(2,0,0) |
| 1,2,4,8,13,9,11,15,20 | (0,0,0)(1,1,1)(2,2,0)<br>(2,0,0)(3,1,1)(4,2,0) |
| 1,2,4,8,13,10 | (0,0,0)(1,1,1)(2,2,0)(2,1,0) |
| 1,2,4,8,13,11 | (0,0,0)(1,1,1)(2,2,0)(2,1,0)(3,2,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,13,11,16,19 | (0,0,0)(1,1,1)(2,2,0)<br>(2,1,0)(3,2,1)(4,3,0) |
| 1,2,4,8,13,12 | (0,0,0)(1,1,1)(2,2,0)(2,1,1) |
| 1,2,4,8,13,12,16 | (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,1,1) |
| 1,2,4,8,13,12,17 | (0,0,0)(1,1,1)(2,2,0)(2,1,1)(3,2,0) |
| 1,2,4,8,13,12,17,16,21 | (0,0,0)(1,1,1)(2,2,0)(2,1,1)<br>(3,2,0)(3,1,1)(4,2,0) |
| 1,2,4,8,13,13 | (0,0,0)(1,1,1)(2,2,0)(2,2,0) |
| 1,2,4,8,13,13,13 | (0,0,0)(1,1,1)(2,2,0)(2,2,0)(2,2,0) |
| 1,2,4,8,13,14 | (0,0,0)(1,1,1)(2,2,0)(3,0,0) |
| 1,2,4,8,13,15 | (0,0,0)(1,1,1)(2,2,0)(3,1,0) |
| 1,2,4,8,13,16 | (0,0,0)(1,1,1)(2,2,0)(3,1,0)<br>(1,1,0)(2,2,1)(3,3,0)(4,2,0) |
| 1,2,4,8,13,17 | (0,0,0)(1,1,1)(2,2,0)(3,1,1) |
| 1,2,4,8,13,18 | (0,0,0)(1,1,1)(2,2,0)(3,2,0) |
| 1,2,4,8,13,18,21,9 | (0,0,0)(1,1,1)(2,2,0)<br>(3,2,0)(4,1,0)(2,0,0) |
| 1,2,4,8,13,18,22 | (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1) |
| 1,2,4,8,13,18,23 | (0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,2,0) |
| 1,2,4,8,13,19 | (0,0,0)(1,1,1)(2,2,0)(3,3,0) |
| 1,2,4,8,13,19,26 | (0,0,0)(1,1,1)(2,2,0)(3,3,0)(4,4,0) |
| 1,2,4,8,13,20 | (0,0,0)(1,1,1)(2,2,0)(3,3,1) |
| 1,2,4,8,13,20,26,20 | (0,0,0)(1,1,1)(2,2,0)<br>(3,3,1)(4,3,0)(3,3,1) |
| 1,2,4,8,13,20,27 | (0,0,0)(1,1,1)(2,2,0)(3,3,1)(4,3,1) |
| 1,2,4,8,13,20,27,33,21 | (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,3,1)(5,3,0)(4,0,0) |
| 1,2,4,8,13,20,28 | (0,0,0)(1,1,1)(2,2,0)(3,3,1)(4,4,0) |
| 1,2,4,8,13,20,28,38 | (0,0,0)(1,1,1)(2,2,0)<br>(3,3,1)(4,4,0)(5,5,1) |
| 1,2,4,8,13,20,28,38,49,62 | (0,0,0)(1,1,1)(2,2,0)(3,3,1)<br>(4,4,0)(5,5,1)(6,6,0)(7,7,1) |
| 1,2,4,8,14 | (0,0,0)(1,1,1)(2,2,1) |
| 1,2,4,8,14,8,14 | (0,0,0)(1,1,1)(2,2,1)(1,1,1)(2,2,1) |
| 1,2,4,8,14,11,<br>8,13,20,29,23,9 | (0,0,0)(1,1,1)(2,2,1)(2,1,0)<br>(1,1,1)(2,2,0)(3,3,1)(4,4,1) |
| 1,2,4,8,14,11,<br>8,13,20,29,23,9 | (0,0,0)(1,1,1)(2,2,1)(2,1,0)(1,1,1)<br>(2,2,0)(3,3,1)(4,4,1)(4,1,0)(2,0,0) |
| 1,2,4,8,14,12 | (0,0,0)(1,1,1)(2,2,1)(2,1,1) |
| 1,2,4,8,14,12,17 | (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,0) |
| 1,2,4,8,14,12,17,24 | (0,0,0)(1,1,1)(2,2,1)<br>(2,1,1)(3,2,0)(4,3,1) |
| 1,2,4,8,14,12,17,24,33 | (0,0,0)(1,1,1)(2,2,1)(2,1,1)<br>(3,2,0)(4,3,1)(5,4,1) |
| 1,2,4,8,14,12,18 | (0,0,0)(1,1,1)(2,2,1)(2,1,1)(3,2,1) |
| 1,2,4,8,14,12,18,9 | (0,0,0)(1,1,1)(2,2,1)<br>(2,1,1)(3,2,1)(2,0,0) |
| 1,2,4,8,14,12,18,12 | (0,0,0)(1,1,1)(2,2,1)<br>(2,1,1)(3,2,1)(2,1,1) |
| 1,2,4,8,14,12,18,12,18 | (0,0,0)(1,1,1)(2,2,1)(2,1,1)<br>(3,2,1)(2,1,1)(3,2,1) |
| 1,2,4,8,14,12,18,13 | (0,0,0)(1,1,1)(2,2,1)<br>(2,1,1)(3,2,1)(3,0,0) |
| 1,2,4,8,14,12,18,15,9 | (0,0,0)(1,1,1)(2,2,1)(2,1,1)<br>(3,2,1)(3,1,0)(2,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,14,12,18,16 | (0,0,0)(1,1,1)(2,2,1)<br>(2,1,1)(3,2,1)(3,1,1) |
| 1,2,4,8,14,12,18,16,22 | (0,0,0)(1,1,1)(2,2,1)(2,1,1)<br>(3,2,1)(3,1,1)(4,2,1) |
| 1,2,4,8,14,12,<br>18,16,22,20,26 | (0,0,0)(1,1,1)(2,2,1)(2,1,1)<br>(3,2,1)(3,1,1)(4,2,1)(4,1,1)(5,2,1) |
| 1,2,4,8,14,13 | (0,0,0)(1,1,1)(2,2,1)(2,2,0) |
| 1,2,4,8,14,13,18 | (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,2,0) |
| 1,2,4,8,14,13,18,23 | (0,0,0)(1,1,1)(2,2,1)<br>(2,2,0)(3,2,0)(4,2,0) |
| 1,2,4,8,14,13,20 | (0,0,0)(1,1,1)(2,2,1)(2,2,0)(3,3,1) |
| 1,2,4,8,14,13,20,24 | (0,0,0)(1,1,1)(2,2,1)<br>(2,2,0)(3,3,1)(4,1,1) |
| 1,2,4,8,14,13,20,25 | (0,0,0)(1,1,1)(2,2,1)<br>(2,2,0)(3,3,1)(4,2,0) |
| 1,2,4,8,14,13,20,26,20 | (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,3,1)(4,3,0)(3,3,1) |
| 1,2,4,8,14,13,20,29 | (0,0,0)(1,1,1)(2,2,1)<br>(2,2,0)(3,3,1)(4,4,1) |
| 1,2,4,8,14,13,20,29,20 | (0,0,0)(1,1,1)(2,2,1)(2,2,0)<br>(3,3,1)(4,4,1)(3,3,1) |
| 1,2,4,8,14,14 | (0,0,0)(1,1,1)(2,2,1)(2,2,1) |
| 1,2,4,8,14,14,12,15,9 | (0,0,0)(1,1,1)(2,2,1)(2,2,1)<br>(2,1,1)(3,1,0)(2,0,0) |
| 1,2,4,8,14,14,14 | (0,0,0)(1,1,1)(2,2,1)(2,2,1)(2,2,1) |
| 1,2,4,8,14,15 | (0,0,0)(1,1,1)(2,2,1)(3,0,0) |
| 1,2,4,8,14,15,13 | (0,0,0)(1,1,1)(2,2,1)(3,0,0)(2,2,0) |
| 1,2,4,8,14,15,14,13 | (0,0,0)(1,1,1)(2,2,1)<br>(3,0,0)(2,2,1)(2,2,0) |
| 1,2,4,8,14,15,14,15 | (0,0,0)(1,1,1)(2,2,1)<br>(3,0,0)(2,2,1)(3,0,0) |
| 1,2,4,8,14,15,15 | (0,0,0)(1,1,1)(2,2,1)(3,0,0)(3,0,0) |
| 1,2,4,8,14,16 | (0,0,0)(1,1,1)(2,2,1)(3,1,0) |
| 1,2,4,8,14,17 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(1,1,0)(2,2,1)(3,3,1)(4,2,0) |
| 1,2,4,8,14,17,8 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)(1,1,1) |
| 1,2,4,8,14,17,8,12,16 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,0)(1,1,1)(2,1,1)(3,1,1) |
| 1,2,4,8,14,17,8,12,16,16 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(1,1,1)(2,1,1)(3,1,1)(3,1,1) |
| 1,2,4,8,14,17,8,14 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,0)(1,1,1)(2,2,1) |
| 1,2,4,8,14,17,8,14,14 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(1,1,1)(2,2,1)(2,2,1) |
| 1,2,4,8,14,17,8,14,15 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(1,1,1)(2,2,1)(3,0,0) |
| 1,2,4,8,14,17,9 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,0,0) |
| 1,2,4,8,14,17,11,15 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,0)(2,1,0)(3,2,0) |
| 1,2,4,8,14,17,13 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,0) |
| 1,2,4,8,14,17,14 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)(2,2,1) |
| 1,2,4,8,14,17,21 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)(4,2,0) |
| 1,2,4,8,14,17,22,29,33 | (0,0,0)(1,1,1)(2,2,1)(3,1,0)<br>(4,2,1)(5,3,1)(6,2,0) |
| 1,2,4,8,14,18 | (0,0,0)(1,1,1)(2,2,1)(3,1,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,14,18,12,18,22 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(2,1,1)(3,2,1)(4,1,1) |
| 1,2,4,8,14,18,13 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)(2,2,0) |
| 1,2,4,8,14,18,14 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)(2,2,1) |
| 1,2,4,8,14,18,14,18 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(2,2,1)(3,1,1) |
| 1,2,4,8,14,18,18 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)(3,1,1) |
| 1,2,4,8,14,18,22 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,1,1) |
| 1,2,4,8,14,18,23 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,0) |
| 1,2,4,8,14,18,23,30 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,0)(5,3,1) |
| 1,2,4,8,14,18,24 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)(4,2,1) |
| 1,2,4,8,14,18,24,14 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,1)(2,2,1) |
| 1,2,4,8,14,18,24,14,18 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(2,2,1)(3,1,1) |
| 1,2,4,8,14,18,24,14,18,24 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(2,2,1)(3,1,1)(4,2,1) |
| 1,2,4,8,14,18,24,18 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,1)(3,1,1) |
| 1,2,4,8,14,18,24,18,24 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(3,1,1)(4,2,1) |
| 1,2,4,8,14,18,24,22,28 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(4,1,1)(5,2,1) |
| 1,2,4,8,14,18,24,23 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,1)(4,2,0) |
| 1,2,4,8,14,18,24,24 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,1)(4,2,1) |
| 1,2,4,8,14,18,24,25 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,0,0) |
| 1,2,4,8,14,18,24,27,9 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,1,0)(2,0,0) |
| 1,2,4,8,14,18,24,28 | (0,0,0)(1,1,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,1,1) |
| 1,2,4,8,14,18,24,28,33 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,1,1)(6,2,0) |
| 1,2,4,8,14,18,24,28,34 | (0,0,0)(1,1,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,1,1)(6,2,1) |
| 1,2,4,8,14,19 | (0,0,0)(1,1,1)(2,2,1)(3,2,0) |
| 1,2,4,8,14,19,8,14,19 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(1,1,1)(2,2,1)(3,2,0) |
| 1,2,4,8,14,19,11,8,14,19 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,1,0)(1,1,1)(2,2,1)(3,2,0) |
| 1,2,4,8,14,19,11,15 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(2,1,0)(3,2,0) |
| 1,2,4,8,14,19,11,16 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(2,1,0)(3,2,1) |
| 1,2,4,8,14,19,11,16,23 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,1,0)(3,2,1)(4,3,1) |
| 1,2,4,8,14,19,11,16,23,24 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,1,0)(3,2,1)(4,3,1)(5,0,0) |
| 1,2,4,8,14,19,<br>11,16,23,27,17 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,1,0)(3,2,1)(4,3,1)(5,2,0)(4,0,0) |
| 1,2,4,8,14,19,11,<br>16,23,28,35,39,17 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,1,0)(3,2,1)(4,3,1)(5,2,1)<br>(6,3,1)(7,2,0)(4,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,14,19,11,16,23,29 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,1,0)(3,2,1)(4,3,1)(5,3,0) |
| 1,2,4,8,14,19,11,<br>16,23,29,20,26,34,41 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,1,0)(3,2,1)(4,3,1)(5,3,0)<br>(4,2,0)(5,3,1)(6,4,1)(7,4,0) |
| 1,2,4,8,14,19,12 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,1,1) |
| 1,2,4,8,14,19,13 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,0) |
| 1,2,4,8,14,19,14,13 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(2,2,1)(2,2,0) |
| 1,2,4,8,14,19,14,15 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(2,2,1)(3,0,0) |
| 1,2,4,8,14,19,14,17,9 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,0)(2,0,0) |
| 1,2,4,8,14,19,14,18,24,29 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0) |
| 1,2,4,8,14,19,14,<br>18,24,29,24,27,9 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,1,0)(2,0,0) |
| 1,2,4,8,14,19,14,<br>18,24,29,24,28,34,39 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0)<br>(4,2,1)(5,1,1)(6,2,1)(7,2,0) |
| 1,2,4,8,14,19,14,19 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(2,2,1)(3,2,0) |
| 1,2,4,8,14,19,14,19,11,15 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,2,0)(2,1,0)(3,2,0) |
| 1,2,4,8,14,19,14,<br>19,11,16,23,29 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(2,2,1)<br>(3,2,0)(2,1,0)(3,2,1)(4,3,1)(5,3,0) |
| 1,2,4,8,14,19,14,<br>19,11,16,23,29,23,29 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,2,0)(2,1,0)(3,2,1)<br>(4,3,1)(5,3,0)(4,3,1)(5,3,0) |
| 1,2,4,8,14,19,<br>14,19,14,18,9 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,2,0)(2,2,1)(3,1,0)(2,0,0) |
| 1,2,4,8,14,19,14,19,14,19 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(2,2,1)(3,2,0)(2,2,1)(3,2,0) |
| 1,2,4,8,14,19,15 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,0,0) |
| 1,2,4,8,14,19,17,9 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(3,1,0)(2,0,0) |
| 1,2,4,8,14,19,17,14,19 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,1,0)(2,2,1)(3,2,0) |
| 1,2,4,8,14,19,18 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,1,1) |
| 1,2,4,8,14,19,18,24,25 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,1,1)(4,2,1)(5,0,0) |
| 1,2,4,8,14,19,18,24,29 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,1,1)(4,2,1)(5,2,0) |
| 1,2,4,8,14,19,<br>18,24,29,32,9 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(3,1,1)(4,2,1)(5,2,0)(5,1,0)(2,0,0) |
| 1,2,4,8,14,19,19 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(3,2,0) |
| 1,2,4,8,14,19,22,9 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(4,1,0)(2,0,0) |
| 1,2,4,8,14,19,<br>22,14,19,22,9 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,1,0)<br>(2,2,1)(3,2,0)(4,1,0)(2,0,0) |
| 1,2,4,8,14,19,22,19,22,9 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,1,0)(3,2,0)(4,1,0)(2,0,0) |
| 1,2,4,8,14,19,22,25,9 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,1,0)(5,1,0)(2,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,14,19,22,26 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(4,1,0)(5,2,0) |
| 1,2,4,8,14,19,23,29,34 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,1,1)(5,2,1)(6,2,0) |
| 1,2,4,8,14,19,<br>23,29,34,37,9 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,1,1)(5,2,1)(6,2,0)(7,1,0)(2,0,0) |
| 1,2,4,8,14,19,24 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,2,0) |
| 1,2,4,8,14,19,25 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,3,0) |
| 1,2,4,8,14,19,25,31,37 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,0)(5,3,0)(6,3,0) |
| 1,2,4,8,14,19,26 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)(4,3,1) |
| 1,2,4,8,14,19,26,27 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(4,3,1)(5,0,0) |
| 1,2,4,8,14,19,26,29,33 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,1,0)(6,2,0) |
| 1,2,4,8,14,19,26,29,34 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,1,0)(6,2,1) |
| 1,2,4,8,14,19,26,29,34,39 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,1,0)(6,2,1)(7,2,1) |
| 1,2,4,8,14,19,26,29,34,41 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,1,0)(6,2,1)(7,3,1) |
| 1,2,4,8,14,19,<br>26,29,34,41,47 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,1,0)(6,2,1)(7,3,1)(8,3,0) |
| 1,2,4,8,14,19,26,30 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(4,3,1)(5,1,1) |
| 1,2,4,8,14,19,26,30,36,41 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,1,1)(6,2,1)(7,2,0) |
| 1,2,4,8,14,19,26,31 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(4,3,1)(5,2,0) |
| 1,2,4,8,14,19,26,32,26 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,3,0)(4,3,1) |
| 1,2,4,8,14,19,26,33 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,0)(4,3,1)(5,4,1) |
| 1,2,4,8,14,19,26,33,34 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,4,1)(6,0,0) |
| 1,2,4,8,14,19,26,33,39 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,4,1)(6,3,0)(5,0,0) |
| 1,2,4,8,14,19,26,33,41 | (0,0,0)(1,1,1)(2,2,1)(3,2,0)<br>(4,3,1)(5,4,1)(6,4,0) |
| 1,2,4,8,14,20 | (0,0,0)(1,1,1)(2,2,1)(3,2,1) |
| 1,2,4,8,14,20,8,12 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(1,1,1)(2,1,1) |
| 1,2,4,8,14,20,8,14 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(1,1,1)(2,2,1) |
| 1,2,4,8,14,20,8,14,19 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(1,1,1)(2,2,1)(3,2,0) |
| 1,2,4,8,14,20,8,14,20 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(1,1,1)(2,2,1)(3,2,1) |
| 1,2,4,8,14,20,13 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,0) |
| 1,2,4,8,14,20,13,18,23 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,0)(3,2,0)(4,2,0) |
| 1,2,4,8,14,20,14 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(2,2,1) |
| 1,2,4,8,14,20,14,15 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,14,20,14,17,9 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,0)(2,0,0) |
| 1,2,4,8,14,20,14,18 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,1,1) |
| 1,2,4,8,14,20,14,18,24 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1) |
| 1,2,4,8,14,20,14,18,24,29 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0) |
| 1,2,4,8,14,20,<br>14,18,24,29,35 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,0)(6,3,0) |
| 1,2,4,8,14,20,<br>14,18,24,29,36 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,1,1)(4,2,1)(5,2,0)(6,3,1) |
| 1,2,4,8,14,20,14,18,24,30 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,1) |
| 1,2,4,8,14,20,<br>14,18,24,30,24,27,9 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,1,1)(4,2,1)(5,2,1)<br>(4,2,1)(5,1,0)(2,0,0) |
| 1,2,4,8,14,20,14,19 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,2,0) |
| 1,2,4,8,14,20,<br>14,19,25,31,37 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,0)(4,3,0)(5,3,0)(6,3,0) |
| 1,2,4,8,14,20,14,20 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,2,1) |
| 1,2,4,8,14,20,<br>14,20,13,18,23 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,1)(2,2,0)(3,2,0)(4,2,0) |
| 1,2,4,8,14,20,<br>14,20,14,17,9 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,1)(2,2,1)(3,1,0)(2,0,0) |
| 1,2,4,8,14,20,<br>14,20,14,18,24,29 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,1,1)(4,2,1)(5,2,0) |
| 1,2,4,8,14,20,14,<br>20,14,18,24,30,24,29 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,1)(2,2,1)(3,1,1)<br>(4,2,1)(5,2,1)(4,2,1)(5,2,0) |
| 1,2,4,8,14,20,14,20,14,19 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,1)(2,2,1)(3,2,0) |
| 1,2,4,8,14,20,14,<br>20,14,19,25,31,37 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,1)(2,2,1)(3,2,0)<br>(4,3,0)(5,3,0)(6,3,0) |
| 1,2,4,8,14,20,<br>14,20,14,20,14,19 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(2,2,1)<br>(3,2,1)(2,2,1)(3,2,1)(2,2,1)(3,2,0) |
| 1,2,4,8,14,20,14,<br>20,14,20,14,19,25,31,37 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,1)(2,2,1)(3,2,1)<br>(2,2,1)(3,2,0)(4,3,0)(5,3,0)(6,3,0) |
| 1,2,4,8,14,20,15 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,0,0) |
| 1,2,4,8,14,20,17,9 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,1,0)(2,0,0) |
| 1,2,4,8,14,20,<br>17,22,29,35 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,1,0)(4,2,1)(5,3,1)(6,3,0) |
| 1,2,4,8,14,20,<br>17,22,29,36,29,35 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(4,2,1)(5,3,1)(6,3,1)(5,3,1)(6,3,0) |
| 1,2,4,8,14,20,<br>17,22,29,36,30 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,0)<br>(4,2,1)(5,3,1)(6,3,1)(6,0,0) |
| 1,2,4,8,14,20,18 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,1,1) |
| 1,2,4,8,14,20,18,24,30 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,1,1)(4,2,1)(5,2,1) |
| 1,2,4,8,14,20,19 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,14,20,19,24,29 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,0)(4,2,0)(5,2,0) |
| 1,2,4,8,14,20,19,25 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,2,0)(4,3,1) |
| 1,2,4,8,14,20,20 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(3,2,1) |
| 1,2,4,8,14,20,20,14,19 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,1)(2,2,1)(3,2,0) |
| 1,2,4,8,14,20,<br>20,14,20,14,19 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(3,2,1)(2,2,1)(3,2,1)(3,2,0) |
| 1,2,4,8,14,20,20,14,20,19 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(3,2,1)(3,2,0) |
| 1,2,4,8,14,20,25 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)(4,2,0) |
| 1,2,4,8,14,20,26,19 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,1)(3,2,0) |
| 1,2,4,8,14,20,26,20,25 | (0,0,0)(1,1,1)(2,2,1)(3,2,1)<br>(4,2,1)(3,2,1)(4,2,0) |
| 1,2,4,8,14,20,26,25 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,1)(4,2,0) |
| 1,2,4,8,14,20,26,31 | (0,0,0)(1,1,1)(2,2,1)<br>(3,2,1)(4,2,1)(5,2,0) |
| 1,2,4,8,14,21 | (0,0,0)(1,1,1)(2,2,1)(3,3,0) |
| 1,2,4,8,14,21,28,35 | (0,0,0)(1,1,1)(2,2,1)<br>(3,3,0)(4,3,0)(5,3,0) |
| 1,2,4,8,14,21,30 | (0,0,0)(1,1,1)(2,2,1)(3,3,0)(4,4,1) |
| 1,2,4,8,14,21,30,41,51 | (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,4,1)(5,5,1)(6,5,0) |
| 1,2,4,8,14,21,30,41,52,42 | (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,4,1)(5,5,1)(6,5,1)(6,0,0) |
| 1,2,4,8,14,21,30,41,53 | (0,0,0)(1,1,1)(2,2,1)(3,3,0)<br>(4,4,1)(5,5,1)(6,6,0) |
| 1,2,4,8,14,22 | (0,0,0)(1,1,1)(2,2,1)(3,3,1) |
| 1,2,4,8,14,22,14 | (0,0,0)(1,1,1)(2,2,1)(3,3,1)(2,2,1) |
| 1,2,4,8,14,22,14,20 | (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(2,2,1)(3,2,1) |
| 1,2,4,8,14,22,14,21 | (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(2,2,1)(3,3,0) |
| 1,2,4,8,14,22,14,22 | (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(2,2,1)(3,3,1) |
| 1,2,4,8,14,22,20 | (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,2,1) |
| 1,2,4,8,14,22,20,28 | (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(3,2,1)(4,3,1) |
| 1,2,4,8,14,22,21 | (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,3,0) |
| 1,2,4,8,14,22,22 | (0,0,0)(1,1,1)(2,2,1)(3,3,1)(3,3,1) |
| 1,2,4,8,14,22,23 | (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,0,0) |
| 1,2,4,8,14,22,29 | (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,3,0) |
| 1,2,4,8,14,22,32 | (0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,4,1) |
| 1,2,4,8,14,22,32,44 | (0,0,0)(1,1,1)(2,2,1)<br>(3,3,1)(4,4,1)(5,5,1) |
| 1,2,4,8,15 | (0,0,0)(1,1,1)(2,2,2) |
| 1,2,4,8,15,22 | (0,0,0)(1,1,1)(2,2,2)(3,2,2) |
| 1,2,4,8,15,22,23 | (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,0,0) |
| 1,2,4,8,15,22,24 | (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,1,0) |
| 1,2,4,8,15,22,25,9 | (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,1,0)(2,0,0) |
| 1,2,4,8,15,22,26 | (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,1,1) |
| 1,2,4,8,15,22,27 | (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,15,22,27,15 | (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,0)(2,2,2) |
| 1,2,4,8,15,22,<br>27,15,22,27,15 | (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,0)<br>(2,2,2)(3,2,2)(4,2,0)(2,2,2) |
| 1,2,4,8,15,22,27,16 | (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,0)(3,0,0) |
| 1,2,4,8,15,22,28 | (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,1) |
| 1,2,4,8,15,22,29 | (0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2) |
| 1,2,4,8,15,22,29,29 | (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,2)(4,2,2) |
| 1,2,4,8,15,22,29,36 | (0,0,0)(1,1,1)(2,2,2)<br>(3,2,2)(4,2,2)(5,2,2) |
| 1,2,4,8,15,23 | (0,0,0)(1,1,1)(2,2,2)(3,3,0) |
| 1,2,4,8,15,24 | (0,0,0)(1,1,1)(2,2,2)(3,3,1) |
| 1,2,4,8,15,24,35 | (0,0,0)(1,1,1)(2,2,2)(3,3,1)(4,4,1) |
| 1,2,4,8,15,24,36 | (0,0,0)(1,1,1)(2,2,2)(3,3,1)(4,4,2) |
| 1,2,4,8,15,25 | (0,0,0)(1,1,1)(2,2,2)(3,3,2) |
| 1,2,4,8,15,25,26 | (0,0,0)(1,1,1)(2,2,2)(3,3,2)(4,0,0) |
| 1,2,4,8,15,25,38 | (0,0,0)(1,1,1)(2,2,2)(3,3,2)(4,4,2) |
| 1,2,4,8,15,26 | (0,0,0)(1,1,1)(2,2,2)(3,3,3) |
| 1,2,4,8,15,26,42 | (0,0,0)(1,1,1)(2,2,2)(3,3,3)(4,4,4) |
| 1,2,4,8,15,26,42,64 | (0,0,0)(1,1,1)(2,2,2)<br>(3,3,3)(4,4,4)(5,5,5) |
| 1,2,4,8,16 | (0,0,0,0)(1,1,1,1) |
| 1,2,4,8,16,8 | (0,0,0,0)(1,1,1,1)(1,1,1,0) |
| 1,2,4,8,16,8,14 | (0,0,0,0)(1,1,1,1)(1,1,1,0)(2,2,1,0) |
| 1,2,4,8,16,8,14,22 | (0,0,0,0)(1,1,1,1)(1,1,1,0)<br>(2,2,1,0)(3,3,1,0) |
| 1,2,4,8,16,8,15 | (0,0,0,0)(1,1,1,1)(1,1,1,0)(2,2,2,0) |
| 1,2,4,8,16,8,15,26 | (0,0,0,0)(1,1,1,1)(1,1,1,0)<br>(2,2,2,0)(3,3,3,0) |
| 1,2,4,8,16,8,16 | (0,0,0,0)(1,1,1,1)(1,1,1,0)(2,2,2,1) |
| 1,2,4,8,16,14 | (0,0,0,0)(1,1,1,1)(1,1,1,0)<br>(2,2,2,1)(2,2,1,0) |
| 1,2,4,8,16,14,24,22 | (0,0,0,0)(1,1,1,1)(1,1,1,0)(2,2,2,1)<br>(2,2,1,0)(3,3,2,1)(3,3,1,0) |
| 1,2,4,8,16,15 | (0,0,0,0)(1,1,1,1)(1,1,1,0)<br>(2,2,2,1)(2,2,2,0) |
| 1,2,4,8,16,15,27 | (0,0,0,0)(1,1,1,1)(1,1,1,0)<br>(2,2,2,1)(2,2,2,0)(3,3,3,1) |
| 1,2,4,8,16,15,27,26 | (0,0,0,0)(1,1,1,1)(1,1,1,0)(2,2,2,1)<br>(2,2,2,0)(3,3,3,1)(3,3,3,0) |
| 1,2,4,8,16,16 | (0,0,0,0)(1,1,1,1)(1,1,1,1) |
| 1,2,4,8,16,16,16 | (0,0,0,0)(1,1,1,1)(1,1,1,1)(1,1,1,1) |
| 1,2,4,8,16,17 | (0,0,0,0)(1,1,1,1)(2,0,0,0) |
| 1,2,4,8,16,18 | (0,0,0,0)(1,1,1,1)(2,1,0,0) |
| 1,2,4,8,16,18,7,12,21 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,0,0)(2,2,1,1) |
| 1,2,4,8,16,18,7,12,21,23 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,0,0)(2,2,1,1)(3,1,0,0) |
| 1,2,4,8,16,18,8 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)(2,2,1,0) |
| 1,2,4,8,16,18,8,16,18 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,0,0)(2,2,1,1)(3,1,0,0)<br>(2,2,1,0)(3,3,2,1)(4,1,0,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,16,18,16 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,0,0)(2,2,1,1) |
| 1,2,4,8,16,18,22 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,0,0)(2,2,1,1)(3,1,1,0) |
| 1,2,4,8,16,18,22,28 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,1,0)(4,2,1,0) |
| 1,2,4,8,16,18,22,29 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,1,0)(4,2,2,0) |
| 1,2,4,8,16,18,22,30 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,0,0)<br>(2,2,1,1)(3,1,1,0)(4,2,2,1) |
| 1,2,4,8,16,18,<br>22,30,32,36,44 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,0,0)(2,2,1,1)(3,1,1,0)<br>(4,2,2,1)(5,1,1,0)(6,2,2,1) |
| 1,2,4,8,16,19,8 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0) |
| 1,2,4,8,16,19,8,16,8 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)(3,1,0,0) |
| 1,2,4,8,16,<br>19,8,16,19,8 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(1,1,1,0) |
| 1,2,4,8,16,19,9 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(2,0,0,0) |
| 1,2,4,8,16,19,16 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(2,2,2,1) |
| 1,2,4,8,16,19,16,19,16 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)(3,1,0,0)<br>(2,2,2,1)(3,1,0,0)(2,2,2,1) |
| 1,2,4,8,16,19,17 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(3,0,0,0) |
| 1,2,4,8,16,19,18 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(3,1,0,0) |
| 1,2,4,8,16,19,19,8 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(3,1,0,0)(1,1,1,0) |
| 1,2,4,8,16,19,19,9 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(3,1,0,0)(2,2,2,1) |
| 1,2,4,8,16,19,23 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(4,2,0,0) |
| 1,2,4,8,16,19,24 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(4,2,1,0) |
| 1,2,4,8,16,19,24,37 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,0,0)(4,2,1,1) |
| 1,2,4,8,16,20 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)(3,1,1,0) |
| 1,2,4,8,16,20,26 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,1,0)(4,2,1,0) |
| 1,2,4,8,16,20,27 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,1,0)(4,2,2,0) |
| 1,2,4,8,16,20,28 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,1,1,0)(4,2,2,1) |
| 1,2,4,8,16,20,28,32,40 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)(3,1,1,0)<br>(4,2,2,1)(5,1,1,0)(6,2,2,1) |
| 1,2,4,8,16,21 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,1,0)(2,2,2,1)(3,2,0,0) |
| 1,2,4,8,16,21,15 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,0)<br>(2,2,2,1)(3,2,0,0)(2,2,2,0) |
| 1,2,4,8,16,21,16 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(1,1,1,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,16,21,16,21,16 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(1,1,1,1)(2,1,0,0)(1,1,1,1) |
| 1,2,4,8,16,21,21,16 | (0,0,0,0)(1,1,1,1)(2,1,0,0)<br>(2,1,0,0)(1,1,1,1) |
| 1,2,4,8,16,21,27 | (0,0,0,0)(1,1,1,1)(2,1,0,0)(3,2,0,0) |
| 1,2,4,8,16,22 | (0,0,0,0)(1,1,1,1)(2,1,1,0) |
| 1,2,4,8,16,22,8,16 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1) |
| 1,2,4,8,16,22,8,16,20 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,1,1,0) |
| 1,2,4,8,16,<br>22,8,16,20,28 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,1,1,0)(4,2,2,1) |
| 1,2,4,8,16,22,8,16,21 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,0,0) |
| 1,2,4,8,16,22,8,16,22 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,1,0) |
| 1,2,4,8,16,22,14,24,32 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,1,0)<br>(2,2,1,0)(3,3,2,1)(4,3,1,0) |
| 1,2,4,8,16,22,15,27 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,1,0)<br>(2,2,2,0)(3,3,3,1)(4,3,1,0) |
| 1,2,4,8,16,22,16 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)(2,2,2,1) |
| 1,2,4,8,16,22,22,16 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)(3,2,1,0)(2,2,2,1) |
| 1,2,4,8,16,22,23 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)(4,0,0,0) |
| 1,2,4,8,16,22,28,16 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)(4,2,1,0)(2,2,2,1) |
| 1,2,4,8,16,22,30 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)(4,3,1,0) |
| 1,2,4,8,16,22,31 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)(4,3,2,0) |
| 1,2,4,8,16,22,32 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,1,0)(4,3,2,1) |
| 1,2,4,8,16,22,32,40 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,1,0)<br>(4,3,2,1)(5,4,1,0)(6,5,2,1) |
| 1,2,4,8,16,23 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,2,0) |
| 1,2,4,8,16,23,14,24,23 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,2,0)<br>(2,2,1,0)(3,3,2,1)(4,3,2,0) |
| 1,2,4,8,16,23,15 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,0)<br>(2,2,2,1)(3,2,2,0)(2,2,2,0) |
| 1,2,4,8,16,23,15,27,38 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,2,0)<br>(2,2,2,0)(3,3,3,1)(4,3,3,0) |
| 1,2,4,8,16,23,16 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(1,1,1,1) |
| 1,2,4,8,16,23,16,20 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0) |
| 1,2,4,8,16,23,16,23,16 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(1,1,1,1)(2,1,1,0)(1,1,1,1) |
| 1,2,4,8,16,23,20 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(2,1,1,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,16,23,23,16 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(2,1,1,0)(1,1,1,1) |
| 1,2,4,8,16,23,30,16 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,1,1,0)(1,1,1,1) |
| 1,2,4,8,16,23,32 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(3,2,1,0) |
| 1,2,4,8,16,23,33 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(3,2,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,2,0)(4,3,2,0) |
| 1,2,4,8,16,23,33,16 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,2,1,0)(1,1,1,1) |
| 1,2,4,8,16,23,33,46,16 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,2,1,0)(4,3,1,0)(1,1,1,1) |
| 1,2,4,8,16,23,34 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(3,2,2,0) |
| 1,2,4,8,16,23,34,50 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,2,2,0)(4,3,3,0) |
| 1,2,4,8,16,23,35 | (0,0,0,0)(1,1,1,1)(2,1,1,0)(3,2,2,1) |
| 1,2,4,8,16,23,35,35 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(3,2,2,1) |
| 1,2,4,8,16,23,35,42,16 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,1,1,0)(1,1,1,1) |
| 1,2,4,8,16,23,35,45,16 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,1,0)(1,1,1,1) |
| 1,2,4,8,16,23,35,46 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,0) |
| 1,2,4,8,16,23,35,46,63 | (0,0,0,0)(1,1,1,1)(2,1,1,0)<br>(3,2,2,1)(4,2,2,0)(5,3,3,1) |
| 1,2,4,8,16,24 | (0,0,0,0)(1,1,1,1)(2,1,1,1) |
| 1,2,4,8,16,24,16,24 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(1,1,1,1)(2,1,1,1) |
| 1,2,4,8,16,24,20 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(2,1,1,0) |
| 1,2,4,8,16,24,23,16,24 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(2,1,1,0)(1,1,1,1)(2,1,1,1) |
| 1,2,4,8,16,24,23,30,16,24 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(2,1,1,0)<br>(3,1,1,0)(1,1,1,1)(2,1,1,1) |
| 1,2,4,8,16,24,23,33,16,24 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(2,1,1,0)<br>(3,2,1,0)(1,1,1,1)(2,1,1,1) |
| 1,2,4,8,16,24,23,34 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(2,1,1,0)(3,2,2,0) |
| 1,2,4,8,16,24,23,35 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(2,1,1,0)(3,2,2,1) |
| 1,2,4,8,16,24,23,35,47 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(2,1,1,0)(3,2,2,1)(4,2,2,1) |
| 1,2,4,8,16,24,24 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(2,1,1,1) |
| 1,2,4,8,16,24,25 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(3,0,0,0) |
| 1,2,4,8,16,24,26 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(3,1,0,0) |
| 1,2,4,8,16,24,27,16 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(1,1,1,1) |
| 1,2,4,8,16,24,27,16,24 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(1,1,1,1)(2,1,1,1) |
| 1,2,4,8,16,24,17,16,24,26 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(1,1,1,1)(2,1,1,1)(3,1,0,0) |
| 1,2,4,8,16,24,27,17 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,0,0,0) |
| 1,2,4,8,16,24,27,18 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,0,0) |
| 1,2,4,8,16,24,27,19 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,0) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,16,24,<br>27,23,16,24,27,18 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(3,1,0,0)<br>(2,1,1,0)(1,1,1,1)(2,1,1,1)(3,1,0,0) |
| 1,2,4,8,16,24,27,23,17 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,0)(2,0,0,0) |
| 1,2,4,8,16,24,<br>27,23,33,16,24,26 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,0)(3,2,1,0)<br>(1,1,1,1)(2,1,1,1)(3,1,0,0) |
| 1,2,4,8,16,24,27,23,34 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,0)(3,2,2,0) |
| 1,2,4,8,16,24,27,23,35 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,0)(3,2,2,1) |
| 1,2,4,8,16,24,27,<br>23,35,47,50,36 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,0)(3,2,2,1)<br>(4,2,2,1)(5,2,0,0)(4,0,0,0) |
| 1,2,4,8,16,24,27,24 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(2,1,1,1) |
| 1,2,4,8,16,24,27,31 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,0,0)(4,2,0,0) |
| 1,2,4,8,16,24,28 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(3,1,1,0) |
| 1,2,4,8,16,24,31 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(3,1,1,0)<br>(1,1,1,0)(2,2,2,1)(3,2,2,1)(4,2,2,0) |
| 1,2,4,8,16,24,31,16 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(1,1,1,1) |
| 1,2,4,8,16,24,31,16,24 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(1,1,1,1)(2,1,1,1) |
| 1,2,4,8,16,<br>24,31,16,24,28 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(3,1,1,0)<br>(1,1,1,1)(2,1,1,1)(3,1,1,0) |
| 1,2,4,8,16,24,31,17 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,0,0,0) |
| 1,2,4,8,16,24,31,18 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,1,0,0) |
| 1,2,4,8,16,24,31,20 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,1,1,0) |
| 1,2,4,8,16,24,31,23,16 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,1,1,0)(1,1,1,1) |
| 1,2,4,8,16,24,<br>31,23,16,24,31,17 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,1,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)(2,0,0,0) |
| 1,2,4,8,16,24,31,23,31 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,1,1,0)(3,2,0,0) |
| 1,2,4,8,16,24,<br>31,23,33,16,24,31,17 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(3,1,1,0)<br>(2,1,1,0)(3,2,1,0)(1,1,1,1)<br>(2,1,1,1)(3,1,1,0)(2,0,0,0) |
| 1,2,4,8,16,24,31,23,34 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,1,1,0)(3,2,2,0) |
| 1,2,4,8,16,24,31,24 | (0,0,0,0)(1,1,1,1)(2,1,1,1)<br>(3,1,1,0)(2,1,1,1) |
| 1,2,4,8,16,24,32 | (0,0,0,0)(1,1,1,1)(2,1,1,1)(3,1,1,1) |
| 1,2,4,8,16,26 | (0,0,0,0)(1,1,1,1)(2,2,1,0) |
| 1,2,4,8,16,27 | (0,0,0,0)(1,1,1,1)(2,2,1,0)<br>(1,1,1,0)(2,2,2,1)(3,3,2,0) |
| 1,2,4,8,16,27,16 | (0,0,0,0)(1,1,1,1)(2,2,1,0)(1,1,1,1) |
| 1,2,4,8,16,28 | (0,0,0,0)(1,1,1,1)(2,2,1,1) |
| 1,2,4,8,16,28,44 | (0,0,0,0)(1,1,1,1)(2,2,1,1)(3,3,1,1) |
| 1,2,4,8,16,28,44,64 | (0,0,0,0)(1,1,1,1)(2,2,1,1)<br>(3,3,1,1)(4,4,1,1) |

| Y sequence | BMS |
|---|---|
| 1,2,4,8,16,29 | (0,0,0,0)(1,1,1,1)(2,2,2,0) |
| 1,2,4,8,16,31 | (0,0,0,0)(1,1,1,1)(2,2,2,2) |
| 1,2,4,8,16,31,57 | (0,0,0,0)(1,1,1,1)(2,2,2,2)(3,3,3,3) |
| 1,2,4,8,16,32 | (0,0,0,0,0)(1,1,1,1,1) |
| 1,2,4,8,16,32,48 | (0,0,0,0,0)(1,1,1,1,1)(2,1,1,1,1) |
| 1,2,4,8,16,32,56 | (0,0,0,0,0)(1,1,1,1,1)(2,2,1,1,1) |
| 1,2,4,8,16,32,60 | (0,0,0,0,0)(1,1,1,1,1)(2,2,2,1,1) |
| 1,2,4,8,16,32,61 | (0,0,0,0,0)(1,1,1,1,1)(2,2,2,2,0) |
| 1,2,4,8,16,32,62 | (0,0,0,0,0)(1,1,1,1,1)(2,2,2,2,1) |
| 1,2,4,8,16,32,63 | (0,0,0,0,0)(1,1,1,1,1)(2,2,2,2,2) |
| 1,2,4,8,16,32,64 | (0,0,0,0,0,0)(1,1,1,1,1,1) |
| 1,2,4,8,16,32,64,128 | (0,0,0,0,0,0,0)(1,1,1,1,1,1,1) |
| 1,2,4,8,16,32,64,128,256 | (0,0,0,0,0,0,0,0)(1,1,1,1,1,1,1,1) |

## A.21 Y sequence vs 0 − Y

The results in this section are primarily cited from [2].

| Y sequence | 0 − Y sequence |
|---|---|
| 1 | 1 |
| 1,1 | 1,1 |
| 1,1,1 | 1,1,1 |
| 1,1,1,1 | 1,1,1,1 |
| 1,2 | 1,2 |
| 1,2,1 | 1,2,1 |
| 1,2,1,1 | 1,2,1,1 |
| 1,2,1,2 | 1,2,1,2 |
| 1,2,1,2,1 | 1,2,1,2,1 |
| 1,2,1,2,1,2 | 1,2,1,2,1,2 |
| 1,2,2 | 1,2,2 |
| 1,2,2,1 | 1,2,2,1 |
| 1,2,2,1,2 | 1,2,2,1,2 |
| 1,2,2,1,2,2 | 1,2,2,1,2,2 |
| 1,2,2,2 | 1,2,2,2 |
| 1,2,2,2,2 | 1,2,2,2,2 |
| 1,2,3 | 1,2,3 |
| 1,2,3,1 | 1,2,3,1 |
| 1,2,3,1,2 | 1,2,3,1,2 |
| 1,2,3,1,2,3 | 1,2,3,1,2,3 |
| 1,2,3,2 | 1,2,3,2 |
| 1,2,3,2,2 | 1,2,3,2,2 |
| 1,2,3,2,3 | 1,2,3,2,3 |
| 1,2,3,2,3,2 | 1,2,3,2,3,2 |
| 1,2,3,2,3,2,3 | 1,2,3,2,3,2,3 |
| 1,2,3,3 | 1,2,3,3 |
| 1,2,3,3,2 | 1,2,3,3,2 |
| 1,2,3,3,2,3 | 1,2,3,3,2,3 |
| 1,2,3,3,2,3,3 | 1,2,3,3,2,3,3 |
| 1,2,3,3,3 | 1,2,3,3,3 |
| 1,2,3,3,3,3 | 1,2,3,3,3,3 |
| 1,2,3,4 | 1,2,3,4 |
| 1,2,3,4,2 | 1,2,3,4,2 |
| 1,2,3,4,2,3,4 | 1,2,3,4,2,3,4 |
| 1,2,3,4,3 | 1,2,3,4,3 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,3,4,3,4 | 1,2,3,4,3,4 |
| 1,2,3,4,4 | 1,2,3,4,4 |
| 1,2,3,4,5 | 1,2,3,4,5 |
| 1,2,3,4,5,4 | 1,2,3,4,5,4 |
| 1,2,3,4,5,4,5 | 1,2,3,4,5,4,5 |
| 1,2,3,4,5,5 | 1,2,3,4,5,5 |
| 1,2,3,4,5,6 | 1,2,3,4,5,6 |
| 1,2,3,4,5,6,7 | 1,2,3,4,5,6,7 |
| 1,2,4 | 1,3 |
| 1,2,4,1 | 1,3,1 |
| 1,2,4,1,2 | 1,3,1,2 |
| 1,2,4,1,2,3 | 1,3,1,2,3 |
| 1,2,4,1,2,4 | 1,3,1,3 |
| 1,2,4,2 | 1,3,2 |
| 1,2,4,2,2 | 1,3,2,2 |
| 1,2,4,2,3 | 1,3,2,3 |
| 1,2,4,2,4 | 1,3,2,4 |
| 1,2,4,3 | 1,3,2,4,3 |
| 1,2,4,3,5 | 1,3,2,4,3,5 |
| 1,2,4,3,5,4,6 | 1,3,2,4,3,5,4,6 |
| 1,2,4,4 | 1,3,3 |
| 1,2,4,4,1,2,4,4 | 1,3,3,1,3,3 |
| 1,2,4,4,2 | 1,3,3,2 |
| 1,2,4,4,2,4,4 | 1,3,3,2,4,4 |
| 1,2,4,4,3 | 1,3,3,2,4,4,3 |
| 1,2,4,4,3,5,5 | 1,3,3,2,4,4,3,5,5 |
| 1,2,4,4,4 | 1,3,3,3 |
| 1,2,4,4,4,4 | 1,3,3,3,3 |
| 1,2,4,5 | 1,3,4 |
| 1,2,4,5,2 | 1,3,4,2 |
| 1,2,4,5,2,4,5 | 1,3,4,2,4,5 |
| 1,2,4,5,3 | 1,3,4,2,4,5,3 |
| 1,2,4,5,3,5,6 | 1,3,4,2,4,5,3,5,6 |
| 1,2,4,5,4 | 1,3,4,3 |
| 1,2,4,5,4,4 | 1,3,4,3,3 |
| 1,2,4,5,4,5 | 1,3,4,3,4 |
| 1,2,4,5,4,5,4,5 | 1,3,4,3,4,3,4 |
| 1,2,4,5,5 | 1,3,4,4 |
| 1,2,4,5,5,4,5 | 1,3,4,4,3,4 |
| 1,2,4,5,5,4,5,5 | 1,3,4,4,3,4,4 |
| 1,2,4,5,5,5 | 1,3,4,4,4 |
| 1,2,4,5,5,5,5 | 1,3,4,4,4,4 |
| 1,2,4,5,6 | 1,3,4,5 |
| 1,2,4,5,6,7 | 1,3,4,5,6 |
| 1,2,4,5,7 | 1,3,4,6 |
| 1,2,4,5,7,7 | 1,3,4,6,6 |
| 1,2,4,5,7,8 | 1,3,4,6,7 |
| 1,2,4,5,7,8,10 | 1,3,4,6,7,9 |
| 1,2,4,6 | 1,3,5 |
| 1,2,4,6,2 | 1,3,5,2 |
| 1,2,4,6,2,4,6 | 1,3,5,2,4,6 |
| 1,2,4,6,3 | 1,3,5,2,4,6,3 |
| 1,2,4,6,3,5,7 | 1,3,5,2,4,6,3,5,7 |
| 1,2,4,6,4 | 1,3,5,3 |
| 1,2,4,6,4,5 | 1,3,5,3,4 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,6,4,5,4 | 1,3,5,3,4,3 |
| 1,2,4,6,4,5,4,5 | 1,3,5,3,4,3,4 |
| 1,2,4,6,4,5,5 | 1,3,5,3,4,4 |
| 1,2,4,6,4,5,6 | 1,3,5,3,4,5 |
| 1,2,4,6,4,5,7 | 1,3,5,3,4,6 |
| 1,2,4,6,4,5,7,8 | 1,3,5,3,4,6,7 |
| 1,2,4,6,4,5,7,8,10 | 1,3,5,3,4,6,7,9 |
| 1,2,4,6,4,5,7,9 | 1,3,5,3,4,6,8 |
| 1,2,4,6,4,5,7,9,7 | 1,3,5,3,4,6,8,6 |
| 1,2,4,6,4,6 | 1,3,5,3,5 |
| 1,2,4,6,4,6,4,6 | 1,3,5,3,5,3,5 |
| 1,2,4,6,5 | 1,3,5,4 |
| 1,2,4,6,5,5 | 1,3,5,4,4 |
| 1,2,4,6,5,6 | 1,3,5,4,5 |
| 1,2,4,6,5,7 | 1,3,5,4,6 |
| 1,2,4,6,5,7,9 | 1,3,5,4,6,8 |
| 1,2,4,6,5,7,9,8,10,12 | 1,3,5,4,6,8,7,9,11 |
| 1,2,4,6,6 | 1,3,5,5 |
| 1,2,4,6,6,4 | 1,3,5,5,3 |
| 1,2,4,6,6,4,5,7,9,9 | 1,3,5,5,3,4,6,8,8 |
| 1,2,4,6,6,4,<br>5,7,9,9,7 | 1,3,5,5,3,4,6,8,8,6 |
| 1,2,4,6,6,4,6 | 1,3,5,5,3,5 |
| 1,2,4,6,6,4,<br>6,5,7,9,9 | 1,3,5,5,3,5,4,6,8,8 |
| 1,2,4,6,6,4,<br>6,5,7,9,9,7,9 | 1,3,5,5,3,5,<br>4,6,8,8,6,8 |
| 1,2,4,6,6,4,6,6 | 1,3,5,5,3,5,5 |
| 1,2,4,6,6,5 | 1,3,5,5,4 |
| 1,2,4,6,6,5,7 | 1,3,5,5,4,6 |
| 1,2,4,6,6,5,7,8 | 1,3,5,5,4,6,7 |
| 1,2,4,6,6,5,7,9 | 1,3,5,5,4,6,8 |
| 1,2,4,6,6,5,7,9,9 | 1,3,5,5,4,6,8,8 |
| 1,2,4,6,6,6 | 1,3,5,5,5 |
| 1,2,4,6,6,6,4,6,6,6 | 1,3,5,5,5,3,5,5,5 |
| 1,2,4,6,6,6,5 | 1,3,5,5,5,4 |
| 1,2,4,6,6,6,6 | 1,3,5,5,5,5 |
| 1,2,4,6,7 | 1,3,5,6 |
| 1,2,4,6,7,2 | 1,3,5,6,2 |
| 1,2,4,6,7,3 | 1,3,5,6,2,4,6,7,3 |
| 1,2,4,6,7,4 | 1,3,5,6,3 |
| 1,2,4,6,7,4,6,7 | 1,3,5,6,3,5,6 |
| 1,2,4,6,7,5 | 1,3,5,6,4 |
| 1,2,4,6,7,5,7,9,10 | 1,3,5,6,4,6,8,9 |
| 1,2,4,6,7,6 | 1,3,5,6,5 |
| 1,2,4,6,7,6,4,6,7 | 1,3,5,6,5,3,5,6 |
| 1,2,4,6,7,6,4,6,7,6 | 1,3,5,6,5,3,5,6,5 |
| 1,2,4,6,7,6,6 | 1,3,5,6,5,5 |
| 1,2,4,6,7,6,7 | 1,3,5,6,5,6 |
| 1,2,4,6,7,7 | 1,3,5,6,6 |
| 1,2,4,6,7,8 | 1,3,5,6,7 |
| 1,2,4,6,7,8,5 | 1,3,5,6,7,4 |
| 1,2,4,6,7,9 | 1,3,5,6,8 |
| 1,2,4,6,7,9,6 | 1,3,5,6,8,5 |
| 1,2,4,6,7,9,6,2,4 | 1,3,5,6,8,5,2,4 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,6,7,9, 6,4,6,7,9 | 1,3,5,6,8,5,3,5,6,8 |
| 1,2,4,6,7,9, 7,4,6,7,9 | 1,3,5,6,8,6,3,5,6,8 |
| 1,2,4,6,7,9,9 | 1,3,5,6,8,8 |
| 1,2,4,6,7,9,9,9 | 1,3,5,6,8,8,8 |
| 1,2,4,6,7,9,10 | 1,3,5,6,8,9 |
| 1,2,4,6,7,9, 10,4,6,7,9 | 1,3,5,6,8,9,3,5,6,8 |
| 1,2,4,6,7,9,11 | 1,3,5,6,8,10 |
| 1,2,4,6,7,9, 11,9,10,12,14 | 1,3,5,6,8,10,8,9,11,13 |
| 1,2,4,6,7,9,11,11 | 1,3,5,6,8,10,10 |
| 1,2,4,6,7,9,11,12 | 1,3,5,6,8,10,11 |
| 1,2,4,6,7,9,11,12,12 | 1,3,5,6,8,10,11,11 |
| 1,2,4,6,7,9,11,12,14 | 1,3,5,6,8,10,11,13 |
| 1,2,4,6,7,9,11,12,14,16 | 1,3,5,6,8,10,11,13,15 |
| 1,2,4,6,8 | 1,3,5,7 |
| 1,2,4,6,8,2 | 1,3,5,7,2 |
| 1,2,4,6,8,2,2 | 1,3,5,7,2,2 |
| 1,2,4,6,8,2,3 | 1,3,5,7,2,3 |
| 1,2,4,6,8,2,4 | 1,3,5,7,2,4 |
| 1,2,4,6,8,2,4,5 | 1,3,5,7,2,4,5 |
| 1,2,4,6,8,2,4,5,7 | 1,3,5,7,2,4,5,7 |
| 1,2,4,6,8,2,4,6 | 1,3,5,7,2,4,6 |
| 1,2,4,6,8,2,4,6,7 | 1,3,5,7,2,4,6,7 |
| 1,2,4,6,8,2,4,6,7,9 | 1,3,5,7,2,4,6,7,9 |
| 1,2,4,6,8,2, 4,6,7,9,11 | 1,3,5,7,2,4,6,7,9,11 |
| 1,2,4,6,8,2,4,6,8 | 1,3,5,7,2,4,6,8 |
| 1,2,4,6,8,2,4,6,8,2 | 1,3,5,7,2,4,6,8,2 |
| 1,2,4,6,8,2, 4,6,8,2,4,6,8 | 1,3,5,7,2,4, 6,8,2,4,6,8 |
| 1,2,4,6,8,3 | 1,3,5,7,2,4,6,8,3 |
| 1,2,4,6,8,3,5,7,9 | 1,3,5,7,2,4, 6,8,3,5,7,9 |
| 1,2,4,6,8,4 | 1,3,5,7,3 |
| 1,2,4,6,8,4,2 | 1,3,5,7,3,2 |
| 1,2,4,6,8,4,2,4 | 1,3,5,7,3,2,4 |
| 1,2,4,6,8,4,2,4,6 | 1,3,5,7,3,2,4,6 |
| 1,2,4,6,8,4,2,4,6,8 | 1,3,5,7,3,2,4,6,8 |
| 1,2,4,6,8,4, 2,4,6,8,3 | 1,3,5,7,3,2,4,6,8,3 |
| 1,2,4,6,8,4, 2,4,6,8,4 | 1,3,5,7,3,2,4,6,8,4 |
| 1,2,4,6,8,4,4 | 1,3,5,7,3,3 |
| 1,2,4,6,8,4,5 | 1,3,5,7,3,4 |
| 1,2,4,6,8,4,5,7,9,11 | 1,3,5,7,3,4,6,8,10 |
| 1,2,4,6,8,4, 5,7,9,11,7 | 1,3,5,7,3,4,6,8,10,6 |
| 1,2,4,6,8,4,6 | 1,3,5,7,3,5 |
| 1,2,4,6,8,4,6,7 | 1,3,5,7,3,5,6 |
| 1,2,4,6,8,4,6,7,9 | 1,3,5,7,3,5,8 |
| 1,2,4,6,8,4,6,7,9,11 | 1,3,5,7,3,5,6,8,10 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,6,8,4,<br>6,7,9,11,13 | 1,3,5,7,3,5,6,8,10,12 |
| 1,2,4,6,8,4,<br>6,7,9,11,13,5 | 1,3,5,7,3,5,<br>6,8,10,12,4 |
| 1,2,4,6,8,4,6,7,9,<br>11,13,5,4,6,7,9,11,13 | 1,3,5,7,3,5,6,8,10,<br>12,4,3,5,6,8,10,12 |
| 1,2,4,6,8,4,<br>6,7,9,11,13,5,5 | 1,3,5,7,3,5,<br>6,8,10,12,4,4 |
| 1,2,4,6,8,4,<br>6,7,9,11,13,5,6 | 1,3,5,7,3,5,<br>6,8,10,12,4,5 |
| 1,2,4,6,8,4,6,<br>7,9,11,13,5,7 | 1,3,5,7,3,5,<br>6,8,10,12,4,6 |
| 1,2,4,6,8,4,6,7,<br>9,11,13,5,7,9,11 | 1,3,5,7,3,5,6,<br>8,10,12,4,6,8,10 |
| 1,2,4,6,8,4,<br>6,7,9,11,13,6 | 1,3,5,7,3,5,<br>6,8,10,12,5 |
| 1,2,4,6,8,4,6,7,<br>9,11,13,6,7,9,11,13 | 1,3,5,7,3,5,6,8,<br>10,12,5,6,8,10,12 |
| 1,2,4,6,8,4,<br>6,7,9,11,13,7 | 1,3,5,7,3,5,<br>6,8,10,12,6 |
| 1,2,4,6,8,4,6,7,<br>9,11,13,7,9,11,13 | 1,3,5,7,3,5,6,8,<br>10,12,6,8,10,12 |
| 1,2,4,6,8,4,<br>6,7,9,11,13,9 | 1,3,5,7,3,5,<br>6,8,10,12,8 |
| 1,2,4,6,8,4,6,7,<br>9,11,13,9,10,12,14,16 | 1,3,5,7,3,5,6,8,10,<br>12,8,10,11,13,15,17 |
| 1,2,4,6,8,4,6,8 | 1,3,5,7,3,5,7 |
| 1,2,4,6,8,5 | 1,3,5,7,4 |
| 1,2,4,6,8,5,7,9,11 | 1,3,5,7,4,6,8,10 |
| 1,2,4,6,8,6 | 1,3,5,7,5 |
| 1,2,4,6,8,6,7 | 1,3,5,7,5,6 |
| 1,2,4,6,8,6,7,9 | 1,3,5,7,5,6,8 |
| 1,2,4,6,8,6,7,9,11,13 | 1,3,5,7,5,6,8,10,12 |
| 1,2,4,6,8,6,8 | 1,3,5,7,5,7 |
| 1,2,4,6,8,6,8,6,8 | 1,3,5,7,5,7,5,7 |
| 1,2,4,6,8,7 | 1,3,5,7,6 |
| 1,2,4,6,8,7,5 | 1,3,5,7,6,4 |
| 1,2,4,6,8,7,6 | 1,3,5,7,6,5 |
| 1,2,4,6,8,7,6,7 | 1,3,5,7,6,5,6 |
| 1,2,4,6,8,7,6,8 | 1,3,5,7,6,4 |
| 1,2,4,6,8,7,6,8,7 | 1,3,5,7,6,5,7 |
| 1,2,4,6,8,7,7 | 1,3,5,7,6,4 |
| 1,2,4,6,8,7,8 | 1,3,5,7,6,6 |
| 1,2,4,6,8,7,9 | 1,3,5,7,6,8 |
| 1,2,4,6,8,7,9,11,13 | 1,3,5,7,6,8,10,12 |
| 1,2,4,6,8,8 | 1,3,5,7,7 |
| 1,2,4,6,8,8,6 | 1,3,5,7,7,5 |
| 1,2,4,6,8,8,6,8 | 1,3,5,7,7,5,7 |
| 1,2,4,6,8,8,6,8,7 | 1,3,5,7,7,5,7,6 |
| 1,2,4,6,8,8,6,8,8 | 1,3,5,7,7,5,7,7 |
| 1,2,4,6,8,8,7 | 1,3,5,7,7,6 |
| 1,2,4,6,8,8,8 | 1,3,5,7,7,7 |
| 1,2,4,6,8,8,8,6,8,8,8 | 1,3,5,7,7,7,5,7,7,7 |
| 1,2,4,6,8,8,8,7 | 1,3,5,7,7,7,6 |
| 1,2,4,6,8,8,8,8 | 1,3,5,7,7,7,7 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,6,8,9 | 1,3,5,7,8 |
| 1,2,4,6,8,9,8 | 1,3,5,7,8,7 |
| 1,2,4,6,8,9,8,9 | 1,3,5,7,8,7,8 |
| 1,2,4,6,8,9,9 | 1,3,5,7,8,8 |
| 1,2,4,6,8,9,11 | 1,3,5,7,8,10 |
| 1,2,4,6,8,9,11,13,15 | 1,3,5,7,8,10,12,14 |
| 1,2,4,6,8,9,11,13,15,16 | 1,3,5,7,8,10,12,14,15 |
| 1,2,4,6,8,9, 11,13,15,16,18 | 1,3,5,7,8,10,12,14,15,17 |
| 1,2,4,6,8,10 | 1,3,5,7,9 |
| 1,2,4,6,8,10,6 | 1,3,5,7,9,5 |
| 1,2,4,6,8,10,8 | 1,3,5,7,9,7 |
| 1,2,4,6,8,10,8,10 | 1,3,5,7,9,7,9 |
| 1,2,4,6,8,10,9 | 1,3,5,7,9,8 |
| 1,2,4,6,8,10,10 | 1,3,5,7,9,9 |
| 1,2,4,6,8,10,10,10 | 1,3,5,7,9,9,9 |
| 1,2,4,6,8,10,11 | 1,3,5,7,9,10 |
| 1,2,4,6,8,10,12 | 1,3,5,7,9,11 |
| 1,2,4,6,8,10,12,14 | 1,3,5,7,9,11,13 |
| 1,2,4,7 | 1,3,6 |
| 1,2,4,7,2 | 1,3,6,2 |
| 1,2,4,7,2,4,7 | 1,3,6,2,4,7 |
| 1,2,4,7,4 | 1,3,6,3 |
| 1,2,4,7,4,6 | 1,3,6,3,5 |
| 1,2,4,7,4,6,8 | 1,3,6,3,5,7 |
| 1,2,4,7,4,7 | 1,3,6,3,6 |
| 1,2,4,7,4,7,4,7 | 1,3,6,3,6,3,6 |
| 1,2,4,7,5 | 1,3,6,4 |
| 1,2,4,7,5,5 | 1,3,6,4,4 |
| 1,2,4,7,5,6 | 1,3,6,4,5 |
| 1,2,4,7,5,7 | 1,3,6,4,6 |
| 1,2,4,7,5,7,10 | 1,3,6,4,6,9 |
| 1,2,4,7,6 | 1,3,6,5 |
| 1,2,4,7,6,7 | 1,3,6,5,6 |
| 1,2,4,7,6,8 | 1,3,6,5,7 |
| 1,2,4,7,6,9 | 1,3,6,5,8 |
| 1,2,4,7,6,9,8 | 1,3,6,5,8,7 |
| 1,2,4,7,6,9,8,11 | 1,3,6,5,8,7,10 |
| 1,2,4,7,7 | 1,3,6,6 |
| 1,2,4,7,7,4,7,7 | 1,3,6,6,3,6,6 |
| 1,2,4,7,7,5 | 1,3,6,6,4 |
| 1,2,4,7,7,6 | 1,3,6,6,5 |
| 1,2,4,7,7,6,9,9 | 1,3,6,6,5,8,8 |
| 1,2,4,7,7,7 | 1,3,6,6,6 |
| 1,2,4,7,8 | 1,3,6,7 |
| 1,2,4,7,8,10 | 1,3,6,7,9 |
| 1,2,4,7,8,10,13 | 1,3,6,7,9,12 |
| 1,2,4,7,9 | 1,3,6,8 |
| 1,2,4,7,9,7,9 | 1,3,6,8,6,8 |
| 1,2,4,7,9,8 | 1,3,6,8,7 |
| 1,2,4,7,9,9 | 1,3,6,8,8 |
| 1,2,4,7,9,12 | 1,3,6,8,11 |
| 1,2,4,7,10 | 1,3,6,9 |
| 1,2,4,7,10,10 | 1,3,6,9,9 |
| 1,2,4,7,10,11 | 1,3,6,9,10 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,7,10,12 | 1,3,6,9,11 |
| 1,2,4,7,10,12,7,10,12 | 1,3,6,9,11,6,9,11 |
| 1,2,4,7,10,12,8 | 1,3,6,9,11,7 |
| 1,2,4,7,10,12,8,10 | 1,3,6,9,11,7,9 |
| 1,2,4,7,10,12,9 | 1,3,6,9,11,8 |
| 1,2,4,7,10,12,9,12,15,17 | 1,3,6,9,11,8,11,14,16,13 |
| 1,2,4,7,10,12,10 | 1,3,6,9,11,9 |
| 1,2,4,7,10,12,10,12 | 1,3,6,9,11,9,11 |
| 1,2,4,7,10,12,11 | 1,3,6,9,11,10 |
| 1,2,4,7,10,12,12 | 1,3,6,9,11,11 |
| 1,2,4,7,10,12,13 | 1,3,6,9,11,12 |
| 1,2,4,7,10,12,14 | 1,3,6,9,11,13 |
| 1,2,4,7,10,12,15 | 1,3,6,9,11,14 |
| 1,2,4,7,10,12,15,18 | 1,3,6,9,11,14,17 |
| 1,2,4,7,10,12,15,18,20 | 1,3,6,9,11,14,17,19 |
| 1,2,4,7,10,13 | 1,3,6,9,12 |
| 1,2,4,7,10,13 ,1,2,4,7,10,13 | 1,3,6,9,12,1,3,6,9,12 |
| 1,2,4,7,10,13,4,7,10,13 | 1,3,6,9,12,3,6,9,12 |
| 1,2,4,7,10,13,6,9,12,15 | 1,3,6,9,12,5,8,11,14 |
| 1,2,4,7,10,13,7,10,13 | 1,3,6,9,12,6,9,12 |
| 1,2,4,7,10,13,10 | 1,3,6,9,12,9 |
| 1,2,4,7,10,13,16 | 1,3,6,9,12,15 |
| 1,2,4,7,11 | 1,3,6,10 |
| 1,2,4,7,11,4,7,11 | 1,3,6,10,3,6,10 |
| 1,2,4,7,11,6,9,13 | 1,3,6,10,5,8,12 |
| 1,2,4,7,11,7 | 1,3,6,10,6 |
| 1,2,4,7,11,7,10,13 | 1,3,6,10,6,9,12 |
| 1,2,4,7,11,7,11 | 1,3,6,10,6,10 |
| 1,2,4,7,11,10,14 | 1,3,6,10,9,13 |
| 1,2,4,7,11,11 | 1,3,6,10,10 |
| 1,2,4,7,11,12 | 1,3,6,10,11 |
| 1,2,4,7,11,15 | 1,3,6,10,14 |
| 1,2,4,7,11,15,19 | 1,3,6,10,14,18 |
| 1,2,4,7,11,16 | 1,3,6,10,15 |
| 1,2,4,7,11,16,21,26 | 1,3,6,10,15,20,25 |
| 1,2,4,7,11,16,22 | 1,3,6,10,15,21 |
| 1,2,4,8 | 1,4 |
| 1,2,4,8,1 | 1,4,1 |
| 1,2,4,8,2 | 1,4,2 |
| 1,2,4,8,2,4,8 | 1,4,2,5 |
| 1,2,4,8,3 | 1,4,2,5,3 |
| 1,2,4,8,4 | 1,4,3 |
| 1,2,4,8,4,6,8 | 1,4,3,5,7 |
| 1,2,4,8,4,7 | 1,4,3,6 |
| 1,2,4,8,4,7,11 | 1,4,3,6,10 |
| 1,2,4,8,4,8 | 1,4,3,7 |
| 1,2,4,8,5 | 1,4,3,7,4 |
| 1,2,4,8,5,4,8,5 | 1,4,3,7,4,3,7,4 |
| 1,2,4,8,5,5 | 1,4,3,7,4,4 |
| 1,2,4,8,5,5,4,8,5 | 1,4,3,7,4,4,3,7,4 |
| 1,2,4,8,6 | 1,4,3,7,5 |
| 1,2,4,8,6,9 | 1,4,3,7,5,8 |
| 1,2,4,8,6,10 | 1,4,3,7,5,9 |
| 1,2,4,8,6,10,7 | 1,4,3,7,5,9,6 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,6,10,8 | 1,4,3,7,5,9,7 |
| 1,2,4,8,6,10,8,12 | 1,4,3,7,5,9,7,11 |
| 1,2,4,8,7 | 1,4,3,7,6 |
| 1,2,4,8,7,12 | 1,4,3,7,6,11 |
| 1,2,4,8,7,12,8 | 1,4,3,7,6,11,7 |
| 1,2,4,8,7,12,11 | 1,4,3,7,6,11,10 |
| 1,2,4,8,8 | 1,4,4 |
| 1,2,4,8,8,4 | 1,4,4,3 |
| 1,2,4,8,8,7 | 1,4,4,3,7,7,6 |
| 1,2,4,8,8,8 | 1,4,4,4 |
| 1,2,4,8,9 | 1,4,5 |
| 1,2,4,8,9,4,8,9 | 1,4,5,3,7,8 |
| 1,2,4,8,9,8 | 1,4,5,4 |
| 1,2,4,8,9,8,9 | 1,4,5,4,5 |
| 1,2,4,8,9,9 | 1,4,5,5 |
| 1,2,4,8,9,11 | 1,4,5,7 |
| 1,2,4,8,9,11,14 | 1,4,5,7,10 |
| 1,2,4,8,9,11,15 | 1,4,5,8 |
| 1,2,4,8,10 | 1,4,6 |
| 1,2,4,8,10,4,8,10 | 1,4,6,3,7,9 |
| 1,2,4,8,10,7 | 1,4,6,3,7,9,6 |
| 1,2,4,8,10,7,12,14 | 1,4,6,3,7,9,6,11,13 |
| 1,2,4,8,10,8 | 1,4,6,3,7,9,7 |
| 1,2,4,8,10,10 | 1,4,6,3,7,9,9 |
| 1,2,4,8,10,12 | 1,4,6,3,7,9,11 |
| 1,2,4,8,10,13 | 1,4,6,3,7,9,12 |
| 1,2,4,8,10,14 | 1,4,6,3,7,9,13 |
| 1,2,4,8,11 | 1,4,6,3,7,10 |
| 1,2,4,8,11,7,12,16 | 1,4,6,3,7,10,6,11,15 |
| 1,2,4,8,11,8 | 1,4,6,4 |
| 1,2,4,8,11,8,4 | 1,4,6,4,3 |
| 1,2,4,8,11,8,4,8 | 1,4,6,4,3,7 |
| 1,2,4,8,11,8,4,8,8 | 1,4,6,4,3,7,7 |
| 1,2,4,8,11,8,4,8,10 | 1,4,6,4,3,7,9 |
| 1,2,4,8,11,8,4,8,10,8 | 1,4,6,4,3,7,9,7 |
| 1,2,4,8,11,8,4,8,11 | 1,4,6,4,3,7,10 |
| 1,2,4,8,11,8,<br>4,8,11,7,12,15,12 | 1,4,6,4,3,7,<br>10,6,11,14,11 |
| 1,2,4,8,11,8,4,8,11,8 | 1,4,6,4,3,7,10,7 |
| 1,2,4,8,11,8,7 | 1,4,6,4,3,7,10,7,6 |
| 1,2,4,8,11,8,8 | 1,4,6,4,4 |
| 1,2,4,8,11,8,10 | 1,4,6,4,6 |
| 1,2,4,8,11,8,11 | 1,4,6,4,6,3,7,10,7,10 |
| 1,2,4,8,11,8,11,8 | 1,4,6,4,6,4 |
| 1,2,4,8,11,9 | 1,4,6,5 |
| 1,2,4,8,11,9,8 | 1,4,6,5,4 |
| 1,2,4,8,11,10 | 1,4,6,6 |
| 1,2,4,8,11,11 | 1,4,6,6,3,7,10,10 |
| 1,2,4,8,11,11,8 | 1,4,6,6,4 |
| 1,2,4,8,11,11,8,11,8 | 1,4,6,6,4,6,4 |
| 1,2,4,8,11,11,8,11,9 | 1,4,6,6,4,6,5 |
| 1,2,4,8,11,11,8,11,11,8 | 1,4,6,6,4,6,6,4 |
| 1,2,4,8,11,11,9 | 1,4,6,6,5 |
| 1,2,4,8,11,11,11,8 | 1,4,6,6,6,4 |
| 1,2,4,8,11,12 | 1,4,6,7 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,11,13 | 1,4,6,8 |
| 1,2,4,8,11,14 | 1,4,6,8,3,7,10,13 |
| 1,2,4,8,11,14,8 | 1,4,6,8,4 |
| 1,2,4,8,11,14,9 | 1,4,6,8,5 |
| 1,2,4,8,11,14,11,8 | 1,4,6,8,6,4 |
| 1,2,4,8,11,14,11,14,8 | 1,4,6,8,6,8,4 |
| 1,2,4,8,11,14,12 | 1,4,6,8,7 |
| 1,2,4,8,11,14,14,8 | 1,4,6,8,8,4 |
| 1,2,4,8,11,14,14,14,8 | 1,4,6,8,8,8,4 |
| 1,2,4,8,11,14,15 | 1,4,6,8,9 |
| 1,2,4,8,11,14,17,8 | 1,4,6,8,10,4 |
| 1,2,4,8,11,14,17,14,8 | 1,4,6,8,10,8,4 |
| 1,2,4,8,11,14,17,14,17,8 | 1,4,6,8,10,8,10,4 |
| 1,2,4,8,11,14,17,14,17,12 | 1,4,6,8,10,8,10,7 |
| 1,2,4,8,11,14,17,17,8 | 1,4,6,8,10,10,4 |
| 1,2,4,8,11,14,17,18 | 1,4,6,8,10,11 |
| 1,2,4,8,11,14,17,20,8 | 1,4,6,8,10,12,4 |
| 1,2,4,8,11,15 | 1,4,6,9 |
| 1,2,4,8,11,15,4,8 | 1,4,6,9,3,7 |
| 1,2,4,8,11,15,4,8,9,11 | 1,4,6,9,3,7,8,10 |
| 1,2,4,8,11,15,4,<br>8,9,11,6,10,11,13 | 1,4,6,9,3,7,8,<br>10,5,9,10,12 |
| 1,2,4,8,11,15,<br>4,8,9,11,7 | 1,4,6,9,3,7,8,10,6 |
| 1,2,4,8,11,15,<br>4,8,9,11,8 | 1,4,6,9,3,7,8,10,7 |
| 1,2,4,8,11,15,<br>4,8,9,11,15 | 1,4,6,9,3,7,8,11 |
| 1,2,4,8,11,15,4,8,10 | 1,4,6,9,3,7,9 |
| 1,2,4,8,11,15,4,8,11,8 | 1,4,6,9,3,7,10,7 |
| 1,2,4,8,11,15,4,8,11,15 | 1,4,6,9,3,7,10,14 |
| 1,2,4,8,11,15,7,12,16,21 | 1,4,6,9,3,7,10,<br>14,6,11,15,20 |
| 1,2,4,8,11,15,8 | 1,4,6,9,4 |
| 1,2,4,8,11,15,8,11,15 | 1,4,6,9,4,6,9 |
| 1,2,4,8,11,15,11,15 | 1,4,6,9,6,9 |
| 1,2,4,8,11,15,13,17 | 1,4,6,9,8,11 |
| 1,2,4,8,11,15,15 | 1,4,6,9,9 |
| 1,2,4,8,11,15,16 | 1,4,6,9,10 |
| 1,2,4,8,11,15,18,8 | 1,4,6,9,11,4 |
| 1,2,4,8,11,15,19 | 1,4,6,9,12 |
| 1,2,4,8,11,15,19,20 | 1,4,6,9,12,13 |
| 1,2,4,8,11,15,19,22,8 | 1,4,6,9,12,14,4 |
| 1,2,4,8,11,15,19,23 | 1,4,6,9,12,15 |
| 1,2,4,8,11,15,20 | 1,4,6,9,13 |
| 1,2,4,8,11,15,20,20 | 1,4,6,9,13,13 |
| 1,2,4,8,11,15,20,21 | 1,4,6,9,13,14 |
| 1,2,4,8,11,15,20,25 | 1,4,6,9,13,17 |
| 1,2,4,8,11,15,20,25,30 | 1,4,6,9,13,17,21 |
| 1,2,4,8,11,15,20,26 | 1,4,6,9,13,18 |
| 1,2,4,8,11,16 | 1,4,6,10 |
| 1,2,4,8,11,16,4,8 | 1,4,6,10,3,7 |
| 1,2,4,8,11,16,4,8,11,8 | 1,4,6,10,3,7,10,7 |
| 1,2,4,8,11,16,<br>4,8,11,11,8 | 1,4,6,10,3,7,10,10,7 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,11,16,4,8,11,15 | 1,4,6,10,3,7,10,14 |
| 1,2,4,8,11,16, 4,8,11,15,20 | 1,4,6,10,3,7,10,14,19 |
| 1,2,4,8,11,16,4,8,11,16 | 1,4,6,10,3,7,10,15 |
| 1,2,4,8,11,16,7,12,15 | 1,4,6,10,3,7, 10,15,6,11,14 |
| 1,2,4,8,11,16,7,12,16,21 | 1,4,6,10,3,7,10, 15,6,11,15,20 |
| 1,2,4,8,11,16,7,12,16,22 | 1,4,6,10,3,7,10, 15,6,11,15,21 |
| 1,2,4,8,11,16,8 | 1,4,6,10,4 |
| 1,2,4,8,11,16,8,8 | 1,4,6,10,4,4 |
| 1,2,4,8,11,16,8,10 | 1,4,6,10,4,6 |
| 1,2,4,8,11,16,8,11 | 1,4,6,10,4,6,3, 7,10,15,7,10 |
| 1,2,4,8,11,16,8,11,8 | 1,4,6,10,4,6,4 |
| 1,2,4,8,11,16,8,11,11,8 | 1,4,6,10,4,6,6,4 |
| 1,2,4,8,11,16,8,11,12 | 1,4,6,10,4,6,7 |
| 1,2,4,8,11,16,8,11,15 | 1,4,6,10,4,6,9 |
| 1,2,4,8,11,16,8,11,16 | 1,4,6,10,4,6,10 |
| 1,2,4,8,11,16,8,11,16,8 | 1,4,6,10,4,6,10,4 |
| 1,2,4,8,11,16, 8,11,16,8,10 | 1,4,6,10,4,6,10,4,6 |
| 1,2,4,8,11,16, 8,11,16,8,11,15 | 1,4,6,10,4,6,10,4,6,9 |
| 1,2,4,8,11,16, 8,11,16,8,11,16 | 1,4,6,10,4,6,10,4,6,10 |
| 1,2,4,8,11,16,9 | 1,4,6,10,5 |
| 1,2,4,8,11,16,10 | 1,4,6,10,6 |
| 1,2,4,8,11,16,11,8 | 1,4,6,10,6,4 |
| 1,2,4,8,11,16,11,15 | 1,4,6,10,6,9 |
| 1,2,4,8,11,16,11,15,8 | 1,4,6,10,6,9,4 |
| 1,2,4,8,11,16,11,16 | 1,4,6,10,6,10 |
| 1,2,4,8,11,16,14,19 | 1,4,6,10,8,12 |
| 1,2,4,8,11,16,15 | 1,4,6,10,9 |
| 1,2,4,8,11,16,15,19,23 | 1,4,6,10,9,12,15 |
| 1,2,4,8,11,16,15,21 | 1,4,6,10,9,14 |
| 1,2,4,8,11,16,15,21,20 | 1,4,6,10,9,14,13 |
| 1,2,4,8,11,16,16 | 1,4,6,10,10 |
| 1,2,4,8,11,16,17 | 1,4,6,10,11 |
| 1,2,4,8,11,16,17,15,21,22 | 1,4,6,10,11,9,14,15 |
| 1,2,4,8,11,16,17,16 | 1,4,6,10,11,10 |
| 1,2,4,8,11,16,17,18 | 1,4,6,10,11,12 |
| 1,2,4,8,11,16,17,19 | 1,4,6,10,11,13 |
| 1,2,4,8,11,16,17,19,23 | 1,4,6,10,11,14 |
| 1,2,4,8,11,16,18 | 1,4,6,10,12 |
| 1,2,4,8,11,16,19,8 | 1,4,6,10,12,4 |
| 1,2,4,8,11,16,19,8,8 | 1,4,6,10,12,4,4 |
| 1,2,4,8,11,16, 19,8,9,11,15,18,23 | 1,4,6,10,12,4,5,8,10,14 |
| 1,2,4,8,11,16,19,8,10 | 1,4,6,10,12,4,6 |
| 1,2,4,8,11,16,19,8,11,8 | 1,4,6,10,12,4,6,4 |
| 1,2,4,8,11,16,19,8,11,11,8 | 1,4,6,10,12,4,6,6,4 |
| 1,2,4,8,11,16,19,8,11,15 | 1,4,6,10,12,4,6,9 |
| 1,2,4,8,11,16,19,8,11,16 | 1,4,6,10,12,4,6,10 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,11,16,<br>19,8,11,16,19,8 | 1,4,6,10,12,4,6,10,12,4 |
| 1,2,4,8,11,16,19,9 | 1,4,6,10,12,5 |
| 1,2,4,8,11,16,19,10 | 1,4,6,10,12,6 |
| 1,2,4,8,11,16,19,11,8 | 1,4,6,10,12,6,4 |
| 1,2,4,8,11,16,<br>19,11,8,11,16 | 1,4,6,10,12,6,4,6,10 |
| 1,2,4,8,11,16,19,11,9 | 1,4,6,10,12,6,5 |
| 1,2,4,8,11,16,19,11,11,8 | 1,4,6,10,12,6,6,4 |
| 1,2,4,8,11,16,19,11,14,8 | 1,4,6,10,12,6,8,4 |
| 1,2,4,8,11,16,19,11,15 | 1,4,6,10,12,6,9 |
| 1,2,4,8,11,16,19,11,16 | 1,4,6,10,12,6,10 |
| 1,2,4,8,11,16,<br>19,11,16,19,8 | 1,4,6,10,12,6,10,12,4 |
| 1,2,4,8,11,16,19,12 | 1,4,6,10,12,7 |
| 1,2,4,8,11,16,19,14,8 | 1,4,6,10,12,8,4 |
| 1,2,4,8,11,16,19,14,17,8 | 1,4,6,10,12,8,10,4 |
| 1,2,4,8,11,16,19,14,18 | 1,4,6,10,12,8,11 |
| 1,2,4,8,11,16,19,14,19 | 1,4,6,10,12,8,12 |
| 1,2,4,8,11,16,19,15 | 1,4,6,10,12,9 |
| 1,2,4,8,11,16,19,15,21 | 1,4,6,10,12,9,14 |
| 1,2,4,8,11,16,<br>19,15,21,24,20 | 1,4,6,10,12,9,14,16,13 |
| 1,2,4,8,11,16,19,16 | 1,4,6,10,12,10 |
| 1,2,4,8,11,16,19,17 | 1,4,6,10,12,11 |
| 1,2,4,8,11,16,19,19,8 | 1,4,6,10,12,12,4 |
| 1,2,4,8,11,16,19,22,8 | 1,4,6,10,12,14,4 |
| 1,2,4,8,11,16,19,23 | 1,4,6,10,12,15 |
| 1,2,4,8,11,16,19,24 | 1,4,6,10,12,16 |
| 1,2,4,8,11,16,19,24,27,8 | 1,4,6,10,12,16,18,4 |
| 1,2,4,8,11,16,20 | 1,4,6,10,13 |
| 1,2,4,8,11,16,20,15,21,26 | 1,4,6,10,13,9,14,18 |
| 1,2,4,8,11,16,20,16 | 1,4,6,10,13,10 |
| 1,2,4,8,11,16,20,24,16 | 1,4,6,10,13,16,10 |
| 1,2,4,8,11,16,20,25 | 1,4,6,10,13,17 |
| 1,2,4,8,11,16,20,25,31 | 1,4,6,10,13,17,22 |
| 1,2,4,8,11,16,20,26 | 1,4,6,10,13,18 |
| 1,2,4,8,11,16,20,26,31,38 | 1,4,6,10,13,18,22,28 |
| 1,2,4,8,12 | 1,4,7 |
| 1,2,4,8,12,4,8 | 1,4,7,3,7 |
| 1,2,4,8,12,4,8,10 | 1,4,7,3,7,9 |
| 1,2,4,8,12,4,8,11 | 1,4,7,3,7,10 |
| 1,2,4,8,12,4,8,11,16 | 1,4,7,3,7,10,15 |
| 1,2,4,8,12,4,8,12 | 1,4,7,3,7,11 |
| 1,2,4,8,12,8 | 1,4,7,4 |
| 1,2,4,8,12,8,11,16 | 1,4,7,4,6,10 |
| 1,2,4,8,12,8,11,16,21 | 1,4,7,4,6,10,14 |
| 1,2,4,8,12,8,11,16,21,15 | 1,4,7,4,6,10,14,9 |
| 1,2,4,8,12,8,11,16,21,16 | 1,4,7,4,6,10,14,10 |
| 1,2,4,8,12,8,12 | 1,4,7,4,7 |
| 1,2,4,8,12,8,12,8,12 | 1,4,7,4,7,4,7 |
| 1,2,4,8,12,9 | 1,4,7,5 |
| 1,2,4,8,12,9,4,8,12,9 | 1,4,7,5,3,7,11,8 |
| 1,2,4,8,12,9,8 | 1,4,7,5,4 |
| 1,2,4,8,12,9,8,11,16 | 1,4,7,5,4,6,10 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,12,9,8,11,16,21 | 1,4,7,5,4,6,10,14 |
| 1,2,4,8,12,9,<br>8,11,16,21,17 | 1,4,7,5,4,6,10,14,11 |
| 1,2,4,8,12,9,8,11,<br>16,21,17,11,16,21,17 | 1,4,7,5,4,6,10,<br>14,11,6,10,14,11 |
| 1,2,4,8,12,9,8,11,<br>16,21,17,14,19,24,20 | 1,4,7,5,4,6,10,<br>14,11,8,12,16,13 |
| 1,2,4,8,12,9,8,<br>11,16,21,17,15 | 1,4,7,5,4,6,10,14,11,9 |
| 1,2,4,8,12,9,8,<br>11,16,21,17,16 | 1,4,7,5,4,6,10,14,11,10 |
| 1,2,4,8,12,9,8,11,<br>16,21,17,16,20,26,32,27 | 1,4,7,5,4,6,10,14,<br>11,10,13,18,23,19 |
| 1,2,4,8,12,9,8,12 | 1,4,7,5,4,7 |
| 1,2,4,8,12,9,8,12,9 | 1,4,7,5,4,7,5 |
| 1,2,4,8,12,9,9 | 1,4,7,5,5 |
| 1,2,4,8,12,10 | 1,4,7,6 |
| 1,2,4,8,12,11,8 | 1,4,7,6,4 |
| 1,2,4,8,12,11,<br>8,11,16,21,20 | 1,4,7,6,4,6,10,14,13 |
| 1,2,4,8,12,11,<br>8,11,16,21,20,16 | 1,4,7,6,4,6,10,14,13,10 |
| 1,2,4,8,12,11,8,12 | 1,4,7,6,4,7 |
| 1,2,4,8,12,11,9 | 1,4,7,6,5 |
| 1,2,4,8,12,11,11,8,12 | 1,4,7,6,6,4,7 |
| 1,2,4,8,12,11,12 | 1,4,7,6,7 |
| 1,2,4,8,12,11,14,8,12 | 1,4,7,6,8,4,7 |
| 1,2,4,8,12,11,14,17,8,12 | 1,4,7,6,8,10,4,7 |
| 1,2,4,8,12,11,15 | 1,4,7,6,9 |
| 1,2,4,8,12,11,15,20 | 1,4,7,6,9,13 |
| 1,2,4,8,12,11,16 | 1,4,7,6,10 |
| 1,2,4,8,12,11,16,20,16 | 1,4,7,6,10,13,10 |
| 1,2,4,8,12,11,16,20,26 | 1,4,7,6,10,13,18 |
| 1,2,4,8,12,11,16,21 | 1,4,7,6,10,14 |
| 1,2,4,8,12,12 | 1,4,7,7 |
| 1,2,4,8,12,12,8,12 | 1,4,7,7,4,7 |
| 1,2,4,8,12,12,8,12,9 | 1,4,7,7,4,7,5 |
| 1,2,4,8,12,12,8,12,10 | 1,4,7,7,4,7,6 |
| 1,2,4,8,12,12,<br>8,12,11,8,12 | 1,4,7,7,4,7,6,4,7 |
| 1,2,4,8,12,12,<br>8,12,11,16,21,21 | 1,4,7,7,4,7,6,10,14,14 |
| 1,2,4,8,12,12,8,12,12 | 1,4,7,7,4,7,7 |
| 1,2,4,8,12,12,10,8,12 | 1,4,7,7,6,3,<br>7,11,11,9,7,11 |
| 1,2,4,8,12,12,11,8 | 1,4,7,7,6,4 |
| 1,2,4,8,12,12,11,8,12 | 1,4,7,7,6,4,7 |
| 1,2,4,8,12,12,11,8,12,10 | 1,4,7,7,6,4,7,6 |
| 1,2,4,8,12,12,<br>11,8,12,11,16,21,21 | 1,4,7,7,6,4,<br>7,6,10,14,14 |
| 1,2,4,8,12,12,11,8,12,<br>11,16,21,21,19,24,29,29 | 1,4,7,7,6,4,7,6,<br>10,14,14,12,16,20,20 |
| 1,2,4,8,12,12,11,8,12,12 | 1,4,7,7,6,4,7,7 |
| 1,2,4,8,12,12,<br>11,14,8,12,12 | 1,4,7,7,6,8,4,7,7 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,12,12,11,15 | 1,4,7,7,6,9 |
| 1,2,4,8,12,12,11,16,21,21 | 1,4,7,7,6,10,14,14 |
| 1,2,4,8,12,12,12 | 1,4,7,7,7 |
| 1,2,4,8,12,12,12,12 | 1,4,7,7,7,7 |
| 1,2,4,8,12,13 | 1,4,7,8 |
| 1,2,4,8,12,13,15 | 1,4,7,8,10 |
| 1,2,4,8,12,13,15,17,19 | 1,4,7,8,10,12,14 |
| 1,2,4,8,12,13,15,18 | 1,4,7,8,10,13 |
| 1,2,4,8,12,13,15,19 | 1,4,7,8,11 |
| 1,2,4,8,12,13,15,19,23 | 1,4,7,8,11,14 |
| 1,2,4,8,12,14 | 1,4,7,9 |
| 1,2,4,8,12,14,4 | 1,4,7,9,3 |
| 1,2,4,8,12,14,4,8 | 1,4,7,9,3,7 |
| 1,2,4,8,12,14,4,8,12 | 1,4,7,9,3,7,11 |
| 1,2,4,8,12,14,4,8,12,13 | 1,4,7,9,3,7,11,12 |
| 1,2,4,8,12,14,<br>4,8,12,13,15,19 | 1,4,7,9,3,7,11,12,15 |
| 1,2,4,8,12,14,4,<br>8,12,13,15,19,23 | 1,4,7,9,3,7,11,12,15,18 |
| 1,2,4,8,12,14,4,<br>8,12,13,15,19,23,25 | 1,4,7,9,3,7,<br>11,12,15,18,20 |
| 1,2,4,8,12,14,4,8,<br>12,13,15,19,23,25,7 | 1,4,7,9,3,7,11,<br>12,15,18,20,6 |
| 1,2,4,8,12,14,4,8,12,<br>13,15,19,23,25,11,15 | 1,4,7,9,3,7,11,<br>12,15,18,20,10,14 |
| 1,2,4,8,12,14,4,8,12,14 | 1,4,7,9,3,7,11,13 |
| 1,2,4,8,12,14,7 | 1,4,7,9,3,7,11,13,6 |
| 1,2,4,8,12,14,8 | 1,4,7,9,3,7,11,13,7 |
| 1,2,4,8,12,14,8,12 | 1,4,7,9,3,7,11,13,7,11 |
| 1,2,4,8,12,14,8,<br>12,13,15,19,23,25 | 1,4,7,9,3,7,11,<br>13,7,11,12,15,18,20 |
| 1,2,4,8,12,14,8,<br>12,13,15,19,23,25,9 | 1,4,7,9,3,7,11,13,<br>7,11,12,15,18,20,8 |
| 1,2,4,8,12,14,8,<br>12,13,15,19,23,25,10 | 1,4,7,9,3,7,11,13,<br>7,11,12,15,18,20,9 |
| 1,2,4,8,12,14,8,12,14 | 1,4,7,9,3,7,<br>11,13,7,11,13 |
| 1,2,4,8,12,14,9 | 1,4,7,9,3,7,11,13,8 |
| 1,2,4,8,12,14,11,8 | 1,4,7,9,3,7,11,13,10,7 |
| 1,2,4,8,12,14,11,8,12,14 | 1,4,7,9,3,7,11,<br>13,10,7,11,13 |
| 1,2,4,8,12,14,<br>11,14,8,12,14 | 1,4,7,9,3,7,11,<br>13,13,10,7,11,13 |
| 1,2,4,8,12,14,11,15 | 1,4,7,9,3,7,11,13,10,14 |
| 1,2,4,8,12,14,11,16 | 1,4,7,9,3,7,11,13,10,15 |
| 1,2,4,8,12,14,11,16,21,23 | 1,4,7,9,3,7,<br>11,13,10,15,20,22 |
| 1,2,4,8,12,14,12 | 1,4,7,9,3,7,11,13,11 |
| 1,2,4,8,12,14,14 | 1,4,7,9,3,7,11,13,11,13 |
| 1,2,4,8,12,14,16 | 1,4,7,9,3,7,11,13,13 |
| 1,2,4,8,12,14,16,18 | 1,4,7,9,3,7,11,13,15 |
| 1,2,4,8,12,14,17 | 1,4,7,9,3,7,11,13,16 |
| 1,2,4,8,12,14,18 | 1,4,7,9,3,7,11,13,17 |
| 1,2,4,8,12,14,18,22 | 1,4,7,9,3,7,11,13,17,21 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,12,14,18,22,24 | 1,4,7,9,3,7,<br>11,13,17,21,23 |
| 1,2,4,8,12,15 | 1,4,7,9,3,7,11,14 |
| 1,2,4,8,12,15,4,8,12,15 | 1,4,7,9,3,7,11,<br>14,3,7,11,14 |
| 1,2,4,8,12,15,7,<br>12,17,20,12,17,19 | 1,4,7,9,3,7,11,14,<br>6,11,16,19,11,16,18 |
| 1,2,4,8,12,15,7,12,<br>17,20,12,17,19,23,27,30 | 1,4,7,9,3,7,11,14,6,11,<br>16,19,11,16,18,22,26,29 |
| 1,2,4,8,12,15,7,12,17,21 | 1,4,7,9,3,7,11,<br>14,6,11,16,20 |
| 1,2,4,8,12,15,8 | 1,4,7,9,4 |
| 1,2,4,8,12,15,<br>8,11,16,21,25 | 1,4,7,9,4,6,10,14,17 |
| 1,2,4,8,12,15,8,12 | 1,4,7,9,4,7 |
| 1,2,4,8,12,15,8,12,14 | 1,4,7,9,4,7,9 |
| 1,2,4,8,12,15,8,12,15,8 | 1,4,7,9,4,7,9,4 |
| 1,2,4,8,12,15,8,<br>12,15,8,12,15,8 | 1,4,7,9,4,7,<br>9,4,7,9,4 |
| 1,2,4,8,12,15,9 | 1,4,7,9,5 |
| 1,2,4,8,12,15,9,4 | 1,4,7,9,5,3 |
| 1,2,4,8,12,15,9,8 | 1,4,7,9,5,4 |
| 1,2,4,8,12,15,<br>9,8,12,15,8 | 1,4,7,9,5,4,7,9,4 |
| 1,2,4,8,12,15,<br>9,8,12,15,9 | 1,4,7,9,5,4,7,9,5 |
| 1,2,4,8,12,15,9,9 | 1,4,7,9,5,5 |
| 1,2,4,8,12,15,10 | 1,4,7,9,6 |
| 1,2,4,8,12,15,11,8 | 1,4,7,9,6,4 |
| 1,2,4,8,12,15,<br>11,8,12,15,9 | 1,4,7,9,6,4,7,9,5 |
| 1,2,4,8,12,15,<br>11,8,12,15,9,9 | 1,4,7,9,6,4,7,9,5,5 |
| 1,2,4,8,12,15,<br>11,8,12,15,10 | 1,4,7,9,6,4,7,9,6 |
| 1,2,4,8,12,15,11,9 | 1,4,7,9,6,5 |
| 1,2,4,8,12,15,11,12 | 1,4,7,9,6,7 |
| 1,2,4,8,12,15,<br>11,14,8,12,15,9 | 1,4,7,9,6,8,4,7,9,5 |
| 1,2,4,8,12,15,11,15 | 1,4,7,9,6,9 |
| 1,2,4,8,12,15,11,16 | 1,4,7,9,6,10 |
| 1,2,4,8,12,15,11,16,21,24 | 1,4,7,9,6,10,14,16 |
| 1,2,4,8,12,15,11,16,<br>21,24,8,12,15,9 | 1,4,7,9,6,10,<br>14,16,4,7,9,5 |
| 1,2,4,8,12,15,11,16,21,25 | 1,4,7,9,6,10,14,17 |
| 1,2,4,8,12,15,<br>11,16,21,25,16 | 1,4,7,9,6,10,14,17,10 |
| 1,2,4,8,12,15,<br>11,16,21,25,17 | 1,4,7,9,6,10,14,17,11 |
| 1,2,4,8,12,15,11,<br>16,21,25,20,26,32,37,27 | 1,4,7,9,6,10,14,<br>17,13,18,23,27,19 |
| 1,2,4,8,12,15,12 | 1,4,7,9,7 |
| 1,2,4,8,12,15,12,8 | 1,4,7,9,7,4 |
| 1,2,4,8,12,15,<br>12,8,12,15,9 | 1,4,7,9,7,4,7,9,5 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,12,15,<br>12,8,12,15,9,9 | 1,4,7,9,7,4,7,9,5,5 |
| 1,2,4,8,12,15,<br>12,8,12,15,10 | 1,4,7,9,7,4,7,9,6 |
| 1,2,4,8,12,15,12,8,<br>12,15,11,16,21,25,21 | 1,4,7,9,7,4,7,<br>9,6,10,14,17,14 |
| 1,2,4,8,12,15,12,8,<br>12,15,11,16,21,25,21,15 | 1,4,7,9,7,4,7,<br>9,6,10,14,17,14,9 |
| 1,2,4,8,12,15,<br>12,8,12,15,12 | 1,4,7,9,7,4,7,9,7 |
| 1,2,4,8,12,15,12,9 | 1,4,7,9,7,5 |
| 1,2,4,8,12,15,<br>12,11,8,12,15,12 | 1,4,7,9,7,6,4,7,9,7 |
| 1,2,4,8,12,15,12,<br>11,14,8,12,15,12 | 1,4,7,9,7,<br>6,8,4,7,9,7 |
| 1,2,4,8,12,15,12,11,15 | 1,4,7,9,7,6,9 |
| 1,2,4,8,12,15,<br>12,11,16,21,25,17 | 1,4,7,9,7,6,10,14,17,11 |
| 1,2,4,8,12,15,<br>12,11,16,21,25,21 | 1,4,7,9,7,6,10,14,17,14 |
| 1,2,4,8,12,15,12,12 | 1,4,7,9,7,7 |
| 1,2,4,8,12,15,12,<br>12,11,16,21,25,17 | 1,4,7,9,7,7,<br>6,10,14,17,11 |
| 1,2,4,8,12,15,12,<br>12,11,16,21,25,21,21 | 1,4,7,9,7,7,6,<br>10,14,17,14,14 |
| 1,2,4,8,12,15,12,12,12 | 1,4,7,9,7,7,7 |
| 1,2,4,8,12,15,12,13 | 1,4,7,9,7,8 |
| 1,2,4,8,12,15,12,14 | 1,4,7,9,7,9 |
| 1,2,4,8,12,15,12,15,8 | 1,4,7,9,7,9,4 |
| 1,2,4,8,12,15,<br>12,15,8,12,14 | 1,4,7,9,7,9,4,7,9 |
| 1,2,4,8,12,15,<br>12,15,8,12,15,8 | 1,4,7,9,7,9,4,7,9,4 |
| 1,2,4,8,12,15,<br>12,15,8,12,15,9 | 1,4,7,9,7,9,4,7,9,5 |
| 1,2,4,8,12,15,<br>12,15,8,12,15,12 | 1,4,7,9,7,9,4,7,9,7 |
| 1,2,4,8,12,15,<br>12,15,8,12,15,12,13 | 1,4,7,9,7,9,<br>4,7,9,7,8 |
| 1,2,4,8,12,15,12,<br>15,8,12,15,12,15,8 | 1,4,7,9,7,9,<br>4,7,9,7,9,4 |
| 1,2,4,8,12,15,12,15,<br>8,12,15,12,15,8,12,15,9 | 1,4,7,9,7,9,4,<br>7,9,7,9,4,7,9,5 |
| 1,2,4,8,12,15,12,15,9 | 1,4,7,9,7,9,5 |
| 1,2,4,8,12,15,12,15,11,15 | 1,4,7,9,7,9,6,9 |
| 1,2,4,8,12,15,12,<br>15,11,16,21,25,17 | 1,4,7,9,7,9,<br>6,10,14,17,11 |
| 1,2,4,8,12,15,12,<br>15,11,16,21,25,21,25,17 | |
| 1,2,4,8,12,15,12,15,12 | 1,4,7,9,7,9,7 |
| 1,2,4,8,12,15,12,15,12,13 | 1,4,7,9,7,9,7,8 |
| 1,2,4,8,12,15,12,15,12,14 | 1,4,7,9,7,9,7,9 |
| 1,2,4,8,12,15,<br>12,15,12,15,8 | 1,4,7,9,7,9,7,9,4 |
| 1,2,4,8,12,15,<br>12,15,12,15,9 | 1,4,7,9,7,9,7,9,5 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,12,15,12, 15,12,15,12,15,9 | 1,4,7,9,7,9, 7,9,7,9,5 |
| 1,2,4,8,12,15,13 | 1,4,7,9,8 |
| 1,2,4,8,12,15, 15,8,12,15,13 | 1,4,7,9,9,4,7,9,8 |
| 1,2,4,8,12,15,15,9 | 1,4,7,9,9,5 |
| 1,2,4,8,12,15,15,10,13 | 1,4,7,9,9,6,9 |
| 1,2,4,8,12,15, 15,11,16,21,25,17 | 1,4,7,9,9,6,10,14,17,11 |
| 1,2,4,8,12,15, 15,11,16,21,25,25,17 | 1,4,7,9,9,6, 10,14,17,17,11 |
| 1,2,4,8,12,15,15,12 | 1,4,7,9,9,7 |
| 1,2,4,8,12,15,15,12,15,9 | 1,4,7,9,9,7,9,5 |
| 1,2,4,8,12,15, 15,12,15,15,9 | 1,4,7,9,9,7,9,9,5 |
| 1,2,4,8,12,15,15,13 | 1,4,7,9,9,8 |
| 1,2,4,8,12,15,15,15,9 | 1,4,7,9,9,9,5 |
| 1,2,4,8,12,15,15, 15,12,15,15,15,9 | 1,4,7,9,9, 9,7,9,9,9,5 |
| 1,2,4,8,12,15,15,15,15,9 | 1,4,7,9,9,9,9,5 |
| 1,2,4,8,12,15,16 | 1,4,7,9,10 |
| 1,2,4,8,12,15,18,9 | 1,4,7,9,11,5 |
| 1,2,4,8,12,15,18,11,15 | 1,4,7,9,11,6,9 |
| 1,2,4,8,12,15,18,12 | 1,4,7,9,11,7 |
| 1,2,4,8,12,15,18,12,15,9 | 1,4,7,9,11,7,9,5 |
| 1,2,4,8,12,15, 18,12,15,18,9 | 1,4,7,9,11,7,9,11,5 |
| 1,2,4,8,12,15,18,15,9 | 1,4,7,9,11,9,5 |
| 1,2,4,8,12,15,18,15,18,9 | 1,4,7,9,11,9,11,5 |
| 1,2,4,8,12,15,18,16 | 1,4,7,9,11,10 |
| 1,2,4,8,12,15,18,18,9 | 1,4,7,9,11,11,5 |
| 1,2,4,8,12,15, 18,18,15,18,18,9 | 1,4,7,9,11,11,9,11,11,5 |
| 1,2,4,8,12,15,18,18,18,9 | 1,4,7,9,11,11,11,5 |
| 1,2,4,8,12,15,18,19 | 1,4,7,9,11,12 |
| 1,2,4,8,12,15,18,21,9 | 1,4,7,9,11,13,5 |
| 1,2,4,8,12,15,18,21,21,9 | 1,4,7,9,11,13,13,5 |
| 1,2,4,8,12,15,18,21,24,9 | 1,4,7,9,11,13,15,5 |
| 1,2,4,8,12,15,19 | 1,4,7,9,12 |
| 1,2,4,8,12,15, 19,11,16,21,25,17 | 1,4,7,9,12,6,10,14,17,11 |
| 1,2,4,8,12,15, 19,11,16,21,25,30 | 1,4,7,9,12,6,10,14,17,21 |
| 1,2,4,8,12,15,19,19 | 1,4,7,9,12,12 |
| 1,2,4,8,12,15,19,23,27 | 1,4,7,9,12,15,18 |
| 1,2,4,8,12,15,19,24 | 1,4,7,9,12,16 |
| 1,2,4,8,12,15,20 | 1,4,7,9,13 |
| 1,2,4,8,12,15,20, 25,28,8,12,15,9 | 1,4,7,9,13, 17,19,4,7,9,5 |
| 1,2,4,8,12,15,20,25,28,9 | 1,4,7,9,13,17,19,5 |
| 1,2,4,8,12,15,20,25,29 | 1,4,7,9,13,17,20 |
| 1,2,4,8,12,15,20,25,29,20 | 1,4,7,9,13,17,20,13 |
| 1,2,4,8,12,15,20,25,29,21 | 1,4,7,9,13,17,20,14 |
| 1,2,4,8,12,15,20, 25,29,21,11,16,21,25,17 | 1,4,7,9,13,17,20, 14,6,10,14,17,11 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,12,15,<br>20,25,29,35,41,46,36 | 1,4,7,9,13,17,<br>20,25,30,34,26 |
| 1,2,4,8,12,16 | 1,4,7,10 |
| 1,2,4,8,12,16,<br>8,12,15,20,25,30 | |
| 1,2,4,8,12,16,8,12,15,20,<br>25,30,20,25,29,35,41,47 | 1,4,7,10,4,7,9,13,17,<br>21,13,17,20,25,30,35 |
| 1,2,4,8,12,16,8,12,16 | 1,4,7,10,4,7,10 |
| 1,2,4,8,12,16,9 | 1,4,7,10,5 |
| 1,2,4,8,12,16,11,8,12,16 | 1,4,7,10,6,4,7,10 |
| 1,2,4,8,12,16,11,15 | 1,4,7,10,6,9 |
| 1,2,4,8,12,16,<br>11,16,21,25,17 | 1,4,7,10,6,10,14,17,11 |
| 1,2,4,8,12,16,11,16,21,26 | 1,4,7,10,6,10,14,18 |
| 1,2,4,8,12,16,12 | 1,4,7,10,7 |
| 1,2,4,8,12,16,<br>12,11,16,21,26,21 | 1,4,7,10,7,6,10,14,18,14 |
| 1,2,4,8,12,16,12,12 | 1,4,7,10,7,7 |
| 1,2,4,8,12,16,12,13 | 1,4,7,10,7,8 |
| 1,2,4,8,12,16,12,14 | 1,4,7,10,7,9 |
| 1,2,4,8,12,16,12,15,8 | 1,4,7,10,7,9,4 |
| 1,2,4,8,12,16,12,15,<br>8,12,15,20,25,30,25,28,9 | 1,4,7,10,7,9,4,7,<br>9,13,17,21,17,19,5 |
| 1,2,4,8,12,16,<br>12,15,8,12,16 | 1,4,7,10,7,9,4,7,10 |
| 1,2,4,8,12,16,12,15,9 | 1,4,7,10,7,9,5 |
| 1,2,4,8,12,16,12,15,9,8 | 1,4,7,10,7,9,5,4 |
| 1,2,4,8,12,16,12,<br>15,9,8,12,15,9 | 1,4,7,10,7,<br>9,5,4,7,9,5 |
| 1,2,4,8,12,16,12,<br>15,9,8,12,15,11,16 | 1,4,7,10,7,9,<br>5,4,7,9,6,10 |
| 1,2,4,8,12,16,12,15,<br>9,8,12,15,11,16,21,25,17 | 1,4,7,10,7,9,5,<br>4,7,9,6,10,14,17,11 |
| 1,2,4,8,12,16,12,15,<br>9,8,12,15,11,16,21,25,21 | 1,4,7,10,7,9,5,4,<br>7,9,6,10,14,17,14 |
| 1,2,4,8,12,16,12,15,<br>9,8,12,15,11,16,21,25,30 | 1,4,7,10,7,9,5,4,<br>7,9,6,10,14,17,21 |
| 1,2,4,8,12,16,12,15,<br>9,8,12,15,11,16,21,25,31 | 1,4,7,10,7,9,5,4,<br>7,9,6,10,14,17,22 |
| 1,2,4,8,12,16,12,15,<br>9,8,12,15,11,16,21,26 | 1,4,7,10,7,9,5,4,<br>7,9,6,10,14,18 |
| 1,2,4,8,12,16,12,15,<br>9,8,12,15,11,16,21,26,21 | 1,4,7,10,7,9,5,4,<br>7,9,6,10,14,18,14 |
| 1,2,4,8,12,16,12,<br>15,9,8,12,15,12 | 1,4,7,10,7,9,<br>5,4,7,9,7 |
| 1,2,4,8,12,16,12,15,<br>9,8,12,15,12,15,12 | 1,4,7,10,7,9,5,<br>4,7,9,7,9,7 |
| 1,2,4,8,12,16,12,<br>15,9,8,12,15,15,12 | 1,4,7,10,7,9,<br>5,4,7,9,9,7 |
| 1,2,4,8,12,16,12,<br>15,9,8,12,15,18,12 | 1,4,7,10,7,9,<br>5,4,7,9,11,7 |
| 1,2,4,8,12,16,12,<br>15,9,8,12,15,19 | 1,4,7,10,7,9,<br>5,4,7,9,12 |
| 1,2,4,8,12,16,<br>12,15,9,8,12,16 | 1,4,7,10,7,9,5,4,7,10 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,12,16,12,<br>15,9,8,12,16,12 | 1,4,7,10,7,9,<br>5,4,7,10,7 |
| 1,2,4,8,12,16,12,<br>15,9,8,12,16,12,15 | 1,4,7,10,7,9,<br>5,4,7,10,7,9 |
| 1,2,4,8,12,16,<br>12,15,9,8,12,16,9 | 1,4,7,10,7,9,5,<br>4,7,10,7,9,5 |
| 1,2,4,8,12,16,12,15,10 | 1,4,7,10,7,9,6 |
| 1,2,4,8,12,16,12,<br>15,11,8,12,16,12,15,9 | 1,4,7,10,7,9,<br>6,4,7,10,7,9,5 |
| 1,2,4,8,12,16,12,15,11,9 | 1,4,7,10,7,9,6,5 |
| 1,2,4,8,12,16,12,15,11,10 | 1,4,7,10,7,9,6,6 |
| 1,2,4,8,12,16,12,15,11,15 | 1,4,7,10,7,9,6,9 |
| 1,2,4,8,12,16,12,<br>15,11,16,21,26,21,25,17 | 1,4,7,10,7,9,6,<br>10,14,18,14,17,11 |
| 1,2,4,8,12,16,12,15,12 | 1,4,7,10,7,9,7 |
| 1,2,4,8,12,16,12,15,12,12 | 1,4,7,10,7,9,7,7 |
| 1,2,4,8,12,16,<br>12,15,12,15,9 | 1,4,7,10,7,9,7,9,5 |
| 1,2,4,8,12,16,12,<br>15,12,15,11,16,21,26 | 1,4,7,10,7,9,<br>7,9,6,10,14,18 |
| 1,2,4,8,12,16,<br>12,15,12,15,12 | 1,4,7,10,7,9,7,9,7 |
| 1,2,4,8,12,16,12,15,15,9 | 1,4,7,10,7,9,9,5 |
| 1,2,4,8,12,16,12,15,15,12 | 1,4,7,10,7,9,9,7 |
| 1,2,4,8,12,16,<br>12,15,15,12,15,9 | 1,4,7,10,7,9,9,7,9,5 |
| 1,2,4,8,12,16,12,15,19 | 1,4,7,10,7,9,12 |
| 1,2,4,8,12,16,12,15,20 | 1,4,7,10,7,9,13 |
| 1,2,4,8,12,16,<br>12,15,20,25,29,21 | 1,4,7,10,7,9,13,17,20,14 |
| 1,2,4,8,12,16,<br>12,15,20,25,29,25 | 1,4,7,10,7,9,13,17,20,17 |
| 1,2,4,8,12,16,<br>12,15,20,25,30 | 1,4,7,10,7,9,13,17,21 |
| 1,2,4,8,12,16,<br>12,15,20,25,30,25 | 1,4,7,10,7,9,13,17,21,17 |
| 1,2,4,8,12,16,12,<br>15,20,25,30,25,29 | 1,4,7,10,7,9,<br>13,17,21,17,20 |
| 1,2,4,8,12,16,12,<br>15,20,25,30,25,29,21 | 1,4,7,10,7,9,13,<br>17,21,17,20,14 |
| 1,2,4,8,12,16,12,16 | 1,4,7,10,7,10 |
| 1,2,4,8,12,16,12,16,12 | 1,4,7,10,7,10,7 |
| 1,2,4,8,12,16,12,16,12,14 | 1,4,7,10,7,10,7,9 |
| 1,2,4,8,12,16,<br>12,16,12,15,9 | 1,4,7,10,7,10,7,9,5 |
| 1,2,4,8,12,16,<br>12,16,12,15,15,9 | 1,4,7,10,7,10,7,9,6,5 |
| 1,2,4,8,12,16,<br>12,16,12,15,12 | 1,4,7,10,7,10,7,9,7 |
| 1,2,4,8,12,16,12,<br>16,12,15,12,15,12 | 1,4,7,10,7,<br>10,7,9,7,9,7 |
| 1,2,4,8,12,16,12,16,12,16 | 1,4,7,10,7,10,7,10 |
| 1,2,4,8,12,16,<br>12,16,12,16,12,15,9 | 1,4,7,10,7,10,<br>7,10,7,9,5 |
| 1,2,4,8,12,16,<br>12,16,12,16,12,16 | 1,4,7,10,7,10,7,10,7,10 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,12,16,13 | 1,4,7,10,8 |
| 1,2,4,8,12,16,13,8 | 1,4,7,10,8,4 |
| 1,2,4,8,12,16,13,8,12,14 | 1,4,7,10,8,4,7,9 |
| 1,2,4,8,12,16,<br>13,8,12,15,9 | 1,4,7,10,8,4,7,9,5 |
| 1,2,4,8,12,16,13,<br>8,12,15,9,8,12,15,9 | 1,4,7,10,8,4,7,<br>9,5,4,7,9,5 |
| 1,2,4,8,12,16,13,<br>8,12,15,11,16,21,25,17 | 1,4,7,10,8,4,7,<br>9,6,10,14,17,11 |
| 1,2,4,8,12,16,13,<br>8,12,15,11,16,21,25,21 | 1,4,7,10,8,4,7,<br>9,6,10,14,17,14 |
| 1,2,4,8,12,16,13,<br>8,12,15,11,16,21,26 | 1,4,7,10,8,4,7,<br>9,6,10,14,18 |
| 1,2,4,8,12,16,13,8,<br>12,15,11,16,21,26,21,26 | 1,4,7,10,8,4,7,<br>9,6,10,14,18,14,18 |
| 1,2,4,8,12,16,13,<br>8,12,15,11,16,21,26,22 | 1,4,7,10,8,4,7,<br>9,6,10,14,18,15 |
| 1,2,4,8,12,16,13,8,12,<br>15,11,16,21,26,22,11,15 | 1,4,7,10,8,4,7,<br>9,6,10,14,18,15,6,9 |
| 1,2,4,8,12,16,13,8,<br>12,15,11,16,21,26,22,13 | 1,4,7,10,8,4,7,<br>9,6,10,14,18,15,8 |
| 1,2,4,8,12,16,13,8,12,<br>15,11,16,21,26,22,14,18 | 1,4,7,10,8,4,7,9,<br>6,10,14,18,15,8,11 |
| 1,2,4,8,12,16,13,8,12,<br>15,11,16,21,26,22,14,19 | 1,4,7,10,8,4,7,9,<br>6,10,14,18,15,8,12 |
| 1,2,4,8,12,16,13,8,<br>12,15,11,16,21,26,22,15 | 1,4,7,10,8,4,7,9,<br>6,10,14,18,15,9 |
| 1,2,4,8,12,16,13,8,<br>12,15,11,16,21,26,22,16 | 1,4,7,10,8,4,7,9,<br>6,10,14,18,15,10 |
| 1,2,4,8,12,16,<br>13,8,12,15,12 | 1,4,7,10,8,4,7,9,7 |
| 1,2,4,8,12,16,<br>13,8,12,15,12,15,12 | 1,4,7,10,8,<br>4,7,9,7,9,7 |
| 1,2,4,8,12,16,<br>13,8,12,15,15,12 | 1,4,7,10,8,4,7,9,9,7 |
| 1,2,4,8,12,16,<br>13,8,12,15,18,12 | 1,4,7,10,8,4,7,9,11,7 |
| 1,2,4,8,12,16,<br>13,8,12,15,19 | 1,4,7,10,8,4,7,9,12 |
| 1,2,4,8,12,16,<br>13,8,12,15,20 | 1,4,7,10,8,4,7,9,13 |
| 1,2,4,8,12,16,13,<br>8,12,15,20,25,30,26 | 1,4,7,10,8,4,<br>7,9,13,17,21,18 |
| 1,2,4,8,12,16,13,8,<br>12,15,20,25,30,26,11,16 | 1,4,7,10,8,4,7,<br>9,13,17,21,18,6,10 |
| 1,2,4,8,12,16,13,8,<br>12,15,20,25,30,26,12 | 1,4,7,10,8,4,7,<br>9,13,17,21,18,7 |
| 1,2,4,8,12,16,13,8,<br>12,15,20,25,30,26,14 | 1,4,7,10,8,4,7,<br>9,13,17,21,18,9 |
| 1,2,4,8,12,16,13,8,<br>12,15,20,25,30,26,15,19 | 1,4,7,10,8,4,7,9,<br>13,17,21,18,9,12 |
| 1,2,4,8,12,16,13,<br>8,12,15,20,25,30,26,17 | 1,4,7,10,8,4,7,<br>9,13,17,21,18,11 |
| 1,2,4,8,12,16,13,<br>8,12,15,20,25,30,26,19 | 1,4,7,10,8,4,7,9,<br>13,17,21,18,12 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,12,16,13,<br>8,12,15,20,25,30,26,20 | 1,4,7,10,8,4,7,9,<br>13,17,21,18,13 |
| 1,2,4,8,12,16,13,8,12,16 | 1,4,7,10,8,4,7,10 |
| 1,2,4,8,12,16,<br>13,8,12,16,12 | 1,4,7,10,8,4,7,10,7 |
| 1,2,4,8,12,16,<br>13,8,12,16,12,14 | 1,4,7,10,8,4,7,10,7,9 |
| 1,2,4,8,12,16,<br>13,8,12,16,12,15,9 | 1,4,7,10,8,4,<br>7,10,7,9,5 |
| 1,2,4,8,12,16,<br>13,8,12,16,12,15,12 | 1,4,7,10,8,4,<br>7,10,7,9,7 |
| 1,2,4,8,12,16,13,<br>8,12,16,12,15,12,15,9 | 1,4,7,10,8,4,<br>7,10,7,9,7,9,5 |
| 1,2,4,8,12,16,13,<br>8,12,16,12,15,19 | 1,4,7,10,8,4,<br>7,10,7,9,12 |
| 1,2,4,8,12,16,13,8,<br>12,16,12,15,20,25,30 | 1,4,7,10,8,4,7,<br>10,7,9,13,17,21 |
| 1,2,4,8,12,16,13,8,<br>12,16,12,15,20,25,30,26 | 1,4,7,10,8,4,7,10,<br>7,9,13,17,21,18 |
| 1,2,4,8,12,16,<br>13,8,12,16,12,16 | 1,4,7,10,8,4,7,10,7,10 |
| 1,2,4,8,12,16,<br>13,8,12,16,13 | 1,4,7,10,8,4,7,10,8 |
| 1,2,4,8,12,16,13,9 | 1,4,7,10,8,5 |
| 1,2,4,8,12,16,<br>13,11,8,12,16,13 | 1,4,7,10,8,6,4,7,10,8 |
| 1,2,4,8,12,16,13,11,15 | 1,4,7,10,8,6,9 |
| 1,2,4,8,12,16,<br>13,11,16,21,25,17 | 1,4,7,10,8,6,10,14,17,11 |
| 1,2,4,8,12,16,<br>13,11,16,21,26,22 | 1,4,7,10,8,6,10,14,18,15 |
| 1,2,4,8,12,16,13,12 | 1,4,7,10,8,7 |
| 1,2,4,8,12,16,13,12,11,15 | 1,4,7,10,8,7,6,9 |
| 1,2,4,8,12,16,<br>13,12,11,16,21,25,17 | 1,4,7,10,8,7,<br>6,10,14,17,11 |
| 1,2,4,8,12,16,<br>13,12,11,16,21,26 | 1,4,7,10,8,7,6,10,14,18 |
| 1,2,4,8,12,16,<br>13,12,11,16,21,26,21 | 1,4,7,10,8,7,<br>6,10,14,18,14 |
| 1,2,4,8,12,16,13,<br>12,11,16,21,26,21,25,17 | 1,4,7,10,8,7,6,<br>10,14,18,14,17,11 |
| 1,2,4,8,12,16,13,<br>12,11,16,21,26,21,26 | 1,4,7,10,8,7,6,<br>10,14,18,14,18 |
| 1,2,4,8,12,16,13,<br>12,11,16,21,26,22 | 1,4,7,10,8,7,<br>6,10,14,18,15 |
| 1,2,4,8,12,16,13,<br>12,11,16,21,26,22,21 | 1,4,7,10,8,7,6,<br>10,14,18,15,14 |
| 1,2,4,8,12,16,13,12,12 | 1,4,7,10,8,7,7 |
| 1,2,4,8,12,16,13,12,13 | 1,4,7,10,8,7,8 |
| 1,2,4,8,12,16,13,12,15,9 | 1,4,7,10,8,7,9,5 |
| 1,2,4,8,12,16,<br>13,12,15,11,16 | 1,4,7,10,8,7,9,6,9 |
| 1,2,4,8,12,16,<br>13,12,15,11,16,21,25,17 | 1,4,7,10,8,7,9,<br>6,10,14,17,11 |
| 1,2,4,8,12,16,<br>13,12,15,11,16,21,26 | 1,4,7,10,8,7,<br>9,6,10,14,18 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,12,16,13,<br>12,15,11,16,21,26,22 | 1,4,7,10,8,7,9,<br>6,10,14,18,15 |
| 1,2,4,8,12,16,13,12,15,12 | 1,4,7,10,8,7,9,7 |
| 1,2,4,8,12,16,13,12,15,19 | 1,4,7,10,8,7,9,12 |
| 1,2,4,8,12,16,<br>13,12,15,20,25,29,21 | 1,4,7,10,8,7,<br>9,13,17,20,14 |
| 1,2,4,8,12,16,<br>13,12,15,20,25,30 | 1,4,7,10,8,<br>7,9,13,17,21 |
| 1,2,4,8,12,16,<br>13,12,15,20,25,30,26 | 1,4,7,10,8,7,<br>9,13,17,21,18 |
| 1,2,4,8,12,16,13,<br>12,15,20,25,30,26,25,29,21 | 1,4,7,10,8,7,9,13,<br>17,21,18,17,20,14 |
| 1,2,4,8,12,16,13,12,16 | 1,4,7,10,8,7,10 |
| 1,2,4,8,12,16,13,12,16,13 | 1,4,7,10,8,7,10,8 |
| 1,2,4,8,12,16,13,13 | 1,4,7,10,8,8 |
| 1,2,4,8,12,16,13,14 | 1,4,7,10,8,9 |
| 1,2,4,8,12,16,13,15 | 1,4,7,10,8,10 |
| 1,2,4,8,12,16,<br>13,15,19,23,27,24 | 1,4,7,10,8,11,14,17,15 |
| 1,2,4,8,12,16,14 | 1,4,7,10,9 |
| 1,2,4,8,12,16,14,2 | 1,4,7,10,9,2 |
| 1,2,4,8,12,16,14,4 | 1,4,7,10,9,3 |
| 1,2,4,8,12,16,<br>14,8,12,16,13 | 1,4,7,10,9,3,7,11,<br>15,13,7,11,15,12 |
| 1,2,4,8,12,16,<br>14,8,12,16,14 | 1,4,7,10,9,3,7,11,<br>15,13,7,11,15,13 |
| 1,2,4,8,12,16,14,9 | 1,4,7,10,9,3,<br>7,11,15,13,8 |
| 1,2,4,8,12,16,14,10 | 1,4,7,10,9,3,<br>7,11,15,13,9 |
| 1,2,4,8,12,16,14,10,9 | 1,4,7,10,9,3,<br>7,11,15,13,9,8 |
| 1,2,4,8,12,16,14,10,11 | 1,4,7,10,9,3,7,<br>11,15,13,9,10 |
| 1,2,4,8,12,16,14,10,12 | 1,4,7,10,9,3,7,<br>11,15,13,9,11 |
| 1,2,4,8,12,16,<br>14,10,14,18,22,19 | 1,4,7,10,9,3,7,11,<br>15,13,9,13,17,21,18 |
| 1,2,4,8,12,16,<br>14,11,8,12,16,14 | 1,4,7,10,9,3,7,11,<br>15,13,10,7,11,15,13 |
| 1,2,4,8,12,16,15 | 1,4,7,10,9,<br>3,7,11,15,14 |
| 1,2,4,8,12,16,15,8 | 1,4,7,10,9,4 |
| 1,2,4,8,12,16,<br>15,8,12,15,9 | 1,4,7,10,9,4,7,9,5 |
| 1,2,4,8,12,16,15,8,12,16 | 1,4,7,10,9,4,7,10 |
| 1,2,4,8,12,16,<br>15,8,12,16,13 | 1,4,7,10,9,4,7,10,8 |
| 1,2,4,8,12,16,<br>15,8,12,16,14 | 1,4,7,10,9,4,7,10,9 |
| 1,2,4,8,12,16,15,9 | 1,4,7,10,9,5 |
| 1,2,4,8,12,16,15,9,8 | 1,4,7,10,9,5,4 |
| 1,2,4,8,12,16,<br>15,9,8,12,16 | 1,4,7,10,9,5,4,7,10 |
| 1,2,4,8,12,16,<br>15,9,8,12,16,14 | 1,4,7,10,9,5,4,7,10,9 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,12,16,15,9,<br>8,12,16,15,8,12,16,14 | 1,4,7,10,9,5,4,<br>7,10,9,4,7,10,9 |
| 1,2,4,8,12,16,<br>15,9,8,12,16,15,9 | 1,4,7,10,9,<br>5,4,7,10,9,5 |
| 1,2,4,8,12,16,15,9,9 | 1,4,7,10,9,5,5 |
| 1,2,4,8,12,16,15,10 | 1,4,7,10,9,6 |
| 1,2,4,8,12,16,<br>15,11,8,12,16,14 | 1,4,7,10,9,6,4,7,10,9 |
| 1,2,4,8,12,16,<br>15,11,8,12,16,15,9 | 1,4,7,10,9,6,<br>4,7,10,9,5 |
| 1,2,4,8,12,16,15,11,13 | 1,4,7,10,9,6,8 |
| 1,2,4,8,12,16,15,11,15 | 1,4,7,10,9,6,9 |
| 1,2,4,8,12,16,15,11,16 | 1,4,7,10,9,6,10 |
| 1,2,4,8,12,16,<br>15,11,16,21,25,17 | 1,4,7,10,9,6,<br>10,14,17,11 |
| 1,2,4,8,12,16,<br>15,11,16,21,26 | 1,4,7,10,9,6,10,14,18 |
| 1,2,4,8,12,16,<br>15,11,16,21,26,21,25,17 | 1,4,7,10,9,6,10,<br>14,18,14,17,11 |
| 1,2,4,8,12,16,<br>15,11,16,21,26,21,26 | 1,4,7,10,9,6,<br>10,14,18,14,18 |
| 1,2,4,8,12,16,<br>15,11,16,21,26,22 | 1,4,7,10,9,<br>6,10,14,18,15 |
| 1,2,4,8,12,16,15,11,<br>16,21,26,24,8,12,16,15,9 | 1,4,7,10,9,6,10,<br>14,18,16,4,7,10,9,5 |
| 1,2,4,8,12,16,15,<br>11,16,21,26,24,9 | 1,4,7,10,9,6,<br>10,14,18,16,5 |
| 1,2,4,8,12,16,15,<br>11,16,21,26,24,15 | 1,4,7,10,9,6,<br>10,14,18,16,9 |
| 1,2,4,8,12,16,15,<br>11,16,21,26,24,20 | 1,4,7,10,9,6,<br>10,14,18,16,13 |
| 1,2,4,8,12,16,15,<br>11,16,21,26,24,21 | 1,4,7,10,9,6,<br>10,14,18,16,14 |
| 1,2,4,8,12,16,15,<br>11,16,21,26,24,21,25,17 | 1,4,7,10,9,6,10,<br>14,18,16,14,17,11 |
| 1,2,4,8,12,16,15,<br>11,16,21,26,24,21,26 | 1,4,7,10,9,6,10,<br>14,18,16,14,18 |
| 1,2,4,8,12,16,15,<br>11,16,21,26,24,21,26,22 | 1,4,7,10,9,6,10,<br>14,18,16,14,18,15 |
| 1,2,4,8,12,16,15,<br>11,16,21,26,24,28 | 1,4,7,10,9,6,<br>10,14,18,16,19 |
| 1,2,4,8,12,16,15,<br>11,16,21,26,25 | 1,4,7,10,9,6,<br>10,14,18,17 |
| 1,2,4,8,12,16,15,<br>11,16,21,26,25,17 | 1,4,7,10,9,6,<br>10,14,18,17,11 |
| 1,2,4,8,12,16,15,12 | 1,4,7,10,9,7 |
| 1,2,4,8,12,16,15,<br>12,11,16,21,26,25,21 | 1,4,7,10,9,7,<br>6,10,14,18,17,14 |
| 1,2,4,8,12,16,15,12,12 | 1,4,7,10,9,7,7 |
| 1,2,4,8,12,16,15,12,13 | 1,4,7,10,9,7,8 |
| 1,2,4,8,12,16,15,12,14 | 1,4,7,10,9,7,9 |
| 1,2,4,8,12,16,15,12,15,8 | 1,4,7,10,9,7,9,4 |
| 1,2,4,8,12,16,15,<br>12,15,8,12,16,15,12,14 | 1,4,7,10,9,7,<br>9,4,7,10,9,7,9 |
| 1,2,4,8,12,16,15,12,15,9 | 1,4,7,10,9,7,9,5 |
| 1,2,4,8,12,16,15,12,15,12 | 1,4,7,10,9,7,9,7 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,12,16,<br>15,12,15,15,12 | 1,4,7,10,9,7,9,9,7 |
| 1,2,4,8,12,16,<br>15,12,15,18,12 | 1,4,7,10,9,7,9,11,7 |
| 1,2,4,8,12,16,15,12,15,19 | 1,4,7,10,9,7,9,12 |
| 1,2,4,8,12,16,15,12,15,20 | 1,4,7,10,9,7,9,13 |
| 1,2,4,8,12,16,<br>15,12,15,20,25,29 | 1,4,7,10,9,7,9,13,17,20 |
| 1,2,4,8,12,16,15,<br>12,15,20,25,29,21 | 1,4,7,10,9,7,<br>9,13,17,20,14 |
| 1,2,4,8,12,16,15,<br>12,15,20,25,29,25 | 1,4,7,10,9,7,<br>9,13,17,20,17 |
| 1,2,4,8,12,16,<br>15,12,15,20,25,30 | 1,4,7,10,9,7,9,13,17,21 |
| 1,2,4,8,12,16,15,<br>12,15,20,25,30,25,29,21 | 1,4,7,10,9,7,9,<br>13,17,21,17,20,14 |
| 1,2,4,8,12,16,15,<br>12,15,20,25,30,25,30 | 1,4,7,10,9,7,<br>9,13,17,21,17,21 |
| 1,2,4,8,12,16,15,<br>12,15,20,25,30,26 | 1,4,7,10,9,7,<br>9,13,17,21,18 |
| 1,2,4,8,12,16,15,<br>12,15,20,25,30,28,9 | 1,4,7,10,9,7,9,<br>13,17,21,19,5 |
| 1,2,4,8,12,16,15,<br>12,15,20,25,30,29 | 1,4,7,10,9,7,<br>9,13,17,21,20 |
| 1,2,4,8,12,16,15,<br>12,15,20,25,30,29,21 | 1,4,7,10,9,7,<br>9,13,17,21,20,14 |
| 1,2,4,8,12,16,15,12,16 | 1,4,7,10,9,7,10 |
| 1,2,4,8,12,16,15,12,<br>16,11,16,21,26,25,21,26 | 1,4,7,10,9,7,10,<br>6,10,14,18,17,14,18 |
| 1,2,4,8,12,16,15,12,16,12 | 1,4,7,10,9,7,10,7 |
| 1,2,4,8,12,16,<br>15,12,16,12,14 | 1,4,7,10,9,7,10,7,9 |
| 1,2,4,8,12,16,<br>15,12,16,12,15,9 | 1,4,7,10,9,7,10,7,9,5 |
| 1,2,4,8,12,16,<br>15,12,16,12,15,10 | 1,4,7,10,9,7,10,7,9,6 |
| 1,2,4,8,12,16,<br>15,12,16,12,15,12 | 1,4,7,10,9,7,10,7,9,7 |
| 1,2,4,8,12,16,15,<br>12,16,12,15,12,14 | 1,4,7,10,9,7,10,7,9,7,9 |
| 1,2,4,8,12,16,15,<br>12,16,12,15,12,15,9 | 1,4,7,10,9,7,<br>10,7,9,7,9,5 |
| 1,2,4,8,12,16,15,<br>12,16,12,15,13 | 1,4,7,10,9,7,10,7,9,8 |
| 1,2,4,8,12,16,15,<br>12,16,12,15,15,12 | 1,4,7,10,9,7,<br>10,7,9,9,7 |
| 1,2,4,8,12,16,15,<br>12,16,12,15,18,12 | 1,4,7,10,9,7,<br>10,7,9,11,7 |
| 1,2,4,8,12,16,15,<br>12,16,12,15,19 | 1,4,7,10,9,7,10,7,9,12 |
| 1,2,4,8,12,16,15,12,<br>16,12,15,20,25,30,29,21 | 1,4,7,10,9,7,10,7,<br>9,13,17,21,20,14 |
| 1,2,4,8,12,16,<br>15,12,16,12,16 | 1,4,7,10,9,7,10,7,10 |
| 1,2,4,8,12,16,15,<br>12,16,12,16,12,15,9 | 1,4,7,10,9,7,<br>10,7,10,7,9,5 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,12,16,15,12,16,13 | 1,4,7,10,9,7,10,8 |
| 1,2,4,8,12,16, 15,12,16,13,9 | 1,4,7,10,9,7,10,8,5 |
| 1,2,4,8,12,16, 15,12,16,13,12 | 1,4,7,10,9,7,10,8,7 |
| 1,2,4,8,12,16, 15,12,16,13,12,14 | 1,4,7,10,9,7,10,8,7,9 |
| 1,2,4,8,12,16, 15,12,16,13,12,15,9 | 1,4,7,10,9,7, 10,8,7,9,5 |
| 1,2,4,8,12,16, 15,12,16,13,12,16 | 1,4,7,10,9,7,10,8,7,10 |
| 1,2,4,8,12,16,15, 12,16,13,12,16,12,15,9 | 1,4,7,10,9,7,10, 8,7,10,7,9,5 |
| 1,2,4,8,12,16,15, 12,16,13,12,16,12,16 | 1,4,7,10,9,7, 10,8,7,10,7,10 |
| 1,2,4,8,12,16,15, 12,16,13,12,16,13 | 1,4,7,10,9,7, 10,8,7,10,8 |
| 1,2,4,8,12,16, 15,12,16,13,13 | 1,4,7,10,9,7,10,8,8 |
| 1,2,4,8,12,16,15,12,16,14 | 1,4,7,10,9,7,10,9 |
| 1,2,4,8,12,16, 15,12,16,15,8 | 1,4,7,10,9,7,10,9,4 |
| 1,2,4,8,12,16, 15,12,16,15,9 | 1,4,7,10,9,7,10,9,5 |
| 1,2,4,8,12,16,15, 12,16,15,12,15,19 | 1,4,7,10,9, 7,10,9,7,9,12 |
| 1,2,4,8,12,16, 15,12,16,15,12,16 | 1,4,7,10,9,7,10,9,7,10 |
| 1,2,4,8,12,16,15, 12,16,15,12,16,12,16 | 1,4,7,10,9,7, 10,9,7,10,7,10 |
| 1,2,4,8,12,16,15, 12,16,15,12,16,13 | 1,4,7,10,9,7, 10,9,7,10,8 |
| 1,2,4,8,12,16,15, 12,16,15,12,16,15,9 | 1,4,7,10,9,7, 10,9,7,10,9,5 |
| 1,2,4,8,12,16,15,12, 16,15,12,16,15,12,16,15,9 | 1,4,7,10,9,7,10, 9,7,10,9,7,10,9,5 |
| 1,2,4,8,12,16,15,13 | 1,4,7,10,9,8 |
| 1,2,4,8,12,16,15,13,8 | 1,4,7,10,9,8,4 |
| 1,2,4,8,12,16, 15,13,8,12,15,9 | 1,4,7,10,9,8,4,7,9,5 |
| 1,2,4,8,12,16,15,13, 8,12,15,11,16,21,26,25,22 | 1,4,7,10,9,8,4,7, 9,6,10,14,18,17,15 |
| 1,2,4,8,12,16, 15,13,8,12,15,12 | 1,4,7,10,9,8,4,7,9,7 |
| 1,2,4,8,12,16,15, 13,8,12,15,20,25,30,29,26 | 1,4,7,10,9,8,4, 7,9,13,17,21,20,18 |
| 1,2,4,8,12,16, 15,13,8,12,16 | 1,4,7,10,9,8,4,7,10 |
| 1,2,4,8,12,16,15, 13,8,12,16,12,15,9 | 1,4,7,10,9,8, 4,7,10,7,9,5 |
| 1,2,4,8,12,16,15, 13,8,12,16,12,15,12 | 1,4,7,10,9,8, 4,7,10,7,9,7 |
| 1,2,4,8,12,16,15, 13,8,12,16,12,16 | 1,4,7,10,9,8, 4,7,10,7,10 |
| 1,2,4,8,12,16, 15,13,8,12,16,13 | 1,4,7,10,9,8,4,7,10,8 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,12,16,<br>15,13,8,12,16,14 | 1,4,7,10,9,8,4,7,10,9 |
| 1,2,4,8,12,16,<br>15,13,8,12,16,15,9 | 1,4,7,10,9,8,<br>4,7,10,9,5 |
| 1,2,4,8,12,16,15,<br>13,8,12,16,15,12 | 1,4,7,10,9,8,<br>4,7,10,9,7 |
| 1,2,4,8,12,16,15,<br>13,8,12,16,15,12,14 | 1,4,7,10,9,8,<br>4,7,10,9,7,9 |
| 1,2,4,8,12,16,15,<br>13,8,12,16,15,12,16 | 1,4,7,10,9,8,<br>4,7,10,9,7,10 |
| 1,2,4,8,12,16,15,<br>13,8,12,16,15,12,16,13 | 1,4,7,10,9,8,4,<br>7,10,9,7,10,8 |
| 1,2,4,8,12,16,15,<br>13,8,12,16,15,12,16,15,9 | 1,4,7,10,9,8,4,7,<br>10,9,7,10,9,5 |
| 1,2,4,8,12,16,15,<br>13,8,12,16,15,13 | 1,4,7,10,9,<br>8,4,7,10,9,8 |
| 1,2,4,8,12,16,15,13,9 | 1,4,7,10,9,8,5 |
| 1,2,4,8,12,16,15,13,10 | 1,4,7,10,9,8,6 |
| 1,2,4,8,12,16,15,<br>13,11,8,12,16,15,13 | 1,4,7,10,9,8,<br>6,4,7,10,9,8 |
| 1,2,4,8,12,16,15,<br>13,11,16,21,26,25,22 | 1,4,7,10,9,8,<br>6,10,14,18,17,15 |
| 1,2,4,8,12,16,15,13,12 | 1,4,7,10,9,8,7 |
| 1,2,4,8,12,16,<br>15,13,12,15,9 | 1,4,7,10,9,8,7,9,5 |
| 1,2,4,8,12,16,15,13,12,16 | 1,4,7,10,9,8,7,10 |
| 1,2,4,8,12,16,<br>15,13,12,16,12 | 1,4,7,10,9,8,7,10,7 |
| 1,2,4,8,12,16,15,<br>13,12,16,12,15,9 | 1,4,7,10,9,8,<br>7,10,7,9,5 |
| 1,2,4,8,12,16,<br>15,13,12,16,12,16 | 1,4,7,10,9,8,7,10,7,10 |
| 1,2,4,8,12,16,<br>15,13,12,16,13 | 1,4,7,10,9,8,7,10,8 |
| 1,2,4,8,12,16,<br>15,13,12,16,14 | 1,4,7,10,9,8,7,10,9 |
| 1,2,4,8,12,16,<br>15,13,12,16,15,9 | 1,4,7,10,9,8,7,10,9,5 |
| 1,2,4,8,12,16,<br>15,13,12,16,15,12 | 1,4,7,10,9,8,7,10,9,7 |
| 1,2,4,8,12,16,<br>15,13,12,16,15,12,16 | 1,4,7,10,9,8,<br>7,10,9,7,10 |
| 1,2,4,8,12,16,15,<br>13,12,16,15,12,16,14 | 1,4,7,10,9,8,<br>7,10,9,7,10,9 |
| 1,2,4,8,12,16,15,<br>13,12,16,15,12,16,15,9 | 1,4,7,10,9,8,7,<br>10,9,7,10,9,5 |
| 1,2,4,8,12,16,<br>15,13,12,16,15,13 | 1,4,7,10,9,8,7,10,9,8 |
| 1,2,4,8,12,16,15,13,13 | 1,4,7,10,9,8,8 |
| 1,2,4,8,12,16,15,14 | 1,4,7,10,9,9 |
| 1,2,4,8,12,16,15,15,8 | 1,4,7,10,9,9,4 |
| 1,2,4,8,12,16,<br>15,15,8,12,16,15,14 | 1,4,7,10,9,9,<br>4,7,10,9,9 |
| 1,2,4,8,12,16,15,15,9 | 1,4,7,10,9,9,5 |
| 1,2,4,8,12,16,15,<br>15,9,8,12,16,15,15,9 | 1,4,7,10,9,9,<br>5,4,7,10,9,9,5 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,12,16,15,15,10 | 1,4,7,10,9,9,6 |
| 1,2,4,8,12,16,15,<br>15,11,8,12,16,15,15,9 | 1,4,7,10,9,9,<br>6,4,7,10,9,9,5 |
| 1,2,4,8,12,16,15,15,11,16 | 1,4,7,10,9,9,6,10 |
| 1,2,4,8,12,16,<br>15,15,11,16,21,26 | 1,4,7,10,9,9,6,10,14,18 |
| 1,2,4,8,12,16,<br>15,15,11,16,21,26,25,17 | 1,4,7,10,9,9,6,<br>10,14,18,17,11 |
| 1,2,4,8,12,16,15,<br>15,11,16,21,26,25,25,17 | 1,4,7,10,9,9,6,<br>10,14,18,17,17,11 |
| 1,2,4,8,12,16,15,15,12 | 1,4,7,10,9,9,7 |
| 1,2,4,8,12,16,<br>15,15,12,15,12 | 1,4,7,10,9,9,7,9,7 |
| 1,2,4,8,12,16,<br>15,15,12,15,20 | 1,4,7,10,9,9,7,9,13 |
| 1,2,4,8,12,16,15,<br>15,12,15,20,25,30 | 1,4,7,10,9,9,<br>7,9,13,17,21 |
| 1,2,4,8,12,16,15,<br>15,12,15,20,25,30,29,21 | 1,4,7,10,9,9,7,<br>9,13,17,21,20,14 |
| 1,2,4,8,12,16,15,15,<br>12,15,20,25,30,29,29,21 | 1,4,7,10,9,9,7,9,<br>13,17,21,20,20,14 |
| 1,2,4,8,12,16,15,15,12,16 | 1,4,7,10,9,9,7,10 |
| 1,2,4,8,12,16,<br>15,15,12,16,12 | 1,4,7,10,9,9,7,10,7 |
| 1,2,4,8,12,16,15,<br>15,12,16,12,15,9 | 1,4,7,10,9,9,<br>7,10,7,9,5 |
| 1,2,4,8,12,16,15,<br>15,12,16,12,16 | 1,4,7,10,9,9,7,10,7,10 |
| 1,2,4,8,12,16,<br>15,15,12,16,13 | 1,4,7,10,9,9,7,10,8 |
| 1,2,4,8,12,16,<br>15,15,12,16,15,9 | 1,4,7,10,9,<br>9,7,10,9,5 |
| 1,2,4,8,12,16,<br>15,15,12,16,15,12 | 1,4,7,10,9,<br>9,7,10,9,7 |
| 1,2,4,8,12,16,15,<br>15,12,16,15,12,16,15,9 | 1,4,7,10,9,9,7,<br>10,9,7,10,9,5 |
| 1,2,4,8,12,16,15,<br>15,12,16,15,13,9 | 1,4,7,10,9,9,7,10,9,8 |
| 1,2,4,8,12,16,15,<br>15,12,16,15,15,9 | 1,4,7,10,9,<br>9,7,10,9,9,5 |
| 1,2,4,8,12,16,15,15,12,<br>16,15,15,12,16,15,15,9 | 1,4,7,10,9,9,7,<br>10,9,9,7,10,9,9,5 |
| 1,2,4,8,12,16,15,15,13 | 1,4,7,10,9,9,8 |
| 1,2,4,8,12,16,15,15,15,9 | 1,4,7,10,9,9,9,5 |
| 1,2,4,8,12,16,15,16 | 1,4,7,10,9,10 |
| 1,2,4,8,12,16,15,16,8 | 1,4,7,10,9,10,4 |
| 1,2,4,8,12,16,<br>15,16,8,12,15,9 | 1,4,7,10,9,10,4,7,9,5 |
| 1,2,4,8,12,16,15,16,8,<br>12,15,11,16,21,26,25,26 | 1,4,7,10,9,10,4,7,<br>9,6,10,14,18,17,18 |
| 1,2,4,8,12,16,<br>15,16,8,12,15,12 | 1,4,7,10,9,10,4,7,9,7 |
| 1,2,4,8,12,16,<br>15,16,8,12,15,19 | 1,4,7,10,9,10,4,7,9,12 |
| 1,2,4,8,12,16,15,<br>16,8,12,15,20,25,29,21 | 1,4,7,10,9,10,4,<br>7,9,13,17,20,14 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,12,16,15,<br>16,8,12,15,20,25,30 | 1,4,7,10,9,10,<br>4,7,9,13,17,21 |
| 1,2,4,8,12,16,15,<br>16,8,12,15,20,25,30,29,30 | 1,4,7,10,9,10,4,<br>7,9,13,17,21,20,21 |
| 1,2,4,8,12,16,15,16,8,<br>12,15,20,25,30,29,30,19 | 1,4,7,10,9,10,4,7,<br>9,13,17,21,20,21,12 |
| 1,2,4,8,12,16,<br>15,16,8,12,16 | 1,4,7,10,9,10,4,7,10 |
| 1,2,4,8,12,16,15,<br>16,8,12,16,12,15,9 | 1,4,7,10,9,10,4,<br>7,10,7,9,5 |
| 1,2,4,8,12,16,15,<br>16,8,12,16,12,16 | 1,4,7,10,9,10,<br>4,7,10,7,10 |
| 1,2,4,8,12,16,15,<br>16,8,12,16,13 | 1,4,7,10,9,10,4,7,10,8 |
| 1,2,4,8,12,16,15,<br>16,8,12,16,15,9 | 1,4,7,10,9,10,<br>4,7,10,9,5 |
| 1,2,4,8,12,16,15,<br>16,8,12,16,15,12 | 1,4,7,10,9,10,<br>4,7,10,9,7 |
| 1,2,4,8,12,16,15,<br>16,8,12,16,15,12,16 | 1,4,7,10,9,10,<br>4,7,10,9,7,10 |
| 1,2,4,8,12,16,15,16,<br>8,12,16,15,12,16,15,9 | 1,4,7,10,9,10,<br>4,7,10,9,7,10,9,5 |
| 1,2,4,8,12,16,15,<br>16,8,12,16,15,13 | 1,4,7,10,9,10,<br>4,7,10,9,8 |
| 1,2,4,8,12,16,15,<br>16,8,12,16,15,15,9 | 1,4,7,10,9,10,<br>4,7,10,9,9,5 |
| 1,2,4,8,12,16,15,<br>16,8,12,16,15,16 | 1,4,7,10,9,<br>10,4,7,10,9,10 |
| 1,2,4,8,12,16,15,16,9 | 1,4,7,10,9,10,5 |
| 1,2,4,8,12,16,15,<br>16,11,8,12,16,15,16 | 1,4,7,10,9,10,<br>6,4,7,10,9,10 |
| 1,2,4,8,12,16,15,16,11,16 | 1,4,7,10,9,10,6,10 |
| 1,2,4,8,12,16,<br>15,16,11,16,21,26 | 1,4,7,10,9,10,6,10,14,18 |
| 1,2,4,8,12,16,15,<br>16,11,16,21,26,25,26 | 1,4,7,10,9,10,6,<br>10,14,18,17,18 |
| 1,2,4,8,12,16,15,16,12 | 1,4,7,10,9,10,7 |
| 1,2,4,8,12,16,<br>15,16,12,15,9 | 1,4,7,10,9,10,7,9,5 |
| 1,2,4,8,12,16,<br>15,16,12,15,12 | 1,4,7,10,9,10,7,9,7 |
| 1,2,4,8,12,16,<br>15,16,12,15,20 | 1,4,7,10,9,10,7,9,13 |
| 1,2,4,8,12,16,15,<br>16,12,15,20,25,30 | 1,4,7,10,9,10,<br>7,9,13,17,21 |
| 1,2,4,8,12,16,15,16,12,16 | 1,4,7,10,9,10,7,10 |
| 1,2,4,8,12,16,<br>15,16,12,16,12,15,9 | 1,4,7,10,9,10,7,10,7,9,5 |
| 1,2,4,8,12,16,<br>15,16,12,16,12,16 | 1,4,7,10,9,10,7,10,7,10 |
| 1,2,4,8,12,16,<br>15,16,12,16,13 | 1,4,7,10,9,10,7,10,8 |
| 1,2,4,8,12,16,<br>15,16,12,16,15,9 | 1,4,7,10,9,10,7,10,9,5 |
| 1,2,4,8,12,16,<br>15,16,12,16,15,16 | 1,4,7,10,9,10,7,10,9,10 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,12,16,15,16,13 | 1,4,7,10,9,10,8 |
| 1,2,4,8,12,16,15,16,15,9 | 1,4,7,10,9,10,9,5 |
| 1,2,4,8,12,16,15,16,15,12 | 1,4,7,10,9,10,9,7 |
| 1,2,4,8,12,16,<br>15,16,15,15,9 | 1,4,7,10,9,10,9,9,5 |
| 1,2,4,8,12,16,15,16,15,16 | 1,4,7,10,9,10,9,10 |
| 1,2,4,8,12,16,15,16,16 | 1,4,7,10,9,10,10 |
| 1,2,4,8,12,16,15,17 | 1,4,7,10,9,11 |
| 1,2,4,8,12,16,15,18,9 | 1,4,7,10,9,11,5 |
| 1,2,4,8,12,16,15,<br>18,9,8,12,16,15,18,9 | 1,4,7,10,9,11,5,<br>4,7,10,9,11,5 |
| 1,2,4,8,12,16,15,18,10 | 1,4,7,10,9,11,6 |
| 1,2,4,8,12,16,15,<br>18,11,16,21,26,25,29,17 | |
| 1,2,4,8,12,16,15,18,12 | 1,4,7,10,9,11,7 |
| 1,2,4,8,12,16,<br>15,18,12,15,9 | 1,4,7,10,9,11,7,9,5 |
| 1,2,4,8,12,16,15,18,<br>12,15,20,25,30,29,33,21 | 1,4,7,10,9,11,7,9,<br>13,17,21,20,23,14 |
| 1,2,4,8,12,16,15,18,12,16 | 1,4,7,10,9,11,7,10 |
| 1,2,4,8,12,16,<br>15,18,12,16,13 | 1,4,7,10,9,11,7,10,8 |
| 1,2,4,8,12,16,<br>15,18,12,16,15,9 | 1,4,7,10,9,11,7,10,9,5 |
| 1,2,4,8,12,16,<br>15,18,12,16,15,16 | 1,4,7,10,9,11,7,10,9,10 |
| 1,2,4,8,12,16,<br>15,18,12,16,15,18,9 | 1,4,7,10,9,11,<br>7,10,9,11,5 |
| 1,2,4,8,12,16,15,18,13 | 1,4,7,10,9,11,8 |
| 1,2,4,8,12,16,15,18,15,9 | 1,4,7,10,9,11,9,5 |
| 1,2,4,8,12,16,15,18,15,16 | 1,4,7,10,9,11,9,10 |
| 1,2,4,8,12,16,<br>15,18,15,18,9 | 1,4,7,10,9,11,9,11,5 |
| 1,2,4,8,12,16,15,18,16 | 1,4,7,10,9,11,10 |
| 1,2,4,8,12,16,15,18,18,9 | 1,4,7,10,9,11,11,5 |
| 1,2,4,8,12,16,15,18,19 | 1,4,7,10,9,11,12 |
| 1,2,4,8,12,16,15,18,21,9 | 1,4,7,10,9,11,13,5 |
| 1,2,4,8,12,16,15,18,21,22 | 1,4,7,10,9,11,13,14 |
| 1,2,4,8,12,16,<br>15,18,21,24,9 | 1,4,7,10,9,11,13,15,5 |
| 1,2,4,8,12,16,15,19 | 1,4,7,10,9,12 |
| 1,2,4,8,12,16,15,19,8 | 1,4,7,10,9,12,4 |
| 1,2,4,8,12,16,<br>15,19,8,12,15,9 | 1,4,7,10,9,12,4,7,9,5 |
| 1,2,4,8,12,16,15,<br>19,8,12,15,11,16,21,25,17 | 1,4,7,10,9,12,4,<br>7,9,6,10,14,17,11 |
| 1,2,4,8,12,16,15,<br>19,8,12,15,11,16,21,26 | 1,4,7,10,9,12,<br>4,7,9,6,10,14,18 |
| 1,2,4,8,12,16,15,<br>19,8,12,15,11,16,21,26,21 | 1,4,7,10,9,12,4,<br>7,9,6,10,14,18,14 |
| 1,2,4,8,12,16,15,19,<br>8,12,15,11,16,21,26,21,26 | 1,4,7,10,9,12,4,7,<br>9,6,10,14,18,14,18 |
| 1,2,4,8,12,16,15,<br>19,8,12,15,11,16,21,26,22 | 1,4,7,10,9,12,4,7,<br>9,6,10,14,18,15 |
| 1,2,4,8,12,16,15,19,<br>8,12,15,11,16,21,26,22,21 | 1,4,7,10,9,12,4,7,<br>9,6,10,14,18,15,14 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,12,16,15,19,<br>8,12,15,11,16,21,26,22,22 | 1,4,7,10,9,12,4,7,<br>9,6,10,14,18,15,15 |
| 1,2,4,8,12,16,15,19,<br>8,12,15,11,16,21,26,25,17 | 1,4,7,10,9,12,4,7,<br>9,6,10,14,18,17,11 |
| 1,2,4,8,12,16,15,19,<br>8,12,15,11,16,21,26,25,22 | 1,4,7,10,9,12,4,7,<br>9,6,10,14,18,17,15 |
| 1,2,4,8,12,16,15,19,<br>8,12,15,11,16,21,26,25,26 | 1,4,7,10,9,12,4,7,<br>9,6,10,14,18,17,18 |
| 1,2,4,8,12,16,<br>15,19,8,12,15,12 | 1,4,7,10,9,12,4,7,9,7 |
| 1,2,4,8,12,16,<br>15,19,8,12,15,12,12 | 1,4,7,10,9,<br>12,4,7,9,7,7 |
| 1,2,4,8,12,16,15,<br>19,8,12,15,12,14 | 1,4,7,10,9,<br>12,4,7,9,7,9 |
| 1,2,4,8,12,16,15,<br>19,8,12,15,12,15,9 | 1,4,7,10,9,12,<br>4,7,9,7,9,5 |
| 1,2,4,8,12,16,<br>15,19,8,12,15,13 | 1,4,7,10,9,12,4,7,9,8 |
| 1,2,4,8,12,16,<br>15,19,8,12,15,14 | 1,4,7,10,9,12,4,7,9,9 |
| 1,2,4,8,12,16,<br>15,19,8,12,15,15,12 | 1,4,7,10,9,12,<br>4,7,9,9,5 |
| 1,2,4,8,12,16,<br>15,19,8,12,15,16 | 1,4,7,10,9,12,4,7,9,10 |
| 1,2,4,8,12,16,<br>15,19,8,12,15,18,12 | 1,4,7,10,9,12,<br>4,7,9,11,5 |
| 1,2,4,8,12,16,<br>15,19,8,12,15,19 | 1,4,7,10,9,12,4,7,9,12 |
| 1,2,4,8,12,16,<br>15,19,8,12,15,20 | 1,4,7,10,9,12,4,7,9,13 |
| 1,2,4,8,12,16,15,<br>19,8,12,15,20,25,29,21 | 1,4,7,10,9,12,4,<br>7,9,13,17,20,14 |
| 1,2,4,8,12,16,<br>15,19,8,12,15,20,25,30 | 1,4,7,10,9,12,<br>4,7,9,13,17,21 |
| 1,2,4,8,12,16,15,<br>19,8,12,15,20,25,30,25 | 1,4,7,10,9,12,4,<br>7,9,13,17,21,17 |
| 1,2,4,8,12,16,15,19,<br>8,12,15,20,25,30,25,29,21 | 1,4,7,10,9,12,4,7,<br>9,13,17,21,17,20,14 |
| 1,2,4,8,12,16,15,19,<br>8,12,15,20,25,30,29,21 | 1,4,7,10,9,12,4,7,<br>9,13,17,21,20,14 |
| 1,2,4,8,12,16,15,19,<br>8,12,15,20,25,30,29,35 | 1,4,7,10,9,12,4,7,<br>9,13,17,21,20,24 |
| 1,2,4,8,12,16,15,19,<br>8,12,15,20,25,30,29,35,13 | 1,4,7,10,9,12,4,7,<br>9,13,17,21,20,24,8 |
| 1,2,4,8,12,16,15,19,<br>8,12,15,20,25,30,29,35,16 | 1,4,7,10,9,12,4,7,<br>9,13,17,21,20,24,10 |
| 1,2,4,8,12,16,15,19,<br>8,12,15,20,25,30,29,35,19 | 1,4,7,10,9,12,4,7,<br>9,13,17,21,20,24,12 |
| 1,2,4,8,12,16,<br>15,19,8,12,16 | 1,4,7,10,9,12,4,7,10 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,8,12,15,9 | 1,4,7,10,9,12,<br>4,7,10,4,7,9,5 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,8,12,15,12 | 1,4,7,10,9,12,<br>4,7,10,4,7,9,7 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,8,12,15,19 | 1,4,7,10,9,12,<br>4,7,10,4,7,9,12 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,12,16,15,<br>19,8,12,16,8,12,16 | 1,4,7,10,9,12,<br>4,7,10,4,7,10 |
| 1,2,4,8,12,16,<br>15,19,8,12,16,9 | 1,4,7,10,9,12,4,7,10,5 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,11,15 | 1,4,7,10,9,<br>12,4,7,10,6,9 |
| 1,2,4,8,12,16,15,19,8,<br>12,16,11,16,21,26,25,30 | 1,4,7,10,9,12,4,7,<br>10,6,10,14,18,17,21 |
| 1,2,4,8,12,16,<br>15,19,8,12,16,12 | 1,4,7,10,9,12,4,7,10,7 |
| 1,2,4,8,12,16,15,19,<br>8,12,16,12,8,12,15,9 | 1,4,7,10,9,12,<br>4,7,10,7,4,7,9,5 |
| 1,2,4,8,12,16,15,19,<br>8,12,16,12,8,12,15,12 | 1,4,7,10,9,12,<br>4,7,10,7,4,7,9,7 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,8,12,16 | 1,4,7,10,9,12,<br>4,7,10,7,4,7,10 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,8,12,16,12 | 1,4,7,10,9,12,4,<br>7,10,7,4,7,10,7 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,12 | 1,4,7,10,9,12,<br>4,7,10,7,7 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,13 | 1,4,7,10,9,12,<br>4,7,10,7,8 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,15,9 | 1,4,7,10,9,12,<br>4,7,10,7,9,5 |
| 1,2,4,8,12,16,15,19,<br>8,12,16,12,15,9,8,12,16 | 1,4,7,10,9,12,4,<br>7,10,7,9,5,4,7,10 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,15,9,9 | 1,4,7,10,9,12,4,<br>7,10,7,9,5,5 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,15,12 | 1,4,7,10,9,12,<br>4,7,10,7,9,7 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,15,12,15,9 | 1,4,7,10,9,12,4,<br>7,10,7,9,7,9,5 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,15,15,9 | 1,4,7,10,9,12,4,<br>7,10,7,9,9,5 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,15,18,9 | 1,4,7,10,9,12,4,<br>7,10,7,9,11,5 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,15,19 | 1,4,7,10,9,12,<br>4,7,10,7,9,12 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,15,20 | 1,4,7,10,9,12,<br>4,7,10,7,9,13 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,16 | 1,4,7,10,9,12,<br>4,7,10,7,10 |
| 1,2,4,8,12,16,15,19,<br>8,12,16,12,16,8,12,15,9 | 1,4,7,10,9,12,4,<br>7,10,7,10,4,7,9,5 |
| 1,2,4,8,12,16,15,19,<br>8,12,16,12,16,8,12,15,12 | 1,4,7,10,9,12,4,7,<br>10,7,10,4,7,9,7 |
| 1,2,4,8,12,16,15,19,<br>8,12,16,12,16,8,12,16 | 1,4,7,10,9,12,4,<br>7,10,7,10,4,7,10 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,16,12 | 1,4,7,10,9,12,<br>4,7,10,7,10,7 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,12,16,12,15,9 | 1,4,7,10,9,12,4,<br>7,10,7,10,7,9,5 |
| 1,2,4,8,12,16,15,19,<br>8,12,16,12,16,12,15,12 | 1,4,7,10,9,12,4,<br>7,10,7,10,7,9,7 |
| 1,2,4,8,12,16,15,19,<br>8,12,16,12,16,12,16 | 1,4,7,10,9,12,4,<br>7,10,7,10,7,10 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,12,16,<br>15,19,8,12,16,13 | 1,4,7,10,9,12,4,7,10,8 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,13,8,12,15,9 | 1,4,7,10,9,12,<br>4,7,10,8,4,7,9,5 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,13,8,12,15,12 | 1,4,7,10,9,12,4,<br>7,10,8,4,7,9,7 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,13,8,12,16 | 1,4,7,10,9,12,<br>4,7,10,8,4,7,10 |
| 1,2,4,8,12,16,15,19,<br>8,12,16,13,8,12,16,12,16 | 1,4,7,10,9,12,4,<br>7,10,8,4,7,10,7,10 |
| 1,2,4,8,12,16,15,19,<br>8,12,16,13,8,12,16,13 | 1,4,7,10,9,12,4,<br>7,10,8,4,7,10,8 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,13,11,15 | 1,4,7,10,9,12,<br>4,7,10,8,6,9 |
| 1,2,4,8,12,16,15,19,8,<br>12,16,13,11,16,21,25,17 | 1,4,7,10,9,12,4,7,<br>10,8,6,10,14,17,11 |
| 1,2,4,8,12,16,15,19,<br>8,12,16,13,11,16,21,26 | 1,4,7,10,9,12,4,7,<br>10,8,6,10,14,18 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,13,12 | 1,4,7,10,9,12,<br>4,7,10,8,7 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,13,12,15,9 | 1,4,7,10,9,12,4,<br>7,10,8,7,9,5 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,13,12,16 | 1,4,7,10,9,12,<br>4,7,10,8,7,10 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,13,13 | 1,4,7,10,9,12,<br>4,7,10,8,8 |
| 1,2,4,8,12,16,<br>15,19,8,12,16,14 | 1,4,7,10,9,12,4,7,10,9 |
| 1,2,4,8,12,16,<br>15,19,8,12,16,15,9 | 1,4,7,10,9,12,<br>4,7,10,9,5 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,11,15 | 1,4,7,10,9,12,<br>4,7,10,9,6,9 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,12 | 1,4,7,10,9,12,<br>4,7,10,9,7 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,12,15,9 | 1,4,7,10,9,12,4,<br>7,10,9,7,9,5 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,12,15,19 | 1,4,7,10,9,12,4,<br>7,10,9,7,9,12 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,12,16 | 1,4,7,10,9,12,<br>4,7,10,9,7,10 |
| 1,2,4,8,12,16,15,19,<br>8,12,16,15,12,16,11,15 | 1,4,7,10,9,12,4,<br>7,10,9,7,10,6,9 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,12,16,12 | 1,4,7,10,9,12,4,<br>7,10,9,7,10,7 |
| 1,2,4,8,12,16,15,19,8,<br>12,16,15,12,16,12,15,9 | 1,4,7,10,9,12,4,<br>7,10,9,7,10,7,9,5 |
| 1,2,4,8,12,16,15,19,8,<br>12,16,15,12,16,12,15,12 | 1,4,7,10,9,12,4,<br>7,10,9,7,10,7,9,7 |
| 1,2,4,8,12,16,15,19,8,<br>12,16,15,12,16,12,15,19 | 1,4,7,10,9,12,4,7,<br>10,9,7,10,7,9,12 |
| 1,2,4,8,12,16,15,19,<br>8,12,16,15,12,16,12,16 | 1,4,7,10,9,12,4,<br>7,10,9,7,10,7,10 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,12,16,13 | 1,4,7,10,9,12,4,<br>7,10,9,7,10,8 |
| 1,2,4,8,12,16,15,19,<br>8,12,16,15,12,16,15,9 | 1,4,7,10,9,12,4,<br>7,10,9,7,10,9,5 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,13 | 1,4,7,10,9,12,<br>4,7,10,9,8 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,15,9 | 1,4,7,10,9,12,<br>4,7,10,9,9,5 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,15,12 | 1,4,7,10,9,12,<br>4,7,10,9,9,7 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,15,13 | 1,4,7,10,9,12,<br>4,7,10,9,9,8 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,15,15,9 | 1,4,7,10,9,12,<br>4,7,10,9,9,9,5 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,16 | 1,4,7,10,9,12,<br>4,7,10,9,10 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,17 | 1,4,7,10,9,12,<br>4,7,10,9,11 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,18,9 | 1,4,7,10,9,12,<br>4,7,10,9,11,5 |
| 1,2,4,8,12,16,15,<br>19,8,12,16,15,19 | 1,4,7,10,9,12,<br>4,7,10,9,12 |
| 1,2,4,8,12,16,15,19,11,15 | 1,4,7,10,9,12,6,9 |
| 1,2,4,8,12,16,15,<br>19,11,16,21,25,17 | 1,4,7,10,9,12,<br>6,10,14,17,11 |
| 1,2,4,8,12,16,15,<br>19,11,16,21,26,25,17 | 1,4,7,10,9,12,<br>6,10,14,18,17,11 |
| 1,2,4,8,12,16,15,<br>19,11,16,21,26,25,29,17 | 1,4,7,10,9,12,6,<br>10,14,18,17,20,11 |
| 1,2,4,8,12,16,15,<br>19,11,16,21,26,25,30 | 1,4,7,10,9,12,6,<br>10,14,18,17,21 |
| 1,2,4,8,12,16,15,19,12 | 1,4,7,10,9,12,7 |
| 1,2,4,8,12,16,<br>15,19,12,15,9 | 1,4,7,10,9,12,7,9,5 |
| 1,2,4,8,12,16,15,19,12,16 | 1,4,7,10,9,12,7,10 |
| 1,2,4,8,12,16,<br>15,19,12,16,12,15,9 | 1,4,7,10,9,<br>12,7,10,7,9,5 |
| 1,2,4,8,12,16,<br>15,19,12,16,15,9 | 1,4,7,10,9,12,7,10,9,5 |
| 1,2,4,8,12,16,<br>15,19,12,16,15,19 | 1,4,7,10,9,12,<br>7,10,9,12 |
| 1,2,4,8,12,16,15,19,15,19 | 1,4,7,10,9,12,9,12 |
| 1,2,4,8,12,16,15,19,18,22 | 1,4,7,10,9,12,11,14 |
| 1,2,4,8,12,16,15,19,19 | 1,4,7,10,9,12,12 |
| 1,2,4,8,12,16,15,19,20 | 1,4,7,10,9,12,13 |
| 1,2,4,8,12,16,15,<br>19,22,8,12,16,15,19 | 1,4,7,10,9,12,<br>14,4,7,10,9,12 |
| 1,2,4,8,12,16,15,19,23 | 1,4,7,10,9,12,15 |
| 1,2,4,8,12,16,<br>15,19,23,26,9 | 1,4,7,10,9,12,15,17,5 |
| 1,2,4,8,12,16,<br>15,19,23,26,20 | 1,4,7,10,9,12,15,17,13 |
| 1,2,4,8,12,16,15,19,23,27 | 1,4,7,10,9,12,15,18 |
| 1,2,4,8,12,16,15,<br>19,23,27,19,23,27 | 1,4,7,10,9,12,<br>15,18,12,15,18 |
| 1,2,4,8,12,16,15,19,24 | 1,4,7,10,9,12,16 |
| 1,2,4,8,12,16,15,20 | 1,4,7,10,9,13 |
| 1,2,4,8,12,16,15,20,25 | 1,4,7,10,9,13,17 |
| 1,2,4,8,12,16,15,20,25,27 | 1,4,7,10,9,13,17,19 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,12,16,<br>15,20,25,28,9 | 1,4,7,10,9,13,17,19,5 |
| 1,2,4,8,12,16,15,20,25,29 | 1,4,7,10,9,13,17,20 |
| 1,2,4,8,12,16,<br>15,20,25,29,21 | 1,4,7,10,9,13,17,20,14 |
| 1,2,4,8,12,16,15,20,25,30 | 1,4,7,10,9,13,17,21 |
| 1,2,4,8,12,16,<br>15,20,25,30,29,34 | 1,4,7,10,9,13,<br>17,21,20,24 |
| 1,2,4,8,12,16,15,<br>20,25,30,29,34,39,44 | 1,4,7,10,9,13,17,<br>21,20,24,28,32 |
| 1,2,4,8,12,16,16 | 1,4,7,10,10 |
| 1,2,4,8,12,16,<br>16,8,12,15,9 | 1,4,7,10,10,4,7,9,5 |
| 1,2,4,8,12,16,<br>16,8,12,15,20,25,30,30 | 1,4,7,10,10,4,<br>7,9,13,17,21,21 |
| 1,2,4,8,12,16,16,<br>8,12,15,20,25,30,30,19 | 1,4,7,10,10,4,7,<br>9,13,17,21,21,12 |
| 1,2,4,8,12,16,16,8,12,16 | 1,4,7,10,10,4,7,10 |
| 1,2,4,8,12,16,<br>16,8,12,16,13 | 1,4,7,10,10,4,7,10,8 |
| 1,2,4,8,12,16,<br>16,8,12,16,15,19 | 1,4,7,10,10,4,7,10,9,12 |
| 1,2,4,8,12,16,16,<br>8,12,16,15,20,25,30,30 | 1,4,7,10,10,4,7,<br>10,9,13,17,21,21 |
| 1,2,4,8,12,16,16,8,12,<br>16,15,20,25,30,30,11,15 | 1,4,7,10,10,4,7,<br>10,9,13,17,21,21,6,9 |
| 1,2,4,8,12,16,16,8,<br>12,16,15,20,25,30,30,19 | 1,4,7,10,10,4,7,<br>10,9,13,17,21,21,12 |
| 1,2,4,8,12,16,<br>16,8,12,16,16 | 1,4,7,10,10,4,7,10,10 |
| 1,2,4,8,12,16,16,9 | 1,4,7,10,10,5 |
| 1,2,4,8,12,16,<br>16,11,8,12,16,16 | 1,4,7,10,10,6,4,7,10,10 |
| 1,2,4,8,12,16,16,11,15 | 1,4,7,10,10,6,9 |
| 1,2,4,8,12,16,16,11,15,20 | 1,4,7,10,10,6,9,13 |
| 1,2,4,8,12,16,16,11,16 | 1,4,7,10,10,6,10 |
| 1,2,4,8,12,16,<br>16,11,16,21,25 | 1,4,7,10,10,6,10,14,17 |
| 1,2,4,8,12,16,<br>16,11,16,21,25,17 | 1,4,7,10,10,<br>6,10,14,17,11 |
| 1,2,4,8,12,16,<br>16,11,16,21,26,26 | 1,4,7,10,10,<br>6,10,14,18,18 |
| 1,2,4,8,12,16,16,12 | 1,4,7,10,10,7 |
| 1,2,4,8,12,16,16,12,14 | 1,4,7,10,10,7,9 |
| 1,2,4,8,12,16,16,<br>12,15,8,12,16,16 | 1,4,7,10,10,<br>7,9,4,7,10,10 |
| 1,2,4,8,12,16,16,12,15,9 | 1,4,7,10,10,7,9,5 |
| 1,2,4,8,12,16,16,12,15,19 | 1,4,7,10,10,7,9,12 |
| 1,2,4,8,12,16,16,<br>12,15,20,25,29,21 | 1,4,7,10,10,7,<br>9,13,17,20,14 |
| 1,2,4,8,12,16,16,12,<br>15,20,25,30,30,25,29,21 | 1,4,7,10,10,7,9,<br>13,17,21,21,17,20,14 |
| 1,2,4,8,12,16,16,12,16 | 1,4,7,10,10,7,10 |
| 1,2,4,8,12,16,<br>16,12,16,12,15,9 | 1,4,7,10,10,7,10,7,9,5 |
| 1,2,4,8,12,16,16,12,16,13 | 1,4,7,10,10,7,10,8 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,12,16,<br>16,12,16,15,9 | 1,4,7,10,10,7,10,9,5 |
| 1,2,4,8,12,16,<br>16,12,16,15,15,9 | 1,4,7,10,10,7,10,9,9,5 |
| 1,2,4,8,12,16,<br>16,12,16,15,16 | 1,4,7,10,10,7,10,9,10 |
| 1,2,4,8,12,16,<br>16,12,16,15,18,9 | 1,4,7,10,10,7,10,9,11,5 |
| 1,2,4,8,12,16,<br>16,12,16,15,19 | 1,4,7,10,10,7,10,9,12 |
| 1,2,4,8,12,16,16,12,16,16 | 1,4,7,10,10,7,10,10 |
| 1,2,4,8,12,16,<br>16,12,16,16,12,15,9 | 1,4,7,10,10,<br>7,10,10,7,9,5 |
| 1,2,4,8,12,16,16,13 | 1,4,7,10,10,8 |
| 1,2,4,8,12,16,<br>16,13,12,15,9 | 1,4,7,10,10,8,7,9,5 |
| 1,2,4,8,12,16,16,14 | 1,4,7,10,10,9 |
| 1,2,4,8,12,16,16,15,9 | 1,4,7,10,10,9,5 |
| 1,2,4,8,12,16,16,15,13 | 1,4,7,10,10,9,8 |
| 1,2,4,8,12,16,16,15,19 | 1,4,7,10,10,9,12 |
| 1,2,4,8,12,16,16,16 | 1,4,7,10,10,10 |
| 1,2,4,8,12,16,16,16,16 | 1,4,7,10,10,10,10 |
| 1,2,4,8,12,16,17 | 1,4,7,10,11 |
| 1,2,4,8,12,16,<br>17,19,23,27,31 | 1,4,7,10,11,14,17,20 |
| 1,2,4,8,12,16,18 | 1,4,7,10,12 |
| 1,2,4,8,12,16,19,9 | 1,4,7,10,12,5 |
| 1,2,4,8,12,16,19,10 | 1,4,7,10,12,6 |
| 1,2,4,8,12,16,19,11,15 | 1,4,7,10,12,6,9 |
| 1,2,4,8,12,16,<br>19,11,16,21,26,26 | 1,4,7,10,12,<br>6,10,14,18,18 |
| 1,2,4,8,12,16,<br>19,11,16,21,26,27 | 1,4,7,10,12,<br>6,10,14,18,19 |
| 1,2,4,8,12,16,<br>19,11,16,21,26,28 | 1,4,7,10,12,<br>6,10,14,18,20 |
| 1,2,4,8,12,16,19,11,<br>16,21,26,29,8,12,16,19,9 | 1,4,7,10,12,6,10,<br>14,18,20,4,7,10,12,5 |
| 1,2,4,8,12,16,19,<br>11,16,21,26,30,22 | 1,4,7,10,12,6,<br>10,14,18,21,15 |
| 1,2,4,8,12,16,19,12 | 1,4,7,10,12,7 |
| 1,2,4,8,12,16,19,12,15,9 | 1,4,7,10,12,7,9,5 |
| 1,2,4,8,12,16,19,12,16 | 1,4,7,10,12,7,10 |
| 1,2,4,8,12,16,<br>19,12,16,19,9 | 1,4,7,10,12,7,10,12,5 |
| 1,2,4,8,12,16,19,13 | 1,4,7,10,12,8 |
| 1,2,4,8,12,16,19,15,9 | 1,4,7,10,12,9,5 |
| 1,2,4,8,12,16,19,16 | 1,4,7,10,12,10 |
| 1,2,4,8,12,16,19,16,19,9 | 1,4,7,10,12,10,12,5 |
| 1,2,4,8,12,16,19,17 | 1,4,7,10,12,11 |
| 1,2,4,8,12,16,19,19,9 | 1,4,7,10,12,12,5 |
| 1,2,4,8,12,16,19,23 | 1,4,7,10,12,15 |
| 1,2,4,8,12,16,19,24,29,34 | 1,4,7,10,12,16,20,24 |
| 1,2,4,8,12,16,20 | 1,4,7,10,13 |
| 1,2,4,8,12,16,20,24 | 1,4,7,10,13,16 |
| 1,2,4,8,12,16,20,24,28 | 1,4,7,10,13,16,19 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,13 | 1,4,8 |
| 1,2,4,8,13,8 | 1,4,8,4 |
| 1,2,4,8,13,8,12 | 1,4,8,4,7 |
| 1,2,4,8,13,8,12,16 | 1,4,8,4,7,10 |
| 1,2,4,8,13,8,13 | 1,4,8,4,8 |
| 1,2,4,8,13,8,13,8,13 | 1,4,8,4,8,4,8 |
| 1,2,4,8,13,9 | 1,4,8,5 |
| 1,2,4,8,13,9,11,15,20 | 1,4,8,5,8,12 |
| 1,2,4,8,13,10 | 1,4,8,6 |
| 1,2,4,8,13,11 | 1,4,8,6,9 |
| 1,2,4,8,13,11,16,19 | 1,4,8,6,10,15 |
| 1,2,4,8,13,12 | 1,4,8,7 |
| 1,2,4,8,13,12,16 | 1,4,8,7,10 |
| 1,2,4,8,13,12,17 | 1,4,8,7,11 |
| 1,2,4,8,13,12,17,16,21 | 1,4,8,7,11,10,14 |
| 1,2,4,8,13,13 | 1,4,8,8 |
| 1,2,4,8,13,13,13 | 1,4,8,8,8 |
| 1,2,4,8,13,14 | 1,4,8,9 |
| 1,2,4,8,13,15 | 1,4,8,10 |
| 1,2,4,8,13,16 | 1,4,8,10,3,7,12,15 |
| 1,2,4,8,13,17 | 1,4,8,11 |
| 1,2,4,8,13,18 | 1,4,8,12 |
| 1,2,4,8,13,18,21,9 | 1,4,8,12,14,5 |
| 1,2,4,8,13,18,22 | 1,4,8,12,15 |
| 1,2,4,8,13,18,23 | 1,4,8,12,16 |
| 1,2,4,8,13,19 | 1,4,8,13 |
| 1,2,4,8,13,19,26 | 1,4,8,13,19 |
| 1,2,4,8,13,20 | 1,4,8,14 |
| 1,2,4,8,13,20,26,20 | 1,4,8,14,19,14 |
| 1,2,4,8,13,20,27 | 1,4,8,14,20 |
| 1,2,4,8,13,20,27,33,21 | 1,4,8,14,20,25,15 |
| 1,2,4,8,13,20,28 | 1,4,8,14,21 |
| 1,2,4,8,13,20,28,38 | 1,4,8,14,21,30 |
| 1,2,4,8,13,20,28,38,49,62 | 1,4,8,14,21,30,40,52 |
| 1,2,4,8,14 | 1,4,9 |
| 1,2,4,8,14,8,14 | 1,4,9,4,9 |
| 1,2,4,8,14,11,<br>8,13,20,29,23,9 | 1,4,9,6,4,8,14,22 |
| 1,2,4,8,14,11,<br>8,13,20,29,23,9 | 1,4,9,6,4,8,14,22,16,5 |
| 1,2,4,8,14,12 | 1,4,9,7 |
| 1,2,4,8,14,12,17 | 1,4,9,7,11 |
| 1,2,4,8,14,12,17,24 | 1,4,9,7,11,17 |
| 1,2,4,8,14,12,17,24,33 | 1,4,9,7,11,17,25 |
| 1,2,4,8,14,12,18 | 1,4,9,7,12 |
| 1,2,4,8,14,12,18,9 | 1,4,9,7,12,5 |
| 1,2,4,8,14,12,18,12 | 1,4,9,7,12,7 |
| 1,2,4,8,14,12,18,12,18 | 1,4,9,7,12,7,12 |
| 1,2,4,8,14,12,18,13 | 1,4,9,7,12,8 |
| 1,2,4,8,14,12,18,15,9 | 1,4,9,7,12,9,5 |
| 1,2,4,8,14,12,18,16 | 1,4,9,7,12,10 |
| 1,2,4,8,14,12,18,16,22 | 1,4,9,7,12,10,15 |
| 1,2,4,8,14,12,<br>18,16,22,20,26 | 1,4,9,7,12,10,15,13,18 |
| 1,2,4,8,14,13 | 1,4,9,8 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,14,13,18 | 1,4,9,8,12 |
| 1,2,4,8,14,13,18,23 | 1,4,9,8,12,16 |
| 1,2,4,8,14,13,20 | 1,4,9,8,14 |
| 1,2,4,8,14,13,20,24 | 1,4,9,8,14,17 |
| 1,2,4,8,14,13,20,25 | 1,4,9,8,14,18 |
| 1,2,4,8,14,13,20,26,20 | 1,4,9,8,14,19,14 |
| 1,2,4,8,14,13,20,29 | 1,4,9,8,14,22 |
| 1,2,4,8,14,13,20,29,20 | 1,4,9,8,14,22,14 |
| 1,2,4,8,14,14 | 1,4,9,9 |
| 1,2,4,8,14,14,12,15,9 | 1,4,9,9,7,9,5 |
| 1,2,4,8,14,14,14 | 1,4,9,9,9 |
| 1,2,4,8,14,15 | 1,4,9,10 |
| 1,2,4,8,14,15,13 | 1,4,9,10,8 |
| 1,2,4,8,14,15,14,13 | 1,4,9,10,9,8 |
| 1,2,4,8,14,15,14,15 | 1,4,9,10,9,10 |
| 1,2,4,8,14,15,15 | 1,4,9,10,10 |
| 1,2,4,8,14,16 | 1,4,9,11 |
| 1,2,4,8,14,17 | 1,4,9,11,3,7,13,16 |
| 1,2,4,8,14,17,8 | 1,4,9,11,4 |
| 1,2,4,8,14,17,8,12,16 | 1,4,9,11,4,7,10 |
| 1,2,4,8,14,17,8,12,16,16 | 1,4,9,11,4,7,10,10 |
| 1,2,4,8,14,17,8,14 | 1,4,9,11,4,9 |
| 1,2,4,8,14,17,8,14,14 | 1,4,9,11,4,9,9 |
| 1,2,4,8,14,17,8,14,15 | 1,4,9,11,4,9,10 |
| 1,2,4,8,14,17,9 | 1,4,9,11,5 |
| 1,2,4,8,14,17,11,15 | 1,4,9,11,6,9 |
| 1,2,4,8,14,17,13 | 1,4,9,11,8 |
| 1,2,4,8,14,17,14 | 1,4,9,11,9 |
| 1,2,4,8,14,17,21 | 1,4,9,11,14 |
| 1,2,4,8,14,17,22,29,33 | 1,4,9,11,15,21,24 |
| 1,2,4,8,14,18 | 1,4,9,12 |
| 1,2,4,8,14,18,12,18,22 | 1,4,9,12,7,12,15 |
| 1,2,4,8,14,18,13 | 1,4,9,12,8 |
| 1,2,4,8,14,18,14 | 1,4,9,12,9 |
| 1,2,4,8,14,18,14,18 | 1,4,9,12,9,12 |
| 1,2,4,8,14,18,18 | 1,4,9,12,12 |
| 1,2,4,8,14,18,22 | 1,4,9,12,15 |
| 1,2,4,8,14,18,23 | 1,4,9,12,16 |
| 1,2,4,8,14,18,23,30 | 1,4,9,12,16,22 |
| 1,2,4,8,14,18,24 | 1,4,9,12,17 |
| 1,2,4,8,14,18,24,14 | 1,4,9,12,17,9 |
| 1,2,4,8,14,18,24,14,18 | 1,4,9,12,17,9,12 |
| 1,2,4,8,14,18,24,14,18,24 | 1,4,9,12,17,9,12,17 |
| 1,2,4,8,14,18,24,18 | 1,4,9,12,17,12 |
| 1,2,4,8,14,18,24,18,24 | 1,4,9,12,17,12,17 |
| 1,2,4,8,14,18,24,22,28 | 1,4,9,12,17,15,20 |
| 1,2,4,8,14,18,24,23 | 1,4,9,12,17,16 |
| 1,2,4,8,14,18,24,24 | 1,4,9,12,17,17 |
| 1,2,4,8,14,18,24,25 | 1,4,9,12,17,18 |
| 1,2,4,8,14,18,24,27,9 | 1,4,9,12,17,19,5 |
| 1,2,4,8,14,18,24,28 | 1,4,9,12,17,20 |
| 1,2,4,8,14,18,24,28,33 | 1,4,9,12,17,20,24 |
| 1,2,4,8,14,18,24,28,34 | 1,4,9,12,17,20,25 |
| 1,2,4,8,14,19 | 1,4,9,13 |
| 1,2,4,8,14,19,8,14,19 | 1,4,9,13,4,9,13 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,14,19,11,8,14,19 | 1,4,9,13,6,4,9,13 |
| 1,2,4,8,14,19,11,15 | 1,4,9,13,6,9 |
| 1,2,4,8,14,19,11,16 | 1,4,9,13,6,10 |
| 1,2,4,8,14,19,11,16,23 | 1,4,9,13,6,10,16 |
| 1,2,4,8,14,19,11,16,23,24 | 1,4,9,13,6,10,16,17 |
| 1,2,4,8,14,19,<br>11,16,23,27,17 | 1,4,9,13,6,10,16,19,11 |
| 1,2,4,8,14,19,11,<br>16,23,28,35,39,17 | 1,4,9,13,6,10,<br>16,20,26,29,11 |
| 1,2,4,8,14,19,11,16,23,29 | 1,4,9,13,6,10,16,21 |
| 1,2,4,8,14,19,11,<br>16,23,29,20,26,34,41 | 1,4,9,13,6,10,16,<br>21,13,18,25,31 |
| 1,2,4,8,14,19,12 | 1,4,9,13,7 |
| 1,2,4,8,14,19,13 | 1,4,9,13,8 |
| 1,2,4,8,14,19,14,13 | 1,4,9,13,9,8 |
| 1,2,4,8,14,19,14,15 | 1,4,9,13,9,10 |
| 1,2,4,8,14,19,14,17,9 | 1,4,9,13,9,11,5 |
| 1,2,4,8,14,19,14,18,24,29 | 1,4,9,13,9,12,17,21 |
| 1,2,4,8,14,19,14,<br>18,24,29,24,27,9 | 1,4,9,13,9,12,<br>17,21,17,19,5 |
| 1,2,4,8,14,19,14,<br>18,24,29,24,28,34,39 | 1,4,9,13,9,12,17,<br>21,17,20,25,29 |
| 1,2,4,8,14,19,14,19 | 1,4,9,13,9,13 |
| 1,2,4,8,14,19,14,19,11,15 | 1,4,9,13,9,13,6,9 |
| 1,2,4,8,14,19,14,<br>19,11,16,23,29 | 1,4,9,13,9,13,6,10,16,21 |
| 1,2,4,8,14,19,14,<br>19,11,16,23,29,23,29 | 1,4,9,13,9,13,<br>6,10,16,21,16,21 |
| 1,2,4,8,14,19,<br>14,19,14,18,9 | 1,4,9,13,9,13,9,11,5 |
| 1,2,4,8,14,19,14,19,14,19 | 1,4,9,13,9,13,9,13 |
| 1,2,4,8,14,19,15 | 1,4,9,13,10 |
| 1,2,4,8,14,19,17,9 | 1,4,9,13,11,5 |
| 1,2,4,8,14,19,17,14,19 | 1,4,9,13,11,9,13 |
| 1,2,4,8,14,19,18 | 1,4,9,13,12 |
| 1,2,4,8,14,19,18,24,25 | 1,4,9,13,12,17,18 |
| 1,2,4,8,14,19,18,24,29 | 1,4,9,13,12,17,21 |
| 1,2,4,8,14,19,<br>18,24,29,32,9 | 1,4,9,13,12,17,21,23,5 |
| 1,2,4,8,14,19,19 | 1,4,9,13,13 |
| 1,2,4,8,14,19,22,9 | 1,4,9,13,15,5 |
| 1,2,4,8,14,19,<br>22,14,19,22,9 | 1,4,9,13,15,9,13,15,5 |
| 1,2,4,8,14,19,22,19,22,9 | 1,4,9,13,15,13,15,5 |
| 1,2,4,8,14,19,22,25,9 | 1,4,9,13,15,17,5 |
| 1,2,4,8,14,19,22,26 | 1,4,9,13,15,18 |
| 1,2,4,8,14,19,23,29,34 | 1,4,9,13,16,21,25 |
| 1,2,4,8,14,19,<br>23,29,34,37,9 | 1,4,9,13,16,21,25,27,5 |
| 1,2,4,8,14,19,24 | 1,4,9,13,17 |
| 1,2,4,8,14,19,25 | 1,4,9,13,18 |
| 1,2,4,8,14,19,25,31,37 | 1,4,9,13,18,23,28 |
| 1,2,4,8,14,19,26 | 1,4,9,13,19 |
| 1,2,4,8,14,19,26,27 | 1,4,9,13,19,20 |
| 1,2,4,8,14,19,26,29,33 | 1,4,9,13,19,21,24 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,14,19,26,29,34 | 1,4,9,13,19,21,25 |
| 1,2,4,8,14,19,26,29,34,39 | 1,4,9,13,19,21,25,29 |
| 1,2,4,8,14,19,26,29,34,41 | 1,4,9,13,19,21,25,31 |
| 1,2,4,8,14,19, 26,29,34,41,47 | 1,4,9,13,19,21,25,31,36 |
| 1,2,4,8,14,19,26,30 | 1,4,9,13,19,22 |
| 1,2,4,8,14,19,26,30,36,41 | 1,4,9,13,19,22,27,31 |
| 1,2,4,8,14,19,26,31 | 1,4,9,13,19,23 |
| 1,2,4,8,14,19,26,32,26 | 1,4,9,13,19,24,19 |
| 1,2,4,8,14,19,26,33 | 1,4,9,13,19,27 |
| 1,2,4,8,14,19,26,33,34 | 1,4,9,13,19,27,28 |
| 1,2,4,8,14,19,26,33,39 | 1,4,9,13,19,27,32,20 |
| 1,2,4,8,14,19,26,33,41 | 1,4,9,13,19,27,34 |
| 1,2,4,8,14,20 | 1,4,9,14 |
| 1,2,4,8,14,20,8,12 | 1,4,9,14,4,7 |
| 1,2,4,8,14,20,8,14 | 1,4,9,14,4,9 |
| 1,2,4,8,14,20,8,14,19 | 1,4,9,14,4,9,13 |
| 1,2,4,8,14,20,8,14,20 | 1,4,9,14,4,9,14 |
| 1,2,4,8,14,20,13 | 1,4,9,14,8 |
| 1,2,4,8,14,20,13,18,23 | 1,4,9,14,8,12,16 |
| 1,2,4,8,14,20,14 | 1,4,9,14,9 |
| 1,2,4,8,14,20,14,15 | 1,4,9,14,9,10 |
| 1,2,4,8,14,20,14,17,9 | 1,4,9,14,9,11,5 |
| 1,2,4,8,14,20,14,18 | 1,4,9,14,9,12 |
| 1,2,4,8,14,20,14,18,24 | 1,4,9,14,9,12,17 |
| 1,2,4,8,14,20,14,18,24,29 | 1,4,9,14,9,12,17,21 |
| 1,2,4,8,14,20, 14,18,24,29,35 | 1,4,9,14,9,12,17,21,26 |
| 1,2,4,8,14,20, 14,18,24,29,36 | 1,4,9,14,9,12,17,21,27 |
| 1,2,4,8,14,20,14,18,24,30 | 1,4,9,14,9,12,17,22 |
| 1,2,4,8,14,20, 14,18,24,30,24,27,9 | 1,4,9,14,9, 12,17,22,17,19,5 |
| 1,2,4,8,14,20,14,19 | 1,4,9,14,9,13 |
| 1,2,4,8,14,20, 14,19,25,31,37 | 1,4,9,14,9,13,18,23,28 |
| 1,2,4,8,14,20,14,20 | 1,4,9,14,9,14 |
| 1,2,4,8,14,20, 14,20,13,18,23 | 1,4,9,14,9,14,8,12,16 |
| 1,2,4,8,14,20, 14,20,14,17,9 | 1,4,9,14,9,14,9,11,5 |
| 1,2,4,8,14,20, 14,20,14,18,24,29 | 1,4,9,14,9,14,9,12,17,21 |
| 1,2,4,8,14,20,14, 20,14,18,24,30,24,29 | 1,4,9,14,9,14,9, 12,17,22,17,21 |
| 1,2,4,8,14,20,14,20,14,19 | 1,4,9,14,9,14,9,13 |
| 1,2,4,8,14,20,14, 20,14,19,25,31,37 | 1,4,9,14,9,14, 9,13,18,23,28 |
| 1,2,4,8,14,20, 14,20,14,20,14,19 | 1,4,9,14,9,14,9,14,9,13 |
| 1,2,4,8,14,20,14, 20,14,20,14,19,25,31,37 | 1,4,9,14,9,14,9, 14,9,13,18,23,28 |
| 1,2,4,8,14,20,15 | 1,4,9,14,10 |
| 1,2,4,8,14,20,17,9 | 1,4,9,14,11,5 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,14,20,<br>17,22,29,35 | 1,4,9,14,11,15,21,26 |
| 1,2,4,8,14,20,<br>17,22,29,36,29,35 | 1,4,9,14,11,<br>15,21,27,21,26 |
| 1,2,4,8,14,20,<br>17,22,29,36,30 | 1,4,9,14,11,15,21,27,22 |
| 1,2,4,8,14,20,18 | 1,4,9,14,12 |
| 1,2,4,8,14,20,18,24,30 | 1,4,9,14,12,17,22 |
| 1,2,4,8,14,20,19 | 1,4,9,14,13 |
| 1,2,4,8,14,20,19,24,29 | 1,4,9,14,13,17,21 |
| 1,2,4,8,14,20,19,25 | 1,4,9,14,13,19 |
| 1,2,4,8,14,20,20 | 1,4,9,14,14 |
| 1,2,4,8,14,20,20,14,19 | 1,4,9,14,14,9,13 |
| 1,2,4,8,14,20,<br>20,14,20,14,19 | 1,4,9,14,14,9,14,13 |
| 1,2,4,8,14,20,20,14,20,19 | 1,4,9,14,14,13 |
| 1,2,4,8,14,20,25 | 1,4,9,14,18 |
| 1,2,4,8,14,20,26,19 | 1,4,9,14,19,13 |
| 1,2,4,8,14,20,26,20,25 | 1,4,9,14,19,14,18 |
| 1,2,4,8,14,20,26,25 | 1,4,9,14,19,18 |
| 1,2,4,8,14,20,26,31 | 1,4,9,14,19,23 |
| 1,2,4,8,14,21 | 1,4,9,15 |
| 1,2,4,8,14,21,28,35 | 1,4,9,15,21,27 |
| 1,2,4,8,14,21,30 | 1,4,9,15,23 |
| 1,2,4,8,14,21,30,41,51 | 1,4,9,15,23,33,42 |
| 1,2,4,8,14,21,30,41,52,42 | 1,4,9,15,23,33,43,34 |
| 1,2,4,8,14,21,30,41,53 | 1,4,9,15,23,33,44 |
| 1,2,4,8,14,22 | 1,4,9,16 |
| 1,2,4,8,14,22,14 | 1,4,9,16,9 |
| 1,2,4,8,14,22,14,20 | 1,4,9,16,9,14 |
| 1,2,4,8,14,22,14,21 | 1,4,9,16,9,15 |
| 1,2,4,8,14,22,14,22 | 1,4,9,16,9,16 |
| 1,2,4,8,14,22,20 | 1,4,9,16,14 |
| 1,2,4,8,14,22,20,28 | 1,4,9,16,14,21 |
| 1,2,4,8,14,22,21 | 1,4,9,16,15 |
| 1,2,4,8,14,22,22 | 1,4,9,16,16 |
| 1,2,4,8,14,22,23 | 1,4,9,16,17 |
| 1,2,4,8,14,22,29 | 1,4,9,16,22 |
| 1,2,4,8,14,22,32 | 1,4,9,16,25 |
| 1,2,4,8,14,22,32,44 | 1,4,9,16,25,36 |
| 1,2,4,8,15 | 1,4,10 |
| 1,2,4,8,15,22 | 1,4,10,16 |
| 1,2,4,8,15,22,23 | 1,4,10,16,17 |
| 1,2,4,8,15,22,24 | 1,4,10,16,18 |
| 1,2,4,8,15,22,25,9 | 1,4,10,16,18,5 |
| 1,2,4,8,15,22,26 | 1,4,10,16,19 |
| 1,2,4,8,15,22,27 | 1,4,10,16,20 |
| 1,2,4,8,15,22,27,15 | 1,4,10,16,20,10 |
| 1,2,4,8,15,22,<br>27,15,22,27,15 | 1,4,10,16,20,10,16,20,10 |
| 1,2,4,8,15,22,27,16 | 1,4,10,16,20,11 |
| 1,2,4,8,15,22,28 | 1,4,10,16,21 |
| 1,2,4,8,15,22,29 | 1,4,10,16,22 |
| 1,2,4,8,15,22,29,29 | 1,4,10,16,22,22 |
| 1,2,4,8,15,22,29,36 | 1,4,10,16,22,28 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,15,23 | 1,4,10,17 |
| 1,2,4,8,15,24 | 1,4,10,18 |
| 1,2,4,8,15,24,35 | 1,4,10,18,28 |
| 1,2,4,8,15,24,36 | 1,4,10,18,29 |
| 1,2,4,8,15,25 | 1,4,10,19 |
| 1,2,4,8,15,25,26 | 1,4,10,19,20 |
| 1,2,4,8,15,25,38 | 1,4,10,19,31 |
| 1,2,4,8,15,26 | 1,4,10,20 |
| 1,2,4,8,15,26,42 | 1,4,10,20,35 |
| 1,2,4,8,15,26,42,64 | 1,4,10,20,35,56 |
| 1,2,4,8,16 | 1,5 |
| 1,2,4,8,16,8 | 1,5,4 |
| 1,2,4,8,16,8,14 | 1,5,4,9 |
| 1,2,4,8,16,8,14,22 | 1,5,4,9,16 |
| 1,2,4,8,16,8,15 | 1,5,4,10 |
| 1,2,4,8,16,8,15,26 | 1,5,4,10,20 |
| 1,2,4,8,16,8,16 | 1,5,4,11 |
| 1,2,4,8,16,14 | 1,5,4,11,9 |
| 1,2,4,8,16,14,24,22 | 1,5,4,11,9,18,16 |
| 1,2,4,8,16,15 | 1,5,4,11,10 |
| 1,2,4,8,16,15,27 | 1,5,4,11,10,21 |
| 1,2,4,8,16,15,27,26 | 1,5,4,11,10,21,20 |
| 1,2,4,8,16,16 | 1,5,5 |
| 1,2,4,8,16,16,16 | 1,5,5,5 |
| 1,2,4,8,16,17 | 1,5,6 |
| 1,2,4,8,16,18 | 1,5,7 |
| 1,2,4,8,16,18,7,12,21 | 1,5,7,3,8 |
| 1,2,4,8,16,18,7,12,21,23 | 1,5,7,3,8,10 |
| 1,2,4,8,16,18,8 | 1,5,7,3,8,10,7 |
| 1,2,4,8,16,18,8,16,18 | 1,5,7,3,8,10,15,17 |
| 1,2,4,8,16,18,16 | 1,5,7,3,8,10,8 |
| 1,2,4,8,16,18,22 | 1,5,7,3,8,11 |
| 1,2,4,8,16,19,8 | 1,5,7,4 |
| 1,2,4,8,16,19,8,16,8 | 1,5,7,4,11,13 |
| 1,2,4,8,16,<br>19,8,16,19,8 | 1,5,7,4,11,13,4 |
| 1,2,4,8,16,19,9 | 1,5,7,4,11,13,5 |
| 1,2,4,8,16,19,16 | 1,5,7,4,11,13,11 |
| 1,2,4,8,16,19,16,19,16 | 1,5,7,4,11,13,11,13,11 |
| 1,2,4,8,16,19,17 | 1,5,7,4,11,13,12 |
| 1,2,4,8,16,19,18 | 1,5,7,4,11,13,13 |
| 1,2,4,8,16,19,19,8 | 1,5,7,4,11,13,13,4 |
| 1,2,4,8,16,19,19,9 | 1,5,7,4,11,13,13,11 |
| 1,2,4,8,16,19,23 | 1,5,7,4,11,13,16 |
| 1,2,4,8,16,19,24 | 1,5,7,4,11,13,17 |
| 1,2,4,8,16,19,24,37 | 1,5,7,4,11,13,18 |
| 1,2,4,8,16,20 | 1,5,7,4,11,14 |
| 1,2,4,8,16,20,26 | 1,5,7,4,11,14,19 |
| 1,2,4,8,16,20,27 | 1,5,7,4,11,14,20 |
| 1,2,4,8,16,20,28 | 1,5,7,4,11,14,21 |
| 1,2,4,8,16,20,28,32,40 | 1,5,7,4,11,14,21,24,31 |
| 1,2,4,8,16,21 | 1,5,7,4,11,15 |
| 1,2,4,8,16,21,15 | 1,5,7,4,11,15,10 |
| 1,2,4,8,16,21,16 | 1,5,7,5 |
| 1,2,4,8,16,21,16,21,16 | 1,5,7,5,7,5 |

| Y sequence | $0 - \mathrm{Y}$ sequence |
|---|---|
| 1,2,4,8,16,21,21,16 | 1,5,7,7,5 |
| 1,2,4,8,16,21,27 | 1,5,7,10 |
| 1,2,4,8,16,22 | 1,5,8 |
| 1,2,4,8,16,22,8,16 | 1,5,8,4,11 |
| 1,2,4,8,16,22,8,16,20 | 1,5,8,4,11,14 |
| 1,2,4,8,16,<br>22,8,16,20,28 | 1,5,8,4,11,14,21 |
| 1,2,4,8,16,22,8,16,21 | 1,5,8,4,11,15 |
| 1,2,4,8,16,22,8,16,22 | 1,5,8,4,11,16 |
| 1,2,4,8,16,22,14,24,32 | 1,5,8,4,11,16,9,18,25 |
| 1,2,4,8,16,22,15,27 | 1,5,8,4,11,16,10,21,29 |
| 1,2,4,8,16,22,16 | 1,5,8,4,11,16,11 |
| 1,2,4,8,16,22,22,16 | 1,5,8,4,11,16,16,11 |
| 1,2,4,8,16,22,23 | 1,5,8,4,11,16,17 |
| 1,2,4,8,16,22,28,16 | 1,5,8,4,11,16,21,11 |
| 1,2,4,8,16,22,30 | 1,5,8,4,11,16,23 |
| 1,2,4,8,16,22,31 | 1,5,8,4,11,16,24 |
| 1,2,4,8,16,22,32 | 1,5,8,4,11,16,25 |
| 1,2,4,8,16,22,32,40 | 1,5,8,4,11,16,25,36,51 |
| 1,2,4,8,16,23 | 1,5,8,4,11,17 |
| 1,2,4,8,16,23,14,24,23 | 1,5,8,4,11,17,9,18,26 |
| 1,2,4,8,16,23,15 | 1,5,8,4,11,17,10 |
| 1,2,4,8,16,23,15,27,38 | 1,5,8,4,11,17,10,21,31 |
| 1,2,4,8,16,23,16 | 1,5,8,5 |
| 1,2,4,8,16,23,16,20 | 1,5,8,5,8 |
| 1,2,4,8,16,23,16,23,16 | 1,5,8,5,8,5 |
| 1,2,4,8,16,23,20 | 1,5,8,8 |
| 1,2,4,8,16,23,23,16 | 1,5,8,8,5 |
| 1,2,4,8,16,23,30,16 | 1,5,8,11,5 |
| 1,2,4,8,16,23,32 | 1,5,8,13 |
| 1,2,4,8,16,23,33 | 1,5,8,13,4,11,17,26 |
| 1,2,4,8,16,23,33,16 | 1,5,8,13,5 |
| 1,2,4,8,16,23,33,46,16 | 1,5,8,13,20,5 |
| 1,2,4,8,16,23,34 | 1,5,8,14 |
| 1,2,4,8,16,23,34,50 | 1,5,8,14,24 |
| 1,2,4,8,16,23,35 | 1,5,8,15 |
| 1,2,4,8,16,23,35,35 | 1,5,8,15,15 |
| 1,2,4,8,16,23,35,42,16 | 1,5,8,15,18,5 |
| 1,2,4,8,16,23,35,45,16 | 1,5,8,15,20,5 |
| 1,2,4,8,16,23,35,46 | 1,5,8,15,21 |
| 1,2,4,8,16,23,35,46,63 | 1,5,8,15,21,32 |
| 1,2,4,8,16,24 | 1,5,9 |
| 1,2,4,8,16,24,16,24 | 1,5,9,5,9 |
| 1,2,4,8,16,24,20 | 1,5,9,8 |
| 1,2,4,8,16,24,23,16,24 | 1,5,9,8,5,9 |
| 1,2,4,8,16,24,23,30,16,24 | 1,5,9,8,11,5,9 |
| 1,2,4,8,16,24,23,33,16,24 | 1,5,9,8,13,5,9 |
| 1,2,4,8,16,24,23,34 | 1,5,9,8,14 |
| 1,2,4,8,16,24,23,35 | 1,5,9,8,15 |
| 1,2,4,8,16,24,23,35,47 | 1,5,9,8,15,22 |
| 1,2,4,8,16,24,24 | 1,5,9,9 |
| 1,2,4,8,16,24,25 | 1,5,9,10 |
| 1,2,4,8,16,24,26 | 1,5,9,11 |
| 1,2,4,8,16,24,27,16 | 1,5,9,11,5 |
| 1,2,4,8,16,24,27,16,24 | 1,5,9,11,5,9 |

| Y sequence | 0 − Y sequence |
|---|---|
| 1,2,4,8,16,24,17,16,24,26 | 1,5,9,11,5,9,11 |
| 1,2,4,8,16,24,27,17 | 1,5,9,11,6 |
| 1,2,4,8,16,24,27,18 | 1,5,9,11,7 |
| 1,2,4,8,16,24,27,19 | 1,5,9,11,8 |
| 1,2,4,8,16,24, 27,23,16,24,27,18 | 1,5,9,11,8,5,9,11 |
| 1,2,4,8,16,24,27,23,17 | 1,5,9,11,8,6 |
| 1,2,4,8,16,24, 27,23,33,16,24,26 | 1,5,9,11,8,13,5,9,11 |
| 1,2,4,8,16,24,27,23,34 | 1,5,9,11,8,14 |
| 1,2,4,8,16,24,27,23,35 | 1,5,9,11,8,15 |
| 1,2,4,8,16,24,27, 23,35,47,50,36 | 1,5,9,11,8,15,22,26,16 |
| 1,2,4,8,16,24,27,24 | 1,5,9,11,9 |
| 1,2,4,8,16,24,27,31 | 1,5,9,11,14 |
| 1,2,4,8,16,24,28 | 1,5,9,12 |
| 1,2,4,8,16,24,31 | 1,5,9,12,4,11,18,24 |
| 1,2,4,8,16,24,31,16 | 1,5,9,12,5 |
| 1,2,4,8,16,24,31,16,24 | 1,5,9,12,5,9 |
| 1,2,4,8,16, 24,31,16,24,28 | 1,5,9,12,5,9,12 |
| 1,2,4,8,16,24,31,17 | 1,5,9,12,6 |
| 1,2,4,8,16,24,31,18 | 1,5,9,12,7 |
| 1,2,4,8,16,24,31,20 | 1,5,9,12,8 |
| 1,2,4,8,16,24,31,23,16 | 1,5,9,12,8,5 |
| 1,2,4,8,16,24, 31,23,16,24,31,17 | 1,5,9,12,8,5,9,12,6 |
| 1,2,4,8,16,24,31,23,31 | 1,5,9,12,8,12 |
| 1,2,4,8,16,24, 31,23,33,16,24,31,17 | 1,5,9,12,8,13,5,9,12,6 |
| 1,2,4,8,16,24,31,23,34 | 1,5,9,12,8,14 |
| 1,2,4,8,16,24,31,24 | 1,5,9,12,9 |
| 1,2,4,8,16,24,32 | 1,5,9,13 |
| 1,2,4,8,16,26 | 1,5,11 |
| 1,2,4,8,16,27 | 1,5,11,4,11,21 |
| 1,2,4,8,16,27,16 | 1,5,11,5 |
| 1,2,4,8,16,28 | 1,5,12 |
| 1,2,4,8,16,28,44 | 1,5,12,22 |
| 1,2,4,8,16,28,44,64 | 1,5,12,22,35 |
| 1,2,4,8,16,29 | 1,5,13 |
| 1,2,4,8,16,31 | 1,5,15 |
| 1,2,4,8,16,31,57 | 1,5,15,35 |
| 1,2,4,8,16,32 | 1,6 |
| 1,2,4,8,16,32,48 | 1,6,11 |
| 1,2,4,8,16,32,56 | 1,6,15 |
| 1,2,4,8,16,32,60 | 1,6,18 |
| 1,2,4,8,16,32,61 | 1,6,19 |
| 1,2,4,8,16,32,62 | 1,6,20 |
| 1,2,4,8,16,32,63 | 1,6,21 |
| 1,2,4,8,16,32,64 | 1,7 |
| 1,2,4,8,16,32,64,128 | 1,8 |
| 1,2,4,8,16,32,64,128,256 | 1,9 |

## A.22 Y sequence (post-SHO)

| 1 − Y sequence |
|---|
| 1, 3 |
| 1, 3, 2, 5 |
| 1, 3, 2, 5, 4 |
| 1, 3, 2, 5, 4, 9 |
| 1, 3, 2, 5, 4, 9, 8 |
| 1, 3, 2, 5, 4, 9, 8, 17 |
| 1, 3, 2, 5, 4, 9, 8, 17, 15 |
| 1, 3, 2, 5, 4, 9, 8, 17, 16 |
| 1, 3, 3 |
| 1, 3, 4 |
| 1, 3, 4, 2, 5, 6, 5 |
| 1, 3, 4, 2, 5, 6, 9 |
| 1, 3, 4, 2, 5, 7 |
| 1, 3, 4, 2, 5, 7, 4, 9, 11 |
| 1, 3, 4, 2, 5, 7, 4, 9, 11, 7, 13, 15 |
| 1, 3, 4, 2, 5, 7, 4, 9, 11, 8 |
| 1, 3, 4, 2, 5, 7, 4, 9, 11, 9 |
| 1, 3, 4, 2, 5, 7, 4, 9, 12 |
| 1, 3, 4, 2, 5, 7, 4, 9, 12, 7, 13, 17 |
| 1, 3, 4, 2, 5, 7, 4, 9, 12, 8 |
| 1, 3, 4, 2, 5, 7, 4, 9, 12, 8, 17, 22, 16 |
| 1, 3, 4, 2, 5, 7, 4, 9, 12, 9 |
| 1, 3, 4, 2, 5, 7, 4, 9, 12, 9, 12 |
| 1, 3, 4, 2, 5, 7, 4, 9, 12, 11 |
| 1, 3, 4, 2, 5, 7, 4, 9, 12, 16 |
| 1, 3, 4, 2, 5, 7, 4, 9, 12, 18 |
| 1, 3, 4, 2, 5, 7, 4, 9, 12, 18, 19, 22 |
| 1, 3, 4, 2, 5, 7, 4, 9, 12, 18, 22 |
| 1, 3, 4, 2, 5, 7, 4, 9, 13 |
| 1, 3, 4, 2, 5, 7, 4, 9, 13, 8, 17, 20, 26, 31 |
| 1, 3, 4, 2, 5, 7, 4, 9, 13, 8, 17, 21 |
| 1, 3, 4, 2, 5, 7, 4, 9, 13, 8, 17, 21, 20, 26, 31 |
| 1, 3, 4, 2, 5, 7, 4, 9, 13, 8, 17, 21, 21 |
| 1, 3, 4, 2, 5, 7, 4, 9, 13, 8, 17, 21, 26 |
| 1, 3, 4, 2, 5, 7, 4, 9, 13, 8, 17, 21, 26, 34, 41 |
| 1, 3, 4, 2, 5, 7, 4, 9, 13, 8, 17, 21, 27 |
| 1, 3, 4, 2, 5, 7, 4, 9, 13, 8, 17, 21, 30 |
| 1, 3, 4, 2, 5, 7, 4, 9, 13, 8, 17, 22 |
| 1, 3, 4, 2, 5, 7, 4, 9, 13, 8, 17, 23 |
| 1, 3, 4, 2, 5, 7, 4, 9, 13, 8, 17, 25 |
| 1, 3, 4, 2, 5, 7, 5 |
| 1, 3, 4, 2, 5, 7, 5, 5 |
| 1, 3, 4, 2, 5, 7, 5, 7 |
| 1, 3, 4, 2, 5, 7, 5, 7, 5 |
| 1, 3, 4, 2, 5, 7, 7, 5 |
| 1, 3, 4, 2, 5, 7, 9, 5 |
| 1, 3, 4, 2, 5, 7, 10 |
| 1, 3, 4, 2, 5, 7, 10, 4 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 12 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 12, 9 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 12, 18 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13 |

| 1 − Y sequence |
|---|
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 9 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 17, 9 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 18 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 18, 8 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 18, 9 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 18, 24 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 18, 26 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 18, 26, 33 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 18, 26, 33, 26 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 18, 26, 33, 41 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 19 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 19, 8, 17, 25, 37 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 19, 9 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 19, 25, 9 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 19, 27, 9 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 20 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 20, 8, 17, 25, 39, 17 |
| 1, 3, 4, 2, 5, 7, 10, 4, 9, 13, 20, 8, 17, 25, 40 |
| 1, 3, 4, 2, 5, 7, 10, 5 |
| 1, 3, 4, 2, 5, 7, 10, 5, 5 |
| 1, 3, 4, 2, 5, 7, 10, 5, 7 |
| 1, 3, 4, 2, 5, 7, 10, 5, 7, 4, 9, 13, 20, 9, 13, 19 |
| 1, 3, 4, 2, 5, 7, 10, 5, 7, 5 |
| 1, 3, 4, 2, 5, 7, 10, 5, 7, 10 |
| 1, 3, 4, 2, 5, 7, 10, 5, 7, 10, 4, 9, 13, 20, 9, 13, 19, 9 |
| 1, 3, 4, 2, 5, 7, 10, 5, 7, 10, 4, 9, 13, 20, 9, 13, 20 |
| 1, 3, 4, 2, 5, 7, 10, 5, 7, 10, 5 |
| 1, 3, 4, 2, 5, 7, 10, 7, 5 |
| 1, 3, 4, 2, 5, 7, 10, 7, 10 |
| 1, 3, 4, 2, 5, 7, 10, 7, 10, 5 |
| 1, 3, 4, 2, 5, 7, 10, 9, 5 |
| 1, 3, 4, 2, 5, 7, 10, 9, 12, 5 |
| 1, 3, 4, 2, 5, 7, 10, 10, 5 |
| 1, 3, 4, 2, 5, 7, 10, 13, 5 |
| 1, 3, 4, 2, 5, 7, 10, 14 |
| 1, 3, 4, 2, 5, 7, 10, 14, 4, 9, 13, 20, 29, 9 |
| 1, 3, 4, 2, 5, 7, 10, 14, 4, 9, 13, 20, 30, 9 |
| 1, 3, 4, 2, 5, 7, 10, 14, 4, 9, 13, 20, 31 |
| 1, 3, 4, 2, 5, 7, 10, 14, 5 |
| 1, 3, 4, 2, 5, 7, 10, 14, 19, 5 |
| 1, 3, 4, 2, 5, 7, 11 |
| 1, 3, 4, 2, 5, 7, 11, 4, 9, 13 |
| 1, 3, 4, 2, 5, 7, 11, 4, 9, 13, 9 |
| 1, 3, 4, 2, 5, 7, 11, 4, 9, 13, 17, 9 |
| 1, 3, 4, 2, 5, 7, 11, 4, 9, 13, 19, 9 |
| 1, 3, 4, 2, 5, 7, 11, 4, 9, 13, 20, 9 |
| 1, 3, 4, 2, 5, 7, 11, 4, 9, 13, 21 |
| 1, 3, 4, 2, 5, 7, 11, 4, 9, 13, 21, 8 |
| 1, 3, 4, 2, 5, 7, 11, 5 |
| 1, 3, 4, 2, 5, 7, 11, 5, 7, 10, 5 |
| 1, 3, 4, 2, 5, 7, 11, 5, 7, 10, 14, 5 |
| 1, 3, 4, 2, 5, 7, 11, 5, 7, 11 |
| 1, 3, 4, 2, 5, 7, 11, 7, 11 |
| 1, 3, 4, 2, 5, 7, 11, 9, 13 |
| 1, 3, 4, 2, 5, 7, 11, 10, 5 |

| 1 − Y sequence |
|---|
| 1, 3, 4, 2, 5, 7, 11, 10, 15 |
| 1, 3, 4, 2, 5, 7, 11, 10, 15, 14, 20 |
| 1, 3, 4, 2, 5, 7, 11, 11 |
| 1, 3, 4, 2, 5, 7, 11, 13 |
| 1, 3, 4, 2, 5, 7, 11, 13, 5 |
| 1, 3, 4, 2, 5, 7, 11, 13, 11 |
| 1, 3, 4, 2, 5, 7, 11, 13, 17 |
| 1, 3, 4, 2, 5, 7, 11, 14, 5 |
| 1, 3, 4, 2, 5, 7, 11, 14, 11 |
| 1, 3, 4, 2, 5, 7, 11, 14, 16 |
| 1, 3, 4, 2, 5, 7, 11, 14, 18 |
| 1, 3, 4, 2, 5, 7, 11, 14, 19 |
| 1, 3, 4, 2, 5, 7, 11, 14, 19, 23, 29 |
| 1, 3, 4, 2, 5, 7, 11, 15 |
| 1, 3, 4, 2, 5, 7, 11, 15, 19 |
| 1, 3, 4, 2, 5, 7, 11, 16 |
| 1, 3, 4, 2, 5, 7, 11, 16, 5 |
| 1, 3, 4, 2, 5, 7, 11, 16, 22 |
| 1, 3, 4, 2, 5, 7, 11, 17 |
| 1, 3, 4, 2, 5, 7, 11, 17, 23 |
| 1, 3, 4, 2, 5, 7, 11, 17, 25 |
| 1, 3, 4, 2, 5, 7, 11, 18 |
| 1, 3, 4, 2, 5, 7, 11, 18, 29 |
| 1, 3, 4, 2, 5, 7, 11, 19 |
| 1, 3, 4, 2, 5, 7, 11, 19, 35 |
| 1, 3, 4, 2, 5, 7, 12 |
| 1, 3, 4, 2, 5, 7, 12, 5 |
| 1, 3, 4, 2, 5, 7, 12, 5, 7, 12 |
| 1, 3, 4, 2, 5, 7, 12, 7, 12 |
| 1, 3, 4, 2, 5, 7, 12, 10, 5 |
| 1, 3, 4, 2, 5, 7, 12, 11 |
| 1, 3, 4, 2, 5, 7, 12, 12 |
| 1, 3, 4, 2, 5, 7, 12, 15 |
| 1, 3, 4, 2, 5, 7, 12, 15, 5 |
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| 1, 3, 4, 2, 5, 7, 12, 15, 12 |
| 1, 3, 4, 2, 5, 7, 12, 16 |
| 1, 3, 4, 2, 5, 7, 12, 16, 12 |
| 1, 3, 4, 2, 5, 7, 12, 16, 21, 5 |
| 1, 3, 4, 2, 5, 7, 12, 16, 22, 5 |
| 1, 3, 4, 2, 5, 7, 12, 16, 23 |
| 1, 3, 4, 2, 5, 7, 12, 16, 23, 12 |
| 1, 3, 4, 2, 5, 7, 12, 16, 24 |
| 1, 3, 4, 2, 5, 7, 12, 16, 25 |
| 1, 3, 4, 2, 5, 7, 12, 16, 25, 33, 50 |
| 1, 3, 4, 2, 5, 8 |
| 1, 3, 4, 2, 5, 8, 5 |
| 1, 3, 4, 2, 5, 8, 5, 7, 12, 17 |
| 1, 3, 4, 2, 5, 8, 5, 7, 12, 17, 12 |
| 1, 3, 4, 2, 5, 8, 5, 8 |
| 1, 3, 4, 2, 5, 8, 7, 5 |
| 1, 3, 4, 2, 5, 8, 7, 5, 7, 12, 17, 16, 12 |
| 1, 3, 4, 2, 5, 8, 7, 5, 8 |
| 1, 3, 4, 2, 5, 8, 7, 10 |
| 1, 3, 4, 2, 5, 8, 7, 10, 5 |

| 1 − Y sequence |
|---|
| 1, 3, 4, 2, 5, 8, 7, 10, 5, 8 |
| 1, 3, 4, 2, 5, 8, 7, 11 |
| 1, 3, 4, 2, 5, 8, 7, 12 |
| 1, 3, 4, 2, 5, 8, 7, 12, 17 |
| 1, 3, 4, 2, 5, 8, 8 |
| 1, 3, 4, 2, 5, 8, 9 |
| 1, 3, 4, 2, 5, 8, 9, 11 |
| 1, 3, 4, 2, 5, 8, 10 |
| 1, 3, 4, 2, 5, 8, 10, 2 |
| 1, 3, 4, 2, 5, 8, 10, 2, 4 |
| 1, 3, 4, 2, 5, 8, 10, 2, 5 |
| 1, 3, 4, 2, 5, 8, 10, 2, 5, 8, 10 |
| 1, 3, 4, 2, 5, 8, 10, 4 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 15, 18, 21, 23 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 15, 18, 21, 23, 13, 21 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 15, 18, 21, 23, 14 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 6, 11, 16, 18 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 7 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 7, 12 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 7, 13 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 7, 13, 17, 19 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 7, 13, 19, 21 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 8 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 9 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 9, 14, 15, 18, 21, 23 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 9, 14, 16 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 9, 14, 16, 9, 14, 16 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 11 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 11, 9, 14, 16 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 11, 13 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 11, 14 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 11, 15 |
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| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 12 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 13 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 13, 9 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 13, 9, 14, 16 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 13, 9, 14, 16, 13, 9, 14, 16 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 13, 11 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 13, 12, 9, 14, 16, 13, 9, 14, 16 |
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| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 13, 17, 9, 14, 16 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 13, 18 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 13, 19 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 13, 19, 9, 14, 16 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 13, 20 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 13, 20, 9 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 13, 20, 9, 14, 16 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 13, 21 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 13, 22 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 13, 22, 31, 33 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 14 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 14, 16 |

| 1 − Y sequence |
|---|
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| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 19 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 20 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 21 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 16, 21, 26, 28 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 7, 13, 19, 23 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 8 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 8, 17, 26, 31, 15 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 8, 17, 26, 31, 16 |
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| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 9, 14, 16 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 10 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 11 |
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| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 12, 9, 14, 16 |
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| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 12, 18, 24, 26 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 12, 18, 24, 27, 17 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 12, 18, 24, 27, 18 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 12, 18, 24, 27, 18, 24, 26 |
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| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 12, 18, 24, 27, 20 |
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| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 12, 18, 24, 27, 24 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 12, 18, 24, 27, 24, 26 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 12, 18, 24, 27, 26 |
| 1, 3, 4, 2, 5, 8, 10, 4, 9, 14, 17, 12, 18, 24, 27, 29 |
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| 1, 3, 7, 13 |
| 1, 3, 7, 14 |
| 1, 3, 7, 15 |
| 1, 3, 8 |
| 1, 3, 8, 17 |
| 1, 3, 8, 18 |
| 1, 3, 8, 19 |
| 1, 3, 8, 20 |
| 1, 3, 9 |
| 1, 3, 9, 25 |
| 1, 3, 9, 26 |
| 1, 3, 9, 27 |
| 1, 3, 9, 27, 81 |
| 1, 4 |
| 1, 4, 4 |
| 1, 4, 5 |
| 1, 4, 5, 3, 10, 18 |
| 1, 4, 5, 4 |
| 1, 4, 6 |
| 1, 4, 7 |
| 1, 4, 10 |
| 1, 4, 12 |
| 1, 4, 13 |
| 1, 4, 14 |
| 1, 4, 15 |
| 1, 4, 16 |
| 1, 4, 16, 64 |
| 1, 5 |
| 1, 5, 24 |
| 1, 5, 25 |
| 1, 6 |
| 1, 7 |

## A.23 $\omega-$Y sequence (post-MHO)

| weak magma $\omega-$ Y sequence |
|---|
| 1, 4 |
| 1, 4, 3 |
| 1, 4, 3, 11 |

| weak magma $\omega - \mathrm{Y}$ sequence |
|---|
| 1, 4, 4 |
| 1, 4, 5 |
| 1, 4, 5, 3 |
| 1, 4, 5, 3, 4 |
| 1, 4, 5, 3, 5 |
| 1, 4, 5, 3, 6 |
| 1, 4, 5, 3, 7 |
| 1, 4, 5, 3, 8 |
| 1, 4, 5, 3, 9 |
| 1, 4, 5, 3, 10 |
| 1, 4, 5, 3, 11 |
| 1, 4, 5, 3, 11, 12 |
| 1, 4, 5, 3, 11, 13 |
| 1, 4, 5, 3, 11, 14 |
| 1, 4, 5, 3, 11, 15 |
| 1, 4, 5, 3, 11, 16 |
| 1, 4, 5, 3, 11, 17 |
| 1, 4, 5, 3, 11, 18 |
| 1, 4, 5, 3, 11, 19 |
| 1, 4, 5, 3, 11, 20 |
| 1, 4, 5, 3, 11, 21 |
| 1, 4, 5, 3, 11, 22 |
| 1, 4, 5, 3, 11, 23 |
| 1, 4, 5, 3, 11, 24 |
| 1, 4, 5, 3, 11, 25 |
| 1, 4, 5, 3, 11, 26 |
| 1, 4, 5, 3, 11, 27 |
| 1, 4, 5, 3, 11, 28 |
| 1, 4, 5, 3, 11, 29 |
| 1, 4, 5, 3, 11, 30 |
| 1, 4, 5, 3, 11, 31 |
| 1, 4, 5, 3, 11, 32 |
| 1, 4, 5, 3, 11, 33 |
| 1, 4, 5, 3, 11, 34 |
| 1, 4, 5, 3, 11, 35 |
| 1, 4, 5, 3, 11, 36 |
| 1, 4, 5, 4 |
| 1, 4, 5, 4, 3 |
| 1, 4, 5, 4, 4 |
| 1, 4, 5, 4, 5 |
| 1, 4, 5, 5 |
| 1, 4, 5, 5, 3 |
| 1, 4, 5, 5, 4 |
| 1, 4, 5, 5, 4, 5, 4 |
| 1, 4, 5, 5, 4, 5, 5, 4 |
| 1, 4, 5, 5, 5 |
| 1, 4, 5, 6 |
| 1, 4, 5, 6, 7 |
| 1, 4, 5, 7 |
| 1, 4, 5, 8 |
| 1, 4, 5, 8, 18 |
| 1, 4, 5, 9 |
| 1, 4, 5, 9, 13 |
| 1, 4, 5, 9, 14, 22 |
| 1, 4, 5, 9, 14, 23 |

| weak magma $\omega - Y$ sequence |
|---|
| 1, 4, 6 |
| 1, 4, 6, 3 |
| 1, 4, 6, 3, 11, 37, 10 |
| 1, 4, 6, 4 |
| 1, 4, 6, 4, 5 |
| 1, 4, 6, 4, 6 |
| 1, 4, 6, 4, 6, 4 |
| 1, 4, 6, 5 |
| 1, 4, 6, 5, 9 |
| 1, 4, 6, 5, 9, 15 |
| 1, 4, 6, 6 |
| 1, 4, 6, 6, 4 |
| 1, 4, 6, 6, 5 |
| 1, 4, 6, 6, 6 |
| 1, 4, 6, 7 |
| 1, 4, 6, 7, 11 |
| 1, 4, 6, 8 |
| 1, 4, 6, 9 |
| 1, 4, 6, 9, 16 |
| 1, 4, 6, 10 |
| 1, 4, 6, 11 |
| 1, 4, 6, 12 |
| 1, 4, 6, 13 |
| 1, 4, 6, 14 |
| 1, 4, 7 |
| 1, 4, 7, 8 |
| 1, 4, 7, 9 |
| 1, 4, 7, 10 |
| 1, 4, 7, 10, 13 |
| 1, 4, 8 |
| 1, 4, 8, 15 |
| 1, 4, 8, 16 |
| 1, 4, 9 |
| 1, 4, 9, 21 |
| 1, 4, 9, 22 |
| 1, 4, 9, 22, 64 |
| 1, 4, 10 |
| 1, 4, 10, 19 |
| 1, 4, 10, 22 |
| 1, 4, 10, 22, 46 |
| 1, 4, 11 |
| 1, 4, 11, 30 |
| 1, 4, 11, 30, 88, 280 |
| 1, 4, 12 |
| 1, 4, 12, 32 |
| 1, 4, 13 |
| 1, 4, 13, 41 |
| 1, 4, 13, 41, 134 |
| 1, 4, 13, 41, 134, 465 |
| 1, 4, 14 |
| 1, 4, 14, 46 |
| 1, 4, 14, 46, 146 |
| 1, 4, 15 |
| 1, 4, 15, 57 |
| 1, 4, 15, 57, 230 |

| weak magma $\omega - \mathrm{Y}$ sequence |
|---|
| 1, 4, 15, 57, 230, 1011 |
| 1, 4, 16 |
| 1, 4, 16, 66 |
| 1, 4, 16, 66, 286 |
| 1, 4, 17 |
| 1, 4, 17, 77 |
| 1, 4, 17, 77, 372 |
| 1, 4, 17, 77, 372, 1915 |
| 1, 4, 18 |
| 1, 4, 18, 90 |
| 1, 4, 18, 90, 494 |
| 1, 4, 19 |
| 1, 4, 19, 106 |
| 1, 4, 19, 106, 683 |
| 1, 4, 20 |
| 1, 4, 20, 126 |
| 1, 4, 20, 126, 992 |
| 1, 5 |
| 1, 6 |
| 1, 7 |

# Appendix B

# Important Notations and Their Limits

The number of googology notations, large ordinal notations, and various derived notations in the world is extremely vast, but the vast majority of them are either ill-defined, overly cumbersome and weak, or lack reference value. This table selects a portion of the more classic and important notations, arranged in ascending order according to the limits of their growth rates, for the reader's reference. The content of this table is taken from[33], and the limit of a googology notation is its maximum FGH growth rate.

It should be noted that not all notations in the table are universally recognized within the googology community, and these notations may also be ill-defined. Currently, only $\omega - Y$ is considered relatively reliable in terms of its well-definedness, while only BMS's well-definedness has been proven with some rigor. Therefore, googology notations beyond this are still in a dangerous zone and may be ruled out by stricter analysis at any time. Furthermore, due to the extreme fragmentation and decentralization of googology resources, many notations are proposed in informal or non-public settings. Consequently, not all notations can be associated with corresponding definitions and sources (especially more recent notations and domestic notations). Some notations for which publicly available definitions can be found have cited corresponding references.

The results in this table are updated to 2023.

## B.1 Part I

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| 后继[34]<br>Successor | 0 |
| 加法[35]<br>Addition | 0 |
| 进位计数制<br>n-base | 1<br>(0) |
| 单位进数<br>（如万进/华严经）<br>Unit Number system | 1<br>(0) |
| 乘法[36]<br>Multiplication | 1<br>(0) |
| 科学计数法[37]<br>Scientific Notation | 2<br>(0)(0) |
| 幂集[38]<br>Power set | 2<br>(0)(0) |
| 指数函数[39]<br>Exponential function | 2<br>(0)(0) |
| 阶乘[40]<br>Factorial | 2<br>(0)(0) |
| -illion 系统[41]<br>-illion system | >2<br>>(0)(0) |

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| -yillion 系统[42]<br>-yillion system | >2<br>>(0)(0) |
| 指数塔[43]<br>Tetration | 3<br>(0)(0)(0) |
| 多边形记号[44]<br>Polygon Notation | 3<br>(0)(0)(0) |
| 第五级运算[45]<br>Pentiration | 4<br>(0)(0)(0)(0) |

## B.2 Part II

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| Moser 多边形记号[44]<br>Moser's Polygon Notation<br>(Moser, 1950) | $\omega$<br>$\varphi(1)$<br>$\psi(1)$ (BOCF)<br>(0)(1) |
| Monafo 超运算[46]<br>Monafo's Hyper operation | $\omega$<br>$\varphi(1)$<br>$\psi(1)$ (BOCF)<br>(0)(1) |
| 下箭头[47]<br>Down-Arraw | $\omega$<br>$\varphi(1)$<br>$\psi(1)$ (BOCF)<br>(0)(1) |
| 超阶乘[48]<br>Hyper Factorial | $\omega$<br>$\varphi(1)$<br>$\psi(1)$ (BOCF)<br>(0)(1) |
| 超运算[46]<br>Hyper operation | $\omega$<br>$\varphi(1)$<br>$\psi(1)$ (BOCF)<br>(0)(1) |
| Knuth 上箭头[49]<br>Knuth's Up-Arraw<br>(Knuth, 1976) | $\omega$<br>$\varphi(1)$<br>$\psi(1)$ (BOCF)<br>(0)(1) |
| 首个超限序数[50]<br>FTO<br>(First Transfinite Ordinal) | $\omega$<br>$\varphi(1)$<br>$\psi(1)$ (BOCF)<br>(0)(1) |
| Ackermann 函数[51]<br>Ackerman's Function<br>(Ackermann, 1919) | $\omega$<br>$\varphi(1)$<br>$\psi(1)$ (BOCF)<br>(0)(1) |
| Graham 函数[52]<br>Graham's Function<br>(Graham, 1980) | $\omega+1$<br>$\varphi(1)+1$<br>$\psi(1)+1$ (BOCF)<br>(0)(1)(0) |

| Notation Name<br>(Chinese Name/English Abbreviation/English Full Name/<br>Proposer/Date Proposed) | Limit |
|---|---|
| Clarkkkkson[53]<br>(1988) | $\omega+1$<br>$\varphi(1)+1$<br>$\psi(1)+1$ (BOCF)<br>(0)(1)(0) |
| Graham 记号[52]<br>Graham Notation<br>(Graham, 1980) | $\omega\cdot 2$<br>$\varphi(1)\cdot 2$<br>$\psi(1)\cdot 2$ (BOCF)<br>(0)(1)(0)(1) |
| Forcal 函数[54]<br>Forcal Function<br>(Aarex) | $\omega^2$<br>$\varphi(2)$<br>$\psi(2)$ (BOCF)<br>(0)(1)(1) |
| Conway 链[55]<br>Conway Chain<br>(Chained Arrow)<br>(Conway, Kenneth, 1971) | $\omega^2$<br>$\varphi(2)$<br>$\psi(2)$ (BOCF)<br>(0)(1)(1) |
| f$\phi$ 函数[56]<br>f$\phi$ function | $\omega^2$<br>$\varphi(2)$<br>$\psi(2)$ (BOCF)<br>(0)(1)(1) |
| CG 函数[55]<br>CG function<br>(Conway, Kenneth, 1971) | $\omega^2$<br>$\varphi(2)$<br>$\psi(2)$ (BOCF)<br>(0)(1)(1) |
| Bowers {} 记号[57]<br>Bowers' {}<br>(Bowers, Bird, Spencer, 2002) | $\omega^2$<br>$\varphi(2)$<br>$\psi(2)$ (BOCF)<br>(0)(1)(1) |
| 扩展 Conway 链[55]<br>Extended chained Arrow<br>(Hurford, 1995) | $\omega^3$<br>$\varphi(3)$<br>$\psi(3)$ (BOCF)<br>(0)(1)(1)(1) |
| C 函数[55]<br>C function<br>(Hurford) | $\omega^3+1$<br>$\varphi(3)+1$<br>$\psi(3)+1$ (BOCF)<br>(0)(1)(1)(1)(0)(1) |
| MGH 首次追平 FGH<br>1st time MGH catches FGH | $\omega^\omega$<br>$\varphi(\varphi(1))$<br>$\psi(\psi(1))$ (BOCF)<br>(0)(1)(2) |
| 线性数阵序数<br>LAO<br>(Linar Array Ord)<br>(1976) | $\omega^\omega$<br>$\varphi(\varphi(1))$<br>$\psi(\psi(1))$ (BOCF)<br>(0)(1)(2) |
| (-2)-Y 序列<br>(-2)-Y sequence<br>(2023) | $\omega^\omega$<br>$\varphi(\varphi(1))$<br>$\psi(\psi(1))$ (BOCF)<br>(0)(1)(2) |
| Aarex 多边形记号[58]<br>Aarex's Polygon Notation<br>(Aarex) | $\omega^\omega$<br>$\varphi(\varphi(1))$<br>$\psi(\psi(1))$ (BOCF)<br>(0)(1)(2) |

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| Bashicu 数阵算符<br>Bashicu Array operator<br>(Bashicu) | $\omega^\omega$<br>$\varphi(\varphi(1))$<br>$\psi(\psi(1))$ (BOCF)<br>(0)(1)(2) |
| 线性数阵[57]<br>Linar Arrays<br>(Bowers, 2003) | $\omega^\omega$<br>$\varphi(\varphi(1))$<br>$\psi(\psi(1))$ (BOCF)<br>(0)(1)(2) |
| 平面数阵[57]<br>Planar Arrays<br>(Bowers, 2003) | $\omega^{\omega^2}$<br>$\varphi(\varphi(2))$<br>$\psi(\psi(2))$ (BOCF)<br>(0)(1)(2)(2) |
| 维度数阵[57]<br>Dimensional Arrays<br>(Bowers, 2003) | $\omega^{\omega^\omega}$<br>$\varphi(\varphi(\varphi(1)))$<br>$\psi(\psi(\psi(1)))$ (BOCF)<br>(0)(1)(2)(3) |

## B.3 Part III

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| PTO $\left(\left(\Pi_0^1 - \mathrm{CA}\right)_0\right)$ | $\varepsilon_0$<br>$\varphi(1,0)$<br>$\psi(0)$ (MOCF)<br>$\psi(\Omega)$ (BOCF)<br>(0,0)(1,1) |
| 小 Cantor 序数[59]<br>SCO<br>(Small Cantor's Ordinal) | $\varepsilon_0$<br>$\varphi(1,0)$<br>$\psi(0)$ (MOCF)<br>$\psi(\Omega)$ (BOCF)<br>(0,0)(1,1) |
| HH 首次追平 FGH<br>1st time HH catches FGH | $\varepsilon_0$<br>$\varphi(1,0)$<br>$\psi(0)$ (MOCF)<br>$\psi(\Omega)$ (BOCF)<br>(0,0)(1,1) |
| 初等序列[60]<br>PrSS<br>(Primitive Sequence System)<br>(Bashicu, 2014) | $\varepsilon_0$<br>$\varphi(1,0)$<br>$\psi(0)$ (MOCF)<br>$\psi(\Omega)$ (BOCF)<br>(0,0)(1,1) |
| Worm 函数[61]<br>Worm Function<br>(2002) | $\varepsilon_0$<br>$\varphi(1,0)$<br>$\psi(0)$ (MOCF)<br>$\psi(\Omega)$ (BOCF)<br>(0,0)(1,1) |
| (-1)-Y 序列<br>(-1)-Y sequence<br>(2022) | $\varepsilon_0$<br>$\varphi(1,0)$<br>$\psi(0)$ (MOCF)<br>$\psi(\Omega)$ (BOCF)<br>(0,0)(1,1) |

| Notation Name<br>(Chinese Name/English Abbreviation/English Full Name/<br>Proposer/Date Proposed) | Limit |
|---|---|
| Brace 数阵[62]<br>Brace Array Notation<br>(HypCos, 2013) | $\varepsilon_0$<br>$\varphi(1,0)$<br>$\psi(0)$ (MOCF)<br>$\psi(\Omega)$ (BOCF)<br>(0,0)(1,1) |
| Hydra 问题[63]<br>Hydra<br>(Kirby, Paris, 1984) | $\varepsilon_0$<br>$\varphi(1,0)$<br>$\psi(0)$ (MOCF)<br>$\psi(\Omega)$ (BOCF)<br>(0,0)(1,1) |
| 多维数阵[57]<br>Multi Dimensional Arrays<br>(Bowers) | $\varepsilon_0$<br>$\varphi(1,0)$<br>$\psi(0)$ (MOCF)<br>$\psi(\Omega)$ (BOCF)<br>(0,0)(1,1) |
| Goodstein 序列[64]<br>Goodstein sequence<br>(Goodstein) | $\varepsilon_0$<br>$\varphi(1,0)$<br>$\psi(0)$ (MOCF)<br>$\psi(\Omega)$ (BOCF)<br>(0,0)(1,1) |
| 燃烧数[65]<br>Fusible Number | $\geq \varepsilon_0$<br>$\geq \varphi(1,0)$<br>$\geq \psi(0)$ (MOCF)<br>$\geq \psi(\Omega)$ (BOCF)<br>$\geq$(0,0)(1,1) |
| marxen.c 函数[66]<br>marxen.c function<br>(Marxen) | $\varepsilon_0 + \omega \cdot 3$<br>$\varphi(1,0) + \varphi(1) \cdot 3$<br>$\psi(0) + \omega \cdot 3$ (MOCF)<br>$\psi(\Omega) + \psi(1) \cdot 3$ (BOCF)<br>(0,0)(1,1)(0,0)(1,0)-<br>-(0,0)(1,0)(0,0)(1,0) |
| PTO $\left(\left(\Pi_0^1 - \mathrm{TR}\right)_0\right)$ | $\varepsilon_\omega$<br>$\varphi(1,\omega)$<br>$\psi(\omega)$ (MOCF)<br>$\psi(\Omega \cdot \omega)$ (BOCF)<br>(0,0)(1,1)(2,0) |
| 含 [ ] 的 Bird 数阵[67]<br>BAN with [ ]<br>(Bird, 2006) | $\varepsilon_\omega$<br>$\varphi(1,\omega)$<br>$\psi(\omega)$ (MOCF)<br>$\psi(\Omega \cdot \omega)$ (BOCF)<br>(0,0)(1,1)(2,0) |
| 0-下降记号[68]<br>0-dropping notation | $\varepsilon_\omega$<br>$\varphi(1,\omega)$<br>$\psi(\omega)$ (MOCF)<br>$\psi(\Omega \cdot \omega)$ (BOCF)<br>(0,0)(1,1)(2,0) |
| Cantor 序数[69]<br>CO<br>(Cantor's Ordinal) | $\zeta_0$<br>$\varphi(2,0)$<br>$\psi(\Omega)$ (MOCF)<br>$\psi(\Omega^2)$ (BOCF)<br>(0,0)(1,1)(2,1) |

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| $\varepsilon$ 函数[59]<br>$\varepsilon$ function | $\zeta_0$<br>$\varphi(2,0)$<br>$\psi(\Omega)$ (MOCF)<br>$\psi(\Omega^2)$ (BOCF)<br>(0,0)(1,1)(2,1) |
| 扩展维度数阵[57]<br>Extended Dimensional Arrays | $\zeta_0$<br>$\varphi(2,0)$<br>$\psi(\Omega)$ (MOCF)<br>$\psi(\Omega^2)$ (BOCF)<br>(0,0)(1,1)(2,1) |
| 大 Cantor 序数[70]<br>LCO<br>(Large Cantor's Ordinal) | $\eta_0$<br>$\varphi(3,0)$<br>$\psi(\Omega^2)$ (MOCF)<br>$\psi(\Omega^3)$ (BOCF)<br>(0,0)(1,1)(2,1)(2,1) |
| $\zeta$ 函数[69]<br>$\zeta$ function | $\eta_0$<br>(0,0)(1,1)(2,1)(2,1) |
| $\eta$ 函数[70]<br>$\eta$ function | $\varphi(4,0)$<br>$\psi(\Omega^3)$ (MOCF)<br>$\psi(\Omega^4)$ (BOCF)<br>(0,0)(1,1)(2,1)(2,1)(2,1) |
| 长初等序列[71]<br>LPrSS<br>(Long Primitive Sequence System) | $\varphi(\omega,0)$<br>$\psi(\Omega^\omega)$ (MOCF)<br>$\psi(\Omega^\omega)$ (BOCF)<br>(0,0)(1,1)(2,1)(3,0) |
| 0-递增元序列[72-74]<br>0-IUN<br>(0-Increase Unit Notation)<br>(318‘4, 2023) | $\varphi(\omega,0)$<br>$\psi(\Omega^\omega)$ (MOCF)<br>$\psi(\Omega^\omega)$ (BOCF)<br>(0)(1,1)(2,1)(3) |
| 超 Cantor 序数[70]<br>HCO<br>(Hyper Cantor's Ordinal) | $\varphi(\omega,0)$<br>$\psi(\Omega^\omega)$ (MOCF)<br>$\psi(\Omega^\omega)$ (BOCF)<br>(0,0)(1,1)(2,1)(3,0) |

## B.4 Part IV

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| Feferman-Schutte 序数[75]<br>FSO<br>(Feferman-Schutte Ordinal) | $\Gamma_0$<br>$\varphi(1,0,0)$<br>$\psi(\Omega^\Omega)$ (MOCF)<br>$\psi(\Omega^\Omega)$ (BOCF)<br>(0,0)(1,1)(2,1)(3,1) |
| 二元 Veblen 函数[76]<br>Binary Veblen Function | $\Gamma_0$<br>$\varphi(1,0,0)$<br>$\psi(\Omega^\Omega)$ (MOCF)<br>$\psi(\Omega^\Omega)$ (BOCF)<br>(0,0)(1,1)(2,1)(3,1) |

| Notation Name<br>(Chinese Name/English Abbreviation/English Full Name/<br>Proposer/Date Proposed) | Limit |
|---|---|
| 弱 Veblen 函数[76]<br>Small Veblen Function<br>$(0-\phi)$ | $\Gamma_0$<br>$\varphi(1,0,0)$<br>$\psi(\Omega^{\Omega})$ (MOCF)<br>$\psi(\Omega^{\Omega})$ (BOCF)<br>(0,0)(1,1)(2,1)(3,1) |
| $\Gamma$ 函数[75]<br>$\Gamma$ Function | $\varphi(1,1,0)$<br>$\psi(\Omega^{\Omega+1})$ (MOCF)<br>$\psi(\Omega^{\Omega+1})$ (BOCF)<br>(0,0)(1,1)(2,1)(3,1)(2,1) |
| $\omega$-下降 Worm[68]<br>$\omega$-dropping Worm | $\varphi(\omega,0,0)$<br>$\psi(\Omega^{\Omega\cdot\omega})$ (MOCF)<br>$\psi(\Omega^{\Omega\cdot\omega})$ (BOCF)<br>(0)(1,1)(2,1)(3,1)(3) |
| Ackermann 序数<br>AO<br>(Ackermann Ordinal) | $\varphi(1,0,0,0)$<br>$\psi(\Omega^{\Omega^2})$ (MOCF)<br>$\psi(\Omega^{\Omega^2})$ (BOCF)<br>(0,0)(1,1)(2,1)(3,1)(3,1) |
| E# 记号[77]<br>E#<br>(Saibian, 2008) | $\varphi(1,0,0,0)$<br>$\psi(\Omega^{\Omega^2})$ (MOCF)<br>$\psi(\Omega^{\Omega^2})$ (BOCF)<br>(0,0)(1,1)(2,1)(3,1)(3,1) |
| tree 函数[78]<br>tree<br>(Friedman, 2000) | $\varphi(1@\omega)$<br>$\psi(\Omega^{\Omega^{\omega}})$ (MOCF)<br>$\psi(\Omega^{\Omega^{\omega}})$ (BOCF)<br>(0,0)(1,1)(2,1)(3,1)(4,0) |
| 小 Veblen 序数[79]<br>SVO<br>(Small Veblen Ordinal) | $\varphi(1@\omega)$<br>$\psi(\Omega^{\Omega^{\omega}})$ (MOCF)<br>$\psi(\Omega^{\Omega^{\omega}})$ (BOCF)<br>(0,0)(1,1)(2,1)(3,1)(4,0) |
| otree 函数[78]<br>otree<br>(Friedman) | $\varphi(1@\omega)$<br>$\psi(\Omega^{\Omega^{\omega}})$ (MOCF)<br>$\psi(\Omega^{\Omega^{\omega}})$ (BOCF)<br>(0,0)(1,1)(2,1)(3,1)(4,0) |
| TREE 函数[78]<br>TREE<br>(Friedman, 2001) | $\geq \varphi(\omega@\omega)$<br>$\geq \psi(\Omega^{\Omega^{\omega}\cdot\omega})$ (MOCF)<br>$\geq \psi(\Omega^{\Omega^{\omega}\cdot\omega})$ (BOCF)<br>$\geq$(0,0)(1,1)(2,1)(3,1)(4,0)(3,0) |
| Veblen 函数[76]<br>Veblen Function<br>(Veblen, 1908) | $\varphi(1@(1,0))$<br>$\psi(\Omega^{\Omega^{\Omega}})$ (MOCF)<br>$\psi(\Omega^{\Omega^{\Omega}})$ (BOCF)<br>(0,0)(1,1)(2,1)(3,1)(4,1) |
| Bowers 数阵[57]<br>BEAF<br>(Bowers' Exploding Array Function)<br>(Chris Bird's Limit) | $\varphi(1@(1,0))$<br>$\psi(\Omega^{\Omega^{\Omega}})$ (MOCF)<br>$\psi(\Omega^{\Omega^{\Omega}})$ (BOCF)<br>(0,0)(1,1)(2,1)(3,1)(4,1) |
| 扩展小 Veblen 序数<br>ESVO<br>(Extended Small Veblen Ordinal) | $\varphi(1@(1@\omega))$<br>$\psi(\Omega^{\Omega^{\Omega^{\omega}}})$ (MOCF)<br>$\psi(\Omega^{\Omega^{\Omega^{\omega}}})$ (BOCF)<br>(0,0)(1,1)(2,1)(3,1)(4,1)(5,0) |

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| 大 Veblen 函数[80]<br>Large Veblen Function<br>$(2-\phi)$<br>(74& 4574, 2021) | $\varphi(1@(1@(1,0)))$<br>$\psi(\Omega^{\Omega^{\Omega^{\Omega}}})$ (MOCF)<br>$\psi(\Omega^{\Omega^{\Omega^{\Omega}}})$ (BOCF)<br>(0,0)(1,1)(2,1)(3,1)(4,1)(5,1) |
| 扩展大 Veblen 序数<br>ELVO<br>(Extended Large Veblen Ordinal) | $\varphi(1@(1@(1,0)))$<br>$\psi(\Omega^{\Omega^{\Omega^{\Omega}}})$ (MOCF)<br>$\psi(\Omega^{\Omega^{\Omega^{\Omega}}})$ (BOCF)<br>(0,0)(1,1)(2,1)(3,1)(4,1)(5,1) |
| Bachmann-Howard 序数[81]<br>BHO<br>(Bachmann-Howard Ordinal) | $\psi(\psi_1(0))$ (MOCF)<br>$\psi(\Omega_2)$ (BOCF)<br>(0,0)(1,1)(2,2) |
| Madore's $\psi$ 函数[82]<br>MPF<br>(Madore's $\psi$ Function) | $\psi(\psi_1(0))$ (MOCF)<br>$\psi(\Omega_2)$ (BOCF)<br>(0,0)(1,1)(2,2) |
| 大 Veblen 系统[76]<br>LVS<br>(Large Veblen System) | $\psi(\psi_1(0))$ (MOCF)<br>$\psi(\Omega_2)$ (BOCF)<br>(0,0)(1,1)(2,2) |
| 扩展 Veblen 系统[76]<br>ExV<br>(Extended Veblen Function) | $\psi(\psi_1(0))$ (MOCF)<br>$\psi(\Omega_2)$ (BOCF)<br>(0,0)(1,1)(2,2) |
| $\Omega$ 记号<br>$\Omega$ Notation | $\psi(\psi_1(0))$ (MOCF)<br>$\psi(\Omega_2)$ (BOCF)<br>(0)(1,1)(2,2) |
| 含 $\sim$ 的 Bird 数阵[83]<br>Bird's $\sim$<br>(Bird) | $\psi(\psi_2(0))$ (MOCF)<br>$\psi(\Omega_3)$ (BOCF)<br>(0,0)(1,1)(2,2)(3,3) |
| 含 ♦ 的 Bird 数阵[83]<br>Bird's ♦ | $\psi(\psi_3(0))$ (MOCF)<br>$\psi(\Omega_4)$ (BOCF)<br>(0,0)(1,1)(2,2)(3,3)(4,4) |

## B.5 Part V

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| PTO $\left(\left(\Pi_1^1-\mathrm{CA}\right)_0\right)$ | $\psi(\Omega_\omega)$ (MOCF)<br>$\psi(\Omega_\omega)$ (BOCF)<br>(0,0,0)(1,1,1) |
| Buchholz's 序数[84]<br>BO<br>(Buchholz's Ordinal) | $\psi(\Omega_\omega)$ (MOCF)<br>$\psi(\Omega_\omega)$ (BOCF)<br>(0,0,0)(1,1,1) |
| SGH 首次追平 FGH<br>1st SF catching | $\psi(\Omega_\omega)$ (MOCF)<br>$\psi(\Omega_\omega)$ (BOCF)<br>(0,0,0)(1,1,1) |
| 单一递增元序列[72-74]<br>SIUN<br>(Single Increase Unit Notation)<br>(318‘4, 2023) | $\psi(\Omega_\omega)$ (MOCF)<br>$\psi(\Omega_\omega)$ (BOCF)<br>(0,0,0)(1,1,1) |
| 超初等序列[85]<br>HPrSS<br>(Hyper Primitive Sequence System) | $\psi(\Omega_\omega)$ (MOCF)<br>$\psi(\Omega_\omega)$ (BOCF)<br>(0,0,0)(1,1,1) |

| Notation Name<br>(Chinese Name/English Abbreviation/English Full Name/<br>Proposer/Date Proposed) | Limit |
|---|---|
| 进制数阵 I<br>ABN I<br>(Array Basic Notation I)<br>(4574, 2021) | $\psi(\Omega_\omega)$ (MOCF)<br>$\psi(\Omega_\omega)$ (BOCF)<br>(0,0,0)(1,1,1) |
| 双行序列系统[86]<br>PSS<br>(Pair Sequence System) | $\psi(\Omega_\omega)$ (MOCF)<br>$\psi(\Omega_\omega)$ (BOCF)<br>(0,0,0)(1,1,1) |
| 含 & 的 BEAF[57]<br>& in BEAF | $\psi(\Omega_\omega)$ (MOCF)<br>$\psi(\Omega_\omega)$ (BOCF)<br>(0,0,0)(1,1,1) |
| 增长链<br>Increasable Chain<br>(4574, 2023) | $>\psi(\Omega_\omega)$ (MOCF)<br>$>\psi(\Omega_\omega)$ (BOCF)<br>$>$(0,0,0)(1,1,1) |
| SCG 函数[87]<br>SCG Function<br>(Subcubic Graph number)<br>(Friedman) | $\psi(\Omega_\omega\cdot\omega)\sim\psi({\Omega_\omega}^\omega)$ (MOCF)<br>$\psi(\Omega_\omega\cdot\omega)\sim\psi({\Omega_\omega}^\omega)$ (BOCF)<br>在 $(0,0,0)(1,1,1)(2,0,0)$<br>与 $(0,0,0)(1,1,1)(2,1)(3,0,0)$ 之间 |
| SSCG 函数[87]<br>SSCG Function<br>(Simple Subcubic Graph number)<br>(Friedman) | $\psi(\Omega_\omega\cdot\omega)\sim\psi({\Omega_\omega}^\omega)$ (MOCF)<br>$\psi(\Omega_\omega\cdot\omega)\sim\psi({\Omega_\omega}^\omega)$ (BOCF)<br>在 $(0,0,0)(1,1,1)(2,0,0)$<br>与 $(0,0,0)(1,1,1)(2,1)(3,0,0)$ 之间 |
| Buchholz's $\psi$ 函数[88]<br>BPF<br>(Buchholz's $\psi$ Function)<br>(Buchholz, 1986) | $\psi(\psi_\omega(0))$ (MOCF)<br>$\psi(\Omega_{\omega+1})$ (BOCF)<br>$(0,0,0)(1,1,1)(2,1)(3,2,0)$ |
| Takeuti-Feferman-Buchholz's 序数[89]<br>TFBO<br>(Takeuti-Feferman-Buchholz's Ordinal) | $\psi(\psi_\omega(0))$ (MOCF)<br>$\psi(\Omega_{\omega+1})$ (BOCF)<br>$(0,0,0)(1,1,1)(2,1)(3,2,0)$ |
| PTO $\left(\Pi^1_1-\mathrm{CA}+\mathrm{BI}\right)$ | $\psi(\psi_\omega(0))$ (MOCF)<br>$\psi(\Omega_{\omega+1})$ (BOCF)<br>$(0,0,0)(1,1,1)(2,1,0)(3,2,0)$ |
| BHydra 函数[90]<br>BHydra<br>(Buchholz's Hydra)<br>(Buchholz, 1987) | $\psi(\psi_\omega(0))$ (MOCF)<br>$\psi(\Omega_{\omega+1})$ (BOCF)<br>$(0,0,0)(1,1,1)(2,1)(3,2,0)$ |
| Bird 数阵[83]<br>BAN<br>(Bird's Array Notation)<br>(Bird, 2014) | $\psi(\Omega_\Omega)$ (MOCF)<br>$\psi(\Omega_\Omega)$ (BOCF)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,0)$ |
| Bird 序数[83]<br>BIO<br>(Bird's Ordinal) | $\psi(\Omega_\Omega)$ (MOCF)<br>$\psi(\Omega_\Omega)$ (BOCF)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,0)$ |
| PTO $\left(\left(\Pi^1_1-\mathrm{TR}\right)_0\right)$ | $\psi(2\ 1-2)$<br>$\psi(\psi_I(0))$ (M-like)<br>$\psi(I)$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0)$ |
| SGH 第 $\omega$ 次追平 FGH<br>$\omega$th SF catching | $\psi(2\ 1-2)$<br>$\psi(\psi_I(0))$ (M-like)<br>$\psi(I)$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0)$ |
| 扩展 Buchholz 序数[91]<br>EBO<br>(Extended Buchholz's Ordinal) | $\psi(2\ 1-2)$<br>$\psi(\psi_I(0))$ (M-like)<br>$\psi(I)$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0)$ |

| Notation Name<br>(Chinese Name/English Abbreviation/English Full Name/<br>Proposer/Date Proposed) | Limit |
|---|---|
| 扩展 Buchholz$\psi$ 函数[92]<br>EBPF<br>(Extended Buchholz's $\psi$ Function)<br>(Maksudov, 1987) | $\psi(2\ 1-2)$<br>$\psi(\psi_I(0))$ (M-like)<br>$\psi(I)$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0)$ |
| 急序列系统<br>SSS<br>(Sudden Sequence System)<br>(Bashicu, 2017) | $\psi(2\ 1-2)$<br>$\psi(\psi_I(0))$ (M-like)<br>$\psi(I)$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0)$ |
| Buchholz's $\Phi$ 函数<br>Buchholz's $\Phi$<br>(Buchholz, 1987) | $\psi(I^{I^I})$ (M-like)<br>$\psi(I^{I^I})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,0)(4,1,0)(5,1,0)$ |
| Jäger 序数<br>JO<br>(Jäger's Ordinal) | $\psi(2\text{ aft }1-2)$<br>$\psi\left(\psi_{\Omega_{I+1}}(0)\right)$ (M-like)<br>$\psi(\Omega_{I+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)$ |
| PTO(KPi) | $\psi(2\text{ aft }1-2)$<br>$\psi\left(\psi_{\Omega_{I+1}}(0)\right)$ (M-like)<br>$\psi(\Omega_{I+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)$ |
| Jäger-Buchholz 函数[93]<br>JBF<br>(Jäger-Buchholz Function)<br>(Jäger, Buchholz) | $\psi(2\text{ aft }1-2)$<br>$\psi\left(\psi_{\Omega_{I+1}}(0)\right)$ (M-like)<br>$\psi(\Omega_{I+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)$ |
| N-like 首次追平 BOCF<br>1st NOCF catches BOCF | $\psi(2\text{ aft }1-2)$<br>$\psi\left(\psi_{\Omega_{I+1}}(0)\right)$ (M-like)<br>$\psi(\Omega_{I+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)$ |
| 小不可达序数<br>SIO<br>(Small Inaccessible Ordinal)<br>(2023) | $\psi(1-2\ 1-2)$<br>$\psi(I_\omega)$ (M-like)<br>$\psi(I_\omega)$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,1)$ |
| 超阶乘数阵[94]<br>HAN<br>(Hyper-factoral Array Notation)<br>(2013) | $\psi((1-)^{(2\ 1-2)}2\ 1-2)$<br>$\psi(I_I)$ (M-like)<br>$\psi(I_I)$ (B-like)<br>(0,0,0)(1,1,1)(2,1,1)(3,1,1)-<br>-(2,1,1)(3,1,0)(1,1,1)(2,1,1)-<br>-(3,1,0)(4,2,1)(5,2,1)(6,2,1)-<br>-(5,2,1)(6,1,0)(2,0,0) |
| 大数入门 OCF[95]<br>(HypCos, 2014) | $\psi(2\text{ aft }2\ 1-2\ 1-2)$<br>$\psi(\psi_{\Omega_{I(1,0)+1}}(0))$ (M-like)<br>$\psi(\Omega_{I(1,0)+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,1)-$<br>$(2,1,1)(3,1,0)(4,2,0)$ |
| PTO(KPH) | $\psi(2\text{ aft }2\ 1-2\ 1-2)$<br>$\psi(\psi_{\Omega_{I(1,0)+1}}(0))$ (M-like)<br>$\psi(\Omega_{I(1,0)+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,1)-$<br>$(2,1,1)(3,1,0)(4,2,0)$ |
| 多重 Buchholz 序数<br>MBO<br>(Mutiply Buchholz's Ordinal) | $\psi((2\ 1-)^\omega\ 2)$<br>$\psi(I(\omega,0))$ (M-like)<br>$\psi(I(\omega,0))$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)$ |

| Notation Name<br>(Chinese Name/English Abbreviation/English Full Name/<br>Proposer/Date Proposed) | Limit |
|---|---|
| 线性 R 函数[96]<br>LRF<br>(Linear R Function)<br>(HypCos, 2013) | $\psi((2\ 1-)^{\omega}\ 2)$<br>$\psi(I(\omega,0))$ (M-like)<br>$\psi(I(\omega,0))$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)$ |
| 简单迭代数阵[97]<br>SIAN<br>(Simple Iteration Array Notation)<br>(aeroplane32, 2018) | $\psi((2-\ 1-)^{\omega}\ 2)$<br>$\psi(I(\omega,0))$ (M-like)<br>$\psi(I(\omega,0))$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)$ |
| 超级数阵<br>UlAN<br>(Ultra Array Notation)<br>(五年高考, 2020) | $\psi((2\ 1-)^{\omega}\ 2)$<br>$\psi(I(\omega,0))$ (M-like)<br>$\psi(I(\omega,0))$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)$ |
| Fire 数阵<br>FAN<br>(Fire Array Notation)<br>(1000°C 的人, 2023) | $\psi((2\ 1-)^{\omega}\ 2)$<br>$\psi(I(\omega,0))$ (M-like)<br>$\psi(I(\omega,0))$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)$ |
| 线性 Zenith 记号[98]<br>LZN<br>(Linear Zenith Notation) | $\psi((2\ 1-)^{\omega}\ 2)$<br>$\psi(I(\omega,0))$ (M-like)<br>$\psi(I(\omega,0))$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)$ |
| a 强数组记号-1[99]<br>aSAN-1<br>(a Strong Array Notation-1) | $\psi((2\ 1-)^{\omega}\ 2)$<br>$\psi(I(\omega,0))$ (M-like)<br>$\psi(I(\omega,0))$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)$ |

## B.6 Part VI

| Notation Name<br>(Chinese Name/English Abbreviation/English Full Name/<br>Proposer/Date Proposed) | Limit |
|---|---|
| 超限 Buchholz 序数<br>TBO<br>(Transfinty Buchholz's Ordinal) | $\psi((2\ 1-)^{1,1}\ 2)$<br>$\psi(I(1,0,0))$ (M-like)<br>$\psi(I(1,0,0))$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(3,1,0)(2,0,0)$ |
| 维度 R 函数[96]<br>Dimensional R Function | $\psi((2\ 1-)^{1,1}\ 2)$<br>$\psi(I(1,0,0))$ (M-like)<br>$\psi(I(1,0,0))$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(3,1,0)(2,0,0)$ |
| 小 Rathjen 序数<br>SRO<br>(Small Rathjen's Ordinal) | $\psi(2 \text{ aft } 2-2)$<br>$\psi(\psi_{\Omega_{M+1}}(0))$ (M-like)<br>$\psi(\Omega_{M+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(3,1,0)(4,2,0)$ |
| Mahlo 序数折叠函数<br>Mahlo OCF | $\psi(2 \text{ aft } 2-2)$<br>$\psi(\psi_{\Omega_{M+1}}(0))$ (M-like)<br>$\psi(\Omega_{M+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(3,1,0)(4,2,0)$ |

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| PTO(KPM) | $\psi(2 \text{ aft } 2-2)$<br>$\psi(\psi_{\Omega_{M+1}}(0))$ (M-like)<br>$\psi(\Omega_{M+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(3,1,0)(4,2,0)$ |
| Rathjen's $\chi$ 函数[100]<br>Rathjen's $\chi$<br>(Rathjen, 1989) | $\psi(1-2-2)$<br>$\psi(M_\omega)$ (M-like)<br>$\psi(M_\omega)$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(3,1,1)$ |
| 字节数阵[101]<br>Byte Notation<br>(Glise229, 2020) | $\psi(1-2-2)$<br>$\psi(M_\omega)$ (M-like)<br>$\psi(M_\omega)$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(3,1,1)$ |
| 小 Mahlo 序数<br>SMO<br>(Small Mahlo Ordinal)<br>(2023) | $\psi(1-2-2)$<br>$\psi(M_\omega)$ (M-like)<br>$\psi(M_\omega)$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(3,1,1)$ |
| 疯狂 Mahlo 记号<br>CMN<br>(Crazy Mahlo Notation)<br>(Bugit, 2022) | $\psi(2 \text{ aft } 2-2-2)$<br>$\psi(\psi_{\Omega_{N+1}}(0))$ (M-like)<br>$\psi(\Omega_{N+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(3,1,1)(3,1,0)(4,2,0)$ |
| 小不可转换序数<br>SNO<br>(Small Nonconvertible Ordinal) | $\psi(1-2-2-2)$<br>$\psi(N_\omega)$ (M-like)<br>$\psi(N_\omega)$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(3,1,1)(3,1,1)$ |
| PTO $(\mathrm{KP}+\Pi_3-$ Ref $)$ | $\psi(2 \text{ aft } 3)$<br>$\psi(\psi_{\Omega_{K+1}}(0))$ (M-like)<br>$\psi(\Omega_{K+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(4,1,0)(5,2,0)$ |
| Rathjen 序数<br>RO<br>(Rathjen's Ordinal)<br>(Rathjen) | $\psi(2 \text{ aft } 3)$<br>$\psi(\psi_{\Omega_{K+1}}(0))$ (M-like)<br>$\psi(\Omega_{K+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(4,1,0)(5,2,0)$ |
| 美元记号[102]<br>Dollar Function<br>(Wythagoras, 2013) | $\psi(2 \text{ aft } 3)$<br>$\psi(\psi_{\Omega_{K+1}}(0))$ (M-like)<br>$\psi(\Omega_{K+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(4,1,0)(5,2,0)$ |
| Rathjen's $\Xi$ 函数[100]<br>Rathjen's $\Xi$<br>(Rathjen) | $\psi(2 \text{ aft } 3)$<br>$\psi(\psi_{\Omega_{K+1}}(0))$ (M-like)<br>$\psi(\Omega_{K+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(4,1,0)(5,2,0)$ |
| 弱 UN 序数折叠函数[103]<br>Weak UNOCF<br>(Weak Username5243 OCF)<br>(Username5243, 2018) | $\psi((3-)^\omega)$<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(4,1,1)(5,0,0)$ |

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|---|---|
| Duchhardt 序数折叠函数[104]<br>Duchhardt's OCF<br>(Duchhardt) | $\psi(2 \text{ aft } 4)$<br>$\psi(\psi_{\Omega_{\Pi_4+1}}(0))$ (M-like)<br>$\psi(\Omega_{\Pi_4+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(4,1,1)(5,1,0)(6,2,0)$ |
| PTO $(\mathrm{KP}+\Pi_4-$ Ref $)$ | $\psi(2 \text{ aft } 4)$<br>$\psi(\psi_{\Omega_{\Pi_4+1}}(0))$ (M-like)<br>$\psi(\Omega_{\Pi_4+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(4,1,1)(5,1,0)(6,2,0)$ |
| Duchhardt 序数[104]<br>DO<br>(Duchhardt's Ordinal) | $\psi(2 \text{ aft } 4)$<br>$\psi(\psi_{\Omega_{\Pi_4+1}}(0))$ (M-like)<br>$\psi(\Omega_{\Pi_4+1})$ (B-like)<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(4,1,1)(5,1,0)(6,2,0)$ |
| 扩展 I 函数<br>EIF<br>(Extend I Function)<br>(Bugit, 2023) | $\psi((3-)^{\omega})$<br>$(0,0,0)(1,1,1)(2,1,1)-$<br>$(3,1,1)(4,1,1)(5,1,1)(6,0,0)$ |
| $\sup\{\mathrm{PTO}(\mathrm{KP}+\Pi_n-\mathrm{Ref}) \mid n \in \omega\}$ | $\psi(\Pi_\omega)$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0)$<br>$(0,0,0)(1,1,1)(2,2,0)$ |
| 小 Stegert 序数<br>SSO<br>(Small Stegert Ordinal) | $\psi(\Pi_\omega)$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0)$<br>$(0,0,0)(1,1,1)(2,2,0)$ |
| 反射数阵记号<br>RAN<br>(Reflecting Array Notation)<br>(Y_cpper) | $\psi(\Pi_\omega)$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0)$<br>$(0,0,0)(1,1,1)(2,2,0)$ |
| 2-转移 $\psi$[105]<br>2-shifted $\psi$<br>(Solar Zone) | $\psi(\Pi_\omega)$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0)$<br>$(0,0,0)(1,1,1)(2,2,0)$ |
| 初级下降数阵[99]<br>PDAN<br>(Pirmary Dropping Array Notation) | $\psi(\Pi_\omega)$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0)$<br>$(0,0,0)(1,1,1)(2,2,0)$ |
| 魔塔数阵[106]<br>MOTAN<br>(MOTA Notation)<br>(Gomen, 2021) | $\psi(\Pi_\omega)$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0)$<br>$(0,0,0)(1,1,1)(2,2,0)$ |
| M 记号<br>M notation<br>(Test_alpha0, 2021) | $\psi(\Pi_\omega)$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0)$<br>$(0,0,0)(1,1,1)(2,2,0)$ |
| C 记号<br>C notation<br>(Y_cpper, 2023) | $\psi(\Pi_\omega)$<br>$\psi(\lambda\alpha.(\alpha+1)-\Pi_0)$<br>$(0,0,0)(1,1,1)(2,2,0)$ |
| PTO(stability) | $\psi(\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$<br>$(0,0,0)(1,1,1)(2,2,0)$<br>$(3,2,0)(4,1,0)(2,0,0)$ |
| 大 Stegert 序数[107]<br>LSO<br>(Large Stegert Ordinal)<br>(Stegert) | $\psi(\Pi_{1,0})$<br>$\psi(\lambda\alpha.(\alpha\cdot 2)-\Pi_0)$<br>$(0,0,0)(1,1,1)(2,2,0)-$<br>$(3,2,0)(4,1,0)(2,0,0)$ |

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|---|---|
| Stegert 序数折叠函数[107]<br>Stegert's OCF<br>(Stegert) | $\psi(\Pi_{1,0})$<br>$\psi\left(\lambda\alpha.(\alpha\cdot 2)-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,0)-$<br>$(3,2,0)(4,1,0)(2,0,0)$ |
| 扩展箭头记号<br>EUAN<br>(Extend Arrow Array Notation)<br>(Y_cpper, 2020) | $\psi(\Pi_{1,0})$<br>$\psi\left(\lambda\alpha.(\alpha\cdot 2)-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,0)-$<br>$(3,2,0)(4,1,0)(2,0,0)$ |
| 容许-非递归分离序数<br>APO<br>(Admissible-parameter free-<br>effective cardinal Ordinal) | $\psi\left(\Pi_1\left(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1\right)\right)$<br>$(0,0,0)(1,1,1)(2,2,0)-$<br>$(3,2,0)(4,1,1)$ |
| 容许初等序列<br>Adm PrSS<br>(Admissible PrSS)<br>(Alpha, Destoria, 2023) | $\psi\left(\Pi_1\left(\lambda\alpha.(\Omega_{\alpha+1})-\Pi_1\right)\right)$<br>$(0,0,0)(1,1,1)(2,2,0)-$<br>$(3,2,0)(4,1,1)$ |
| 投影首次追平 pfec 稳定<br>1s time projection catches-<br>with pfec stable | 可能为<br>$\psi(\lambda\alpha.\Gamma(\Omega_{\alpha+1}+1)-\Pi_0)$<br>$(0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,2,0)$ |

## B.7 Part VII

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| 首个返回齿轮序数<br>BGO<br>(1st Back Gear Ordinal) | $\psi\left(\Pi_1\left(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1\right)\right)$<br>$(0,0,0)(1,1,1)(2,2,1)$ |
| 扩展 M 记号<br>extended-M notation<br>(Test_alpha0, 2021) | $\psi\left(\Pi_1\left(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1\right)\right)$<br>$(0,0,0)(1,1,1)(2,2,1)$ |
| 进制数阵 III<br>ABNIII<br>(Array Basic Notation III)<br>(2022) | $\psi\left(\Pi_1\left(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1\right)\right)$<br>$(0,0,0)(1,1,1)(2,2,1)$ |
| 链接序数折叠函数<br>LkOCF<br>(Linked Ordinal Collapsing Function)<br>(74(Nonconvertible)) | $\psi\left(\Pi_1\left(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1\right)\right)$<br>$(0,0,0)(1,1,1)(2,2,1)$ |
| 次阶下降数阵[99]<br>SDAN<br>(Secondry Strong Array Notation)<br>(HypCos, 2015) | $\psi\left(\Pi_1\left(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1\right)^{\omega}\right)$<br>$(0,0,0)(1,1,1)(2,2,1)-$<br>$(2,1,1)(3,2,1)(3,0,0)$ |
| K 记号<br>K Notation<br>(Test_alpha0, 2021) | $\psi\left(\lambda\alpha.(\alpha+1)-\Pi_0\left(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1\right)\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(2,2,0)$ |
| 二阶下降 C 记号<br>2-Dropping C Notation<br>(Y_cpper, 2023) | $\psi\left(\lambda\alpha.(\alpha+1)-\Pi_0\left(\lambda\alpha.(\Omega_{\alpha+2})-\Pi_1\right)\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(2,2,0)$ |
| BMS4.1 首次追平 BMS4<br>1st time BM4.1 catches BM4 | $\psi\left(\lambda\alpha.\Omega_{\alpha+\omega}-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(3,0,0)$ |

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| 小下降序数<br>SDO<br>(Small Dropping Ordinal)<br>(Username5243) | $\psi\left(\lambda\alpha.\Omega_{\alpha+\omega}-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(3,0,0)$ |
| 扩展魔塔数阵[108]<br>Extend MOTAN<br>(Gomen, 2021) | $\psi\left(\lambda\alpha.\Omega_{\alpha+\omega}-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(3,0,0)$ |
| 强数阵<br>SAN(DAN)<br>(Strong Array Notation)<br>(HypCos, 2015) | $\psi\left(\lambda\alpha.\Omega_{\alpha+\omega}-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(3,0,0)$ |
| $\omega$ 记号<br>$\omega$ notation<br>(Test_alpha0, 2021) | $\psi\left(\lambda\alpha.\Omega_{\alpha+\omega}-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(3,0,0)$ |
| R 函数[99]<br>R function<br>(HypCos, 2014) | $\psi\left(\lambda\alpha.\Omega_{\alpha+\omega}-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(3,0,0)$ |
| Taranovsky 序数记号[109]<br>TON(main)<br>(Taranovsky's Ordinal Notation)<br>(Taranovsky, 2015) | $\psi\left(\lambda\alpha.\Omega_{\alpha+\omega}-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(3,0,0)$ |
| 长初等序列 $\psi$<br>LPrSS$\psi$<br>(Long Primary Sequence System $\psi$) | $\psi\left(\lambda\alpha.\Omega_{\alpha+\omega}-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(3,0,0)$ |
| Tar 函数[110]<br>n-intar c's<br>(Taranovsky Function) | $\psi\left(\lambda\alpha.\Omega_{\alpha+\omega}-\Pi_0\right)+1$<br>$(0,0,0)(1,1,1)(2,2,1)(3,0,0)(0,0,0)$ |
| S 记号<br>S Notation<br>(Test_alpha0, 2021) | $\psi\left(\lambda\alpha.\Omega_{\alpha+\omega+1}-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)$<br>$-(3,0,0)(2,2,1)$ |
| 大下降序数<br>LDO<br>(Large Dropping Ordinal)<br>(Username5243) | $\psi\left(\lambda\alpha.\Phi(1,\alpha+1)-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(3,2,0)$ |
| 多逗号系统<br>MCS<br>(Multiple Comma System)<br>(74(Nonconvertable)) | $\psi\left(\lambda\alpha.\Phi(1,\alpha+1)-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(3,2,0)$ |
| 大型 X 系统<br>LXN<br>(Large X Notation)<br>(Test_alpha0) | $\psi\left(\lambda\alpha.\Phi(1,\alpha+1)-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(3,2,0)$ |
| 嵌套下降数阵[99]<br>NDAN<br>(Nested Down Arrow Notation)<br>(HypCos, 2017) | $\psi\left(\lambda\alpha.\Phi(1,\alpha+1)-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(3,2,0)$ |
| UN 序数折叠函数[103]<br>UNOCF<br>(Username5243's-<br>Ordinal Collapsing Function)<br>(Username5243, 2018) | $\psi\left(\lambda\alpha.\Phi(1,\alpha+1)-\Pi_0\right)$<br>$(0)(1,1,1)(2,2,1)(3,2,0)$ |

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| 弱下降子扩张数阵[99]<br>WDEN<br>(Weak Dropping-Exrtended Notation)<br>(HypCos, 2017) | $\psi\left(\lambda\alpha.\Phi(1,\alpha+1)-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(3,2,0)$ |
| 多重弱下降子扩张数阵[99]<br>MWDEN<br>(Multy Weak Dropping-Exrtended Notation)<br>(HypCos, 2017) | $\psi\left(\lambda\alpha.\Phi(1,\alpha+1)-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(3,2,0)$ |
| 双重 +1 稳定序数<br>DSO<br>(Doubly +1 stable) | $\psi\left(\lambda\alpha.\left(\lambda\beta.(\beta+1)-\Pi_0\right)-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(3,3,0)$ |
| E 记号<br>E Notation<br>(Y_cpper, 2023) | $\psi\left(\lambda\alpha.\left(\lambda\beta.(\beta+1)-\Pi_0\right)-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,1)(3,3,0)$ |
| 三重 +1 稳定序数<br>TSO<br>(Triply +1 stable) | $\psi(\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\gamma+1)$<br>$-\Pi_0)-\Pi_0)-\Pi_0)$<br>$(0,0,0)(1,1,1)(2,2,1)-$<br>$(3,3,1)(4,4,0)$ |
| 反射度<br>DOR<br>(Degrees of Reflection)<br>(Taranovsky, 2015) | $\leq\psi\left(\omega-\pi-\Pi_0\right)$<br>$\leq(0,0,0)(1,1,1)(2,2,2)$ |
| IBMS 首次追平 BMS<br>1st time IBMs catches BMs | $\psi\left(\omega-\pi-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,2)$ |
| 赝大 Rathjen 序数<br>pLRO<br>(pseudo Large Rathjen's Ordinal) | $\psi\left(\omega-\pi-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,2)$ |
| 弱提升型差序数函数<br>weak DLON<br>(weak $\delta$-Lifted $\delta$ON)<br>(Aarex) | $\psi\left(\omega-\pi-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,2)$ |
| 提升型 M 记号<br>LMN<br>(Lifting M-Notation)<br>(Test_alpha0, 2021) | $\psi\left(\omega-\pi-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,2)$ |
| 提升型 $\omega$ 记号<br>WMN<br>(Lifting $\omega$-Notation) | $\psi\left(\omega-\pi-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,2)$ |
| Lumi 提升型双行序列系统<br>Lumi's LPSS<br>(Lumi) | $\psi\left(\omega-\pi-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,2)$ |
| 递增元序列[72-74]<br>IUN<br>(Increase Unit Notation)<br>(318'4, 2023) | $\psi\left(\omega-\pi-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,2)$ |
| 急 Hydra 序列<br>Sudden Hydra<br>(Bashicu, 2018) | $\psi\left(\omega-\pi-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,2)$ |
| 2-投影记号<br>2-Projection<br>(Test_alpha0, 2020) | $\psi\left(\omega-\pi-\Pi_0\right)$<br>$(0,0,0)(1,1,1)(2,2,2)$ |

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| 追平函数[111] Catching Function (HypCos, 2014) | $\psi(\omega-\pi-\Pi_0)$ $(0)(1,1,1)(2,2,2)$ |
| pfec $\Sigma_1$ 分离 pfec $\Sigma_1$-Separation | $(0,0,0)(1,1,1)(2,2,2)-$ $(3,2,1)(4,3,2)$ |
| pfec $\Sigma_1$ 稳定 pfec $\Sigma_1$- Stb (pfec $\Sigma_1$- Stable) (76(Nonconvertible)) | $\psi((\omega+1)-\pi-(+1)-\Pi_0)$ $(0,0,0)(1,1,1)(2,2,2)-$ $(3,2,2)(4,2,0)(3,0,0)$ |
| Rathjen 序数折叠函数[112] Rathjen's OCF (Rathjen, 1991) | $(0,0,0)(1,1,1)(2,2,2)-$ $(3,2,2)(4,2,0)(3,0,0)$ |
| 超级急序列 Ultra SSS (Ultra Sudden Sequence System) | $(0,0,0)(1,1,1)(2,2,2)-$ $(3,2,2)(4,2,0)(3,0,0)$ |
| 超强 Aarex 数阵 3+[113] aSAN3+ (Aarex's Super Strong Array Notation) (Aarex) | 约 $(0,0,0)(1,1,1)(2,2,2)-$ $(3,2,2)(4,2,2)(4,0,0)$ |
| 方括号稳定 Bracket Stable (HypCos, 2014) | 约 $(0,0,0)(1,1,1)(2,2,2)-$ $(3,2,2)(4,2,2)(4,2,0)(3,0,0)$ |
| 提升型 K 记号 LKN (Lifting K-Notation) (Test_alpha0, 2020) | $(0,0,0)(1,1,1)(2,2,2)(3,3,0)$ |
| 初级下降算符下降数阵[99] pDDN (primary Dropper Dropping Notation) (HypCos, 2017) | $(0,0,0)(1,1,1)(2,2,2)(3,3,0)$ |
| 3-投影记号 3-Projection (Test_alpha0, 2020) | $(0,0,0)(1,1,1)(2,2,2)(3,3,3)$ |

## B.8 Part VIII

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| PTO $\left(\left(\Pi_2^1-\mathrm{CA}\right)_0\right)$ | 可能 $\leq(0,0,0,0)(1,1,1,1)$ |
| $\Sigma_1$ 稳定 (至 $\omega.\pi$ ) $\Sigma_1$ stb. (up to $\omega.\pi$ ) | 可能 $\leq(0,0,0,0)(1,1,1,1)$ |
| 三行矩阵系统序数 TSSO (Trio Sequence System Ordinal) | $(0,0,0,0)(1,1,1,1)$ |
| 简单投影 Simple Projection (Test_alpha0, 2020) | $(0,0,0,0)(1,1,1,1)$ |
| 锁定 OCF Locked OCF (Bugit) | $(0,0,0,0)(1,1,1,1)$ |

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| 三行矩阵系统[86]<br>TSS<br>(Trio Sequence System)<br>(Bashicu, 2014) | $(0,0,0,0)(1,1,1,1)$ |
| 嵌套函数<br>nestf<br>(NEST-Function)<br>(2020) | $(0,0,0,0)(1,1,1,1)$ |
| 超初等序列 $\psi$<br>HPrSS$\psi$<br>(Hyper Primaritive-<br>Sequence System psi) | $(0,0,0,0)(1,1,1,1)$ |
| PTO $\left(\Pi^1{}_2 - \mathrm{CA} + \mathrm{BI}\right)$ | 可能 $\geq (0,0,0,0)(1,1,1,1)-$<br>$(2,1,1,0)(3,2,2,0)$ |
| $\Sigma_1$ 分离<br>$\Sigma_1$ separation | 可能 $\geq (0,0,0,0)(1,1,1,1)-$<br>$(2,1,1,0)(3,2,2,0)$ |
| 大常规投影序数<br>LSPO<br>(Large Simple Projection Ordinal) | $(0,0,0,0)(1,1,1,1)-$<br>$(2,1,1,1)(3,1,0,0)(2,0,0,0)$ |
| 弱 MCS 投影<br>Weak MCS Projection<br>(Test_alpha0, 2021) | $(0,0,0,0)(1,1,1,1)-$<br>$(2,1,1,1)(3,1,1,1)(4,0,0,0)$ |
| +n 超越序数<br>+nTO<br>((+n)-Transcendental ordinal) | $(0,0,0,0)(1,1,1,1)-$<br>$(2,2,1,1)(3,0,0,0)$ |
| 小超投影记号<br>SHPN<br>(Small Hyper Projection Notation) | $(0,0,0,0)(1,1,1,1)-$<br>$(2,2,1,1)(3,0,0,0)$ |
| 非递归 TON[114]<br>Non-recursive TON<br>(Non-recursive<br>Taranovsky's Ordinal Notation) | $(0,0,0,0)(1,1,1,1)-$<br>$(2,2,1,1)(3,0,0,0)$ |
| Eveog 序数<br>EGO<br>(Eveog's Ordinal)<br>(Eveog, 2023) | $(0,0,0,0)(1,1,1,1)-$<br>$(2,2,1,1)(3,0,0,0)$ |
| Aarex 强 exUNOCF<br>Aarex's Strong exUNOCF<br>(Aarex) | $(0,0,0,0)(1,1,1,1)(2,2,2,0)$ |
| 四行矩阵系统-<br>首个返回齿轮序数<br>Q1BGO<br>(Quadro Sequence System<br>1st Back Gear Ordinal) | $(0,0,0,0)(1,1,1,1)(2,2,2,0)$ |
| Trange Ink 的 ExUNOCF<br>Trange Ink's ExUNOCF<br>(Trangeg Ink, 2023) | $(0,0,0,0)(1,1,1,1)(2,2,2,1)$ |
| 弱 $\alpha$ 序数记号<br>weak $\alpha$ON<br>(weak alhpa Ordinal Notation)<br>(Bugit, 2023) | 可能为<br>$(0,0,0,0)(1,1,1,1)(2,2,2,1)-$<br>$(2,1,1,1)(3,2,2,1)(3,0,0,0)$ |

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| 强提升型差序数函数[115]<br>Strong DLON<br>(Strong $\delta$-Lifted $\delta$ON)<br>(Aarex) | 可能为<br>$(0,0,0,0)(1,1,1,1)-$<br>$(2,2,2,1)(3,0,0,0)$ |
| 大 $\Omega$ 返回序数<br>BOBO<br>(Big Omega Back Ordinal) | $(0,0,0,0)(1,1,1,1)(2,2,2,2)$ |
| 四行矩阵系统[86]<br>QSS<br>(Quadro Sequence System)<br>(Bashicu, 2018) | $(0,0,0,0,0)(1,1,1,1,1)$ |
| 四行矩阵系统序数<br>QSSO<br>(Quadro Sequence System Ordinal) | $(0,0,0,0,0)(1,1,1,1,1)$ |
| $\Sigma_2$ 稳定<br>$\Sigma_2$ Stablilty<br>(Yto, 2021) | $\geq (0,0,0,0,0)(1,1,1,1,1)$ |
| 三重内涵公理序数<br>TCAO<br>(Trio Comprehension Axiom Ordinal) | $\geq (0,0,0,0,0)(1,1,1,1,1)$ |

## B.9 Part IX

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| 理想 BMS[86]<br>IBMS(BM3.3)<br>(Idealized Bashicu Martix System)<br>(Bashicu, 2017) | 可能为 $\mathrm{Y}(1,3)$ |
| 小 Hydra 序数<br>SHO<br>(Small Hydra Ordinal) | $\mathrm{Y}(1,3)$ |
| 扩展 Hydra<br>Ex-Hydra<br>(Extended Hydra) | $\mathrm{Y}(1,3)$<br>(Gomen, 2021) |
| $0-\mathrm{Y}$ 序列[116]<br>$0-\mathrm{Y}$ Sequence<br>(Yukito, 2020) | $\mathrm{Y}(1,3)$ |
| Bashicu 矩阵系统[86]<br>BMS<br>(Bashicu Matrix System) | $\mathrm{Y}(1,3)$<br>(Bashicu, 2014) |
| Carne 矩阵系统<br>CMS<br>(Carne Matrix System)<br>(Test_alpha0) | $\mathrm{Y}(1,3)$ |
| 级层递增元序列[72-74]<br>HIUN<br>(Hierarchial Increase Unit Notation)<br>(318‘4, 2023) | $\mathrm{Y}(1,3)$ |
| 0-基本序数序列[117]<br>0-FOS<br>(0-Fundamental Ordinal Sequence) | $\mathrm{Y}(1,3)$ |

| Notation Name<br>(Chinese Name/English Abbreviation/English Full Name/<br>Proposer/Date Proposed) | Limit |
|---|---|
| 强 (n,0) 投影<br>Strong(n,0)-projection | $\mathrm{Y}(1,3)$ |
| 弱强 OCF<br>Weak Strong OCF | $\mathrm{Y}(1,3)$ |
| 急 BMS[118]<br>BSM<br>(Bashicu Sudden Matrix)<br>(Bashicu, 2018) | 可能为 $\mathrm{Y}(1,3)$ |
| 超 BMS[118]<br>BHM<br>(Bashicu Hyper Matrix)<br>(Bashicu, 2018) | 可能为 $\mathrm{Y}(1,3)$ |
| 相似模式[119]<br>PoR<br>(Patterns of Resemblance)<br>(Clarson, 2001) | $\geq \mathrm{Y}(1,3)$ |
| Arai 序数折叠函数[120]<br>AOCF<br>(Arai's Ordinal Collapsing Function)<br>(Arai, 2023) | $\geq \mathrm{Y}(1,3)$ |
| $\Sigma_n$ 稳定<br>$\Sigma_n$-Stb<br>($\Sigma_n$ Stability) | $\geq \mathrm{Y}(1,3)$ |
| 二阶宇宙序数<br>$\beta$O<br>Beta Universe Ordinal | $\geq \mathrm{Y}(1,3)$ |
| PTO($\mathrm{Z}_2$) | $\geq \mathrm{Y}(1,3)$ |
| 疯狂 Hydra<br>CHN<br>(Crazy Hydra Notation) | $\mathrm{Y}(1,3,4)$<br>(Gomen，2021) |
| K 原始序列<br>KPrSS<br>(摆烂的小猫, 2023) | $\mathrm{Y}(1,3,4,2,5,8,10)$ |
| 超限 BMS[121]<br>TBMS<br>(Transfinite Bashicu Matrix System)<br>(Bubby3) | $\mathrm{Y}(1,3,4,2,5,8,10)$ |
| Ω 行矩阵系统序数<br>ΩSSO<br>(ΩSequence System Ordinal) | $\mathrm{Y}(1,3,4,2,5,8,10)$ |
| TBMS 首次追平 OCF<br>1st time TBMS catches OCF | $\mathrm{Y}(1,3,4,2,5,8,10,-4,9,14,17,10)$ |
| 不可数 TBMS<br>Uncountable TBMS | $\mathrm{Y}(1,3,4,2,5,8,10,6)$ |
| 无降格 Keidonxi 多项式序列[122-123]<br>KPnD<br>Keidonxi's Polynomial Sequence<br>with no debasing<br>(318‘4, 2023) | 可能为<br>$\mathrm{Y}(1,3,4,2,5,8,10,14)$ |
| 弱溅射 TBMS<br>Weak Splatium TBMS<br>(Bubby3) | 可能为<br>$\mathrm{Y}(1,3,4,2,5,8,10,14)$ |

| Notation Name<br>(Chinese Name/English Abbreviation/English Full Name/<br>Proposer/Date Proposed) | Limit |
|---|---|
| Hassium TBMS[122-123]<br>Hassium's TBMS<br>(Hassium, 2020) | $\mathrm{Y}(1,3,4,2,5,8,11)$ |
| Keidonxi 对角化多项式序列<br>KDP<br>(Keidonxi's Diagonalized<br>Polynomial Sequence)<br>(318'4, 2023) | $\mathrm{Y}(1,3,4,2,5,8,11)$ |
| 不可升级 TBMS<br>Upgradeless TBMS | 可能为<br>$\mathrm{Y}(1,3,4,2,5,9)$ |
| 禁戒 Hydra 序数<br>GHO<br>No-go Hydra Ordinal<br>(Asheep, 2023) | $\mathrm{Y}(1,3,4,3)$ |
| 普通 Bubby3 TBMS[121]<br>Bubby3 TBMS Normal<br>(Bubby3, 2018) | $\mathrm{Y}(1,3,4,6)$ |
| 推广 Bubby3 TBMS[121]<br>Bubby3 TBMS Extended<br>(Bubby3, 2018) | $\mathrm{Y}(1,3,5)$ |
| 循环不动点<br>LFP<br>(Loop Fixed Point)<br>(4574) | 可能为 $\mathrm{Y}(1,3,6)$ |
| 无 (1,3,4,2,5,7,5) 升级的<br>Y 序列首次追平 1-Y<br>Y w/o(1,3,4,2,5,7,5)<br>upgrading catches 1-Y | 可能为 $\mathrm{Y}(1,3,7)$ |
| IY 首次追平 1-Y 追平强 Y<br>1st time IY catches 1-Y<br>catches strong Y | 可能为 $\mathrm{Y}(1,3,8)$ |
| $\omega+1$ 行 Y<br>$\omega+1$ row Y | $\mathrm{Y}(1,3,9)$ |
| Discord 猜想的 fffz<br>discord's sus fffz | $\mathrm{Y}(1,3,9)$ |
| VZ 序列<br>VZ Sequence | $\leq \mathrm{Y}(1,\omega)$ |
| Y 序列[116]<br>Y Sequence<br>(Yukito, 2020) | $\mathrm{Y}(1,\omega)$ |
| 小 Y 序列序数<br>SYO<br>(Small Y Sequence Ordinal)<br>(Yukito) | $\mathrm{Y}(1,\omega)$ |
| 强 Y 序列[116]<br>Strong Magma Y Sequence | $\geq \mathrm{Y}(1,\omega)$ |
| $2-\mathrm{Y}$ 序列<br>$2-\mathrm{Y}$ Sequence<br>(Gomen, 2023) | $\omega-\mathrm{Y}(1,5)$ |
| $\omega-\mathrm{Y}$ 序列[124]<br>$\omega-\mathrm{Y}$ Sequence<br>(Yukito, 2020) | $\omega-\mathrm{Y}(1,\omega)$ |

| Notation Name<br>(Chinese Name/English Abbreviation/English Full Name/<br>Proposer/Date Proposed) | Limit |
|---|---|
| 多维 BMS<br>DBMS<br>(Dimensional Bashicu Martix System)<br>(2021) | $\omega - Y(1,\omega)$ |
| 中等 Hydra 序数<br>MHO<br>(Medium Hydra Ordinal) | $\omega - Y(1,\omega)$ |

## B.10 Part X

本部分记号尚未完善，排名不分先后。

| Notation Name<br>(Chinese Name/English Abbreviation/English Full Name/<br>Proposer/Date Proposed) | Limit |
|---|---|
| 超限 DBMS<br>TDBMS<br>(Transfinty Dimensional<br>Bashicu Matrix System)<br>(Y_cpper, 2023) | 理想极限<br>可能在此 |
| Y 矩阵<br>YMS<br>(Yukito Matrix System)<br>(ProjectCF, 2023) | 理想极限<br>可能在此 |
| Crater BMS 矩阵<br>CTBMS<br>(Crater Bashicu Matrix System) | 理想极限<br>可能在此 |
| Yto $Y-Y$ 序列<br>Yto's $Y-Y$<br>(Yto) | 理想极限<br>可能在此 |
| 山脉记号系列<br>MN<br>(Mountain Notation)<br>(HypCos, 2024) | 理想极限<br>可能在此 |
| $X-Y$ 序列[125]<br>$X-Y$ Sequence<br>(Gomen, 2023) | 理想极限<br>可能在此 |
| 变异矩阵系统[126]<br>MM3<br>(Mutant Martix System)<br>(HypCos, 2024) | 理想极限<br>可能在此 |
| $\Omega-Y$ 序列<br>$\Omega-Y$ sequence<br>（未理想） | 理想极限<br>可能在此 |
| 保留 Y 序列<br>RY<br>(Remaining Y Sequence)<br>（未理想） | 理想极限<br>可能在此 |
| 基本序数序列[117,127-128]<br>FOS<br>(Fundamental Ordinal Sequence)<br>(318‘4, 2024)<br>（未理想） | 理想极限<br>可能在此 |

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| 伪伪伪 z（兼容系统）<br>fffz<br>(Fake Fake Fake Zeta)<br>（夏夜星空, 2024）<br>（未理想） | 理想极限<br>可能在此 |

## B.11 Part XI

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| PTO $\left(\left(\Pi_1^2-\mathrm{CA}\right)_0\right)$ | |
| PTO($Z_3$) | |
| Loader 数[129]<br>Loader's Number<br>(Loader, 2001) | 上界为 PTO($Z_\infty$) |
| PTO($Z_\infty$) | |
| PTO(KP + P) | |
| PTO(ZFC) | |
| PTO(ZFC $+n$ 不可达基数)<br>PTO(ZFC+$n$-Inaccessible cardinal) | |
| PTO(ZFC $+\Sigma_\omega$ 反射)<br>PTO(ZFC $+\Sigma_\omega$ Reflecting) | |
| PTO(ZFC + 强 $x$ Mahlo 基数)<br>PTO(ZFC + strongly $x$ -Mahlo cardinal) | |
| 有限承诺游戏[130]<br>(Finite polynomial copy/invert games,FPCI)<br>(Friedman) | 上界为 PTO(ZFC + strongly $x$ -Mahlo cardinal) |
| PTO(ZFC $+\Pi_2^1-$ 不可描述基数)<br>PTO(ZFC $+\Pi_2^1-$ Indescribable) | |
| PTO(ZFC + 完全不可描述基数)<br>PTO(ZFC + Totally-Indescribable) | |
| Friedman 有限树函数[131]<br>Friedman Finite Tree<br>(Friedman) | 上界为 PTO(ZFC+n-subtle cardinal)<br>(Friedman) |
| PTO(ZFC $+n$ 微妙基数)<br>PTO(ZFC $+n$ -subtle cardinal) | |
| PTO(ZFC $+0^\sharp$) | |
| PTO(ZFC $+\omega_1-$ Erdös) | |
| PTO(ZFC + SRP) | |
| 贪心团序列[132]<br>USGDCS<br>(Upper Shift Greedy Down Clique Sequence)<br>(Friedman) | PTO(ZFC+SRP) |
| PTO(ZFC + 可测基数)<br>PTO(ZFC+Measurable cardinal) | |
| PTO(Z2 + PD) | |
| PTO(ZF + ADR) | |
| PTO(ZFC + 超强基数)<br>PTO(ZFC + Superstrong cardinal) | |

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| PTO(ZFC + 超紧基数) PTO(ZFC + Supercompact cardinal) | |
| PTO(ZFC + 巨大基数 +) PTO(ZFC + HUGE + ) | |
| 贪心团序列 2[132] USGDCS2 (Upper Shift Greedy Down Clique Sequence2) | 上界为 PTO(ZFC + HUGE + ) |
| Laver Table[133] (Laver) | 上界为 PTO(ZFC+I3) |
| PTO(ZFC + I3) | |
| PTO(ZFC + I2) | |
| PTO(ZFC + I1) | |
| Laver Table Yarn[134] (test_alpha0,2023) | 上界为 PTO(ZFC + I1) |
| PTO(ZFC + I0) | |

## B.12 Part XII

| Notation Name (Chinese Name/English Abbreviation/English Full Name/ Proposer/Date Proposed) | Limit |
|---|---|
| Church-Kleene 序数[135] CKO (Church-Kleene Ordinal) | $\omega_1^{\mathrm{CK}}$ |
| 忙碌海狸函数[136] BB (Busy Beaver) (Rado, 1962) | $\omega_1^{\mathrm{CK}}$ |
| 调用次数 ToC (Times of Calls) (74(Nonconvertable), 2023) | $(\omega_1^{\mathrm{CK}})^2$ |
| n 阶忙碌海狸函数[136] Level-n BB (Level-n Busy Beaver) (Kleene) | $\omega_n^{\mathrm{CK}}$ |
| Ξ 函数[137] Ξ Function | $\varphi(1,0)^{\mathrm{CK}}$ |
| Aarex 函数[138] Aarex's Function (Aarex) | $\varphi(1,0)^{\mathrm{CK}} + \psi(\Omega_\omega) \cdot \omega$ |
| 无限时间 Turing 机[139] ITTM (Infinite Time Turing Machine) (Hamkins, Lewis, 1998) | $\lambda$ |
| Rayo 函数[140] Rayo's Function (Rayo, 2007) (可能未理想) | 理想极限 可能在此 |

| Notation Name<br>(Chinese Name/English Abbreviation/English Full Name/<br>Proposer/Date Proposed) | Limit |
|---|---|
| 大数花园数[141]<br>LNG<br>Large Number Garden Number<br>(P 進大好き bot, 2019)<br>（可能未理想） | 理想极限<br>可能在此 |
| Davinci 数[142]<br>Davinci103's Number<br>(Davinci103, 2024)<br>（可能未理想） | 理想极限<br>可能在此 |

# Appendix C

# Table of Countable Non-Recursive Ordinals

The content of this section is taken from [143]. This table is updated to 2024.

## C.1 Reflection Ordinals

| Reflection Ordinal |
|---|
| $\omega_1^{\mathrm{CK}} = \Omega$ |
| $\Omega_2$ |
| $\Omega_\omega$ |
| $\Omega_{\omega+1}$ |
| $\Omega_{\omega\cdot 2}$ |
| $\Omega_{\varepsilon_0}$ |
| $\Omega_{\mathrm{Y}(1,3)}$ |
| $\Omega_\Omega$ |
| $\Omega_{\Omega_2}$ |
| $\Omega_{\Omega_\omega}$ |
| $\Omega_{\Omega_\Omega}$ |
| $\Phi(1,0) = \Omega_{\Omega_{\cdots}}$ |
| $\Phi(1,1)$ |
| $\Phi(1,\omega)$ |
| $\Phi(1,\Omega)$ |
| $\Phi(1,\Phi(1,0))$ |
| $\Phi(2,0)$ |
| $\Phi(3,0)$ |
| $\Phi(\omega,0)$ |
| $\Phi(\Omega,0)$ |
| $\Phi(\Phi(1,0),0)$ |
| $\Phi(1,0,0)$ |
| $\Phi(1,0,1)$ |
| $\Phi(1,1,0)$ |
| $\Phi(2,0,0)$ |
| $\Phi(1,0,0,0)$ |
| $\Phi(1,0,0,0,0)$ |
| $\Phi(1@\omega)$ |
| $\Phi(1@\Omega)$ |
| $\Phi(1@\Phi(1,0))$ |
| $\Phi(1@(1,0))$ |
| $\Phi(1@(1@(1,0)))$ |
| $I$ |

| Reflection Ordinal |
|---|
| $\Omega_{I+1}$ |
| $\Omega_{\Omega_{I+1}}$ |
| $\Phi(1, I+1)$ |
| $\Phi(1, 0, I+1)$ |
| $\Phi(1@(I+1))$ |
| $\Phi(1@(1@(I+1)))$ |
| $I_2$ |
| $\Omega_{I_2+1}$ |
| $\Phi\left(1, I_2+1\right)$ |
| $I_2$ |
| $\Omega_{I_2+1}$ |
| $\Phi\left(1, I_2+1\right)$ |
| $I_3$ |
| $I_4$ |
| $I_\omega$ |
| $I_\Omega$ |
| $I_{\Phi(1,0)}$ |
| $I_I$ |
| $I_{I_2}$ |
| $I_{I_\omega}$ |
| $I_{I_I}$ |
| $I_{I_{I_I}}$ |
| $I - \varphi(1, 0)$ |
| $I - \varphi(2, 0)$ |
| $I - \varphi(\omega, 0)$ |
| $I - \varphi(1, 0, 0)$ |
| $I - \varphi(1@\omega)$ |
| $I - \varphi(1@(1, 0))$ |
| $I(1, 0)$ |
| $\Omega_{I(1,0)+1}$ |
| $\Phi(1, I(1, 0)+1)$ |
| $I_{I(1,0)+1}$ |
| $I - \varphi(1, I(1, 0)+1)$ |
| $I - \varphi(1, 0, I(1, 0)+1)$ |
| $I - \varphi(1@(I(1, 0)+1))$ |
| $I(1, 1)$ |
| $I(1, 2)$ |
| $I(1, \omega)$ |
| $I(1, \Omega)$ |
| $I(1, I)$ |
| $I(1, I(1, 0))$ |
| $I(1, 0) - \varphi(1, 0)$ |
| $I(1, 0) - \varphi(1, 0, 0)$ |
| $I(1, 0) - \varphi(1@\omega)$ |
| $I(1, 0) - \varphi(1@(1, 0))$ |
| $I(2, 0)$ |
| $I(2, 1)$ |
| $I(3, 0)$ |
| $I(\omega, 0)$ |
| $I(\Omega, 0)$ |
| $I(I, 0)$ |
| $I(1, 0, 0)$ |
| $I(1, 0, 0, 0)$ |
| $I(1@\omega)$ |
| $I(1@\Omega)$ |

| Reflection Ordinal |
|---|
| $I(1@I)$ |
| $I(1@I(1,0))$ |
| $I(1@(1,0))$ |
| $M$ |
| $\Omega_{M+1}$ |
| $I_{M+1}$ |
| $I(1,M+1)$ |
| $I(1@(M+1))$ |
| $I(1@(1@(M+1)))$ |
| $M_2$ |
| $M_3$ |
| $M_\omega$ |
| $M_\Omega$ |
| $M_I$ |
| $M_M$ |
| $M_{M_M}$ |
| $M-\varphi(1,0)$ |
| $M-\varphi(1,0,0)$ |
| $M-\varphi(1@\omega)$ |
| $M-\varphi(1@(1,0))$ |
| $M-I(1,0)$ |
| $M-I(1,0,0)$ |
| $M-I(1@\omega)$ |
| $M-I(1@(1,0))$ |
| $M(1,0)$ |
| $\Omega_{M(1,0)+1}$ |
| $I_{M(1,0)+1}$ |
| $M_{M(1,0)+1}$ |
| $M-\varphi(1,M(1,0)+1)$ |
| $M-I(1,M(1,0)+1)$ |
| $M(1,1)$ |
| $M(1,2)$ |
| $M(1,\omega)$ |
| $M(1,M)$ |
| $M(1,0)-\varphi(1,0)$ |
| $M(1,0)-I(1,0)$ |
| $M(2,0)$ |
| $M(2,1)$ |
| $M(3,0)$ |
| $M(\omega,0)$ |
| $M(1,0,0)$ |
| $M(1,0,0,0)$ |
| $M(1@\omega)$ |
| $M(1@M)$ |
| $M(1@M(1,0))$ |
| $M(1@(1,0))$ |
| $M(1@(1@(1,0)))$ |
| $N=(1\text{ st})\ 2-2-2$ |
| $N_2=2\text{ nd }2-2-2$ |
| $1-(2-2-2)$ |
| $1-1-(2-2-2)$ |
| $(1-)^{(2)}\ 2-2-2$ |
| $(1-)^{(2\ 1-2)}\ 2-2-2$ |
| $(1-)^{(2-2)}\ 2-2-2$ |
| $(1-)^{(2-2-2)}\ 2-2-2$ |

| Reflection Ordinal |
|---|
| $(1-)^{1,0}\ 2-2-2$ |
| $(1-)^{1,0,0}\ 2-2-2$ |
| $2\ 1-2-2-2$ |
| $(1-)^{1,0}\ 2\ 1-2-2-2$ |
| $2\ 1-(2\ 1-2-2-2)$ |
| $(2\ 1-)^{\omega}\ 2-2-2$ |
| $(2\ 1-)^{1,0}\ 2-2-2$ |
| $2-2\ 1-2-2-2$ |
| $1-(2-2\ 1-2-2-2)$ |
| $(1-)^{1,0}\ 1-(2-2\ 1-2-2-2)$ |
| $2\ 1-(2-2\ 1-2-2-2)$ |
| $(2\ 1-)^{1,0}\ 2-2\ 1-2-2-2$ |
| $2-2\ 1-(2-2\ 1-2-2-2)$ |
| $(2-2\ 1-)^{\omega}\ 2-2-2$ |
| $(2-2\ 1-)^{1,0}\ 2-2-2$ |
| $2-2-2\ 1-2-2-2$ |
| $1-(2-2-2\ 1-2-2-2)$ |
| $2\ 1-(2-2-2\ 1-2-2-2)$ |
| $2-2\ 1-(2-2-2\ 1-2-2-2)$ |
| $2-2-2\ 1-(2-2-2\ 1-2-2-2)$ |
| $(2-2-2\ 1-)^{\omega}$ |
| $(2-2-2\ 1-)^{1,0}$ |
| $2-2-2-2=(2-)^{4}$ |
| $(2-)^{5}$ |
| $(2-)^{6}$ |
| psd. $(2-)^{\omega}$ |
| $(2-)^{\omega}$ |
| $(2-)^{\omega+1}$ |
| $(2-)^{(2)}$ |
| $(2-)^{(2\ 1-2)}$ |
| $(2-)^{(2-2)}$ |
| $(2-)^{(2-2-2)}$ |
| $(2-)^{(2-)^{\omega}}$ |
| $(2-)^{1,0}$ |
| $(2-)^{1,0,0}$ |
| $(2-)^{1@\omega}$ |
| $(2-)^{1@(1,0)}$ |
| $K=(1\text{ st })\Pi_3-$ reflection |
| $K_2$ |
| $K_3$ |
| $K_{\omega}=1-3$ |
| $(1-)^{1,0}\ 3$ |
| $2\ 1-3$ |
| $2-2\ 1-3$ |
| $2-2-2\ 1-3$ |
| $(2-)^{\omega}\ 1-3$ |
| $(2-)^{1,0}\ 1-3$ |
| $3\ 1-3$ |
| $2\ 1-(3\ 1-3)$ |
| $2-2\ 1-(3\ 1-3)$ |
| $2-2-2\ 1-(3\ 1-3)$ |
| $(2-)^{\omega}\ 1-(3\ 1-3)$ |
| $(2-)^{1,0}\ 1-(3\ 1-3)$ |
| $3\ 1-(3\ 1-3)$ |
| $(3\ 1-)^{3}$ |

| Reflection Ordinal |
|---|
| $(3\ 1-)^{\omega}$ |
| $(3\ 1-)^{1,0}$ |
| $(3\ 1-)^{1@\omega}$ |
| $(3\ 1-)^{1@(1,0)}$ |
| $2-3$ |
| $2\ 1-2-3$ |
| $2-2\ 1-2-3$ |
| $(2-)^{\omega}\ 1-2-3$ |
| $(2-)^{1,0}\ 1-2-3$ |
| $3\ 1-2-3$ |
| $(3\ 1-)^{2}\ 2-3$ |
| $(3\ 1-)^{3}\ 2-3$ |
| $(3\ 1-)^{\omega}\ 2-3$ |
| $(3\ 1-)^{1,0}\ 2-3$ |
| $(2-3)\ 1-2-3$ |
| $2\ 1-((2-3)\ 1-2-3)$ |
| $3\ 1-((2-3)\ 1-2-3)$ |
| $3\ 1-(3\ 1-((2-3)\ 1-2-3))$ |
| $(3\ 1-)^{\omega}((2-3)\ 1-2-3)$ |
| $(3\ 1-)^{1,0}((2-3)\ 1-2-3)$ |
| $(2-3\ 1-)^{3}$ |
| $(2-3\ 1-)^{4}$ |
| $(2-3\ 1-)^{\omega}$ |
| $(2-3\ 1-)^{1,0}$ |
| $2-2-3$ |
| $2\ 1-2-2-3$ |
| $3\ 1-2-2-3$ |
| $(2-3)\ 1-2-2-3$ |
| $(2-2-3)\ 1-2-2-3$ |
| $(2-2-3\ 1-)^{\omega}$ |
| $(2-2-3\ 1-)^{1,0}$ |
| $2-2-2-3$ |
| $(2-)^{4}\ 3$ |
| $(2-)^{\omega}\ 3$ |
| $(2-)^{1,0}\ 3$ |
| $3\ 2-3$ |
| $2\ 1-(3\ 2-3)$ |
| $3\ 1-(3\ 2-3)$ |
| $(2-3)\ 1-(3\ 2-3)$ |
| $(2-2-3)\ 1-(3\ 2-3)$ |
| $((2-)^{\omega}\ 3)\ 1-(3\ 2-3)$ |
| $\big((2-)^{1,0}\ 3\big)\ 1-(3\ 2-3)$ |
| $(3\ 2-3)\ 1-(3\ 2-3)$ |
| $(3\ 2-3\ 1-)^{3}$ |
| $(3\ 2-3\ 1-)^{\omega}$ |
| $(3\ 2-3\ 1-)^{1,0}$ |
| $2-(3\ 2-3)$ |
| $(2-(3\ 2-3)\ 1-)^{2}$ |
| $(2-(3\ 2-3)\ 1-)^{3}$ |
| $(2-(3\ 2-3)\ 1-)^{\omega}$ |
| $(2-(3\ 2-3)\ 1-)^{1,0}$ |
| $2-2-(3\ 2-3)$ |
| $2-2-2-(3\ 2-3)$ |
| $(2-)^{\omega}\ (3\ 2-3)$ |
| $(2-)^{1,0}\ (3\ 2-3)$ |

| Reflection Ordinal |
|---|
| $3\ 2-(3\ 2-3)$ |
| $(3\ 2-)^3$ |
| $(3\ 2-)^4$ |
| $(3\ 2-)^\omega$ |
| $(3\ 2-)^{1,0}$ |
| $3-3$ |
| $2\ 1-3-3$ |
| $3\ 1-3-3$ |
| $2-3\ 1-3-3$ |
| $(3\ 2-3)\ 1-3-3$ |
| $((32-)^\omega)\ 1-3-3$ |
| $\left((32-)^{1,0}\right)\ 1-3-3$ |
| $3-3\ 1-3-3$ |
| $(3-3\ 1-)^3$ |
| $(3-3\ 1-)^\omega$ |
| $(3-3\ 1-)^{1,0}$ |
| $2-3-3$ |
| $2-2-3-3$ |
| $(2-)^\omega\ 3-3$ |
| $(2-)^{1,0}\ 3-3$ |
| $3\ 2-3-3$ |
| $(3\ 2-)^2\ 3-3$ |
| $(3\ 2-)^\omega\ 3-3$ |
| $(3\ 2-)^{1,0}\ 3-3$ |
| $3-3\ 2-3-3$ |
| $(3-3\ 2-)^3$ |
| $(3-3\ 2-)^\omega$ |
| $(3-3\ 2-)^{1,0}$ |
| $3-3-3$ |
| $3-3-3-3$ |
| $(3-)^5$ |
| $(3-)^\omega$ |
| $(3-)^{1,0}$ |
| $(3-)^{1@\omega}$ |
| $(3-)^{1@(1,0)}$ |
| $(3-)^{1@(1@(1,0))}$ |
| $\kappa = \Pi_4 -$ reflection |
| $\Pi_5$ |
| $\Pi_6$ |

## C.2 p.f.e.c. $\Sigma_1$-Stable Ordinals

| p.f.e.c. $\Sigma_1$-Stable Ordinal |
|---|
| psd. $\Pi_\omega = \lambda\alpha.(\alpha+1) - \Pi_0$ |
| $\Pi_\omega = \lambda\alpha.(\alpha+1) - \Pi_1$ |
| $\Pi_{\omega+1_{\text{Ideedlized}}} = \lambda\alpha.(\alpha+1) - \Pi_2$ |
| $\Pi_{\omega\cdot 2} = \lambda\alpha.(\alpha+2) - \Pi_1$ |
| $\Pi_{\omega^2} = \lambda\alpha.(\alpha+\omega) - \Pi_1$ |
| $\Pi_\Omega = \lambda\alpha.(\alpha+\Omega) - \Pi_1$ |
| $\Pi_I = \lambda\alpha.(\alpha+I) - \Pi_1$ |
| $\Pi_M = \lambda\alpha.(\alpha+M) - \Pi_1$ |
| $\Pi_K = \lambda\alpha.(\alpha+K) - \Pi_1$ |
| $\Pi_{\Pi_\omega} = \lambda\alpha.\,(\alpha+\Pi_\omega) - \Pi_1 = \lambda\alpha.\,(\alpha+\lambda\alpha.(\alpha+1) - \Pi_1) - \Pi_1$ |

| p.f.e.c. $\Sigma_1$-Stable Ordinal |
|---|
| $\Pi_{1,0} = \lambda\alpha.(\alpha \cdot 2) - \Pi_1$ |
| $\lambda\alpha.(\alpha \cdot 3) - \Pi_1$ |
| $\lambda\alpha.(\alpha \cdot \omega) - \Pi_1$ |
| $\lambda\alpha.(\alpha \cdot \Omega) - \Pi_1$ |
| $\lambda\alpha.\left(\alpha \cdot \lambda\alpha.(\alpha+1) - \Pi_1\right) - \Pi_1$ |
| $\lambda\alpha.\left(\alpha^2\right) - \Pi_1$ |
| $\lambda\alpha.\left(\alpha^3\right) - \Pi_1$ |
| $\lambda\alpha.\left(\alpha^\omega\right) - \Pi_1$ |
| $\lambda\alpha.\left(\alpha^\alpha\right) - \Pi_1$ |
| $\lambda\alpha.\left(\alpha^{\alpha^\alpha}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\varepsilon_{\alpha+1}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\zeta_{\alpha+1}\right) - \Pi_1$ |
| $\lambda\alpha.(\varphi(\omega, \alpha+1)) - \Pi_1$ |
| $\lambda\alpha.(\mathrm{BHO}[\alpha+1]) - \Pi_1$ |
| $\lambda\alpha.(\mathrm{BO}[\alpha+1]) - \Pi_1$ |
| $\lambda\alpha.(\mathrm{SHO}[\alpha+1]) - \Pi_1$ |
| $\lambda\alpha.(\mathrm{PTO(ZFC)}[\alpha+1]) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\alpha+1}\right) - \Pi_1 =$ Non - Gandy |
| $\lambda\alpha.\left(\Omega_{\alpha+1} + 1\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\alpha+1} + \alpha\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\alpha+1} \cdot 2\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\alpha+1} \cdot \omega\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\alpha+1} \cdot \alpha\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\alpha+1}^2\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\alpha+1}^\omega\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\alpha+1}^{\Omega_{\alpha+1}}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\varepsilon_{\Omega_{\alpha+1}+1}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\zeta_{\Omega_{\alpha+1}+1}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\alpha+2}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\alpha+3}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\alpha+\omega}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\alpha\cdot 2}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\alpha^2}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\alpha^\alpha}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\varepsilon_{\alpha+1}}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\Omega_{\alpha+1}}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\Omega_{\Omega_{\alpha+1}}}\right) - \Pi_1$ |
| $\lambda\alpha.(\Phi(1, \alpha+1)) - \Pi_1$ |
| $\lambda\alpha.(\Phi(1, 0, \alpha+1)) - \Pi_1$ |
| $\lambda\alpha.(\Phi(1@(\alpha+1))) - \Pi_1$ |
| $\lambda\alpha.\left(I_{\alpha+1}\right) - \Pi_1$ |
| $\lambda\alpha.\left(M_{\alpha+1}\right) - \Pi_1$ |
| $\lambda\alpha.\left(K_{\alpha+1}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Pi_\omega[\alpha+1]\right) - \Pi_1 = \lambda\alpha.\left(\lambda\beta.(\beta+1) - \Pi_1\right) - \Pi_1$ |
| $\lambda\alpha.\left(\lambda\beta.(\beta+1) - \Pi_1 + \alpha\right) - \Pi_1$ |
| $\lambda\alpha.\left(\lambda\beta.(\beta+1) - \Pi_1 + \Omega_{\alpha+1}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\lambda\beta.(\beta+1) - \Pi_1 + K_{\alpha+1}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\lambda\beta.(\beta+1) - \Pi_1 \cdot 2\right) - \Pi_1$ |
| $\lambda\alpha.\left(\left(\lambda\beta.(\beta+1) - \Pi_1\right)^2\right) - \Pi_1$ |
| $\lambda\alpha.\left(\left(\lambda\beta.(\beta+1) - \Pi_1\right)^{\lambda\beta.(\beta+1)-\Pi_1}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\varepsilon_{\lambda\beta.(\beta+1)-\Pi_1+1}\right) - \Pi_1$ |
| $\lambda\alpha.\left(\Omega_{\lambda\beta.(\beta+1)-\Pi_1+1}\right) - \Pi_1$ |
| $\lambda\alpha.\left(K_{\lambda\beta.(\beta+1)-\Pi_1+1}\right) - \Pi_1$ |

| p.f.e.c. $\Sigma_1$-Stable Ordinal |
|---|
| $\lambda\alpha.\,(2\mathrm{nd}\lambda\beta.(\beta+1)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(3\mathrm{rd}\lambda\beta.(\beta+1)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(1-\lambda\beta.(\beta+1)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(2-\lambda\beta.(\beta+1)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\omega-\lambda\beta.(\beta+1)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,\left((\lambda\beta.(\beta+1)-\Pi_1-)^3\right)-\Pi_1$ |
| $\lambda\alpha.\,((\lambda\beta.(\beta+1)-\Pi_1-)^\omega)-\Pi_1$ |
| $\lambda\alpha.\,\left((\lambda\beta.(\beta+1)-\Pi_1-)^{1,0}\right)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.(\beta+2)-\Pi_0)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.(\beta+2)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.(\beta+2)-\Pi_2)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.(\beta+3)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.(\beta+\omega)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\beta+\lambda\alpha.\,(\lambda\beta.(\beta+1)-\Pi_1)-\Pi_1)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.(\beta+\alpha)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\beta+\Omega_{\alpha+1})-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\beta+K_{\alpha+1})-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\beta+\lambda\beta.(\beta+1)-\Pi_1)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.(\beta\cdot 2)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.(\beta\cdot\omega)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.(\beta\cdot\alpha)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,\left(\lambda\beta.\,\left(\beta^2\right)-\Pi_1\right)-\Pi_1$ |
| $\lambda\alpha.\,\left(\lambda\beta.\,\left(\beta^\beta\right)-\Pi_1\right)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\varepsilon_{\beta+1})-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\Omega_{\beta+1})-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(K_{\beta+1})-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\lambda\gamma.(\gamma+1)-\Pi_1)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\lambda\gamma.\,(\lambda\delta.(\delta+1)-\Pi_1)-\Pi_1)-\Pi_1)-\Pi_1$ |
| $\omega-\pi-\Pi_0=\lambda\alpha.\,(\omega-\pi-\Pi_0)-\Pi_0$ |
| $2-\lambda\alpha.\,(\omega-\pi-\Pi_0)-\Pi_0$ |
| $\omega-\lambda\alpha.\,(\omega-\pi-\Pi_0)-\Pi_0$ |
| $\lambda\alpha.(\alpha+1)-\Pi_0-(\lambda\alpha.\,(\omega-\pi-\Pi_0)-\Pi_0)$ |
| $\lambda\alpha.\,(\omega-\pi-\Pi_0)-\Pi_0-(\lambda\alpha.\,(\omega-\pi-\Pi_0)-\Pi_0)$ |
| $(\lambda\alpha.\,(\omega-\pi-\Pi_0)-\Pi_0-)^3$ |
| $(\lambda\alpha.\,(\omega-\pi-\Pi_0)-\Pi_0-)^\omega$ |
| $(\lambda\alpha.\,(\omega-\pi-\Pi_0)-\Pi_0-)^\alpha$ |
| $(\lambda\alpha.\,(\omega-\pi-\Pi_0)-\Pi_0-)^{\Omega_{\alpha+1}}$ |
| $\lambda\alpha.\,(\omega-\pi-\Pi_0)-\Pi_1$ |
| $\lambda\alpha.\,(\omega-\pi-\Pi_0)-\Pi_2$ |
| $\lambda\alpha.\,(\omega-\pi-\Pi_0+1)-\Pi_1$ |
| $\lambda\alpha.\,(\omega-\pi-\Pi_0+\alpha)-\Pi_1=\lambda\alpha.\,(\lambda\beta.\,(\omega-\pi-\Pi_0)-\Pi_0+\alpha)-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\omega-\pi-\Pi_0)-\Pi_0+\Omega_{\alpha+1})-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\omega-\pi-\Pi_0)-\Pi_0\cdot 2)-\Pi_1$ |
| $\lambda\alpha.\,(2\mathrm{nd}\,\lambda\beta.\,(\omega-\pi-\Pi_0)-\Pi_0)-\Pi_1$ |
| $\lambda\alpha.\,(1-\lambda\beta.\,(\omega-\pi-\Pi_0)-\Pi_0)-\Pi_1$ |
| $\lambda\alpha.\,(\omega-\lambda\beta.\,(\omega-\pi-\Pi_0)-\Pi_0)-\Pi_1$ |
| $\lambda\alpha.\,\left((\lambda\beta.\,(\omega-\pi-\Pi_0)-\Pi_0-)^2\right)-\Pi_1$ |
| $\lambda\alpha.\,\left((\lambda\beta.\,(\omega-\pi-\Pi_0)-\Pi_0-)^3\right)-\Pi_1$ |
| $\lambda\alpha.\,((\lambda\beta.\,(\omega-\pi-\Pi_0)-\Pi_0-)^\omega)-\Pi_1$ |
| $\lambda\alpha.\,((\lambda\beta.\,(\omega-\pi-\Pi_0)-\Pi_0-)^\alpha)-\Pi_1$ |
| $\lambda\alpha.\,\left((\lambda\beta.\,(\omega-\pi-\Pi_0)-\Pi_0-)^{\Omega_{\alpha+1}}\right)-\Pi_1$ |
| $\lambda\alpha.\,\left((\lambda\beta.\,(\omega-\pi-\Pi_0)-\Pi_0-)^{\lambda\beta.(\omega-\pi-\Pi_0)-\Pi_0[\alpha+1]}\right)-\Pi_1$ |

| p.f.e.c. $\Sigma_1$-Stable Ordinal |
|---|
| $\lambda\alpha.\left((\lambda\beta.(\omega-\pi-\Pi_0)-\Pi_0-)^{\beta}\right)-\Pi_1$ |
| $\lambda\alpha.\left((\lambda\beta.(\omega-\pi-\Pi_0)-\Pi_0-)^{\Omega_{\beta+1}}\right)-\Pi_1$ |
| $\lambda\alpha.(\lambda\beta.(\omega-\pi-\Pi_0)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.(\lambda\beta.(\lambda\gamma.(\omega-\pi-\Pi_0)-\Pi_1)-\Pi_1)-\Pi_1$ |
| $\omega-\pi-\Pi_0$ onto $\omega-\pi-\Pi_0$ |
| $\lambda\alpha.(\omega-\pi-\Pi_0$ onto $\omega-\pi-\Pi_0)-\Pi_1$ |
| $\lambda\alpha.\left(\omega-\pi-\Pi_0 \text{ onto } {}^{3}\right)-\Pi_1$ |
| $\lambda\alpha.(\omega-\pi-\Pi_0 \text{ onto } {}^{\omega})-\Pi_1$ |
| $\lambda\alpha.(\omega-\pi-\Pi_0 \text{ onto } {}^{\alpha})-\Pi_1$ |
| $\lambda\alpha.\left(\omega-\pi-\Pi_0 \text{ onto } {}^{\beta}\right)-\Pi_1$ |
| $\lambda\alpha.(\omega-\pi-\Pi_0 \text{ onto } {}^{\gamma})-\Pi_1$ |
| $\lambda\alpha.\left(\omega-\pi-\Pi_0 \text{ onto } {}^{\alpha(\omega)}\right)-\Pi_1$ |
| $\omega-\pi-\Pi_1$ |
| $\omega-\pi-\Pi_0$ onto $\omega-\pi-\Pi_1$ |
| $\omega-\pi-\Pi_0$ onto ${}^{2}\omega-\pi-\Pi_1$ |
| $\omega-\pi-\Pi_0$ onto ${}^{\omega}\omega-\pi-\Pi_1$ |
| $\omega-\pi-\Pi_0$ onto ${}^{\alpha}\omega-\pi-\Pi_1$ |
| $\omega-\pi-\Pi_0$ onto ${}^{\alpha(\omega)}\omega-\pi-\Pi_1$ |
| $\omega-\pi-\Pi_1\omega-\pi-\Pi_0$ onto $\omega-\pi-\Pi_1$ |
| $(\omega-\pi-\Pi_1\omega-\pi-\Pi_0$ onto $)^{2}\,\omega-\pi-\Pi_1$ |
| $(\omega-\pi-\Pi_1\omega-\pi-\Pi_0$ onto $)^{\omega}\,\omega-\pi-\Pi_1$ |
| $(\omega-\pi-\Pi_1\omega-\pi-\Pi_0$ onto $)^{\alpha(\omega)}\,\omega-\pi-\Pi_1$ |
| $\omega-\pi-\Pi_1$ onto $\omega-\pi-\Pi_1$ |
| $\omega-\pi-\Pi_1$ onto ${}^{3}$ |
| $\omega-\pi-\Pi_1$ onto ${}^{\omega}$ |
| $\omega-\pi-\Pi_1$ onto ${}^{\alpha(\omega)}$ |
| $\omega-\pi-\Pi_2$ |
| $\omega-\pi-\Pi_3$ |
| $(\omega+1)-\pi-(+1)-\Pi_0$ |
| $\omega-\pi-\Pi_0$ onto $(\omega+1)-\pi-(+1)-\Pi_0$ |
| $\omega-\pi-\Pi_1$ onto $(\omega+1)-\pi-(+1)-\Pi_0$ |
| $(\omega+1)-\pi-(+1)-\Pi_0$ onto ${}^{2}$ |
| $(\omega+1)-\pi-(+1)-\Pi_0$ onto ${}^{\omega}$ |
| $(\omega+1)-\pi-(+1)-\Pi_0$ onto ${}^{\alpha(\omega)}$ |
| $(\omega+1)-\pi-(+1)-\Pi_1$ |
| $(\omega+1)-\pi-(+1)-\Pi_2$ |
| $(\omega+1)-\pi-(+2)-\Pi_0$ |
| $(\omega+1)-\pi-(+2)-\Pi_1$ |
| $(\omega+1)-\pi-(+\omega)-\Pi_1$ |
| $(\omega+1)-\pi-(+\alpha)-\Pi_1$ |
| $(\omega+1)-\pi-(+\beta)-\Pi_1$ |
| $(\omega+1)-\pi-(\cdot 2)-\Pi_1$ |
| $(\omega+1)-\pi-(\cdot 3)-\Pi_1$ |
| $(\omega+1)-\pi-\left(\Omega_{\alpha(\omega)+1}\right)-\Pi_1$ |
| $(\omega+2)-\pi-(+1)-\Pi_1$ |
| $(\omega+3)-\pi-(+1)-\Pi_1$ |
| $(\omega\cdot 2)-\pi-\Pi_1$ |
| $\omega^2-\pi-\Pi_1$ |
| $\omega^{\omega}-\pi-\Pi_1$ |
| $\Omega-\pi-\Pi_1$ |
| $(\lambda\alpha.(\alpha+1)-\Pi_1)-\pi-\Pi_1$ |
| $(\lambda\alpha.(\lambda\beta.(\beta+1)-\Pi_1)-\pi-\Pi_1)-\pi-\Pi_1$ |
| $\alpha-\pi-\Pi_1$ |
| $\alpha\cdot 2-\pi-\Pi_1$ |

| p.f.e.c. $\Sigma_1$-Stable Ordinal |
|---|
| $\alpha^2 - \pi - \Pi_1$ |
| $\Omega_{\alpha+1} - \pi - \Pi_1$ |
| $(\lambda\beta.(\beta+1) - \Pi_1[\alpha+1]) - \pi - \Pi_1$ |
| $(\lambda\beta.\,(\omega - \pi - \Pi_1) - \Pi_1[\alpha+1]) - \pi - \Pi_1$ |
| $(\lambda\beta.\,(\alpha - \pi - \Pi_1) - \Pi_1[\alpha+1]) - \pi - \Pi_1$ |
| $(\lambda\gamma.\,(\alpha - \pi - \Pi_1) - \Pi_1[\alpha+1]) - \Pi_1[\alpha+1]) - \pi - \Pi_1$ |
| $\beta - \pi - \Pi_1$ |
| $\gamma - \pi - \Pi_1$ |
| $\alpha(\omega) - \pi - \Pi_1$ |
| $\alpha(\Omega) - \pi - \Pi_1$ |
| $\alpha\,(\lambda\alpha.\,(\omega - \pi - \Pi_1) - \Pi_1) - \pi - \Pi_1$ |
| $\alpha\,(\lambda\alpha.\,(\alpha - \pi - \Pi_1) - \Pi_1) - \pi - \Pi_1$ |
| $\alpha\,(\lambda\alpha.\,(\alpha\,(\lambda\alpha.\,(\alpha - \pi - \Pi_1) - \Pi_1) - \pi - \Pi_1) - \Pi_1) - \pi - \Pi_1$ |
| $\alpha(\alpha(0)) - \pi - \Pi_1$ |
| $\alpha(\alpha(0)+1) - \pi - \Pi_1$ |
| $\alpha\left(\Omega_{\alpha(0)+1}\right) - \pi - \Pi_1$ |
| $\alpha\,(\lambda\beta.\,(\alpha(\alpha(0)) - \pi - \Pi_1)\,[\alpha(0)+1]) - \pi - \Pi_1$ |
| $\alpha(\alpha(1)) - \pi - \Pi_1$ |
| $\alpha(\alpha(2)) - \pi - \Pi_1$ |
| $\alpha(\alpha(\omega)) - \pi - \Pi_1$ |
| $\alpha(\alpha(\alpha(0))) - \pi - \Pi_1$ |
| $\alpha(1,0)$ |
| $\alpha(1,1)$ |
| $\alpha(1,\omega)$ |
| $\alpha(1,\alpha)$ |
| $\alpha(1,\beta)$ |
| $\alpha(1,\alpha(\omega))$ |
| $\alpha(1,\alpha(1,0))$ |
| $\alpha(2,0)$ |
| $\alpha(\omega,0)$ |
| $\alpha(1,0,0)$ |
| $\alpha(1,0,0,0)$ |
| $\alpha(1@\omega)$ |
| $\alpha(1@(1,0))$ |
| $\alpha(1@(1@(1,0)))$ |

## C.3 Bracket Stability

| Bracket Stability |
|---|
| $\lambda\alpha.\,(\Pi_3[2]) - \Pi_{1[1\text{--stable}}$ |
| $\lambda\alpha.\,(\Pi_3[2]\Pi_0 \text{ onto } \Pi_3[2]) - \Pi_1$ |
| $\lambda\alpha.\,\left((\Pi_3[2]\Pi_0 \text{ onto })^2\,\Pi_3[2]\right) - \Pi_1$ |
| $\lambda\alpha.\,((\Pi_3[2]\Pi_0 \text{ onto })^\omega\,\Pi_3[2]) - \Pi_1$ |
| $\lambda\alpha.\,((\Pi_3[2]\Pi_0 \text{ onto })^\alpha\,\Pi_3[2]) - \Pi_1$ |
| $\lambda\alpha.\,\left((\Pi_3[2]\Pi_0 \text{ onto })^{\alpha(1,0)}\,\Pi_3[2]\right) - \Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[2]\Pi_2 \text{ onto } \Pi_3[2]) - \Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[2]\Pi_3 \text{ onto } \Pi_3[2]) - \Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[2]\Pi_4) - \Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[2]\Pi_5) - \Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[2]\lambda\beta.(\beta+1) - \Pi_1) - \Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[2]\lambda\beta.\,(\omega - \pi - \Pi_1) - \Pi_1) - \Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[2]\lambda\beta.\,(\Pi_3[2]) - \Pi_1) - \Pi_1$ |

| Bracket Stability |
|---|
| $\lambda\alpha.\,(\Pi_3[2]\lambda\beta.\,(\Pi_3[2]\lambda\gamma.\,(\Pi_3[2])-\Pi_1)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_{0-3}[2])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_{0-{}^2 3}[2])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_{0-{}^w 3}[2])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_{0-{}^\alpha 3}[2])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3 0-3[2])-\Pi_1$ |
| $\lambda\alpha.\,\big(\Pi_{3\ 0-(3\ 0-3)}[2]\big)-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_{3-3}[2])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_{3-3-3}[2])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_{3-\omega}[2])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_{3-\alpha}[2])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_4[2])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_5[2])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_\omega[2])-\Pi_0=\lambda\alpha.\,(\lambda\beta.(\beta+1)-\Pi_0[2])-\Pi_0$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\omega-\pi-\Pi_0)-\Pi_0[2])-\Pi_0$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\lambda\gamma.\,(\omega-\pi-\Pi_0)-\Pi_0[2])-\Pi_0[2])-\Pi_0$ |
| $\omega-\pi-[2]-\Pi_0$ |
| $\omega-\pi-[2]-\Pi_1$ |
| $\omega-\pi-[2]-\Pi_2$ |
| $(\omega+1)-\pi-(+1)-[2]-\Pi_1$ |
| $\Omega-\pi-[2]-\Pi_1$ |
| $(\lambda\alpha.\,(\omega-\pi-\Pi_1)-\Pi_1)-\pi-[2]-\Pi_1$ |
| $(\lambda\alpha.\,(\Pi_3[2])-\Pi_1)-\pi-[2]-\Pi_1$ |
| $(\lambda\alpha.\,(\omega-\pi-[2]-\Pi_1)-\Pi_1)-\pi-[2]-\Pi_1$ |
| $\alpha-\pi-[2]-\Pi_1$ |
| $\alpha(\omega)-\pi-[2]-\Pi_1$ |
| $\alpha(\alpha(0))-\pi-[2]-\Pi_1$ |
| $\alpha(1,0)-\pi-[2]-\Pi_1$ |
| $\alpha(1,0,0)-\pi-[2]-\Pi_1$ |
| $\alpha(1@\omega)-\pi-[2]-\Pi_1$ |
| $\alpha(1@(1,0))-\pi-[2]-\Pi_1$ |
| $\alpha(1@(1@(1,0)))-\pi-[2]-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3]\Pi_0 \text{ onto } \Pi_3[3])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3]\Pi_2 \text{ onto } \Pi_3[3])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3]\Pi_4)-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3]\lambda\beta.(\beta+1)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3]\lambda\beta.\,(\omega-\pi-\Pi_1)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3]\lambda\beta.\,(\Pi_3[2])-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3]\lambda\beta.\,(\omega-\pi-[2]-\Pi_1)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3]\lambda\beta.\,(\Pi_3[3])-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3]\lambda\beta.\,(\Pi_3[3]\lambda\gamma.\,(\Pi_3[3])-\Pi_1)-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3]\Pi_0[2] \text{ onto } \Pi_3[3])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3]\Pi_3[2] \text{ onto } \Pi_3[3])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3]\Pi_4[2])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3]\lambda\beta.(\beta+1)-\Pi_1[2])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3]\omega-\pi-[2]-\Pi_1)-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3]\lambda\beta.\,(\Pi_3[3])-\Pi_1[2])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[3]\lambda\beta.\,(\Pi_3[3]\lambda\gamma.\,(\Pi_3[3])-\Pi_1[2])-\Pi_1[2])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_{0-3}[3])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3 0-3[3])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_{3-3}[3])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_{3-3-3}[3])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_4[3])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_5[3])-\Pi_1$ |

| Bracket Stability |
|---|
| $\lambda\alpha.\,(\lambda\beta.(\beta+1)-\Pi_1[3])-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\omega-\pi-\Pi_1)-\Pi_1[3])-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\omega-\pi-[2]-\Pi_1)-\Pi_1[3])-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\Pi_3[3])-\Pi_1[3])-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\lambda\gamma.\,(\Pi_3[3])-\Pi_1[3])-\Pi_1[3])-\Pi_1$ |
| $\omega-\pi-[3]-\Pi_1$ |
| $\omega-\pi-[4]-\Pi_1$ |
| $\omega-\pi-[5]-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_0[\omega])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_{0-0}[\omega])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_{0-\omega}[\omega])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_{0-\alpha}[\omega])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_3[\omega])-\Pi_1$ |
| $\lambda\alpha.\,(\Pi_4[\omega])-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.(\beta+1)-\Pi_1[\omega])-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\Pi_0[\omega])-\Pi_1[\omega])-\Pi_1$ |
| $\lambda\alpha.\,(\lambda\beta.\,(\lambda\gamma.\,(\Pi_0[\omega])-\Pi_1[\omega])-\Pi_1[\omega])-\Pi_1$ |
| $\omega-\pi-[\omega]-\Pi_0$ |
| $\lambda\alpha.\,(\Pi_0[\omega+1])-\Pi_1$ |
| $\omega-\pi-[\omega+1]-\Pi_0$ |
| $\lambda\alpha.\,(\Pi_0[\omega+2])-\Pi_1$ |
| $\omega-\pi-[\Omega]-\Pi_0$ |
| $\omega-\pi-[\alpha]-\Pi_0$ |
| $\omega-\pi-[\beta]-\Pi_0$ |
| $\omega-\pi-[\alpha(\omega)]-\Pi_0$ |
| $\omega-\pi-[\alpha(1,0)]-\Pi_0$ |
| $\omega-\pi-[\Pi_3[2]]-\Pi_0$ |
| $\omega-\pi-[\Pi_\omega[2]]-\Pi_0$ |
| $\omega-\pi-[\omega-\pi-[2]-\Pi_0]-\Pi_0$ |
| $\omega-\pi-[\omega-\pi-[3]-\Pi_0]-\Pi_0$ |
| $\omega-\pi-[\omega-\pi-[\omega]-\Pi_0]-\Pi_0$ |
| $\omega-\pi-[\omega-\pi-[\omega-\pi-[\omega]-\Pi_0]-\Pi_0]-\Pi_0$ |

## C.4 Projective Ordinals

| Projective Ordinal |
|---|
| $\lambda\alpha\cdot(\lambda\beta\cdot(\Pi_0[\beta])-\Pi_1)-\Pi_1=\psi_\alpha\left(\psi_\beta\left({\alpha_{\beta+1}}^{\beta}\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left({\alpha_{\beta+1}}^{\Omega_{\beta+1}}\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left({\alpha_{\beta+1}}^{\alpha_{\beta+1}}\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\varepsilon_{\alpha_{\beta+1}+1}\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\Omega_{\alpha_{\beta+1}+1}\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\alpha_{\alpha_{\beta+1}}\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\alpha_{\alpha_{\alpha_{\beta+1}}}\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_{\beta_2}\left(\alpha_{\beta_2+1}\cdot\beta\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_{\beta_2}\left({\alpha_{\beta_2+1}}^{2}\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_{\beta_2}\left({\alpha_{\beta_2+1}}^{\alpha_{\beta_2+1}}\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_{\beta_2}\left(\varepsilon_{\alpha_{\beta_2+1}+1}\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_{\beta_2}\left(\Omega_{\alpha_{\beta_2+1}+1}\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_{\beta_2}\left(\alpha_{\alpha_{\beta_2+1}+1}\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_{\beta_3}\left(\alpha_{\beta_3+1}\cdot\beta\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\beta_\omega\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\beta_\alpha\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\beta_\beta\right)\right)$ |

| Projective Ordinal |
|---|
| $\psi_\alpha\left(\psi_\beta\left(\beta_{\beta_\beta}\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_\gamma\left(\beta_{\gamma+1}\cdot\gamma\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_\gamma\left({\beta_{\gamma+1}}^2\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_\gamma\left({\beta_{\gamma+1}}^{\beta_{\gamma+1}}\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_\gamma\left(\varepsilon_{\beta_{\gamma+1}+1}\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_\gamma\left(\Omega_{\beta_{\gamma+1}+1}\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_\gamma\left(\alpha_{\beta_{\gamma+1}+1}\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_\gamma\left(\beta_{\beta_{\gamma+1}+1}\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_\gamma\left(\beta_{\beta_{\beta_{\gamma+1}+1}}\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_\gamma\left(\psi_{\gamma_2}\left(\beta_{\gamma_2+1}\cdot\gamma\right)\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_\gamma\left(\psi_{\gamma_3}\left(\beta_{\gamma_3+1}\cdot\gamma\right)\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_\gamma\left(\gamma_\omega\right)\right)\right)$ |
| $\psi_\alpha\left(\psi_\beta\left(\psi_\gamma\left(\psi_\delta\left(\delta_\omega\right)\right)\right)\right)$ |

## C.5 Admissible Stability and $\Sigma_n$-Stable Ordinals

| Admissible Stability and $\Sigma_n$-Stable Ordinal |
|---|
| $\omega-$ ply $-$ stable $_{\text{Admissible Stability}}$ |
| $\omega-$ ply $-$ stable onto $^2$ |
| $\omega-$ ply $-$ stable onto $^\alpha$ |
| $\omega-$ ply $-$ stable onto $^{\alpha(\omega)}$ |
| $(\omega+1)-$ ply $-(+1)-$ stable |
| $(\omega\cdot 2)-$ ply $-$ stable |
| $\Omega-$ ply $-$ stable |
| $\alpha-$ ply $-$ stable |
| $\alpha(1,0)-$ ply $-$ stable |
| (next $\Pi_3[2]$) $-$ stable $=\Sigma_2-$ Admissible Ordinal |
| ( next $\Pi_4[2]$) $-$ stable |
| ( next $\Pi_\omega[2]$) $-$ stable |
| ( next $\omega-$ ply $-[2]-$ stable ) $-$ stable |
| ( next $\omega-$ ply $-[3]-$ stable ) $-$ stable |
| ( next $\omega-$ ply $-[\omega]-$ stable ) $-$ stable |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\Pi_0[\alpha]$ reflecting. |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\Pi_0[\beta]$ reflecting. |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable . |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and $\alpha$ is $\alpha(1)+1-$ stable . |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and $\alpha$ is $\alpha(\omega)+1-$ stable . |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and satisfies $\lambda\alpha.(\beta)-\Pi_2$ . |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and satisfies $\lambda\alpha\cdot(\alpha(\beta+1))-\Pi_2$ . |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and satisfies $\lambda\alpha.(\alpha(\beta+(\ldots)))-\Pi_2$ . |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and the top of the stability chain is $\Pi_0[\gamma]1-\Pi_0[\gamma]$ . |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and the top of the stability chain is $\Pi_0[\gamma]2-\Pi_0[\gamma]$. |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and the top of the stability chain is $\Pi_0[\gamma]3$ . |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and $\alpha$ is $\gamma+1-$ stable . |

| Admissible Stability and $\Sigma_n$-Stable Ordinal |
|---|
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable, and $\alpha$ is $\beta_1-$ stable, $\beta_1$ is $\beta_1+1-$ stable . |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable, and $\alpha$ is $\beta_1-$ stable, $\beta_1$ is $\gamma_1-\Sigma_2-$ stable. |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable, and $\alpha$ is $\beta_2-$ stable, $\beta_2$ is $\gamma_2-\Sigma_2-$ stable . |
| There exist 3 pairs $\{\beta_n,\gamma_n\}$ such that $\alpha$ is $\beta_n-$ stable, $\beta_n$ is $\gamma_n-\Sigma_2-$ stable. |
| There exist 4 pairs $\{\beta_n,\gamma_n\}$ such that $\alpha$ is $\beta_n-$ stable, $\beta_n$ is $\gamma_n-\Sigma_2-$ stable. |
| There exist $\omega$ pairs $\{\beta_n,\gamma_n\}$ such that $\alpha$ is $\beta_n-$ stable, $\beta_n$ is $\gamma_n-\Sigma_2-$ stable. |
| There exist $\alpha$ pairs $\{\beta_n,\gamma_n\}$ such that $\alpha$ is $\beta_n-$ stable, $\beta_n$ is $\gamma_n-\Sigma_2-$ stable. |
| $n$th $X$ i.e. there exist $n$ pairs $\{\beta_n,\gamma_n\}$ such that $\alpha$ is $\beta_n-$ stable, $\beta_n$ is $\gamma_n-\Sigma_2-$ stable. |
| $\Pi_3$ onto $X$ |
| $\omega-$ ply $-$ stable onto $X$ |
| next $\Sigma_2-$ stable onto $X$ |
| $\alpha'$ onto $X$ , where $\alpha'$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable, and $\alpha'$ is $\alpha'(1)+1-$ stable . |
| $\alpha'$ onto $X$ , where $\alpha'$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable, and $\alpha'$ is $\alpha'(\omega)+1-$ stable . |
| $\alpha'$ onto $X$ , where $\alpha'$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable, and satisfies $\alpha'$ is $\lambda\alpha'(\beta)-\Pi_2$ . |
| $\alpha'$ onto $X$ , where $\alpha'$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable, and the top of the stability chain is $\Pi_0[\gamma]1-\Pi_0[\gamma]$ . |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable, and $\alpha(1)$ is $\gamma+1-$ stable . |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable, and $\alpha(1)$ is $\beta_1-$ stable, $\beta_1$ is $\beta_1+1-$ stable . |
| There exist $\omega$ pairs $\{\beta_n,\gamma_n\}$ such that $\alpha(1)$ is $\beta_n-$ stable, $\beta_n$ is $\gamma_n-\Sigma_2-$ stable. |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable, and $\alpha(2)$ is $\gamma+1-$ stable. |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable, and $\alpha(3)$ is $\gamma+1-$ stable. |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable, and $\beta$ is $\gamma+1-$ stable . |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable, and $\beta$ is $\beta_1-$ stable, $\beta_1$ is $\gamma_1-\Sigma_2-$ stable . |
| $\alpha$ is $\beta-$ stable, and $\beta$ is $\gamma-\Sigma_2-$ stable, and $\beta$ is $\beta_1-$ stable, $\beta_1$ is $\gamma_1-\Sigma_2-$ stable, and $\beta_1$ is $\beta_2-$ stable, $\beta_2$ is $\gamma_2-\Sigma_2-$ stable. |
| There exist 3 triples $\{\beta_n,\beta_{n+1},\gamma_{n+1}\}$ , $\gamma_n\in\beta_{n+1}$ , such that $\alpha$ is $\beta_0-$ stable, $\beta_n$ is $\beta_{n+1}-$ stable, $\beta_{n+1}$ is $\gamma_{n+1}-\Sigma_2-$ stable . |
| There exist 4 triples $\{\beta_n,\beta_{n+1},\gamma_{n+1}\}$ , $\gamma_n\in\beta_{n+1}$ , such that $\alpha$ is $\beta_0-$ stable, $\beta_n$ is $\beta_{n+1}-$ stable, $\beta_{n+1}$ is $\gamma_{n+1}-\Sigma_2-$ stable . |
| There exist $\omega$ triples $\{\beta_n,\beta_{n+1},\gamma_{n+1}\}$, $\gamma_n\in\beta_{n+1}$ , such that $\alpha$ is $\beta_0-$ stable, $\beta_n$ is $\beta_{n+1}-$ stable, $\beta_{n+1}$ is $\gamma_{n+1}-\Sigma_2-$ stable . |
| There exist $(1,0)$ triples $\{\beta_n,\beta_{n+1},\gamma_{n+1}\}$ , $\gamma_n\in\beta_{n+1}$ , such that $\alpha$ is $\beta_0-$ stable, $\beta_n$ is $\beta_{n+1}-$ stable, $\beta_{n+1}$ is $\gamma_{n+1}-\Sigma_2-$ stable . |

| Admissible Stability and $\Sigma_n$-Stable Ordinal |
|---|
| $\alpha$ is $\beta-$ stable, $\beta$ is $(\gamma,\gamma')-2-o-\Sigma_2-$ stable. |
| $\alpha$ is $\beta-$ stable, $\beta$ is $(\gamma,\gamma')-2-o-\Sigma_2-$ stable, and $\gamma'\in\beta_1$ , $\beta$ is $\beta_1-$ stable, $\beta_1$ is $(\gamma_1,\gamma_1')-2-o-\Sigma_2-$ stable . |
| There exist 3 triples $\left\{\beta_n,\beta_{n+1},\gamma_{n+1}'\right\}$ ,$\gamma_n'\in\beta_{n+1}$ , such that $\alpha$ is $\beta_0-$ stable, $\beta_n$ is $\beta_{n+1}-$ stable, $\beta_{n+1}$ is $\left(\gamma_n,\gamma_{n+1}'\right)-2-o-\Sigma_2-$ stable . |
| There exist 4 triples $\left\{\beta_n,\beta_{n+1},\gamma_{n+1}'\right\}$ , $\gamma_n'\in\beta_{n+1}$ , such that $\alpha$ is $\beta_0-$ stable, $\beta_n$ is $\beta_{n+1}-$ stable, $\beta_{n+1}$ is $\left(\gamma_n,\gamma_{n+1}'\right)-2-o-\Sigma_2-$ stable . |
| There exist $\omega$ triples $\left\{\beta_n,\beta_{n+1},\gamma_{n+1}'\right\}$ ,$\gamma_n'\in\beta_{n+1}$ , such that $\alpha$ is $\beta_0-$ stable, $\beta_n$ is $\beta_{n+1}-$ stable, $\beta_{n+1}$ is $\left(\gamma_n,\gamma_{n+1}'\right)-2-o-\Sigma_2-$ stable . |
| There exist $(1,0)$ triples $\left\{\beta_n,\beta_{n+1},\gamma_{n+1}'\right\}$ ,$\gamma_n'\in\beta_{n+1}$ , such that $\alpha$ is $\beta_0-$ stable, $\beta_n$ is $\beta_{n+1}-$ stable, $\beta_{n+1}$ is $\left(\gamma_n,\gamma_{n+1}'\right)-2-o-\Sigma_2-$ stable . |
| $\alpha$ is $\beta$ - stable, $\beta$ is $3-o-\Sigma_2-$ stable. |
| $\alpha$ is $\beta-$ stable, $\beta$ is $4-o-\Sigma_2-$ stable . |
| $\alpha$ is $\beta$ - stable, $\beta$ is $\omega-o-\Sigma_2-$ stable. |
| $\alpha$ is $\beta-$ stable, $\beta$ is $\alpha-o-\Sigma_2-$ stable. |
| $\alpha$ is $\beta-$ stable, $\beta$ is $(1\text{ st }\Sigma_2-\tau)-\mathrm{o}-\Sigma_2-$ stable . |
| $\alpha$ is $\beta-$ stable, $\beta$ is $(2\text{ nd }\Sigma_2-\tau)-\mathrm{o}-\Sigma_2-$ stable . |
| Let $\gamma_n$ be the $1+n$th $\Sigma_2$ stable target. $\alpha$ is $\beta-$ stable, $\beta$ is $(1,0\text{ th }\Sigma_2-\tau)-\mathrm{o}-\Sigma_2-$ stable, i.e. $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-o-\Sigma_2-$ stable. |
| $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_2-$ stable, and $\gamma\in\Pi_3$ . |
| $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_2-$ stable, and $\gamma\in\Pi_4$ . |
| $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_2-$ stable, and $\gamma$ is $\gamma+1-$ stable . |
| $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_2-$ stable, and $\gamma$ is $\beta_1-$ stable, $\beta_1$ is $\gamma_1-\Sigma_2-$ stable . |
| There exist 3 pairs $\{\beta_n,\gamma_n\}$ , such that $\alpha$ is $\beta_0-$ stable, $\beta_n$ is $\gamma_n-\Sigma_2-$ stable, $\gamma_n$ is $\beta_{n+1}-$ stable . |
| There exist $\omega$ pairs $\{\beta_n,\gamma_n\}$ , such that $\alpha$ is $\beta_0-$ stable, $\beta_n$ is $\gamma_n-\Sigma_2-$ stable, $\gamma_n$ is $\beta_{n+1}-$ stable . |
| There exist $(1,0)$ pairs $\{\beta_n,\gamma_n\}$ , such that $\alpha$ is $\beta_0-$ stable, $\beta_n$ is $\gamma_n-\Sigma_2-$ stable, $\gamma_n$ is $\beta_{n+1}-$ stable . |
| $\alpha$ is $\beta-$ stable, $\beta$ is $(\gamma,\gamma')-\Sigma_2-$ stable, and $\gamma$ is $\gamma'-$ stable, i.e. $\alpha\to\beta\to_2(\gamma,\gamma'),\gamma\to\gamma'$ . |
| $\alpha$ is $\beta-$ stable, $\beta$ is $(\gamma,\gamma',\gamma'')-\Sigma_2-$ stable, and $\gamma$ is $\gamma''-$ stable, i.e. $\alpha\to\beta\to_2(\gamma,\gamma',\gamma''),\gamma\to\gamma'\to\gamma''$ . |
| $\omega-$ ply of the above $\Sigma_2-\tau$ stability hierarchy. |
| $\alpha-$ ply of the above $\Sigma_2-\tau$ stability hierarchy. |
| $\beta-$ ply of the above $\Sigma_2-\tau$ stability hierarchy. |
| $\gamma-$ ply of the above $\Sigma_2-\tau$ stability hierarchy. |
| $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_2-$ stable, and $\gamma$ is $\delta-\Sigma_2-$ stable . i.e. $\alpha\to\beta\to_2\gamma\to_2\delta$ . |
| $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_2-$ stable, and $\gamma$ is $2-\mathrm{o}-\Sigma_2-$ stable. $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_2-$ stable, and $\gamma$ is $\omega-o-\Sigma_2-$ stable. |
| $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_2-$ stable, and $\gamma$ is $\delta-\Sigma_2-$ stable, and $\delta\in\Pi_3$ . |
| $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_2-$ stable, and $\gamma$ is $\delta-\Sigma_2-$ stable, and $\delta\in\Pi_4$ . |

| Admissible Stability and $\Sigma_n$-Stable Ordinal |
|---|
| $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and $\gamma$ is $\delta-\Sigma_2-$ stable, and $\delta$ is $\delta+1-$ stable . |
| There exists 1 pair $\{\delta_n,\varepsilon_n\}$ such that $\alpha$ is $\beta-$ stable,<br>$\beta$ is $\gamma-\Sigma_2-$ stable, and $\gamma$ is $\delta_0-\Sigma_2-$ stable. |
| $\delta_n$ is $\varepsilon_n-$ stable, $\varepsilon_n$ is $\delta_{n+1}-$ stable .<br>i.e. $\alpha\to\beta\to_2\gamma\to_2\delta\to\varepsilon\to_2\delta_1$ . |
| There exist $\omega$ pairs $\{\delta_n,\varepsilon_n\}$ such that $\alpha$ is $\beta-$ stable,<br>$\beta$ is $\gamma-\Sigma_2-$ stable, and $\gamma$ is $\delta_0-\Sigma_2-$ stable. |
| $\delta_n$ is $\varepsilon_n-$ stable, $\varepsilon_n$ is $\delta_{n+1}-$ stable . |
| There exist $(1,0)$ pairs $\{\delta_n,\varepsilon_n\}$ such that $\alpha$ is $\beta-$ stable,<br>$\beta$ is $\gamma-\Sigma_2-$ stable, and $\gamma$ is $\delta_0-\Sigma_2-$ stable. |
| $\delta_n$ is $\varepsilon_n-$ stable, $\varepsilon_n$ is $\delta_{n+1}-$ stable . |
| $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and $\gamma$ is $\delta-\Sigma_2-$ stable, and $\delta$ is $\varepsilon-$ stable,<br>$\varepsilon$ is $(\delta_1,\delta_1')-\Sigma_2-$ stable, and $\delta_1$ is $\delta_1'-$ stable ,<br>i.e. $\alpha\to\beta\to_2\gamma\to_2\delta\to\varepsilon\to_2(\delta_1,\delta_1'),\delta_1\to\delta_1'$ . |
| $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and $\gamma$ is $\delta-\Sigma_2-$ stable, and $\delta$ is $\varepsilon-$ stable,<br>$\varepsilon$ is $(\delta_1,\delta_1',\delta_1'')-\Sigma_2-$ stable, and $\delta_1$ is $\delta_1''-$ stable,<br>i.e. $\alpha\to\beta\to_2\gamma\to_2\delta\to\varepsilon\to_2(\delta_1,\delta_1',\delta_1''),\delta_1\to\delta_1'\to\delta_1''$ . |
| $\omega-$ ply of the above $\Sigma_2-\tau$ stability hierarchy. |
| $\gamma-$ ply of the above $\Sigma_2-\tau$ stability hierarchy. |
| $\delta-$ ply of the above $\Sigma_2-\tau$ stability hierarchy. |
| $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and $\gamma$ is $\delta-\Sigma_2-$ stable, and $\delta$ is $\varepsilon-$ stable,<br>$\varepsilon$ is $\delta_1-\Sigma_2-$ stable, and $\delta_1$ is $\delta_1'-\Sigma_2-$ stable.<br>i.e. $\alpha\to\beta\to_2\gamma\to_2\delta\to\varepsilon\to_2\delta_1\to_2\delta_1'$ . |
| There exist 2 triples $\left\{\varepsilon_n,\delta_{n+1},\delta_{n+1}'\right\}$ ,<br>such that $\alpha$ is $\beta$-stable, $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and $\gamma$ is $\delta_0-\Sigma_2-$ stable, $\delta_0$ is $\varepsilon_0-$ stable,<br>$\varepsilon_n$ is $\delta_{n+1}-\Sigma_2-$ stable, $\delta_{n+1}$ is $\delta_{n+1}'-\Sigma_2-$ stable . |
| There exist $\omega$ triples $\left\{\varepsilon_n,\delta_{n+1},\delta_{n+1}'\right\}$ such that $\alpha$ is $\beta-$ stable,<br>$\beta$ is $\gamma-\Sigma_2-$ stable, and $\gamma$ is $\delta_0-\Sigma_2-$ stable,<br>$\delta_0$ is $\varepsilon_0-$ stable, $\varepsilon_n$ is $\delta_{n+1}-\Sigma_2-$ stable,<br>$\delta_{n+1}$ is $\delta_{n+1}'-\Sigma_2-$ stable . |
| There exist $(1,0)$ triples $\left\{\varepsilon_n,\delta_{n+1},\delta_{n+1}'\right\}$ ,<br>such that $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and $\gamma$ is $\delta_0-\Sigma_2-$ stable, $\delta_0$ is $\varepsilon_0-$ stable,<br>$\varepsilon_n$ is $\delta_{n+1}-\Sigma_2-$ stable, $\delta_{n+1}$ is $\delta_{n+1}'-\Sigma_2-$ stable. |
| $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and $\gamma$ is $(\delta,\delta')-\Sigma_2-$ stable, and $\delta$ is $\delta'-$ stable .<br>i.e. $\alpha\to\beta\to_2\gamma\to_2(\delta,\delta'),\delta\to\delta'$ . |
| $\alpha$ is $\beta$ - stable, $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and $\gamma$ is $(\delta,\delta',\delta'')-\Sigma_2-$ stable, and $\delta$ Continuing translation from the last line $\alpha$ is $\beta$ - stable, $\beta$ is $\gamma-\Sigma_2-$ stable,<br>and $\gamma$ is $(\delta,\delta',\delta'')-\Sigma_2-$ stable, and $\delta$ is $\delta''-$ stable .<br>i.e. $\alpha\to\beta\to_2\gamma\to_2(\delta,\delta',\delta''),\delta\to\delta'\to\delta''$ . |
| $\omega-$ ply of the above $\Sigma_2-\tau$ stability hierarchy. |
| $\alpha-$ ply of the above $\Sigma_2-\tau$ stability hierarchy. |
| $\beta-$ ply of the above $\Sigma_2-\tau$ stability hierarchy. |
| $\gamma-$ ply of the above $\Sigma_2-\tau$ stability hierarchy. |
| $\delta-$ ply of the above $\Sigma_2-\tau$ stability hierarchy. |
| $4-$ ply $-\Sigma_2-$ stable |
| $5-$ ply $-\Sigma_2-$ stable |
| $6-$ ply $-\Sigma_2-$ stable |
| $\omega-$ ply $-\Sigma_2-$ stable |
| $(\omega+1)-$ ply $-\Sigma_2-$ stable |

| Admissible Stability and $\Sigma_n$-Stable Ordinal |
|---|
| $\alpha-$ ply $-\Sigma_2-$ stable |
| $\alpha(\alpha(0))-$ ply $-\Sigma_2-$ stable |
| $\alpha(1,0)-$ ply $-\Sigma_2-$ stable |
| $\alpha(1,0,0)-$ ply $-\Sigma_2-$ stable |
| $\alpha(1@\omega)-$ ply $-\Sigma_2-$ stable |
| $\alpha(1@(1,0))-$ ply $-\Sigma_2-$ stable |
| $\alpha(1@(1@(1,0)))-$ ply $-\Sigma_2-$ stable |
| $\lambda\alpha\cdot(\Pi_4[2]\,[\Sigma_2])-\Pi_1$ |
| $\lambda\alpha\cdot(\Pi_5[2]\,[\Sigma_2])-\Pi_1$ |
| $\lambda\alpha\cdot(\Pi_\omega[2]\,[\Sigma_2])-\Pi_1$ |
| $\omega-\pi-\Sigma_2-[2]-\Pi_1$ |
| $\omega-\pi-\Sigma_2-[3]-\Pi_1$ |
| $\omega-\pi-\Sigma_2-[\omega]-\Pi_1$ |
| $\omega-\pi-\Sigma_2-[\omega-\pi-\Sigma_2-[\omega]-\Pi_1]-\Pi_1$ |
| $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_3-$ stable. |
| $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_4-$ stable. |
| $\alpha$ is $\beta-$ stable, $\beta$ is $\gamma-\Sigma_5-$ stable. |

## C.6 Gap Ordinals and Elementary Embeddings

| Gap Ordinals and Elementary Embeddings |
|---|
| $\sup\{\Sigma_n-\text{adm}\mid n\in\omega\}$ |
| real. $\Sigma_\omega-\text{adm}$,<br>i.e. $\forall n\in\omega,\alpha\in\Sigma_n-\text{adm}$ .<br>It satisfies $L_{\alpha+1}\models(\alpha=\omega_1)$ ,<br>thus it is $\omega_1$ in $L_{\alpha+1}$ , denoted as $\omega_1^{L_{\alpha+1}}$ .<br>$\omega_1^{L_{\alpha+1}}$ is a Gap Ordinal of length 1. |
| Elementary embedding $j:L_\alpha\to L_\beta$ ,<br>slightly larger than $\omega_1^{L_{\alpha+1}}$ . |
| Elementary embeddings $j:L_\alpha\to L_\beta$ and $j_1:L_\alpha\to L_{\beta_1}$ ,<br>i.e. there exist elementary embeddings from $L_\alpha$ to two different $L_\beta$.<br>Such embeddings correspond to $\Sigma_{\omega\times 2}$ stability. |
| Elementary embeddings $j:L_\alpha\to L_{\beta_n},n<4$ . |
| Elementary embeddings $j:L_\alpha\to L_{\beta_n},n<\omega$ . |
| Elementary embedding $j_0:L_\alpha\to L_\beta$ ,<br>and for $\beta$ there exists an elementary embedding $j_1:L_\beta\to L_\gamma$. $2-$ply. |
| Elementary embedding $j_0:L_\alpha\to L_\beta$ ,<br>and elementary embedding $j_1:L_\beta\to L_\gamma$ ,<br>and elementary embedding $j_2:L_\gamma\to L_\delta$ . $3-$ ply . |
| $\omega-$ ply of elementary embeddings $j:L_\alpha\to L_\beta$ . |
| $(1,0)-$ ply of elementary embeddings $j:L_\alpha\to L_\beta$ . |
| $\Pi_3[2]$ of elementary embeddings $j:L_\alpha\to L_\beta$ . |
| $\omega-[2]-$ ply of elementary embeddings $j:L_\alpha\to L_\beta$ . |
| ..... |
| $\alpha$ satisfies $L_{\alpha+2}\models(\alpha=\omega_1)$ ,<br>i.e. $\omega_1^{L_{\alpha+2}}$ ,<br>and is a Gap Ordinal of length 2. |
| Elementary embedding $j:L_{\alpha+1}\to L_{\beta+1}$ , slightly larger than $\omega_1^{L_{\alpha+2}}$ . |
| Elementary embedding $j_0:L_{\alpha+1}\to L_{\beta+1}$ ,<br>and elementary embedding $j_1:L_\beta\to L_\gamma$ ,<br>similar to the stability chain $\alpha\to_2\beta\to\gamma$ . |
| Elementary embeddings $j:L_{\alpha+1}\to L_{\beta_n+1},n<\omega$ . |
| $2-$ ply of elementary embeddings $j:L_{\alpha+1}\to L_{\beta+1}$ . |

| Gap Ordinals and Elementary Embeddings |
|---|
| $\omega-$ ply of elementary embeddings $j : L_{\alpha+1} \to L_{\beta+1}$ . |
| $\omega - [2]-$ ply of elementary embeddings $j : L_{\alpha+1} \to L_{\beta+1}$ . |
| ..... |
| $\omega_1^{L_{\alpha+3}}$ ,<br>a Gap Ordinal of length 3. |
| Elementary embedding $j : L_{\alpha+2} \to L_{\beta+2}$ . |
| $\omega - [2]-$ ply of elementary embeddings $j : L_{\alpha+2} \to L_{\beta+2}$ . |
| ..... |
| $\omega_1^{L_{\alpha+\omega}}$ |
| $\omega_1^{L_{\alpha+\varepsilon_0}}$ |
| $\omega_1^{L_{\alpha+\Omega}}$ |
| $\omega_1^{L_{\alpha+\Sigma_\omega-\mathrm{adm}}}$ |
| $\omega_1^{L_{\alpha\times 2}}$ ,<br>a Gap Ordinal of length $\alpha$.<br>Here $L_{\alpha+\alpha} \models \alpha = \omega_1$ ,<br>the elementary embedding changes accordingly: $j : L_\alpha \to L_{\omega_1}$ |
| $\omega_1^{L_{\alpha\times 2+1}}$ |
| Elementary embedding $j : L_{\alpha+1} \to L_{\omega_1+1}$ |
| $\omega_1^{L_{\alpha\times 2+2}}$ |
| Elementary embedding $j : L_{\alpha+2} \to L_{\omega_1+2}$ |
| ..... |
| $\omega_1^{L_{\alpha\times\omega}}$ |
| $\omega_1^{L_{\alpha\times\Omega}}$ |
| $\omega_1^{L_{\alpha^2}}$ |
| $\omega_1^{L_{\alpha^\alpha}}$ |
| $\omega_1^{L_{\varepsilon_{\alpha+1}}}$ |
| $\omega_1^{L_{\Omega_{\alpha+\omega}}}$ .<br>Let $\beta = \Omega_{\alpha+\omega}$ ,<br>then $L_\beta \models \alpha = \omega_1$<br>and $\beta \in \Pi_1$ onto $\Sigma_1 -$ adm.<br>Corresponds to proof-theoretic ordinal: PTO $\left(\Pi_1^2 - \mathrm{CA}_0\right)$ . |
| $L_\beta \models \alpha = \omega_1$ and $\beta$ is $\omega-$ ply - stable,<br>corresponds to proof-theoretic ordinal: PTO $\left(\Pi_2^2 - \mathrm{CA}_0\right)$ . |
| $L_\beta \models \alpha = \omega_1$ and $\beta \in \Sigma_2 -$ adm |
| $L_\beta \models \alpha = \omega_1$ and $\beta$ is $\omega-$ ply $-\Sigma_2-$ stable,<br>corresponds to proof-theoretic ordinal: PTO $\left(\Pi_3^2 - \mathrm{CA}_0\right)$ . |
| $L_\beta \models \alpha = \omega_1$ and $\beta \in \Sigma_3 -$ adm |
| ..... |
| $L_\beta \models \alpha = \omega_1$ and $\beta \in \Sigma_\omega -$ adm.<br>This $\beta$ witnesses the existence of an $\omega_1$ below it,<br>and itself is $\Sigma_\omega -$ adm, hence $L_{\beta+1} \models \beta = \omega_2$.<br>(Using $\alpha$ again) Thus we get $\omega_2^{L_{\alpha+1}}$ . |
| $\omega_2^{L_{\alpha+1}}$ ,<br>corresponds to proof-theoretic ordinal PTO $(\mathrm{Z}_3)$ . |
| Elementary embedding $j : L_\alpha \to L_\beta$ and $\alpha$ is $\omega_2^{L_{\alpha+1}}$ .<br>Such an embedding is denoted as $j : L_\alpha \to_{\omega_2} L_\beta$. |
| Elementary embedding $j : L_\alpha \to_{\omega_2} L_\beta$ and elementary embedding $j : L_\beta \to L_\gamma$,<br>$2-$ ply in the $\omega_2$ sense. |
| $\omega-$ ply of elementary embeddings $j : L_\alpha \to_{\omega_2} L_\beta$ |
| ..... |
| $\omega_2^{L_{\alpha+2}}$ |
| $\omega_2^{L_{\alpha+\omega}}$ |
| $\omega_2^{L_{\alpha\times 2}}$ |

| Gap Ordinals and Elementary Embeddings |
|---|
| Elementary embedding $j : L_\alpha \to L_{\omega_2}$ |
| $\omega_2^{L_{\alpha\times 2+1}}$ |
| $\omega_2^{L_{\alpha^2}}$ |
| $\omega_2^{L_{\Omega_{\alpha+1}}}$ |
| $L_\beta \models \alpha = \omega_2$ and $\beta \in \Sigma_1 - \mathrm{adm}$ |
| $L_\beta \models \alpha = \omega_2$ and $\beta \in \Sigma_2 - \mathrm{adm}$ |
| $L_\beta \models \alpha = \omega_2$ and $\beta \in \Sigma_3 - \mathrm{adm}$ |
| $\omega_3^{L_{\alpha+1}}$.<br>Corresponds to proof-theoretic ordinal PTO $(\mathrm{Z}_4)$ |
| $\omega_4^{L_{\alpha+1}}$.<br>Corresponds to proof-theoretic ordinal PTO $(\mathrm{Z}_5)$ |
| $\omega_\omega^{L_{\alpha+1}}$.<br>Corresponds to proof-theoretic ordinal PTO $(\mathrm{Z}_\omega)$ |
| $\omega_{\omega_1}^{L_{\alpha+1}}$ |
| $\omega_{\omega_2}^{L_{\alpha+1}}$ |
| $\omega_{\omega_\omega}^{L_{\alpha+1}}$ |
| $\omega_{(1,0)}^{L_{\alpha+1}}$ |
| $\omega_{(1,0,0)}^{L_{\alpha+1}}$ |
| $\omega_{(1@\omega)}^{L_{\alpha+1}}$ |
| ..... |
| The power admissible cardinal in $L_{\alpha+1}$,<br>i.e. $(\Sigma_1 - \mathrm{WC})^{L_{\alpha+1}}$ |
| The cardinal stability hierarchy in $L_{\alpha+1}$,<br>i.e. $(V_\alpha \to_{\Sigma_1} V_\beta)^{L_\alpha+1}$ |
| The $\Sigma_2$ worldly cardinal in $L_{\alpha+1}$,<br>i.e. $(\Sigma_2 - \mathrm{WC})^{L_{\alpha+1}}$ |
| The cardinal $\Sigma_2$ stability hierarchy in $L_{\alpha+1}$,<br>i.e. $(V_\alpha \to_{\Sigma_2} V_\beta)^{L_{\alpha+1}}$ |
| ..... |
| The worldly cardinal in $L_{\alpha+1}$.<br>Corresponds to proof-theoretic ordinal: PTO(ZFC) .<br>At the same time, it is also an inaccessible cardinal in $L_{\alpha+1}$. |
| $\Sigma_1-$ totally stable |
| $\Sigma_2-$ totally stable |
| $\Sigma_3-$ totally stable |
| $\Sigma_\omega-$ totally stable |
| ..... |
| $\omega_1^L$, $\omega_1$ in $L$. |
| ..... |
| 2 worldly cardinals in $L_{\alpha+1}$.<br>Corresponds to proof-theoretic ordinal:<br>PTO(ZFC+ there is a worldly cardinal ) . |
| $\omega$ worldly cardinals in $L_{\alpha+1}$.<br>Corresponds to proof-theoretic ordinal:<br>PTO(ZFC+ there is a proper class of worldly cardinals). |
| ..... |
| Worldly cardinal in $L_{\alpha+2}$. |
| Worldly cardinal in $L_{\alpha\times 2}$. |
| Elementary embedding $j : V_\alpha \to V_\beta$ in $L_{\alpha+1}$ |
| $\omega-$ ply of elementary embeddings $j : V_\alpha \to V_\beta$ in $L_{\alpha+1}$ |
| Inaccessible cardinal in $L_{\alpha+2}$ |
| Elementary embedding $j : V_{\alpha+1} \to V_{\beta+1}$ in $L_{\alpha+2}$ |
| Inaccessible cardinal in $L_{\alpha+\omega}$ |

| Gap Ordinals and Elementary Embeddings |
|---|
| Elementary embedding $j : V_{\alpha+1} \to V_{\beta+1}$ in $L_{\alpha+2}$ |
| Elementary embedding $j : V_{\alpha+1} \to V_{\beta+1}$ in $L_{\alpha+2}$ |
| Mahlo cardinal in $L_{\alpha+1}$. |
| Weakly compact cardinal in $L_{\alpha+n}$.<br>(It at least belongs to $L_{\alpha+\omega}$). |
| Various indescribable cardinals in $L_{\alpha+n}$. |
| ..... |
| Various large cardinals in $L_{\alpha+n}$. |
| ..... |
| The critical point of a non-trivial elementary embedding $j : L \to L$. |

# Appendix D

# Table of Proof-Theoretic Ordinals

The content of this table is taken from [33,144-145]. The first column lists ordinals expressed using OCFs or other notations; the parts outside the square brackets are the notations from the original table, as conventions vary greatly across papers; the parts inside the square brackets are conversions to the MOCF notation commonly used in googology, and may be imprecise. Due to space limitations, sometimes we need to split a single longer axiom to write it. If the end of the previous line and the beginning of the next line in the same cell are connected by $-$ , they represent the same name of the axiomatic system. In the table, $n$ is a natural number, $\nu$ is any nonzero ordinal, and $\gamma$ is a limit ordinal.

## D.1 Proof-Theoretic Ordinals below ZFC

| Proof-Theoretic Ordinal | Arithmetic System | Set Theory System | Other Systems |
|---|---|---|---|
| $-$ | $Q$ | $\mathrm{KP}^-$ | |
| $\omega^2$ | RFA<br>$\mathrm{I}\Delta_0$ | | |
| $\omega^3$ | $\mathrm{RCA}_0^*$<br>$\mathrm{WKL}_0^*$<br>$\mathrm{I}\Delta_0 + \exp$ | | |
| $\omega^n$ | $\mathrm{I}\ \Delta_0 + \mathcal{E}_n$ is total | | |
| $\omega^\omega$<br>[146-147] | $\mathrm{RCA}_0$<br>$\mathrm{WKL}_0$<br>PRA<br>$\mathrm{RCA}_0^2$ | CPRC<br>$\mathrm{KP}^- + \Pi_1^{\mathrm{set}} -$<br>$-\mathrm{Foundation} + \mathrm{IND}$ | |
| $\omega^{\omega^{\omega^\omega}}$<br>[148] | $\mathrm{RCA}_0 + \left(\Pi_2^0\right)^- - \mathrm{IND}$ | | |
| $\omega \uparrow\uparrow (n+2)$<br>[149] | $\mathrm{I}\Sigma_{n+1}$ | | |
| $\varepsilon_0$<br>[150-151] | PA<br>$\mathrm{ACA}_0$<br>$\Delta_1^1 - \mathrm{CA}_0$<br>$\Sigma_1^1 - \mathrm{AC}_0$ | $\mathrm{KP}^{-\infty}$ | $\mathrm{EM}_0$ |
| $\varepsilon_1$ | $\mathrm{ACA}_0 + \mathrm{KPHT}$ | | |
| $\varepsilon_\omega$<br>[152-153] | $\mathrm{ACA}_0 + \mathrm{iRT}$<br>$\mathrm{RCA}_0 + \forall Y \forall n \exists X -$<br>$-(\mathrm{TJ}(n, X, Y))$ | | |
| $\varepsilon_{\varepsilon_0}$<br>[154] | ACA<br>$FP_n - \mathrm{ACA}_0'$<br>$FP_n - \mathrm{ACA}'$ | | |

| Proof-Theoretic Ordinal | Arithmetic System | Set Theory System | Other Systems |
|---|---|---|---|
| $\zeta_0$<br>$\psi_{\Omega_1}(\Omega)$<br>$[\psi(\Omega)]$<br>[155,153,156] | $\mathrm{ACA}_0 + \forall X \exists Y -$<br>$-(\mathrm{TJ}(\omega, X, Y))$<br>$\mathrm{ACA}_0 + (\mathrm{BR})$<br>$\mathrm{p}_1\,(\mathrm{ACA}_0)$ | | |
| $\varphi(2, \varepsilon_0)$<br>[155] | $\mathrm{ACA} + \forall X \exists Y -$<br>$-(\mathrm{TJ}(\omega, X, Y))$<br>RFN | | |
| $\varphi(\omega, 0)$<br>$\psi_{\Omega_1}(\Omega^\omega)$<br>$[\psi(\Omega^\omega)]$<br>[153,157] | $\Delta^1_1 - \mathrm{CR}$<br>$\mathrm{RCA}^*_0 + \Pi^1_1 - \mathrm{CA}^-$<br>$\Sigma^1_1 - \mathrm{DC}_0$ | | $\mathrm{ID}^\#_1$<br>$\mathrm{EM}_0 + \mathrm{JR}$<br>PID<br>$Acc - \mathrm{ID}(\mathrm{Acc})$<br>$(\Pi^0_0(\mathrm{P})\mathrm{P} \cup \mathrm{N}) -$<br>$-\mathrm{ID})$<br>$(\Pi^0_0(\mathrm{P}), \mathrm{P} \wedge \mathrm{N}) -$<br>$-\mathrm{ID}(\ \mathrm{Acc}\ )$ |
| $\varphi(\nu + 1, 0)$<br>[153] | $\mathrm{ACA}_0 + \forall X \exists Y -$<br>$-\,(\mathrm{TJ}\,(\omega^\nu, X, Y))$ | | |
| $\psi_{\Omega_1}\,(\Omega^{\varepsilon_0})$<br>$[\psi(\Omega^{\varepsilon_0})]$<br>[157-158] | $\Delta^1_1 - \mathrm{CA}$<br>$\Sigma^1_1 - \mathrm{AC}$<br>$\left(\Pi^0_1 - \mathrm{CA}\right)_{<\varepsilon_0}$ | | |
| $\psi_{\Omega_1}(\Omega^{\psi_{\Omega_1}(\Omega^\omega)})$<br>$[\psi(\Omega^{\psi(\Omega^\omega)})]$<br>[159] | | PRS $\omega$ | |
| $\Gamma_0$<br>$\psi_{\Omega_1}\left(\Omega^\Omega\right)$<br>$[\psi(\Omega^\Omega)]$<br>[146,160,151]<br>[161-162] | $\mathrm{ATR}_0$<br>$\Delta^1_1 - \mathrm{CA} + \mathrm{BR}$<br>$\mathrm{RCA}_0 + \Sigma^0_1 - \mathrm{RT}$<br>$\mathrm{RCA}_0 + \Delta^0_1$-RT<br>$\mathrm{ACA}_0 + \Delta^0_1$-det.<br>$\mathrm{ACA}_0 + \Sigma^0_1$-det.<br>$\mathrm{FP}_0$ | $\mathrm{KPi}^-$<br>$\mathrm{CZF}^- + \mathrm{INAC}$ | $\widehat{\mathrm{ID}}_{<\omega}$<br>$\widehat{\mathrm{ID}}^*$<br>$\mathrm{ML}_{<\omega}$<br>MLU<br>U(PA) |
| $\varphi(1, 0, \omega^\omega)$<br>[163] | | $\mathrm{KPl}^0 + (\Sigma_1 - \mathrm{I}_\omega)$ | |
| $\varphi(1, 0, \varepsilon_0)$<br>[156,164] | ATR | | $\widehat{\mathrm{ID}}_\omega$ |
| $\psi_{\Omega_1}\left(\Omega^{\Omega+1}\right)$<br>$[\psi\left(\Omega^{\Omega+1}\right)]$ | $\mathrm{RCA}_0 + \forall X \exists M -$<br>$-(X \in M \wedge M -$<br>$- \models_\omega \mathrm{ATR}_0)$ | | |
| $\psi_{\Omega_1}\left(\Omega^{\Omega+\omega}\right)$<br>$[\psi\left(\Omega^{\Omega+\omega}\right)]$<br>[156] | $\mathrm{ATR}_0 + \Sigma^1_1 - \mathrm{DC}$ | | $\widehat{\mathrm{ID}}_{<\omega^\omega}$ |
| $\psi_{\Omega_1}\left(\Omega^{\Omega+\varepsilon_0}\right)$<br>$[\psi\left(\Omega^{\Omega+\varepsilon_0}\right)]$<br>[156] | $\mathrm{ATR} + \Sigma^1_1 - \mathrm{DC}$ | | $\widehat{\mathrm{ID}}_{<\varepsilon_0}$ |
| $\psi_{\Omega_1}\left(\Omega^{\Omega+\Gamma_0}\right)$<br>$[\psi\left(\Omega^{\Omega+\Gamma_0}\right)]$ | | | $\widehat{\mathrm{ID}}_{<\Gamma_0}$<br>MLS |
| $\varphi(2, 0, 0)$<br>[163] | $\mathrm{FTR}_0$ | $\mathrm{KPh}^-$ | Aut $(\widehat{\mathrm{ID}})$ |
| $\varphi(2, 0, \varepsilon_0)$<br>[163] | FTR | | |

| Proof-Theoretic Ordinal | Arithmetic System | Set Theory System | Other Systems |
|---|---|---|---|
| $\varphi(2,\varepsilon_0,0)$ [163] | | $\mathrm{KPh}^0 + (\mathrm{F}-\mathrm{I}_\omega)$ | |
| $\psi_{\Omega_1}\left(\Omega^{\Omega\cdot\omega}\right)$ $[\psi\left(\Omega^{\Omega\cdot\omega}\right)]$ | | $\mathrm{KPM}^-$ | |
| $\varphi(\varepsilon_0,0,0)$ [156] | $\Sigma^1_1-\mathrm{TDC}$ | | |
| $\varphi(1,0,0,0)$ [156] | $\mathrm{p}_1\left(\Sigma^1_1-\mathrm{TDC}_0\right)$ | | |
| $\psi_{\Omega_1}\left(\Omega^{\Omega^\omega}\right)$ $[\psi\left(\Omega^{\Omega^\omega}\right)]$ [165,156] | $\mathrm{RCA}^*_0+\Pi^1_1-\mathrm{CA}^-$ $\mathrm{p}_3\left(\mathrm{ACA}_0\right)$ | | FIT TID |
| $\vartheta\left(\Omega^\Omega\right)$ [156] | $\mathrm{p}_1\left(\mathrm{p}_3\left(\mathrm{ACA}_0\right)\right)$ | | |
| $\theta_{(n+2)(\Omega^\omega)}0$ [165,146,166] | $\mathrm{ACA}_0+\Pi^1_{n+2}-\mathrm{BI}$ $\Pi^1_{n+1}-\mathrm{RFN}$ $\left(\Pi^1_{n+2}-\mathrm{BI}\right)_0$ $\left(\Pi^1_{n+2}-\mathrm{BI}\right)^-_0$ | $\mathrm{KP}\omega^-+\Pi^{\mathrm{set}}_{n+2}-$ $-\mathrm{Foundation}$ | |
| $\theta_{(n+2)(\Omega^{\varepsilon_0})}0$ [146,166] | $\mathrm{ACA}+\Pi^1_{n+2}-\mathrm{BI}$ $\left(\Pi^1_{n+2}-\mathrm{BI}^-\right)$ | $\mathrm{KP}\omega^-+\mathrm{IND}+\Pi^{\mathrm{set}}_{n+2}-$ $-\mathrm{Foundation}$ | |
| $\psi_{\Omega_1}\left(\varepsilon_{\Omega+1}\right)$ $[\psi(\psi_1(0))]$ [158,148] [167-168] | $\mathrm{ACA}+\mathrm{BI}$ $\mathrm{ACA}_0+\Pi^1_1-\mathrm{CA}^-$ $\Pi^0_1-\mathrm{FXP}_0$ | $\mathrm{KP}$ $\mathrm{KP}+\Pi^{\mathrm{set}}_2-$ $-\mathrm{Reflection}$ $\mathrm{KP}+(\mathrm{BI}^*)$ $\mathrm{KP}+(\mathrm{ATR}^*_0)$ $\mathrm{CZF}$ $\mathrm{KP}\omega_2\restriction+\Delta_1-$ $-\mathrm{CA}+s\Pi^1_1-\mathrm{ref}$ | $\mathrm{ID}_1$ $\mathrm{ID}^2_1$ $\mathrm{ML}_1\ \mathrm{V}$ |
| $\psi_{\Omega_1}\left(\zeta_{\Omega+1}\right)$ $[\psi(\Omega_2)]$ | $\mathrm{RCA}_0+\forall X\exists M-$ $-(X\in M\wedge M\models_\omega-$ $-\mathrm{ACA}+\mathrm{BI})$ | | |
| $\psi_{\Omega_1}\left(\Gamma_{\Omega+1}\right)$ $[\psi({\Omega_2}^{\Omega_2})]$ [169] | $\mathrm{ATR}^\bullet_0$ $\mathrm{FP}^\bullet_0$ $\Sigma^1_1-\mathrm{DC}^\bullet_0+(\mathrm{SUB}^\bullet)$ $\Sigma^1_1-\mathrm{AC}^\bullet_0+(\mathrm{SUB}^\bullet)$ | | $\widehat{\mathrm{ID}}^\bullet_{<\omega}$ $\mathcal{U}\left(\mathrm{ID}_1\right)$ |
| $\psi_{\Omega_1}\left(\varepsilon_{\Omega_2+1}\right)$ $[\psi(\psi_2(0))]$ | | $\mathrm{KP}+\exists\omega^{\mathrm{ck}}_1$ | $\mathrm{ID}_2$ $\mathrm{ID}^2_2$ |
| $\psi_{\Omega_1}\left(\Omega_\omega\right)$ $[\psi(\Omega_\omega)]$ [158,151] | $\Pi^1_1-\mathrm{CA}_0$ $\Delta^1_2-\mathrm{CA}_0$ $\mathrm{RCA}_0+\Sigma^0_1\wedge\Pi^0_1-\mathrm{det.}$ $\mathrm{RCA}_0+\Delta^0_2-\mathrm{RT}$ | $\mathrm{KPl}^r$ $\mathrm{KPi}^r$ $\mathrm{KP}\beta^r$ | $\mathrm{ID}_{<\omega}$ $\left(\mathrm{ID}^2_{<\omega}\right)_0$ |
| $\psi_0(\Omega_\omega\cdot\omega^\omega)$ $[\psi(\Omega_\omega\cdot\omega^\omega)]$ [170-171] | $\Pi^1_1-\mathrm{CA}_0+\Pi^1_2-\mathrm{IND}$ | | |
| $\psi_{\Omega_1}(\Omega_\omega\cdot\varepsilon_0)$ $[\psi(\Omega_\omega\cdot\varepsilon_0)]$ [172] | $\Pi^1_1-\mathrm{CA}$ | W-KPl | $\mathrm{W}-\mathrm{ID}_\omega$ $\mathrm{ID}^2_{<\omega}$ |
| $\psi_{\Omega_1}(\Omega_\omega\cdot\Omega)$ $[\psi(\Omega_\omega\cdot\Omega)]$ [172] | $\Pi^1_1-\mathrm{CA}+\mathrm{BR}$ | | |

| Proof-Theoretic Ordinal | Arithmetic System | Set Theory System | Other Systems |
|---|---|---|---|
| $\psi_0({\Omega_\omega}^\omega)$<br>$[\psi({\Omega_\omega}^\omega)]$<br>[170-171] | $\Pi^1_1 - \mathrm{CA}_0 + \Pi^1_2 - \mathrm{BI}$ | | |
| $\psi_0({\Omega_\omega}^{\omega^\omega})$<br>$[\psi({\Omega_\omega}^{\omega^\omega})]$<br>[170-171] | $\Pi^1_1 - \mathrm{CA}_0 + \Pi^1_2 -$<br>$-\mathrm{BI} + \Pi^1_3 - \mathrm{IND}$ | | |
| $\psi_{\Omega_1}(\varepsilon_{\Omega_\omega+1})$<br>$[\psi(\psi_\omega(0))]$<br>[172-173] | $\Pi^1_1 - \mathrm{CA} + \mathrm{BI}$ | KPl | $\mathrm{ID}_\omega$<br>$\mathrm{BID}^2_\omega$ |
| $\psi_{\Omega_1}(\Omega_{\omega^\omega})$<br>$[\psi(\Omega_{\omega^\omega})]$<br>[173] | $\Delta^1_2 - \mathrm{CR}$<br>$\left(\Pi^1_1 - \mathrm{CA}_{<\omega^\omega}\right)$ | $\mathrm{KPl}^r_{\omega^\omega}$ | $\mathrm{ID}_{<\omega^\omega}$ |
| $\psi_{\Omega_1}(\Omega_{\varepsilon_0})$<br>$[\psi(\Omega_{\varepsilon_0})]$<br>[158] | $\Delta^1_2 - \mathrm{CA}$<br>$\Sigma^1_2 - \mathrm{AC}$<br>$\left(\Pi^1_1 - \mathrm{CA}_{<\varepsilon_0}\right)$ | $\mathrm{KPl}^r_{\varepsilon_0}$<br>$\mathrm{W} - \mathrm{KPi}$<br>$\mathrm{W} - \mathrm{KP}\beta$ | $\mathrm{ID}_{<\varepsilon_0}$<br>$\mathrm{ID}^2_{<\varepsilon_0}$<br>$\mathrm{BID}^2_{<\varepsilon_0}$ |
| $\psi_\Omega(\Omega_{\nu\cdot\omega})$<br>$[\psi(\Omega_{\nu\cdot\omega})]$<br>[174] | $\left(\Pi^1_1 - \mathrm{CA}^+_\nu\right)_0$ | $\mathrm{KPl}^r_{\nu+}$ | $\mathrm{ID}_{<\nu\cdot\omega}$<br>$\left(\mathrm{PID}^2_\nu\right)_0$ |
| $\psi_\Omega(\Omega_\gamma \cdot \omega)$<br>$[\psi(\Omega_\gamma \cdot \omega)]$<br>[174] | $\left(\Pi^1_1 - \mathrm{CA}_{\gamma-}\right)_0$ | $\mathrm{KPl}^r_\gamma$ | $\left(\mathrm{NUID}^2_\gamma\right)_0$ |
| $\psi_\Omega(\Omega_{\nu\cdot\omega} \cdot \varepsilon_0)$<br>$[\psi(\Omega_{\nu\cdot\omega} \cdot \varepsilon_0)]$<br>[174] | $\Pi^1_1 - \mathrm{CA}^+_\nu$ | $\mathrm{W} - \mathrm{KPl}_{\nu+}$ | $\mathrm{W} - \mathrm{ID}_{\nu\omega}$<br>$\mathrm{PID}^2_\nu$ |
| $\psi_{\Omega_1}(\Omega_\gamma \cdot \varepsilon_0)$<br>$[\psi(\Omega_\gamma \cdot \varepsilon_0)]$<br>[158,174] | $\left(\Pi^1_1 - \mathrm{CA}_\gamma\right)$<br>$\Pi^1_1 - \mathrm{CA}_{\gamma-}$ | $\mathrm{W} - \mathrm{KPl}_\gamma$ | $\mathrm{W} - \mathrm{ID}_\gamma$<br>$\mathrm{ID}^2_\gamma$<br>$\mathrm{NUID}^2_\gamma$ |
| $\psi_\Omega(\Omega_{\nu\cdot\omega} \cdot \Omega)$<br>$[\psi(\Omega_{\nu\cdot\omega} \cdot \Omega)]$<br>[174] | $\Pi^1_1 - \mathrm{CA}^+_\nu + \mathrm{BR}$ | | $\mathrm{PID}^2_\nu + \mathrm{BR}$ |
| $\psi_\Omega(\Omega_\gamma \cdot \Omega)$<br>$[\psi(\Omega_\gamma \cdot \Omega)]$<br>[174] | $\Pi^1_1 - \mathrm{CA}_{\gamma-} + \mathrm{BR}$ | | $\mathrm{NUID}^2_\gamma + \mathrm{BR}$ |
| $\psi_{\Omega_1}(\Omega_{\omega^\gamma})$<br>$[\psi(\Omega_{\omega^\gamma})]$<br>[158] | $\left(\Pi^1_1 - \mathrm{CA}_{\omega\gamma}\right)_0$<br>$\left(\Pi^1_1 - \mathrm{CA}_{<\omega\gamma}\right)$<br>$\left(\Pi^1_1 - \mathrm{CA}_{<\omega\gamma}\right) + \mathrm{BI}$ | | $\left(\mathrm{ID}^2_{\omega^\gamma}\right)_0 \mathrm{ID}_{<\omega^\gamma}$<br>$\mathrm{BID}^2_{<\omega^\gamma}$<br>$\left(\mathrm{ID}^2_{<\nu}\right) + \mathrm{BI}$ |
| $\psi_\Omega(\varepsilon_{\Omega_\nu+1})$<br>$[\psi(\psi_\nu(0))]$<br>[174] | $\left(\Pi^1_1 - \mathrm{CA}_\nu\right)_0$ | $\mathrm{KPl}_\nu$ | $\mathrm{ID}_\nu$<br>$\left(\mathrm{ID}^2_\nu\right)_0$ |
| $\psi_\Omega(\varepsilon_{\Omega_\nu+\varepsilon_0})$<br>$[\psi(\psi_\nu(\varepsilon_0))]$<br>[174] | $\Pi^1_1 - \mathrm{CA}_\nu$ | | $\mathrm{ID}^2_\nu$ |
| $\psi_\Omega(\varepsilon_{\Omega_\nu+\Omega})$<br>$[\psi(\psi_\nu(\Omega))]$<br>[174] | $\Pi^1_1 - \mathrm{CA}_\nu + \mathrm{BR}$ | | $\mathrm{ID}^2_\nu + \mathrm{BR}$ |
| $\psi_\Omega\left(\varepsilon_{\varepsilon_{\Omega_\nu+1}}\right)$<br>$[\psi(\psi_\nu(\psi_\nu(0)))]$<br>[174] | | | $\mathrm{BID}^2_\nu$ |
| $\psi_\Omega\left(\varepsilon_{\Omega_{\nu+1}+1}\right)$<br>$[\psi(\psi_{\nu+1}(0))]$<br>[174] | $\Pi^1_1 - \mathrm{CA}_\nu + \mathrm{BI}$ | $\mathrm{KPl}_{\nu+1}$ | $\mathrm{ID}_{\nu+1}$<br>$\mathrm{ID}^2_\nu + \mathrm{BI}$ |

| Proof-Theoretic Ordinal | Arithmetic System | Set Theory System | Other Systems |
|---|---|---|---|
| $\psi_\Omega\left(\varepsilon_{\Omega_{\nu\cdot\omega}+1}\right)$<br>$[\psi\left(\psi_{\nu\cdot\omega}(0)\right)]$<br>[174] | $\Pi^1_1 - \mathrm{CA}^+_\nu + \mathrm{BI}$ | $\mathrm{KPl}_{\nu+}$ | $\mathrm{ID}_{\nu\omega}$<br>$\mathrm{PID}^2_\nu + \mathrm{BI}$<br>$\mathrm{PBID}^2_\nu$ |
| $\psi_{\Omega_1}\left(\varepsilon_{\Omega_\gamma+1}\right)$<br>$[\psi\left(\psi_\gamma(0)\right)]$<br>[158,174-175] | $\left(\Pi^1_1 - \mathrm{CA}_\gamma\right)_0$<br>$\left(\Pi^1_1 - \mathrm{CA}_\gamma\right) + \mathrm{BI}$<br>$\Pi^1_1 - \mathrm{CA}_{\gamma-} + \mathrm{BI}$ | $\mathrm{KPl}_\gamma$ | $\mathrm{ID}_\gamma$<br>$\left(\mathrm{ID}^2_\gamma\right)_0$<br>$\mathrm{ID}^i_\gamma(\mathcal{O})\mathrm{BID}^2_\nu$<br>$\mathrm{ID}^2_\gamma + \mathrm{BI}$<br>$\mathrm{NUID}^2_\gamma + \mathrm{BI}$<br>$\mathrm{NUBID}^2_\gamma$ |
| $\psi_{\Omega_1}\left(\varepsilon_{\Omega_\Omega+1}\right)$<br>$[\psi\left(\psi_\Omega(0)\right)]$<br>[176] | | $\mathrm{KPl}^*$<br>$\mathrm{KPl}^r_\Omega$ | $\mathrm{ID}_{\prec^*}$<br>$\mathrm{BID}^{2*}$<br>$\mathrm{ID}^{2*} + \mathrm{BI2}$ |
| $\psi_{\Omega_1}\left(\psi_I(0)\right)$<br>[176] | $\Pi^1_1 - \mathrm{TR}_0$<br>$\Pi^1_1 - \mathrm{TR}_0 + \Delta^1_2 - \mathrm{CA}_0$<br>$\Delta^1_2$ - $\mathrm{CA} + \mathrm{BI}-$<br>$-\left(\mathrm{impl} - \Sigma^1_2\right)$<br>$\Delta^1_2 - \mathrm{CA} + \mathrm{BR}-$<br>$-\left(\mathrm{impl} - \Sigma^1_2\right)$<br>$\mathrm{RCA}_0 + \Delta^0_2$-det.<br>$\mathrm{RCA}_0 + \Delta^1_1 - \mathrm{RT}$ | $\mathrm{Aut} - \mathrm{KPl}^r$<br>$\mathrm{Aut} - \mathrm{KPl}^r + \mathrm{KPi}^r$<br>$\mathrm{KPi}^w + \mathrm{FOUNDR}-$<br>$-(\mathrm{impl} - \Sigma)$<br>$\mathrm{KPi}^w + \mathrm{FOUND}-$<br>$-(\mathrm{impl} - \Sigma)$ | $\mathrm{Aut} - \mathrm{ID}^{pos}_0$<br>$\mathrm{Aut}-\mathrm{ID}^{mon}_0$ |
| $\psi_{\Omega_1}\left(\psi_I(0)\cdot\varepsilon_0\right)$<br>[176] | $\Pi^1_1 - \mathrm{TR}$ | $\mathrm{W} - \mathrm{Aut} - \mathrm{KPl}$ | $\mathrm{Aut} - \mathrm{ID}^{pos}$<br>$\mathrm{Aut} - \mathrm{ID}^{mon}$<br>$\mathrm{Aut} - \mathrm{KPl}^w$ |
| $\psi_{\Omega_1}\left(\psi_{\Omega_{\psi_I(0)+1}}(0)\right)$<br>[176] | $\Pi^1_1 - \mathrm{TR} + \mathrm{BI}$ | $\mathrm{Aut} - \mathrm{KPl}$ | $\mathrm{Aut} - \mathrm{ID}^{pos}_2$<br>$\mathrm{Aut}-\mathrm{ID}^{mon}_2$<br>$\mathrm{Aut} - \mathrm{BID}$ |
| $\psi_{\Omega_1}\left(\psi_I\left(I^\omega\right)\right)$<br>[176] | $\Delta^1_2 - \mathrm{TR}_0$<br>$\Sigma^1_2 - \mathrm{TRDC}_0$<br>$\Delta^1_2 - \mathrm{CA}_0-$<br>$+\Sigma^1_2 - \mathrm{BI}$ | | $\mathrm{KPi}^r + (\Sigma-$<br>$-\mathrm{FOUND})$<br>$\mathrm{KPi}^r + (\Sigma-$<br>$-\mathrm{REC})$ |
| $\psi_{\Omega_1}\left(\psi_I\left(I^{\varepsilon_0}\right)\right)$<br>[176] | $\Delta^1_2 - \mathrm{TR}$<br>$\Sigma^1_2 - \mathrm{TRDC}$<br>$\Delta^1_2 - \mathrm{CA}-$<br>$+\Sigma^1_2 - \mathrm{BI}$ | | $\mathrm{KPi}^w + (\Sigma-$<br>$-\mathrm{FOUND})$<br>$\mathrm{KPi}^w + (\Sigma-$<br>$-\mathrm{REC})$ |
| $\psi_{\Omega_1}\left(\varepsilon_{I+1}\right)$<br>[176,158] | $\Delta^1_2 - \mathrm{CA} + \mathrm{BI}$<br>$\Sigma^1_2 - \mathrm{AC} + \mathrm{BI}$ | $\mathrm{KPi}$<br>$\mathrm{KP}\beta$<br>$\mathrm{CZF} + \mathrm{REA}$ | $\mathrm{T}_0$ |
| $\psi_{\Omega_1}\left(\Omega_{I+\omega}\right)$ | | $\mathrm{KPi}^+$ | $\mathrm{ML}_1$ W<br>$\mathrm{KP}_1$ W<br>IARI |
| $\psi_{\Omega_1}\left(\varepsilon_{M+1}\right)$<br>[177] | $\Delta^1_2 - \mathrm{CA} + \mathrm{BI} + (\mathrm{M})$ | KPM<br>CZFM | |
| $\psi_{\Omega_1}\left(\Omega_{M+\omega}\right)$ | | $\mathrm{KPM}^+$ | MLM<br>Agda |
| $\Psi^0_{\Omega_1}\left(\varepsilon_{K+1}\right)$<br>[178] | $\mathrm{ACA} + \mathrm{BI} + \Pi^1_4-$<br>$-\beta-\mathrm{model} - \mathrm{Reflection}$ | $\mathrm{KP} + \Pi^{\mathrm{set}}_3 -$<br>$-\mathrm{Reflection}$ | |

| Proof-Theoretic Ordinal | Arithmetic System | Set Theory System | Other Systems |
|---|---|---|---|
| $\Psi_{\mathbb{X}}^{\varepsilon_{\xi_n+1}}$ [178-179] | $\mathrm{ACA}+\mathrm{BI}+\Pi^1_{n+5}-$ $-\beta-\mathrm{model}-\mathrm{Reflection}$ | $\mathrm{KP}+\Pi^{\mathrm{set}}_{n+4}-$ $-\mathrm{Reflection}$ | |
| $\Psi_{\mathbb{X}}^{\varepsilon_{\Xi+1}}$ [179] | $\mathrm{ACA}+\mathrm{BI}-$ $-\beta-\mathrm{model}-\mathrm{Reflection}$ | $\mathrm{KP}+\Pi^{\mathrm{set}}_{\omega}-$ $-\mathrm{Reflection}$ | |
| $\Psi_{\mathbb{H}}^{\varepsilon_{\Upsilon}+1}$ [179] | | $\mathrm{KPi}+\forall\alpha\exists\kappa-$ $-(L_\kappa \prec_1 L_{\kappa+\alpha})$ | |
| $\psi(\Omega_{\mathbb{S}+\omega})$ [180] | $\Pi^1_1-\mathrm{CA}_0+\Pi^1_2-\mathrm{CA}^-$ | $\mathrm{KPl}^r+\exists M-$ $-(\mathrm{Trans}(M)\wedge M\prec_1 V)$ | |
| $\Psi_{\mathbb{K}}^{\varepsilon_{\mathrm{I}+1}}$ [181] | $\Delta^1_2-\mathrm{CA}+\mathrm{BI}-$ $+\Pi^1_2-\mathrm{CA}^-$ | $\mathrm{KPi}+\exists M-$ $-(\mathrm{Trans}(M)\wedge M\prec_1 V)$ | |
| $\mathcal{I}_\omega\cap\omega_1^{\mathrm{CK}}$ [181] | $\Pi^1_2-\mathrm{CA}_0$ $\Delta^1_3-\mathrm{CA}_0$ | | |
| $\mathcal{I}_{\omega+1}\cap\omega_1^{\mathrm{CK}}$ [181] | $\Pi^1_2-\mathrm{CA}+\mathrm{BI}$ | $\mathrm{KP}+\Sigma^{\mathrm{set}}_1-$ $-\mathrm{Separation}$ $\mathrm{KPi}+\forall\alpha\exists\beta-$ $-\beta>\alpha(\beta\text{ stable })$ | |
| $\mathcal{I}_{\varepsilon_0}\cap\omega_1^{\mathrm{CK}}$ [181] | $\Delta^1_3-\mathrm{CA}$ $\Sigma^1_3-\mathrm{AC}$ | | |
| $\psi_\Omega(\varepsilon_{\mathbb{I}_n+1})$ [182] | $\Delta^1_{n+2}-\mathrm{CA}+\mathrm{BI}$ $\Sigma^1_{n+2}-\mathrm{AC}+\mathrm{BI}$ $\Sigma^1_{n+2}-\mathrm{DC}+\mathrm{BI}$ | $\mathrm{KP}+\Pi^{\mathrm{set}}_n-$ $-\mathrm{Collection}$ | |
| $\psi_\Omega(\mathbb{I}_\omega)$ [182] | $\mathrm{Z}_2=\Pi^1_\infty-\mathrm{CA}$ $\Delta^2_1-\mathrm{CA}_0$ $\mathrm{Z}_2+\Sigma^1_\infty-\mathrm{AC}$ | $\mathrm{KP}+\Sigma^{\mathrm{set}}_\omega-$ $-\mathrm{Separation}$ $\mathrm{KP}+\Sigma^{\mathrm{set}}_\omega-$ $-\mathrm{Separation}$ $+\Sigma^{\mathrm{set}}_\omega-$ $-\mathrm{Collection}$ $\mathrm{ZFC}^-:=\mathrm{ZFC}-$ $-\mathrm{Powerset}$ | |
| | $\mathrm{Z}_{n+3}=\Pi^{n+2}_\infty-\mathrm{CA}$ $\Delta^{n+3}_1-\mathrm{CA}_0$ | $\mathrm{ZFC}^-+V=L-$ $+\exists\omega_{n+1}$ | |
| | $\mathrm{Z}_\infty=\Pi^\infty_0-\mathrm{CA}$ | Z ZC IZ | |
| | | $\mathrm{IZF}=\mathrm{CZF}-$ $+\mathrm{Powerset}+\Pi^{\mathrm{set}}_\omega-$ $-\mathrm{Separation}$ $\mathrm{ZF}=\mathrm{CZF}+\mathrm{LEM}-$ $=\mathrm{IZF}+\mathrm{LEM}$ ZFC $\mathrm{ZFC}+V=L$ AST IST NBG = GBC GB | |

## D.2 Proof-Theoretic Ordinals related to ZFC

For this part of the ordinals, we do not yet know their exact values, and the order among the various ordinals is also relatively coarse. The content of this section can serve as a reference for the large cardinal table.

| Proof-Theoretic Ordinal |
|---|
| $\mathrm{S}_0 = (\text{ Ext }) + (\text{ Null }) + (\text{ Pair }) + (\text{ Union }) + (\text{ Diff })$ ("Rudimentary set theory") |
| $\mathrm{S}_1 = \mathrm{S}_0 + (\text{ Powerset })$ |
| $\mathrm{M}_0 = \mathrm{S}_1 + (\Delta_0^{\mathrm{set}} \text{ −Separation })$ |
| $\mathrm{M}_1 = \mathrm{M}_0 + (\text{ Regularity }) + (\text{ Transitive Containement })$ |
| $\mathrm{KP}^- = \mathrm{S}_0 + (\text{ Infinity }) + (\Delta_0^{\mathrm{set}} \text{ -Separation }) + (\Delta_0^{\mathrm{set}} \text{ -Collection })$ |
| $\mathrm{KP}^{-\infty} = \mathrm{S}_0 + (\text{ Foundation }) + (\Delta_0^{\mathrm{set}} \text{ -Separation }) + (\Delta_0^{\mathrm{set}} \text{ -Collection })$ |
| $\mathrm{KP} = \mathrm{KP}^{-\infty} + (\text{ Infinity }) = \mathrm{KP}^- + (\text{Foundation })$ |
| $\mathrm{KPl} = \mathrm{KP} + (\text{ universe limit of admissible sets })$ |
| $\mathrm{KPi} = \mathrm{KP} + (\text{ recursively inaccessible universe })$ |
| $\mathrm{KPh} = \mathrm{KP} + (\text{ recursively hyperinaccessible universe})$ |
| $\mathrm{KPM} = \mathrm{KP} +$ (recursively Mahlo universe) |
| $\mathrm{ZBQC} = \mathrm{M}_0 + (\text{ Regularity }) + (\text{ Infinity }) + (\text{ Choice })$<br>$\mathrm{NFU} + (\text{ Infinity }) + (\text{ Choice })$ |
| $\mathrm{MAC} = \mathrm{M}_1 + (\text{ Infinity }) + (\text{ Choice })$<br>$= \mathrm{ZBQC} + (\text{ Transitive Containement })$ |
| $\mathrm{MOST} = \mathrm{MAC} + (\Delta_0^{\mathrm{set}} - \text{ Collection })$<br>$= \mathrm{ZBQC} + \mathrm{KP} + (\Sigma_1^{\mathrm{set}} - \text{ Separation })$ |
| $\mathrm{Z} = \mathrm{S}_1 + (\text{ Regularity }) + (\text{ Infinity }) + (\Sigma_\omega^{\mathrm{set}} \text{ -Separation })$ |
| $\mathrm{ZC} = \mathrm{Z} + (\text{ Choice })$<br>$= \mathrm{ZBQC} + (\sigma_\omega^{\mathrm{set}} \text{ -Separation })$ |
| $\mathrm{MAC} + \forall m\, (\beth_{\beth_m} \text{ exists })$<br>NFU+(Infinity) +( Choice) |
| $\mathrm{Z} + (\Pi_2^{\mathrm{set}} \text{ -Replacement })$<br>$\mathrm{NFU}^* = \mathrm{NFU} + (\text{ Counting }) + (\text{ Strongly Cantorian Separation })$ |
| $\mathrm{Z} + (\Pi_m^{\mathrm{set}} \text{ -Replacement })$ |
| $\mathrm{ZF} = \mathrm{Z} + (\Pi_\omega^{\mathrm{set}} \text{ -Replacement })$<br>AST<br>GB |
| $\mathrm{ZFC} = \mathrm{ZF} + (\text{ Choice })$<br>$\mathrm{NBG} = \mathrm{GBC} = \mathrm{GB} + (\text{ Global Choice })$ |
| $\mathrm{ZFC} + (\text{ there is a worldly cardinal })$ |
| NBG+ (there is a stationary proper class of worldly cardinals) |
| NBG+(Class Forcing Theorem)<br>NBG+(Clopen Class Game Determinacy) |
| $\mathrm{MK} = \mathrm{NBG} + \left(\Pi_\infty^{\mathrm{class}} - \mathrm{CA}\right)$ |
| ZFC+ (there is an inaccessible cardinal)<br>$\mathrm{ZFC} + \left(\Pi_1^1 \text{ Perfect Set Property }\right)$<br>$\mathrm{ZFC} + \left(\Sigma_3^1 \text{ Lebesgue measurability }\right)$ |
| $\mathrm{ZFC} + (\text{ there are } \omega \text{ inaccessible cardinals})$<br>$\mathrm{ZFC} + (\forall \alpha\, (\omega \leq \alpha \leq \aleph_\omega \Rightarrow \lvert V_\alpha \cap L\rvert = \lvert\alpha\rvert))$ |
| ZFC+ (there is a proper class of inaccessible cardinals)<br>ZFC+(Grothiendieck Universe Axiom) |
| $\mathrm{ZFC} + (\text{ there is a } \Sigma_n^{\mathrm{set}} \text{ -reflecting cardinal })$ |
| ZFC+ (there is a $\sigma_\omega^{\mathrm{set}}$ -reflecting cardinal)<br>ZFC+ (Ord is Mahlo) |
| $\mathrm{ZFC} + (\text{ there is an uplifting cardinal})$<br>ZFC+(Resurrection Axioms) |
| $\mathrm{ZFC} + (\text{ there is a Mahlo cardinal })$ |
| $\mathrm{SMAH} = \mathrm{ZFC} + (\text{ there is a } n\text{-Mahlo cardinal })_{n\in\mathbb{N}}$<br>$\mathrm{NFUA} = \mathrm{NFU} + (\text{ Infinity }) + (\text{ Cantorian Sets })$ |
| $\mathrm{SMAH}^+ = \mathrm{ZFC} + \forall n (\text{ there is a } n\text{-Mahlo cardinal })$ |

| Proof-Theoretic Ordinal |
|---|
| MK + ( Ord is weakly compact )<br>$\mathrm{GPK}^+_\infty = \mathrm{GPK}^+ +$ (Infinity )<br>NFUB =NFU +(Infinity ) + (Cantorian Sets ) + (Small Ordinals ) |
| ZFC + ( there is a weakly compact cardinal )<br>ZFC + ($\omega_2$ has the tree property ) |
| ZFC + ( there is a totally indescribable cardinal) |
| ZFC + ( there is a subtle cardinal ) |
| ZFC + ( there is an ineffable cardinal) |
| ZFC + $\forall\alpha\,(\alpha < \omega_1 \Rightarrow$ there is a $\alpha$-Erdös cardinal ) |
| ZFC + $\left(0^\sharp\right.$ exists ) |
| ZFC + ($L \models \aleph_\omega$ is regular ) |
| ZFC + $\forall\alpha\,(\alpha \geq \omega \Longrightarrow |V_\alpha \cap L| = |\alpha|)$ |
| ZFC+ (parameter-free $\Sigma^1_1$-determinacy) |
| ZFC + $\forall x\left(x \in \mathbb{R} \Longrightarrow x^\sharp\right.$ exists ) |
| ZFC + $\left(\Sigma^1_1\right.$-determinacy ) |
| ZFC + $\forall x$ ( $x^\sharp$ exists ) |
| ZFC + $\left(\Sigma^1_2\right.$ universal Baireness ) |
| ZFC + ( there is an $\omega_1$-Erdős cardinal) |
| ZFC+(Chang's Conjecture) |
| SRP = ZFC + ( there is cardinal with the $n$-stationary Ramsey property )$_{n\in\mathbb{N}}$ |
| $\mathrm{SRP}^+$ = ZFC + $\forall n$ (there is a cardinal with the $n$-stationary Ramsey property) |
| MK + ( Ord is measurable )<br>NFUM = NFU +( Infinity ) + ( Large Ordinals ) + ( Small Ordinals ) |
| ZFM = ZFC + ( there is a measurable cardinal )<br>ZFC + ($\mathrm{NS}_{\omega_1}$ is precipitous )<br>ZF + ($\omega_1$ is measurable ) |
| ZFC + ( there is a measurable cardinal $\kappa$ such that $o(\kappa) = 2$)<br>ZFC + ($\mathrm{NS}_{\omega_2}$ is precipitous ) |
| ZFC + ( there is a measurable cardinal $\kappa$ such that $o(\kappa) = \kappa^{++}$)<br>ZFC + $\neg$SCH<br>ZFC + $\left(2^{\aleph_\omega} = \aleph_{\omega+2}\right)$ |
| ZFC + ( Ord is Woodin)<br>ZFC + $\neg$SCH<br>$\mathrm{Z}_2$ + $\left(\Delta^1_2\right.$-determinacy) (conjectural) |
| MK + ( Ord is Woodin)<br>ZFC + $\neg$SCH<br>$\mathrm{Z}_3$+ (lightface $\Delta^1_2$-determinacy) |
| NBG + ( Ord is Woodin)<br>ZFC + $\neg$SCH<br>$\mathrm{Z}_3$+ ($\Delta^1_2$-determinacy) |
| ZFC + ( there is a Woodin cardinal )<br>ZFC + $\left(\Delta^1_2\right.$-determinacy )<br>ZFC + (OD $\models$ AD)<br>ZFC + ($\mathrm{NS}_{\omega_1}$ is $\omega_2$-saturated ) |
| ZFC + ( there are $n$ Woodin cardinals )$_{n\in\mathbb{N}}$<br>$\mathrm{Z}_2$ + (PD) |
| ZFC+ (there are $\omega$ Woodin cardinals)<br>ZF + (AD)<br>ZFC + ($L(\mathbb{R}) \models$ AD)<br>ZFC + (OD($\mathbb{R}$) $\models$ AD) |
| ZF + DC + ($\omega_1$ is $\mathcal{P}(\omega_1)$-strongly compact)<br>ZFC + ($\mathrm{NS}_{\omega_1}$ is $\omega_1$-dense ) |
| ZF + DC + ($\omega_1$ is $\mathcal{P}(\mathbb{R})$-strongly compact )<br>ZF + DC + ($\mathrm{AD}_\mathbb{R}$) |
| ZFC+ (there is a superstrong cardinal) |

| Proof-Theoretic Ordinal |
|---|
| ZFC + ( there is a subcompact cardinal)<br>ZFC + $(V = L[\vec{E}]) + \exists\kappa\,(\neg\square_\kappa)$ |
| ZFC + ( there is a strongly compact cardinal)<br>ZFC + ( Proper Forcing Axiom) |
| ZFC + ( there is a supercompact cardinal)<br>ZFC+(Martin's Maximum) |
| ZFC + $\forall n$ (there is a proper class of $C^{(n)}$-extendible cardinals)<br>ZFC+(Vopěnka's Principle) |
| ZFC + ( there is a high-jump cardinal) |
| HUGE = ZFC + ( there is a $n$-huge cardinal $)_{n\in\mathbb{N}}$ |
| ZFC + ( Wholeness Axiom $\mathrm{WA}_n$) |
| ZFC + I3 = ZFC + $\exists\lambda\,(E_0(\lambda))$ |
| ZFC + I2 = ZFC + $\exists\lambda\,(E_1(\lambda))$ |
| ZFC + I1 = ZFC + $\exists\lambda\,(E_\omega(\lambda))$ |
| ZFC + I0 |
| ZF + DC + $\exists\lambda\exists j : V_{\lambda+2} \prec_{\Sigma^{\mathrm{set}}_\omega} V_{\lambda+2}$ |
| $\mathrm{ZF}_j$ + DC + ( there is a Reinhardt cardinal ) |
| ZF + DC + ( there is a Berkeley cardinal) |

# Appendix E

# Named Ordinals

The content of this appendix is taken from[183,33], with some modifications. The content of this appendix is updated to 2025.

| Name | Value |
|---|---|
| 首个超限序数<br>(First Transfinite Ordinal)<br>(FTO) | $(0)(1)$<br>$\omega$ |
| 线性数阵序数<br>(Linar Array Ord)<br>(LAO) | $(0)(1)(2)$<br>$\omega^\omega$ |
| 小 Cantor 序数<br>(Small Cantor's Ordinal)<br>(SCO) | $(0,0)(1,1)$<br>$\varepsilon_0$<br>$\varphi(1,0)$<br>$\psi(0)$ (MOCF)<br>$\psi(\Omega)$ (BOCF) |
| Cantor 序数<br>(Cantor's Ordinal)<br>(CO) | $(0,0)(1,1)(2,1)$<br>$\zeta_0$<br>$\varphi(2,0)$<br>$\psi(\Omega)$ (MOCF)<br>$\psi(\Omega^2)$ (BOCF) |
| 大 Cantor 序数<br>(Large Cantor's Ordinal)<br>(LCO) | $(0,0)(1,1)(2,1)(2,1)$<br>$\eta_0$<br>$\varphi(3,0)$<br>$\psi(\Omega^2)$ (MOCF)<br>$\psi(\Omega^3)$ (BOCF) |
| 超 Cantor 序数<br>(Hyper Cantor's Ordinal)<br>(HCO) | $(0,0)(1,1)(2,1)(3,0)$<br>$\varphi(\omega,0)$<br>$\psi(\Omega^\omega)$ (MOCF)<br>$\psi(\Omega^\omega)$ (BOCF) |
| Feferman-Schütte 序数<br>(Feferman-Schütte Ordinal)<br>(FSO) | $(0,0)(1,1)(2,1)(3,1)$<br>$\varphi(1,0,0)$<br>$\Gamma_0$<br>$\psi(\Omega^\Omega)$ (MOCF)<br>$\psi(\Omega^\Omega)$ (BOCF) |

| Name | Value |
|---|---|
| Ackermann 序数<br>(Ackermann Ordinal)<br>(ACO) | $(0,0)(1,1)(2,1)(3,1)(3,1)$<br>$\varphi(1,0,0,0)$<br>$\psi(\Omega^{\Omega^2})$ (MOCF)<br>$\psi(\Omega^{\Omega^2})$ (BOCF) |
| 小 Veblen 序数<br>(Small Veblen Ordinal)<br>(SVO) | $(0,0)(1,1)(2,1)(3,1)(4,0)$<br>$\varphi(1@\omega)$<br>$\psi(\Omega^{\Omega^\omega})$ (MOCF)<br>$\psi(\Omega^{\Omega^\omega})$ (BOCF) |
| 大 Veblen 序数<br>(Large Veblen Ordinal)<br>(LVO) | $(0,0)(1,1)(2,1)(3,1)(4,1)$<br>$\varphi(1@(1,0))$<br>$\psi(\Omega^{\Omega^\Omega})$ (MOCF)<br>$\psi(\Omega^{\Omega^\Omega})$ (BOCF) |
| 扩展小 Veblen 序数<br>(Extended Small Veblen Ordinal)<br>(ESVO) | $(0,0)(1,1)(2,1)(3,1)(4,1)(5,0)$<br>$\psi(\Omega^{\Omega^{\Omega^\omega}})$ (MOCF)<br>$\psi(\Omega^{\Omega^\omega})$ (BOCF) |
| 扩展大 Veblen 序数<br>(Extended Large Veblen Ordinal)<br>(ELVO) | $(0,0)(1,1)(2,1)(3,1)(4,1)(5,1)$<br>$\psi(\Omega^{\Omega^{\Omega^\Omega}})$ (MOCF)<br>$\psi(\Omega^{\Omega^\Omega})$ (BOCF) |
| Bachmann-Howard 序数<br>(Bachmann-Howard Ordinal)<br>(BHO) | $(0,0)(1,1)(2,2)$<br>$\psi(\psi_1(0))$ (MOCF)<br>$\psi(\Omega_2)$ (BOCF) |
| Buchholz 序数<br>(Buchholz's Ordinal)<br>(BO) | $(0,0,0)(1,1,1)$<br>$\psi(\Omega_\omega)$ (MOCF)<br>$\psi(\Omega_\omega)$ (BOCF) |
| Takeuti-Feferman-Buchholz 序数<br>(Takeuti-Feferman-Buchholz's Ordinal)<br>(TFBO) | $(0,0,0)(1,1,1)(2,1,0)(3,2,0)$<br>$\psi(\psi_\omega(0))$ (MOCF)<br>$\psi(\Omega_{\omega+1})$ (BOCF) |
| Bird 序数<br>(Bird's Ordinal)<br>(BIO) | $(0,0,0)(1,1,1)(2,1,1)(3,1,0)$<br>$\psi(\Omega_\Omega)$ (MOCF)<br>$\psi(\Omega_\Omega)$ (BOCF) |
| 扩展 Buchholz 序数<br>(Extended Buchholz's Ordinal)<br>(EBO) | $(0,0,0)(1,1,1)(2,1,1)(3,1,0)(2,0,0)$<br>$\psi(\psi_I(0))$ (M-like)<br>$\psi(I)$ (B-like) |
| Jäger 序数<br>(Jäger's Ordinal)<br>(JO) | $(0,0,0)(1,1,1)(2,1,1)(3,1,0)(4,2,0)$<br>$\psi\left(\psi_{\Omega_{I+1}}(0)\right)$ (M-like)<br>$\psi(\Omega_{I+1})$ (B-like) |
| 小不可达序数<br>(Small Inaccessible Ordinal)<br>(SIO) | $(0,0,0)(1,1,1)(2,1,1)(3,1,1)$<br>$\psi(I_\omega)$ (M-like)<br>$\psi(I_\omega)$ (B-like) |
| 多重 Buchholz 序数<br>(Mutiply Buchholz's Ordinal)<br>(MBO) | $(0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,0,0)$<br>$\psi(I(\omega,0))$ (M-like)<br>$\psi(I(\omega,0))$ (B-like) |
| 超限 Buchholz 序数<br>(Transfinty Buchholz's Ordinal)<br>(TBO) | $(0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)(2,0,0)$<br>$\psi\left(\psi_{I(1,0,0)}(0)\right)$ (M-like)<br>$\psi(I(1,0,0))$ (B-like) |

| Name | Value |
|---|---|
| 小 Rathjen 序数<br>(Small Rathjen's Ordinall)<br>(SRO) | $(0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,0)(4,2,0)$<br>$\psi\left(\psi_{\Omega_{M+1}}(0)\right)$ (M-like)<br>$\psi\left(\Omega_{M+1}\right)$ (B-like) |
| 小 Mahlo 序数<br>(Small Mahlo Ordinall)<br>(SMO) | $(0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)$<br>$\psi\left(M_\omega\right)$ (M-like)<br>$\psi\left(M_\omega\right)$ (B-like) |
| 小不可交换序数<br>(Small Nonconvertible Ordinal)<br>(SNO) | $(0,0,0)(1,1,1)(2,1,1)(3,1,1)(3,1,1)(3,1,1)$<br>$\psi(1-2-2-2)$<br>$\psi\left(N_\omega\right)$ (M-like)<br>$\psi\left(N_\omega\right)$ (B-like) |
| Rathjen 序数<br>(Rathjen's Ordinal)<br>(RO) | $(0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,0)(5,2,0)$<br>$\psi(2 \text{ aft } 3)$<br>$\psi\left(\psi_{\Omega_{K+1}}(0)\right)$ (M-like)<br>$\psi\left(\Omega_{K+1}\right)$ (B-like) |
| 小弱紧致序数<br>(Small Weakly Compect Ordinal)<br>(SKO) | $(0,0,0)(1,1,1)(2,1,1)(3,1,1)(4,1,1)$<br>$\psi(1-3)$<br>$\psi\left(K_\omega\right)$ (M-like)<br>$\psi\left(K_\omega\right)$ (B-like) |
| Duchhardt 序数<br>(Duchhardt's Ordinal)<br>(DO) | $(0,0,0)(1,1,1)(2,1,1)(3,1,1)-$<br>$-(4,1,1)(5,1,0)(6,2,0)$<br>$\psi(2 \text{ aft } 4)$<br>$\psi\left(\psi_{\Omega_{\Pi_4+1}}(0)\right)$ (M-like)<br>$\psi\left(\Omega_{\Pi_4+1}\right)$ (B-like) |
| 小 Stegert 序数<br>(Small Stegert Ordinal)<br>(SSO) | $(0,0,0)(1,1,1)(2,2,0)$<br>$\psi(\Pi_\omega)$<br>$\psi\left(\lambda\alpha.(\alpha+1)-\Pi_0\right)$ |
| 大 Stegert 序数<br>(Large Stegert Ordinal)<br>(LSO) | $(0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,0)(2,0,0)$<br>$\psi\left(\lambda\alpha.(\alpha\cdot 2)-\Pi_0\right)$ |
| 容许-非递归分离序数<br>(Admissible-parameter free effective cardinal Ordinal)<br>(APO) | $(0,0,0)(1,1,1)(2,2,0)(3,2,0)(4,1,1)$<br>$\psi\left(\Pi_1-(\lambda\alpha.\Omega_{\alpha+1}-\Pi_1)\right)$ |
| 首个返回序数<br>(1st Back Gear Ordinal)<br>(BGO) | $(0,0,0)(1,1,1)(2,2,1)$<br>$\psi\left(\Pi_1-(\lambda\alpha.\Omega_{\alpha+2}-\Pi_1)\right)$ |
| 小下降序数<br>(Small Dropping Ordinal)<br>(SDO) | $(0,0,0)(1,1,1)(2,2,1)(3,0,0)$<br>$\psi\left(\lambda\alpha.\Omega_{\alpha+\omega}-\Pi_0\right)$ |
| 大下降序数<br>(Large Dropping Ordinal)<br>(LDO) | $(0,0,0)(1,1,1)(2,2,1)(3,2,0)$<br>$\psi\left(\lambda\alpha.\Phi(1,\alpha+1)-\Pi_0\right)$ |
| 双重 +1 稳定序数<br>(Doubly +1 Stable Ordinal)<br>(DSO) | $(0,0,0)(1,1,1)(2,2,1)(3,3,0)$<br>$\psi\left(\lambda\alpha.\left(\lambda\beta.(\beta+1)-\Pi_0\right)-\Pi_0\right)$ |
| 三重 +1 稳定序数<br>(Triply +1 Stable Ordinal)<br>(TSO) | $(0,0,0)(1,1,1)(2,2,1)(3,3,1)(4,4,0)$<br>$\psi\left(\lambda\alpha.\left(\lambda\beta.\left(\lambda\gamma.(\gamma+1)-\Pi_0\right)-\Pi_0\right)-\Pi_0\right)$ |

| Name | Value |
|---|---|
| 大 Rathjen 序数<br>(Large Rathjen's Ordinal)<br>(LRO)<br>或称为<br>赝大 Rathjen 序数<br>(pseudo Large Rathjen's Ordinal)<br>(pseudo LRO)<br>(pLRO)<br>或称为<br>小 Bashicu 序数<br>(Small Bashicu Ordinal)<br>(SBO) | $(0,0,0)(1,1,1)(2,2,2)$<br>$\psi\left(\omega-\pi-\Pi_0\right)$ |
| 最小 $\Sigma_2$ 稳定序数<br>(min $\Sigma_2$ Ordinal)<br>(M2O) | $(0,0,0)(1,1,1)(2,2,2)(3,2,2)(4,2,2)(4,2,1)$<br>$\psi\left(\psi_a\left(\psi_b\left(a_{b+1}{}^{\Omega_{b+1}}\cdot\omega\right)\right)\right)$ |
| 三行矩阵系统序数<br>(Trio Sequence System Ordinal)<br>(TSSO) | $(0,0,0,0)(1,1,1,1)$<br>$\psi($ pseudo. $\omega-$ projection $)$ |
| 大常规投影序数<br>(Large Simple Projection Ordinal)<br>(LSPO) | $(0,0,0,0)(1,1,1,1)(2,1,1,1)(3,1,0,0)(2,0,0,0)$<br>$\psi(\min\alpha$ is $\alpha-$ projection $)$ |
| 大 Omega 返回序数<br>(Big Omega Back Ordinal)<br>(BOBO) | $(0,0,0,0)(1,1,1,1)(2,2,2,2)$ |
| 四行矩阵系统序数<br>(Quadro Sequence System Ordinal)<br>(QSSO) | $(0,0,0,0,0)(1,1,1,1,1)$ |
| 小 Hydra 序数<br>(Small Hydra Ordinal)<br>(SHO) | limit of BMS<br>$\mathrm{Y}(1,3)$ |
| $\Omega$ 行矩阵系统序数<br>($\Omega$ Sequence System Ordinal)<br>($\Omega$SSO) | $\mathrm{Y}(1,3,4,2,5,8,10)$ |
| 禁戒 Hydra 序数<br>(No-Go Hydra Ordinal)<br>(GHO) | $\mathrm{Y}(1,3,4,3)$ |
| 小 Y 序列序数<br>(Small Y-Sequence Ordinal)<br>(SYO) | $\mathrm{Y}(1,\omega)$ |
| 中等 Hydra 序数<br>(Medium Hydra Ordinal)<br>(MHO) | $\omega-\mathrm{Y}(1,\omega)$ |
| 三重 CA 序数<br>(Tribly CA Ordinal)<br>(TCAO) | $\mathrm{PTO}\left(\left(\Pi_3^1-\mathrm{CA}\right)_0\right)$ |
| $\mathrm{Z}_2$ 序数<br>(Beta Universe Ordinal)<br>($\beta$O) | $\mathrm{PTO}\left(\mathrm{Z}_2\right)$ |
| Church-Kleene 序数<br>(Church-Kleene Ordinal)<br>(CKO) | $\omega_1^{\mathrm{CK}}$<br>$\Omega$ |
| 首个不可数序数<br>(First Uncountable Ordinal)<br>(FUO) | $\omega_1$ |

# Appendix F

# Large Cardinal Axioms Table

This table is quoted from [184-185], with slight modifications. An arrow indicates either a direct implication or a relative consistency implication between two cardinal axioms, or both.

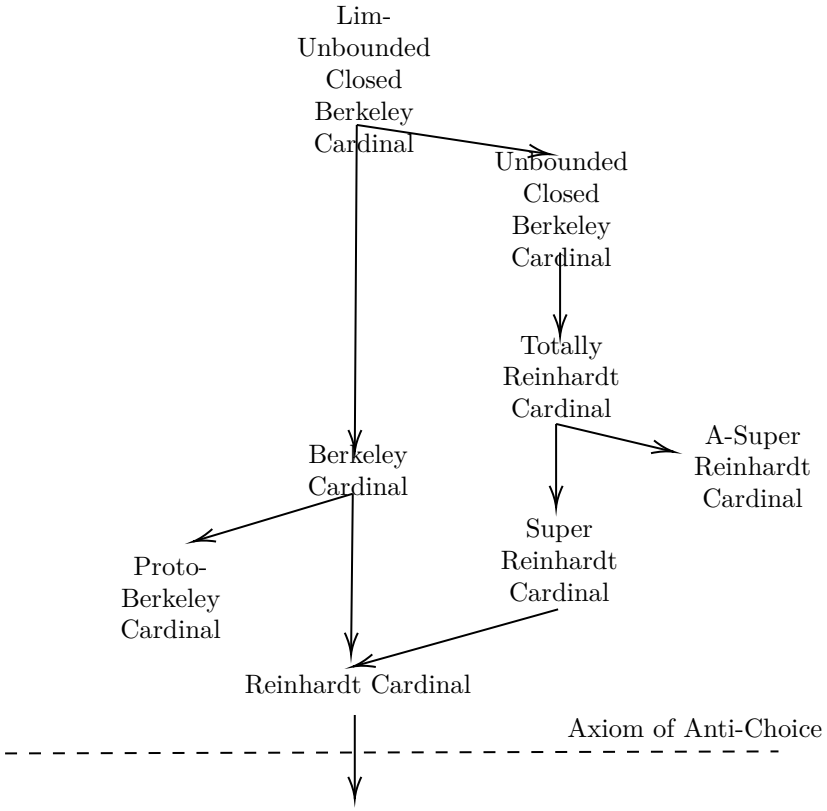


Figure F.1: The large cardinals diagram (1).

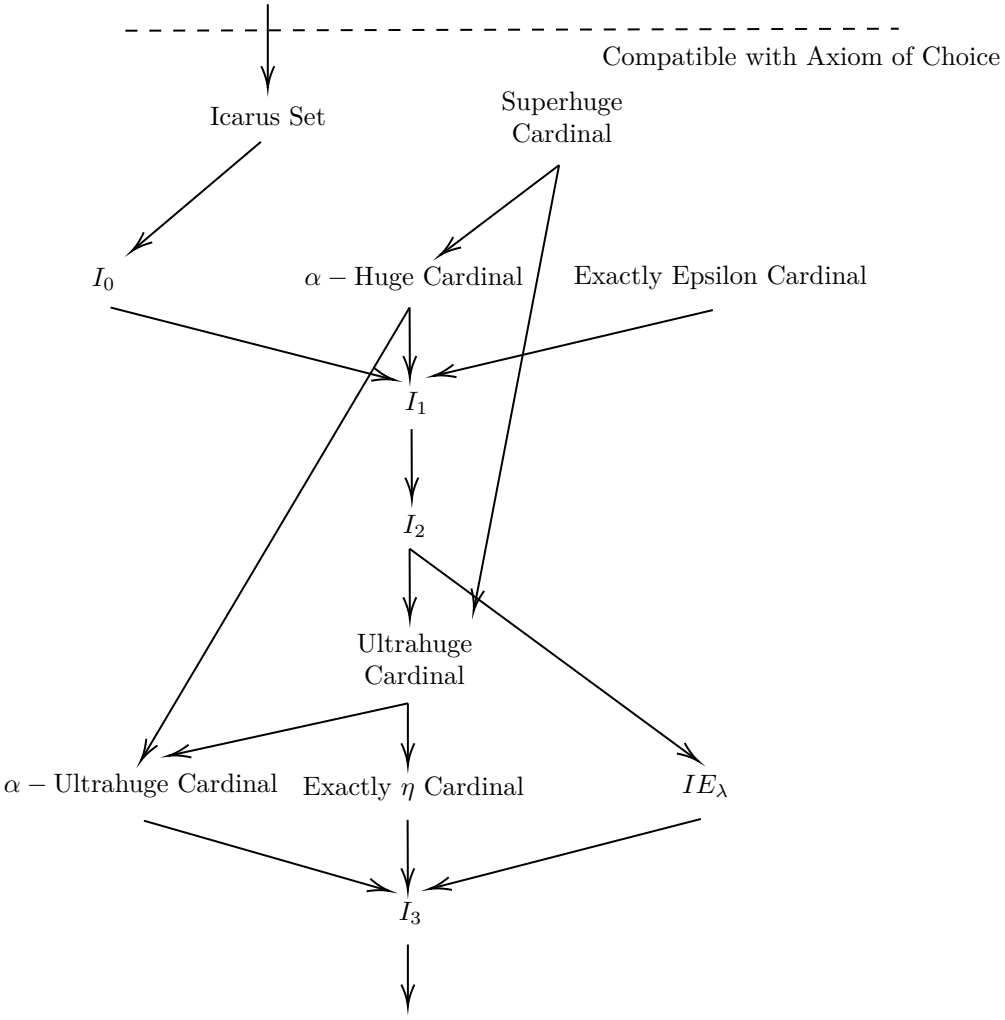


Figure F.2: The large cardinals diagram (2).

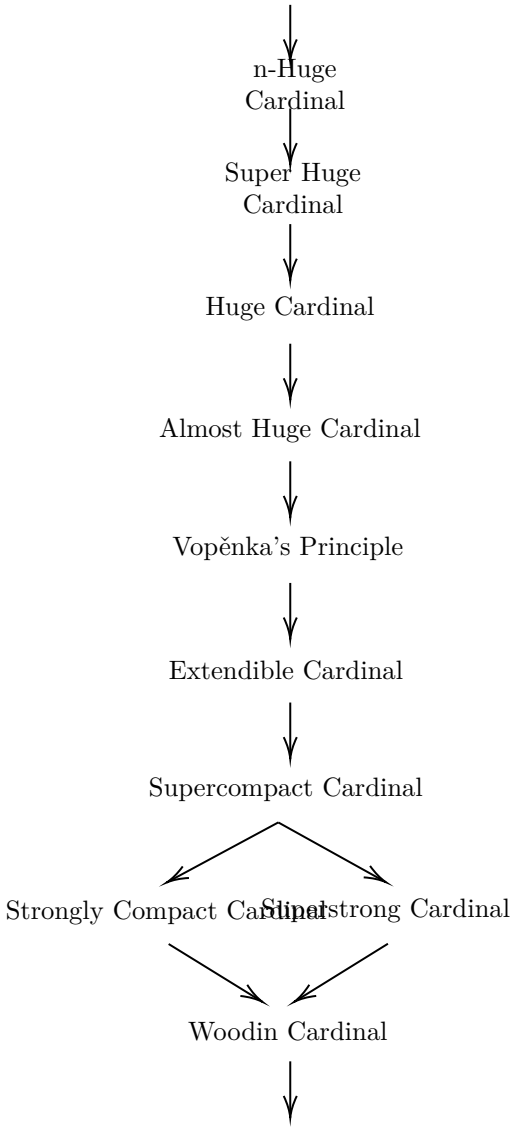


Figure F.3: The large cardinals diagram (3).

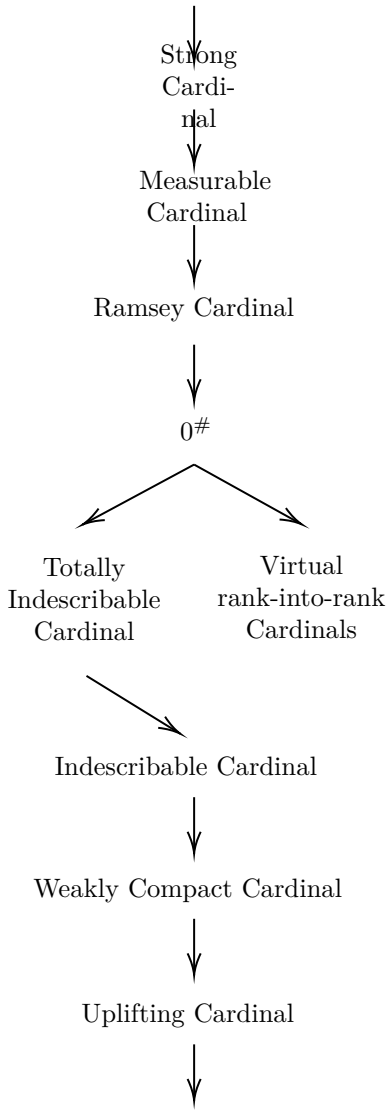


Figure F.4: The large cardinals diagram (4).

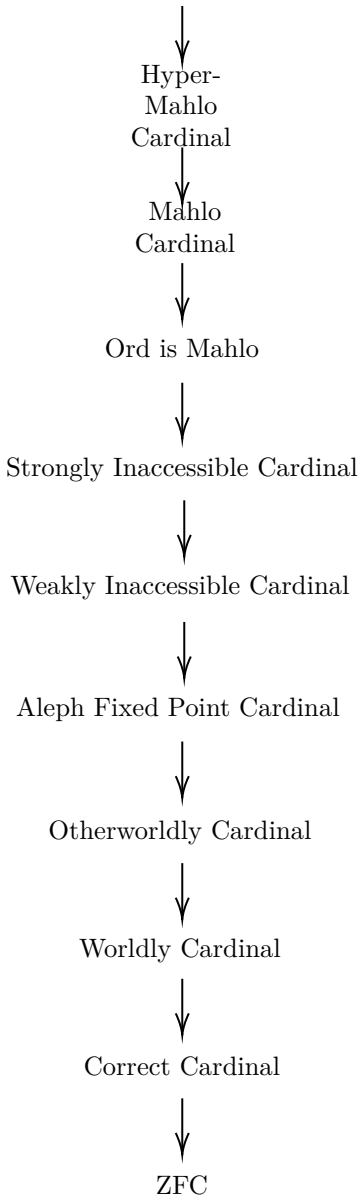


Figure F.5: The large cardinals diagram (5).

# Appendix G

# Ranking of Notations in Different Eras

The content of this table is adapted from[186], updated to 2024. Notations before 2014 are ranked: the top 30 for large-number notations and the top 15 for ordinal notations. After 2014, no distinction is made between large-number notations and ordinal notations, with the top 40 being selected. After 2024, the top 50 are selected. The ranking in the table is based on historical strength, i.e., "the relative strength of each notation as perceived in that year." If discrepancies in the strength of notations are discovered over time, previous records are not altered; instead, updates are made only in subsequent records. Notations previously considered well-defined that are later found to be ill-defined are removed from the list. If multiple versions of the same notation exist, only the strongest version from that year is recorded. The table does not include non-recursive notations, non-computable functions, or proof-theoretic ordinals. For some important notations, additional remarks are provided only in the year they were introduced.

## G.1 1980

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Graham's Function $G(n)$ | Ronald Graham |
| 2 | Graham's Function $g(n)$ | Ronald Graham |
| 3 | Hyperlicious Function | - |
| 4 | Ackermann's Function | Wilhelm Ackermann |
| 5 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 6 | Hyper Operation | - |
| 7 | Down-Arraw | Donald Ervin Knuth |
| 8 | Sudan function | Sudan |
| 9 | Gödel numbers | Gödel |
| 10 | Grzegorczyk's hierarchy | Grzegorczyk |
| 11 | $G_n + 2(n)$ function | Milton Green |
| 12 | $B_n(n)$ function | Milton Green |
| 13 | $M_n + 2(n)$ function | Milton Green |
| 14 | Peter's function | Rosza Peter |
| 15 | Robinsion's function | Raphael M. Robinson |
| 16 | Buck's function | Buck |
| 17 | Robert's function | Robert Ritchie |
| 18 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 19 | Mixed Factorial | - |
| 20 | Moser's Polygon Notation | Leo Moser |
| 21 | Pentiration | - |
| 22 | Old Polygon Notation | Leo Moser |
| 23 | Left tower | - |
| 24 | Tetration | Hans Maurer |
| 25 | Iterative Factorial | - |
| 26 | Power Towers | - |

| Ranking | Name | Proposer |
|---|---|---|
| 27 | -yillion System | Donald Ervin Knuth |
| 28 | -illion System | - |
| 29 | HuaYan Sutra | - |
| 30 | Hyper Factorial | - |
| | Ordinal | |
| 1 | Ordinal diagrams | Gaisi Takeuti |
| 2 | Feferman's $\theta$ | Fefermann |
| 3 | Bachmann's $\psi$ | Bachmann |
| 4 | Veblen Function | Oswald Veblen |
| 5 | $\Gamma$ function | - |
| 6 | Doubly variables Veblen Function | Oswald Veblen |
| 7 | $\eta$ function | - |
| 8 | $\zeta$ function | - |
| 9 | $\varepsilon$ function | - |
| 10 | Hardy Hierarchy | Stanley S.Wainer |
| 11 | $\omega$ with operation | - |
| 12 | $\omega^n$ | - |
| 13 | $\omega \cdot n$ | - |
| 14 | $\omega + n$ | - |
| 15 | $n$ | - |

## G.2 1981

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Graham's Function $G(n)$ | Ronald Graham |
| 2 | Graham's Function $g(n)$ | Ronald Graham |
| 3 | Hyperlicious Function | - |
| 4 | Ackermann's Function | Wilhelm Ackermann |
| 5 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 6 | Hyper Operation | - |
| 7 | Down-Arraw | Donald Ervin Knuth |
| 8 | Sudan function | Sudan |
| 9 | Gödel numbers | Gödel |
| 10 | Grzegorczyk's hierarchy | Grzegorczyk |
| 11 | $G_n + 2(n)$ function | Milton Green |
| 12 | $B_n(n)$ function | Milton Green |
| 13 | $M_n + 2(n)$ function | Milton Green |
| 14 | Peter's function | Rosza Peter |
| 15 | Robinsion's function | Raphael M. Robinson |
| 16 | Buck's function | Buck |
| 17 | Robert's function | Robert Ritchie |
| 18 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 19 | Mixed Factorial | - |
| 20 | Moser's Polygon Notation | Leo Moser |
| 21 | Pentiration | - |
| 22 | Old Polygon Notation | Leo Moser |
| 23 | Left tower | - |
| 24 | Tetration | Hans Maurer |
| 25 | Iterative Factorial | - |
| 26 | Power Towers | - |
| 27 | -yillion System | Donald Ervin Knuth |
| 28 | -illion System | - |
| 29 | HuaYan Sutra | - |

| Ranking | Name | Proposer |
|---|---|---|
| 30 | Hyper Factorial | - |
| | Ordinal | |
| 1 | Ordinal diagrams | Gaisi Takeuti |
| 2 | Feferman's $\theta$ | Fefermann |
| 3 | 1st Catching Point of G/F | Girard |
| 4 | Bachmann's $\psi$ | Bachmann |
| 5 | Veblen Function | Oswald Veblen |
| 6 | $\Gamma$ function | - |
| 7 | Doubly variables Veblen Function | Oswald Veblen |
| 8 | $\eta$ function | - |
| 9 | $\zeta$ function | - |
| 10 | $\varepsilon$ function | - |
| 11 | Hardy Hierarchy | Stanley S.Wainer |
| 12 | $\omega$ with operation | - |
| 13 | $\omega^n$ | - |
| 14 | $\omega \cdot n$ | - |
| 15 | $\omega + n$ | - |

## G.3 1982

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Goodstein Sequence | Goodstein |
| 2 | Graham's Function $G(n)$ | Ronald Graham |
| 3 | Graham's Function $g(n)$ | Ronald Graham |
| 4 | Hyperlicious Function | - |
| 5 | Ackermann's Function | Wilhelm Ackermann |
| 6 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 7 | Hyper Operation | - |
| 8 | Down-Arraw | Donald Ervin Knuth |
| 9 | Sudan function | Sudan |
| 10 | Gödel numbers | Gödel |
| 11 | Grzegorczyk's hierarchy | Grzegorczyk |
| 12 | $G_n + 2(n)$ function | Milton Green |
| 13 | $B_n(n)$ function | Milton Green |
| 14 | $M_n + 2(n)$ function | Milton Green |
| 15 | Peter's function | Rosza Peter |
| 16 | Robinsion's function | Raphael M. Robinson |
| 17 | Buck's function | Buck |
| 18 | Robert's function | Robert Ritchie |
| 19 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 20 | Mixed Factorial | - |
| 21 | Moser's Polygon Notation | Leo Moser |
| 22 | Pentiration | - |
| 23 | Old Polygon Notation | Leo Moser |
| 24 | Left tower | - |
| 25 | Tetration | Hans Maurer |
| 26 | Iterative Factorial | - |
| 27 | Power Towers | - |
| 28 | -yillion System | Donald Ervin Knuth |
| 29 | -illion System | - |
| 30 | HuaYan Sutra | - |
| | Ordinal | |
| 1 | Ordinal diagrams | Gaisi Takeuti |

| Ranking | Name | Proposer |
|---|---|---|
| 2 | Feferman's $\theta$ | Fefermann |
| 3 | 1st Catching Point of G/F | Girard |
| 4 | Bachmann's $\psi$ | Bachmann |
| 5 | Veblen Function | Oswald Veblen |
| 6 | $\Gamma$ function | - |
| 7 | Doubly variables Veblen Function | Oswald Veblen |
| 8 | $\eta$ function | - |
| 9 | $\zeta$ function | - |
| 10 | $\varepsilon$ function | - |
| 11 | Hardy Hierarchy | Stanley S.Wainer |
| 12 | $g$ Hierarchy | Girard |
| 13 | $\omega$ with operation | - |
| 14 | $\omega^n$ | - |
| 15 | $\omega \cdot n$ | - |

## G.4 1983

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Goodstein Sequence | Goodstein |
| 2 | Graham's Function $G(n)$ | Ronald Graham |
| 3 | Graham's Function $g(n)$ | Ronald Graham |
| 4 | Hyperlicious Function | - |
| 5 | Ackermann's Function | Wilhelm Ackermann |
| 6 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 7 | Hyper Operation | - |
| 8 | Down-Arraw | Donald Ervin Knuth |
| 9 | Sudan function | Sudan |
| 10 | Gödel numbers | Gödel |
| 11 | Grzegorczyk's hierarchy | Grzegorczyk |
| 12 | $G_n + 2(n)$ function | Milton Green |
| 13 | $B_n(n)$ function | Milton Green |
| 14 | $M_n + 2(n)$ function | Milton Green |
| 15 | Peter's function | Rosza Peter |
| 16 | Robinsion's function | Raphael M. Robinson |
| 17 | Buck's function | Buck |
| 18 | Robert's function | Robert Ritchie |
| 19 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 20 | Mixed Factorial | - |
| 21 | Moser's Polygon Notation | Leo Moser |
| 22 | Pentiration | - |
| 23 | Old Polygon Notation | Leo Moser |
| 24 | Left tower | - |
| 25 | Tetration | Hans Maurer |
| 26 | Iterative Factorial | - |
| 27 | Power Towers | - |
| 28 | -yillion System | Donald Ervin Knuth |
| 29 | -illion System | - |
| 30 | HuaYan Sutra | - |
| | Ordinal | |
| 1 | Ordinal diagrams | Gaisi Takeuti |
| 2 | Feferman's $\theta$ | Fefermann |
| 3 | 1st Catching Point of G/F | Girard |
| 4 | Bachmann's $\psi$ | Bachmann |

| Ranking | Name | Proposer |
|---|---|---|
| 5 | Veblen Function | Oswald Veblen |
| 6 | $\Gamma$ function | - |
| 7 | Doubly variables Veblen Function | Oswald Veblen |
| 8 | $\eta$ function | - |
| 9 | $\zeta$ function | - |
| 10 | $\varepsilon$ function | - |
| 11 | Hardy Hierarchy | Stanley S.Wainer |
| 12 | $g$ Hierarchy | Girard |
| 13 | $\omega$ with operation | - |
| 14 | $\omega^n$ | - |
| 15 | $\omega \cdot n$ | - |

## G.5 1984

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Goodstein Sequence | Goodstein |
| 2 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 3 | Graham's Function $G(n)$ | Ronald Graham |
| 4 | Graham's Function $g(n)$ | Ronald Graham |
| 5 | Hyperlicious Function | - |
| 6 | Ackermann's Function | Wilhelm Ackermann |
| 7 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 8 | Hyper Operation | - |
| 9 | Down-Arraw | Donald Ervin Knuth |
| 10 | Sudan function | Sudan |
| 11 | Gödel numbers | Gödel |
| 12 | Grzegorczyk's hierarchy | Grzegorczyk |
| 13 | $G_n + 2(n)$ function | Milton Green |
| 14 | $B_n(n)$ function | Milton Green |
| 15 | $M_n + 2(n)$ function | Milton Green |
| 16 | Peter's function | Rosza Peter |
| 17 | Robinsion's function | Raphael M. Robinson |
| 18 | Buck's function | Buck |
| 19 | Robert's function | Robert Ritchie |
| 20 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 21 | Mixed Factorial | - |
| 22 | Moser's Polygon Notation | Leo Moser |
| 23 | Pentiration | - |
| 24 | Old Polygon Notation | Leo Moser |
| 25 | Left tower | - |
| 26 | Tetration | Hans Maurer |
| 27 | Iterative Factorial | - |
| 28 | Power Towers | - |
| 29 | -yillion System | Donald Ervin Knuth |
| 30 | -illion System | - |
| | Ordinal | |
| 1 | Ordinal diagrams | Gaisi Takeuti |
| 2 | Feferman's $\theta$ | Fefermann |
| 3 | 1st Catching Point of G/F | Girard |
| 4 | Bachmann's $\psi$ | Bachmann |
| 5 | Veblen Function | Oswald Veblen |
| 6 | $\Gamma$ function | - |
| 7 | Doubly variables Veblen Function | Oswald Veblen |

| Ranking | Name | Proposer |
|---|---|---|
| 8 | $\eta$ function | - |
| 9 | $\zeta$ function | - |
| 10 | $\varepsilon$ function | - |
| 11 | FGH below $\varepsilon_0$ | Rose |
| 12 | Hardy Hierarchy | Stanley S.Wainer |
| 13 | $g$ Hierarchy | Girard |
| 14 | $\omega$ with operation | - |
| 15 | $\omega^n$ | - |

## G.6 1985

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Goodstein Sequence | Goodstein |
| 2 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 3 | Graham's Function $G(n)$ | Ronald Graham |
| 4 | Graham's Function $g(n)$ | Ronald Graham |
| 5 | Hyperlicious Function | - |
| 6 | Ackermann's Function | Wilhelm Ackermann |
| 7 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 8 | Hyper Operation | - |
| 9 | Down-Arraw | Donald Ervin Knuth |
| 10 | Sudan function | Sudan |
| 11 | Gödel numbers | Gödel |
| 12 | Grzegorczyk's hierarchy | Grzegorczyk |
| 13 | $G_n + 2(n)$ function | Milton Green |
| 14 | $B_n(n)$ function | Milton Green |
| 15 | $M_n + 2(n)$ function | Milton Green |
| 16 | Peter's function | Rosza Peter |
| 17 | Robinsion's function | Raphael M. Robinson |
| 18 | Buck's function | Buck |
| 19 | Robert's function | Robert Ritchie |
| 20 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 21 | Mixed Factorial | - |
| 22 | Moser's Polygon Notation | Leo Moser |
| 23 | Pentiration | - |
| 24 | Old Polygon Notation | Leo Moser |
| 25 | Left tower | - |
| 26 | Tetration | Hans Maurer |
| 27 | Iterative Factorial | - |
| 28 | Power Towers | - |
| 29 | -yillion System | Donald Ervin Knuth |
| 30 | -illion System | - |
| | Ordinal | |
| 1 | Ordinal diagrams | Gaisi Takeuti |
| 2 | Feferman's $\theta$ | Fefermann |
| 3 | 1st Catching Point of G/F | Girard |
| 4 | Bachmann's $\psi$ | Bachmann |
| 5 | Veblen Function | Oswald Veblen |
| 6 | $\Gamma$ function | - |
| 7 | Doubly variables Veblen Function | Oswald Veblen |
| 8 | $\eta$ function | - |
| 9 | $\zeta$ function | - |
| 10 | $\varepsilon$ function | - |

| Ranking | Name | Proposer |
|---|---|---|
| 11 | FGH below $\varepsilon_0$ | Rose |
| 12 | Hardy Hierarchy | Stanley S.Wainer |
| 13 | $g$ Hierarchy | Girard |
| 14 | $\omega$ with operation | - |
| 15 | $\omega^n$ | - |

## G.7 1986

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Goodstein Sequence | Goodstein |
| 2 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 3 | Graham's Function $G(n)$ | Ronald Graham |
| 4 | Graham's Function $g(n)$ | Ronald Graham |
| 5 | Hyperlicious Function | - |
| 6 | Ackermann's Function | Wilhelm Ackermann |
| 7 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 8 | Hyper Operation | - |
| 9 | Down-Arraw | Donald Ervin Knuth |
| 10 | Sudan function | Sudan |
| 11 | Gödel numbers | Gödel |
| 12 | Grzegorczyk's hierarchy | Grzegorczyk |
| 13 | $G_n + 2(n)$ function | Milton Green |
| 14 | $B_n(n)$ function | Milton Green |
| 15 | $M_n + 2(n)$ function | Milton Green |
| 16 | Peter's function | Rosza Peter |
| 17 | Robinsion's function | Raphael M. Robinson |
| 18 | Buck's function | Buck |
| 19 | Robert's function | Robert Ritchie |
| 20 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 21 | Mixed Factorial | - |
| 22 | Moser's Polygon Notation | Leo Moser |
| 23 | Pentiration | - |
| 24 | Old Polygon Notation | Leo Moser |
| 25 | Left tower | - |
| 26 | Tetration | Hans Maurer |
| 27 | Iterative Factorial | - |
| 28 | Power Towers | - |
| 29 | -yillion System | Donald Ervin Knuth |
| 30 | -illion System | - |
| | Ordinal | |
| 1 | Ordinal diagrams | Gaisi Takeuti |
| 2 | Feferman's $\theta$ | Fefermann |
| 3 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 4 | 1st Catching Point of G/F | Girard |
| 5 | Madore's $\psi$ Function | Madore |
| 6 | Bachmann's $\psi$ | Bachmann |
| 7 | Veblen Function | Oswald Veblen |
| 8 | $\Gamma$ function | - |
| 9 | Doubly variables Veblen Function | Oswald Veblen |
| 10 | $\eta$ function | - |
| 11 | $\zeta$ function | - |
| 12 | $\varepsilon$ function | - |
| 13 | FGH below $\varepsilon_0$ | Rose |

| Ranking | Name | Proposer |
|---|---|---|
| 14 | Hardy Hierarchy | Stanley S.Wainer |
| 15 | $g$ Hierarchy | Girard |

## G.8 1987

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Goodstein Sequence | Goodstein |
| 2 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 3 | Graham's Function $G(n)$ | Ronald Graham |
| 4 | Graham's Function $g(n)$ | Ronald Graham |
| 5 | Hyperlicious Function | - |
| 6 | Ackermann's Function | Wilhelm Ackermann |
| 7 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 8 | Hyper Operation | - |
| 9 | Down-Arraw | Donald Ervin Knuth |
| 10 | Sudan function | Sudan |
| 11 | Gödel numbers | Gödel |
| 12 | Grzegorczyk's hierarchy | Grzegorczyk |
| 13 | $G_n + 2(n)$ function | Milton Green |
| 14 | $B_n(n)$ function | Milton Green |
| 15 | $M_n + 2(n)$ function | Milton Green |
| 16 | Peter's function | Rosza Peter |
| 17 | Robinsion's function | Raphael M. Robinson |
| 18 | Buck's function | Buck |
| 19 | Robert's function | Robert Ritchie |
| 20 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 21 | Edelsbrunner-Herbert's function | Edelsbrunner, Herbert |
| 22 | Mixed Factorial | - |
| 23 | Moser's Polygon Notation | Leo Moser |
| 24 | Pentiration | - |
| 25 | Old Polygon Notation | Leo Moser |
| 26 | Left tower | - |
| 27 | Tetration | Hans Maurer |
| 28 | Iterative Factorial | - |
| 29 | Power Towers | - |
| 30 | -yillion System | Donald Ervin Knuth |
| | Ordinal | |
| 1 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 2 | Ordinal diagrams | Gaisi Takeuti |
| 3 | Feferman's $\theta$ | Fefermann |
| 4 | Buchholz's Hydra | Wilfried Buchholz |
| 5 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 6 | 1st Catching Point of G/F | Girard |
| 7 | Madore's $\psi$ Function | Madore |
| 8 | Bachmann's $\psi$ | Bachmann |
| 9 | Veblen Function | Oswald Veblen |
| 10 | $\Gamma$ function | - |
| 11 | Doubly variables Veblen Function | Oswald Veblen |
| 12 | $\eta$ function | - |
| 13 | $\zeta$ function | - |
| 14 | $\varepsilon$ function | - |
| 15 | FGH below $\varepsilon_0$ | Rose |

## G.9 1988

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Goodstein Sequence | Goodstein |
| 2 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 3 | Graham's Function $G(n)$ | Ronald Graham |
| 4 | Graham's Function $g(n)$ | Ronald Graham |
| 5 | Hyperlicious Function | - |
| 6 | Ackermann's Function | Wilhelm Ackermann |
| 7 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 8 | Hyper Operation | - |
| 9 | Down-Arraw | Donald Ervin Knuth |
| 10 | Sudan function | Sudan |
| 11 | Gödel numbers | Gödel |
| 12 | Grzegorczyk's hierarchy | Grzegorczyk |
| 13 | $G_n + 2(n)$ function | Milton Green |
| 14 | $B_n(n)$ function | Milton Green |
| 15 | $M_n + 2(n)$ function | Milton Green |
| 16 | Peter's function | Rosza Peter |
| 17 | Robinsion's function | Raphael M. Robinson |
| 18 | Buck's function | Buck |
| 19 | Robert's function | Robert Ritchie |
| 20 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 21 | Edelsbrunner-Herbert's function | Edelsbrunner, Herbert |
| 22 | Alteration of M/R's Ackf | - |
| 23 | Mixed Factorial | - |
| 24 | Moser's Polygon Notation | Leo Moser |
| 25 | Pentiration | - |
| 26 | Old Polygon Notation | Leo Moser |
| 27 | Left tower | - |
| 28 | Tetration | Hans Maurer |
| 29 | Iterative Factorial | - |
| 30 | Power Towers | - |
| | Ordinal | |
| 1 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 2 | Ordinal diagrams | Gaisi Takeuti |
| 3 | Feferman's $\theta$ | Fefermann |
| 4 | Buchholz's Hydra | Wilfried Buchholz |
| 5 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 6 | 1st Catching Point of G/F | Girard |
| 7 | Madore's $\psi$ Function | Madore |
| 8 | Bachmann's $\psi$ | Bachmann |
| 9 | Veblen Function | Oswald Veblen |
| 10 | $\Gamma$ function | - |
| 11 | Doubly variables Veblen Function | Oswald Veblen |
| 12 | $\eta$ function | - |
| 13 | $\zeta$ function | - |
| 14 | $\varepsilon$ function | - |
| 15 | FGH below $\varepsilon_0$ | Rose |

## G.10 1989

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Goodstein Sequence | Goodstein |
| 2 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 3 | Graham's Function $G(n)$ | Ronald Graham |
| 4 | Graham's Function $g(n)$ | Ronald Graham |
| 5 | Hyperlicious Function | - |
| 6 | Ackermann's Function | Wilhelm Ackermann |
| 7 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 8 | Hyper Operation | - |
| 9 | Down-Arraw | Donald Ervin Knuth |
| 10 | Sudan function | Sudan |
| 11 | Gödel numbers | Gödel |
| 12 | Grzegorczyk's hierarchy | Grzegorczyk |
| 13 | $G_n + 2(n)$ function | Milton Green |
| 14 | $B_n(n)$ function | Milton Green |
| 15 | $M_n + 2(n)$ function | Milton Green |
| 16 | Peter's function | Rosza Peter |
| 17 | Robinsion's function | Raphael M. Robinson |
| 18 | Buck's function | Buck |
| 19 | Robert's function | Robert Ritchie |
| 20 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 21 | Edelsbrunner-Herbert's function | Edelsbrunner, Herbert |
| 22 | Alteration of M/R's Ackf | - |
| 23 | Mixed Factorial | - |
| 24 | Moser's Polygon Notation | Leo Moser |
| 25 | Pentiration | - |
| 26 | Old Polygon Notation | Leo Moser |
| 27 | Left tower | - |
| 28 | Tetration | Hans Maurer |
| 29 | Iterative Factorial | - |
| 30 | Power Towers | - |
| # | Laver Table 被研究出 | |
| | Ordinal | |
| 1 | Rathjen's $\chi$ | Rathjen |
| 2 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 3 | Ordinal diagrams | Gaisi Takeuti |
| 4 | Feferman's $\theta$ | Fefermann |
| 5 | Buchholz's Hydra | Wilfried Buchholz |
| 6 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 7 | 1st Catching Point of G/F | Girard |
| 8 | Madore's $\psi$ Function | Madore |
| 9 | Bachmann's $\psi$ | Bachmann |
| 10 | Veblen Function | Oswald Veblen |
| 11 | $\Gamma$ function | - |
| 12 | Doubly variables Veblen Function | Oswald Veblen |
| 13 | $\eta$ function | - |
| 14 | $\zeta$ function | - |
| 15 | $\varepsilon$ function | - |

## G.11 1990

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Goodstein Sequence | Goodstein |

| Ranking | Name | Proposer |
|---|---|---|
| 2 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 3 | Graham's Function $G(n)$ | Ronald Graham |
| 4 | Graham's Function $g(n)$ | Ronald Graham |
| 5 | Hyperlicious Function | - |
| 6 | Ackermann's Function | Wilhelm Ackermann |
| 7 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 8 | Hyper Operation | - |
| 9 | Down-Arraw | Donald Ervin Knuth |
| 10 | Sudan function | Sudan |
| 11 | Gödel numbers | Gödel |
| 12 | Grzegorczyk's hierarchy | Grzegorczyk |
| 13 | $G_n + 2(n)$ function | Milton Green |
| 14 | $B_n(n)$ function | Milton Green |
| 15 | $M_n + 2(n)$ function | Milton Green |
| 16 | Peter's function | Rosza Peter |
| 17 | Robinsion's function | Raphael M. Robinson |
| 18 | Buck's function | Buck |
| 19 | Robert's function | Robert Ritchie |
| 20 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 21 | Edelsbrunner-Herbert's function | Edelsbrunner, Herbert |
| 22 | Alteration of M/R's Ackf | - |
| 23 | Mixed Factorial | - |
| 24 | Moser's Polygon Notation | Leo Moser |
| 25 | Pentiration | - |
| 26 | Old Polygon Notation | Leo Moser |
| 27 | Left tower | - |
| 28 | Tetration | Hans Maurer |
| 29 | Iterative Factorial | - |
| 30 | Power Towers | - |
| | Ordinal | |
| 1 | Rathjen's $\Xi$ | Rathjen |
| 2 | Rathjen's $\chi$ | Rathjen |
| 3 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 4 | Ordinal diagrams | Gaisi Takeuti |
| 5 | Feferman's $\theta$ | Fefermann |
| 6 | Buchholz's Hydra | Wilfried Buchholz |
| 7 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 8 | 1st Catching Point of G/F | Girard |
| 9 | Madore's $\psi$ Function | Madore |
| 10 | Bachmann's $\psi$ | Bachmann |
| 11 | Veblen Function | Oswald Veblen |
| 12 | $\Gamma$ function | - |
| 13 | Doubly variables Veblen Function | Oswald Veblen |
| 14 | $\eta$ function | - |
| 15 | $\zeta$ function | - |

## G.12 1991

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Goodstein Sequence | Goodstein |
| 2 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 3 | Graham's Function $G(n)$ | Ronald Graham |
| 4 | Graham's Function $g(n)$ | Ronald Graham |

| Ranking | Name | Proposer |
|---|---|---|
| 5 | Hyperlicious Function | - |
| 6 | Ackermann's Function | Wilhelm Ackermann |
| 7 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 8 | Hyper Operation | - |
| 9 | Down-Arraw | Donald Ervin Knuth |
| 10 | Sudan function | Sudan |
| 11 | Gödel numbers | Gödel |
| 12 | Grzegorczyk's hierarchy | Grzegorczyk |
| 13 | $G_n + 2(n)$ function | Milton Green |
| 14 | $B_n(n)$ function | Milton Green |
| 15 | $M_n + 2(n)$ function | Milton Green |
| 16 | Peter's function | Rosza Peter |
| 17 | Robinsion's function | Raphael M. Robinson |
| 18 | Buck's function | Buck |
| 19 | Robert's function | Robert Ritchie |
| 20 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 21 | Edelsbrunner-Herbert's function | Edelsbrunner, Herbert |
| 22 | Alteration of M/R's Ackf | - |
| 23 | Mixed Factorial | - |
| 24 | Moser's Polygon Notation | Leo Moser |
| 25 | Pentiration | - |
| 26 | Old Polygon Notation | Leo Moser |
| 27 | Left tower | - |
| 28 | Tetration | Hans Maurer |
| 29 | Iterative Factorial | - |
| 30 | Power Towers | - |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 5 | Ordinal diagrams | Gaisi Takeuti |
| 6 | Feferman's $\theta$ | Fefermann |
| 7 | Buchholz's Hydra | Wilfried Buchholz |
| 8 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 9 | 1st Catching Point of G/F | Girard |
| 10 | Madore's $\psi$ Function | Madore |
| 11 | Bachmann's $\psi$ | Bachmann |
| 12 | Veblen Function | Oswald Veblen |
| 13 | $\Gamma$ function | - |
| 14 | Doubly variables Veblen Function | Oswald Veblen |
| 15 | $\eta$ function | - |

## G.13 1992

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Goodstein Sequence | Goodstein |
| 2 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 3 | Graham's Function $G(n)$ | Ronald Graham |
| 4 | Graham's Function $g(n)$ | Ronald Graham |
| 5 | Hyperlicious Function | - |
| 6 | Ackermann's Function | Wilhelm Ackermann |

| Ranking | Name | Proposer |
|---|---|---|
| 7 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 8 | Hyper Operation | - |
| 9 | Down-Arraw | Donald Ervin Knuth |
| 10 | Sudan function | Sudan |
| 11 | Gödel numbers | Gödel |
| 12 | Grzegorczyk's hierarchy | Grzegorczyk |
| 13 | $G_n + 2(n)$ function | Milton Green |
| 14 | $B_n(n)$ function | Milton Green |
| 15 | $M_n + 2(n)$ function | Milton Green |
| 16 | Peter's function | Rosza Peter |
| 17 | Robinsion's function | Raphael M. Robinson |
| 18 | Buck's function | Buck |
| 19 | Robert's function | Robert Ritchie |
| 20 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 21 | Edelsbrunner-Herbert's function | Edelsbrunner, Herbert |
| 22 | Alteration of M/R's Ackf | - |
| 23 | Mixed Factorial | - |
| 24 | Moser's Polygon Notation | Leo Moser |
| 25 | Pentiration | - |
| 26 | Old Polygon Notation | Leo Moser |
| 27 | Left tower | - |
| 28 | Tetration | Hans Maurer |
| 29 | Iterative Factorial | - |
| 30 | Power Towers | - |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 5 | Ordinal diagrams | Gaisi Takeuti |
| 6 | Feferman's $\theta$ | Fefermann |
| 7 | Buchholz's Hydra | Wilfried Buchholz |
| 8 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 9 | 1st Catching Point of G/F | Girard |
| 10 | Madore's $\psi$ Function | Madore |
| 11 | Bachmann's $\psi$ | Bachmann |
| 12 | Veblen Function | Oswald Veblen |
| 13 | $\Gamma$ function | - |
| 14 | Doubly variables Veblen Function | Oswald Veblen |
| 15 | $\eta$ function | - |

## G.14 1993

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Goodstein Sequence | Goodstein |
| 2 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 3 | Graham's Function $G(n)$ | Ronald Graham |
| 4 | Graham's Function $g(n)$ | Ronald Graham |
| 5 | Hyperlicious Function | - |
| 6 | Ackermann's Function | Wilhelm Ackermann |
| 7 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 8 | Hyper Operation | - |

| Ranking | Name | Proposer |
|---|---|---|
| 9 | Down-Arraw | Donald Ervin Knuth |
| 10 | Sudan function | Sudan |
| 11 | Gödel numbers | Gödel |
| 12 | Grzegorczyk's hierarchy | Grzegorczyk |
| 13 | $G_n + 2(n)$ function | Milton Green |
| 14 | $B_n(n)$ function | Milton Green |
| 15 | $M_n + 2(n)$ function | Milton Green |
| 16 | Peter's function | Rosza Peter |
| 17 | Robinsion's function | Raphael M. Robinson |
| 18 | Buck's function | Buck |
| 19 | Robert's function | Robert Ritchie |
| 20 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 21 | Edelsbrunner-Herbert's function | Edelsbrunner, Herbert |
| 22 | Alteration of M/R's Ackf | - |
| 23 | Mixed Factorial | - |
| 24 | Moser's Polygon Notation | Leo Moser |
| 25 | Pentiration | - |
| 26 | Old Polygon Notation | Leo Moser |
| 27 | Left tower | - |
| 28 | Tetration | Hans Maurer |
| 29 | Iterative Factorial | - |
| 30 | Power Towers | - |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 5 | Ordinal diagrams | Gaisi Takeuti |
| 6 | Feferman's $\theta$ | Fefermann |
| 7 | Buchholz's Hydra | Wilfried Buchholz |
| 8 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 9 | 1st Catching Point of G/F | Girard |
| 10 | Madore's $\psi$ Function | Madore |
| 11 | Bachmann's $\psi$ | Bachmann |
| 12 | Veblen Function | Oswald Veblen |
| 13 | $\Gamma$ function | - |
| 14 | Doubly variables Veblen Function | Oswald Veblen |
| 15 | $\eta$ function | - |

## G.15 1994

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Goodstein Sequence | Goodstein |
| 2 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 3 | Graham's Function $G(n)$ | Ronald Graham |
| 4 | Graham's Function $g(n)$ | Ronald Graham |
| 5 | Hyperlicious Function | - |
| 6 | Ackermann's Function | Wilhelm Ackermann |
| 7 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 8 | Hyper Operation | - |
| 9 | Down-Arraw | Donald Ervin Knuth |
| 10 | Sudan function | Sudan |

| Ranking | Name | Proposer |
|---|---|---|
| 11 | Gödel numbers | Gödel |
| 12 | Grzegorczyk's hierarchy | Grzegorczyk |
| 13 | $G_n + 2(n)$ function | Milton Green |
| 14 | $B_n(n)$ function | Milton Green |
| 15 | $M_n + 2(n)$ function | Milton Green |
| 16 | Peter's function | Rosza Peter |
| 17 | Robinsion's function | Raphael M. Robinson |
| 18 | Buck's function | Buck |
| 19 | Robert's function | Robert Ritchie |
| 20 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 21 | Edelsbrunner-Herbert's function | Edelsbrunner, Herbert |
| 22 | Alteration of M/R's Ackf | - |
| 23 | Baxter's Derivation | Lew Baxter |
| 24 | Mixed Factorial | - |
| 25 | Moser's Polygon Notation | Leo Moser |
| 26 | Pentiration | - |
| 27 | Old Polygon Notation | Leo Moser |
| 28 | Left tower | - |
| 29 | Tetration | Hans Maurer |
| 30 | Iterative Factorial | - |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 5 | Ordinal diagrams | Gaisi Takeuti |
| 6 | Feferman's $\theta$ | Fefermann |
| 7 | Buchholz's Hydra | Wilfried Buchholz |
| 8 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 9 | 1st Catching Point of G/F | Girard |
| 10 | Madore's $\psi$ Function | Madore |
| 11 | Bachmann's $\psi$ | Bachmann |
| 12 | Veblen Function | Oswald Veblen |
| 13 | $\Gamma$ function | - |
| 14 | Doubly variables Veblen Function | Oswald Veblen |
| 15 | $\eta$ function | - |

## G.16 1995

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Goodstein Sequence | Goodstein |
| 2 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 3 | CG Function | John Horton Conway, Richard Kenneth Guy |
| 4 | Chained Arrow | John Horton Conway, Richard Kenneth Guy |
| 5 | Graham's Function $G(n)$ | Ronald Graham |
| 6 | Graham's Function $g(n)$ | Ronald Graham |
| 7 | Hyperlicious Function | - |
| 8 | Ackermann's Function | Wilhelm Ackermann |
| 9 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 10 | Hyper Operation | - |

| Ranking | Name | Proposer |
|---|---|---|
| 11 | Down-Arraw | Donald Ervin Knuth |
| 12 | Sudan function | Sudan |
| 13 | Gödel numbers | Gödel |
| 14 | Grzegorczyk's hierarchy | Grzegorczyk |
| 15 | $G_n + 2(n)$ function | Milton Green |
| 16 | $B_n(n)$ function | Milton Green |
| 17 | $M_n + 2(n)$ function | Milton Green |
| 18 | Peter's function | Rosza Peter |
| 19 | Robinsion's function | Raphael M. Robinson |
| 20 | Buck's function | Buck |
| 21 | Robert's function | Robert Ritchie |
| 22 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 23 | Edelsbrunner-Herbert's function | Edelsbrunner, Herbert |
| 24 | Alteration of M/R's Ackf | - |
| 25 | Baxter's Derivation | Lew Baxter |
| 26 | Mixed Factorial | - |
| 27 | Moser's Polygon Notation | Leo Moser |
| 28 | Pentiration | - |
| 29 | Old Polygon Notation | Leo Moser |
| 30 | Left tower | - |
| # | Laver Table<br>在 ZFC+I0 中被证明存在 | |
| | Ordinal | |
| 1 | Rathjen's<br>Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 5 | Ordinal diagrams | Gaisi Takeuti |
| 6 | Feferman's $\theta$ | Fefermann |
| 7 | Buchholz's Hydra | Wilfried Buchholz |
| 8 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 9 | 1st Catching Point of G/F | Girard |
| 10 | Madore's $\psi$ Function | Madore |
| 11 | Bachmann's $\psi$ | Bachmann |
| 12 | Veblen Function | Oswald Veblen |
| 13 | $\Gamma$ function | - |
| 14 | Doubly variables Veblen Function | Oswald Veblen |
| 15 | $\eta$ function | - |

## G.17 1996

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Goodstein Sequence | Goodstein |
| 2 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 3 | CG Function | John Horton Conway,<br>Richard Kenneth Guy |
| 4 | Chained Arrow | John Horton Conway,<br>Richard Kenneth Guy |
| 5 | Graham's Function $G(n)$ | Ronald Graham |
| 6 | Graham's Function $g(n)$ | Ronald Graham |
| 7 | Hyperlicious Function | - |
| 8 | Ackermann's Function | Wilhelm Ackermann |

| Ranking | Name | Proposer |
|---|---|---|
| 9 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 10 | Hyper Operation | - |
| 11 | Down-Arraw | Donald Ervin Knuth |
| 12 | Munafo's Hyper Operation | Munafo |
| 13 | Sudan function | Sudan |
| 14 | Gödel numbers | Gödel |
| 15 | Grzegorczyk's hierarchy | Grzegorczyk |
| 16 | $G_n + 2(n)$ function | Milton Green |
| 17 | $B_n(n)$ function | Milton Green |
| 18 | $M_n + 2(n)$ function | Milton Green |
| 19 | Peter's function | Rosza Peter |
| 20 | Robinsion's function | Raphael M. Robinson |
| 21 | Buck's function | Buck |
| 22 | Robert's function | Robert Ritchie |
| 23 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 24 | Edelsbrunner-Herbert's function | Edelsbrunner, Herbert |
| 25 | Alteration of M/R's Ackf | - |
| 26 | Baxter's Derivation | Lew Baxter |
| 27 | E# (Old) | - |
| 28 | Mixed Factorial | - |
| 29 | Moser's Polygon Notation | Leo Moser |
| 30 | Pentiration | - |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 5 | Ordinal diagrams | Gaisi Takeuti |
| 6 | Feferman's $\theta$ | Fefermann |
| 7 | Buchholz's Hydra | Wilfried Buchholz |
| 8 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 9 | 1st Catching Point of G/F | Girard |
| 10 | Madore's $\psi$ Function | Madore |
| 11 | Bachmann's $\psi$ | Bachmann |
| 12 | Veblen Function | Oswald Veblen |
| 13 | $\Gamma$ function | - |
| 14 | Doubly variables Veblen Function | Oswald Veblen |
| 15 | $\eta$ function | - |

## G.18 1997

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Goodstein Sequence | Goodstein |
| 2 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 3 | CG Function | John Horton Conway, Richard Kenneth Guy |
| 4 | Chained Arrow | John Horton Conway, Richard Kenneth Guy |
| 5 | Graham's Function $G(n)$ | Ronald Graham |
| 6 | Graham's Function $g(n)$ | Ronald Graham |
| 7 | Hyperlicious Function | - |
| 8 | Ackermann's Function | Wilhelm Ackermann |

| Ranking | Name | Proposer |
|---|---|---|
| 9 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 10 | Hyper Operation | - |
| 11 | Down-Arraw | Donald Ervin Knuth |
| 12 | Munafo's Hyper Operation | Munafo |
| 13 | Mythical tree problem | Harvey Friedman |
| 14 | Sudan function | Sudan |
| 15 | Gödel numbers | Gödel |
| 16 | Grzegorczyk's hierarchy | Grzegorczyk |
| 17 | $G_n + 2(n)$ function | Milton Green |
| 18 | $B_n(n)$ function | Milton Green |
| 19 | $M_n + 2(n)$ function | Milton Green |
| 20 | Peter's function | Rosza Peter |
| 21 | Robinsion's function | Raphael M. Robinson |
| 22 | Buck's function | Buck |
| 23 | Robert's function | Robert Ritchie |
| 24 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 25 | Edelsbrunner-Herbert's function | Edelsbrunner, Herbert |
| 26 | Alteration of M/R's Ackf | - |
| 27 | Baxter's Derivation | Lew Baxter |
| 28 | $E\#$ (Old) | - |
| 29 | Mixed Factorial | - |
| 30 | Moser's Polygon Notation | Leo Moser |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 5 | Ordinal diagrams | Gaisi Takeuti |
| 6 | Feferman's $\theta$ | Fefermann |
| 7 | Buchholz's Hydra | Wilfried Buchholz |
| 8 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 9 | 1st Catching Point of G/F | Girard |
| 10 | Madore's $\psi$ Function | Madore |
| 11 | Bachmann's $\psi$ | Bachmann |
| 12 | Veblen Function | Oswald Veblen |
| 13 | $\Gamma$ function | - |
| 14 | Doubly variables Veblen Function | Oswald Veblen |
| 15 | $\eta$ function | - |

## G.19 1998

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Friedman's Sequence | Harvey Friedman |
| 2 | Goodstein Sequence | Goodstein |
| 3 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 4 | CG Function | John Horton Conway, Richard Kenneth Guy |
| 5 | Chained Arrow | John Horton Conway, Richard Kenneth Guy |
| 6 | Clarkkkkson | - |
| 7 | Graham's Function $G(n)$ | Ronald Graham |
| 8 | Graham's Function $g(n)$ | Ronald Graham |

| Ranking | Name | Proposer |
|---|---|---|
| 9 | Hyperlicious Function | - |
| 10 | Ackermann's Function | Wilhelm Ackermann |
| 11 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 12 | Hyper Operation | - |
| 13 | Down-Arraw | Donald Ervin Knuth |
| 14 | Munafo's Hyper Operation | Munafo |
| 15 | Mythical tree problem | Harvey Friedman |
| 16 | Munafo's function | Munafo |
| 17 | Sudan function | Sudan |
| 18 | Gödel numbers | Gödel |
| 19 | Grzegorczyk's hierarchy | Grzegorczyk |
| 20 | $G_n + 2(n)$ function | Milton Green |
| 21 | $B_n(n)$ function | Milton Green |
| 22 | $M_n + 2(n)$ function | Milton Green |
| 23 | Peter's function | Rosza Peter |
| 24 | Robinsion's function | Raphael M. Robinson |
| 25 | Buck's function | Buck |
| 26 | Robert's function | Robert Ritchie |
| 27 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 28 | Edelsbrunner-Herbert's function | Edelsbrunner, Herbert |
| 29 | Alteration of M/R's Ackf | - |
| 30 | Baxter's Derivation | Lew Baxter |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 5 | Ordinal diagrams | Gaisi Takeuti |
| 6 | Feferman's $\theta$ | Fefermann |
| 7 | Buchholz's Hydra | Wilfried Buchholz |
| 8 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 9 | 1st Catching Point of G/F | Girard |
| 10 | Madore's $\psi$ Function | Madore |
| 11 | Bachmann's $\psi$ | Bachmann |
| 12 | Veblen Function | Oswald Veblen |
| 13 | $\Gamma$ function | - |
| 14 | Doubly variables Veblen Function | Oswald Veblen |
| 15 | $\eta$ function | - |

## G.20 1999

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Friedman's Sequence | Harvey Friedman |
| 2 | Goodstein Sequence | Goodstein |
| 3 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 4 | CG Function | John Horton Conway, Richard Kenneth Guy |
| 5 | Chained Arrow | John Horton Conway, Richard Kenneth Guy |
| 6 | Clarkkkkson | - |
| 7 | Graham's Function $G(n)$ | Ronald Graham |
| 8 | Graham's Function $g(n)$ | Ronald Graham |

| Ranking | Name | Proposer |
|---|---|---|
| 9 | Hyperlicious Function | - |
| 10 | Ackermann's Function | Wilhelm Ackermann |
| 11 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 12 | Hyper Operation | - |
| 13 | Bowers' Operators | Jonathan Bowers |
| 14 | Nambir's Hyper Operation | Nambir |
| 15 | R-R Hyper operation | Ruzorbov Romolio |
| 16 | Down-Arraw | Donald Ervin Knuth |
| 17 | Munafo's Hyper Operation | Munafo |
| 18 | Mythical tree problem | Harvey Friedman |
| 19 | Sudan function | Sudan |
| 20 | Gödel numbers | Gödel |
| 21 | Grzegorczyk's hierarchy | Grzegorczyk |
| 22 | $G_n + 2(n)$ function | Milton Green |
| 23 | $B_n(n)$ function | Milton Green |
| 24 | $M_n + 2(n)$ function | Milton Green |
| 25 | Peter's function | Rosza Peter |
| 26 | Robinsion's function | Raphael M. Robinson |
| 27 | Buck's function | Buck |
| 28 | Robert's function | Robert Ritchie |
| 29 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| 30 | Edelsbrunner-Herbert's function | Edelsbrunner, Herbert |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 5 | Ordinal diagrams | Gaisi Takeuti |
| 6 | Feferman's $\theta$ | Fefermann |
| 7 | Buchholz's Hydra | Wilfried Buchholz |
| 8 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 9 | 1st Catching Point of G/F | Girard |
| 10 | Madore's $\psi$ Function | Madore |
| 11 | Bachmann's $\psi$ | Bachmann |
| 12 | Veblen Function | Oswald Veblen |
| 13 | $\Gamma$ function | - |
| 14 | Doubly variables Veblen Function | Oswald Veblen |
| 15 | $\eta$ function | - |

## G.21 2000

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Friedman's Sequence | Harvey Friedman |
| 2 | Goodstein Sequence | Goodstein |
| 3 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 4 | CG Function | John Horton Conway, Richard Kenneth Guy |
| 5 | Bowers' {} | Jonathan Bowers |
| 6 | Chained Arrow | John Horton Conway, Richard Kenneth Guy |
| 7 | Clarkkkkson | - |
| 8 | Graham's Function $G(n)$ | Ronald Graham |

| Ranking | Name | Proposer |
|---|---|---|
| 9 | Graham's Function $g(n)$ | Ronald Graham |
| 10 | Hyperlicious Function | - |
| 11 | Ackermann's Function | Wilhelm Ackermann |
| 12 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 13 | Hyper Operation | - |
| 14 | Bowers' Operators | Jonathan Bowers |
| 15 | Nambir's Hyper Operation | Nambir |
| 16 | R-R Hyper Operation | Ruzorbov Romolio |
| 17 | Down-Arraw | Donald Ervin Knuth |
| 18 | Munafo's Hyper Operation | Munafo |
| 19 | Mythical tree problem | Harvey Friedman |
| 20 | Sudan function | Sudan |
| 21 | Gödel numbers | Gödel |
| 22 | Grzegorczyk's hierarchy | Grzegorczyk |
| 23 | $G_n + 2(n)$ function | Milton Green |
| 24 | $B_n(n)$ function | Milton Green |
| 25 | $M_n + 2(n)$ function | Milton Green |
| 26 | Peter's function | Rosza Peter |
| 27 | Robinsion's function | Raphael M. Robinson |
| 28 | Buck's function | Buck |
| 29 | Robert's function | Robert Ritchie |
| 30 | Meyer-Ritchie function | Meyer, Robert Ritchie |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 5 | Ordinal diagrams | Gaisi Takeuti |
| 6 | Feferman's $\theta$ | Fefermann |
| 7 | Buchholz's Hydra | Wilfried Buchholz |
| 9 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 9 | 1st Catching Point of G/F | Girard |
| 10 | Madore's $\psi$ Function | Madore |
| 11 | Bachmann's $\psi$ | Bachmann |
| 12 | Veblen Function | Oswald Veblen |
| 13 | otree function | Harvey Friedman |
| 14 | tree function | Harvey Friedman |
| 15 | $\Gamma$ function | - |

## G.22 2001

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | marxen.c function | Heiner Marxen |
| 2 | Friedman's Sequence | Harvey Friedman |
| 3 | Goodstein Sequence | Goodstein |
| 4 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 5 | Extended Bowers'Array Notation | Jonathan Bowers |
| 6 | Bowers'Array Notation | Jonathan Bowers |
| 7 | Pete.C function series | Pete |
| 8 | CG Function | John Horton Conway, Richard Kenneth Guy |
| 9 | Bowers' {} | Jonathan Bowers |

| Ranking | Name | Proposer |
|---|---|---|
| 10 | Chained Arrow | John Horton Conway, Richard Kenneth Guy |
| 11 | Clarkkkkson | - |
| 12 | Graham's Function $G(n)$ | Ronald Graham |
| 13 | Graham's Function $g(n)$ | Ronald Graham |
| 14 | Hyperlicious Function | - |
| 15 | Ackermann's Function | Wilhelm Ackermann |
| 16 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 17 | Hyper Operation | - |
| 18 | Bowers' Operators | Jonathan Bowers |
| 19 | Nambir's Hyper Operation | Nambir |
| 20 | R-R Hyper Operation | Ruzorbov Romolio |
| 21 | Down-Arraw | Donald Ervin Knuth |
| 22 | Munafo's Hyper Operation | Munafo |
| 23 | Mythical tree problem | Harvey Friedman |
| 24 | Sudan function | Sudan |
| 25 | Gödel numbers | Gödel |
| 26 | Grzegorczyk's hierarchy | Grzegorczyk |
| 27 | $G_n + 2(n)$ function | Milton Green |
| 28 | $B_n(n)$ function | Milton Green |
| 29 | $M_n + 2(n)$ function | Milton Green |
| 30 | Peter's function | Rosza Peter |
| # | Loader.C 被 Loader 提出 | |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 5 | Ordinal diagrams | Gaisi Takeuti |
| 6 | Feferman's $\theta$ | Fefermann |
| 7 | Buchholz's Hydra | Wilfried Buchholz |
| 8 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 9 | Patterns of Resemblance | Carlson |
| 10 | 1st Catching Point of G/F | Girard |
| 11 | Madore's $\psi$ Function | Madore |
| 12 | Bachmann's $\psi$ | Bachmann |
| 13 | Veblen Function | Oswald Veblen |
| 14 | TREE function | Harvey Friedman |
| 15 | otree function | Harvey Friedman |

## G.23 2002

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | marxen.c function | Heiner Marxen |
| 2 | Multi Dimensional Arrays | Jonathan Bowers |
| 3 | Friedman's Sequence | Harvey Friedman |
| 4 | Worm function | - |
| 5 | Goodstein Sequence | Goodstein |
| 6 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 7 | Dimensional Arrays | Jonathan Bowers |
| 8 | Linar Arrays | Jonathan Bowers |
| 9 | Extended Bowers'Array Notation | Jonathan Bowers |

| Ranking | Name | Proposer |
|---|---|---|
| 10 | Bowers'Array Notation | Jonathan Bowers |
| 11 | Pete.C function series | Pete |
| 12 | CG Function | John Horton Conway, Richard Kenneth Guy |
| 13 | Bowers' {} | Jonathan Bowers |
| 14 | Chained Arrow | John Horton Conway, Richard Kenneth Guy |
| 15 | Clarkkkkson | - |
| 16 | Graham's Function $G(n)$ | Ronald Graham |
| 17 | Graham's Function $g(n)$ | Ronald Graham |
| 18 | Hyperlicious Function | - |
| 19 | Ackermann's Function | Wilhelm Ackermann |
| 20 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 21 | Hyper Operation | - |
| 22 | Bowers' Operators | Jonathan Bowers |
| 23 | Nambir's Hyper Operation | Nambir |
| 24 | R-R Hyper Operation | Ruzorbov Romolio |
| 25 | Down-Arraw | Donald Ervin Knuth |
| 26 | Munafo's Hyper Operation | Munafo |
| 27 | Mythical tree problem | Harvey Friedman |
| 28 | Sudan function | Sudan |
| 29 | Gödel numbers | Gödel |
| 30 | Grzegorczyk's hierarchy | Grzegorczyk |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 5 | Ordinal diagrams | Gaisi Takeuti |
| 6 | Feferman's $\theta$ | Fefermann |
| 7 | Buchholz's Hydra | Wilfried Buchholz |
| 8 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 9 | Patterns of Resemblance | Carlson |
| 10 | 1st Catching Point of G/F | Girard |
| 11 | Madore's $\psi$ Function | Madore |
| 12 | Bachmann's $\psi$ | Bachmann |
| 13 | Veblen Function | Oswald Veblen |
| 14 | TREE function | Harvey Friedman |
| 15 | otree function | Harvey Friedman |

## G.24 2003

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Bowers' Dimensional Arrays with [ ] | - |
| 2 | marxen.c function | Heiner Marxen |
| 3 | Multi Dimensional Arrays | Jonathan Bowers |
| 4 | Friedman's Sequence | Harvey Friedman |
| 5 | Worm function | - |
| 6 | Goodstein Sequence | Goodstein |
| 7 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 8 | Dimensional Arrays | Jonathan Bowers |
| 9 | Linar Arrays | Jonathan Bowers |

| Ranking | Name | Proposer |
|---|---|---|
| 10 | Extended Bowers'Array Notation | Jonathan Bowers |
| 11 | Bowers'Array Notation | Jonathan Bowers |
| 12 | Pete.C function series | Pete |
| 13 | CG Function | John Horton Conway, Richard Kenneth Guy |
| 14 | Bowers' {} | Jonathan Bowers |
| 15 | Chained Arrow | John Horton Conway, Richard Kenneth Guy |
| 16 | Clarkkkkson | - |
| 17 | Graham's Function $G(n)$ | Ronald Graham |
| 18 | Graham's Function $g(n)$ | Ronald Graham |
| 19 | Hyperlicious Function | - |
| 20 | Ackermann's Function | Wilhelm Ackermann |
| 21 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 22 | Hyper Operation | - |
| 23 | Bowers' Operators | Jonathan Bowers |
| 24 | Nambir's Hyper Operation | Nambir |
| 25 | R-R Hyper Operation | Ruzorbov Romolio |
| 26 | Down-Arraw | Donald Ervin Knuth |
| 27 | Munafo's Hyper Operation | Munafo |
| 28 | Mythical tree problem | Harvey Friedman |
| 29 | Sudan function | Sudan |
| 30 | Gödel numbers | Gödel |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 5 | Ordinal diagrams | Gaisi Takeuti |
| 6 | Feferman's $\theta$ | Fefermann |
| 7 | Buchholz's Hydra | Wilfried Buchholz |
| 8 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 9 | Patterns of Resemblance | Carlson |
| 10 | 1st Catching Point of G/F | Girard |
| 11 | Madore's $\psi$ Function | Madore |
| 12 | Bachmann's $\psi$ | Bachmann |
| 13 | Veblen Function | Oswald Veblen |
| 14 | TREE function | Harvey Friedman |
| 15 | otree function | Harvey Friedman |

## G.25 2004

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | E# with # Hyper Operation | Sbiis Saibian |
| 2 | Bowers' Dimensional Arrays with [ ] | - |
| 3 | marxen.c function | Heiner Marxen |
| 4 | Multi Dimensional Arrays | Jonathan Bowers |
| 5 | E# with # Operation | Sbiis Saibian |
| 6 | Friedman's Sequence | Harvey Friedman |
| 7 | Worm function | - |
| 8 | Goodstein Sequence | Goodstein |
| 9 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |

| Ranking | Name | Proposer |
|---|---|---|
| 10 | Dimensional Arrays | Jonathan Bowers |
| 11 | Linar Arrays | Jonathan Bowers |
| 12 | E# with Multiple # | Sbiis Saibian |
| 13 | Extended Bowers'Array Notation | Jonathan Bowers |
| 14 | Bowers'Array Notation | Jonathan Bowers |
| 15 | Pete.C function series | Pete |
| 16 | CG Function | John Horton Conway, Richard Kenneth Guy |
| 17 | Bowers' {} | Jonathan Bowers |
| 18 | Chained Arrow | John Horton Conway, Richard Kenneth Guy |
| 19 | Clarkkkkson | - |
| 20 | Graham's Function $G(n)$ | Ronald Graham |
| 21 | Graham's Function $g(n)$ | Ronald Graham |
| 22 | Hyperlicious Function | - |
| 23 | Ackermann's Function | Wilhelm Ackermann |
| 24 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 25 | Hyper Operation | - |
| 26 | Bowers' Operators | Jonathan Bowers |
| 27 | Nambir's Hyper Operation | Nambir |
| 28 | R-R Hyper Operation | Ruzorbov Romolio |
| 29 | Down-Arraw | Donald Ervin Knuth |
| 30 | Munafo's Hyper Operation | Munafo |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 5 | Ordinal diagrams | Gaisi Takeuti |
| 6 | Feferman's $\theta$ | Fefermann |
| 7 | Buchholz's Hydra | Wilfried Buchholz |
| 8 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 9 | Patterns of Resemblance | Carlson |
| 10 | 1st Catching Point of G/F | Girard |
| 11 | Madore's $\psi$ Function | Madore |
| 12 | Bachmann's $\psi$ | Bachmann |
| 13 | Veblen Function | Oswald Veblen |
| 14 | TREE function | Harvey Friedman |
| 15 | otree function | Harvey Friedman |

## G.26 2005

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | E# with # Hyper Operation | Sbiis Saibian |
| 2 | Bowers' Dimensional Arrays with [ ] | - |
| 3 | marxen.c function | Heiner Marxen |
| 4 | Multi Dimensional Arrays | Jonathan Bowers |
| 5 | E# with # Operation | Sbiis Saibian |
| 6 | Friedman's Sequence | Harvey Friedman |
| 7 | Worm function | - |
| 8 | Goodstein Sequence | Goodstein |
| 9 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |

| Ranking | Name | Proposer |
|---|---|---|
| 10 | Dimensional Arrays | Jonathan Bowers |
| 11 | Linar Arrays | Jonathan Bowers |
| 12 | E# with Multiple # | Sbiis Saibian |
| 13 | Extended Bowers'Array Notation | Jonathan Bowers |
| 14 | Bowers'Array Notation | Jonathan Bowers |
| 15 | Pete.C function series | Pete |
| 16 | CG Function | John Horton Conway, Richard Kenneth Guy |
| 17 | Bowers' {} | Jonathan Bowers |
| 18 | Chained Arrow | John Horton Conway, Richard Kenneth Guy |
| 19 | Clarkkkkson | - |
| 20 | Graham's Function $G(n)$ | Ronald Graham |
| 21 | Graham's Function $g(n)$ | Ronald Graham |
| 22 | Hyperlicious Function | - |
| 23 | Ackermann's Function | Wilhelm Ackermann |
| 24 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 25 | Hyper Operation | - |
| 26 | Bowers' Operators | Jonathan Bowers |
| 27 | Nambir's Hyper Operation | Nambir |
| 28 | R-R Hyper Operation | Ruzorbov Romolio |
| 29 | Down-Arraw | Donald Ervin Knuth |
| 30 | Munafo's Hyper Operation | Munafo |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 5 | Ordinal diagrams | Gaisi Takeuti |
| 6 | Feferman's $\theta$ | Fefermann |
| 7 | Buchholz's Hydra | Wilfried Buchholz |
| 8 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 9 | Patterns of Resemblance | Carlson |
| 10 | 1st Catching Point of G/F | Girard |
| 11 | Madore's $\psi$ Function | Madore |
| 12 | Bachmann's $\psi$ | Bachmann |
| 13 | Veblen Function | Oswald Veblen |
| 14 | TREE function | Harvey Friedman |
| 15 | otree function | Harvey Friedman |

## G.27 2006

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Bird's Array (V1) | Chris Bird |
| 2 | E# with # Hyper Operation | Sbiis Saibian |
| 3 | Bowers' Dimensional Arrays with [ ] | - |
| 4 | Bird's multiple [ ] | Chris Bird |
| 5 | marxen.c function | Heiner Marxen |
| 6 | Multi Dimensional Arrays | Jonathan Bowers |
| 7 | E# with # Operation | Sbiis Saibian |
| 8 | Bird's [ ] | Chris Bird |
| 9 | Friedman's Sequence | Harvey Friedman |

| Ranking | Name | Proposer |
|---|---|---|
| 10 | Worm function | - |
| 11 | Goodstein Sequence | Goodstein |
| 12 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 13 | Dimensional Arrays | Jonathan Bowers |
| 14 | Linar Arrays | Jonathan Bowers |
| 15 | E# with Multiple # | Sbiis Saibian |
| 16 | Extended Bowers'Array Notation | Jonathan Bowers |
| 17 | Bowers'Array Notation | Jonathan Bowers |
| 18 | Bowers'Array Notation | Jonathan Bowers |
| 19 | Pete.C function series | Pete |
| 20 | CG Function | John Horton Conway, Richard Kenneth Guy |
| 21 | Bowers' {} | Jonathan Bowers |
| 22 | Chained Arrow | John Horton Conway, Richard Kenneth Guy |
| 23 | Clarkkkkson | - |
| 24 | Graham's Function $G(n)$ | Ronald Graham |
| 25 | Graham's Function $g(n)$ | Ronald Graham |
| 26 | Hyperlicious Function | - |
| 27 | Ackermann's Function | Wilhelm Ackermann |
| 28 | Knuth's Up-Arraw | Donald Ervin Knuth |
| 29 | Hyper Operation | - |
| 30 | Bowers' Operators | Jonathan Bowers |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 5 | Ordinal diagrams | Gaisi Takeuti |
| 6 | Feferman's $\theta$ | Fefermann |
| 7 | Buchholz's Hydra | Wilfried Buchholz |
| 8 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 9 | SCG Function | - |
| 10 | SSCG Function | - |
| 11 | Patterns of Resemblance | Carlson |
| 12 | 1st Catching Point of G/F | Girard |
| 13 | Madore's $\psi$ Function | Madore |
| 14 | Bachmann's $\psi$ | Bachmann |
| 15 | Veblen Function | Oswald Veblen |

## G.28 2007

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Bird's Array (V1) | Chris Bird |
| 2 | Bowers' & | Jonathan Bowers |
| 3 | E# with # Hyper Operation | Sbiis Saibian |
| 4 | Bowers' Dimensional Arrays with [ ] | - |
| 5 | Bird's multiple [ ] | Chris Bird |
| 6 | marxen.c function | Heiner Marxen |
| 7 | Multi Dimensional Arrays | Jonathan Bowers |
| 8 | E# with # Operation | Sbiis Saibian |
| 9 | Bird's [ ] | Chris Bird |

| Ranking | Name | Proposer |
|---|---|---|
| 10 | Friedman's Sequence | Harvey Friedman |
| 11 | Worm function | - |
| 12 | Goodstein Sequence | Goodstein |
| 13 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 14 | Dimensional Arrays | Jonathan Bowers |
| 15 | Linar Arrays | Jonathan Bowers |
| 16 | Taro's multivariable Ackermann function | Taro |
| 17 | E# with Multiple # | Sbiis Saibian |
| 18 | Extended Bowers'Array Notation | Jonathan Bowers |
| 19 | Bowers'Array Notation | Jonathan Bowers |
| 20 | Bowers'Array Notation | Jonathan Bowers |
| 21 | Pete.C function series | Pete |
| 22 | CG Function | John Horton Conway, Richard Kenneth Guy |
| 23 | Bowers' {} | Jonathan Bowers |
| 24 | Chained Arrow | John Horton Conway, Richard Kenneth Guy |
| 25 | Clarkkkkson | - |
| 26 | Graham's Function $G(n)$ | Ronald Graham |
| 27 | Graham's Function $g(n)$ | Ronald Graham |
| 28 | Hyperlicious Function | - |
| 29 | Ackermann's Function | Wilhelm Ackermann |
| 30 | Knuth's Up-Arraw | Donald Ervin Knuth |
| # | Rayo 函数被 Rayo 提出 | |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Rathjen's $\Xi$ | Rathjen |
| 3 | Rathjen's $\chi$ | Rathjen |
| 4 | Mahlo OCF | - |
| 5 | $I$ OCF | Gerhard Jäger, Wilfried Buchholz |
| 6 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 7 | Ordinal diagrams | Gaisi Takeuti |
| 8 | Feferman's $\theta$ | Fefermann |
| 9 | Buchholz's Hydra | Wilfried Buchholz |
| 10 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 11 | SCG Function | - |
| 12 | SSCG Function | - |
| 13 | Patterns of Resemblance | Carlson |
| 14 | Wilken's $\theta$ | Gunnar Wilken |
| 15 | 1st Catching Point of G/F | Girard |

## G.29 2008

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Bird's Array (V1) | Chris Bird |
| 2 | Bowers' Exploding Array Function | Jonathan Bowers |
| 3 | Bowers' & | Jonathan Bowers |
| 4 | Hyper-E Notation | Sbiis Saibian |
| 5 | E# with # Hyper Operation | Sbiis Saibian |
| 6 | Bowers' Dimensional Arrays with [ ] | - |

| Ranking | Name | Proposer |
|---|---|---|
| 7 | Bird's multiple [ ] | Chris Bird |
| 8 | marxen.c function | Heiner Marxen |
| 9 | Multi Dimensional Arrays | Jonathan Bowers |
| 10 | E# with # Operation | Sbiis Saibian |
| 11 | Bird's [ ] | Chris Bird |
| 12 | Friedman's Sequence | Harvey Friedman |
| 13 | Worm function | - |
| 14 | Goodstein Sequence | Goodstein |
| 15 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 16 | Dimensional Arrays | Jonathan Bowers |
| 17 | Linar Arrays | Jonathan Bowers |
| 18 | Taro's multivariable Ackermann function | Taro |
| 19 | E# with Multiple # | Sbiis Saibian |
| 20 | Extended Bowers'Array Notation | Jonathan Bowers |
| 21 | Bowers'Array Notation | Jonathan Bowers |
| 22 | Bowers'Array Notation | Jonathan Bowers |
| 23 | Pete.C function series | Pete |
| 24 | CG Function | John Horton Conway, Richard Kenneth Guy |
| 25 | Bowers' {} | Jonathan Bowers |
| 26 | Chained Arrow | John Horton Conway, Richard Kenneth Guy |
| 27 | Clarkkkkson | - |
| 28 | Graham's Function $G(n)$ | Ronald Graham |
| 29 | Graham's Function $g(n)$ | Ronald Graham |
| 30 | Hyperlicious Function | - |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Duchhardt's OCF | Duchhardt |
| 3 | Rathjen's $\Xi$ | Rathjen |
| 4 | Rathjen's $\chi$ | Rathjen |
| 5 | Mahlo OCF | - |
| 6 | $I$ OCF | Gerhard Jäger, Wilfried Buchholz |
| 7 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 8 | Ordinal diagrams | Gaisi Takeuti |
| 9 | Feferman's $\theta$ | Fefermann |
| 10 | Buchholz's Hydra | Wilfried Buchholz |
| 11 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 12 | SCG Function | - |
| 13 | SSCG Function | - |
| 14 | Patterns of Resemblance | Carlson |
| 15 | Wilken's $\theta$ | Gunnar Wilken |

## G.30 2009

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Bird's Array (V1) | Chris Bird |
| 2 | Bowers' Exploding Array Function | Jonathan Bowers |
| 3 | Bowers' & | Jonathan Bowers |
| 4 | Hyper-E Notation | Sbiis Saibian |

| Ranking | Name | Proposer |
|---|---|---|
| 5 | E# with # Hyper Operation | Sbiis Saibian |
| 6 | Bower's Dimensional Arrays with [ ] | - |
| 7 | Bird's multiple [ ] | Chris Bird |
| 8 | marxen.c function | Heiner Marxen |
| 9 | Multi Dimensional Arrays | Jonathan Bowers |
| 10 | E# with # Operation | Sbiis Saibian |
| 11 | Bird's [ ] | Chris Bird |
| 12 | Friedman's Sequence | Harvey Friedman |
| 13 | Worm function | - |
| 14 | Goodstein Sequence | Goodstein |
| 15 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 16 | Dimensional Arrays | Jonathan Bowers |
| 17 | Linar Arrays | Jonathan Bowers |
| 18 | Taro's multivariable Ackermann function | Taro |
| 19 | E# with Multiple # | Sbiis Saibian |
| 20 | Extended Bowers'Array Notation | Jonathan Bowers |
| 21 | Bowers'Array Notation | Jonathan Bowers |
| 22 | Bowers'Array Notation | Jonathan Bowers |
| 23 | Pete.C function series | Pete |
| 24 | CG Function | John Horton Conway, Richard Kenneth Guy |
| 25 | Bowers' {} | Jonathan Bowers |
| 26 | Chained Arrow | John Horton Conway, Richard Kenneth Guy |
| 27 | Clarkkkkson | - |
| 28 | Graham's Function $G(n)$ | Ronald Graham |
| 29 | Graham's Function $g(n)$ | Ronald Graham |
| 30 | Hyperlicious Function | - |
| # | FPCI 被发明<br>同 Carlson 记号被提出 | |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Duchhardt's OCF | Duchhardt |
| 3 | Rathjen's $\Xi$ | Rathjen |
| 4 | Rathjen's $\chi$ | Rathjen |
| 5 | Mahlo OCF | - |
| 6 | $I$ OCF | Gerhard Jäger, Wilfried Buchholz |
| 7 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 8 | Ordinal diagrams | Gaisi Takeuti |
| 9 | Feferman's $\theta$ | Fefermann |
| 10 | Buchholz's Hydra | Wilfried Buchholz |
| 11 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 12 | SCG Function | - |
| 13 | SSCG Function | - |
| 14 | Patterns of Resemblance | Carlson |
| 15 | Wilken's $\theta$ | Gunnar Wilken |

## G.31 2010

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |

| Ranking | Name | Proposer |
|---|---|---|
| 1 | Bird's Array (V1) | Chris Bird |
| 2 | Bowers' Exploding Array Function | Jonathan Bowers |
| 3 | Bowers' & | Jonathan Bowers |
| 4 | Hyper-E Notation | Sbiis Saibian |
| 5 | E# with # Hyper Operation | Sbiis Saibian |
| 6 | Bower's Dimensional Arrays with [ ] | - |
| 7 | Bird's multiple [ ] | Chris Bird |
| 8 | marxen.c function | Heiner Marxen |
| 9 | Multi Dimensional Arrays | Jonathan Bowers |
| 10 | E# with # Operation | Sbiis Saibian |
| 11 | Bird's [ ] | Chris Bird |
| 12 | Friedman's Sequence | Harvey Friedman |
| 13 | Worm function | - |
| 14 | Goodstein Sequence | Goodstein |
| 15 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 16 | Dimensional Arrays | Jonathan Bowers |
| 17 | Linar Arrays | Jonathan Bowers |
| 18 | Taro's multivariable Ackermann function | Taro |
| 19 | E# with Multiple # | Sbiis Saibian |
| 20 | Extended Bowers'Array Notation | Jonathan Bowers |
| 21 | Bowers'Array Notation | Jonathan Bowers |
| 22 | Bowers'Array Notation | Jonathan Bowers |
| 23 | Pete.C function series | Pete |
| 24 | CG Function | John Horton Conway, Richard Kenneth Guy |
| 25 | Bowers' {} | Jonathan Bowers |
| 26 | Chained Arrow | John Horton Conway, Richard Kenneth Guy |
| 27 | Clarkkkkson | - |
| 28 | Graham's Function $G(n)$ | Ronald Graham |
| 29 | Graham's Function $g(n)$ | Ronald Graham |
| 30 | Hyperlicious Function | - |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Duchhardt's OCF | Duchhardt |
| 3 | Rathjen's $\Xi$ | Rathjen |
| 4 | Rathjen's $\chi$ | Rathjen |
| 5 | Mahlo OCF | - |
| 6 | $I$ OCF | Gerhard Jäger, Wilfried Buchholz |
| 7 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 8 | Ordinal diagrams | Gaisi Takeuti |
| 9 | Feferman's $\theta$ | Fefermann |
| 10 | Buchholz's Hydra | Wilfried Buchholz |
| 11 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 12 | SCG Function | - |
| 13 | SSCG Function | - |
| 14 | Patterns of Resemblance | Carlson |
| 15 | Wilken's $\theta$ | Gunnar Wilken |

## G.32 2011

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Bird's Array (V2) | Chris Bird |
| 2 | Bird's $\sim$ | Chris Bird |
| 3 | Bird's Array (V1) | Chris Bird |
| 4 | Bowers' Exploding Array Function | Jonathan Bowers |
| 5 | Bowers' & | Jonathan Bowers |
| 6 | Hyper-E Notation | Sbiis Saibian |
| 7 | E# with # Hyper Operation | Sbiis Saibian |
| 8 | Bower's Dimensional Arrays with [ ] | - |
| 9 | Bird's multiple [ ] | Chris Bird |
| 10 | marxen.c function | Heiner Marxen |
| 11 | Multi Dimensional Arrays | Jonathan Bowers |
| 12 | E# with # Operation | Sbiis Saibian |
| 13 | Bird's [ ] | Chris Bird |
| 14 | Friedman's Sequence | Harvey Friedman |
| 15 | Worm function | - |
| 16 | Goodstein Sequence | Goodstein |
| 17 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 18 | Dimensional Arrays | Jonathan Bowers |
| 19 | Linar Arrays | Jonathan Bowers |
| 20 | Taro's multivariable Ackermann function | Taro |
| 21 | E# with Multiple # | Sbiis Saibian |
| 22 | Bird's Linar Arrays | Chris Bird |
| 23 | C Function Peter | Hurford |
| 24 | Extended Bowers'Array Notation | Jonathan Bowers |
| 25 | Bowers'Array Notation | Jonathan Bowers |
| 26 | Bowers'Array Notation | Jonathan Bowers |
| 27 | Pete.C function series | Pete |
| 28 | CG Function | John Horton Conway, Richard Kenneth Guy |
| 29 | Bowers' {} | Jonathan Bowers |
| 30 | Chained Arrow | John Horton Conway, Richard Kenneth Guy |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Stegart's Large OCF | Jan-Carl Stegart |
| 3 | Stegart's Small OCF | Jan-Carl Stegart |
| 4 | Duchhardt's OCF | Duchhardt |
| 5 | Rathjen's $\Xi$ | Rathjen |
| 6 | Rathjen's $\chi$ | Rathjen |
| 7 | Mahlo OCF | - |
| 8 | $I$ OCF | Gerhard Jäger, Wilfried Buchholz |
| 9 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 10 | Ordinal diagrams | Gaisi Takeuti |
| 11 | Feferman's $\theta$ | Fefermann |
| 12 | Buchholz's Hydra | Wilfried Buchholz |
| 13 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 14 | SCG Function | - |
| 15 | SSCG Function | - |

## G.33 2012

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Bird's Array (V2) | Chris Bird |
| 2 | Bird's $\sim$ | Chris Bird |
| 3 | Bird's Array (V1) | Chris Bird |
| 4 | Bowers' Exploding Array Function | Jonathan Bowers |
| 5 | Bowers' & | Jonathan Bowers |
| 6 | Hyper-E Notation | Sbiis Saibian |
| 7 | E# with # Hyper Operation | Sbiis Saibian |
| 8 | Bower's Dimensional Arrays with [ ] | - |
| 9 | Bird's multiple [ ] | Chris Bird |
| 10 | marxen.c function | Heiner Marxen |
| 11 | Fusible number | - |
| 12 | Multi Dimensional Arrays | Jonathan Bowers |
| 13 | E# with # Operation | Sbiis Saibian |
| 14 | Bird's [ ] | Chris Bird |
| 15 | Friedman's Sequence | Harvey Friedman |
| 16 | Worm function | - |
| 17 | Goodstein Sequence | Goodstein |
| 18 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 19 | Dimensional Arrays | Jonathan Bowers |
| 20 | Linar Arrays | Jonathan Bowers |
| 21 | Taro's multivariable Ackermann function | Taro |
| 22 | E# with Multiple # | Sbiis Saibian |
| 23 | Bird's Linar Arrays | Chris Bird |
| 24 | Linar R Function (Old) | HypCos |
| 25 | C Function Peter | Hurford |
| 26 | Extended Bowers'Array Notation | Jonathan Bowers |
| 27 | Bowers'Array Notation | Jonathan Bowers |
| 28 | Bowers'Array Notation | Jonathan Bowers |
| 29 | Pete.C function series | Pete |
| 30 | CG Function | John Horton Conway, Richard Kenneth Guy |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Stegart's Large OCF | Jan-Carl Stegart |
| 3 | Stegart's Small OCF | Jan-Carl Stegart |
| 4 | Duchhardt's OCF | Duchhardt |
| 5 | Rathjen's $\Xi$ | Rathjen |
| 6 | Rathjen's $\chi$ | Rathjen |
| 7 | Mahlo OCF | - |
| 8 | $I$ OCF | Gerhard Jäger, Wilfried Buchholz |
| 9 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 10 | Ordinal diagrams | Gaisi Takeuti |
| 11 | Feferman's $\theta$ | Fefermann |
| 12 | Buchholz's Hydra | Wilfried Buchholz |
| 13 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 14 | SCG Function | - |
| 15 | SSCG Function | - |

## G.34 2013

| Ranking | Name | Proposer |
|---|---|---|
| | Large number | |
| 1 | Dollar Function | Wythagoras |
| 2 | Linar R Function | HypCos |
| 3 | Hyper-factoral Array Notation | Lawerence Hollom |
| 4 | Bowers' Exploding Array Function | Jonathan Bowers |
| 5 | Bird's $U(n)$ | function Chris Bird |
| 6 | Bowers' & | Jonathan Bowers |
| 7 | Bird's Array ($V2$) | Chris Bird |
| 8 | Bird's $\sim$ | Chris Bird |
| 9 | Bird's $S(n)$ function | Chris Bird |
| 10 | Bird's Array (V1) | Chris Bird |
| 11 | Bird's $H(n)$ function | Chris Bird |
| 12 | Extended Hyper-E | - |
| 13 | Hyper-E Notation | Sbiis Saibian |
| 14 | E# with # Hyper Operation | Sbiis Saibian |
| 15 | Bower's Dimensional | Arrays with [ ] - |
| 16 | Bird's multiple [ ] | Chris Bird |
| 17 | marxen.c function | Heiner Marxen |
| 18 | Fusible number | - |
| 19 | Multi Dimensional Arrays | Jonathan Bowers |
| 20 | E# with # | Operation Sbiis Saibian |
| 21 | Bird's [ ] | Chris Bird |
| 22 | Friedman's Sequence | Harvey Friedman |
| 23 | Worm function | - |
| 24 | Goodstein Sequence | Goodstein |
| 25 | Kirby-Paris Hydra | Laurie Kirby, Jeff Paris |
| 26 | Cascading-E Notation | Sbiis Saibian |
| 27 | Dimensional Arrays | Jonathan Bowers |
| 28 | Linar Arrays | Jonathan Bowers |
| 29 | Taro's multivariable Ackermann function | Taro |
| 30 | E# with multiple # | Sbiis Saibian |
| | Ordinal | |
| 1 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 2 | Stegart's Large OCF | Jan-Carl Stegart |
| 3 | Stegart's Small OCF | Jan-Carl Stegart |
| 4 | Duchhardt's OCF | Duchhardt |
| 5 | Rathjen's $\Xi$ | Rathjen |
| 6 | $K$ OCF | - |
| 7 | Rathjen's $\chi$ | Rathjen |
| 8 | Mahlo OCF | - |
| 9 | $I$ OCF | Gerhard Jäger, Wilfried Buchholz |
| 10 | Buchholz's $\Phi$ | Wilfried Buchholz |
| 11 | Ordinal diagrams | Gaisi Takeuti |
| 12 | Feferman's $\theta$ | Fefermann |
| 13 | Buchholz's Hydra | Wilfried Buchholz |
| 14 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 15 | SCG Function | - |

## G.35 2014

| Ranking | Name | Proposer |
|---|---|---|
| 1 | $[\,]-$Stb | HypCos |
| 2 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 3 | Catching Function | HypCos |
| 4 | $R$ Function | HypCos |
| 5 | Bowers' Exploding Array Function | Jonathan Bowers |
| 6 | Stegart's Large OCF | Jan-Carl Stegart |
| 7 | Stegart's Small OCF | Jan-Carl Stegart |
| 8 | Duchhardt's OCF | Duchhardt |
| 9 | Dollar Function | Wythagoras |
| 10 | Rathjen's $\Xi$ | Rathjen |
| 11 | $K$ OCF | - |
| 12 | Rathjen's $\chi$ | Rathjen |
| 13 | Mahlo OCF | - |
| 14 | Linar $R$ Function | HypCos |
| 15 | Hyper-factoral Array Notation | Lawerence Hollom |
| 16 | $I$ OCF | Gerhard Jäger<br>Wilfried Buchholz |
| 17 | Buchhoz's $\Phi$ | Wilfried Buchholz |
| 18 | Bird's Array | Chris Bird |
| 19 | Ordinal diagrams | Gaisi Takeuti |
| 20 | Feferman's $\theta$ | Fefermann |
| 21 | Buchholz's Hydra | Wilfried Buchholz |
| 22 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 23 | SCG Function | - |
| 24 | SSCG Function | - |
| 25 | Bowers' & | Jonathan Bowers |
| 26 | Taranovsky's Ordinal Notation (Old) | Taranovsky |
| 27 | Patterns of Resemblance | Carlson |
| 28 | Wilken's $\theta$ | Gunnar Wilken |
| 29 | Extended Hyper-E | - |
| 30 | 1st Catching Point of G/F | Girard |
| 31 | Bird's $U(n)$ function | Chris Bird |
| 32 | Bird's Array (V2) | Chris Bird |
| 33 | Bird's $\sim$ | Chris Bird |
| 34 | Madore's $\psi$ Function | Madore |
| 35 | Bachmann's $\psi$ | Bachmann |
| 36 | Bird's $S(n)$ function | Chris Bird |
| 37 | Bird's Array (V1) | Chris Bird |
| 38 | Bird's $H(n)$ function | Chris Bird |
| 39 | Veblen Function | Oswald Veblen |
| 40 | TREE function | Harvey Friedman |

## G.36 2015

| Ranking | Name | Proposer |
|---|---|---|
| 1 | Taranovsky's Ordinal Notation | Taranovsky |
| 2 | Trio Sequence System | Bashicu |
| 3 | $[\,]-$Stb | HypCos |
| 4 | Catching Function | HypCos |
| 5 | Strong Array Notation | HypCos |
| 6 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 7 | $R$ Function | HypCos |
| 8 | Bowers' Exploding Array Function | Jonathan Bowers |
| 9 | Secondry Dropping Array Notation | HypCos |

| Ranking | Name | Proposer |
|---|---|---|
| 10 | Degrees of Reflection | Taranovsky |
| 11 | Stegart's Large OCF | Jan-Carl Stegart |
| 12 | Pirmary Dropping Array Notation | HypCos |
| 13 | Stegart's Small OCF | Jan-Carl Stegart |
| 14 | Duchhardt's OCF | Duchhardt |
| 15 | Dollar Function | Wythagoras |
| 16 | Rathjen's $\Xi$ | Rathjen |
| 17 | $K$ OCF | - |
| 18 | Rathjen's $\chi$ | Rathjen |
| 19 | Mahlo OCF | - |
| 20 | Linar $R$ Function | HypCos |
| 21 | Hyper-factoral Array Notation | Lawerence Hollom |
| 22 | $I$ OCF | Gerhard Jäger<br>Wilfried Buchholz |
| 23 | Buchhoz's $\Phi$ | Wilfried Buchholz |
| 24 | Bird's $H(n)$ function | Chris Bird |
| 25 | Bird's Array | Chris Bird |
| 26 | Ordinal diagrams | Gaisi Takeuti |
| 27 | Feferman's $\theta$ | Fefermann |
| 28 | Buchholz's Hydra | Wilfried Buchholz |
| 29 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 30 | SCG Function | - |
| 31 | SSCG Function | - |
| 32 | Bowers' & | Jonathan Bowers |
| 33 | Taranovsky's Ordinal Notation (Old) | Taranovsky |
| 34 | Patterns of Resemblance | Carlson |
| 35 | Wilken's $\theta$ | Gunnar Wilken |
| 36 | Pair Sequence System | Bashicu |
| 37 | Extended Hyper-E | - |
| 38 | Mulptily Expanding Array Notation | HypCos |
| 39 | 1st Catching Point of G/F | Girard |
| 40 | Bird's $U(n)$ function | Chris Bird |

## G.37 2016

| Ranking | Name | Proposer |
|---|---|---|
| 1 | Taranovsky's Ordinal Notation | Taranovsky |
| 2 | Trio Sequence System | Bashicu |
| 3 | $[\,]-$Stb | HypCos |
| 4 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 5 | Iteration of n-built from below | Taranovsky |
| 6 | Built-from-below | Taranovsky |
| 7 | Degrees of Reflection | Taranovsky |
| 8 | Reflection configuration | Taranovsky |
| 9 | Catching Function | HypCos |
| 10 | Strong Array Notation | HypCos |
| 11 | Dropping hydra | HypCos |
| 12 | $R$ Function | HypCos |
| 13 | 3-Dropping hydra | HypCos |
| 14 | Bowers' Exploding Array Function | Jonathan Bowers |
| 15 | Secondry Dropping Array Notation | HypCos |
| 16 | Stegart's Large OCF | HypCos |
| 17 | Pirmary Dropping Array Notation | Jan-Carl Stegart |
| 18 | Stegart's Small OCF | HypCos |

| Ranking | Name | Proposer |
|---|---|---|
| 19 | 2-Dropping hydra | Jan-Carl Stegart |
| 20 | Duchhardt's OCF | Wythagoras |
| 21 | Dollar Function | Duchhardt |
| 22 | Rathjen's $\Xi$ | Rathjen |
| 23 | $K$ OCF | - |
| 24 | Rathjen's $\chi$ | Rathjen |
| 25 | Mahlo OCF | - |
| 26 | Linar $R$ Function | HypCos |
| 27 | Hyper-factoral Array Notation | Lawerence Hollom |
| 28 | $I$ OCF | Gerhard Jäger<br>Wilfried Buchholz |
| 29 | Buchhoz's $\Phi$ | Wilfried Buchholz |
| 30 | Extended Buchholz's $\psi$ Function | Denis Maksudov |
| 31 | Bird's $H(n)$ function | Chris Bird |
| 32 | Bird's Array | Chris Bird |
| 33 | Ordinal diagrams | Fefermann |
| 34 | Feferman's $\theta$ | Gaisi Takeuti |
| 35 | Buchholz's Hydra | Wilfried Buchholz |
| 36 | Buchholz's $\psi$ Function | Wilfried Buchholz |
| 37 | SCG Function | - |
| 38 | SSCG Function | - |
| 39 | Bowers' & | Jonathan Bowers |
| 40 | Taranovsky's Ordinal Notation (Old) | Taranovsky |

## G.38 2017

| Ranking | Name | Proposer |
|---|---|---|
| 1 | Taranovsky's Ordinal Notation | Taranovsky |
| 2 | idealized Bashicu Martix System | Bashicu |
| 3 | Trio Sequence System | Bashicu |
| 4 | primary Dropper Dropping Notation | HypCos |
| 5 | [ ]−Stb | HypCos |
| 6 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 7 | WDmEN | HypCos |
| 8 | Iteration of n-built from below | Taranovsky |
| 9 | Built-from-below | Taranovsky |
| 10 | Degrees of Reflection | Taranovsky |
| 11 | Reflection configuration | Taranovsky |
| 12 | Catching Function | HypCos |
| 13 | Multiple Weak Declinator Expanding | HypCos |
| 14 | Weak Declinator Expanding | HypCos |
| 15 | Nested Dropping Array Notation | HypCos |
| 16 | Strong Array Notation | HypCos |
| 17 | Dropping hydra | HypCos |
| 18 | $R$ Function | HypCos |
| 19 | 3-Dropping hydra | HypCos |
| 20 | Bowers' Exploding Array Function | Jonathan Bowers |
| 21 | Secondry Dropping Array Notation | HypCos |
| 22 | Stegart's Large OCF | Jan-Carl Stegart |
| 23 | Pirmary Dropping Array Notation | HypCos |
| 24 | Stegart's Small OCF | Jan-Carl Stegart |
| 25 | 2-Dropping hydra | HypCos |
| 26 | Duchhardt's OCF | Duchhardt |
| 27 | Dollar Function | Wythagoras |

| Ranking | Name | Proposer |
|---|---|---|
| 28 | Rathjen's $\Xi$ | Rathjen |
| 29 | $K$ OCF | - |
| 30 | Rathjen's $\chi$ | Rathjen |
| 31 | Mahlo OCF | - |
| 32 | Linar $R$ Function | HypCos |
| 33 | Hyper-factoral Array Notation | Lawerence Hollom |
| 34 | $I$ OCF | Gerhard Jäger<br>Wilfried Buchholz |
| 35 | Buchhoz's $\Phi$ | Wilfried Buchholz |
| 36 | Sudden Sequence System | Bashicu |
| 37 | Extended Buchholz's $\psi$ Function | Denis Maksudov |
| 38 | Bird's $H(n)$ function | Chris Bird |
| 39 | Bird's Array | Chris Bird |
| 40 | Ordinal diagrams | Gaisi Takeuti |
| # | Little Bigeddon<br>被 Emlightened 提出 | |

## G.39 2018

| Ranking | Name | Proposer |
|---|---|---|
| 1 | Bashicu Matrix System | Bashicu |
| 2 | BMOCF | P 進大好き bot |
| 3 | idealized Bashicu Martix System | Bashicu |
| 4 | Quadro Sequence System | Bashicu |
| 5 | Trio Sequence System | Bashicu |
| 6 | primary Dropper Dropping Notation | HypCos |
| 7 | [ ]−Stb | HypCos |
| 8 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 9 | Sudden Hydra | Bashicu |
| 10 | WDmEN | HypCos |
| 11 | Iteration of n-built from below | Taranovsky |
| 12 | Built-from-below | Taranovsky |
| 13 | Degrees of Reflection | Taranovsky |
| 14 | Reflection configuration | Taranovsky |
| 15 | Catching Function | HypCos |
| 16 | Multiple Weak Declinator Expanding | HypCos |
| 17 | Weak Declinator Expanding | HypCos |
| 18 | Nested Dropping Array Notation | HypCos |
| 19 | Username5243's OCF | Username5243 |
| 20 | Taranovsky's Ordinal Notation | Taranovsky |
| 21 | Strong Array Notation | HypCos |
| 22 | Dropping hydra | HypCos |
| 23 | R Function | HypCos |
| 24 | NICE Function | Naroyuko |
| 25 | 3-Dropping hydra | HypCos |
| 26 | Bowers' Exploding Array Function | Jonathan Bowers |
| 27 | Secondry Dropping Array Notation | HypCos |
| 28 | Stegart's Large OCF | Jan-Carl Stegart |
| 29 | Pirmary Dropping Array Notation | HypCos |
| 30 | Stegart's Small OCF | Jan-Carl Stegart |
| 31 | 2-Dropping hydra | HypCos |
| 32 | $\pi$ Notation | Username5243 |
| 33 | Duchhardt's OCF | Duchhardt |
| 34 | Dollar Function | Wythagoras |

| Ranking | Name | Proposer |
|---|---|---|
| 35 | Weak Username5243's OCF | Username5243 |
| 36 | Rathjen's $\Xi$ | Rathjen |
| 37 | $K$ OCF | - |
| 38 | Rathjen's $\chi$ | Rathjen |
| 39 | Mahlo OCF | - |
| 40 | Linar $R$ Function | HypCos |
| # | Sasquatch<br>被 Emlightened 提出 | |

## G.40 2019

| Ranking | Name | Proposer |
|---|---|---|
| 1 | Bubby3 TBMS Extended | Bubby3 |
| 2 | Bubby3 TBMS Normal | Bubby3 |
| 3 | Bashicu Sudden Matrix | Bashicu |
| 4 | Bashicu Hyper Matrix | Bashicu |
| 5 | Bashicu Matrix System | Bashicu |
| 6 | Bashicu Matrix V4.1 | Bashicu |
| 7 | BMOCF | P 進大好き bot |
| 8 | idealized Bashicu Martix System | Bashicu |
| 9 | Quadro Sequence System | Bashicu |
| 10 | Trio Sequence System | Bashicu |
| 11 | primary Dropper Dropping Notation | HypCos |
| 12 | [ ]−Stb | HypCos |
| 13 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 14 | Sudden Hydra | Bashicu |
| 15 | WDmEN | HypCos |
| 16 | Iteration of n-built from below | Taranovsky |
| 17 | Built-from-below | Taranovsky |
| 18 | Degrees of Reflection | Taranovsky |
| 19 | Reflection configuration | Taranovsky |
| 20 | Multiple Weak Declinator Expanding | HypCos |
| 21 | Weak Declinator Expanding | HypCos |
| 22 | Nested Dropping Array Notation | HypCos |
| 23 | Username5243's OCF | Username5243 |
| 24 | Taranovsky's Ordinal Notation | Taranovsky |
| 25 | Strong Array Notation | HypCos |
| 26 | Dropping hydra | HypCos |
| 27 | R Function | HypCos |
| 28 | NICE Function | Naroyuko |
| 29 | 3-Dropping hydra | HypCos |
| 30 | Secondry Dropping Array Notation | HypCos |
| 31 | Catching Function | HypCos |
| 32 | Stegart's Large OCF | Jan-Carl Stegart |
| 33 | Pirmary Dropping Array Notation | HypCos |
| 34 | Stegart's Small OCF | Jan-Carl Stegart |
| 35 | 2-Dropping hydra | HypCos |
| 36 | Order Level Array Notation V3 | ych |
| 37 | $\pi$ Notation | Username5243 |
| 38 | Duchhardt's OCF | Duchhardt |
| 39 | Dollar Function | Wythagoras |
| 40 | Weak Username5243's OCF | Username5243 |

## G.41 2020

| Ranking | Name | Proposer |
|---|---|---|
| 1 | Y Sequence | Yukito |
| 2 | Bubby3 TBMS Extended | Bubby3 |
| 3 | Bubby3 TBMS Normal | Bubby3 |
| 4 | Hassium's TBMS | Hassium |
| 5 | Weak Splatium TBMS | Bubby3 |
| 6 | Bashicu Sudden Matrix | Bashicu |
| 7 | Bashicu Hyper Matrix | Bashicu |
| 8 | Bashicu Matrix System | Bashicu |
| 9 | Bashicu Matrix V4.1 | Bashicu |
| 10 | 0−Y Sequence | Yukito |
| 11 | BMOCF | P 進大好き bot |
| 12 | idealized Bashicu Martix System | Bashicu |
| 13 | Quadro Sequence System | Bashicu |
| 14 | Small Hyper Projection Notation | test_alpha0 |
| 15 | Trio Sequence System | Bashicu |
| 16 | Simple Projection | test_alpha0 |
| 17 | ex-UNOCF | P 進大好き bot |
| 18 | 3−Projection | test_alpha0 |
| 19 | primary Dropper Dropping Notation | HypCos |
| 20 | [ ]−Stb | HypCos |
| 21 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 22 | Sudden Hydra | Bashicu |
| 23 | WDmEN | HypCos |
| 24 | Iteration of n-built from below | Taranovsky |
| 25 | Built-from-below | Taranovsky |
| 26 | Degrees of Reflection | Taranovsky |
| 27 | Reflection configuration | Taranovsky |
| 28 | 2−Projection | test_alpha0 |
| 29 | Multiple Weak Declinator Expanding | HypCos |
| 30 | Weak Declinator Expanding | HypCos |
| 31 | Nested Dropping Array Notation | HypCos |
| 32 | Username5243's OCF | Username5243 |
| 33 | Taranovsky's Ordinal Notation | Taranovsky |
| 34 | Strong Array Notation | HypCos |
| 35 | Dropping hydra | HypCos |
| 36 | $R$ Function | HypCos |
| 37 | Order Level Array Notation V4 | ych |
| 38 | NICE Function | Naroyuko |
| 39 | 3−Dropping hydra | HypCos |
| 40 | Secondry Dropping Array Notation | HypCos |

## G.42 2021

| Ranking | Name | Proposer |
|---|---|---|
| 1 | Crater BMS | Bubby3, Aarex |
| 2 | $\omega$−Y Sequence | Yukito, naruyoko |
| 3 | 2−Y Sequence | Yukito, naruyoko |
| 4 | Y Sequence | Yukito |
| 5 | Dimensional Bashicu Martix System | - |
| 6 | $\omega$+1 row Y | - |
| 7 | Bubby3 TBMS Extended | Bubby3 |

| Ranking | Name | Proposer |
|---|---|---|
| 8 | Bubby3 TBMS Normal | Bubby3 |
| 9 | Aarex's strong ex-UNOCF with $ | Aarex |
| 10 | Hassium's TBMS | Hassium |
| 11 | Weak Splatium TBMS | Bubby3 |
| 12 | Uncountable TBMS | - |
| 13 | Apotheosis Ordinal Notation | - |
| 14 | Bashicu Sudden Matrix | Bashicu |
| 15 | Bashicu Hyper Matrix | Bashicu |
| 16 | Bashicu Matrix System | Bashicu |
| 17 | Bashicu Matrix V4.1 | Bashicu |
| 18 | 0−Y Sequence | Yukito |
| 19 | BMOCF | P 進大好き bot |
| 20 | idealized Bashicu Martix System | Bashicu |
| 21 | $\Sigma_2$ Stb System | Yukito |
| 22 | Quadro Sequence System | Bashicu |
| 23 | Small Hyper Projection Notation | test_alpha0 |
| 24 | Non-recursive TON | Taranovsky |
| 25 | Weak MCS projection | test_alpha0 |
| 26 | Trio Sequence System | Bashicu |
| 27 | HPrSS $\psi$ | - |
| 28 | Simple Projection | test_alpha0 |
| 29 | ex-UNOCF | P 進大好き bot |
| 30 | 3−Projection | test_alpha0 |
| 31 | primary Dropper Dropping Notation | HypCos |
| 32 | Lifting $K$−Notation | test_alpha0 |
| 33 | [ ]−Stb | HypCos |
| 34 | Rathjen's Ordinal Collapsing Function | Rathjen |
| 35 | Sudden Hydra | Bashicu |
| 36 | WDmEN | HypCos |
| 37 | Lifting Ω Notation | test_alpha0 |
| 38 | Iteration of n-built from below | Taranovsky |
| 39 | Built-from-below | Taranovsky |
| 40 | Degrees of Reflection | Taranovsky |

## G.43 2022

| Ranking | Name | Proposer |
|---|---|---|
| 1 | fake fake fake Z function | yahtzee |
| 2 | Strong Ω−Y | - |
| 3 | Crater BMS | Bubby3, Aarex |
| 4 | Yto's Y−Y | - |
| 5 | Ω−Y Sequence | CIF, HypCos |
| 6 | $\omega$−Y Sequence | Yukito, naruyoko |
| 7 | 2−Y Sequence | Yukito, naruyoko |
| 8 | Patterns of Resemblance | Carlson |
| 9 | Y Sequence | Yukito |
| 10 | VZ-Sequense | - |
| 11 | Dimensional Bashicu Martix System | - |
| 12 | $\omega$+1 row Y | - |
| 13 | Bubby3 TBMS Extended | Bubby3 |
| 14 | Bubby3 TBMS Normal | Bubby3 |
| 15 | Aarex's TBMS | Aarex |
| 16 | Aarex's strong ex-UNOCF with $ | Aarex |
| 17 | Aarex's Redirection | Aarex |

| Ranking | Name | Proposer |
|---|---|---|
| 18 | Hassium's TBMS | Hassium |
| 19 | Weak Splatium TBMS | Bubby3 |
| 20 | Uncountable TBMS | - |
| 21 | Crazy-Hydra Notation | Gomen |
| 22 | Apotheosis Ordinal Notation | - |
| 23 | Bashicu Sudden Matrix | Bashicu |
| 24 | Bashicu Hyper Matrix | Bashicu |
| 25 | Bashicu Matrix System | Bashicu |
| 26 | Bashicu Matrix V4.1 | Bashicu |
| 27 | 0−Y Sequence | Yukito |
| 28 | BMOCF | P 進大好き bot |
| 29 | idealized Bashicu Martix System | Bashicu |
| 30 | $\Sigma_2$ Stb System | Yukito |
| 31 | Quadro Sequence System | Bashicu |
| 32 | strong DLON | Aarex |
| 33 | Small Hyper Projection Notation | test_alpha0 |
| 34 | Strong ex-UNOCF | - |
| 35 | Non-recursive TON | Taranovsky |
| 36 | Weak MCS projection | test_alpha0 |
| 37 | Trio Sequence System | Bashicu |
| 38 | HPrSS $\psi$ | - |
| 39 | Simple Projection | test_alpha0 |
| 40 | ex-UNOCF | P 進大好き bot |

## G.44 2023

| Ranking | Name | Proposer |
|---|---|---|
| 1 | Mutant Martix System | Aarex |
| 2 | Nested Crater Y | Bubby3 |
| 3 | Crater Y | Bubby3, Aarex |
| 4 | Strong Ω−Y | - |
| 5 | Crater BMS | Bubby3, Aarex |
| 6 | Yto's Y−Y | - |
| 7 | Ω−Y Sequence | CIF, HypCos |
| 8 | X−Y Sequence | Gomen |
| 9 | $\omega$−Y Sequence | Yukito, naruyoko |
| 10 | Dementional $n$−Y | series Gomen |
| 11 | 2−Y Sequence | Yukito, naruyoko |
| 12 | Strong Y Sequence | - |
| 13 | Y Sequence | Yukito |
| 14 | VZ-Sequense | - |
| 15 | Dimensional Bashicu Martix System | - |
| 16 | Simpleness Admissble Mark | 夏夜星空 |
| 17 | fake fake fake Z function | yahtzee |
| 18 | $\omega$+1 row Y | - |
| 19 | Bubby3 TBMS Extended | Bubby3 |
| 20 | Strong Splatium TBMS | - |
| 21 | Bubby3 TBMS Normal | Bubby3 |
| 22 | Aarex's TBMS | Aarex |
| 23 | Aarex's strong ex-UNOCF with $ | Aarex |
| 24 | Aarex's Redirection | Aarex |
| 25 | Hassium's TBMS | Hassium |
| 26 | Weak Splatium TBMS | Bubby3 |
| 27 | Strong ex-UNOCF Redirection+rows | - |

| Ranking | Name | Proposer |
|---|---|---|
| 28 | Strong ex-UNOCF Defection+1-plus rows | - |
| 29 | Uncountable TBMS | - |
| 30 | Crazy-Hydra Notation | Gomen |
| 31 | Arai's OCF | Toshiyasu Arai |
| 32 | Apotheosis Ordinal Notation | - |
| 33 | Patterns of Resemblance | Carlson |
| 34 | Bashicu Sudden Matrix | Bashicu |
| 35 | Bashicu Hyper Matrix | Bashicu |
| 36 | Crane Matrix System | test_alpha0 |
| 37 | Bashicu Matrix System | Bashicu |
| 38 | Bashicu Matrix V4.1 | Bashicu |
| 39 | 0−Y Sequence | Yukito |
| 40 | Hierarchial Increase Unit Notation | 318‘4 |

## G.45 first half of 2024

| Ranking | Name | Proposer |
|---|---|---|
| 1 | Fake Fake Fake Z rules | Asheep233 & 夏夜星空 |
| 2 | Remaining Y System | - |
| 3 | $a$−Y Description | - |
| 4 | Basic Ordinal Sequence | qwerty |
| 5 | $b$−FOS 318‘4 | |
| 6 | X−Y Sequence | Gomen |
| 7 | Nested Crater Y | Bubby3 |
| 8 | Crater Y | Bubby3 & Aarex |
| 9 | Mutant Martix System | Aarex |
| 10 | Strong Ω−Y | - |
| 11 | Crater BMS | Bubby3 & Aarex |
| 12 | Yto's Y−Y | - |
| 13 | CIF's Ω−Y Sequence | CIF & Hyp_cos |
| 14 | Transfinite DBMS | - |
| 15 | $\omega$−Y Sequence | Yukito & naruyoko |
| 16 | Dementional $n$−Y series | Gomen |
| 17 | abc Notation | - |
| 18 | 2−Y Sequence | Yukito & naruyoko |
| 19 | Strong Y Sequence | - |
| 20 | Y Sequence | Yukito |
| 21 | Apotheosis Ordinal Notation | - |
| 22 | VZ-Sequense | - |
| 23 | Proportional Difference Martix | 318‘4 |
| 24 | fake fake fake Z function | yahtzee |
| 25 | $\omega + 1$ row Y | - |
| 26 | Bracket Sequence System | 🅕娘 |
| 27 | Bubby3 TBMS Extended | Bubby3 |
| 28 | Strong Splatium TBMS | - |
| 29 | Bubby3 TBMS Normal | Bubby3 |
| 30 | Aarex's TBMS | Aarex |
| 31 | Aarex's strong ex-UNOCF with $ | Aarex |
| 32 | Aarex's Redirection | Aarex |
| 33 | Hassium's TBMS | Hassium |
| 34 | Weak Splatium TBMS | Bubby3 |
| 35 | $\alpha$−Ordinaal Notation | Bugit |
| 36 | Strong ex-UNOCF Redirection+rows | - |
| 37 | Strong ex-UNOCF Defection+1-plus rows | - |

| Ranking | Name | Proposer |
|---|---|---|
| 38 | Uncountable TBMS | - |
| 39 | KPrSS | 摆烂的小猫 |
| 40 | Crazy-Hydra Notation | Gomen |
| # | Simpleness Admissble Mark | 夏夜星空 |

## G.46 second half of 2024

| Ranking | Name | Proposer |
|---|---|---|
| 1 | FOS 911 | 318‘4 |
| 2 | Fake Fake Fake Z rules | Asheep233 & 夏夜星空 |
| 3 | Fake Fake Fake Z actions | Yathzee & Aarex |
| 4 | X-P | 最菜萌新 |
| 5 | $a-$Y Description | - |
| 6 | $\omega \cdot 2$ Mountain Notation | Hyp_cos |
| 7 | Mutant Martix System | Aarex & Hyp_cos |
| 8 | X$-$Y Sequence | Gomen |
| 9 | Nested Crater Y | Bubby3 |
| 10 | High Elevate System | 夏夜星空 |
| 11 | Crater Y | Bubby3 & Aarex |
| 12 | Differential Matrix System | 夏夜星空 & Asheep233 |
| 13 | SFSS | waffle3z |
| 14 | Strong $\Omega-$Y | - |
| 15 | Experimental Remaning Matrix | Asheep233 |
| 16 | Crater BMS | Bubby3 & Aarex |
| 17 | Yto's Y$-$Y | - |
| 18 | Transfinite $\omega$ Mountain Notation | Hyp_cos |
| 19 | CIF's $\Omega-$Y Sequence | CIF & Hyp_cos |
| 20 | Transfinite DBMS | - |
| 21 | $\omega-$Y Sequence | Yukito & naruyoko |
| 22 | $\omega$ Mountain Notation | Hyp_cos |
| 23 | $\omega \sim$Y Sequence (Dimensional) | Gomen |
| 24 | $\omega-$Dimension Multi Layer BMS Relation | Asheep233 |
| 25 | abc Notation | - |
| 26 | 2$-$Y Sequence | Yukito & naruyoko |
| 27 | Strong Y Sequence | - |
| 28 | Remaining Y System | - |
| 29 | Y Sequence | Yukito |
| 30 | Apotheosis Ordinal Notation | - |
| 31 | VZ-Sequense | - |
| 32 | Proportional Difference Martix | 318‘4 |
| 33 | $\omega + 1$ row Y | - |
| 34 | Basic Laver Pattern | Test_alpha0 |
| 35 | Bracket Sequence System | $\boxed{\text{F}}$娘 |
| 36 | Bubby3 TBMS Extended | Bubby3 |
| 37 | Strong Splatium TBMS | - |
| 38 | Bubby3 TBMS Normal | Bubby3 |
| 39 | Aarex's TBMS | Aarex |
| 40 | Aarex's strong ex-UNOCF with $ | Aarex |

## G.47 2025

| Ranking | Name | Proposer |
|---|---|---|
| 1 | Fake Fake Fake Z rules | Asheep233 & xyxk |
| 2 | Fake Fake Fake Z actions | Yathzee & Aarex |
| 3 | Fundamental Ordinal Sequence (911) | 318‘4 |
| 4 | $\alpha$-Y Description | - |
| 5 | $\Omega$ row DBMS | qwerty |
| 6 | X $\Omega$-Y Sequence | 周海志 |
| 7 | Mutant Martix System | Aarex & Hyp_cos |
| 8 | Transfinite $\omega^\omega$ Mountain Notation | Hyp_cos |
| 9 | Defective $\omega^\omega$ Mountain Notation | Hyp_cos |
| 10 | Defective $\omega^2$ Mountain Notation | Hyp_cos |
| 11 | Defective $\omega 2$ Mountain Notation | Hyp_cos |
| 12 | Astral $\omega 2$ Mountain Notation | Hyp_cos |
| 13 | X-Y Sequence | Gomen |
| 14 | Nested Crater Y | Bubby3 |
| 15 | High Elevate System | xyxk |
| 16 | Crater Y | Bubby3 & Aarex |
| 17 | Differential Matrix System | xyxk & Asheep233 |
| 18 | Strong $\Omega$-Y | - |
| 19 | Experimental Remaning Matrix | Asheep233 |
| 20 | Crater BMS | Bubby3 & Aarex |
| 21 | Yto's Y-Y | Yto |
| 22 | Transfinite $\omega$ Mountain Notation | Hyp_cos |
| 23 | Transfinite DBMS | - |
| 24 | $\omega$-Y Sequence | Yukito & naruyoko |
| 25 | $\omega$ Mountain Notation | Hyp_cos |
| 26 | $\omega \sim$Y Sequence (Dimensional) | Gomen |
| 27 | $\omega$-Dimension Multi Layer BMS Relation | Asheep233 |
| 28 | abc Notation | - |
| 29 | 2-Y Sequence | Yukito & naruyoko |
| 30 | Remaining Y System | - |
| 31 | Strong Y Sequence | - |
| 32 | Y Sequence | Yukito |
| 33 | Idealized Y Sequence | - |
| 34 | Apotheosis Ordinal Notation | - |
| 35 | Vulcaniz Sequense | Vulcaniz |
| 36 | Proportional Difference Martix | 318‘4 |
| 37 | $\omega + 1$ row Y | - |
| 38 | Defective Embeeding Notation | Hyp_cos |
| 39 | Basic Laver Pattern | Renrui Qi |
| 40 | Bracket Sequence System | waffle3z |

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